Rockchip
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Chapter 1 Introduction

1.1 Overview

PX5 is a low power, high performance processor for mobile phones, personal mobile internet device and other digital multimedia applications, and integrates octa-core Cortex-A53 with separately NEON coprocessor.
Many embedded powerful hardware engines provide optimized performance for high-end application. PX5 supports almost full-format H.264 decoder by 4Kx2K@30fps, H.265 decoder by 4Kx2K@60fps, also support H.264/MVC/VP8 encoder by 1080p@30fps, high-quality JPEG encoder/decoder, and special image preprocessor and postprocessor.
Embedded 3D GPU makes PX5 completely compatible with OpenGL ES3.1, OpenCLES3 and DirectX 9.3. Special 2D hardware engine with MMU will maximize display performance and provide very smoothly operation.
PX5 has high-performance external memory interface (DDR3/DDR3L /LPDDR2/LPDDR3) capable of sustaining demanding memory bandwidths, also provides a complete set of peripheral interface to support very flexible applications.

1.2 Features

The features listed below which may or may not be present in actual product, may be subject to the third party licensing requirements. Please contact Rockchip for actual product feature configurations and licensing requirements.

1.2.1 MicroProcessor

- Octa-core ARM Cortex-A53 MPCore processor, a high-performance, low-power and cached application processor
- Two CPU clusters, with four CPU core for each cluster, One cluster is optimized for high-performance(big cluster) and the other is optimized for low power(little cluster)
- Full implementation of the ARM architecture v8-A instruction set, ARM Neon Advanced SIMD (single instruction, multiple data) support for accelerated media and signal processing computation
- ARMv8 Cryptography Extensions
- In-order pipeline with symmetric dual-issue of most instructions.
- Harvard Level 1 (L1) memory system with a Memory Management Unit (MMU).
- Level 2 (L2) memory system providing cluster memory coherency, including an L2 cache.
- Include VFP v3 hardware to support single and double-precision add, subtract, divide, multiply and accumulate, and square root operations
- SCU ensures memory coherency between the four CPUs for each cluster
- CCI400 ensures the memory coherency between the two clusters
- Integrated 32KB L1 instruction cache , 32KB L1 data cache with 4-way set associative
- 512KB unified L2 Cache for big cluster, 256KB unified L2 Cache for little cluster
- Trustzone technology support
- Full coresight debug solution
  - Debug and trace visibility of whole systems
  - ETM trace support
  - Invasive and non-invasive debug
- Ten separate power domains for CPU core system to support internal power switch and externally turn on/off based on different application scenario
  - PD_A53_L0: 1st Cortex-A53 + Neon + FPU + L1 I/D Cache of little cluster
  - PD_A53_L1: 2nd Cortex-A53 + Neon + FPU + L1 I/D Cache of little cluster
  - PD_A53_L2: 3rd Cortex-A53 + Neon + FPU + L1 I/D Cache of little cluster
  - PD_A53_L3: 4th Cortex-A53 + Neon + FPU + L1 I/D Cache of little cluster
  - PD_SCU_L: SCU + L2 Cache controller, and including PD_A53_L0, PD_A53_L1, PD_A53_L2, PD_A53_L3, debug logic of little cluster

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- PD_A53_B0: 1st Cortex-A53 + Neon + FPU + L1 I/D Cache of big cluster
- PD_A53_B1: 2nd Cortex-A53 + Neon + FPU + L1 I/D Cache of big cluster
- PD_A53_B2: 3rd Cortex-A53 + Neon + FPU + L1 I/D Cache of big cluster
- PD_A53_B3: 4th Cortex-A53 + Neon + FPU + L1 I/D Cache of big cluster
- PD_SCU_B: SCU + L2 Cache controller, and including PD_A53_B0, PD_A53_B1, PD_A53_B2, PD_A53_B3, debug logic of of big cluster

- One isolated voltage domain to support DVFS

1.2.2 Memory Organization

- Internal on-chip memory
  - 20KB BootRom
  - 68KB internal SRAM for security and non-security access, detailed size is programmable
- External off-chip memory①
  - DDR3-1600/DDR3L-1600, with 16/32bits data widths, 2 ranks, totally 4GB(max) address space
  - LPDDR2-1066, with 32bits data width, 2 ranks, totally 4GB(max) address space
  - LPDDR3-1600, with 32bits data width, 2 ranks, totally 4GB(max) address space
  - Async Nand Flash(include LBA Nand), 8bits data width, 4 banks, 60bits ECC
  - Sync ONFI/toggle Nand Flash , 8bits data width, 4 banks, 60bits ECC

1.2.3 Internal Memory

- Internal BootRom
  - Size : 20KB
  - Support system boot from the following device :
    - 8bits Async Nand Flash
    - 8bits toggle Nand Flash
    - SFC interface
    - eMMC interface
    - SDMMC interface
  - Support system code download by the following interface:
    - USB OTG interface
- Internal SRAM
  - Size : 68KB
  - Support security and non-security access
  - Security or non-security space is software programmable
  - Security space can be 0KB,4KB,8KB,12KB,16KB, … up to 64KB by 4KB step

1.2.4 External Memory or Storage device

- Dynamic Memory Interface (DDR3/DDR3L/LPDDR2/LPDDR3)
  - Compatible with JEDEC standard DDR3/DDR3L/LPDDR2/LPDDR3 SDRAM
  - Support one channel, with 16 or 32bits data widths
  - Support up to 2 ranks (chip selects), totally 4GB(max) address space
  - 16bits/32bits data width is software programmable
  - 7 host ports with 64bits/128bits AXI bus interface for system access, AXI bus clock is asynchronous with DDR clock
  - Programmable timing parameters to support DDR3/DDR3L/LPDDR2/LPDDR3 SDRAM from various vendor
  - Advanced command reordering and scheduling to maximize bus utilization
  - Low power modes, such as power-down and self-refresh for DDR3/LPDDR2/LPDDR3 SDRAM; clock stop and deep power-down for LPDDR2/LPDDR3 SDRAM
  - Embedded dynamic drift detection in the PHY to get dynamic drift compensation with the controller
  - Programmable output and ODT impedance with dynamic PVT compensation
  - Support one low-power work mode: power down DDR PHY digital logic power, make

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SDRAM still in self-refresh state to prevent data missing.

- Nand Flash Interface
  - Support async nand flash, each channel 8bits, up to 4 banks
  - Support sync DDR nand flash, each channel 8bits, up to 4 banks
  - Support LBA nand flash in async or sync mode
  - Up to 60bits hardware ECC
  - For async nand flash, support configurable interface timing , maximum data rate is 16bit/cycle
  - Embedded special DMA interface to do data transfer
  - Also support data transfer together with general PERI_DMAC in SoC system

- eMMC Interface
  - Compatible with standard iNAND interface
  - Support MMC4.5.1 protocol
  - Provide eMMC boot sequence to receive boot data from external eMMC device
  - Support FIFO over-run and under-run prevention by stopping card clock automatically
  - Support CRC generation and error detection
  - Embedded clock frequency division control to provide programmable baud rate
  - Support block size from 1 to 65535Bytes
  - 8bits data bus width

- SD/MMC Interface
  - Compatible with SD3.0, MMC ver4.5
  - Support FIFO over-run and under-run prevention by stopping card clock automatically
  - Support CRC generation and error detection
  - Embedded clock frequency division control to provide programmable baud rate
  - Support block size from 1 to 65535Bytes
  - Data bus width is 4bits

1.2.5 System Component

- CRU (clock & reset unit)
  - Support clock gating control for individual components inside PX5
  - One oscillator with 24MHz clock input and 6 embedded PLLs
  - Up to 2.2GHz clock output for all PLLs
  - Support global soft-reset control for whole SOC, also individual soft-reset for every components

- PMU(power management unit)
  - Multiple configurable work modes to save power by different frequency or automatical clock gating control or power domain on/off control
  - Lots of wakeup sources in different mode
  - 3 separate voltage domains
  - 16 separate power domains, which can be power up/down by software based on different application scenes

- Timer
  - 12 on-chip 64bits Timers in SoC with interrupt-based operation for non-secure application
  - 2 on-chip 64bits Timers in SoC with interrupt-based operation for secure application
  - Provide two operation modes: free-running and user-defined count
  - Support timer work state checkable
  - Fixed 24MHz clock input

- PWM
  - Four on-chip PWMs with interrupt-based operation
  - Programmable pre-scaled operation to bus clock and then further scaled

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- Embedded 32-bit timer/counter facility
- Support capture mode
- Support continuous mode or one-shot mode
- Provides reference mode and output various duty-cycle waveform

- WatchDog
  - 32 bits watchdog counter width
  - Counter clock is from apb bus clock
  - Counter counts down from a preset value to 0 to indicate the occurrence of a timeout
  - WDT can perform two types of operations when timeout occurs:
    - Generate a system reset
    - First generate an interrupt and if this is not cleared by the service routine by the time a second timeout occurs then generate a system reset
  - Programmable reset pulse length
  - Totally 16 defined-ranges of main timeout period

- Bus Architecture
  - 128bit/64-bit/32-bit multi-layer AXI/AHB/APB composite bus architecture
  - CCI400 embedded to support two Cortex-A53 cluster cache coherency
  - 5 embedded AXI interconnect
    - CPU interconnect with four 64-bits AXI masters, one 64-bits AXI slaves, one 32-bits AHB master and lots of 32-bits AHB/APB slaves
    - PERI interconnect with two 64-bits AXI masters, one 64-bits AXI slave, five 32-bits AHB masters and lots of 32-bits AHB/APB slaves
    - Display interconnect with three 128-bits AXI master, four 64-bits AXI masters and one 32-bits AHB slave
    - GPU interconnect with one 128-bits AXI master with point-to-point AXI-lite architecture and 32-bits APB slave
    - VCODEC interconnect also with two 64-bits AXI master and two 32-bits AHB slave, they are point-to-point AXI-lite architecture
  - For each interconnect with AXI/AHB/APB composite bus, clocks for AXI/AHB/APB domains are always synchronous, and different integer ratio is supported for them.
  - Flexible different QoS solution to improve the utility of bus bandwidth

- Interrupt Controller
  - Support 3 PPI interrupt source and 160 SPI interrupt sources input from different components inside PX5
  - Support 16 softwre-triggered interrupts
  - Input interrupt level is fixed , only high-level sensitive
  - Two interrupt outputs (nFIQ and nIRQ) separatelyfor each Cortex-A53, both are low-level sensitive
  - Support different interrupt priority for each interrupt source, and they are always software-programmable

- DMAC
  - Micro-code programming based DMA
  - The specific instruction set provides flexibility for programming DMA transfers
  - Linked list DMA function is supported to complete scatter-gather transfer
  - Support internal instruction cache
  - Embedded DMA manager thread
  - Support data transfer types with memory-to-memory, memory-to-peripheral, peripheral-to-memory
  - Signals the occurrence of various DMA events using the interrupt output signals
  - Mapping relationship between each channel and different interrupt outputs is software-programmable
  - Two embedded DMA controller , BUS_DMAC is for bus system, PERI_DMAC is for peripheral system

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- BUS_DMAC features:
  - 6 channels totally
  - 8 hardware request from peripherals
  - 2 interrupt output
  - Dual APB slave interface for register config, designated as secure and non-secure
  - Support trustzone technology and programmable secure state for each DMA channel
- PERI_DMAC features:
  - 7 channels totally
  - 17 hardware request from peripherals
  - 2 interrupt output
  - Not support trustzone technology

- Security system
  - Support trustzone technology for the following components inside PX5
    - Cortex-A53, support security and non-security mode, switch by software
    - BUS_DMAC, support some dedicated channels work only in security mode
    - eFuse, only accessed by Cortex-A53 in security mode
    - Internal memory , part of space is addressed only in security mode, detailed size is software-programmable together with TZMA(trustzone memory adapter)
  - Embedded encryption and decryption engine
    - Support AES-128/192/256 with ECB, CBC, OFB, CTR, CBC-MAC, CMAC, XCBC-MAC, XTS and CCM modes
    - Supports the DES (ECB and CBC modes) and TDES (EDE and DED) algorithms
    - Supports SHA-1, SHA-256 and SHA-512 modes, as well as HMAC
    - Support all mathematical operations required to implement the PKA supported cryptosystems between 128 bits and 3136 bits in size (in steps of 32 bits)
    - Support random bits generator from the ring oscillator
    - Control the AIB interface to the OTP memory and providing an interface for the CPU to access to the non-confidential trusted data
    - Set the device's security lifecycle state according to the values of various flag words in the OTP memory
    - Provide an firmware interface for secure boot, secure debug
    - Provide a security processor sub-system based on an internal 32-bit CPU
  - Support security boot
  - Support security debug

1.2.6 Video CODEC

- Shared internal memory and bus interface for video decoder and encoder②
- Embedded memory management unit(MMU)
- Video Decoder
  - Real-time video decoder of MPEG-1, MPEG-2, MPEG-4, H.263, H.264, AVS, VC-1, VP8, MVC
  - Error detection and concealment support for all video formats
  - Output data format is YUV420 semi-planar, and YUV400(monochrome) is also supported for H.264
  - H.264 up to HP level 5.2 : 4Kx2K@30fps (4096x2304)③
  - MPEG-4 up to ASP level 5 : 1080p@60fps (1920x1088)
  - MPEG-2 up to MP : 2160p@30fps (3840x2160)
  - MPEG-1 up to MP : 1080p@60fps (1920x1088)
  - H.263 : 576p@60fps (720x576)
  - VC-1 up to AP level 3 : 1080p@30fps (1920x1088)
  - VP8 : 2160p@24fps (3840x2160)
  - AVS : 1080p@60fps (1920x1088)
  - MVC : 2160p@30fps (3840x2160)
  - For AVS, 4:4:4 sampling not supported
  - For H.264, image cropping not supported

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- For MPEG-4, GMC(global motion compensation) not supported
- For VC-1, upscaling and range mapping are supported in image post-processor
- For MPEG-4 SP/H.263, using a modified H.264 in-loop filter to implement deblocking filter in post-processor unit

- Video Encoder
  - Support video encoder for H.264 (BP@level4.0, MP@level4.0, HP@level4.0), MVC and VP8
  - Only support I and P slices, not B slices
  - Support error resilience based on constrained intra prediction and slices
  - Input data format:
    - YCbCr 4:2:0 planar
    - YCbCr 4:2:0 semi-planar
    - YCbYCr 4:2:2
    - CbYCrY 4:2:2 interleaved
    - RGB444 and BGR444
    - RGB555 and BGR555
    - RGB565 and BGR565
    - RGB888 and BRG888
    - RGB101010 and BRG101010
  - Image size is from 96x96 to 1920x1088(Full HD)
  - Maximum frame rate is up to 30fps@1920x1080[3]
  - Bit rate supported is from 10Kbps to 20Mbps

1.2.7 HEVC Decoder

- Main/Main10 HEVC/H.265 decoder by 4k@60FPS
- Support up to 4096x2304 resolution
- Support up to 100Mbps bit rate
- Embedded memory management unit(MMU)
- Stream error detector (28 IDs)
- Internal 128k cache for bandwidth reduction
- Multi-clock domains and auto clock-gating design for power saving

1.2.8 JPEG CODEC

- JPEG decoder
  - Input JPEG file : YCbCr 4:0:0, 4:2:0, 4:2:2, 4:4:0, 4:1:1 and 4:4:4 sampling formats
  - Output raw image : YCbCr 4:0:0, 4:2:0, 4:2:2, 4:4:0, 4:1:1 and 4:4:4 semi-planar
  - Decoder size is from 48x48 to 8176x8176(66.8Mpixels)
  - Support JPEG ROI(region of image) decode
  - Maximum data rate[4] is up to 76million pixels per second
  - Embedded memory management unit(MMU)

- JPEG encoder
  - Input raw image :
    - YCbCr 4:2:0 planar
    - YCbCr 4:2:0 semi-planar
    - YCbYCr 4:2:2
    - CbYCrY 4:2:2 interleaved
    - RGB444 and BGR444
    - RGB555 and BGR555
    - RGB565 and BGR565
    - RGB888 and BRG888
    - RGB101010 and BRG101010
  - Output JPEG file : JFIF file format 1.02 or Non-progressive JPEG
  - Encoder image size up to 8192x8192(64million pixels) from 96x32
  - Maximum data rate[4] up to 90million pixels per second

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- Embedded memory management unit(MMU)

1.2.9 Image Enhancement

- Image pre-processor
  - Only used together with HD video encoder inside PX5, not support stand-alone mode
  - Provides RGB to YCbCr 4:2:0 color space conversion, compatible with BT601, BT709 or user defined coefficients
  - Provides YCbCr4:2:2 to YCbCr4:2:0 color space conversion
  - Support cropping operation from 8192x8192 to any supported encoding size
  - Support rotation with 90 or 270 degrees
- Video stabilization
  - Work in combined mode with HD video encoder inside PX5 and stand-alone mode
  - Adaptive motion compensation filter
  - Support scene detection from video sequence, encodes key frame when scene change noticed
- Image Post-Processor (embedded inside video decoder)
  - Combined with HD video decoder and JPEG decoder, post-processor can read input data directly from decoder output to reduce bus bandwidth
  - Also work as a stand-alone mode, its input data is from image data stored in external memory
  - Input data format:
    - Any format generated by video decoder in combined mode
    - YCbCr 4:2:0 semi-planar
    - YCbCr 4:2:0 planar
    - YCbYCr 4:2:2
    - YCrYCb 4:2:2
    - CbYCrY 4:2:2
    - CrYCbY 4:2:2
  - Output data format:
    - YCbCr 4:2:0 semi-planar
    - YCbYCr 4:2:2
    - YCrYCb 4:2:2
    - CbYCrY 4:2:2
    - CrYCbY 4:2:2
    - Fully configurable ARGB channel lengths and locations inside 32bits, such as ARGB8888, RGB565, ARGB4444 etc.
  - Input image size:
    - Combined mode: from 48x48 to 8176x8176 (66.8Mpixels)
    - Stand-alone mode: width from 48 to 8176,height from 48 to 8176, and maximum size limited to 16.7Mpixels
    - Step size is 16 pixels
  - Output image size: from 16x16 to 1920x1088 (horizontal step size 8,vertical step size 2)
  - Support image up-scaling:
    - Bicubic polynomial interpolation with a four-tap horizontal kernel and a two-tap vertical kernel
    - Arbitrary non-integer scaling ratio separately for both dimensions
    - Maximum output width is 3x input width
    - Maximum output height is 3x input height
  - Support image down-scaling:
    - Arbitrary non-integer scaling ratio separately for both dimensions
    - Unlimited down-scaling ratio
  - Support YUV to RGB color conversion, compatible with BT.601-5, BT.709 and user definable conversion coefficient
  - Support dithering (2x2 ordered spatial dithering) for 4/5/6bit RGB channel precision
  - Support programmable alpha channel and alpha blending operation with the following overlay input formats:

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- 8bit alpha + YUV444, big endian channel order with AYUV8888
- 8bit alpha + 24bit RGB, big endian channel order with ARGB8888

- Support deinterlacing with conditional spatial deinterlace filtering, only compatible with YUV420 input format
- Support RGB image contrast/brightness/color saturation adjustment
- Support image cropping & digital zoom only for JPEG or stand-alone mode
- Support picture in picture
- Support image rotation (horizontal flip, vertical flip, rotation 90,180 or 270 degrees)

- Image Enhancement-Processor (IEP)
- Image format
- Input data: XRGB/RGB565/YUV420/YUV422
- Output data: ARGB/RGB565/YUV420/YUV422
- The format ARGB/XRGB/RGB565/YUV support swap
- Support YUV semi-planar/planar
- Support BT601_l/BT601_f/BT709_l/BT709_f color space conversion
- Support RGB dither up/down conversion
- Support YUV up/down sampling conversion
- Max source image resolution: 8192x8192
- Max scaled image resolution: 4096x4096
  - Enhancement
- Gamma adjustment with programmable mapping table
- Hue/Saturation/Brightness/Contrast enhancement
- Color enhancement with programmable coefficient
- Detail enhancement with filter matrix up to 9x9
- Edge enhancement with filter matrix up to 9x9
- Programmable difference table for detail enhancement
- Programmable distance table for detail and edge enhancement
  - Noise reduction
- Compression noise reduction with filter matrix up to 9x9
- Programmable difference table for compression noise reduction
- Programmable distance table for compression noise reduction
- Spatial sampling noise reduction
- Temporal sampling noise reduction
- Optional coefficient for sampling noise reduction
  - Scaling
- Horizontal down-scaling with vertical down-scaling
- Horizontal down-scaling with vertical up-scaling
- Horizontal up-scaling with vertical down-scaling
- Horizontal up-scaling with vertical up-scaling
- Arbitrary non-integer scaling ratio, from 1/16 to 16
  - Deinterlace
- Input 4 fields, output 2 frames mode
- Input 4 fields, output 1 frames mode
- Input 2 fields, output 1 frames mode
- Programmable motion detection coefficient
- Programmable high frequency factor
- Programmable edge interpolation parameter
- Source width up to 1920
- Max resolution for dynamic image
- Deinterlace: 1920x1080
- Sampling noise reduction: 1920x1080
- Compression noise reduction: 4096x2304
- Enhancement: 4096x2304
  - Interface
- Programmable direct path to VOP

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- Embedded memory management unit(MMU)

1.2.10 Graphics Engine

- 3D Graphics Engine :
  - Base handheld architecture fully Microsoft® DirectX™ 9.3, OpenGL® 3.1, and OpenGL ES 3 compliant
  - Support for pull-model attribute evaluation
  - Tile-based deferred rendering architecture with concurrent processing of multiple tiles
  - Multi-threaded Unified Shading Cluster (USC) engine incorporating pixel shader, vertex shader, and GP-GPU (compute shader) functionality
  - USC incorporates an ALU architecture with high SIMD efficiency
  - Fully virtualised memory addressing (up to 1 TB address space), supporting unified memory architecture
  - Fine-grained task switching, workload balancing and power management
  - Dedicated processor for Rogue core firmware execution

- 2D Graphics Engine :
  - BitBlit with Stretch Blit, Simple Blit and Filter Blit
  - Color fill with gradient fill, and pattern fill
  - Line drawing with anti-aliasing and specified width
  - High-performance stretch and shrink
  - Monochrome expansion for text rendering
  - ROP2, ROP3, ROP4
  - Alpha blending modes including global alpha, per pixel alpha, porter-duff and fading
  - 8K x 8K input and 2K x 2K output raster 2D coordinate system
  - Arbitrary degrees rotation with anti-aliasing on every 2D primitive
  - Blending, scaling and rotation are supported in one pass for Bitbilt
  - Source format:
    - ABGR8888, XBGR888, ARGB8888, XRGB888
    - RGB888, RGB565
    - RGBA5551, RGBA4444
    - YUV420 planar, YUV420 semi-planar
    - YUV422 planar, YUV422 semi-planar
    - BPP8, BPP4, BPP2, BPP1
  - Destination formats:
    - ABGR8888, XBGR888, ARGB8888, XRGB888
    - RGB888, RGB565
    - RGBA5551, RGBA4444
    - YUV420 planar, YUV420 semi-planar only in filter and pre-scale mode
    - YUV422 planar, YUV422 semi-planar only in filter and pre-scale mode

1.2.11 Video IN/OUT

- Camera Interface(interface only)
  - Support up to 5M pixels
  - 8bits BT656(PAL/NTSC) interface
  - 16bits BT601 DDR interface
  - 8bits/10bits/12bits raw data interface
  - YUV422 data input format with adjustable YUV sequence
  - YUV422,YUV420 output format with separately Y and UV space
  - Support picture in picture (PIP)
  - Support simple image effects such as Arbitrary(sepia), Negative, Art freeze, Embossing etc.
  - Support static histogram statistics and white balance statistics
  - Support image crop with arbitrary windows
  - Support scale up/down from 1/8 to 8 with arbitrary non-integer ratio

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- Camera Interface and Image Processer(Interface and Image Processing)
  - Maximum input resolution of 8M(3264x2448) pixels
  - Main scaler with pixel-accurate up- and down-scaling to any resolution between 3264x2448 and 32x16 pixel in processing mode
  - Self scaler with pixel-accurate up- and down-scaling to any resolution between 1920x1080 and 32x16 pixel in processing mode
  - support of semiplanar NV21 color storage format
  - support of independent image cropping on main and self path
  - ITU-R BT 601/656 compliant video interface supporting YCbCr or RGB Bayer data
  - 12 bit camera interface
  - 12 bit resolution per color component internally
  - YCbCr 4:2:2 processing
  - quantization and Huffman tables
  - Windowing and frame synchronization
  - Macro block line, frame end, capture error, data loss interrupts and sync. (h_start, v_start) interrupts
  - Luminance/chrominance and chrominance blue/red swapping for YUV input signals
  - Continuous resize support
  - Color processing (contrast, saturation, brightness, hue, offset, range)
  - Display-ready RGB output in self-picture path (RGB888, RGB666 and RGB565)
  - Rotation unit in self-picture path (90°, 180°, 270° and h/v flipping) for RGB output
  - Read port provided to read back a picture from system memory
  - Simultaneous picture read back, resizing and storing through self path while main
  - path captures the camera picture
  - Black level compensation
  - Four channel Lens shade correction (Vignetting)
  - Auto focus measurement
  - White balancing and black level measurement
  - Auto exposure support by brightness measurement in 5x5 sub windows
  - Defect pixel cluster correction unit (DPCC) supports on the fly and table based pixel correction
  - De-noising pre filter (DPF)
  - Enhanced color interpolation (RGB Bayer demosaicing)
  - Chromatic aberration correction
  - Combined edge sensitive Sharpening / Blurring filter (Noise filter)
  - Color correction matrix (cross talk matrix)
  - Global Tone Mapping with wide dynamic range unit (WDR)
  - Image Stabilization support and Video Stabilization Measurement
  - Flexible Histogram calculation
  - Digital image effects (Emboss, Sketch, Sepia, B/W (Grayscale), Color Selection, Negative image, sharpening)
  - Solarize effect through gamma correction

- Display Interface
  - Embedded one channel display interfacesT.
  - Parallel Display interface
    - Parallel RGB LCD Interface:
      - 24-bit(RGB888),18-bit(RGB666), 15-bit(RGB565)
    - dither down:
      - allegro, FRC
      - gamma after dither
    - Max output resolution: 4096x2304
    - Scaning timing 8192x4096
  - IFDBC
    - decompress FB generated by GPU FBC
    - support 2048x1536 UI
    - support ARGB888,RGB888,RGB565

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- output for one layer among WIN0/1/2/3
- only support one IFDBC block which can used for WIN0/1/2/3 by configuration

- Display process
  - Background layer:
    - programmable 24-bit color
  - Win0 (Video0) layer:
    - RGB888, ARGB888, RGB565, YCbCr422, YCbCr420, YCbCr444
    - Support virtual display
    - 1/8 to 8 scaling-down and scaling-up engine:
      - Scale up using bicubic or bilinear;
      - Scale down using bilinear or average;
      - 4 Bicubic tables : precise,spline,catrom,mitchell;
      - coord 8bit, coe 8bit signed
    - x-mirror,y-mirror
  - Win1 (Video1) layer:
    - RGB888, ARGB888, RGB565, YCbCr422, YCbCr420, YCbCr444
    - Support virtual display
    - 1/8 to 8 scaling-down and scaling-up engine
      - Scale up using bicubic or bilinear;
      - Scale down using bilinear otraverage;
      - 4 Bicubic tables : precise,spline,catrom,mitchell;
      - coord 8bit, coe 8bit signed
    - x-mirror,y-mirror
  - Win2 (UI 0) layer:
    - RGB888, ARGB888, RGB565, 1/2/4/8bpp
    - Support virtual display
    - 4 display regions
    - x-mirror,y-mirror
  - Win3 (UI 1) layer:
    - RGB888, ARGB888, RGB565, 1/2/4/8bpp
    - Support virtual display
    - 4 display regions
    - x-mirror,y-mirror
  - Hardware cursor:
    - RGB888, ARGB888, RGB565, 1/2/4/8bpp
    - Support two size: 32x32,64x64,or 128x128
  - Overlay:
    - Win0/Win1/Win2/Win3 256 level alpha blending (support pre-multiplied alpha)
    - Win0/Win1/Win2/Win3 overlay position exchangeable
    - Win0/Win1/Win2/Win3 Transparency color key
    - Win0/Win1/Win2/Win3 global/per-pixel alpha
    - HWC 256 level alpha blending
    - HWC global/per-pixel alpha
- Others
  - 3 x 256 x 8 bits display LUTs
  - YcbCr2RGB(rec601-mpeg/rec601-jpeg/rec709/BT2020)and RGB2YcbCr
  - Support BCSH function
  - Support CABC function
  - QoS request signals
  - Gather transfer (Max 8)
  - Y/UV scheduler
  - Addr alignment
  - Support IEP direct path(win0/1/2/3)
  - Embedded memory management unit(MMU)
  - Support MIPI flow control

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1.2.12 HDMI

- Single Physical Layer PHY with support for HDMI 1.4 and 2.0 operation
- For HDMI operation, support for the following:
  - Up to 1080p at 120 Hz and 4k x 2k at 60 Hz HDTV display resolutions and up to QXGA graphic display resolutions
  - 3-D video formats
  - Up to 10-bit Deep Color modes
  - Up to 18 Gbps aggregate bandwidth
  - 13.5–600 MHz input reference clock
  - HPD input analog comparator
- Link controller flexible interface with 30-, 60- or 120-bit SDR data access
- Support HDCP 1.4/HDCP2.2

1.2.13 MIPI_DSI/LVDS/TTL combo PHY

- MIPI_TX
  - Support 4 data lane, providing up to 4Gbps data rate
  - Support 1080p @ 60fps output
- LVDS
  - Comply with the TIA/EIA-644-A LVDS standard
  - Combine LVTTL IO, support LVDS/LVTTL data output
  - Support reference clock frequency range from 10Mhz to 148.5Mhz
  - Support LVDS RGB 24/18bits color data transfer
  - Support VESA/JEIDA LVDS data format transfer
- TTL
  - Combine LVTTL IO, support LVDS/LVTTL data output

1.2.14 MIPI_CSI PHY

- Embedded one MIPI_CSI PHY
- Support 4 data lane, providing up to 4Gbps data rate

1.2.15 eDP PHY

- Support 4Kx2K @ 30fps
- Compliant with eDP TM Specification, version 1.1
- Up to 4 physical lanes of 2.7/1.62 Gbps/lane(HBR2/HBR/RBR)
- RGB, YCbCr 4:4:4, YCbCr 4:2:2 and 8/10/12 bit per component video format
- Encoded bit stream (Dolby Digital, or DTS) – IEC61937 compliant
- Support VESA DMT and CVT timing standards
- Fully support EIA/CEA-861D video timing and Info Frame structure
- Hot plug and unplug detection and link status monitor
- Support DDC/CI and MCCS command transmission when the monitor includes a display controller.

1.2.16 Audio Interface

- I2S/PCM with 8ch
  - Up to 8 channels (4xTX, 2xRX)
  - Audio resolution from 16bits to 32bits
  - Sample rate up to 192KHz
  - Provides master and slave work mode, software configurable
  - Support 3 I2S formats (normal, left-justified, right-justified)
  - Support 4 PCM formats(early, late1, late2, late3)
  - I2S and PCM mode cannot be used at the same time

- I2S/PCM with 2ch
  - Up to 2 channels (2xTX, 2xRX)

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- Audio resolution from 16bits to 32bits
- Sample rate up to 192KHz
- Provides master and slave work mode, software configurable
- Support 3 I2S formats (normal, left-justified, right-justified)
- Support 4 PCM formats(early, late1, late2, late3)
- I2S and PCM mode cannot be used at the same time

- SPDIF
  - Support two 16-bit audio data store together in one 32-bit wide location
  - Support biphase format stereo audio data output
  - Support 16 to 31 bit audio data left or right justified in 32-bit wide sample data buffer
  - Support 16, 20, 24 bits audio data transfer in linear PCM mode
  - Support non-linear PCM transfer

1.2.17 Connectivity

- SDIO interface
  - Compatible with SDIO 3.0 protocol
  - 4bits data bus widths

- High-speed ADC stream interface
  - Support single-channel 8bits/10bits interface
  - DMA-based and interrupt-based operation
  - Support 8bits TS stream interface

- TS interface
  - Supports one TS input channels.
  - Supports 4 TS Input Mode: sync/valid mode in the case of serial TS input; nosync/valid mode, sync/valid, sync/burst mode in the case of parallel TS input.
  - Supports 2 TS sources: demodulators and local memory.
  - Supports 2 Built-in PTIs(Programmable Transport Interface) to process TS simultaneously, and Each PTI supports:
    - 64 PID filters.
    - TS descrambling with 16 sets of Control Word under CSA v2.0 standard, up to 104Mbps
    - 16 PES/ES filters with PTS/DTS extraction and ES start code detection.
    - 4/8 PCR extraction channels
    - 64 Section filters with CRC check, and three interrupt mode: stop per unit, full-stop, recycle mode with version number check
    - PID done and error interrupts for each channel
    - PCR/DTS/PTS extraction interrupt for each channel
  - Supports 1 PVR(Personal Video Recording) output channel.
  - 1 built-in multi-channel DMA Controller.

- Smart Card
- support card activation and deactivation
- support cold/warm reset
- support Answer to Reset (ATR) response reception
- support T0 for asynchronous half-duplex character transmission
- support T1 for asynchronous half-duplex block transmission
- support automatic operating voltage class selection
- support adjustable clock rate and bit (baud) rate
- support configurable automatic byte repetition

- GMAC 10/100/1000M Ethernet Controller
- Supports 10/100/1000-Mbps data transfer rates with the RGMII interfaces
- Supports 10/100-Mbps data transfer rates with the RMII interfaces
- Supports both full-duplex and half-duplex operation

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- Preamble and start-of-frame data (SFD) insertion in Transmit, and deletion in Receive paths
- Automatic CRC and pad generation controllable on a per-frame basis
- Options for Automatic Pad/CRC Stripping on receive frames
- Programmable Inter Frame Gap (40-96 bit times in steps of 8)
- Supports a variety of flexible address filtering modes
- Separate 32-bit status returned for transmission and reception packets
- Supports IEEE 802.1Q VLAN tag detection for reception frames
- Support detection of LAN wake-up frames and AMD Magic Packet frames
- Support checksum off-load for received IPv4 and TCP packets encapsulated by the Ethernet frame
- Support checking IPv4 header checksum and TCP, UDP, or ICMP checksum encapsulated in IPv4 or IPv6 datagrams
- Comprehensive status reporting for normal operation and transfers with errors
- Automatic generation of PAUSE frame control or backpressure signal to the GMAC core based on Receive FIFO-fill (threshold configurable) level
- Handles automatic retransmission of Collision frames for transmission
- Discards frames on late collision, excessive collisions, excessive deferral and under run conditions

- SPI Controller
  - 3 on-chip SPI controller inside PX5
  - Support serial-master and serial-slave mode, software-configurable
  - DMA-based or interrupt-based operation
  - Embedded two 32x16bits FIFO for TX and RX operation respectively
  - Support 2 chip-selects output in serial-master mode

- Uart Controller
  - 5 on-chip uart controller inside PX5
  - DMA-based or interrupt-based operation
  - For all UART, two 64Bytes FIFOs are embedded for TX/RX operation respectively
  - Support 5bit,6bit,7bit,8bit serial data transmit or receive
  - Standard asynchronous communication bits such as start, stop and parity
  - Support different input clock for uart operation to get up to 4Mbps or other special baud rate
  - Support non-integer clock divides for baud clock generation
  - Auto flow control mode is for all UART, except UART_DBG

- I2C controller
  - 6 on-chip I2C controller in PX5
  - Multi-master I2C operation
  - Support 7bits and 10bits address mode
  - Software programmable clock frequency and transfer rate up to 400Kbit/s in the fast mode
  - Serial 8bits oriented and bidirectional data transfers can be made at up to 100Kbit/s in the standard mode

- GPIO
  - Totally 128 GPIOs
  - All of GPIOs can be used to generate interrupt to Cortex-A53
  - GPIO0 can be used to wakeup system from low-power mode
  - The pull direction(pullup or pulldown) for all of GPIOs are software-programmable
  - All of GPIOs are always in input direction in default after power-on-reset
  - The drive strength for all of GPIOs is software-programmable

- USB Host2.0
  - Embedded 2 USB Host2.0 interfaces

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- Compatible with USB Host2.0 specification
- Supports high-speed(480Mbps), full-speed(12Mbps) and low-speed (1.5Mbps) mode
- Provides 16 host mode channels
- Support periodic out channel in host mode

- USB OTG2.0
  - Compatible with USB OTG2.0 specification
  - Supports high-speed(480Mbps), full-speed(12Mbps) and low-speed (1.5Mbps) mode
  - Support up to 9 device mode endpoints in addition to control endpoint 0
  - Support up to 6 device mode IN endpoints including control endpoint 0
  - Endpoints 1/3/5/7 can be used only as data IN endpoint
  - Endpoints 2/4/6 can be used only as data OUT endpoint
  - Endpoints 8/9 can be used as data OUT and IN endpoint
  - Provides 9 host mode channels

1.2.18 Others

- Temperature Sensor(TS-ADC)
- 2 bipolar-based temperature-sensing cell embedded
- 1-channel 8-bits SAR ADC
- Temperature accuracy sensed is ±5 degree

- SAR-ADC(Successive Approximation Register)
  - 3-channel single-ended 10-bit SAR analog-to-digital converter
  - Conversion speed range is up to 1 MSPS
  - SAR-ADC clock must be less than 1MHz

- eFuse
  - Two high-density electrical Fuse is integrated: 256bits (32x8) / 1024bits (32x32)
  - Support standby mode

*Notes:①**:** DDR3/LPDDR2/LPDDR3 are not used simultaneously as well as async and sync ddr nand flash*
*②**:** In PX5, Video decoder and encoder are not used simultaneously because of shared internal buffer*
*③**:** Actual maximum frame rate will depend on the clock frequency and system bus performance*
*④**:** Actual maximum data rate will depend on the clock frequency and JPEG **c**ompression rate*

1.3 Block Diagram

The following diagram shows the basic block diagram for PX5.

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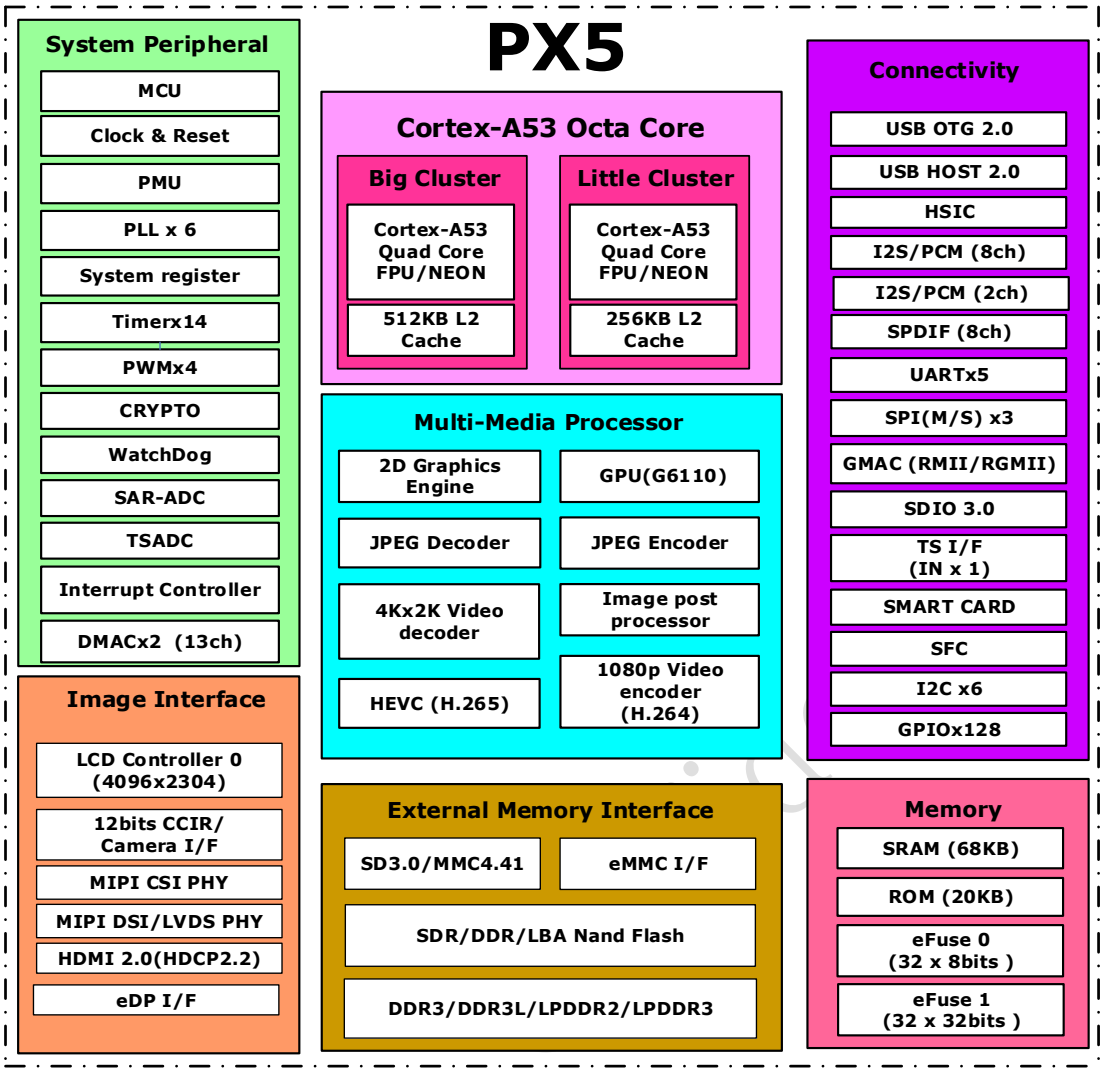

Fig. 1-1 PX5 Block Diagram

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Chapter 2 System Overview

2.1 Address Mapping for Cortex-A53

PX5 support to boot from internal bootrom, which support remap function by software programming. Remap is controlled by SGRF_SOC_CON3 bit[6]. The following figure show the address mapping for cortex-A53.

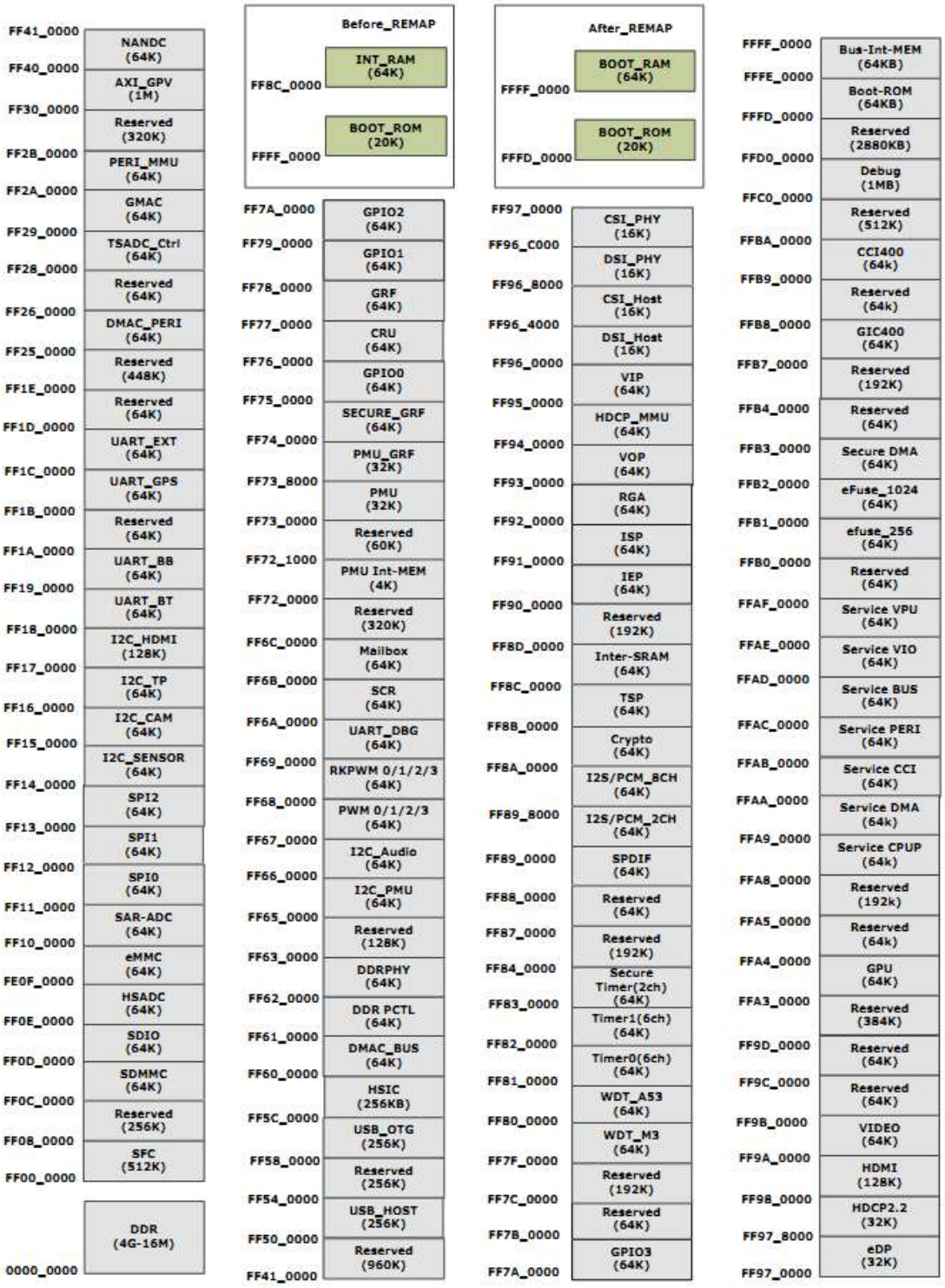

Fig. 2-1 PX5 Address Mapping

2.2 Address Mapping of MCU

There is a fixed address offset for the address mapping of MCU compared to the cortex-A53, which is described as the following formula.

ADDR_MCU = ADDR_CA53 – 0xBF00_0000

2.3 System Boot

PX5 provides system boot from off-chip devices such as SDMMC card, 8bits async nand flash or toggle nand flash, SPI nor or nand, and eMMC memory. When boot code is not ready in these devices, also provide system code download into them by USB OTG interface. All of the boot code will be stored in internal bootrom. The following is the whole boot procedure for boot code, which will be stored in bootrom in advance.
The following features are supports.

- Support secure boot mode and non-secure boot mode
- Support system boot from the following device:
- 8bits Async Nand Flash
- 8bits Toggle Nand Flash
- SFC interface
- eMMC interface
- SDMMC Card
- Support system code download by USB OTG

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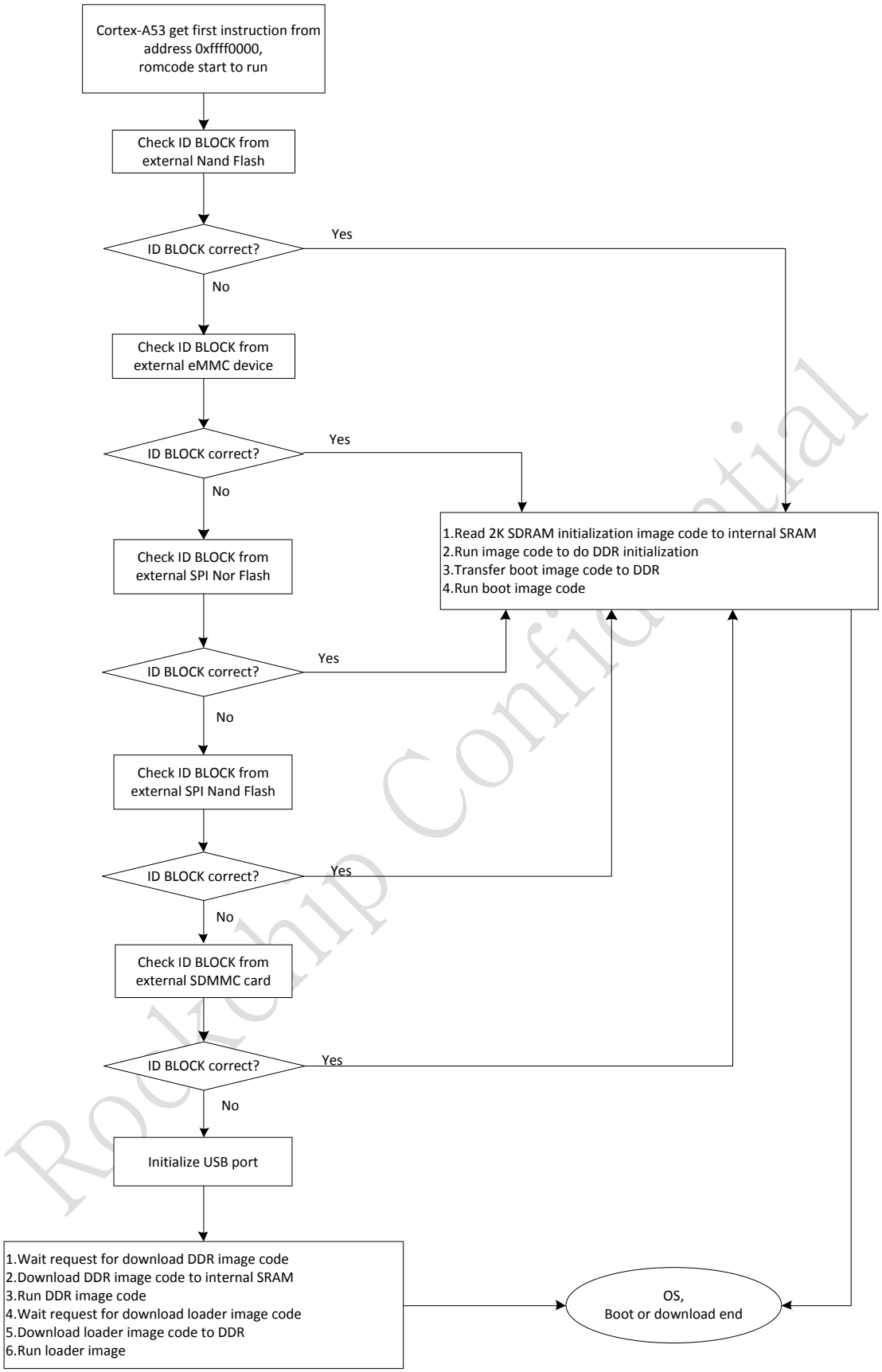

Fig. 2-2 PX5 boot procedure flow

2.4 System Interrupt connection for Cortex-A53

PX5 provides an general interrupt controller(GIC) for Cortex-A53 MPCore processor, which has 160 SPI (shared peripheral interrupts) interrupt sources and 4 PPI(Private peripheral interrupt)

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interrupt source and separately generates one nIRQ and one nFIQ to CPU. The triggered type for each interrupts is high level sensitive, not programmable. The detailed interrupt sources connection is in the following table. For detailed GIC setting, please refer to Chapter 15.

Table 2-1 PX5 Interrupt connection list

IRQ Type	IRQ ID	Source(spi)	Polarity
PPI	26	HYPERVISOR TIMER	High level
	27	VIRTUAL TIMER	High level
	29	SECURE PHYSICAL TIMER	High level
	30	NON-SECURE PHY TIMER	High level
SPI	32	DMAC_BUS (0)	High level
	33	DMAC_BUS (1)	High level
	34	DMAC_PERI (0)	High level
	35	DMAC_PERI (1)	High level
	36	UPCTL 0	High level
	37	hsadc	High level
	38	NA	High level
	39	NA	High level
	40	GPU_IRQ	High level
	41	Video encoder	High level
	42	Video decoder	High level
	43	NA	High level
	44	HEVC	High level
	45	VIP	High level
	46	ISP Mini	High level
	47	VOP0	High level
	48	HDCP	High level
	49	IEP	High level
	50	RGA	High level
	51	DSI 0 HOST	High level
	52	NA	High level
	53	NA	High level
	54	NA	High level
	55	USB OTG	High level
	56	USB Host 0 ehci	High level
	57	USB Host0 ohci	High level
	58	HSIC	High level
	59	GMAC	High level
	60	GMAC PMT	High level
	61	SFC	High level
	62	NA	High level
	63	NA	High level
	64	SD/MMC	High level
	65	SDIO 0	High level
	66	NA	High level
	67	eMMC	High level
	68	SARADC	High level

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	69	TSADC	High level
	70	NandC 0	High level
	71	PERI MMU	High level
	72	I2S_2CH	High level
	73	SPI2	High level
	74	TSP	High level
	75	SCR	High level
	76	SPI0	High level
	77	SPI1	High level
	78	Timer6	High level
	79	Timer7	High level
	80	Crypto	High level
	81	Timer8	High level
	82	Timer9	High level
	83	Timer10	High level
	84	Timer11	High level
	85	I2S/PCM (8ch)	High level
	86	SPDIF(8ch)	High level
	87	UART0_BT	High level
	88	UART1_BB	High level
	89	UART2_DBG	High level
	90	UART3_GPS	High level
	91	UART4_EXP	High level
	92	I2C_PMU	High level
	93	I2C_AUDIO	High level
	94	I2C_SENSOR	High level
	95	I2C_CAM	High level
	96	I2C_TP	High level
	97	I2C_HDMI	High level
	98	Timer 0	High level
	99	Timer 1	High level
	100	Timer 2	High level
	101	Timer 3	High level
	102	Timer 4	High level
	103	Timer 5	High level
	104	Secure Timer 0	High level
	105	Secure Timer 1	High level
	106	PWM0	High level
	107	PWM1	High level
	108	PWM2	High level
	109	PWM3	High level
	110	RK_PWM	High level
	111	WDT_A53	High level
	112	PMU	High level
	113	GPIO0	High level

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	114	GPIO1	High level
	115	GPIO2	High level
	116	GPIO3	High level
	117	NA	High level
	118	WDT_M3	High level
	119	NA	High level
	120	NA	High level
	121	NA	High level
	122	AHB arbiter0 (usb)	High level
	123	AHB arbiter1 (emem)	High level
	124	AHB arbiter1 (mmc)	High level
	125	usbotg_id	High level
	126	usbotg_bvalid	High level
	127	usbotg_linestate	High level
	128	usbhost0_linestate	High level
	129	NA	High level
	130	NA	High level
	131	sdmmc_detect_n	High level
	132	sdio0_detect_n	High level
	133	mipi_csi_host	High level
	134	hdmi wakeup	High level
	135	hdmi	High level
	136	edp_hdmi	High level
	137	edp_dp	High level
	138	sdmmc detect dual edge	High level
	139	NA	High level
	140	NA	High level
	141	cci_nevntcntoverflow	High level
	142	CCI400	High level
	143	NA	High level
	144	npmuirq_pd_core_l[0]	High level
	145	npmuirq_pd_core_l[1]	High level
	146	npmuirq_pd_core_l[2]	High level
	147	npmuirq_pd_core_l[3]	High level
	148	npmuirq_pd_core_b[0]	High level
	149	npmuirq_pd_core_b[1]	High level
	150	npmuirq_pd_core_b[2]	High level
	151	npmuirq_pd_core_b[3]	High level
	152	nvcpumntirq_pd_core_l[0]	High level
	153	nvcpumntirq_pd_core_l[1]	High level
	154	nvcpumntirq_pd_core_l[2]	High level
	155	nvcpumntirq_pd_core_l[3]	High level
	156	nvcpumntirq_pd_core_b[0]	High level
	157	nvcpumntirq_pd_core_b[1]	High level
	158	nvcpumntirq_pd_core_b[2]	High level
	159	nvcpumntirq_pd_core_b[3]	High level
	160	ncomirq_pd_core_l[0]	High level
	161	ncomirq_pd_core_l[1]	High level

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	162	ncomirq_pd_core_l[2]	High level
	163	ncomirq_pd_core_l[3]	High level
	164	ncomirq_pd_core_b[0]	High level
	165	ncomirq_pd_core_b[1]	High level
	166	ncomirq_pd_core_b[2]	High level
	167	ncomirq_pd_core_b[3]	High level
	168	nexterrirq_pd_core_l	High level
	169	nexterrirq_pd_core_b	High level
	170	ctiirq_pd_core_l[0]	High level
	171	ctiirq_pd_core_l[1]	High level
	172	ctiirq_pd_core_l[2]	High level
	173	ctiirq_pd_core_l[3]	High level
	174	ctiirq_pd_core_b[0]	High level
	175	ctiirq_pd_core_b[1]	High level
	176	ctiirq_pd_core_b[2]	High level
	177	ctiirq_pd_core_b[3]	High level
	178	mailbox_mcu_int0	High level
	179	mailbox_mcu_int1	High level
	180	mailbox_mcu_int2	High level
	181	mailbox_mcu_int3	High level

2.5 System Interrupt connection for MCU

PX5 provides an interrupt controller for MCU, which have 240 interrupt source which generates one nIRQ and one nFIQ to CPU. The triggered type for each interrupts is high level sensitive, not programmable.

Table 2-2 PX5 Interrupt connection list

IRQ Type	IRQ ID	Source(spi)	Polarity
SPI	0	DMAC_BUS (0)	High level
	1	DMAC_BUS (1)	High level
	2	DMAC_PERI (0)	High level
	3	DMAC_PERI (1)	High level
	4	UPCTL 0	High level
	5	hsadc	High level
	6	NA	High level
	7	NA	High level
	8	GPU_IRQ	High level
	9	Video encoder	High level
	10	Video decoder	High level
	11	NA	High level
	12	HEVC	High level
	13	VIP	High level
	14	ISP Mini	High level
	15	VOP0	High level
	16	HDCP	High level
	17	IEP	High level
	18	RGA	High level
	19	DSI 0 HOST	High level
	20	NA	High level
	21	NA	High level

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	22	NA	High level
	23	USB OTG	High level
	24	USB Host 0 ehci	High level
	25	USB Host0 ohci	High level
	26	HSIC	High level
	27	GMAC	High level
	28	GMAC PMT	High level
	29	SFC	High level
	30	NA	High level
	31	NA	High level
	32	SD/MMC	High level
	33	SDIO 0	High level
	34	NA	High level
	35	eMMC	High level
	36	SARADC	High level
	37	TSADC	High level
	38	NandC 0	High level
	39	PERI MMU	High level
	40	I2S_2CH	High level
	41	SPI2	High level
	42	TSP	High level
	43	SCR	High level
	44	SPI0	High level
	45	SPI1	High level
	46	Timer6	High level
	47	Timer7	High level
	48	Crypto	High level
	49	Timer8	High level
	50	Timer9	High level
	51	Timer10	High level
	52	Timer11	High level
	53	I2S/PCM (8ch)	High level
	54	SPDIF(8ch)	High level
	55	UART0_BT	High level
	56	UART1_BB	High level
	57	UART2_DBG	High level
	58	UART3_GPS	High level
	59	UART4_EXP	High level
	60	I2C_PMU	High level
	61	I2C_AUDIO	High level
	62	I2C_SENSOR	High level
	63	I2C_CAM	High level
	64	I2C_TP	High level
	65	I2C_HDMI	High level
	66	Timer 0	High level

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	67	Timer 1	High level
	68	Timer 2	High level
	69	Timer 3	High level
	70	Timer 4	High level
	71	Timer 5	High level
	72	Secure Timer 0	High level
	73	Secure Timer 1	High level
	74	PWM0	High level
	75	PWM1	High level
	76	PWM2	High level
	77	PWM3	High level
	78	RK_PWM	High level
	79	WDT_A53	High level
	80	PMU	High level
	81	GPIO0	High level
	82	GPIO1	High level
	83	GPIO2	High level
	84	GPIO3	High level
	85	NA	High level
	86	WDT_M3	High level
	87	NA	High level
	88	NA	High level
	89	NA	High level
	90	AHB arbiter0 (usb)	High level
	91	AHB arbiter1 (emem)	High level
	92	AHB arbiter1 (mmc)	High level
	93	usbotg_id	High level
	94	usbotg_bvalid	High level
	95	usbotg_linestate	High level
	96	usbhost0_linestate	High level
	97	NA	High level
	98	NA	High level
	99	sdmmc_detect_n	High level
	100	sdio0_detect_n	High level
	101	mipi_csi_host	High level
	102	hdmi wakeup	High level
	103	hdmi	High level
	104	edp_hdmi	High level
	105	edp_dp	High level
	106	sdmmc detect dual edge	High level
	107	NA	High level
	108	NA	High level
	109	cci_nevntcntoverflow	High level
	110	CCI400	High level
	111	NA	High level
	112	npmuirq_pd_core_l[0]	High level

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	113	npmuirq_pd_core_l[1]	High level
	114	npmuirq_pd_core_l[2]	High level
	115	npmuirq_pd_core_l[3]	High level
	116	npmuirq_pd_core_b[0]	High level
	117	npmuirq_pd_core_b[1]	High level
	118	npmuirq_pd_core_b[2]	High level
	119	npmuirq_pd_core_b[3]	High level
	120	nvcpumntirq_pd_core_l[0]	High level
	121	nvcpumntirq_pd_core_l[1]	High level
	122	nvcpumntirq_pd_core_l[2]	High level
	123	nvcpumntirq_pd_core_l[3]	High level
	124	nvcpumntirq_pd_core_b[0]	High level
	125	nvcpumntirq_pd_core_b[1]	High level
	126	nvcpumntirq_pd_core_b[2]	High level
	127	nvcpumntirq_pd_core_b[3]	High level
	128	ncomirq_pd_core_l[0]	High level
	129	ncomirq_pd_core_l[1]	High level
	130	ncomirq_pd_core_l[2]	High level
	131	ncomirq_pd_core_l[3]	High level
	132	ncomirq_pd_core_b[0]	High level
	133	ncomirq_pd_core_b[1]	High level
	134	ncomirq_pd_core_b[2]	High level
	135	ncomirq_pd_core_b[3]	High level
	136	nexterrirq_pd_core_l	High level
	137	nexterrirq_pd_core_b	High level
	138	mailbox_ca53_int0	High level
	139	mailbox_ca53_int1	High level
	140	mailbox_ca53_int2	High level
	141	mailbox_ca53_int3	High level

2.6 System DMA hardware request connection

PX5 provides 2 DMA controllers: DMAC_BUS inside bus system and DMAC_PERI inside peripheral system. As for DMAC_BUS, there are 8 hardware request ports. And 17 hardware request ports are used in DMAC_PERI, the trigger type for each of them is high level, not programmable. For detailed descriptions of DMAC_BUS and DMAC_PERI, please refer to Chapter 12 and Chapter 13.

Table 2-3 PX5 DMAC_BUS Hardware request connection list

Req Number	Source	Polarity
0	I2S/PCM(8ch) tx	High level
1	I2S/PCM(8ch) rx	High level
2	PWM	High level
3	SPDIF 8CH tx	High level
4	UART_DBG tx	High level
5	UART_DBG rx	High level
6	I2S/PCM(2ch) tx	High level
7	I2S/PCM(2ch) rx	High level

Table 2-4 PX5 DMAC_PERI Hardware request connection list

Req Number	Source	Polarity
0	HS-ADC/TSI	High level

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1	UART_BT TX	High level
2	UART_BT RX	High level
3	UART_BB TX	High level
4	UART_BB RX	High level
5	N/A	N/A
6	N/A	N/A
7	UART_GPS TX	High level
8	UART_GPS RX	High level
9	UART_EXP TX	High level
10	UART_EXP RX	High level
11	SPI0 TX	High level
12	SPI0 RX	High level
13	SPI1 TX	High level
14	SPI1 RX	High level
15	SPI2 TX	High level
16	SPI2 RX	High level

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Chapter 3 Clock & Reset Unit (CRU)

3.1 Overview

The CRU is an APB slave module that is designed for generating all of the system clocks, resets of chip. CRU generates system clock from PLL output clock or external clock source, and generates system reset from external power-on-reset, watchdog timer reset or software reset.

CRU supports the following features:

- Compliance to the AMBA APB interface
- Embedded six PLLs
- Flexible selection of clock source
- Supports the respectivegating of all clocks
- Supports the respective software reset of all modules

3.2 Block Diagram

The CRU comprises with:

- PLL
- Register configuration unit
- Clock generate unit
- Reset generate unit

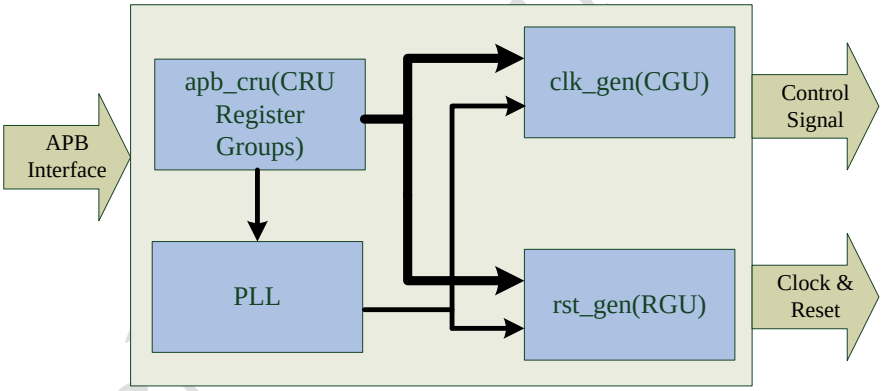

Fig. 3-1 CRU Architecture

3.3 System Clock Solution

3.3.1 CRU architecture

The following diagrams show CRU clock architecture (mux and divider information).

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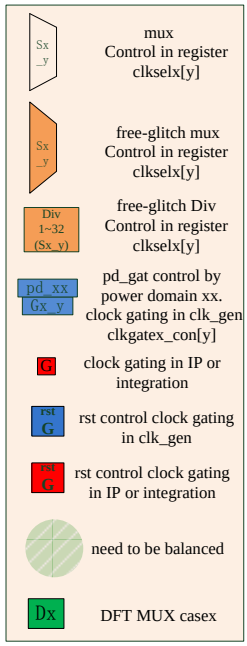

Please check the symbols on the left firstly before reading the following diagram

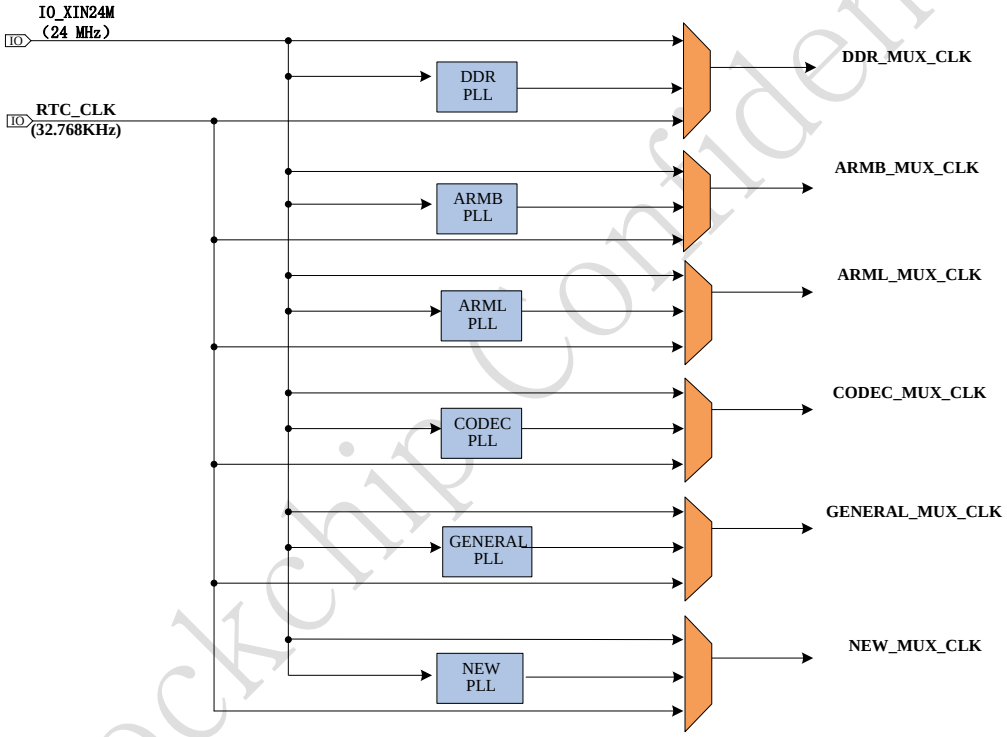

Fig.3-2 CRU Clock Architecture Diagram 1

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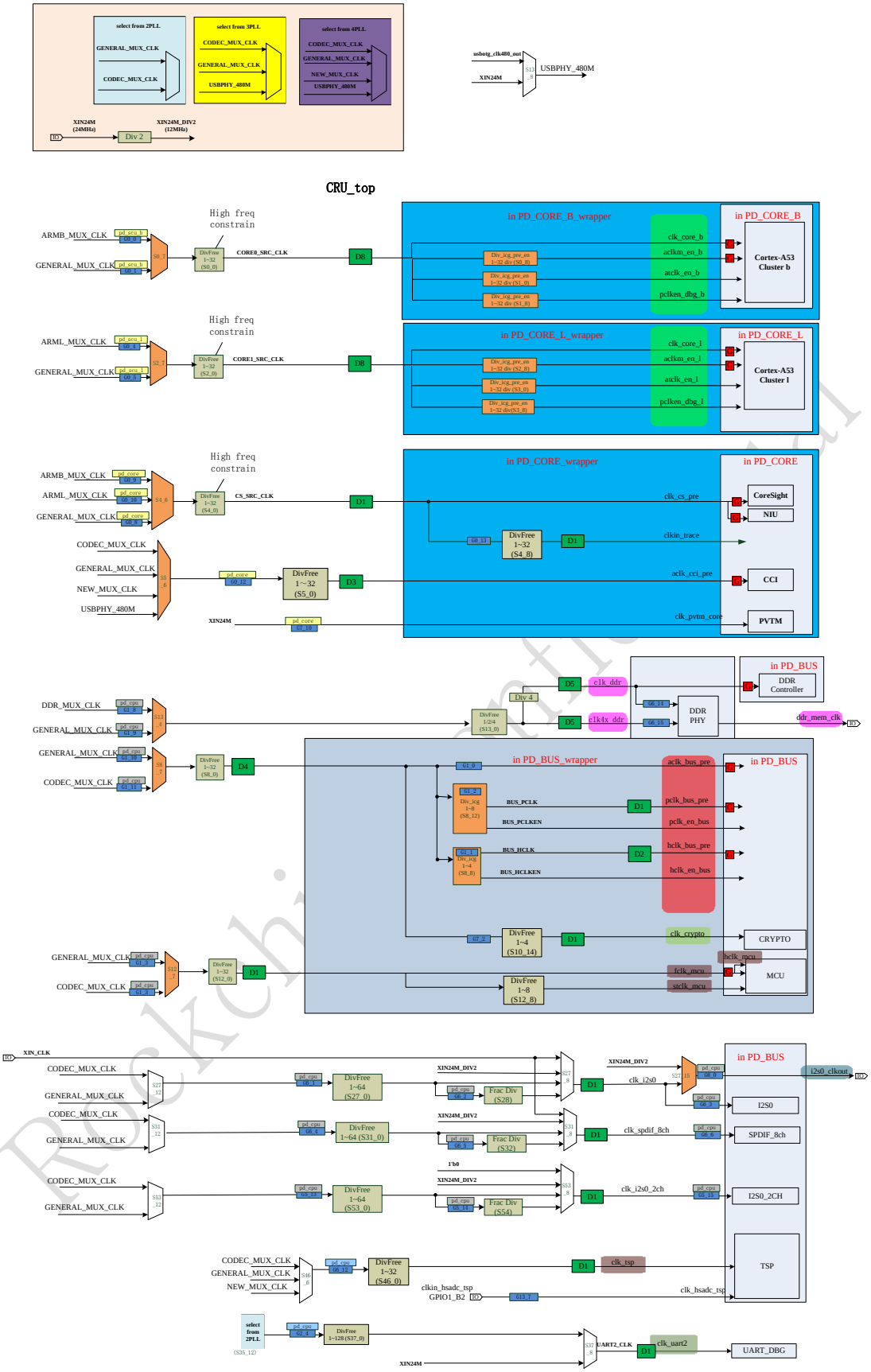

Fig. 3-2 CRU Clock Architecture Diagram 2

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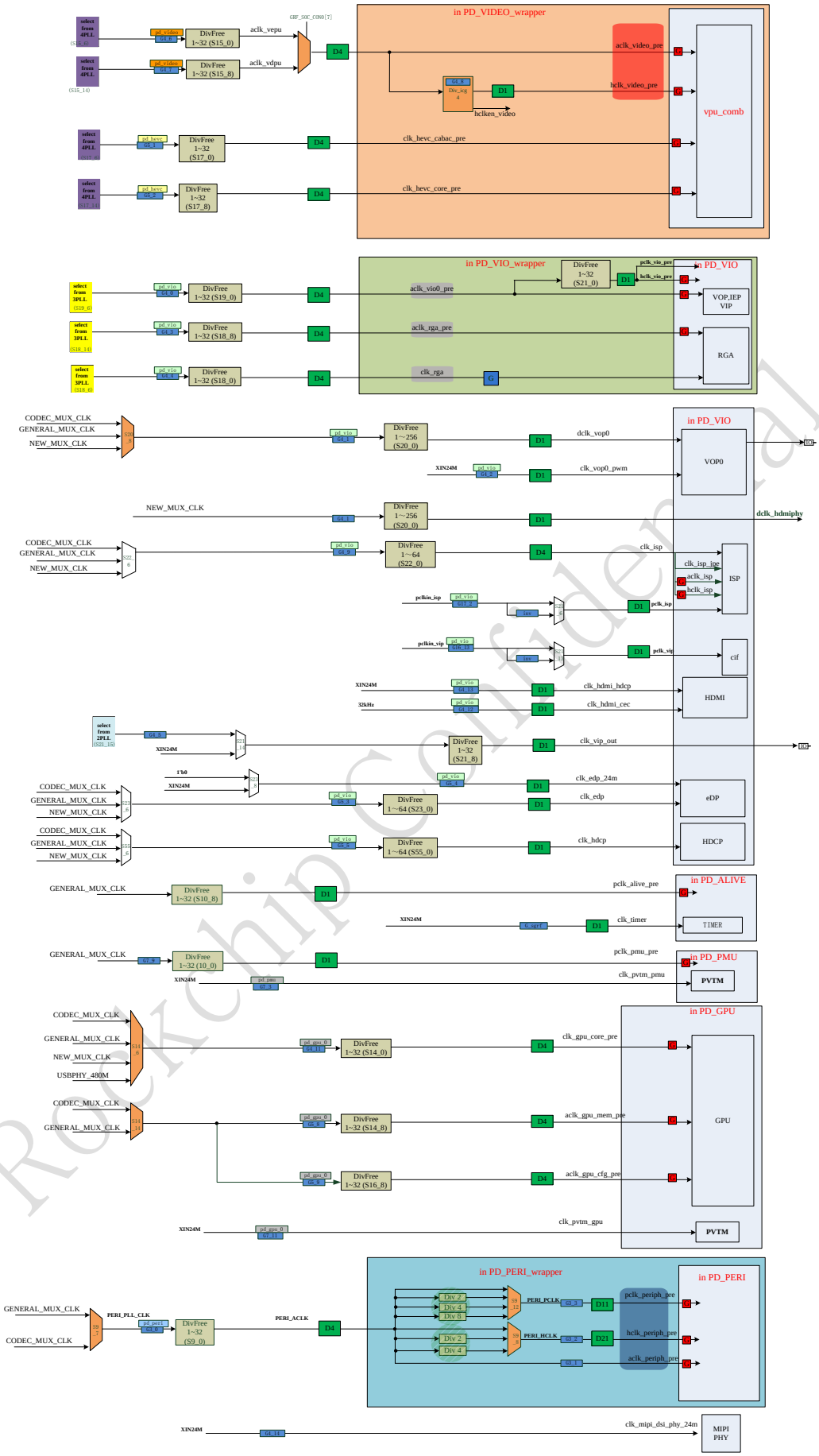

Fig. 3-3 CRU Clock Architecture Diagram 3

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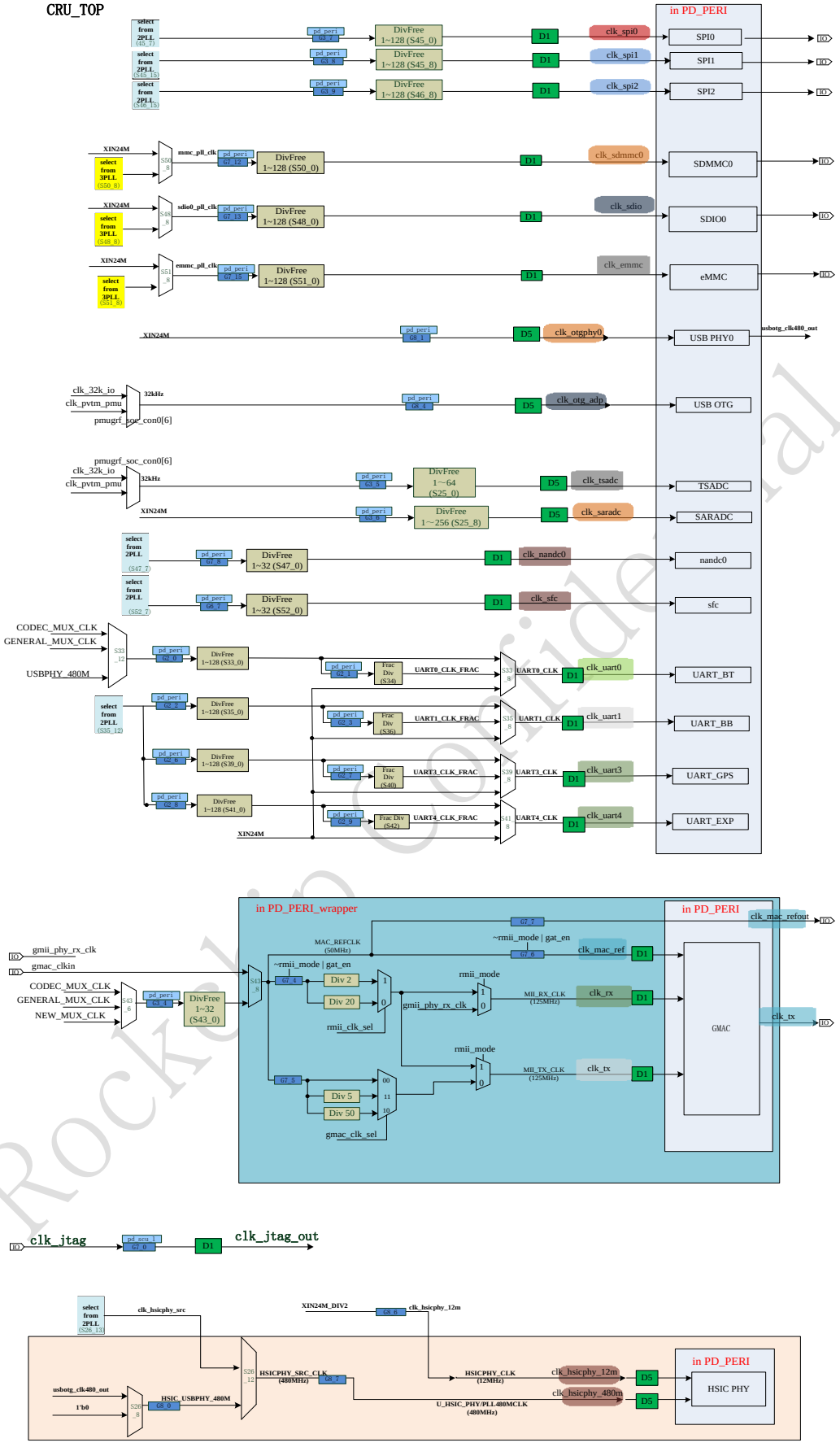

Fig. 3-4 CRU Clock Architecture Diagram 4

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PD	src clk	gating num	out clk
PD_CORE	aclk_cci_pre	G11_4	aclk_core_niu_cpup
		G11_3	aclk_core_niu_cci
		G11_2	aclk_cci400
		G11_1	aclk_adb400m_pd_core_b
		G11_0	aclk_adb400m_pd_core_l
	aclkm_core_b	G10_0	aclk_adb400s_pd_core_b
	aclkm_core_l	G9_0	aclk_adb400s_pd_core_l
	clk_core_b	G10_1	dbg_pd_core_b_clk
	clk_core_l	G9_1	dbg_pd_core_l_clk
	clk_cs_pre	G11_7	clk_dbg
		G11_6	pclk_core_niu_sdbg
		G11_5	hclk_core_niu_dbg
PD_BUS	aclk_bus_pre	G12_12	aclk_strc_sys
		G12_11	aclk_dmac
		G12_6	clk_intmem1
		G12_5	clk_intmem0
		G12_4	aclk_intmem
		G13_9	aclk_gic400
	clk_ddr	G13_2	nclk_ddrupctl
	clk_hsadc_tsp	G13_7	clk_hsadc_0_tsp
	fclk_mcu	G13_14	hclk_noc_mcu
		G13_13	stclk_mcu
		G13_12	fclk_mcu
		G13_11	hclk_mcu
	hclk_bus_pre	G12_10	hclk_spdif
		G12_9	hclk_rom
		G12_8	hclk_i2s_2ch
		G12_7	hclk_i2s_8ch
		G13_10	hclk_tsp
		G13_4	sclk_crypto
		G13_3	mclk_crypto
	pclk_bus_pre	G12_14	pclk_ddrphy
		G12_13	pclk_ddrupctl
		G12_3	pclk_i2caudio
		G12_2	pclk_i2cpmu
		G12_1	pclk_mailbox
		G12_0	pclk_pwm0
		G13_8	pclk_sim
		G13_6	pclk_pwm1
		G13_5	pclk_uartdbg
		G13_1	pclk_efuse_256
		G13_0	pclk_efuse_1024
PD_VIDEO	aclk_video_pre	G15_0	aclk_video
	clk_hevc_ca_pre	G15_3	clk_hevc_ca
	clk_hevc_core_pre	G15_2	clk_hevc_core
	hclk_video_pre	G15_1	hclk_video
PD_VIO	aclk_rga_pre	G16_10	aclk_vio1_noc
		G16_0	aclk_rga
		G17_10	aclk_hdcp
	aclk_vio0_pre	G16_11	aclk_vip
		G16_9	aclk_vio0_noc
		G16_5	aclk_vop
		G16_4	aclk_vop_iep
		G16_2	aclk_iep
	clk_isp	G16_14	hclk_isp
		G17_0	aclk_isp
	hclk_vio_pre	G16_12	hclk_vip
		G16_8	hclk_vio_noc
		G16_7	hclk_vio_ahb_arbi
		G16_6	hclk_vop
		G16_3	hclk_iep
		G16_1	hclk_rga
		G17_12	hclk_hdcpmmu
		G17_7	hclk_vio_h2p
	pclk_vio_pre	G17_11	pclk_hdcp
		G17_9	pclk_edp_ctrl
		G17_8	pclk_vio_h2p
		G17_6	pclk_hdmi_ctrl
		G17_4	pclk_mipi_csi
		G17_3	pclk_mipi_dsi0
	pclkin_vip	G16_13	pclkin_vip
		G17_2	pclkin_isp
PD_GPU	clk_gpu_core_pre	G18_2	clk_gpu_core
	aclk_gpu_mem_pre	G18_1	aclk_gpu_mem
	aclk_gpu_cfg_pre	G18_0	aclk_gpu_cfg

PD	src clk	gating num	out clk
PD_PERI	aclk_periph_pre	G19_3	aclk_dmac_peri
		G19_2	aclk_peri_axi_matrix
		G20_15	hclk_sfc
		G20_13	aclk_gmac
		G20_8	aclk_peri_niu
		G21_4	aclk_peri_mmu
	hclk_periph_pre	G19_0	hclk_peri_axi_matrix
		G20_11	hclk_nand0
		G20_10	hclk_mmc_peri
		G20_9	hclk_emem_peri
		G20_7	hclk_peri_ahb_arbi
		G20_6	hclk_usb_peri
		G20_5	hclk_hsic
		G20_4	hclk_host1
		G20_3	hclk_host0
		G20_2	pmu_hclk_otg0
		G20_1	hclk_otg0
		G21_3	hclk_hsadc
		G21_2	hclk_emmc
		G21_1	hclk_sdio0
		G21_0	hclk_sdmmc
	pclk_periph_pre	G19_15	pclk_saradc
		G19_14	pclk_i2c5
		G19_13	pclk_i2c4
		G19_12	pclk_i2c3
		G19_11	pclk_i2c2
		G19_10	pclk_uart4
		G19_9	pclk_uart3
		G19_8	pclk_uart1
		G19_7	pclk_uart0
		G19_6	pclk_spi2
		G19_5	pclk_spi1
		G19_4	pclk_spi0
		G19_1	pclk_peri_axi_matrix
		G20_14	pclk_emmc
		G20_0	pclk_tsadc
PD_ALIVE	pclk_alive_pre	G22_13	pclk_timer1
		G22_12	pclk_timer0
		G22_9	pclk_alive_niu
		G22_8	pclk_grf
		G22_3	pclk_gpio3
		G22_2	pclk_gpio2
		G22_1	pclk_gpio1
		sgrf_soc_con3[7]	pclk_wdt
		sgrf_soc_con3[9]	pclk_wdt_m3
		sgrf_soc_con3[5]	pclk_stimer
	pclk_vio_pre	G22_11	pclk_dphyrx_phy
		G22_10	pclk_dphytx0_phy
PD_PMU	pclk_pmu_pre	G23_5	pclk_pmugrf
		G23_4	pclk_gpio0
		G23_3	pclk_sgrf
		G23_2	pclk_pmu_noc
		G23_1	pclk_intmem1
		G23_0	pclk_pmu
	xin24M	G24_11	timer15_clk
		G24_10	timer14_clk
		G24_9	timer13_clk
		G24_8	timer12_clk
		G24_7	timer11_clk
		G24_6	timer10_clk
		G24_5	timer05_clk
		G24_4	timer04_clk
		G24_3	timer03_clk
		G24_2	timer02_clk
		G24_1	timer01_clk
		G24_0	timer00_clk

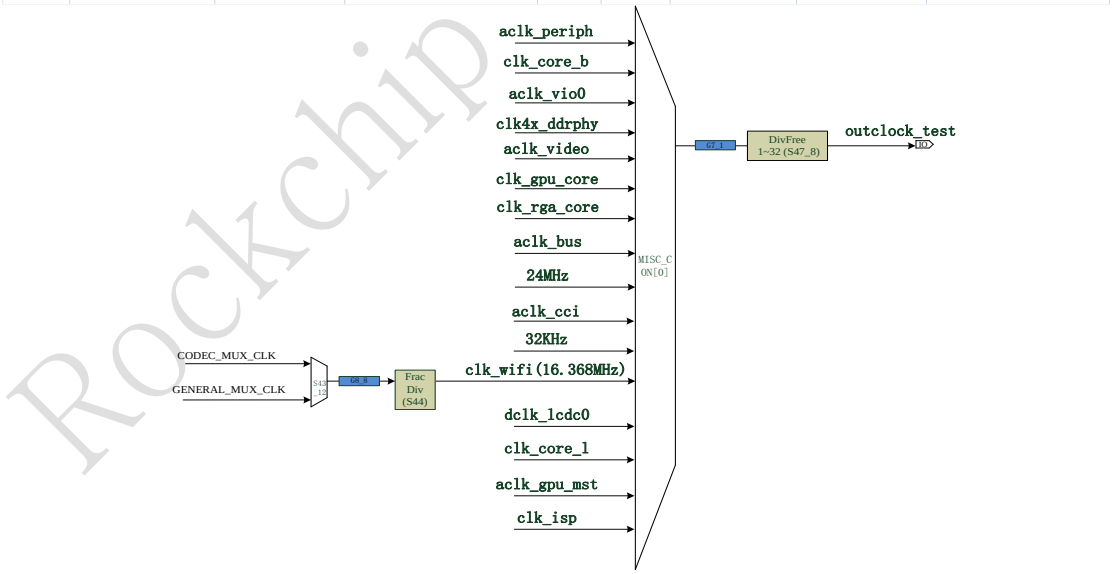

Fig. 3-5 CRU Clock Architecture Diagram 5

3.4 System Reset Solution

The following diagrams show reset architecture in this block.

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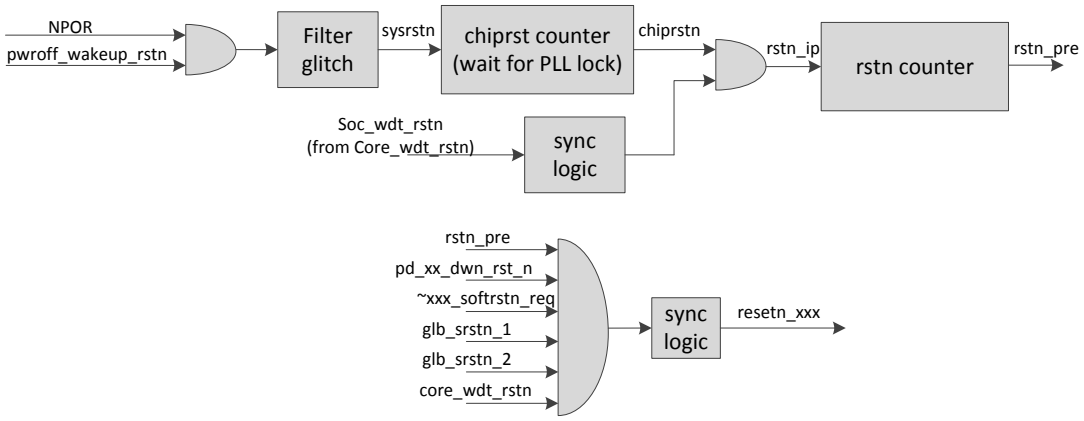

Fig. 3-6 Reset Architecture Diagram

Reset source of each reset signal includes hardware reset (NPOR), power-off mode wakeup reset(pwroff_wakeup_rstn), soc watch dog reset (soc_wdt_rstn), power domain power down reset (pd_xx_dwn_rst_n), software reset request (xxx_softrstn_req), global software reset1(glb_srstn_1), global software reset2(glb_srstn_2) and A53 core watch dog reset(core_wdt_rstn).

The 'xx' of pd_xx_dwn_rst_n represents a53_b_0, a53_b_1, a53_b_2,a53_b_3, a53_l_0, a53_l_1, a53_l_2,a53_l_3,scu_b, scu_l, cpu, peri, vio, video or gpu_0, gpu_1. The 'xxx' of resetn_xxx and xxx_softrstn_req is the module name.

Pwroff_wakeup_rstn is the reset when wakeup from the power-off mode, it will reset the all SOC logic except internal PMU.

Soc_wdt_rstn is the reset from watch-dog IP in the SoC, but core_wdt_rstn is the reset from A53 core watch-dog block.

Glb_srstn_1 and glb_srstn_2 are the global software reset by programming CRU register. When writing register CRU_GLB_SRST_FST_VALUE as 0xfdb9, glb_srstn_1 will be asserted, and when writing register CRU_GLB_SRST_SND_VALUE as 0xeca8, glb_srstn_2 will be asserted. The two software resets will be self-clear by hardware. Glb_srstn_1 will reset the all logic except PMU_SYS_REG0~3. And Glb_srstn_2 will reset the all logic except PMU_SYS_REG0~3, GRF and all GPIOs.

3.5 Function Description

There are six PLLs:

ARM PLL BIG, ARM PLL LITTLE, DDR PLL, CODEC PLL, GENERAL PLL and NEW PLL in CRU.

PLLs all can be set to slow mode or deep slow mode, directly output selectable 24MHz or 32.768kHz. When power on or changing PLL setting, we must force PLL into slow mode to ensure output stable clock.

To maximize the flexibility, some of clocks can select divider source from three PLLs (CODEC PLL, GENERAL PLLand NEW PLL).

To provide some specific frequency, another solution is integrated: fractional divider. In order to be sure the performance for divided clock, there is some usage limit, we can only get low frequency and divider factor must be larger than 20.

All clocks can be software gated and all reset can be software generated.

3.6 PLL Introduction

3.6.1 Overview

This chip uses 2.2GHz PLL for all six PLLs.The 2.2GHz PLL is a general purpose, high-performance PLL-based clock generator. The VCO operates from 440 MHz to 2200MHz. It has a programmable output frequency, which ranges from 27.5 MHz to 2200 MHz configured through a 6-bit input divider, a 13-bit feedback divider and a 4-bit output divider. Around 50% duty cycle of output clocks can be achieved by enabling the output divider. It can also be used as a clock buffer through a bypass mode that bypasses and powers down the PLL. A full power-down mode is also available.

2.2GHz PLL supports the following features:

- Fully integrated, including loop filter

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- Power supply: 1.0V single power supply
- VCO operating range: 440MHz – 2200MHz
- Output frequency range: 27.5MHz – 2200MHz
- Input frequency range: 269kHz – 2200MHz
- PFD comparison frequency range: 269kHz – 2200MHz
- Low power consumption: 3mA @ 1100MHz during normal operation
- Contains 6-bit input, 13-bit feedback and 4-bit output dividers
- Input divider value range: 1–64
- Feedback divider value range: 1–4096
- Output divider value range: 1, 2-16 (even only)
- Bandwidth adjustment of div. reference: 1–4096
- Output duty cycle: +/-5% (/1), +/-2% (/N)
- Period jitter (P-P) (max): +/-2.5% output cycle
- Reset pulse width (min): 5us
- Lock time (min allowed): 500 div. reference cycles
- Freq. overshot (full-~/half-~) (max): 40%/50%
- Ref. input jitter (long-term, P-P) (max): 2% div. reference cycle
- Reference H/L pulse width (min) : 230ps
- Bypass and Power-down mode
- Lock detector

3.6.2 Block diagram

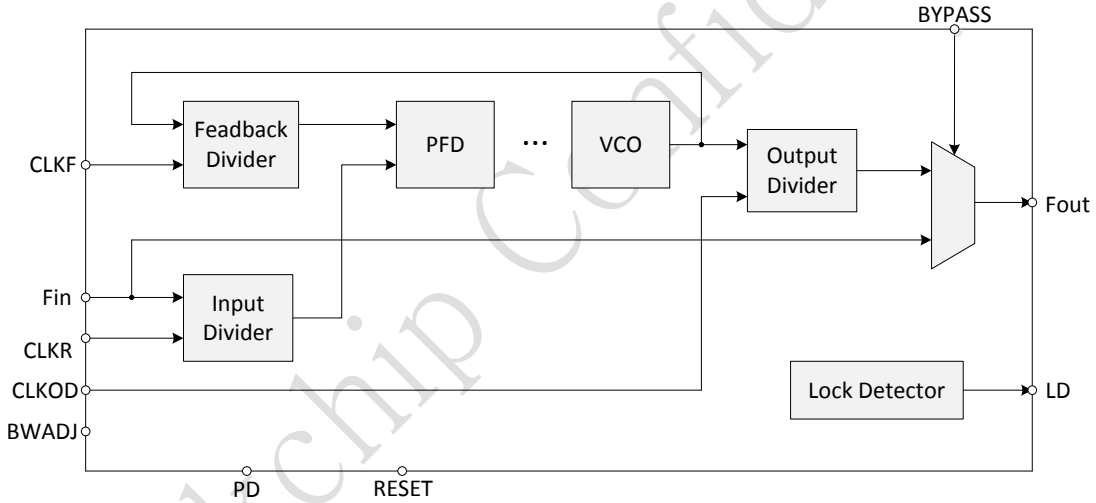

Fig. 3-7 PLL Block Diagram

3.6.3 Operation mode

A. Locked

The positive edges of the PLL feedback and reference signals are phase aligned in normal operation. Because the feedback signal is internal, NO phase relationship is guaranteed between RCLK and CLKOUT. The output clock frequency is programmable through the divider setting of CLKR[5:0], CLKF[12:0] and CLKOD[3:0].

B. Reset (RESET=1)

The PLL outputs a fixed free-running frequency in the range of 20MHz to 200MHz for a divide by 1 output depending on the specific PLL type.

C. Power-down (PWRDN=1)

All analog circuitry in the PLL is turned off so as to only dissipate leakage current. The digital dividers are not affected.

D. Bypass (BYPASS=1)

The reference input is bypassed directly to the outputs.

E. Test (TEST=1)

The reference input drives all dividers cascaded one after the other for production testing.

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3.6.4 PLL Bandwidth Adjustment

The loop bandwidth (BW) of the PLL can be adjusted using BWADJ[11:0]. The bandwidth is given by:BW = nom_BW*sqrt(NF / 2 / NB), where nom_BW is approximately given by:nom_BW = Fref / (NR*20), and Fref is the reference clock frequency. The damping factor (D) is approximately given by:D = nom_D*sqrt(NF / 2 / NB), where nom_D is approximately 1. Because the damping factor changes with bandwidth settings, the bandwidth is practically limited to:nom_BW/sqrt(2) < BW < nom_BW*sqrt(2), in order to limit the damping factor range to 0.7 - 1.4.The -3dB bandwidth (Fbw_3dB) is approximately given by:Fbw_3dB = 2.4 * nom_BW * (NF / 2 / NB). The recommended setting for NB is NF / 2, which will yield the nominal bandwidth. Note that nom_BW and nom_D are chosen to result in optimal PLL loop dynamics.

3.7 Register Description

This section describes the control/status registers of the design.

3.7.1 Register Summary

Name	Offset	Size	Reset Value	Description
CRU_APLLB_CON0	0x00000	W	0x00000b01	ARM big cluster PLL configuration register0
CRU_APLLB_CON1	0x00004	W	0x0000031f	ARM big cluster PLL configuration register1
CRU_APLLB_CON2	0x00008	W	0x0000018f	ARM big cluster PLL configuration register2
CRU_APLLB_CON3	0x0000c	W	0x00000008	ARM big cluster PLL configuration register3
CRU_APLLL_CON0	0x00010	W	0x00000b01	ARM little cluster PLL configuration register0
CRU_APLLL_CON1	0x00014	W	0x0000031f	ARM little cluster PLL configuration register1
CRU_APLLL_CON2	0x00018	W	0x0000018f	ARM little cluster PLL configuration register2
CRU_APLLL_CON3	0x0001c	W	0x00000008	ARM little cluster PLL configuration register3
CRU_DPLL_CON0	0x00020	W	0x00000b00	DDR PLL configuration register0
CRU_DPLL_CON1	0x00024	W	0x0000031f	DDR PLL configuration register1
CRU_DPLL_CON2	0x00028	W	0x0000018f	DDR PLL configuration register2
CRU_DPLL_CON3	0x0002c	W	0x00000008	DDR PLL configuration register3
CRU_CPLL_CON0	0x00030	W	0x00000101	CODEC PLL configuration register0
CRU_CPLL_CON1	0x00034	W	0x0000007c	CODEC PLL configuration register1
CRU_CPLL_CON2	0x00038	W	0x0000003e	CODEC PLL configuration register2
CRU_CPLL_CON3	0x0003c	W	0x00000008	CODEC PLL configuration register3
CRU_GPLL_CON0	0x00040	W	0x00000b01	GENERAL PLL configuration register0
CRU_GPLL_CON1	0x00044	W	0x00000251	GENERAL PLL configuration register1
CRU_GPLL_CON2	0x00048	W	0x00000128	GENERAL PLL configuration register2
CRU_GPLL_CON3	0x0004c	W	0x00000008	GENERAL PLL configuration register3
CRU_NPLL_CON0	0x00050	W	0x00000b01	NEW PLL configuration register0
CRU_NPLL_CON1	0x00054	W	0x00000251	NEW PLL configuration register1
CRU_NPLL_CON2	0x00058	W	0x00000128	NEW PLL configuration register2

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Name	Offset	Size	Reset Value	Description
CRU_NPLL_CON3	0x0005c	W	0x00000008	NEW PLL configuration register3
CRU_CLKSEL0_CON	0x00100	W	0x00000101	Internal clock select and divide register0
CRU_CLKSEL1_CON	0x00104	W	0x00000303	Internal clock select and divide register1
CRU_CLKSEL2_CON	0x00108	W	0x00000101	Internal clock select and divide register2
CRU_CLKSEL3_CON	0x0010c	W	0x00000303	Internal clock select and divide register3
CRU_CLKSEL4_CON	0x00110	W	0x00000383	Internal clock select and divide register4
CRU_CLKSEL5_CON	0x00114	W	0x00000041	Internal clock select and divide register5
CRU_CLKSEL6_CON	0x00118	W	0x00000000	Internal clock select and divide register6
CRU_CLKSEL7_CON	0x0011c	W	0x00000000	Internal clock select and divide register7
CRU_CLKSEL8_CON	0x00120	W	0x00003182	Internal clock select and divide register8
CRU_CLKSEL9_CON	0x00124	W	0x00002182	Internal clock select and divide register9
CRU_CLKSEL10_CON	0x00128	W	0x0000c707	Internal clock select and divide register10
CRU_CLKSEL11_CON	0x0012c	W	0x00000000	Internal clock select and divide register11
CRU_CLKSEL12_CON	0x00130	W	0x00000303	Internal clock select and divide register12
CRU_CLKSEL13_CON	0x00134	W	0x00000000	Internal clock select and divide register13
CRU_CLKSEL14_CON	0x00138	W	0x00000202	Internal clock select and divide register14
CRU_CLKSEL15_CON	0x0013c	W	0x00000202	Internal clock select and divide register15
CRU_CLKSEL16_CON	0x00140	W	0x00000300	Internal clock select and divide register16
CRU_CLKSEL17_CON	0x00144	W	0x00000202	Internal clock select and divide register17
CRU_CLKSEL18_CON	0x00148	W	0x00000101	Internal clock select and divide register18
CRU_CLKSEL19_CON	0x0014c	W	0x00000002	Internal clock select and divide register19
CRU_CLKSEL20_CON	0x00150	W	0x00000207	Internal clock select and divide register20
CRU_CLKSEL21_CON	0x00154	W	0x0000470f	Internal clock select and divide register21

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Name	Offset	Size	Reset Value	Description
CRU_CLKSEL22_CON	0x00158	W	0x00000002	Internal clock select and divide register22
CRU_CLKSEL23_CON	0x0015c	W	0x00000002	Internal clock select and divide register23
CRU_CLKSEL24_CON	0x00160	W	0x00000000	Internal clock select and divide register24
CRU_CLKSEL25_CON	0x00164	W	0x00001720	Internal clock select and divide register25
CRU_CLKSEL26_CON	0x00168	W	0x00002000	Internal clock select and divide register26
CRU_CLKSEL27_CON	0x0016c	W	0x00000300	Internal clock select and divide register27
CRU_CLKSEL28_CON	0x00170	W	0x0bb8ea60	Internal clock select and divide register28
CRU_CLKSEL29_CON	0x00174	W	0x00000000	Internal clock select and divide register29
CRU_CLKSEL30_CON	0x00178	W	0x00000000	Internal clock select and divide register30
CRU_CLKSEL31_CON	0x0017c	W	0x00000300	Internal clock select and divide register31
CRU_CLKSEL32_CON	0x00180	W	0x0bb8ea60	Internal clock select and divide register32
CRU_CLKSEL33_CON	0x00184	W	0x00000200	Internal clock select and divide register33
CRU_CLKSEL34_CON	0x00188	W	0x0bb8ea60	Internal clock select and divide register34
CRU_CLKSEL35_CON	0x0018c	W	0x00000200	Internal clock select and divide register35
CRU_CLKSEL36_CON	0x00190	W	0x0bb8ea60	Internal clock select and divide register36
CRU_CLKSEL37_CON	0x00194	W	0x00000100	Internal clock select and divide register37
CRU_CLKSEL38_CON	0x00198	W	0x00000000	Internal clock select and divide register38
CRU_CLKSEL39_CON	0x0019c	W	0x00000200	Internal clock select and divide register39
CRU_CLKSEL40_CON	0x001a0	W	0x0bb8ea60	Internal clock select and divide register40
CRU_CLKSEL41_CON	0x001a4	W	0x00000200	Internal clock select and divide register41
CRU_CLKSEL42_CON	0x001a8	W	0x0bb8ea60	Internal clock select and divide register42
CRU_CLKSEL43_CON	0x001ac	W	0x0000008e	Internal clock select and divide register43

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Name	Offset	Size	Reset Value	Description
CRU_CLKSEL44_CON	0x001b0	W	0x001f05dc	Internal clock select and divide register44
CRU_CLKSEL45_CON	0x001b4	W	0x00000707	Internal clock select and divide register45
CRU_CLKSEL46_CON	0x001b8	W	0x00000707	Internal clock select and divide register46
CRU_CLKSEL47_CON	0x001bc	W	0x00000083	Internal clock select and divide register47
CRU_CLKSEL48_CON	0x001c0	W	0x00000203	Internal clock select and divide register48
CRU_CLKSEL49_CON	0x001c4	W	0x00000000	reserved
CRU_CLKSEL50_CON	0x001c8	W	0x00000203	Internal clock select and divide register50
CRU_CLKSEL51_CON	0x001cc	W	0x00000203	Internal clock select and divide register51
CRU_CLKSEL52_CON	0x001d0	W	0x00000083	Internal clock select and divide register52
CRU_CLKSEL53_CON	0x001d4	W	0x00000300	Internal clock select and divide register53
CRU_CLKSEL54_CON	0x001d8	W	0x0bb8ea60	Internal clock select and divide register54
CRU_CLKSEL55_CON	0x001dc	W	0x00000002	Internal clock select and divide register55
CRU_CLKGATE0_CON	0x00200	W	0x00000000	Internal clock gating control register0
CRU_CLKGATE1_CON	0x00204	W	0x00000000	Internal clock gating control register1
CRU_CLKGATE2_CON	0x00208	W	0x00000000	Internal clock gating control register2
CRU_CLKGATE3_CON	0x0020c	W	0x00000000	Internal clock gating control register3
CRU_CLKGATE4_CON	0x00210	W	0x00000000	Internal clock gating control register4
CRU_CLKGATE5_CON	0x00214	W	0x00000000	Internal clock gating control register5
CRU_CLKGATE6_CON	0x00218	W	0x00000000	Internal clock gating control register6
CRU_CLKGATE7_CON	0x0021c	W	0x00000000	Internal clock gating control register7
CRU_CLKGATE8_CON	0x00220	W	0x00000000	Internal clock gating control register8
CRU_CLKGATE9_CON	0x00224	W	0x00000000	
CRU_CLKGATE10_CON	0x00228	W	0x00000000	
CRU_CLKGATE11_CON	0x0022c	W	0x00000000	
CRU_CLKGATE12_CON	0x00230	W	0x00000000	
CRU_CLKGATE13_CON	0x00234	W	0x00000000	
CRU_CLKGATE14_CON	0x00238	W	0x00000000	

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Name	Offset	Size	Reset Value	Description
CRU_CLKGATE15_CON	0x0023c	W	0x00000000	
CRU_CLKGATE16_CON	0x00240	W	0x00000000	
CRU_CLKGATE17_CON	0x00244	W	0x00000000	
CRU_CLKGATE18_CON	0x00248	W	0x00000000	
CRU_CLKGATE19_CON	0x0024c	W	0x00000000	
CRU_CLKGATE20_CON	0x00250	W	0x00000000	
CRU_CLKGATE21_CON	0x00254	W	0x00000000	
CRU_CLKGATE22_CON	0x00258	W	0x00000000	
CRU_CLKGATE23_CON	0x0025c	W	0x00000000	
CRU_CLKGATE24_CON	0x00260	W	0x00000000	
CRU_GLB_SRST_FST_VALUE	0x00280	W	0x00000000	The first global software reset config value
CRU_GLB_SRST_SND_VALUE	0x00284	W	0x00000000	The second global software reset config value
CRU_SOFTRST0_CON	0x00300	W	0x00000000	Internal software reset control register0
CRU_SOFTRST1_CON	0x00304	W	0x00003000	Internal software reset control register1
CRU_SOFTRST2_CON	0x00308	W	0x00000000	Internal software reset control register2
CRU_SOFTRST3_CON	0x0030c	W	0x00000000	Internal software reset control register3
CRU_SOFTRST4_CON	0x00310	W	0x00000000	Internal software reset control register4
CRU_SOFTRST5_CON	0x00314	W	0x00000000	Internal software reset control register5
CRU_SOFTRST6_CON	0x00318	W	0x00000000	Internal software reset control register6
CRU_SOFTRST7_CON	0x0031c	W	0x00000000	Internal software reset control register7
CRU_SOFTRST8_CON	0x00320	W	0x00000000	Internal software reset control register8
CRU_SOFTRST9_CON	0x00324	W	0x00000000	Internal software reset control register9

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Name	Offset	Size	Reset Value	Description
CRU_SOFTRST10_CON	0x00328	W	0x00000000	Internal software reset control register10
CRU_SOFTRST11_CON	0x0032c	W	0x00000000	Internal software reset control register11
CRU_SOFTRST12_CON	0x00330	W	0x00000000	Internal software reset control register0
CRU_SOFTRST13_CON	0x00334	W	0x00000000	Internal software reset control register0
CRU_SOFTRST14_CON	0x00338	W	0x00000000	Internal software reset control register0
CRU_MISC_CON	0x00380	W	0x00000000	SCU control register
CRU_GLB_CNT_TH	0x00384	W	0x00000064	global reset wait counter threshold
CRU_GLB_RST_CON	0x00388	W	0x00000000	global reset trigger select
CRU_GLB_RST_ST	0x0038c	W	0x00000000	global reset status
CRU_SDMMC_CON0	0x00400	W	0x00000004	sdmmc control0
CRU_SDMMC_CON1	0x00404	W	0x00000000	sdmmc control1
CRU_SDIO0_CON0	0x00408	W	0x00000004	sdio0 control0
CRU_SDIO0_CON1	0x0040c	W	0x00000000	sdio0 control1
CRU_SDIO1_CON0	0x00410	W	0x00000004	sdio1 control0
CRU_SDIO1_CON1	0x00414	W	0x00000000	sdio1 control1
CRU_EMMC_CON0	0x00418	W	0x00000004	emmc control0
CRU_EMMC_CON1	0x0041c	W	0x00000000	emmc control1

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

3.7.2 Detail Register Description

CRU_APLLB_CON0
Address: Operational Base + offset (0x00000)
ARM big cluster PLL configuration register0

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RO	0x0	reserved
13:8	RW	0x0b	clkr PLL CLKR factor control NR = CLKR + 1 NR: 1-64
7:4	RO	0x0	reserved
3:0	RW	0x1	clkod PLL CLKOD factor control NO = CLKOD + 1 NO: 1, 2-16 (even only)

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CRU_APLLB_CON1
Address: Operational Base + offset (0x00004)
ARM big cluster PLL configuration register1

Bit	Attr	Reset Value	Description
31	RO	0x0	lock PLL lock status 1'b0: unlock 1'b1: lock
30:13	RO	0x0	reserved
12:0	RW	0x031f	clkf PLL CLKF factor control NF = CLKF + 1 NF: 1-4096

CRU_APLLB_CON2
Address: Operational Base + offset (0x00008)
ARM big cluster PLL configuration register2

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x18f	bwadj PLL loop bandwidth adjust NB = BWADJ + 1

CRU_APLLB_CON3
Address: Operational Base + offset (0x0000c)
ARM big cluster PLL configuration register3

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved
9:8	RW	0x0	apllb_work_mode ARM big cluster PLL work mode select 2'b00: Slow mode, clock from external 24MHz OSC (default) 2'b01: Normal mode, clock from PLL output 2'b10: Deep slow mode, clock from external 32.768kHz
7:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5	RW	0x0	reset PLL reset control 1'b0: normal 1'b1: reset
4	RW	0x0	test PLL test control 1'b0: normal 1'b1: test mode
3	RW	0x1	ensat PLL saturation behavior enable 1'b0: disable 1'b1: enable
2	RW	0x0	fasten PLL enable fast locking circuit 1'b0: disable 1'b1: enable
1	RW	0x0	power_down PLL power down control 1'b0: no power down 1'b1: power down
0	RW	0x0	bypass PLL bypass mode control 1'b0: no bypass 1'b1: bypass

CRU_APLLL_CON0
Address: Operational Base + offset (0x00010)
ARM little cluster PLL configuration register0

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RO	0x0	reserved
13:8	RW	0x0b	clkr PLL CLKR factor control NR = CLKR + 1 NR: 1-64
7:4	RO	0x0	reserved
3:0	RW	0x1	clkod PLL CLKOD factor control NO = CLKOD + 1 NO: 1, 2-16 (even only)

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CRU_APLLL_CON1
Address: Operational Base + offset (0x00014)
ARM little cluster PLL configuration register1

Bit	Attr	Reset Value	Description
31	RO	0x0	lock PLL lock status 1'b0: unlock 1'b1: lock
30:13	RO	0x0	reserved
12:0	RW	0x031f	clkf PLL CLKF factor control NF = CLKF + 1 NF: 1-4096

CRU_APLLL_CON2
Address: Operational Base + offset (0x00018)
ARM little cluster PLL configuration register2

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x18f	bwadj PLL loop bandwidth adjust NB = BWADJ + 1

CRU_APLLL_CON3
Address: Operational Base + offset (0x0001c)
ARM little cluster PLL configuration register3

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved
9:8	RW	0x0	aplll_work_mode ARM little cluster PLL work mode select 2'b00: Slow mode, clock from external 24MHz OSC (default) 2'b01: Normal mode, clock from PLL output 2'b10: Deep slow mode, clock from external 32.768kHz
7:6	RO	0x0	reserved
5	RW	0x0	reset PLL reset control 1'b0: normal 1'b1: reset

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Bit	Attr	Reset Value	Description
4	RW	0x0	test PLL test control 1'b0: normal 1'b1: test mode
3	RW	0x1	ensat PLL saturation behavior enable 1'b0: disable 1'b1: enable
2	RW	0x0	fasten PLL enable fast locking circuit 1'b0: disable 1'b1: enable
1	RW	0x0	power_down PLL power down control 1'b0: no power down 1'b1: power down
0	RW	0x0	bypass PLL bypass mode control 1'b0: no bypass 1'b1: bypass

CRU_DPLL_CON0
Address: Operational Base + offset (0x00020)
DDR PLL configuration register0

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RO	0x0	reserved
13:8	RW	0x0b	clkr PLL CLKR factor control NR = CLKR + 1 NR: 1-64
7:4	RO	0x0	reserved
3:0	RW	0x0	clkod PLL CLKOD factor control NO = CLKOD + 1 NO: 1, 2-16 (even only)

CRU_DPLL_CON1
Address: Operational Base + offset (0x00024)
DDR PLL configuration register1

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31	RO	0x0	lock PLL lock status 1'b0: unlock 1'b1: lock
30:13	RO	0x0	reserved
12:0	RW	0x031f	clkf PLL CLKF factor control NF = CLKF + 1 NF: 1-4096

CRU_DPLL_CON2
Address: Operational Base + offset (0x00028)
DDR PLL configuration register2

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x18f	bwadj PLL loop bandwidth adjust NB = BWADJ + 1

CRU_DPLL_CON3
Address: Operational Base + offset (0x0002c)
DDR PLL configuration register3

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved
9:8	RW	0x0	dpll_work_mode DDR PLL work mode select 2'b00: Slow mode, clock from external 24MHz OSC (default) 2'b01: Normal mode, clock from PLL output 2'b10: Deep slow mode, clock from external 32.768kHz
7:6	RO	0x0	reserved
5	RW	0x0	reset PLL reset control 1'b0: normal 1'b1: reset
4	RW	0x0	test PLL test control 1'b0: normal 1'b1: test mode

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Bit	Attr	Reset Value	Description
3	RW	0x1	ensat PLL saturation behavior enable 1'b0: disable 1'b1: enable
2	RW	0x0	fasten PLL enable fast locking circuit 1'b0: disable 1'b1: enable
1	RW	0x0	power_down PLL power down control 1'b0: no power down 1'b1: power down
0	RW	0x0	bypass PLL bypass mode control 1'b0: no bypass 1'b1: bypass

CRU_CPLL_CON0
Address: Operational Base + offset (0x00030)
CODEC PLL configuration register0

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RO	0x0	reserved
13:8	RW	0x01	clkr PLL CLKR factor control NR = CLKR + 1 NR: 1-64
7:4	RO	0x0	reserved
3:0	RW	0x1	clkod PLL CLKOD factor control NO = CLKOD + 1 NO: 1, 2-16 (even only)

CRU_CPLL_CON1
Address: Operational Base + offset (0x00034)
CODEC PLL configuration register1

Bit	Attr	Reset Value	Description
31	RO	0x0	lock PLL lock status 1'b0: unlock 1'b1: lock

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Bit	Attr	Reset Value	Description
30:13	RO	0x0	reserved
12:0	RW	0x007c	clkf PLL CLKF factor control NF = CLKF + 1 NF: 1-4096

CRU_CPLL_CON2
Address: Operational Base + offset (0x00038)
CODEC PLL configuration register2

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x03e	bwadj PLL loop bandwidth adjust NB = BWADJ + 1

CRU_CPLL_CON3
Address: Operational Base + offset (0x0003c)
CODEC PLL configuration register3

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved
9:8	RW	0x0	cpll_work_mode CODEC PLL work mode select 2'b00: Slow mode, clock from external 24MHz OSC (default) 2'b01: Normal mode, clock from PLL output 2'b10: Deep slow mode, clock from external 32.768kHz
7:6	RO	0x0	reserved
5	RW	0x0	reset PLL reset control 1'b0: normal 1'b1: reset
4	RW	0x0	test PLL test control 1'b0: normal 1'b1: test mode
3	RW	0x1	ensat PLL saturation behavior enable 1'b0: disable 1'b1: enable

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Bit	Attr	Reset Value	Description
2	RW	0x0	fasten PLL enable fast locking circuit 1'b0: disable 1'b1: enable
1	RW	0x0	power_down PLL power down control 1'b0: no power down 1'b1: power down
0	RW	0x0	bypass PLL bypass mode control 1'b0: no bypass 1'b1: bypass

CRU_GPLL_CON0
Address: Operational Base + offset (0x00040)
GENERAL PLL configuration register0

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RO	0x0	reserved
13:8	RW	0x0b	clkr PLL CLKR factor control NR = CLKR + 1 NR: 1-64
7:4	RO	0x0	reserved
3:0	RW	0x1	clkod PLL CLKOD factor control NO = CLKOD + 1 NO: 1, 2-16 (even only)

CRU_GPLL_CON1
Address: Operational Base + offset (0x00044)
GENERAL PLL configuration register1

Bit	Attr	Reset Value	Description
31	RO	0x0	lock
30:13	RO	0x0	reserved
12:0	RW	0x0251	clkf PLL CLKF factor control NF = CLKF + 1 NF: 1-4096

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CRU_GPLL_CON2
Address: Operational Base + offset (0x00048)
GENERAL PLL configuration register2

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x128	bwadj PLL loop bandwidth adjust NB = BWADJ + 1

CRU_GPLL_CON3
Address: Operational Base + offset (0x0004c)
GENERAL PLL configuration register3

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved
9:8	RW	0x0	gpll_work_mode GENERAL PLL work mode select 2'b00: Slow mode, clock from external 24MHz OSC (default) 2'b01: Normal mode, clock from PLL output 2'b10: Deep slow mode, clock from external 32.768kHz
7:6	RO	0x0	reserved
5	RW	0x0	reset PLL reset control 1'b0: normal 1'b1: reset
4	RW	0x0	test PLL test control 1'b0: normal 1'b1: test mode
3	RW	0x1	ensat PLL saturation behavior enable 1'b0: disable 1'b1: enable
2	RW	0x0	fasten PLL enable fast locking circuit 1'b0: disable 1'b1: enable
1	RW	0x0	power_down PLL power down control 1'b0: no power down 1'b1: power down

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Bit	Attr	Reset Value	Description
0	RW	0x0	bypass PLL bypass mode control 1'b0: no bypass 1'b1: bypass

CRU_NPLL_CON0
Address: Operational Base + offset (0x00050)
NEW PLL configuration register0

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RO	0x0	reserved
13:8	RW	0x0b	clkr PLL CLKR factor control NR = CLKR + 1 NR: 1-64
7:4	RO	0x0	reserved
3:0	RW	0x1	clkod PLL CLKOD factor control NO = CLKOD + 1 NO: 1, 2-16 (even only)

CRU_NPLL_CON1
Address: Operational Base + offset (0x00054)
NEW PLL configuration register1

Bit	Attr	Reset Value	Description
31	RO	0x0	lock
30:13	RO	0x0	reserved
12:0	RW	0x0251	clkf PLL CLKF factor control NF = CLKF + 1 NF: 1-4096

CRU_NPLL_CON2
Address: Operational Base + offset (0x00058)
NEW PLL configuration register2

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x128	bwadj PLL loop bandwidth adjust NB = BWADJ + 1

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CRU_NPLL_CON3
Address: Operational Base + offset (0x0005c)
NEW PLL configuration register3

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved
9:8	RW	0x0	npll_work_mode NEW PLL work mode select 2'b00: Slow mode, clock from external 24MHz OSC (default) 2'b01: Normal mode, clock from PLL output 2'b10: Deep slow mode, clock from external 32.768kHz
7:6	RO	0x0	reserved
5	RW	0x0	reset PLL reset control 1'b0: normal 1'b1: reset
4	RW	0x0	test PLL test control 1'b0: normal 1'b1: test mode
3	RW	0x1	ensat PLL saturation behavior enable 1'b0: disable 1'b1: enable
2	RW	0x0	fasten PLL enable fast locking circuit 1'b0: disable 1'b1: enable
1	RW	0x0	power_down PLL power down control 1'b0: no power down 1'b1: power down
0	RW	0x0	bypass PLL bypass mode control 1'b0: no bypass 1'b1: bypass

CRU_CLKSEL0_CON
Address: Operational Base + offset (0x00100)
Internal clock select and divide register0

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved
12:8	RW	0x01	aclkm_core_b_div_con Control A53 big cluster AXI clock divider frequency clk=clk_src/(div_con+1)
7	RW	0x0	core_b_clk_pll_sel A53 big cluster clock pll source selection 1'b0: select ARMB PLL 1'b1: select GENERAL PLL
6:5	RO	0x0	reserved
4:0	RW	0x01	a53_core_b_div_con Control A53 big cluster clock divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL1_CON
Address: Operational Base + offset (0x00104)
Internal clock select and divide register1

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved
12:8	RW	0x03	pclk_dbg_b_div_con Control A53 big cluster external APB clock divider frequency clk=clk_src/(div_con+1)
7:5	RO	0x0	reserved
4:0	RW	0x03	atclk_core_b_div_con Control A53 big cluster ATB clock divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL2_CON
Address: Operational Base + offset (0x00108)
Internal clock select and divide register2

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved
12:8	RW	0x01	aclkm_core_l_div_con Control A53 little cluster AXI clock divider frequency clk=clk_src/(div_con+1)
7	RW	0x0	core_l_clk_pll_sel A53 little cluster clock pll source selection 1'b0: select ARML PLL 1'b1: select GENERAL PLL
6:5	RO	0x0	reserved
4:0	RW	0x01	a53_core_l_div_con Control A53 little cluster clock divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL3_CON
Address: Operational Base + offset (0x0010c)
Internal clock select and divide register3

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved
12:8	RW	0x03	pclk_dbg_l_div_con Control A53 little cluster external APB clock divider frequency clk=clk_src/(div_con+1)
7:5	RO	0x0	reserved
4:0	RW	0x03	atclk_core_l_div_con Control A53 little cluster ATB clock divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL4_CON
Address: Operational Base + offset (0x00110)
Internal clock select and divide register4

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved
12:8	RW	0x03	clkin_trace_div_con Control trace clock in divider frequency clk=clk_src/(div_con+1)
7:6	RW	0x2	cs_clk_pll_sel coresight clock pll source selection 2'b00 : select ARMB PLL 2'b01 : select ARML PLL 2'b10 : select GENERAL PLL
5	RO	0x0	reserved
4:0	RW	0x03	a53_cs_div_con Control coresight clock divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL5_CON
Address: Operational Base + offset (0x00114)
Internal clock select and divide register5

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	RO	0x0	reserved
7:6	RW	0x1	cci_aclk_pll_sel CCI clock pll source selection 2'b00 : select CODEC PLL 2'b01 : select GENERAL PLL 2'b10 : select USBPHY PLL 2'b11 : select NEW PLL
5	RO	0x0	reserved
4:0	RW	0x01	cci_aclk_div_con Control CCI clock divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL6_CON
Address: Operational Base + offset (0x00118)
Internal clock select and divide register6

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	reserved

CRU_CLKSEL7_CON
Address: Operational Base + offset (0x0011c)
Internal clock select and divide register7

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	reserved

CRU_CLKSEL8_CON
Address: Operational Base + offset (0x00120)
Internal clock select and divide register8

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RO	0x0	reserved
14:12	RW	0x3	cpu_pclk_div_con Control pd_bus APB clock divider frequency clk=clk_src/(div_con+1)
11:10	RO	0x0	reserved
9:8	RW	0x1	cpu_hclk_div_con Control pd_bus AHB clock divider frequency clk=clk_src/(div_con+1)
7	RW	0x1	cpu_clk_pll_sel pd_bus AXI clock pll source selection 1'b0 : select CODEC PLL 1'b1 : select GENERAL PLL
6:5	RO	0x0	reserved
4:0	RW	0x02	cpu_div_con Control pd_bus AXI clock divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL9_CON
Address: Operational Base + offset (0x00124)
Internal clock select and divide register9

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit

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Bit	Attr	Reset Value	Description
15:14	RO	0x0	reserved
13:12	RW	0x2	periph_pclk_div_con Control pd_peri APB clock divider frequency 2'b00: aclk_periph:pclk_periph = 1:1 2'b01: aclk_periph:pclk_periph = 2:1 2'b10: aclk_periph:pclk_periph = 4:1 2'b11: aclk_periph:pclk_periph = 8:1
11:10	RO	0x0	reserved
9:8	RW	0x1	periph_hclk_div_con Control pd_peri AHB clock divider frequency 2'b00: aclk_periph:hclk_periph = 1:1 2'b01: aclk_periph:hclk_periph = 2:1 2'b10: aclk_periph:hclk_periph = 4:1
7	RW	0x1	periph_pll_sel pd_peri AXI clock pll source selection 1'b0 : select CODEC PLL 1'b1 : select GENERAL PLL
6:5	RO	0x0	reserved
4:0	RW	0x02	periph_aclk_div_con Control pd_peri AXI clock divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL10_CON
Address: Operational Base + offset (0x00128)
Internal clock select and divide register10

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RW	0x3	crypto_clk_div_con Control crypto clock divider frequency clk=clk_src/(div_con+1)
13	RO	0x0	reserved
12:8	RW	0x07	alive_pclk_div_con Control pd_alive APB clock divider frequency clk=clk_src/(div_con+1)
7:5	RO	0x0	reserved
4:0	RW	0x07	pmu_pclk_div_con Control pd_pmu APB clock divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL11_CON
Address: Operational Base + offset (0x0012c)

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Internal clock select and divide register11

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	reserved

CRU_CLKSEL12_CON
Address: Operational Base + offset (0x00130)
Internal clock select and divide register12

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:11	RO	0x0	reserved
10:8	RW	0x3	mcustclk_div_con Control MCU STCLK clock divider frequency clk=clk_src/(div_con+1)
7	RW	0x0	mcu_clk_pll_sel MCU clock pll source selection 1'b0 : select CODEC PLL 1'b1 : select GENERAL PLL
6:5	RO	0x0	reserved
4:0	RW	0x03	mcu_div_con Control MCU clock divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL13_CON
Address: Operational Base + offset (0x00134)
Internal clock select and divide register13

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:9	RO	0x0	reserved
8	RW	0x0	usbphy_480m_sel USBPHY 480MHz clock select 1'b0 : select 24MHz clock 1'b1 : select 480MHz clock
7:5	RO	0x0	reserved
4	RW	0x0	ddr_clk_pll_sel DDR clock pll source selection 1'b0 : select DDR PLL 1'b1 : select GENERAL PLL
3:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	ddr_div_con Control DDR clock divider frequency clk=clk_src/(div_con+1)(0,1,3 only)

CRU_CLKSEL14_CON
Address: Operational Base + offset (0x00138)
Internal clock select and divide register14

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RO	0x0	reserved
14	RW	0x0	gpu_aclk_pll_sel GPU AXI clock pll source selection 1'b0 : select CODEC PLL 1'b1 : select GENERAL PLL
13	RO	0x0	reserved
12:8	RW	0x02	gpu_aclk_mst_div_con Control GPU master aclk(mem_clk->gpu_mem) divider frequency clk=clk_src/(div_con+1)
7:6	RW	0x0	gpu_clk_core_pll_sel GPU core clock pll source selection 2'b00 : select CODEC PLL 2'b01 : select GENERAL PLL 2'b10 : select USBPHY PLL 2'b11 : select NEW PLL
5	RO	0x0	reserved
4:0	RW	0x02	gpu_clk_core_div_con Control GPU core clock divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL15_CON
Address: Operational Base + offset (0x0013c)
Internal clock select and divide register15

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit

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Bit	Attr	Reset Value	Description
15:14	RW	0x0	vdpu_aclk_pll_sel VDPU AXI clock pll source selection 2'b00 : select CODEC PLL 2'b01 : select GENERAL PLL 2'b10 : select NEW PLL 2'b11 : select USBPHY PLL
13	RO	0x0	reserved
12:8	RW	0x02	vdpu_aclk_div_con Control VDPU AXI clock divider frequency clk=clk_src/(div_con+1)
7:6	RW	0x0	vepu_aclk_pll_sel VEPU AXI clock pll source selection 2'b00 : select CODEC PLL 2'b01 : select GENERAL PLL 2'b10 : select NEW PLL 2'b11 : select USBPHY PLL
5	RO	0x0	reserved
4:0	RW	0x02	vepu_aclk_div_con Control VEPU AXI clock divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL16_CON
Address: Operational Base + offset (0x00140)
Internal clock select and divide register16

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved
12:8	RW	0x03	gpu_aclk_slv_div_con Control GPU slave aclk(sys_clk->gpu_cfg) divider frequency clk=clk_src/(div_con+1)
7:0	RO	0x0	reserved

CRU_CLKSEL17_CON
Address: Operational Base + offset (0x00144)
Internal clock select and divide register17

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RW	0x0	hevc_core_pll_sel HEVC core clock pll source selection 2'b00 : select CODEC PLL 2'b01 : select GENERAL PLL 2'b10 : select NEW PLL 2'b11 : select USBPHY PLL
13	RO	0x0	reserved
12:8	RW	0x02	hevc_core_clk_div_con Control HEVC core clock divider frequency clk=clk_src/(div_con+1)
7:6	RW	0x0	hevc_cabac_pll_sel HEVC cabac clock pll source selection 2'b00 : select CODEC PLL 2'b01 : select GENERAL PLL 2'b10 : select NEW PLL 2'b11 : select USBPHY PLL
5	RO	0x0	reserved
4:0	RW	0x02	hevc_cabac_clk_div_con Control HEVC cabac clock divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL18_CON

Address: Operational Base + offset (0x00148)

Internal clock select and divide register18

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RW	0x0	rga_aclk_pll_sel RGA AXI clock pll source selection 2'b00 : select CODEC PLL 2'b01 : sleect GENERAL PLL 2'b1x : select USBPHY PLL
13	RO	0x0	reserved
12:8	RW	0x01	rga_aclk_div_con Control RGA AXI clock divider frequency clk=clk_src/(div_con+1)

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Bit	Attr	Reset Value	Description
7:6	RW	0x0	rga_clk_pll_sel RGA clock pll source selection 2'b00 : select CODEC PLL 2'b01 : sleect GENERAL PLL 2'b1x : select USBPHY PLL
5	RO	0x0	reserved
4:0	RW	0x01	rga_clk_div_con Control RGA clock divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL19_CON
Address: Operational Base + offset (0x0014c)
Internal clock select and divide register19

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	RO	0x0	reserved
7:6	RW	0x0	vio0_aclk_pll_sel VOP AXI clock pll source selection 2'b00 : select CODEC PLL 2'b01 : sleect GENERAL PLL 2'b1x : select USBPHY PLL
5	RO	0x0	reserved
4:0	RW	0x02	vio0_alk_div_con Control VOP AXI clock divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL20_CON
Address: Operational Base + offset (0x00150)
Internal clock select and divide register20

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved
9:8	RW	0x2	vop_dclk_pll_sel VOP clock pll source selection 2'b00 : select CODEC PLL 2'b01 : select GENERAL PLL 2'b10 : select NEW PLL

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Bit	Attr	Reset Value	Description
7:0	RW	0x07	vop0_dclk_div_con Control VOP0 clock divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL21_CON
Address: Operational Base + offset (0x00154)
Internal clock select and divide register21

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	cif_pll_sel CIF clock pll source selection 1'b0: select codec PLL 1'b1: select general PLL
14	RW	0x1	cif_clk_sel CIF clock output selection 1'b0: select PLL divout 1'b1: select 24MHz
13	RW	0x0	cif_clkin_inv_sel CIF clkin invert selection 1'b0: normal 1'b1: invert
12:8	RW	0x07	cif_div_con cif clock divider frequency clk=clk_src/(div_con+1)
7	RO	0x0	reserved
6	RW	0x0	isp_clkin_inv_sel ISP clkin invert selection 1'b0: normal 1'b1: invert
5	RO	0x0	reserved
4:0	RW	0x0f	hclk_vio_div_con Control pd_vio AHB clock divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL22_CON
Address: Operational Base + offset (0x00158)
Internal clock select and divide register22

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	RO	0x0	reserved
7:6	RW	0x0	isp_pll_sel ISP clock pll source selection 2'b00 : select CODEC PLL 2'b01 : select GENERAL PLL 2'b1x : select NEW PLL
5:0	RW	0x02	isp_div_con Control isp divider freuency clk=clk_src/(div_con+1)

CRU_CLKSEL23_CON
Address: Operational Base + offset (0x0015c)
Internal clock select and divide register23

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:9	RO	0x0	reserved
8	RW	0x0	edp_24m_sel EDP 24M clock source selection 1'b0 : select xin24m 1'b1 : 1'b0
7:6	RW	0x0	edp_pll_sel EDP clock pll source selection 2'b00 : select CODEC PLL 2'b01 : select GENERAL PLL 2'b1x : select NEW PLL
5:0	RW	0x02	edp_div_con Control edp divider freuency clk=clk_src/(div_con+1)

CRU_CLKSEL24_CON
Address: Operational Base + offset (0x00160)
Internal clock select and divide register24

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	reserved

CRU_CLKSEL25_CON

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Address: Operational Base + offset (0x00164)
Internal clock select and divide register25

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	RW	0x17	saradc_div_con Control SARADC divider freuency clk=clk_src/(div_con+1)
7:6	RO	0x0	reserved
5:0	RW	0x20	tsadc_div_con Control TSADC divider freuency clk=clk_src/(div_con+1)

CRU_CLKSEL26_CON
Address: Operational Base + offset (0x00168)
Internal clock select and divide register26

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RO	0x0	reserved
13:12	RW	0x2	hsicphy_pll_sel HSIC PHY 480Mhz clock pll source selection 2'b00 : select CODEC PLL 2'b01 : select GENERAL PLL 2'b1x : select USB PHY PLL
11:10	RO	0x0	reserved
9:8	RW	0x0	hsic_usbphy_480m_sel usb480m for hsic 2'bx0 : select usbotg_clk480_out PLL clock 2'bx1 : 1'b0
7:0	RO	0x0	reserved

CRU_CLKSEL27_CON
Address: Operational Base + offset (0x0016c)
Internal clock select and divide register27

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	i2s_out_sel Control I2S output clock frequency selection 1'b0: select clk_i2s 1'b1: select 12MHz
14:13	RO	0x0	reserved
12	RW	0x0	i2s_pll_sel I2S clock pll source selection 1'b0 : select CODEC PLL 1'b1 : select GENERAL PLL
11:10	RO	0x0	reserved
9:8	RW	0x3	i2s_clk_sel Control I2S clock work frequency selection 2'b00: select divider ouput from pll divider 2'b01: select divider ouput from fraction divider 2'b10: select clock from IO input 2'b11: select 12MHz from osc input
7	RO	0x0	reserved
6:0	RW	0x00	i2s_pll_div_con Control I2S divider freuency clk=clk_src/(div_con+1)

CRU_CLKSEL28_CON
Address: Operational Base + offset (0x00170)
Internal clock select and divide register28

Bit	Attr	Reset Value	Description
31:0	RW	0x0bb8ea60	i2s_frac_div_con Control I2S fraction divider frequency High 16-bit for numerator Low 16-bit for denominator

CRU_CLKSEL29_CON
Address: Operational Base + offset (0x00174)
Internal clock select and divide register29

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	reserved

CRU_CLKSEL30_CON
Address: Operational Base + offset (0x00178)
Internal clock select and divide register30

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	reserved

CRU_CLKSEL31_CON
Address: Operational Base + offset (0x0017c)
Internal clock select and divide register31

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved
12	RW	0x0	spdif_8ch_pll_sel SPDIF clock pll source selection 1'b0 : select CODEC PLL 1'b1 : select GENERAL PLL
11:10	RO	0x0	reserved
9:8	RW	0x3	spdif_8ch_clk_sel Control SPDIF clock work frequency selection 2'b00: select divider ouput from pll divider 2'b01: select divider ouput from fraction divider 2'b10: select clock from IO input 2'b11: select 12MHz from osc input
7	RO	0x0	reserved
6:0	RW	0x00	spdif_8ch_pll_div_con Control SPDIF divider freuency clk=clk_src/(div_con+1)

CRU_CLKSEL32_CON
Address: Operational Base + offset (0x00180)
Internal clock select and divide register32

Bit	Attr	Reset Value	Description
31:0	RW	0x0bb8ea60	spdif_8ch_frac_div_con Control SPDIF fraction divider frequency High 16-bit for numerator Low 16-bit for denominator

CRU_CLKSEL33_CON
Address: Operational Base + offset (0x00184)
Internal clock select and divide register33

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RO	0x0	reserved
13:12	RW	0x0	uart0_src_sel UART0 clock pll source selection 2'b00 : select CODEC PLL 2'b01 : select GENERAL PLL 2'b1x : select USBPHY PLL
11:10	RO	0x0	reserved
9:8	RW	0x2	uart0_clk_sel Control UART0 clock work frequency selection 2'b00: select divider ouput from pll divider 2'b01: select divider ouput from fraction divider 2'b1x: select 24MHz from osc input
7	RO	0x0	reserved
6:0	RW	0x00	uart0_div_con Control UART0 divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL34_CON
Address: Operational Base + offset (0x00188)
Internal clock select and divide register34

Bit	Attr	Reset Value	Description
31:0	RW	0x0bb8ea60	uart0_frac_div_con Control UART0 fraction divider frequency High 16-bit for numerator Low 16-bit for denominator

CRU_CLKSEL35_CON
Address: Operational Base + offset (0x0018c)
Internal clock select and divide register35

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
12	RW	0x0	uart_pll_sel UART1 / UART2 / UART3 / UART4 clock pll source selection 1'b0 : select CODEC PLL 1'b1 : select GENERAL PLL
11:10	RO	0x0	reserved
9:8	RW	0x2	uart1_clk_sel Control UART1 clock work frequency selection 2'b00: select divider ouput from pll divider 2'b01: select divider ouput from fraction divider 2'b1x: select 24MHz from osc input
7	RO	0x0	reserved
6:0	RW	0x00	uart1_div_con Control UART1 divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL36_CON
Address: Operational Base + offset (0x00190)
Internal clock select and divide register36

Bit	Attr	Reset Value	Description
31:0	RW	0x0bb8ea60	uart1_frac_div_con Control UART1 fraction divider frequency High 16-bit for numerator Low 16-bit for denominator

CRU_CLKSEL37_CON
Address: Operational Base + offset (0x00194)
Internal clock select and divide register37

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:9	RO	0x0	reserved
8	RW	0x1	uart2_clk_sel Control UART2 clock work frequency selection 1'b0: select divider ouput from pll divider 1'b1: select 24MHz from osc input
7	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
6:0	RW	0x00	uart2_div_con Control UART2 divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL38_CON
Address: Operational Base + offset (0x00198)
Internal clock select and divide register38

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	reserved

CRU_CLKSEL39_CON
Address: Operational Base + offset (0x0019c)
Internal clock select and divide register39

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved
9:8	RW	0x2	uart3_clk_sel Control UART3 clock work frequency selection 2'b00: select divider ouput from pll divider 2'b01: select divider ouput from fraction divider 2'b1x: select 24MHz from osc input
7	RO	0x0	reserved
6:0	RW	0x00	uart3_div_con Control UART3 divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL40_CON
Address: Operational Base + offset (0x001a0)
Internal clock select and divide register40

Bit	Attr	Reset Value	Description
31:0	RW	0x0bb8ea60	uart3_frac_div_con Control UART3 fraction divider frequency High 16-bit for numerator Low 16-bit for denominator

CRU_CLKSEL41_CON
Address: Operational Base + offset (0x001a4)
Internal clock select and divide register41

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved
9:8	RW	0x2	uart4_clk_sel Control UART4 clock work frequency selection 2'b00: select divider ouput from pll divider 2'b01: select divider ouput from fraction divider 2'b1x: select 24MHz from osc input
7	RO	0x0	reserved
6:0	RW	0x00	uart4_div_con Control UART4 divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL42_CON
Address: Operational Base + offset (0x001a8)
Internal clock select and divide register42

Bit	Attr	Reset Value	Description
31:0	RW	0x0bb8ea60	uart4_frac_div_con Control UART4 fraction divider frequency High 16-bit for numerator Low 16-bit for denominator

CRU_CLKSEL43_CON
Address: Operational Base + offset (0x001ac)
Internal clock select and divide register43

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved
12	RW	0x0	wifi_pll_sel Control wifi clock PLL source selection 1'b0: select codec pll clock 1'b1: select general pll clock
11:9	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
8	RW	0x0	rmii_extclk_sel Control RMII external clock selection 1'b0: select internal divider clock 1'b1: select external input clock
7:6	RW	0x2	mac_pll_sel Control EMAC clock PLL source selection 2'b00: select new pll clock 2'b01: select codec pll clock 2'b1x : select general pll clock
5	RO	0x0	reserved
4:0	RW	0x0e	mac_div_con Control EMAC divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL44_CON
Address: Operational Base + offset (0x001b0)
Internal clock select and divide register44

Bit	Attr	Reset Value	Description
31:0	RW	0x001f05dc	wifi_frac_div_con Control WIFI fraction divider frequency High 16-bit for numerator Low 16-bit for denominator

CRU_CLKSEL45_CON
Address: Operational Base + offset (0x001b4)
Internal clock select and divide register45

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	spi1_pll_sel Control SPI1 clock PLL source selection 1'b0 : select CODEC PLL 1'b1 : select GENERAL PLL
14:8	RW	0x07	spi1_div_con Control SPI1 divider frequency clk=clk_src/(div_con+1)
7	RW	0x0	spi0_pll_sel Control SPI0 clock PLL source selection 1'b0 : select CODEC PLL 1'b1 : select GENERAL PLL

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Bit	Attr	Reset Value	Description
6:0	RW	0x07	spi0_div_con Control SPI0 divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL46_CON
Address: Operational Base + offset (0x001b8)
Internal clock select and divide register46

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	spi2_pll_sel Control SPI2 clock PLL source selection 1'b0 : select CODEC PLL 1'b1 : select GENERAL PLL
14:8	RW	0x07	spi2_div_con Control SPI2 divider frequency clk=clk_src/(div_con+1)
7:6	RW	0x0	tsp_pll_sel Control tsp clock PLL source selection 2'b00: select codec pll clock 2'b01: select general pll clock 2'b1x: select new pll clock
5	RO	0x0	reserved
4:0	RW	0x07	tsp_div_con Control tsp clock divider frequency clk=clk_src/(clk_div_con+1)

CRU_CLKSEL47_CON
Address: Operational Base + offset (0x001bc)
Internal clock select and divide register47

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved
12:8	RW	0x00	test_div_con test out clk divider frequency clk=clk_src/(clk_div_con+1)

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Bit	Attr	Reset Value	Description
7	RW	0x1	nandc0_pll_sel Control nandc0 clock PLL source selection 1'b0: select codec pll clock 1'b1: select general pll clock
6:5	RO	0x0	reserved
4:0	RW	0x03	nandc0_div_con Control nandc0 clock divider frequency clk=clk_src/(clk_div_con+1)

CRU_CLKSEL48_CON
Address: Operational Base + offset (0x001c0)
Internal clock select and divide register48

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved
9:8	RW	0x2	sdio0_pll_sel Control SDIO0 clock PLL source selection 2'b00: select codec pll clock 2'b01: select general pll clock 2'b10: select usbphy 480m 2'b11: select 24MHz
7	RO	0x0	reserved
6:0	RW	0x03	sdio0_div_con Control SDIO0 divider frequency clk=clk_src/(clk_div_con+1)

CRU_CLKSEL49_CON
Address: Operational Base + offset (0x001c4)
reserved

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	reserved

CRU_CLKSEL50_CON
Address: Operational Base + offset (0x001c8)
Internal clock select and divide register50

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit

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Bit	Attr	Reset Value	Description
15:10	RO	0x0	reserved
9:8	RW	0x2	mmc0_pll_sel Control SDMMC0 clock PLL source selection 2'b00: select codec pll clock 2'b01: select general pll clock 2'b10: select usbphy 480m 2'b11: select 24MHz
7	RO	0x0	reserved
6:0	RW	0x03	mmc0_div_con Control SDMMC0 divider frequency clk=clk_src/(clk_div_con+1)

CRU_CLKSEL51_CON
Address: Operational Base + offset (0x001cc)
Internal clock select and divide register51

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved
9:8	RW	0x2	emmc_pll_sel Control EMMC clock PLL source selection 2'b00: select codec pll clock 2'b01: select general pll clock 2'b10: select usbphy 480m 2'b11: select 24MHz
7	RO	0x0	reserved
6:0	RW	0x03	emmc_div_con Control EMMC divider frequency clk=clk_src/(clk_div_con+1)

CRU_CLKSEL52_CON
Address: Operational Base + offset (0x001d0)
Internal clock select and divide register52

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7	RW	0x1	sfc_pll_sel Control sfc clock PLL source selection 1'b0: select codec pll clock 1'b1: select general pll clock
6:5	RO	0x0	reserved
4:0	RW	0x03	sfc_div_con Control sfc clock divider frequency clk=clk_src/(clk_div_con+1)

CRU_CLKSEL53_CON
Address: Operational Base + offset (0x001d4)
Internal clock select and divide register53

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved
12	RW	0x0	i2s_2ch_pll_sel I2S 2CH clock pll source selection 1'b0 : select CODEC PLL 1'b1 : select GENERAL PLL
11:10	RO	0x0	reserved
9:8	RW	0x3	i2s_2ch_clk_sel Control I2S 2CH clock work frequency selection 2'b00: select divider ouput from pll divider 2'b01: select divider ouput from fraction divider 2'b10: 1'b0 2'b11: select 12MHz from osc input
7	RO	0x0	reserved
6:0	RW	0x00	i2s_2ch_pll_div_con Control I2S 2CH divider freuency clk=clk_src/(div_con+1)

CRU_CLKSEL54_CON
Address: Operational Base + offset (0x001d8)
Internal clock select and divide register54

Bit	Attr	Reset Value	Description
31:0	RW	0x0bb8ea60	i2s_2ch_frac_div_con Control I2S 2CH fraction divider frequency High 16-bit for numerator Low 16-bit for denominator

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CRU_CLKSEL55_CON
Address: Operational Base + offset (0x001dc)
Internal clock select and divide register55

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	RO	0x0	reserved
7:6	RW	0x0	hdcp_pll_sel HDCP clock pll source selection 2'b00 : select CODEC PLL 2'b01 : select GENERAL PLL 2'b1x : select NEW PLL
5:0	RW	0x02	hdcp_div_con Control hdcp divider freuency clk=clk_src/(div_con+1)

CRU_CLKGATE0_CON
Address: Operational Base + offset (0x00200)
Internal clock gating control register0

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RO	0x0	reserved
13	RW	0x0	clkin_trace_gate_en trace clock in disable When HIGH, disable clock
12	RW	0x0	aclk_cci_src_gate_en CCI400 AXI clock disable When HIGH, disable clock
11	RO	0x0	reserved
10	RW	0x0	cs_aplll_clk_gate_en Coresight APLLL path clock disable When HIGH, disable clock
9	RW	0x0	cs_apllb_clk_gate_en Coresight APLLB path clock disable When HIGH, disable clock
8	RW	0x0	cs_gpll_clk_gate_en Coresight GPLL path clock disable When HIGH, disable clock

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Bit	Attr	Reset Value	Description
7:6	RO	0x0	reserved
5	RW	0x0	core_l_gpll_clk_gate_en A53 little cluster GPLL path clock disable When HIGH, disable clock
4	RW	0x0	core_l_apll_clk_gate_en A53 little cluster APLL path clock disable When HIGH, disable clock
3:2	RO	0x0	reserved
1	RW	0x0	core_b_gpll_clk_gate_en A53 big cluster GPLL path clock disable When HIGH, disable clock
0	RW	0x0	core_b_apll_clk_gate_en A53 big cluster APLL path clock disable When HIGH, disable clock

CRU_CLKGATE1_CON
Address: Operational Base + offset (0x00204)
Internal clock gating control register1

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:12	RO	0x0	reserved
11	RW	0x0	cpu_cpll_clk_gate_en PD_BUS CPLL path clock disable When HIGH, disable clock
10	RW	0x0	cpu_gpll_clk_gate_en PD_BUS GPLL path clock disable When HIGH, disable clock
9	RW	0x0	ddr_gpll_clk_gate_en DDR GPLL path clock disable When HIGH, disable clock
8	RW	0x0	ddr_dpll_clk_gate_en DDR DPLL path clock disable When HIGH, disable clock
7:4	RO	0x0	reserved
3	RW	0x0	mcu_pll_clk_gate_en MCU GPLL / CPLL path clock disable When HIGH, disable clock
2	RW	0x0	pclk_cpu_gate_en PD_BUS APB clock disable When HIGH, disable clock

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Bit	Attr	Reset Value	Description
1	RW	0x0	hclk_cpu_gate_en PD_BUS AHB clock disable When HIGH, disable clock
0	RW	0x0	aclk_cpu_gate_en PD_BUS AXI clock disable When HIGH, disable clock

CRU_CLKGATE2_CON

Address: Operational Base + offset (0x00208)

Internal clock gating control register2

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved
9	RW	0x0	clk_uart4_frac_src_gate_en UART4 fraction divider source clock disable When HIGH, disable clock
8	RW	0x0	clk_uart4_src_gate_en UAR4 source clock disable When HIGH, disable clock
7	RW	0x0	clk_uart3_frac_src_gate_en UART3 fraction divider source clock disable When HIGH, disable clock
6	RW	0x0	clk_uart3_src_gate_en UART3 source clock disable When HIGH, disable clock
5	RO	0x0	reserved
4	RW	0x0	clk_uart2_src_gate_en UART2 source clock disable When HIGH, disable clock
3	RW	0x0	clk_uart1_frac_src_gate_en UART1 fraction divider source clock disable When HIGH, disable clock
2	RW	0x0	clk_uart1_src_gate_en UART1 source clock disable When HIGH, disable clock
1	RW	0x0	clk_uart0_frac_src_gate_en UART0 fraction divider source clock disable When HIGH, disable clock
0	RW	0x0	clk_uart0_src_gate_en UART0 source clock disable When HIGH, disable clock

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CRU_CLKGATE3_CON
Address: Operational Base + offset (0x0020c)
Internal clock gating control register3

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved
9	RW	0x0	clk_spi2_src_gate_en SPI2 source clock disable When HIGH, disable clock
8	RW	0x0	clk_spi1_src_gate_en SPI1 source clock disable When HIGH, disable clock
7	RW	0x0	clk_spi0_src_gate_en SPI0 source clock disable When HIGH, disable clock
6	RW	0x0	clk_saradc_src_gate_en SARADC source clock disable When HIGH, disable clock
5	RW	0x0	clk_tsadc_src_gate_en TSADC source clock disable When HIGH, disable clock
4	RW	0x0	clk_mac_src_gate_en MAC source clock disable When HIGH, disable clock
3	RW	0x0	pclk_periph_gate_en PERIPH system APB clock(pclk_periph) disable When HIGH, disable clock
2	RW	0x0	hclk_periph_gate_en PERIPH system AHB clock(hclk_periph) disable When HIGH, disable clock
1	RW	0x0	aclk_periph_gate_en PERIPH system AXI clock(aclk_periph) disable When HIGH, disable clock
0	RW	0x0	clk_periph_src_gate_en PERIPH system source clock disable When HIGH, disable clock

CRU_CLKGATE4_CON
Address: Operational Base + offset (0x00210)

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Internal clock gating control register4

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RO	0x0	reserved
14	RW	0x0	clk_dsiphy_24m_gate_en DSI PHY clock disable When HIGH, disable clock
13	RW	0x0	clk_hdmi_hdcp_gate_en HDMI HDCP clock disable When HIGH, disable clock
12	RW	0x0	clk_hdmi_cec_gate_en HDMI CEC clock disable When HIGH, disable clock
11	RW	0x0	clk_gpu_core_src_gate_en gpu clock disable When HIGH, disable clock
10	RO	0x0	reserved
9	RW	0x0	clk_isp_gate_en ISP clock clock disable When HIGH, disable clock
8	RW	0x0	hclk_video_gate_en VIDEO AHB source clock disable When HIGH, disable clock
7	RW	0x0	aclk_vdpu_src_gate_en VDPU AXI source clock disable When HIGH, disable clock
6	RW	0x0	aclk_vepu_src_gate_en VEPU AXI source clock disable When HIGH, disable clock
5	RW	0x0	clk_cif0_out_src_gate_en CIF output clock disable When HIGH, disable clock
4	RW	0x0	clk_rga_src_gate_en RGA func souce clock disable When HIGH, disable clock
3	RW	0x0	aclk_rga_src_gate_en RGA AXI souce clock disable When HIGH, disable clock
2	RW	0x0	clk_vop0_pwm_gate_en VOP0 PWM clock disable When HIGH, disable clock

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Bit	Attr	Reset Value	Description
1	RW	0x0	dclk_vop0_src_gate_en VOP0 DCLK source clock disable When HIGH, disable clock
0	RW	0x0	aclk_vio0_src_gate_en VIO0 AXI source clock disable When HIGH, disable clock

CRU_CLKGATE5_CON

Address: Operational Base + offset (0x00214)

Internal clock gating control register5

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	clk_i2s0_2ch_gate_en I2S 2CH clock disable When HIGH, disable clock
14	RW	0x0	clk_i2s0_2ch_frac_src_gate_en I2S 2CH fraction divider source clock disable When HIGH, disable clock
13	RW	0x0	clk_i2s0_2ch_src_gate_en I2S 2CH source clock disable When HIGH, disable clock
12:10	RO	0x0	reserved
9	RW	0x0	aclk_gpu_cfg_src_gate_en aclk_gpu_cfg clock disable When HIGH, disable clock
8	RW	0x0	aclk_gpu_mem_src_gate_en aclk_gpu_mem clock disable When HIGH, disable clock
7:6	RO	0x0	reserved
5	RW	0x0	clk_hdcp_gate_en HDCP clock disable When HIGH, disable clock
4	RW	0x0	clk_edp_24m_gate_en EDP 24m clock disable When HIGH, disable clock
3	RW	0x0	clk_edp_gate_en EDP clock disable When HIGH, disable clock
2	RW	0x0	clk_hevc_core_src_gate_en HEVC core source clock disable When HIGH, disable clock

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Bit	Attr	Reset Value	Description
1	RW	0x0	clk_hevc_cabac_src_gate_en HEVC cabac source clock disable When HIGH, disable clock
0	RO	0x0	reserved

CRU_CLKGATE6_CON

Address: Operational Base + offset (0x00218)

Internal clock gating control register6

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	clk4x_ddrphy_gate_en DDRPHY0 clock4x disable When HIGH, disable clock
14	RW	0x0	clk_ddrphy_gate_en DDRPHY0 clock disable When HIGH, disable clock
13	RO	0x0	reserved
12	RW	0x0	clk_tsp_src_gate_en TSP clock disable When HIGH, disable clock
11:8	RO	0x0	reserved
7	RW	0x0	clk_sfc_src_gate_en SFC clock disable When HIGH, disable clock
6	RW	0x0	clk_spdif_8ch_gate_en SPDIF clock disable When HIGH, disable clock
5	RW	0x0	clk_spdif_8ch_frac_src_gate_en SPDIF fraction divider source clock disable When HIGH, disable clock
4	RW	0x0	clk_spdif_8ch_src_gate_en SPDIF source clock disable When HIGH, disable clock
3	RW	0x0	clk_i2s0_gate_en I2S clock disable When HIGH, disable clock
2	RW	0x0	clk_i2s0_frac_src_gate_en I2S fraction divider source clock disable When HIGH, disable clock

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Bit	Attr	Reset Value	Description
1	RW	0x0	clk_i2s0_src_gate_en I2S source clock disable When HIGH, disable clock
0	RW	0x0	clk_i2s0_out_gate_en I2S output clock disable When HIGH, disable clock

CRU_CLKGATE7_CON

Address: Operational Base + offset (0x0021c)

Internal clock gating control register7

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	clk_emmc_src_gate_en EMMC work clock disable When HIGH, disable clock
14	RO	0x0	reserved
13	RW	0x0	clk_sdio0_src_gate_en SDIO0 work clock disable When HIGH, disable clock
12	RW	0x0	clk_mmc0_src_gate_en SDMMC0 work clock disable When HIGH, disable clock
11	RW	0x0	pvtm_gpu_gate_en pd_gpu PVTM clock disable When HIGH, disable clock
10	RW	0x0	pvtm_core_gate_en pd_core PVTM clock disable When HIGH, disable clock
9	RW	0x0	pclk_pmu_gate_en pd_pmu APB bus clock disable When HIGH, disable clock
8	RW	0x0	clk_nandc0_src_gate_en nandc0 clock disable When HIGH, disable clock
7	RW	0x0	clk_mac_refout_gate_en MAC ref output clock clock disable When HIGH, disable clock
6	RW	0x0	clk_mac_ref_gate_en MAC ref clock clock disable When HIGH, disable clock

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Bit	Attr	Reset Value	Description
5	RW	0x0	clk_mac_tx_gate_en MAC tx clock clock disable When HIGH, disable clock
4	RW	0x0	clk_mac_rx_gate_en MAC rx clock clock disable When HIGH, disable clock
3	RW	0x0	pvtm_pmu_gate_en PD_PMU pvtm clock disable When HIGH, disable clock
2	RW	0x0	clk_crypto_gate_en crypto clock disable When HIGH, disable clock
1	RW	0x0	testclk_gate_en Test output clock disable When HIGH, disable clock
0	RW	0x0	clk_jtag_gate_en JTAG clock disable When HIGH, disable clock

CRU_CLKGATE8_CON
Address: Operational Base + offset (0x00220)
Internal clock gating control register8

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:9	RO	0x0	reserved
8	RW	0x0	clk_wifi_gate_en TEST CLK: WIFI clock disable When HIGH, disable clock
7	RW	0x0	clk_hsicphy_src_gate_en HSICPHY src clock disable When HIGH, disable clock
6	RW	0x0	clk_hsic_12m_gate_en HSIC 12MHz clock disable When HIGH, disable clock
5	RO	0x0	reserved
4	RW	0x0	clk_otg_adp_gate_en OTG adp clock disable When HIGH, disable clock
3:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1	RW	0x0	clk_otgphy0_gate_en OTGPHY0 clock(clk_otgphy0) disable When HIGH, disable clock
0	RW	0x0	clk_hsic_usbphy480m_gate_en HSIC usbphy480M clock disable When HIGH, disable clock

CRU_CLKGATE9_CON
Address: Operational Base + offset (0x00224)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:2	RO	0x0	reserved
1	RW	0x0	dbg_pd_core_l_clk_gate_en pd core l debug clock disable When HIGH, disable clock
0	RW	0x0	aclk_adb400s_pd_core_l_gate_en pd core l adb400s aclk disable When HIGH, disable clock

CRU_CLKGATE10_CON
Address: Operational Base + offset (0x00228)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:2	RO	0x0	reserved
1	RW	0x0	dbg_pd_core_b_clk_gate_en pd core b debug clock disable When HIGH, disable clock
0	RW	0x0	aclk_adb400s_pd_core_b_gate_en pd core b adb400s aclk disable When HIGH, disable clock

CRU_CLKGATE11_CON
Address: Operational Base + offset (0x0022c)

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	RO	0x0	reserved
7	RW	0x0	clk_dbg_gate_en debug clock disable When HIGH, disable clock
6	RW	0x0	pclk_core_niu_sdbg_gate_en core niu debug pclk disable When HIGH, disable clock
5	RW	0x0	hclk_core_niu_dbg_gate_en core niu debug hclk disable When HIGH, disable clock
4	RW	0x0	aclk_core_niu_cpup_gate_en core niu cpup aclk disable When HIGH, disable clock
3	RW	0x0	aclk_core_niu_cci_gate_en core niu cci aclk disable When HIGH, disable clock
2	RW	0x0	aclk_cci400_gate_en CCI400 aclk disable When HIGH, disable clock
1	RW	0x0	aclk_adb400m_pd_core_b_gate_en pd core b adb400m aclk disable When HIGH, disable clock
0	RW	0x0	aclk_adb400m_pd_core_l_gate_en pd core l adb400m aclk disable When HIGH, disable clock

CRU_CLKGATE12_CON

Address: Operational Base + offset (0x00230)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RO	0x0	reserved
14	RW	0x0	pclk_ddrphy_gate_en ddrphy pclk disable When HIGH, disable clock

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Bit	Attr	Reset Value	Description
13	RW	0x0	pclk_ddrupctl_gate_en ddrupctl pclk disable When HIGH, disable clock
12	RW	0x0	aclk_strc_sys_gate_en structure system aclk disable When HIGH, disable clock
11	RW	0x0	aclk_dmac_gate_en dmac aclk disable When HIGH, disable clock
10	RW	0x0	hclk_spdif_gate_en spdif hclk disable When HIGH, disable clock
9	RW	0x0	hclk_rom_gate_en rom hclk disable When HIGH, disable clock
8	RW	0x0	hclk_i2s_2ch_gate_en i2s 2ch hclk disable When HIGH, disable clock
7	RW	0x0	hclk_i2s_8ch_gate_en i2s 8ch hclk disable When HIGH, disable clock
6	RW	0x0	clk_intmem1_gate_en intmem1 clk disable When HIGH, disable clock
5	RW	0x0	clk_intmem0_gate_en intmem0 clk disable When HIGH, disable clock
4	RW	0x0	aclk_intmem_gate_en intmem aclk disable When HIGH, disable clock
3	RW	0x0	pclk_i2caudio_gate_en i2caudio pclk disable When HIGH, disable clock
2	RW	0x0	pclk_i2cpmu_gate_en i2cpmu pclk disable When HIGH, disable clock
1	RW	0x0	pclk_mailbox_gate_en mailbox pclk disable When HIGH, disable clock
0	RW	0x0	pclk_pwm0_gate_en pwm0 pclk disable When HIGH, disable clock

CRU_CLKGATE13_CON
Address: Operational Base + offset (0x00234)

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RO	0x0	reserved
14	RW	0x0	hclk_noc_mcu_gate_en noc mcu htclk disable When HIGH, disable clock
13	RW	0x0	stclk_mcu_gate_en mcu stclk disable When HIGH, disable clock
12	RW	0x0	fclk_mcu_gate_en mcu fclk disable When HIGH, disable clock
11	RW	0x0	hclk_mcu_gate_en mcu hclk disable When HIGH, disable clock
10	RW	0x0	hclk_tsp_gate_en tsp hclk disable When HIGH, disable clock
9	RW	0x0	aclk_gic400_gate_en gic400 aclk disable When HIGH, disable clock
8	RW	0x0	pclk_sim_gate_en sim pclk disable When HIGH, disable clock
7	RW	0x0	clk_hsadc_0_tsp_gate_en hsadc_0_tsp clk disable When HIGH, disable clock
6	RW	0x0	pclk_pwm1_gate_en pwm1 pclk disable When HIGH, disable clock
5	RW	0x0	pclk_uartdbg_gate_en uartdbg pclk disable When HIGH, disable clock
4	RW	0x0	sclk_crypto_gate_en crypto sclk disable When HIGH, disable clock
3	RW	0x0	mclk_crypto_gate_en crypto mclk disable When HIGH, disable clock

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Bit	Attr	Reset Value	Description
2	RW	0x0	nclk_ddrupctl_gate_en ddrupctl nclk disable When HIGH, disable clock
1	RW	0x0	pclk_efuse_256_gate_en efuse 256 pclk disable When HIGH, disable clock
0	RW	0x0	pclk_efuse_1024_gate_en efuse 1024 pclk disable When HIGH, disable clock

CRU_CLKGATE14_CON
Address: Operational Base + offset (0x00238)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:0	RO	0x0	reserved

CRU_CLKGATE15_CON
Address: Operational Base + offset (0x0023c)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:4	RO	0x0	reserved
3	RW	0x0	clk_hevc_ca_gate_en hevc ca clock disable When HIGH, disable clock
2	RW	0x0	clk_hevc_core_gate_en hevc core clock disable When HIGH, disable clock
1	RW	0x0	hclk_video_gate_en video hclk disable When HIGH, disable clock
0	RW	0x0	aclk_video_gate_en video aclk disable When HIGH, disable clock

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CRU_CLKGATE16_CON
Address: Operational Base + offset (0x00240)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RO	0x0	reserved
14	RW	0x0	hclk_isp_gate_en isp hclk disable When HIGH, disable clock
13	RW	0x0	pclkin_vip_gate_en vip pclkin disable When HIGH, disable clock
12	RW	0x0	hclk_vip_gate_en vip hclk disable When HIGH, disable clock
11	RW	0x0	aclk_vip_gate_en vip aclk disable When HIGH, disable clock
10	RW	0x0	aclk_vio1_noc_gate_en vio1 noc aclk disable When HIGH, disable clock
9	RW	0x0	aclk_vio0_noc_gate_en vio0 noc aclk disable When HIGH, disable clock
8	RW	0x0	hclk_vio_noc_gate_en vio noc hclk disable When HIGH, disable clock
7	RW	0x0	hclk_vio_ahb_arbi_gate_en vio ahb arbi hclk disable When HIGH, disable clock
6	RW	0x0	hclk_vop_gate_en vop hclk disable When HIGH, disable clock
5	RW	0x0	aclk_vop_gate_en vop aclk disable When HIGH, disable clock
4	RW	0x0	aclk_vop_iep_gate_en vop_iep aclk disable When HIGH, disable clock
3	RW	0x0	hclk_iep_gate_en iep hclk disable When HIGH, disable clock

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Bit	Attr	Reset Value	Description
2	RW	0x0	aclk_iep_gate_en iep aclk disable When HIGH, disable clock
1	RW	0x0	hclk_rga_gate_en rga hclk disable When HIGH, disable clock
0	RW	0x0	aclk_rga_gate_en rga aclk disable When HIGH, disable clock

CRU_CLKGATE17_CON
Address: Operational Base + offset (0x00244)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved
12	RW	0x0	hclk_hdcpmmu_gate_en hdcpmmu hclk disable When HIGH, disable clock
11	RW	0x0	pclk_hdcp_gate_en hdcp pclk disable When HIGH, disable clock
10	RW	0x0	aclk_hdcp_gate_en hdcp aclk disable When HIGH, disable clock
9	RW	0x0	pclk_edp_ctrl_gate_en edp ctrl pclk disable When HIGH, disable clock
8	RW	0x0	pclk_vio_h2p_gate_en vio h2p pclk disable When HIGH, disable clock
7	RW	0x0	hclk_vio_h2p_gate_en vio h2p hclk disable When HIGH, disable clock
6	RW	0x0	pclk_hdmi_ctrl_gate_en hdmi ctrl pclk disable When HIGH, disable clock
5	RO	0x0	reserved
4	RW	0x0	pclk_mipi_csi_gate_en csi mipi pclk disable When HIGH, disable clock

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Bit	Attr	Reset Value	Description
3	RW	0x0	pclk_mipi_dsi0_gate_en dsi0 mipi pclk disable When HIGH, disable clock
2	RW	0x0	pclkin_isp_gate_en isp pclkin disable When HIGH, disable clock
1	RO	0x0	reserved
0	RW	0x0	aclk_isp_gate_en isp aclk disable When HIGH, disable clock

CRU_CLKGATE18_CON
Address: Operational Base + offset (0x00248)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:3	RO	0x0	reserved
2	RW	0x0	clk_gpu_core_gate_en gpu core clk disable When HIGH, disable clock
1	RW	0x0	aclk_gpu_mem_gate_en gpu mem aclk disable When HIGH, disable clock
0	RW	0x0	aclk_gpu_cfg_gate_en gpu cfg aclk disable When HIGH, disable clock

CRU_CLKGATE19_CON
Address: Operational Base + offset (0x0024c)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	pclk_saradc_gate_en saradc pclk disable When HIGH, disable clock

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Bit	Attr	Reset Value	Description
14	RW	0x0	pclk_i2c5_gate_en i2c5 pclk disable When HIGH, disable clock
13	RW	0x0	pclk_i2c4_gate_en i2c4 pclk disable When HIGH, disable clock
12	RW	0x0	pclk_i2c3_gate_en i2c3 pclk disable When HIGH, disable clock
11	RW	0x0	pclk_i2c2_gate_en i2c2 pclk disable When HIGH, disable clock
10	RW	0x0	pclk_uart4_gate_en uart4 pclk disable When HIGH, disable clock
9	RW	0x0	pclk_uart3_gate_en uart3 pclk disable When HIGH, disable clock
8	RW	0x0	pclk_uart1_gate_en uart1 pclk disable When HIGH, disable clock
7	RW	0x0	pclk_uart0_gate_en uart0 pclk disable When HIGH, disable clock
6	RW	0x0	pclk_spi2_gate_en spi2 pclk disable When HIGH, disable clock
5	RW	0x0	pclk_spi1_gate_en spi1 pclk disable When HIGH, disable clock
4	RW	0x0	pclk_spi0_gate_en spi0 pclk disable When HIGH, disable clock
3	RW	0x0	aclk_dmac_peri_gate_en peri dmac aclk disable When HIGH, disable clock
2	RW	0x0	aclk_peri_axi_matrix_gate_en peri axi matrix aclk disable When HIGH, disable clock
1	RW	0x0	pclk_peri_axi_matrix_gate_en peri axi matrix pclk disable When HIGH, disable clock
0	RW	0x0	hclk_peri_axi_matrix_gate_en peri axi matrix hclk disable When HIGH, disable clock

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CRU_CLKGATE20_CON
Address: Operational Base + offset (0x00250)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	hclk_sfc_gate_en sfc hclk disable When HIGH, disable clock
14	RW	0x0	pclk_gmac_gate_en gmac pclk disable When HIGH, disable clock
13	RW	0x0	aclk_gmac_gate_en gmac aclk disable When HIGH, disable clock
12	RO	0x0	reserved
11	RW	0x0	hclk_nand0_gate_en nand0 hclk disable When HIGH, disable clock
10	RW	0x0	hclk_mmc_peri_gate_en peri mmc hclk disable When HIGH, disable clock
9	RW	0x0	hclk_emem_peri_gate_en peri emem hclk disable When HIGH, disable clock
8	RW	0x0	aclk_peri_niu_gate_en peri niu aclk disable When HIGH, disable clock
7	RW	0x0	hclk_peri_ahb_arbi_gate_en peri ahb arbi hclk disable When HIGH, disable clock
6	RW	0x0	hclk_usb_peri_gate_en peri usb hclk disable When HIGH, disable clock
5	RW	0x0	hclk_hsic_gate_en hsic hclk disable When HIGH, disable clock
4	RW	0x0	hclk_host1_gate_en host1 hclk disable When HIGH, disable clock

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Bit	Attr	Reset Value	Description
3	RW	0x0	hclk_host0_gate_en host0 hclk disable When HIGH, disable clock
2	RW	0x0	pmu_hclk_otg0_gate_en otg0 pmu hclk disable When HIGH, disable clock
1	RW	0x0	hclk_otg0_gate_en otg0 hclk disable When HIGH, disable clock
0	RW	0x0	pclk_tsadc_gate_en tsadc pclk disable When HIGH, disable clock

CRU_CLKGATE21_CON
Address: Operational Base + offset (0x00254)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:5	RO	0x0	reserved
4	RW	0x0	aclk_peri_mmu_gate_en peri mmu aclk disable When HIGH, disable clock
3	RW	0x0	hclk_hsadc_gate_en hsadc hclk disable When HIGH, disable clock
2	RW	0x0	hclk_emmc_gate_en emmc hclk disable When HIGH, disable clock
1	RW	0x0	hclk_sdio0_gate_en sdio0 hclk disable When HIGH, disable clock
0	RW	0x0	hclk_sdmmc_gate_en sdmmc hclk disable When HIGH, disable clock

CRU_CLKGATE22_CON
Address: Operational Base + offset (0x00258)

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RO	0x0	reserved
13	RW	0x0	pclk_timer1_gate_en timer1 pclk disable When HIGH, disable clock
12	RW	0x0	pclk_timer0_gate_en timer0 pclk disable When HIGH, disable clock
11	RW	0x0	pclk_dphyrx_phy_gate_en dphyrx phy pclk disable When HIGH, disable clock
10	RW	0x0	pclk_dphytx0_phy_gate_en dphytx0 phy pclk disable When HIGH, disable clock
9	RW	0x0	pclk_alive_niu_gate_en alive niu pclk disable When HIGH, disable clock
8	RW	0x0	pclk_grf_gate_en grf pclk disable When HIGH, disable clock
7:4	RO	0x0	reserved
3	RW	0x0	pclk_gpio3_gate_en gpio3 pclk disable When HIGH, disable clock
2	RW	0x0	pclk_gpio2_gate_en gpio2 pclk disable When HIGH, disable clock
1	RW	0x0	pclk_gpio1_gate_en gpio1 pclk disable When HIGH, disable clock
0	RO	0x0	reserved

CRU_CLKGATE23_CON
Address: Operational Base + offset (0x0025c)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit

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Bit	Attr	Reset Value	Description
15:6	RO	0x0	reserved
5	RW	0x0	pclk_pmugrf_gate_en pmugrf pclk disable When HIGH, disable clock
4	RW	0x0	pclk_gpio0_gate_en gpio0 pclk disable When HIGH, disable clock
3	RW	0x0	pclk_sgrf_gate_en sgrf pclk disable When HIGH, disable clock
2	RW	0x0	pclk_pmu_noc_gate_en pmu noc pclk disable When HIGH, disable clock
1	RW	0x0	pclk_intmem1_gate_en intmem1 pclk disable When HIGH, disable clock
0	RW	0x0	pclk_pmu_gate_en pmu pclk disable When HIGH, disable clock

CRU_CLKGATE24_CON
Address: Operational Base + offset (0x00260)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:12	RO	0x0	reserved
11	RW	0x0	timer15_clk_gate_en timer15 clock disable When HIGH, disable clock
10	RW	0x0	timer14_clk_gate_en timer14 clock disable When HIGH, disable clock
9	RW	0x0	timer13_clk_gate_en timer13 clock disable When HIGH, disable clock
8	RW	0x0	timer12_clk_gate_en timer12 clock disable When HIGH, disable clock
7	RW	0x0	timer11_clk_gate_en timer11 clock disable When HIGH, disable clock

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Bit	Attr	Reset Value	Description
6	RW	0x0	timer10_clk_gate_en timer10 clock disable When HIGH, disable clock
5	RW	0x0	timer05_clk_gate_en timer05 clock disable When HIGH, disable clock
4	RW	0x0	timer04_clk_gate_en timer04 clock disable When HIGH, disable clock
3	RW	0x0	timer03_clk_gate_en timer03 clock disable When HIGH, disable clock
2	RW	0x0	timer02_clk_gate_en timer02 clock disable When HIGH, disable clock
1	RW	0x0	timer01_clk_gate_en timer01 clock disable When HIGH, disable clock
0	RW	0x0	timer00_clk_gate_en timer00 clock disable When HIGH, disable clock

CRU_GLB_SRST_FST_VALUE
Address: Operational Base + offset (0x00280)
The first global software reset config value

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	glb_srst_fst_value The first global software reset config value If config 0xfdb9, it will generate first global software reset.

CRU_GLB_SRST_SND_VALUE
Address: Operational Base + offset (0x00284)
The second global software reset config value

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	glb_srst_snd_value The second global software reset config value If config 0xeca8, it will generate second global software reset.

CRU_SOFTRST0_CON
Address: Operational Base + offset (0x00300)
Internal software reset control register0

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	core_b_dbg_srstn_req core big cluster debug software reset request. When HIGH, reset relative logic
14	RW	0x0	socdbg_b_srstn_req SOC debug reset request When HIGH, reset relative logic
13:12	RO	0x0	reserved
11	R/WSC	0x0	strc_sys_asrstn_req PD BUS NOC AXI software reset request When HIGH, reset relative logic
10	R/WSC	0x0	pd_core_b_niu_asrstn_req PD COREB NIU software reset request When HIGH, reset relative logic
9	R/WSC	0x0	adb_b_srstn_req ADB between A53 big clustert and CCI software reset request When HIGH, reset relative logic
8	R/WSC	0x0	l2_b_srstn_req A53 big cluster L2 software reset request When HIGH, reset relative logic
7	RW	0x0	core_b_po3_srstn_req Core big cluster CPU3 PO software reset request. When HIGH, reset relative logic
6	RW	0x0	core_b_po2_srstn_req Core little cluster CPU2 PO software reset request. When HIGH, reset relative logic
5	RW	0x0	core_b_po1_srstn_req Core big cluster CPU1 PO software reset request. When HIGH, reset relative logic
4	R/WSC	0x0	core_b_po0_srstn_req Core big cluster CPU 0 PO software reset request. When HIGH, reset relative logic
3	RW	0x0	core_b_3_srstn_req Core big cluster CPU3 software reset request. When HIGH, reset relative logic

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Bit	Attr	Reset Value	Description
2	RW	0x0	core_b_2_srstn_req Core big cluster CPU2 software reset request. When HIGH, reset relative logic
1	RW	0x0	core_b_1_srstn_req Core big cluster CPU1 software reset request. When HIGH, reset relative logic
0	R/WSC	0x0	core_b_0_srstn_req core big cluster CPU0 software reset request. When HIGH, reset relative logic

CRU_SOFTRST1_CON

Address: Operational Base + offset (0x00304)

Internal software reset control register1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	efuse_psrstn_req EFUSE APB software reset request. When HIGH, reset relative logic
14	RW	0x0	mcu_noc_hsrstn_req MCU NOC AHB software reset request When HIGH, reset relative logic
13	RW	0x1	mcupo_srstn_req MCU PO software reset request When HIGH, reset relative logic
12	RW	0x1	mcusys_srstn_req MCU SYS software reset request When HIGH, reset relative logic
11	RO	0x0	reserved
10	RW	0x0	efuse_256_psrstn_req EFUSE 256 APB software reset request. When HIGH, reset relative logic
9	RW	0x0	i2s_2ch_srstn_req I2S 2CH software reset request. When HIGH, reset relative logic
8	RW	0x0	mailbox_srstn_req MAILBOX software reset request. When HIGH, reset relative logic
7	RW	0x0	i2s_srstn_req I2S software reset request. When HIGH, reset relative logic

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Bit	Attr	Reset Value	Description
6	RO	0x0	reserved
5	RW	0x0	spdif_8ch_srstn_req SPDIF 8ch software reset request. When HIGH, reset relative logic
4	RW	0x0	rom_srstn_req ROM software reset request. When HIGH, reset relative logic
3	RW	0x0	intmem_srstn_req Internal memory software reset request. When HIGH, reset relative logic
2	RW	0x0	dma1_srstn_req DMA1 software reset request. When HIGH, reset relative logic
1:0	RO	0x0	reserved

CRU_SOFTRST2_CON
Address: Operational Base + offset (0x00308)
Internal software reset control register2

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	i2c5_srstn_req I2C5 software reset request. When HIGH, reset relative logic
14	RW	0x0	i2c4_srstn_req I2C4 software reset request. When HIGH, reset relative logic
13	RW	0x0	i2c3_srstn_req I2C3 software reset request. When HIGH, reset relative logic
12	RW	0x0	i2c2_srstn_req I2C2 software reset request. When HIGH, reset relative logic
11	RW	0x0	i2c1_srstn_req I2C1 software reset request. When HIGH, reset relative logic
10	RW	0x0	i2c0_srstn_req I2C0 software reset request. When HIGH, reset relative logic

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Bit	Attr	Reset Value	Description
9	RW	0x0	pmugrf_psrstn_req PMUGRF software reset request. When HIGH, reset relative logic
8:5	RO	0x0	reserved
4	RW	0x0	gpio4_srstn_req GPIO4 software reset request. When HIGH, reset relative logic
3	RW	0x0	gpio3_srstn_req GPIO3 software reset request. When HIGH, reset relative logic
2	RW	0x0	gpio2_srstn_req GPIO2 software reset request. When HIGH, reset relative logic
1	RW	0x0	gpio1_srstn_req GPIO1 software reset request. When HIGH, reset relative logic
0	RW	0x0	gpio0_srstn_req GPIO0 software reset request. When HIGH, reset relative logic

CRU_SOFTRST3_CON
Address: Operational Base + offset (0x0030c)
Internal software reset control register3

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	usb_peri_srstn_req USB PERIPH software reset request. When HIGH, reset relative logic
14	RW	0x0	emem_peri_srstn_req EMEM ahb bus software reset request. When HIGH, reset relative logic
13	RW	0x0	pd_peri_ahb_arbitor_srstn_req pd_peri ahb arbitor software reset request. cypro, nandc, hsic, otg, uhost AHB arbitor reset control When HIGH, reset relative logic
12	RW	0x0	periph_niu_srstn_req PERIPH NIU software reset request. When HIGH, reset relative logic

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Bit	Attr	Reset Value	Description
11	RW	0x0	periphsys_psrstn_req PERIPH APB software reset request. pd_peri bus matrix apb softreset When HIGH, reset relative logic
10	RW	0x0	periphsys_hsrstn_req PERIPH AHB software reset request. pd_peri bus matrix ahb softreset When HIGH, reset relative logic
9	RW	0x0	periphsys_asrstn_req PERIPH AXI software reset request. pd_peri bus matrix axi softreset When HIGH, reset relative logic
8	RW	0x0	pmu_srstn_req PMU software reset request. When HIGH, reset relative logic
7	RW	0x0	grf_srstn_req GRF software reset request. When HIGH, reset relative logic
6:3	RO	0x0	reserved
2	RW	0x0	periph_mmu_srstn_req PERIPH MMU software reset request. When HIGH, reset relative logic
1	RW	0x0	mmc_peri_srstn_req pd_peri mmc AHB bus software reset request. emmc, sdio, sdmmc AHB arbitor reset control When HIGH, reset relative logic
0	RW	0x0	dw_pwm_srstn_req DW PWM software reset request. When HIGH, reset relative logic

CRU_SOFTRST4_CON

Address: Operational Base + offset (0x00310)

Internal software reset control register4

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	sfc_srstn_req SFC software reset request. When HIGH, reset relative logic
14	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
13	RW	0x0	nandc0_srstn_req NANDC0 software reset request. When HIGH, reset relative logic
12	RW	0x0	hsadc_hsrstn_req HSADC AHB software reset request. When HIGH, reset relative logic
11	RW	0x0	hsicphy_srstn_req HSICPHY software reset request. When HIGH, reset relative logic
10	RW	0x0	hsic_aux_srstn_req HSIC AUX AHB software reset request. When HIGH, reset relative logic
9	RW	0x0	hsic_srstn_req HSIC AHB software reset request. When HIGH, reset relative logic
8	RW	0x0	host0_hsrstn_req USB HOST0 software reset request. When HIGH, reset relative logic. It asserts "host0_aux_resetn" and "phy_rst_i_n" of USB HOST0.
7:6	RO	0x0	reserved
5	RW	0x0	rk_pwm_srstn_req RK_PWM software reset request. When HIGH, reset relative logic
4	RO	0x0	reserved
3	RW	0x0	gps_srstn_req GPS software reset request. When HIGH, reset relative logic
2	RW	0x0	mac_srstn_req MAC software reset request. When HIGH, reset relative logic
1	RO	0x0	reserved
0	RW	0x0	dma2_srstn_req DMA2 software reset request. When HIGH, reset relative logic

CRU_SOFTRST5_CON

Address: Operational Base + offset (0x00314)

Internal software reset control register5

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:12	RO	0x0	reserved
11	RW	0x0	security_grf_psrstn_req security GRF APB software reset request. When HIGH, reset relative logic
10	RW	0x0	pd_pmu_niu_psrstn_req pd_pmu niu APB software reset request. When HIGH, reset relative logic
9	RW	0x0	pd_pmu_intmem_psrstn_req pd_pmu internal memory apb software reset request. When HIGH, reset relative logic
8	RW	0x0	pd_alive_niu_psrstn_req pd_alive niu APB software reset request. When HIGH, reset relative logic
7	RW	0x0	saradc_srstn_req SARADC software reset request. When HIGH, reset relative logic
6	RO	0x0	reserved
5	RW	0x0	spi2_srstn_req SPI2 software reset request. When HIGH, reset relative logic
4	RW	0x0	spi1_srstn_req SPI1 software reset request. When HIGH, reset relative logic
3	RW	0x0	spi0_srstn_req SPI0 software reset request. When HIGH, reset relative logic
2:0	RO	0x0	reserved

CRU_SOFTRST6_CON

Address: Operational Base + offset (0x00318)

Internal software reset control register6

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit

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Bit	Attr	Reset Value	Description
15	RW	0x0	edp_24msrstn_req EDP 24m clock domain software reset request. When HIGH, reset relative logic
14	RW	0x0	isp_srstn_req ISP software reset request. When HIGH, reset relative logic
13	RW	0x0	rga_hsrstn_req RGA AHB software reset request. When HIGH, reset relative logic
12	RW	0x0	rga_asrstn_req RGA AXI software reset request. When HIGH, reset relative logic
11	RW	0x0	iep_hsrstn_req IEP AHB software reset request. When HIGH, reset relative logic
10	RW	0x0	iep_asrstn_req IEP AXI software reset request. When HIGH, reset relative logic
9	RW	0x0	rga_core_srstn_req RGA func software reset request. When HIGH, reset relative logic
8	RW	0x0	vip_srstn_req VIP software reset request. IEP ISP VOP's NIU software reset. When HIGH, reset relative logic
7	RO	0x0	reserved
6	RW	0x0	lcdc0_dsrstn_req LCDC0 DCLK software reset request. When HIGH, reset relative logic
5	RW	0x0	lcdc0_hsrstn_req LCDC0 AHB software reset request. When HIGH, reset relative logic
4	RW	0x0	lcdc0_asrstn_req LCDC0 AXI software reset request. When HIGH, reset relative logic
3	RW	0x0	vio0_bus_hsrstn_req VIO NIU AHB software reset request. When HIGH, reset relative logic
2	RW	0x0	vio0_niu_asrstn_req VIO0 NIU AXI software reset request. IEP ISP VOP's NIU software reset. When HIGH, reset relative logic

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Bit	Attr	Reset Value	Description
1	RW	0x0	rga_niu_asrstn_req RGA NIU AXI software reset request. IEP ISP VOP's NIU software reset. When HIGH, reset relative logic
0	RW	0x0	vio_arbi_hsrstn_req VIO arbitor AHB software reset request. When HIGH, reset relative logic

CRU_SOFTRST7_CON
Address: Operational Base + offset (0x0031c)
Internal software reset control register7

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	gpu_mem_niu_asrstn_req NOC mem AXI in PD_GPU software reset request. When HIGH, reset relative logic
14	RW	0x0	gpu_sys_asrstn_req GPU sys software reset request. When HIGH, reset relative logic
13	RW	0x0	gpu_pvtm_srstn_req gpu pvtm software reset request. When HIGH, reset relative logic
12	RW	0x0	core_pvtm_srstn_req core pvtm software reset request. When HIGH, reset relative logic
11	RW	0x0	pmu_pvtm_srstn_req PVTM in PD_PMU software reset request. When HIGH, reset relative logic
10	RW	0x0	edp_psrstn_req edp ctrl apb bus software reset request. When HIGH, reset relative logic
9	RW	0x0	hdmi_srstn_req HDMI software reset request. When HIGH, reset relative logic
8	RW	0x0	gpu_core_srstn_req GPU core software reset request. When HIGH, reset relative logic
7:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5	RW	0x0	mipicsi_psrstn_req MIPi CSI APB software reset request. When HIGH, reset relative logic
4	RW	0x0	mipidphyrx_psrstn_req MIPI CSI PHY APB software reset request. When HIGH, reset relative logic
3	RW	0x0	mipidsi0_psrstn_req MIPi DSI0 APB software reset request. When HIGH, reset relative logic
2	RW	0x0	mipidphytx_psrstn_req MIPI DSI PHY APB software reset request. When HIGH, reset relative logic
1	RW	0x0	video_hsrstn_req VIDEO AHB software reset request. When HIGH, reset relative logic
0	RW	0x0	video_asrstn_req VIDEO AXI software reset request. When HIGH, reset relative logic

CRU_SOFTRST8_CON
Address: Operational Base + offset (0x00320)
Internal software reset control register8

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RO	0x0	reserved
13	RW	0x0	usb_adp_srstn_req OTG adp clock software reset request. When HIGH, reset relative logic
12	RO	0x0	reserved
11	RW	0x0	usbphy1_utmirstn_req USBHOST0 PHY utmi software reset request. When HIGH, reset relative logic
10	RW	0x0	usbphy0_utmirstn_req USBOTG PHY utmi software reset request. When HIGH, reset relative logic
9	RW	0x0	usbhostc0_srstn_req USBHOST0 controller software reset request. When HIGH, reset relative logic It asserts "utmi_resetn_i_n" and "hresetn_host0" of USB HOST0.

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Bit	Attr	Reset Value	Description
8	RW	0x0	usbotgphy1_srstn_req USBOTG PHY software reset request. When HIGH, reset relative logic
7	RW	0x0	usbhostc0_hsrstn_req USBHOST0 AHB BUS software reset request. When HIGH, reset relative logic
6	RW	0x0	usbotgc0_srstn_req USBOTG controller software reset request. When HIGH, reset relative logic
5	RW	0x0	usbotgphy0_srstn_req USBOTG PHY software reset request. When HIGH, reset relative logic
4	RW	0x0	usbotg0_hsrstn_req USBOTG AHB BUS software reset request. When HIGH, reset relative logic
3	RW	0x0	emmc_srstn_req EMMC software reset request. When HIGH, reset relative logic
2	RO	0x0	reserved
1	RW	0x0	sdio0_srstn_req SDIO0 software reset request. When HIGH, reset relative logic
0	RW	0x0	mmc0_srstn_req SDMMC0 software reset request. When HIGH, reset relative logic

CRU_SOFTRST9_CON

Address: Operational Base + offset (0x00324)

Internal software reset control register9

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	tsadc_psrstn_req TSADC APB software reset request. When HIGH, reset relative logic
14	RO	0x0	reserved
13	RW	0x0	gpu_cfg_niu_asrstn_req NOC cfg AXI in PD_GPU software reset request. When HIGH, reset relative logic
12:11	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
10	RW	0x0	video_srstn_req VIDEO software reset request. When HIGH, reset relative logic
9	RW	0x0	rga_h2p_brg_srstn_req RGA AHB to APB bridge software reset request. When HIGH, reset relative logic
8:6	RO	0x0	reserved
5	RW	0x0	lcdcpwm0_srstn_req lcdc_pwm0 software reset request. When HIGH, reset relative logic
4	RW	0x0	gic_srstn_req GIC software reset request. When HIGH, reset relative logic
3	RO	0x0	reserved
2	RW	0x0	pd_core_apb_noc_srstn_req pd_core APB software reset request. When HIGH, reset relative logic
1	RW	0x0	pd_core_ahb_noc_srstn_req PD_CORE AHB software reset request. When HIGH, reset relative logic
0	RW	0x0	dbg_resetn Coresight software reset request. When HIGH, reset relative logic

CRU_SOFTRST10_CON

Address: Operational Base + offset (0x00328)

Internal software reset control register10

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RO	0x0	reserved
14	RW	0x0	crypto_srstn_req crypto working clk domain software reset request. When HIGH, reset relative logic
13	RW	0x0	sys_bus_srst_req AHB Arbiter logic in PD_BUS software reset request. When HIGH, reset relative logic
12:11	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
10	RW	0x0	ddrmsch0_srstn_req DDR0 memory scheduler software reset request. When HIGH, reset relative logic
9:8	RO	0x0	reserved
7	RW	0x0	video_niu_hsrstn_req NOC for VIDEO AHB bus software reset request. When HIGH, reset relative logic
6	RO	0x0	reserved
5	RW	0x0	video_niu_arstn_req NOC for VIDEO software reset request. When HIGH, reset relative logic
4	RO	0x0	reserved
3	RW	0x0	ddrctrl0_psrstn_req DDR controller0 APB software reset request. When HIGH, reset relative logic
2	RW	0x0	ddrctrl0_srstn_req DDR controller0 software reset request. When HIGH, reset relative logic
1	RW	0x0	ddrphy0_psrstn_req DDR PHY0 APB software reset request. When HIGH, reset relative logic
0	RW	0x0	ddrphy0_srstn_req DDR PHY0 software reset request. When HIGH, reset relative logic

CRU_SOFTRST11_CON

Address: Operational Base + offset (0x0032c)

Internal software reset control register11

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RO	0x0	reserved
13	RW	0x0	tsp_clkin0_srstn_req TSP clockin 0 software reset request. When HIGH, reset relative logic
12	RW	0x0	tsp_hsrstn_req tsp software reset request. When HIGH, reset relative logic
11	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
10	RW	0x0	simc_prstn_req SIMC APB software reset request. When HIGH, reset relative logic
9:8	RO	0x0	reserved
7	RW	0x0	uart4_srstn_req UART4 software reset request. When HIGH, reset relative logic
6	RW	0x0	uart3_srstn_req UART3 software reset request. When HIGH, reset relative logic
5	RW	0x0	uart2_srstn_req UART2 software reset request. When HIGH, reset relative logic
4	RW	0x0	uart1_srstn_req UART1 software reset request. When HIGH, reset relative logic
3	RW	0x0	uart0_srstn_req UART0 software reset request. When HIGH, reset relative logic
2:0	RO	0x0	reserved

CRU_SOFTRST12_CON
Address: Operational Base + offset (0x00330)
Internal software reset control register0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	core_l_dbg_srstn_req core little cluster debug software reset request. When HIGH, reset relative logic
14	RW	0x0	socdbg_l_srstn_req SOC debug reset request When HIGH, reset relative logic
13	R/WSC	0x0	cci400_srstn_req CCI software reset request. When HIGH, reset relative logic
12	R/WSC	0x0	cci400_ddr_srstn_req CCI NIU software reset request. When HIGH, reset relative logic

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Bit	Attr	Reset Value	Description
11	R/WSC	0x0	cci400_sys_srstn_req CCI NIU software reset request. When HIGH, reset relative logic
10	R/WSC	0x0	pd_core_l_niu_asrstn_req PD COREL NIU software reset request When HIGH, reset relative logic
9	R/WSC	0x0	adb_l_srstn_req ADB between A53 little cluster and CCI software reset request. When HIGH, reset relative logic
8	R/WSC	0x0	l2_l_srstn_req A53 little cluster L2 software reset request. When HIGH, reset relative logic
7	RW	0x0	core_l_po3_srstn_req Core little cluster CPU3 PO software reset request. When HIGH, reset relative logic
6	RW	0x0	core_l_po2_srstn_req Core little cluster CPU2 PO software reset request. When HIGH, reset relative logic
5	RW	0x0	core_l_po1_srstn_req Core little cluster CPU1 PO software reset request. When HIGH, reset relative logic
4	R/WSC	0x0	core_l_po0_srstn_req Core little cluster CPU 0 PO software reset request. When HIGH, reset relative logic
3	RW	0x0	core_l_3_srstn_req Core little cluster CPU3 software reset request. When HIGH, reset relative logic
2	RW	0x0	core_l_2_srstn_req Core little cluster CPU2 software reset request. When HIGH, reset relative logic
1	RW	0x0	core_l_1_srstn_req Core little cluster CPU1 software reset request. When HIGH, reset relative logic
0	R/WSC	0x0	core_l_0_srstn_req Core little cluster CPU0 software reset request. When HIGH, reset relative logic

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CRU_SOFTRST13_CON
Address: Operational Base + offset (0x00334)
Internal software reset control register0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RO	0x0	reserved
14	RW	0x0	trace_srstn_req trace software reset request. When HIGH, reset relative logic. This reset will not self-cleared.
13	RW	0x0	cci400_srstn_req CCI software reset request. When HIGH, reset relative logic. This reset will not self-cleared.
12	RW	0x0	cci400_ddr_srstn_req CCI NIU software reset request. When HIGH, reset relative logic. This reset will not self-cleared.
11	RW	0x0	cci400_sys_srstn_req CCI NIU software reset request. When HIGH, reset relative logic. This reset will not self-cleared.
10	RW	0x0	pd_core_l_niu_asrstn_req PD COREL NIU software reset request When HIGH, reset relative logic
9	RW	0x0	adb_l_srstn_req ADB between A53 little cluster and CCI software reset request. When HIGH, reset relative logic. This reset will not self-cleared.
8	RW	0x0	l2_l_srstn_req A53 little cluster L2 software reset request. When HIGH, reset relative logic. This reset will not self-cleared.
7	RW	0x0	core_l_po0_srstn_req Core little cluster CPU 0 PO software reset request. When HIGH, reset relative logic. This reset will not self-cleared.

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Bit	Attr	Reset Value	Description
6	RW	0x0	core_l_0_srstn_req Core little cluster CPU0 software reset request. When HIGH, reset relative logic. This reset will not self-cleared.
5	RW	0x0	strc_sys_asrstn_req PD BUS NOC AXI software reset request When HIGH, reset relative logic. This reset will not self-cleared.
4	RW	0x0	pd_core_b_niu_asrstn_req PD COREB NIU software reset request When HIGH, reset relative logic
3	RW	0x0	adb_b_srstn_req ADB between A53 big clustert and CCI software reset request When HIGH, reset relative logic. This reset will not self-cleared.
2	RW	0x0	l2_b_srstn_req A53 big cluster L2 software reset request When HIGH, reset relative logic. This reset will not self-cleared.
1	RW	0x0	core_b_po0_srstn_req Core big cluster CPU 0 PO software reset request. When HIGH, reset relative logic. This reset will not self-cleared.
0	RW	0x0	core_b_0_srstn_req core big cluster CPU0 software reset request. When HIGH, reset relative logic. This reset will not self-cleared.

CRU_SOFTRST14_CON
Address: Operational Base + offset (0x00338)
Internal software reset control register0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
13	RW	0x0	timer1_psrstn_req timer1 APB software reset request. When HIGH, reset relative logic. This reset will not self-cleared.
12	RW	0x0	timer0_psrstn_req timer0 APB software reset request. When HIGH, reset relative logic. This reset will not self-cleared.
11	RW	0x0	timer15_srstn_req timer15 software reset request. When HIGH, reset relative logic. This reset will not self-cleared.
10	RW	0x0	timer14_srstn_req timer14 software reset request. When HIGH, reset relative logic. This reset will not self-cleared.
9	RW	0x0	timer13_srstn_req timer13 software reset request. When HIGH, reset relative logic. This reset will not self-cleared.
8	RW	0x0	timer12_srstn_req timer12 software reset request. When HIGH, reset relative logic. This reset will not self-cleared.
7	RW	0x0	timer11_srstn_req timer11 software reset request. When HIGH, reset relative logic. This reset will not self-cleared.
6	RW	0x0	timer10_srstn_req timer10 software reset request. When HIGH, reset relative logic. This reset will not self-cleared.
5	RW	0x0	timer05_srstn_req timer05 software reset request. When HIGH, reset relative logic. This reset will not self-cleared.
4	RW	0x0	timer04_srstn_req timer04 software reset request. When HIGH, reset relative logic. This reset will not self-cleared.
3	RW	0x0	timer03_srstn_req timer03 software reset request. When HIGH, reset relative logic. This reset will not self-cleared.

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Bit	Attr	Reset Value	Description
2	RW	0x0	timer02_srstn_req timer02 software reset request. When HIGH, reset relative logic. This reset will not self-cleared.
1	RW	0x0	timer01_srstn_req timer01 software reset request. When HIGH, reset relative logic. This reset will not self-cleared.
0	RW	0x0	timer00_srstn_req timer00 software reset request. When HIGH, reset relative logic. This reset will not self-cleared.

CRU_MISC_CON
Address: Operational Base + offset (0x00380)
SCU control register

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	npll_lockout_mask
14	RW	0x0	gpll_lockout_mask
13	RW	0x0	cpll_lockout_mask
12	RW	0x0	dpll_lockout_mask
11	RW	0x0	aplll_lockout_mask
10	RW	0x0	apllb_lockout_mask
9	RO	0x0	reserved
8	RW	0x0	core_dbgrst_wfien A53 dbg reset wait for STANDBYWFI enable 1'b0 : A53 dbg reset has no relation to STANDBYWFI status 1'b1 : A53 dgb reset is asserted after STANDBYWFI valid
7	RW	0x0	core_wrst_wfien A53 warm reset wait for STANDBYWFI enable 1'b0 : A53 warm reset has no relation to STANDBYWFI status 1'b1 : A53 warm reset is asserted after STANDBYWFI valid

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Bit	Attr	Reset Value	Description
6	RW	0x0	core_srst_wfien A53 software reset wait for STANDBYWFI enable 1'b0 : A53 software reset has no relation to STANDBYWFI status 1'b1 : A53 software reset is asserted after STANDBYWFI valid
5	RW	0x0	dbgrstn_en A53 DBGRSTN reset enable 1'b0 : disable A53 DBGRSTN reset 1'b1 : enable A53 DBGRSTN reset
4	RW	0x0	warmrstn_en A53 warm reset enable 1'b0 : disable A53 warm reset 1'b1 : enable A53 warm reset
3:0	RW	0x0	testclk_sel Output clock selection for test 4'b0000: aclk_periph 4'b0001: clk_core_b 4'b0010: aclk_vio0 4'b0011: clk4x_ddrphy 4'b0100: aclk_video 4'b0101: clk_gpu_core 4'b0110: clk_rga_core 4'b0111: aclk_cpu 4'b1000: 24MHz 4'b1001: aclk_cci 4'b1010: 32KHz 4'b1011: clk_wifi(16.368MHz) 4'b1100: dclk_lcdc0 4'b1101: clk_core_l 4'b1110: aclk_gpu_mst 4'b1111: clk_isp

CRU_GLB_CNT_TH
Address: Operational Base + offset (0x00384)
global reset wait counter threshold

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:0	RW	0x064	glb_rst_cnt_th Global soft reset counter threshold

CRU_GLB_RST_CON
Address: Operational Base + offset (0x00388)
global reset trigger select

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Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:2	RW	0x0	pmu_glb_srst_ctrl pmu reset by global soft reset select 2'b00: pmu reset by first global soft reset 2'b01: pmu reset by second global soft reset 2'b10: pmu not reset by any global soft reset
1	RW	0x0	wdt_glb_srst_ctrl watch_dog trigger global soft reset select 1'b0: watch_dog trigger second global reset 1'b1: watch_dog trigger first global reset
0	RW	0x0	tsadc_glb_srst_ctrl TSADC trigger global soft reset select 1'b0: tsadc trigger second global reset 1'b1: tsadc trigger first global reset

CRU_GLB_RST_ST
Address: Operational Base + offset (0x0038c)
global reset status

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	W1C	0x0	snd_glb_wdt_rst_st second global watch_dog triggered reset flag 1'b0: last hot reset is not second global watch_dog triggered reset 1'b1: last hot reset is second global watch_dog triggered reset
4	W1C	0x0	fst_glb_wdt_rst_st first global watch_dog triggered reset flag 1'b0: last hot reset is not first global watch_dog triggered reset 1'b1: last hot reset is first global watch_dog triggered reset
3	W1C	0x0	snd_glb_tsadc_rst_st second global TSADC triggered reset flag 1'b0: last hot reset is not second global TSADC triggered reset 1'b1: last hot reset is second global TSADC triggered reset
2	W1C	0x0	fst_glb_tsadc_rst_st first global TSADC triggered reset flag 1'b0: last hot reset is not first global TSADC triggered reset 1'b1: last hot reset is first global TSADC triggered reset

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Bit	Attr	Reset Value	Description
1	W1C	0x0	snd_glb_rst_st second global rst flag 1'b0: last hot reset is not second global rst 1'b1: last hot reset is second global rst
0	W1C	0x0	fst_glb_rst_st first global rst flag 1'b0: last hot reset is not first global rst 1'b1: last hot reset is first global rst

CRU_SDMMC_CON0
Address: Operational Base + offset (0x00400)
sdmmc control0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:12	RO	0x0	reserved
11	WO	0x0	sdmmc_drv_sel sdmmc drive select sdmmc drive select
10:3	WO	0x00	sdmmc_drv_delaynum sdmmc drive delay number sdmmc drive delay number
2:1	WO	0x2	sdmmc_drv_degree sdmmc drive degree sdmmc drive degree
0	WO	0x0	sdmmc_init_state sdmmc initial state sdmmc initial state

CRU_SDMMC_CON1
Address: Operational Base + offset (0x00404)
sdmmc control1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:11	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
10	WO	0x0	sdmmc_sample_sel sdmmc sample select sdmmc sample select
9:2	WO	0x00	sdmmc_sample_delaynum sdmmc sample delay number sdmmc sample delay number
1:0	WO	0x0	sdmmc_sample_degree sdmmc sample degree sdmmc sample degree

CRU_SDIO0_CON0
Address: Operational Base + offset (0x00408)
sdio0 control0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:12	RO	0x0	reserved
11	WO	0x0	sdio0_drv_sel sdio0 drive select sdio0 drive select
10:3	WO	0x00	sdio0_drv_delaynum sdio0 drive delay number sdio0 drive delay number
2:1	WO	0x2	sdio0_drv_degree sdio0 drive degree sdio0 drive degree
0	WO	0x0	sdio0_init_state sdio0 initial state sdio0 initial state

CRU_SDIO0_CON1
Address: Operational Base + offset (0x0040c)
sdio0 control1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:11	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
10	WO	0x0	sdio0_sample_sel sdio0 sample select sdio0 sample select
9:2	WO	0x00	sdio0_sample_delaynum sdio0 sample delay number sdio0 sample delay number
1:0	WO	0x0	sdio0_sample_degree sdio0 sample degree sdio0 sample degree

CRU_SDIO1_CON0
Address: Operational Base + offset (0x00410)
sdio1 control0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:12	RO	0x0	reserved
11	WO	0x0	sdio1_drv_sel sdio1 drive select sdio1 drive select
10:3	WO	0x00	sdio1_drv_delaynum sdio1 drive delay number sdio1 drive delay number
2:1	WO	0x2	sdio1_drv_degree sdio1 drive degree sdio1 drive degree
0	WO	0x0	sdio1_init_state sdio1 initial state sdio1 initial state

CRU_SDIO1_CON1
Address: Operational Base + offset (0x00414)
sdio1 control1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:11	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
10	WO	0x0	sdio1_sample_sel sdio1 sample select sdio1 sample select
9:2	WO	0x00	sdio1_sample_delaynum sdio1 sample delay number sdio1 sample delay number
1:0	WO	0x0	sdio1_sample_degree sdio1 sample degree sdio1 sample degree

CRU_EMMC_CON0
Address: Operational Base + offset (0x00418)
emmc control0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:12	RO	0x0	reserved
11	WO	0x0	emmc_drv_sel emmc drive select emmc drive select
10:3	WO	0x00	emmc_drv_delaynum emmc drive delay number emmc drive delay number
2:1	WO	0x2	emmc_drv_degree emmc drive degree emmc drive degree
0	WO	0x0	emmc_init_state emmc initial state emmc initial state

CRU_EMMC_CON1
Address: Operational Base + offset (0x0041c)
emmc control1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask. When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:11	RO	0x0	reserved
10	WO	0x0	emmc_sample_sel emmc sample select emmc sample select
9:2	WO	0x00	emmc_sample_delaynum emmc sample delay number emmc sample delay number

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Bit	Attr	Reset Value	Description
1:0	WO	0x0	emmc_sample_degree emmc sample degree emmc sample degree

3.8 Timing Diagram

Power on reset timing is shown as follow:

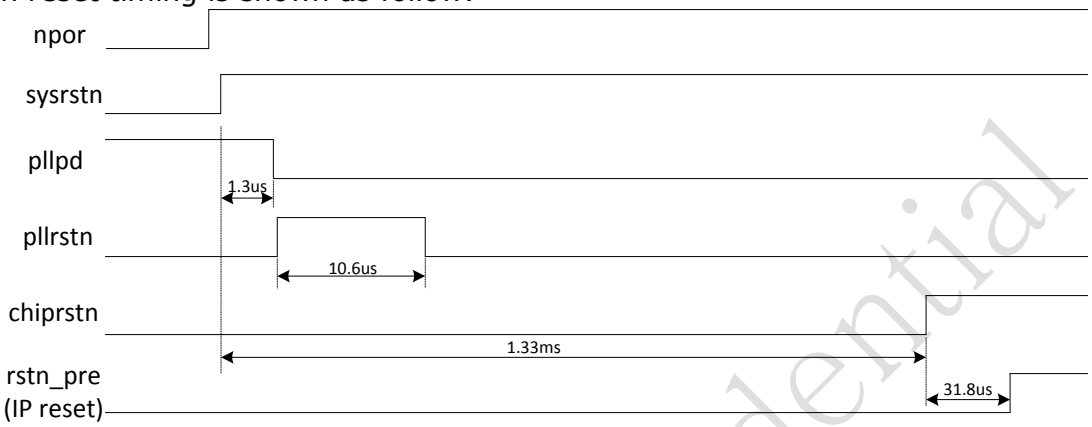

Fig. 3-8 Chip Power On Reset Timing Diagram

Npor is hardware reset signal from out-chipand power-off mode wakeup reset from PMU, which is filtered glitch to obtain signal sysrstn. To make PLLs work normally, the power down signal (pllpd) must be high when reset, and maintains high for more then 1us when sysrstn de-active. Then PLL reset signals (pllrstn) are asserted for about 10.6us, and PLLs start to lock when pllrstn de-assert, and consume about 1330us to lock. So the system will wait about 1330us, then de-active reset signal chiprstn. The signal chiprstn is used to generate output clocks in CRU. After CRU start output clocks, the system waits again for 256cycles (31.8us) to de-active signal rstn_pre, which is used to generate power on reset of all IP.

3.9 Application Notes

3.9.1 PLL usage

A. PLL output frequency configuration

The output frequency Fout is related to the input frequency Fin by:

Fout = ((Fin /NR) * NF)/ NO

Fout is clock output of PLL, and Fin is clock input of PLL from external oscillators (24MHz). Another, other factors such as NF, NR, NO can be configured by programming CRU_APLLB_CONi,CRU_APLLL_CONi,CRU_DPLL_CONi, CRU_CPLL_CONi, CRU_GPLL_CONi, CRU_NPLL_CONi registers (i=0,1,2), and their value will affect Fout as follows.

(1) CLKR: A 6-bit bus that selects the values 1-64 for the reference divider (NR)

NR = CLKR[5:0] + 1

Example:

/1 pgm 000000

/4 pgm 000011

/8 pgm 000111

(2) CLKF: A 13-bit bus that selects the values 1-4096 for the PLL multiplication factor (NF)

NF = CLKF[12:0] + 1

Example:

X1 pgm0000000000000

X2 pgm 0000000000001

X4096 pgm0111111111111

(3) CLKOD: A 4-bit bus that selects the value1,2-16(even only) for the PLL post VCO divider

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(NO)
NO = CLKOD[3:0] + 1
Example:
/1 pgm 0000
/2 pgm 0001
/4 pgm 0011
/8 pgm 0111

- BWADJ: A 12-bit bus that selects the values 1-4096 for the bandwidth divider (NB)

NB = BWADJ[11:0] + 1
Example:
/1 pgm 000000000000
/4 pgm 000000000011
/8 pgm 000000000111

The recommended setting of NB: NB = NF / 2.

B. PLL frequency range requirement
If different CLKR, CLKF and CLKOD configuration value cause internal out of range, unpredicted result will be caused.

Fin value range requirement:	269kHz – 2200MHz
Fref = Fin/NR value range requirement:	269kHz – 2200MHz
Fvco = (Fin/NR)*NF value range requirement:	440MHz – 2200MHz
Fout = ((Fin/NR)*NF)/NO value range requirement:	27.5MHz – 2200MHz

C. PLL setting consideration
Optimization of the PLL settings for jitter < +/- 2.5% of the output period/sq rt(NO) require running the VCO at maximum frequency and dividing down using the NO divider to get the required Fout, i.e. maximum NO.
Optimization for minimum power (Fvco/1100MHz * 3.3 mA) requires setting the VCO frequency at the minimum frequency and using the lowest NO setting.
These two values, minimum jitter or minimum power will determine your choice of settings. A larger value of input divider NR gives a longer lock time, and higher long term as well as period jitter. It is better to use a lower value of NR where possible.

3.9.2 PLL frequency change method

When the PLL settings are changed, it has to reset PLL by programming registers CRU_APLLB_CON3, CRU_APLLL_CON3, CRU_DPLL_CON3, CRU_CPLL_CON3, CRU_GPLL_CON3, CRU_NPLL_CON3, and reserve at least 5us after valid settings, referring to the following figure.

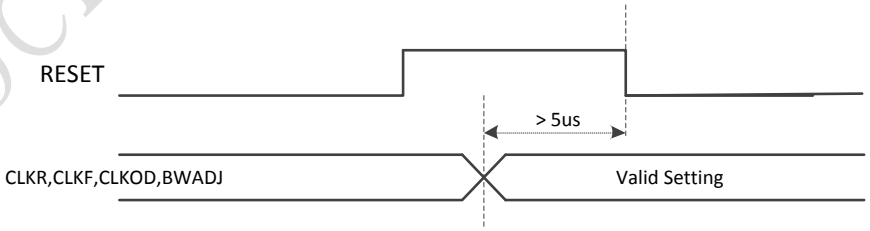

Fig. 3-9 PLL setting change timing

Before set some factors such as NR/NF/NO/BS to change PLL output frequency, you must change chip from normal to slow mode by programming CRU_xPLL_CON3[9:8]. Then until PLL is lock state by checking CRU_xPLL_CON1[15]register, or after delay about (NR * 500) / Fin, you can change PLL into normal mode.

3.9.3 Fractional divider usage

To get specific frequency, clocks of I2S, SPDIF, UARTcan be generated by fractional divider. Generally you must set that denominator is **20 times larger** than numerator to generate precise clock frequency. So the fractional divider applies only to generate low frequency clock like I2S, UART.

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All the fractional divider has auto-gating control. When fractional divider is not selected, the divider clock is gated. So fractional divider must be selected before changing configuration.

3.9.4 Global software reset

Two global software resets are designed in this chip, you can program CRU_GLB_SRST_FST_VALUE[15:0] as 0xfdb9 to assert the first global software reset glb_srstn_1 and program CRU_GLB_SRST_SND_VALUE[15:0] as 0xeca8 to assert the second global software reset glb_srstn_2. These two software resets are self-deasserted by hardware.
TSADC, WDT and PMU also can trigger glb_srstn_1 or glb_srstn_2.s
CRU_GLB_RST_CON controls which global soft-reset will be triggered.
After global reset, the reset trigger source can be check in CRU_GLB_RST_ST.
glb_srstn_1 resets almost all chip logic except PMU_SYS_REG0~3, which can be used to store something when reset.
glb_srstn_2 resets almost all chip logic except PMU_SYS_REG0~3, GRF and GPIOs.

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Chapter 4 Power Management Unit (PMU)

4.1 Overview

In order to meet low power requirements, a power management unit (PMU) is designed for controlling power resources in PX5. The PX5 PMU is dedicated for managing the power of the whole chip.

4.2 Features

- Support 3 voltage domains including VD_CORE, VD_LOGIC and VD_PMU
- Support 15 separate power domains in the whole chip, which can be power up/down by software based on different application scenes
- In power mode, PMU could power up/down pd_a53_l0, pd_scu_l, vd_core, and pd_bus by hardware
- Support Cortex-A53 core source clock gating in power mode
- Support Cortex-A53 L2 flush request by hardware in power mode
- Support power down/up pd_vio, pd_gpu_0/pd_gpu_1, pd_peri, pd_video by software
- Support power down/up pd_scu_l, pd_a53_l0/1/2/3 by software
- Support power down/up pd_scu_b, pd_a53_b0/1/2/3 by software

4.3 Block Diagram

4.3.1 power domain partition

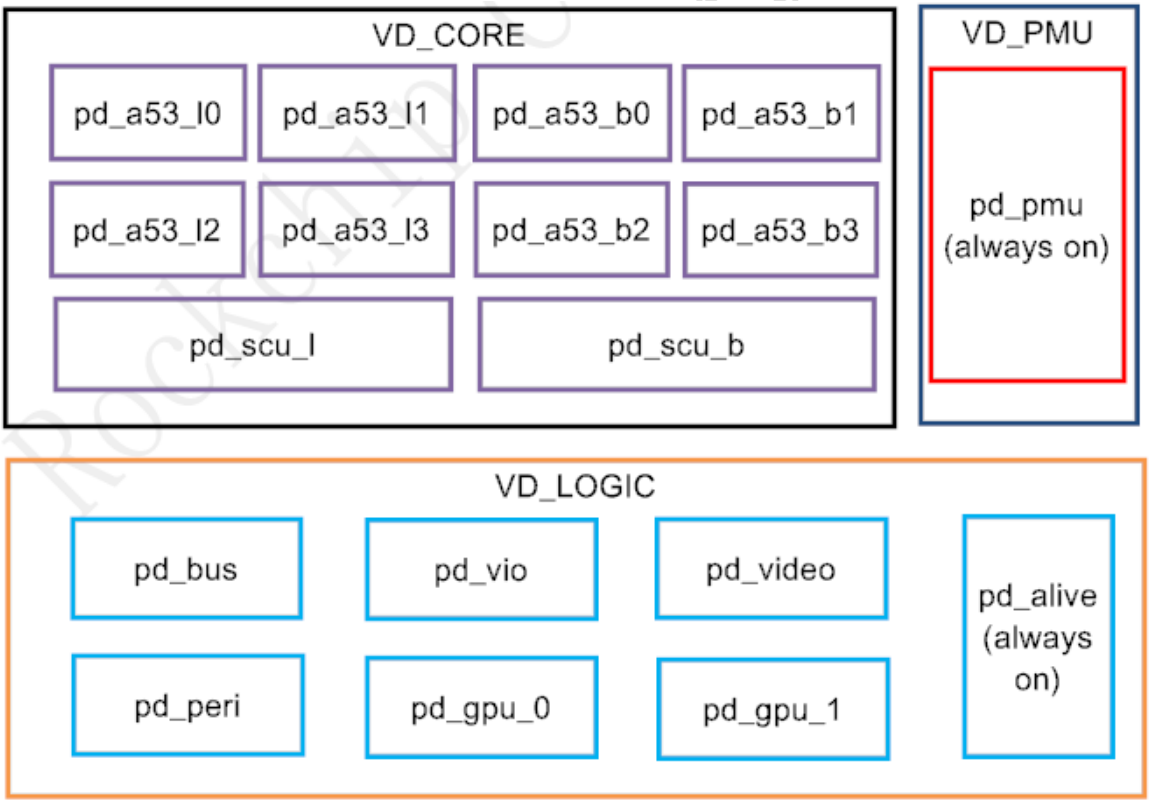

Fig. 4-1 PX5 Power Domain Partition

The above diagram describes the power domain and voltage domain partition, and the following table lists all the power domains.

Table 4-1 PX5 Power Domain and Voltage Domain Summary

Voltage Domain	Power Domain	Description
VD_CORE (PD_CORE system)	pd_a53_l0	Cortex-A53 little cluster core 0, L1C and Neon
	pd_a53_l1	Cortex-A53 little cluster core 1, L1C and Neon
	pd_a53_l2	Cortex-A53 little cluster core 2, L1C and Neon

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	pd_a53_l3	Cortex-A53 little cluster core 3, L1C and Neon
	pd_scu_l	Cortex-A53 little cluster SCU, L2C and L2 MEM
	pd_a53_b0	Cortex-A53 big cluster core 0, L1C and Neon
	pd_a53_b1	Cortex-A53 big cluster core 1, L1C and Neon
	pd_a53_b2	Cortex-A53 big cluster core 2, L1C and Neon
	pd_a53_b3	Cortex-A53 big cluster core 3, L1C and Neon
	pd_scu_b	Cortex-A53 big cluster SCU, L2C and L2 MEM
VD_LOGIC	pd_bus	Soc architecture subsystem, include soc architecture (NOC) ,Internal Memory, TZMA, GIC400, MCU, CRYPTO, I2S, SPDIF, TSP, BUS_DMA, DDR controller, PWM 0/1/2/3, I2C, SCU, Mailbox
	pd_peri	Peripheral subsystem , include DMAC_PERI, GMAC, NANDC0, HSIC/USB Host0/USB OTG, SDMMC/SDIO0/SDIO1/eMMC，HSADC, TSADC, UART, I2C, SPI, TSP
	pd_vio	Video input/output system, include VOP, ISP, IEP, RGA, MIPI-CSI, MIPI-DSI, LVDS, HDMI, eDP
	pd_alive	CRU，GRF， GPIO 1-3， TIMER, WDT
	pd_video	Video Encode&Decode , include VEPU，VDPU, HEVC
	pd_gpu_0	Include bus logic of GPU
	pd_gpu_1	Include GPU core
VD_PMU	pd_pmu	PMU, SRAM(4K), Secure GRF, GPIO0, PVTM

Notes: "Always on" means that their power supply can be switched off only by external PMIC module. Only one "always on" power domain is in a voltage domain.

4.3.2 PMU block diagram

The following figure is the PMU block diagram. The PMU includes the 3 following sections:

- APB interface and register, which can accept the system configuration
- Low Power State Control, which generate low power control signals.
- Power Switch Control, which control all power domain switch

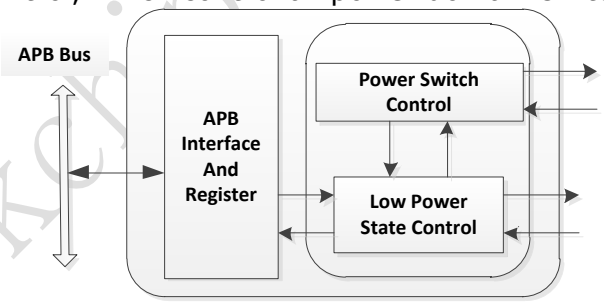

Fig. 4-2 PMU Bock Diagram

4.4 Function Description

First of all, we define two operation modes of PMU, normal mode and power mode.
When operating at normal mode, that means software can manage power sources directly by accessing PMU register.
For example, Cortex-A53 CPU can write PMU_PWRDN_CON register to determine that power off/on which power domain independently.
When operating at power mode, software manages power sources indirectly through FSM (Finite States Machine) in PMU.
That means software also can configure PMU registers to power down/up some power resources, but these setting will not be executed immediately after configuration. They will delay to execute after FSM running in particular phase. The power mode can support some functions that cannot support in normal mode. For example, some components inside PX5 (e.g. Cortex-A53 little cluster core 0) can shut down itself through power mode.
To entering power mode, after setting some power configurations, the PMU_POWER_MODE[0]

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bit must be set 1 to enable PMU FSM. Then Cortex-A53 CPU needs to execute a WFI command to perform ready signal. After PMU detects all Cortex-A53 CPUs in WFI status, then the FSM will be fetched. And the specific power sources will be controlled during specific status in FSM. So the power mode is a "delay affect" way to handle power sources inside the PX5 chip.

4.5 Register Description

4.5.1 Register Summary

Name	Offset	Size	Reset Value	Description
PMU_PMU_WAKEUP_CFG0	0x0000	W	0x00000000	pmu wakeup configure register 0
PMU_PMU_WAKEUP_CFG1	0x0004	W	0x00000000	pmu wakeup configure register 1
PMU_PMU_WAKEUP_CFG2	0x0008	W	0x00000000	pmu wakeup configure register 2
PMU_PMU_PWRDN_CON	0x000c	W	0x00000000	pmu power down configure register
PMU_PMU_PWRDN_ST	0x0010	W	0x00000000	pmu power down status register
PMU_PMU_PWRMODE_CORE_CON	0x0014	W	0x00000000	pmu power mode configure register of core resource
PMU_PMU_PWRMODE_COMMON_CON	0x0018	W	0x00000000	pmu power mode configure register of common resource
PMU_PMU_SFT_CON	0x001c	W	0x00000000	pmu software configure register
PMU_PMU_INT_CON	0x0020	W	0x00000000	pmu interrupt configure register
PMU_PMU_INT_ST	0x0024	W	0x00000000	pmu interrupt status register
PMU_PMU_GPIO_POS_INT_CON	0x0028	W	0x00000000	pmu gpio posedge interrupt configure register
PMU_PMU_GPIO_NEG_INT_CON	0x002c	W	0x00000000	pmu gpio negedge interrupt configure register
PMU_PMU_GPIO_POS_INT_ST	0x0030	W	0x00000000	pmu gpio posedge interrupt status register
PMU_PMU_GPIO_NEG_INT_ST	0x0034	W	0x00000000	pmu gpio negedge interrupt status register
PMU_PMU_CORE_PWR_ST	0x0038	W	0x00000000	pmu core power status register
PMU_PMU_BUS_IDLE_REQ	0x003c	W	0x00000000	pmu bus idle request register
PMU_PMU_BUS_IDLE_ST	0x0040	W	0x00000000	pmu bus idle status register
PMU_PMU_POWER_ST	0x0044	W	0x00000000	pmu power status register
PMU_PMU_OSC_CNT	0x0048	W	0x00000000	pmu osc count register
PMU_PMU_PLLLOCK_CNT	0x004c	W	0x00000000	pmu pll lock count register
PMU_PMU_PLLRST_CNT	0x0050	W	0x00000000	pmu pll reset count register
PMU_PMU_STABLE_CNT	0x0054	W	0x00000000	pmu power stable count register
PMU_PMU_DDRIO_PWRON_CNT	0x0058	W	0x00000000	pmu ddrio power on count register
PMU_PMU_WAKEUP_RST_CLR_CNT	0x005c	W	0x00000000	pmu wakeup reset clear count register
PMU_PMU_DDR_SREF_ST	0x0060	W	0x00000000	pmu ddr self refresh status register
PMU_PMU_SYS_REG0	0x0064	W	0x00000000	pmu system register 0
PMU_PMU_SYS_REG1	0x0068	W	0x00000000	pmu system register 1
PMU_PMU_SYS_REG2	0x006c	W	0x00000000	pmu system register 2

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Name	Offset	Size	Reset Value	Description
PMU_PMU_SYS_REG3	0x0070	W	0x00000000	pmu system register 3
PMU_PMU_GPU_APM_CON	0x0074	W	0x00000000	pmu gpu apm power configure register
PMU_PMU_GPU_APM_ST	0x0078	W	0x00000000	pmu gpu apm power status register
PMU_PMU_TIMEOUT_CNT	0x007c	W	0x00000000	pmu timeout count register

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

4.5.2 Detail Register Description

PMU_PMU_WAKEUP_CFG0

Address: Operational Base + offset (0x0000)

pmu wakeup configure register 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	gpio0d_posedge_en gpio0d posedge pulse wakeup enable 0: disable 1: enable
23:16	RW	0x00	gpio0c_posedge_en gpio0c posedge pulse wakeup enable 0: disable 1: enable
15:8	RW	0x00	gpio0b_posedge_en gpio0b posedge pulse wakeup enable 0: disable 1: enable
7:0	RW	0x00	gpio0a_posedge_en gpio0a posedge pulse wakeup enable 0: disable 1: enable

PMU_PMU_WAKEUP_CFG1

Address: Operational Base + offset (0x0004)

pmu wakeup configure register 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	gpio0d_negedge_en gpio0d negedge pulse wakeup enable 0: disable 1: enable
23:16	RW	0x00	gpio0c_negedge_en gpio0c negedge pulse wakeup enable 0: disable 1: enable
15:8	RW	0x00	gpio0b_negedge_en gpio0b negedge pulse wakeup enable 0: disable 1: enable
7:0	RW	0x00	gpio0a_negedge_en gpio0a negedge pulse wakeup enable 0: disable 1: enable

PMU_PMU_WAKEUP_CFG2

Address: Operational Base + offset (0x0008)

pmu wakeup configure register 2

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10	RW	0x0	tiemout_en pmu time out wakeup enable 0: disable 1: enable
9	RW	0x0	wdt_mcu_en mcu watch dog wakeup enable 0: disable 1: enable
8	RW	0x0	sft_en software wakeup enable 0: disable 1: enable
7	RW	0x0	usbdev_en usb device detect wakeup enable 0: disable 1: enable
6	RW	0x0	timer_en timer wakeup enable 0: disable 1: enable
5	RW	0x0	sim_en sim detect wakeup enable 0: disable 1: enable
4	RW	0x0	sdmmc_en sdmmc detect wakeup enable 0: disable 1: enable
3	RW	0x0	sdio_en sdio detect wakeup enable 0: disable 1: enable
2	RW	0x0	gpio_int_en gpio interrupt wakeup enable 0: disable 1: enable
1	RW	0x0	int_cluster_b_en cluster_b interrupt wakeup enable 0: disable 1: enable
0	RW	0x0	int_cluster_l_en cluster_l interrupt wakeup enable 0: disable 1: enable

PMU_PMU_PWRDN_CON

Address: Operational Base + offset (0x000c)

pmu power down configure register

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18	RW	0x0	chip_pwrdwn_en chip power down enable 0: disable 1: enable

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Bit	Attr	Reset Value	Description
17	RW	0x0	pd_gpu_1_pwrdwn_en pd_gpu_1 power down enable 0: disable 1: enable
16	RW	0x0	pd_gpu_0_pwrdwn_en pd_gpu_0 power down enable 0: disable 1: enable
15	RW	0x0	pd_vio_pwrdwn_en pd_vio power down enable 0: disable 1: enable
14	RW	0x0	pd_video_pwrdwn_en pd_video power down enable 0: disable 1: enable
13	RW	0x0	pd_peri_pwrdwn_en pd_peri power down enable 0: disable 1: enable
12	RW	0x0	pd_bus_pwrdwn_en pd_bus power down enable 0: disable 1: enable
11	RO	0x0	reserved
10	RW	0x0	pd_core_pwrdwn_en pd_core power down enable 0: disable 1: enable
9	RW	0x0	pd_scu_b_pwrdwn_en pd_scu_b power down enable 0: disable 1: enable
8	RW	0x0	pd_a53_b3_pwrdwn_en pd_a53_b3 power down enable 0: disable 1: enable
7	RW	0x0	pd_a53_b2_pwrdwn_en pd_a53_b2 power down enable 0: disable 1: enable
6	RW	0x0	pd_a53_b1_pwrdwn_en pd_a53_l2 power down enable 0: disable 1: enable
5	RW	0x0	pd_a53_b0_pwrdwn_en pd_a53_l0 power down enable 0: disable 1: enable
4	RW	0x0	pd_scu_l_pwrdwn_en pd_scu_l power down enable 0: disable 1: enable

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Bit	Attr	Reset Value	Description
3	RW	0x0	pd_a53_l3_pwrdwn pd_a53_l3 power down enable 0: disable 1: enable
2	RW	0x0	pd_a53_l2_pwrdwn pd_a53_l2 power down enable 0: disable 1: enable
1	RW	0x0	pd_a53_l1_pwrdwn pd_a53_l1 power down enable 0: disable 1: enable
0	RW	0x0	pd_a53_l0_pwrdwn_en pd_a53_l0 power down enable 0: disable 1: enable

PMU_PMU_PWRDN_ST
Address: Operational Base + offset (0x0010)
pmu power down status register

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RW	0x0	pd_gpu_1_pwr_stat pd_gpu_1 power state 0: powered up 1: powered down
15	RW	0x0	pd_gpu_0_pwr_stat pd_gpu_0 power state 0: powered up 1: powered down
14	RW	0x0	pd_vio_pwr_stat pd_vio power state 0: powered up 1: powered down
13	RW	0x0	pd_video_pwr_stat pd_video power state 0: powered up 1: powered down
12	RW	0x0	pd_peri_pwr_stat pd_peri power state 0: powered up 1: powered down
11	RW	0x0	pd_bus_pwr_stat pd_bus power stat 0: powered up 1: powered down
10	RW	0x0	pd_core_pwr_stat pd_core power state 0: powered up 1: powered down
9	RW	0x0	pd_scu_b_pwr_stat pd_scu_b power state 0: powered up 1: powered down

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Bit	Attr	Reset Value	Description
8	RW	0x0	pd_a53_b3_pwr_stat pd_a53_b3 power state 0: powered up 1: powered down
7	RW	0x0	pd_a53_b2_pwr_stat pd_a53_b2 power state 0: powered up 1: powered down
6	RW	0x0	pd_a53_b1_pwr_stat pd_a53_b1 power state 0: powered up 1: powered down
5	RW	0x0	pd_a53_b0_pwr_stat pd_a53_b0 power state 0: powered up 1: powered down
4	RW	0x0	pd_scu_l_pwr_stat pd_scu_l power state 0: powered up 1: powered down
3	RW	0x0	pd_a53_l3_pwr_stat pd_a53_l3 power state 0: powered up 1: powered down
2	RW	0x0	pd_a53_l2_pwr_stat pd_a53_l2 power state 0: powered up 1: powered down
1	RW	0x0	pd_a53_l1_pwr_stat pd_a53_l1 power state 0: powered up 1: powered down
0	RW	0x0	pd_a53_l0_pwr_stat pd_a53_l0 power state 0: powered up 1: powered down

PMU_PMU_PWRMODE_CORE_CON

Address: Operational Base + offset (0x0014)

pmu power mode configure register of core resource

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9	RW	0x0	l2_flush_en flush l2 by hardware when in power mode 0: disable 1: enable
8	RW	0x0	l2_idle_en wait l2 idle when in power mode 0: disable 1: enable
7	RW	0x0	core_pd_en power down pd_core when in power mode 0: disable 1: enable

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Bit	Attr	Reset Value	Description
6	RW	0x0	scu_l_pd_en power down scu_l when in power mode 0: disable 1: enable
5	RW	0x0	clr_core send cluster_l access peri idle request to niu in power mode 0: disable 1: enable
4	RW	0x0	clr_cluster_l idle request to adb400 in the cluster_l enable 0: disable 1: enable
3	RW	0x0	cpu0_pd_en power down core0 of cluster_l in power mode 0: disable 1: enable
2	RW	0x0	clr_cci send core niu access memory schedule idle request when in power mode 0: disable 1: enable
1	RW	0x0	clk_core_src_gate_en cluster_l clock gate enable when in power mode 0: disable 1: enable
0	RW	0x0	global_int_disable_l_cfg global interrupt disable configure for cluster_l 0: enable spi to gic400 1: disable spi to gic400(will be work if global_int_disable_b_cfg = 1)

PMU_PMU_PWRMODE_COMMON_CON

Address: Operational Base + offset (0x0018)

pmu power mode configure register of common resource

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29	RW	0x0	main_cluster cluster remap when power mode when enable, cluster_l/cluster_b power control signal will switch to each other 0: main cluster is cluster_l, power mode will power down cluster_l and core_l0 by hardware 1: main cluster is cluster_b, power mode will power down cluster_b and core_b0 by hardware
28:26	RO	0x0	reserved
25	RW	0x0	clr_cci400 send cci400 idle request when in power mode 0: disable 1: enable

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Bit	Attr	Reset Value	Description
24	RW	0x0	clr_cxcs send cxcs idle request when in power mode 0: disable 1: enable
23	RW	0x0	clr_mcu wait mcu sleeping when in power mode 0: disable 1: enable
22	RW	0x0	clr_gpu send gpu niu idle request when in power mode 0: disable 1: enable
21	RW	0x0	clr_vio send vio niu idle request when in power mode 0: disable 1: enable
20	RW	0x0	clr_video send video niu idle request when in power mode 0: disable 1: enable
19	RW	0x0	clr_peri send peri niu idle request when in power mode 0: disable 1: enable
18	RW	0x0	clr_msch send memory schedule idle request when in power mode 0: disable 1: enable
17	RW	0x0	clr_dma send dma niu idle request when in power mode 0: disable 1: enable
16	RW	0x0	clr_bus send bus niu idle request when in power mode 0: disable 1: enable
15	RW	0x0	clr_alive send alive niu idle request when in power mode 0: disable 1: enable
14	RW	0x0	clr_pmu send pmu niu idle request when in power mode 0: disable 1: enable
13	W1C	0x0	ddrio_ret_de_req ddrio retention de-assert request write one clear

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Bit	Attr	Reset Value	Description
12	RW	0x0	ddrio_ret_en ddrio retention enable when in power mode 0: disable 1: enable
11	RW	0x0	ddrc_gating_en ddr controller auto gating when in power mode 0: disable 1: enable
10	RW	0x0	sref_enter_en ddr self_refresh by hardware when in power mode 0: disable 1: enable
9	RW	0x0	input_clamp_en clamp vd_logic when in power mode 0: disable 1: enable
8	RW	0x0	osc_disable osc disable when in power mode 1: disable 0: enable
7	RW	0x0	alive_use_lf alive low frequency mode when in power mode 0: disable 1: enable
6	RW	0x0	pmu_use_lf pmu low frequency mode enable when in power mode 0: disable 1: enable
5	RW	0x0	power_off_req_cfg send power off request to PMIC when in power mode 0: disable 1: enable
4	RW	0x0	pll_pd_en power down pll when in power mode 0: disable 1: enable
3	RW	0x0	wakeup_reset_en wakeup reset enable when in power mode 0: disable 1: enable
2	RW	0x0	bus_pd_en power down pd_bus when power mode 0: disable 1: enable
1	RW	0x0	reserved
0	RW	0x0	power_mode_en enter power mode enable, will auto self-clear when in power mode 0: disable 1: enable

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PMU_PMU_SFT_CON
Address: Operational Base + offset (0x001c)
pmu software configure register

Bit	Attr	Reset Value	Description
31:28	RW	0x0	dbgpwrdup_b dbg powered up of cluster_b status by software 0: dbgpwrdup of cluster_b is 0 1: dbgpwrdup of cluster_b is 1
27	RW	0x0	dbgpwrupreq_b_en dbg pwrup requst of cluster_b function enable 0: disable 1: enable
26	RW	0x0	dbgpwrupreq_l_en dbg powered up request function of cluster_l enable 0: disable 1: enable
25	RW	0x0	dbgnopwrdwn_gpu_1_enable dbg no power down of pd_gpu_1 function enable 0: disable 1: enable
24:21	RW	0x0	dbgnopwrdwn_b_enable dbg nopower down of cluster_b function enable 0: disable 1: enable
20:17	RW	0x0	dbgnopwrdwn_l_enable dbgnopowerdown function of cluster_l enable 0: disable 1: enable
16	RW	0x0	gpu_apm_reject gpu apm request reject enable bit 0: do not reject apm power request 1: reject apm power request
15	RW	0x0	osc_disable_cfg osc disable configure by software 1: disable osc 0: enable psc
14	RW	0x0	pmu_lf_ena_cfg pd_pmu low frequency mode configure by software 0: disable 1: enable
13	RW	0x0	alive_lf_ena_cfg pd_alive low frequency mode configure by software 0: disable 1: enable
12	RW	0x0	power_off_ddrio_cfg power off ddrio configure by software 0: disable 1: enable

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Bit	Attr	Reset Value	Description
11	RW	0x0	ddr_io_ret_cfg ddr io retention configure by software 0: disable 1: enable
10	RW	0x0	upctl_c_sysreq_cfg upctl idle request configure 0: disable 1: enable
9	RW	0x0	global_int_disable_b global interrupt disable send by cluster_b will work when global_int_disable_l is enabled
8	RW	0x0	cluster_l_clk_src_gating_cfg cluster_l clock source gating configure 0: disable 1: enable
7	RW	0x0	cluster_b_clk_src_gating_cfg cluster_b clock source gating configure 0: disable 1: enable
6	RW	0x0	l2flushreq_cluster_b l2flush request to cluster_b by software 0: disable 1: enable
5	RW	0x0	acinactm_cluster_b send acinactm to cluster_b by software 1: disable gpu apm interface 0: enable gpu apm interface
4	RW	0x0	l2flushreq_cluster_l send l2 flush request to cluster_l by software 0: disable 1: enable
3	RW	0x0	acinactm_cluster_l_cfg acinactm indicate to cluster_l 0: acinactm to cluster_l is 0 1: acinactm to cluster_l is 1
2	RW	0x0	gpu_apm_dis gpu apm interface disable 1: disable gpu apm interface 0: enable gpu apm interface
1	RW	0x0	dbgpwrdup_l0_cfg dbg powered up of pd_a53_l0 enable when in power mode 0: disable 1: enable
0	RW	0x0	wakeup_sft software wakeup request bit A 0 to 1 pulse posedge will wakeup pmu when in low power mode

PMU_PMU_INT_CON

Address: Operational Base + offset (0x0020)

pmu interrupt configure register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31	RW	0x0	pd_gpu_1_pwr_switch_int_en pd_gpu_1 power switch interrupt enable 0: disable 1: enable
30	RW	0x0	pd_gpu_0_pwr_switch_int_en pd_gpu_0 power switch interrupt enable 0: disable 1: enable
29	RW	0x0	pd_vio_pwr_switch_int_en pd_vio power switch interrupt enable 0: disable 1: enable
28	RW	0x0	pd_video_pwr_switch_int_en pd_video power switch interrupt enable 0: disable 1: enable
27	RW	0x0	pd_peri_pwr_switch_int_en pd_peri power switch interrupt enable 0: disable 1: enable
26	RW	0x0	pd_bus_pwr_switch_int_en pd_bus power switch interrupt enable 0: disable 1: enable
25	RW	0x0	pd_core_pwr_switch_int_en pd_core power switch interrupt enable 0: disable 1: enable
24	RW	0x0	pd_scu_b_pwr_switch_int_en pd_scu_b power switch interrupt enable 0: disable 1: enable
23	RW	0x0	pd_a53_b3_pwr_switch_int_en pd_a53_b3 power switch interrupt enable 0: disable 1: enable
22	RW	0x0	pd_a53_b2_pwr_switch_int_en pd_a53_b2 power switch interrupt enable 0: disable 1: enable
21	RW	0x0	pd_a53_b1_pwr_switch_int_en pd_a53_b1 power switch interrupt enable 0: disable 1: enable
20	RW	0x0	pd_a53_b0_pwr_switch_int_en pd_a53_b0 power switch interrupt enable 0: disable 1: enable
19	RW	0x0	pd_scu_l_pwr_switch_int_en pd_scu_l power switch interrupt enable 0: disable 1: enable

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Bit	Attr	Reset Value	Description
18	RW	0x0	pd_a53_l3_pwr_switch_int_en pd_a53_l3 power switch interrupt enable 0: disable 1: enable
17	RW	0x0	pd_a53_l2_pwr_switch_int_en pd_a53_l2 power switch interrupt enable 0: disable 1: enable
16	RW	0x0	pd_a53_l1_pwr_switch_int_en pd_a53_l1 power switch interrupt enable 0: disable 1: enable
15	RW	0x0	pd_a53_l0_pwr_switch_int_en pd_a53_l0 power switch interrupt enable 0: disable 1: enable
14	RW	0x0	wakeup_sft_int_en software wakeup interrupt enable 0: disable 1: enable
13	RW	0x0	wakeup_timeout_int_en timeout wakeup interrupt enable 0: disable 1: enable
12	RW	0x0	wakeup_wdt_mcu_int_en mcu watch dog wakeup interrupt enable 0: disable 1: enable
11	RW	0x0	wakeup_sdmmc_int_en sdmmc wakeup interrupt enable 0: disable 1: enable
10	RW	0x0	wakeup_sdio_int_en sdio wakeup interrupt enable 0: disable 1: enable
9	RW	0x0	wakeup_gpio_pos_int_en gpio posedge wakeup interrupt enable 0: disable 1: enable
8	RW	0x0	wakeup_gpio_neg_int_en gpio negedge wakeup interrupt enable 0: disable 1: enable
7	RW	0x0	wakeup_gpio_int_int_en gpio interrupt wakeup interrupt enable 0: disable 1: enable
6	RW	0x0	wakeup_usbdev_int_en usbdev wakeup interrupt enable 0: disable 1: enable

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Bit	Attr	Reset Value	Description
5	RW	0x0	wakeup_timer_int_en timer interrupt wakeup enable 0: disable 1: enable
4	RW	0x0	wakeup_int_cluster_l_int_en cluster_l interrupt wakeup enable 0: disable 1: enable
3	RW	0x0	wakeup_int_clsuter_b_int_en cluster_b wakeup interrupt enable 0: disable 1: enable
2	RW	0x0	wakeup_sim_int_en sim wakeup interrupt enable 0: disable 1: enable
1	RW	0x0	pwrmode_wakeup_int_en power mode wakeup interrupt enable 0: disable 1: enable
0	RW	0x0	pmu_int_en global interrupt enable 0: disable 1: enable

PMU_PMU_INT_ST

Address: Operational Base + offset (0x0024)

pmu interrupt status register

Bit	Attr	Reset Value	Description
31	RW	0x0	pd_gpu_1_pwr_switch_status pd_gpu_1 power switch status 0: the power of pd_gpu_1 is not switched 1: the power of pd_gpu_1 is switched
30	RW	0x0	pd_gpu_0_switch_status pd_gpu_0 power switch status 0: the power of pd_gpu_0 is not switched 1: the power of pd_gpu_0 is switched
29	RW	0x0	pd_vio_pwr_switch_status pd_vio power switch status 0: the power of pd_vio is not switched 1: the power of pd_vio is switched
28	RW	0x0	pd_video_pwr_switch_status pd_video power switch status 0: the power of pd_video is not switched 1: the power of pd_video is switched
27	RW	0x0	pd_peri_pwr_switch_status pd_peri power switch status 0: the power of pd_peri is not switched 1: the power of pd_peri is switched
26	RW	0x0	pd_bus_pwr_switch_status pd_bus power switch status 0: the power of pd_bus is not switched 1: the power of pd_bus is switched

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Bit	Attr	Reset Value	Description
25	RW	0x0	pd_core_pwr_switch_status pd_core power switch status 0: the power of pd_core is not switched 1: the power of pd_core is switched
24	RW	0x0	pd_scu_b_pwr_switch_status pd_scu_b power switch status 0: the power of pd_scu_b is not switched 1: the power of pd_scu_b is switched
23	RW	0x0	pd_a53_b3_pwr_switch_status pd_a53_b3 power switch status 0: the power of pd_a53_b3 is not switched 1: the power of pd_a53_b3 is switched
22	RW	0x0	pd_a53_b2_pwr_switch_status pd_a53_b2 power switch status 0: the power of pd_a53_b2 is not switched 1: the power of pd_a53_b2 is switched
21	RW	0x0	pd_a53_b1_pwr_switch_status pd_a53_b1 power switch status 0: the power of pd_a53_b1 is not switched 1: the power of pd_a53_b1 is switched
20	RW	0x0	pd_a53_b0_pwr_switch_status pd_a53_b0 power switch status 0: the power of pd_a53_b0 is not switched 1: the power of pd_a53_b0 is switched
19	RW	0x0	pd_scu_l_pwr_switch_status pd_scu_l power switch status 0: the power of pd_scu_l is not switched 1: the power of pd_scu_l is switched
18	RW	0x0	pd_a53_l3_pwr_switch_status pd_a53_l3 power switch status 0: the power of pd_a53_l3 is not switched 1: the power of pd_a53_l3 is switched
17	RW	0x0	pd_a53_l2_pwr_switch_status pd_a53_l2 power switch status 0: the power of pd_a53_l2 is not switched 1: the power of pd_a53_l2 is switched
16	RW	0x0	pd_a53_l1_pwr_switch_status pd_a53_l1 power switch status 0: the power of pd_a53_l1 is not switched 1: the power of pd_a53_l1 is switched
15	RW	0x0	pd_a53_l0_pwr_switch_status pd_a53_l0 power switch status 0: the power of pd_a53_l0 is not switched 1: the power of pd_a53_l0 is switched
14	RW	0x0	wakeup_sft_status software wakeup status 0: not wakeup by software 1: wakeup by software
13	RW	0x0	wakeup_timeout_status timeout wakeup status 0: not wakeup by timeout 1: wakeup by timeout

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Bit	Attr	Reset Value	Description
12	RW	0x0	wakeup_wdt_mcu_status mcu watch dog wakeup status 0: not wakeup by mcu watchdog 1: wakeup by mcu watchdog
11	RW	0x0	wakeup_sdmmc_status sdmmc wakeup status 0: not wakeup by sdmmc detect 1: wakeup by sdmmc detect
10	RW	0x0	wakeup_sdio_status sdio wakeup status 0: not wakeup by sdio detect 1: wakeup by sdio detect
9	RW	0x0	wakeup_gpio_pos_status gpio posedge pulse wakeup status 0: not wakeup by gpio posedge pulse 1: wakeup by gpio posedge pulse
8	RW	0x0	wakeup_gpio_neg_status gpio negedge pulse wakeup status 0: not wakeup by gpio negedge pulse 1: wakeup by gpio negedge pulse
7	RW	0x0	wakeup_gpio_int_status gpio interrupt wakeup status 0: not wakeup by gpio int 1: wakeup by gpio int
6	RW	0x0	wakeup_usbdev_status usbdev detect wakeup status 0: not wakeup by usbdev detect 1: wakeup by usbdev detect
5	RW	0x0	wakeup_timer_status timer wakeup status 0: not wakeup by timer 1: wakeup by timer
4	RW	0x0	wakeup_int_cluster_l_status cluster_l interrupt wakeup status 0: not wakeup by interrupt cluster_l 1: wakeup by interrupt cluster_l
3	RW	0x0	wakeup_int_cluster_b_status cluster_b interrupt wakeup status 0: not wakeup by interrupt cluster_b 1: wakeup by interrupt cluster_b
2	RW	0x0	wakeup_sim_status sim wakeup status 0: not wakeup by sim 1: wakeup by sim
1	RW	0x0	pwrmode_wakeup_status power mode wakeup status 0: not wakeup from power mode 1: wakeup from power mode
0	RO	0x0	reserved

PMU_PMU_GPIO_POS_INT_CON

Address: Operational Base + offset (0x0028)

pmu gpio posedge interrupt configure register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	gpio0d_pos_int_en gpio0d posedge pulse interrupt enable 0: disable 1: enable
23:16	RW	0x00	gpio0c_pos_int_en gpio0c posedge pulse interrupt enable 0: disable 1: enable
15:8	RW	0x00	gpio0b_pos_int_en gpio0b posedge pulse interrupt enable 0: disable 1: enable
7:0	RW	0x00	gpio0a_pos_int_en gpio0a posedge pulse interrupt enable 0: disable 1: enable

PMU_PMU_GPIO_NEG_INT_CON

Address: Operational Base + offset (0x002c)

pmu gpio negedge interrupt configure register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	gpio0d_neg_int_en gpio0d negedge pulse interrupt enable 0: disable 1: enable
23:16	RW	0x00	gpio0c_neg_int_en gpio0c negedge pulse interrupt enable 0: disable 1: enable
15:8	RW	0x00	gpio0b_neg_int_en gpio0b negedge pulse interrupt enable 0: disable 1: enable
7:0	RW	0x00	gpio0a_neg_int_en gpio0b negedge pulse interrupt enable 0: disable 1: enable

PMU_PMU_GPIO_POS_INT_ST

Address: Operational Base + offset (0x0030)

pmu gpio posedge interrupt status register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	gpio0d_pos_int_status gpio0d posedge pulse interrupt status 0: not wakeup by gpio0d posedge pulse 1: wakeup by gpio0d posedge pulse
23:16	RW	0x00	gpio0c_pos_int_status gpio0c posedge pulse interrupt status 0: not wakeup by gpio0c posedge pulse 1: wakeup by gpio0c posedge pulse
15:8	RW	0x00	gpio0b_pos_int_status gpio0b posedge pulse interrupt status 0: not wakeup by gpio0b posedge pulse 1: wakeup by gpio0b posedge pulse

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Bit	Attr	Reset Value	Description
7:0	RW	0x00	gpio0a_pos_int_status gpio0a posedge pulse interrupt status 0: not wakeup by gpio0a posedge pulse 1: wakeup by gpio0a posedge pulse

PMU_PMU_GPIO_NEG_INT_ST
Address: Operational Base + offset (0x0034)
pmu gpio negedge interrupt status register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	gpio0d_neg_int_status gpio0d negedge pulse interrupt status 0: not wakeup by gpio0d negedge pulse 1: wakeup by gpio0d negedge pulse
23:16	RW	0x00	gpio0c_neg_int_status gpio0c negedge pulse interrupt status 0: not wakeup by gpio0c negedge pulse 1: wakeup by gpio0c negedge pulse
15:8	RW	0x00	gpio0b_neg_int_status gpio0b negedge pulse interrupt status 0: not wakeup by gpio0b negedge pulse 1: wakeup by gpio0b negedge pulse
7:0	RW	0x00	gpio0a_neg_int_status gpio0a negedge pulse interrupt status 0: not wakeup by gpio0a negedge pulse 1: wakeup by gpio0a negedge pulse

PMU_PMU_CORE_PWR_ST
Address: Operational Base + offset (0x0038)
pmu core power status register

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved
20	RW	0x0	mcu_sleeping mcu sleeping status 0: mcu sleeping status is 0 1: mcu sleeping status is 1
19:16	RW	0x0	standbywfi_cluster_b standbywfi status of cluster_b 0: cluster_b standbywfi status is 0 1: cluster_b standbywfi status is 1
15:12	RW	0x0	standbywfe_cluster_b standbywfe status of cluster_b 0: cluster_b standbywfe status is 0 1: cluster_b standbywfe status is 1
11	RW	0x0	standbywfil2_cluster_b standbywfil2 status of cluster_b 0: cluster_b standbywfil2 status is 0 1: cluster_b standbywfil2 status is 1
10	RW	0x0	l2flushdone_cluster_b l2flushdone status of cluster_b 0: cluster_b l2flushdone status is 0 1: cluster_b l2flushdone status is 1
9:6	RW	0x0	standbywfi_cluster_l standbywfi status of cluster_l 0: cluster_l standbywfi status is 0 1: cluster_l standbywfi status is 1

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Bit	Attr	Reset Value	Description
5:2	RW	0x0	standbywfe_cluster_l standbywfe status of cluster_l 0: cluster_l standbywfe status is 0 1: cluster_l standbywfe status is 1
1	RW	0x0	standbywfil2_cluster_l standbywfil2 status of cluster_l 0: cluster_l standbywfil2 status is 0 1: cluster_l standbywfil2 status is 1
0	RW	0x0	l2flushdone_cluster_l l2flushdone status of cluster_l 0: cluster_l l2flushdone status is 0 1: cluster_l l2flushdone status is 1

PMU_PMU_BUS_IDLE_REQ

Address: Operational Base + offset (0x003c)

pmu bus idle request register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RW	0x0	idle_req_cci400 send idle request to cci400 low power interface 0: disable 1: enable
14	RW	0x0	idle_req_cci send idle request to cci niu 0: disable 1: enable
13	RW	0x0	idle_req_msch send idle request to memory schedule 0: disable 1: enable
12	RW	0x0	idle_req_pmu send idle request to pmu niu 0: disable 1: enable
11	RW	0x0	idle_req_alive send idle request to alive niu 0: disable 1: enable
10	RW	0x0	idle_req_cxcs send idle request to cxcs power interface 0: disable 1: enable
9	RO	0x0	reserved
8	RW	0x0	idle_req_vio send idle request to vio niu 0: disable 1: enable
7	RW	0x0	idle_req_video send idle request to video niu 0: disable 1: enable

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Bit	Attr	Reset Value	Description
6	RW	0x0	idle_req_peri send idle request to peri niu 0: disable 1: enable
5	RW	0x0	idle_req_dma send idle request to dma niu 0: disable 1: enable
4	RW	0x0	idle_req_bus send idle request to bus niu 0: disable 1: enable
3	RW	0x0	idle_req_core send idle request to core niu(cpu access periphral register path) 0: disable 1: enable
2	RW	0x0	idle_req_gpu send idle request to gpu niu 0: disable 1: enable
1	RW	0x0	idle_req_cluster_b send idle request to adb400 of cluster_b 0: disable 1: enable
0	RW	0x0	idle_req_cluster_l send idle request to adb400 of cluster_l 0: disable 1: enable

PMU_PMU_BUS_IDLE_ST

Address: Operational Base + offset (0x0040)

pmu bus idle status register

Bit	Attr	Reset Value	Description
31	RW	0x0	cactives_cci cactives status of cci low power interface 0: active status of cci is 0 (inactive) 1: active status of cci is 1 (active)
30	RW	0x0	cactives_cxcs active status of cxcs low power interface 0: active status of cxcs is 0 (inactive) 1: active status of cxcs is 1 (active)
29	RW	0x0	idle_pmu idle status of pmu niu 0: idle status of pmu_niu is 0 1: idle status of pmu_niu is 1
28	RW	0x0	idle_alive idle status of alive niu 0: idle status of alive_niu is 0 1: idle status of alive_niu is 1
27	RW	0x0	idle_msch idle status of msch niu 0: idle status of msch_niu is 0 1: idle status of msch_niu is 1

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Bit	Attr	Reset Value	Description
26	RW	0x0	idle_cci idle status of cci niu 0: idle status of cci_niu is 0 1: idle status of cci_niu is 1
25	RO	0x0	reserved
24	RW	0x0	idle_vio idle status of vio niu 0: idle status of vio_niu is 0 1: idle status of vio_niu is 1
23	RW	0x0	idle_video idle status of video niu 0: idle status of video_niu is 0 1: idle status of video_niu is 1
22	RW	0x0	idle_peri idle status of peri niu 0: idle status of peri_niu is 0 1: idle status of peri_niu is 1
21	RW	0x0	idle_dma idle status of dma niu 0: idle status of dma_niu is 0 1: idle status of dma_niu is 1
20	RW	0x0	idle_bus idle status of bus niu 0: idle status of bus_niu is 0 1: idle status of bus_niu is 1
19	RW	0x0	idle_core idle status of core niu 0: idle status of core_niu is 0 1: idle status of core_niu is 1
18	RW	0x0	idle_gpu idle status of gpu niu 0: idle status of gpu_niu is 0 1: idle status of gpu_niu is 1
17	RW	0x0	idle_cluster_b inverted actives status of cluster_b adb400 0: cluster_l adb400 is active 1: cluster_l adb400 is inactive
16	RW	0x0	idle_cluster_l inverted actives status of cluster_l adb400 0: cluster_l adb400 is active 1: cluster_l adb400 is inactive
15	RW	0x0	pwrdnack_cci400 idle acknowledge status from cci400 low power interface, active low 0: idle acknowledge status of cci400 is 0 (active) 1: idle acknowledge status of cci400 is 1 (inactive)
14	RW	0x0	pwrdnack_cxcs idle acknowledge status from cxcs low power interface, active low 0: idle acknowledge status of cxcs is 0 (active) 1: idle acknowledge status of cxcs is 1 (inactive)

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Bit	Attr	Reset Value	Description
13	RW	0x0	idle_ack_pmu idle acknowledge status from pmu niu 0: idle acknowledge status of pmu_niu is 0 1: idle acknowledge status of pmu_niu is 1
12	RW	0x0	idle_ack_alive idle acknowledge status from alive niu 0: idle acknowledge status of alive_niu is 0 1: idle acknowledge status of alive_niu is 1
11	RW	0x0	idle_ack_msch idle acknowledge status from msch niu 0: idle acknowledge status of msch_niu is 0 1: idle acknowledge status of msch_niu is 1
10	RW	0x0	idle_ack_cci idle acknowledge status from cci niu 0: idle acknowledge status of cci_niu is 0 1: idle acknowledge status of cci_niu is 1
9	RO	0x0	reserved
8	RW	0x0	idle_ack_vio idle acknowledge status from vio niu 0: idle acknowledge status of vio_niu is 0 1: idle acknowledge status of vio_niu is 1
7	RW	0x0	idle_ack_video idle acknowledge status from video niu 0: idle acknowledge status of video_niu is 0 1: idle acknowledge status of video_niu is 1
6	RW	0x0	idle_ack_peri idle acknowledge status from peri niu 0: idle acknowledge status of peri_niu is 0 1: idle acknowledge status of peri_niu is 1
5	RW	0x0	idle_ack_dma idle acknowledge status from dma niu 0: idle acknowledge status of dma_niu is 0 1: idle acknowledge status of dma_niu is 1
4	RW	0x0	idle_ack_bus idle acknowledge status from bus niu 0: idle acknowledge status of bus_niu is 0 1: idle acknowledge status of bus_niu is 1
3	RW	0x0	idle_ack_core idle acknowledge status from core_niu(cpu access register path) 0: idle acknowledge status of core_niu is 0 1: idle acknowledge status of core_niu is 1
2	RW	0x0	idle_ack_gpu idle acknowledge status from gpu niu 0: idle acknowledge status of gpu_niu is 0 1: idle acknowledge status of gpu_niu is 1
1	RW	0x0	idle_ack_cluster_b idle acknowledge status from cluster_b adb400 0: idle acknowledge status of cluster_b is 0 1: idle acknowledge status of cluster_b is 1

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Bit	Attr	Reset Value	Description
0	RW	0x0	idle_ack_cluster_l idle acknowledge status from cluster_l adb400 0: idle acknowledge status of cluster_l is 0 1: idle acknowledge status of cluster_l is 1

PMU_PMU_POWER_ST
Address: Operational Base + offset (0x0044)
pmu power status register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:0	RW	0x00	power_state power state of pmu FSM

PMU_PMU_OSC_CNT
Address: Operational Base + offset (0x0048)
pmu osc count register

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:0	RW	0x00000	pmu_osc_cnt pmu osc stable counter value

PMU_PMU_PLLLOCK_CNT
Address: Operational Base + offset (0x004c)
pmu pll lock count register

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:0	RW	0x00000	pmu_plllock_cnt pmu pll lock counter value

PMU_PMU_PLLRST_CNT
Address: Operational Base + offset (0x0050)
pmu pll reset count register

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:0	RW	0x00000	pmu_pllrst_cnt pmu pll reset counter value

PMU_PMU_STABLE_CNT
Address: Operational Base + offset (0x0054)
pmu power stable count register

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:0	RW	0x00000	pmu_stable_cnt pmu PMIC stable counter value

PMU_PMU_DDRIO_PWRON_CNT
Address: Operational Base + offset (0x0058)
pmu ddrio power on count register

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
19:0	RW	0x00000	pmu_ddrio_pwron_cnt pmu ddrio power on counter value

PMU_PMU_WAKEUP_RST_CLR_CNT
Address: Operational Base + offset (0x005c)
pmu wakeup reset clear count register

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:0	RW	0x00000	pmu_wakeup_rst_cnt pmu wakeup reset counter value

PMU_PMU_DDR_SREF_ST
Address: Operational Base + offset (0x0060)
pmu ddr self refresh status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	upctl_c_sysack acknowledge status of upctl 0: acknowledge status is 0 (active) 1: acknowledge status is 1 (inactive)
0	RW	0x0	upctl_c_active active status of upctl 0: active status is 0 (inactive) 1: active status is 1 (active)

PMU_PMU_SYS_REG0
Address: Operational Base + offset (0x0064)
pmu system register 0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pmu_sys_reg0 system register 0

PMU_PMU_SYS_REG1
Address: Operational Base + offset (0x0068)
pmu system register 1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pmu_sys_reg1 system register 1

PMU_PMU_SYS_REG2
Address: Operational Base + offset (0x006c)
pmu system register 2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pmu_sys_reg2 system register 2

PMU_PMU_SYS_REG3
Address: Operational Base + offset (0x0070)
pmu system register 3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pmu_sys_reg3 system register 3

PMU_PMU_GPU_APM_CON

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Address: Operational Base + offset (0x0074)
pmu gpu apm power configure register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	pmu_apm_event_complete software configure gpu apm complete bit 0: apm complete bit is controlled by pmu apm hardware 1: apm complete bit is 1
0	RW	0x0	pmu_apm_abort_cfg software configure apm abort bit 0: pmu apm abort bit is controlled by apm interface 1: pmu apm abort bit is 1

PMU_PMU_GPU_APM_ST
Address: Operational Base + offset (0x0078)
pmu gpu apm power status register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:2	RW	0x0	apm_power_event_domain pmu apm power event domain status
1	RW	0x0	apm_power_event_type pmu apm power event type status 0: apm power event type is 0 1: apm power event type is 1
0	RW	0x0	apm_power_event_req pmu apm power event request status 0: do not have apm power event request 1: have apm power event request

PMU_PMU_TIMEOUT_CNT
Address: Operational Base + offset (0x007c)
pmu timeout count register

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:0	RW	0x00000	timeout_count timeout wakeup counter value

4.6 Timing Diagram

4.6.1 Each domain power switch timing

The following figure is the each domain power down and power up timing.

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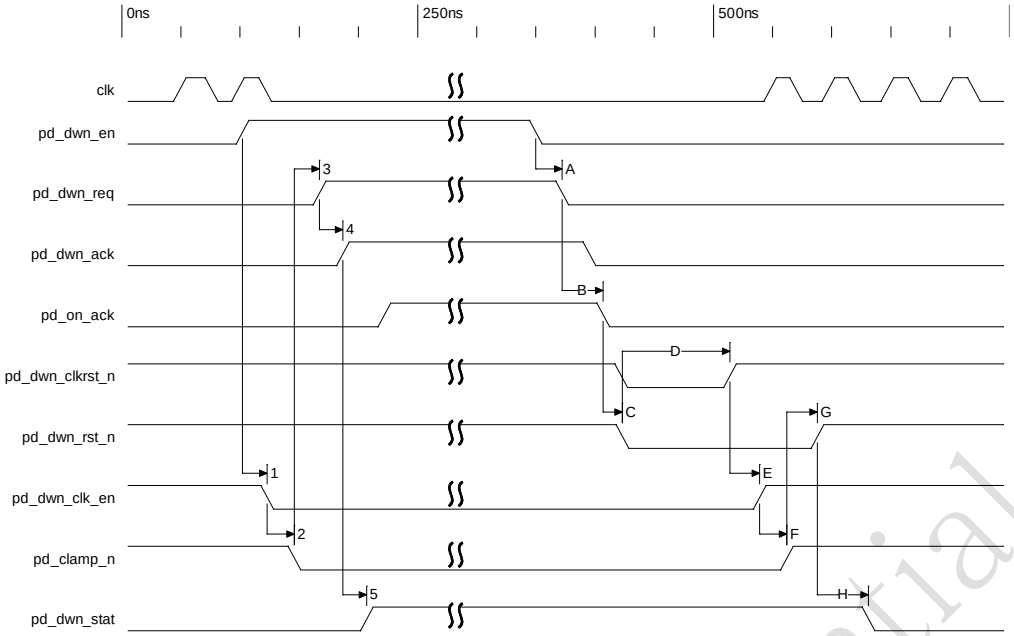

Fig. 4-3 Each Domain Power Switch Timing

4.6.2 External wakeup PAD timing

The PMU supports a lot of external wakeup sources, such as SD/MMDC, USBDEV, SIM detect wakeup, GPIO0 wakeup source and so on. All these external wakeup sources must meet the timing requirement (at least 200us) when the wakeup event is asserted. The following figure gives the timing information.

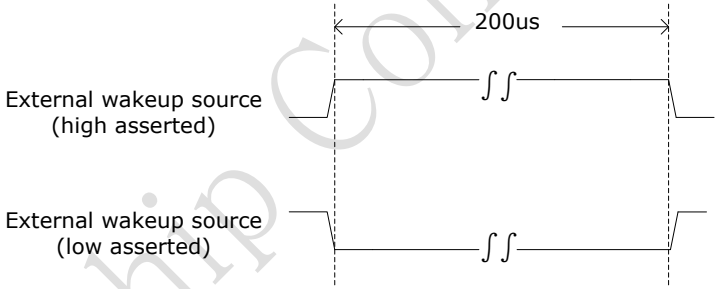

Fig. 4-4 External Wakeup Source PAD Timing

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Chapter 5 General Register Files (GRF)

5.1 Overview

The general register file will be used to do static set by software, which is composed of many registers for system control. The GRF is divided into two sections, one is GRF, used for general non-secure system, the other is PMUGRF, used for always on sysyem

5.2 Function Description

The function of general register file is：

- IOMUX control
- Control the state of GPIO in power-down mode
- GPIO PAD pull down and pull up control
- Used for common system control
- Used to record the system state

5.3 GRF Register Description

5.3.1 Register Summary

Name	Offset	Size	Reset Value	Description
GRF_GPIO1A_IOMUX	0x00000	W	0x00000000	GPIO1A iomux control
GRF_GPIO1B_IOMUX	0x00004	W	0x00000000	GPIO1B iomux control
GRF_GPIO1C_IOMUX	0x00008	W	0x00000000	GPIO1C iomux control
GRF_GPIO1D_IOMUX	0x0000c	W	0x00000000	GPIO1D iomux control
GRF_GPIO2A_IOMUX	0x00010	W	0x00000000	GPIO2A iomux control
GRF_GPIO2B_IOMUX	0x00014	W	0x00000040	GPIO2B iomux control
GRF_GPIO2C_IOMUX	0x00018	W	0x00000000	GPIO2C iomux control
GRF_GPIO2D_IOMUX	0x0001c	W	0x00000000	GPIO2D iomux control
GRF_GPIO3A_IOMUX	0x00020	W	0x00000000	GPIO3A iomux control
GRF_GPIO3B_IOMUX	0x00024	W	0x00000000	GPIO3B iomux control
GRF_GPIO3C_IOMUX	0x00028	W	0x00000000	GPIO3C iomux control
GRF_GPIO3D_IOMUX	0x0002c	W	0x00000000	GPIO3D iomux control
GRF_GPIO1A_P	0x00100	W	0x0000aaaa	GPIO1A PU/PD control
GRF_GPIO1B_P	0x00104	W	0x0000aaaa	GPIO1B PU/PD control
GRF_GPIO1C_P	0x00108	W	0x00005555	GPIO1C PU/PD control
GRF_GPIO1D_P	0x0010c	W	0x00006995	GPIO1D PU/PD control
GRF_GPIO2A_P	0x00110	W	0x00005555	GPIO2A PU/PD control
GRF_GPIO2B_P	0x00114	W	0x0000aa59	GPIO2B PU/PD control
GRF_GPIO2C_P	0x00118	W	0x000096aa	GPIO2C PU/PD control
GRF_GPIO2D_P	0x0011c	W	0x00005559	GPIO2D PU/PD control
GRF_GPIO3A_P	0x00120	W	0x00005a99	GPIO3A PU/PD control
GRF_GPIO3B_P	0x00124	W	0x000099aa	GPIO3B PU/PD control
GRF_GPIO3C_P	0x00128	W	0x000069a5	GPIO3C PU/PD control
GRF_GPIO3D_P	0x0012c	W	0x00005655	GPIO3D PU/PD control
GRF_GPIO1A_E	0x00200	W	0x00000000	GPIO1A drive strength control
GRF_GPIO1B_E	0x00204	W	0x00000800	GPIO1B drive strength control
GRF_GPIO1C_E	0x00208	W	0x0000aaa0	GPIO1C drive strength control

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Name	Offset	Size	Reset Value	Description
GRF_GPIO1D_E	0x0020c	W	0x00000002	GPIO1D drive strength control
GRF_GPIO2A_E	0x00210	W	0x00005655	GPIO2A drive strength control
GRF_GPIO2B_E	0x00214	W	0x00000119	GPIO2B drive strength control
GRF_GPIO2C_E	0x00218	W	0x00000000	GPIO2C drive strength control
GRF_GPIO2D_E	0x0021c	W	0x00005500	GPIO2D drive strength control
GRF_GPIO3A_E	0x00220	W	0x00000009	GPIO3A drive strength control
GRF_GPIO3B_E	0x00224	W	0x0000682a	GPIO3B drive strength control
GRF_GPIO3C_E	0x00228	W	0x00003115	GPIO3C drive strength control
GRF_GPIO3D_E	0x0022c	W	0x00000304	GPIO3D drive strength control
GRF_GPIO1L_SR	0x00300	W	0x00000800	GPIO1A/B SR control
GRF_GPIO1H_SR	0x00304	W	0x00000000	GPIO1C/D SR control
GRF_GPIO2L_SR	0x00308	W	0x00000200	GPIO2A/B SR control
GRF_GPIO2H_SR	0x0030c	W	0x00000000	GPIO2C/D SR control
GRF_GPIO3L_SR	0x00310	W	0x00004702	GPIO3A/B SR control
GRF_GPIO3H_SR	0x00314	W	0x00000040	GPIO3C/D SR control
GRF_GPIO_SMT	0x00380	W	0x00000315	GPIO smitter control register
GRF_SOC_CON0	0x00400	W	0x00000038	SoC control register 0
GRF_SOC_CON1	0x00404	W	0x000001c0	SoC control register 1
GRF_SOC_CON2	0x00408	W	0x000007f0	SoC control register 2
GRF_SOC_CON3	0x0040c	W	0x00002020	SoC control register 3
GRF_SOC_CON4	0x00410	W	0x00002001	SoC control register 4
GRF_SOC_CON5	0x00414	W	0x00000000	SoC control register 5
GRF_SOC_CON6	0x00418	W	0x00000000	SoC control register 6
GRF_SOC_CON7	0x0041c	W	0x00000000	SoC control register 7
GRF_SOC_CON8	0x00420	W	0x00007538	SoC control register 8
GRF_SOC_CON9	0x00424	W	0x00000100	SoC control register 9
GRF_SOC_CON10	0x00428	W	0x00000000	SoC control register 10
GRF_SOC_CON11	0x0042c	W	0x00002000	SoC control register 11
GRF_SOC_CON12	0x00430	W	0x00006000	SoC control register 12
GRF_SOC_CON13	0x00434	W	0x00000000	SoC control register 13
GRF_SOC_CON14	0x00438	W	0x00000000	SoC control register 14
GRF_SOC_CON15	0x0043c	W	0x00002000	SoC control register 15
GRF_SOC_CON16	0x00440	W	0x00000000	SoC control register 16
GRF_SOC_CON17	0x00444	W	0x00000000	SoC control register 17
GRF_SOC_STATUS0	0x00480	W	0x00000000	SoC status register 0
GRF_SOC_STATUS1	0x00484	W	0x00000000	SoC status register 1
GRF_SOC_STATUS2	0x00488	W	0x00000000	SoC status register 2
GRF_SOC_STATUS3	0x0048c	W	0x00000000	SoC status register 3
GRF_SOC_STATUS4	0x00490	W	0x00000000	SoC status register 4
GRF_SOC_STATUS5	0x00494	W	0x00000000	SoC status register 5
GRF_SOC_STATUS6	0x00498	W	0x00000000	SoC status register 6
GRF_SOC_STATUS7	0x0049c	W	0x00000000	SoC status register 7
GRF_SOC_STATUS8	0x004a0	W	0x00000000	SoC status register 8

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Name	Offset	Size	Reset Value	Description
GRF_SOC_STATUS9	0x004a4	W	0x00000000	SoC status register 9
GRF_SOC_STATUS10	0x004a8	W	0x00000000	SoC status register 10
GRF_SOC_STATUS11	0x004ac	W	0x00000000	SoC status register 11
GRF_SOC_STATUS12	0x004b0	W	0x00000000	SoC status register 12
GRF_SOC_STATUS13	0x004b4	W	0x00000000	SoC status register 13
GRF_SOC_STATUS14	0x004b8	W	0x00000000	SoC status register 14
GRF_SOC_STATUS15	0x004bc	W	0x00000000	SoC status register 15
GRF_CPU_CON0	0x00500	W	0x0000000b	CPU little cluster control register 0
GRF_CPU_CON1	0x00504	W	0x00000000	CPU little cluster control register 1
GRF_CPU_CON2	0x00508	W	0x0000000b	CPU big cluster control register 2
GRF_CPU_CON3	0x0050c	W	0x00000000	CPU big cluster control register 1
GRF_CPU_STATUS0	0x00520	W	0x00000000	CPU status register 0
GRF_CPU_STATUS1	0x00524	W	0x00000000	CPU status register 1
GRF_CCI_STATUS0	0x00540	W	0x00000000	CCI status register 0
GRF_CCI_STATUS1	0x00544	W	0x00000000	CCI status register 1
GRF_CCI_STATUS2	0x00548	W	0x00000000	CCI status register 2
GRF_CCI_STATUS3	0x0054c	W	0x00000000	CCI status register 3
GRF_CCI_STATUS4	0x00550	W	0x00000000	CCI status register 4
GRF_CCI_STATUS5	0x00554	W	0x00000000	CCI status register 5
GRF_CCI_STATUS6	0x00558	W	0x00000000	CCI status register 6
GRF_CCI_STATUS7	0x0055c	W	0x00000000	CCI status register 7
GRF_CCI_STATUS8	0x00560	W	0x00000000	CCI status register 8
GRF_CCI_STATUS9	0x00564	W	0x00000000	CCI status register 9
GRF_CCI_STATUS10	0x00568	W	0x00000000	CCI status register 10
GRF_CCI_STATUS11	0x0056c	W	0x00000000	CCI status register 11
GRF_CCI_STATUS12	0x00570	W	0x00000000	CCI status register 12
GRF_CCI_STATUS13	0x00574	W	0x00000000	CCI status register 13
GRF_CCI_STATUS14	0x00578	W	0x00000000	CCI status register 14
GRF_CCI_STATUS15	0x0057c	W	0x00000000	CCI status register 15
GRF_DDRC0_CON0	0x00600	W	0x00000000	DDRC0 control register 0
GRF_SIG_DETECT_CON	0x00680	W	0x00000000	external signal detect configue register
GRF_SIG_DETECT_STATUS	0x00690	W	0x00000000	external signal detect configue register
GRF_SIG_DETECT_CLR	0x006a0	W	0x00000000	external signal detect configue register
GRF_UOC0_CON0	0x00700	W	0x00000000	OTG control register
GRF_UOC1_CON1	0x00718	W	0x00000000	usb host control register
GRF_UOC1_CON2	0x0071c	W	0x0000c820	UOC1 control register 2
GRF_UOC1_CON3	0x00720	W	0x00000b40	UOC1 control register 3
GRF_UOC1_CON4	0x00724	W	0x0000001c	USB HOST 2.0 control register
GRF_UOC1_CON5	0x00728	W	0x00000000	USB HOST 2.0 control register
GRF_UOC3_CON0	0x00738	W	0x000030eb	UOC3 control register 0

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Name	Offset	Size	Reset Value	Description
GRF_UOC3_CON1	0x0073c	W	0x00000003	UOC3 control register 1
GRF_UOC4_CON0	0x00740	W	0x00000080	UOC4 control register 0
GRF_UOC4_CON1	0x00744	W	0x00000820	UOC4 control register 1
GRF_USBPHY_CON0	0x00780	W	0x00008518	USB PHY control register
GRF_USBPHY_CON1	0x00784	W	0x0000e007	USB PHY control register
GRF_USBPHY_CON2	0x00788	W	0x000002e7	USB PHY control register
GRF_USBPHY_CON3	0x0078c	W	0x00000200	USB PHY control register
GRF_USBPHY_CON4	0x00790	W	0x00005556	USB PHY control register
GRF_USBPHY_CON5	0x00794	W	0x00004555	USB PHY control register
GRF_USBPHY_CON6	0x00798	W	0x00000005	USB PHY control register
GRF_USBPHY_CON7	0x0079c	W	0x000068c0	USB PHY control register
GRF_USBPHY_CON8	0x007a0	W	0x00008518	USB PHY control register
GRF_USBPHY_CON9	0x007a4	W	0x0000e007	USB PHY control register
GRF_USBPHY_CON10	0x007a8	W	0x000002e7	USB PHY control register
GRF_USBPHY_CON11	0x007ac	W	0x00000200	USB PHY control register
GRF_USBPHY_CON12	0x007b0	W	0x00005556	USB PHY control register
GRF_USBPHY_CON13	0x007b4	W	0x00004555	USB PHY control register
GRF_USBPHY_CON14	0x007b8	W	0x00000005	USB PHY control register
GRF_USBPHY_CON15	0x007bc	W	0x000068c0	USB PHY control register
GRF_PVTM_CON0	0x00800	W	0x00000000	PVT monitor control register 0
GRF_PVTM_CON1	0x00804	W	0x016e3600	PVT monitor control register 1
GRF_PVTM_CON2	0x00808	W	0x016e3600	PVT monitor control register 2
GRF_PVTM_STATUS0	0x0080c	W	0x00000000	PVT monitor status register 0
GRF_PVTM_STATUS1	0x00810	W	0x00000000	PVT monitor status register 1
GRF_PVTM_STATUS2	0x00814	W	0x00000000	PVT monitor status register 2
GRF_IO_VSEL	0x00900	W	0x00000004	IO voltage select
GRF_SARADC_TESTBIT	0x00904	W	0x00000000	SARADC Test bit register
GRF_FAST_BOOT_ADDR	0x00f80	W	0x00000000	Fast boot address

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

5.3.2 Detail Register Description

GRF_GPIO1A_IOMUX

Address: Operational Base + offset (0x00000)

GPIO1A iomux control

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio1a7_sel GPIO1A[7] iomux select 2'b00: gpio 2'b01: cif_data9 2'b10: ts_data7 2'b11: reserved
13:12	RW	0x0	gpio1a6_sel GPIO1A[6] iomux select 2'b00: gpio 2'b01: cif_data8 2'b10: ts_data6 2'b11: reserved
11:10	RW	0x0	gpio1a5_sel GPIO1A[5] iomux select 2'b00: gpio 2'b01: cif_data7 2'b10: ts_data5 2'b11: reserved
9:8	RW	0x0	gpio1a4_sel GPIO1A[4] iomux select 2'b00: gpio 2'b01: cif_data6 2'b10: ts_data4 2'b11: reserved
7:6	RW	0x0	gpio1a3_sel GPIO1A[3] iomux select 2'b00: gpio 2'b01: cif_data5 2'b10: ts_data3 2'b11: reserved
5:4	RW	0x0	gpio1a2_sel GPIO1A[2] iomux select 2'b00: gpio 2'b01: cif_data4 2'b10: ts_data2 2'b11: reserved

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Bit	Attr	Reset Value	Description
3:2	RW	0x0	gpio1a1_sel GPIO1A[1] iomux select 2'b00: gpio 2'b01: cif_data3 2'b10: ts_data1 2'b11: reserved
1:0	RW	0x0	gpio1a0_sel GPIO1A[0] iomux select 2'b00: gpio 2'b01: cif_data2 2'b10: ts_data0 2'b11: reserved

GRF_GPIO1B_IOMUX
Address: Operational Base + offset (0x00004)
GPIO1B iomux control

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio1b7_sel GPIO1B[7] iomux select 2'b00: gpio 2'b01: cif_data11 2'b10: spi1_csn0 2'b11: reserved
13:12	RW	0x0	gpio1b6_sel GPIO1B[6] iomux select 2'b00: gpio 2'b01: cif_data10 2'b10: spi1_clk 2'b11: reserved
11:10	RW	0x0	gpio1b5_sel GPIO1B[5] iomux select 2'b00: gpio 2'b01: cif_data1 2'b10: reserved 2'b11: reserved

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Bit	Attr	Reset Value	Description
9:8	RW	0x0	gpio1b4_sel GPIO1B4] iomux select 2'b00: gpio 2'b01: cif_data0 2'b10: reserved 2'b11: reserved
7:6	RW	0x0	gpio1b3_sel GPIO1B[3] iomux select 2'b00: gpio 2'b01: cif_clkout 2'b10: ts_fail 2'b11: reserved
5:4	RW	0x0	gpio1b2_sel GPIO1B[2] iomux select 2'b00: gpio 2'b01: cif_clkin 2'b10: ts_clk 2'b11: reserved
3:2	RW	0x0	gpio1b1_sel GPIO1B[1] iomux select 2'b00: gpio 2'b01: cif_href 2'b10: ts_valid 2'b11: reserved
1:0	RW	0x0	gpio1b0_sel GPIO1B[0] iomux select 2'b00: gpio 2'b01: cif_vsync 2'b10: ts_sync 2'b11: reserved

GRF_GPIO1C_IOMUX
Address: Operational Base + offset (0x00008)
GPIO1C iomux control

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:14	RW	0x0	gpio1c7_sel GPIO1C[7] iomux select 2'b00: gpio 2'b01: flash_data5 2'b10: emmc_data5 2'b11: spi0_txd
13:12	RW	0x0	gpio1c6_sel GPIO1C[6] iomux select 2'b00: gpio 2'b01: flash_data4 2'b10: emmc_data4 2'b11: spi0_rxd
11:10	RW	0x0	gpio1c5_sel GPIO1C[5] iomux select 2'b00: gpio 2'b01: flash_data3 2'b10: emmc_data3 2'b11: sfc_sio3
9:8	RW	0x0	gpio1c4_sel GPIO1C4] iomux select 2'b00: gpio 2'b01: flash_data2 2'b10: emmc_data2 2'b11: sfc_sio2
7:6	RW	0x0	gpio1c3_sel GPIO1C[3] iomux select 2'b00: gpio 2'b01: flash_data1 2'b10: emmc_data1 2'b11: sfc_sio1
5:4	RW	0x0	gpio1c2_sel GPIO1C[2] iomux select 2'b00: gpio 2'b01: flash_data0 2'b10: emmc_data0 2'b11: sfc_sio0
3:2	RW	0x0	gpio1c1_sel GPIO1C[1] iomux select 2'b00: gpio 2'b01: i2c3cam_sda 2'b10: spi1_txd 2'b11: reserved
1:0	RW	0x0	gpio1c0_sel GPIO1C[0] iomux select 2'b00: gpio 2'b01: i2c3cam_scl 2'b10: spi1_rxd 2'b11: reserved

GRF_GPIO1D_IOMUX
Address: Operational Base + offset (0x0000c)
GPIO1D iomux control

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio1d7_sel GPIO1D[7] iomux select 2'b00: gpio 2'b01: flash_wrn 2'b10: sfc_csn0 2'b11: reserved
13:12	RW	0x0	gpio1d6_sel GPIO1D[6] iomux select 2'b00: gpio 2'b01: flash_cle 2'b10: reserved 2'b11: reserved
11:10	RW	0x0	gpio1d5_sel GPIO1D[5] iomux select 2'b00: gpio 2'b01: flash_ale 2'b10: spi0_clk 2'b11: reserved
9:8	RW	0x0	gpio1d4_sel GPIO1D[4] iomux select 2'b00: gpio 2'b01: flash_rdn 2'b10: sfc_csn1 2'b11: reserved
7:6	RW	0x0	gpio1d3_sel GPIO1D[3] iomux select 2'b00: gpio 2'b01: flash_wp 2'b10: emmc_pwren 2'b11: reserved
5:4	RW	0x0	gpio1d2_sel GPIO1D[2] iomux select 2'b00: gpio 2'b01: flash_rdy 2'b10: emmc_cmd 2'b11: sfc_clk

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Bit	Attr	Reset Value	Description
3:2	RW	0x0	gpio1d1_sel GPIO1D[1] iomux select 2'b00: gpio 2'b01: flash_data7 2'b10: emmc_data7 2'b11: spi0_csn1
1:0	RW	0x0	gpio1d0_sel GPIO1D[0] iomux select 2'b00: gpio 2'b01: flash_data6 2'b10: emmc_data6 2'b11: spi0_csn0

GRF_GPIO2A_IOMUX
Address: Operational Base + offset (0x00010)
GPIO2A iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio2a7_sel GPIO2A[7] iomux select 2'b00: gpio 2'b01: sdmmc0_data2 2'b10: jtag_tck 2'b11: reserved
13:12	RW	0x0	gpio2a6_sel GPIO2A[6] iomux select 2'b00: gpio 2'b01: sdmmc0_data1 2'b10: uart2dbg_sin 2'b11: reserved
11:10	RW	0x0	gpio2a5_sel GPIO2A[5] iomux select 2'b00: gpio 2'b01: sdmmc0_data0 2'b10: uart2dbg_sout 2'b11: reserved

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Bit	Attr	Reset Value	Description
9:8	RW	0x0	gpio2a4_sel GPIO2A[4] iomux select 2'b00: gpio 2'b01: flash_dqs 2'b10: emmc_clkout 2'b11: reserved
7:6	RW	0x0	gpio2a3_sel GPIO2A[3] iomux select 2'b00: gpio 2'b01: flash_csn3 2'b10: emmc_rstnout 2'b11: reserved
5:4	RW	0x0	gpio2a2_sel GPIO2A[2] iomux select 2'b00: gpio 2'b01: flash_csn2 2'b10: reserved 2'b11: reserved
3:2	RW	0x0	gpio2a1_sel GPIO2A[1] iomux select 2'b00: gpio 2'b01: flash_csn1 2'b10: reserved 2'b11: reserved
1:0	RW	0x0	gpio2a0_sel GPIO2A[0] iomux select 2'b00: gpio 2'b01: flash_csn0 2'b10: reserved 2'b11: reserved

GRF_GPIO2B_IOMUX
Address: Operational Base + offset (0x00014)
GPIO2B iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:14	RW	0x0	gpio2b7_sel GPIO2B[7] iomux select 2'b00: gpio 2'b01: i2s_sdi 1'b10: reserved 1'b11: reserved
13:12	RW	0x0	gpio2b6_sel GPIO2B[6] iomux select 2'b00: gpio 2'b01: i2s_lrcktx 2'b10: reserved 2'b11: reserved
11:10	RW	0x0	gpio2b5_sel GPIO2B[5] iomux select 2'b00: gpio 2'b01: i2s_lrckrx 2'b10: pcm_sync 2'b11: reserved
9:8	RW	0x0	gpio2b4_sel GPIO2B[4] iomux select 2'b00: gpio 2'b01: i2s_sclk 2'b10: reserved 2'b11: reserved
7:6	RW	0x1	gpio2b3_sel GPIO2B[3] iomux select 2'b00: gpio 2'b01: sdmmc0_dtectn 2'b10: reserved 2'b11: reserved
5:4	RW	0x0	gpio2b2_sel GPIO2B[2] iomux select 2'b00: gpio 2'b01: sdmmc0_cmd 2'b10: mcujtag_tms 2'b11: reserved
3:2	RW	0x0	gpio2b1_sel GPIO2B[1] iomux select 2'b00: gpio 2'b01: sdmmc0_clkout 2'b10: mcujtag_tck 2'b11: reserved
1:0	RW	0x0	gpio2b0_sel GPIO2B[0] iomux select 2'b00: gpio 2'b01: sdmmc0_data3 2'b10: jtag_tms 2'b11: reserved

GRF_GPIO2C_IOMUX
Address: Operational Base + offset (0x00018)
GPIO2C iomux control

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio2c7_sel GPIO2C[7] iomux select 2'b00: gpio 2'b01: spdif_tx 2'b10: edp_hpd 2'b11: reserved
13:12	RW	0x0	gpio2c6_sel GPIO2C[6] iomux select 2'b00: gpio 2'b01: i2c1audio_scl 2'b10: reserved 2'b11: reserved
11:10	RW	0x0	gpio2c5_sel GPIO2C[5] iomux select 2'b00: gpio 2'b01: i2c1audio_sda 2'b10: reserved 2'b11: reserved
9:8	RW	0x0	gpio2c4_sel GPIO2C[4] iomux select 2'b00: gpio 2'b01: i2s_clk 2'b10: reserved 2'b11: reserved
7:6	RW	0x0	gpio2c3_sel GPIO2C[3] iomux select 2'b00: gpio 2'b01: i2s_sdo3 2'b10: pcm_in 2'b11: reserved
5:4	RW	0x0	gpio2c2_sel GPIO2C[2] iomux select 2'b00: gpio 2'b01: i2s_sdo2 2'b10: pcm_clk 2'b11: reserved

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Bit	Attr	Reset Value	Description
3:2	RW	0x0	gpio2c1_sel GPIO2C[1] iomux select 2'b00: gpio 2'b01: i2s_sdo1 2'b10: pcm_out 2'b11: reserved
1:0	RW	0x0	gpio2c0_sel GPIO2C[0] iomux select 2'b00: gpio 2'b01: i2s_sdo0 2'b10: reserved 2'b11: reserved

GRF_GPIO2D_IOMUX
Address: Operational Base + offset (0x0001c)
GPIO2D iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio2d7_sel GPIO2D[7] iomux select 2'b00: gpio 2'b01: sdio0_data3 2'b10: reserved 2'b11: reserved
13:12	RW	0x0	gpio2d6_sel GPIO2D[6] iomux select 2'b00: gpio 2'b01: sdio0_data2 2'b10: reserved 2'b11: reserved
11:10	RW	0x0	gpio2d5_sel GPIO2D[5] iomux select 2'b00: gpio 2'b01: sdio0_data1 2'b10: reserved 2'b11: reserved

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Bit	Attr	Reset Value	Description
9:8	RW	0x0	gpio2d4_sel GPIO2D[4] iomux select 2'b00: gpio 2'b01: sdio0_data0 2'b10: reserved 2'b11: reserved
7:6	RW	0x0	gpio2d3_sel GPIO2D[3] iomux select 2'b00: gpio 2'b01: uart0bt_rtsn 2'b10: reserved 2'b11: reserved
5:4	RW	0x0	gpio2d2_sel GPIO2D[2] iomux select 2'b00: gpio 2'b01: uart0bt_ctsn 2'b10: reserved 2'b11: reserved
3:2	RW	0x0	gpio2d1_sel GPIO2D[1] iomux select 2'b00: gpio 2'b01:uart0bt_sout 2'b10: reserved 2'b11: reserved
1:0	RW	0x0	gpio2d0_sel GPIO2D[0] iomux select 2'b00: gpio 2'b01: uart0bt_sin 2'b10: reserved 2'b11: reserved

GRF_GPIO3A_IOMUX
Address: Operational Base + offset (0x00020)
GPIO3A iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:14	RW	0x0	gpio3a7_sel GPIO3A[7] iomux select 2'b00: gpio 2'b01: reserved 2'b10: reserved 2'b11: reserved
13:12	RW	0x0	gpio3a6_sel GPIO3A[6] iomux select 2'b00: gpio 2'b01: sdio0_intn 2'b10: reserved 2'b11: reserved
11:10	RW	0x0	gpio3a5_sel GPIO3A[5] iomux select 2'b00: gpio 2'b01: sdio0_bkpwr 2'b10: reserved 2'b11: reserved
9:8	RW	0x0	gpio3a4_sel GPIO3A[4] iomux select 2'b00: gpio 2'b01: sdio0_pwren 2'b10: reserved 2'b11: reserved
7:6	RW	0x0	gpio3a3_sel GPIO3A[3] iomux select 2'b00: gpio 2'b01: sdio0_wrprt 2'b10: reserved 2'b11: reserved
5:4	RW	0x0	gpio3a2_sel GPIO3A[2] iomux select 2'b00: gpio 2'b01: sdio0_detectn 2'b10: reserved 2'b11: reserved
3:2	RW	0x0	gpio3a1_sel GPIO3A[1] iomux select 2'b00: gpio 2'b01: sdio0_clkout 2'b10: reserved 2'b11: reserved
1:0	RW	0x0	gpio3a0_sel GPIO3A[0] iomux select 2'b00: gpio 2'b01: sdio0_cmd 2'b10: reserved 2'b11: reserved

GRF_GPIO3B_IOMUX
Address: Operational Base + offset (0x00024)
GPIO3B iomux control

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio3b7_sel GPIO3B[7] iomux select 2'b00: gpio 2'b01: max_rxd0 2'b10: gps_sig 2'b11: reserved
13:12	RW	0x0	gpio3b6_sel GPIO3B[6] iomux select 2'b00: gpio 2'b01: mac_txd3 2'b10: gps_mag 2'b11: reserved
11:10	RW	0x0	gpio3b5_sel GPIO3B[5] iomux select 2'b00: gpio 2'b01: mac_txen 2'b10: reserved 2'b11: reserved
9:8	RW	0x0	gpio3b4_sel GPIO3B[4] iomux select 2'b00: gpio 2'b01: mac_col 2'b10: reserved 2'b11: reserved
7:6	RW	0x0	gpio3b3_sel GPIO3B[3] iomux select 2'b00: gpio 2'b01: mac_crs 2'b10: reserved 2'b11: reserved
5:4	RW	0x0	gpio3b2_sel GPIO3B[2] iomux select 2'b00: gpio 2'b01: mac_txd2 2'b10: reserved 2'b11: reserved

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Bit	Attr	Reset Value	Description
3:2	RW	0x0	gpio3b1_sel GPIO3B[1] iomux select 2'b00: gpio 2'b01: mac_txd1 2'b10: reserved 2'b11: reserved
1:0	RW	0x0	gpio3b0_sel GPIO3B[0] iomux select 2'b00: gpio 2'b01: mac_txd0 2'b10: pwm_0 2'b11: vop_pwm

GRF_GPIO3C_IOMUX
Address: Operational Base + offset (0x00028)
GPIO3C iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio3c7_sel GPIO3C[7] iomux select 2'b00: gpio 2'b01: edphdmi_cecinout 2'b10: isp_flashtrigin 2'b11: reserved
13:12	RW	0x0	gpio3c6_sel GPIO3C[6] iomux select 2'b00: gpio 2'b01: mac_clk 2'b10: isp_shuttertrig 2'b11: reserved
11:10	RW	0x0	gpio3c5_sel GPIO3C[5] iomux select 2'b00: gpio 2'b01: mac_rxer 2'b10: isp_prelighttrig 2'b11: reserved

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Bit	Attr	Reset Value	Description
9:8	RW	0x0	gpio3c4_sel GPIO3C[4] iomux select 2'b00: gpio 2'b01: mac_rxdv 2'b10: isp_flashtrigout 2'b11: reserved
7:6	RW	0x0	gpio3c3_sel GPIO3C[3] iomux select 2'b00: gpio 2'b01: mac_mdc 2'b10: isp_shutteren 2'b11: reserved
5:4	RW	0x0	gpio3c2_sel GPIO3C[2] iomux select 2'b00: gpio 2'b01: mac_rxd3 2'b10: usb_drvvbus1 2'b11: reserved
3:2	RW	0x0	gpio3c1_sel GPIO3C[1] iomux select 2'b00: gpio 2'b01: mac_rxd2 2'b10: uart3gps_rtsn 2'b11: usb_drvvbus0
1:0	RW	0x0	gpio3c0_sel GPIO3C[0] iomux select 2'b00: gpio 2'b01: mac_rxd1 2'b10: uart3gps_ctsn 2'b11: gps_rfclk

GRF_GPIO3D_IOMUX
Address: Operational Base + offset (0x0002c)
GPIO3D iomux control

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:14	RW	0x0	gpio3d7_sel GPIO3D[7] iomux select 2'b00: gpio 2'b01: sc_vcc18v 2'b10: i2c2sensor_sda 2'b11: gpujtag_tck
13:12	RW	0x0	gpio3d6_sel GPIO3D[6] iomux select 2'b00: gpio 3'b01: ir_tx 3'b10: uart3gps_sout 3'b11: pwm3
11:10	RW	0x0	gpio3d5_sel GPIO3D[5] iomux select 2'b00: gpio 3'b01: ir_rx 3'b10: uart3gsp_sin 3'b11: reserved
9:8	RW	0x0	gpio3d4_sel GPIO3D[4] iomux select 2'b00: gpio 3'b01: mac_txclkout 3'b10: spi1_csn1 3'b11: reserved
7:6	RW	0x0	gpio3d3_sel GPIO3D[3] iomux select 2'b00: gpio 3'b01: hdmii2c_scl 3'b10: i2c5hdmi_scl 3'b11: reserved
5:4	RW	0x0	gpio3d2_sel GPIO3D[2] iomux select 2'b00: gpio 3'b01: hdmii2c_sda 3'b10: i2c5hdmi_sda 3'b11: reserved
3:2	RW	0x0	gpio3d1_sel GPIO3D[1] iomux select 2'b00: gpio 3'b01: mac_rxclkin 3'b10: i2c4tp_scl 3'b11: reserved
1:0	RW	0x0	gpio3d0_sel GPIO3D[0] iomux select 2'b00: gpio 3'b01: mac_mdio 3'b10: i2c4tp_sda 3'b11: reserved

GRF_GPIO1A_P

Address: Operational Base + offset (0x00100)

GPIO1A PU/PD control

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0xaaaa	gpio1a_p GPIO1A PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Noraml operaton); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull-down); 2'b11: Repeater(Bus keeper)

GRF_GPIO1B_P
Address: Operational Base + offset (0x00104)
GPIO1B PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0xaaaa	gpio1b_p GPIO1B PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Noraml operaton); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull-down); 2'b11: Repeater(Bus keeper)

GRF_GPIO1C_P
Address: Operational Base + offset (0x00108)
GPIO1C PU/PD control

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x5555	gpio1c_p GPIO1C PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Noraml operaton); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull-down); 2'b11: Repeater(Bus keeper)

GRF_GPIO1D_P
Address: Operational Base + offset (0x0010c)
GPIO1D PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x6995	gpio1d_p GPIO1D PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Noraml operaton); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull-down); 2'b11: Repeater(Bus keeper)

GRF_GPIO2A_P
Address: Operational Base + offset (0x00110)
GPIO2A PU/PD control

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x5555	gpio2a_p GPIO2A PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Noraml operaton); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull-down); 2'b11: Repeater(Bus keeper)

GRF_GPIO2B_P
Address: Operational Base + offset (0x00114)
GPIO2B PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0xaa59	gpio2b_p GPIO2B PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Noraml operaton); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull-down); 2'b11: Repeater(Bus keeper)

GRF_GPIO2C_P
Address: Operational Base + offset (0x00118)
GPIO2C PU/PD control

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x96aa	gpio2c_p GPIO2C PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Noraml operaton); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull-down); 2'b11: Repeater(Bus keeper)

GRF_GPIO2D_P
Address: Operational Base + offset (0x0011c)
GPIO2D PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x5559	gpio2d_p GPIO2D PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Noraml operaton); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull-down); 2'b11: Repeater(Bus keeper)

GRF_GPIO3A_P
Address: Operational Base + offset (0x00120)
GPIO3A PU/PD control

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x5a99	gpio3a_p GPIO3A PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Noraml operaton); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull-down); 2'b11: Repeater(Bus keeper)

GRF_GPIO3B_P
Address: Operational Base + offset (0x00124)
GPIO3B PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x99aa	gpio3b_p GPIO3B PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Noraml operaton); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull-down); 2'b11: Repeater(Bus keeper)

GRF_GPIO3C_P
Address: Operational Base + offset (0x00128)
GPIO3C PU/PD control

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x69a5	gpio3c_p GPIO3C PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Noraml operaton); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull-down); 2'b11: Repeater(Bus keeper)

GRF_GPIO3D_P
Address: Operational Base + offset (0x0012c)
GPIO3D PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x5655	gpio3d_p GPIO3D PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Noraml operaton); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull-down); 2'b11: Repeater(Bus keeper)

GRF_GPIO1A_E
Address: Operational Base + offset (0x00200)
GPIO1A drive strength control

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0000	gpio1a_e GPIO1D drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO1B_E
Address: Operational Base + offset (0x00204)
GPIO1B drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0800	gpio1b_e GPIO1B drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO1C_E
Address: Operational Base + offset (0x00208)
GPIO1C drive strength control

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0xaaa0	gpio1c_e GPIO1C drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO1D_E

Address: Operational Base + offset (0x0020c)

GPIO1D drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio1d7_e GPIO1D_7 drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

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Bit	Attr	Reset Value	Description
13:12	RW	0x0	gpio1d6_e GPIO1D_6 drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA
11:10	RW	0x0	gpio1d5_e GPIO1D_5 drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA
9:8	RW	0x0	gpio1d4_e GPIO1D_4 drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA
7:6	RW	0x0	gpio1d3_e GPIO1D_3 drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA
5:4	RW	0x0	gpio1d2_e GPIO1D_2 drive strength control, every GPIO bit corresponding to 2bits 2'b00: 12mA 2'b01: 8mA 2'b10: 4mA 2'b11: 2mA
3:2	RW	0x0	gpio1d1_e GPIO1D_1 drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA
1:0	RW	0x2	gpio1d0_e GPIO1D_0 drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO2A_E

Address: Operational Base + offset (0x00210)

GPIO2A drive strength control

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x5655	gpio2a_e GPIO2A drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO2B_E
Address: Operational Base + offset (0x00214)
GPIO2B drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0119	gpio2b_e GPIO2B drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO2C_E
Address: Operational Base + offset (0x00218)
GPIO2C drive strength control

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0000	gpio2c_e GPIO2C drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO2D_E
Address: Operational Base + offset (0x0021c)
GPIO2D drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x5500	gpio2d_e GPIO2D drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO3A_E
Address: Operational Base + offset (0x00220)
GPIO3A drive strength control

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0009	gpio3a_e GPIO3A drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO3B_E
Address: Operational Base + offset (0x00224)
GPIO3B drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x682a	gpio3b_e GPIO3B drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO3C_E
Address: Operational Base + offset (0x00228)
GPIO3C drive strength control

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x3115	gpio3c_e GPIO3C drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO3D_E
Address: Operational Base + offset (0x0022c)
GPIO3D drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0304	gpio3d_e GPIO3D drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO1L_SR
Address: Operational Base + offset (0x00300)
GPIO1A/B SR control

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:8	RW	0x08	gpio1b_sr GPIO1B slew rate control for each bit 1'b0: slow (half frequency) 1'b1: fast
7:0	RW	0x00	gpio1a_sr GPIO1A slew rate control for each bit 1'b0: slow (half frequency) 1'b1: fast

GRF_GPIO1H_SR
Address: Operational Base + offset (0x00304)
GPIO1C/D SR control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:8	RW	0x00	gpio1d_sr GPIO1D slew rate control for each bit 1'b0: slow (half frequency) 1'b1: fast
7:0	RW	0x00	gpio1c_sr GPIO1C slew rate control for each bit 1'b0: slow (half frequency) 1'b1: fast

GRF_GPIO2L_SR

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Address: Operational Base + offset (0x00308)
GPIO2A/B SR control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:8	RW	0x02	gpio2b_sr GPIO2B slew rate control for each bit 1'b0: slow (half frequency) 1'b1: fast
7:0	RW	0x00	gpio2a_sr GPIO2A slew rate control for each bit 1'b0: slow (half frequency) 1'b1: fast

GRF_GPIO2H_SR
Address: Operational Base + offset (0x0030c)
GPIO2C/D SR control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:8	RW	0x00	gpio2d_sr GPIO2D slew rate control for each bit 1'b0: slow (half frequency) 1'b1: fast
7:0	RW	0x00	gpio2c_sr GPIO2C slew rate control for each bit 1'b0: slow (half frequency) 1'b1: fast

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GRF_GPIO3L_SR
Address: Operational Base + offset (0x00310)
GPIO3A/B SR control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:8	RW	0x47	gpio3b_sr GPIO3B slew rate control for each bit 1'b0: slow (half frequency) 1'b1: fast
7:0	RW	0x02	gpio3a_sr GPIO3A slew rate control for each bit 1'b0: slow (half frequency) 1'b1: fast

GRF_GPIO3H_SR
Address: Operational Base + offset (0x00314)
GPIO3C/D SR control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:8	RW	0x00	gpio3d_sr GPIO3D slew rate control for each bit 1'b0: slow (half frequency) 1'b1: fast

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Bit	Attr	Reset Value	Description
7:0	RW	0x40	gpio3c_sr GPIO3C slew rate control for each bit 1'b0: slow (half frequency) 1'b1: fast

GRF_GPIO_SMT

Address: Operational Base + offset (0x00380)

GPIO smitter control register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:10	RO	0x0	reserved
9	RW	0x1	gpio0b_smt_ctrl GPIO0B1 Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.
8	RW	0x1	gpio3d7_smt_ctrl GPIO3D7 Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.
7:4	RW	0x1	gpio3d_smt_ctl GPIO3D[3:0] Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.
3:2	RW	0x1	gpio2c_smt_ctrl GPIO2C[6:5] Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.
1:0	RW	0x1	gpio1c_smt_ctrl GPIO1C[1:0] Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.

GRF_SOC_CON0

Address: Operational Base + offset (0x00400)

SoC control register 0

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:14	RO	0x0	reserved
13:9	RW	0x00	cci_qosoverride CCI400 QOSOVERRIDE input control. If HIGH, internally generated values override the ARQOS and AWQOS inputs. One bit exists for each slave interface.
8:6	RW	0x0	cci_bufferableoverride CCI400 BUFFERABLEOVERRIDE input control. If HIGH, then all transactions from a master interface are made non-bufferable by modifying AWCACHE[0] and ARCACHE[0]. One bit exists for each master interface.
5:3	RW	0x7	cci_barrierterminate CCI BARRIERTERMINAGE input control. If HIGH, then barriers are terminated in the master interface and not propagated downstream. Set this HIGH if the downstream slave does not support barriers. One bit exists for each master interface.
2:0	RW	0x0	cci_broadcastcachemaint CCI400 BROADCASTCACHEMAINT input control. If HIGH, then cache maintenance operations are sent downstream. Only set if there is a downstream cache with an ACE-Lite interface. One bit exists for each master interface.

GRF_SOC_CON1

Address: Operational Base + offset (0x00404)

SoC control register 1

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:9	RO	0x0	reserved
8:4	RW	0x1c	cci_acchannelen CCI400 ACCHANNELEN input control. Slave interface supports DVM messages. This is overridden to 0x0 if you set the Control Override Register[1].
3:0	RW	0x0	cci_ecorevnum CCI400 ECOREVNUM input control. Used to ease the update of the revision field register in case of an ECO.

GRF_SOC_CON2

Address: Operational Base + offset (0x00408)

SoC control register 2

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:12	RO	0x0	reserved
11	RW	0x0	gpu_timer_en SoC timer input enable for GPU. 0: tie TSVALUE of GPU to low. 1: use system timer for TSVALUE of GPU.
10	RW	0x1	gpu_resetn_jtag gpu jtag port resetn control
9	RW	0x1	gpu_jtag_sys_power_ctrl gpu system jtag power control

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Bit	Attr	Reset Value	Description
8	RW	0x1	gpu_aclk_mem_en gpu aclk master clock enable
7	RW	0x1	gpu_aclk_cfg_en gpu axi slave clock enable
6	RW	0x1	gpu_clk_core_en gpu core clock enable
5	RW	0x1	gpu_dxt_bc_enable gpu dxt_bc_enable port control
4	RW	0x1	gpu_dusts_enable gpu dusts_enable port cotrol
3:2	RW	0x0	gpu_mem_aw_qos GPU AXI master write channel QoS control.
1:0	RW	0x0	gpu_mem_ar_qos GPU AXI master read channel QoS control.

GRF_SOC_CON3

Address: Operational Base + offset (0x0040c)

SoC control register 3

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15	RW	0x0	saradc_tsen_pd_1 GPU TSADC temperature sensor output buffer power down 0: enable buffer 1: power down
14	RW	0x0	saradc_tsen_buf_pd_1 GPU TSADC temperature sensor output buffer power down 0: enable buffer 1: power down
13:8	RW	0x20	saradc_tsen_cal_1 GPU TSADC calibration control for temperature sensor output, increase register value to shift down temperature sensor output.
7	RW	0x0	saradc_tsen_pd_0 Core TSADC temperature sensor power down 0: enable sensor 1: power down

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Bit	Attr	Reset Value	Description
6	RW	0x0	saradc_tsen_buf_pd_0 Core TSADC temperature sensor output buffer power down 0: enable buffer 1: power down
5:0	RW	0x20	saradc_tsen_cal_0 Core TSADC calibration control for temperature sensor output, increase register value to shift down temperature sensor output.

GRF_SOC_CON4
Address: Operational Base + offset (0x00410)
SoC control register 4

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15	RO	0x0	reserved
14:13	RW	0x1	drtype_dmac_peri peri dma drtype bit control, indicates the type of acknowledgement, or request, that the peripheral signals: b00 = single level request b01 = burst level request b10 = acknowledging a flush request that the DMAC requested b11 = reserved.
12	RW	0x0	edp_dc_tp_i eDP PHY analog DC test point input
11:3	RW	0x000	edp_frq_vid_ck_in eDP PHY frequency information of vid_ck_in frq_vid_ck_in<8:0>/8 = freq(vid_ck_in)/10
2	RW	0x0	edp_vid_lock eDP PHY input video PLL stable indicator 1'b1: stable 1'b0: unstable
1	RW	0x0	edp_iddq_en eDP PHY IDDQ enable 1'b0: disable 1'b1: enable, all circuits are power down, all IO are high-z

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Bit	Attr	Reset Value	Description
0	RW	0x1	edp_ref_clk_sel eDP PHY reference clock source selection 1'b0: from PAD(IO_EDP_OSC_CLK_24M) 1'b1: from internal 24MHz or 27MHz clock

GRF_SOC_CON5
Address: Operational Base + offset (0x00414)
SoC control register 5

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:8	RO	0x0	reserved
7	RW	0x0	edp_mem_ctrl_sel eDP memory control selection 1'b1: controlled by eDP controller internal logic 1'b0: controlled by APB BUS
6	RW	0x0	edp_bist_en eDP memory control selection 1'b1: controlled by eDP controller internal logic 1'b0: controlled by APB BUS
5	RW	0x0	edp_hdcp_protect eDP HDCP function protection 1'b1: protect 1'b0: not protect
4	RW	0x0	edp_tx_bscan_en DP TX boundary enable 1'b0: disable 1'b1: enable
3:0	RW	0x0	dp_tx_bscan_data eDP TX boundary data bit0: boundary data to ch0 bit1: boundary data to ch1 bit2: boundary data to ch2 bit3: boundary data to ch3

GRF_SOC_CON6
Address: Operational Base + offset (0x00418)
SoC control register 6

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:13	RO	0x0	reserved
12	RW	0x0	hevc_vcodec_sel hevc_vcodec_sel bit select 0: vcodec 1: hevc
11:8	RW	0x0	dphy_rx_forcerxmode MIPI DPHY RX force lane into receive mode/wait for stop stat. Every bit for one lane, bit3 is for lane3, bit2 is for lane2, bit1 is for lane1, bit0 is for lane0.
7	RW	0x0	dsi0_dpiupdatecfg DSI host0 dpiupdatecfg bit control
6	RW	0x0	dphy_rx_clk_inv_sel dphy_rx clock invertor select 1'b0: with invertor 1'b1: without invertor
5	RW	0x0	clk0_lvds_inv_sel Invertion of VOP dclk for LVDS selection 1'b1: invert 1'b0: not invert
4	RW	0x0	dclk_lvds_div2_sel 2 divide frequency of VOP dclk for LVDS selection 1'b1: 2 divide frequency 1'b0: no divide frequency
3	RW	0x0	dsi_dpicolorm DSI dpicolorm bit control
2	RW	0x0	dsi0_dpishutd DSI host dpishutdn bit control
1	RW	0x0	isp_mipi_csi_host_sel ISP or MIPI_CSI host slelect 1'b0: mipi csi host 1'b1: ISP
0	RW	0x0	disable_isp ISP disable bit control

GRF_SOC_CON7
Address: Operational Base + offset (0x0041c)
SoC control register 7

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15	RW	0x0	mipi_phy_ttl_mode mipi_phy_ttl_mode bit control 0: dsi/lvds mode 1: ttl mode
14:13	RW	0x0	lvds_select lvds_select control bit
12	RW	0x0	lvds_mode lvds_mode bit control
11	RW	0x0	lvds_msbsel LVDS controller msbsel signal control
10:7	RW	0x0	forcetxstopmode MIPI DPHY TX force lane into transmit mode and generate stop sate. Every bit for one lane, bit3 is for lane3, bit2 is for lane2, bit1 is for lane1, bit0 is for lane0.
6	RW	0x0	dphy_tx0_forcerxmode MIPI DPHY TX force lane into receive mode/wait for stop stat. Every bit for one lane, bit3 is for lane3, bit2 is for lane2, bit1 is for lane1, bit0 is for lane0.
5	RW	0x0	lane0_turndisable MIPI DPHY TX0 disable turn around control Every bit for one lane, bit3 is for lane3, bit2 is for lane2, bit1 is for lane1, bit0 is for lane0.
4:1	RO	0x0	reserved
0	RW	0x0	vcodec_sel vdpu vepu clock select 1'b0: select vepu aclk as vcodec main clock 1'b1: select vdpu aclk as vcodec main clock

GRF_SOC_CON8

Address: Operational Base + offset (0x00420)

SoC control register 8

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x7538	mcu_stcalib_0 mcu_stcalib[15:0] bit control

GRF_SOC_CON9
Address: Operational Base + offset (0x00424)
SoC control register 9

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:10	RO	0x0	reserved
9:0	RW	0x100	mcu_stcalib_1 mcu_stcalib[25:16] bit control

GRF_SOC_CON10
Address: Operational Base + offset (0x00428)
SoC control register 10

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0000	mcu_code_addr_start MCU Code address start bit27 ~ bit12

GRF_SOC_CON11
Address: Operational Base + offset (0x0042c)
SoC control register 11

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x2000	mcu_sram_addr_start MCU SRAM address start bit27 ~ bit12

GRF_SOC_CON12
Address: Operational Base + offset (0x00430)
SoC control register 12

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x6000	mcu_exsram_addr_start MCU external sram start address bit27 ~ bit12

GRF_SOC_CON13
Address: Operational Base + offset (0x00434)
SoC control register 13

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0000	mcu_experi_addr_start MCU external periphral start address bit27 ~ bit12

GRF_SOC_CON14
Address: Operational Base + offset (0x00438)
SoC control register 14

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:12	RW	0x0	experi_addr_start experi_addr_start bit31~bit28
11:8	RW	0x0	exsram_addr_start exsram_addr_start bit31~bit28
7:4	RW	0x0	sram_addr_start sram_addr_start bit31~bit28
3:0	RW	0x0	code_addr_start code_addr_start bit31~bit28

GRF_SOC_CON15
Address: Operational Base + offset (0x0043c)
SoC control register 15

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15	RO	0x0	reserved
14	RW	0x0	flash_poc_ctrol flash IO domain poc control selection 0: controled by gpio_0b5 pad 1: controled by bit 2 of IO_VSEL
13	RW	0x1	force_jtag Force select jtag function from sdmmc0 IO
12	RW	0x0	pwm_sel PWM solution selection 1'b1: RK design PWM 1'b0: DW PWM

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Bit	Attr	Reset Value	Description
11:9	RW	0x0	gmac_phy_intf_sel PHY interface select 3'b001: RGMII 3'b100: RMII All others: Reserved
8	RW	0x0	gmac_flowctrl GMAC transmit flow control When set high, instructs the GMAC to transmit PAUSE Control frames in Full-duplex mode. In Half-duplex mode, the GMAC enables the Back-pressure function until this signal is made low again
7	RW	0x0	gmac_speed MAC speed 1'b1: 100-Mbps 1'b0: 10-Mbps
6	RW	0x0	rmii_mode RMII mode selection 1'b1: RMII mode
5:4	RW	0x0	gmac_clk_sel RGMII clock selection 2'b00: 125MHz 2'b11: 25MHz 2'b10: 2.5MHz
3	RW	0x0	rmii_clk_sel RMII clock selection 1'b1: 25MHz 1'b0: 2.5MHz
2	RW	0x0	pause_mmc_peri PERI MMC AHB bus arbiter pause control
1	RW	0x0	pause_emem_peri PERI EMEM AHB bus arbiter pause control
0	RW	0x0	pause_usb_peri PERI USB AHB bus arbiter pause control

GRF_SOC_CON16

Address: Operational Base + offset (0x00440)

SoC control register 16

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15	RW	0x0	rxclk_dly_ena_gmac RGMII TX clock delayline enable 1'b1: enable 1'b0: disable
14:8	RW	0x00	clk_rx_dl_cfg_gmac RGMII RX clock delayline value
7	RW	0x0	txclk_dly_ena_gmac RGMII TX clock delayline enable 1'b1: enable 1'b0: disable
6:0	RW	0x00	clk_tx_dl_cfg_gmac RGMII TX clock delayline value

GRF_SOC_CON17
Address: Operational Base + offset (0x00444)
SoC control register 17

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:10	RO	0x0	reserved
9:5	RW	0x00	uart_rts_sel uart_rts_sel bit control
4	RW	0x0	uart4_cts_sel UART polarity selection for cts port Every bit for one UART, bit4 is for UART_EXP, bit3 is for UART_GPS, bit2 is for UART_DBG, bit1 is for UART_BB, bit0 is for UART_BT. 1'b1: high asserted 1'b0: low asserted
3	RW	0x0	uart3_cts_sel UART polarity selection for cts port Every bit for one UART, bit4 is for UART_EXP, bit3 is for UART_GPS, bit2 is for UART_DBG, bit1 is for UART_BB, bit0 is for UART_BT. 1'b1: high asserted 1'b0: low asserted
2	RW	0x0	hsadc_clk_sel hsadc_clk source select 1'b0: with clock invetor 1'b1: without clock invertor

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Bit	Attr	Reset Value	Description
1	RW	0x0	uart1_cts_sel UART polarity selection for cts port Every bit for one UART, bit4 is for UART_EXP, bit3 is for UART_GPS, bit2 is for UART_DBG, bit1 is for UART_BB, bit0 is for UART_BT. 1'b1: high asserted 1'b0: low asserted
0	RW	0x0	uart0_cts_sel UART polarity selection for cts port Every bit for one UART, bit4 is for UART_EXP, bit3 is for UART_GPS, bit2 is for UART_DBG, bit1 is for UART_BB, bit0 is for UART_BT. 1'b1: high asserted 1'b0: low asserted

GRF_SOC_STATUS0

Address: Operational Base + offset (0x00480)

SoC status register 0

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RW	0x0	ddrphy_lock
14	RW	0x0	sdmmc_idle sdmmc_idle status
13	RW	0x0	gpu_idle gpu idle status
12	RO	0x0	gmac_portselect gmac_portselect status
11	RO	0x0	newpll_clk NEW PLL clock status
10	RO	0x0	generalpll_clk GENERAL PLL clock output
9	RO	0x0	codecpll_clk CODEC PLL clock output
8	RO	0x0	ddrpll_clk DDR PLL clock output
7	RO	0x0	arm_b_pll_clk ARM_B PLL clock output
6	RO	0x0	arm_l_pll_clk ARM_L PLL clock output
5	RO	0x0	newpll_lock NEW PLL lock
4	RO	0x0	general_pll_lock General PLL lock
3	RO	0x0	codecpll_lock CODEC PLL lock status
2	RO	0x0	ddrpll_lock DDR PLL lock status
1	RO	0x0	arm_b_pll_lock ARM_B PLL lock status
0	RO	0x0	arm_l_pll_lock ARM_L PLL lock status

GRF_SOC_STATUS1

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Address: Operational Base + offset (0x00484)
SoC status register 1

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	nif0_fifo0 DDR channel0 NIF interface FIFO0 status

GRF_SOC_STATUS2
Address: Operational Base + offset (0x00488)
SoC status register 2

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	nif0_fifo1 DDR channel0 NIF interface FIFO1 status

GRF_SOC_STATUS3
Address: Operational Base + offset (0x0048c)
SoC status register 3

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	nif0_fifo2 DDR channel0 NIF interface FIFO2 status

GRF_SOC_STATUS4
Address: Operational Base + offset (0x00490)
SoC status register 4

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	nif0_fifo3 DDR channel0 NIF interface FIFO3 status

GRF_SOC_STATUS5
Address: Operational Base + offset (0x00494)
SoC status register 5

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	dfi0_eff_wr_num DDR channel 0 DFI interface write command number

GRF_SOC_STATUS6
Address: Operational Base + offset (0x00498)
SoC status register 6

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	dfi0_eff_rd_num DDR channel 0 DFI interface read command number

GRF_SOC_STATUS7
Address: Operational Base + offset (0x0049c)
SoC status register 7

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	dfi0_eff_act_num DDR channel 0 DFI interface active command number

GRF_SOC_STATUS8
Address: Operational Base + offset (0x004a0)
SoC status register 8

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	dfi0_timer_val DDR channel 0 DFI interface timer value

GRF_SOC_STATUS9
Address: Operational Base + offset (0x004a4)
SoC status register 9

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	dfi1_eff_wr_num DDR channel 1 DFI interface write command number

GRF_SOC_STATUS10
Address: Operational Base + offset (0x004a8)
SoC status register 10

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	dfi1_eff_rd_num DDR channel 1 DFI interface read command number

GRF_SOC_STATUS11
Address: Operational Base + offset (0x004ac)
SoC status register 11

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	dfi1_eff_act_num DDR channel 1 DFI interface active command number

GRF_SOC_STATUS12
Address: Operational Base + offset (0x004b0)
SoC status register 12

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	dfi1_timer_val DDR channel 1 DFI interface timer value

GRF_SOC_STATUS13
Address: Operational Base + offset (0x004b4)
SoC status register 13

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
24	RO	0x0	dphy_rx_rxclkactivehs dphy_rx_rxclkactivehs bit status
23	RO	0x0	dphy_rx_direction dphy_rx_direction bit status
22	RO	0x0	dphy_rx_ulpsactivenot_0 dphy_rx_ulpsactivenot_0 bit status
21	RO	0x0	dphy_rx_ulpsactivenot_1 dphy_rx_ulpsactivenot_1 bit status
20	RO	0x0	dphy_rx_ulpsactivenot_2 dphy_rx_ulpsactivenot_2 bit status
19	RO	0x0	dphy_rx_ulpsactivenot_3 dphy_rx_ulpsactivenot3 bit status
18:16	RO	0x0	ddrupctl0_stat ddrupctl0 stat status bit
15:0	RO	0x0000	ddrupctl0_bbflags ddrupctl0 bbflags status bit

GRF_SOC_STATUS14

Address: Operational Base + offset (0x004b8)

SoC status register 14

Bit	Attr	Reset Value	Description
31	RW	0x0	dp_detected_usbphy usb phy dp_detected_usbphy status bit status
30	RW	0x0	cp_detected_usbphy usb phy cp_detected_usbphy status bit status
29	RW	0x0	dcp_detected_usbphy usb phy dcp_detected_usbphy status bit status
28:23	RO	0x00	ehci_usbsts_host0 usb host 0 ehci_usbsts_host0 bit status
22:12	RO	0x000	ehci_xfer_cnt_host0 usb host 0 ehci_xfer_cnt_host0 bit status
11:8	RO	0x0	ehci_lpsmc_state_host0 usb host 0 ehci_lpsmc_state_host0 bit status
7	RO	0x0	ehci_xfer_prdc_host0 usb host 0 ehci_xfer_prdc_host0 bit status
6	RO	0x0	ehci_bufacc_host0 usb host 0 ehci_bufacc_host0 bit status
5	RO	0x0	ohci_globalsuspend_host0 usb host 0ohci_globalsuspend_host0 bit status
4	RO	0x0	ohci_drwe_host0 usb host 0ohci_drwe_host0 bit status
3	RO	0x0	ohci_rwe_host0 usb host 0ohci_rwe_host0 bit status
2	RO	0x0	ohci_ccs_host0 usb host 0 ohci_ccs_host0 bit status
1	RO	0x0	ohci_bufacc_host0 usb host 0 ohci_bufacc bit status
0	RO	0x0	ohci_rmtwkp_host0 usb host 0 ohci_rmtwkp_host0 bit status

GRF_SOC_STATUS15

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Address: Operational Base + offset (0x004bc)
SoC status register 15

Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved
26	RO	0x0	otg0_utmiotg_iddig usb otg utmiotg_iddig bit status
25:24	RO	0x0	otg0_utmi_linestate usb otg utmi_linestate bit status
23	RO	0x0	otg0_utmisrp_bvalid usb otg utmisrp_bvalid bit status
22	RO	0x0	otg0_utmiotg_vbusvalid usb otg utmiotg_vbusvalid bit status
21:16	RO	0x00	grf_stat_ehci_usbsts hsic ehci_usbsts bit status bit status
15:5	RO	0x000	grf_stat_ehci_xfer_cnt hsic ehci_xfer_cnt bit status
4	RO	0x0	grf_stat_ehci_xfer_prdc hsic ehci_xfer_prdc bit status
3:0	RO	0x0	grf_stat_ehci_lpsmc_state hsic ehci_lpsmc_state bit status

GRF_CPU_CON0
Address: Operational Base + offset (0x00500)
CPU little cluster control register 0

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:12	RW	0x0	cfgte pd_core_l ca53 cpu cfgte bit control
11:8	RW	0x0	cfgend Field0008 Abstract pd_core_l ca53 cpu cfgend bit control
7	RW	0x0	l2rstdisable Field0007 Abstract pd_core_l ca53 l2rstdisable
6	RW	0x0	dbgl1rstdisable Field0006 Abstract pd_core_l ca53 cpu dbgl1rstdisable
5	RO	0x0	reserved
4	RW	0x0	clrexmonreq_pd_core_l pd_core_l ca53 cpu clrexmonreq_pd_core_l bit control

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Bit	Attr	Reset Value	Description
3	RW	0x1	sysbardisable pd_core_l ca53 cpu sysbardisable bit control 0: disable 1: enable
2	RW	0x0	broadcastcachemaint Field0002 Abstract pd_core_l ca53 cpu broadcastcachemaint bit control 0: disable 1: enable
1	RW	0x1	broadcastouter pd_core_l ca53 cpu broadcastouter bit control 0: disable 1: enable
0	RW	0x1	broadcastinner pd_core_l ca53 cpu broadcastinner bit control 1: enable 0: disable

GRF_CPU_CON1
Address: Operational Base + offset (0x00504)
CPU little cluster control register 1

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:10	RO	0x0	reserved
9:8	RW	0x0	clusteridaff1 pd_core_l ca53 cpu clusteridaff1 bit control
7:4	RW	0x0	pd_core_arqos pd_core_l pd_core_arqos bit control
3:0	RW	0x0	pd_core_awqos pd_core_l pd_core_awqos bit control

GRF_CPU_CON2
Address: Operational Base + offset (0x00508)
CPU big cluster control register 2

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:12	RW	0x0	cfgte pd_core_b ca53 cpu cfgte bit control
11:8	RW	0x0	cfgend Field0008 Abstract pd_core_b ca53 cpu cfgend bit control
7	RW	0x0	l2rstdisable pd_core_b ca53 cpu l2rstdisable bit control
6	RW	0x0	dbgl1rstdisable Field0006 Abstract pd_core_b ca53 cpu dbgl1rstdisable bit control
5	RO	0x0	reserved
4	RW	0x0	clrexmonreq_pd_core_b pd_core_b clrexmonreq_pd_core_b bit control
3	RW	0x1	sysbardisable Field0003 Abstract pd_core_b ca53 cpu sysbardisable bit control
2	RW	0x0	broadcastcachemaint pd_core_b ca53 cpu broadcastcachemaint bit control
1	RW	0x1	broadcastouter Field0001 Abstract pd_core_b ca53 cpu broadcastouter bit control
0	RW	0x1	broadcastinner pd_core_b ca53 cpu broadcastinner bit control

GRF_CPU_CON3

Address: Operational Base + offset (0x0050c)

CPU big cluster control register 1

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:10	RO	0x0	reserved
9:8	RW	0x0	clusteridaff1 Field0000 Abstract clusteridaff1 bit control
7:4	RW	0x0	pd_core_arqos pd_core_b_arqos bit control
3:0	RW	0x0	pd_core_awqos pd_core_b_awqos bit control

GRF_CPU_STATUS0

Address: Operational Base + offset (0x00520)

CPU status register 0

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:24	RO	0x0	ncorereset_pd_core_b ncorereset_pd_core_b status
23:20	RO	0x0	ncorereset_pd_core_l ncorereset_pd_core_l status
19	RO	0x0	mcu_sleeping mcu_sleeping status
18	RO	0x0	mcu_sleepdeep mcu_sleepdeep status
17	RO	0x0	mcu_core_halted mcu_core_halted status
16	RO	0x0	mcu_core_lockup mcu_core_lockup status
15:13	RO	0x0	reserved
12	RO	0x0	clrexnonack_pd_core_b clrexnonack_pd_core_b status
11:8	RO	0x0	smpen_pd_core_b smpen_pd_core_b status
7:5	RO	0x0	reserved
4	RO	0x0	clrextmonack_pd_core_l clrextmonack_pd_core_l status
3:0	RO	0x0	smpen_pd_core_l smpen_pd_core_l status

GRF_CPU_STATUS1

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Address: Operational Base + offset (0x00524)
CPU status register 1

Bit	Attr	Reset Value	Description
31:24	RO	0x00	wrmemattr_pd_core_b wrmemattr of pd_core_b status
23:16	RO	0x00	rdmemattr_pd_core_b rdmemattr of pd_core_b status
15:8	RO	0x00	wrmemattr_pd_core_l wrmemattr of pd_core_l status
7:0	RO	0x00	rdmemattr_pd_core_l rdmemattr of pd_core_l status

GRF_CCI_STATUS0
Address: Operational Base + offset (0x00540)
CCI status register 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18	RO	0x0	cci400_activem0 Indicates that the master interface has active transactions. You can use it to gate the clock to downstream components.
17	RO	0x0	cci400_activem1 Indicates that the master interface has active transactions. You can use it to gate the clock to downstream components.
16	RO	0x0	cci400_activem2 Indicates that the master interface has active transactions. You can use it to gate the clock to downstream components.
15:12	RO	0x0	evntawqos4 Value of the QoS value regulator of slave interface 4 last applied to AWQOS.
11:8	RO	0x0	evntarqos4 Value of the QoS value regulator of slave interface 4 last applied to ARQOS.
7:4	RO	0x0	evntawqos3 Value of the QoS value regulator of slave interface 3 last applied to AWQOS.
3:0	RO	0x0	evntarqos3 Value of the QoS value regulator of slave interface 3 last applied to ARQOS.

GRF_CCI_STATUS1
Address: Operational Base + offset (0x00544)
CCI status register 1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	evntbus Bits 31~0 of CCI400 EVNTBUS output. CCI-400 events, exported if enabled in the PMCR.

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GRF_CCI_STATUS2
Address: Operational Base + offset (0x00548)
CCI status register 2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	evntbus Bits 63~32 of CCI400 EVNTBUS output. CCI-400 events, exported if enabled in the PMCR.

GRF_CCI_STATUS3
Address: Operational Base + offset (0x0054c)
CCI status register 3

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	evntbus Bits 95~64 of CCI400 EVNTBUS output. CCI-400 events, exported if enabled in the PMCR.

GRF_CCI_STATUS4
Address: Operational Base + offset (0x00550)
CCI status register 4

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	evntbus Bits 127~96 of CCI400 EVNTBUS output. CCI-400 events, exported if enabled in the PMCR.

GRF_CCI_STATUS5
Address: Operational Base + offset (0x00554)
CCI status register 5

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:0	RO	0x00000000	evntbus Bits 158~128 of CCI400 EVNTBUS output. CCI-400 events, exported if enabled in the PMCR.

GRF_CCI_STATUS6
Address: Operational Base + offset (0x00558)
CCI status register 6

Bit	Attr	Reset Value	Description
31:24	RO	0x00	arcount_m0 The counter value of CCI 400 master 0 port AR channel.
23:16	RO	0x00	rcount_m0 The counter value of CCI 400 master 0 port R channel.

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Bit	Attr	Reset Value	Description
15:8	RO	0x00	awcount_m0 The counter value of CCI 400 master 0 interface AW channel.
7:0	RO	0x00	bcount_m0 The counter value of CCI 400 master 0 interface B channel.

GRF_CCI_STATUS7
Address: Operational Base + offset (0x0055c)
CCI status register 7

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RO	0x0	arcount_m0 The difference between AR channel and R channel of CCI 400 master 0 interface.
3:0	RO	0x0	awdiff_m0 The difference between AW channel and B channel of CCI 400 master 0 interface.

GRF_CCI_STATUS8
Address: Operational Base + offset (0x00560)
CCI status register 8

Bit	Attr	Reset Value	Description
31:24	RO	0x00	arcount_m2 The counter value of CCI 400 master 2 port AR channel.
23:16	RO	0x00	rcount_m2 The counter value of CCI 400 master 2 port R channel.
15:8	RO	0x00	awcount_m2 The counter value of CCI 400 master 2 interface AW channel.
7:0	RO	0x00	bcount_m2 The counter value of CCI 400 master 2 interface B channel.

GRF_CCI_STATUS9
Address: Operational Base + offset (0x00564)
CCI status register 9

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RO	0x0	arcount_m2 The difference between AR channel and R channel of CCI 400 master 2 interface.
3:0	RO	0x0	awdiff_m2 The difference between AW channel and B channel of CCI 400 master 2 interface.

GRF_CCI_STATUS10
Address: Operational Base + offset (0x00568)

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CCI status register 10

Bit	Attr	Reset Value	Description
31:24	RO	0x00	arcount_s2 The counter value of CCI 400 slave 2 port AR channel.
23:16	RO	0x00	rcount_s2 The counter value of CCI 400 slave 2 port R channel.
15:8	RO	0x00	awcount_s2 The counter value of CCI 400 slave 2 interface AW channel.
7:0	RO	0x00	bcount_s2 The counter value of CCI 400 slave 2 interface B channel.

GRF_CCI_STATUS11

Address: Operational Base + offset (0x0056c)

CCI status register 11

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RO	0x0	arcount_s2 The difference between AR channel and R channel of CCI 400 slave 2 interface.
3:0	RO	0x0	awdiff_s2 The difference between AW channel and B channel of CCI 400 slave 2 interface.

GRF_CCI_STATUS12

Address: Operational Base + offset (0x00570)

CCI status register 12

Bit	Attr	Reset Value	Description
31:24	RO	0x00	arcount_s3 The counter value of CCI 400 slave 3 port AR channel.
23:16	RO	0x00	rcount_s3 The counter value of CCI 400 slave 3 port R channel.
15:8	RO	0x00	awcount_s3 The counter value of CCI 400 slave 3 interface AW channel.
7:0	RO	0x00	bcount_s3 The counter value of CCI 400 slave 3 interface B channel.

GRF_CCI_STATUS13

Address: Operational Base + offset (0x00574)

CCI status register 13

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:20	RO	0x00	account_s3 The counter value of CCI 400 slave 3 interface AC channel.

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Bit	Attr	Reset Value	Description
19:12	RO	0x00	crcount_s3 The counter value of CCI 400 slave 3 interface CR channel.
11:8	RO	0x0	arcount_s3 The difference between AR channel and R channel of CCI 400 slave 3 interface.
7:4	RO	0x0	awdiff_s3 The difference between AW channel and B channel of CCI 400 slave 3 interface.
3:0	RO	0x0	acdiff_s3 The difference between AC channel and CR channel of CCI 400 slave 3 interface.

GRF_CCI_STATUS14
Address: Operational Base + offset (0x00578)
CCI status register 14

Bit	Attr	Reset Value	Description
31:24	RO	0x00	arcount_s4 The counter value of CCI 400 slave 4 port AR channel.
23:16	RO	0x00	rcount_s4 The counter value of CCI 400 slave 4 port R channel.
15:8	RO	0x00	awcount_s4 The counter value of CCI 400 slave 4 interface AW channel.
7:0	RO	0x00	bcount_s4 The counter value of CCI 400 slave 4 interface B channel.

GRF_CCI_STATUS15
Address: Operational Base + offset (0x0057c)
CCI status register 15

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:20	RO	0x00	account_s4 The counter value of CCI 400 slave 4 interface AC channel.
19:12	RO	0x00	crcount_s4 The counter value of CCI 400 slave 4 interface CR channel.
11:8	RO	0x0	arcount_s4 The difference between AR channel and R channel of CCI 400 slave 4 interface.
7:4	RO	0x0	awdiff_s4 The difference between AW channel and B channel of CCI 400 slave 4 interface.
3:0	RO	0x0	acdiff_s4 The difference between AC channel and CR channel of CCI 400 slave 4 interface.

GRF_DDRC0_CON0

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Address: Operational Base + offset (0x00600)
DDRC0 control register 0

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:12	RO	0x0	reserved
11	RW	0x0	upctl_lp_reset_mode upctl_lp_reset_mode bit control
10	RO	0x0	reserved
9	RW	0x0	noc_rsp_err_stall noc_rsp_err_stall bit control 0: stall response 1: error response
8	RW	0x0	upctl_anfifo upctl_anfifo bit control
7	RW	0x0	upctl_aburstint upctl_aburstint bit control
6	RW	0x0	dfi_eff_stat_en1 dfi_eff_stat_en1 bit control
5	RW	0x0	dfi_eff_stat_en0 dfi_eff_stat_en0 bit control
4	RW	0x0	mobile_ddr_sel mobile_ddr_sel bit control
3	RW	0x0	ddr0_16bit_en ddr0_16bit_en bit control
2	RW	0x0	mash_mainddr3 mash_mainddr3 bit control
1	RW	0x0	msch0_mainpartialpop msch0_mainpartialpop bit control
0	RW	0x0	upctl_c_active_in upctl_c_active_in bit control

GRF_SIG_DETECT_CON
Address: Operational Base + offset (0x00680)
external signal detect configue register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	sd_detect_filter_time_sel Field0000 Abstract sd_detect_filter time sel 00: 5ms 01: 15ms 10: 35ms 11: 50ms
13:12	RW	0x0	host_ls_filter_time_sel Field0000 Abstract host_ls_filter time select 00: 100us 01: 500us 10: 1ms 11:10ms
11:10	RW	0x0	otg_ls_filter_time_sel otg_ls_filter time select 00: 100us 01: 500us 10: 1ms 11:10ms
9:8	RW	0x0	otg_id_filter_time_sel otg_id_filter time select 00: 5ms 01:15ms 10: 35ms
7	RO	0x0	reserved
6	RW	0x0	otg_id_fall_edge_detect_en otg_id_fall_edge_detect enable 0: disable 1: enable
5	RW	0x0	otg_id_rise_edge_detect_en otg_id_detect enable 0: disable 1: enable
4	RW	0x0	host_line_state_detect_en host_line_state_detect enable 0: disable 1: enable

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Bit	Attr	Reset Value	Description
3	RW	0x0	otg_bvalid_detect_en otg_bvalid detect enable 0: disable 1: enable
2	RW	0x0	otg_linestate_detect_en otg_linestate_detect enable 0: disable 1: enable
1	RW	0x0	sd_detect_fall_edge_detect_en Field0000 Abstract sd_detect_falling_edge enable 0: disable 1: enable
0	RW	0x0	sd_detect_rising_edge_dectect_en sd_detect_rising_edge enable 0: disable 1:enable

GRF_SIG_DETECT_STATUS
Address: Operational Base + offset (0x00690)
external signal detect configue register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:7	RO	0x0	reserved
6	RO	0x0	otg_id_fall_edge_detect_status otg_id_fall_edge_detect status 0: disable 1: enable
5	RO	0x0	otg_id_rise_edge_detect_status otg_id_detect status 0: disable 1: enable
4	RO	0x0	host_line_state_detect_status host_line_state_detect status 0: disable 1: enable
3	RO	0x0	otg_bvalid_detect_status otg_bvalid detect status 0: disable 1: enable

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Bit	Attr	Reset Value	Description
2	RO	0x0	otg_linestate_detect_status otg_linestate_detect status 0: disable 1: enable
1	RO	0x0	sd_detect_fall_edge_detect_status sd_detect_falling_edge status 0: disable 1: enable
0	RO	0x0	sd_detect_rising_edge_dectect_status sd_detect_rising_edge status 0: disable 1:enable

GRF_SIG_DETECT_CLR
Address: Operational Base + offset (0x006a0)
external signal detect configue register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:7	RO	0x0	reserved
6	WO	0x0	otg_id_fall_edge_detect_clr otg_id_fall_edge_detect enable 0: disable 1: enable
5	WO	0x0	otg_id_rise_edge_detect_clr otg_id_detect enable 0: disable 1: enable
4	WO	0x0	host_line_state_detect_clr host_line_state_detect enable 0: disable 1: enable
3	WO	0x0	otg_bvalid_detect_clr otg_bvalid detect enable 0: disable 1: enable
2	WO	0x0	otg_linestate_detect_clr otg_linestate_detect enable 0: disable 1: enable

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Bit	Attr	Reset Value	Description
1	WO	0x0	sd_detect_fall_edge_detect_clr Field0000 Abstract sd_detect_falling_edge enable 0: disable 1: enable
0	WO	0x0	sd_detect_rising_edge_dectect_clr sd_detect_rising_edge enable 0: disable 1:enable

GRF_UOC0_CON0
Address: Operational Base + offset (0x00700)
OTG control register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit 0 ~ bit 15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15	RW	0x0	usbotg_dbnce_fltr_bypass USB OTG debounce filter for input signals bypass enable. This bit is only used in simulation mode.
14:11	RO	0x0	reserved
10	RW	0x0	iddig_status control software iddig value 0:host 1:device
9	RW	0x0	iddig_sft_sel 0 : iddig to otg controller select usbphy output 1: iddig to otg controller select grf_uoc0_con0[10]
8	RW	0x0	utmi_dmpulldown 0: DM 15 KOhm pull down disabled 1: DM 15 Kohm pull down enable
7	RW	0x0	utmi_dppulldown 0: DP 15 KOhm pull down disabled 1: DP 15 Kohm pull down enable
6	RW	0x0	utmi_termselect USB Termination Select 1: Full-speed terminations are enabled. 0: High-speed terminations are enabled.

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Bit	Attr	Reset Value	Description
5:4	RW	0x0	utmi_xcvrselect Transceiver Select 11: Sends an LS packet on an FS bus or receives an LS packet. 10: LS Transceiver 01: FS Transceiver 00: HS Transceiver
3:2	RW	0x0	utmi_opmode UTMI+ Operational Mode Function: This controller bus selects the UTMI+ operational mode. 11: Normal operation without SYNC or EOP generation. If the XCVRSEL bus is not set to 00 while OPMODE[1:0] is set to 11, USB PHY behavior is undefined. 10: Disable bit stuffing and NRZI encoding 01: Non-Driving 00: Normal
1	RW	0x0	utmi_suspend_n Suspend Assertion 1: Normal operating mode 0: Suspend mode
0	RW	0x0	usbphy_soft_con_sel 0: software control usb phy disable 1 : software control usb phy enable

GRF_UOC1_CON1
Address: Operational Base + offset (0x00718)
usb host control register

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~bit15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:13	RO	0x0	reserved
12	RW	0x0	usbphy1_vdm_src_en open dm voltage source
11	RW	0x0	usbphy1_vdp_src_en open dp voltage source
10	RW	0x0	usbphy1_rdm_pdwn_en open dm pull down resistor

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Bit	Attr	Reset Value	Description
9	RW	0x0	usbphy1_idp_src_en open dp source current
8	RW	0x0	usbphy1_idm_sink_en open dm sink current enable
7	RW	0x0	usbphy1_idp_sink_en open dp sink current enable
6:0	RO	0x0	reserved

GRF_UOC1_CON2
Address: Operational Base + offset (0x0071c)
UOC1 control register 2

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15	RW	0x1	usbhost0_incr4_en USB HOST0 incr4_en bit control
14	RW	0x1	usbhost0_incr16_en USB HOST0 incr16_en bit control
13	RW	0x0	usbhost0_hubsetup_min USB HOST0 hubsetup_min bit control
12	RW	0x0	usbhost0_app_start_clk USB HOST0 app_start_clk bit control
11:6	RW	0x20	usbhost0_fladj_val_common USB HOST0 fladj_val_common bit control
5:0	RW	0x20	usbhost0_fladj USB HOST0 fladj bit control

GRF_UOC1_CON3
Address: Operational Base + offset (0x00720)
UOC1 control register 3

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15	RW	0x0	usbhost0_ohci_susp_lgcy USB HOST0 ohci_susp_lgcy bit control
14	RW	0x0	usbhost0_ohci_cntsel USB HOST0 ohci_cntsel bit control
13	RW	0x0	usbhost0_app_prt_ovrcur USB HOST0 app_prt_ovrcur bit control
12	RO	0x0	reserved
11	RW	0x1	usbhost0_word_if USB HOST0 word_if bit control
10	RW	0x0	usbhost0_sim_mode USB HOST0 sim_mode bit control
9	RW	0x1	usbhost0_incrx_en USB HOST0 incrx_en bit control
8	RW	0x1	usbhost0_incr8_en USB HOST0 incr8_en bit control
7	RO	0x0	reserved
6	RW	0x1	usbhost0_ohci_clkcktrst USB HOST0 ohci_clkcktrst bit conrol
5:2	RO	0x0	reserved
1	RW	0x0	usbhost_sel usb host controller select 1: choose old usb controller. 0: choose EHCI controller.
0	RW	0x0	usbhost0_autoppd_on_overcur USB HOST0 autoppd_on_overcur bit control

GRF_UOC1_CON4

Address: Operational Base + offset (0x00724)

USB HOST 2.0 control register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~bit15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15	RW	0x0	usbphy_commonon USBPHY common on
14	RO	0x0	reserved
13	RW	0x0	reserved
12	RW	0x0	reserved
11	RW	0x0	reserved
10	RW	0x0	otgphydisable when 1, otgphy is disabled.
9:8	RW	0x0	utmiotg_scaledown_mode OTG controller scaledown mode. This bit is used only in simulation mode.
7:6	RW	0x0	utmihost_scaledown_mode utmihost scaledown mode. This bit is used only in simulation mode.
5	RW	0x0	utmiotg_idpullup Analog ID Input Sample Enable Function: This controller signal controls ID line sampling. 1: ID pin sampling is enabled, and the IDDIG output is valid. 0: ID pin sampling is disabled, and the IDDIG output is not valid.
4	RW	0x1	utmiotg_dppulldown D+ Pull-Down Resistor Enable
3	RW	0x1	utmiotg_dmpulldown D- Pull-Down Resistor Enable
2	RW	0x1	utmiotg_drvvbus Drive VBUS 1: The VBUS Valid comparator is enabled. 0: The VBUS Valid comparator is disabled.
1	RW	0x0	utmisrp_chrgvbus VBUS Input Charge Enable
0	RW	0x0	utmisrp_dischrgvbus VBUS Input Discharge Enable

GRF_UOC1_CON5

Address: Operational Base + offset (0x00728)

USB HOST 2.0 control register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit 0 ~ bit 15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:9	RO	0x0	reserved
8	RW	0x0	utmi_dmpulldown 0: DM 15 KOhm pull down disabled 1: DM 15 Kohm pull down enable
7	RW	0x0	utmi_dppulldown 0: DP 15 KOhm pull down disabled 1: DP 15 Kohm pull down enable
6	RW	0x0	utmi_termselect USB Termination Select 1: Full-speed terminations are enabled. 0: High-speed terminations are enabled.
5:4	RW	0x0	utmi_xcvrselect Transceiver Select 11: Sends an LS packet on an FS bus or receives an LS packet. 10: LS Transceiver 01: FS Transceiver 00: HS Transceiver
3:2	RW	0x0	utmi_opmode UTMI+ Operational Mode Function: This controller bus selects the UTMI+ operational mode. 11: Normal operation without SYNC or EOP generation. If the XCVRSEL bus is not set to 00 while OPMODE[1:0] is set to 11, USB PHY behavior is undefined. 10: Disable bit stuffing and NRZI encoding 01: Non-Driving 00: Normal
1	RW	0x0	utmi_suspend_n Suspend Assertion 1: Normal operating mode 0: Suspend mode
0	RW	0x0	usbphy_soft_con_sel 0: software control usb phy disable 1 : software control usb phy enable

GRF_UOC3_CON0

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Address: Operational Base + offset (0x00738)
UOC3 control register 0

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15	RO	0x0	reserved
14	RW	0x0	hsicphy_soft_con_sel HSIC PHY software control enale
13	RW	0x1	hsicphy_txbitstuffenh HSIC high byte transmit bit-stuffing enable this controller signal controls biy stuffing on DATAIN[15:8] when OPMODE[1:0]=2'b11 1'b1: bit stuffing is enabled 1'b0: bit stuffing is disabled
12	RW	0x1	hsicphy_txbitstuffen HSIC low byte transmit bit-stuffing enable this controller signal controls biy stuffing on DATAIN[7:0] when OPMODE[1:0]=2'b11 1'b1: bit stuffing is enabled 1'b0: bit stuffing is disabled
11	RW	0x0	hsichhy_siddq HSIC SIDDQ test enable 1'b1: the analog blocks are power down 1'b0: the analog blocks are power up
10	RW	0x0	hsicphy_port_reset HSIC per-port reset when asserted, this customer-specific signal reset the corresponding prot's transmit and receive logic without disabling the clocks within the HSIC PHY 1'b1: the transmit and receive FSMs are reset 1'b0: tjhe transmit and receive FSMs are operational
9:6	RW	0x3	hsicphy_txsrtune drive slew rate adjustment 4'b1111: +20% 4'b0111: +10% 4'b0011: design default 4'b0001: -10% 4'b0000: -20%

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Bit	Attr	Reset Value	Description
5:4	RW	0x2	hsicphy_txrpdtune HSIC driver pull-down impedance adjustment 2'b11: -5% 2'b10: design default 2'b01: +5% 2'b00: +11%
3:2	RW	0x2	hsicphy_txrputune HSIC driver pull-up impedance adjustment 2'b11: -5% 2'b10: design default 2'b01: +5% 2'b00: +11%
1	RW	0x1	hsicphy_dmpulldown HSIC bus keepers resistor enable This control signal selects the HSIC PHY to operate as a host or deivce.
0	RW	0x1	hsicphy_dppulldown HSIC bus keepers resistor enable This control signal detects that the HSIC PHY is being used as a host.

GRF_UOC3_CON1

Address: Operational Base + offset (0x0073c)

UOC3 control register 1

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:5	RO	0x0	reserved
4	RW	0x0	hsic_utmi_xcvrselect HSIC PHY transceiver select 1'b1: transceiver is in suspend, resume or connect mode 1'b0: transceiver is in HS mode
3:2	RW	0x0	hsic_utmi_opmode HSIC PHY operation mode 2'b11: normal mode without SYNC or EOP generation 2'b10: disable bit stuffing and NRZI encodeing 2'b01: Non-driving 2'b00: normal

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Bit	Attr	Reset Value	Description
1	RW	0x1	hsic_utmi_suspend_n HSIC PHY suspend mode enable Asserting this signal places the HSIC PHY in suspend mode. 1'b1: normal mode 1'b0: suspend mode
0	RW	0x1	hsic_utmi_sleep_n HSIC PHY sleep mode enable Asserting this signal places the HSIC PHY in sleep mode. 1'b1: normal mode 1'b0: sleep mode

GRF_UOC4_CON0

Address: Operational Base + offset (0x00740)

UOC4 control register 0

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:10	RO	0x0	reserved
9	RW	0x0	hsic_app_prt_ovrcur HSIC app_prt_ovrcur bit control
8	RW	0x0	hsic_autoppd_on_overcur HSIC autoppd_on_overcur bit control
7	RW	0x1	hsic_word_if HSIC word_if bit control
6	RW	0x0	hsic_sim_mode HSIC sim_mode bit control
5	RW	0x0	hsic_incrx_en HSIC incrx_en bit control
4	RW	0x0	hsic_incr8_en HSIC incr8_en bit control
3	RW	0x0	hsic_incr4_en HSIC incr4_en bit control
2	RW	0x0	hsic_incr16_en HSIC incr16_en bit control
1	RW	0x0	hsic_hubsetup_min HSIC hubsetup_min bit control
0	RW	0x0	hsic_app_start_clk HSIC app_start_clk bit control

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GRF_UOC4_CON1
Address: Operational Base + offset (0x00744)
UOC4 control register 1

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:12	RO	0x0	reserved
11:6	RW	0x20	hsic_fladj_val_common HSIC fladj_val_common bit control
5:0	RW	0x20	hsic_fladj HSIC fladj bit control

GRF_USBPHY_CON0
Address: Operational Base + offset (0x00780)
USB PHY control register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x8518	usbphy_con usbphy control register bit 15~0

GRF_USBPHY_CON1
Address: Operational Base + offset (0x00784)
USB PHY control register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0xe007	usbphy_con usbphy control register bit 31~16

GRF_USBPHY_CON2
Address: Operational Base + offset (0x00788)
USB PHY control register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x02e7	usbphy_con usbphy control register bit 47~32

GRF_USBPHY_CON3
Address: Operational Base + offset (0x0078c)
USB PHY control register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0200	usbphy_con usbphy control register bit 63~48

GRF_USBPHY_CON4
Address: Operational Base + offset (0x00790)
USB PHY control register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x5556	usbphy_con usbphy control register bit 79~64

GRF_USBPHY_CON5
Address: Operational Base + offset (0x00794)
USB PHY control register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x4555	usbphy_con usbphy control register bit 95~80

GRF_USBPHY_CON6
Address: Operational Base + offset (0x00798)
USB PHY control register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0005	usbphy_con usbphy control register bit 111~96

GRF_USBPHY_CON7
Address: Operational Base + offset (0x0079c)
USB PHY control register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x68c0	usbphy_con usbphy control register bit 127~112

GRF_USBPHY_CON8
Address: Operational Base + offset (0x007a0)
USB PHY control register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x8518	usbphy_con usbphy control register bit 143~128

GRF_USBPHY_CON9
Address: Operational Base + offset (0x007a4)
USB PHY control register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0xe007	usbphy_con usbphy control register bit 159~144

GRF_USBPHY_CON10
Address: Operational Base + offset (0x007a8)
USB PHY control register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x02e7	usbphy_con usbphy control register bit 175~160

GRF_USBPHY_CON11
Address: Operational Base + offset (0x007ac)
USB PHY control register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0200	usbphy_con usbphy control register bit 191~176

GRF_USBPHY_CON12
Address: Operational Base + offset (0x007b0)
USB PHY control register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x5556	usbphy_con usbphy control register bit 207~192

GRF_USBPHY_CON13
Address: Operational Base + offset (0x007b4)
USB PHY control register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x4555	usbphy_con usbphy control register bit 223~208

GRF_USBPHY_CON14
Address: Operational Base + offset (0x007b8)
USB PHY control register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0005	usbphy_con usbphy control register bit 239~224

GRF_USBPHY_CON15
Address: Operational Base + offset (0x007bc)
USB PHY control register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x68c0	usbphy_con usbphy control register bit 255~240

GRF_PVTM_CON0

Address: Operational Base + offset (0x00800)

PVT monitor control register 0

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:10	RO	0x0	reserved
9	RW	0x0	pvtm_gpu_osc_en pd_gpu PVT monitor oscilator enable 1'b1: enable 1'b0: disable
8	RW	0x0	pvtm_gpu_start pd_gpu PVT monitor start control
7:2	RO	0x0	reserved
1	RW	0x0	pvtm_core_osc_en pd_core PVT monitor oscilator enable 1'b1: enable 1'b0: disable
0	RW	0x0	pvtm_core_start pd_core PVT monitor start control

GRF_PVTM_CON1

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Address: Operational Base + offset (0x00804)
PVT monitor control register 1

Bit	Attr	Reset Value	Description
31:0	RW	0x016e3600	pvtm_core_cal_cnt pd_core pvtm calculator counter

GRF_PVTM_CON2
Address: Operational Base + offset (0x00808)
PVT monitor control register 2

Bit	Attr	Reset Value	Description
31:0	RW	0x016e3600	pvtm_gpu_cal_cnt pd_gpu pvtm calculator counter

GRF_PVTM_STATUS0
Address: Operational Base + offset (0x0080c)
PVT monitor status register 0

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	pvtm_gpu_freq_done pd_gpu pvtm frequency calculate done stutus
0	RW	0x0	pvtm_core_freq_done pd_core pvtm frequency calculate done stutus

GRF_PVTM_STATUS1
Address: Operational Base + offset (0x00810)
PVT monitor status register 1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pvtm_core_freq_cnt pd_core pvtm frequency count

GRF_PVTM_STATUS2
Address: Operational Base + offset (0x00814)
PVT monitor status register 2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pvtm_gpu_freq_cnt pd_gpu pvtm frequency count

GRF_IO_VSEL
Address: Operational Base + offset (0x00900)
IO voltage select

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:9	RO	0x0	reserved
8	RW	0x0	gpio1830_v18sel GPIO1830 IO domain 1.8V voltage selection 1'b0: 3.3V/2.5V 1'b1: 1.8V
7	RW	0x0	gpio30_v18sel GPIO30 IO domain 1.8V voltage selection 1'b0: 3.3V/2.5V 1'b1: 1.8V
6	RW	0x0	sdcard_v18sel SDCARD IO domain 1.8V voltage selection 1'b0: 3.3V/2.5V 1'b1: 1.8V
5	RW	0x0	audio_v18sel AUDIO IO domain 1.8V voltage selection 1'b0: 3.3V/2.5V 1'b1: 1.8V
4	RO	0x0	reserved
3	RW	0x0	wifi_v18sel WIFI IO domain 1.8V voltage selection 1'b0: 3.3V/2.5V 1'b1: 1.8V
2	RW	0x1	flash0_v18sel FLASH0 IO domain 1.8V voltage selection 1'b0: 3.3V/2.5V 1'b1: 1.8V
1	RW	0x0	dvp_v18sel DVP IO domain 1.8V voltage selection 1'b0: 3.3V/2.5V 1'b1: 1.8V
0	RO	0x0	reserved

GRF_SARADC_TESTBIT

Address: Operational Base + offset (0x00904)

SARADC Test bit register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:8	RO	0x0	reserved
7:0	RW	0x00	saradc_testbit SARADC test bit

GRF_FAST_BOOT_ADDR
Address: Operational Base + offset (0x00f80)
Fast boot address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	fast_boot_addr fast boot address control

5.4 PMUGRF Register Description

5.4.1 Register Summary

Name	Offset	Size	Reset Value	Description
PMUGRF_GPIO0A_IOMUX	0x00000	W	0x00000000	GPIO0A iomux control
PMUGRF_GPIO0B_IOMUX	0x00004	W	0x00000000	GPIO0B iomux control
PMUGRF_GPIO0C_IOMUX	0x00008	W	0x00000000	GPIO0B iomux control
PMUGRF_GPIO0D_IOMUX	0x0000c	W	0x00000000	GPIO0D iomux control
PMUGRF_GPIO0A_P	0x00010	W	0x0000aaaa	GPIO0A PU/PD control
PMUGRF_GPIO0B_P	0x00014	W	0x0000aaaa	GPIO0B PU/PD control
PMUGRF_GPIO0C_P	0x00018	W	0x0000aaaa	GPIO0C PU/PD control
PMUGRF_GPIO0D_P	0x0001c	W	0x0000aaaa	GPIO0D PU/PD control
PMUGRF_GPIO0A_E	0x00020	W	0x00005555	GPIO0A drive strength control
PMUGRF_GPIO0B_E	0x00024	W	0x00005555	GPIO0D drive strength control
PMUGRF_GPIO0C_E	0x00028	W	0x00000009	GPIO0C drive strength control
PMUGRF_GPIO0D_E	0x0002c	W	0x00000400	GPIO0D drive strength control
PMUGRF_GPIO0L_SR	0x00030	W	0x00000000	GPIO0 A/B SR control

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Name	Offset	Size	Reset Value	Description
PMUGRF_GPIO0H_SR	0x00034	W	0x0000000f	GPIO0C/D SR control
PMUGRF_SOC_CON0	0x00100	W	0x00000380	SoC control register 0
PMUGRF_PMUPVTM_CON0	0x00180	W	0x00000000	pmu pvtm configuration register
PMUGRF_PMUPVTM_CON1	0x00184	W	0x00000000	pmu pvtm configuration register
PMUGRF_PMUPVTM_STATUS0	0x00190	W	0x00000000	pmu pvtm status register
PMUGRF_PMUPVTM_STATUS1	0x00194	W	0x00000000	pmu pvtm status register
PMUGRF_OS_REG0	0x00200	W	0x00000000	os register
PMUGRF_OS_REG1	0x00204	W	0x00000000	os register
PMUGRF_OS_REG2	0x00208	W	0x00000000	os register
PMUGRF_OS_REG3	0x0020c	W	0x00000000	os register

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

5.4.2 Detail Register Description

PMUGRF_GPIO0A_IOMUX

Address: Operational Base + offset (0x00000)

GPIO0A iomux control

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio0a7_sel GPIO0A[7] iomux select 2'b00: gpio 2'b01: i2c0pmu_scl 2'b10: reserved 2'b11: reserved
13:12	RW	0x0	gpio0a6_sel GPIO0A[6] iomux select 2'b00: gpio 2'b01: i2c0pmu_sda 2'b10: reserved 2'b11: reserved

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Bit	Attr	Reset Value	Description
11:10	RW	0x0	gpio0a5_sel GPIO0A[5] iomux select 2'b00: gpio 2'b01: reserved 2'b10: reserved 2'b11: reserved
9:8	RW	0x0	gpio0a4_sel GPIO0A[4] iomux select 2'b00: gpio 2'b01: reserved 2'b10: reserved 2'b11: reserved
7:6	RW	0x0	gpio0a3_sel GPIO0A[3] iomux select 2'b00: gpio 2'b01: tsadc_int 2'b10: pmu_debug0 2'b11: reserved
5:4	RW	0x0	gpio0a2_sel GPIO0A[2] iomux select 2'b00: gpio 2'b01: reserved 2'b10: reserved 2'b11: reserved
3:2	RW	0x0	gpio0a1_sel GPIO0A[1] iomux select 2'b00: gpio 2'b01: reserved 2'b10: reserved 2'b11: reserved
1:0	RW	0x0	gpio0a0_sel GPIO0A[0] iomux select 2'b00: gpio 2'b01: global_pwroff 2'b10: pmic_sleep 2'b11: reserved

PMUGRF_GPIO0B_IOMUX
Address: Operational Base + offset (0x00004)
GPIO0B iomux control

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio0b7_sel GPIO0B[7] iomux select 2'b00: gpio 2'b01: lcdc_data11 2'b10: trace_data1 2'b11: jtag_tdi
13:12	RW	0x0	gpio0b6_sel GPIO0B[6] iomux select 2'b00: gpio 2'b01: lcdcdata10 2'b10: trace_data0 2'b11 :jtag_trstn
11:10	RW	0x0	gpio0b5_sel GPIO0B[5] iomux select 2'b00: gpio 2'b01: sc_detect 2'b10: spi2_csn0 2'b11: reserved
9:8	RW	0x0	gpio0b4_sel GPIO0B4] iomux select 2'b00: gpio 2'b01: sc_io 2'b10: spi2_clk 2'b11: gpujtag_tdo
7:6	RW	0x0	gpio0b3_sel GPIO0B[3] iomux select 2'b00: gpio 2'b01: sc_clk 2'b10: spi2_txd 2'b11: gpujtag_tdi
5:4	RW	0x0	gpio0b2_sel GPIO0B[2] iomux select 2'b00: gpio 2'b01: sc_rst 2'b10: spi2_rxd 2'b11: gpujtag_tms

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Bit	Attr	Reset Value	Description
3:2	RW	0x0	gpio0b1_sel GPIO0B[1] iomux select 2'b00: gpio 2'b01: sc_vcc33v 2'b10: i2c2sensor_scl 2'b11: gpujtag_trstn
1:0	RW	0x0	gpio0b0_sel GPIO0B[0] iomux select 2'b00: gpio 2'b01: test_clkout 2'b10: pwm_1 2'b11: pmu_debug_1

PMUGRF_GPIO0C_IOMUX
Address: Operational Base + offset (0x00008)
GPIO0B iomux control

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio0c7_sel GPIO0C[7] iomux select 2'b00: gpio 2'b01: lcdc_data19 2'b10: trace_data9 2'b11: uart1bb_rtsn
13:12	RW	0x0	gpio0c6_sel GPIO0C[6] iomux select 2'b00: gpio 2'b01: lcdc_data18 2'b10: trace_data8 2'b11: uart1bb_ctsn
11:10	RW	0x0	gpio0c5_sel GPIO0C[5] iomux select 2'b00: gpio 2'b01: lcdc_data17 2'b10: trace_data7 2'b11: uart1bb_sout

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Bit	Attr	Reset Value	Description
9:8	RW	0x0	gpio0c4_sel GPIO0C[4] iomux select 2'b00: gpio 2'b01: lcdc_data16 2'b10: trace_data6 2'b11: uart1bb_sin
7:6	RW	0x0	gpio0c3_sel GPIO0C[3] iomux select 2'b00: gpio 2'b01: lcdc_data15 2'b10: trace_data5 2'b11: mcujtag_tdo
5:4	RW	0x0	gpio0c2_sel GPIO0C[2] iomux select 2'b00: gpio 2'b01: lcdc_data14 2'b10: trace_data4 2'b11: mcujtag_tdi
3:2	RW	0x0	gpio0c1_sel GPIO0C[1] iomux select 2'b00: gpio 2'b01: lcdc_data13 2'b10: trace_data3 2'b11: mcujtag_trstn
1:0	RW	0x0	gpio0c0_sel GPIO0C[0] iomux select 2'b00: gpio 2'b01: lcdc_data12 2'b10: trace_data2 2'b11: jtag_tdo

PMUGRF_GPIO0D_IOMUX
Address: Operational Base + offset (0x0000c)
GPIO0D iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:14	RW	0x0	gpio0d7_sel GPIO0D[7] iomux select 2'b00: gpio 2'b01: lcdc_dclk 2'b10: trace_ctl 2'b11: pmu_debug5
13:12	RW	0x0	gpio0d6_sel GPIO0D[6] iomux select 2'b00: gpio 2'b01: lcdc_den 2'b10: trace_clk 2'b11: pmu_debug4
11:10	RW	0x0	gpio0d5_sel GPIO0D[5] iomux select 2'b00: gpio 2'b01: lcdc_vsync 2'b10: trace_data15 2'b11: pmu_debug3
9:8	RW	0x0	gpio0d4_sel GPIO0D[4] iomux select 2'b00: gpio 2'b01: lcdc_hsync 2'b10: trace_data14 2'b11: pmu_debug2
7:6	RW	0x0	gpio0d3_sel GPIO0D[3] iomux select 2'b00: gpio 2'b01: lcdc_data23 2'b10: trace_data13 2'b11: uart4exp_sin
5:4	RW	0x0	gpio0d2_sel GPIO0D[2] iomux select 2'b00: gpio 2'b01: lcdc_data22 2'b10: trace_data12 2'b11: uart4exp_sout
3:2	RW	0x0	gpio0d1_sel GPIO0D[1] iomux select 2'b00: gpio 2'b01: lcdc_data21 2'b10: trace_data11 2'b11: uart4exp_rtsn
1:0	RW	0x0	gpio0d0_sel GPIO0D[0] iomux select 2'b00: gpio 2'b01: lcdc_data20 2'b10: trace_data10 2'b11: uart4exp_ctsn

PMUGRF_GPIO0A_P

Address: Operational Base + offset (0x00010)

GPIO0A PU/PD control

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0xaaaa	gpio0a_p GPIO0A PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Noraml operaton); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull-down); 2'b11: Repeater(Bus keeper)

PMUGRF_GPIO0B_P
Address: Operational Base + offset (0x00014)
GPIO0B PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0xaaaa	gpio0b_p GPIO0B PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Noraml operaton); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull-down); 2'b11: Repeater(Bus keeper)

PMUGRF_GPIO0C_P
Address: Operational Base + offset (0x00018)
GPIO0C PU/PD control

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0xaaaa	gpio0c_p GPIO0C PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Noraml operaton); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull-down); 2'b11: Repeater(Bus keeper)

PMUGRF_GPIO0D_P
Address: Operational Base + offset (0x0001c)
GPIO0D PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0xaaaa	gpio0d_p GPIO0D PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Noraml operaton); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull-down); 2'b11: Repeater(Bus keeper)

PMUGRF_GPIO0A_E
Address: Operational Base + offset (0x00020)
GPIO0A drive strength control

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x5555	gpio0a_e GPIO0A drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

PMUGRF_GPIO0B_E

Address: Operational Base + offset (0x00024)

GPIO0D drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x5555	gpio0b_e GPIO0B drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

PMUGRF_GPIO0C_E

Address: Operational Base + offset (0x00028)

GPIO0C drive strength control

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0009	gpio0c_e GPIO0C drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

PMUGRF_GPIO0D_E
Address: Operational Base + offset (0x0002c)
GPIO0D drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0400	gpio0d_e GPIO0D drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

PMUGRF_GPIO0L_SR
Address: Operational Base + offset (0x00030)
GPIO0 A/B SR control

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:8	RW	0x00	gpio0b_sr GPIO0B slew rate control for each bit 1'b0: slow (half frequency) 1'b1: fast
7:0	RW	0x00	gpio0a_sr GPIO0A slew rate control for each bit 1'b0: slow (half frequency) 1'b1: fast

PMUGRF_GPIO0H_SR
Address: Operational Base + offset (0x00034)
GPIO0C/D SR control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:8	RW	0x00	gpio0d_sr GPIO0D slew rate control for each bit 1'b0: slow (half frequency) 1'b1: fast
7:0	RW	0x0f	gpio0c_sr GPIO0C slew rate control for each bit 1'b0: slow (half frequency) 1'b1: fast

PMUGRF_SOC_CON0

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Address: Operational Base + offset (0x00100)
SoC control register 0

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable bit0~15 write enable When bit 16=1, bit 0 can be written by software . When bit 16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software . When bit 17=0, bit 1 cannot be written by software; ..... When bit 31=1, bit 15 can be written by software . When bit 31=0, bit 15 cannot be written by software;
15:11	RO	0x0	reserved
10	RW	0x0	pd_pmu_rst_hold pd_pmu IP reset hold control, including gpio0, pmu_grf, secure_grf block. 1'b0: don't hold the reset 1'b1: keep the reset as de-asserted even if the reset source is asserted
9	RW	0x1	ddrphy_bufferen_io_en ddr phy bufferen controlled by IO 0: enable 1: disable
8	RW	0x1	ddrphy_bufferen_core_en ddr phy bufferen controlled by core 0: enable 1: disable
7	RW	0x1	pwm2_sel 0: 32K clock source 1: pwm2
6	RW	0x0	chip_32k_src chip 32K clock source select 0: from external 1: from internal, pvtm
5	RW	0x0	poc_vop_sel18 VOP IO domain 1.8V voltage selection 0:3.3V/2.5V 1:1.8V
4	RW	0x0	poc_pmu_sel18 PMU IO domain 1.8V voltage selection 0:3.3V/2.5V 1:1.8V
3	RW	0x0	gpio_0a7_smt gpio_0b7 schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.

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Bit	Attr	Reset Value	Description
2	RW	0x0	gpio_0a6_smt Field0002 Abstract gpio_0a6 schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.
1:0	RW	0x0	x32k_crnt_ctrl x32k pad drive strength control bit 00: 2mA 01: 4mA 10: 8mA 11: 12mA

PMUGRF_PMUPVTM_CON0
Address: Operational Base + offset (0x00180)
pmu pvtm configuration register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	pvtm_core_osc_en pd_core PVT monitor oscilator enable 1'b1: enable 1'b0: disable
0	RW	0x0	pvtm_core_start pd_core PVT monitor start control

PMUGRF_PMUPVTM_CON1
Address: Operational Base + offset (0x00184)
pmu pvtm configuration register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pvtm_core_cal_cnt pd_core pvtm calculator counter

PMUGRF_PMUPVTM_STATUS0
Address: Operational Base + offset (0x00190)
pmu pvtm status register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	pvtm_freq_done pvtm frequency calculate done status

PMUGRF_PMUPVTM_STATUS1
Address: Operational Base + offset (0x00194)
pmu pvtm status register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pvtm_freq_cnt pvtm frequency count

PMUGRF_OS_REG0
Address: Operational Base + offset (0x00200)
os register

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	os_reg0 os register

PMUGRF_OS_REG1
Address: Operational Base + offset (0x00204)
os register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	os_reg1 os register

PMUGRF_OS_REG2
Address: Operational Base + offset (0x00208)
os register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	os_reg2 os register

PMUGRF_OS_REG3
Address: Operational Base + offset (0x0020c)
os register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	os_reg3 os register

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Chapter 6 System Debug

6.1 Overview

The PX5 uses the Coresight-SOC Technology to support real-time debug access and trace for the multi-core. Software can access debug components and control of debug behavior through a DAP (Debug Access Port). A standard infrastructure is implemented for the capture and transmission of trace data, combination of multiple data streams by funneling together, and then output of data to a trace port. The cross-triggering components are also implemented in PX5 for debug component broadcast events to each other.

6.1.1 Features

- Access to debug features and on-chip AHB, APB, and JTAG buses through a JTAG or Single Wire Debug (SWD) interface.
- Merging of multiple trace sources into a single trace stream.
- Capture of trace streams on-chip or off-chip.
- Cross-triggering between different debug and trace components.
- Support Timestamp generation.

6.1.2 Debug components address map

The following table shows the debug components address in memory map, the DEBUG system base address is 0xffc00000:

Module	Offset	Size
DAP_ROM	0x00000	4KB
Trace Funnel	0x01000	4KB
CTI_TPIU	0x03000	4KB
Timestamp	0x04000	4KB
TPIU	0x05000	4KB
CLUSTERL_ROM	0x20000	4KB
CLUSTERL_DBG0	0x30000	4KB
CLUSTERL_PMU0	0x31000	4KB
CLUSTERL_DBG1	0x32000	4KB
CLUSTERL_PMU1	0x33000	4KB
CLUSTERL_DBG2	0x34000	4KB
CLUSTERL_PMU2	0x35000	4KB
CLUSTERL_DBG3	0x36000	4KB
CLUSTERL_PMU3	0x37000	4KB
CLUSTERL_CTI0	0x38000	4KB
CLUSTERL_CTI1	0x39000	4KB
CLUSTERL_CTI2	0x3A000	4KB
CLUSTERL_CTI3	0x3B000	4KB
CLUSTERL_ETM0	0x3C000	4KB
CLUSTERL_ETM1	0x3D000	4KB
CLUSTERL_ETM2	0x3E000	4KB
CLUSTERL_ETM3	0x3F000	4KB
CLUSTERB_ROM	0x40000	4KB
CLUSTERB_DBG0	0x50000	4KB
CLUSTERB_PMU0	0x51000	4KB
CLUSTERB_DBG1	0x52000	4KB
CLUSTERB_PMU1	0x53000	4KB
CLUSTERB_DBG2	0x54000	4KB
CLUSTERB_PMU2	0x55000	4KB
CLUSTERB_DBG3	0x56000	4KB
CLUSTERB_PMU3	0x57000	4KB
CLUSTERB_CTI0	0x58000	4KB
CLUSTERB_CTI1	0x59000	4KB

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CLUSTERB_CTI2	0x5A000	4KB
CLUSTERB_CTI3	0x5B000	4KB
CLUSTERB_ETM0	0x5C000	4KB
CLUSTERB_ETM1	0x5D000	4KB
CLUSTERB_ETM2	0x5E000	4KB
CLUSTERB_ETM3	0x5F000	4KB

6.2 Block Diagram

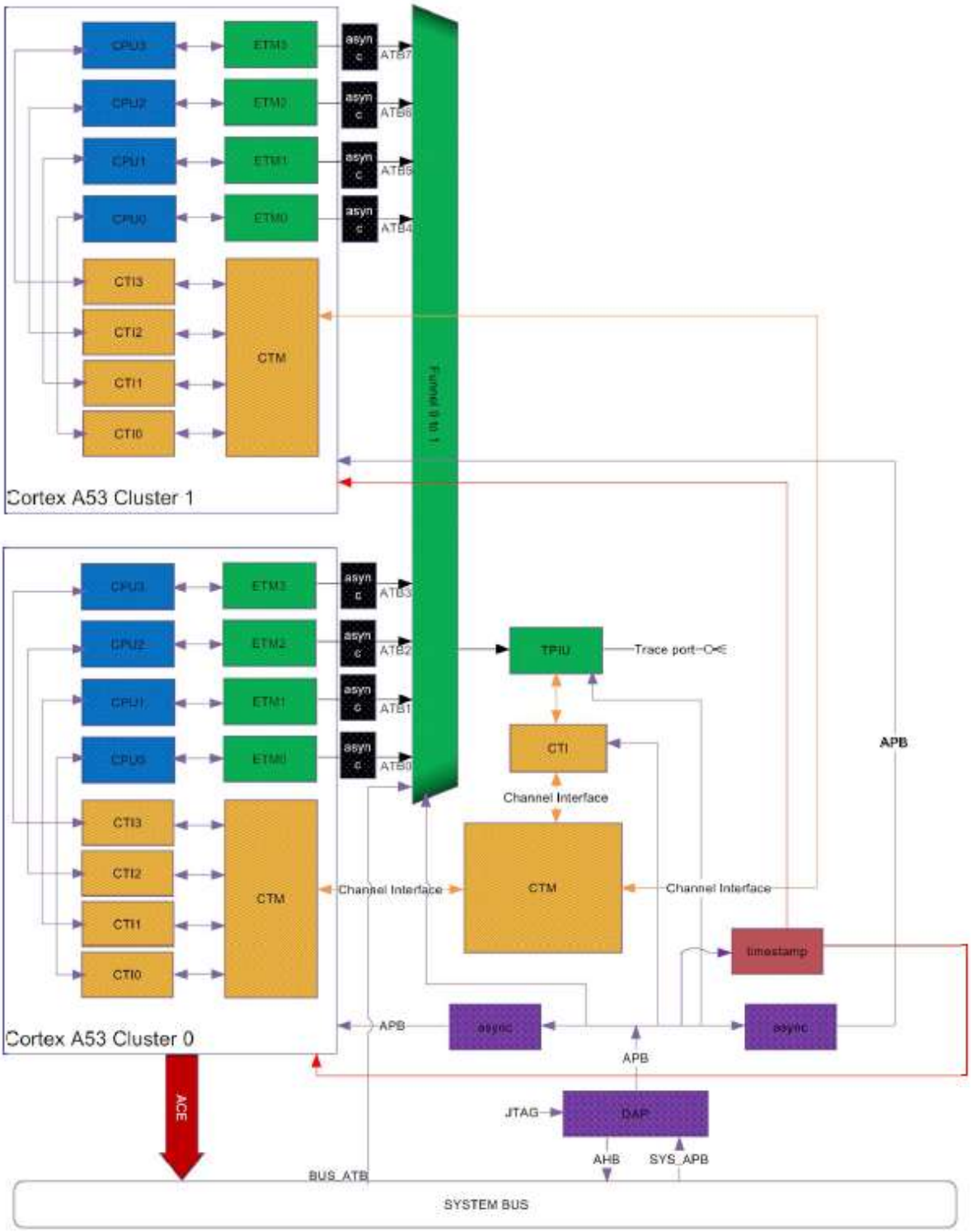

Fig. 6-1 PX5 Debug system structure

The PX5 DEBUG system is designed for a dual-cluster processor system, as shown in figure

above. The JTAG/SW controller can access all debug system components and system memory through DAP which is a package of APs(access port).
The data flow of trace can be linked together through a "trace funnel component" and be sent to IO by a TPIU component.
ECT (embedded cross trigger) is supported by CTM/CTIs in DEBUG system and CTM/CTIs in the Cortex-A53 clusters. And a timestamp generator in the DEBUG system can generate a 64bit binary count and be shared and synchronized to Cortex-A53 clusters by asynchronous bridges.

6.3 Function description

6.3.1 DAP (Debug Access Port)

DAP in PX5 is designed base on ARM coresight400 technology, which do not provide a "universal" DAP component as the previous version. So the real DAP in PX5 is consist of a group of coresight400 components which are integrated by ourselves.
The PX5 DAP architecture is illustrated in figure 6-2, and the be described in the following sections.

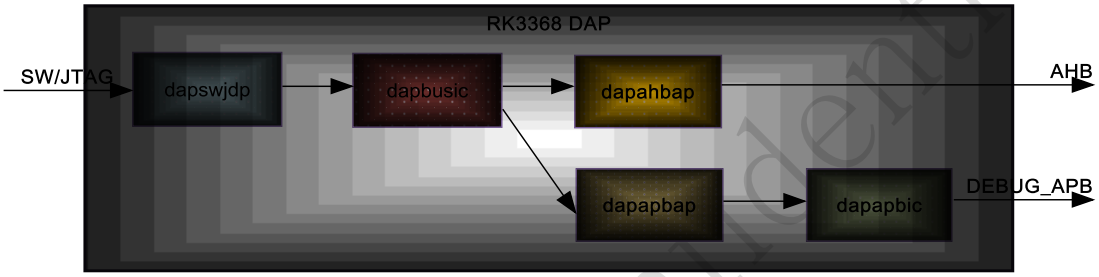

Fig. 6-2 PX5 Debug system structure

SWJ-DP:
The SWJ-DP is a combined JTAG-DP and SW-DP that enables you to connect either an SWD or JTAG probe to a target. It is the standard CoreSight debug port, and enables access either to the JTAG-DP or SW-DP blocks. To make efficient use of package pins, serial wire shares, or overlays, the JTAG pins use an auto-detect mechanism that switches between JTAG-DP and SW-DP depending on which probe is connected. A special sequence on the swdiotms pin switches between JTAG-DP and SW-DP.
The SWJ-DP consists of a wrapper around the JTAG-DP and SW-DP. It selects JTAG or SWD as the connection mechanism and enables either JTAG-DP or SW-DP as the interface to the DAP. Figure 6-3 shows the structure of the SWJ-DP.

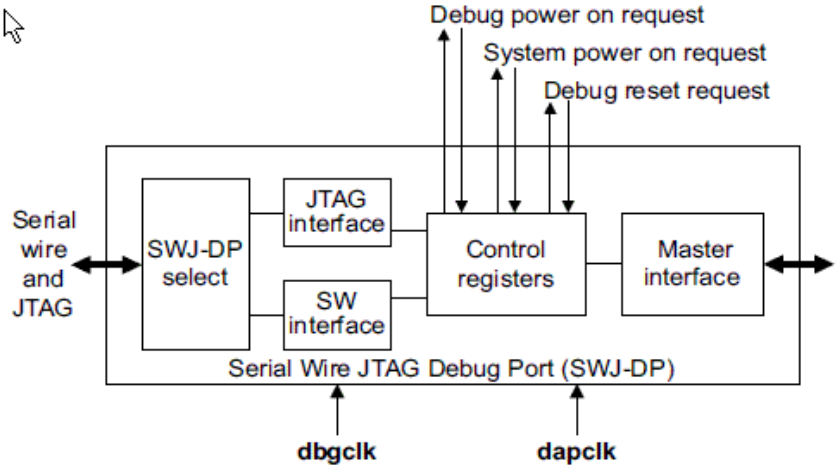

Fig. 6-3 PX5 SWJ-DP structure

When operating as a JTAG-DP this follows the JTAG-DP as defined in the ARM Debug interface Architecture Specification, ADIv5.0 to ADIv5.2. It also contains an explanation of its programmer model, capabilities, and features.
The JTAG-DP contains a debug port state machine that controls the JTAG-DP mode operation, including controlling the scan chain interface that provides the external physical interface to the JTAG-DP. It is based closely on the JTAG TAP State Machine. See IEEE std 1149.1-2001.

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When operating as an SW-DP Interface, this implementation is taken from the ARM debug Interface Architecture Specification, ADIv5.0 to ADIv5.2, and operates with a synchronous serial interface. This uses a single bidirectional data signal and a clock signal.
The SW-DP provides a low pin count, bidirectional serial connection to the DAP with a reference clock signal from synchronous operation.
Communications with the SW-DP use a 3-phase protocol:

- A host-to-target packet request
- A target-to-host acknowledge response
- A data transfer phase, if required. This can be target-to-host or host-to-target, depending on the request made in the first phase.

DAPBUSIC:
The DAPBUS interconnect is a combinational component for connecting the DP to the APs which are AHB-AP and APB-AP in PX5.
To address a particular AP, the DP uses the eight MSBs of its address bus, dapcaddrs[15:0]. The value driven on these address lines is determined by the APSEL[7:0] field in the AP Select register.
AHB-AP:
The AHB-AP implements the MEM-AP architecture to directly connect to an AHB-based memory system. Connection to other memory systems is possible through suitable bridging.
APB-AP:
The APB-AP implements the MEM-AP architecture to connect directly to an APB based system. This bus is normally dedicated to CoreSight and other debug components.
APBIC:
The APB interconnect connects one or more APB bus masters, for example an APB-AP and an APB interface driven by an on-chip processor. APB interconnects can be cascaded. The PX5 APBIC implements a ROM table at address 0x00000000, which identifies the locations of the CoreSight compoents accessed through it.

6.3.2 ETM (Embedded Trace Macro)

There are eight ETMs in the PX5 along with each Cortex-A53 processor separately in little and big clusters. The ETM trace unit is a module that performs real-time instruction flow tracing based on the Embedded Trace Macrocell(ETM), architecture ETMv4. ETM is a CoreSight component, and is an integral part of the ARM Real-time Debug solution, DS-5 Development Studio.
The figure 6-4 shows the main function blocks of the ETM trace unit.

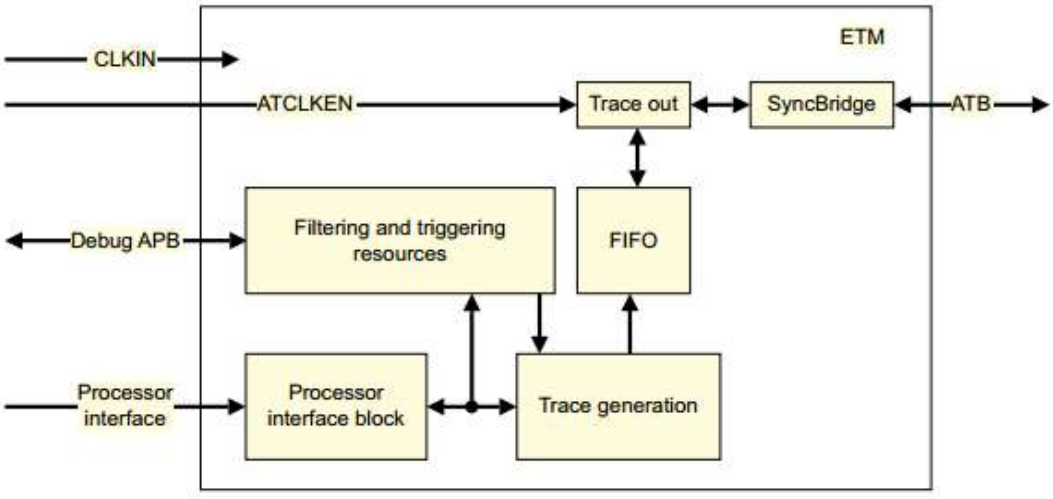

Fig. 6-4 PX5 SWJ-DP structure

6.3.3 Trace funnel

The ATB funnel component merges multiple ATB buses into a single ATB buses into a single ATB bus. If the optional APB interface is implemented, a debugger can also control the arbitration scheme and selectively enable the ATB slave interfaces for tracing.
The following figure shows ATB funnel block architecture.

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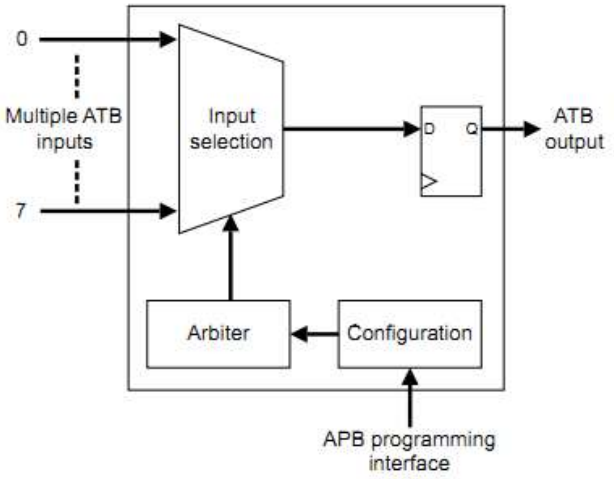

Fig. 6-5 Trace funnel architecture

6.3.4 TPIU

The TPIU acts as a bridge between the on-chip trace data, with separate IDs, to a data stream, encapsulating IDs where required, that is then captured by a Trace Port Analyzer (TPA). Fig. 5-1 shows the main blocks of the TPIU and the clock domains.

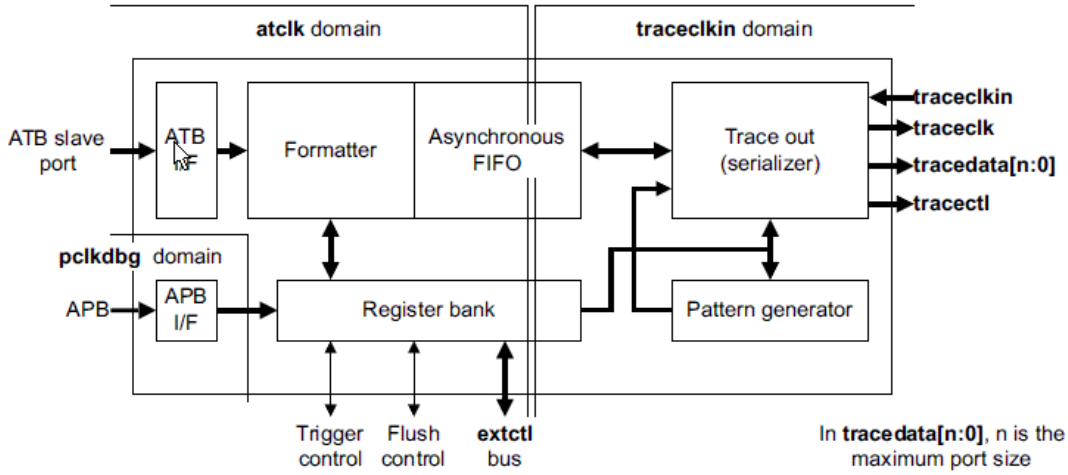

Fig. 6-6 PX5 TPIU structure

The TPIU contains the following components:

- Formatter
- Inserts source ID signals into the data packet stream so that trace data can be re-associated with its trace source. See TPIU formatter and FIFO.
- Asynchronous FIFO

Enables trace data to be driven out at a speed that is not dependent on the on-chip bus clock.

- Register bank

Contains the management, control and status registers for triggers, flushing behavior and external control.

- Trace out

The trace out block serializes formatted data before it goes off-chip.

- Pattern Generator

The pattern generator unit provides a simple set of defined bit sequences or patterns that can be output over the Trace Port and be detected by the TPA or other associated Trace Capture Device (TCD). The TCD can use these patterns to indicate if it is possible to increase or to decrease the trace port clock speed.

- ATB interface
- The TPIU accepts trace data from a trace source, either direct from a trace source or using a Trace Funnel.
- APB interface

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The APB interface is the programming interface for the TPIU.
Software must consider the following when programming the TPIU registers for trace capture:

- TPAs that are only capable of operation with tracectl must only use the formatter in either bypass or normal mode, not in continuous mode.
- ARM recommends that following a trigger event within a multi-trace source configuration, a flush is performed to ensure that all historical information related to the trigger is output.
- If flush on trigger event and stop on trigger event options are chosen then any data after the trigger is not captured by the TPA. When the TPIU is instructed to stop, it discards any subsequent trace data, including data returned by the flush. Select Stop on Flush completion instead.
- Although multiple flushes can be scheduled using flush on trigger event, flush on flushin, and manual flush, when one of these requests are made, it masks additional requests of the same type. This means repeated writing to the manual flush bit does not schedule multiple manual requests unless each is permitted to complete first.
- Unless multiple triggers are required, it is not advisable to set both trigger event and trigger on flush completion, if flush on trigger event is also enabled. In addition, if trigger on trigin is enabled with this configuration, it can also cause multiple trigger markers grom on trigger request.

6.3.5 ECT (CTI & CTM)

The ECT for CoreSight consists of a number of CTIs and CTMs connected together. This enables ARM/ETM subsystems to interact. That is cross trigger, with each other. The debug system enables debug support for multiple cores, together with cross triggering between the cores and their respective ETMs.
The main function of the ECT (CTI and CTM) is to pass debug events from one core to another. For example, the ECT can communicate debug state information from one core to another, so that program execution on both processors can be stopped at the same time if required.

- CTI (Cross Trigger Interface)
- The CTI combines and maps the trigger requests, and broadcasts them to all other interfaces on the ECT as channel events. When the CTI receives a channel event it maps this onto a trigger output. This enables subsystems to cross trigger with each other. The receiving and transmitting of triggers is performed through the trigger interface.
- CTM (Cross Trigger Matrix)
- This block controls the distribution of channel events. It provides Channel Interfaces (CIs) for connection to either CTIs or CTMs. This enables multiple CTIs to be linked together.

6.3.6 Timestamp

- The timestamp components generate and distribute a consistent time value to multiple processors in PX5.
- The timestamp system is shown in the following figure:

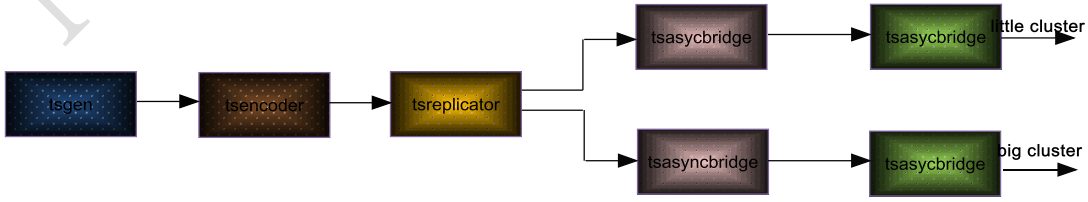

Fig. 6-7 PX5 Timestamp structure

The timestamp interconnect provides a mechanism for efficiently distributing a timestamp value across a potentially large system in a way that is cost-effective to implement. It has the following features:

- Uses a master timing reference with a fixed frequency of typically 10-50 MHz
- Time always counts forward
- Time available as a natural binary number to software
- Writeable and readable count value

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- Distributed synchronization of timestamp
- Time value presented as a 64-bit binary count.

The interconnect ensures that any components that uses the distributed timestamp are synchronized to the distributed count value with minimal skew while the timestamp interconnect is clocked.
The CoreSight timestamp generator can be used in one of the following ways:

- To generate the time reported by ARM processors that implements the Generic Timer specification. Software expects that this time does not count backwards, and so it is important that only secure software can change the timestamp value. The programmers model of the timestamp generator has been designed to enable non-secure software to read the timestamp value while only permitting secure software to change the timestamp value.
- To generate the time used to align traces and other debug information in the CoreSight system. The timestamp generator is controlled by debug software and connected to the debug APB interconnect. The read-only interface is not used.

6.4 Register description

For details of DEBUG system components (such as DAP, FUNNEL, TPIU, TIMESTAMP and CTI) registers, please reference ARM document "DDI0480F_soc_r3p1_trm" chapter 3.
For details of debug components inside Cortex-A53 cluster (such as ETM and CA53 CTI), please reference ARM document "DDI0500C_cortex_a53_r0p2_trm" chapter 11.

6.5 Interface description

6.5.1 DAP SWJ-DP interface

The following figure is the DAP SWJ-DP interface, the SWJ-DP is a combined JTAG-DP and SW-DP that enable you connect either a Serial Write Debug(SWJ) to JTAG probe to a target.

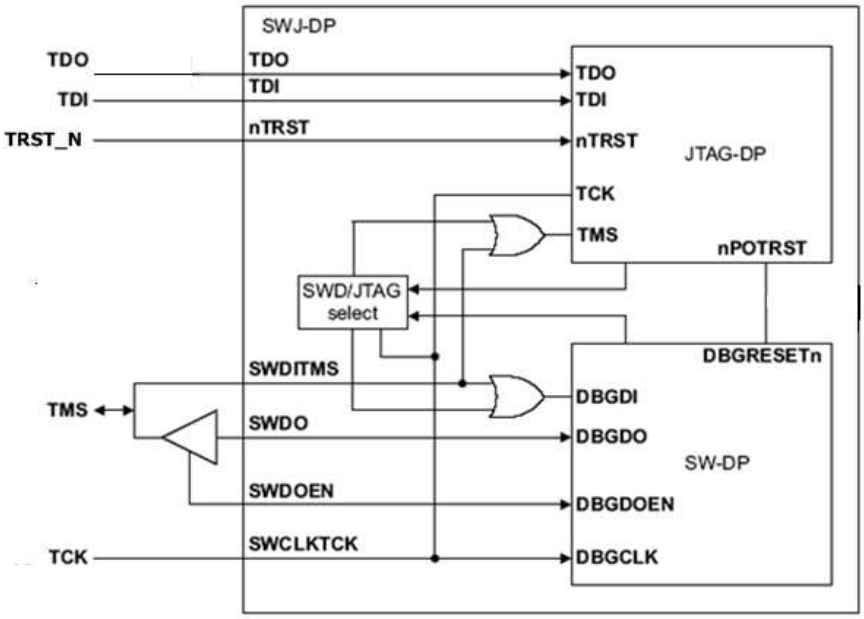

Fig. 6-8 DAP SWJ interface

Table 6-1 SWJ interface

Module Pin	Direction	PAD Name	IOMUX Setting
TRST_N	I	IO_LCDCdata10_TRACEdata0_JTAGtrstn_LCDCgpio0b6	GRF_GPIO0B_IOMUX[13:12]= 0x3
TCK	I	IO_SDMMC0data2_JTAGtck_SDCARDgpio2a7	GRF_GPIO2A_IOMUX[15:14]= 0x2 0x2aa
TDI	I	IO_LCDCdata11_TRACEdata1_JTAGtdi_LCDCgpio0b7	GRF_GPIO0B_IOMUX[15:14]= 0x3

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TMS	IO	IO_SDMMC0data3_JTAGtms_SDCARDgpio2b0	GRF_GPIO2B_IOMUX[1:0]= 0x2
TDO	O	IO_LCDCdata12_TRACEdata2_JTAGtdo_LCDCgpio0c0	GRF_GPIO0C_IOMUX[1:0]= 0x3

6.5.2 TPIU trace port interface

Table 6-2 TPIU interface

Module Pin	Direction	PAD Name	IOMUX Setting
trace_data[0]	O	IO_LCDCdata10_TRACEdata0_JTAGtrstn_LCDCgpio0b6	GRF_GPIO0B_IOMUX[13:12]= 0x2
trace_data[1]	O	IO_LCDCdata11_TRACEdata1_JTAGtdi_LCDCgpio0b7	GRF_GPIO0B_IOMUX[15:14]= 0x2
trace_data[2]	O	IO_LCDCdata12_TRACEdata2_JTAGtdo_LCDCgpio0c0	GRF_GPIO0C_IOMUX[1:0]= 0x2
trace_data[3]	O	IO_LCDCdata13_TRACEdata3_M3JTAGtrstn_LCDCgpio0c1	GRF_GPIO0C_IOMUX[3:2]= 0x2
trace_data[4]	O	IO_LCDCdata14_TRACEdata4_M3JTAGtdi_LCDCgpio0c2	GRF_GPIO0C_IOMUX[5:4]= 0x2
trace_data[5]	O	IO_LCDCdata15_TRACEdata5_M3JTAGtdo_LCDCgpio0c3	GRF_GPIO0C_IOMUX[7:6]= 0x2
trace_data[6]	O	IO_LCDCdata16_TRACEdata6_UART1BBsin_LCDCgpio0c4	GRF_GPIO0C_IOMUX[9:8]= 0x2
trace_data[7]	O	IO_LCDCdata17_TRACEdata7_UART1BBsout_LCDCgpio0c5	GRF_GPIO0C_IOMUX[11:10]= 0x2
trace_data[8]	O	IO_LCDCdata18_TRACEdata8_UART1BBctsn_LCDCgpio0c6	GRF_GPIO0C_IOMUX[13:12]= 0x2
trace_data[9]	O	IO_LCDCdata19_TRACEdata9_UART1BBrtsn_LCDCgpio0c7	GRF_GPIO0C_IOMUX[15:14]= 0x2
trace_data[10]	O	IO_LCDCdata20_TRACEdata10_UART4EXPctsn_LCDCgpio0d0	GRF_GPIO0D_IOMUX[1:0]= 0x2
trace_data[11]	O	IO_LCDCdata21_TRACEdata11_UART4EXPrtsn_LCDCgpio0d1	GRF_GPIO0D_IOMUX[3:2]= 0x2
trace_data[12]	O	IO_LCDCdata22_TRACEdata12_UART4EXPsout_LCDCgpio0d2	GRF_GPIO0D_IOMUX[5:4]= 0x2
trace_data[13]	O	IO_LCDCdata23_TRACEdata13_UART4EXPsin_LCDCgpio0d3	GRF_GPIO0D_IOMUX[7:6]= 0x2
trace_data[14]	O	IO_LCDChsync_TRACEdata14_PMUdebug2_LCDCgpio0d4	GRF_GPIO0D_IOMUX[9:8]= 0x2
trace_data[15]	O	IO_LCDCvsync_TRACEdata15_PMUdebug3_LCDCgpio0d5	GRF_GPIO0D_IOMUX[11:10]= 0x2
trace_clk	O	IO_LCDCden_TRACEclk_PMUdebug4_LCDCgpio0d6	GRF_GPIO0D_IOMUX[13:12]= 0x2
trace_ctl	O	IO_LCDCdclk_TRACEctl_PMUdebug5_LCDCgpio0d7	GRF_GPIO0D_IOMUX[15:14]= 0x2

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Chapter 7 System Security

7.1 Overview

The PX5 support the system security application requirement base on the TrustZone access control scheme. The following secure feature are supported

- Secure control of JTAG access
- Secure boot
- eight secure address space in DDR device, the start address and end address for each address scope is configurable, maximum 4GB secure address are supported
- 64K secure internal SRAM
- Many devices are configurable to act as secure or non-secure

7.2 Block Diagram

The following figure show the system security architecture. All the devices which support security access are demonstrated in this figure.

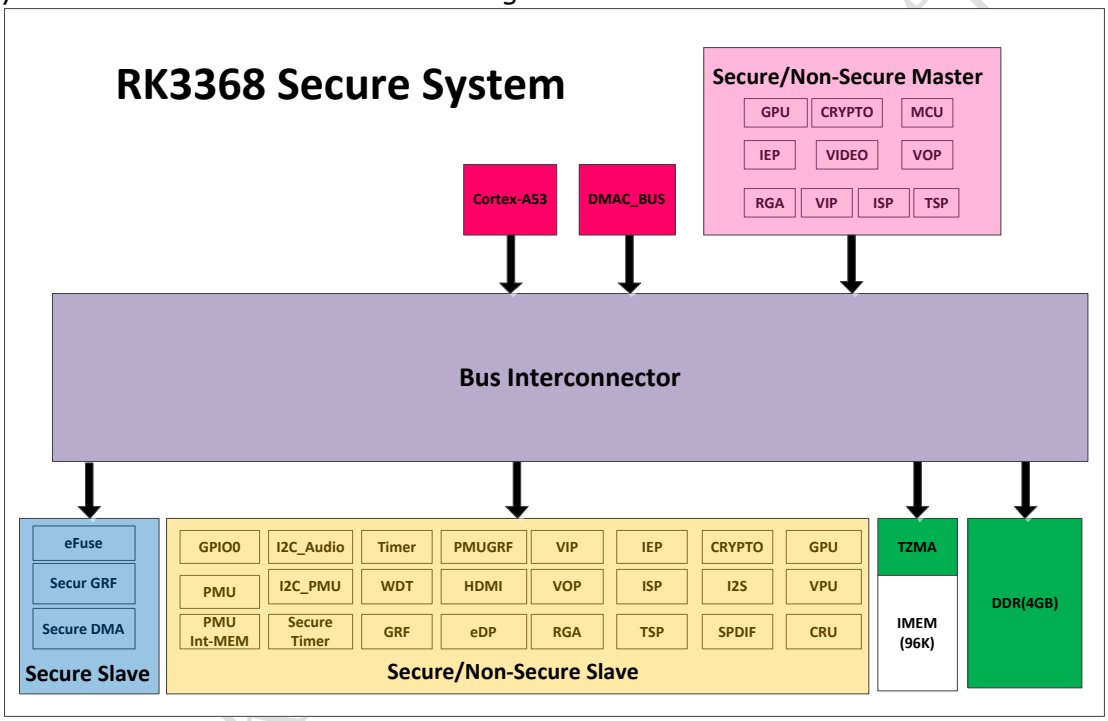

Fig. 7-1 PX5 security architecture

7.3 Function Description

7.3.1 Cortex-A53 Security Extension architecture

The processor implements the TrustZone Security Extensions architecture to facilitate the development of secure applications.
Security Extensions are based on these fundamental principles:

- The extensions define a class of core operation that you can switch between secure and non-secure state. Most code runs in non-secure state. Only trusted code runs in secure state
- The extensions define some memory as secure memory. When the core is in secure state, it can access secure memory
- Entry into secure state is strictly controlled
- Exit from secure state can only occur at programmed points
- Debug is strictly controlled
- The processor enters secure state on reset

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7.3.2 TZMA

The TZMA (TrustZone Memory Adapter) is a bridge between AXI bus and Embedded SRAM, which support the flexible secure access by controlling R0SIZE port.

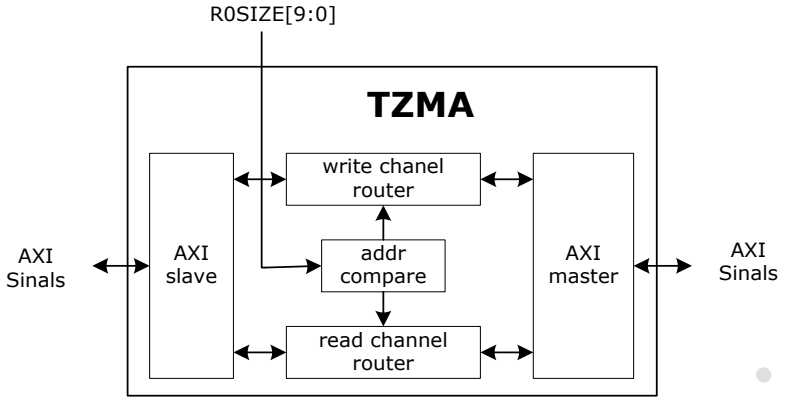

Fig. 7-2 TZMA block diagram

TZMA can support 0KB, 4KB, 8KB, 12KB… up to 96KB by 4KB step(the whole Embedded SRAM space) secure access by setting SGRF_SOC_CON4[9:0].

7.3.3 DMAC_BUS secure access

The DMAC_BUS is an AMBA compliant peripheral, which provides an AXI interface to perform the DMA transfer and two APB interfaces that control its operation. The DMAC_BUS implements TrustZone secure technology with one APB interface operating in the secure state and the other operating in the Non-secure state. For the detailed description for DMAC_BUS, please refer to Chapter 10.

The following diagram shows the interface of DMAC_BUS.

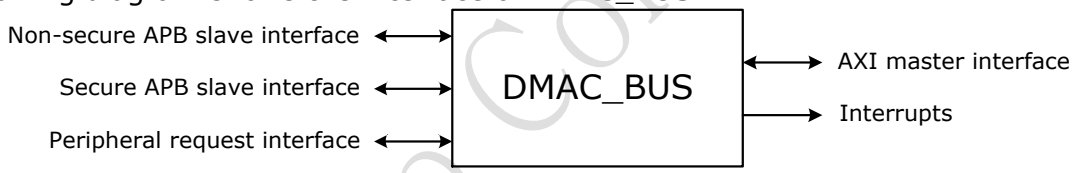

Fig. 7-3 DMAC_BUS interface

The DMAC_BUS support the secure feature in the following section:

- DMA manager thread
- DMA channel thread
- Event and interrupts
- Peripheral request interface

1. The security of DMA manager thread is controlled by input port boot_manager_ns.
0=assign DMA manager to the Secure state
1=assign DMA manager to the Non-secure state
2. The security of DMA channel thread is controlled by instruction DMAGO ns bit. If ns is present, DMA channel operation is in the non-secure state. Otherwise, the execution of the instruction depends on the security of the state of DMA manager:
DMA manager is in the secure state, DMA channel operates in the secure state
DMA manager is in the non-secure state, DMA abort
3. The security state of the event-interrupt source is controlled by the input port boot_irq_ns[x:0], if boot_irq_ns[x] is LOW, the DMAC_BUS assign event<x> or irq[x] to the secure state, otherwise the DMAC_BUS assign event<x> or irq[x] to the non-secure state.
4. The security state of peripheral request interface is controlled by the input port boot_irq_ns[x], if boot_irq_ns[x] is lOW, the DMAC_BUS assign peripheral interface x to secure state, otherwise the DMAC_BUS assign peripheral interface x to non-secure state.

7.3.4 Bus components security setting

The following table describes the security support on bus components which have the master interface or slave interface.

Table 7-1 bus components security setting

	secure	All transactions originating from this master interface are

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AXI master		flagged as secure transaction and can access both secure and non-secure component.
	non-secure	All transactions originating from this master interface are flagged as non-secure transactions and cannot access secure component
	per access	The AxPROTx signal determines the security setting of each transaction, and the slave that it can access
	per configure	Can be configured act as secure or non-secure by Secure GRF register
AHB-Lite master	secure	All transactions originating from this master interface are flagged as secure transaction and can access both secure and non-secure component.
	non-secure	All transactions originating from this master interface are flagged as non-secure transactions and cannot access secure component
	per configure	Can be configured act as secure or non-secure by Secure GRF register
AXI slave	secure	Only secure transactions can access this component
	non-secure	Both secure and non-secure transactions can access this components
	boot-secure	You can use software to configure whether it permits secure and non-secure transactions to access component. When boot up, this component only can be accessed by secure transactions.
	per configure	Can be configured act as secure or non-secure by Secure GRF register
AHB slave	secure	Only secure transactions can access this components
	non-secure	Both secure and non-secure transactions can access this components
	boot-secure	You can use software to configure whether it permit secure and non-secure transactions to access this components. When boot up, this component only can be accessed by secure transactions.
	per configure	Can be configured act as secure or non-secure by Secure GRF register
APB slave	secure	Only secure transactions can access this components
	non-secure	Both secure and non-secure transactions can access this components
	boot-secure	You can use software to configure whether it permit secure and non-secure transactions to access this components. When boot up, this component only can be accessed by secure transactions.
	per configure	Can be configured act as secure or non-secure by Secure GRF register

Notes: When a non-secure master tries to access a secure slave, an error response will be returned. In PX5, Cortex-A53 non-secure access a secure slave will cause a data-abort, and dmac_bus non-secure access a secure slave will cause an interrupt for access error.

7.3.5 PX5 secure device setting

Table 7-2 PX5 secure device setting

Cortex-A53	AXI master	Per access
secure_grf	APB slave	Secure
eFuse	APB slave	Secure
Embedded SRAM	AXI slave	Controlled by TZMA, the secure space can be set to 0, 4KB, 8KB, 12KB …up to 64KB
DMAC_BUS	AXI master	Per access

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	Secure APB slave	Secure
	Non-secure APB slave	Non-secure
GPU	AXI slave	per configure
	AXI master	per configure
Crypto	AHB slave	per configure
	AHB master	per configure
MCU	AHB master	per configure
IEP	AXI master	per configure
	AHB slave	per configure
VIDEO	AHB slave	per configure
	AXI master	per configure
VOP	AHB slave	per configure
	AXI master	per configure
RGA	AHB slave	per configure
	AXI master	per configure
VIP	AHB slave	per configure
	AXI master	per configure
ISP	AHB slave	per configure
	AXI master	per configure
TSP	AHB slave	per configure
	AHB master	per configure
GPIO0	APB slave	per configure
PMU	APB slave	per configure
PMU-Int-MEM	APB slave	per configure
I2C_Audio	APB slave	per configure
I2C_PMU	APB slave	per configure
Secure Timer	APB slave	per configure
Timer	APB slave	per configure
WDT	APB slave	per configure
GRF	APB slave	per configure
PMUGRF	APB slave	per configure
HDMI	AHB slave	per configure
eDP	AHB slave	per configure
I2S_8CH	AHB slave	per configure
I2S_2CH	AHB slave	per configure
SPDIF	AHB slave	per configure
CRU	APB slave	per configure
DDR	AXI slave	Support eight secure address scope, the start address and end address for each address scope is configurable

7.3.6 PX5 device secure input port setting

The following table lists all the secure input ports for the secure device. These secure input ports could be set by configuring SGRF registers.

Table 7-3 PX5 device secure input port setting

Input Port	Module	Function description
boot_manager_ns	DMAC_BUS	When the DMAC exits from reset, this signal controls the security state of the DMAmanager thread: 1'b0: assigns DMA manager to the secure state 1'b1: assigns DMA manager to the non-secure state
boot_perih_ns[2:0]	DMAC_BUS	Controls the security state of a peripheral request interface, when the DMAC exits

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		from reset: boot_periph_ns[x] is LOW The DMAC assigns peripheral request interface x to the secure state boot_periph_ns[x] is HIGH The DMAC assigns peripheral request interface x to the non-secure state.
boot_irq_ns[7:0]	DMAC_BUS	Controls the security state of an event-interrupt resource, when the DMAC exits fromreset: boot_irq_ns[x] is LOW The DMAC assigns event<x> or irq[x] to the secure state. boot_irq_ns[x] is HIGH The DMAC assigns event<x> or irq[x] to the non-secure state.

7.3.7 Secure JTAG

The JTAG access is controlled by system secure control register, the following table show the detailed information about the secure control to JTAG.

JTAG Type	Register
CA53 JTAG	SGRF_SOC_CON0[3] SGRF_SOC_CON0[4] SGRF_SOC_CON0[5] SGRF_SOC_CON0[6]
MCU_JTAG	SGRF_SOC_CON5[14]
GPU_JTAG	SGRF_SOC_CON5[11]

7.4 Application Notes

Secure software conception

The basis of the security extensions model is that the computing environment splits into two isolated states, the secure state and the non-secure state, with no leakage of secure data to the non-secure state. Software secure monitor code, running in the monitor mode, links the two states and acts as a gatekeeper to manage program flow. The system can have both secure and non-secure peripherals that are suitable to secure and non-secure device driver control. Following figure shows the relationship between the secure and non-secure states. The operating system (OS) splits into the secure OS, that includes the secure kernel, and the non-secure OS, that includes the non-secure kernel.

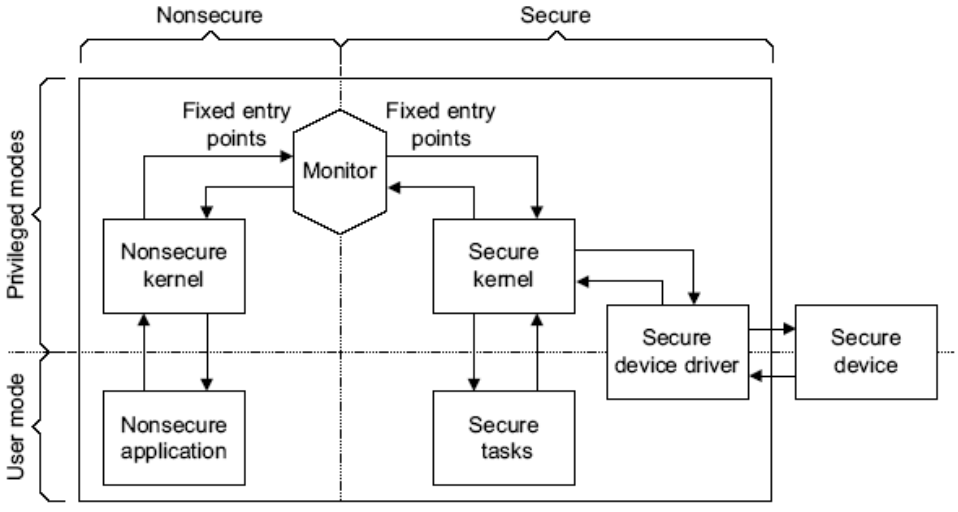

Fig. 7-4 Software Diagram of Secure and Non-secure

In normal non-secure operation, the OS runs tasks in the usual way. When a user process

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requires secure execution it makes a request to the secure kernel, that operates in privileged mode. This then calls the secure monitor to transfer execution to the secure state.
This approach to secure systems means that the platform OS that works in the non-secure state, has only a few fixed entry points into the secure state through the secure monitor. The trusted code base for the secure state, that includes the secure kernel and secure device drivers, is small and therefore much easier to maintain and verify.

Secure/Non-secure memory space for Embedded SRAM

The following figure gives an example of embedded SRAM secure/non-secure memory space setting. The software configure 4KB secure space. The bottom 4KB space will act as secure space and the other 60K space will be non-secure space.

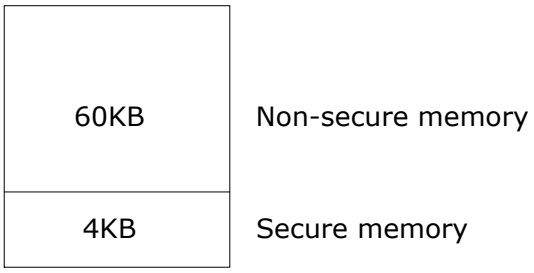

Fig. 7-5 Embedded SRAM secure memory space setting

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Chapter 8 Cortex-A53

8.1 Overview

The PX5 has two Cortex-A53 clusters, which are named little cluster and big cluster. Each cluster contains quad Cortex-A53 processor, which is a mid-range, low-power processor that implements the ARMv8-A architecture.
The Cortex-A53[①] processor includes following features:

- Full implementation of the ARMv8-A architecture instruction set
- Support for both AArch32 and AArch64 Execution status.
- Support for all exception levels, EL0, EL1, EL2, and EL3, in each execution states.
- Support A32 instruction set, previously called the ARM instruction set.
- Support T32 instruction set, previously called the Thumb instruction set.
- Support A63 instruction set.
- In-order pipeline with symmetric dual-issue of most instructions.
- Harvard Level 1(L1) memory system with a Memory Management Unit (MMU).
- Level 2(L2) memory system providing cluster memory coherency, with L2 cache.
- Support advanced SIMD and Floating-point Extension for integer and floating-point vector operations.
- Support ARMv8 Cryptography Extensions.
- Support AMBA 4 ACE bus architecture.

Note: The Revision of CA53 in PX5 is r0p3-00eac0

The configuration details of little cluster and big cluster are shown in following tables

Table 8-1 Big Cluster Configuration

Number of CPU	4
L1 I cache size	32K
L1 D cache size	32K
L2 cache size	512K
L2 data RAM output latency	3 cycles
L2 data RAM input latency	2 cycles
CPU cache protection	No
SCU L2 cache protection	No
BUS master interface	ACE
NEON and floating point support	Yes
Cryptography extension	Yes

Table 8-2 Little Cluster Configuration

Number of CPU	4
L1 I cache size	32K
L1 D cache size	32K
L2 cache size	512K
L2 data RAM output latency	3 cycles
L2 data RAM input latency	2 cycles
CPU cache protection	No
SCU L2 cache protection	No
BUS master interface	ACE
NEON and floating point support	Yes

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Cryptography extension	Yes

8.2 Block Diagram

The Cortex-A53 sub system is shown in Figure 8-1. As illustrated, little cluster and big cluster connect to a CCI400 coherency bus through asynchronous bridges which can handle with CDC (clock domain crossing) issue.
An event asynchronous bridge is placed between 2 clusters, which can generate event trigger signal to the other cluster when CPU executes a SEV instruction.
The little cluster and big cluster are sharing a same system counter, which can run under a constant frequency clock. That can guarantee the PMU inside Cortex-A53 little cluster and big cluster user the same reference clock and so on.
The Cortex-A53 sub system in PX5 supports cache coherency, which means little cluster and big cluster can snoop sharable cache data with each other. This feature can be enabled by configure registers of CCI400. When it comes up from system reset, cache coherency is DISABLED.

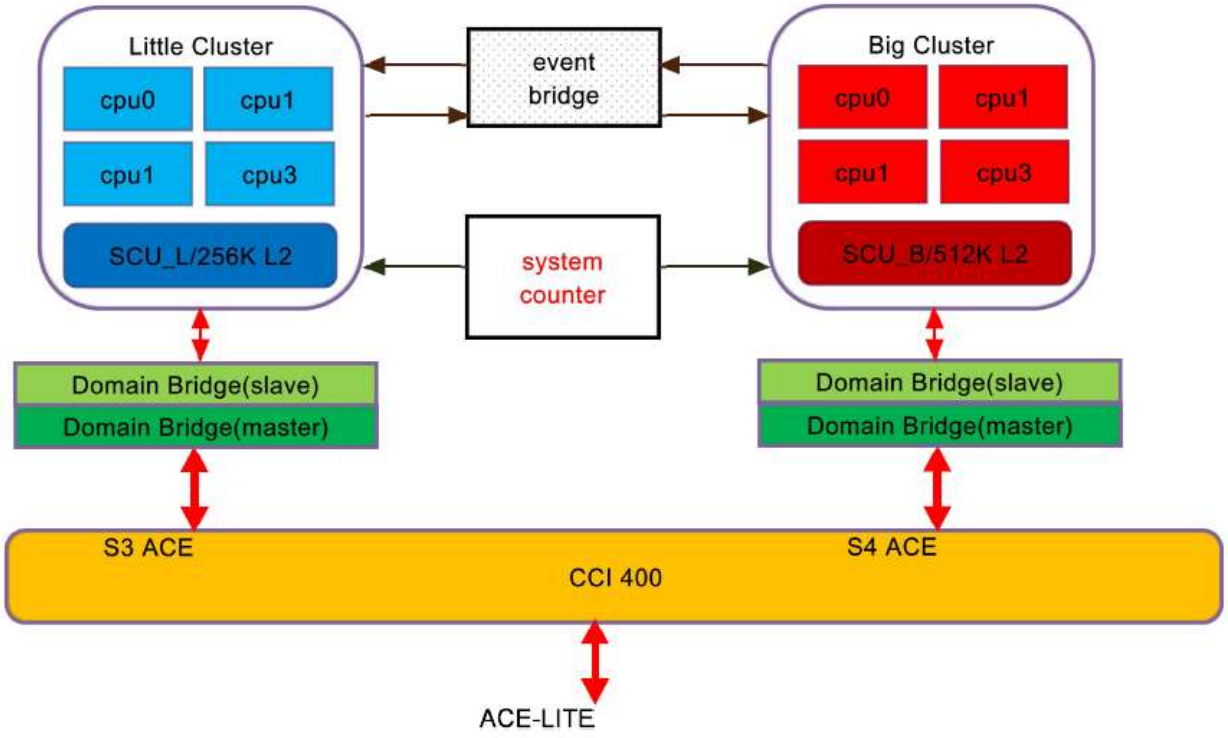

Fig. 8-1 Cortex-A53 Block Diagram

8.3 Function Description

Please refer to the document cortex_a53_r0p2_trm.pdf for the detail function description.

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Chapter 9 Micro Control Unit (MCU)

9.1 Overview

The MCU processor is a low-power processor that features low gate count, low interrupt latency, and low-cost debug. It is intended for deeply embedded applications that require fast interrupt response features.
The key features of the MCU include:

- High Performance
- Many instructions, including multiply, are single cycle.
- Separate data and instruction buses allow simultaneous data and instruction accesses to be performed.
- Low Power Consumption
- It has power-saving mode support. The processor can enter sleep mode using instructions. The design has separated clocks for essential blocks, so clocking circuits for most parts of the processor can be stopped during sleep.

9.2 Block Diagram

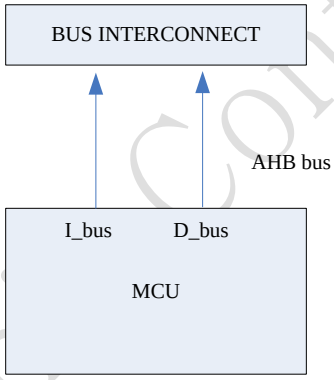

Fig. 9-1 MCU Connection diagram

9.3 Function description

As a co-processor, the MCU can be used to implement access to peripheral or other devices. Also, when CPU is power down, the MCU would remain working and decrease power consumption.

9.4 Interface Description

Table 9-1 MCU jtag interface description

Module Pin	Direction	Pad Name	IOMUX Setting
mcujtag_tck	I/O	IO_SDMMC0clkout_MCUJTAGtck_SDCARDgpio2b1	GRF_GPIO2B_IOMUX[3:2]=10
mcujtag_trstn	I/O	IO_LCDCdata13_TRACEdata3_MCUJTAGtrstn_LCDCgpio0c1	PMUGRF_GPIO0C_IOMUX[3:2]=11
mcujtag_tms	O	IO_SDMMC0cmd_MCUJTAGtms_SDCARDgpio2b2	GRF_GPIO2B_IOMUX[5: 4]=10
mcujtag_tdi	I	IO_LCDCdata14_TRACEdata4_MCUJTAGtdi_LCDCgpio0c2	PMUGRF_GPIO0C_IOMUX[5:4]=11

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mcujtag_tdo	O	IO_LCDCdata15_TRACEdata5_MCUJTAGtdo_LCDCgpio0c3	GRF_GPIO0B_IOMUX[7:6]=11

9.5 Application Notes

9.5.1 Clock and Reset control

We can conifig CRU_CLKSEL12_CON[7] to select MCU clock source between CPLL and GPLL, and set CRU_CLKSEL12_CON[4:0] to divide clock source from 1:1 to 1:32.
We can config CRU_CLKGATE13_CON[11] to gate the mcu hclk, and config CRU_CLKGATE13_CON[12] to gate the mcu fclk clock.
The MCU has two reset signals: sysresetn and poresetn, and we can config CRU_SOFTRST1_CON[13] and CRU_SOFTRST1_CON[12] to reset or unreset MCU.

9.5.2 MCU Interrupt

The MCU can handle up to 240 external interrupts.

9.5.3 Memory and Address Map

The MCU processor is addressed by different memory map regions.

Address space 0x00000000 ~ 0x20000000 is for MCU CODE, and we can config GRF_SOC_CON14[3:0] and GRF_SOC_CON10[15:0] to map the mcu_code_start_addr[31:12].

Address space 0x20000000 ~ 0x40000000 is for MCU SRAM, and we can config GRF_SOC_CON14[7:4] and GRF_SOC_CON11[15:0] to map the mcu_sram_start_addr[31:12].

Address space 0x40000000 ~ 0x60000000 is for MCU PERI, and it is by default mapped to system peri start address: 0xff000000.

Address space 0x60000000 ~ 0xa0000000 is for MCU EXSRAM, and we can config GRF_SOC_CON14[11:8] and GRF_SOC_CON12[15:0] to map the mcu_exsram_start_addr[31:12].

Address space 0xa0000000 ~ 0xe0000000 is for MCU EXPERI, and we can config GRF_SOC_CON14[15:12] and GRF_SOC_CON13[15:0] to map the mcu_experi_start_addr[31:12].

For example, if we put MCU boot code at 0x00100000 in DDR, the recommended configurations are as below:
GRF_SOC_CON10 = 0xffff0100;
GRF_SOC_CON11 = 0xffff0000;
GRF_SOC_CON12 = 0xffff0000;
GRF_SOC_CON13 = 0xffff0000;
GRF_SOC_CON14 = 0xffff0000;
CRU_SOFTRST1_CON = 0x30000000;

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Chapter 10 Interconnect

10.1 Overview

The chip-level interconnect consists of main interconnect, peri interconnects and cci400. It enables communication among the modules and subsystems in the device.
The main interconnect supports the following features:

- Cross-bar exchange network
- A special internal slave for accessing the configuration register
- Little-endian platform
- Embedded memory scheduler for ddr transaction generation
- QoS management for optimizing the transaction flow
- Transaction statistics for analyzing the transaction flow
- Security protection mechanism to compatible with the TrustZone technology
- The peri interconnect belong to peripheral system which is responsible for peripheral devices control such as usb device, flash device, uart, spi etc.
- CCI400 is used for cache coherency management between Cortex-A53 big cluster and little cluster

10.2 Block Diagram

The interconnect comprises with:

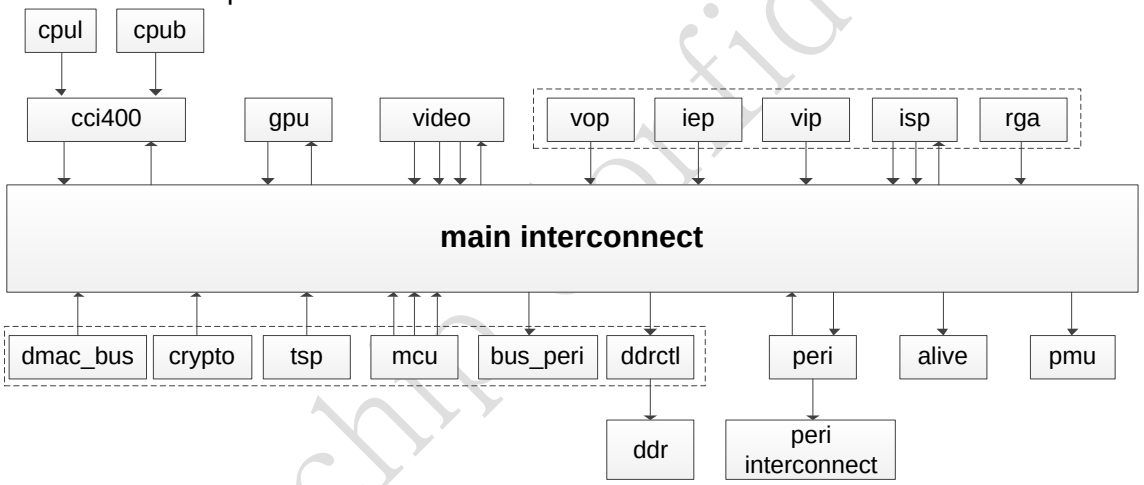

Fig. 10-1 Secury System Block Diagram

10.3 Function Description(main interconnect)

10.3.1 Master & Slave

The main interconnect is connected with all the related IPs of the system, the interface between the IP and the interconnect is called as NIU(native interface unit). All the connected NIU are list as bellowing:

Table 10-1 Master NIU

Master NIU	Description
cpup	Cortex-A53 P port, access to any peripheral device
cci	Cortex-A53 M port, access to main memory
dbg	Jtag Debug master , access to any peripheral
bus_dmac	DMAC in pd_bus power domain
crypto	Crypto , used in trustzone technolgy

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tsp	TSP in pd_bus power domain
mcu	MCU in pd_bus power domain
gpu	gpu of pd_gpu power domain
video	Video codec of pd_video power domain
vop	VOP of pd_vio power domain
iep	IEP of pd_vio power domain
vip	VIP of pd_vio power domain
isp	ISP of pd_vio power domain
rga	RGA of pd_vio power domain
peri	peri interconnects of pd_peri power domain

Table 10-2 slave NIU

Slave NIU	Description
msch	memory scheduler in pd_bus power domain
bus_ahb	ahb slave in pd_bus power domain
imem	internal memory in pd_bus power domain
bus_apb	apb slave in pd_bus power domain
pmu_apb	pmu apb slave in pd_pmu power domain
alive_apb	alive apb slave in pd_alive power domain
vio_ahb	vio ahb slave in pd_vio power domain
video_ahb	video ahb slave in pd_video power domain
gpu_axi	gpu axi slave in pd_gpu power domain
service_bus	service module inside the interconnect in pd_bus power domain, used as register configuration
service_core	service module in pd_core power domain
service_dmac	service module in pd_bus power domain
service_gpu	service module in pd_gpu power domain
service_video	service module in pd_video power domain
service_peri	service module in pd_peri power domain
service_vio	service module in pd_vio power domain

10.3.2 Clock & Power domain

The interconnect is divided to several clock domain and power domain. Each domain contains different IPs.
The clock/power domain and IP combination is list as bellowing:

Table 10-3 Clock and Power domain

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Clock domain	Clock	Power domain	IP
alive_clk_dm	alive_pclk	alive_pwr	CRU/PLL/GRF/GPIO/TIMER/WDT
bus_clk_dm	bus_aclk	bus_pwr	IMEM
	bus_hclk		I2S/SPDIF/CRYPTO/TSP
	bus_pclk		PWM/I2C/UART/SCR/MAILBOX/DMAC/PCTL
core_clk_dm	cci_aclk cpup_aclk	core_pwr	Cortex-A53
gpu_clk_dm	gpu_aclk	gpu_pwr	IMG G6110
hevc_vcodec_aclk_dm	hevc_aclk	video_pwr	HEVC_VCODEC
hevc_vcodec_hclk_dm	hevc_hclk		
msch_clk_dm	msch_clk	bus_pwr	Memory scheduler
peri_clk_dm	peri_aclk	peri_pwr	Peri-PL301
pmu_clk_dm	pmu_pclk	pmu_pwr	PMU/PMU_IMEM/SGRF/GPIO
vio0_clk_dm	vio0_aclk	vio_pwr	VOP/IEP/VIP
vio1_clk_dm	vio1_aclk		RGA/HDCP2.2
vio_isp_clk_dm	isp_aclk		ISP
vio_hclk_dm	vio_hclk		

10.3.3 QoS management

The interconnect offers 4 modes of qos management:

- None, QoSGenerator is disabled, and priority information are stuck at 0.
- Fixed, QoSGenerator drives applies a fixed urgency to read transactions, and a (possibly different) urgency to write transactions.
- Limiter, QoSGenerator behaves as in fixed mode, but limits the traffic bandwidth coming from that socket, possibly stalling requests if the initiator attempts to exceed its budget.
- Regulator, QoSGenerator promotes are demotes hurry, depending the bandwidth obtained by the initiator is below or beyond a bandwidth budget. As transactions exceeding the bandwidth limit are sent (even though demoted), the regulator mode may be considered as a softer version of the limiter mode.

Limiter Behavior

When configured in bandwidth limiter, the unit uses a 23 bit counter to measure the average bandwidth. This counter has a 1/256 byte resolution and works as follows:

- Adds the number of byte rounded up to 16 (1 -> 16) and then multiplied by 256 to the current value, each time a request is sent.
- Subtracts the Bandwidth register value every cycle. If the Counter becomes negative, force it to 0.
- If the Counter value is greater than the Saturation register value multiplied by 16*256, any incoming request is stalled until this condition disappears. Note that the Counter cannot wrap-around because the maximum value it can reach is: SaturationMax*16*256 + BurstMax*256 = 1023*4K+4K*256 = 5116K or 223 = 8192K.

The following example will show the Counter behavior: 32 byte bursts, F=400MHz, BW=200MB/s, T=0.32us. The Bandwidth register will be set to 256*200/400 = 128, and the Saturation register to 128*0.32*400/4096 = 4 (which corresponds to 64 bytes).

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Regulator Behavior
When configured in bandwidth regulator, the unit uses a 23 bit counter to measure the average bandwidth. This counter has a 1/256 byte resolution and works as follows:

- Adds the number of byte rounded up to 16 and then multiplied by 256 to the current value, each time a response is received. If the result is greater than the Saturation register value multiplied by 16*256, saturation to this value is applied.
- Subtracts the Bandwidth register value every cycle. If the Counter becomes negative, force it to 0.
- If the Counter value is less than or equal to the Saturation register value multiplied by 16*256/2, the SocketMst Hurry signal will be set to the HurryHigh register, and HurryLow otherwise. Note that Urgency and Press will be also set to the same value.

The following example will show the Counter behavior: 1Kbyte bursts, F=500MHz, BW=2GB/s,T=2.048us. The Bandwidth register will be set to 256*2000/500 = 1024, and the Saturation register to 1024*2.048*500/4096 = 256 (which corresponds to 4 Kbytes).

QoS Generator Programming
Bandwidth: This log2(socket.wData/8)+8 bits register defines the bandwidth in 1/256th byte per cycle unit. This allows a 2 MByte/s resolution at 500MHz. When thebandwidth is given in MByte/s, the value of this register will
be equal to 256*BWMB/s / FMHz.

Saturation: This 10 bits register defines the number of byte used for bandwidth measurement. It is expressed in 16bytes unit (up to 16 Kbyte). Usually the integration window is given in us or in cycle: the value of this register will beequal to Bandwidh*Tus*FMHz/(256*16)or Bandwidh*Ncycle/(256*16).

10.3.4 Memory Scheduler

Memory scheduler is a special NIU of the interconnect, it mainly deal with the transaction inside the interconnect and convert it to the transaction which the ddr protocol controller can recognize.
Following table shows the software configurable setting for the memory scheduler when the system connected to different size of ddr device.
The DDRCONF[3:0] is a configurable register inside the interconnect.
R: indicates Row bits
B: indicates Bank bits
C: indicates Column bits
D: indicates Chip selects bits

Table 10-4 DDR configuration

DDR CONF[3:0]	
0	C RRRD RRRR RRRR RRRR RBBB CCCC CCCC ----
1	C RRDR RRRR RRRR RRRR RBBB CCCC CCCC ----
2	C RDRR RRRR RRRR RRRR RBBB CCCC CCCC ----
3	C DRRR RRRR RRRR RRRR RBBB CCCC CCCC ----
4	R RDRR RRRR RRRR RRRR BBBC CCCC CCCC ----
5	R DRRR RRRR RRRR RRRR BBBC CCCC CCCC ----
6	D RRRR RRRR RRRR RRRR BBBC CCCC CCCC ----
7	C CRRR DRRR RRRR RRRR RRBB BCCC CCCC ----
8	C CRRD RRRR RRRR RRRR RRBB BCCC CCCC ----
9	C CRDR RRRR RRRR RRRR RRBB BCCC CCCC ----

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10	C CDRR RRRR RRRR RRRR RRBB BCCC CCCC ----
11	C RDBB BRRR RRRR RRRR RRRR CCCC CCCC ----
12	R RDBB BRRR RRRR RRRR RRRC CCCC CCCC ----
13	C RRDB BBRR RRRR RRRR RRRR CCCC CCCC ----
14	C RRRR RRRR RRRR RRRR DBBB CCCC CCCC ----
15	C CRRR RRRR RRRR RRRR RDBB BCCC CCCC ----

10.4 Register Description(main interconnect)

10.4.1 Internal Address Mapping

Table 10-5 Service module

service Module	Base address
dmac_qos	0xffa8_0000
crypto	0xffa8_0080
cpup_qos	0xffa9_0000
cci_r_qos	0xffaa_0000
cci_w_qos	0xffaa_0080
peri_qos	0xffab_0000
msch_qos	0xffac_0000
iep	0xffad_0000
isp_r0	0xffad_0080
isp_r1	0xffad_0100
isp_w0	0xffad_0180
isp_w1	0xffad_0200
vip	0xffad_0280
vop	0xffad_0300
rga_r	0xffad_0380
rga_w	0xffad_0400
hevc_r	0xffae_0000
vpu_r	0xffae_0080
vpu_w	0xffae_0100

10.4.2 Registers Summary

service_bus: memory scheduler register summary

Name	Offset	Size	Reset Value	Description
coreid	0x0000	W	0x624F8202	core id of memory scheduler 0
revisionid	0x0004	W	0x0126f200	revisionid
ddrconf	0x0008	W	0x00000000	ddr configuration register
ddrtiming	0x000c	W	0x2475931c	ddr timing register
ddrmode	0x0010	W	0x00000000	ddr mode register
readlatency	0x0014	W	0x00000032	read latency register
activate	0x0038	W	0x00000400	activate register
devtodev	0x003c	W	0x00000000	devtodev register

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service_core: cpup_qos register summary

Name	Offset	Size	Reset Value	Description
priority	0x0008	W	0x00000101	priority register
mode	0x000c	W	0x00000003	qos mode register
bandwidth	0x0010	W	0x00000005	qos bandwidth register
saturation	0x0014	W	0x00000040	qos saturation register
extcontrol	0x0018	W	0x00000000	qos extcontrol register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*
other master's qos register is same with service_core:cpup_qos

10.4.3 Detail Register Description

service_bus: memory scheduler coreid
Address: Operational Base + offset (0x0000)

Bit	Attr	Reset Value	Description
31:0	RO	0x624F8202	msch0' core id

service_bus: memory scheduler revisionid
Address: Operational Base + offset (0x0004)

Bit	Attr	Reset Value	Description
31:0	RO	0x0126f200	msch0' revision id

service_bus: memory scheduler ddrconf
Address: Operational Base + offset (0x0008)
Memory scheduler configuration register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x0	DdrConf select the ddr rank,row,bank,col sequence

service_bus: memory scheduler ddrtiming
Address: Operational Base + offset (0x000c)
Memory scheduler timing register

Bit	Attr	Reset Value	Description
31	RW	0x0	BwRatio Bandwidth determines , in conjunction with field BwRatioExtended of register ddrmode, the number of DRAM data bus cycles required to process a acheduler data bus word. When BwRatio and BwRatioExtended are set to 0, the bandwidth ratio is 1:1, that is one DRAM cycles per scheduler word. When BwRatio is set to 1, and BwRatioExtended is set to 0, the bandwidth ratio is 2:1, that is ,two DRAM cycles per scheduler word.
30:26	RW	0x00	WrToRd Minimum time between the last DRAM Write command and a Read command.

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Bit	Attr	Reset Value	Description
25:21	RW	0x00	RdToWr Minimum time between the last DRAM Read command and a Write command.
20:18	RW	0x0	BurstLen DRAM burst duration on the DRAM data bus. Also equal to minimum time between two DRAM commands.
17:12	RW	0x00	WrToMiss Minimum time between the last DRAM Write command and a new Read or Write command in another page of the same bank.
11:6	RW	0x00	RdToMiss Minimum time between the last DRAM Read command and a new Read or Write command in another page of the same bank.
5:0	RW	0x00	ActToAct Minimum time between two consecutive DRAM Activate commands on the same bank.

service_bus: memory scheduler ddrmode
Address: Operational Base + offset (0x0010)
Memory scheduler mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	BwRatioExtended When 0, bandwidth ratio is determined by BwRatio. When 1, the bandwidth ratio is 4:1, that is four DRAM cycles per scheduler word, and BwRatio must be set to 0.
0	RW	0x0	AutoPrecharge When set to 1, pages are automatically closed after each access(no page hit). When set to 0, pages are left open until an access in a different page occurs.

service_bus: memory scheduler readlatency
Address: Operational Base + offset (0x0014)
Memory scheduler read latency register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:0	RW	0x32	ReadLatency Stores a user-defined read response latency ceiling, expressed in scheduler clock cycles. When the scheduler determines that the latency of a read transaction to the controller exceeded this value, the scheduler stalls further transactions to reduce the number of commands in the memory controller queue.

service_bus: memory scheduler activate
Address: Operational Base + offset (0x0038)
Memory scheduler activate register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10	RW	0x01	Fawbank Indicates the number of banks of a given device involved in the FAW period during which four banks can be activate.
9:4	RW	0x00	Faw The length of the FAW period, in cycles.
3:0	RW	0x00	Rrd The minimum number of scheduler clock cycles between two consecutive DRAM activate commands on different banks on the same device.

service_bus: memory scheduler devtodev
Address: Operational Base + offset (0x003c)
Memory scheduler devtodev register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:4	RW	0x00	Buswrtord The time delay, expressed in cycles, between a write and a read operation on different devices.
3:2	RW	0x00	Busrdtowr The time delay, expressed in cycles, between a read and a write operation on different devices.
1:0	RW	0x00	Busrdtord The time delay, expressed in cycles, between a read and a read operation on different devices.

Following is cpup port's QoS register detail description. Other ports have the same register. The only different is the base address's offset. The offset information is descripted in Table10-6 ~~ Table10-12.

service_core: cpup_qos Priority
Address: Operational Base + offset (0x0008)
CPU master0 priority register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:2	RW	0x2	P1 sets the high hurry level (i.e. when the measured bandwidth does not exceed the setting) when in Regulator mode, or read urgency level when in fixed or limiter mode.
1:0	RW	0x0	P0 sets the low hurry level, that is, when the measured bandwidth exceeds the setting, when in regulator mode, or write urgency level when in fixed or limiter mode.

service_core: cpup_qos Mode
Address: Operational Base + offset (0x000c)
CPU master0 QoS mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x3	Mode determines which of the following modes the QoSGenerator will adopt at reset: 0 : None 1 : Fixed 2 : Limiter 3 : Regulator

service_core: cpup_qos Bandwidth
Address: Operational Base + offset (0x0010)
CPU master0 QoS bandwidth register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:0	RW	0x007	Bandwidth Parameter bandwidth determines the bandwidth triggering of the limiter or the regulator. It is expressed in bytes per second. This parameter becomes available when limiter or regulator hardware is implemented.

service_core: cpup_qos Saturation
Address: Operational Base + offset (0x0014)
CPU master0 QoS saturation register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:0	RW	0x3ff	Saturation Parameter saturation determines the excursion of the payload counter, used to estimate the bandwidth, expressed in bytes. This parameter becomes available when limiter or regulator hardware is implemented.

service_core: cpup_qos ExtControl
Address: Operational Base + offset (0x0018)
CPU master0 QoS saturation register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
2	RW	0x0	Replace the External reference by the local clock.
1	RW	0x0	ExtThr input controls Low/High priority instead of bandwidth threshold.
0	W	0x0	Combines the Socket QoS with Regulator QoS

10.5 Function Description(peri interconnect)

The peri interconnect contain the following master:
peri,dmac,usb,nandc,mac,sdmmc,sfc.
A Global Programmers View (GPV) module exists to configure some properties of the interconnect. The arbitration scheme of this interconnect is configurable through GPV.
The priority from high to low is as follow:

- peri/sfc
- dmac
- usb
- nandc/mac/sdmmc/sfc

Customers can configure the Qos value through the GPV to change this priority. If you config them to same priority, then the interconnect uses a Least Recently Used (LRU) algorithm.

10.6 Register Description(peri interconnect)

10.6.1 Registers Summary

Name	Offset	Size	Reset Value	Description
PERI_RQos_M0	0x42100	4bits	0x0003	peri AXI Port Read channel QoS value.
PERI_WQos_M0	0x42104	4bits	0x0003	peri AXI Port Write channel quality value.
PERI_RQos_M1	0x43100	4bits	0x0002	dmac Port Read channel QoS value.
PERI_WQos_M1	0x43104	4bits	0x0002	dmac Port Write channel quality value.
PERI_RQos_M2	0x44100	4bits	0x0001	nandc Port Read channel QoS value.
PERI_WQos_M2	0x44104	4bits	0x0001	nandc Port Write channel quality value.
PERI_RQos_M3	0x45100	4bits	0x0000	usb Port Read channel QoS value.
PERI_WQos_M3	0x45104	4bits	0x0000	usb Port Write channel quality value.

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PERI_RQos_M4	0x46100	4bits	0x0000	sfc Port Read channel QoS value.
PERI_WQos_M4	0x46104	4bits	0x0000	sfc Port Write channel quality value.

*Notes: Size: **B** – Byte (8 bits) access, **HW** – Half WORD (16 bits) access, **W** –WORD (32 bits) access*

10.6.2 Detail Register Description

PERI_RQos_M0
Address:Operational Base+0x42100
PERI_AXI Port Read Qos register

Bit	Attr	Reset Value	Description
3:0	RW	0x1	Read channel QoS value.Higher value indicates higher priority.

PERI_WQos_M0
Address:Operational Base+0x42104
PERI_AXI Port Write Qos register

Bit	Attr	Reset Value	Description
3:0	RW	0x1	Write channel QoS value.Higher value indicates higher priority.

PERI_RQos_M1
Address:Operational Base+0x43100
DMAC Port Read Qos register

Bit	Attr	Reset Value	Description
3:0	RW	0x0	Read channel QoS value.Higher value indicates higher priority.

PERI_WQos_M1
Address:Operational Base+0x43104
DMAC Port Write Qos register

Bit	Attr	Reset Value	Description
3:0	RW	0x0	Write channel QoS value.Higher value indicates higher priority.

PERI_RQos_M2
Address:Operational Base+0x44100
NANDC Port Read Qos register

Bit	Attr	Reset Value	Description
3:0	RW	0x0	Read channel QoS value.Higher value indicates higher priority.

PERI_WQos_M2
Address:Operational Base+0x44104
NANDC Port Write Qos register

Bit	Attr	Reset Value	Description
3:0	RW	0x0	Write channel QoS value.Higher value indicates higher priority.

PERI_RQos_M3
Address: Operational Base+0x45100
USB Port Read Qos register

Bit	Attr	Reset Value	Description
3:0	RW	0x0	Read channel QoS value.Higher value indicates higher priority.

PERI_WQos_M3

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Address: Operational Base+0x45104
USB Port Write Qos register

Bit	Attr	Reset Value	Description
3:0	RW	0x0	Write channel QoS value.Higher value indicates higher priority.

PERI_RQos_M4
Address: Operational Base+0x46100
GPS Port Read Qos register

Bit	Attr	Reset Value	Description
3:0	RW	0x0	Read channel QoS value.Higher value indicates higher priority.

PERI_WQos_M4
Address: Operational Base+0x46104
GPS Port Write Qos register

Bit	Attr	Reset Value	Description
3:0	RW	0x0	Write channel QoS value.Higher value indicates higher priority.

10.7 Function Description(CCI-400)

The CCI-400 Cache Coherent Interconnect is an infrastructure component that supports the following:

- Data coherency between up to two ACE masters, and three ACE-Lite masters with three independent Points-of-Serialization (PoS) and full barrier support.
- High-bandwidth, cross-bar interconnect functionality between the masters and up to three slaves.
- DVM message transport between masters.
- QoS Virtual Networks (QVN).
- Quality-of-Service (QoS) regulation for shaping traffic profiles.

Note: The Revision of CCI-400 in PX5 is r1p4-00rel0

10.7.1 Snoop connectivity and control

The CCI-400 has a fully-connected snoop topology, so if they are enabled:

- Each ACE slave interface snoops the other ACE slave interface.
- ACE-Lite slave interfaces snoop both ACE slave interfaces.
- DVM messages are broadcast through all enabled slave interfaces.

Snooping and DVM message broadcast are disabled at reset, so you must enable the appropriate masters for snooping and DVM messages using the Snoop Control Registers before shareable or DVM messages are sent to the CCI-400.
Each ACE slave interface has programmable bits in the Snoop Control Registers. These bits control the issuing of snoop and DVM message requests on that interface.

10.7.2 Speculative fetch

For an application where the probability of a miss is high, then the snoop request and response time adds directly to the latency for each transaction labelled as shareable. To mitigate this, you can program each master interface to issue a fetch downstream in parallel with issuing a snoop request. This is known as a speculative fetch.
In the event that the snoop associated with a speculative fetch hits in a cache, then the data from the snoop is returned in preference to the data from the speculative fetch.
A speculative fetch is issued before all hazards that had arisen from the corresponding snoop have been resolved. Therefore, it is sometimes necessary to discard the data returned from memory and retry the fetch. These cases are:

- When a hazarding write transaction is detected. This hazarding write transaction must be

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ordered before the speculative fetch.
- When data from the speculative fetch returns before the snoop response for that transaction, and the read data buffer is already occupied by data waiting for a snoop response. control the issuing of snoop and DVM message requests on that interface.

10.7.3 Cache maintenance operations

The CCI-400 supports the snooping of cache maintenance operations based on the Snoop Control Register. See Snoop Control Registers. You can use snooping and cache maintenance to manage level 1 and 2 caches within the same domain as the CCI-400. You can optionally send cache maintenance operations downstream to manage a level 3 cache, depending on the BROADCASTCACHEMAINT inputs.

10.7.4 QoS value

Each CCI-400 slave interface has ARQOS and AWQOS input signals that transport a transaction-based QoS value. This determines the relative priority between transactions on that interface, or between interfaces. The CCI-400 uses the QoS value when it chooses between transaction requests at arbitration points and within queues. Transaction requests with the highest QoS value are prioritized. The CCI-400 uses a Least Recently Granted (LRG) scheme when two or more transactions share the highest value.
QoS values are propagated by CCI-400. If downstream interconnect and slave devices are sensitive to the QoS value, then the service rate is dependent on this value.

10.8 Register Description(peri interconnect)

10.8.1 Registers Summary

Name	Offset	Size	Reset Value	Description
CCI400_SYS_CTRL	0x00000	W	0x00000000	Control Override Register
CCI400_SPECU_CTRL	0x00004	W	0x00000000	Speculation Control Register
CCI400_SECURE_CTRL	0x00008	W	0x00000000	Secure Access Register
CCI400_PEND_STATUS	0x0000c	W	0x00000000	Status Register
CCI400_ERROR_STATUS	0x00010	W	0x00000000	Imprecise Error Register
CCI400_PFMMON_CTRL	0x00100	W	0x00000000	Performance Monitor Control Register (PMCR)
CCI400_SNOOP_CTRL	0x04000	W	0x00000000	Snoop Control Register for slave interface x
CCI400_RD_CHAN_QOS_CTRL	0x04100	W	0x00000000	Read Channel QoS Value Override Register for slave interface x
CCI400_WR_CHAN_QOS_CTRL	0x04104	W	0x00000000	Write Channel QoS Value Override slave interface x
CCI400_QOS_CTRL	0x0410c	W	0x00000000	QoS Control Register for slave interface x
CCI400_REGUL_TARGET_CTRL	0x04130	W	0x00000000	Regulator Target Register for slave interface x
CCI400_QOS_REGUL_SCAL	0x04134	W	0x00000000	QoS Regulator Scale Factor Register for slave interface 1
CCI400_OQS_RANGE_CTRL	0x04138	W	0x00000000	QoS Range Register for slave interface x

10.8.2 Detail Register Description

CCI400_SYS_CTRL
Address: Operational Base + offset (0x00000)

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Control Override Register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	RW	0x0	retry_reduce_en Disable retry reduction buffers for speculative fetches 0: Retry reduction buffers enabled 1: Retry reduction buffers disabled
4	RW	0x0	priority_promotion_dis Disable priority promotion 0: The CCI-400 uses ARQOSARBS inputs to promote the priority of earlier requests 1: The CCI-400 ignores ARQOSARBS inputs.
3	RW	0x0	termin_en Terminate all barrier transactions. 0: Master interfaces terminate barriers according to the BARRIERTERMINATE inputs. 1: All master interfaces terminate barriers
2	RW	0x0	specu_disable Disable speculative fetche 0: Send speculative fetches according to the Speculation Control Registe 1: Disable speculative fetches from all master interfaces
1	RW	0x0	dvm_disable DVM message disable 0: Send speculative fetches according to the Speculation Control Register 1: Disable speculative fetches from all master interfaces
0	RW	0x0	snoop_disable snoop_disable control 0: Snoop masters according to the Snoop Control Registers 1: Disable all snoops, but not DVM messages

CCI400_SPECU_CTRL

Address: Operational Base + offset (0x00004)

Speculation Control Register

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
20:16	RW	0x00	slv_specu_dis Disable speculative fetches for transactions through a slave interface. One bit for each slave interface: S4, S3, S2, S1, and S0 0: Enable speculative fetches 1: Disable speculative fetches
15:3	RO	0x0	reserved
2:0	RW	0x0	mst_specu_dis Disable speculative fetches from a master interface. One bit for each master interface: M2, M1, and M0. 0b0 Enable speculative fetches. 0b1 Disable speculative fetches

CCI400_SECURE_CTRL
Address: Operational Base + offset (0x00008)
Secure Access Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	non_secure_override Non-secure register access override 0: Disable Non-secure access to CCI-400 registers 1: Enable Non-secure access to CCI-400 registers

CCI400_PEND_STATUS
Address: Operational Base + offset (0x0000c)
Status Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	pending_status Indicates whether any changes to the snoop or DVM enables is pending in the CCI-400 0: No change pending 1: Change pending

CCI400_ERROR_STATUS
Address: Operational Base + offset (0x00010)
Imprecise Error Register

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
20	RW	0x0	imprecise_err_slv4 Imprecise error indicator for slave interface 4
19	RW	0x0	imprecise_err_slv3 Imprecise error indicator for slave interface 3
18	RW	0x0	imprecise_err_slv2 Imprecise error indicator for slave interface 2
17	RW	0x0	imprecise_err_slv1 Imprecise error indicator for slave interface 1
16	RW	0x0	imprecise_err_slv0 Imprecise error indicator for slave interface 0.
15:3	RO	0x0	reserved
2	RW	0x0	imprecise_err_mst2 Imprecise error indicator for master interface 2
1	RW	0x0	imprecise_err_mst1 Imprecise error indicator for master interface 1
0	RW	0x0	imprecise_err_mst0 Imprecise error indicator for master interface 0: 0: No error from the time this bit was last reset. 1: An error response has been received, but not signalled precisely.

CCI400_PFMMON_CTRL
Address: Operational Base + offset (0x00100)
Performance Monitor Control Register (PMCR)

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:11	RW	0x00	counter_num Specifies the number of counters implemented.
10:6	RO	0x0	reserved
5	RW	0x0	DP Disables cycle counter, CCNT, if non-invasive debug is prohibited: 0b0 Count is not disabled when NIDEN input is LOW. 0b1 Count is disabled when NIDEN input is LOW.
4	RW	0x0	EX Enable export of the events to the event bus, EVNTBUS, for an external monitoring block to trace events: 0b0 Do not export EVNTBUS. 0b1 Export EVNTBUS.
3	RW	0x0	CCD Cycle count divider: 0b0 Count every clock cycle when enabled. 0b1 Count every 64th clock cycle when enabled.

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Bit	Attr	Reset Value	Description
2	RW	0x0	CCR Cycle counter reset: 0b0 No action. 0b1 Reset cycle counter, CCNT, to zero.
1	RW	0x0	RST Performance counter reset: 0b0 No action. 0b1 Reset all performance counters to zero, not including CCNT.
0	RW	0x0	CEN Enable bit: 0b0 Disable all counters, including CCNT. 0b1 Enable all counters, including CCNT.

CCI400_SNOOP_CTRL
Address: Operational Base + offset (0x04000)
Snoop Control Register for slave interface x

Bit	Attr	Reset Value	Description
31	RW	0x0	slv_dvm_en Slave interface supports DVM messages. This is overridden to 0x0 if you set the Control Override Register [1].
30	RW	0x0	slv_snoop_en Slave interface supports snoops. This is overridden to 0x0 if you set the Control Override Register [0].
29:2	RO	0x0	reserved
1	RW	0x0	slv_dvm_issue_en Enable issuing of DVM message requests from this slave interface. RAZ/WI for interfaces not supporting DVM messages: 0b0 Disable DVM message requests. 0b1 Enable DVM message requests.
0	RW	0x0	slv_snoop_issue_en Enable issuing of snoop requests from this slave interface. RAZ/WI for interfaces not supporting snoops: 0b0 Disable snoop requests. 0b1 Enable snoop requests.

CCI400_RD_CHAN_QOS_CTRL
Address: Operational Base + offset (0x04100)
Read Channel QoS Value Override Register for slave interface x

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:8	RW	0x0	arqos_read Field0000 Abstract Reads what value is currently applied to transactions with ARQOS=0, provided QOSOVERRIDE is HIGH and the QoS value regulator is enabled.
7:4	RO	0x0	reserved
3:0	RW	0x0	arqos_override ARQOS value override for the slave interface

CCI400_WR_CHAN_QOS_CTRL
Address: Operational Base + offset (0x04104)
Write Channel QoS Value Override slave interface x

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:8	RW	0x0	awqos_read Reads what value is currently applied to transactions with AWQOS=0, provided QOSOVERRIDE is HIGH and the QoS value regulator is enabled.
7:4	RO	0x0	reserved
3:0	RW	0x0	awqos_override AWQOS value override for the slave interface

CCI400_QOS_CTRL
Address: Operational Base + offset (0x0410c)
QoS Control Register for slave interface x

Bit	Attr	Reset Value	Description
31	RW	0x0	qos_regul_en Determines whether this CCI-400 implementation supports QoS regulation. 0b0 QoS regulation fully supported as described in this document. See Quality of Service on page 2-17. 0b1 QoS regulation not supported, reads and writes to this register have no effect.
30:22	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
21	RW	0x0	bandwidth_regul_mode Sets the mode for bandwidth regulation: 0b0 Normal mode. The QoS value is stable when the master is idle. 0b1 Quiesce High mode. The QoS value tends to the maximum when the master is idle.
20	RW	0x0	rd_qos_regul_cfg Configures the mode of the QoS value regulator for read transactions. 0b0 Latency mode. 0b1 Period mode, for bandwidth regulation
19:17	RO	0x0	reserved
16	RW	0x0	wr_qos_regul_cfg Configures the mode of the QoS value regulator for write transactions: 0b0 Latency mode. 0b1 Period mode, for bandwidth regulation
15:4	RO	0x0	reserved
3	RW	0x0	slv_rd_outstanding_regul_en Enable regulation of outstanding read transactions for slave interfaces: ?ACE-Lite interfaces only, for example S0, S1, and S2. ?RAZ/WI for ACE interfaces, for example S3 and S4
2	RW	0x0	slv_wr_outstanding_regul_en Enable regulation of outstanding write transactions for slave interfaces. 旳CE-Lite interfaces only, for example S0, S1, and S2. ?RAZ/WI for ACE interfaces, for example S3 and S4.
1	RW	0x0	slave_rd_qos_regul_en Enable QoS value regulation on reads for slave interfaces.
0	RW	0x0	slave_wr_qos_regul_en Enable QoS value regulation on writes for slave interfaces.

CCI400_REGUL_TARGET_CTRL
Address: Operational Base + offset (0x04130)
Regulator Target Register for slave interface x

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
27:16	RW	0x000	ar_chan_regul_target AR channel regulator target.
15:12	RO	0x0	reserved
11:0	RW	0x000	aw_chan_regul_target AW channel regulator target

CCI400_QOS_REGUL_SCAL
Address: Operational Base + offset (0x04134)
QoS Regulator Scale Factor Register for slave interface 1

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:8	RW	0x0	arqos_scal_fact ARQOS scale factor, power of 2 in the range 2^-5 -2^-12
7:3	RO	0x0	reserved
2:0	RW	0x0	awqos_scal_fact AWQOS scale factor, power of 2 in the range 2^-5 -2^-12

CCI400_OQS_RANGE_CTRL
Address: Operational Base + offset (0x04138)
QoS Range Register for slave interface x

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:24	RW	0x0	max_arqos Maximum ARQOS value.
23:20	RO	0x0	reserved
19:16	RW	0x0	min_arqos Minimum ARQOS value
15:12	RO	0x0	reserved
11:8	RW	0x0	max_awqos Maximum AWQOS value.
7:4	RO	0x0	reserved
3:0	RW	0x0	min_awqos Minimum AWQOS value.

10.9 Application Notes

10.9.1 QoS setting

The cpum read channel, vop read channel , gpu write channel have the external QoS control. After reset each master port both have priority setting as 1. It’s recommended that field 0 of qos.ExtControl set to 1 to enable the external qos control. And prioirty setting of each master kept at 1.

10.9.2 Idle request

The main interconnect supports flushing the ongoing transaction when the software needed to

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do so.
If the GPU power domain need to disconnect from the main interconnect, Idle request has to be sent to GPU NIU, the NIU will respond a ack, and when it's ready to be disconnect, one Idle signal will be send out . Then, if GPU still have transaction to be sent to the memory scheduler, it will be stalled by the NIU.
If the GPU system power domain is disconnected as the above flow, then CPU want to access to the GPU system, it will response error to CPU.

The sequence is like following figure shows:

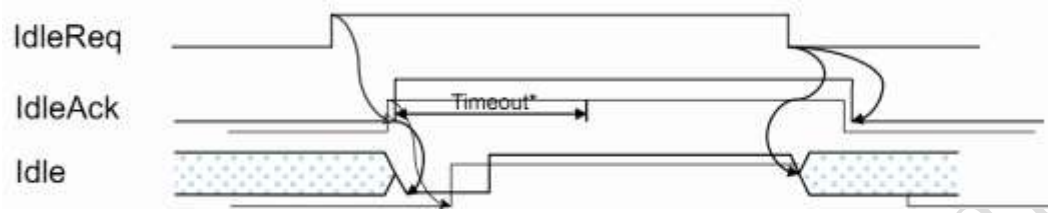

Fig. 10-2 Idle request

The idle request is set by PMU register.

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Chapter 11 Mailbox

11.1 Overview

The Mailbox module is a simple APB peripheral that allows both the Cortex-A53 MCU system to communicate by writing operation to generate interrupt. The registers are accessible by both CPU via APB interface.
The Mailbox has the following main features:

- Support dual-core system: Cortex-A53 and MCU
- Support APB interface
- Support four mailbox elements, each element includes one data word, one command word register and one flag bit that can represent one interrupt
- Four interrupts to Cortex-A53
- Four interrupts to MCU
- Provide 32 lock registers for software to use to indicate whether mailbox is occupied

11.2 Block Diagram

The figure below shows Mailbox block diagram.

Fig. 11-1 Mailbox Block Diagram

11.3 Function Description

11.3.1 Interrupt to Cortex-A53

The interrupt to Cortex-A53(CA53_IRQ[i]) is generated when B2A_INTEN[i] equals to 1 and there are writing operation to B2A_CMD_i and B2A_DATA_i orderly.(i=0~3)
The interrupt to Cortex-A53(CA53_IRQ[i],i=0~3) is cleared when writing 1 to B2A_STATUS[i]. (i=0~3)

11.3.2 Interrupt to MCU

The interrupt to MCU(MCU_IRQ[i]) is generated when A2B_INTEN[i] equals to 1 and there are writing operation to A2B_CMD_i and A2B_DATA_i orderly.(i=0~3)
The interrupt to MCU(MCU_IRQ[i],i=0~3) is cleared when writing 1 to A2B_STATUS[i]. (i=0~3)

11.4 Register Description

11.4.1 Register Summary

Name	Offset	Size	Reset Value	Description
MAILBOX_A2B_INTEN	0x00000	W	0x00000000	Cortex-A53 to MCU interrupt enable register
MAILBOX_A2B_STATUS	0x00004	W	0x00000000	Cortex-A53 to MCU interrupt status register
MAILBOX_A2B_CMD_0	0x00008	W	0x00000000	Cortex-A53 to MCU command 0
MAILBOX_A2B_DAT_0	0x0000c	W	0x00000000	Cortex-A53 to MCU data 0

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Name	Offset	Size	Reset Value	Description
MAILBOX_A2B_CMD_1	0x00010	W	0x00000000	Cortex-A53 to MCU command 1
MAILBOX_A2B_DAT_1	0x00014	W	0x00000000	Cortex-A53 to MCU data 1
MAILBOX_A2B_CMD_2	0x00018	W	0x00000000	Cortex-A53 to MCU command 2
MAILBOX_A2B_DAT_2	0x0001c	W	0x00000000	Cortex-A53 to MCU data 2
MAILBOX_A2B_CMD_3	0x00020	W	0x00000000	Cortex-A53 to MCU command 3
MAILBOX_A2B_DAT_3	0x00024	W	0x00000000	Cortex-A53 to MCU data 3
MAILBOX_B2A_INTEN	0x00028	W	0x00000000	Cortex-A53 to MCU interrupt enable register
MAILBOX_B2A_STATUS	0x0002c	W	0x00000000	MCU to Cortex-A53 interrupt status register
MAILBOX_B2A_CMD_0	0x00030	W	0x00000000	MCU to Cortex-A53 command 0
MAILBOX_B2A_DAT_0	0x00034	W	0x00000000	MCU to Cortex-A53 data 0
MAILBOX_B2A_CMD_1	0x00038	W	0x00000000	MCU to Cortex-A53 command 1
MAILBOX_B2A_DAT_1	0x0003c	W	0x00000000	MCU to Cortex-A53 data 1
MAILBOX_B2A_CMD_2	0x00040	W	0x00000000	MCU to Cortex-A53 command 2
MAILBOX_B2A_DAT_2	0x00044	W	0x00000000	MCU to Cortex-A53 data 2
MAILBOX_B2A_CMD_3	0x00048	W	0x00000000	MCU to Cortex-A53 command 3
MAILBOX_B2A_DAT_3	0x0004c	W	0x00000000	MCU to Cortex-A53 data 3
MAILBOX_ATOMIC_LOCK_00	0x00100	W	0x00000000	Lock flag register 00
MAILBOX_ATOMIC_LOCK_01	0x00104	W	0x00000000	Lock flag register 01
MAILBOX_ATOMIC_LOCK_02	0x00108	W	0x00000000	Lock flag register 02
MAILBOX_ATOMIC_LOCK_03	0x0010c	W	0x00000000	Lock flag register 03
MAILBOX_ATOMIC_LOCK_04	0x00110	W	0x00000000	Lock flag register 04
MAILBOX_ATOMIC_LOCK_05	0x00114	W	0x00000000	Lock flag register 05
MAILBOX_ATOMIC_LOCK_06	0x00118	W	0x00000000	Lock flag register 06
MAILBOX_ATOMIC_LOCK_07	0x0011c	W	0x00000000	Lock flag register 07
MAILBOX_ATOMIC_LOCK_08	0x00120	W	0x00000000	Lock flag register 08
MAILBOX_ATOMIC_LOCK_09	0x00124	W	0x00000000	Lock flag register 09
MAILBOX_ATOMIC_LOCK_10	0x00128	W	0x00000000	Lock flag register 10
MAILBOX_ATOMIC_LOCK_11	0x0012c	W	0x00000000	Lock flag register 11

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Name	Offset	Size	Reset Value	Description
MAILBOX_ATOMIC_LOCK_12	0x00130	W	0x00000000	Lock flag register 12
MAILBOX_ATOMIC_LOCK_13	0x00134	W	0x00000000	Lock flag register 13
MAILBOX_ATOMIC_LOCK_14	0x00138	W	0x00000000	Lock flag register 14
MAILBOX_ATOMIC_LOCK_15	0x0013c	W	0x00000000	Lock flag register 15
MAILBOX_ATOMIC_LOCK_16	0x00140	W	0x00000000	Lock flag register 16
MAILBOX_ATOMIC_LOCK_17	0x00144	W	0x00000000	Lock flag register 17
MAILBOX_ATOMIC_LOCK_18	0x00148	W	0x00000000	Lock flag register 18
MAILBOX_ATOMIC_LOCK_19	0x0014c	W	0x00000000	Lock flag register 19
MAILBOX_ATOMIC_LOCK_20	0x00150	W	0x00000000	Lock flag register 20
MAILBOX_ATOMIC_LOCK_21	0x00154	W	0x00000000	Lock flag register 21
MAILBOX_ATOMIC_LOCK_22	0x00158	W	0x00000000	Lock flag register 22
MAILBOX_ATOMIC_LOCK_23	0x0015c	W	0x00000000	Lock flag register 23
MAILBOX_ATOMIC_LOCK_24	0x00160	W	0x00000000	Lock flag register 24
MAILBOX_ATOMIC_LOCK_25	0x00164	W	0x00000000	Lock flag register 25
MAILBOX_ATOMIC_LOCK_26	0x00168	W	0x00000000	Lock flag register 26
MAILBOX_ATOMIC_LOCK_27	0x0016c	W	0x00000000	Lock flag register 27
MAILBOX_ATOMIC_LOCK_28	0x00170	W	0x00000000	Lock flag register 28
MAILBOX_ATOMIC_LOCK_29	0x00174	W	0x00000000	Lock flag register 29
MAILBOX_ATOMIC_LOCK_30	0x00178	W	0x00000000	Lock flag register 30
MAILBOX_ATOMIC_LOCK_31	0x0017c	W	0x00000000	Lock flag register 31

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

11.4.2 Detail Register Description

MAILBOX_A2B_INTEN

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Address: Operational Base + offset (0x00000)
Cortex-A53 to MCU interrupt enable register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	int3 interrupt enable for int3
2	RW	0x0	int2 interrupt enable for int2
1	RW	0x0	int1 interrupt enable for int1
0	RW	0x0	int0 interrupt enable for int0

MAILBOX_A2B_STATUS
Address: Operational Base + offset (0x00004)
Cortex-A53 to MCU interrupt status register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	int3 interrupt status for int3. when writte 1, int is cleared.
2	RW	0x0	int2 interrupt status for int2. when writte 1, int is cleared.
1	RW	0x0	int1 interrupt status for int1. when writte 1, int is cleared.
0	RW	0x0	int0 interrupt status for int0. when writte 1, int is cleared.

MAILBOX_A2B_CMD_0
Address: Operational Base + offset (0x00008)
Cortex-A53 to MCU command 0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	command command of Cortex-A53 to MCU

MAILBOX_A2B_DAT_0
Address: Operational Base + offset (0x0000c)
Cortex-A53 to MCU data 0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	data data of Cortex-A53 to MCU

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MAILBOX_A2B_CMD_1
Address: Operational Base + offset (0x00010)
Cortex-A53 to MCU command 1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	command command of Cortex-A53 to MCU

MAILBOX_A2B_DAT_1
Address: Operational Base + offset (0x00014)
Cortex-A53 to MCU data 1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	data data of Cortex-A53 to MCU

MAILBOX_A2B_CMD_2
Address: Operational Base + offset (0x00018)
Cortex-A53 to MCU command 2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	command command of Cortex-A53 to MCU

MAILBOX_A2B_DAT_2
Address: Operational Base + offset (0x0001c)
Cortex-A53 to MCU data 2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	data data of Cortex-A53 to MCU

MAILBOX_A2B_CMD_3
Address: Operational Base + offset (0x00020)
Cortex-A53 to MCU command 3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	command command of Cortex-A53 to MCU

MAILBOX_A2B_DAT_3
Address: Operational Base + offset (0x00024)
Cortex-A53 to MCU data 3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	data data of Cortex-A53 to MCU

MAILBOX_B2A_INTEN
Address: Operational Base + offset (0x00028)
Cortex-A53 to MCU interrupt enable register

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Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	int3 interrupt enable for int3
2	RW	0x0	int2 interrupt enable for int2
1	RW	0x0	int1 interrupt enable for int1
0	RW	0x0	int0 interrupt enable for int0

MAILBOX_B2A_STATUS
Address: Operational Base + offset (0x0002c)
MCU to Cortex-A53 interrupt status register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	int3 interrupt status for int3. when writte 1, int is cleared.
2	RW	0x0	int2 interrupt status for int2. when writte 1, int is cleared.
1	RW	0x0	int1 interrupt status for int1. when writte 1, int is cleared.
0	RW	0x0	int0 interrupt status for int0. when writte 1, int is cleared.

MAILBOX_B2A_CMD_0
Address: Operational Base + offset (0x00030)
MCU to Cortex-A53 command 0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	command command of MCU to Cortex-A53

MAILBOX_B2A_DAT_0
Address: Operational Base + offset (0x00034)
MCU to Cortex-A53 data 0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	data data of MCU to Cortex-A53

MAILBOX_B2A_CMD_1
Address: Operational Base + offset (0x00038)
MCU to Cortex-A53 command 1

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	command command of MCU to Cortex-A53

MAILBOX_B2A_DAT_1
Address: Operational Base + offset (0x0003c)
MCU to Cortex-A53 data 1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	data data of MCU to Cortex-A53

MAILBOX_B2A_CMD_2
Address: Operational Base + offset (0x00040)
MCU to Cortex-A53 command 2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	command command of MCU to Cortex-A53

MAILBOX_B2A_DAT_2
Address: Operational Base + offset (0x00044)
MCU to Cortex-A53 data 2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	data data of MCU to Cortex-A53

MAILBOX_B2A_CMD_3
Address: Operational Base + offset (0x00048)
MCU to Cortex-A53 command 3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	command command of MCU to Cortex-A53

MAILBOX_B2A_DAT_3
Address: Operational Base + offset (0x0004c)
MCU to Cortex-A53 data 3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	data data of MCU to Cortex-A53

MAILBOX_ATOMIC_LOCK_00
Address: Operational Base + offset (0x00100)
Lock flag register 00

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
0	RW	0x0	atomic_lock_00 lock flag bit 00

MAILBOX_ATOMIC_LOCK_01
Address: Operational Base + offset (0x00104)
Lock flag register 01

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_01 lock flag bit 01

MAILBOX_ATOMIC_LOCK_02
Address: Operational Base + offset (0x00108)
Lock flag register 02

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_02 lock flag bit 02

MAILBOX_ATOMIC_LOCK_03
Address: Operational Base + offset (0x0010c)
Lock flag register 03

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_03 lock flag bit 03

MAILBOX_ATOMIC_LOCK_04
Address: Operational Base + offset (0x00110)
Lock flag register 04

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_04 lock flag bit 04

MAILBOX_ATOMIC_LOCK_05
Address: Operational Base + offset (0x00114)
Lock flag register 05

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_05 lock flag bit 05

MAILBOX_ATOMIC_LOCK_06

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Address: Operational Base + offset (0x00118)
Lock flag register 06

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_06 lock flag bit 06

MAILBOX_ATOMIC_LOCK_07
Address: Operational Base + offset (0x0011c)
Lock flag register 07

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_07 lock flag bit 07

MAILBOX_ATOMIC_LOCK_08
Address: Operational Base + offset (0x00120)
Lock flag register 08

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_08 lock flag bit 08

MAILBOX_ATOMIC_LOCK_09
Address: Operational Base + offset (0x00124)
Lock flag register 09

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_09 lock flag bit 09

MAILBOX_ATOMIC_LOCK_10
Address: Operational Base + offset (0x00128)
Lock flag register 10

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_10 lock flag bit 10

MAILBOX_ATOMIC_LOCK_11
Address: Operational Base + offset (0x0012c)
Lock flag register 11

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_11 lock flag bit 11

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MAILBOX_ATOMIC_LOCK_12
Address: Operational Base + offset (0x00130)
Lock flag register 12

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_12 lock flag bit 12

MAILBOX_ATOMIC_LOCK_13
Address: Operational Base + offset (0x00134)
Lock flag register 13

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_13 lock flag bit 13

MAILBOX_ATOMIC_LOCK_14
Address: Operational Base + offset (0x00138)
Lock flag register 14

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_14 lock flag bit 14

MAILBOX_ATOMIC_LOCK_15
Address: Operational Base + offset (0x0013c)
Lock flag register 15

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_15 lock flag bit 15

MAILBOX_ATOMIC_LOCK_16
Address: Operational Base + offset (0x00140)
Lock flag register 16

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_16 lock flag bit 16

MAILBOX_ATOMIC_LOCK_17
Address: Operational Base + offset (0x00144)
Lock flag register 17

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_17 lock flag bit 17

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MAILBOX_ATOMIC_LOCK_18
Address: Operational Base + offset (0x00148)
Lock flag register 18

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_18 lock flag bit 18

MAILBOX_ATOMIC_LOCK_19
Address: Operational Base + offset (0x0014c)
Lock flag register 19

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_19 lock flag bit 19

MAILBOX_ATOMIC_LOCK_20
Address: Operational Base + offset (0x00150)
Lock flag register 20

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_20 lock flag bit 20

MAILBOX_ATOMIC_LOCK_21
Address: Operational Base + offset (0x00154)
Lock flag register 21

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_21 lock flag bit 21

MAILBOX_ATOMIC_LOCK_22
Address: Operational Base + offset (0x00158)
Lock flag register 22

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_22 lock flag bit 22

MAILBOX_ATOMIC_LOCK_23
Address: Operational Base + offset (0x0015c)
Lock flag register 23

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_23 lock flag bit 23

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MAILBOX_ATOMIC_LOCK_24
Address: Operational Base + offset (0x00160)
Lock flag register 24

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_24 lock flag bit 24

MAILBOX_ATOMIC_LOCK_25
Address: Operational Base + offset (0x00164)
Lock flag register 25

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_25 lock flag bit 25

MAILBOX_ATOMIC_LOCK_26
Address: Operational Base + offset (0x00168)
Lock flag register 26

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_26 lock flag bit 26

MAILBOX_ATOMIC_LOCK_27
Address: Operational Base + offset (0x0016c)
Lock flag register 27

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_27 lock flag bit 27

MAILBOX_ATOMIC_LOCK_28
Address: Operational Base + offset (0x00170)
Lock flag register 28

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_28 lock flag bit 28

MAILBOX_ATOMIC_LOCK_29
Address: Operational Base + offset (0x00174)
Lock flag register 29

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_29 lock flag bit 29

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MAILBOX_ATOMIC_LOCK_30
Address: Operational Base + offset (0x00178)
Lock flag register 30

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_30 lock flag bit 30

MAILBOX_ATOMIC_LOCK_31
Address: Operational Base + offset (0x0017c)
Lock flag register 31

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	atomic_lock_31 lock flag bit 31

11.5 Interface Description

N/A.

11.6 Application Notes

- The order for writing MAILBOX_B2A_CMD_i/ MAILBOX_A2B_CMD_i and MAILBOX_B2A_DATA_i/MAILBOX_A2B_DATA_i registers is limited: MAILBOX_B2A_CMD_i/MAILBOX_A2B_CMD_ifirstly, then MAILBOX_B2A_DATA_i/ MAILBOX_A2B_DATA_i. If wrong order is used, then the interrupt cannot be generated successfully.
- If you want to clear the interrupt, you can read out the STATUS register and writing 1 to corresponding bit.
- When using mailbox, software should read MAILBOX_ATOMIC_LOCK_i first. That the reading value is 0 means that it is available, and 1 means it has been automatically locked. Writing MAILBOX_ATOMIC_LOCK_i will clear this bit.

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Chapter 12 DMA Controller for Bus System (DMAC_BUS)

12.1 Overview

This device supports 2 Direct Memory Access (DMA) tops, one for bus system (DMAC_BUS), and the other one for Peripheral system (DMAC_PERI).Both of these two dma support transfers between memory and memory, peripheral and memory.
DMAC_BUS supports is under secure state after reset. The secure state can be changed by configuring SGRF module.
DMAC_BUS is mainly used for data transfer of the following slaves: I2S_2CH/ I2S_8CH/SPDIF/UART_DBG/Embedded SRAM and transfer data from/to external DDR SDRAM.
Following table shows the DMAC_BUS peripheral request mapping scheme.

Table 12-1 DMAC_BUS Request Mapping Table

Req number	Source	Polarity
0	I2S_8CH tx	High level
1	I2S_8CH rx	High level
2	PWM	High level
3	SPDIF	High level
4	UART_DBG tx	High level
5	UART_DBG rx	High level
6	I2S_2CH tx	High level
7	I2S_2CH rx	High level

DMAC_BUS supports the following features:

- Supports Trustzone technology
- Supports 8 peripheral request
- Up to 64bits data size
- 6 channel at the same time
- Up to burst 16
- 12 interrupt output and 1 abort output
- Supports 32 MFIFO depth

12.2 Block Diagram

Figure 12-1 shows the block diagram of DMAC_BUS

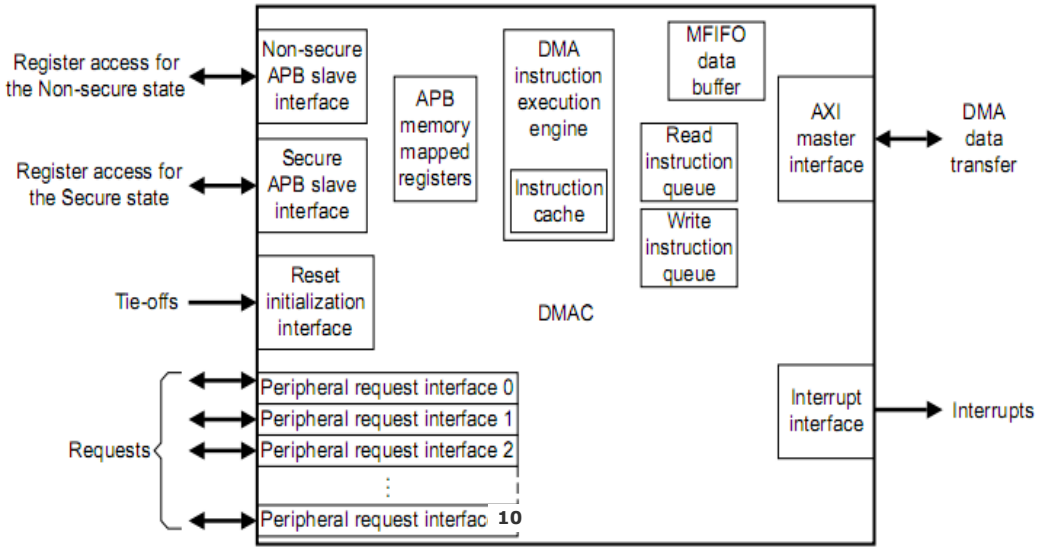

Fig. 12-1 Block diagram of DMAC_BUS

As the DMAC_BUS supports Trustzone technology, so dual APB interfaces enable the operation of the DMAC_BUS to be partitioned into the secure state and Non-secure state. You can use

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the APB interfaces to access status registers and also directly execute instructions in the DMAC_BUS.The default interface after reset is secure apb interface.

12.3 Function Description

12.3.1 Introduction

The DMAC contains an instruction processing block that enables it to process program code that controls a DMA transfer. The program code is stored in a region of system memory that the DMAC accesses using its AXI interface. The DMAC stores instructions temporarily in a cache.
DMAC_BUS supports 6 channels, each channel capable of supporting a single concurrent thread of DMA operation. In addition, a single DMA manager thread exists, and you can use it to initialize the DMA channel threads. The DMAC executes up to one instruction for each AXI clock cycle. To ensure that it regularly executes each active thread, it alternates by processing the DMA manager thread and then a DMA channel thread. It uses a round-robin process when selecting the next active DMA channel thread to execute.
The DMAC uses variable-length instructions that consist of one to six bytes. It provides a separate Program Counter (PC) register for each DMA channel. When a thread requests an instruction from an address, the cache performs a look-up. If a cache hit occurs, then the cache immediately provides the data. Otherwise, the thread is stalled while the DMAC uses the AXI interface to perform a cache line fill. If an instruction is greater than 4 bytes, or spans the end of a cache line, the DMAC performs multiple cache accesses to fetch the instruction. When a cache line fill is in progress, the DMAC enables other threads to access the cache, but if another cache miss occurs, this stalls the pipeline until the first line fill is complete.
When a DMA channel thread executes a load or store instruction, the DMAC adds the instruction to the relevant read or write queue. The DMAC uses these queues as an instruction storage buffer prior to it issuing the instructions on the AXI bus. The DMAC also contains a Multi First-In-First-Out (MFIFO) data buffer that it uses to store data that it reads, or writes, during a DMA transfer.

12.3.2 Operating states

Figure shows the operating states for the DMA manager thread and DMA channel threads.

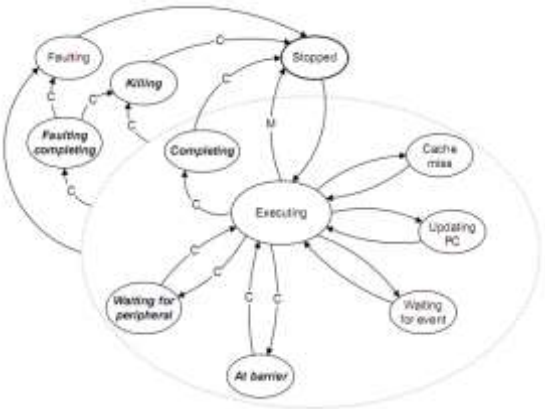

Fig. 12-2 DMAC_BUS operation states

Note:
arcs with no letter designator indicate state transitions for the DMA manager and DMA channel threads, otherwise use is restricted as follows:
C DMA channel threads only.
M DMA manager thread only.

After the DMAC exits from reset, it sets all DMA channel threads to the stopped state, and the status of boot_from_pc(tie-off interface of dmac) controls the DMA manager thread state:
boot_from_pc is LOW :DMA manager thread moves to the Stopped state.
boot_from_pc is HIGH :DMA manager thread moves to the Executing state.

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12.4 Register Description

12.4.1 Register summary

Name	Offset	Size	Reset Value	Description
DMAC_BUS_DSR	0x0000	W	0x0	DMA Status Register.
DMAC_BUS_DPC	0x0004	W	0x0	DMA Program Counter Register.
-	-	-	-	reserved
DMAC_BUS_INTEN	0x0020	W	0x0	Interrupt Enable Register
DMAC_BUS_EVENT_RIS	0x0024	W	0x0	Event Status Register.
DMAC_BUS_INTMIS	0x0028	W	0x0	Interrupt Status Register
DMAC_BUS_INTCLR	0x002C	W	0x0	Interrupt Clear Register
DMAC_BUS_FSRD	0x0030	W	0x0	Fault Status DMA Manager Register.
DMAC_BUS_FSRC	0x0034	W	0x0	Fault Status DMA Channel Register.
DMAC_BUS_FTRD	0x0038	W	0x0	Fault Type DMA Manager Register.
-	-	-	-	reserved
DMAC_BUS_FTR0	0x0040	W	0x0	Fault type for DMA Channel 0
DMAC_BUS_FTR1	0x0044	W	0x0	Fault type for DMA Channel 1
DMAC_BUS_FTR2	0x0048	W	0x0	Fault type for DMA Channel 2
DMAC_BUS_FTR3	0x004C	W	0x0	Fault type for DMA Channel 3
DMAC_BUS_FTR4	0x0050	W	0x0	Fault type for DMA Channel 4
DMAC_BUS_FTR5	0x0054	W	0x0	Fault type for DMA Channel 5
-	-	-	-	reserved
DMAC_BUS_CSR0	0x0100	W	0x0	Channel Status for DMA Channel 0
DMAC_BUS_CSR1	0x0108	W	0x0	Channel Status for DMA Channel 1
DMAC_BUS_CSR2	0x0110	W	0x0	Channel Status for DMA Channel 2
DMAC_BUS_CSR3	0x0118	W	0x0	Channel Status for DMA Channel 3
DMAC_BUS_CSR4	0x0120	W	0x0	Channel Status for DMA Channel 4
DMAC_BUS_CSR5	0x0128	W	0x0	Channel Status for DMA Channel 5
DMAC_BUS_CPC0	0x0104	W	0x0	Channel PC for DMA Channel 0
DMAC_BUS_CPC1	0x010c	W	0x0	Channel PC for DMA Channel 1
DMAC_BUS_CPC2	0x0114	W	0x0	Channel PC for DMA Channel 2
DMAC_BUS_CPC3	0x011c	W	0x0	Channel PC for DMA Channel 3
DMAC_BUS_CPC4	0x0124	W	0x0	Channel PC for DMA Channel 4
DMAC_BUS_CPC5	0x012c	W	0x0	Channel PC for DMA Channel 5
DMAC_BUS_SAR0	0x0400	W	0x0	Source Address for DMA Channel 0
DMAC_BUS_SAR1	0x0420	W	0x0	Source Address for DMA Channel 1
DMAC_BUS_SAR2	0x0440	W	0x0	Source Address for DMA Channel 2
DMAC_BUS_SAR3	0x0460	W	0x0	Source Address for DMA Channel 3
DMAC_BUS_SAR4	0x0480	W	0x0	Source Address for DMA Channel 4
DMAC_BUS_SAR5	0x04a0	W	0x0	Source Address for DMA Channel 5
DMAC_BUS_DAR0	0x0404	W	0x0	Dest Address for DMAChannel 0
DMAC_BUS_DAR1	0x0424	W	0x0	Dest Address for DMAChannel 1
DMAC_BUS_DAR2	0x0444	W	0x0	Dest Address for DMAChannel 2
DMAC_BUS_DAR3	0x0464	W	0x0	Dest Address for DMAChannel 3
DMAC_BUS_DAR4	0x0484	W	0x0	Dest Address for DMAChannel 4
DMAC_BUS_DAR5	0x04a4	W	0x0	Dest Address for DMAChannel 5
DMAC_BUS_CCR0	0x0408	W	0x0	Channel Control for DMA Channel 0
DMAC_BUS_CCR1	0x0428	W	0x0	Channel Control for DMA Channel 1
DMAC_BUS_CCR2	0x0448	W	0x0	Channel Control for DMA Channel 2
DMAC_BUS_CCR3	0x0468	W	0x0	Channel Control for DMA Channel 3
DMAC_BUS_CCR4	0x0488	W	0x0	Channel Control for DMA Channel 4
DMAC_BUS_CCR5	0x04a8	W	0x0	Channel Control for DMA Channel 5

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DMAC_BUS_LC0_0	0x040C	W	0x0	Loop Counter 0 for DMA Channel 0
DMAC_BUS_LC0_1	0x042C	W	0x0	Loop Counter 0 for DMA Channel 1
DMAC_BUS_LC0_2	0x044C	W	0x0	Loop Counter 0 for DMA Channel 2
DMAC_BUS_LC0_3	0x046C	W	0x0	Loop Counter 0 for DMA Channel 3
DMAC_BUS_LC0_4	0x048C	W	0x0	Loop Counter 0 for DMA Channel 4
DMAC_BUS_LC0_5	0x04aC	W	0x0	Loop Counter 0 for DMA Channel 5
DMAC_BUS_LC1_0	0x0410	W	0x0	Loop Counter 1 for DMA Channel 0
DMAC_BUS_LC1_1	0x0430	W	0x0	Loop Counter 1 for DMA Channel 1
DMAC_BUS_LC1_2	0x0450	W	0x0	Loop Counter 1 for DMA Channel 2
DMAC_BUS_LC1_3	0x0470	W	0x0	Loop Counter 1 for DMA Channel 3
DMAC_BUS_LC1_4	0x0490	W	0x0	Loop Counter 1 for DMA Channel 4
DMAC_BUS_LC1_5	0x04b0	W	0x0	Loop Counter 1 for DMA Channel 5
-	-	-	-	reserved
DMAC_BUS_DBGSTATUS	0x0D00	W	0x0	Debug Status Register.
DMAC_BUS_DBGCMD	0x0D04	W	0x0	Debug Command Register.
DMAC_BUS_DBGINST0	0x0D08	W	0x0	Debug Instruction-0 Register.
DMAC_BUS_DBGINST1	0x0D0C	W	0x0	Debug Instruction-1 Register.
DMAC_BUS_CR0	0x0E00	W		Configuration Register 0.
DMAC_BUS_CR1	0x0E04	W		Configuration Register 1.
DMAC_BUS_CR2	0x0E08	W		Configuration Register 2.
DMAC_BUS_CR3	0x0E0C	W		Configuration Register 3.
DMAC_BUS_CR4	0x0E10	W		Configuration Register 4.
DMAC_BUS_CRDn	0x0E14	W		Configuration Register Dn.
DMAC_BUS_WD	0x0E80	W	0x0	Watchdog Register.

Notes:
*Size: **B** – Byte (8 bits) access, **HW** – Half WORD (16 bits) access, **W** –WORD (32 bits) access*

12.4.2 Detail Register Description

DMAC_BUS_DSR
Address:Operational Base+0x0
DMA Manager Status Register

Bit	Attr	Reset Value	Description
31:10	-	-	Reserved
9	R	0x0	Provides the security status of the DMA manager thread: 0 = DMA manager operates in the Secure state 1 = DMA manager operates in the Non-secure state.
8:4	R	0x0	When the DMA manager thread executes a DMAWFE instruction, it waits for the following event to occur: b00000 = event[0] b00001 = event[1] b00010 = event[2] … b11111 = event[31].
3:0	R	0x0	The operating state of the DMA manager: b0000 = Stopped b0001 = Executing b0010 = Cache miss b0011 = Updating PC b0100 = Waiting for event b0101-b1110 = reserved b1111 = Faulting.

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DMAC_BUS_DPC
Address:Operational Base+0x4
DMA Program Counter Register

Bit	Attr	Reset Value	Description
31:0	R	0x0	Program counter for the DMA manager thread

DMAC_BUS_INTEN
Address:Operational Base+0x20
Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:0	RW	0x0	Program the appropriate bit to control how the DMAC responds when it executes DMASEV: Bit [N] = 0 If the DMAC executes DMASEV for the event-interrupt resource N then the DMAC signals event N to all of the threads. Set bit [N] to 0 if your system design does not use irq[N] to signal an interrupt request. Bit [N] = 1 If the DMAC executes DMASEV for the event-interrupt resource N then the DMAC sets irq[N] HIGH. Set bit [N] to 1 if your system designer requires irq[N] to signal an interrupt request.

DMAC_BUS_EVENT_RIS
Address:Operational Base+0x24
Event-Interrupt Raw Status Register

Bit	Attr	Reset Value	Description
31:0	R	0x0	Returns the status of the event-interrupt resources: Bit [N] = 0 Event N is inactive or irq[N] is LOW. Bit [N] = 1 Event N is active or irq[N] is HIGH.

DMAC_BUS_INTMIS
Address:Operational Base+0x28
Interrupt Status Register

Bit	Attr	Reset Value	Description
31:0	R	0x0	Provides the status of the interrupts that are active in the DMAC: Bit [N] = 0 Interrupt N is inactive and therefore irq[N] is LOW. Bit [N] = 1 Interrupt N is active and therefore irq[N] is HIGH

DMAC_BUS_INTCLR
Address:Operational Base+0x2c
Interrupt Clear Register

Bit	Attr	Reset Value	Description
31:0	W	0x0	Controls the clearing of the irq outputs: Bit [N] = 0 The status of irq[N] does not change. Bit [N] = 1 The DMAC sets irq[N] LOW if the INTEN Register programs the DMAC to signal an interrupt. Otherwise, the status of irq[N] does not change.

DMAC_BUS_FSRD

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Address:Operational Base+0x30
Fault Status DMA Manager Register

Bit	Attr	Reset Value	Description
31:0	R	0x0	Provides the fault status of the DMA manager. Read as: 0 = the DMA manager thread is not in the Faulting state 1 = the DMA manager thread is in the Faulting state.

DMAC_BUS_FSRC
Address:Operational Base+0x34
Fault Status DMA Channel Register

Bit	Attr	Reset Value	Description
31:0	R	0x0	Each bit provides the fault status of the corresponding channel. Read as: Bit [N] = 0 No fault is present on DMA channel N. Bit [N] = 1 DMA channel N is in the Faulting or Faulting completing state.

DMAC_BUS_FTRD
Address:Operational Base+0x38
Fault Type DMA Manager Register

Bit	Attr	Reset Value	Description
31	-	-	reserved
30	R	0x0	If the DMA manager aborts, this bit indicates if the erroneous instruction was read from the system memory or from the debug interface: 0 = instruction that generated an abort was read from system memory 1 = instruction that generated an abort was read from the debug interface.
29:17	-	-	reserved
16	R	0x0	Indicates the AXI response that the DMAC receives on the RRESP bus, after the DMA manager performs an instruction fetch: 0 = OKAY response 1 = EXOKAY, SLVERR, or DECERR response
15:6	-	-	reserved
5	R	0x0	Indicates if the DMA manager was attempting to execute DMAWFE or DMASEV with inappropriate security permissions: 0 = DMA manager has appropriate security to execute DMAWFE or DMASEV 1 = a DMA manager thread in the Non-secure state attempted to execute either: • DMAWFE to wait for a secure event • DMASEV to create a secure event or secure interrupt
4	R	0x0	ndicates if the DMA manager was attempting to execute DMAGO with inappropriate security permissions: 0 = DMA manager has appropriate security to execute DMAGO 1 = a DMA manager thread in the Non-secure state attempted to execute DMAGO to create a

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			DMA channel operating in the Secure state.
3:2	-	-	reserved
1	R	0x0	Indicates if the DMA manager was attempting to execute an instruction operand that was not valid for the configuration of the DMAC: 0 = valid operand 1 = invalid operand.
0	R	0x0	Indicates if the DMA manager was attempting to execute an undefined instruction: 0 = defined instruction 1 = undefined instruction.

DMAC_BUS_FTR0~DMAC_BUS_FTR5

Address:Operational Base+0x40
Operational Base+0x44
Operational Base+0x48
Operational Base+0x4c
Operational Base+0x50
Operational Base+0x54

Fault Type DMA Channel Register

Bit	Attr	Reset Value	Description
31	R	0x0	Indicates if the DMA channel has locked-up because of resource starvation: 0 = DMA channel has adequate resources 1 = DMA channel has locked-up because of insufficient resources. This fault is an imprecise abort
30	R	0x0	If the DMA channel aborts, this bit indicates if the erroneous instruction was read from the system memory or from the debug interface: 0 = instruction that generated an abort was read from system memory 1 = instruction that generated an abort was read from the debug interface. This fault is an imprecise abort but the bit is only valid when a precise abort occurs.
29:19	-	-	reserved
18	R	0x0	Indicates the AXI response that the DMAC receives on the RRESP bus, after the DMA channel thread performs a data read: 0 = OKAY response 1 = EXOKAY, SLVERR, or DECERR response. This fault is an imprecise abort
17	R	0x0	Indicates the AXI response that the DMAC receives on the BRESP bus, after the DMA channel thread performs a data write: 0 = OKAY response 1 = EXOKAY, SLVERR, or DECERR response. This fault is an imprecise abort.
16	R	0x0	Indicates the AXI response that the DMAC receives on the RRESP bus, after the DMA channel thread performs an instruction fetch: 0 = OKAY response 1 = EXOKAY, SLVERR, or DECERR response. This fault is a precise abort.
15:14	-	-	reserved

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13	R	0x0	Indicates if the MFIFO did not contain the data to enable the DMAC to perform the DMAST: 0 = MFIFO contains all the data to enable the DMAST to complete 1 = previous DMALDs have not put enough data in the MFIFO to enable the DMAST to complete. This fault is a precise abort.
12	R	0x0	Indicates if the MFIFO prevented the DMA channel thread from executing DMALD or DMAST. Depending on the instruction: DMALD 0 = MFIFO contains sufficient space 1 = MFIFO is too small to hold the data that DMALD requires. DMAST 0 = MFIFO contains sufficient data 1 = MFIFO is too small to store the data to enable DMAST to complete. This fault is an imprecise abort
11:8	-	-	reserved
7	R	0x0	Indicates if a DMA channel thread, in the Non-secure state, attempts to program the CCRn Register to perform a secure read or secure write: 0 = a DMA channel thread in the Non-secure state is not violating the security permissions 1 = a DMA channel thread in the Non-secure state attempted to perform a secure read or secure write. This fault is a precise abort
6	R	0x0	Indicates if a DMA channel thread, in the Non-secure state, attempts to execute DMAWFP, DMALDP, DMASTP, or DMAFLUSHP with inappropriate security permissions: 0 = a DMA channel thread in the Non-secure state is not violating the security permissions 1 = a DMA channel thread in the Non-secure state attempted to execute either: • DMAWFP to wait for a secure peripheral • DMALDP or DMASTP to notify a secure peripheral • DMAFLUSHP to flush a secure peripheral. This fault is a precise abort.
5	R	0x0	Indicates if the DMA channel thread attempts to execute DMAWFE or DMASEV with inappropriate security permissions: 0 = a DMA channel thread in the Non-secure state is not violating the security permissions 1 = a DMA channel thread in the Non-secure state attempted to execute either: • DMAWFE to wait for a secure event • DMASEV to create a secure event or secure interrupt. This fault is a precise abort.
4:2	-	-	reserved
1	R	0x0	Indicates if the DMA channel thread was attempting to execute an instruction operand that was not valid for the configuration of the DMAC:

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			0 = valid operand 1 = invalid operand. This fault is a precise abort.
0	R	0x0	Indicates if the DMA channel thread was attempting to execute an undefined instruction: 0 = defined instruction 1 = undefined instruction. This fault is a precise abort

DMAC_BUS_CSR0~DMAC_BUS_CSR5

Address:Operational Base+0x100
Operational Base+0x108
Operational Base+0x110
Operational Base+0x118
Operational Base+0x120
Operational Base+0x128

Channel Status Registers

Bit	Attr	Reset Value	Description
31:22	-	-	reserved
21	R	0x0	The channel non-secure bit provides the security of the DMA channel: 0 = DMA channel operates in the Secure state 1 = DMA channel operates in the Non-secure state
20:16	-	-	reserved
15	R	0x0	When the DMA channel thread executes DMAWFP this bit indicates if the periph operand was set: 0 = DMAWFP executed with the periph operand not set 1 = DMAWFP executed with the periph operand set
14	R	0x0	When the DMA channel thread executes DMAWFP this bit indicates if the burst or single operand were set: 0 = DMAWFP executed with the single operand set 1 = DMAWFP executed with the burst operand set.
13:9	-	-	reserved
8:4	R	0x0	If the DMA channel is in the Waiting for event state or the Waiting for peripheral state then these bits indicate the event or peripheral number that the channel is waiting for: b00000 = DMA channel is waiting for event, or peripheral, 0 b00001 = DMA channel is waiting for event, or peripheral, 1 b00010 = DMA channel is waiting for event, or peripheral, 2 ... b11111 = DMA channel is waiting for event, or peripheral, 31
3:0	R	0x0	The channel status encoding is: b0000 = Stopped b0001 = Executing

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			b0010 = Cache miss b0011 = Updating PC b0100 = Waiting for event b0101 = At barrier b0110 = reserved b0111 = Waiting for peripheral b1000 = Killing b1001 = Completing b1010-b1101 = reserved b1110 = Faulting completing b1111 = Faulting

DMAC_BUS_CPC0~DMAC_BUS_CPC5

Address:Operational Base+0x104
Operational Base+0x10c
Operational Base+0x114
Operational Base+0x11c
Operational Base+0x124
Operational Base+0x12c

Channel Program Counter Registers

Bit	Attr	Reset Value	Description
31:0	R	0x0	Program counter for the DMA channel n thread

DMAC_BUS_SAR0~DMAC_BUS_SAR5

Address:Operational Base+0x400
Operational Base+0x420
Operational Base+0x440
Operational Base+0x460
Operational Base+0x480
Operational Base+0x4a0

Source Address Registers

Bit	Attr	Reset Value	Description
31:0	R	0x0	Address of the source data for DMA channel n

DMAC_BUS_DAR0~DMAC_BUS_DAR5

Address:Operational Base+0x404
Operational Base+0x424
Operational Base+0x444
Operational Base+0x464
Operational Base+0x484
Operational Base+0x4a4

DestinationAddress Registers

Bit	Attr	Reset Value	Description
31:0	R	0x0	Address of the Destinationdata for DMA channel n

DMAC_BUS_CCR0~DMAC_BUS_CCR5

Address:Operational Base+0x408
Operational Base+0x428
Operational Base+0x448
Operational Base+0x468
Operational Base+0x488
Operational Base+0x4a8

Channel Control Registers

Bit	Attr	Reset Value	Description
31:28	-	-	reserved
27:25	R	0x0	Programs the state of AWCACHE[3,1:0]a when the DMAC writes the destination data.

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			Bit [27] 0 = AWCACHE[3] is LOW 1 = AWCACHE[3] is HIGH. Bit [26] 0 = AWCACHE[1] is LOW 1 = AWCACHE[1] is HIGH. Bit [25] 0 = AWCACHE[0] is LOW 1 = AWCACHE[0] is HIGH
24:22	R	0x0	Programs the state of AWPROT[2:0]a when the DMAC writes the destination data. Bit [24] 0 = AWPROT[2] is LOW 1 = AWPROT[2] is HIGH. Bit [23] 0 = AWPROT[1] is LOW 1 = AWPROT[1] is HIGH. Bit [22] 0 = AWPROT[0] is LOW 1 = AWPROT[0] is HIGH
21:18	R	0x0	For each burst, these bits program the number of data transfers that the DMAC performs when it writes the destination data: b0000 = 1 data transfer b0001 = 2 data transfers b0010 = 3 data transfers ... b1111 = 16 data transfers. The total number of bytes that the DMAC writes out of the MFIFO when it executes a DMAST instruction is the product of dst_burst_len and dst_burst_size
17:15	R	0x0	For each beat within a burst, it programs the number of bytes that the DMAC writes to the destination: b000 = writes 1 byte per beat b001 = writes 2 bytes per beat b010 = writes 4 bytes per beat b011 = writes 8 bytes per beat b100 = writes 16 bytes per beat b101-b111 = reserved. The total number of bytes that the DMAC writes out of the MFIFO when it executes a DMAST instruction is the product of dst_burst_len and dst_burst_size.
14	R	0x0	Programs the burst type that the DMAC performs when it writes the destination data: 0 = Fixed-address burst. The DMAC signals AWBURST[0] LOW. 1 = Incrementing-address burst. The DMAC signals AWBURST[0] HIGH.
13:11	R	0x0	Set the bits to control the state of ARCACHE[2:0]a when the DMAC reads the source data. Bit [13] 0 = ARCACHE[2] is LOW 1 = ARCACHE[2] is HIGH. Bit [12] 0 = ARCACHE[1] is LOW 1 = ARCACHE[1] is HIGH. Bit [11] 0 = ARCACHE[0] is LOW 1 = ARCACHE[0] is HIGH.
10:8	R	0x0	Programs the state of ARPROT[2:0]a when the DMAC reads the source data. Bit [10] 0 = ARPROT[2] is LOW 1 = ARPROT[2] is HIGH. Bit [9] 0 = ARPROT[1] is LOW 1 = ARPROT[1] is HIGH. Bit [8] 0 = ARPROT[0] is LOW

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			1 = ARPROT[0] is HIGH.
7:4	R	0x0	For each burst, these bits program the number of data transfers that the DMAC performs when it reads the source data: b0000 = 1 data transfer b0001 = 2 data transfers b0010 = 3 data transfers ... b1111 = 16 data transfers. The total number of bytes that the DMAC reads into the MFIFO when it executes a DMALD instruction is the product of src_burst_len and src_burst_size
3:1	R	0x0	For each beat within a burst, it programs the number of bytes that the DMAC reads from the source: b000 = reads 1 byte per beat b001 = reads 2 bytes per beat b010 = reads 4 bytes per beat b011 = reads 8 bytes per beat b100 = reads 16 bytes per beat b101-b111 = reserved. The total number of bytes that the DMAC reads into the MFIFO when it executes a DMALD instruction is the product of src_burst_len and src_burst_size
0	R	0x0	Programs the burst type that the DMAC performs when it reads the source data: 0 = Fixed-address burst. The DMAC signals ARBURST[0] LOW. 1 = Incrementing-address burst. The DMAC signals ARBURST[0] HIGH

DMAC_BUS_LC0_0~DMAC_BUS_LC0_5

Address:Operational Base+0x40c
Operational Base+0x42c
Operational Base+0x44c
Operational Base+0x46c
Operational Base+0x48c
Operational Base+0x4ac

Loop Counter 0 Registers

Bit	Attr	Reset Value	Description
31:8	-	-	reserved
7:0	R	0x0	Loop counter 0 iterations

DMAC_BUS_LC1_0~DMAC_BUS_LC1_5

Address:Operational Base+0x410
Operational Base+0x430
Operational Base+0x450
Operational Base+0x470
Operational Base+0x490
Operational Base+0x4b0

Loop Counter 1 Registers

Bit	Attr	Reset Value	Description
31:8	-	-	reserved
7:0	R	0x0	Loop counter 1 iterations

DMAC_BUS_DBGSTATUS

Address:Operational Base+0xd00

Debug Status Register

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Bit	Attr	Reset Value	Description
31:2	-	-	reserved
1:0	R	0x0	The debug encoding is as follows: b00 = execute the instruction that the DBGINST [1:0] Registers contain b01 = reserved b10 = reserved b11 = reserved.

DMAC_BUS_DBGCMD
Address:Operational Base+0xd04
Debug Command Register

Bit	Attr	Reset Value	Description
31:2	-	-	reserved
1:0	W	0x0	The debug encoding is as follows: b00 = execute the instruction that the DBGINST [1:0] Registers contain b01 = reserved b10 = reserved b11 = reserved

DMAC_BUS_DBGINST0
Address:Operational Base+0xd08
Debug Instruction-0 Register

Bit	Attr	Reset Value	Description
31:24	W	0x0	Instruction byte 1
23:16	W	0x0	Instruction byte 0
15:11	-	-	reserved
10:8	W	0x0	DMA channel number: b000 = DMA channel 0 b001 = DMA channel 1 b010 = DMA channel 2 … b111 = DMA channel 7
7:1	-	-	reserved
0	W	0x0	The debug thread encoding is as follows: 0 = DMA manager thread 1 = DMA channel.

DMAC_BUS_DBGINST1
Address:Operational Base+0xd0c
Debug Instruction-1 Register

Bit	Attr	Reset Value	Description
31:24	W	0x0	Instruction byte 5
23:16	W	0x0	Instruction byte 4
15:8	W	0x0	Instruction byte 3
7:0	W	0x0	Instruction byte 2

DMAC_BUS_CR0
Address:Operational Base+0xe00
Configuration Register 0

Bit	Attr	Reset Value	Description
31:22	-	-	reserved
21:17	R	0x2	Number of interrupt outputs that the DMAC provides: b00000 = 1 interrupt output, irq[0]

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			b00001 = 2 interrupt outputs, irq[1:0] b00010 = 3 interrupt outputs, irq[2:0] ... b11111 = 32 interrupt outputs, irq[31:0].
16:12	R	0x7	Number of peripheral request interfaces that the DMAC provides: b00000 = 1 peripheral request interface b00001 = 2 peripheral request interfaces b00010 = 3 peripheral request interfaces ... b11111 = 32 peripheral request interfaces.
11:7	-	-	reserved
6:4	R	0x5	Number of DMA channels that the DMAC supports: b000 = 1 DMA channel b001 = 2 DMA channels b010 = 3 DMA channels ... b111 = 8 DMA channels.
3	-	-	reserved
2	R	0x0	Indicates the status of the boot_manager_ns signal when the DMAC exited from reset: 0 = boot_manager_ns was LOW 1 = boot_manager_ns was HIGH.
1	R	0x0	Indicates the status of the boot_from_pc signal when the DMAC exited from reset: 0 = boot_from_pc was LOW 1 = boot_from_pc was HIGH
0	R	0x1	Supports peripheral requests: 0 = the DMAC does not provide a peripheral request interface 1 = the DMAC provides the number of peripheral request interfaces that the num_periph_req field specifies.

DMAC_BUS_CR1

Address:Operational Base+0xe04

Configuration Register 1

Bit	Attr	Reset Value	Description
31:8	-	-	reserved
7:4	R	0x5	[7:4] num_i-cache_lines Number of i-cache lines: b0000 = 1 i-cache line b0001 = 2 i-cache lines b0010 = 3 i-cache lines … b1111 = 16 i-cache lines.
3	-	-	reserved
2:0	R	0x7	The length of an i-cache line: b000-b001 = reserved b010 = 4 bytes b011 = 8 bytes b100 = 16 bytes b101 = 32 bytes b110-b111 = reserved

DMAC_BUS_CR2

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Address:Operational Base+0xe08
Configuration Register 2

Bit	Attr	Reset Value	Description
31:0	R	0x0	Provides the value of boot_addr[31:0] when the DMAC exited from reset

DMAC_BUS_CR3
Address:Operational Base+0xe0c
Configuration Register 3

Bit	Attr	Reset Value	Description
31:0	R	0x0	Provides the security state of an event-interrupt resource: Bit [N] = 0 Assigns event<N> or irq[N] to the Secure state. Bit [N] = 1 Assigns event<N> or irq[N] to the Non-secure state.

DMAC_BUS_CR4
Address:Operational Base+0xe10
Configuration Register 4

Bit	Attr	Reset Value	Description
31:0	R	0x6	Provides the security state of the peripheral request interfaces: Bit [N] = 0 Assigns peripheral request interface N to the Secure state. Bit [N] = 1 Assigns peripheral request interface N to the Non-secure state

DMAC_BUS_CRDn
Address:Operational Base+0xe14
DMA Configuration Register

Bit	Attr	Reset Value	Description
31:30	-	-	reserved
29:20	R	0x20	The number of lines that the data buffer contains: b000000000 = 1 line b000000001 = 2 lines … b111111111 = 1024 lines
19:16	R	0x9	The depth of the read queue: b0000 = 1 line b0001 = 2 lines ... b1111 = 16 lines.
15	-	-	reserved
14:12	R	0x4	Read issuing capability that programs the number of outstanding read transactions: b000 = 1 b001 = 2 … b111 = 8
11:8	R	0x7	The depth of the write queue: b0000 = 1 line b0001 = 2 lines … b1111 = 16 lines.
7	-	-	reserved

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6:4	R	0x3	Write issuing capability that programs the number of outstanding write transactions: b000 = 1 b001 = 2 … b111 = 8
3	-	-	reserved
2:0		0x3	The data bus width of the AXI interface: b000 = reserved b001 = reserved b010 = 32-bit b011 = 64-bit b100 = 128-bit b101-b111 = reserved.

DMAC_BUS_WD
Address:Operational Base+0xe80
DMA Watchdog Register

Bit	Attr	Reset Value	Description
-	-	-	reserved
0	RW	0x0	Controls how the DMAC responds when it detects a lock-up condition: 0 = the DMAC aborts all of the contributing DMA channels and sets irq_abort HIGH 1 = the DMAC sets irq_abort HIGH.

12.5 Timing Diagram

Following picture shows the relationship between dma_req and dma_ack.

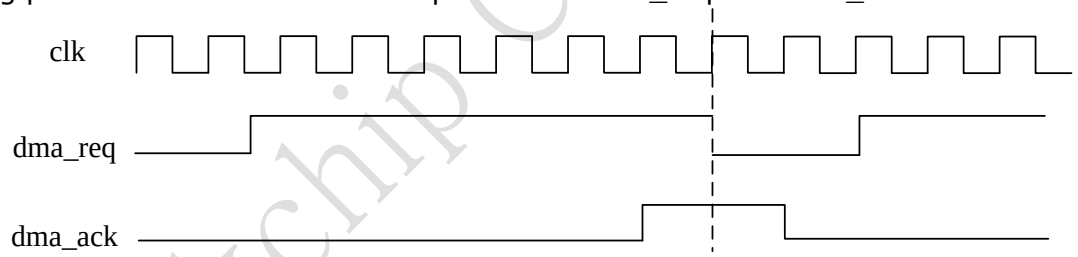

Fig. 12-3 DMAC_BUS request and acknowledge timing

12.6 Interface Description

DMAC_BUS has the following tie-off signals. It can be configured by SGRF register. (Please refer to SGRF chapters to find how to configure)

interface	Reset value	Control source
boot_addr	0x0	Secure SGRF
boot_from_pc	0x0	Secure SGRF
boot_manager_ns	0x0	Secure SGRF
boot_irq_ns	0x0	Secure SGRF
boot_periph_ns	0x0	SGRF

boot_addr
Configures the address location that contains the first instruction the DMAC executes, when it exits from reset.
boot_from_pc
Controls the location in which the DMAC executes its initial instruction, after it exits from

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reset:
0 = DMAC waits for an instruction from either APB interface
1 = DMA manager thread executes the instruction that is located at the boot_addr

boot_manager_ns

When the DMAC exits from reset, this signal controls the security state of the DMA manager thread:
0 = assigns DMA manager to the Secure state
1 = assigns DMA manager to the Non-secure state.

boot_irq_ns

Controls the security state of an event-interrupt resource, when the DMAC exits from reset:
boot_irq_ns[x] is LOW
The DMAC assigns event<x> or irq[x] to the Secure state.
boot_irq_ns[x] is HIGH
The DMAC assigns event<x> or irq[x] to the Non-secure state.

boot_periph_ns

Controls the security state of a peripheral request interface, when the DMAC exits from reset:
boot_periph_ns[x] is LOW
The DMAC assigns peripheral request interface x to the Secure state.
boot_periph_ns[x] is HIGH
The DMAC assigns peripheral request interface x to the Non-secure state.

12.7 Application Notes

12.7.1 Using the APB slave interfaces

You must ensure that you use the appropriate APB interface, depending on the security state in which the boot_manager_ns initializes the DMAC to operate. For example, if the DMAC is in the secure state, you must issue the instruction using the secure APB interface, otherwise the DMAC ignores the instruction. You can use the secure APB interface, or the non-secure APB interface, to start or restart a DMA channel when the DMAC is in the Non-secure state.
The necessary steps to start a DMA channel thread using the debug instruction registers as following:
1. Create a program for the DMA channel.
2. Store the program in a region of system memory.
3. Poll the DBGSTATUS Register to ensure that debug is idle, that is, the dbgstatus bit is 0.
4. Write to the DBGINST0 Register and enter the:
• Instruction byte 0 encoding for DMAGO.
• Instruction byte 1 encoding for DMAGO.
• Debug thread bit to 0. This selects the DMA manager thread.
5. Write to the DBGINST1 Register with the DMAGO instruction byte [5:2] data, see Debug Instruction-1 Register o. You must set these four bytes to the address of the first instruction in the program, that was written to system memory in step 2.
6. Writing zero to the DBGCMD Register. The DMAC starts the DMA channel thread and sets the dbgstatus bit to 1.

12.7.2 Security usage

When the DMAC exits from reset, the status of the configuration signals that tie-off signals which descripted in chapter 12.6.

DMA manager thread is in the secure state

If the DNS bit is 0, the DMA manager thread operates in the secure state and it only performs secure instruction fetches. When a DMA manager thread in the secure state processes:

DMAGO

It uses the status of the ns bit, to set the security state of the DMA channel thread by writing to the CNS bit for that channel.

DMAWFE

It halts execution of the thread until the event occurs. When the event occurs, the DMAC continues execution of the thread, irrespective of the security state of the corresponding INS bit.

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DMASEV
It sets the corresponding bit in the INT_EVENT_RIS Register, irrespective of the security state of the corresponding INS bit.
DMA manager thread is in the Non-secure state
If the DNS bit is 1, the DMA manager thread operates in the Non-secure state, and it only performs non-secure instruction fetches. When a DMA manager thread in the Non-secure state processes:
DMAGO
The DMAC uses the status of the ns bit, to control if it starts a DMA channel thread. If:
ns = 0
The DMAC does not start a DMA channel thread and instead it:
1. Executes a NOP.
2. Sets the FSRD Register, see Fault Status DMA Manager
3. Sets the dmago_err bit in the FTRD Register, see Fault Type DMA Manager Register.
4. Moves the DMA manager to the Faulting state.
ns = 1
The DMAC starts a DMA channel thread in the Non-secure state and programs the CNS bit to be non-secure.
DMAWFE
The DMAC uses the status of the corresponding INS bit, in the CR3 Register, to control if it waits for the event. If:
INS = 0
The event is in the Secure state. The DMAC:
1. Executes a NOP.
2. Sets the FSRD Register, see Fault Status DMA Manager Register.
3. Sets the mgr_evnt_err bit in the FTRD Register, see Fault Type DMA Manager Register.
4. Moves the DMA manager to the Faulting state.
INS = 1
The event is in the Non-secure state. The DMAC halts execution of the thread and waits for the event to occur.
DMASEV
The DMAC uses the status of the corresponding INS bit, in the CR3Register, to control if it creates the event-interrupt. If:
INS = 0
The event-interrupt resource is in the Secure state. The DMAC:
1. Executes a NOP.
2. Sets the FSRD Register, see Fault Status DMA Manager Register.
3. Sets the mgr_evnt_err bit in the FTRD Register, see Fault Type DMA Manager Register.
4. Moves the DMA manager to the Faulting state.
INS = 1
The event-interrupt resource is in the Non-secure state. The DMAC creates the event-interrupt.
DMA channel thread is in the secure state
When the CNS bit is 0, the DMA channel thread is programmed to operate in the Secure state and it only performs secure instruction fetches.
When a DMA channel thread in the secure state processes the following instructions:
DMAWFE
The DMAC halts execution of the thread until the event occurs. When the event occurs, the DMAC continues execution of the thread, irrespective of the security state of the corresponding INS bit, in the CR3 Register.
DMASEV
The DMAC creates the event-interrupt, irrespective of the security state of the corresponding INS bit, in the CR3 Register.
DMAWFP
The DMAC halts execution of the thread until the peripheral signals a DMA request. When this occurs, the DMAC continues execution of the thread, irrespective of the security state of the

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corresponding PNS bit, in the CR4 Register.

DMALDP, DMASTP

The DMAC sends a message to the peripheral to communicate that data transfer is complete, irrespective of the security state of the corresponding PNS bit, in the CR4 Register.

DMAFLUSHP

The DMAC clears the state of the peripheral and sends a message to the peripheral to resend its level status, irrespective of the security state of the corresponding PNS bit, in the CR4 Register.

When a DMA channel thread is in the Secure state, it enables the DMAC to perform secure and non-secure AXI accesses

DMA channel thread is in the Non-secure state

When the CNS bit is 1, the DMA channel thread is programmed to operate in the Non-secure state and it only performs non-secure instruction fetches.

When a DMA channel thread in the Non-secure state processes the following instructions:

DMAWFE

The DMAC uses the status of the corresponding INS bit, in the CR3 Register, to control if it waits for the event. If:

INS = 0

The event is in the Secure state. The DMAC:

1. Executes a NOP.
2. Sets the appropriate bit in the FSRC Register that corresponds to the DMA channel number. See Fault Status DMA Channel Register.
3. Sets the ch_evnt_err bit in the FTRn Register, see Fault Type DMA Channel Registers.
4. Moves the DMA channel to the Faulting completing state.

INS = 1

The event is in the Non-secure state. The DMAC halts execution of the thread and waits for the event to occur.

DMASEV

The DMAC uses the status of the corresponding INS bit, in the CR3 Register, to control if it creates the event. If:

INS = 0

The event-interrupt resource is in the Secure state. The DMAC:

1. Executes a NOP.
2. Sets the appropriate bit in the FSRC Register that corresponds to the DMA channel number. See Fault Status DMA Channel Register.
3. Sets the ch_evnt_err bit in the FTRn Register, see Fault Type DMA Channel Registers .
4. Moves the DMA channel to the Faulting completing state.

INS = 1

The event-interrupt resource is in the Non-secure state. The DMAC creates the event-interrupt.

DMAWFP

The DMAC uses the status of the corresponding PNS bit, in the CR4 Register, to control if it waits for the peripheral to signal a request. If:

PNS = 0

The peripheral is in the Secure state. The DMAC:

1. Executes a NOP.
2. Sets the appropriate bit in the FSRC Register that corresponds to the DMA channel number. See Fault Status DMA Channel Register.
3. Sets the ch_periph_err bit in the FTRn Register, see Fault Type DMA Channel Registers.
4. Moves the DMA channel to the Faulting completing state.

PNS = 1

The peripheral is in the Non-secure state. The DMAC halts execution of the thread and waits for the peripheral to signal a request.

DMALDP, DMASTP

The DMAC uses the status of the corresponding PNS bit, in the CR4 Register, to control if it sends an acknowledgement to the peripheral. If:

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PNS = 0
The peripheral is in the Secure state. The DMAC:
1. Executes a NOP.
2. Sets the appropriate bit in the FSRC Register that corresponds to the DMA channel number. See Fault Status DMA Channel Register.
3. Sets the ch_periph_err bit in the FTRn Register, see Fault Type DMA Channel Registers.
4. Moves the DMA channel to the Faulting completing state.
PNS = 1
The peripheral is in the Non-secure state. The DMAC sends a message to the peripheral to communicate when the data transfer is complete.

DMAFLUSHP

The DMAC uses the status of the corresponding PNS bit, in the CR4 Register, to control if it sends a flush request to the peripheral. If:
PNS = 0
The peripheral is in the Secure state. The DMAC:
1. Executes a NOP.
2. Sets the appropriate bit in the FSRC Register that corresponds to the DMA channel number. See Fault Status DMA Channel Registe.
3. Sets the ch_periph_err bit in the FTRn Register, see Fault Type DMA Channel Registers.
4. Moves the DMA channel to the Faulting completing state.
PNS = 1
The peripheral is in the Non-secure state. The DMAC clears the state of the peripheral and sends a message to the peripheral to resend its level status.
When a DMA channel thread is in the Non-secure state, and a DMAMOV CCR instruction attempts to program the channel to perform a secure AXI transaction, the DMAC:
1. Executes a DMANOP.
2. Sets the appropriate bit in the FSRC Register that corresponds to the DMA channel number. See Fault Status DMA Channel Registe.
3. Sets the ch_rdwr_err bit in the FTRn Register, see Fault Type DMA Channel Registers.
4. Moves the DMA channel thread to the Faulting completing state.

12.7.3 Programming restrictions

Fixed unaligned bursts

The DMAC does not support fixed unaligned bursts. If you program the following conditions, the DMAC treats this as a programming error:
Unaligned read
• src_inc field is 0 in the CCRn Register
• the SARn Register contains an address that is not aligned to the size of data that the src_burst_size field contain
Unaligned write
• dst_inc field is 0 in the CCRn Register
• the DARn Register contains an address that is not aligned to the size of data that the dst_burst_size field contains

Endian swap size restrictions

If you program the endian_swap_size field in the CCRn Register, to enable a DMA channel to perform an endian swap then you must set the corresponding SARn Register and the corresponding DARn Register to contain an address that is aligned to the value that the endian_swap_size field contains.

Updating DMA channel control registers during a DMA cyclerestrictions

Prior to the DMAC executing a sequence of DMALD and DMAST instructions, the values you program in to the CCRn Register, SARn Register, and DARn Register control the data byte lane manipulation that the DMAC performs when it transfers the data from the source address to the destination address. You'd better not update these registers during a DMA cycle.

Resource sharing between DMA channels

DMA channel programs share the MFIFO data storage resource. You must not start a set ofconcurrently running DMA channel programs with a resource requirement that exceeds theconfigured size of the MFIFO. If you exceed this limit then the DMAC might lock up and generate a Watchdog abort.

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12.7.4 Unaligned transfers may be corrupted

For a configuration with more than one channel, if any of channels 1 to 7 is performing transfers between certain types of misaligned source and destination addresses, then the output data may be corrupted by the action of channel 0.
Data corruption might occur if all of the following are true:
1. Two beats of AXI read data are received for one of channels 1 to 7.
2. Source and destination address alignments mean that each read data beat is split across two lines in the data buffer (see Splitting data, below).
3. There is one idle cycle between the two read data beats .
4. Channel 0 performs an operation that updates channel control information during this idle cycle (see Updates to channel control information, below)

Splitting data

Depending upon the programmed values for the DMA transfer, one beat of read data from the AXI interface mneed to be split across two lines in the internal data buffer. This occurs when the read data beat contains datbytes which will be written to addresses that wrap around at the AXI interface data width, so that these bytes could not be transferred by a single AXI write data beat of the full interface width.
Most applications of DMA-330 do not split data in this way, so are NOT vulnerable to data corruption from thisdefect.
The following cases are NOT vulnerable to data corruption because they do not split data:

- Byte lane offset between source and destination addresses is 0 When source and destination addresses have the same byte lane alignment, the offset is 0 and a wrap operation that splits data cannot occur.
- Byte lane offset between source and destination addresses is a multiple of source size

Source size in CCRn	Allowed offset between SARn and DARn
SS8	any offset allowed.
SS16	0,2,4,6,8,10,12,14
SS32	0,4,8,12
SS64	0,8

12.7.5 Interrupt shares between channel.

As the DMAC_BUS does not record which channel (or list of channels) have asserted an interrupt. So it will depend on your program and whether any of the visible information for that program can be used to determine progress, and help identify the interrupt source.
There are 4 likely information sources that can be used to determine the progress made by a program:

- Program counter (PC)
- Source address
- Destination address
- Loop counters (LC)

For example, a program might emit an interrupt each time that it iterates around a loop. In this case, the interrupt service routine (ISR) would need to store the loop value of each channel when it is called, and then compare against the new value when it is next called. A change in value would indicate that the program has progressed.
The ISR must be carefully written to ensure that no interrupts are lost. The sequence of operations is as follows:
1. Disable interrupts
2. Immediately clear the interrupt in DMA-330
3. Check the relevant registers for both channels to determine which must be serviced
4. Take appropriate action for the channels
5. Re-enable interrupts and exit ISR

12.7.6 Instruction sets

Table 12-2 DMAC Instruction sets

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Mnemonic	Instruction	Thread usage: ● M = DMA manager ● C = DMA channel
DMAADDH	Add Halfword	C
DMAEND	End	M/C
DMAFLUSHP	Flush and notify Peripheral	C
DMAGO	Go	M
DMAKILL	Kill	C
DMALD	Load	C
DMALDP	Load Peripheral	C
DMALP	Loop	C
DMALPEND	Loop End	C
DMALPFE	Loop Forever	C
DMAMOV	Move	C
DMANOP	No operation	M/C
DMARMB	Read Memory Barrier	C
DMASEV	Send Event	M/C
DMAST	Store	C
DMASTP	Store and notify Peripheral	C
DMASTZ	Store Zero	C
DMAWFE	Wai t For Event M	M/C
DMAWFP	Wait For Peripheral	C
DMAWMB	Write Memory Barrier	C
DMAADNH	Add Negative Halfword	C

12.7.7 Assembler directives

In this document, only DMMADNH instruction is took as an example to show the way the instruction assembl*ed. For the other instructions, please refer to DDI0424A_dmac_pl330_r0p0_trm.pdf .*

DMAADNH

Add Negative Halfword adds an immediate negative 16-bit value to the SARn Register or D ARn Register, for the DMA channel thread. This enables the DMAC to support 2D DMA oper ations, or reading or writing an area of memory in adifferent order to naturally incrementin g addresses. See Source Address Registers and Destination Address Registers.

The immediate unsigned 16-bit value is one-extended to 32 bits, to create a value that is t he two's complement representation of a negative number between -65536 and -1, before the DMAC adds it to the address using 32-bit addition. The DMAC discards the carry bit so t hat addresses wrap from 0xFFFFFFFF to 0x00000000. The net effect is to subtract between 65536 and 1 from the current value in the Source or Destination Address Register.

Following table shows the instruction encoding.

Imm[15:8]	Imm[7:0]	0	1	0	1	1	1	ra	0

Assembler syntax

DMAADNH <address_register>, <16-bit immediate>

where:

<address_register>

Selects the address register to use. It must be either:

SAR

SARn Register and sets ra to 0.

DAR

DARn Register and sets ra to 1.

<16-bit immediate>

The immediate value to be added to the <address_register>.

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You should specify the 16-bit immediate as the number that is to be represented in the inst ruction encoding. For example, DMAADNH DAR, 0xFFF0 causes the value 0xFFFFFFF0 to be added to the current value of the Destination Address Register, effectively subtracting 16 fr om the DAR.
You can only use this instruction in a DMA channel thread.

12.7.8 MFIFO usage

For MFIFO usage , please refer to pl330_trm.pdf

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Chapter 13 DMA Controller for Peripheral System (DMAC_PERI)

13.1 Overview

DMAC_PERI does not support TrustZone technology and work under non-secure state only.
DMAC_PERI is mainly used for data transfer of the following slaves: HSADC, UART_BT, UART_BB, UART_GPS, UART_EXP, SPI0, SPI1, SPI2.
Following table shows the DMAC_PERI request mapping scheme.

Table 13-1 DMAC_PERI Request Mapping Table

Req number	Source	Polarity
0	HSADC/TSI	High level
1	UART_BT tx	High level
2	UART_BT rx	High level
3	UART_BB tx	High level
4	UART_BB rx	High level
5	Reserved	
6	Reserved	
7	UART_GPS tx	High level
8	UART_GPS rx	High level
9	UART_EXP tx	High level
10	UART_EXP rx	High level
11	SPI0_tx	High level
12	SPI0_rx	High level
13	SPI1_tx	High level
14	SPI1_rx	High level
15	SPI2_tx	High level
16	SPI2_rx	High level
17	Reserved	
18	Reserved	
19	Reserved	

DMAC_PERI supports the following features:

- Supports 20 peripheral request.
- Up to 64bits data size.
- 8 channel at the same time.
- Up to burst 16.
- 16 interrupts output and 1 abort output.
- Supports 64 MFIFO depth.

13.2 Block Diagram

Figure 13-1 shows the block diagram of DMAC_PERI.

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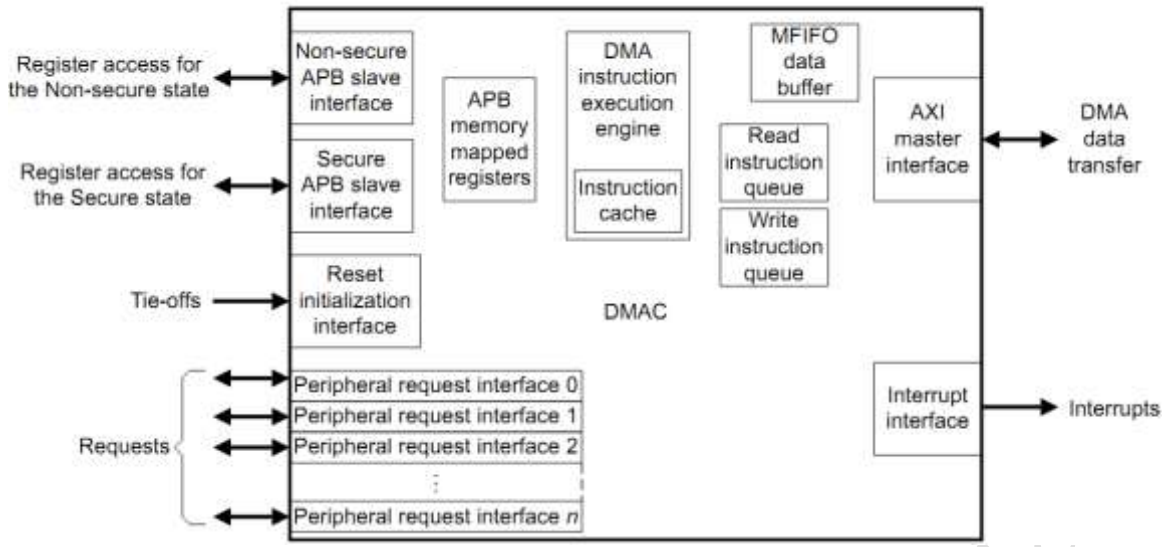

Fig. 13-1 Block diagram of DMAC_PERI

13.3 Function Description

Please refer to chapter 12.3 for the similar description.

13.4 Register Description

13.4.1 Register summary

Name	Offset	Size	Reset Value	Description
DMAC_PERI_DSR	0x0000	W	0x0	DMA Status Register.
DMAC_PERI_DPC	0x0004	W	0x0	DMA Program Counter Register.
-	-	-	-	reserved
DMAC_PERI_INTEN	0x0020	W	0x0	Interrupt Enable Register
DMAC_PERI_EVENT_RIS	0x0024	W	0x0	Event Status Register.
DMAC_PERI_INTMIS	0x0028	W	0x0	Interrupt Status Register
DMAC_PERI_INTCLR	0x002C	W	0x0	Interrupt Clear Register
DMAC_PERI_FSRD	0x0030	W	0x0	Fault Status DMA Manager Register.
DMAC_PERI_FSRC	0x0034	W	0x0	Fault Status DMA Channel Register.
DMAC_PERI_FTRD	0x0038	W	0x0	Fault Type DMA Manager Register.
-	-	-	-	reserved
DMAC_PERI_FTR0	0x0040	W	0x0	Fault type for DMA Channel 0
DMAC_PERI_FTR1	0x0044	W	0x0	Fault type for DMA Channel 1
DMAC_PERI_FTR2	0x0048	W	0x0	Fault type for DMA Channel 2
DMAC_PERI_FTR3	0x004C	W	0x0	Fault type for DMA Channel 3
DMAC_PERI_FTR4	0x0050	W	0x0	Fault type for DMA Channel 4
DMAC_PERI_FTR5	0x0054	W	0x0	Fault type for DMA Channel 5
DMAC_PERI_FTR6	0x0058	W	0x0	Fault type for DMA Channel 6
-	-	-	-	reserved
DMAC_PERI_CSR0	0x0100	W	0x0	Channel Status for DMA Channel 0
DMAC_PERI_CSR1	0x0108	W	0x0	Channel Status for DMA Channel 1
DMAC_PERI_CSR2	0x0110	W	0x0	Channel Status for DMA Channel 2
DMAC_PERI_CSR3	0x0118	W	0x0	Channel Status for DMA Channel 3
DMAC_PERI_CSR4	0x0120	W	0x0	Channel Status for DMA Channel 4
DMAC_PERI_CSR5	0x0128	W	0x0	Channel Status for DMA Channel 5
DMAC_PERI_CSR6	0x0130	W	0x0	Channel Status for DMA Channel 6
DMAC_PERI_CPC0	0x0104	W	0x0	Channel PC for DMA Channel 0
DMAC_PERI_CPC1	0x010c	W	0x0	Channel PC for DMA Channel 1

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DMAC_PERI_CPC2	0x0114	W	0x0	Channel PC for DMA Channel 2
DMAC_PERI_CPC3	0x011c	W	0x0	Channel PC for DMA Channel 3
DMAC_PERI_CPC4	0x0124	W	0x0	Channel PC for DMA Channel 4
DMAC_PERI_CPC5	0x012c	W	0x0	Channel PC for DMA Channel 5
DMAC_PERI_CPC6	0x0134	W	0x0	Channel PC for DMA Channel 6
DMAC_PERI_SAR0	0x0400	W	0x0	Source Address for DMA Channel 0
DMAC_PERI_SAR1	0x0420	W	0x0	Source Address for DMA Channel 1
DMAC_PERI_SAR2	0x0440	W	0x0	Source Address for DMA Channel 2
DMAC_PERI_SAR3	0x0460	W	0x0	Source Address for DMA Channel 3
DMAC_PERI_SAR4	0x0480	W	0x0	Source Address for DMA Channel 4
DMAC_PERI_SAR5	0x04a0	W	0x0	Source Address for DMA Channel 5
DMAC_PERI_SAR6	0x04c0	W	0x0	Source Address for DMA Channel 6
DMAC_PERI_DAR0	0x0404	W	0x0	Dest Address for DMAChannel 0
DMAC_PERI_DAR1	0x0424	W	0x0	Dest Address for DMAChannel 1
DMAC_PERI_DAR2	0x0444	W	0x0	Dest Address for DMAChannel 2
DMAC_PERI_DAR3	0x0464	W	0x0	Dest Address for DMAChannel 3
DMAC_PERI_DAR4	0x0484	W	0x0	Dest Address for DMAChannel 4
DMAC_PERI_DAR5	0x04a4	W	0x0	Dest Address for DMAChannel 5
DMAC_PERI_DAR6	0x04c4	W	0x0	Dest Address for DMAChannel 6
DMAC_PERI_CCR0	0x0408	W	0x0	Channel Control for DMA Channel 0
DMAC_PERI_CCR1	0x0428	W	0x0	Channel Control for DMA Channel 1
DMAC_PERI_CCR2	0x0448	W	0x0	Channel Control for DMA Channel 2
DMAC_PERI_CCR3	0x0468	W	0x0	Channel Control for DMA Channel 3
DMAC_PERI_CCR4	0x0488	W	0x0	Channel Control for DMA Channel 4
DMAC_PERI_CCR5	0x04a8	W	0x0	Channel Control for DMA Channel 5
DMAC_PERI_CCR6	0x04c8	W	0x0	Channel Control for DMA Channel 6
DMAC_PERI_LC0_0	0x040C	W	0x0	Loop Counter 0 for DMA Channel 0
DMAC_PERI_LC0_1	0x042C	W	0x0	Loop Counter 0 for DMA Channel 1
DMAC_PERI_LC0_2	0x044C	W	0x0	Loop Counter 0 for DMA Channel 2
DMAC_PERI_LC0_3	0x046C	W	0x0	Loop Counter 0 for DMA Channel 3
DMAC_PERI_LC0_4	0x048C	W	0x0	Loop Counter 0 for DMA Channel 4
DMAC_PERI_LC0_5	0x04aC	W	0x0	Loop Counter 0 for DMA Channel 5
DMAC_PERI_LC0_6	0x04cC	W	0x0	Loop Counter 0 for DMA Channel 6
DMAC_PERI_LC1_0	0x0410	W	0x0	Loop Counter 1 for DMA Channel 0
DMAC_PERI_LC1_1	0x0430	W	0x0	Loop Counter 1 for DMA Channel 1
DMAC_PERI_LC1_2	0x0450	W	0x0	Loop Counter 1 for DMA Channel 2
DMAC_PERI_LC1_3	0x0470	W	0x0	Loop Counter 1 for DMA Channel 3
DMAC_PERI_LC1_4	0x0490	W	0x0	Loop Counter 1 for DMA Channel 4
DMAC_PERI_LC1_5	0x04b0	W	0x0	Loop Counter 1 for DMA Channel 5
DMAC_PERI_LC1_6	0x04d0	W	0x0	Loop Counter 1 for DMA Channel 6
-	-	-	-	reserved
DMAC_PERI_DBGSTATUS	0x0D00	W	0x0	Debug Status Register.
DMAC_PERI_DBGCMD	0x0D04	W	0x0	Debug Command Register.
DMAC_PERI_DBGINST0	0x0D08	W	0x0	Debug Instruction-0 Register.
DMAC_PERI_DBGINST1	0x0D0C	W	0x0	Debug Instruction-1 Register.
DMAC_PERI_CR0	0x0E00	W		Configuration Register 0.
DMAC_PERI_CR1	0x0E04	W		Configuration Register 1.
DMAC_PERI_CR2	0x0E08	W		Configuration Register 2.
DMAC_PERI_CR3	0x0E0C	W		Configuration Register 3.
DMAC_PERI_CR4	0x0E10	W		Configuration Register 4.
DMAC_PERI_CRDn	0x0E14	W		Configuration Register Dn.

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DMAC_PERI_WD	0X0E80	W		Watchdog Register

Notes:
*Size: **B** – Byte (8 bits) access, **HW** – Half WORD (16 bits) access, **W** –WORD (32 bits) access*

13.4.2 Detail Register Description

DMAC_PERI_DSR
Address:Operational Base+0x0
DMA Manager Status Register

Bit	Attr	Reset Value	Description
31:10	-	-	Reserved
9	R	0x0	Provides the security status of the DMA manager thread: 0 = DMA manager operates in the Secure state 1 = DMA manager operates in the Non-secure state.
8:4	R	0x0	When the DMA manager thread executes a DMAWFE instruction, it waits for the following event to occur: b00000 = event[0] b00001 = event[1] b00010 = event[2] … b11111 = event[31].
3:0	R	0x0	The operating state of the DMA manager: b0000 = Stopped b0001 = Executing b0010 = Cache miss b0011 = Updating PC b0100 = Waiting for event b0101-b1110 = reserved b1111 = Faulting.

DMAC_PERI_DPC
Address:Operational Base+0x4
DMA Program Counter Register

Bit	Attr	Reset Value	Description
31:0	R	0x0	Program counter for the DMA manager thread

DMAC_PERI_INTEN
Address:Operational Base+0x20
Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:0	RW	0x0	Program the appropriate bit to control how the DMAC responds when it executes DMASEV: Bit [N] = 0 If the DMAC executes DMASEV for the event-interrupt resource N then the DMAC signals event N to all of the threads. Set bit [N] to 0 if your system design does not use irq[N] to signal an interrupt request. Bit [N] = 1 If the DMAC executes DMASEV for the event-interrupt resource N then the DMAC sets irq[N] HIGH. Set bit [N] to 1 if your system designer requires irq[N] to signal an interrupt request.

DMAC_PERI_EVENT_RIS
Address:Operational Base+0x24
Event-Interrupt Raw Status Register

Bit	Attr	Reset Value	Description

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31:0	R	0x0	Returns the status of the event-interrupt resources: Bit [N] = 0 Event N is inactive or irq[N] is LOW. Bit [N] = 1 Event N is active or irq[N] is HIGH.

DMAC_PERI_INTMIS
Address:Operational Base+0x28
Interrupt Status Register

Bit	Attr	Reset Value	Description
31:0	R	0x0	Provides the status of the interrupts that are active in the DMAC: Bit [N] = 0 Interrupt N is inactive and therefore irq[N] is LOW. Bit [N] = 1 Interrupt N is active and therefore irq[N] is HIGH

DMAC_PERI_INTCLR
Address:Operational Base+0x2c
Interrupt Clear Register

Bit	Attr	Reset Value	Description
31:0	W	0x0	Controls the clearing of the irq outputs: Bit [N] = 0 The status of irq[N] does not change. Bit [N] = 1 The DMAC sets irq[N] LOW if the INTEN Register programs the DMAC to signal an interrupt. Otherwise, the status of irq[N] does not change.

DMAC_PERI_FSRD
Address:Operational Base+0x30
Fault Status DMA Manager Register

Bit	Attr	Reset Value	Description
31:0	R	0x0	Provides the fault status of the DMA manager. Read as: 0 = the DMA manager thread is not in the Faulting state 1 = the DMA manager thread is in the Faulting state.

DMAC_PERI_FSRC
Address:Operational Base+0x34
Fault Status DMA Channel Register

Bit	Attr	Reset Value	Description
31:0	R	0x0	Each bit provides the fault status of the corresponding channel. Read as: Bit [N] = 0 No fault is present on DMA channel N. Bit [N] = 1 DMA channel N is in the Faulting or Faulting completing state.

DMAC_PERI_FTRD
Address:Operational Base+0x38
Fault Type DMA Manager Register

Bit	Attr	Reset Value	Description
31	-	-	reserved
30	R	0x0	If the DMA manager aborts, this bit indicates if the erroneous instruction was read from the system memory or from the debug interface: 0 = instruction that generated an abort was read from system memory 1 = instruction that generated an abort was read from the debug interface.
29:17	-	-	reserved

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16	R	0x0	Indicates the AXI response that the DMAC receives on the RRESP bus, after the DMA manager performs an instruction fetch: 0 = OKAY response 1 = EXOKAY, SLVERR, or DECERR response
15:6	-	-	reserved
5	R	0x0	Indicates if the DMA manager was attempting to execute DMAWFE or DMASEV with inappropriate security permissions: 0 = DMA manager has appropriate security to execute DMAWFE or DMASEV 1 = a DMA manager thread in the Non-secure state attempted to execute either: • DMAWFE to wait for a secure event • DMASEV to create a secure event or secure interrupt
4	R	0x0	ndicates if the DMA manager was attempting to execute DMAGO with inappropriate security permissions: 0 = DMA manager has appropriate security to execute DMAGO 1 = a DMA manager thread in the Non-secure state attempted to execute DMAGO to create a DMA channel operating in the Secure state.
3:2	-	-	reserved
1	R	0x0	Indicates if the DMA manager was attempting to execute an instruction operand that was not valid for the configuration of the DMAC: 0 = valid operand 1 = invalid operand.
0	R	0x0	Indicates if the DMA manager was attempting to execute an undefined instruction: 0 = defined instruction 1 = undefined instruction.

DMAC_PERI_FTR0~DMAC_PERI_FTR6

Address:Operational Base+0x40
Operational Base+0x44
Operational Base+0x48
Operational Base+0x4c
Operational Base+0x50
Operational Base+0x54
Operational Base+0x58

Fault Type DMA Channel Register

Bit	Attr	Reset Value	Description
31	R	0x0	Indicates if the DMA channel has locked-up because of resource starvation: 0 = DMA channel has adequate resources 1 = DMA channel has locked-up because of insufficient resources. This fault is an imprecise abort
30	R	0x0	If the DMA channel aborts, this bit indicates if the erroneous instruction was read from the system memory or from the debug interface: 0 = instruction that generated an abort was read from system memory 1 = instruction that generated an abort was read from the debug interface.

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			This fault is an imprecise abort but the bit is only valid when a precise abort occurs.
29:19	-	-	reserved
18	R	0x0	Indicates the AXI response that the DMAC receives on the RRESP bus, after the DMA channel thread performs a data read: 0 = OKAY response 1 = EXOKAY, SLVERR, or DECERR response. This fault is an imprecise abort
17	R	0x0	Indicates the AXI response that the DMAC receives on the BRESP bus, after the DMA channel thread performs a data write: 0 = OKAY response 1 = EXOKAY, SLVERR, or DECERR response. This fault is an imprecise abort.
16	R	0x0	Indicates the AXI response that the DMAC receives on the RRESP bus, after the DMA channel thread performs an instruction fetch: 0 = OKAY response 1 = EXOKAY, SLVERR, or DECERR response. This fault is a precise abort.
15:14	-	-	reserved
13	R	0x0	Indicates if the MFIFO did not contain the data to enable the DMAC to perform the DMAST: 0 = MFIFO contains all the data to enable the DMAST to complete 1 = previous DMALDs have not put enough data in the MFIFO to enable the DMAST to complete. This fault is a precise abort.
12	R	0x0	Indicates if the MFIFO prevented the DMA channel thread from executing DMALD or DMAST. Depending on the instruction: DMALD 0 = MFIFO contains sufficient space 1 = MFIFO is too small to hold the data that DMALD requires. DMAST 0 = MFIFO contains sufficient data 1 = MFIFO is too small to store the data to enable DMAST to complete. This fault is an imprecise abort
11:8	-	-	reserved
7	R	0x0	Indicates if a DMA channel thread, in the Non-secure state, attempts to program the CCRn Register to perform a secure read or secure write: 0 = a DMA channel thread in the Non-secure state is not violating the security permissions 1 = a DMA channel thread in the Non-secure state attempted to perform a secure read or secure write. This fault is a precise abort
6	R	0x0	Indicates if a DMA channel thread, in the Non-secure state, attempts to execute DMAWFP, DMALDP, DMASTP, or DMAFLUSHP with inappropriate security permissions: 0 = a DMA channel thread in the Non-secure state is not violating the security permissions 1 = a DMA channel thread in the Non-secure state attempted to execute either: • DMAWFP to wait for a secure peripheral • DMALDP or DMASTP to notify a secure peripheral

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			• DMAFLUSHP to flush a secure peripheral. This fault is a precise abort.
5	R	0x0	Indicates if the DMA channel thread attempts to execute DMAWFE or DMASEV with inappropriate security permissions: 0 = a DMA channel thread in the Non-secure state is not violating the security permissions 1 = a DMA channel thread in the Non-secure state attempted to execute either: • DMAWFE to wait for a secure event • DMASEV to create a secure event or secure interrupt. This fault is a precise abort.
4:2	-	-	reserved
1	R	0x0	Indicates if the DMA channel thread was attempting to execute an instruction operand that was not valid for the configuration of the DMAC: 0 = valid operand 1 = invalid operand. This fault is a precise abort.
0	R	0x0	Indicates if the DMA channel thread was attempting to execute an undefined instruction: 0 = defined instruction 1 = undefined instruction. This fault is a precise abort

DMAC_PERI_CSR0~DMAC_PERI_CSR6

Address:Operational Base+0x100
Operational Base+0x108
Operational Base+0x110
Operational Base+0x118
Operational Base+0x120
Operational Base+0x128
Operational Base+0x130

Channel Status Registers

Bit	Attr	Reset Value	Description
31:22	-	-	reserved
21	R	0x0	The channel non-secure bit provides the security of the DMA channel: 0 = DMA channel operates in the Secure state 1 = DMA channel operates in the Non-secure state
20:16	-	-	reserved
15	R	0x0	When the DMA channel thread executes DMAWFP this bit indicates if the periph operand was set: 0 = DMAWFP executed with the periph operand not set 1 = DMAWFP executed with the periph operand set
14	R	0x0	When the DMA channel thread executes DMAWFP this bit indicates if the burst or single operand were set: 0 = DMAWFP executed with the single operand set 1 = DMAWFP executed with the burst operand set.
13:9	-	-	reserved
8:4	R	0x0	If the DMA channel is in the Waiting for event state or the Waiting for peripheral state then these bits indicate the event or peripheral number that the channel is waiting for: b00000 = DMA channel is waiting for event, or peripheral, 0 b00001 = DMA channel is waiting for event, or peripheral, 1 b00010 = DMA channel is waiting for event, or peripheral, 2 ...

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			b11111 = DMA channel is waiting for event, or peripheral, 31
3:0	R	0x0	The channel status encoding is: b0000 = Stopped b0001 = Executing b0010 = Cache miss b0011 = Updating PC b0100 = Waiting for event b0101 = At barrier b0110 = reserved b0111 = Waiting for peripheral b1000 = Killing b1001 = Completing b1010-b1101 = reserved b1110 = Faulting completing b1111 = Faulting

DMAC_PERI_CPC0~DMAC_PERI_CPC6

Address:Operational Base+0x104
Operational Base+0x10c
Operational Base+0x114
Operational Base+0x11c
Operational Base+0x124
Operational Base+0x12c
Operational Base+0x134

Channel Program Counter Registers

Bit	Attr	Reset Value	Description
31:0	R	0x0	Program counter for the DMA channel n thread

DMAC_PERI_SAR0~DMAC_PERI_SAR6

Address:Operational Base+0x400
Operational Base+0x420
Operational Base+0x440
Operational Base+0x460
Operational Base+0x480
Operational Base+0x4a0
Operational Base+0x4c0

Source Address Registers

Bit	Attr	Reset Value	Description
31:0	R	0x0	Address of the source data for DMA channel n

DMAC_PERI_DAR0~DMAC_PERI_DAR5

Address:Operational Base+0x404
Operational Base+0x424
Operational Base+0x444
Operational Base+0x464
Operational Base+0x484
Operational Base+0x4a4

DestinationAddress Registers

Bit	Attr	Reset Value	Description
31:0	R	0x0	Address of the Destinationdata for DMA channel n

DMAC_PERI_CCR0~DMAC_PERI_CCR6

Address:Operational Base+0x408
Operational Base+0x428
Operational Base+0x448
Operational Base+0x468

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Operational Base+0x488
Operational Base+0x4a8
Operational Base+0x4c8

Channel Control Registers

Bit	Attr	Reset Value	Description
31:28	-	-	reserved
27:25	R	0x0	Programs the state of AWCACHE[3,1:0]a when the DMAC writes the destination data. Bit [27] 0 = AWCACHE[3] is LOW 1 = AWCACHE[3] is HIGH. Bit [26] 0 = AWCACHE[1] is LOW 1 = AWCACHE[1] is HIGH. Bit [25] 0 = AWCACHE[0] is LOW 1 = AWCACHE[0] is HIGH
24:22	R	0x0	Programs the state of AWPROT[2:0]a when the DMAC writes the destination data. Bit [24] 0 = AWPROT[2] is LOW 1 = AWPROT[2] is HIGH. Bit [23] 0 = AWPROT[1] is LOW 1 = AWPROT[1] is HIGH. Bit [22] 0 = AWPROT[0] is LOW 1 = AWPROT[0] is HIGH
21:18	R	0x0	For each burst, these bits program the number of data transfers that the DMAC performs when it writes the destination data: b0000 = 1 data transfer b0001 = 2 data transfers b0010 = 3 data transfers ... b1111 = 16 data transfers. The total number of bytes that the DMAC writes out of the MFIFO when it executes a DMAST instruction is the product of dst_burst_len and dst_burst_size
17:15	R	0x0	For each beat within a burst, it programs the number of bytes that the DMAC writes to the destination: b000 = writes 1 byte per beat b001 = writes 2 bytes per beat b010 = writes 4 bytes per beat b011 = writes 8 bytes per beat b100 = writes 16 bytes per beat b101-b111 = reserved. The total number of bytes that the DMAC writes out of the MFIFO when it executes a DMAST instruction is the product of dst_burst_len and dst_burst_size.
14	R	0x0	Programs the burst type that the DMAC performs when it writes the destination data: 0 = Fixed-address burst. The DMAC signals AWBURST[0] LOW. 1 = Incrementing-address burst. The DMAC signals AWBURST[0] HIGH.
13:11	R	0x0	Set the bits to control the state of ARCACHE[2:0]a when the DMAC reads the source data. Bit [13] 0 = ARCACHE[2] is LOW 1 = ARCACHE[2] is HIGH. Bit [12] 0 = ARCACHE[1] is LOW 1 = ARCACHE[1] is HIGH. Bit [11] 0 = ARCACHE[0] is LOW

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			1 = ARCACHE[0] is HIGH.
10:8	R	0x0	Programs the state of ARPROT[2:0]a when the DMAC reads the source data. Bit [10] 0 = ARPROT[2] is LOW 1 = ARPROT[2] is HIGH. Bit [9] 0 = ARPROT[1] is LOW 1 = ARPROT[1] is HIGH. Bit [8] 0 = ARPROT[0] is LOW 1 = ARPROT[0] is HIGH.
7:4	R	0x0	For each burst, these bits program the number of data transfers that the DMAC performs when it reads the source data: b0000 = 1 data transfer b0001 = 2 data transfers b0010 = 3 data transfers ... b1111 = 16 data transfers. The total number of bytes that the DMAC reads into the MFIFO when it executes a DMALD instruction is the product of src_burst_len and src_burst_size
3:1	R	0x0	For each beat within a burst, it programs the number of bytes that the DMAC reads from the source: b000 = reads 1 byte per beat b001 = reads 2 bytes per beat b010 = reads 4 bytes per beat b011 = reads 8 bytes per beat b100 = reads 16 bytes per beat b101-b111 = reserved. The total number of bytes that the DMAC reads into the MFIFO when it executes a DMALD instruction is the product of src_burst_len and src_burst_size
0	R	0x0	Programs the burst type that the DMAC performs when it reads the source data: 0 = Fixed-address burst. The DMAC signals ARBURST[0] LOW. 1 = Incrementing-address burst. The DMAC signals ARBURST[0] HIGH

DMAC_PERI_LC0_0~DMAC_PERI_LC0_6

Address:Operational Base+0x40c
Operational Base+0x42c
Operational Base+0x44c
Operational Base+0x46c
Operational Base+0x48c
Operational Base+0x4ac
Operational Base+0x4cc

Loop Counter 0 Registers

Bit	Attr	Reset Value	Description
31:8	-	-	reserved
7:0	R	0x0	Loop counter 0 iterations

DMAC_PERI_LC1_0~DMAC_PERI_LC1_6

Address:Operational Base+0x410
Operational Base+0x430
Operational Base+0x450
Operational Base+0x470
Operational Base+0x490

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Operational Base+0x4b0
Operational Base+0x4e0

Loop Counter 1 Registers

Bit	Attr	Reset Value	Description
31:8	-	-	reserved
7:0	R	0x0	Loop counter 1 iterations

DMAC_PERI_DBGSTATUS
Address:Operational Base+0xd00
Debug Status Register

Bit	Attr	Reset Value	Description
31:2	-	-	reserved
1:0	R	0x0	The debug encoding is as follows: b00 = execute the instruction that the DBGINST [1:0] Registers contain b01 = reserved b10 = reserved b11 = reserved.

DMAC_PERI_DBGCMD
Address:Operational Base+0xd04
Debug Command Register

Bit	Attr	Reset Value	Description
31:2	-	-	reserved
1:0	W	0x0	The debug encoding is as follows: b00 = execute the instruction that the DBGINST [1:0] Registers contain b01 = reserved b10 = reserved b11 = reserved

DMAC_PERI_DBGINST0
Address:Operational Base+0xd08
Debug Instruction-0 Register

Bit	Attr	Reset Value	Description
31:24	W	0x0	Instruction byte 1
23:16	W	0x0	Instruction byte 0
17:11	-	-	reserved
10:8	W	0x0	DMA channel number: b000 = DMA channel 0 b001 = DMA channel 1 b010 = DMA channel 2 … b111 = DMA channel 7
7:1	-	-	reserved
0	W	0x0	The debug thread encoding is as follows: 0 = DMA manager thread 1 = DMA channel.

DMAC_PERI_DBGINST1
Address:Operational Base+0xd0c
Debug Instruction-1 Register

Bit	Attr	Reset Value	Description
31:24	W	0x0	Instruction byte 5
23:16	W	0x0	Instruction byte 4

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15:8	W	0x0	Instruction byte 3
7:0	W	0x0	Instruction byte 2

DMAC_PERI_CR0
Address:Operational Base+0xe00
Configuration Register 0

Bit	Attr	Reset Value	Description
31:22	-	-	reserved
21:17	R	0x2	Number of interrupt outputs that the DMAC provides: b00000 = 1 interrupt output, irq[0] b00001 = 2 interrupt outputs, irq[1:0] b00010 = 3 interrupt outputs, irq[2:0] ... b11111 = 32 interrupt outputs, irq[31:0].
16:12	R	0x7	Number of peripheral request interfaces that the DMAC provides: b00000 = 1 peripheral request interface b00001 = 2 peripheral request interfaces b00010 = 3 peripheral request interfaces ... b11111 = 32 peripheral request interfaces.
11:7	-	-	reserved
6:4	R	0x5	Number of DMA channels that the DMAC supports: b000 = 1 DMA channel b001 = 2 DMA channels b010 = 3 DMA channels ... b111 = 8 DMA channels.
3	-	-	reserved
2	R	0x0	Indicates the status of the boot_manager_ns signal when the DMAC exited from reset: 0 = boot_manager_ns was LOW 1 = boot_manager_ns was HIGH.
1	R	0x0	Indicates the status of the boot_from_pc signal when the DMAC exited from reset: 0 = boot_from_pc was LOW 1 = boot_from_pc was HIGH
0	R	0x1	Supports peripheral requests: 0 = the DMAC does not provide a peripheral request interface 1 = the DMAC provides the number of peripheral request interfaces that the num_periph_req field specifies.

DMAC_PERI_CR1
Address:Operational Base+0xe04
Configuration Register 1

Bit	Attr	Reset Value	Description
31:8	-	-	reserved
7:4	R	0x5	[7:4] num_i-cache_lines Number of i-cache lines: b0000 = 1 i-cache line b0001 = 2 i-cache lines b0010 = 3 i-cache lines …

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			b1111 = 16 i-cache lines.
3	-	-	reserved
2:0	R	0x7	The length of an i-cache line: b000-b001 = reserved b010 = 4 bytes b011 = 8 bytes b100 = 16 bytes b101 = 32 bytes b110-b111 = reserved

DMAC_PERI_CR2
Address:Operational Base+0xe08
Configuration Register 2

Bit	Attr	Reset Value	Description
31:0	R	0x0	Provides the value of boot_addr[31:0] when the DMAC exited from reset

DMAC_PERI_CR3
Address:Operational Base+0xe0c
Configuration Register 3

Bit	Attr	Reset Value	Description
31:0	R	0x0	Provides the security state of an event-interrupt resource: Bit [N] = 0 Assigns event<N> or irq[N] to the Secure state. Bit [N] = 1 Assigns event<N> or irq[N] to the Non-secure state.

DMAC_PERI_CR4
Address:Operational Base+0xe10
Configuration Register 4

Bit	Attr	Reset Value	Description
31:0	R	0x6	Provides the security state of the peripheral request interfaces: Bit [N] = 0 Assigns peripheral request interface N to the Secure state. Bit [N] = 1 Assigns peripheral request interface N to the Non-secure state

DMAC_PERI_CRDn
Address:Operational Base+0xe14
DMA Configuration Register

Bit	Attr	Reset Value	Description
31:30	-	-	reserved
29:20	R	0x20	The number of lines that the data buffer contains: b000000000 = 1 line b000000001 = 2 lines … b111111111 = 1024 lines
19:16	R	0x9	The depth of the read queue: b0000 = 1 line b0001 = 2 lines ... b1111 = 16 lines.
15	-	-	reserved
14:12	R	0x4	Read issuing capability that programs the

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			number of outstanding read transactions: b000 = 1 b001 = 2 … b111 = 8
11:8	R	0x7	The depth of the write queue: b0000 = 1 line b0001 = 2 lines … b1111 = 16 lines.
7	-	-	reserved
6:4	R	0x3	Write issuing capability that programs the number of outstanding write transactions: b000 = 1 b001 = 2 … b111 = 8
3	-	-	reserved
2:0		0x3	The data bus width of the AXI interface: b000 = reserved b001 = reserved b010 = 32-bit b011 = 64-bit b100 = 128-bit b101-b111 = reserved.

DMAC_PERI_WD
Address:Operational Base+0xe80
DMA Watchdog Register

Bit	Attr	Reset Value	Description
31:1	-	-	reserved
0	RW	0x0	Controls how the DMAC responds when it detects a lock-up condition: 0 = the DMAC aborts all of the contributing DMA channels and sets irq_abort HIGH 1 = the DMAC sets irq_abort HIGH.

13.5 Timing Diagram

Please refer to chapter 12.5 for the similar description.

13.6 Interface Description

DMAC_PERI has the following tie-off signals. They are configured as constant in the design and can't be changed by software.
DMAC_PERI

interface	Constant value
boot_addr	0x0
boot_from_pc	0x0
boot_manager_ns	0x1
boot_irq_ns	0xffff
boot_periph_ns	0xfffff

boot_addr
Configures the address location that contains the first instruction the DMAC executes, when it exits from reset.

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boot_from_pc
Controls the location in which the DMAC executes its initial instruction, after it exits from reset:
0 = DMAC waits for an instruction from either APB interface
1 = DMA manager thread executes the instruction that is located at the boot_addr
boot_manager_ns
When the DMAC exits from reset, this signal controls the security state of the DMA manager thread:
0 = assigns DMA manager to the Secure state
1 = assigns DMA manager to the Non-secure state.
boot_irq_ns
Controls the security state of an event-interrupt resource, when the DMAC exits from reset:
boot_irq_ns[x] is LOW
The DMAC assigns event<x> or irq[x] to the Secure state.
boot_irq_ns[x] is HIGH
The DMAC assigns event<x> or irq[x] to the Non-secure state.
boot_periph_ns
Controls the security state of a peripheral request interface, when the DMAC exits from reset:
boot_periph_ns[x] is LOW
The DMAC assigns peripheral request interface x to the Secure state.
boot_periph_ns[x] is HIGH
The DMAC assigns peripheral request interface x to the Non-secure state.

13.7 Application Notes

Please refer to chapter 12.7 for the similar description.

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Chapter 14 Memory-Management-Unit (MMU)

14.1 Overview

An MMU controls address translation, access permissions, memory attribute determination, and checking at a memory system level.
There are two typs of MMU in PX5.
The MMU used in pd_peri can be refered to the document DDI0472A_corelink_mmu_400_r0p0_trm.pdf.
The MMU used in other modules will be introduced below.

14.2 Block Diagram

The MMU divides memory into 4KB pages, where each page can be individually configured.
The MMU uses a 2-level page table structure:
1. The first level, the Page Directory consists of 1024 Directory Table Entries (DTEs), each pointing to a Page Table.
2. The second level, the Page Table consists of 1024 Page Table Entries (PTEs), each pointing to a page in memory
Fig. 14-1 shows the structure of the two-level page table.

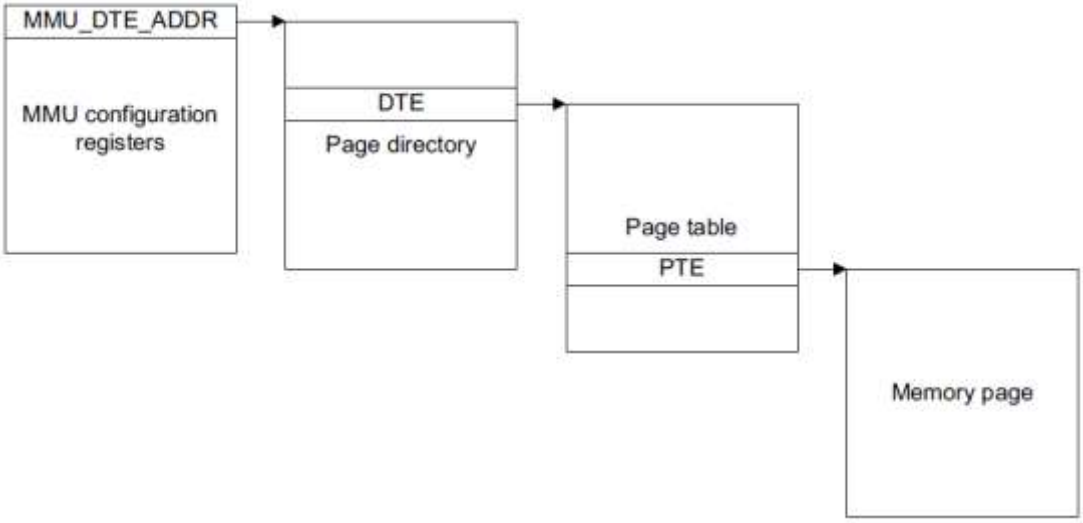

Fig. 14-1 Power Domain Partition

Fig. 4-2 shows the arrangement of the MMU address bits.

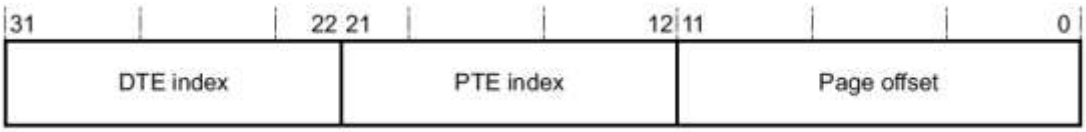

Fig. 14-2 Power Domain Partition

The MMU uses the following algorithm to translate an address:
1. Find the DTE at address given by:
MMU_DTE_ADDR + (4 * DTE index)
2. Find the PTE at address given by:
Page table address from DTE + (4 * PTE index)
3. Calculate effective address as follows:
Page address from PTE + Page offset.

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The page directory is a 4KB data structure that contains 1024 32-bit DTEs. The page directory must align at a 4KB boundary in memory.
Each DTE contains the address of a page table and a page table present bit. The system:
• initializes the entire page directory before use
• clears the page table present bit for any DTE that does not point to a valid page table.

Fig. 14-3 shows the page bit assignments.

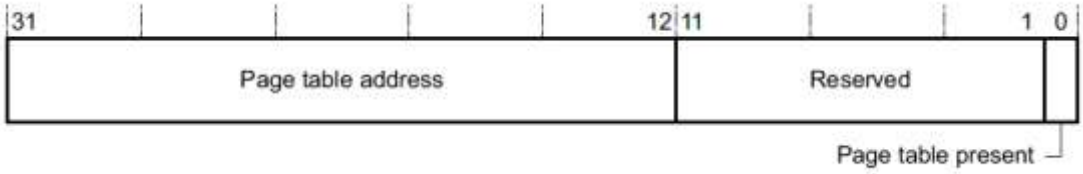

Fig. 14-3 Page directory entry bit assignments

Bits	Name	Function
[31:12]	Page table address	This field stores bits [31:12] of the address for a page table
[11:1]	Reserved	Reserved, write as zero
[0]	Page table present	This bit indicates when the page table address points to a valid page table. 0 = page table not valid 1= page table valid.

The page table is a 4KB data structure containing 1024 32-bit PTEs. The page table must be aligned at a 4KB boundary in memory.
Each PTE contains the address of a page of memory, a Page Table present bit, and Read/Write Permission bits. The entire Page Table must be initialized before use, and any PTE not pointing to a valid page must clear the Page Present bit.
Fig.14-4 shows the page table entry bit assignments.

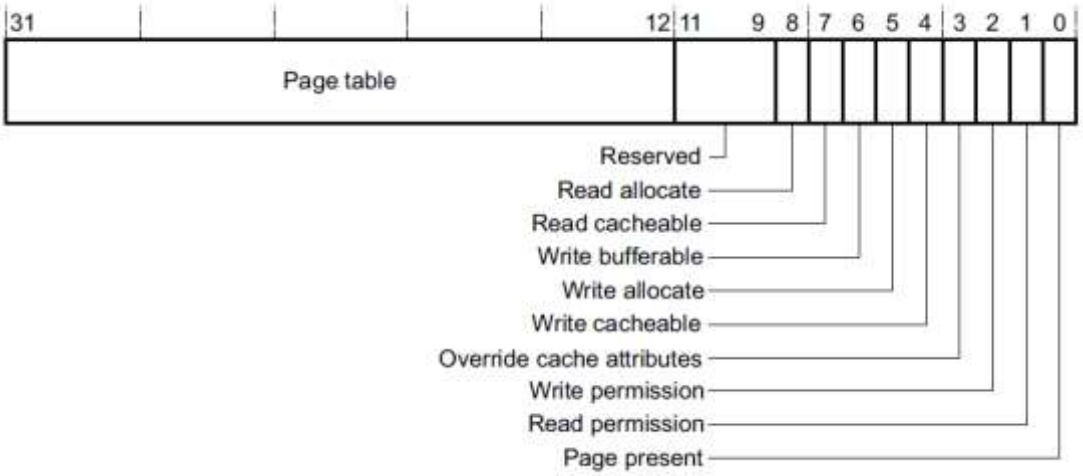

Fig. 14-4 Page directory entry bit assignments

Bits	Name	Function
[31:12]	Page table address	This field stores bits [31:12] of the address for a page table
[11:9]	Reserved	Reserved, write as zero

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[8]	Read allocate	If set, allocate cache space on read misses. Must not be set if the Read cacheable bit is not set. Only used for reads, if the Override cache attributes bit is set.
[7]	Read cacheable	If set, enable caching or prefetching of data. Only used for reads, if the Override cache attributes bit is set.
[6]	Write bufferable	If set, enable write to be delayed on their way to memory. Only used for writes, if the Override cache attributes bit is set.
[5]	Write allocate	If set, allocate cache space on write misses. Must not be set if the Write cachecable bit is not set. Only used for writes, if the Override cache attributes is set.
[4]	Write cachecable	If set, enable different writes to be merged together. Only used for writes, if the Override cache attributes bit is set.
[3]	Override cache attributes	If set, the cacheability attributes specified in bits [8:4] are used to control the cache attribytes used on the memory bus. If cleared, the default cacheability attributes from the specific processors are used on the system bus.
[2]	Write permission	Enable write accesses to the page, if present.
[1]	Read permission	Enable read accesses from the page, if present.
[0]	Page present	This bit indicates when the page table field points to a valid page. 0 = page not valid 1 = page valid.

14.3 Register Description

14.3.1 Register Summary

Name	Offset	Size	Reset Value	Description
MMU_DTE_ADDR	0x0000	W	0x00000000	MMU current page table address
MMU_STATUS	0x0004	W	0x00000018	MMU status register
MMU_CMD	0x0008	W	0x00000000	MMU command register
MMU_PAGE_FAULT_ADDR	0x000c	W	0x00000000	MMU logic address of last page fault register
MMU_ZAP_ONE_LINE	0x0010	W	0x00000000	MMU zap cache line register
MMU_INT_RAWSTAT	0x0014	W	0x00000000	MMU raw interrupt status register
MMU_INT_CLEAR	0x0018	W	0x00000000	MMU interrupt clear register
MMU_INT_MASK	0x001c	W	0x00000000	MMU interrupt mask register
MMU_INT_STATUS	0x0020	W	0x00000000	MMU interrupt status register
MMU_AUTO_GATING	0x0024	W	0x00000000	clock atuo gating register

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

14.3.2 Detail Register Description

MMU_DTE_ADDR
Address: Operational Base + offset (0x0000)

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MMU current page table address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mmu_dte_addr page table address

MMU_STATUS

Address: Operational Base + offset (0x0004)

MMU status register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:6	RO	0x00	mmu_page_fault_bus_id Index of master responsible for the last page fault
5	RO	0x0	mmu_page_fault_is_write The direction of access for last page fault: 0: read 1:write
4	RO	0x1	mmu_replay_buffer_empty The MMU replay buffer is empty.
3	RO	0x1	mmu_idle the MMu is idle when accesses are being translated and there are no unfinished translated access. The MMU_IDLE signal only reports idle when the MMU processor is idle and accesses are active on the external bus. Note: the MMU can be idle in page fault mode.
2	RO	0x0	mmu_stall_active MMU stall mode currently enabled. The mode is enabled by command.
1	RO	0x0	mmu_page_fault_active MMU page fault mode currently enabled.The mode is enabled by command
0	RO	0x0	mmu_paging_enabled mmu paging is enabled.

MMU_CMD

Address: Operational Base + offset (0x0008)

MMU command register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
2:0	RW	0x0	mmu_cmd 0: MMU_ENABLE_PAGING. enable paging. 1: MMU_DISABLE_PAGING. disable paging. 2: MMU_ENABLE_STALL. turn on stall mode. 3: MMU_DISABLE_STALL. turn off stall mode. 4: MMU_ZAP_CACHE. zap the entire page table cache. 5: MMU_PAGE_FAULT_DONE. leave page fault mode. 6: MMU_FORCE_RESET. reset the mmu. The MMU_ENABLE_STALL command can always be issued. Other commands are ignored unless the MMU is idle or stalled.

MMU_PAGE_FAULT_ADDR
Address: Operational Base + offset (0x000c)
MMU logic address of last page fault register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	mmu_page_fault_addr address of last page fault

MMU_ZAP_ONE_LINE
Address: Operational Base + offset (0x0010)
MMU zap cache line register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mmu_zap_one_line address to be invalidated from the page table cache.

MMU_INT_RAWSTAT
Address: Operational Base + offset (0x0014)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RO	0x0	read_bus_error read bus error
0	RO	0x0	page_fault page fault

MMU_INT_CLEAR
Address: Operational Base + offset (0x0018)
MMU interrupt clear register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1	RW	0x0	read_bus_error_clear read bus error interrupt clear. write 1 to this register can clear read bus error interrupt.
0	RW	0x0	page_fault_clear page fault interrupt clear, write 1 to this register can clear page fault interrupt.

MMU_INT_MASK
Address: Operational Base + offset (0x001c)
MMU interrupt mask register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	read_bus_error_int_en read bus error interrupt enable
0	RW	0x0	page_fault_int_en page fault interrupt enable

MMU_INT_STATUS
Address: Operational Base + offset (0x0020)
MMU interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	read_bus_error read bus error interrupt
0	RO	0x0	page_fault page fault interrupt

MMU_AUTO_GATING
Address: Operational Base + offset (0x0024)
clock atuo gating register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	mmu_atuo_gating mmu clock auto gating when it is 1, the mmu will auto gating itself

14.4 MMU Base address

The table below shows the MMU base address in different modules.

ISP_MMU0_BASE	ISP_BASEADDR + 0x4000
ISP_MMU1_BASE	ISP_BASEADDR + 0x5000
VOP_MMU_BASE	VOP_BASEADDR + 0x300
IEP_MMU_BASE	IEP_BASE + 0x800
VIP_MMU_BASE	VIP_BASE + 0x800
HDCP_MMU_BASE	0xff940000

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Chapter 15 Generic Interrupt Controller (GIC)

15.1 Overview

There is a generic interrupt controller (GIC400) in PX5 which generates physical interrupts to both Cortex-A53 clusters. It has two interfaces, the distributor interface connects to the interrupt source, and the CPU interface connects to Cortex-A53 little cluster and big cluster. The details of CPU interface connectivity are shown in the following table.

Table 15-1 CPU interface connectivity

CPU Interface Number	Connectivity
CPU interface 0	Little Cluster CPU0
CPU interface 1	Little Cluster CPU1
CPU interface 2	Little Cluster CPU2
CPU interface 3	Little Cluster CPU3
CPU interface 4	Big Cluster CPU0
CPU interface 5	Big Cluster CPU1
CPU interface 6	Big Cluster CPU2
CPU interface 7	Big Cluster CPU3

It supports the following features:

- Supports 160 hardware interrupt inputs
- Masking of any interrupts
- Prioritization of interrupts
- Distribution of the interrupts to the target Cortex-A53 processor(s)
- Generation of interrupts by software
- Supports Security Extensions

Note: The Revision of GIC-400 in PX5 is r0p1-00rel0

15.2 Block Diagram

The generic interrupt controller comprises with:

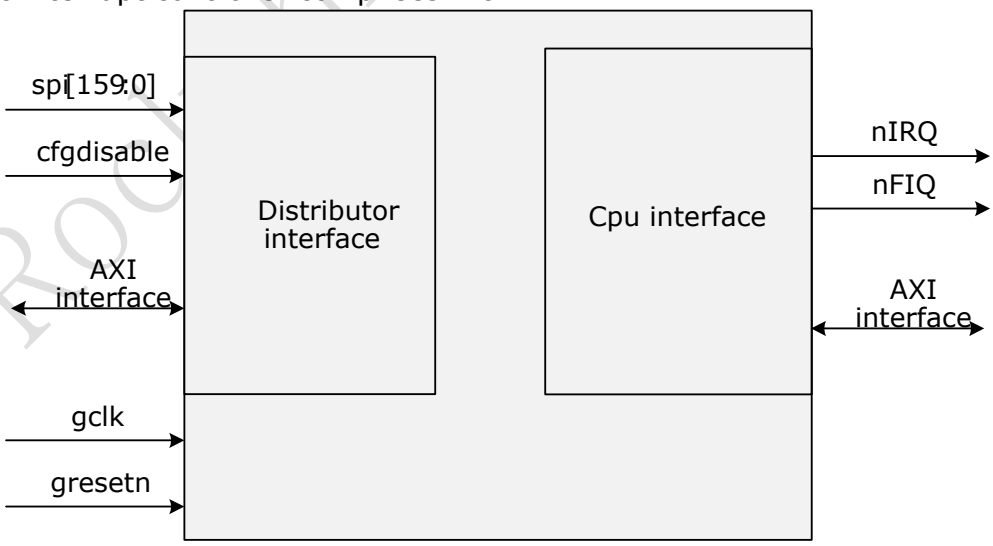

Fig. 15-1 Block Diagram

15.3 Function Description

Please refer to the document IHI0048B_gic_architecture_specification.pdf from ARM for the detail function description.

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Chapter 16 Dynamic Memory Interface (DMC)

16.1 Overview

The DMC includes two section: dynamic ram protocol controller(PCTL) and PHY.
The PCTL SoC application bus interface supports a lowest-latency native application interface (NIF). To maximize data transfer efficiency, NIF commands transfer data without flow control. To simplify command processing, the NIF accepts addresses in rank, bank, row, column format.
The DDR PHY provides control features to ease the customer implementation of digitally controlled features of the PHY such as initialization, DQS gate training, write leveling training and programmable configuration controls. The DDR PHY has built-in self test features to provide support for production testing of the compatible PHY. It also provides a DFI 2.1.1 interface to the PCTL.
The DMC supports the following features:

- Complete, integrated DDR3, LPDDR2 , LPDDR3 solution
- DFI 2.1.1 interface compatibility
- Up to 1600 Mbps in 1:2 frequency ratio for DDR3 and LPDDR3, using a 400MHz controller clock and 800MHz memory clock.
- Up to 1066 Mbps in 1:2 frequency ratio for LPDDR2, using a 266MHz controller clock and 533MHz memory clock.
- Support for x16, x32 DDR3 memories, for a total memory data path width of 32 bits
- Support for x16, x32 LPDDR2 memories, for a total memory data path width of 32 bits
- Support for x16, x32 LPDDR3 memories, for a total memory data path width of 32 bits
- Up to 2 memory ranks; devices within a rank tie to a common chip select
- Up to 8 open memory banks, maximum of eight per rank
- Per-NIF transaction controllable bank management policies: open-page, close-page
- Low area, low power architecture with minimal buffering on the data, avoiding duplication of storage resources within the system
- PCTL NIF slave interface facilitates easy integration with an external scheduler or standard on-chip buses
- Efficient DDR protocol implementation with in-order column (Read and Write) commands and out-of-order Activate and Precharge commands
- Three clock cycles best case command latency (best case is when a command is to an open page and the shift array in the PCTL is empty)
- 1T or 2T memory command timing
- Automatic power-down and self-refresh entry and exit
- Software and hardware driven self-refresh entry and exit
- Programmable memory initialization
- Partial population of memories, where not all DDR byte lanes are populated with memory chips
- Programmable per rank memory ODT (On-Die Termination) support for reads and writes
- APB interface for PCTL and PHY software-accessible registers
- Automatic DQS gate training
- Automatic Write Leveling training
- At-speed built-in-self-test (BIST) loopback testing on both the address and data channels for DDR PHYs

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16.2 Block Diagram

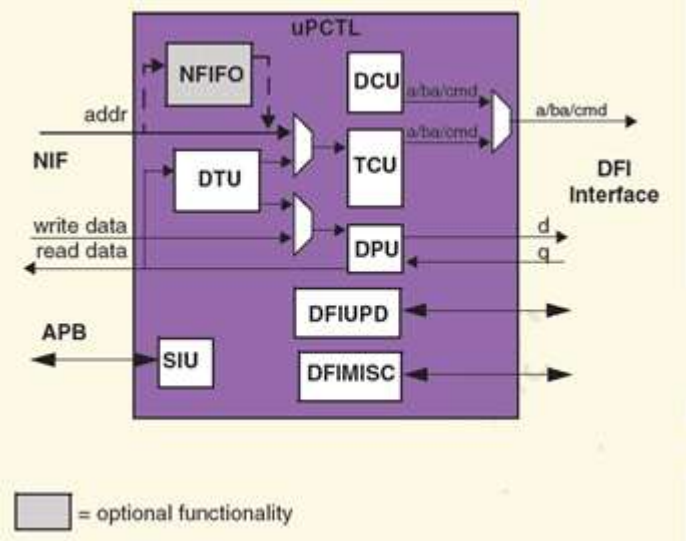

Fig. 16-1 Protocol controller architecture

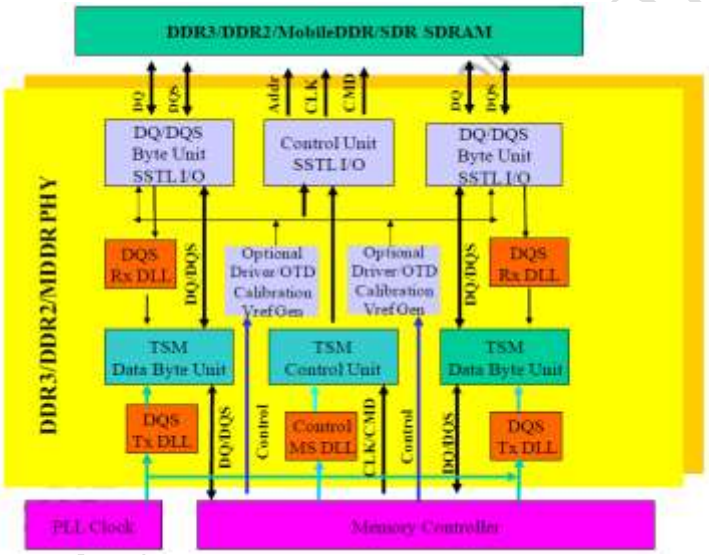

Fig. 16-2 PHY architecture

16.3 Function Description

16.3.1 Protocol controller(PCTL)

PCTL operations are defined in terms of the current state of the Operational State Machine. Software can move PCTL in any of the operational states by issuing commands via the SCTL register. Transitions from one operational state to the other occur pass through a "transitional" state. Transitional states are exited automatically by the PCTL after all the necessary actions required to change operational state have been completed. The current operational state of PCTL is reported by the STAT register and is also available from the p_ctl_stat output.
PCTL supports the following operational states:

- Init_mem - This state is the default state entered after reset. All writable registers can be programmed. While in this state software can program PCTL and initialize the PHY and the memories. The memories are not refreshed and data that has previously been written to the memories may be lost as a result. The Init_mem state is also used when it is desirable to stop any automatic PCTL function that directly affects the memories, like Power Down and Refresh, or when a software reset of the memory subsystem has to be executed.
- Config - This state is used to suspend temporarily the normal NIF traffic and allow software to reprogram PCTL and memories if necessary, while still keeping active the

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periodic generation of Refresh cycles to the memories. Power Down entry and exit sequences are possible while in Config state.

- Access - This is the operational state where NIF transactions are accepted by the PCTL and converted into memory read and writes. None of the registers can be programmed except SCFG, SCTL, ECCCLR and DTU* registers.
- Low_power - Memories are in self refresh mode. The PCTL does not generate refresh cycles while in this state.

Access and Low_power states can also be entered and exited by the hardware low power signals (c_*). In case of conflicting software and hardware low-power commands, the resulting operational state taken by the controller can be either one of the two conflicting requests.

Figure 16-3 illustrates the operational and transitional states.

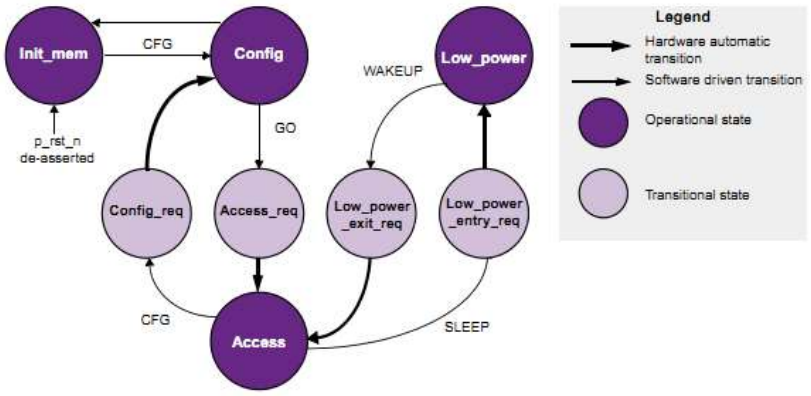

Fig. 16-3 Protocol controller state diagram

The controller clock is half frequency of the SDRAM clock (ck). The configuration clock can run at a frequency equal to or less than the controller clock. The configuration clock drives all non-DDR timing logic, such as configuration registers, PHY initialization, output impedance, and so on.

16.3.2 DDR PHY

DDR PHY provides turn key physical interface solutions for chip requiring access to DDR3/LPDDR2/LPDDR3 SDRAM device. It is optimized for low power and high speed (up to 1600Mbps for DDR3 and LPDDR3, up to 1066Mbps for LPDDR2) applications with robust timing and small silicon area. It supports DDR3, LPDDR2 and LPDDR3 SDRAM components in the market. The PHY components contain DDR specialized functional and utility SSTL I/Os up to 800MHz, critical timing synchronization module (TSM) and a low power/jitter DLLs with programmable fine-grain control for any SDRAM interface.

DDR PHY uses a DFI digital interface to connect the memory controller. All interface timing is in (1/2)XCK(SDRAM) clock domain. The controller to PHY interface is running at (1/2)XCK(SDRAM) therefore read/write bus is four times width. DDR muxing is done in the PHY block together with all related per-byte lane timing adjustment. The interface is fairly generic and support high performance input and output data flow gearing toward 200Mbps to 1600Mbps DDR3, LPDDR2 and LPDDR3 SDRAM speed in wide range.

With configurable timing and driving strength and ODT parameters to interface to the wide variety of SDRAMs, the PHY is very flexible with advanced command capability to increase SDRAM operation efficiency.

16.4 Register Description

16.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
DDR_PCTL_SCFG	0x0000	W	0x00000300	State Configuration Register
DDR_PCTL_SCTL	0x0004	W	0x00000000	Operational State Control Register
DDR_PCTL_STAT	0x0008	W	0x00000000	Operational State Status Register

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Name	Offset	Size	Reset Value	Description
DDR_PCTL_INTRSTAT	0x000c	W	0x00000000	Interrupt Status Register
DDR_PCTL_MCMD	0x0040	W	0x00100000	Memory Command Register
DDR_PCTL_POWCTL	0x0044	W	0x00000000	Power Up Control Register
DDR_PCTL_POWSTAT	0x0048	W	0x00000000	Power Up Status Register
DDR_PCTL_CMDTSTAT	0x004c	W	0x00000000	Command Timers Status Register
DDR_PCTL_CMDTSTATEN	0x0050	W	0x00000000	Command Timers Status Enable Register
DDR_PCTL_MRRCFG0	0x0060	W	0x00000000	Mode Register Read Configuration 0
DDR_PCTL_MRRSTAT0	0x0064	W	0x00000000	Mode Register Read Status 0 Register
DDR_PCTL_MRRSTAT1	0x0068	W	0x00000000	Mode Register Read Status 0 Register
DDR_PCTL_MCFG1	0x007c	W	0x00000000	Memory Configuration 1 Register
DDR_PCTL_MCFG	0x0080	W	0x00040020	Memory Configuration Register
DDR_PCTL_PPCFG	0x0084	W	0x00000000	Partially Populated Memories Configuration Register
DDR_PCTL_MSTAT	0x0088	W	0x00000000	Memory Status Register
DDR_PCTL_LPDDR2ZQCFG	0x008c	W	0xab0a560a	LPDDR2 ZQ Configuration Register
DDR_PCTL_DTUPDES	0x0094	W	0x00000000	DTU Status Register
DDR_PCTL_DTUNA	0x0098	W	0x00000000	DTU Number of Addresses Created Register
DDR_PCTL_DTUNE	0x009c	W	0x00000000	DTU Number of Errors Register
DDR_PCTL_DTUPRD0	0x00a0	W	0x00000000	DTU Parallel Read 0 Register
DDR_PCTL_DTUPRD1	0x00a4	W	0x00000000	DTU Parallel Read 1 Register
DDR_PCTL_DTUPRD2	0x00a8	W	0x00000000	DTU Parallel Read 2 Register
DDR_PCTL_DTUPRD3	0x00ac	W	0x00000000	DTU Parallel Read 3 Register
DDR_PCTL_DTUAWDT	0x00b0	W	0x00000290	DTU Address Width Register
DDR_PCTL_TOGCNT1U	0x00c0	W	0x00000064	Toggle Counter 1us Register
DDR_PCTL_TINIT	0x00c4	W	0x000000c8	t_init Timing Register
DDR_PCTL_TRSTH	0x00c8	W	0x00000000	t_rsth Timing Register
DDR_PCTL_TOGCNT100N	0x00cc	W	0x00000001	Toggle Counter 100ns
DDR_PCTL_TREFI	0x00d0	W	0x0000004e	t_refi Timing Register
DDR_PCTL_TMRD	0x00d4	W	0x00000001	t_mrd Timing Register
DDR_PCTL_TRFC	0x00d8	W	0x00000002	
DDR_PCTL_TRP	0x00dc	W	0x00010006	t_trp Timing Register
DDR_PCTL_TRTW	0x00e0	W	0x00000002	t_rtw Timing Register
DDR_PCTL_TAL	0x00e4	W	0x00000000	AL Register
DDR_PCTL_TCL	0x00e8	W	0x00000004	CL Timing Register
DDR_PCTL_TCWL	0x00ec	W	0x00000003	CWL Timing Register
DDR_PCTL_TRAS	0x00f0	W	0x00000010	t_ras Timing Register
DDR_PCTL_TRC	0x00f4	W	0x00000016	t_rc Timing Register
DDR_PCTL_TRCD	0x00f8	W	0x00000006	t_rcd Timing Register

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Name	Offset	Size	Reset Value	Description
DDR_PCTL_TRRD	0x00fc	W	0x00000004	t_rrd Timing Register
DDR_PCTL_TRTP	0x0100	W	0x00000003	t_rtp Timing Register
DDR_PCTL_TWR	0x0104	W	0x00000006	t_wr Register
DDR_PCTL_TWTR	0x0108	W	0x00000004	t_wtr Timing Register
DDR_PCTL_TEXSR	0x010c	W	0x00000001	t_exsr Timing Register
DDR_PCTL_TXP	0x0110	W	0x00000001	t_xp Timing Register
DDR_PCTL_TXPDLL	0x0114	W	0x00000000	t_xpdll Timing Register
DDR_PCTL_TZQCS	0x0118	W	0x00000000	t_zqcs Timing Register
DDR_PCTL_TZQCSI	0x011c	W	0x00000000	t_zqcsi Timing Register
DDR_PCTL_TDQS	0x0120	W	0x00000001	t_dqs Timing Register
DDR_PCTL_TCKSRE	0x0124	W	0x00000000	t_cksre Timing Register
DDR_PCTL_TCKSRX	0x0128	W	0x00000000	t_cksrx Timing Register
DDR_PCTL_TCKE	0x012c	W	0x00000003	t_cke Timing Register
DDR_PCTL_TMOD	0x0130	W	0x00000000	t_mod Timing Register
DDR_PCTL_TRSTL	0x0134	W	0x00000000	Reset Low Timing Register
DDR_PCTL_TZQCL	0x0138	W	0x00000000	t_zqcl Timing Register
DDR_PCTL_TMRR	0x013c	W	0x00000002	t_mrr Timing Register
DDR_PCTL_TCKESR	0x0140	W	0x00000004	t_ckesr Timing Register
DDR_PCTL_TDPD	0x0144	W	0x00000000	t_dpd Timing Register
DDR_PCTL_TREFI_MEM_DDR3	0x0148	W	0x00000000	t_refi_mem_ddr3 Timing Register
DDR_PCTL_DTUWACTL	0x0200	W	0x00000000	DTU Write Address Control
DDR_PCTL_DTURACTL	0x0204	W	0x00000000	DTU Read Address Control Register
DDR_PCTL_DTUCFG	0x0208	W	0x00000000	DTU Configuration Control Register
DDR_PCTL_DTUECTL	0x020c	W	0x00000000	DTU Execute Control Register
DDR_PCTL_DTUWD0	0x0210	W	0x00000000	DTU Write Data #0 Register
DDR_PCTL_DTUWD1	0x0214	W	0x00000000	DTU Write Data #1 Register
DDR_PCTL_DTUWD2	0x0218	W	0x00000000	DTU Write Data #2 Register
DDR_PCTL_DTUWD3	0x021c	W	0x00000000	DTU Write Data #3 Register
DDR_PCTL_DTUWDM	0x0220	W	0x00000000	DTU Write Data Mask Register
DDR_PCTL_DTURD0	0x0224	W	0x00000000	DTU Read Data #0 Register
DDR_PCTL_DTURD1	0x0228	W	0x00000000	DTU Read Data #1 Register
DDR_PCTL_DTURD2	0x022c	W	0x00000000	DTU Read Data #2 Register
DDR_PCTL_DTURD3	0x0230	W	0x00000000	DTU Read Data #3 Register
DDR_PCTL_DTULFSRWD	0x0234	W	0x00000000	DTU LFSR Seed for Write Data Generation Register
DDR_PCTL_DTULFSRRD	0x0238	W	0x00000000	DTU LFSR Seed for Read Data Generation Register
DDR_PCTL_DTUEAF	0x023c	W	0x00000000	DTU Error Address FIFO Register
DDR_PCTL_DFITCTRLDELAY	0x0240	W	0x00000002	DFI tctrl_delay Register
DDR_PCTL_DFIODTCFG	0x0244	W	0x00000000	DFI ODT Configuration

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Name	Offset	Size	Reset Value	Description
DDR_PCTL_DFIODTCFG1	0x0248	W	0x06060000	DFI ODT Timing Configuration 1 (for Latency and Length)
DDR_PCTL_DFIODTRANKMAP	0x024c	W	0x00008421	DFI ODT Rank Mapping
DDR_PCTL_DFITPHYWRDATA	0x0250	W	0x00000001	DFI tphy_wrdata Register
DDR_PCTL_DFITPHYWRLAT	0x0254	W	0x00000001	DFI tphy_wrlat Register
DDR_PCTL_DFITPHYWRDATALAT	0x0258	W	0x00000000	DFI tphy_wrdata_lat Register
DDR_PCTL_DFITRDDATAEN	0x0260	W	0x00000001	DFI trddata_en Register
DDR_PCTL_DFITPHYRDLAT	0x0264	W	0x0000000f	DFI tphy_rdlat Register
DDR_PCTL_DFITPHYUPDTYPE0	0x0270	W	0x00000010	DFI tphyupd_type0 Register
DDR_PCTL_DFITPHYUPDTYPE1	0x0274	W	0x00000010	DFI tphyupd_type1 Register
DDR_PCTL_DFITPHYUPDTYPE2	0x0278	W	0x00000010	DFI tphyupd_type2 Register
DDR_PCTL_DFITPHYUPDTYPE3	0x027c	W	0x00000010	DFI tphyupd_type3 Register
DDR_PCTL_DFITCTRLUPDMIN	0x0280	W	0x00000010	DFI tctrlupd_min Register
DDR_PCTL_DFITCTRLUPDMAX	0x0284	W	0x00000040	DFI tctrlupd_max Register
DDR_PCTL_DFITCTRLUPDDLY	0x0288	W	0x00000008	DFI tctrlupddly Register
DDR_PCTL_DFIUPDCFG	0x0290	W	0x00000003	DFI Update Configuration Register
DDR_PCTL_DFITREFMSKI	0x0294	W	0x00000000	DFI Masked Refresh Interval
DDR_PCTL_DFITCTRLUPDI	0x0298	W	0x00000000	DFI tctrlupd_interval Register
DDR_PCTL_DFITRCFG0	0x02ac	W	0x00000000	DFI Training Configuration 0 Register
DDR_PCTL_DFITRSTAT0	0x02b0	W	0x00000000	DFI Training Status 0 Register
DDR_PCTL_DFITRWRLVLEN	0x02b4	W	0x00000000	DFI Training dfi_wrlvl_en Register
DDR_PCTL_DFITRRDLVLEN	0x02b8	W	0x00000000	DFI Training dfi_rdlvl_en Register
DDR_PCTL_DFITRRDLVLGATEEN	0x02bc	W	0x00000000	DFI Training dfi_rdlvl_gate_en Register
DDR_PCTL_DFISTSTAT0	0x02c0	W	0x00000000	DFI Status Status 0 Register
DDR_PCTL_DFISTCFG0	0x02c4	W	0x00000000	DFI Status Configuration 0 Register
DDR_PCTL_DFISTCFG1	0x02c8	W	0x00000000	DFI Status Configuration 1 Register

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Name	Offset	Size	Reset Value	Description
DDR_PCTL_DFITDRAMCLKEN	0x02d0	W	0x00000002	DFI tdram_clk_enable Register
DDR_PCTL_DFITDRAMCLKDIS	0x02d4	W	0x00000002	DFI tdram_clk_disable Register
DDR_PCTL_DFISTCFG2	0x02d8	W	0x00000000	DFI Status Configuration 2 Register
DDR_PCTL_DFISTPARCLR	0x02dc	W	0x00000000	DFI Status Parity Clear Register
DDR_PCTL_DFISTPARLOG	0x02e0	W	0x00000000	DFI Status Parity Log Register
DDR_PCTL_DFILPCFG0	0x02f0	W	0x00070000	DFI Low Power Configuration 0 Register
DDR_PCTL_DFITRWRLVLRESP0	0x0300	W	0x00000000	DFI Training dfi_wrlvl_resp Status 0 Register
DDR_PCTL_DFITRWRLVLRESP1	0x0304	W	0x00000000	DFI Training dfi_wrlvl_resp Status 1 Register
DDR_PCTL_DFITRWRLVLRESP2	0x0308	W	0x00000000	DFI Training dfi_wrlvl_resp Status 2 Register
DDR_PCTL_DFITRRDLVLRESP0	0x030c	W	0x00000000	DFI Training dfi_rdlvl_resp Status 0 Register
DDR_PCTL_DFITRRDLVLRESP1	0x0310	W	0x00000000	DFI Training dfi_rdlvl_resp Status 1 Register
DDR_PCTL_DFITRRDLVLRESP2	0x0314	W	0x00000000	DFI Training dfi_rdlvl_resp Status 2 Register
DDR_PCTL_DFITRWRLVLDELAY0	0x0318	W	0x00000000	DFI Training dfi_wrlvl_delay Configuration 0 Register
DDR_PCTL_DFITRWRLVLDELAY1	0x031c	W	0x00000000	DFI Training dfi_wrlvl_delay Configuration 1 Register
DDR_PCTL_DFITRWRLVLDELAY2	0x0320	W	0x00000000	DFI Training dfi_wrlvl_delay Configuration 2 Register
DDR_PCTL_DFITRRDLVLDELAY0	0x0324	W	0x00000000	DFI Training dfi_rdlvl_delay Configuration 0 Register
DDR_PCTL_DFITRRDLVLDELAY1	0x0328	W	0x00000000	DFI Training dfi_rdlvl_delay Configuration 1 Register
DDR_PCTL_DFITRRDLVLDELAY2	0x032c	W	0x00000000	DFI Training dfi_rdlvl_delay Configuration 2 Register
DDR_PCTL_DFITRRDLVLGATEDELAY0	0x0330	W	0x00000000	DFI Training dfi_rdlvl_gate_delay Configuration 0
DDR_PCTL_DFITRRDLVLGATEDELAY1	0x0334	W	0x00000000	DFI Training dfi_rdlvl_gate_delay Configuration 1
DDR_PCTL_DFITRRDLVLGATEDELAY2	0x0338	W	0x00000000	DFI Training dfi_rdlvl_gate_delay Configuration 2
DDR_PCTL_DFITRCMD	0x033c	W	0x00000000	DFI Training Command Register
DDR_PCTL_IPVR	0x03f8	W	0x00000000	IP Version Register
DDR_PCTL_IPTR	0x03fc	W	0x44574300	IP Type Register

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

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Name	Offset	Size	Reset Value	Description
DDRPHY_REG0	0x0000	W	0x0000000c	DDR PHY register 00
DDRPHY_REG1	0x0004	W	0x00000000	DDR PHY register 01
DDRPHY_REG2	0x0008	W	0x00000000	DDR PHY register 02
DDRPHY_REG3	0x000c	W	0x00000022	DDR PHY register 03
DDRPHY_REG4	0x0010	W	0x00000022	DDR PHY register 04
DDRPHY_REG5	0x0014	W	0x00000002	DDR PHY register 05
DDRPHY_REG6	0x0018	W	0x00000002	DDR PHY register 06
DDRPHY_REGB	0x002c	W	0x00000060	DDR PHY register 0B
DDRPHY_REGC	0x0030	W	0x00000000	DDR PHY register 0C
DDRPHY_REG11	0x0044	W	0x000000aa	DDR PHY register 11
DDRPHY_REG12	0x0048	W	0x00000002	DDR PHY register 12
DDRPHY_REG13	0x004c	W	0x0000000c	DDR PHY register 13
DDRPHY_REG14	0x0050	W	0x00000008	DDR PHY register 13
DDRPHY_REG16	0x0058	W	0x000000aa	DDR PHY register 16
DDRPHY_REG20	0x0080	W	0x000000aa	DDR PHY register 20
DDRPHY_REG21	0x0084	W	0x000000aa	DDR PHY register 21
DDRPHY_REG26	0x0098	W	0x0000000c	DDR PHY register 26
DDRPHY_REG27	0x009c	W	0x00000000	DDR PHY register 27
DDRPHY_REG28	0x00a0	W	0x00000001	DDR PHY register 28
DDRPHY_REG70	0x01c0	W	0x00000077	DDR PHY register 70
DDRPHY_REGB0	0x02c0	W	0x00000077	DDR PHY register B0
DDRPHY_REGB1	0x02c4	W	0x00000077	DDR PHY register B1
DDRPHY_REGB2	0x02c8	W	0x00000077	DDR PHY register B2
DDRPHY_REGB3	0x02cc	W	0x00000077	DDR PHY register B3
DDRPHY_REGB4	0x02d0	W	0x00000077	DDR PHY register B4
DDRPHY_REGB5	0x02d4	W	0x00000077	DDR PHY register B5
DDRPHY_REGB6	0x02d8	W	0x00000077	DDR PHY register B6
DDRPHY_REGB7	0x02dc	W	0x00000077	DDR PHY register B7
DDRPHY_REGB8	0x02e0	W	0x00000077	DDR PHY register B8
DDRPHY_REGB9	0x02e4	W	0x00000077	DDR PHY register B9
DDRPHY_REGBA	0x02e8	W	0x00000077	DDR PHY register BA
DDRPHY_REGBB	0x02ec	W	0x00000077	DDR PHY register BB
DDRPHY_REGBC	0x02f0	W	0x00000077	DDR PHY register BC
DDRPHY_REGBD	0x02f4	W	0x00000077	DDR PHY register BD
DDRPHY_REGBE	0x02f8	W	0x00000077	DDR PHY register BE
DDRPHY_REGEC	0x03b0	W	0x00000064	DDR PHY register EC
DDRPHY_REGED	0x03b4	W	0x00000012	DDR PHY register ED
DDRPHY_REGEE	0x03b8	W	0x00000002	DDR PHY register EE
DDRPHY_REGEF	0x03bc	W	0x00000000	DDR PHY register EF
DDRPHY_REGF0	0x03c0	W	0x00000000	DDR PHY register F0
DDRPHY_REGF1	0x03c4	W	0x00000000	DDR PHY register F1
DDRPHY_REGF2	0x03c8	W	0x00000000	DDR PHY register F2

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Name	Offset	Size	Reset Value	Description
DDRPHY_REGFA	0x03e8	W	0x00000000	DDR PHY register FA
DDRPHY_REGFB	0x03ec	W	0x00000000	DDR PHY register FB
DDRPHY_REGFC	0x03f0	W	0x00000000	DDR PHY register FC
DDRPHY_REGFD	0x03f4	W	0x00000000	DDR PHY register FD
DDRPHY_REGFE	0x03f8	W	0x00000000	DDR PHY register FE
DDRPHY_REGFF	0x03fc	W	0x00000000	DDR PHY register FF

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

16.4.2 Detail Registers Description

DDR_PCTL_SCFG
Address: Operational Base + offset (0x0000)
State Configuration Register

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:8	RW	0x3	bbflags_timing The n_bbflags is a NIF output vector which provides combined information about the status of each memory bank. The de-assertion is based on when precharge, activates,reads/writes are scheduled by the TCU block. It may be possible to de-assert n_bbflags earlier than calculated by the TCU block. Programming bbflags_timing is used to achieve this.The maximum recommended value is: UPCTL_TCU_SED_P - TRP.t_rp. The programmed value is the maximum number of "early" cycles that n_bbflags maybe de-asserted. The actual achieved de-assertion depends on the traffic profile. In 1:2 mode the maximum allowed programmable value is 4'b0111 In 1:1 mode the value can be 4'b1111
7	RW	0x0	ac_pdd_en Enables assertion of ac_pdd to indicate to the PHY of an opportunity to switch to a Low power mode
6	RW	0x0	nfifo_nif1_dis For Synopsys internal use only for NFIFO testing. 1'b0 = Only supported setting. 1'b1 = For Synopsys internal use only.
5:1	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
0	RW	0x0	hw_low_power_en Enables the hardware low-power interface. Allows the system to request via hardware (c_sysreq input) to enter the memories into Self-Refresh. The handshaking between the request and acknowledge hardware low power signals (c_sysreq and c_sysack, respectively) is always performed, but the uPCTL response depends on the value set on this register field and by the value driven on the c_active_in input pin. 1'b0 = Disabled. Requests are always denied and uPCTL is unaffected by c_sysreq 1'b1 = Enabled. Requests are accepted or denied, depending on the current operational state of uPCTL and on the value of c_active_in.

DDR_PCTL_SCTL
Address: Operational Base + offset (0x0004)
Operational State Control Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2:0	RW	0x0	state_cmd Issues an operational state transition request to the uPCTL. 3'b000 = INIT (move to Init_mem from Config) 3'b001 = CFG (move to Config from Init_mem or Access) 3'b010 = GO (move to Access from Config) 3'b011 = SLEEP (move to Low_power from Access) 3'b100 = WAKEUP (move to Access from Low_power) Others = Reserved

DDR_PCTL_STAT
Address: Operational Base + offset (0x0008)
Operational State Status Register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
6:4	RO	0x0	lp_trig Reports the status of what triggered an entry to Low_power state. Is only set if in Low_power state. The individual bits report the following: - lp_trig[2]: Software driven due to SCTL.state_cmd==SLEEP. - lp_trig[1]: Hardware driven due to Hardware Low Power Interface. - lp_trig[0]: Hardware driven due to Auto Self Refresh (MCFG1.sr_idle>0). Note, if more than one trigger happens at the exact same time, more than one bit of lp_trig may be asserted high.
3	RO	0x0	reserved
2:0	RO	0x0	ctl_stat Returns the current operational state of the uPCTL. 3'b000 = Init_mem 3'b001 = Config 3'b010 = Config_req 3'b011 = Access 3'b100 = Access_req 3'b101 = Low_power 3'b110 = Low_power_entry_req 3'b111 = Low_power_exit_req Others = Reserved

DDR_PCTL_INTRSTAT
Address: Operational Base + offset (0x000c)
Interrupt Status Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RO	0x0	parity_intr Indicates that a DFI parity error has been detected 1'b0 = No error 1'b1 = Parity error
0	RO	0x0	ecc_intr Indicates that an ECC error has been detected 1'b0 = No error 1'b1 = Parity error

DDR_PCTL_MCMD
Address: Operational Base + offset (0x0040)

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Memory Command Register

Bit	Attr	Reset Value	Description
31	R/WSC	0x0	start_cmd Start command. When this bit is set to 1, the command operation defined in the cmd_opcode field is started. This bit is automatically cleared by the uPCTL after the command is finished. The application can poll this bit to determine when uPCTL is ready to accept another command. This bit cannot be cleared to 1'b0 by software.
30:28	RO	0x0	reserved
27:24	RW	0x0	cmd_add_del Set the additional delay associated with each command to 2^n internal timers clock cycles, where n is the bit field value. If n=0, the delay is 0. Max value is n=10.
23:20	RW	0x1	rank_sel Rank select for the command to be executed. 4'b0001 = Rank 0 4'b0010 = Rank 1 4'b0100 = Rank 2 4'b1000 = Rank 3 4'b0000 = Reserved Multiple 1'b1s in rank_sel mean multiple ranks are selected, which is useful broadcasting commands in parallel to multiple ranks during initialization and configuration of the memories. If MCMD.cmd_opcode=RSTL, all ranks should be selected as it cannot be performed to individual ranks
19:17	RW	0x0	bank_addr refer to cmd_addr field description.

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Bit	Attr	Reset Value	Description
16:4	RW	0x0000	cmd_addr If LPDDR2/LPDDR3, cmd_addr field and bank_addr field are merged into one field - lpddr23_addr: ---------- lpddr23_addr[19:4]: Mode register address and value driven on the relevant memory address bits ([19:4]), during a Mode Register Write operation, defined by cmd_opcode=MRS or a Mode Register Read operation, defined by cmd_opcode=MRR. For other values of cmd_opcode, this field is ignored. bits [19:12] correspond to OP7 .. OP0 and bits [11:4] correspond to MA7 .. MA0 ---------- If mDDR/DDR2/DDR3, consider as two separate register fields - bank_addr and cmd_addr: ---------- bank_addr[19:17]: Mode Register address driven on the memory bank address bits, BA1, BA0, during a Mode Register Set operation, defined by cmd_opcode=MRS. For other values of cmd_opcode, this field is ignored. 3'b000 = MR0 (MR in DDR2) 3'b001 = MR1 (EMR in DDR2) 3'b010 = MR2 (EMR(2) in DDR2) 3'b011 = MR3 (EMR(3) in DDR2) Others = Reserved ---------- cmd_addr[16:14]: Mode Register value driven on the memory address bits, A12 to A0, during a Mode Register Set operation defined by cmd_opcode=MRS. For other values of cmd_opcode this field is ignored. Refer to the memory specification for the correct settings of the various bits of this field during a MRS operation.

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Bit	Attr	Reset Value	Description
3:0	RW	0x0	cmd_opcode Command to be issued to the memory. 4'b0000 = Deselect. This is only used for timing purposes, no actual direct Deselect command is passed to the memories. 4'b0001 = Precharge All (PREA) 4'b0010 = Refresh (REF) 4'b0011 = Mode Register Set (MRS) - is MRW in LPDDR2/LPDDR3, MRS otherwise 4'b0100 = ZQ Calibration Short (ZQCS, only applies to LPDDR2/LPDDR3/DDR3) 4'b0101 = ZQ Calibration Long (ZQCL, only applies to LPDDR2/LPDDR3/DDR3) 4'b0110 = Software Driven Reset (RSTL, only applies to DDR3) 4'b0111 = Reserved 4'b1000 - Mode Register Read (MRR) - is MRR in LPDDR2/LPDDR3, is SRR in mDRR and is MPR in DDR3 4'b1001 - Deep Power Down Entry (DPDE, only applies to mDDR/LPDDR2/LPDDR3) 4'b1010 - DFI Controller Update (DFICTRLUPD) Others - Reserved It is recommended that DPDE and RSTL commands using the MCMD register should not be performed in Config State as automatic Refresh logic runs in Config state and may cause these commands to operate incorrectly.

DDR_PCTL_POWCTL
Address: Operational Base + offset (0x0044)
Power Up Control Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
0	R/WSC	0x0	power_up_start Start the memory power up sequence. When this bit is set to 1'b1, uPCTL starts the CKE and RESET# power up sequence to the memories. This bit is automatically cleared by uPCTL after the sequence is completed. This bit cannot be cleared to 1'b0 by software.

DDR_PCTL_POWSTAT
Address: Operational Base + offset (0x0048)
Power Up Status Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RO	0x0	power_up_done Returns the status of the memory power-up sequence. 1'b0 = Power-up sequence has not been performed. 1'b1 = Power-up sequence has been performed.

DDR_PCTL_CMDTSTAT
Address: Operational Base + offset (0x004c)
Command Timers Status Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RO	0x0	cmd_tstat Returns the status of the timers for memory commands. This ANDs all the command timers together. 1'b0 = One or more command timers has not expired. 1'b1 = All command timers have expired.

DDR_PCTL_CMDTSTATEN
Address: Operational Base + offset (0x0050)
Command Timers Status Enable Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
0	RW	0x0	cmd_tstat_en Enables the generation of the status of the timers for memory commands. Is enabled before CMDTSTAT register is read. 1'b0 - Disabled 1'b1 - Enabled

DDR_PCTL_MRRCFG0
Address: Operational Base + offset (0x0060)
Mode Register Read Configuration 0

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x0	mrr_byte_sel Selects which byte's data to store when performing an MRR command via MCMD. LegalValues: 0 .. 8

DDR_PCTL_MRRSTAT0
Address: Operational Base + offset (0x0064)
Mode Register Read Status 0 Register

Bit	Attr	Reset Value	Description
31:24	RO	0x00	mrrstat_beat3 MRR/MPR read data beat 3
23:16	RO	0x00	mrrstat_beat2 MRR/MPR read data beat 2
15:8	RO	0x00	mrrstat_beat1 MRR/MPR read data beat 1
7:0	RO	0x00	mrrstat_beat0 MRR/MPR read data beat 0

DDR_PCTL_MRRSTAT1
Address: Operational Base + offset (0x0068)
Mode Register Read Status 0 Register

Bit	Attr	Reset Value	Description
31:24	RO	0x00	mrrstat_beat7 MRR/MPR read data beat 7
23:16	RO	0x00	mrrstat_beat6 MRR/MPR read data beat 6
15:8	RO	0x00	mrrstat_beat5 MRR/MPR read data beat 5
7:0	RO	0x00	mrrstat_beat4 MRR/MPR read data beat 4

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DDR_PCTL_MCFG1
Address: Operational Base + offset (0x007c)
Memory Configuration 1 Register

Bit	Attr	Reset Value	Description
31	RW	0x0	hw_exit_idle_en When this bit is programmed to 1'b1 the c_active_in pin can be used to exit from the automatic clock stop , power down or self-refresh modes.
30:25	RO	0x0	reserved
24	RW	0x0	zq_resistor_shared - 1 Denotes that ZQ resistor is shared between ranks. Means ZQinit/ZQCL/ZQCS commands are sent to one rank at a time with tZQinit/tZQCL/tZQCS timing met between commands so that commands to different ranks do not overlap. - 0 ZQ resistor is not shared.
23:16	RW	0x00	hw_idle Hardware idle period. The c_active output is driven high if the NIF is idle in Access state for hw_idle * 32 * n_clk cycles. The hardware idle function is disabled when hw_idle=0.
15:11	RO	0x0	reserved
10:8	RW	0x0	tfaw_cfg_offset Used to fine tune the tFAW setting as programmed in MCFG.tfaw_cfg tFAW= (4 + MCFG.tfaw_cfg)*tRRD - tfaw_cfg_offset
7:0	RW	0x00	sr_idle Self Refresh idle period. Memories are placed into Self-Refresh mode if the NIF is idle in Access state for sr_idle * 32 * n_clk cycles. The automatic self refresh function is disabled when sr_idle=0.

DDR_PCTL_MCFG
Address: Operational Base + offset (0x0080)
Memory Configuration Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	mddr_lpddr23_clock_stop_idle Clock stop idle period in n_clk cycles. Memories are placed into clock stop mode if the NIF is idle for mddr_lpddr2_clkstop_idle n_clk cycles. The automatic clock stop function is disabled when mddr_lpddr2_clkstop_idle=0. Clock stop mode is only applicable in mDDR/LPDDR2/LPDDR3.
23:22	RW	0x0	mddr_lpddr23_en mDDR/LPDDR2 Enable. Enables support for mDDR or LPDDR2 or LPDDR3. 2'b00 = mDDR/LPDDR2/LPDDR3 Disabled 2'b01 = LPDDR3 Enabled 2'b10 = mDDR Enabled 2'b11 = LPDDR2 Enabled
21:20	RW	0x0	mddr_lpddr23_bl Field0000 Abstract mDDR/LPDDR2 Burst Length. The BL setting must be consistent with the value programmed into the BL field of MR. 2'b00 = BL2, Burst length of 2 (MR.BL=3'b001, mDDR only) 2'b01 = BL4, Burst length of 4 (MR.BL=3'b010, for mDDR and LPDDR2 only) 2'b10 = BL8, Burst length of 8 (MR.BL=3'b011, for mDDR, LPDDR2 and LPDDR3) 2'b11 = BL16, Burst length of 16 (MR.BL=3'b100, for mDDR and LPDDR2 only) This value is effective only if MCFG.mddr_lpddr23_en[1:0]!=2'b00. Otherwise, MCFG.mem_bl is used to define uPCTL's Burst Length (for DDR2/DDR3).
19:18	RW	0x1	tfaw_cfg Field0000 Abstract Sets tFAW to be 4, 5 or 6 times tRRD. 2'b00 = set tFAW=4*tRRD 2'b01 = set tFAW=5*tRRD 2'b10 = set tFAW=6*tRRD

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Bit	Attr	Reset Value	Description
17	RW	0x0	pd_exit_mode Selects the mode for Power Down Exit. For DDR2/DDR3, the power down exit mode setting in uPCTL must be consistent with the value programmed into the power down exit mode bit of MR0. For mDDR/LPDDR2/LPDDR3, only fast exit mode is valid. 1'b0 = slow exit 1'b1 = fast exit
16	RW	0x0	pd_type Sets the Power down type. 1'b0 = Precharge Power Down 1'b1 = Active Power Down
15:8	RW	0x00	pd_idle Power-down idle period in n_clk cycles. Memories are placed into power-down mode if the NIF is idle for pd_idle n_clk cycles. The automatic power down function is disabled when pd_idle=0.
7	RW	0x0	mddr_lpddr2_bst_even Ensures BST commands only placed an even no. of cycles from previous column command (Read or Write). 1'b0 = Disabled 1'b1 = Enabled If using the "Gen 2 DDR MultiPHYs" in LPDDR2/LPDDR3, set this to 1'b1. Otherwise, set it to 1'b0.
6	RW	0x0	lpddr2_s4 Enables LPDDR2-S4 support. 1'b0 = LPDDR2-S4 disabled (LPDDR2-S2 enabled) 1'b1 = LPDDR2-S4 enabled
5	RW	0x1	ddr3_en Select DDR2 or DDR3 protocol. Ignored, if mDDR or LPDDR2 or LPDDR3 support is enabled. 1'b0 = DDR2 Protocol Rules 1'b1 = DDR3 Protocol Rules

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Bit	Attr	Reset Value	Description
4	RW	0x0	stagger_cs For multi-rank commands from the DCU, stagger the assertion of CS_N to odd and even ranks by one n_clk cycle. This is useful when using RDIMMs, when multi-rank commands may be interpreted as writes to control words in the register chip. 1'b0 = Do not stagger CS_N 1'b1 = Stagger CS_N
3	RW	0x0	two_t_en Enables 2T timing for memory commands. 1'b0= Disabled 1'b1 = Enabled
2	RW	0x0	bl8int_en Setting this bit enables the BL8 interrupt function of DDR2. This is the capability to early terminate a BL8 after only 4 DDR beats by issuing the next command two cycles earlier. This functionality is only available for DDR2 memories and this setting is ignored for mDDR/LPDDR2/LPDDR3 and DDR3. 1'b0 = Disabled 1'b1 = Enabled
1	RW	0x0	cke_or_en This bit is intended to be set for 4-rank RDIMMs, which have a 2-bit CKE input. If set, dfi_cke[0] is asserted to enable either of the even ranks (0 and 2), while dfi_cke[1] is asserted to enable either of the odd ranks (1 and 3). dfi_cke[3:2] are inactive (0) 1'b0: Disabled 1'b1: Enabled
0	RW	0x0	mem_bl DDR Burst Length. The BL setting in DDR2 / DDR3 must be consistent with the value programmed into the BL field of MR0. 1'b0 = BL4, Burst length of 4 (MR0.BL=3'b010, DDR2 only) 1'b1 = BL8, Burst length of 8 (MR0.BL=3'b011 for DDR2, MR0.BL=2'b00 for DDR3)

DDR_PCTL_PPCFG

Address: Operational Base + offset (0x0084)

Partially Populated Memories Configuration Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8:1	RW	0x00	rpmem_dis Reduced Population Disable bits. Setting these bits disables the corresponding NIF/DDR data lanes from writing or reading data. Lane 0 is always present, hence only 8 bits are required for the remaining lanes including the ECC lane. In 1:2 mode bit 0 of rpmem_dis covers n_wdata/n_rdata/m_ctl_d/m_phy_q[63:32], bit 1 [95:64] etc. In 1:1 mode bit 0 of rpmem_dis covers n_wdata/n_rdata/m_ctl_d/m_phy_q[31:16], bit 2 [47:32] etc. There are no restrictions on which byte lanes can be disabled, other than byte lane 0 is required. Gaps between enabled byte lanes are allowed For each bit: 1'b0 = lane exists 1'b1 = lane is disabled
0	RW	0x0	ppmem_en Partially Population Enable bit. Setting this bit enables the partial population of external memories where the entire application bus is routed to a reduced size memory system. The lower half of the SDRAM data bus, bit 0 up to bit UPCTL_M_DW/2-1, is the active portion when Partially Populated memories are enabled. An example of this is shown in Example 8-1 on page 263. 1'b0 = Disabled 1'b1 = Enabled

DDR_PCTL_MSTAT
Address: Operational Base + offset (0x0088)
Memory Status Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
2	RO	0x0	self_refresh Indicates if uPCTL, through auto self refresh, has placed the memories in Self Refresh. 1'b0 = Memory is not in Self Refresh 1'b1 = Memory is in Self Refresh
1	RO	0x0	clock_stop Indicates if uPCTL has placed the memories in Clock Stop. 1'b0 = Memory is not in Clock Stop 1'b1 = Memory is in Clock Stop
0	RO	0x0	power_down Indicates if uPCTL has placed the memories in Power Down. 1'b0 = Memory is not in Power Down 1'b1 = Memory is in Power-Down

DDR_PCTL_LPDDR2ZQCFG
Address: Operational Base + offset (0x008c)
LPDDR2 ZQ Configuration Register

Bit	Attr	Reset Value	Description
31:24	RW	0xab	zqcl_op Value to drive on memory address bits [19:12] for an automatic hardware generated ZQCL command (LPDDR2/LPDDR3). Corresponds to OP7 .. OP0 of Mode Register Write (MRW) command which is used to send ZQCL command to memory.
23:16	RW	0x0a	zqcl_ma Value to drive on memory address bits [11:4] for an automatic hardware generated ZQCL command (LPDDR2/LPDDR3). Corresponds to MA7 .. MA0 of Mode Register Write (MRW) command which is used to send ZQCL command to memory.

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Bit	Attr	Reset Value	Description
15:8	RW	0x56	zqcs_op Value to drive on memory address bits [19:12] for an automatic hardware generated ZQCS command (LPDDR2/LPDDR3). Corresponds to OP7 .. OP0 of Mode Register Write (MRW) command which is used to send ZQCS command to memory.
7:0	RW	0x0a	zqcs_ma Value to drive on memory address bits [11:4] for an automatic hardware generated ZQCS command (LPDDR2/LPDDR3). Corresponds to MA7 .. MA0 of Mode Register Write (MRW) command which is used to send ZQCS command to memory.

DDR_PCTL_DTUPDES
Address: Operational Base + offset (0x0094)
DTU Status Register

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13	RO	0x0	dtu_rd_missing Indicates if one or more read beats of data did not return from memory.
12:9	RO	0x0	dtu_eaffl Indicates the number of entries in the FIFO that is holding the log of error addresses for data comparison
8	RO	0x0	dtu_random_error Indicates that the random data generated had some failures when written and read to the memories
7	RO	0x0	dtu_err_b7 Detected at least 1 bit error for bit 7 in the programmable data buffers
6	RO	0x0	dtu_err_b6 Detected at least 1 bit error for bit 6 in the programmable data buffers
5	RO	0x0	dtu_err_b5 Detected at least 1 bit error for bit 5 in the programmable data buffers
4	RO	0x0	dtu_err_b4 Detected at least 1 bit error for bit 4 in the programmable data buffers

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Bit	Attr	Reset Value	Description
3	RO	0x0	dtu_err_b3 Detected at least 1 bit error for bit 3 in the programmable data buffers
2	RO	0x0	dtu_err_b2 Detected at least 1 bit error for bit 2 in the programmable data buffers
1	RO	0x0	dtu_err_b1 Detected at least 1 bit error for bit 1 in the programmable data buffers
0	RO	0x0	dtu_err_b0 Detected at least 1 bit error for bit 0 in the programmable data buffers

DDR_PCTL_DTUNA
Address: Operational Base + offset (0x0098)
DTU Number of Addresses Created Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	dtu_num_address Indicates the number of addresses that were created on the NIF interface during random data generation.

DDR_PCTL_DTUNE
Address: Operational Base + offset (0x009c)
DTU Number of Errors Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	dtu_num_errors Indicates the number of errors that were detected on the readback of the NIF data during random data generation.

DDR_PCTL_DTUPRD0
Address: Operational Base + offset (0x00a0)
DTU Parallel Read 0 Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	dtu_allbits_1 Allows all the bit ones from each of the 16 received read bytes to be read in parallel. Used as part of read data eye training where a transition is required to be monitored to train the eye.

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Bit	Attr	Reset Value	Description
15:0	RO	0x0000	dtu_allbits_0 Allows all the bit zeros from each of the 16 received read bytes to be read in parallel. Used as part of read data eye training where a transition is required to be monitored to train the eye.

DDR_PCTL_DTUPRD1
Address: Operational Base + offset (0x00a4)
DTU Parallel Read 1 Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	dtu_allbits_3 Allows all the bit threes from each of the 16 received read bytes to be read in parallel. Used as part of read data eye training where a transition is required to be monitored to train the eye.
15:0	RO	0x0000	dtu_allbits_2 Allows all the bit twos from each of the 16 received read bytes to be read in parallel. Used as part of read data eye training where a transition is required to be monitored to train the eye.

DDR_PCTL_DTUPRD2
Address: Operational Base + offset (0x00a8)
DTU Parallel Read 2 Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	dtu_allbits_5 Allows all the bit fives from each of the 16 received read bytes to be read in parallel. Used as part of read data eye training where a transition is required to be monitored to train the eye.
15:0	RO	0x0000	dtu_allbits_4 Allows all the bit fours from each of the 16 received read bytes to be read in parallel. Used as part of read data eye training where a transition is required to be monitored to train the eye.

DDR_PCTL_DTUPRD3
Address: Operational Base + offset (0x00ac)
DTU Parallel Read 3 Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RO	0x0000	dtu_allbits_7 Allows all the bit sevens from each of the 16 received read bytes to be read in parallel. Used as part of read data eye training where a transition is required to be monitored to train the eye.
15:0	RO	0x0000	dtu_allbits_6 Allows all the bit sixes from each of the 16 received read bytes to be read in parallel. Used as part of read data eye training where a transition is required to be monitored to train the eye.

DDR_PCTL_DTUAWDT
Address: Operational Base + offset (0x00b0)
DTU Address Width Register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:9	RW	0x1	number_ranks Number of supported memory ranks. 2'b00 = 1 rank 2'b01 = 2 ranks 2'b10 = 3 ranks 2'b11 = 4 ranks
8	RO	0x0	reserved
7:6	RW	0x2	row_addr_width Width of the memory row address bits. 2'b00 = 13 bits wide 2'b01 = 14 bits wide 2'b10 = 15 bits wide 2'b11 = 16 bits wide
5	RO	0x0	reserved
4:3	RW	0x2	bank_addr_width Field0000 Abstract Width of the memory bank address bits. 2'b00 = 2 bits wide (4 banks) 2'b01 = 3 bits wide (8 banks) Others = Reserved
2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	column_addr_width Field0000 Abstract Width of the memory column address bits. 2'b00 = 7 bits wide 2'b01 = 8 bits wide 2'b10 = 9 bits wide 2'b11 = 10 bits wide

DDR_PCTL_TOGCNT1U
Address: Operational Base + offset (0x00c0)
Toggle Counter 1us Register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:0	RW	0x064	toggle_counter_1u The number of internal timers clock cycles

DDR_PCTL_TINIT
Address: Operational Base + offset (0x00c4)
t_init Timing Register

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8:0	RW	0x0c8	t_init Defines the time period (in us) to hold dfi_cke and dfi_reset_n stable during the memory power up sequence. The value programmed must correspond to at least 200us. The actual time period defined is TINIT * TOGCNT1U * internal timers clock .period

DDR_PCTL_TRSTH
Address: Operational Base + offset (0x00c8)
t_rsth Timing Register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:0	RW	0x000	t_rsth Defines the time period (in us) to hold the dfi_reset_n signal high after it is de-asserted during the DDR3 Power Up/Reset sequence. The value programmed for DDR3 must correspond to minimum 500us of delay. For mDDR and DDR2, this register should be programmed to 0.The actual time period defined is TRSTH * TOGCNT1U * internal timers clock period.

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DDR_PCTL_TOGCNT100N
Address: Operational Base + offset (0x00cc)
Toggle Counter 100ns

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:0	RW	0x01	toggle_counter_100n The number of internal timers clock cycles.

DDR_PCTL_TREFI
Address: Operational Base + offset (0x00d0)
t_refi Timing Register

Bit	Attr	Reset Value	Description
31	R/WSC	0x0	upd_ref Update Refresh counter values. Write 1'b1 to this bit whenever num_add_ref and/or t_refi register bit fields are updated. When this is set to 1'b1, the internal counters for Refresh interval and for how many refreshes performed each refresh interval get updated to use latest values of t_refi and num_add_ref respectively. It is cleared to 1'b0 automatically by hardware within a couple of clock cycles. This bit cannot be cleared to 1'b0 by software. If not set to 1'b1 when num_add_ref/t_refi are changed, then the previous values are continued to be used by the internal logic.
30:19	RO	0x0	reserved
18:16	RW	0x0	num_add_ref Number of additional Refreshes per t_refi interval. 0 - 1 refresh performed every t_refi interval 1 - 2 refreshes performed every t_refi interval 2 - 3 refreshes performed every t_refi interval 3 - 4 refreshes performed every t_refi interval 4 - 5 refreshes performed every t_refi interval 5 - 6 refreshes performed every t_refi interval 6 - 7 refreshes performed every t_refi interval 7 - 8 refreshes performed every t_refi interval For mDDR, max supported value is 6, as tRFCmax is 8*tREFI.
15:13	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
12:0	RW	0x004e	t_refi Defines the time period (in 100ns units) of the Refresh interval. The actual time period defined is TREFI * TOGCNT100N * internal timers clock period.

DDR_PCTL_TMRD
Address: Operational Base + offset (0x00d4)
t_mrd Timing Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2:0	RW	0x1	t_mrd Mode Register Set command cycle time in memory clock cycles. mDDR: Time from MRS to any valid command. LPDDR2/LPDDR3: Time from MRS (MRW) to any valid command. DDR2: Time from MRS to any valid command. DDR3: Time from MRS to MRS command. mDDR Legal Values: 2 LPDDR2 Legal Values: 5 LPDDR3 Legal Values: 5 DDR2 Legal Values: 2..3 DDR3 Legal Values: 2..4

DDR_PCTL_TRFC
Address: Operational Base + offset (0x00d8)

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8:0	RW	0x002	t_rfc Refresh to Active/Refresh command time in memory clock cycles. mDDR Legal Values: 7..28 LPDDR2 Legal Values: 15..112 LPDDR3 Legal Values: 22..168 DDR2 Legal Values: 15..131 DDR3 Legal Values: 36.. 374

DDR_PCTL_TRP
Address: Operational Base + offset (0x00dc)
t_trp Timing Register

Bit	Attr	Reset Value	Description
31:18	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
17:16	RW	0x1	prea_extra Additional cycles required for a Precharge All (PREA) command - in addition to t_rp. In terms of memory clock cycles mDDR Value: 0 LPDDR2 Value: Value that corresponds (tRPab -tRPpb). Rounded up in terms of memory clock cycles. Values can be 0, 1, 2. LPDDR3 Value: Value that corresponds (tRPab -tRPpb). Rounded up in terms of memory clock cycles. Values can be 0, 1, 2, 3. DDR2 Value: 1 if 8 Banks, 0 otherwise DDR3 Value: 0
15:5	RO	0x0	reserved
4:0	RW	0x06	t_rp Precharge period in memory clock cycles. For LPDDR2/LPDDR3, this should be set to TRPpb. mDDR Legal Values: 2..3 LPDDR2 Legal Values: 3..13 LPDDR3 Legal Values: 3..20 DDR2 Legal Values: 3..7 DDR3 Legal Values: 5..14

DDR_PCTL_TRTW
Address: Operational Base + offset (0x00e0)
t_rtw Timing Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x2	t_rtw Read to Write turnaround time in memory clock cycle. mDDR Legal Values: 3..11 LPDDR2 Legal Values: 1..11 LPDDR3 Legal Values: 1..12 DDR2 Legal Values: 2..10 DDR3 Legal Values: 2..10

DDR_PCTL_TAL
Address: Operational Base + offset (0x00e4)
AL Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x0	t_al Additive Latency in memory clock cycles. For DDR2 this must match the value programmed into the AL field of MR1. For DDR3 this must be 0, CL-1, CL-2 depending weather the AL value in MR1 is 0,1, or 2 respectively. CL is the CAS latency programmed into MR0. For mDDR, LPDDR2 and LPDDR3, there is no AL field in the mode registers, and this setting should be set to 0 mDDR Legal Values: 0 LPDDR2 Legal Values: 0 LPDDR3 Legal Values: 0 DDR2 Legal Values: AL DDR3 Legal Values: 0, CL-1, CL-2 (depending on AL=0,1,2 in MR1)

DDR_PCTL_TCL
Address: Operational Base + offset (0x00e8)
CL Timing Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x4	t_cl CAS Latency in memory clock cycles. If mDDR/DDR2/DDR3, the uPCTL setting must match the value programmed into the CL field of MR0. If LPDDR2/LPDDR3, the uPCTL setting must match RL (Read Latency), where RL is the value programmed into the "RL & WL"field of MR2 mDDR/DDR2/3 Legal Value: CL LPDDR2/LPDDR3 Legal Value: RL

DDR_PCTL_TCWL
Address: Operational Base + offset (0x00ec)
CWL Timing Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3:0	RW	0x3	t_cwl CAS Write Latency in memory clock cycles. For mDDR, the setting must be 1. For LPDDR2/LPDDR3 the setting must match WL (Write Latency), where WL is the value programmed into the "RL & WL"field of MR2. For DDR2 the setting must match CL-1, where CL is the value programmed into the CL field of MR0. For DDR3, the setting must match the value programmed in the memory CWL field of MR2. mDDR Legal Value: 1 LPDDR2/LPDDR3 Legal Values: WL DDR2 Legal Value: CL-1 DDR3 Legal Value: CWL

DDR_PCTL_TRAS
Address: Operational Base + offset (0x00f0)
t_ras Timing Register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RW	0x10	t_ras Activate to Precharge command time in memory clock cycles. mDDR Legal Values: 4..8 LPDDR2 Legal Values: 7..23 LPDDR3 Legal Values: 7..34 DDR2 Legal Values: 8..24 DDR3 Legal Values: 15..38

DDR_PCTL_TRC
Address: Operational Base + offset (0x00f4)
t_rc Timing Register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5:0	RW	0x16	t_rc Row Cycle time in memory clock cycles. Specifies the minimum Activate to Activate distance for accesses to same bank. mDDR Legal Values: 5..11 LPDDR2 Legal Values: 10..36 LPDDR3 Legal Values: 11..53 DDR2 Legal Values: 11..31 DDR3 Legal Values: 20..52

DDR_PCTL_TRCD
Address: Operational Base + offset (0x00f8)
t_rcd Timing Register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x06	t_rcd Row to Column delay in memory clock cycles. Specifies the minimum Activate to Column distance. mDDR Legal Values: 2..3 LPDDR2 Legal Values: 3..13 LPDDR3 Legal Values: 4..20 DDR2 Legal Values: 3..7 DDR3 Legal Values: 5..14

DDR_PCTL_TRRD
Address: Operational Base + offset (0x00fc)
t_rrd Timing Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x4	t_rrd Row-to-Row delay in memory clock cycles. Specifies the minimum Activate-to-Activate distance for consecutive accesses to different banks in the same rank. mDDR Legal Values: 1..2 LPDDR2 Legal Values: 2..6 LPDDR3 Legal Values: 2..8 DDR2 Legal Values: 2..6 DDR3 Legal Values: 4..8

DDR_PCTL_TRTP
Address: Operational Base + offset (0x0100)
t_rtp Timing Register

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Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x3	t_rtp Read to Precharge time in memory clock cycles. Specifies the minimum distance Read to Precharge for consecutive accesses to same bank. mDDR Value: 0 LPDDR2 Legal Values: 2..4 LPDDR3 Legal Values: 2..6 DDR2 Legal Values: 2..4 DDR3 Legal Values: 3..8

DDR_PCTL_TWR
Address: Operational Base + offset (0x0104)
t_wr Register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x06	t_wr Write recovery time in memory clock cycles. When using close page the uPCTL setting must be consistent with the WR field setting of MR0. mDDR Legal Values: 2..3 LPDDR2 Legal Values: 3..8 LPDDR3 Legal Values: 3..12 DDR2 Legal Values: 3..8 DDR3 Legal Values: 6..16

DDR_PCTL_TWTR
Address: Operational Base + offset (0x0108)
t_wtr Timing Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x4	t_wtr Write to Read turnaround time, in memory clock cycles. mDDR Legal Values: 1..2 LPDDR2 Legal Values: 2..4 LPDDR3 Legal Values: 4..6 DDR2 Legal Values: 2..4 DDR3 Legal Values: 3..8

DDR_PCTL_TEXSR
Address: Operational Base + offset (0x010c)
t_exsr Timing Register

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Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:0	RW	0x001	t_exsr Exit Self Refresh to first valid command delay, in memory clock cycles. For mDDR, this should be programmed to match tXSR. For LPDDR2/LPDDR3, this should be programmed to match tXSR. For DDR2, this should be programmed to match tXSRD (SRE to read-related command) as defined by the memory device specification. For DDR3, this should be programmed to match tXSDLL (SRE to a command requiring DLL locked) as defined by the memory device specification. mDDR Legal Values: 17..40 LPDDR2 Legal Values: 17..117 LPDDR3 Legal Values: 24..176 DDR2 Typical Value: 200 DDR3 Typical Value: 512

DDR_PCTL_TXP
Address: Operational Base + offset (0x0110)
t_xp Timing Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2:0	RW	0x1	t_xp Exit Power Down to first valid command delay when DLL is on (fast exit), measured in memory clock cycles. Legal Values: 1..7

DDR_PCTL_TXPDLL
Address: Operational Base + offset (0x0114)
t_xpdll Timing Register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RW	0x00	t_xpdll Exit Power Down to first valid command delay when DLL is off (slow exit), measured in memory clock cycles. mDDR/LPDDR2/LPDDR3 Value: 0 DDR2/DDR3 Legal Values: 3..63

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DDR_PCTL_TZQCS
Address: Operational Base + offset (0x0118)
t_zqcs Timing Register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:0	RW	0x00	t_zqcs SDRAM ZQ Calibration Short period, in memory clock cycles. Should be programmed to match the tZQCS timing value as defined in the memory specification. mDDR Value: 0 LPDDR2 Legal Values: 15..48 LPDDR3 Legal Values: 15..72 DDR2 Value: 0 DDR3 Typical Value: 64

DDR_PCTL_TZQCSI
Address: Operational Base + offset (0x011c)
t_zqcsi Timing Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	t_zqcsi W SDRAM ZQCS interval, measured in Refresh interval units. The total time period defined is TZQCSI*TREFI * TOGCNT100N * internal timers clock period. Programming a value of 0 in t_zqcsi disables the auto-ZQCS functionality in uPCTL. mDDR Value: 0 LPDDR2 Legal Values: 0..4294967295 LPDDR3 Legal Values: 0..4294967295 DDR2 Value: 0 DDR3 Legal Values: 0..4294967295 DDR3 Legal Values: 0..4294967295

DDR_PCTL_TDQS
Address: Operational Base + offset (0x0120)
t_dqs Timing Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3:0	RW	0x1	t_dqs Additional data turnaround time in memory clock cycles for accesses to different ranks. Used to increase the distance between column commands to different ranks, allowing more tolerance as the driver source changes on the bidirectional DQS and/or DQ signals. mDDR Legal Values: 1..12 LPDDR2 Legal Values: 1..12 LPDDR3 Legal Values: 1..12 DDR2 Legal Values: 1..12 DDR3 Legal Values: 1..12

DDR_PCTL_TCKSRE
Address: Operational Base + offset (0x0124)
t_cksre Timing Register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x00	t_cksre This is the time after Self Refresh Entry that CKE is held high before going low. In memory clock cycles. Specifies the clock disable delay after SRE. In DDR3, this should be programmed to match the greatest value between 10ns and 5 memory clock periods. In LPDDR3, this should be programmed to 2. mDDR Value: 0 LPDDR2 Value: 0 LPDDR3 Value: 2 DDR2 Value: 0 DDR3 Legal Values: 5..15

DDR_PCTL_TCKSRX
Address: Operational Base + offset (0x0128)
t_cksrx Timing Register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4:0	RW	0x00	t_cksrx This is the time (before Self Refresh Exit) that CKE is maintained high before issuing SRX. In memory clock cycles. Specifies the clock stable time before SRX. In DDR3, This should be programmed to match the greatest value between 10ns and 5 memory clock periods. In LPDDR2/LPDDR3, this should be programmed to 2. mDDR Value: 0 LPDDR2 Value: 2 LPDDR3 Legal Values: 2 DDR2 Value: 0 DDR3 Legal Values: 5..15

DDR_PCTL_TCKE
Address: Operational Base + offset (0x012c)
t_cke Timing Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2:0	RW	0x3	t_cke CKE minimum pulse width in memory clock cycles. mDDR Legal Value: 2 LPDDR2 Legal Values: 3 LPDDR3 Legal Values: 3 DDR2 Legal Value: 3 DDR3 Legal Values: 3..6

DDR_PCTL_TMOD
Address: Operational Base + offset (0x0130)
t_mod Timing Register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4:0	RW	0x00	t_mod In DDR3 mode, this is the time from MRS to any valid non-MRS command (except DESELECT or NOP) in memory clock cycles. Default Value: 0 mDDR Value: 0 LPDDR2 Value: 0 LPDDR3 Legal Values: 0 DDR2 Value: 0 DDR3 Legal Values: 0..31

DDR_PCTL_TRSTL
Address: Operational Base + offset (0x0134)
Reset Low Timing Register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:0	RW	0x00	t_rstl Memory Reset Low time, in memory clock cycles. Defines the time period to hold dfi_reset_n signal low during a software driven DDR3 Reset Operation. The value programmed must correspond to at least 100ns of delay. mDDR Value: 0 LPDDR2 Value: 0 LPDDR3 Value: 0 DDR2 Value: 0 DDR3 Legal Values: 1..127

DDR_PCTL_TZQCL
Address: Operational Base + offset (0x0138)
t_zqcl Timing Register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:0	RW	0x000	t_zqcl SDRAM ZQ Calibration Long period in memory clock cycles. If LPDDR2/LPDDR3, should be programmed to tZQCL. If DDR3, should be programmed to match the memory tZQinit timing value for the first ZQCL command during memory initialization; should be programmed to match tZQoper timing value after reset and initialization. mDDR Value: 0 LPDDR2 Legal Values: 60..192 LPDDR3 Legal Values: 60..288 DDR2 Value: 0 DDR3 Legal Values: 0..1023

DDR_PCTL_TMRR
Address: Operational Base + offset (0x013c)
t_mrr Timing Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x02	t_mrr Time for a Mode Register Read (MRR command from MCMD).

DDR_PCTL_TCKESR
Address: Operational Base + offset (0x0140)
t_ckesr Timing Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3:0	RW	0x4	t_ckesr Minimum CKE low width for Self Refresh entry to exit timing in memory clock cycles. Recommended settings: - mDDR: t_ckesr = 0 - LPDDR2/LPDDR3: t_ckesr = tCKESR setting from memories, rounded up in terms of memory cycles. - DDR2: t_ckesr = 0 - DDR3: t_ckesr = t_cke + 1 mDDR Value: 0 LPDDR2 Legal Values: 3..8 LPDDR3 Legal Values: 3..12 DDR2 Value: 0 DDR3 Legal Values: 4..7

DDR_PCTL_TDPD

Address: Operational Base + offset (0x0144)

t_dpd Timing Register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:0	RW	0x000	t_dpd Minimum Deep Power Down time. Is in terms of us. When a MCMD.DPDE command occurs, TDPD time is waited before MCMD.start_cmd can be cleared. MCMD_cmd_add_del (if any) does not start until TDPD has completed. This ensures TDPD requirement for the memory is not violated. The actual time period defined is TDPD* TOGCNT1U * internal timers clock period. Only applies for mDDR and LPDDR2 as Deep Power Down (DPD) is only valid for these memory types. For mDDR, tDPD=0, while for LPDDR2, tDPD=500 us. For LPDDR2/LPDDR3, if 500 us is waited externally by system, then set tDPD=0. mDDR Value: 0 LPDDR2 Legal Values: 0 or 500 LPDDR3 Legal Values: 0 or 500 DDR2 Legal Value: 0 DDR3 Legal Values: 0

DDR_PCTL_TREFI_MEM_DDR3
Address: Operational Base + offset (0x0148)
t_refi_mem_ddr3 Timing Register

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14:0	RW	0x0000	t_refi_mem_ddr3 DDR3 tREFI (7.8us/3.9us) time, in memory clock cycles. In DDR3 mode the value used to ensure that a maximum of 16 Refresh commands can be issued within 2 x tREFI, which is a DDR3 restriction mDDR Value: 0 LPDDR2 Values: 0 LPDDR3 Values: 0 DDR2 Value: 0

DDR_PCTL_DTUWACTL

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Address: Operational Base + offset (0x0200)
DTU Write Address Control

Bit	Attr	Reset Value	Description
31:30	RW	0x0	dtu_wr_rank Write rank to where data is to be targeted
29	RO	0x0	reserved
28:13	RW	0x0000	dtu_wr_row Write row to where data is to be targeted
12:10	RW	0x0	dtu_wr_bank Write bank to where data is to be targeted
9:0	RW	0x000	dtu_wr_col FWrite column to where data is to be targeted

DDR_PCTL_DTURACTL
Address: Operational Base + offset (0x0204)
DTU Read Address Control Register

Bit	Attr	Reset Value	Description
31:30	RW	0x0	dtu_rd_rank Read rank from where data comes
29	RO	0x0	reserved
28:13	RW	0x0000	dtu_rd_row Read row from where data comes
12:10	RW	0x0	dtu_rd_bank Read bank from where data comes
9:0	RW	0x000	dtu_rd_col Read column from where data comes

DDR_PCTL_DTUCFG
Address: Operational Base + offset (0x0208)
DTU Configuration Control Register

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:16	RW	0x00	dtu_row_increments Number of times to increment the row address when generating random data, up to a maximum of 127 times.
15	RW	0x0	dtu_wr_multi_rd When set puts the DTU into write once multiple reads mode.
14	RW	0x0	dtu_data_mask_en Controls whether random generated data masks are transmitted. Unless enabled all data bytes are written to memory and expected to be read from memory.

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Bit	Attr	Reset Value	Description
13:10	RW	0x0	dtu_target_lane Selects one of the byte lanes for data comparison into the programmable read data buffer.
9	RW	0x0	dtu_generate_random Generate transfers using random data, otherwise generate transfers from the programmable write data buffers.
8	RW	0x0	dtu_incr_banks When the column address rolls over increment the bank address until we reach and conclude bank 7.
7	RW	0x0	dtu_incr_cols Increment the column address until we saturate. Return to zero if DTUCFG.dtu_incr_banks is set to 1 and we are not at bank 7.
6:1	RW	0x00	dtu_nalen Length of the NIF transfer sequence that is passed through the uPCTL for each created address.
0	RW	0x0	dtu_enable When set, allows the DTU module to take ownership of the NIF interface: 1: DTU enabled 0: DTU disabled

DDR_PCTL_DTUECTL
Address: Operational Base + offset (0x020c)
DTU Execute Control Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	R/WSC	0x0	wr_multi_rd_rst When set, resets the DTU in write once multiple reads mode, to allow a new write to be performed. This bit automatically clears.
1	R/WSC	0x0	run_error_reports When set, initiates the calculation of the error status bits. This bit automatically clears when the re-calculation is done. This is only used in debug mode to verify the comparison logic.

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Bit	Attr	Reset Value	Description
0	R/WSC	0x0	run_dtu When set, initiates the running of the DTU read and write transfer. This bit automatically clears when the transfers are completed

DDR_PCTL_DTUWD0
Address: Operational Base + offset (0x0210)
DTU Write Data #0 Register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	dtu_wr_byte3 Write data byte
23:16	RW	0x00	dtu_wr_byte2 Write data byte
15:8	RW	0x00	dtu_wr_byte1 Write data byte
7:0	RW	0x00	dtu_wr_byte0 Write data byte

DDR_PCTL_DTUWD1
Address: Operational Base + offset (0x0214)
DTU Write Data #1 Register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	dtu_wr_byte7 Write data byte
23:16	RW	0x00	dtu_wr_byte6 Write data byte
15:8	RW	0x00	dtu_wr_byte5 Write data byte
7:0	RW	0x00	dtu_wr_byte4 Write data byte

DDR_PCTL_DTUWD2
Address: Operational Base + offset (0x0218)
DTU Write Data #2 Register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	dtu_wr_byte11 Write data byte
23:16	RW	0x00	dtu_wr_byte10 Write data byte
15:8	RW	0x00	dtu_wr_byte9 Write data byte
7:0	RW	0x00	dtu_wr_byte8 Write data byte

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DDR_PCTL_DTUWD3
Address: Operational Base + offset (0x021c)
DTU Write Data #3 Register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	dtu_wr_byte15 Write data byte
23:16	RW	0x00	dtu_wr_byte14 Write data byte
15:8	RW	0x00	dtu_wr_byte13 Write data byte
7:0	RW	0x00	dtu_wr_byte12 Write data byte

DDR_PCTL_DTUWDM
Address: Operational Base + offset (0x0220)
DTU Write Data Mask Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	dm_wr_byte0 Write data mask bit, one bit for each byte. Each bit should be 0 for a byte lane that contains valid write data.

DDR_PCTL_DTURD0
Address: Operational Base + offset (0x0224)
DTU Read Data #0 Register

Bit	Attr	Reset Value	Description
31:24	RO	0x00	dtu_rd_byte3 Read byte
23:16	RO	0x00	dtu_rd_byte2 Read byte
15:8	RO	0x00	dtu_rd_byte1 Read byte
7:0	RO	0x00	dtu_rd_byte0 Read byte

DDR_PCTL_DTURD1
Address: Operational Base + offset (0x0228)
DTU Read Data #1 Register

Bit	Attr	Reset Value	Description
31:24	RO	0x00	dtu_rd_byte7 Read byte
23:16	RO	0x00	dtu_rd_byte6 Read byte

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Bit	Attr	Reset Value	Description
15:8	RO	0x00	dtu_rd_byte5 Read byte
7:0	RO	0x00	dtu_rd_byte4 Read byte

DDR_PCTL_DTURD2
Address: Operational Base + offset (0x022c)
DTU Read Data #2 Register

Bit	Attr	Reset Value	Description
31:24	RO	0x00	dtu_rd_byte11 Read byte
23:16	RO	0x00	dtu_rd_byte10 Read byte
15:8	RO	0x00	dtu_rd_byte9 Read byte
7:0	RO	0x00	dtu_rd_byte8 Read byte

DDR_PCTL_DTURD3
Address: Operational Base + offset (0x0230)
DTU Read Data #3 Register

Bit	Attr	Reset Value	Description
31:24	RO	0x00	dtu_rd_byte15 Read byte
23:16	RO	0x00	dtu_rd_byte14 Read byte
15:8	RO	0x00	dtu_rd_byte13 Read byte
7:0	RO	0x00	dtu_rd_byte12 Read byte

DDR_PCTL_DTULFSRWD
Address: Operational Base + offset (0x0234)
DTU LFSR Seed for Write Data Generation Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dtu_lfsr_wseed This is the initial seed for the random write data generation LFSR (linear feedback shift register), shared with the write mask generation.

DDR_PCTL_DTULFSRRD
Address: Operational Base + offset (0x0238)
DTU LFSR Seed for Read Data Generation Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dtu_lfsr_rseed This is the initial seed for the random read data generation LFSR (linear feedback shift register), this is shared with the read mask generation. The read data mask is reconstructed the same as the write data mask was created, allowing the " on the fly comparison" ignore bytes which were not written.

DDR_PCTL_DTUEAF
Address: Operational Base + offset (0x023c)
DTU Error Address FIFO Register

Bit	Attr	Reset Value	Description
31:30	RO	0x0	ea_rank Indicates the rank that the error occurred in during random data generation. There could be a number of entries in this FIFO. If FIFO is empty one reads zeroes.
29	RO	0x0	reserved
28:13	RO	0x0000	ea_row Indicates the row that the error occurred in during random data generation. There could be a number of entries in this FIFO. If FIFO is empty one reads zeroes.
12:10	RO	0x0	ea_bank Indicates the bank that the error occurred in during random data generation. There could be a number of entries in this FIFO. If FIFO is empty one reads zeroes
9:0	RO	0x000	ea_column Indicates the column address that the error occurred in during random data generation. There could be a number of entries in this FIFO. If FIFO is empty one reads zeroes.

DDR_PCTL_DFITCTRLDELAY
Address: Operational Base + offset (0x0240)
DFI tctrl_delay Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3:0	RW	0x2	tctrl_delay Specifies the number of DFI clock cycles after an assertion or deassertion of the DFI control signals that the control signals at the PHY-DRAM interface reflect the assertion or de-assertion. If the DFI clock and the memory clock are not phase-aligned, this timing parameter should be rounded up to the next integer value.

DDR_PCTL_DFIODTCFG
Address: Operational Base + offset (0x0244)
DFI ODT Configuration

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28	RW	0x0	rank3_odt_default Default ODT value of rank 3 when there is no read/write activity
27	RW	0x0	rank3_odt_write_sel Enable/disable ODT for rank 3 when a write access is occurring on this rank
26	RW	0x0	rank3_odt_write_nse Enable/disable ODT for rank 3 when a write access is occurring on a different rank
25	RW	0x0	rank3_odt_read_sel Enable/disable ODT for rank 3 when a read access is occurring on this rank
24	RW	0x0	rank3_odt_read_nsel Enable/disable ODT for rank 3 when a read access is occurring on a different rank
23:21	RO	0x0	reserved
20	RW	0x0	rank2_odt_default Default ODT value of rank 2 when there is no read/write activity
19	RW	0x0	rank2_odt_write_sel Enable/disable ODT for rank 2 when a write access is occurring on this rank
18	RW	0x0	rank2_odt_write_nse Enable/disable ODT for rank 2 when a write access is occurring on a different rank
17	RW	0x0	rank2_odt_read_sel Enable/disable ODT for rank 2 when a read access is occurring on this rank

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Bit	Attr	Reset Value	Description
16	RW	0x0	rank2_odt_read_nsel Enable/disable ODT for rank 2 when a read access is occurring on a different rank
15:13	RO	0x0	reserved
12	RW	0x0	rank1_odt_default Default ODT value of rank 1 when there is no read/write activity
11	RW	0x0	rank1_odt_write_sel Enable/disable ODT for rank 1 when a write access is occurring on this rank
10	RW	0x0	rank1_odt_write_nse Enable/disable ODT for rank 1 when a write access is occurring on a different rank
9	RW	0x0	rank1_odt_read_sel Enable/disable ODT for rank 1 when a read access is occurring on this rank
8	RW	0x0	rank1_odt_read_nsel Enable/disable ODT for rank 1 when a read access is occurring on a different rank
7:5	RO	0x0	reserved
4	RW	0x0	rank0_odt_default Default ODT value of rank 0 when there is no read/write activity
3	RW	0x0	rank0_odt_write_sel Enable/disable ODT for rank 0 when a write access is occurring on this rank
2	RW	0x0	rank0_odt_write_nse Enable/disable ODT for rank 0 when a write access is occurring on a different rank
1	RW	0x0	rank0_odt_read_sel Enable/disable ODT for rank 0 when a read access is occurring on this rank
0	RW	0x0	rank0_odt_read_nsel Enable/disable ODT for rank 0 when a read access is occurring on a different rank

DDR_PCTL_DFIODTCFG1

Address: Operational Base + offset (0x0248)

DFI ODT Timing Configuration 1 (for Latency and Length)

Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
26:24	RW	0x6	odt_len_bl8_r ODT length for BL8 read transfers Length of dfi_odt signal for BL8 reads. This is in terms of SDR cycles. For BL4 reads, the length of dfi_odt is always 2 cycles shorter than the value in this register field.
23:19	RO	0x0	reserved
18:16	RW	0x6	odt_len_bl8_w ODT length for BL8 write transfers Length of dfi_odt signal for BL8 writes. This is in terms of SDR cycles. For BL4 writes, the length of dfi_odt is always 2 cycles shorter than the value in this register field.
15:13	RO	0x0	reserved
12:8	RW	0x00	odt_lat_r Field0000 Abstract ODT latency for reads Latency after a read command that dfi_odt is set. This is in terms of SDR cycles.
7:5	RO	0x0	reserved
4:0	RW	0x00	odt_lat_w ODT latency for writes Latency after a write command that dfi_odt is set. This is in terms of SDR cycles

DDR_PCTL_DFIODTRANKMAP

Address: Operational Base + offset (0x024c)

DFI ODT Rank Mapping

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
15:12	RW	0x8	odt_rank_map3 Rank mapping for dfi_odt[3] Determines whether dfi_odt[3] should be asserted when the uPCTL requires to terminate each rank Bit 15 = 1: dfi_odt[3] will be asserted to terminate rank 3 Bit 14 = 1: dfi_odt[3] will be asserted to terminate rank 2 Bit 13 = 1: dfi_odt[3] will be asserted to terminate rank 1 Bit 12 = 1: dfi_odt[3] will be asserted to terminate rank 0 This field exists only if UPCTL_M_NRANKS = 4
11:8	RW	0x4	odt_rank_map2 Rank mapping for dfi_odt[2] Determines which rank access(es) will cause dfi_odt[2] to be asserted Bit 11 = 1: dfi_odt[2] will be asserted to terminate rank 3 Bit 10 = 1: dfi_odt[2] will be asserted to terminate rank 2 Bit 9 = 1: dfi_odt[2] will be asserted to terminate rank 1 Bit 8 = 1: dfi_odt[2] will be asserted to terminate rank 0 This field exists only if UPCTL_M_NRANKS = 4
7:4	RW	0x2	odt_rank_map1 Rank mapping for dfi_odt[1] Determines which rank access(es) will cause dfi_odt[1] to be asserted Bit 7= 1: dfi_odt[1] will be asserted to terminate rank 3 Bit 6= 1: dfi_odt[1] will be asserted to terminate rank 2 Bit 5= 1: dfi_odt[1] will be asserted to terminate rank 1 Bit 4= 1: dfi_odt[1] will be asserted to terminate rank 0 This field exists only if UPCTL_M_NRANKS >

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Bit	Attr	Reset Value	Description
3:0	RW	0x1	odt_rank_map0 Rank mapping for dfi_odt[0] Determines which rank access(es) will cause dfi_odt[0] to be asserted Bit 3= 1: dfi_odt[0] will be asserted to terminate rank 3 Bit 2= 1: dfi_odt[0] will be asserted to terminate rank 2 Bit 1= 1: dfi_odt[0] will be asserted to terminate rank 1 Bit 0= 1: dfi_odt[0] will be asserted to terminate rank 0

DDR_PCTL_DFITPHYWRDATA
Address: Operational Base + offset (0x0250)
DFI tphy_wrdata Register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RW	0x01	tphy_wrdata Specifies the number of DFI clock cycles between when the dfi_wrdata_en signal is asserted to when the associated write data is driven on the dfi_wrdata signal. This has no impact on performance, only adjusts the relative time between enable and data transfer.

DDR_PCTL_DFITPHYWRLAT
Address: Operational Base + offset (0x0254)
DFI tphy_wrlat Register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RW	0x01	tphy_wrlat Specifies the number of DFI clock cycles between when a write command is sent on the DFI control interface and when the dfi_wrdata_en signal is asserted.

DDR_PCTL_DFITPHYWRDATALAT
Address: Operational Base + offset (0x0258)
DFI tphy_wrdata_lat Register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5:0	RW	0x00	tphy_wrdata_lat Specifies the number of DFI clock cycles of latency of the DFI write data through the PHY

DDR_PCTL_DFITRDDATAEN
Address: Operational Base + offset (0x0260)
DFI trddata_en Register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RW	0x01	trddata_en Specifies the number of DFI clock cycles from the assertion of a read command on the DFI to the assertion of the dfi_rddata_en signal.

DDR_PCTL_DFITPHYRDLAT
Address: Operational Base + offset (0x0264)
DFI tphy_rdlat Register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RW	0x0f	tphy_rdlat Specifies the maximum number of DFI clock cycles allowed from the assertion of the dfi_rddata_en signal to the assertion of the dfi_rddata_valid signal.

DDR_PCTL_DFITPHYUPDTYPE0
Address: Operational Base + offset (0x0270)
DFI tphyupd_type0 Register

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x010	tphyupd_type0 Specifies the maximum number of DFI clock cycles that the dfi_phyupd_req signal may remain asserted after the assertion of the dfi_phyupd_ack signal for dfi_phyupd_type = 0x0. The dfi_phyupd_req signal may de-assert at any cycle after the assertion of the dfi_phyupd_ack signal.

DDR_PCTL_DFITPHYUPDTYPE1
Address: Operational Base + offset (0x0274)
DFI tphyupd_type1 Register

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
11:0	RW	0x010	tphyupd_type1 Specifies the maximum number of DFI clock cycles that the dfi_phyupd_req signal may remain asserted after the assertion of the dfi_phyupd_ack signal for dfi_phyupd_type = 0x1. The dfi_phyupd_req signal may de-assert at any cycle after the assertion of the dfi_phyupd_ack signal.

DDR_PCTL_DFITPHYUPDTYPE2
Address: Operational Base + offset (0x0278)
DFI tphyupd_type2 Register

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x010	tphyupd_type2 Specifies the maximum number of DFI clock cycles that the dfi_phyupd_req signal may remain asserted after the assertion of the dfi_phyupd_ack signal for dfi_phyupd_type = 0x2. The dfi_phyupd_req signal may de-assert at any cycle after the assertion of the dfi_phyupd_ack signal.

DDR_PCTL_DFITPHYUPDTYPE3
Address: Operational Base + offset (0x027c)
DFI tphyupd_type3 Register

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x010	tphyupd_type3 Specifies the maximum number of DFI clock cycles that the dfi_phyupd_req signal may remain asserted after the assertion of the dfi_phyupd_ack signal for dfi_phyupd_type = 0x3. The dfi_phyupd_req signal may de-assert at any cycle after the assertion of the dfi_phyupd_ack signal.

DDR_PCTL_DFITCTRLUPDMIN
Address: Operational Base + offset (0x0280)
DFI tctrlupd_min Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
15:0	RW	0x0010	tctrlupd_min Specifies the minimum number of DFI clock cycles that the dfi_ctrlupd_req signal must be asserted.

DDR_PCTL_DFITCTRLUPDMAX
Address: Operational Base + offset (0x0284)
DFI tctrlupd_max Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0040	tctrlupd_max Specifies the maximum number of DFI clock cycles that the dfi_ctrlupd_req signal can assert.

DDR_PCTL_DFITCTRLUPDDLY
Address: Operational Base + offset (0x0288)
DFI tctrlupddly Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x8	tctrlupd_dly Delay in DFI clock cycles between time a uPCTL-initiated update could be started and time uPCTL-initated update actually starts (dfi_ctrlupd_req going high).

DDR_PCTL_DFIUPDCFG
Address: Operational Base + offset (0x0290)
DFI Update Configuration Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x1	dfi_phyupd_en Enables the support for acknowledging PHY-initiated updates: 1'b0 = Disabled 1'b1 = Enabled
0	RW	0x1	dfi_ctrlupd_en Enables the generation of uPCTL-initiated updates: 1'b0 = Disabled 1'b1 = Enabled

DDR_PCTL_DFITREFMSKI
Address: Operational Base + offset (0x0294)
DFI Masked Refresh Interval

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Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	trefmski Time period of the masked Refresh interva This value is only used if TREFI==0. Defines the time period (in 100ns units) of the masked Refresh (REFMSK) interval. The actual time period defined is DFITREFMSKI* TOGCNT100N * internal timers clock period.

DDR_PCTL_DFITCTRLUPDI
Address: Operational Base + offset (0x0298)
DFI tctrlupd_interval Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	tctrlupd_interval DFI uPCTL-initiated updates interval, measured in terms of Refresh interval units. If TREFI!=0, the time period is defined as DFITCTRLUPDI*TREFI * TOGCNT100N * internal timers clock period. If TREFI==0 and DFITREFMSKI!=0, the period changes to DFITCTRLUPDI*DFITREFMSKI* * TOGCNT100N * internal timers clock period. Programming a value of 0 is the same as programming a value of 1; for instance, a uPCTL-initiated update occurs every Refresh interval.

DDR_PCTL_DFITRCFG0
Address: Operational Base + offset (0x02ac)
DFI Training Configuration 0 Register

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	dfi_wrlvl_rank_sel Determines the value to drive on the output signal dfi_wrlvl_cs_n. The value on dfi_wrlvl_cs_n is the inverse of the setting in this field.
15:13	RO	0x0	reserved
12:4	RW	0x000	dfi_rdlvl_edge Determines the value to drive on the output signal dfi_rdlvl_edge. The value on dfi_rdlvl_edge is the same as the setting in this field.

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Bit	Attr	Reset Value	Description
3:0	RW	0x0	dfi_rdlvl_rank_sel Determines the value to drive on the output signal dfi_rdlvl_cs_n. The value on dfi_rdlvl_cs_n is the inverse of the setting in this field.

DDR_PCTL_DFITRSTAT0
Address: Operational Base + offset (0x02b0)
DFI Training Status 0 Register

Bit	Attr	Reset Value	Description
31:18	RO	0x0	reserved
17:16	RO	0x0	dfi_wrlvl_mode Reports the value of the input signal dfi_wrlvl_mode.
15:10	RO	0x0	reserved
9:8	RO	0x0	dfi_rdlvl_gate_mode Reports the value of the input signal dfi_rdlvl_gate_mode.
7:2	RO	0x0	reserved
1:0	RO	0x0	dfi_rdlvl_mode Reports the value of the input signal dfi_rdlvl_mode.

DDR_PCTL_DFITRWRLVLEN
Address: Operational Base + offset (0x02b4)
DFI Training dfi_wrlvl_en Register

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8:0	RW	0x000	dfi_wrlvl_en Determines the value to drive on the output signal dfi_wrlvl_en.

DDR_PCTL_DFITRRDLVLEN
Address: Operational Base + offset (0x02b8)
DFI Training dfi_rdlvl_en Register

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8:0	RW	0x000	dfi_rdlvl_en Determines the value to drive on the output signal dfi_rdlvl_en.

DDR_PCTL_DFITRRDLVLGATEEN
Address: Operational Base + offset (0x02bc)
DFI Training dfi_rdlvl_gate_en Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8:0	RW	0x000	dfi_rdlvl_gate_en Determines the value to drive on the output signal dfi_rdlvl_gate_en.

DDR_PCTL_DFISTSTAT0
Address: Operational Base + offset (0x02c0)
DFI Status Status 0 Register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24:16	RO	0x000	dfi_data_byte_disable Reports the value of the output signal dfi_data_byte_disable.
15:6	RO	0x0	reserved
5:4	RO	0x0	dfi_freq_ratio Reports the value of the output signal dfi_freq_ratio.
3:2	RO	0x0	reserved
1	RO	0x0	dfi_init_start Reports the value of the output signal dfi_init_start.
0	RO	0x0	dfi_init_complete Reports the value of the input signal dfi_init_complete.

DDR_PCTL_DFISTCFG0
Address: Operational Base + offset (0x02c4)
DFI Status Configuration 0 Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	dfi_data_byte_disable_en Enables the driving of the dfi_data_byte_disable signal. The value driven on dfi_data_byte_disable is dependent on the setting of PPCFG register. 1'b0 - Drive dfi_data_byte_disable to default value of all zeroes. 1'b1 - Drive dfi_data_byte_disable according to value as defined by PPCFG register setting. Note: should be set to 1only after PPCFG is correctly set.

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Bit	Attr	Reset Value	Description
1	RW	0x0	dfi_freq_ratio_en Enables the driving of the dfi_freq_ratio signal. When enabled, the dfi_freq_ratio value driven is dependent on configuration parameter UPCTL_FREQ_RATIO: 2纰is driven when UPCTL_FREQ_RATIO=1; 2'b01 is driven when UPCTL_FREQ_RATIO=2. 1'b0 - Drive dfi_freq_ratio to default value of 2'b00. 1'b1 - Drive dfi_freq_ratio value according to how configuration parameter is set.
0	RW	0x0	dfi_init_start Sets the value of the dfi_init_start signal. 1'b0 - dfi_init_start is driven low 1'b1 - dfi__init_start is driven high

DDR_PCTL_DFISTCFG1
Address: Operational Base + offset (0x02c8)
DFI Status Configuration 1 Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	dfi_dram_clk_disable_en_dpd Enables support of the dfi_dram_clk_disable signal with Deep Power Down (DPD). DPD is only for mDDR/LPDDR2. 1'b0 - Disable dfi_dram_clk_disable support in relation to DPD 1'b1 - Enable dfi_dram_clk_disable support in relation to DPD
0	RW	0x0	dfi_dram_clk_disable_en Enables support of the dfi_dram_clk_disable signal with Self Refresh (SR). 1'b0 - Disable dfi_dram_clk_disable support in relation to SR 1'b1 - Enable dfi_dram_clk_disable support in relation to SR

DDR_PCTL_DFITDRAMCLKEN
Address: Operational Base + offset (0x02d0)
DFI tdram_clk_enable Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3:0	RW	0x2	tdram_clk_enable Specifies the number of DFI clock cycles from the de-assertion of the dfi_dram_clk_disable signal on the DFI until the first valid rising edge of the clock to the DRAM memory devices, at the PHY-DRAM boundary. If the DFI clock and the memory clock are not phasealigned, this timing parameter should be rounded up to the next integer value.

DDR_PCTL_DFITDRAMCLKDIS
Address: Operational Base + offset (0x02d4)
DFI tdram_clk_disable Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x2	tdram_clk_disable Specifies the number of DFI clock cycles from the assertion of the dfi_dram_clk_disable signal on the DFI until the clock to the DRAM memory devices, at the PHY-DRAM boundary, maintains a low value. If the DFI clock and the memory clock are not phasealigned, this timing parameter should be rounded up to the next integer value.

DDR_PCTL_DFISTCFG2
Address: Operational Base + offset (0x02d8)
DFI Status Configuration 2 Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	parity_en Enables the DFI parity generation feature (driven on output signal dfi_parity_in) 1'b0 - Disable DFI parity generation 1'b1 - Enable DFI parity generation
0	RW	0x0	parity_intr_en Enable interrupt generation for DFI parity error (from input signal dfi_parity_error). 1'b0 - Disable interrupt 1'b1 - Enable interrupt

DDR_PCTL_DFISTPARCLR
Address: Operational Base + offset (0x02dc)
DFI Status Parity Clear Register

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Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	R/WSC	0x0	parity_log_clr Set this bit to 1'b1 to clear the DFI Status Parity Log register (DFISTPARLOG). 1'b0 = Do not clear DFI status Parity Log register 1'b1 = Clear DFI status Parity Log register
0	R/WSC	0x0	parity_intr_clr Set this bit to 1'b1 to clear the interrupt generated by an DFI parity error (as enabled by DFISTCFG2.parity_intr_en). It also clears the INTRSTAT.parity_intr register field. It is automatically cleared by hardware when the interrupt has been cleared.

DDR_PCTL_DFISTPARLOG
Address: Operational Base + offset (0x02e0)
DFI Status Parity Log Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	parity_err_cnt Increments any time the DFI parity logic detects a parity error(s) (on dfi_parity_error).

DDR_PCTL_DFILPCFG0
Address: Operational Base + offset (0x02f0)
DFI Low Power Configuration 0 Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:28	RW	0x0	dfi_lp_wakeup_dpd Value to drive on dfi_lp_wakeup signal when Deep Power Down mode is entered. Determines the DFI's tlp_wakeup time: 4'b0000 - 16 cycles 4'b0001 - 32 cycles 4'b0010 - 64 cycles 4'b0011 - 128 cycles 4'b0100 - 256 cycles 4'b0101 - 512 cycles 4'b0110 - 1024 cycles 4'b0111 - 2048 cycles 4'b1000 - 4096 cycles 4'b1001 - 8192 cycles 4'b1010 - 16384 cycles 4'b1011 - 32768 cycles 4'b1100 - 65536 cycles 4'b1101 - 131072 cycles 4'b1110 - 262144 cycles 4'b1111 - Unlimited
27:25	RO	0x0	reserved
24	RW	0x0	dfi_lp_en_dpd Enables DFI Low Power interface handshaking during Deep Power Down Entry/Exit. 1'b0 - Disabled 1'b1 - Enabled
23:20	RO	0x0	reserved
19:16	RW	0x7	dfi_tlp_resp Setting for tlp_resp time. Same value is used for both Power Down and Self refresh and Deep Power Down modes. DFI 2.1 specification, recommends using value of 7 always.

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Bit	Attr	Reset Value	Description
15:12	RW	0x0	dfi_lp_wakeup_sr Value to drive on dfi_lp_wakeup signal when Self Refresh mode is entered. Determines the DFI's tlp_wakeup time: 4'b0000 - 16 cycles 4'b0001 - 32 cycles 4'b0010 - 64 cycles 4'b0011 - 128 cycles 4'b0100 - 256 cycles 4'b0101 - 512 cycles 4'b0110 - 1024 cycles 4'b0111 - 2048 cycles 4'b1000 - 4096 cycles 4'b1001 - 8192 cycles 4'b1010 - 16384 cycles 4'b1011 - 32768 cycles 4'b1100 - 65536 cycles 4'b1101 - 131072 cycles 4'b1110 - 262144 cycles 4'b1111 - Unlimited
11:9	RO	0x0	reserved
8	RW	0x0	dfi_lp_en_sr Enables DFI Low Power interface handshaking during Self Refresh Entry/Exit. 1'b0 - Disabled 1'b1 - Enabled

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Bit	Attr	Reset Value	Description
7:4	RW	0x0	dfi_lp_wakeup_pd Value to drive on dfi_lp_wakeup signal when Power Down mode is entered. Determines the DFI's tlp_wakeup time: 4'b0000 - 16 cycles 4'b0001 - 32 cycles 4'b0010 - 64 cycles 4'b0011 - 128 cycles 4'b0100 - 256 cycles 4'b0101 - 512 cycles 4'b0110 - 1024 cycles 4'b0111 - 2048 cycles 4'b1000 - 4096 cycles 4'b1001 - 8192 cycles 4'b1010 - 16384 cycles 4'b1011 - 32768 cycles 4'b1100 - 65536 cycles 4'b1101 - 131072 cycles 4'b1110 - 262144 cycles 4'b1111 - Unlimited
3:1	RO	0x0	reserved
0	RW	0x0	dfi_lp_en_pd Enables DFI Low Power interface handshaking during Power Down Entry/Exit. 1'b0 - Disabled 1'b1 - Enabled

DDR_PCTL_DFITRWRLVLRESP0
Address: Operational Base + offset (0x0300)
DFI Training dfi_wrlvl_resp Status 0 Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	dfi_wrlvl_resp0 Reports the status of the dif_wrlvl_resp[31:0] signal.

DDR_PCTL_DFITRWRLVLRESP1
Address: Operational Base + offset (0x0304)
DFI Training dfi_wrlvl_resp Status 1 Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	dfi_wrlvl_resp1 Reports the status of the dif_wrlvl_resp[63:32] signal.

DDR_PCTL_DFITRWRLVLRESP2
Address: Operational Base + offset (0x0308)

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DFI Training dfi_wrlvl_resp Status 2 Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RO	0x00	dfi_wrlvl_resp2 Reports the status of the dif_wrlvl_resp[71:64] signal.

DDR_PCTL_DFITRRDLVLRESP0

Address: Operational Base + offset (0x030c)

DFI Training dfi_rdlvl_resp Status 0 Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	dfi_rdlvl_resp0 Reports the status of the dif_rdlvl_resp[31:0] signal.

DDR_PCTL_DFITRRDLVLRESP1

Address: Operational Base + offset (0x0310)

DFI Training dfi_rdlvl_resp Status 1 Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	dfi_rdlvl_resp1 Reports the status of the dif_rdlvl_resp[63:32] signal.

DDR_PCTL_DFITRRDLVLRESP2

Address: Operational Base + offset (0x0314)

DFI Training dfi_rdlvl_resp Status 2 Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RO	0x00	dfi_rdlvl_resp2 Reports the status of the dif_rdlvl_resp[71:64] signal.

DDR_PCTL_DFITRWRLVLDELAY0

Address: Operational Base + offset (0x0318)

DFI Training dfi_wrlvl_delay Configuration 0 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dfi_wrlvl_delay0 Sets the value to be driven on the signal dfi_wrlvl_delay_x[31:0].

DDR_PCTL_DFITRWRLVLDELAY1

Address: Operational Base + offset (0x031c)

DFI Training dfi_wrlvl_delay Configuration 1 Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dfi_wrlvl_delay1 Sets the value to be driven on the signal dfi_wrlvl_delay_x[63:32].

DDR_PCTL_DFITRWRLVLDELAY2
Address: Operational Base + offset (0x0320)
DFI Training dfi_wrlvl_delay Configuration 2 Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	dfi_wrlvl_delay2 Sets the value to be driven on the signal dfi_wrlvl_delay_x[71:64].

DDR_PCTL_DFITRRDLVLDELAY0
Address: Operational Base + offset (0x0324)
DFI Training dfi_rdlvl_delay Configuration 0 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dfi_rdlvl_delay0 Sets the value to be driven on the signal dfi_rdlvl_delay_x[31:0].

DDR_PCTL_DFITRRDLVLDELAY1
Address: Operational Base + offset (0x0328)
DFI Training dfi_rdlvl_delay Configuration 1 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dfi_rdlvl_delay1 Sets the value to be driven on the signal dfi_rdlvl_delay_x[63:32].

DDR_PCTL_DFITRRDLVLDELAY2
Address: Operational Base + offset (0x032c)
DFI Training dfi_rdlvl_delay Configuration 2 Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	dfi_rdlvl_delay2 Sets the value to be driven on the signal dfi_rdlvl_delay_x[71:64].

DDR_PCTL_DFITRRDLVLGATEDELAY0
Address: Operational Base + offset (0x0330)
DFI Training dfi_rdlvl_gate_delay Configuration 0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dfi_rdlvl_gate_delay0 Sets the value to be driven on the signal dfi_rdlvl_gate_delay_x[31:0].

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DDR_PCTL_DFITRRDLVLGATEDELAY1
Address: Operational Base + offset (0x0334)
DFI Training dfi_rdlvl_gate_delay Configuration 1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dfi_rdlvl_gate_delay1 Sets the value to be driven on the signal dfi_rdlvl_gate_delay_x[63:32].

DDR_PCTL_DFITRRDLVLGATEDELAY2
Address: Operational Base + offset (0x0338)
DFI Training dfi_rdlvl_gate_delay Configuration 2

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	dfi_rdlvl_gate_delay2 Sets the value to be driven on the signal dfi_rdlvl_gate_delay_x[71:64].

DDR_PCTL_DFITRCMD
Address: Operational Base + offset (0x033c)
DFI Training Command Register

Bit	Attr	Reset Value	Description
31	R/WSC	0x0	dfitrcmd_start DFI Training Command Start. When this bit is set to 1, the command operation defined in the dfitrcmd_opcode field is started. This bit is automatically cleared by the uPCTL after the command is finished. The application can poll this bit to determine when uPCTL is ready to accept another command. This bit cannot be cleared to 1b0 by software.
30:13	RO	0x0	reserved
12:4	RW	0x000	dfitrcmd_en DFI Training Command Enable. Selects which bits of chosen DFI Training command to drive to 1'b1.
3:2	RO	0x0	reserved
1:0	RW	0x0	dfitrcmd_opcode DFI Training Command Opcode. Select which DFI Training command to generate for one n_clk cycle: 2'b00 - dfi_wrlvl_load 2'b01 - dfi_wrlvl_strobe 2'b10 - dfi_rdlvl_load 2'b11 - Reserved.

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DDR_PCTL_IPVR
Address: Operational Base + offset (0x03f8)
IP Version Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	ip_version ASCII value for each number in the version, followed by a *.

DDR_PCTL_IPTR
Address: Operational Base + offset (0x03fc)
IP Type Register

Bit	Attr	Reset Value	Description
31:0	RO	0x44574300	ip_type Contains the IP's identification code, which is an ASCII value to identify the component and it is currently set to the string "DWC". This value never changes.

DDRPHY_REG0
Address: Operational Base + offset (0x0000)
DDR PHY register 00

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x1	soft reset 1, active low
2	RW	0x1	soft reset 0, active low
1:0	RO	0x0	reserved

DDRPHY_REG1
Address: Operational Base + offset (0x0004)
DDR PHY register 01

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x0	PHY working condition select 0x0: ddr3 PHY mode 0x1: ddr2 PHY mode 0x2: lpddr2/3 PHY mode 0x3: reserved

DDRPHY_REG2
Address: Operational Base + offset (0x0008)
DDR PHY register 02

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:6	RW	0x0	Write leveling CS select signal 2'b00: select CS0 and CS1 2'b01: select CS1 2'b10: select CS0
5:4	RW	0x0	DQS gating calibration CS select signal 2'b00: select CS0 and CS1 2'b01: select CS1 2'b10: select CS0
3	RW	0x0	Write leveling calibration bypass mode, active high
2	RW	0x0	Write leveling calibration control, active high
1	RW	0x0	DQS gating calibration bypass mode, active high
0	RW	0x0	DQS gating calibration control, active high

DDRPHY_REG3
Address: Operational Base + offset (0x000c)
DDR PHY register 03

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:4	RW	0x2	right channel A read ODT delay by read odt configure bypass mode
3	RO	0x0	reserved
2:0	RW	0x2	left channel A read ODT delay by read odt configure bypass mode

DDRPHY_REG4
Address: Operational Base + offset (0x0010)
DDR PHY register 04

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:4	RW	0x2	right channel B read ODT delay by read odt configure bypass mode
3	RO	0x0	reserved
2:0	RW	0x2	left channel B read ODT delay by read odt configure bypass mode

DDRPHY_REG5
Address: Operational Base + offset (0x0014)
DDR PHY register 05

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x02	Write leveling load mode[7:0]

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DDRPHY_REG6
Address: Operational Base + offset (0x0018)
DDR PHY register 06

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:6	RW	0x0	Write leveling load mode select[1:0]
5:0	RW	0x02	Write leveling load mode[13:8]

DDRPHY_REGB
Address: Operational Base + offset (0x002c)
DDR PHY register 0B

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x6	CL value DDR2 /DDR3 CAS Latency LPDDR2/3 RL value
3:0	RW	0x0	AL value DDR2/DDR3 additive latency

DDRPHY_REGC
Address: Operational Base + offset (0x0030)
DDR PHY register 0C

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x0	CWL value DDR3 WRITE CAS Latency LPDDR2/3 WL value

DDRPHY_REG11
Address: Operational Base + offset (0x0044)
DDR PHY register 11

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0xa	CMD PRCOMP, except for CK/CKB. The larger the value, the stronger the drive strength.
3:0	RW	0xa	CMD NRCOMP, except for CK/CKB. The larger the value, the stronger the drive strength.

DDRPHY_REG12
Address: Operational Base + offset (0x0048)
DDR PHY register 12

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x1	CMD weak pull up enable, active low

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Bit	Attr	Reset Value	Description
0	RW	0x0	CMD weak pull down enable, active high

DDRPHY_REG13
Address: Operational Base + offset (0x004c)
DDR PHY register 13

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	CMD DLL clock phase select in bypass mode 0: no delay 1: 90°delay
3	RW	0x1	CMD DLL enable 0: disable 1: enable
2:0	RW	0x4	CMD AND ADDRESS DLL delay 0: no delay 1: 22.5°delay 2: 45°delay 3: 67.5°delay 4: 90°delay 5: 112.5°delay 6: 135°delay 7: 157.5°delay

DDRPHY_REG14
Address: Operational Base + offset (0x0050)
DDR PHY register 13

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x1	CK DLL clock phase select in bypass mode 0: no delay 1: 90°delay
2:0	RW	0x0	CK DLL delay 0: no delay 1: 22.5°delay 2: 45°delay 3: 67.5°delay 4: 90°delay 5: 112.5°delay 6: 135°delay 7: 157.5°delay

DDRPHY_REG16
Address: Operational Base + offset (0x0058)
DDR PHY register 16

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0xa	CK/CKB PRCOMP. The larger the value, the stronger the drive strength.
3:0	RW	0xa	CK/CKB NRCOMP. The larger the value, the stronger the drive strength.

DDRPHY_REG20
Address: Operational Base + offset (0x0080)
DDR PHY register 20

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:4	RW	0xa	Left channel A PRCOMP. The larger the value, the stronger the drive strength in the scope from A_DQ0 to A_DQ7. Note: Right channel A is the same, and the register offset is 0xc0; Left channel B is the same, and the register offset is 0x100; Right channel B is the same, and the register offset is 0x140;
3:0	RW	0xa	Left channel A NRCOMP. The larger the value, the stronger the drive strength in the scope from A_DQ0 to A_DQ7. Note: Right channel A is the same, and the register offset is 0xc0; Left channel B is the same, and the register offset is 0x100; Right channel B is the same, and the register offset is 0x140;

DDRPHY_REG21
Address: Operational Base + offset (0x0084)
DDR PHY register 21

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0xa	Left channel A PRCOMP. The larger the value, the stronger the drive strength in the scope from A_DQ0 to A_DQ7. Note: Right channel A is the same, and the register offset is 0xc4; Left channel B is the same, and the register offset is 0x104; Right channel B is the same, and the register offset is 0x144;

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Bit	Attr	Reset Value	Description
3:0	RW	0xa	Left channel A read pull-up ODT. The larger the value, the smaller the pull-up resistance in the scope from A_DQ0 to A_DQ7. Note: Right channel A is the same, and the register offset is 0xc4; Left channel B is the same, and the register offset is 0x104; Right channel B is the same, and the register offset is 0x144;

DDRPHY_REG26
Address: Operational Base + offset (0x0098)
DDR PHY register 26

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	Left channel A write DQ DLL phase select in bypass mode. 0: no delay 1: 90°delay Note: Right channel A is the same, and the register offset is 0xd8; Left channel B is the same, and the register offset is 0x118; Right channel B is the same, and the register offset is 0x158;
3	RW	0x1	Left channel A write DQ DLL enable, active HIGH. Note: Right channel A is the same, and the register offset is 0xd8; Left channel B is the same, and the register offset is 0x118; Right channel B is the same, and the register offset is 0x158;

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Bit	Attr	Reset Value	Description
2:0	RW	0x4	Left channel A write DQ DLL delay 0: no delay 1: 22.5°delay 2: 45°delay 3: 67.5°delay 4: 90°delay 5: 112.5°delay 6: 135°delay 7: 157.5°delay Note: Right channel A is the same, and the register offset is 0xd8; Left channel B is the same, and the register offset is 0x118; Right channel B is the same, and the register offset is 0x158;

DDRPHY_REG27
Address: Operational Base + offset (0x009c)
DDR PHY register 27

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	Left channel A write DQS DLL phase select in bypass mode. 0: no delay 1: 90°delay Note: Right channel A is the same, and the register offset is 0xdc; Left channel B is the same, and the register offset is 0x11c; Right channel B is the same, and the register offset is 0x15c;
2:0	RW	0x0	Left channel A write DQS DLL delay 0: no delay 1: 22.5°delay 2: 45°delay 3: 67.5°delay 4: 90°delay 5: 112.5°delay 6: 135°delay 7: 157.5°delay Note: Right channel A is the same, and the register offset is 0xdc; Left channel B is the same, and the register offset is 0x11c; Right channel B is the same, and the register offset is 0x15c;

DDRPHY_REG28
Address: Operational Base + offset (0x00a0)
DDR PHY register 28

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x1	Left channel A read DQS DLL delay 0: no delay 1: 22.5°delay 2: 45°delay 3: 67.5°delay Note: Right channel A is the same, and the register offset is 0xe0; Left channel B is the same, and the register offset is 0x120; Right channel B is the same, and the register offset is 0x160;

DDRPHY_REG70
Address: Operational Base + offset (0x01c0)
DDR PHY register 70

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 A_DM0 RX de-skew Note: The 0x01c4 offset register is for CS0 A_DQ0 RX de-skew ... The 0x01e4 offset register is for CS0 A_DQS0 RX de-skew The 0x01e8 offset register is for CS0 A_DQSB0 RX de-skew The 0x01ec offset register is for CS0 A_DM1 RX de-skew The 0x01f0 offset register is for CS0 A_DQ8 RX de-skew ... The 0x0210 offset register is for CS0 A_DQS1 RX de-skew The 0x0214 offset register is for CS0 A_DQSB1 RX de-skew ... The following offset register is for CS0 B channel The start offset register for CS1 is 0x0300

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Bit	Attr	Reset Value	Description
3:0	RW	0x7	CS0 A_DM0 TX de-skew Note: The 0x01c4 offset register is for CS0 A_DQ0 TX de-skew ... The 0x01e4 offset register is for CS0 A_DQS0 TX de-skew The 0x01e8 offset register is for CS0 A_DQSB0 TX de-skew The 0x01ec offset register is for CS0 A_DM1 TX de-skew The 0x01f0 offset register is for CS0 A_DQ8 TX de-skew ... The 0x0210 offset register is for CS0 A_DQS1 TX de-skew The 0x0214 offset register is for CS0 A_DQSB1 TX de-skew ... The following offset register is for CS0 B channel The start offset register for CS1 is 0x0300

DDRPHY_REGB0
Address: Operational Base + offset (0x02c0)
DDR PHY register B0

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	A1 de-skew
3:0	RW	0x7	A0 de-skew

DDRPHY_REGB1
Address: Operational Base + offset (0x02c4)
DDR PHY register B1

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	A3 de-skew
3:0	RW	0x7	A2 de-skew

DDRPHY_REGB2
Address: Operational Base + offset (0x02c8)
DDR PHY register B2

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	A5 de-skew

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Bit	Attr	Reset Value	Description
3:0	RW	0x7	A4 de-skew

DDRPHY_REGB3
Address: Operational Base + offset (0x02cc)
DDR PHY register B3

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	A7 de-skew
3:0	RW	0x7	A6 de-skew

DDRPHY_REGB4
Address: Operational Base + offset (0x02d0)
DDR PHY register B4

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	A9 de-skew
3:0	RW	0x7	A8 de-skew

DDRPHY_REGB5
Address: Operational Base + offset (0x02d4)
DDR PHY register B5

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	A11 de-skew
3:0	RW	0x7	A10 de-skew

DDRPHY_REGB6
Address: Operational Base + offset (0x02d8)
DDR PHY register B6

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	A13 de-skew
3:0	RW	0x7	A12 de-skew

DDRPHY_REGB7
Address: Operational Base + offset (0x02dc)
DDR PHY register B7

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	A15 de-skew
3:0	RW	0x7	A14 de-skew

DDRPHY_REGB8
Address: Operational Base + offset (0x02e0)

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DDR PHY register B8

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	B1 de-skew
3:0	RW	0x7	B0 de-skew

DDRPHY_REGB9
Address: Operational Base + offset (0x02e4)
DDR PHY register B9

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	RAS# de-skew
3:0	RW	0x7	B2 de-skew

DDRPHY_REGBA
Address: Operational Base + offset (0x02e8)
DDR PHY register BA

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	WE# de-skew
3:0	RW	0x7	CAS# de-skew

DDRPHY_REGBB
Address: Operational Base + offset (0x02ec)
DDR PHY register BB

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CKB de-skew
3:0	RW	0x7	CK de-skew

DDRPHY_REGBC
Address: Operational Base + offset (0x02f0)
DDR PHY register BC

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CKE de-skew
3:0	RW	0x7	ODT0 de-skew

DDRPHY_REGBD
Address: Operational Base + offset (0x02f4)
DDR PHY register BD

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CSB0 de-skew
3:0	RW	0x7	RESETN de-skew

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DDRPHY_REGBE
Address: Operational Base + offset (0x02f8)
DDR PHY register BE

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CSB1 de-skew
3:0	RW	0x7	ODT1 de-skew

DDRPHY_REGEC
Address: Operational Base + offset (0x03b0)
DDR PHY register EC

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x64	PLL Feedback divide [7:0]

DDRPHY_REGED
Address: Operational Base + offset (0x03b4)
DDR PHY register ED

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x1	PLL clock out enable 1: out enable 0: close clock out
3:2	RO	0x0	reserved
1	RW	0x1	PLL power down enable 1: power down 0: power up
0	RW	0x0	PLL Feedback divide [8]

DDRPHY_REGEE
Address: Operational Base + offset (0x03b8)
DDR PHY register EE

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x02	PLL Pre-divide [4:0]

DDRPHY_REGEF
Address: Operational Base + offset (0x03bc)
DDR PHY register EF

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	Select clock source 1: select DDR PHY internal PLL 0: select System PLL

DDRPHY_REGF0

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Address: Operational Base + offset (0x03c0)
DDR PHY register F0

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RO	0x0	Channel B High 8bit write leveling done
2	RO	0x0	Channel B Low 8bit write leveling done
1	RO	0x0	Channel A High 8bit write leveling done
0	RO	0x0	Channel A Low 8bit write leveling done

DDRPHY_REGF1
Address: Operational Base + offset (0x03c4)
DDR PHY register F1

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RO	0x0	Channel A High 8bit write leveling dqs value
3:0	RO	0x0	Channel A Low 8bit write leveling dqs value

DDRPHY_REGF2
Address: Operational Base + offset (0x03c8)
DDR PHY register F2

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RO	0x0	Channel B High 8bit write leveling dqs value
3:0	RO	0x0	Channel B Low 8bit write leveling dqs value

DDRPHY_REGFA
Address: Operational Base + offset (0x03e8)
DDR PHY register FA

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RO	0x0	Channel B High 8bit dqs gate sample dqs value(idqs) (3)
2	RO	0x0	Channel B Low 8bit dqs gate sample dqs value(idqs) (3)
1	RO	0x0	Channel A High 8bit dqs gate sample dqs value(idqs) (3)
0	RO	0x0	Channel A Low 8bit dqs gate sample dqs value(idqs) (3)

DDRPHY_REGFB
Address: Operational Base + offset (0x03ec)
DDR PHY register FB

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:4	RO	0x0	Calibration get the dll configure channel A low 8bit(3)

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Bit	Attr	Reset Value	Description
3	RO	0x0	Calibration get the ophsel configure channel A low 8bit(3)
2:0	RO	0x0	Calibration get the cyclesel configure channel A low 8bit(3)

DDRPHY_REGFC
Address: Operational Base + offset (0x03f0)
DDR PHY register FC

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:4	RO	0x0	Calibration get the dll configure channel A high 8bit(3)
3	RO	0x0	Calibration get the ophsel configure channel A high 8bit(3)
2:0	RO	0x0	Calibration get the cyclesel configure channel A high 8bit(3)

DDRPHY_REGFD
Address: Operational Base + offset (0x03f4)
DDR PHY register FD

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:4	RO	0x0	Calibration get the dll configure channel B low 8bit(3)
3	RO	0x0	Calibration get the ophsel configure channel B low 8bit(3)
2:0	RO	0x0	Calibration get the cyclesel configure channel B low 8bit(3)

DDRPHY_REGFE
Address: Operational Base + offset (0x03f8)
DDR PHY register FE

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:4	RO	0x0	Calibration get the dll configure channel B high 8bit(3)
3	RO	0x0	Calibration get the ophsel configure channel B high 8bit(3)
2:0	RO	0x0	Calibration get the cyclesel configure channel B high 8bit(3)

DDRPHY_REGFF

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Address: Operational Base + offset (0x03fc)
DDR PHY register FF

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RO	0x0	Channel B High 8bit Calibration done
2	RO	0x0	Channel B Low 8bit Calibration done
1	RO	0x0	Channel A High 8bit Calibration done
0	RO	0x0	Channel A Low 8bit Calibration done

Note:
left channel A signals: A_DQS0, A_DQSB0, A_DQ7~A_DQ0, A_DM0
right channel A signals: A_DQS1, A_DQSB1, A_DQ15~A_DQ8, A_DM1
left channel B signals: B_DQS0, B_DQSB0, B_DQ7~B_DQ0, B_DM0
right channel B signals: B_DQS1, B_DQSB1, B_DQ15~B_DQ8, B_DM1

The delayed phase is in SDRAM CK clock domain, whose frequency is equivalent to SDRAM clock.

16.5 Interface Description

DDR IOs are listed as following Table.

Table 16-1 DDR IO description

Pin Name	Description
CK	Active-high clock signal to the memory device.
CK_B	Active-low clock signal to the memory device.
CKE	Active-high clock enable signal to the memory device for two chip select.
CS_Bi (i=0,1)	Active-low chip select signal to the memory device. There are two chip select.
RAS_B	Active-low row address strobe to the memory device.
CAS_B	Active-low column address strobe to the memory device.
WE_B	Active-low write enable strobe to the memory device.
BA[2:0]	Bank address signal to the memory device.
A[15:0]	Address signal to the memory device.
DQ[31:0]	Bidirectional data line to the memory device.
DQS[3:0]	Active-high bidirectional data strobes to the memory device.
DQS_B[3:0]	Active-low bidirectional data strobes to the memory device.
DM[3:0]	Active-low data mask signal to the memory device.
ODTi (i=0,1)	On-Die Termination output signal for two chip select.
RESETN	DRAM reset signal.
DDR_BUFFER_EN	The PHY I/O enable, active high.

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16.6 Application Notes

16.6.1 State transition of PCTL

To operate PCTL, the programmer must be familiar with the available operational states and how to transition to each state from the current state.
Every software programmable register is accessible only during certain operational states. The general rule is that the PCTL must be in the Init_mem or Config states to successfully write most of the registers.
The following tables provide the programming sequences for moving to the various states of the state machine.

Moving to the Init_mem State

Step	Application	PCTL
1	Read STAT register	Returns the current PCTL state.
2	If STAT.ctl_stat = Init_mem, go to END.	
3	If STAT.ctl_stat =Config, go to Step9.	
4	If STAT.ctl_stat =Access, go to Step8.	
5	If STAT.ctl_stat = Low_power, go to Step7.	
6	Goto Step1.	PCTL is in a Transitional state and not in any of the previous operational states.
7	Write WAKEUP to SCTL.state_cmd and poll STAT.ctl_stat= Access.	Issues SRX, moves to the Access state, updates STAT.ctl_stat =Access when complete.
8	Write CFG to SCTL.state_cmd and poll STAT.ctl_stat= Config.	PCTL stalls the NIF; completes any pending transaction; issues PREA if required; moves into the Config state; updates STAT.ctl_stat =Config whencomplete.
9	Write INIT to SCTL.state_cmd and poll STAT.ctl_stat=Init_mem	Moves into the Init_mem state and updates STAT.ctl_stat =Init_mem.
END		PCTL is in Init_mem state.

Moving to Config State

Step	Application	PCTL
1	Read STAT register.	Returns the current PCTL state.
2	If STAT.ctl_stat= Config, goto END.	
3	If STAT.ctl_stat= Low_power, go toStep6.	
4	If STAT.ctl_stat= Init_mem or Access, go toStep7.	
5	Go to Step1.	PCTL is in a transitional state and is not in any of the previous operational states.
6	Write WAKEUP to CTL.state_cmd and poll STAT.ctl_stat= Access.	Issues SRX, moves to the Access state, and updates STAT.ctl_stat= Access when complete.
7	Write CFG to SCTL.state_cmd and poll STAT.ctl_stat= Config.	PCTL stalls the NIF; completes any pending transaction; issues PREA if required; moves into the Config state; and updates STAT.ctl_stat = Config when complete.
END		PCTL is in Config state.

Moving to Access State

Step	Application	PCTL
1	Read STATregister	Returns the current PCTL state.
2	IfSTAT.ctl_stat= Access, go to END.	
3	IfSTAT.ctl_stat= Config, go to Step9	
4	IfSTAT.ctl_stat= Init_mem, go to Step8	
5	IfSTAT.ctl_stat= Low_power, go to Step7.	
6	Goto Step1.	PCTL is in a transitional state and is not in any of the previous operational states.
7	Write WAKEUP to SCTL.state_cmd and poll STAT.ctl_stat= Access. Goto END	Issues SRX, moves to the Access state, updates STAT.ctl_stat= Access when complete.

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8	Write CFG to SCTL.state_cmd and pollSTAT.ctl_stat= Config.	Moves into the Config state, updates STAT.ctl_stat= Config when complete.
9	Write GO to SCTL.state_cmd and poll STAT.ctl_stat= Access.	Moves into the Access state, updates STAT.ctl_stat= Access when complete.
END		PCTL is in Access state.

Moving to Low Power State

Step	Application	PCTL
1	Read STAT register.	Returns current PCTL state.
2	If STAT.ctl_stat =Low_power, go to END.	
3	If STAT.ctl_stat = Access, go to Step9	
4	If STAT.ctl_stat = Config, go to Step8	
5	If STAT.ctl_stat = Init_mem, go to Step7.	
6	Goto Step1.	PCTL is in transitional state and is not in any of the previous operational states.
7	Write CFG to SCTL.state_cmd and poll STAT.ctl_stat= Config.	Moves into the Config state, updates STAT.ctl_stat = Config when complete.
8	Write GO to SCTL.state_cmd and poll STAT.ctl_stat= Access.	Moves into the Access state, updates STAT.ctl_stat =Access when complete.
9	Write SLEEP to SCTL.state_cmd and poll STAT.ctl_stat= Low_power.	Issues PDX if necessary; completes any pending transactions; issues PREA command; finally, issues SRE and updates STAT.ctl_stat = Low_power.
END		PCTL is in Low Power state

16.6.2 Initialization

DMC Initialization

1. Configure relative system information of grf_ddrc0_con0 register;
2. Configure AL, CL … information of PHY registers.
3. Configure timing registers of PCTL.
4. Start PHY initialization with DDR_PCTL_DFISTCFG0[0] register of PCTL and wait PHY initialization finish with DDR_PCTL_DFISTSTAT0[0] register of PCTL.
5. Power up DRAM by DDR_PCTL_POWCTL[0] register of PCTL and wait power up DRAM finish with DDR_PCTL_POWSATA[0] register of PCTL.
6. Initiate DRAM with MRS command sent by PCTL.
7. After ddr sdram initialization done, start PHY dqs calibration with PHYREG02 register and wait calibration finish with PHYREGFF register.
8. (optional) After dqs calibration, start write leveling traning with PHYREG02 register and wait write leveling training finish with PHYJREGF0 register.
9. Configure PCTL to move to Access State.
10. Start Write and Read.

DDR3 Initialization Sequence

The initialization steps for DDR3 SDRAMs are as follows:

1. Write Deselect command to MCMD, specifying an additional delay in terms of internal timers clock cycles of at least tXPR.
2. Poll MCMD.start_cmd = 0.
3. Write MR2 command to MCMD and poll MCMD.start_cmd = 0.
4. Write MR3 command to MCMD and poll MCMD.start_cmd = 0.
5. Write MR1 command to MCMD and poll MCMD.start_cmd = 0.

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6. Write MR0 command to MCMD, for "DLL Reset", and poll MCMD.start_cmd = 0.
7. Write ZQCL command to MCMD and poll MCMD.start_cmd = 0. If MCFG1.zq_resistor_shared =1 ensure that ZQCL commands are sent to one rank at a time.

LPDDR2/LPDDR3 Initialization Sequence

The initialization steps for LPDDR2/LPDDR3 SDRAMs are as follows:
1. Write MRW(Reset) command to MCMD and poll MCMD.start_cmd = 0. MRW(Reset) is to Mode Register Address 0x3F and the value should be 0x00. So MCMD.lpddr23_addr = 0x003F.
2. Write Deselect command to MCMD, specifying an additional delay in terms of internal timers clock cycles of at least tINIT5 = 10 us. May need to send more than 1 Deselect command to satisfy tINIT5 = 10 us.
3. If ZQ Calibration command is supported (LPDDR2-S4 or LPDDR3), Write MRW(ZQ initialization calibration) command to MCMD and poll MCMD.start_cmd = 0. MRW(ZQ initialization calibration) is to Mode Register Address 0x0A and the value should be 0xFF. So MCMD.lpddr23_addr = 0xFF0A. Set MCMD.cmd_add_del such that tZQINIT=1us is met before next step. If MCFG1.zq_resistor_shared=1 ensure that ZQINIT commands are sent to one rank at a time.
4. Write MR2 command to MCMD and poll MCMD.start_cmd = 0.
5. Write MR1 command to MCMD and poll MCMD.start_cmd = 0.
6. Write MR3 command to MCMD and poll MCMD.start_cmd = 0.

16.6.3 Low Power Operation

Low_power state can be entered/exited via following ways:
- Software control of PCTL State machine (highest priority)
- Hardware Low Power Interface (middle priority)
- Auto Self Refresh feature (lowest priority)

Note the priority of requests from Access to Low_power is highlighted above. The STAT.lp_trig register field reports which of the 3 requests caused the entry to Low_power state.

Software control of PCTL State

The application can request via software to enter the memories into Self Refresh state by issuing the SLEEP command by programming SCTL. PCTL responds to the software request by moving into the Low_power operational state and issuing the SRE command to the memories. Note that the Low_power state can only be reached from the Access state.
In a similar fashion, the application requests to exit the memories from Self Refresh by issuing a WAKEUP command by programming SCTL. PCTL responds to the WAKEUP command issuing SRX and restoring normal NIF address channel operation.

Hardware Low Power Interface

The hardware low power interface can also be used to enter/exit Self Refresh. The functionality is enabled by setting SCFG.hw_low_power_en=1. Once that bit is set, the input c_sysreq has the ability to trigger entry into the Low Power configuration state just like the software methodology (SCTL.state_cmd = SLEEP). A hardware Low Power entry trigger will be ignored/denied if the input c_active_in=1 or n_avalid=1. It may be accepted if c_active_in=0 and n_avalid=0, depending on the current state of the PCTL. When SCFG.hw_low_power_en=1, the outputs c_sysack and c_active provide feedback as required by the AXI low power interface specification (this interface's operation is defined by the AXI specification). c_sysack acknowledges the request to go into the Low_power state, and c_active indicates when the PCTL is actually in the Low_power state.
The c_active output could also be used by an external Low Power controller to decide when to request a transistion to low power. When MCFG1.hw_idle > 0, c_active = 1'b0 indicates that the NIF has been idle for at least MCFG1.hw_idle * 32 * n_clk cycles while in the Access state. When in low power the c_active output can used by an external Low Power controller to trigger a low power exit. c_active will be driven high when either c_active_in or n_avalid are high. The path from c_active_in and n_valid to c_active is asynhronous so even if the clocks have been removed c_active will assert. The Low Power controller should re-enable the clocks when c_active is driven high while in the Low_power state.

Auto Power Down/Self Refresh

The Power Down and/or Self Refresh sequence is automatically started by PCTL when the NIF address channel is idle for a number of cycles, depending on the programmed value in

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MCFG.pd_idle and MCFG1.sr_idle.
Following table outlines the effect of these settings in conjunction with NIF being idle.

pd_idle	sr_idle	Memory modes	Memory Type
0	0	none	All
>0	0	Power Down	All
0	>0	Self Refresh	All
>0	>0	Power Down -> Self Refresh[3]	All

Note:
1. Power Down is entered if NIF is idle for pd_idle.
*2. Following on from that, if NIF continues to be idle for a further sr_idle*32 cycles, Power Down is exited and Self Refresh is entered.*

Removing PCTL's n_clk

In DDR3, LPDDR2 and LPDDR3, the relationship between SRE/SRX and stopping/starting the memory clock (CK) are formalized and are accounted for automatically by PCTL. With DDR3, LPDDR2 and LPDDR3, CK should only be stopped after PCTL has reached the Low_power state. The current operational state can be verified byreading STAT.ctl_stat. The CK must be started and stable before the Software or Hardware Low Power Interface attempts to take the memory out of Self Refresh.
PCTL's n_clk can be safely removed when PCTL is in Low Power state. The sequences outlined in following two tables should be followed for safe operation:

Step	Application	PCTL
1	Write SLEEP to SCTL.state_cmd and poll STAT.ctl_stat = LOW_POWER.	Tells PCTL to move memories into Self Refresh and waits until this completes.
2	Write TREFI=0. Also, write DFITCRLUPDI=0 and DFIREFMSKI=0, if they are not already 0.	Stops any MC-driven DFI updates occurring internally with PCTL
3	Wait a minimum interval which is equivalent to the PCTL's Refresh Interval (previous value of TREFI*TOGCNT100N*internal timers clock period;	Ensures any already scheduled PHY/PVT updates have completed successfully.
4	Stop toggling n_clk to PCTL.	n_clk logic inside PCTL is stopped.
end		

Step	Application	PCTL
1	Drive c_active_in low	Confirms that system external to PCTL can accept a Low- power request
2	Drive c_sysreq low	System Low-power request
3	Wait for PCTL to drive c_sysack low	PCTL Low-power request acknowledgement
4	Check value of c_active when Step 3 occurs. - if c_active=1, request denied. Cannot remove n_clk. Go to END. - if c_active=0, request accepted.	PCTL low-power request status response
5	Stop toggling n_clk to PCTL	n_clk logic inside PCTL is stopped
end		

16.6.4 TX DLLs

All high speed IO signals' phase can be adjusted by TX DLLs. Table 16-2 illustrates these DLLs.

Table 166-2 DDR PHY TX DLLs Delay Step

Offset	Bit	Control Signal Phase	Default	Description
0x4c	2~0	CMD	0x4	CMD DLL delay step
0x50	2~0	CK	0x0	CK DLL delay step

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0x98	2~0	A_DM0, A_DQ7~A_DQ0	0x4	DM and DQ DLL Signal delay step
0xd8	2~0	A_DM1, A_DQ15~A_DQ8	0x4	
0x118	2~0	B_DM0, B_DQ7~B_DQ0	0x4	
0x158	2~0	B_DM1, B_DQ15~B_DQ8	0x4	
0x9c	2~0	A_DQS0, A_DQSB0	0x0	TX DQS DLL Signal delay step
0xdc	2~0	A_DQS1, A_DQSB1	0x0	
0x11c	2~0	B_DQS0, B_DQSB0	0x0	
0x15c	2~0	B_DQS1, B_DQSB1	0x0	

Step 0x0 values means no phase delay, and 0x4 increases delay phase to 90 deg, 0x7 values corresponds to maximum phase delay. All DLLs having 8 delay steps which can get 90 deg phase delay by setting 0x4.

16.6.5 RX DLLs

The RX DLLs are used for sample RX DQS signals with proper phase delay and pulse edges. The DQS squelch (RXMEM) signal opens a window for passing RX DQS pulses, both RX DQS and DQS squelch signal phase can be adjusted by separate DLLs.

Table 16-3 DDR PHY RX DQS Delay Step

Offset	Bit	Control Signal Phase	Default	Description
0x0a0	1~0	A_DQS0,A_DQSB0	0x1	Read DQS DLL delay phase
0x0e0	1~0	A_DQS1,A_DQSB1	0x1	
0x120	1~0	A_DQS0,A_DQSB0	0x1	
0x160	1~0	A_DQS1,A_DQSB1	0x1	
0x0b0	7~5	left channel A DQS gating	0x1	CS0 DQS gating delay, unit x1 clock cycle
0x0f0	7~5	right channel A DQS gating	0x1	
0x130	7~5	left channel B DQS gating	0x1	
0x170	7~5	reft channel B DQS gating	0x1	
0x0b4	7~5	left channel A DQS gating	0x1	CS1 DQS gating delay, unit x1 clock cycle
0x0f4	7~5	right channel A DQS gating	0x1	
0x134	7~5	left channel B DQS gating	0x1	
0x174	7~5	reft channel B DQS gating	0x1	
0x0b0	4~3	left channel A DQS gating	0x1	CS0 additive and accumulative DQS gating delay, unit 4x clock cycle
0x0f0	4~3	right channel A DQS gating	0x1	
0x130	4~3	left channel B DQS gating	0x1	
0x170	4~3	reft channel B DQS gating	0x1	
0x0b4	4~3	left channel A DQS gating	0x1	CS1 additive and accumulative DQS gating delay, unit 4x clock cycle
0x0f4	4~3	right channel A DQS gating	0x1	
0x134	4~3	left channel B DQS gating	0x1	
0x174	4~3	reft channel B DQS gating	0x1	
0x0b0	2~0	left channel A DQS gating	0x4	CS0 additive and accumulative DQS gating delay, unit per DLL step
0x0f0	2~0	right channel A DQS gating	0x4	
0x130	2~0	left channel B DQS	0x4	

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		gating		
0x170	2~0	reft channel B DQS gating	0x4	
0x0b4	2~0	left channel A DQS gating	0x4	CS1 additive and accumulative DQS gating delay, unit per DLL step
0x0f4	2~0	right channel A DQS gating	0x4	
0x134	2~0	left channel B DQS gating	0x4	
0x174	2~0	reft channel B DQS gating	0x4	

16.6.6 High Speed IO Drive Strength

The tuning rage of driver resistance is 28ohm to 138ohm. By default, 0x5 is 46ohm for DDR3 CMD driver. When the control bit is set to be larger, the drive strength becomes stronger. The value is larger, the drive strength is stronger.

Table 16-2 DM/DQ/DQS/CMD Driver output resistance with control bit

Control bit	**4'b0000**	**4'b0001**	**4'b0010**	**4'b0011**	**4'b0100**	**4'b0101**	**4'b0110**	**4'b0111**
Pull-up resistance	+∞	272ohm	135ohm	91ohm	68ohm	54ohm	45ohm	38ohm
Pull-down resistance	+∞	272ohm	135ohm	91ohm	68ohm	54ohm	45ohm	38ohm
Control bit	**4'b1000**	**4'b1001**	**4'b1010**	**4'b1011**	**4'b1100**	**4'b1101**	**4'b1110**	**4'b1111**
Pull-up resistance	68ohm	54ohm	45ohm	39ohm	34ohm	30ohm	27ohm	25ohm
Pull-down resistance	68ohm	54ohm	45ohm	39ohm	34ohm	30ohm	27ohm	25ohm

Table 16-3 DM/DQ/DQS RX ODT resistance with control bit

Control bit	**4'b0000**	**4'b0001**	**4'b0010**	**4'b0011**	**4'b0100**	**4'b0101**	**4'b0110**	**4'b0111**
Pull-up resistance	+∞	1116ohm	558ohm	372ohm	279ohm	223ohm	186ohm	159ohm
Pull-down resistance	+∞	1116ohm	558ohm	372ohm	279ohm	223ohm	186ohm	159ohm
Control bit	**4'b1000**	**4'b1001**	**4'b1010**	**4'b1011**	**4'b1100**	**4'b1101**	**4'b1110**	**4'b1111**
Pull-up resistance	139ohm	124ohm	112ohm	101ohm	93ohm	86ohm	80ohm	74ohm
Pull-down resistance	139ohm	124ohm	112ohm	101ohm	93ohm	86ohm	80ohm	74ohm

16.6.7 PHY Low Speed Mode (200MHz)

DDR PHY supports low speed to high speed DDR3, LPDDR2 and LPDDR3 by using two operating mode: normal delay line mode up to 1600Mbps or more, low power mode where we support any speed up to 533Mbps. If all TX DLLs are bypassed, the PHY will enter low power state.

The DDR PHY entries low power mode when setting DLLs into Bypass mode. Table 16-8 illustrates related register settings.

Table 16-4 Low Power DLL Setting

Offset	**Bit**	**Default**	**Low power Setting**	**Description**
0x290	4	0x0	0x1	right channel B TX DQ DLL in bypass mode
	3	0x0	0x1	left channel B TX DQ DLL in bypass mode
	2	0x0	0x1	right channel A TX DQ DLL in bypass mode
	1	0x0	0x1	left channel A TX DQ DLL in bypass mode
	0	0x0	0x1	CMD/CK DLL in bypass mode
0x4c	4	0x0	0x1	CMD DLL phase select
0x50	3	0x0	0x0	CK DLL phase select
0x98	4	0x0	0x1	A_DQ0~A_DQ7 TX DLL phase select
0x9c	3	0x0	0x0	A_DQS0/A_DQSB0 TX DLL phase select
0xd8	4	0x0	0x1	A_DQ8~A_DQ15 TX DLL phase select
0xdc	3	0x0	0x0	A_DQS1/A_DQSB1 TX DLL phase select
0x118	4	0x0	0x1	B_DQ0~B_DQ7 TX DLL phase select

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0x11c	3	0x0	0x0	B_DQS0/B_DQSB0 TX DLL phase select
0x158	4	0x0	0x1	B_DQ8~B_DQ15 TX DLL phase select
0x15c	3	0x0	0x0	B_DQS1/B_DQSB1 TX DLL phase select

Those bits lead to command DLL in bypass mode. Both DQS0 DLL and DQ7~DQ0 DLLs are bypassed, but only DQS0 DLL should be in inversion mode because DQS0 step delay is 0x00 and DQ7~DQ0 DLL's step delay is 0x4.

16.6.8 Per bit de-skew tuning

Per-bit de-skew is designed for compensating PCB trace mismatch, DDR PHY support skew individually adjustable for all PHY signals. There are eight steps for each bit de-skew adjusting, and the adjust resolution under different corners is shown below:

Table 16-5 per-bit de-skew tuning resolution

	ff	tt	ss
de-skew resolution	15ps	22ps	32ps

Pre-bit de-skew is realized with inverter chain delay, per-bit de-skew control signals select how much inverters are connected to data path, the minimum resolution is determined by the two inverters minimum delay.

16.6.9 DDR PHY Calibration

DDR PHY auto dqs calibration function has been implemented in the PHY. The entire training processes only need to configure the register to start and wait for finish。

The entire training process is as follows:

1. PHY's register is reset, the setup is complete.
2. Send the initial command to dram and complete dram initialization.
3. Set the PHY's register to begin calibration.

Offset	Bit	Default	Description
0x8	5~4	0x0	DQS gating calibration CS select signal 2'b00: select CS0 and CS1 2'b01: select CS1 2'b10: select CS0
	1	0x0	set calibration bypass mode(1:bypass mode; 0:nomal)
	0	0x0	set calibration start (1: start; 0: stop)

4. Wait for the calibration finish by PHYREGFF.
5. Normal read and writes operation can begin.

16.6.10 DDR PHY Write Leveling Training

DDR PHY auto write leveling training function has been implemented in the PHY. The entire training processes only need to configure the register to start and wait for finish。

The entire training process is as follows:

1. PHY's register is reset, the setup is complete.
2. Send the initial command to dram and complete dram initialization.
3. Set the PHYREG05 and PHYREG06 to configure the dram mode register which used to enable dram write leveling training function.
4. Set the PHY's register to begin training.

Offset	Bit	Default	Description
0x8	7~6	0x0	Write leveling CS select signal 2'b00: select CS0 and CS1 2'b01: select CS1 2'b10: select CS0
	3	0x0	Write leveling calibration bypass mode, active high
	2	0x0	Write leveling calibration control, active high

5. Wait for the calibration finish by PHYREGF0.
6. Normal read and writes operation can begin.

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Chapter 17 NandC(Nand Flash Controller)

17.1 Overview

Nand Flash Controller (NandC) is used to control data transmission from host to flash device or from flash device to host. NandC is connected to AHB BUS through an AHB Master and an AHB Slave. The data transmission between host and external memory can be done through AHB Master interface or AHB Slave interface.
NandC supports the following features:

- NandC AHB bus clock (hclk) is asynchronous to NandC working clock (nclk)
- Software Interface Type
- Support directly mode
- Support LLP(Linked List Pointer) mode
- Flash Interface Type
- Support Asynchronous Flash Interface with 8bits datawidth ("Asyn8x" for short)
- Support Asynchronous Flash Interface with 16bits data width ("Asyn16x" for short)
- Support ONFI Synchronous Flash Interface ("ONFI Syn" for short)
- Support Toggle Flash Interface ("Toggle" for short)
- Support 8 flash devices at most
- Flash Type
- Support Managed NAND Flash(LBA) and Raw NAND Flash(NO-LBA)
- Support SLC/MLC/TLC Flash
- Flash Interface Timing
- Asyn8x: configurable timing, one byte per two Nandc working clocks at the fastest speed
- Asyn16x: configurable timing, two bytes per two Nandc working clocks at the fastest speed
- ONFI Syn: configurable timing, two bytes per two Nandc working clocks at the fastest speed
- Toggle: configurable timing, two byte per two Nandc working clocks at the fastest speed
- Randomizer Ability
- Support three randomizer mode with different polynomial
- Support two randomizer width, 8bit and 16bit parallel
- BCH/ECC Ability
- 16bit/1KB BCH/ECC: support 16bitBCH/ECC, which can detect and correct up to 16 error bits in every 1K bytes data
- 24bit/1KB BCH/ECC: support 24bitBCH/ECC, which can detect and correct up to 24 error bits in every 1K bytes data
- 40bit/1KB BCH/ECC: support 40bitBCH/ECC, which can detect and correct up to 40 error bits in every 1K bytes data
- 60bit/1KB BCH/ECC: support 60bitBCH/ECC, which can detect and correct up to 60 error bits in every 1K bytes data
- 8bit/512B BCH/ECC: support 8bitBCH/ECC, which can detect and correct up to 8 error bits in every 512 bytes data
- 12bit/512B BCH/ECC: support 12bitBCH/ECC, which can detect and correct up to 12 error bits in every 512 bytes data
- 20bit/512B BCH/ECC: support 20bitBCH/ECC, which can detect and correct up to 20 error bits in every 512 bytes data
- 30bit/512B BCH/ECC: support 30bitBCH/ECC, which can detect and correct up to 30 error bits in every 512 bytes data
- 16bit/512B BCH/ECC: support 16bitBCH/ECC, which can detect and correct up to 16 error bits in every 512 bytes data
- 24bit/512B BCH/ECC: support 24bitBCH/ECC, which can detect and correct up to 24 error bits in every 512 bytes data
- 40bit/512B BCH/ECC: support 40bitBCH/ECC, which can detect and correct up to 40 error bits in every 512 bytes data

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- 60bit/512B BCH/ECC: support 60bitBCH/ECC, which can detect and correct up to 60 error bits in every 512 bytes data
- Transmission Ability
- Support 32K bytes data transmission at a time at most
- Support two transfer working modes: Bypass or DMA
- Support two transfer codeword size for Managed NAND Flash: 1024 bytes/codeword or 512 bytes/codeword
- Internal Memory
- 2 built-in srams, and the size is 1k bytes respectively
- Can be accessed by other masters
- Can be operated in pingpang mode by other masters

17.2 Block Diagram

NandC comprises with:

- MIF: AHB Master Interface
- SIF : AHB Slave Interface
- SRIF : Sram Interface
- TRANSC : Transfer Controller
- LLPC : LLP Controller
- BCHENC : BCH Encoder
- BCHDEC : BCH Decoder
- RANDMZ : Randomizer
- FIF_GEN : Flash Interface Generation
- DLC : Delay Line Controller

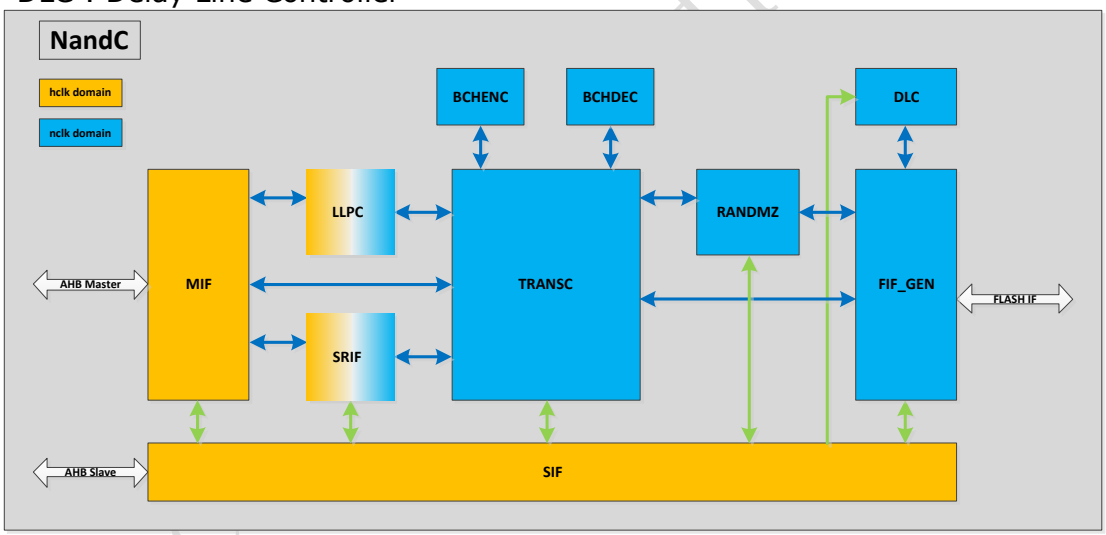

Fig. 17-1 NandC Block Diagram

17.3 Function Description

17.3.1 AHB Interface

There is an AHB master interface in NandC, which is selectable and configurable. It is responsible for transferring data from external memory to internal memory when flash program, or inverse when flash read; and transferring LLP data from external memory to internal register file when LLP is active.
There is an AHB slave interface in NandC. It is responsible for accessing registers and internal memories. The addresses of these registers and memories are listed in "Internal Address Mapping" section.

17.3.2 Flash Type/Flash Interface

Flash device with different types of interfaces is supported. These interfaces include: asynchronous 8bits flash interface, asynchronous 16bits flash interface, ONFI synchronous flash interface, toggle flash interface, and so on. You can select one of them by software (configure FMCTL) to suit for these devices. Also you can configure their timing parameters by

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software (configure FMWAIT_ASYN/FMWAIT_SYN) to have your desired rate.

17.3.3 Linked List Pointer Mode (LLP)

To save the software resource and improve the performance, a LLP is add, which is selectable. When LLP is selected, the flash operation instructions stored in external memory with specific format should be loaded for flash working. The detailed format and working flow are referred to "LLP Application" section.

17.3.4 BCH Encoder/BCH Decoder

The BCH Encoder is responsible for encoding data to be written into flash device. The max encoded length is 1133bytes, in which the data length is 1024bytes, system information is 4bytes, BCH code is 105bytes.
The BCH Decoder is responsible for decoding data read from flash device. The max decoded length is 1133bytes, in which the data length is 1024bytes, spare length is 109bytes.

17.3.5 Randomizer

To improve device lifetime, a randomizer is added in NandC. It includes two parts: Scrambler and Descrambler, which is responsible for scrambling data to be written into flash after bch encoding, and descramblingdata read from flash before bch decoding.

17.3.6 Delay Line Controller

For ONFI Synchronous Flash or Toggle Flash, the data read from flash follows with a strobe signal: DQS, where a skew between them exists. To remove the skew and improve the timing between data and DQS, a Delay Line Controller is needed. It is responsible for detecting the phase of the signal similar to DQS, determining the element number to be shifted, and then shifting the DQS with the determined number.

17.4 Register Description

17.4.1 Internal Address Mapping

Slave address can be divided into different length for different usage, which is shown as follows.

Table 17-1 NandC Address Mapping

Base Address[12:8]	Device	Address Length	Offset Address Range
5'b00_00x(x=0, 1)	FLR	512 BYTE	0x0000 ~ 0x01ff
5'b00_01x(x=0, 1)	SPR	512 BYTE	0x0200 ~ 0x03ff
5'b00_10x(x=0, 1)	FLR1	512 BYTE	0x0400 ~ 0x05ff
5'b00_11x(x=0,1)	FLR2	512 BYTE	0x0600 ~ 0x07ff
5'b01_000	Flash0	256 BYTE	0x0800 ~ 0x08ff
5'b01_001	Flash1	256 BYTE	0x0900 ~ 0x09ff
5'b01_010	Flash2	256 BYTE	0x0a00 ~ 0x0aff
5'b01_011	Flash3	256 BYTE	0x0b00 ~ 0x0bff
5'b01_100	Flash4	256 BYTE	0x0c00 ~ 0x0cff
5'b01_101	Flash5	256 BYTE	0x0d00 ~ 0x0dff
5'b01_110	Flash6	256 BYTE	0x0e00 ~ 0x0eff
5'b01_111	Flash7	256 BYTE	0x0f00 ~ 0x0fff
5'b10_0xx(x=0, 1)	Sram0	1K BYTE	0x1000 ~ 0x13ff
5'b10_1xx(x=0, 1)	Sram1	1K BYTE	0x1400 ~ 0x17ff

17.4.2 Registers Summary

Name	Offset	Size	Reset Value	Description
NANDC_FMCTL	0x0000	W	0x00000200	Flash Interface Control Register
NANDC_FMWAIT_ASYN	0x0004	W	0x3f03f7ff	Flash Timing Control Register For Asynchronous Timing
NANDC_FLCTL	0x0008	W	0x00100000	Internal Transfer Control Register

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Name	Offset	Size	Reset Value	Description
NANDC_BCHCTL	0x000c	W	0x00000008	BCH Control Register
NANDC_MTRANS_CFG	0x0010	W	0x000001d0	Bus Transfer Configuration Register
NANDC_MTRANS_SADDR0	0x0014	W	0x00000000	Start Address Register For Page Data Transmission
NANDC_MTRANS_SADDR1	0x0018	W	0x00000000	Start Address Register For Spare Data Transmission
NANDC_MTRANS_STAT	0x001c	W	0x00000000	Bus Transfer Status Register
NANDC_BCHST0	0x0020	W	0x04000000	BCH Status Register For Codeword 0~1
NANDC_BCHST1	0x0024	W	0x00000000	BCH Status Register For Codeword 2~3
NANDC_BCHST2	0x0028	W	0x00000000	BCH Status Register For Codeword 4~5
NANDC_BCHST3	0x002c	W	0x00000000	BCH Status Register For Codeword 6~7
NANDC_BCHST4	0x0030	W	0x00000000	BCH Status Register For Codeword 8~9
NANDC_BCHST5	0x0034	W	0x00000000	BCH Status Register For Codeword 10~11
NANDC_BCHST6	0x0038	W	0x00000000	BCH Status Register For Codeword 12~13
NANDC_BCHST7	0x003c	W	0x00000000	BCH Status Register For Codeword 14~15
NANDC_BCHLOC0	0x0040	W	0x00000000	BCH Error Bit Location Number Register For Codeword 0~5
NANDC_BCHLOC1	0x0044	W	0x00000000	BCH Error Bit Location Number Register For Codeword 6~11
NANDC_BCHLOC2	0x0048	W	0x00000000	BCH Error Bit Location Number Register For Codeword 12~17
NANDC_BCHLOC3	0x004c	W	0x00000000	BCH Error Bit Location Number Register For Codeword 24~29
NANDC_BCHLOC4	0x0050	W	0x00000000	BCH Error Bit Location Number Register For Codeword 24~29
NANDC_BCHLOC5	0x0054	W	0x00000000	BCH Error Bit Location Number Register For Codeword 30~31
NANDC_BCHLOC6	0x0058	W	0x00000000	Highest Bit For BCH Error Bit Location Number Register
NANDC_BCHDE0_0	0x0070	W	0x00000000	BCH decode result of 0th error bit for codeword 0
NANDC_BCHDE0_1	0x0074	W	0x00000000	BCH decode result of 1th error bit for codeword 0
NANDC_BCHDE0_2	0x0078	W	0x00000000	BCH decode result of 2th error bit for codeword 0
NANDC_BCHDE0_3	0x007c	W	0x00000000	BCH decode result of 3th error bit for codeword 0
NANDC_BCHDE0_4	0x0080	W	0x00000000	BCH decode result of 4th error bit for codeword 0

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Name	Offset	Size	Reset Value	Description
NANDC_BCHDE0_5	0x0084	W	0x00000000	BCH decode result of 5th error bit for codeword 0
NANDC_BCHDE0_6	0x0088	W	0x00000000	BCH decode result of 6th error bit for codeword 0
NANDC_BCHDE0_7	0x008c	W	0x00000000	BCH decode result of 7th error bit for codeword 0
NANDC_BCHDE0_8	0x0090	W	0x00000000	BCH decode result of 8th error bit for codeword 0
NANDC_BCHDE0_9	0x0094	W	0x00000000	BCH decode result of 9th error bit for codeword 0
NANDC_BCHDE0_10	0x0098	W	0x00000000	BCH decode result of 10th error bit for codeword 0
NANDC_BCHDE0_11	0x009c	W	0x00000000	BCH decode result of 11th error bit for codeword 0
NANDC_BCHDE0_12	0x00a0	W	0x00000000	BCH decode result of 12th error bit for codeword 0
NANDC_BCHDE0_13	0x00a4	W	0x00000000	BCH decode result of 13th error bit for codeword 0
NANDC_BCHDE0_14	0x00a8	W	0x00000000	BCH decode result of 14th error bit for codeword 0
NANDC_BCHDE0_15	0x00ac	W	0x00000000	BCH decode result of 15th error bit for codeword 0
NANDC_BCHDE0_16	0x00b0	W	0x00000000	BCH decode result of 16th error bit for codeword 0
NANDC_BCHDE0_17	0x00b4	W	0x00000000	BCH decode result of 17th error bit for codeword 0
NANDC_BCHDE0_18	0x00b8	W	0x00000000	BCH decode result of 18th error bit for codeword 0
NANDC_BCHDE0_19	0x00bc	W	0x00000000	BCH decode result of 19th error bit for codeword 0
NANDC_BCHDE0_20	0x00c0	W	0x00000000	BCH decode result of 20th error bit for codeword 0
NANDC_BCHDE0_21	0x00c4	W	0x00000000	BCH decode result of 21th error bit for codeword 0
NANDC_BCHDE0_22	0x00c8	W	0x00000000	BCH decode result of 22th error bit for codeword 0
NANDC_BCHDE0_23	0x00cc	W	0x00000000	BCH decode result of 23th error bit for codeword 0
NANDC_BCHDE1_0	0x00d0	W	0x00000000	BCH decode result of 0th error bit for codeword 1
NANDC_BCHDE1_1	0x00d4	W	0x00000000	BCH decode result of 1th error bit for codeword 1
NANDC_BCHDE1_2	0x00d8	W	0x00000000	BCH decode result of 2th error bit for codeword 1

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Name	Offset	Size	Reset Value	Description
NANDC_BCHDE1_3	0x00dc	W	0x00000000	BCH decode result of 3th error bit for codeword 1
NANDC_BCHDE1_4	0x00e0	W	0x00000000	BCH decode result of 4th error bit for codeword 1
NANDC_BCHDE1_5	0x00e4	W	0x00000000	BCH decode result of 5th error bit for codeword 1
NANDC_BCHDE1_6	0x00e8	W	0x00000000	BCH decode result of 6th error bit for codeword 1
NANDC_BCHDE1_7	0x00ec	W	0x00000000	BCH decode result of 7th error bit for codeword 1
NANDC_BCHDE1_8	0x00f0	W	0x00000000	BCH decode result of 8th error bit for codeword 1
NANDC_BCHDE1_9	0x00f4	W	0x00000000	BCH decode result of 9th error bit for codeword 1
NANDC_BCHDE1_10	0x00f8	W	0x00000000	BCH decode result of 10th error bit for codeword 1
NANDC_BCHDE1_11	0x00fc	W	0x00000000	BCH decode result of 11th error bit for codeword 1
NANDC_BCHDE1_12	0x0100	W	0x00000000	BCH decode result of 12th error bit for codeword 1
NANDC_BCHDE1_13	0x0104	W	0x00000000	BCH decode result of 13th error bit for codeword 1
NANDC_BCHDE1_14	0x0108	W	0x00000000	BCH decode result of 14th error bit for codeword 1
NANDC_BCHDE1_15	0x010c	W	0x00000000	BCH decode result of 15th error bit for codeword 1
NANDC_BCHDE1_16	0x0110	W	0x00000000	BCH decode result of 16th error bit for codeword 1
NANDC_BCHDE1_17	0x0114	W	0x00000000	BCH decode result of 17th error bit for codeword 1
NANDC_BCHDE1_18	0x0118	W	0x00000000	BCH decode result of 1th error bit for codeword 1
NANDC_BCHDE1_19	0x011c	W	0x00000000	BCH decode result of 19th error bit for codeword 1
NANDC_BCHDE1_20	0x0120	W	0x00000000	BCH decode result of 20th error bit for codeword 1
NANDC_BCHDE1_21	0x0124	W	0x00000000	BCH decode result of 21th error bit for codeword 1
NANDC_BCHDE1_22	0x0128	W	0x00000000	BCH decode result of 22th error bit for codeword 1
NANDC_BCHDE1_23	0x012c	W	0x00000000	BCH decode result of 23th error bit for codeword 1
NANDC_DLL_CTL_REG0	0x0130	W	0x007f7f05	DLL Control Register 0

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Name	Offset	Size	Reset Value	Description
NANDC_DLL_CTL_REG1	0x0134	W	0x00000022	DLL Control Register 1
NANDC_DLL_OBS_REG0	0x0138	W	0x00000200	DLL Status Register
NANDC_RANDMZ_CFG	0x0150	W	0x00000000	Randomizer Configure Register
NANDC_FMWAIT_SYN	0x0158	W	0x00000000	Flash Timing Control Register For Synchronous Timing
NANDC_MTRANS_STAT2	0x015c	W	0x00000000	Bus Transfer Status Register2
NANDC_NANDC_VER	0x0160	W	0x56363232	Nandc Version Register
NANDC_LLP_CTL	0x0164	W	0x00000000	LLP Control Register
NANDC_LLP_STAT	0x0168	W	0x00000001	LLP Status Register
NANDC_INTEN	0x016c	W	0x00000000	NandC Interrupt Enable Register
NANDC_INTCLR	0x0170	W	0x00000000	NandC Interrupt Clear Register
NANDC_INTST	0x0174	W	0x00000000	NandC Interrupt Status Register
NANDC_SPARE0_0	0x0200	W	0xffffffff	System Information for codeword 0
NANDC_SPARE0_1	0x0204	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_2	0x0208	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_3	0x020c	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_4	0x0210	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_5	0x0214	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_6	0x0218	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_7	0x021c	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_8	0x0220	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_9	0x0224	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_10	0x0228	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_11	0x022c	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE1_0	0x0230	W	0xffffffff	System Information for codeword 1
NANDC_SPARE1_1	0x0234	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_2	0x0238	W	0x00000000	Spare Data and BCH Encode Information for codeword 1

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Name	Offset	Size	Reset Value	Description
NANDC_SPARE1_3	0x023c	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_4	0x0240	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_5	0x0244	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_6	0x0248	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_7	0x024c	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_8	0x0250	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_9	0x0254	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_10	0x0258	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_11	0x025c	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE0_12	0x0260	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_13	0x0264	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_14	0x0268	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_15	0x026c	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_16	0x0270	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_17	0x0274	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_18	0x0278	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_19	0x027c	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_20	0x0280	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_21	0x0284	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_22	0x0288	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_23	0x028c	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_24	0x0290	W	0x00000000	Spare Data and BCH Encode Information for codeword 0

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Name	Offset	Size	Reset Value	Description
NANDC_SPARE0_25	0x0294	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_26	0x0298	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE0_27	0x029c	W	0x00000000	Spare Data and BCH Encode Information for codeword 0
NANDC_SPARE1_12	0x02a0	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_13	0x02a4	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_14	0x02a8	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_15	0x02ac	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_16	0x02b0	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_17	0x02b4	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_18	0x02b8	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_19	0x02bc	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_20	0x02c0	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_21	0x02c4	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_22	0x02c8	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_23	0x02cc	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_24	0x02d0	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_25	0x02d4	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_26	0x02d8	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_SPARE1_27	0x02dc	W	0x00000000	Spare Data and BCH Encode Information for codeword 1
NANDC_BCHDE0_24	0x0400	W	0x00000000	BCH decode result of 24th error bit for codeword 0
NANDC_BCHDE0_25	0x0404	W	0x00000000	BCH decode result of 25th error bit for codeword 0
NANDC_BCHDE0_26	0x0408	W	0x00000000	BCH decode result of 26th error bit for codeword 0

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Name	Offset	Size	Reset Value	Description
NANDC_BCHDE0_27	0x040c	W	0x00000000	BCH decode result of 27th error bit for codeword 0
NANDC_BCHDE0_28	0x0410	W	0x00000000	BCH decode result of 28th error bit for codeword 0
NANDC_BCHDE0_29	0x0414	W	0x00000000	BCH decode result of 29th error bit for codeword 0
NANDC_BCHDE0_30	0x0418	W	0x00000000	BCH decode result of 30th error bit for codeword 0
NANDC_BCHDE0_31	0x041c	W	0x00000000	BCH decode result of 31th error bit for codeword 0
NANDC_BCHDE0_32	0x0420	W	0x00000000	BCH decode result of 32th error bit for codeword 0
NANDC_BCHDE0_33	0x0424	W	0x00000000	BCH decode result of 33th error bit for codeword 0
NANDC_BCHDE0_34	0x0428	W	0x00000000	BCH decode result of 3th error bit for codeword 0
NANDC_BCHDE0_35	0x042c	W	0x00000000	BCH decode result of 35th error bit for codeword 0
NANDC_BCHDE0_36	0x0430	W	0x00000000	BCH decode result of 36th error bit for codeword 0
NANDC_BCHDE0_37	0x0434	W	0x00000000	BCH decode result of 37th error bit for codeword 0
NANDC_BCHDE0_38	0x0438	W	0x00000000	BCH decode result of 38th error bit for codeword 0
NANDC_BCHDE0_39	0x043c	W	0x00000000	BCH decode result of 39th error bit for codeword 0
NANDC_BCHDE0_40	0x0440	W	0x00000000	BCH decode result of 40th error bit for codeword 0
NANDC_BCHDE0_41	0x0444	W	0x00000000	BCH decode result of 41th error bit for codeword 0
NANDC_BCHDE0_42	0x0448	W	0x00000000	BCH decode result of 42th error bit for codeword 0
NANDC_BCHDE0_43	0x044c	W	0x00000000	BCH decode result of 43th error bit for codeword 0
NANDC_BCHDE0_44	0x0450	W	0x00000000	BCH decode result of 44th error bit for codeword 0
NANDC_BCHDE0_45	0x0454	W	0x00000000	BCH decode result of 45th error bit for codeword 0
NANDC_BCHDE0_46	0x0458	W	0x00000000	BCH decode result of 46th error bit for codeword 0
NANDC_BCHDE0_47	0x045c	W	0x00000000	BCH decode result of 47th error bit for codeword 0
NANDC_BCHDE0_48	0x0460	W	0x00000000	BCH decode result of 48th error bit for codeword 0

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Name	Offset	Size	Reset Value	Description
NANDC_BCHDE0_49	0x0464	W	0x00000000	BCH decode result of 49th error bit for codeword 0
NANDC_BCHDE0_50	0x0468	W	0x00000000	BCH decode result of 50th error bit for codeword 0
NANDC_BCHDE0_51	0x046c	W	0x00000000	BCH decode result of 51th error bit for codeword 0
NANDC_BCHDE0_52	0x0470	W	0x00000000	BCH decode result of 52th error bit for codeword 0
NANDC_BCHDE0_53	0x0474	W	0x00000000	BCH decode result of 53th error bit for codeword 0
NANDC_BCHDE0_54	0x0478	W	0x00000000	BCH decode result of 54th error bit for codeword 0
NANDC_BCHDE0_55	0x047c	W	0x00000000	BCH decode result of 55th error bit for codeword 0
NANDC_BCHDE0_56	0x0480	W	0x00000000	BCH decode result of 56th error bit for codeword 0
NANDC_BCHDE0_57	0x0484	W	0x00000000	BCH decode result of 57th error bit for codeword 0
NANDC_BCHDE0_58	0x0488	W	0x00000000	BCH decode result of 58th error bit for codeword 0
NANDC_BCHDE0_59	0x048c	W	0x00000000	BCH decode result of 59th error bit for codeword 0
NANDC_BCHDE1_24	0x0490	W	0x00000000	BCH decode result of 24th error bit for codeword 1
NANDC_BCHDE1_25	0x0494	W	0x00000000	BCH decode result of 25th error bit for codeword 1
NANDC_BCHDE1_26	0x0498	W	0x00000000	BCH decode result of 26th error bit for codeword 1
NANDC_BCHDE1_27	0x049c	W	0x00000000	BCH decode result of 27th error bit for codeword 1
NANDC_BCHDE1_28	0x04a0	W	0x00000000	BCH decode result of 28th error bit for codeword 1
NANDC_BCHDE1_29	0x04a4	W	0x00000000	BCH decode result of 29th error bit for codeword 1
NANDC_BCHDE1_30	0x04a8	W	0x00000000	BCH decode result of 30th error bit for codeword 1
NANDC_BCHDE1_31	0x04ac	W	0x00000000	BCH decode result of 31th error bit for codeword 1
NANDC_BCHDE1_32	0x04b0	W	0x00000000	BCH decode result of 32th error bit for codeword 1
NANDC_BCHDE1_33	0x04b4	W	0x00000000	BCH decode result of 33th error bit for codeword 1
NANDC_BCHDE1_34	0x04b8	W	0x00000000	BCH decode result of 34th error bit for codeword 1

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Name	Offset	Size	Reset Value	Description
NANDC_BCHDE1_35	0x04bc	W	0x00000000	BCH decode result of 35th error bit for codeword 1
NANDC_BCHDE1_36	0x04c0	W	0x00000000	BCH decode result of 3th error bit for codeword 1
NANDC_BCHDE1_37	0x04c4	W	0x00000000	BCH decode result of 37th error bit for codeword 1
NANDC_BCHDE1_38	0x04c8	W	0x00000000	BCH decode result of 38th error bit for codeword 1
NANDC_BCHDE1_39	0x04cc	W	0x00000000	BCH decode result of 39th error bit for codeword 1
NANDC_BCHDE1_40	0x04d0	W	0x00000000	BCH decode result of 40th error bit for codeword 1
NANDC_BCHDE1_41	0x04d4	W	0x00000000	BCH decode result of 41th error bit for codeword 1
NANDC_BCHDE1_42	0x04d8	W	0x00000000	BCH decode result of 42th error bit for codeword 1
NANDC_BCHDE1_43	0x04dc	W	0x00000000	BCH decode result of 43th error bit for codeword 1
NANDC_BCHDE1_44	0x04e0	W	0x00000000	BCH decode result of 44th error bit for codeword 1
NANDC_BCHDE1_45	0x04e4	W	0x00000000	BCH decode result of 45th error bit for codeword 1
NANDC_BCHDE1_46	0x04e8	W	0x00000000	BCH decode result of 46th error bit for codeword 1
NANDC_BCHDE1_47	0x04ec	W	0x00000000	BCH decode result of 47th error bit for codeword 1
NANDC_BCHDE1_48	0x04f0	W	0x00000000	BCH decode result of 48th error bit for codeword 1
NANDC_BCHDE1_49	0x04f4	W	0x00000000	BCH decode result of 49th error bit for codeword 1
NANDC_BCHDE1_50	0x04f8	W	0x00000000	BCH decode result of 50th error bit for codeword 1
NANDC_BCHDE1_51	0x04fc	W	0x00000000	BCH decode result of 51th error bit for codeword 1
NANDC_BCHDE1_52	0x0500	W	0x00000000	BCH decode result of 52th error bit for codeword 1
NANDC_BCHDE1_53	0x0504	W	0x00000000	BCH decode result of 53th error bit for codeword 1
NANDC_BCHDE1_54	0x0508	W	0x00000000	BCH decode result of 54th error bit for codeword 1
NANDC_BCHDE1_55	0x050c	W	0x00000000	BCH decode result of 55th error bit for codeword 1
NANDC_BCHDE1_56	0x0510	W	0x00000000	BCH decode result of 56th error bit for codeword 1

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Name	Offset	Size	Reset Value	Description
NANDC_BCHDE1_57	0x0514	W	0x00000000	BCH decode result of 57th error bit for codeword 1
NANDC_BCHDE1_58	0x0518	W	0x00000000	BCH decode result of 58th error bit for codeword 1
NANDC_BCHDE1_59	0x051c	W	0x00000000	BCH decode result of 59th error bit for codeword 1
NANDC_BCHST8	0x0520	W	0x00000000	BCH Status Register For Codeword 16~17
NANDC_BCHST9	0x0524	W	0x00000000	BCH Status Register For Codeword 18~19
NANDC_BCHST10	0x0528	W	0x00000000	BCH Status Register For Codeword 20~21
NANDC_BCHST11	0x052c	W	0x00000000	BCH Status Register For Codeword 22~23
NANDC_BCHST12	0x0530	W	0x00000000	BCH Status Register For Codeword 24~25
NANDC_BCHST13	0x0534	W	0x00000000	BCH Status Register For Codeword 26~27
NANDC_BCHST14	0x0538	W	0x00000000	BCH Status Register For Codeword 28~29
NANDC_BCHST15	0x053c	W	0x00000000	BCH Status Register For Codeword 30~31
NANDC_RANDMZ_SEED13_0	0x0600	W	0x00000000	Seed 0 for Toshiba 13 Power Polynomial Randomizer
NANDC_RANDMZ_SEED13_1	0x0604	W	0x00000000	Seed 1 for Toshiba 13 Power Polynomial Randomizer
NANDC_RANDMZ_SEED13_2	0x0608	W	0x00000000	Seed 2 for Toshiba 13 Power Polynomial Randomizer
NANDC_RANDMZ_SEED13_3	0x060c	W	0x00000000	Seed 3 for Toshiba 13 Power Polynomial Randomizer
NANDC_RANDMZ_SEED13_4	0x0610	W	0x00000000	Seed 4 for Toshiba 13 Power Polynomial Randomizer
NANDC_RANDMZ_SEED13_5	0x0614	W	0x00000000	Seed 5 for Toshiba 13 Power Polynomial Randomizer
NANDC_RANDMZ_SEED13_6	0x0618	W	0x00000000	Seed 6 for Toshiba 13 Power Polynomial Randomizer
NANDC_RANDMZ_SEED13_7	0x061c	W	0x00000000	Seed 7 for Toshiba 13 Power Polynomial Randomizer
NANDC_RANDMZ_SEED13_8	0x0620	W	0x00000000	Seed 8 for Toshiba 13 Power Polynomial Randomizer
NANDC_RANDMZ_SEED13_9	0x0624	W	0x00000000	Seed 9 for Toshiba 13 Power Polynomial Randomizer
NANDC_RANDMZ_SEED13_10	0x0628	W	0x00000000	Seed 10 for Toshiba 13 Power Polynomial Randomizer

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Name	Offset	Size	Reset Value	Description
NANDC_RANDMZ_SEED13_11	0x062c	W	0x00000000	Seed 11 for Toshiba 13 Power Polynomial Randomizer
NANDC_RANDMZ_SEED13_12	0x0630	W	0x00000000	Seed 12 for Toshiba 13 Power Polynomial Randomizer
NANDC_RANDMZ_SEED13_13	0x0634	W	0x00000000	Seed 13 for Toshiba 13 Power Polynomial Randomizer
NANDC_RANDMZ_SEED13_14	0x0638	W	0x00000000	Seed 14 for Toshiba 13 Power Polynomial Randomizer
NANDC_RANDMZ_SEED13_15	0x063c	W	0x00000000	Seed 15 for Toshiba 13 Power Polynomial Randomizer
NANDC_RANDMZ_SEED17_0	0x0640	W	0x00000000	Seed 0 for Toshiba 17 Power Polynomial Randomizer
NANDC_RANDMZ_SEED17_1	0x0644	W	0x00000000	Seed 1 for Toshiba 17 Power Polynomial Randomizer
NANDC_RANDMZ_SEED17_2	0x0648	W	0x00000000	Seed 2 for Toshiba 17 Power Polynomial Randomizer
NANDC_RANDMZ_SEED17_3	0x064c	W	0x00000000	Seed 3 for Toshiba 17 Power Polynomial Randomizer
NANDC_RANDMZ_SEED17_4	0x0650	W	0x00000000	Seed 4 for Toshiba 17 Power Polynomial Randomizer
NANDC_RANDMZ_SEED17_5	0x0654	W	0x00000000	Seed 5 for Toshiba 17 Power Polynomial Randomizer
NANDC_RANDMZ_SEED17_6	0x0658	W	0x00000000	Seed 6 for Toshiba 17 Power Polynomial Randomizer
NANDC_RANDMZ_SEED17_7	0x065c	W	0x00000000	Seed 7 for Toshiba 17 Power Polynomial Randomizer
NANDC_RANDMZ_SEED17_8	0x0660	W	0x00000000	Seed 8 for Toshiba 17 Power Polynomial Randomizer
NANDC_RANDMZ_SEED17_9	0x0664	W	0x00000000	Seed 9 for Toshiba 17 Power Polynomial Randomizer
NANDC_RANDMZ_SEED17_10	0x0668	W	0x00000000	Seed 10 for Toshiba 17 Power Polynomial Randomizer
NANDC_RANDMZ_SEED17_11	0x066c	W	0x00000000	Seed 11 for Toshiba 17 Power Polynomial Randomizer
NANDC_RANDMZ_SEED17_12	0x0670	W	0x00000000	Seed 12 for Toshiba 17 Power Polynomial Randomizer
NANDC_RANDMZ_SEED17_13	0x0674	W	0x00000000	Seed 13 for Toshiba 17 Power Polynomial Randomizer
NANDC_RANDMZ_SEED17_14	0x0678	W	0x00000000	Seed 14 for Toshiba 17 Power Polynomial Randomizer
NANDC_RANDMZ_SEED17_15	0x067c	W	0x00000000	Seed 15 for Toshiba 17 Power Polynomial Randomizer
NANDC_RANDMZ_SEED19_0	0x0680	W	0x00000000	Seed 0 for Toshiba 19 Power Polynomial Randomizer

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Name	Offset	Size	Reset Value	Description
NANDC_RANDMZ_SEED19_1	0x0684	W	0x00000000	Seed 1 for Toshiba 19 Power Polynomial Randomizer
NANDC_RANDMZ_SEED19_2	0x0688	W	0x00000000	Seed 2 for Toshiba 19 Power Polynomial Randomizer
NANDC_RANDMZ_SEED19_3	0x068c	W	0x00000000	Seed 3 for Toshiba 19 Power Polynomial Randomizer
NANDC_RANDMZ_SEED19_4	0x0690	W	0x00000000	Seed 4 for Toshiba 19 Power Polynomial Randomizer
NANDC_RANDMZ_SEED19_5	0x0694	W	0x00000000	Seed 5 for Toshiba 19 Power Polynomial Randomizer
NANDC_RANDMZ_SEED19_6	0x0698	W	0x00000000	Seed 6 for Toshiba 19 Power Polynomial Randomizer
NANDC_RANDMZ_SEED19_7	0x069c	W	0x00000000	Seed 7 for Toshiba 19 Power Polynomial Randomizer
NANDC_RANDMZ_SEED19_8	0x06a0	W	0x00000000	Seed 8 for Toshiba 19 Power Polynomial Randomizer
NANDC_RANDMZ_SEED19_9	0x06a4	W	0x00000000	Seed 9 for Toshiba 19 Power Polynomial Randomizer
NANDC_RANDMZ_SEED19_10	0x06a8	W	0x00000000	Seed 10 for Toshiba 19 Power Polynomial Randomizer
NANDC_RANDMZ_SEED19_11	0x06ac	W	0x00000000	Seed 11 for Toshiba 19 Power Polynomial Randomizer
NANDC_RANDMZ_SEED19_12	0x06b0	W	0x00000000	Seed 12 for Toshiba 19 Power Polynomial Randomizer
NANDC_RANDMZ_SEED19_13	0x06b4	W	0x00000000	Seed 13 for Toshiba 19 Power Polynomial Randomizer
NANDC_RANDMZ_SEED19_14	0x06b8	W	0x00000000	Seed 14 for Toshiba 19 Power Polynomial Randomizer
NANDC_RANDMZ_SEED19_15	0x06bc	W	0x00000000	Seed 15 for Toshiba 19 Power Polynomial Randomizer
NANDC_RANDMZ_SEED23_0	0x06c0	W	0x00000000	Seed 0 for Toshiba 23 Power Polynomial Randomizer
NANDC_RANDMZ_SEED23_1	0x06c4	W	0x00000000	Seed 1 for Toshiba 23 Power Polynomial Randomizer
NANDC_RANDMZ_SEED23_2	0x06c8	W	0x00000000	Seed 2 for Toshiba 23 Power Polynomial Randomizer
NANDC_RANDMZ_SEED23_3	0x06cc	W	0x00000000	Seed 3 for Toshiba 23 Power Polynomial Randomizer
NANDC_RANDMZ_SEED23_4	0x06d0	W	0x00000000	Seed 4 for Toshiba 23 Power Polynomial Randomizer
NANDC_RANDMZ_SEED23_5	0x06d4	W	0x00000000	Seed 5 for Toshiba 23 Power Polynomial Randomizer
NANDC_RANDMZ_SEED23_6	0x06d8	W	0x00000000	Seed 6 for Toshiba 23 Power Polynomial Randomizer

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Name	Offset	Size	Reset Value	Description
NANDC_RANDMZ_SEED23_7	0x06dc	W	0x00000000	Seed 7 for Toshiba 23 Power Polynomial Randomizer
NANDC_RANDMZ_SEED23_8	0x06e0	W	0x00000000	Seed 8 for Toshiba 23 Power Polynomial Randomizer
NANDC_RANDMZ_SEED23_9	0x06e4	W	0x00000000	Seed 9 for Toshiba 23 Power Polynomial Randomizer
NANDC_RANDMZ_SEED23_10	0x06e8	W	0x00000000	Seed 10 for Toshiba 23 Power Polynomial Randomizer
NANDC_RANDMZ_SEED23_11	0x06ec	W	0x00000000	Seed 11 for Toshiba 23 Power Polynomial Randomizer
NANDC_RANDMZ_SEED23_12	0x06f0	W	0x00000000	Seed 12 for Toshiba 23 Power Polynomial Randomizer
NANDC_RANDMZ_SEED23_13	0x06f4	W	0x00000000	Seed 13 for Toshiba 23 Power Polynomial Randomizer
NANDC_RANDMZ_SEED23_14	0x06f8	W	0x00000000	Seed 14 for Toshiba 23 Power Polynomial Randomizer
NANDC_RANDMZ_SEED23_15	0x06fc	W	0x00000000	Seed 15 for Toshiba 23 Power Polynomial Randomizer

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

17.4.3 Detail Register Description

NANDC_FMCTL

Address: Operational Base + offset (0x0000)

Flash Interface Control Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RW	0x0	syn_mode Toggle enable signal, 1 active. 0: ONFI synchronous flash. 1: Toggle synchronous flash.
14	RW	0x0	syn_clken Synchronous flash clock enable signal, 1 active. Only available in Synchronous Mode. 0: flash clock is disabled. 1: flash clock is enabled.
13	RW	0x0	tm Timing mode indication. 0: Asynchronous Mode. 1: Synchronous Mode (Toggle or ONFI Synchronous).

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Bit	Attr	Reset Value	Description
12	RW	0x0	dwidth Flash data bus width indication. 0: 8bits, active in both Asynchronous Mode flash and Synchronous Mode flash. 1: 16bits, active only in Asynchronous Mode flash.
11:10	RO	0x0	reserved
9	RO	0x1	frdy Flash ready/busy indicate signal. 0: flash is busy. 1: flash is ready. This bit is the sample of the pin of R/Bn.
8	RW	0x0	wp Flash write protect. 0: flash program/erase disabled. 1: flash program/erase enabled. This bit is output to the pin of WPn.
7	RW	0x0	fcs7 Flash memory chip 7 select control. 1: hold flash memory chip select activity. 0: flash memory chip select activity free.
6	RW	0x0	fcs6 Flash memory chip 6 select control. 1: hold flash memory chip select activity. 0: flash memory chip select activity free.
5	RW	0x0	fcs5 Flash memory chip 5 select control. 1: hold flash memory chip select activity. 0: flash memory chip select activity free.
4	RW	0x0	fcs4 Flash memory chip 4 select control. 1: hold flash memory chip select activity. 0: flash memory chip select activity free.
3	RW	0x0	fcs3 Flash memory chip 3 select control. 1: hold flash memory chip select activity. 0: flash memory chip select activity free.
2	RW	0x0	fcs2 Flash memory chip 2 select control. 1: hold flash memory chip select activity. 0: flash memory chip select activity free.
1	RW	0x0	fcs1 Flash memory chip 1 select control. 1: hold flash memory chip select activity. 0: flash memory chip select activity free.

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Bit	Attr	Reset Value	Description
0	RW	0x0	fcs0 Flash memory chip 0 select control. 1: hold flash memory chip select activity. 0: flash memory chip select activity free.

NANDC_FMWAIT_ASYN
Address: Operational Base + offset (0x0004)
Flash Timing Control Register For Asynchronous Timing

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30	RW	0x0	fmw_dly_en fmw_dly enable signal,1 active.
29:24	RW	0x3f	fmw_dly The number of delay cycle between two codeword transmission.
23:18	RO	0x0	reserved
17:12	RW	0x3f	csrw When in Asynchronous mode or Toggle address/command mode, this field specifies the number of processor clock cycles from the falling edge of CSn to the falling edge of RDn or WRn. The min value of csrw is 0.
11	RO	0x0	reserved
10:5	RW	0x3f	rwpw When in Asynchronous mode or Toggle address/command mode, this field specifies the width of RDn or WRn in processor clock cycles, 0x0<=rwpw<=0x3f.
4:0	RW	0x1f	rwcs When in Asynchronous mode or Toggle address/command mode, this field specifies the number of processor clock cycles from the rising edge of RDn or WRn to the rising edge of CSn, 0x0<=rwcs<=0x1f.

NANDC_FLCTL
Address: Operational Base + offset (0x0008)
Internal Transfer Control Register

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28	RW	0x0	low_power Nandc low power control 0: normal mode 1: low power mode

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Bit	Attr	Reset Value	Description
27:22	RW	0x00	page_num Transmission codeword number in internal DMA mode when bus-mode is master-mode 1~32: 1~32 codeword. default: not support. Notes: a. Only active in internal DMA mode b. Only active when bus-mode is master-mode
21	RW	0x0	page_size Transmission codeword size in internal DMA mode 0: 1024bytes/codeword 1: 512bytes/codeword
20	RO	0x1	tr_rdy Internal DMA transmission ready indication. 0: internal DMA transmission is busy 1: internal DMA transmission is ready When reading flash, tr_rdy should not be set to 1 until all data transmission and correct finished. When programing flash, tr_rdy should not be set to 1 until all data transmission finished. Notes: Only active in internal DMA mode.
19	RO	0x0	reserved
18:12	RW	0x00	spare_size Spare byte number when lba_en=1. 0<= spare_size<=109. When spare_size>=109, it is treated as 0. Notes: The spare_size must be even number when flash is ONFI Synchronous Flash or Aynchronous Flash with 16bits data width.

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Bit	Attr	Reset Value	Description
11	RW	0x0	lba_en LBA mode indication, 1 active. 0: NO-LBA mode, NandC should transfer both page data and spare data in every codeword, and the page size is 1024 bytes or 512 bytes determined by BCHCTL[16](bchpage), spare size is 32/46/74 bytes or 109 bytes determined by BCHCTL[4] and BCHCTL[18]. 1: LBA mode, NandC should transfer both page data and spare data in every codeword, and the page size is 1024 bytes or 512 bytes determined by FLCTL[21](page_size), spare size is determined by FLCTL[17:12](spare_size). Notes: a. When lba_en is active, BCH CODEC should be disabled, spare_size and page_size are configurable. b. When lba_en is active, cor_able is inactive.
10	RW	0x0	cor_able Auto correct enable indication, 1 active. 0: auto correct disable 1: auto correct enable Notes: a. Only active in internal DMA mode. b. lba_en is prior to cor_able. When lba_en=1, cor_able is ignored.
9:8	RO	0x0	reserved
7	RW	0x0	flash_st_mod Mode for NandC to start internal data transmission in internal DMA mode. 0: busy mode: hardware should not start internal data transmission until flash is ready even flash_st is asserted. 1: ready mode: hardware should start internal data transmission directly when flash_st is asserted. Notes: Only active in internal DMA mode.

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Bit	Attr	Reset Value	Description
6:5	RW	0x0	tr_count Transmission codeword number in internal DMA mode when bus-mode is slave-mode 00: 0 codeword need transferred 01: 1 codeword need transferred 10: 2 codeword need transferred 11: not supported Notes: a. Only active in internal DMA mode. b. Only active when bus-mode is slave-mode.
4	RW	0x0	st_addr Start buffer address. 0: start transfer from sram0 1: start transfer from sram1 Notes: Only active in internal DMA mode.
3	RW	0x0	bypass NandC internal DMA bypass indication. 0: bypass the internal DMA, data are transferred to/from flash by direct path. 1: internal DMA active, data are transferred to/from flash by internal DMA.
2	RW	0x0	flash_st Start signal for NandC to transfer data between flash and internal buffer in internal DMA mode. When asserted, it will auto cleared. 0: not start transmission 1: start transmission Notes: Only active in internal DMA mode
1	RW	0x0	flash_rdn Indicate data flow direction. 0: NandC read data from flash. 1: NandC write data to flash
0	RW	0x0	flash_rst NandC software reset indication. When asserted, it will auto cleared. 0: not software reset 1: software reset Notes: flash_rst is prior to flash_st

NANDC_BCHCTL
Address: Operational Base + offset (0x000c)

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BCH Control Register

Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved
26:19	RW	0x00	bchthres BCH error number threshold
18	RW	0x0	bchmode1 High bit of BCH mode selection for 40bitBCH or 60bitBCH. BchMode=bchmode1, bchmode0: 00: 16bitBCH 01: 24bitBCH 10: 40bitBCH 11: 60bitBCH
17	RO	0x0	reserved
16	RW	0x0	bchpage The data size indication when BCH is active. 0: 1024 bytes, all the 1024 bytes data in codeword are valid data to be transferred. 1: 512 bytes, higher 512bytes are valid, and lower 512bytes are invalid and stuffed with 0xff. Notes: a. Only active when data transferred in internal DMA mode. b. Only active for asynchronous flash.
15:8	RW	0x00	addr BCH active range selection. BCH should be active when access in range address.
7:5	RW	0x0	region BCH active region selection indication. 000: Flash memory 0 region (flash 0) 001: Flash memory 1 region (flash 1) 010: Flash memory 2 region (flash 2) 011: Flash memory 3 region (flash 3) 100: Flash memory 4 region (flash 4) 101: Flash memory 5 region (flash 5) 110: Flash memory 6 region (flash 6) 111: Flash memory 7 region (flash 7)
4	RW	0x0	bchmode0 BCH mode selection indication. BCH mode is determined by both bchmode0 and bchmode1,detailed information is showed in BCHCTL[18].

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Bit	Attr	Reset Value	Description
3	RW	0x1	bchepd BCH encoder/decoder power down indication. 0: BCH encoder/decoder working. 1: BCH encoder/decoder not working.
2	RW	0x0	mode_addrcare BCH address care mode selection indication. 0: address care. 1: address not care. Notes: This bit is just active for data transmission in bypass mode, but not for command and address transmission.
1	RO	0x0	reserved
0	RW	0x0	bchrst BCH software reset indication, When asserted, it will auto cleared. 0: not software reset 1: software reset Notes: a. BCH Decoder should be software reset before decode begin. b. bch software reset should be used with nandc software reset at the same time.

NANDC_MTRANS_CFG
Address: Operational Base + offset (0x0010)
Bus Transfer Configuration Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	W1C	0x0	ahb_rst ahb master interface software reset, auto cleared
14	RW	0x0	fl_pwd Flash power down indication, 1 active. 0: Flash power on, data transferred through master interface is data that to be written into or read from flash. 1: Flash power down, data transferred through master interface is not data that to be written into or read from flash. NandC is just used as DMA for external memory and internal memory.

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Bit	Attr	Reset Value	Description
13:9	RW	0x00	incr_num AHB Master incr num indication. incr_num=1~16. When burst=001, software should configure incr_num. Notes: Only active for master-mode.
8:6	RW	0x7	burst AHB Master burst type indication: 000 : Single transfer 011 : 4-beat burst 101 : 8-beat Burst 111 : 16-beat burst default : not supported Notes: Only active for master-mode.
5:3	RW	0x2	hsize AHB Master data size indication: 000 : 8 bits 001 : 16 bits 010 : 32 bits default : not supported Notes: Only active for master-mode.
2	RW	0x0	bus_mode Bus interface selection. 0: Slave interface, flash data is transferred through slave interface 1: Master interface, flash data is transferred through master interface
1	RW	0x0	ahb_wr Data transfer direction through master interface. 0: read direction(internal memory ->external memory) 1: write direction (internal memory->external memory) Notes: a. Only active for master-mode. b. When read flash(flash_rdn=0), ahb_wr=1; when program flash(flash_rdn=1), ahb_wr=0.

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Bit	Attr	Reset Value	Description
0	W1C	0x0	ahb_wr_st Start indication for loading data from external memory to internal memory or storing data from internal memory to external memory through master. When asserted, it will auto cleared. Notes: a. Only active for master-mode and fl_pwd=1. b. When fl_pwd=0, flash is active, NandC start to transfer data through master interface if flash_st=1 c. When fl_pwd=1, flash is not active, NandC start to transfer data through master interface if ahb_wr_st=1

NANDC_MTRANS_SADDR0
Address: Operational Base + offset (0x0014)
Start Address Register For Page Data Transmission

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	saddr0 Start address for page data transmision. Notes: a. Only active for master-mode. b. Should be aligned with hsize in MTRANS_CFG[5:3].

NANDC_MTRANS_SADDR1
Address: Operational Base + offset (0x0018)
Start Address Register For Spare Data Transmission

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	saddr1 Start address for spare data. Notes: a. Only active for master-mode. b. Should be aligned with hsize in MTRANS_CFG[5:3].

NANDC_MTRANS_STAT
Address: Operational Base + offset (0x001c)
Bus Transfer Status Register

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
21:16	RO	0x00	mtrans_cnt finished counter for codeword transmission through Master interface Notes: Only active for master-mode.
15:0	RO	0x0000	bus_err Bus error indication for codeword0~15. [0] : bus error for codeword 0 …… [15] : bus error for codeword 15 Notes: Only active for master-mode.

NANDC_BCHST0
Address: Operational Base + offset (0x0020)
BCH Status Register For Codeword 0~1

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30	RO	0x0	err_hnum1_h1 Highest bit of err_hnum1
29	RO	0x0	err_tnum1_h1 Highest bit of err_tnum1
28	RO	0x0	err_hnum0_h1 Highest bit of err_hnum0
27	RO	0x0	err_tnum0_h1 Highest bit of err_tnum0
26	RO	0x1	bchrdy Ready indication for bch encoder/decoder, 1 active. 0: bch encoder/decoder is busy 1: bch encoder/decoder is ready
25:21	RO	0x00	err_hnum1_l5 Lower 5 bits of number of error bits found in first 512bytes of 1st backup codeword
20:16	RO	0x00	err_tnum1_l5 Lower 5 bits of number of error bits found in 1st backup codeword
15	RO	0x0	fail1 Indication for the 1st backup codeword decoded failed or not. 0: decode successfully 1: decode fail

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Bit	Attr	Reset Value	Description
14	RO	0x0	done1 Indication for finishing decoding the 1st backup codeword 0: not finished 1: finished
13	RO	0x0	errf1 Indication for error found in 1st backup codeword. 0: no error 1: error found
12:8	RO	0x00	err_hnum0_l5 Lower 5 bits of number of error bits found in first 512bytes of current backup codeword
7:3	RO	0x00	err_tnum0_l5 Lower 5 bits of number of error bits found in current backup codeword
2	RO	0x0	fail0 Indication for current backup codeword decode failed or not 0: decode successfully 1: decode fail
1	RO	0x0	done0 Indication for finishing decoding the current backup codeword. 0: not finished 1: finished
0	RO	0x0	errf0 Indication for error found in current backup codeword. 0: no error 1: error found

NANDC_BCHST1
Address: Operational Base + offset (0x0024)
BCH Status Register For Codeword 2~3

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30	RO	0x0	err_hnum3_h1 Highest bit of err_hnum3
29	RO	0x0	err_tnum3_h1 Highest bit of err_tnum3
28	RO	0x0	err_hnum2_h1 Highest bit of err_hnum2
27	RO	0x0	err_tnum2_h1 Highest bit of err_tnum2

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Bit	Attr	Reset Value	Description
26	RO	0x0	reserved
25:21	RO	0x00	err_hnum3_l5 Lower 5 bits of number of error bits found in first 512bytes of 3th backup codeword
20:16	RO	0x00	err_tnum3_l5 Lower 5 bits of number of error bits found in 3th backup codeword
15	RO	0x0	fail3 Indication for the 3th backup codeword decoded failed or not. 0: decode successfully 1: decode fail
14	RO	0x0	done3 Indication for finishing decoding the 3th backup codeword 0: not finished 1: finished
13	RO	0x0	errf3 Indication for error found in 3th backup codeword. 0: no error 1: error found
12:8	RO	0x00	err_hnum2_l5 Lower 5 bits of number of error bits found in first 512bytes of 2th backup codeword
7:3	RO	0x00	err_tnum2_l5 Lower 5 bits of number of error bits found in 2th backup codeword
2	RO	0x0	fail2 Indication for 2th backup codeword decode failed or not 0: decode successfully 1: decode fail
1	RO	0x0	done2 Indication for finishing decoding the 2th backup codeword. 0: not finished 1: finished
0	RO	0x0	errf2 Indication for error found in 2th backup codeword. 0: no error 1: error found

NANDC_BCHST2

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Address: Operational Base + offset (0x0028)
BCH Status Register For Codeword 4~5

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	bchst_cwd4_cwd5 BCHST information for 4th and 5th backup codeword. Bit assignment is similar to BCHST1 register. For more description, please refer to BCHST1 register.

NANDC_BCHST3
Address: Operational Base + offset (0x002c)
BCH Status Register For Codeword 6~7

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	bchst_cwd6_cwd7 BCHST information for 6th and 7th backup codeword. Bit assignment is similar to BCHST1 register. For more description, please refer to BCHST1 register.

NANDC_BCHST4
Address: Operational Base + offset (0x0030)
BCH Status Register For Codeword 8~9

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	bchst_cwd8_cwd9 BCHST information for 8th and 9th backup codeword. Bit assignment is similar to BCHST1 register. For more description, please refer to BCHST1 register.

NANDC_BCHST5
Address: Operational Base + offset (0x0034)
BCH Status Register For Codeword 10~11

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	bchst_cwd10_cwd11 BCHST information for 10th and 11th backup codeword. Bit assignment is similar to BCHST1 register. For more description, please refer to BCHST1 register.

NANDC_BCHST6
Address: Operational Base + offset (0x0038)
BCH Status Register For Codeword 12~13

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	bchst_cwd12_cwd13 BCHST information for 12th and 13th backup codeword. Bit assignment is similar to BCHST1 register. For more description, please refer to BCHST1 register.

NANDC_BCHST7
Address: Operational Base + offset (0x003c)
BCH Status Register For Codeword 14~15

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	bchst_cwd14_cwd15 BCHST information for 14th and 15th backup codeword. Bit assignment is similar to BCHST1 register. For more description, please refer to BCHST1 register.

NANDC_BCHLOC0
Address: Operational Base + offset (0x0040)
BCH Error Bit Location Number Register For Codeword 0~5

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:25	RO	0x00	err_loc5_l5 Lower 5 bits of number of 8bit error location in 5th backup codeword
24:20	RO	0x00	err_loc4_l5 Lower 5 bits of number of 8bit error location in 4th backup codeword
19:15	RO	0x00	err_loc3_l5 Lower 5 bits of number of 8bit error location in 3rd backup codeword
14:10	RO	0x00	err_loc2_l5 Lower 5 bits of number of 8bit error location in 2nd backup codeword
9:5	RO	0x00	err_loc1_l5 Lower 5 bits of number of 8bit error location in 1st backup codeword
4:0	RO	0x00	err_loc0_l5 Lower 5 bits of number of 8bit error location in current backup codeword

NANDC_BCHLOC1
Address: Operational Base + offset (0x0044)
BCH Error Bit Location Number Register For Codeword 6~11

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Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:25	RO	0x00	err_loc11_l5 Lower 5 bits of number of 8bit error location in 11th backup codeword
24:20	RO	0x00	err_loc10_l5 Lower 5 bits of number of 8bit error location in 10th backup codeword
19:15	RO	0x00	err_loc9_l5 Lower 5 bits of number of 8bit error location in 9th backup codeword
14:10	RO	0x00	err_loc8_l5 Lower 5 bits of number of 8bit error location in 8th backup codeword
9:5	RO	0x00	err_loc7_l5 Lower 5 bits of number of 8bit error location in 7th backup codeword
4:0	RO	0x00	err_loc6_l5 Lower 5 bits of number of 8bit error location in 6th backup codeword

NANDC_BCHLOC2

Address: Operational Base + offset (0x0048)

BCH Error Bit Location Number Register For Codeword 12~17

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:25	RO	0x00	err_loc17_l5 Lower 5 bits of number of 8bit error location in 17th backup codeword
24:20	RO	0x00	err_loc16_l5 Lower 5 bits of number of 8bit error location in 16th backup codeword
19:15	RO	0x00	err_loc15_l5 Lower 5 bits of number of 8bit error location in 15th backup codeword
14:10	RO	0x00	err_loc14_l5 Lower 5 bits of number of 8bit error location in 14th backup codeword
9:5	RO	0x00	err_loc13_l5 Lower 5 bits of number of 8bit error location in 13th backup codeword
4:0	RO	0x00	err_loc12_l5 Lower 5 bits of number of 8bit error location in 12th backup codeword

NANDC_BCHLOC3

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Address: Operational Base + offset (0x004c)
BCH Error Bit Location Number Register For Codeword 24~29

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:25	RO	0x00	err_loc23_l5 Lower 5 bits of number of 8bit error location in 23th backup codeword
24:20	RO	0x00	err_loc22_l5 Lower 5 bits of number of 8bit error location in 22th backup codeword
19:15	RO	0x00	err_loc21_l5 Lower 5 bits of number of 8bit error location in 21th backup codeword
14:10	RO	0x00	err_loc20_l5 Lower 5 bits of number of 8bit error location in 20th backup codeword
9:5	RO	0x00	err_loc19_l5 Lower 5 bits of number of 8bit error location in 19th backup codeword
4:0	RO	0x00	err_loc18_l5 Lower 5 bits of number of 8bit error location in 18th backup codeword

NANDC_BCHLOC4
Address: Operational Base + offset (0x0050)
BCH Error Bit Location Number Register For Codeword 24~29

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:25	RO	0x00	err_loc29_l5 Lower 5 bits of number of 8bit error location in 23th backup codeword
24:20	RO	0x00	err_loc28_l5 Lower 5 bits of number of 8bit error location in 22th backup codeword
19:15	RO	0x00	err_loc27_l5 Lower 5 bits of number of 8bit error location in 21th backup codeword
14:10	RO	0x00	err_loc26_l5 Lower 5 bits of number of 8bit error location in 20th backup codeword
9:5	RO	0x00	err_loc25_l5 Lower 5 bits of number of 8bit error location in 19th backup codeword
4:0	RO	0x00	err_loc24_l5 Lower 5 bits of number of 8bit error location in 18th backup codeword

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NANDC_BCHLOC5
Address: Operational Base + offset (0x0054)
BCH Error Bit Location Number Register For Codeword 30~31

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:5	RO	0x00	err_loc31_l5 Lower 5 bits of number of 8bit error location in 31th backup codeword
4:0	RO	0x00	err_loc30_l5 Lower 5 bits of number of 8bit error location in 30th backup codeword

NANDC_BCHLOC6
Address: Operational Base + offset (0x0058)
Highest Bit For BCH Error Bit Location Number Register

Bit	Attr	Reset Value	Description
31	RO	0x0	err_loc31_h1 High bit for numbers of 8bit error location in 31th codeword
30	RO	0x0	err_loc30_h1 High bit for numbers of 8bit error location in 30th codeword
29	RO	0x0	err_loc29_h1 High bit for numbers of 8bit error location in 29th codeword
28	RO	0x0	err_loc28_h1 High bit for numbers of 8bit error location in 28th codeword
27	RO	0x0	err_loc27_h1 High bit for numbers of 8bit error location in 27th codeword
26	RO	0x0	err_loc26_h1 High bit for numbers of 8bit error location in 26th codeword
25	RO	0x0	err_loc25_h1 High bit for numbers of 8bit error location in 25th codeword
24	RO	0x0	err_loc24_h1 High bit for numbers of 8bit error location in 24th codeword
23	RO	0x0	err_loc23_h1 High bit for numbers of 8bit error location in 23th codeword
22	RO	0x0	err_loc22_h1 High bit for numbers of 8bit error location in 22th codeword

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Bit	Attr	Reset Value	Description
21	RO	0x0	err_loc21_h1 High bit for numbers of 8bit error location in 21th codeword
20	RO	0x0	err_loc20_h1 High bit for numbers of 8bit error location in 20th codeword
19	RO	0x0	err_loc19_h1 High bit for numbers of 8bit error location in 19th codeword
18	RO	0x0	err_loc18_h1 High bit for numbers of 8bit error location in 18th codeword
17	RO	0x0	err_loc17_h1 High bit for numbers of 8bit error location in 17th codeword
16	RO	0x0	err_loc16_h1 High bit for numbers of 8bit error location in 16th codeword
15	RO	0x0	err_loc15_h1 High bit for numbers of 8bit error location in 15th codeword
14	RO	0x0	err_loc14_h1 High bit for numbers of 8bit error location in 14th codeword
13	RO	0x0	err_loc13_h1 High bit for numbers of 8bit error location in 13th codeword
12	RO	0x0	err_loc12_h1 High bit for numbers of 8bit error location in 12th codeword
11	RO	0x0	err_loc11_h1 High bit for numbers of 8bit error location in 11th codeword
10	RO	0x0	err_loc10_h1 High bit for numbers of 8bit error location in 10th codeword
9	RO	0x0	err_loc9_h1 High bit for numbers of 8bit error location in 9th codeword
8	RO	0x0	err_loc8_h1 High bit for numbers of 8bit error location in 8th codeword
7	RO	0x0	err_loc7_h1 High bit for numbers of 8bit error location in 7th codeword

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Bit	Attr	Reset Value	Description
6	RO	0x0	err_loc6_h1 High bit for numbers of 8bit error location in 6th codeword
5	RO	0x0	err_loc5_h1 High bit for numbers of 8bit error location in 5th codeword
4	RO	0x0	err_loc4_h1 High bit for numbers of 8bit error location in 4th codeword
3	RO	0x0	err_loc3_h1 High bit for numbers of 8bit error location in 3th codeword
2	RO	0x0	err_loc2_h1 High bit for numbers of 8bit error location in 2th codeword
1	RO	0x0	err_loc1_h1 High bit for numbers of 8bit error location in 1th codeword
0	RO	0x0	err_loc0_h1 High bit for numbers of 8bit error location in 0th codeword

NANDC_BCHDE0_0
Address: Operational Base + offset (0x0070)
BCH decode result of 0th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:8	RO	0x000	offset The offset byte address of the error bit. The value is 11bit, which is the byte offset address in the codeword. The address can be divided into different part for different use, showed as follows. 0 ~1023: page data 1024~1027: system information 1028~1055: bch information for 16bitBCH 1028~1069: bch information for 24bitBCH 1028~1097: bch information for 40bitBCH 1028~1132: bch information for 60bitBCH
7:0	RO	0x00	err_val The error value of corresponding error byte

NANDC_BCHDE0_1
Address: Operational Base + offset (0x0074)
BCH decode result of 1th error bit for codeword 0

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Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_1 decode result of 1th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_2
Address: Operational Base + offset (0x0078)
BCH decode result of 2th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_2 decode result of 2th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_3
Address: Operational Base + offset (0x007c)
BCH decode result of 3th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_3 decode result of 3th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_4
Address: Operational Base + offset (0x0080)
BCH decode result of 4th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_4 decode result of 4th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

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NANDC_BCHDE0_5
Address: Operational Base + offset (0x0084)
BCH decode result of 5th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_5 decode result of 5th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_6
Address: Operational Base + offset (0x0088)
BCH decode result of 6th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_6 decode result of 6th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_7
Address: Operational Base + offset (0x008c)
BCH decode result of 7th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_7 decode result of 7th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_8
Address: Operational Base + offset (0x0090)
BCH decode result of 8th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
18:0	RO	0x00000	bchde0_8 decode result of 8th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_9
Address: Operational Base + offset (0x0094)
BCH decode result of 9th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_9 decode result of 9th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_10
Address: Operational Base + offset (0x0098)
BCH decode result of 10th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_10 decode result of 10th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_11
Address: Operational Base + offset (0x009c)
BCH decode result of 11th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_11 decode result of 11th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_12
Address: Operational Base + offset (0x00a0)

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BCH decode result of 12th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_12 decode result of 12th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_13
Address: Operational Base + offset (0x00a4)
BCH decode result of 13th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_13 decode result of 13th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_14
Address: Operational Base + offset (0x00a8)
BCH decode result of 14th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_14 decode result of 14th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_15
Address: Operational Base + offset (0x00ac)
BCH decode result of 15th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_15 decode result of 15th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

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NANDC_BCHDE0_16
Address: Operational Base + offset (0x00b0)
BCH decode result of 16th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_16 decode result of 16th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_17
Address: Operational Base + offset (0x00b4)
BCH decode result of 17th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_17 decode result of 17th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_18
Address: Operational Base + offset (0x00b8)
BCH decode result of 18th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_18 decode result of 18th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_19
Address: Operational Base + offset (0x00bc)
BCH decode result of 19th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
18:0	RO	0x00000	bchde0_19 decode result of 1th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_20
Address: Operational Base + offset (0x00c0)
BCH decode result of 20th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_20 decode result of 20th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_21
Address: Operational Base + offset (0x00c4)
BCH decode result of 21th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_21 decode result of 21th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_22
Address: Operational Base + offset (0x00c8)
BCH decode result of 22th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_22 decode result of 22th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_23
Address: Operational Base + offset (0x00cc)

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BCH decode result of 23th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_23 decode result of 23th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE1_0

Address: Operational Base + offset (0x00d0)

BCH decode result of 0th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:8	RO	0x000	offset The offset byte address of the error bit. The value is 11bit, which is the byte offset address in the codeword. The address can be divided into different part for different use, showed as follows. 0 ~1023: page data 1024~1027: system information 1028~1055: bch information for 16bitBCH 1028~1069: bch information for 24bitBCH 1028~1097: bch information for 40bitBCH 1028~1132: bch information for 60bitBCH
7:0	RO	0x00	err_val The error value of corresponding error byte

NANDC_BCHDE1_1

Address: Operational Base + offset (0x00d4)

BCH decode result of 1th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_1 decode result of 1th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_2

Address: Operational Base + offset (0x00d8)

BCH decode result of 2th error bit for codeword 1

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_2 decode result of 2th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_3
Address: Operational Base + offset (0x00dc)
BCH decode result of 3th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_3 decode result of 3th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_4
Address: Operational Base + offset (0x00e0)
BCH decode result of 4th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_4 decode result of 4th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_5
Address: Operational Base + offset (0x00e4)
BCH decode result of 5th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_5 decode result of 5th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

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NANDC_BCHDE1_6
Address: Operational Base + offset (0x00e8)
BCH decode result of 6th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_6 decode result of 6th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_7
Address: Operational Base + offset (0x00ec)
BCH decode result of 7th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_7 decode result of 7th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_8
Address: Operational Base + offset (0x00f0)
BCH decode result of 8th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_8 decode result of 8th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_9
Address: Operational Base + offset (0x00f4)
BCH decode result of 9th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
18:0	RO	0x00000	bchde1_9 decode result of 9th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_10
Address: Operational Base + offset (0x00f8)
BCH decode result of 10th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_10 decode result of 10th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_11
Address: Operational Base + offset (0x00fc)
BCH decode result of 11th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_11 decode result of 11th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_12
Address: Operational Base + offset (0x0100)
BCH decode result of 12th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_12 decode result of 12th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_13
Address: Operational Base + offset (0x0104)

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BCH decode result of 13th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_13 decode result of 13th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_14
Address: Operational Base + offset (0x0108)
BCH decode result of 14th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_14 decode result of 14th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_15
Address: Operational Base + offset (0x010c)
BCH decode result of 15th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_15 decode result of 15th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_16
Address: Operational Base + offset (0x0110)
BCH decode result of 16th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_16 decode result of 16th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

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NANDC_BCHDE1_17
Address: Operational Base + offset (0x0114)
BCH decode result of 17th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_17 decode result of 17th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_18
Address: Operational Base + offset (0x0118)
BCH decode result of 1th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_18 decode result of 18th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_19
Address: Operational Base + offset (0x011c)
BCH decode result of 19th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_19 decode result of 19th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_20
Address: Operational Base + offset (0x0120)
BCH decode result of 20th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
18:0	RO	0x00000	bchde1_20 decode result of 20th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_21
Address: Operational Base + offset (0x0124)
BCH decode result of 21th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_21 decode result of 21th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_22
Address: Operational Base + offset (0x0128)
BCH decode result of 22th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_22 decode result of 22th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_23
Address: Operational Base + offset (0x012c)
BCH decode result of 23th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_23 decode result of 23th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_DLL_CTL_REG0
Address: Operational Base + offset (0x0130)

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DLL Control Register 0

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:16	RW	0x7f	dll_dqs_dly_bypass Holds the read DQS delay setting when the DLL is operating in bypass mode.
15:8	RW	0x7f	dll_dqs_dly Holds the read DQS delay setting when the DLL is operating in normal mode. Typically, this value is 1/4 of a clock cycle. Each increment of this field represents 1/128th of a clock cycle.
7:0	RW	0x05	dll_start_point DLL Start Point Control. This value is loaded into the DLL at initialization and is the value at which the DLL will begin searching for a lock. Each increment of this field represents 1/128th of a clock cycle.

NANDC_DLL_CTL_REG1

Address: Operational Base + offset (0x0134)

DLL Control Register 1

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:4	RW	0x02	dll_incr DLL Increment Value. This sets the increment used by the DLL when searching for a lock. It is recommended keeping this field small (around 0x4) to keep the steps gradual
3:2	RW	0x0	dll_qtren Quarter flag of DLL, active in no-bypass mode. 01:1/4 fclk, dqs_dly=128. 10:1/8 fclk, dqs_dly=64. Default: dqs_dly=dll_dqs_dly(DLL_CTL_REG0[15:8]). When dll_qtr='b01 or 'b10, software not need to configure dll_dqs_dly , and hardware should delay the input signal for 1/4 or 1/8 fclk cycle time; When dll_qtr=0, software need to configure dll_dqs_dly.
1	RW	0x1	dll_bypass DLL Bypass Control, 1active 0: dll not bypass, dll_dqs_dleay= dqs_dly 1: dll bypass, dll_dqs_dleay= dll_dqs_dly_bypass

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Bit	Attr	Reset Value	Description
0	RW	0x0	dll_start Start signal for DLL, 1 active. Notes: It will keep high until dll disabled.

NANDC_DLL_OBS_REG0
Address: Operational Base + offset (0x0138)
DLL Status Register

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:9	RO	0x01	dll_dqs_delay_value Report the delay value for the read DQS signal
8:1	RO	0x00	dll_lock_value Reports the DLL encoder value from the master DLL to the slave DLL's. The slaves use this value to set up their delays for the clk_wr and read DQS signals.
0	RO	0x0	dll_lock DLL Lock indication: 0: DLL has not locked 1: DLL is locked.

NANDC_RANDMZ_CFG
Address: Operational Base + offset (0x0150)
Randomizer Configure Register

Bit	Attr	Reset Value	Description
31	RW	0x0	randmz_en Randomizer enable indication, 1 active. 0: Randomizer active 1: Randomizer not active Notes: a. Not active when data transmission in bypass mode. b. Just active for data, but not for address and command. c. Not active when BchPage=1.
30:29	RW	0x0	randmz_mode Randomizer mode: 00- Samsung randomizer Polynomial=1+x+x^15 10- Samsung randomizer Polynomial=1+x^14+x^15 01-TOSHIBA randomizer
28:24	RW	0x00	page_offset basic seed rotation bits for every 16page

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Bit	Attr	Reset Value	Description
23:20	RW	0x0	cwd_offset basic seed start point for every page
19:0	RW	0x00000	randmz_seed when Samsung randomizer: The seed for randomizer(initial value); when Toshiba randomizer: Seed Agitation Register.

NANDC_FMWAIT_SYN
Address: Operational Base + offset (0x0158)
Flash Timing Control Register For Synchronous Timing

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14:9	RW	0x00	pst Write/Read Postamble time for ONFI synchronous mode or Toggle data mode. This field specifies the number of processor clock cycle for Postamb- le time.
8:3	RW	0x00	pre Write/Read Preamble time for ONFI synchronous mode or Toggle data mode. This field specifies the number of processor clock cycle for preamble time.
2:0	RW	0x0	fclk Half hclk cycle number for flash clock for ONFI synchronous mode or Toggle data mode

NANDC_MTRANS_STAT2
Address: Operational Base + offset (0x015c)
Bus Transfer Status Register2

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RO	0x0000	bus_err2 Bus error indication for codeword16~31. [0] : bus error for codeword 16 …… [15] : bus error for codeword 31 Notes: Only active for master-mode.

NANDC_NANDC_VER
Address: Operational Base + offset (0x0160)
Nandc Version Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RO	0x56363232	version Version indication for NANDC

NANDC_LLP_CTL
Address: Operational Base + offset (0x0164)
LLP Control Register

Bit	Attr	Reset Value	Description
31:6	RW	0x0000000	llp_loc Starting address for LLI0, 64byte align
5	RW	0x0	llp_frdy Working time for FOP_WAIT_FRDY for all FOP in first LLP group: 0: FOP_WAIT_FRDY begin working when started 1: FOP_WAIT_FRDY not begin working until 16 cycles later after started
4:3	RO	0x0	reserved
2	RW	0x0	llp_rst Reset signal for LLP. When asserted, it will auto cleared.
1	RW	0x0	llp_mode 0- current LLI only has FOP 1-current LLI has both CFG and FOP
0	RW	0x0	llp_en Enable signal for LLP 0-LLP disable 1-LLP enable

NANDC_LLP_STAT
Address: Operational Base + offset (0x0168)
LLP Status Register

Bit	Attr	Reset Value	Description
31:6	RO	0x0000000	llp_stat latest LLI_LOC finished, 64byte align
5:2	RO	0x0	reserved
1	RO	0x0	llp_err error status for llp load or execute 0-llp is correct 1-llp is error
0	RO	0x1	llp_rdy ready status for all llp load 0-llp load is busy 1-llp load is ready

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NANDC_INTEN
Address: Operational Base + offset (0x016c)
NandC Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	llp_int_en Enable for LLP finished interrupt. 0: interrupt disable 1: interrupt enable When llp_en_en is active, an interrupt is generated if LLP operation is finished
3	RW	0x0	bchfail_int_en Enable for bch fail interrupt. 0-interrupt disable 1-interrupt enable When bchfail_int_en is active, an interrupt is generated if bch decode failed
2	RW	0x0	bcherr_int_en Enable for bch error interrupt. 0-interrupt disable 1-interrupt enable When bcherr_int_en is active, an interrupt is generated if bch decode error bit is larger than bchthres(BCHCTL[26:19])
1	RW	0x0	frdy_int_en Enable for flash_rdy interrupt 0-interrupt disable 1-interrupt enable When frdy_int_en is active, an interrupt is generated if flash R/B# changes from 0 to 1
0	RW	0x0	dma_int_en Enable for internal DMA transfer finished interrupt 0-interrupt disable 1-interrupt enable When dma_int_en is active, an interrupt is generated if page_num(FLCTL[27:22]) of flash data transfer in DMA mode is finished

NANDC_INTCLR
Address: Operational Base + offset (0x0170)
NandC Interrupt Clear Register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4	RW	0x0	llp_int_clr Clear for LLP finished interrupt. When asserted, this bit will be auto cleared. 0: interrupt not cleared 1: interrupt cleared
3	RW	0x0	bchfail_int_clr Clear for bch decode fail interrupt. When asserted, this bit will be auto cleared. 0-interrupt cleared 1-interrupt not cleared
2	RW	0x0	bcherr_int_clr Clear for bch error interrupt. When asserted, this bit will be auto cleared. 0-interrupt cleared 1-interrupt not cleared
1	RW	0x0	frdy_int_clr Clear for flash_rdy interrupt. When asserted, this bit will be auto cleared. 0-interrupt cleared 1-interrupt not cleared
0	RW	0x0	dma_int_clr Clear for internal DMA transfer finished interrupt. When asserted, this bit will be auto cleared. 0-interrupt cleared 1-interrupt not cleared

NANDC_INTST
Address: Operational Base + offset (0x0174)
NandC Interrupt Status Register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RO	0x0	llp_int_stat Status for LLP finished interrupt, high active
3	RO	0x0	bchfail_int_stat Status for bch decode fail interrupt, high active
2	RO	0x0	bcherr_int_stat Status for bch error interrupt, high active
1	RO	0x0	frdy_int_stat Status for flash_rdy interrupt, high active
0	RO	0x0	dma_int_stat Status for internal DMA transfer finished interrupt, high active

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NANDC_SPARE0_0
Address: Operational Base + offset (0x0200)
System Information for codeword 0

Bit	Attr	Reset Value	Description
31:24	RW	0xff	system_3 the 4th system byte of codeword 0
23:16	RW	0xff	system_2 the 3rd system byte of codeword 0
15:8	RW	0xff	system_1 the 2nd system byte of codeword 0
7:0	RW	0xff	system_0 the 1st system byte of codeword 0

NANDC_SPARE0_1
Address: Operational Base + offset (0x0204)
Spare Data and BCH Encode Information for codeword 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_2
Address: Operational Base + offset (0x0208)
Spare Data and BCH Encode Information for codeword 0

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_3
Address: Operational Base + offset (0x020c)
Spare Data and BCH Encode Information for codeword 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_4
Address: Operational Base + offset (0x0210)
Spare Data and BCH Encode Information for codeword 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_5
Address: Operational Base + offset (0x0214)
Spare Data and BCH Encode Information for codeword 0

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_6
Address: Operational Base + offset (0x0218)
Spare Data and BCH Encode Information for codeword 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_7
Address: Operational Base + offset (0x021c)
Spare Data and BCH Encode Information for codeword 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_8
Address: Operational Base + offset (0x0220)
Spare Data and BCH Encode Information for codeword 0

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_9
Address: Operational Base + offset (0x0224)
Spare Data and BCH Encode Information for codeword 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_10
Address: Operational Base + offset (0x0228)
Spare Data and BCH Encode Information for codeword 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_11
Address: Operational Base + offset (0x022c)
Spare Data and BCH Encode Information for codeword 0

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_0
Address: Operational Base + offset (0x0230)
System Information for codeword 1

Bit	Attr	Reset Value	Description
31:24	RW	0xff	system_3 the 4th system byte of codeword 1
23:16	RW	0xff	system_2 the 3rd system byte of codeword 1
15:8	RW	0xff	system_1 the 2nd system byte of codeword 1
7:0	RW	0xff	system_0 the 1st system byte of codeword 1

NANDC_SPARE1_1
Address: Operational Base + offset (0x0234)
Spare Data and BCH Encode Information for codeword 1

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_2
Address: Operational Base + offset (0x0238)
Spare Data and BCH Encode Information for codeword 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_3
Address: Operational Base + offset (0x023c)
Spare Data and BCH Encode Information for codeword 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_4
Address: Operational Base + offset (0x0240)
Spare Data and BCH Encode Information for codeword 1

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_5
Address: Operational Base + offset (0x0244)
Spare Data and BCH Encode Information for codeword 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_6
Address: Operational Base + offset (0x0248)
Spare Data and BCH Encode Information for codeword 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_7
Address: Operational Base + offset (0x024c)
Spare Data and BCH Encode Information for codeword 1

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_8
Address: Operational Base + offset (0x0250)
Spare Data and BCH Encode Information for codeword 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_9
Address: Operational Base + offset (0x0254)
Spare Data and BCH Encode Information for codeword 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_10
Address: Operational Base + offset (0x0258)
Spare Data and BCH Encode Information for codeword 1

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_11
Address: Operational Base + offset (0x025c)
Spare Data and BCH Encode Information for codeword 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_12
Address: Operational Base + offset (0x0260)
Spare Data and BCH Encode Information for codeword 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_13
Address: Operational Base + offset (0x0264)
Spare Data and BCH Encode Information for codeword 0

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_14
Address: Operational Base + offset (0x0268)
Spare Data and BCH Encode Information for codeword 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_15
Address: Operational Base + offset (0x026c)
Spare Data and BCH Encode Information for codeword 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_16
Address: Operational Base + offset (0x0270)
Spare Data and BCH Encode Information for codeword 0

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_17
Address: Operational Base + offset (0x0274)
Spare Data and BCH Encode Information for codeword 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_18
Address: Operational Base + offset (0x0278)
Spare Data and BCH Encode Information for codeword 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_19
Address: Operational Base + offset (0x027c)
Spare Data and BCH Encode Information for codeword 0

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_20
Address: Operational Base + offset (0x0280)
Spare Data and BCH Encode Information for codeword 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_21
Address: Operational Base + offset (0x0284)
Spare Data and BCH Encode Information for codeword 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_22
Address: Operational Base + offset (0x0288)
Spare Data and BCH Encode Information for codeword 0

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_23
Address: Operational Base + offset (0x028c)
Spare Data and BCH Encode Information for codeword 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_24
Address: Operational Base + offset (0x0290)
Spare Data and BCH Encode Information for codeword 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_25
Address: Operational Base + offset (0x0294)
Spare Data and BCH Encode Information for codeword 0

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_26
Address: Operational Base + offset (0x0298)
Spare Data and BCH Encode Information for codeword 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE0_27
Address: Operational Base + offset (0x029c)
Spare Data and BCH Encode Information for codeword 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH0_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH0_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH0_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH0_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_12
Address: Operational Base + offset (0x02a0)
Spare Data and BCH Encode Information for codeword 1

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_13
Address: Operational Base + offset (0x02a4)
Spare Data and BCH Encode Information for codeword 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_14
Address: Operational Base + offset (0x02a8)
Spare Data and BCH Encode Information for codeword 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_15
Address: Operational Base + offset (0x02ac)
Spare Data and BCH Encode Information for codeword 1

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_16
Address: Operational Base + offset (0x02b0)
Spare Data and BCH Encode Information for codeword 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_17
Address: Operational Base + offset (0x02b4)
Spare Data and BCH Encode Information for codeword 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_18
Address: Operational Base + offset (0x02b8)
Spare Data and BCH Encode Information for codeword 1

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_19
Address: Operational Base + offset (0x02bc)
Spare Data and BCH Encode Information for codeword 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_20
Address: Operational Base + offset (0x02c0)
Spare Data and BCH Encode Information for codeword 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_21
Address: Operational Base + offset (0x02c4)
Spare Data and BCH Encode Information for codeword 1

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_22
Address: Operational Base + offset (0x02c8)
Spare Data and BCH Encode Information for codeword 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_23
Address: Operational Base + offset (0x02cc)
Spare Data and BCH Encode Information for codeword 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_24
Address: Operational Base + offset (0x02d0)
Spare Data and BCH Encode Information for codeword 1

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_25
Address: Operational Base + offset (0x02d4)
Spare Data and BCH Encode Information for codeword 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_26
Address: Operational Base + offset (0x02d8)
Spare Data and BCH Encode Information for codeword 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_SPARE1_27
Address: Operational Base + offset (0x02dc)
Spare Data and BCH Encode Information for codeword 1

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	BCH1_4x_3 (4x+3)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
23:16	RW	0x00	BCH1_4x_2 (4x+2)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~9 40bitBCH: x=0~16 60bitBCH: x=0~25
15:8	RW	0x00	BCH1_4x_1 (4x+1)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~25
7:0	RW	0x00	BCH1_4x_0 (4x+0)th byte of bch spare bits 16bitBCH: x=0~6 24bitBCH: x=0~10 40bitBCH: x=0~17 60bitBCH: x=0~26

NANDC_BCHDE0_24
Address: Operational Base + offset (0x0400)
BCH decode result of 24th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_24 decode result of 24th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_25
Address: Operational Base + offset (0x0404)
BCH decode result of 25th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
18:0	RO	0x00000	bchde0_25 decode result of 25th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_26
Address: Operational Base + offset (0x0408)
BCH decode result of 26th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_26 decode result of 26th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_27
Address: Operational Base + offset (0x040c)
BCH decode result of 27th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_27 decode result of 27th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_28
Address: Operational Base + offset (0x0410)
BCH decode result of 28th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_28 decode result of 28th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_29
Address: Operational Base + offset (0x0414)

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BCH decode result of 29th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_29 decode result of 29th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_30

Address: Operational Base + offset (0x0418)

BCH decode result of 30th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_30 decode result of 30th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_31

Address: Operational Base + offset (0x041c)

BCH decode result of 31th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_31 decode result of 31th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_32

Address: Operational Base + offset (0x0420)

BCH decode result of 32th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_3 decode result of 3th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

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NANDC_BCHDE0_33
Address: Operational Base + offset (0x0424)
BCH decode result of 33th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_33 decode result of 33th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_34
Address: Operational Base + offset (0x0428)
BCH decode result of 3th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_34 decode result of 34th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_35
Address: Operational Base + offset (0x042c)
BCH decode result of 35th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_35 decode result of 35th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_36
Address: Operational Base + offset (0x0430)
BCH decode result of 36th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
18:0	RO	0x00000	bchde0_36 decode result of 36th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_37
Address: Operational Base + offset (0x0434)
BCH decode result of 37th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_37 decode result of 37th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_38
Address: Operational Base + offset (0x0438)
BCH decode result of 38th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_38 decode result of 38th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_39
Address: Operational Base + offset (0x043c)
BCH decode result of 39th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_39 decode result of 39th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_40
Address: Operational Base + offset (0x0440)

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BCH decode result of 40th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_40 decode result of 40th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_41
Address: Operational Base + offset (0x0444)
BCH decode result of 41th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_41 decode result of 41th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_42
Address: Operational Base + offset (0x0448)
BCH decode result of 42th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_42 decode result of 4th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_43
Address: Operational Base + offset (0x044c)
BCH decode result of 43th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_43 decode result of 43th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

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NANDC_BCHDE0_44
Address: Operational Base + offset (0x0450)
BCH decode result of 44th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_44 decode result of 44th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_45
Address: Operational Base + offset (0x0454)
BCH decode result of 45th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_45 decode result of 45th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_46
Address: Operational Base + offset (0x0458)
BCH decode result of 46th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_46 decode result of 46th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_47
Address: Operational Base + offset (0x045c)
BCH decode result of 47th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
18:0	RO	0x00000	bchde0_47 decode result of 47th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_48
Address: Operational Base + offset (0x0460)
BCH decode result of 48th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_48 decode result of 48th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_49
Address: Operational Base + offset (0x0464)
BCH decode result of 49th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_49 decode result of 49th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_50
Address: Operational Base + offset (0x0468)
BCH decode result of 50th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_50 decode result of 50th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_51
Address: Operational Base + offset (0x046c)

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BCH decode result of 51th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_51 decode result of 51th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_52
Address: Operational Base + offset (0x0470)
BCH decode result of 52th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_52 decode result of 52th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_53
Address: Operational Base + offset (0x0474)
BCH decode result of 53th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_53 decode result of 53th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_54
Address: Operational Base + offset (0x0478)
BCH decode result of 54th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_54 decode result of 54th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

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NANDC_BCHDE0_55
Address: Operational Base + offset (0x047c)
BCH decode result of 55th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_55 decode result of 55th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_56
Address: Operational Base + offset (0x0480)
BCH decode result of 56th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_56 decode result of 56th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_57
Address: Operational Base + offset (0x0484)
BCH decode result of 57th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_57 decode result of 57th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_58
Address: Operational Base + offset (0x0488)
BCH decode result of 58th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
18:0	RO	0x00000	bchde0_58 decode result of 58th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE0_59
Address: Operational Base + offset (0x048c)
BCH decode result of 59th error bit for codeword 0

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde0_59 decode result of 59th error bit for codeword 0. Bit assignment is similar to BCHDE0_0 register. For more description, please refer to BCHDE0_0 register.

NANDC_BCHDE1_24
Address: Operational Base + offset (0x0490)
BCH decode result of 24th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_24 decode result of 24th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_25
Address: Operational Base + offset (0x0494)
BCH decode result of 25th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_25 decode result of 25th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_26
Address: Operational Base + offset (0x0498)

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BCH decode result of 26th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_26 decode result of 26th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_27
Address: Operational Base + offset (0x049c)
BCH decode result of 27th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_27 decode result of 27th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_28
Address: Operational Base + offset (0x04a0)
BCH decode result of 28th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_28 decode result of 28th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_29
Address: Operational Base + offset (0x04a4)
BCH decode result of 29th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_29 decode result of 2th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

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NANDC_BCHDE1_30
Address: Operational Base + offset (0x04a8)
BCH decode result of 30th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_30 decode result of 30th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_31
Address: Operational Base + offset (0x04ac)
BCH decode result of 31th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_31 decode result of 31th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_32
Address: Operational Base + offset (0x04b0)
BCH decode result of 32th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_32 decode result of 32th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_33
Address: Operational Base + offset (0x04b4)
BCH decode result of 33th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
18:0	RO	0x00000	bchde1_33 decode result of 33th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_34
Address: Operational Base + offset (0x04b8)
BCH decode result of 34th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_34 decode result of 34th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_35
Address: Operational Base + offset (0x04bc)
BCH decode result of 35th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_35 decode result of 35th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_36
Address: Operational Base + offset (0x04c0)
BCH decode result of 3th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_36 decode result of 36th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_37
Address: Operational Base + offset (0x04c4)

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BCH decode result of 37th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_37 decode result of 37th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_38
Address: Operational Base + offset (0x04c8)
BCH decode result of 38th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_38 decode result of 38th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_39
Address: Operational Base + offset (0x04cc)
BCH decode result of 39th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_39 decode result of 39th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_40
Address: Operational Base + offset (0x04d0)
BCH decode result of 40th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_40 decode result of 40th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

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NANDC_BCHDE1_41
Address: Operational Base + offset (0x04d4)
BCH decode result of 41th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_41 decode result of 41th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_42
Address: Operational Base + offset (0x04d8)
BCH decode result of 42th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_42 decode result of 42th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_43
Address: Operational Base + offset (0x04dc)
BCH decode result of 43th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_43 decode result of 43th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_44
Address: Operational Base + offset (0x04e0)
BCH decode result of 44th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
18:0	RO	0x00000	bchde1_44 decode result of 44th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_45
Address: Operational Base + offset (0x04e4)
BCH decode result of 45th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_45 decode result of 45th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_46
Address: Operational Base + offset (0x04e8)
BCH decode result of 46th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_46 decode result of 46th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_47
Address: Operational Base + offset (0x04ec)
BCH decode result of 47th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_47 decode result of 47th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_48
Address: Operational Base + offset (0x04f0)

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BCH decode result of 48th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_48 decode result of 48th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_49
Address: Operational Base + offset (0x04f4)
BCH decode result of 49th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_49 decode result of 49th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_50
Address: Operational Base + offset (0x04f8)
BCH decode result of 50th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_50 decode result of 50th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_51
Address: Operational Base + offset (0x04fc)
BCH decode result of 51th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_51 decode result of 51th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

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NANDC_BCHDE1_52
Address: Operational Base + offset (0x0500)
BCH decode result of 52th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_52 decode result of 52th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_53
Address: Operational Base + offset (0x0504)
BCH decode result of 53th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_53 decode result of 53th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_54
Address: Operational Base + offset (0x0508)
BCH decode result of 54th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_54 decode result of 54th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_55
Address: Operational Base + offset (0x050c)
BCH decode result of 55th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
18:0	RO	0x00000	bchde1_55 decode result of 55th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_56
Address: Operational Base + offset (0x0510)
BCH decode result of 56th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_56 decode result of 56th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_57
Address: Operational Base + offset (0x0514)
BCH decode result of 57th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_57 decode result of 57th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_58
Address: Operational Base + offset (0x0518)
BCH decode result of 58th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_58 decode result of 58th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHDE1_59
Address: Operational Base + offset (0x051c)

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BCH decode result of 59th error bit for codeword 1

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RO	0x00000	bchde1_59 decode result of 59th error bit for codeword 1. Bit assignment is similar to BCHDE1_0 register. For more description, please refer to BCHDE1_0 register.

NANDC_BCHST8
Address: Operational Base + offset (0x0520)
BCH Status Register For Codeword 16~17

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	bchst_cwd16_cwd17 BCHST information for 16th and 17th backup codeword. Bit assignment is similar to BCHST1 register. For more description, please refer to BCHST1 register.

NANDC_BCHST9
Address: Operational Base + offset (0x0524)
BCH Status Register For Codeword 18~19

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	bchst_cwd18_cwd19 BCHST information for 18th and 19th backup codeword. Bit assignment is similar to BCHST1 register. For more description, please refer to BCHST1 register.

NANDC_BCHST10
Address: Operational Base + offset (0x0528)
BCH Status Register For Codeword 20~21

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	bchst_cwd20_cwd21 BCHST information for 20th and 21th backup codeword. Bit assignment is similar to BCHST1 register. For more description, please refer to BCHST1 register.

NANDC_BCHST11
Address: Operational Base + offset (0x052c)
BCH Status Register For Codeword 22~23

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	bchst_cwd22_cwd23 BCHST information for 22th and 23th backup codeword. Bit assignment is similar to BCHST1 register. For more description, please refer to BCHST1 register.

NANDC_BCHST12
Address: Operational Base + offset (0x0530)
BCH Status Register For Codeword 24~25

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	bchst_cwd24_cwd25 BCHST information for 24th and 25th backup codeword. Bit assignment is similar to BCHST1 register. For more description, please refer to BCHST1 register.

NANDC_BCHST13
Address: Operational Base + offset (0x0534)
BCH Status Register For Codeword 26~27

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	bchst_cwd26_cwd27 BCHST information for 26th and 27th backup codeword. Bit assignment is similar to BCHST1 register. For more description, please refer to BCHST1 register.

NANDC_BCHST14
Address: Operational Base + offset (0x0538)
BCH Status Register For Codeword 28~29

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	bchst_cwd28_cwd29 BCHST information for 28th and 29th backup codeword. Bit assignment is similar to BCHST1 register. For more description, please refer to BCHST1 register.

NANDC_BCHST15
Address: Operational Base + offset (0x053c)
BCH Status Register For Codeword 30~31

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	bchst_cwd30_cwd31 BCHST information for 30th and 31th backup codeword. Bit assignment is similar to BCHST1 register. For more description, please refer to BCHST1 register.

NANDC_RANDMZ_SEED13_0
Address: Operational Base + offset (0x0600)
Seed 0 for Toshiba 13 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	randmz_seed13_0 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED13_1
Address: Operational Base + offset (0x0604)
Seed 1 for Toshiba 13 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	randmz_seed13_1 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED13_2
Address: Operational Base + offset (0x0608)
Seed 2 for Toshiba 13 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	randmz_seed13_2 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED13_3
Address: Operational Base + offset (0x060c)
Seed 3 for Toshiba 13 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	randmz_seed13_3 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED13_4
Address: Operational Base + offset (0x0610)
Seed 4 for Toshiba 13 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
12:0	RW	0x0000	randmz_seed13_4 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED13_5
Address: Operational Base + offset (0x0614)
Seed 5 for Toshiba 13 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	randmz_seed13_5 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED13_6
Address: Operational Base + offset (0x0618)
Seed 6 for Toshiba 13 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	randmz_seed13_6 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED13_7
Address: Operational Base + offset (0x061c)
Seed 7 for Toshiba 13 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	randmz_seed13_7 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED13_8
Address: Operational Base + offset (0x0620)
Seed 8 for Toshiba 13 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	randmz_seed13_8 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED13_9
Address: Operational Base + offset (0x0624)
Seed 9 for Toshiba 13 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	randmz_seed13_9 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED13_10

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Address: Operational Base + offset (0x0628)
Seed 10 for Toshiba 13 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	randmz_seed13_10 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED13_11
Address: Operational Base + offset (0x062c)
Seed 11 for Toshiba 13 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	randmz_seed13_11 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED13_12
Address: Operational Base + offset (0x0630)
Seed 12 for Toshiba 13 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	randmz_seed13_12 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED13_13
Address: Operational Base + offset (0x0634)
Seed 13 for Toshiba 13 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	randmz_seed13_13 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED13_14
Address: Operational Base + offset (0x0638)
Seed 14 for Toshiba 13 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	randmz_seed13_14 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED13_15
Address: Operational Base + offset (0x063c)
Seed 15 for Toshiba 13 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	randmz_seed13_15 The seed for randomizer(initial value);

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NANDC_RANDMZ_SEED17_0
Address: Operational Base + offset (0x0640)
Seed 0 for Toshiba 17 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:0	RW	0x00000	randmz_seed17_0 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED17_1
Address: Operational Base + offset (0x0644)
Seed 1 for Toshiba 17 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:0	RW	0x00000	randmz_seed17_1 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED17_2
Address: Operational Base + offset (0x0648)
Seed 2 for Toshiba 17 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:0	RW	0x00000	randmz_seed17_2 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED17_3
Address: Operational Base + offset (0x064c)
Seed 3 for Toshiba 17 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:0	RW	0x00000	randmz_seed17_3 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED17_4
Address: Operational Base + offset (0x0650)
Seed 4 for Toshiba 17 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:0	RW	0x00000	randmz_seed17_4 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED17_5
Address: Operational Base + offset (0x0654)
Seed 5 for Toshiba 17 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:0	RW	0x00000	randmz_seed17_5 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED17_6
Address: Operational Base + offset (0x0658)
Seed 6 for Toshiba 17 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:0	RW	0x00000	randmz_seed17_6 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED17_7
Address: Operational Base + offset (0x065c)
Seed 7 for Toshiba 17 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:0	RW	0x00000	randmz_seed17_7 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED17_8
Address: Operational Base + offset (0x0660)
Seed 8 for Toshiba 17 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:0	RW	0x00000	randmz_seed17_8 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED17_9
Address: Operational Base + offset (0x0664)
Seed 9 for Toshiba 17 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:0	RW	0x00000	randmz_seed17_9 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED17_10
Address: Operational Base + offset (0x0668)
Seed 10 for Toshiba 17 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:0	RW	0x00000	randmz_seed17_10 The seed for randomizer(initial value);

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NANDC_RANDMZ_SEED17_11
Address: Operational Base + offset (0x066c)
Seed 11 for Toshiba 17 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:0	RW	0x00000	randmz_seed17_11 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED17_12
Address: Operational Base + offset (0x0670)
Seed 12 for Toshiba 17 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:0	RW	0x00000	randmz_seed17_12 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED17_13
Address: Operational Base + offset (0x0674)
Seed 13 for Toshiba 17 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:0	RW	0x00000	randmz_seed17_13 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED17_14
Address: Operational Base + offset (0x0678)
Seed 14 for Toshiba 17 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:0	RW	0x00000	randmz_seed17_14 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED17_15
Address: Operational Base + offset (0x067c)
Seed 15 for Toshiba 17 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:0	RW	0x00000	randmz_seed17_15 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED19_0
Address: Operational Base + offset (0x0680)
Seed 0 for Toshiba 19 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
18:0	RW	0x00000	randmz_seed19_0 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED19_1
Address: Operational Base + offset (0x0684)
Seed 1 for Toshiba 19 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RW	0x00000	randmz_seed19_1 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED19_2
Address: Operational Base + offset (0x0688)
Seed 2 for Toshiba 19 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RW	0x00000	randmz_seed19_2 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED19_3
Address: Operational Base + offset (0x068c)
Seed 3 for Toshiba 19 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RW	0x00000	randmz_seed19_3 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED19_4
Address: Operational Base + offset (0x0690)
Seed 4 for Toshiba 19 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RW	0x00000	randmz_seed19_4 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED19_5
Address: Operational Base + offset (0x0694)
Seed 5 for Toshiba 19 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RW	0x00000	randmz_seed19_5 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED19_6

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Address: Operational Base + offset (0x0698)
Seed 6 for Toshiba 19 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RW	0x00000	randmz_seed19_6 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED19_7
Address: Operational Base + offset (0x069c)
Seed 7 for Toshiba 19 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RW	0x00000	randmz_seed19_7 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED19_8
Address: Operational Base + offset (0x06a0)
Seed 8 for Toshiba 19 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RW	0x00000	randmz_seed19_8 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED19_9
Address: Operational Base + offset (0x06a4)
Seed 9 for Toshiba 19 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RW	0x00000	randmz_seed19_9 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED19_10
Address: Operational Base + offset (0x06a8)
Seed 10 for Toshiba 19 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RW	0x00000	randmz_seed19_10 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED19_11
Address: Operational Base + offset (0x06ac)
Seed 11 for Toshiba 19 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RW	0x00000	randmz_seed19_11 The seed for randomizer(initial value);

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NANDC_RANDMZ_SEED19_12
Address: Operational Base + offset (0x06b0)
Seed 12 for Toshiba 19 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RW	0x00000	randmz_seed19_12 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED19_13
Address: Operational Base + offset (0x06b4)
Seed 13 for Toshiba 19 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RW	0x00000	randmz_seed19_13 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED19_14
Address: Operational Base + offset (0x06b8)
Seed 14 for Toshiba 19 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RW	0x00000	randmz_seed19_14 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED19_15
Address: Operational Base + offset (0x06bc)
Seed 15 for Toshiba 19 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:0	RW	0x00000	randmz_seed19_15 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED23_0
Address: Operational Base + offset (0x06c0)
Seed 0 for Toshiba 23 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:0	RW	0x000000	randmz_seed23_0 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED23_1
Address: Operational Base + offset (0x06c4)
Seed 1 for Toshiba 23 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:0	RW	0x000000	randmz_seed23_1 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED23_2
Address: Operational Base + offset (0x06c8)
Seed 2 for Toshiba 23 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:0	RW	0x000000	randmz_seed23_2 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED23_3
Address: Operational Base + offset (0x06cc)
Seed 3 for Toshiba 23 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:0	RW	0x000000	randmz_seed23_3 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED23_4
Address: Operational Base + offset (0x06d0)
Seed 4 for Toshiba 23 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:0	RW	0x000000	randmz_seed23_4 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED23_5
Address: Operational Base + offset (0x06d4)
Seed 5 for Toshiba 23 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:0	RW	0x000000	randmz_seed23_5 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED23_6
Address: Operational Base + offset (0x06d8)
Seed 6 for Toshiba 23 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:0	RW	0x000000	randmz_seed23_6 The seed for randomizer(initial value);

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NANDC_RANDMZ_SEED23_7
Address: Operational Base + offset (0x06dc)
Seed 7 for Toshiba 23 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:0	RW	0x000000	randmz_seed23_7 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED23_8
Address: Operational Base + offset (0x06e0)
Seed 8 for Toshiba 23 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:0	RW	0x000000	randmz_seed23_8 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED23_9
Address: Operational Base + offset (0x06e4)
Seed 9 for Toshiba 23 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:0	RW	0x000000	randmz_seed23_9 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED23_10
Address: Operational Base + offset (0x06e8)
Seed 10 for Toshiba 23 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:0	RW	0x000000	randmz_seed23_10 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED23_11
Address: Operational Base + offset (0x06ec)
Seed 11 for Toshiba 23 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:0	RW	0x000000	randmz_seed23_11 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED23_12
Address: Operational Base + offset (0x06f0)
Seed 12 for Toshiba 23 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
22:0	RW	0x000000	randmz_seed23_12 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED23_13
Address: Operational Base + offset (0x06f4)
Seed 13 for Toshiba 23 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:0	RW	0x000000	randmz_seed23_13 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED23_14
Address: Operational Base + offset (0x06f8)
Seed 14 for Toshiba 23 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:0	RW	0x000000	randmz_seed23_14 The seed for randomizer(initial value);

NANDC_RANDMZ_SEED23_15
Address: Operational Base + offset (0x06fc)
Seed 15 for Toshiba 23 Power Polynomial Randomizer

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:0	RW	0x000000	randmz_seed23_15 The seed for randomizer(initial value);

17.5 Interface Description

Table 17-2 NandC0 Interface Description

Module Pin	IO	Pad Name	IOMUX Setting
flash_wp	O	GPIO1_D3	GPIO1D_IOMUX[7:6]= 2'b01
flash_ale	O	GPIO1_D5	GPIO1D_IOMUX[11:10]= 2'b01
flash_cle	O	GPIO1_D6	GPIO1D_IOMUX[13:12]= 2'b01
flash_wrn	O	GPIO1_D7	GPIO1D_IOMUX[15:14]= 2'b01
flash_rdn	O	GPIO1_D4	GPIO1D_IOMUX[9:8]= 2'b01
flash_dqs	I/O	GPIO2_A4	GPIO2A_IOMUX[9:8]= 2'b01
flash_rdy	I	GPIO1_D2	GPIO1D_IOMUX[5:4]= 2'b01
flash_csn0	O	GPIO2_A0	GPIO2A_IOMUX[1:0]= 1'b1
flash_csn1	O	GPIO2_A1	GPIO2A_IOMUX[3:2]= 1'b1
flash_csn2	O	GPIO2_A2	GPIO2A_IOMUX[5:4]= 2'b01
flash_csn3	O	GPIO2_A3	GPIO2A_IOMUX[7:6]= 2'b01
flash_data0	I/O	GPIO1_C2	GPIO1C_IOMUX[5:4]= 2'b01
flash_data1	I/O	GPIO1_C3	GPIO1C_IOMUX[7:6]= 2'b01
flash_data2	I/O	GPIO1_C4	GPIO1C_IOMUX[9:8]= 2'b01
flash_data3	I/O	GPIO1_C5	GPIO1C_IOMUX[11:10]= 2'b01
flash_data4	I/O	GPIO1_C6	GPIO1C_IOMUX[13:12]= 2'b01
flash_data5	I/O	GPIO1_C7	GPIO1C_IOMUX[15:14]= 2'b01

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flash_data6	I/O	GPIO1_D0	GPIO1D_IOMUX[1:0]= 2'b01
flash_data7	I/O	GPIO1_D1	GPIO1D_IOMUX[3:2]= 2'b01

Notes: I=input, O=output, I/O=input/output, bidirectional
Notes: I=input, O=output, I/O=input/output, bidirectional

Furthermore, different IOs are selected and connected to different flash interface, which is shown as follows.

Table 17-3 NandC Interface Connection

Module Pin	Direction	Flash Interface			
		Asyn8x	Asyn16x	ONFI	Toggle
flash_csni(i=0~7)	O	√	√	√	√
flash_wp	O	√	√	√	√
flash_ale	O	√	√	√	√
flash_cle	O	√	√	√	√
flash_wrn	O	√	√	√	√
flash_rdn	O	√	√	√	√
flash_data[7:0]	I/O	√	√	√	√
flash_data[15:8]	I/O	-	√	-	-
flash_dqs	I/O	-	-	√	√
flash_rdy	I	√	√	√	√

17.6 Application Notes

17.6.1 Clock Description

There are two clock domains in the NandC. One is hclk, and the other is nclk.

- AHB slave bus uses the hclk to configure the NandC registers.
- AHB master bus uses the hclk to transmit the data between the external memory and internal sram.
- NandC use nclk to transmit the data between the nand flash and internal sram.
- All the flash interface timing configuration is relative to the nclk.

17.6.2 BCHST/BCHLOC/BCHDE/SPARE Application

1. BCHST

There are 16 BCHST-registers in NandC to store 32 codeword's BCH decode status(bchst) information. Every register stores 2 codeword's bchst information except BCHST0, which not only includes bchst information, but also includes one bit for *bchrdy*.

Let bchst_cwd0~bchst_cwd31 be the bchst information for 32 codewords. In BCHST-registers, the latest codeword's bchst is stored into bchst_cwd0, and the former is shifted into bchst_cwd1. That is, bchst_cwd0→ bchst_cwd1 →……→bchst_cwd31. Therefore, for example, if 32 codewords are decoded, then bchst_cwd0 is the bch decode status for codeword31, and bchst_cwd31 is the bch decode status for codeword0.

bchst_cwd0 = {BCHST0[28],BCHST0[12:8],BCHST0[27],BCHST0[7:3],BCHST0[2:0]}
bchst_cwd1 = {BCHST0[30],BCHST0[25:21],BCHST0[29],BCHST0[20:16],BCHST0[15:13]}
bchst_cwd2 = {BCHST1[28],BCHST1[12:8],BCHST1[27],BCHST1[7:3],BCHST1[2:0]}
bchst_cwd3 = {BCHST1[30],BCHST1[25:21],BCHST1[29],BCHST1[20:16],BCHST1[15:13]}
bchst_cwd4 = {BCHST2[28],BCHST2[12:8] ,BCHST2[27],BCHST2[7:3],BCHST2[2:0]}
bchst_cwd5 = {BCHST2[30],BCHST2[25:21],BCHST2[29],BCHST2[20:16],BCHST2[15:13]}
bchst_cwd6 = {BCHST3[28],BCHST3[12:8] ,BCHST3[27],BCHST3[7:3],BCHST3[2:0]}
bchst_cwd7 = {BCHST3[30],BCHST3[25:21],BCHST3[29],BCHST3[20:16],BCHST3[15:13]}
bchst_cwd8 = {BCHST4[28],BCHST4[12:8] ,BCHST4[27],BCHST4[7:3],BCHST4[2:0]}
bchst_cwd9 = {BCHST4[30],BCHST4[25:21],BCHST4[29],BCHST4[20:16],BCHST4[15:13]}
bchst_cwd10 = {BCHST5[28],BCHST5[12:8] ,BCHST5[27],BCHST5[7:3],BCHST5[2:0]}
bchst_cwd11 = {BCHST5[30],BCHST5[25:21],BCHST5[29],BCHST5[20:16],BCHST5[15:13]}
bchst_cwd12 = {BCHST6[28],BCHST6[12:8] ,BCHST6[27],BCHST6[7:3],BCHST6[2:0]}
bchst_cwd13 = {BCHST6[30],BCHST6[25:21],BCHST6[29],BCHST6[20:16],BCHST6[15:13]}
bchst_cwd14 = {BCHST7[28],BCHST7[12:8] ,BCHST7[27],BCHST7[7:3],BCHST7[2:0]}
bchst_cwd15 =

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{BCHST7[30],BCHST7[25:21],BCHST7[29],BCHST7[20:16],BCHST7[15:13]}
bchst_cwd16 = {BCHST8[28],BCHST8[12:8],BCHST8[27],BCHST8[7:3],BCHST8[2:0]}
bchst_cwd17 = {BCHST8[30],BCHST8[25:21],BCHST8[29],BCHST8[20:16],BCHST8[15:13]}
bchst_cwd18 = {BCHST9[28],BCHST9[12:8],BCHST9[27],BCHST9[7:3],BCHST9[2:0]}
bchst_cwd19 = {BCHST9[30],BCHST9[25:21],BCHST9[29],BCHST9[20:16],BCHST9[15:13]}
bchst_cwd20 =
{BCHST10[28],BCHST10[12:8] ,BCHST10[27],BCHST10[7:3],BCHST10[2:0]}
bchst_cwd21 =
{BCHST10[30],BCHST10[25:21],BCHST10[29],BCHST10[20:16],BCHST10[15:13]}
bchst_cwd22 =
{BCHST11[28],BCHST11[12:8] ,BCHST11[27],BCHST11[7:3],BCHST11[2:0]}
bchst_cwd23 =
{BCHST11[30],BCHST11[25:21],BCHST11[29],BCHST11[20:16],BCHST11[15:13]}
bchst_cwd24 =
{BCHST12[28],BCHST12[12:8] ,BCHST12[27],BCHST12[7:3],BCHST12[2:0]}
bchst_cwd25 =
{BCHST12[30],BCHST12[25:21],BCHST12[29],BCHST12[20:16],BCHST12[15:13]}
bchst_cwd26 =
{BCHST13[28],BCHST13[12:8] ,BCHST13[27],BCHST13[7:3],BCHST13[2:0]}
bchst_cwd27 =
{BCHST13[30],BCHST13[25:21],BCHST13[29],BCHST13[20:16],BCHST13[15:13]}
bchst_cwd28 =
{BCHST14[28],BCHST14[12:8] ,BCHST14[27],BCHST14[7:3],BCHST14[2:0]}
bchst_cwd29 =
{BCHST14[30],BCHST14[25:21],BCHST14[29],BCHST14[20:16],BCHST14[15:13]}
bchst_cwd30 =
{BCHST15[28],BCHST15[12:8] ,BCHST15[27],BCHST15[7:3],BCHST15[2:0]}
bchst_cwd31 =
{BCHST15[30],BCHST15[25:21],BCHST15[29],BCHST15[20:16],BCHST15[15:13]}

2. BCHLOC

There are 7 BCHLOC-registers in NandC to store 32 codeword's bch decode location(bchloc) information.

Let bchloc_cwd0~bchloc_cwd31 be the bchloc information for the 32 codeword. In BCHLOC registers, the latest codeword's bchloc is stored into bchloc_cwd0, and the former is shifted into bchloc_cwd1. That is, bchloc_cwd0→bchloc_cwd1→……→bchloc_cwd31. Therefore, for example, if 32 codeword are decoded, then bchloc_cwd0 is the bch decode status for codeword31, and bchloc_cwd31 is the bch decode status for codeword0.

bchloc_cwd0 = {BCHLOC6[0], BCHLOC0[4:0]}
bchloc_cwd1 = {BCHLOC6[1], BCHLOC0[9:5]}
bchloc_cwd2 = {BCHLOC6[2], BCHLOC0[14:10]}
bchloc_cwd3 = {BCHLOC6[3], BCHLOC0[19:15]}
bchloc_cwd4 = {BCHLOC6[4], BCHLOC0[24:20]}
bchloc_cwd5 = {BCHLOC6[5], BCHLOC0[29:25]}
bchloc_cwd6 = {BCHLOC6[6], BCHLOC1[4:0]}
bchloc_cwd7 = {BCHLOC6[7], BCHLOC1[9:5]}
bchloc_cwd8 = {BCHLOC6[8], BCHLOC1[14:10]}
bchloc_cwd9 = {BCHLOC6[9], BCHLOC1[19:15]}
bchloc_cwd10 = {BCHLOC6[10], BCHLOC1[24:20]}
bchloc_cwd11 = {BCHLOC6[11], BCHLOC1[29:25]}
bchloc_cwd12 = {BCHLOC6[12], BCHLOC2[4:0]}
bchloc_cwd13 = {BCHLOC6[13], BCHLOC2[9:5]}
bchloc_cwd14 = {BCHLOC6[14], BCHLOC2[14:10]}
bchloc_cwd15 = {BCHLOC6[15], BCHLOC2[19:15]}
bchloc_cwd16 = {BCHLOC6[16], BCHLOC2[24:20]}
bchloc_cwd17 = {BCHLOC6[17], BCHLOC2[29:25]}
bchloc_cwd18 = {BCHLOC6[18], BCHLOC3[4:0]}
bchloc_cwd19 = {BCHLOC6[19], BCHLOC3[9:5]}

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bchloc_cwd20 = {BCHLOC6[20], BCHLOC3[14:10]}
bchloc_cwd21 = {BCHLOC6[21], BCHLOC3[19:15]}
bchloc_cwd22 = {BCHLOC6[22], BCHLOC3[24:20]}
bchloc_cwd23 = {BCHLOC6[23], BCHLOC3[29:25]}
bchloc_cwd24 = {BCHLOC6[24], BCHLOC4[4:0]}
bchloc_cwd25 = {BCHLOC6[25], BCHLOC4[9:5]}
bchloc_cwd26 = {BCHLOC6[26], BCHLOC4[14:10]}
bchloc_cwd27 = {BCHLOC6[27], BCHLOC4[19:15]}
bchloc_cwd28 = {BCHLOC6[28], BCHLOC4[24:20]}
bchloc_cwd29 = {BCHLOC6[29], BCHLOC4[29:25]}
bchloc_cwd30 = {BCHLOC6[30], BCHLOC5[4:0]}
bchloc_cwd31 = {BCHLOC6[31], BCHLOC5[9:5]}

3. BCHDE

BCHDE includes two register-groups, BCHDE0 and BCHDE1. Each group has 60 registers: BCHDE0_0~BCHDE0_59 and BCHDE1_0~BCHDE1_59. BCHDE0_n(n=0~59) is the decode information of the nth error bit for codeword in sram0, and BCHDE1_n(n=0~59) is the decode information of the nth error bit for codeword in sram1.

The needed number of BCHDE registers is determined by bchmode. That is:

a. When 16bitBCH selected, BCHDEm_0 ~ BCHDEm_15 are available
b. When 24bitBCH selected, BCHDEm_0 ~ BCHDEm_23 are available
c. When 40bitBCH selected, BCHDEm_0 ~ BCHDEm_39 are available
d. When 60bitBCH selected, BCHDEm_0 ~ BCHDEm_59 are available

4. SPARE

SPARE includes two register-groups, SPARE0 and SPARE1. Each group has 28 registers: SPARE0_0~SPARE0_27 and SPARE1_0~SPARE1_27.

When in bch encoding, SPARE0_0 stores system information for codeword in sram0, SPARE0_n(n=1~27) stores encode information for codeword in sram0; SPARE1_0 stores system information for codeword in sram1, SPARE1_n(n=1~27) stores encode information for codeword in sram1.

When in bch decoding, SPARE0_n(n=0~27) stores the spare data read from flash for codeword in sram0; SPARE1_n(n=0~27) stores the spare data read from flash for codeword in sram1.

The needed number of BCHDE registers is determined by bchmode. That is:

a. When 16bitBCH selected, spare data=28bytes, SPAREm_0~SPAREm_7 are available
b. When 24bitBCH selected, spare data=42bytes, SPAREm_0~SPAREm_10 and SPAREm_11[15:0] are available
c. When 40bitBCH selected, spare data=70bytes, SPAREm_0~SPAREm_17 and SPAREm_18[15:0] are available
d. When 60bitBCH selected, spare data=105bytes, SPAREm_0~SPAREm_26 and SPAREm_27[7:0] are available

17.6.3 Bus Mode Application

MTRANS_CFG[2] determines whether the data load/store between internal memory and external memory is through slave interface or master interface.

1. Slave Mode

When MTRANS_CFG[2]=0, slave is selected. i. e. , flash data load/store between internal memory and external memory is through slave interface by cpu or external DMA.

In this mode, software should store page data into internal memory and spare data into SPARE registers before starting flash program operation; and should load page data from internal memory and spare data from SPARE registers after finishing flash read operation.

In this mode, MTRANS_CFG, MTRANS_SADDR0 and MTRANS_SADDR1 are unused. The transfer codeword number is determined by FLCTL[6:5], and the maximum number is 2.

The judgment condition for finishing data transfer is FLCTL[20]. When FLCTL[20] is high, it means that data transfer is finished.

2. Master Mode

When MTRANS_CFG[2]=1, master is selected. i. e. , flash data load/store between internal memory and external memory is through master interface.

In this mode, software should initialize page data and spare data into external memory, and

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set their addresses in MTRANS_SADDR0 and MTRANS_SADDR1 respectively before starting flash program operation. Similarly, software should configure MTRANS_SADDR0 and MTRANS_SADDR1 respectively before starting flash read operation and could read data from addresses in MTRANS_SADDR0 and MTRANS_SADDR1 after NandC transfer finish.
In this mode, MTRANS_CFG, MTRANS_SADDR0 and MTRANS_SADDR1 are used. The transfer codeword number is determined by FLCTL[26:22], and the maximum number is 16.
The judgment condition for finishing data transfer is FLCTL[20]. When FLCTL[20] is high, it means that data transmission is finished.
When MTRANS_CFG[2]=1, page data and spare data are stored in the continuous space of external memory respectively.
For page data, source address is named Saddr0, specified in MTRANS_SADDR0. The space can be divided into many continuous units, and the unit size(named PUnit) is 1024 bytes or 512 bytes determined by FLCTL[21] and FLCTL[11]:

- when FLCTL[11]=0, PUnit is always equal to 1024 bytes
- when FLCTL[11]=1 and FLCTL[21]=0, PUnit is equal to 1024 bytes
- when FLCTL[11]=1 and FLCTL[21]=1, PUnit is equal to 512 bytes

For spare data, source address is named Saddr1, specified in MTRANS_SADDR1. The space can be divided into many continuous units, and the unit size(named SUnit) is 64 bytes or 128 bytes determined by BCHCTL[18], FLCTL[11] and FLCTL[21]:

- When FLCTL[11]=0 and BCHCTL[18]=0, SUnit is equal to 64 bytes
- When FLCTL[11]=0 and BCHCTL[18]=1, SUnit is equal to 128 bytes
- When FLCTL[11]=1 and FLCTL[21]=0, SUnit is equal to 128 bytes
- When FLCTL[11]=1 and FLCTL[21]=1, SUnit is equal to 64 bytes

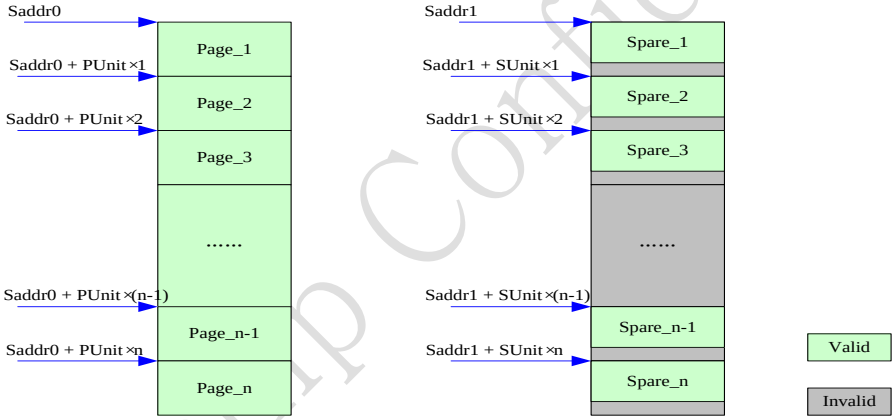

Fig. 17-2 NandC Address Assignment

The detailed format for page data and spare data in every unit is shown in following figures.

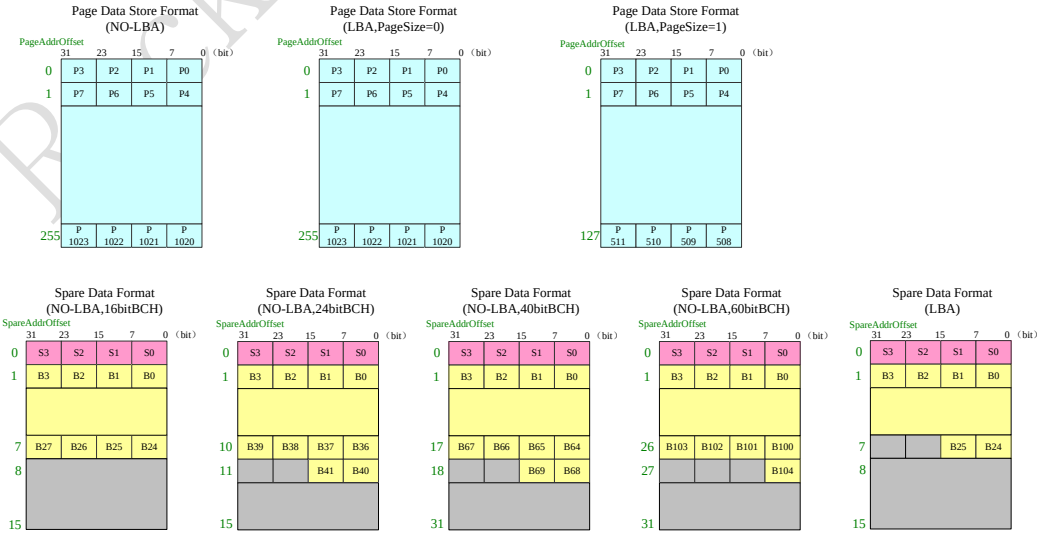

Fig. 17-3 NandC DataFormat

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17.6.4 BchPage Application

BCHCTL[16] determines whether codeword size for page data is 1024 bytes or 512 bytes when FLCTL[11] is 0.

1. 1024bytes

When BCHCTL[16]=0, BchPage=0, hardware needs to write 1024 bytes page data and spare data into flash or read 1024 bytes page data and spare data from flash. All the 1024 bytes page data and spare data are encoded when writing or decoded when reading.

2. 512bytes

When BCHCTL[16]=1, BchPage=1, hardware needs to write 512 bytes page data and spare data into flash or read 512 bytes page data and spare data from flash.

In this mode, the page data unit size for BCH encoder and BCH decoder still is 1024byte. So to support BCH encoder and decoder, software should configure page data as follows:

1th~512th bytes are invalid data which must be stuffed with 0xff, 513th~1024th bytes arevalid page data.

However, Randomizer function is not supported under this condition.

17.6.5 PageSize/SpareSize Application

FLCTL[21] determines whether the codeword size is 1024 bytes or 512 bytes when FLCTL[11] is 1.

1. Big Page

When FLCTL[11]=0(LbaEn=0), the flash to be operated is Raw NAND Flash. Every codeword size is 1024 bytes and FLCTL[21] should always be set to 0, and the PageStep in external memory is 1024 bytes if bus mode is master mode.

At this mode, the spare size and SpareStep in external memory are determined by BCH Mode as follows:

BCH Mode=16bitBCH: spare size=(28+4)bytes , SpareStep=64bytes
BCH Mode=24bitBCH: spare size=(42+4)bytes , SpareStep=64bytes
BCH Mode=40bitBCH: spare size=(70+4)bytes , SpareStep=128bytes
BCH Mode=60bitBCH: spare size=(105+4)bytes, SpareStep=128bytes

2. Small Page

When FLCTL[11]=1, LbaEn=1, the flash to be operated is Managed NAND Flash. Every codeword size could be 1024 bytes or 512 bytes according to FLCTL[21]. If FLCTL[21]=0, codeword size is 1024 bytes, PageStep in external memory is 1024 bytes, and SpareStep is 128bytes. If FLCTL[21]=1, codeword size is 512 bytes, PageStep in external memory is 512 bytes, and SpareStep is 64 bytes.

At this mode, the spare size is configured in FLCTL[18:12], and the max available number is 109.

In the summary, the total data size in every codeword for flash or for software including page data and spare data, is determined by BCHCTL[16], FLCTL[11], FLCTL[21], BCHCTL[4], BCHCTL[18]. Their relationship is shown as follows.

Table 17-4 NandC Page/Spare size for flash

page/spare size for software		page size /codeword	spare size /codeword
FLCTL[11]=0	16bitECC	1024 byte	(4+28)byte
	24bitECC	1024 byte	(4+42)byte
	40bitECC	1024 byte	(4+70)byte
	60bitECC	1024 byte	(4+105)byte
FLCTL[11]=1	FLCTL[21]=0	1024 byte	FLCTL[18:12]
	FLCTL[21]=1	512 byte	FLCTL[18:12]

Notes: that "page/spare size for flash" means that hardware should transfer these numbers of bytes in every codeword to or from flash.

17.6.6 Randomizer Application

RANDMZ_CFG[31] determines whether randomizer is enable or not. When RANDMZ_CFG[31] equals to 1, randomizer is active. Data should be scrambled before written into flash, and descrambled after read from flash.

RANDMZ_CFG[30:29] determines the randomizer polynomial.

When RANDMZ_CFG[30:29]=2'b00, Samsung randomizer, Polynomial = 1+x+x^15.

RANDMZ_CFG[14:0] is the seed for randomizer.

When RANDMZ_CFG[30:29]=2'b01, TOSHIBA randomizer. RANDMZ_CFG[19:0] is the seed

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agitation register for randomizer. RANDMZ_CFG[23:20] is the basic seed start point for every page. RANDMZ_CFG[28:24] is the basic seed rotation bits for every 16 page.
When RANDMZ_CFG[30:29]=2'b10, Samsung randomizer, Polynomial = 1+x^14+x^15. RANDMZ_CFG[14:0] is the seed for randomizer.
The data unit for randomizer is one codeword(data+spare).
However, Randomizer is just available for data transfer by internal DMA mode, but not by for bypass mode. Furthermore, it should not be enable if BCHCTL[16]=0 (BchPage=512bytes).

17.6.7 DLL Application

When Toggle Flash or ONFI Synchronous Flash interface is active, DLL should be used to adjust DQS input with DQ when reading flash.
There are 2 registers for DLL configuration(DLL_CFG_REG0 and DLL_CFG_REG1), and 1 register for DLL status(DLL_OBS_REG0).
The usage guide is as follows:
If bypass mode is used, you should set *dll_bypass* in DLL_CFG_REG1[1] to 1, and set *dll_dqs_dly_bypass* in DLL_CFG_REG0[23:16] to determine the dll element number needed. And then set *dll_start* in DLL_CFG_REG1[0] to 1 to start the DLL.
If auto adjusting is used, you should set *dll_bypass* in DLL_CFG_REG1[1] to 0, and set the *dll_start_point* in DLL_CFG_REG0[7:0] and *dll_incr* in DLL_CFG_REG1[11:4]. You also should set the adjusting mode *dll_qtren* in DLL_CFG_REG1[3:2] to compute the dll element number needed. If *dll_qtren*=2'b00, the dll element number is determined by *dll_dqs_dly* in DLL_CFG_REG0[15:8]; otherwise, it is 1/4 or 1/8 of the total number of dll elements used for *dll_qtren*=2'b01 or *dll_qtren*=2'b10 separately. The last step is to set *dll_start* in DLL_CFG_REG1[0] to 1 to start the DLL.
If you want to monitor the dll working status, you could read DLL_OBS_REG0. If DLL_OBS_REG0[0]=0, it means that DLL is not locked, and still in detecting status. Otherwise, it means that DLL is locked, and *dll_lock_value* in DLL_OBS_REG0[8:1] is the total number of dll elements used, *dll_dqs_delay_value* in DLL_OBS_REG0[16:9] is the total number of DQS delay used.

17.6.8 NandC Interrupt Application

NandC has 1 interrupt output signal and 4 interrupt sources: dma finish interrupt source, flash ready interrupt source, bch error interrupt source, bchfail interrupt source. When one or more of these interrupt source are enabled, NandC interrupt is asserted if one or more interrupt source is high. Software can determine the interrupt source by reading INTST and clear interrupt by writing corresponding bit in INTCLR.

17.6.9 LLP Application

LLP is used in NandC to store and execute instruction groups configured in external memory by software. When LLPCTL[0]=1, LLP is active, NandC will load instruction groups stored in {LLPCTL[31:6], 6'h0} and execute them. Next instruction groups should not be loaded until current instruction execution finished.

1. LLP Structure

The structure of LLP is shown as follows:

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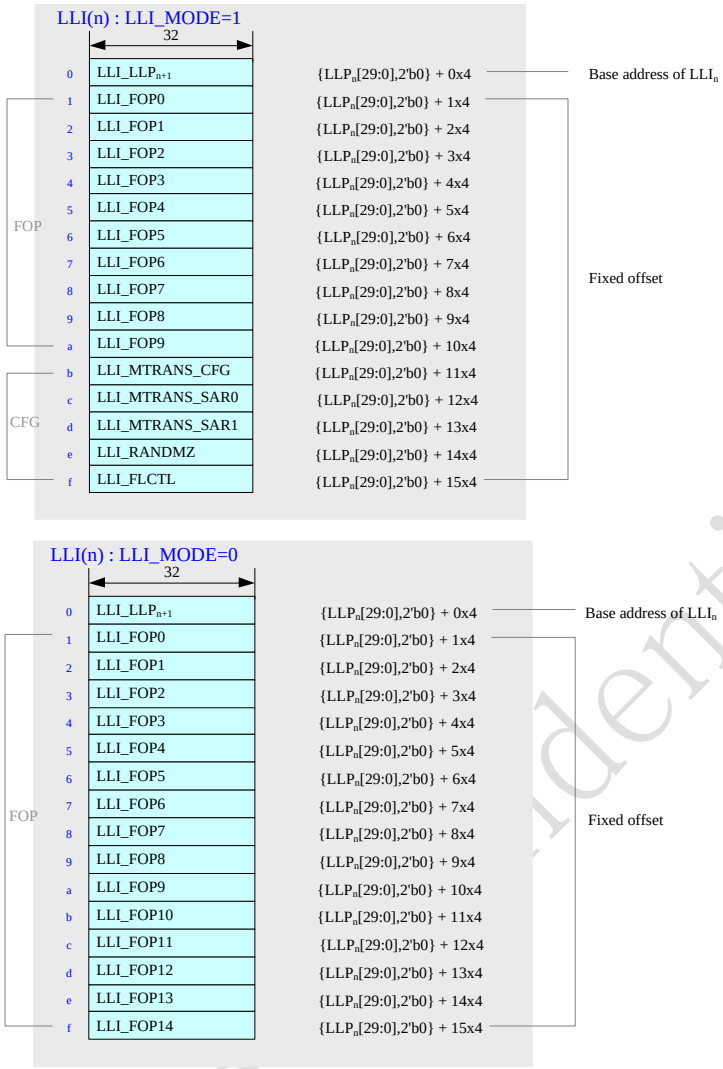


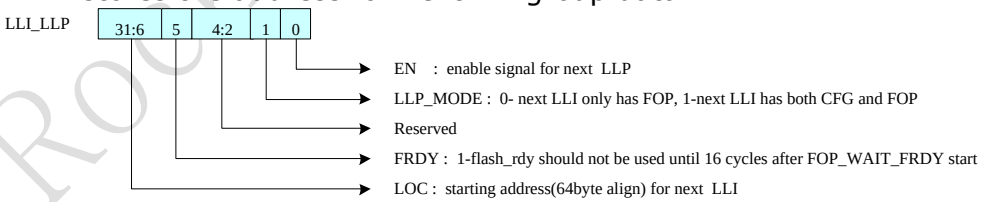
Fig. 17-4 NandC LLP Data Format

LLI_MODE is determined by LLPCTL[1]. If current operation is flash program or flash read, then LLI_MODE=1 is need; otherwise, LLI_MODE=0 is workable.

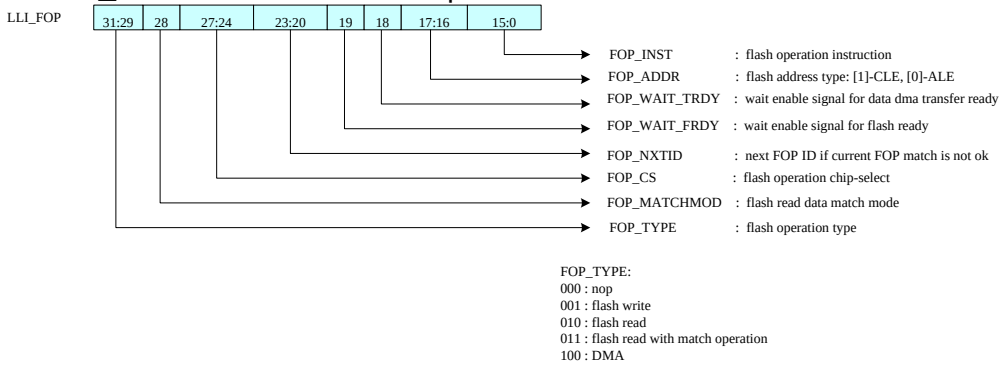
In addition, you could do more than one flash operation in one LLP group, but you should not separate one flash operation into two LLI groups.

2. LLI Format

a. LLI_LLPn+1 stores the address for next LLI group data



b. LLI_FOP0~LLI_FOP14 store the flash operation instruction



FOP_TYPE:
000 : nop
001 : flash write
010 : flash read
011 : flash read with match operation
100 : DMA

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When FOP_TYPE=3'b011, match operation is active, and the PATTERN is LLI_FOP[15:0]. It is matched when "RDATA|PATTERN=PATTERN" with FOP_MATCHMOD=0, or when "RDATA&PATTERN=PATTERN" with FOP_MATCHMOD=1.
c. LLI_MTRANS_CFG/LLI_MTRANS_SADDR0/LLI_MTRANS_SADDR1/ LLI_RANDMZ/ LLI_FLCTL store the configuration for MTRANS_CFG/ MTRANS_SADDR0/MTRANS_SADDR1/RANDMZ/FLCTL.

3. LLP Working Mode

There are two working modes for LLP:
a. Normal mode: LLPCTL[0] is kept to 1 until all LLP loading and executing finished. Software can monitor the progress by LLPSTAT[31:6], LLPSTAT[0].
b. Pause mode: LLPCTL[0] is changed from 1 to 0 during LLP loading or LLP executing. NandC should not stop working until current LLP executing finished. Software can monitor the progress by LLPSTAT[31:6], LLPSTAT[0].

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Chapter 18 Mobile Storage Host Controller

18.1 Overview

The Mobile Storage Host Controller is designed to support Secure Digital memory (SD- max version 3.01) with 1 bits or 4 bits data width, Multimedia Card(MMC-max version 4.5) with 1 bits or 4 bits or 8 bits data width.
The Host Controller supports following features:

- Bus Interface Features:
  - Supports AMBA AHB interface for master and slave
  - Supports internal DMA interface(IDMAC)
    - Supports 16/32-bit data transfers
    - Single-channel; single engine used for Transmit and Receive, which are mutually exclusive
    - Dual-buffer and chained descriptor linked list
    - Each descriptor can transfer up to 4KB of data in chained mode and 8KB of data in dual-buffer mode
    - Programmable burst size for optimal host bus utilization
  - Supports combined single FIFO for both transmit and receive operations
  - Supports FIFO size of 256x32
  - Supports FIFO over-run and under-run prevention by stopping card clock
- Card Interface Features:
  - Supports Secure Digital memory protocol commands
  - Supports Secure Digital I/O protocol commands
  - Supports Multimedia Card protocol commands
  - Supports Command Completion Signal and interrupts to host
  - Supports CRC generation and error detection
  - Supports programmable baud rate
  - Supports power management and power switch
  - Supports card detection and initialization
  - Supports write protection
  - Supports hardware reset
  - Supports SDIO interrupts in 1-bit and 4-bit modes
  - Supports 4-bit mode in SDIO3.0
  - Supports SDIO suspend and resume operation
  - Supports SDIO read wait
  - Supports block size of 1 to 65,535 bytes
  - Supports 1-bit, 4-bit and 8-bit SDR modes
  - Supports 4-bit DDR,8-bit DDR, as defined by SD3.0 and MMC4.41
  - Supports boot in 1-bit, 4-bit and 8-bit SDR modes
  - Supports Packed Commands, CMD21, CMD49
- Clock Interface Features:
  - Supports 0/90/180/270-degree phase shift operation for sample clock (cclk_in_sample) and drive clock(cclk_in_drv) relative to function clock(cclk_in) respectively
  - Supports phase tuning using delay line for sample clock(cclk_in_sample) and drive clock(cclk_in_drv) relative to function clock (cclk_in) respectively. The max number of delay element number is 256.

The Host Controller is instantiated for SDMMC, SDIO, EMMC. The interface difference between these instances is shown in "Interface Description".

18.2 Block Diagram

The Host Controller consists of the following main functional blocks.

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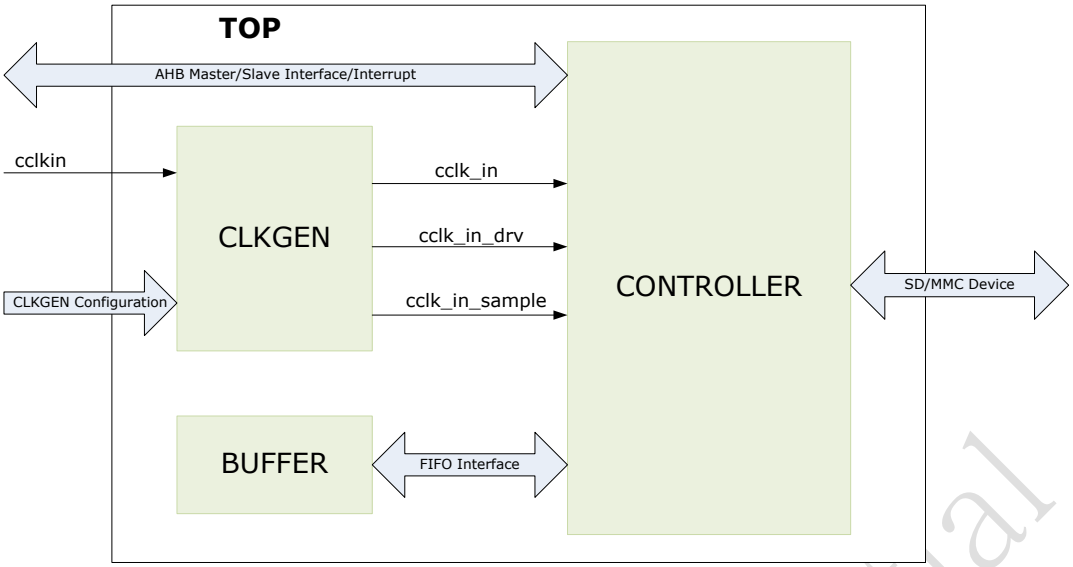

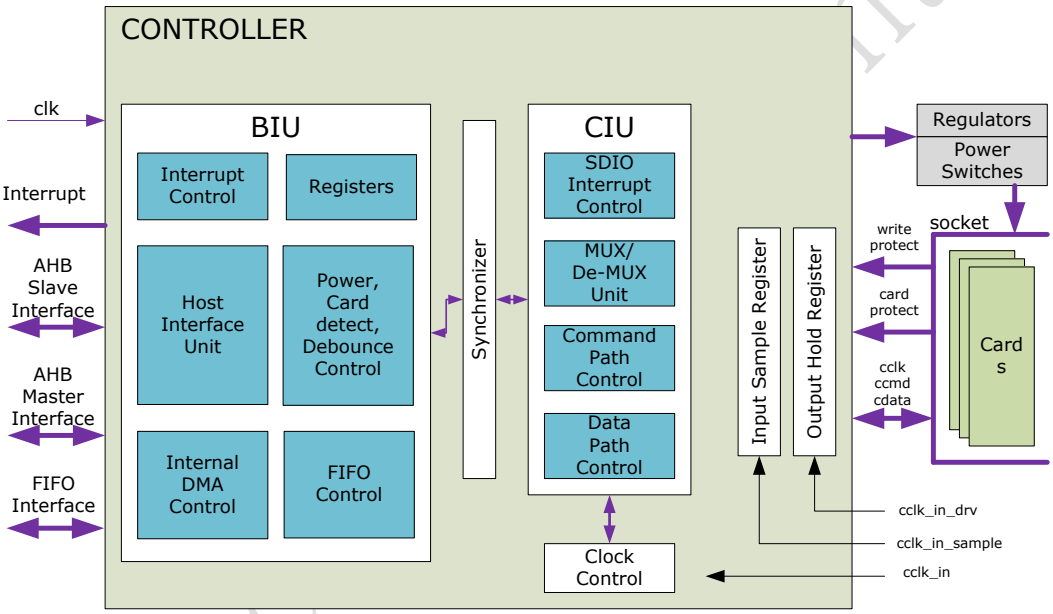

Fig. 18-1 Host Controller Block Diagram

- Clock Generate Unit(CLKGEN): generates card interface clock cclk_in/ cclk_sample/cclk_drv based on cclkin and configuration information.
- Asynchronous dual-port memory(BUFFER): Uses a two-clock synchronous read and synchronous write dual-port RAM. One of the ports is connected to the host clock, and the second port is connected to the card clock.
- Bus Interface Unit (BIU): Provides AMBA AHB interfaces for register and data read/writes.
- Card Interface Unit (CIU): Takes care of the SD/MMC protocols and provides clock management.

18.3 Function Description

18.3.1 Bus Interface Unit

The Bus Interface Unit provides the following functions:

- Host interface
- Interrupt control
- Register access
- External FIFO access
- Power control and card detection

1. Host Interface Unit

The Host Interface Unit is an AHB slave interface, which provides the interface between the

SD/MMC card and the host bus. You can configure the host interface as either an AHB.

2. Register Unit

The register unit is part of the bus interface unit; it provides read and write access to the registers.

All registers reside in the Bus Interface Unit clock domain. When a command is sent to a card by setting the start_bit, which is bit[31] of the CMD register, all relevant registers needed for the CIU operation are transferred to the CIU block. During this time, the registers that are transferred from the BIU to the CIU should not be written. The software should wait for the hardware to clear the start bit before writing to these registers again. The register unit has a hardware locking feature to prevent illegal writes to registers. The lock is necessary in order to avoid metastability violations, both because the host and card clock domains are different and to prevent illegal software operations.

Once a command start is issued by setting the start_bit of the CMD register, the following registers cannot be reprogrammed until the command is accepted by the card interface unit:

- CMD – Command
- CMDARG – Command Argument
- BYTCNT – Byte Count
- BLKSIZ – Block Size
- CLKDIV – Clock Divider
- CLKENA – Clock Enable
- CLKSRC – Clock Source
- TMOUT – Timeout
- CTYPE – Card Type

The hardware resets the start_bit once the CIU accepts the command. If a host write to any of these registers is attempted during this locked time, then the write is ignored and the hardware lock error bit is set in the raw interrupt status register. Additionally, if the interrupt is enabled and not masked for a hardware lock error, then an interrupt is sent to the host.

When the Card Interface Unit is in an idle state, it typically takes the following number of clocks for the command handshake, where clk is the BIU clock and cclk_in is the CIU clock:

3 (clk) + 3 (cclk_in)

Once a command is accepted, you can send another command to the CIU-which has a one-deep command queue-under the following conditions:

- If the previous command was not a data transfer command, the new command is sent to the SD/MMC card once the previous command completes.
- If the previous command is a data transfer command and if wait_prvdata_complete (bit[13]) of the Command register is set for the new command, the new command is sent to the SD/MMC card only when the data transfer completes.
- If the wait_prvdata_complete is 0, then the new command is sent to the SD/MMC card as soon as the previous command is sent. Typically, you should use this only to stop or abort a previous data transfer or query the card status in the middle of a data transfer.

3. Interrupt Controller Unit

The interrupt controller unit generates an interrupt that depends on the controller raw interrupt status, the interrupt-mask register, and the global interrupt-enable register bit. Once an interrupt condition is detected, it sets the corresponding interrupt bit in the raw interrupt status register. The raw interrupt status bit stays on until the software clears the bit by writing a 1 to the interrupt bit; a 0 leaves the bit untouched.

The interrupt port, int, is an active-high, level-sensitive interrupt. The interrupt port is active only when any bit in the raw interrupt status register is active, the corresponding interrupt mask bit is 1, and the global interrupt enable bit is 1. The interrupt port is registered in order to avoid any combinational glitches.

The int_enable is reset to 0 on power-on, and the interrupt mask bits are set to 32'h0, which masks all the interrupts.

Notes:

Before enabling the interrupt, it is always recommended that you write 32'hffff_ffff to the raw interrupt status register in order to clear any pending unserviced interrupts. When clearing interrupts during normal operation, ensure that you clear only the interrupt bits that you serviced.

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The SDIO Interrupts, Receive FIFO Data Request (RXDR), and Transmit FIFO Data Request (TXDR) are set by level-sensitive interrupt sources. Therefore, the interrupt source should be first cleared before you can clear the interrupt bit of the Raw Interrupt register. For example, on seeing the Receive FIFO Data Request (RXDR) interrupt, the FIFO should be emptied so that the "FIFO count greater than the RX-Watermark" condition, which triggers the interrupt, becomes inactive. The rest of the interrupts are triggered by a single clock-pulse-width source.

Table 18-1 Bits in Interrupt Status Register

Bits	Interrupt	Description
24	sdio_interrupt	Interrupt from SDIO card. In MMC-Ver3.3-only mode, these bits are always 0
16	Card no-busy	If card exit busy status, the interrupt happened
15	End Bit Error (read) /Write no CRC (EBE)	Error in end-bit during read operation, or no data CRC or negative CRC received during write operation. *Notes: For MMC CMD19, there may be no CRC status returned by the card. Hence, EBE is set for CMD19. The application should not treat this as an error.*
14	Auto Command Done (ACD)	Stop/abort commands automatically sent by card unit and not initiated by host; similar to Command Done (CD) interrupt.
13	Start Bit Error (SBE)	Error in data start bit when data is read from a card. In 4-bit mode, if all data bits do not have start bit, then this error is set.
12	Hardware Locked write Error (HLE)	During hardware-lock period, write attempted to one of locked registers.
11	FIFO Underrun/ Overrun Error (FRUN)	Host tried to push data when FIFO was full, or host tried to read data when FIFO was empty. Typically this should not happen, except due to error in software. Card unit never pushes data into FIFO when FIFO is full, and pop data when FIFO is empty.
10	Data Starvation by Host Timeout (HTO)	To avoid data loss, card clock out (cclk_out) is stopped if FIFO is empty when writing to card, or FIFO is full when reading from card. Whenever card clock is stopped to avoid data loss, data-starvation timeout counter is started with data-timeout value. This interrupt is set if host does not fill data into FIFO during write to card, or does not read from FIFO during read from card before timeout period. Even after timeout, card clock stays in stopped state, with CIU state machines waiting. It is responsibility of host to push or pop data into FIFO upon interrupt, which automatically restarts cclk_out and card state machines. Even if host wants to send stop/abort command, it still needs to ensure it has to push or pop FIFO so that clock starts in order for stop/abort command to send on cmd signal along with data that is sent or received on data line.
9	Data Read Timeout (DRTO)	Data timeout occurred. Data Transfer Over (DTO) also set if data timeout occurs.
8	Response Timeout (RTO)	Response timeout occurred. Command Done (CD) also set if response timeout occurs. If command involves data transfer and when response times out, no data transfer is attempted by Host Controller.
7	Data CRC Error (DCRC)	Received Data CRC does not match with locally-generated CRC in CIU.
6	Response CRC Error (RCRC)	Response CRC does not match with locally-generated CRC in CIU.
5	Receive FIFO Data	Interrupt set during read operation from card when FIFO

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	Request (RXDR)	level is greater than Receive-Threshold level.
4	Transmit FIFO Data Request (TXDR)	Interrupt set during write operation to card when FIFO level reaches less than or equal to Transmit-Threshold level.
3	Data Transfer Over (DTO)	Data transfer completed, even if there is Start Bit Error or CRC error. This bit is also set when "read data-timeout" occurs. *Notes: DTO bit is set at the end of the last data block, even if the device asserts MMC busy after the last data block.*
2	Command Done(CD)	Command sent to card and got response from card, even if Response Error or CRC error occurs. Also set when response timeout occurs
1	Response Error (RE)	Error in received response set if one of following occurs: ● Transmission bit != 0 ● Command index mismatch ● End-bit != 1
0	Card-Detect (CDT)	When card inserted or removed, this interrupt occurs. Software should read card-detect register (CDETECT, 0x50) to determine current card status.

4. FIFO Controller Unit

The FIFO controller interfaces the external FIFO to the host interface and the card controller unit. When FIFO overrun and under-run conditions occur, the card clock stops in order to avoid data loss.

The FIFO uses a two-clock synchronous read and synchronous write dual-port RAM. One of the ports is connected to the host clock, clk, and the second port is connected to the card clock, cclk_in.

Notes: The FIFO controller does not support simultaneous read/write access from the same port. For debugging purposes, the software may try to write into the FIFO and read back the data; results are indeterminate, since the design does not support read/write access from the same port.

5. Power Control and Card Detection Unit

The register unit has registers that control the power. Power to each card can be selectively turned on or off.

The card detection unit looks for any changes in the card-detect signals for card insertion or card removal. It filters out the debounces associated with mechanical insertion or removal, and generates one interrupt to the host. You can program the debounce filter value.

On power-on, the controller should read in the card_detect port and store the value in the memory. Upon receiving a card-detect interrupt, it should again read the card_detect port and XOR with the previous card-detect status to find out which card has interrupted. If more than one card is simultaneously removed or inserted, there is only one card-detect interrupt; the XOR value indicates which cards have been disturbed. The memory should be updated with the new card-detect value.

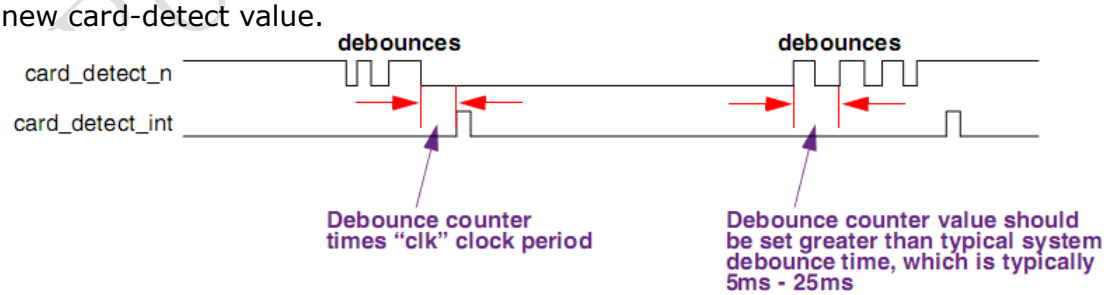

Fig. 18-2 SD/MMC Card-Detect Signal

18.3.2 Card Interface Unit

The Card Interface Unit (CIU) interfaces with the Bus Interface Unit (BIU) and the devices. The host writes command parameters to the BIU control registers, and these parameters are then passed to the CIU. Depending on control register values, the CIU generates SD/MMC

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command and data traffic on a selected card bus according to SD/MMC protocol. The Host Controller accordingly controls the command and data path.
The following software restrictions should be met for proper CIU operation:

- Only one data transfer command can be issued at a time.
- During an open-ended card write operation, if the card clock is stopped because the FIFO is empty, the software must first fill the data into the FIFO and start the card clock. It can then issue only a stop/abort command to the card.
- When issuing card reset commands (CMD0, CMD15 or CMD52_reset) while a card data transfer is in progress, the software must set the stop_abort_cmd bit in the Command register so that the Host Controller can stop the data transfer after issuing the card reset command.
- When the data end bit error is set in the RINTSTS register, the Host Controller does not guarantee SDIO interrupts. The software should ignore the SDIO interrupts and issue the stop/abort command to the card, so that the card stops sending the read data.
- If the card clock is stopped because the FIFO is full during a card read, the software should read at least two FIFO locations to start the card clock.

The CIU block consists of the following primary functional blocks:

- Command path
- Data path
- SDIO interrupt control
- Clock control
- Mux/demux unit
    - Command Path

The command path performs the following functions:

- Loads clock parameters
- Loads card command parameters
- Sends commands to card bus (ccmd_out line)
- Receives responses from card bus (ccmd_in line)
- Sends responses to BIU
- Drives the P-bit on command line

A new command is issued to the Host Controller by programming the BIU registers and setting the start_cmd bit in the Command register. The BIU asserts start_cmd, which indicates that a new command is issued to the SD/MMC device. The command path loads this new command (command, command argument, timeout) and sends acknowledge to the BIU by asserting cmd_taken.
Once the new command is loaded, the command path state machine sends a command to the device bus-including the internally generated CRC7-and receives a response, if any. The state machine then sends the received response and signals to the BIU that the command is done, and then waits for eight clocks before loading a new command.

Load Command Parameters
One of the following commands or responses is loaded in the command path:

- New command from BIU – When start_cmd is asserted, then the start_cmd bit is set in the Command register.
- Internally-generated auto-stop command – When the data path ends, the stop command request is loaded.
- IRQ response with RCA 0x000 – When the command path is waiting for an IRQ response from the MMC card and a "send irq response" request is signaled by the BIU, then the send_irq_response bit is set in the control register.

Loading a new command from the BIU in the command path depends on the following Command register bit settings:

- update_clock_registers_only – If this bit is set in the Command register, the command path updates only the clock enable, clock divider, and clock source registers. If this bit is not set, the command path loads the command, command argument, and timeout registers; it then starts processing the new command.

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- wait_prvdata_complete – If this bit is set, the command path loads the new command under one of the following conditions:
- Immediately, if the data path is free (that is, there is no data transfer in progress), or if an open-ended data transfer is in progress (byte_count = 0).
- After completion of the current data transfer, if a predefined data transfer is in progress.

Send Command and Receive Response

Once a new command is loaded in the command path, update_clock_registers_only bit is unset – the command path state machine sends out a command on the device bus; the command path state machine is illustrated in following figure.

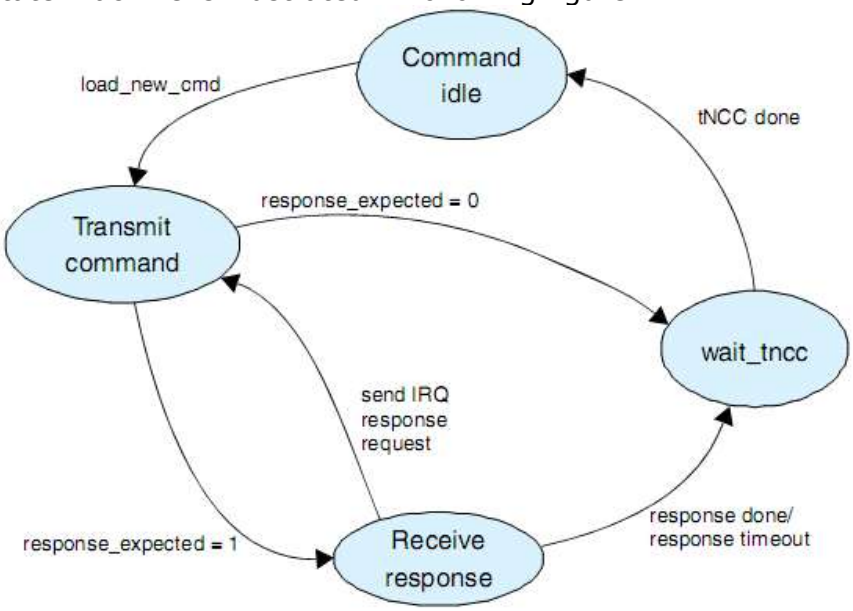

Fig. 18-3 Host Controller Command Path State Machine

The command path state machine performs the following functions, according to Command register bit values:

- send_initialization – Initialization sequence of 80 clocks is sent before sending the command.
- response_expected – Response is expected for the command. After the command is sent out, the command path state machine receives a 48-bit or 136-bit response and sends it to the BIU. If the start bit of the card response is not received within the number of clocks programmed in the timeout register, then the response timeout and command done bit is set in the Raw Interrupt Status register as a signal to the BIU. If the response-expected bit is not set, the command path sends out a command and signals a response done to the BIU; that is, the command done bit is set in the Raw Interrupt Status register.
- response_length – If this bit is set, a 136-bit response is received; if it is not set, a 48-bit response is received.
- check_response_crc – If this bit is set, the command path compares CRC7 received in the response with the internally-generated CRC7. If the two do not match, the response CRC error is signaled to the BIU; that is, the response CRC error bit is set in the Raw Interrupt Status register.

Send Response to BIU

If the response_expected bit is set in the Command register, the received response is sent to the BIU. The Response0 register is updated for a short response, and the Response3, Response2, Response1, and Response0 registers are updated on a long response, after which the Command Done bit is set. If the response is for an auto_stop command sent by the CIU, the response is saved in the Response1 register, after which the Auto Command Done bit is set.

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Additionally, the command path checks for the following:

- Transmission bit = 0
- Command index matches command index of the sent command
- End bit = 1 in received card response

The command index is not checked for a 136-bit response or if the check_response_crc bit is unset. For a 136-bit response and reserved CRC 48-bit responses, the command index is reserved-that is, 111111.

Polling Command Completion Signal

The device generates the Command Completion Signal in order to notify the host controller of the normal command completion or command termination.

Command Completion Signal Detection and Interrupt to Host Processor

If the ccs_expected bit is set in the Command register, the Command Completion Signal (CCS) from the device is indicated by setting the Data Transfer Over (DTO) bit in the RINTSTS register. The Host Controller generates a Data Transfer Over (DTO) interrupt if this interrupt is not masked.

Command Completion Signal Timeout

If the command expects a CCS from the device—if the ccs_expected bit is set in the Command register—the command state machine waits for the CCS and remains in a wait_CCSS state. If the device fails to send out the CCS, the host software should implement a timeout mechanism to free the command and data path. The host controller does not implement a hardware timer; it is the responsibility of the host software to maintain a software timer.
In the event of a CCS timeout, the host should issue a CCSD by setting the send_ccsd bit in the CTRL register. The host controller command state machine sends the CCSD to the device and exits to an idle state. After sending the CCSD, the host should also send a CMD12 to the device in order to abort the outstanding command.

Send Command Completion Signal Disable

If the send_ccsd bit is set in the CTRL register, the host sends a Command Completion Signal Disable (CCSD) pattern on the CMD line. The host can send the CCSD while waiting for the CCS or after a CCS timeout happens.
After sending the CCSD pattern, the host sets the Command Done (CD) bit in RINTSTS and also generates an interrupt to the host if the Command Done interrupt is not masked.

- Data Path

The data path block pops the data FIFO and transmits data on cdata_out during a write data transfer, or it receives data on cdata_in and pushes it into the FIFO during a read data transfer. The data path loads new data parameters—that is, data expected, read/write data transfer, stream/block transfer, block size, byte count, card type, timeout registers—whenever a data transfer command is not in progress.
If the data_expected bit is set in the Command register, the new command is a data transfer command and the data path starts one of the following:

- Transmit data if the read/write bit = 1
- Data receive if read/write bit = 0

Data Transmit

The data transmit state machine, illustrated in following figure, starts data transmission two clocks after a response for the data write command is received; this occurs even if the command path detects a response error or response CRC error. If a response is not received from the card because of a response timeout, data is not transmitted. Depending upon the value of the transfer_mode bit in the Command register, the data transmit state machine puts data on the card data bus in a stream or in block(s).

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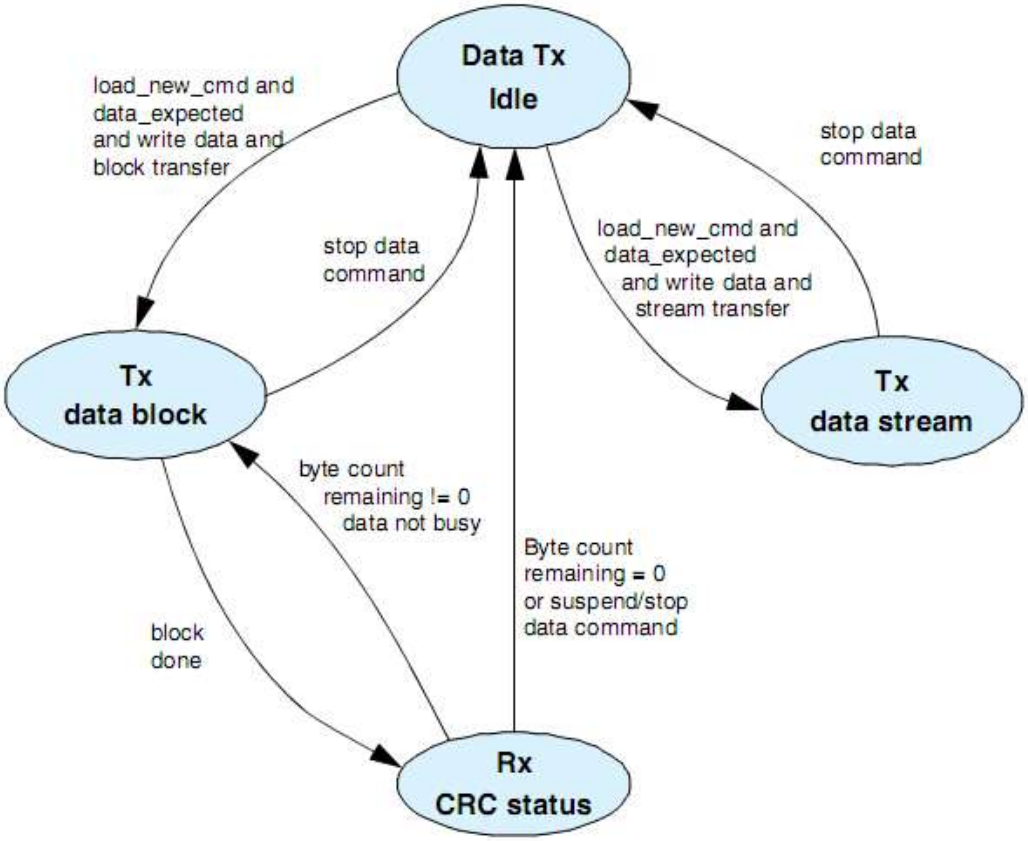

Fig. 18-4 Host Controller Data Transmit State Machine

Stream Data Transmit

If the transfer_mode bit in the Command register is set to 1, it is a stream-write data transfer. The data path pops the FIFO from the BIU and transmits in a stream to the card data bus. If the FIFO becomes empty, the card clock is stopped and restarted once data is available in the FIFO.

If the byte_count register is programmed to 0, it is an open-ended stream-write data transfer. During this data transfer, the data path continuously transmits data in a stream until the host software issues a stop command. A stream data transfer is terminated when the end bit of the stop command and end bit of the data match over two clocks.

If the byte_count register is programmed with a non-zero value and the send_auto_stop bit is set in the Command register, the stop command is internally generated and loaded in the command path when the end bit of the stop command occurs after the last byte of the stream write transfer matches.

This data transfer can also terminate if the host issues a stop command before all the data bytes are transferred to the card bus.

Single Block Data

If the transfer_mode bit in the Command register is set to 0 and the byte_count register value is equal to the value of the block_size register, a single-block write-data transfer occurs. The data transmit state machine sends data in a single block, where the number of bytes equals the block size, including the internally-generated CRC16.

If the CTYPE register bit for the selected card – indicated by the card_num value in the Command register – is set for a 1-bit, 4-bit, or 8-bit data transfer, the data is transmitted on 1, 4, or 8 data lines, respectively, and CRC16 is separately generated and transmitted for 1, 4, or 8 data lines, respectively.

After a single data block is transmitted, the data transmit state machine receives the CRC status from the card and signals a data transfer to the BIU; this happens when the data-transfer-over bit is set in the RINTSTS register.

If a negative CRC status is received from the card, the data path signals a data CRC error to the BIU by setting the data CRC error bit in the RINTSTS register.

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Additionally, if the start bit of the CRC status is not received by two clocks after the end of the data block, a CRC status start bit error is signaled to the BIU by setting the write-no-CRC bit in the RINTSTS register.

Multiple Block Data

A multiple-block write-data transfer occurs if the transfer_mode bit in the Command register is set to 0 and the value in the byte_count register is not equal to the value of the block_size register. The data transmit state machine sends data in blocks, where the number of bytes in a block equals the block size, including the internally-generated CRC16.

If the CTYPE register bit for the selected card – indicated by the card_num value in the Command register – is set to 1-bit, 4-bit, or 8-bit data transfer, the data is transmitted on 1, 4, or 8 data lines, respectively, and CRC16 is separately generated and transmitted on 1, 4, or 8 data lines, respectively.

After one data block is transmitted, the data transmit state machine receives the CRC status from the card. If the remaining byte_count becomes 0, the data path signals to the BIU that the data transfer is done; this happens when the data-transfer-over bit is set in the RINTSTS register.

If the remaining data bytes are greater than 0, the data path state machine starts to transmit another data block.

If a negative CRC status is received from the card, the data path signals a data CRC error to the BIU by setting the data CRC error bit in the RINTSTS register, and continues further data transmission until all the bytes are transmitted.

Additionally, if the CRC status start bit is not received by two clocks after the end of a data block, a CRC status start bit error is signaled to the BIU by setting the write-no-CRC bit in the RINTSTS register; further data transfer is terminated.

If the send_auto_stop bit is set in the Command register, the stop command is internally generated during the transfer of the last data block, where no extra bytes are transferred to the card. The end bit of the stop command may not exactly match the end bit of the CRC status in the last data block.

If the block size is less than 4, 16, or 32 for card data widths of 1 bit, 4 bits, or 8 bits, respectively, the data transmit state machine terminates the data transfer when all the data is transferred, at which time the internally generated stop command is loaded in the command path.

If the byte_count is 0 – the block size must be greater than 0 – it is an open-ended block transfer. The data transmit state machine for this type of data transfer continues the block-write data transfer until the host software issues a stop or abort command.

Data Receive

The data-receive state machine, illustrated in following figure, receives data two clock cycles after the end bit of a data read command, even if the command path detects a response error or response CRC error. If a response is not received from the card because a response timeout occurs, the BIU does not receive a signal that the data transfer is complete; this happens if the command sent by the Host Controller is an illegal operation for the card, which keeps the card from starting a read data transfer.

If data is not received before the data timeout, the data path signals a data timeout to the BIU and an end to the data transfer done. Based on the value of the transfer_mode bit in the Command register, the data-receive state machine gets data from the card data bus in a stream or block(s).

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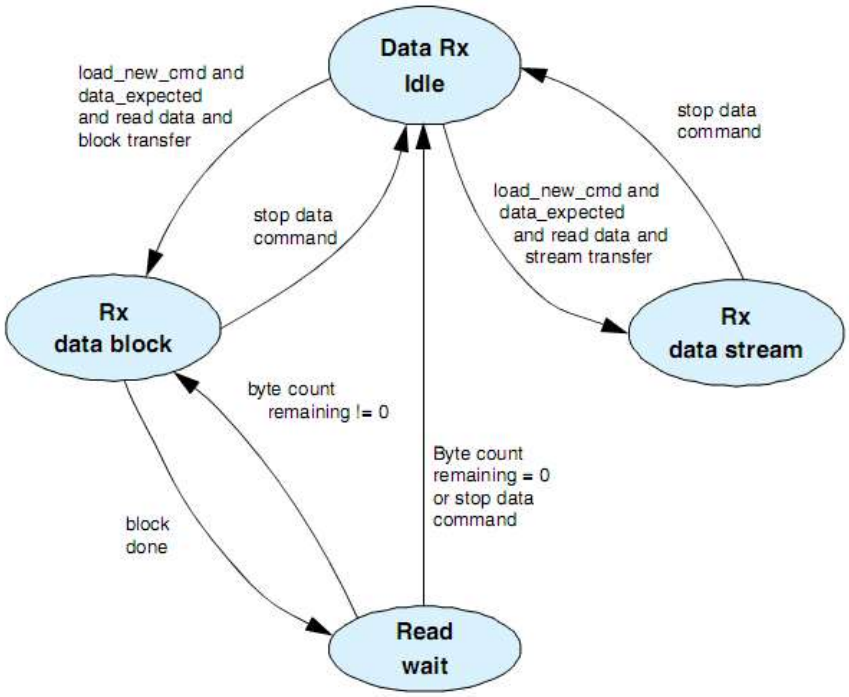

Fig. 18-5 Host Controller Data Receive State Machine

Stream Data Read

A stream-read data transfer occurs if the transfer_mode bit in the Command register equals 1, at which time the data path receives data from the card and pushes it to the FIFO. If the FIFO becomes full, the card clock stops and restarts once the FIFO is no longer full.

An open-ended stream-read data transfer occurs if the byte_count register equals 0. During this type of data transfer, the data path continuously receives data in a stream until the host software issues a stop command. A stream data transfer terminates two clock cycles after the end bit of the stop command.

If the byte_count register contains a non-zero value and the send_auto_stop bit is set in the Command register, a stop command is internally generated and loaded into the command path, where the end bit of the stop command occurs after the last byte of the stream data transfer is received. This data transfer can terminate if the host issues a stop or abort command before all the data bytes are received from the card.

Single-Block Data Read

A single-block read-data transfer occurs if the transfer_mode bit in the Command register is set to 0 and the value of the byte_count register is equal to the value of the block_size register. When a start bit is received before the data times out, data bytes equal to the block size and CRC16 are received and checked with the internally-generated CRC16.

If the CTYPE register bit for the selected card – indicated by the card_num value in the Command register – is set to a 1-bit, 4-bit, or 8-bit data transfer, data is received from 1, 4, or 8 data lines, respectively, and CRC16 is separately generated and checked for 1, 4, or 8 data lines, respectively. If there is a CRC16 mismatch, the data path signals a data CRC error to the BIU. If the received end bit is not 1, the BIU receives an end-bit error.

Multiple-Block Data Read

If the transfer_mode bit in the Command register is set to 0 and the value of the byte_count register is not equal to the value of the block_size register, it is a multiple-block read-data transfer. The data-receive state machine receives data in blocks, where the number of bytes in a block is equal to the block size, including the internally-generated CRC16.

If the CTYPE register bit for the selected card – indicated by the card_num value in the Command register – is set to a 1-bit, 4-bit, or 8-bit data transfer, data is received from 1, 4, or 8 data lines, respectively, and CRC16 is separately generated and checked for 1, 4, or 8 data lines, respectively.

After a data block is received, if the remaining byte_count becomes 0, the data path signals a

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data transfer to the BIU.
If the remaining data bytes are greater than 0, the data path state machine causes another data block to be received. If CRC16 of a received data block does not match the internally-generated CRC16, a data CRC error to the BIU and data reception continue further data transmission until all bytes are transmitted.
Additionally, if the end of a received data block is not 1, data on the data path signals terminate the bit error to the CIU and the data-receive state machine terminates data reception, waits for data timeout, and signals to the BIU that the data transfer is complete.
If the send_auto_stop bit is set in the Command register, the stop command is internally generated when the last data block is transferred, where no extra bytes are transferred from the card; the end bit of the stop command may not exactly match the end bit of the last data block.
If the requested block size for data transfers to cards is less than 4, 16, or 32 bytes for 1-bit, 4-bit, or 8-bit data transfer modes, respectively, the data-transmit state machine terminates the data transfer when all data is transferred, at which point the internally-generated stop command is loaded in the command path. Data received from the card after that are then ignored by the data path.
If the byte_count is 0—the block size must be greater than 0—it is an open-ended block transfer. For this type of data transfer, the data-receive state machine continues the block-read data transfer until the host software issues a stop or abort command.

Auto-Stop
The Host Controller internally generates a stop command and is loaded in the command path when the send_auto_stop bit is set in the Command register.
The software should set the send_auto_stop bit according to details listed in following table.

Table 18-2 Auto-Stop Generation

Card type	Transfer type	Byte Count	send_auto_stop bit set	Comments
MMC	Stream read	0	No	Open-ended stream
MMC	Stream read	>0	Yes	Auto-stop after all bytes transfer
MMC	Stream write	0	No	Open-ended stream
MMC	Stream write	>0	Yes	Auto-stop after all bytes transfer
MMC	Single-block read	>0	No	Byte count =0 is illegal
MMC	Single-block write	>0	No	Byte count =0 is illegal
MMC	Multiple-block read	0	No	Open-ended multiple block
MMC	Multiple-block read	>0	Yes①	Pre-defined multiple block
MMC	Multiple-block write	0	No	Open-ended multiple block
MMC	Multiple-block write	>0	Yes①	Pre-defined multiple block
SDMEM	Single-block read	>0	No	Byte count =0 is illegal
SDMEM	Single-block write	>0	No	Byte count =0 illegal
SDMEM	Multiple-block read	0	No	Open-ended multiple block
SDMEM	Multiple-block read	>0	Yes	Auto-stop after all bytes transfer
SDMEM	Multiple-block write	0	No	Open-ended multiple block
SDMEM	Multiple-block write	>0	Yes	Auto-stop after all bytes transfer
SDIO	Single-block read	>0	No	Byte count =0 is illegal
SDIO	Single-block write	>0	No	Byte count =0 illegal
SDIO	Multiple-block read	0	No	Open-ended multiple block
SDIO	Multiple-block read	>0	No	Pre-defined multiple block
SDIO	Multiple-block write	0	No	Open-ended multiple block
SDIO	Multiple-block write	>0	No	Pre-defined multiple block

①:*The condition under which the transfer mode is set to block transfer and byte_count is equal to block size*

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*is treated as a single-block data transfer command for both MMC and SD cards. If byte_count = n*block_size (n = 2, 3, …), the condition is treated as a predefined multiple-block data transfer command. In the case of an MMC card, the host software can perform a predefined data transfer in two ways: 1) Issue the CMD23 command before issuing CMD18/CMD25 commands to the card – in this case, issue MD18/CMD25 commands without setting the send_auto_stop bit. 2) Issue CMD18/CMD25 commands without issuing CMD23 command to the card, with the send_auto_stop bit set. In this case, the multiple-block data transfer is terminated by an internally-generated auto-stop command after the programmed byte count.*

The following list conditions for the auto-stop command.

- Stream read for MMC card with byte count greater than 0 – The Host Controller generates an internal stop command and loads it into the command path so that the end bit of the stop command is sent out when the last byte of data is read from the card and no extra data byte is received. If the byte count is less than 6 (48 bits), a few extra data bytes are received from the card before the end bit of the stop command is sent.
- Stream write for MMC card with byte count greater than 0 - The Host Controller generates an internal stop command and loads it into the command path so that the end bit of the stop command is sent when the last byte of data is transmitted on the card bus and no extra data byte is transmitted. If the byte count is less than 6 (48 bits), the data path transmits the data last in order to meet the above condition.
- Multiple-block read memory for SD card with byte count greater than 0 – If the block size is less than 4 (single-bit data bus), 16 (4-bit data bus), or 32 (8-bit data bus), the auto-stop command is loaded in the command path after all the bytes are read. Otherwise, the top command is loaded in the command path so that the end bit of the stop command is sent after the last data block is received.
- Multiple-block write memory for SD card with byte count greater than 0 – If the block size is less than 3 (single-bit data bus), 12 (4-bit data bus), or 24 (8-bit data bus), the auto-stop command is loaded in the command path after all data blocks are transmitted. Otherwise, the stop command is loaded in the command path so that the end bit of the stop command is sent after the end bit of the CRC status is received.
- Precaution for host software during auto-stop – Whenever an auto-stop command is issued, the host software should not issue a new command to the SD/MMC device until the auto-stop is sent by the Host Controller and the data transfer is complete. If the host issues a new command during a data transfer with the auto-stop in progress, an auto-stop command may be sent after the new command is sent and its response is received; this can delay sending the stop command, which transfers extra data bytes. For a stream write, extra data bytes are erroneous data that can corrupt the card data. If the host wants to terminate the data transfer before the data transfer is complete, it can issue a stop or abort command, in which case the Host Controller does not generate an auto-stop command.
-
- Non-Data Transfer Commands that Use Data Path

Some non-data transfer commands (non-read/write commands) also use the data path. Following table lists the commands and register programming requirements for them.

Table 18-3 Non-data Transfer Commands and Requirements

	CMD27	CMD30	CMD42	ACMD13	ACMD22	ACMD51
Command register programming						
cmd_index	6'h1B	6'h1E	6'h2A	6'h0D	6'h16	6'h33
response_expect	1	1	1	1	1	1
rResponse_length	0	0	0	0	0	0
check_response_crc	1	1	1	1	1	1
data_expected	1	1	1	1	1	1
read/write	1	0	1	0	0	0
transfer_mode	0	0	0	0	0	0
send_auto_stop	0	0	0	0	0	0
wait_prevdata_comp	0	0	0	0	0	0

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lete						
stop_abort_cmd	0	0	0	0	0	0
Command Argument register programming						
	stuff bits	32-bit writeprotect dataaddress	stuff bits	stuff bits	stuff bits	stuff bits
Block Size register programming						
	16	4	Num_bytes①	64	4	8
Byte Count register programming						
	16	4	Num_bytes①	64	4	8

①: *Num_bytes = No. of bytes specified as per the lock card data structure (Refer to the SD specification and the MMC specification)*

- SDIO Interrupt Control

Interrupts for SD cards are reported to the BIU by asserting an interrupt signal for two clock cycles. SDIO cards signal an interrupt by asserting cdata_in low during the interrupt period; an interrupt period for the selected card is determined by the interrupt control state machine. An interrupt period is always valid for non-active or non-selected cards, and 1-bit data mode for the selected card. An interrupt period for a wide-bus active or selected card is valid for the following conditions:

- Card is idle
- Non-data transfer command in progress
- Third clock after end bit of data block between two data blocks
- From two clocks after end bit of last data until end bit of next data transfer command

Bear in mind that, in the following situations, the controller does not sample the SDIO interrupt of the selected card when the card data width is 4 bits. Since the SDIO interrupt is level-triggered, it is sampled in a further interrupt period and the host does not lose any SDIO interrupt from the card.

- Read/Write Resume – The CIU treats the resume command as a normal data transfer command. SDIO interrupts during the resume command are handled similarly to other data commands. According to the SDIO specification, for the normal data command the interrupt period ends after the command end bit of the data command; for the resume command, it ends after the response end bit. In the case of the resume command, the Controller stops the interrupt sampling period after the resume command end bit, instead of stopping after the response end bit of the resume command.
- Suspend during read transfer – If the read data transfer is suspended by the host, the host sets the abort_read_data bit in the controller to reset the data state machine. In the CIU, the SDIO interrupts are handled such that the interrupt sampling starts after the abort_read_data bit is set by the host. In this case the controller does not sample SDIO interrupts between the period from response of the suspend command to setting the abort_read_data bit, and starts sampling after setting the abort_read_data bit.

- Clock Control

The clock control block provides different clock frequencies required for SD/MMC cards. The cclk_in signal is the source clock (cclk_in >= card max operating frequency) for clock divider of the clock control block. This source clock (cclk_in) is used to generate different card clock frequencies (cclk_out). The card clock can have different clock frequencies, since the card can be a low-speed card or a full-speed card. The Host Controller provides one clock signal (cclk_out).

The clock frequency of a card depends on the following clock control registers:

- Clock Divider register – Internal clock dividers are used to generate different clock frequencies required for card. The division factor for each clock divider can be

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programmed by writing to the Clock Divider register. The clock divider is an 8-bit value that provides a clock division factor from 1 to 510; a value of 0 represents a clock-divider bypass, a value of 1 represents a divide by 2, a value of 2 represents a divide by 4, and so on.

- Clock Control register – cclk_out can be enabled or disabled for each card under the following conditions:
- clk_enable – cclk_out for a card is enabled if the clk_enable bit for a card in the Clock Control register is programmed (set to 1) or disabled (set to 0).
- Low-power mode – Low-power mode of a card can be enabled by setting the low-power mode bit of the Clock Control register to 1. If low-power mode is enabled to save card power, the cclk_out is disabled when the card is idle for at least 8 card clock cycles. It is enabled when a new command is loaded and the command path goes to a non-idle state.

Additionally, cclk_out is disabled when an internal FIFO is full – card read (no more data can be received from card) – or when the FIFO is empty – card write (no data is available for transmission). This helps to avoid FIFO overrun and underrun conditions. It is used by the command and data path to qualify cclk_in for driving outputs and sampling inputs at the programmed clock frequency for the selected card, according to the Clock Divider and Clock Source register values.

Under the following conditions, the card clock is stopped or disabled, along with the active clk_en, for the selected card:

- Clock can be disabled by writing to Clock Enable register (clk_en bit = 1).
- If low-power mode is selected and card is idle, or not selected for 8 clocks.
- FIFO is full and data path cannot accept more data from the card and data transfer is incomplete –to avoid FIFO overrun.
- FIFO is empty and data path cannot transmit more data to the card and data transfer is incomplete – to avoid FIFO underrun.
- Error Detection
- Response
- Response timeout – Response expected with response start bit is not received within programmed number of clocks in timeout register.
- Response CRC error – Response is expected and check response CRC requested; response CRC7 does not match with the internally-generated CRC7.
- Response error – Response transmission bit is not 0, command index does not match with the command index of the send command, or response end bit is not 1.
- Data transmit
- No CRC status – During a write data transfer, if the CRC status start bit is not received two clocks after the end bit of the data block is sent out, the data path does the following:
- Signals no CRC status error to the BIU
- Terminates further data transfer
- Signals data transfer done to the BIU
- Negative CRC – If the CRC status received after the write data block is negative (that is, not 010), a data CRC error is signaled to the BIU and further data transfer is continued.
- Data starvation due to empty FIFO – If the FIFO becomes empty during a write data transmission, or if the card clock is stopped and the FIFO remains empty for data timeout clocks, then a data-starvation error is signaled to the BIU and the data path continues to wait for data in the FIFO.
- Data receive
- Data timeout – During a read-data transfer, if the data start bit is not received before the number of clocks that were programmed in the timeout register, the data path does the following:
- Signals data-timeout error to the BIU
- Terminates further data transfer
- Signals data transfer done to BIU
- Data start bit error – During a 4-bit or 8-bit read-data transfer, if the all-bit data line

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does not have a start bit, the data path signals a data start bit error to the BIU and waits for a data timeout, after which it signals that the data transfer is done.
- Data CRC error – During a read-data-block transfer, if the CRC16 received does not match with the internally generated CRC16, the data path signals a data CRC error to the BIU and continues further data transfer.
- Data end-bit error – During a read-data transfer, if the end bit of the received data is not 1, the data path signals an end-bit error to the BIU, terminates further data transfer, and signals to the BIU that the data transfer is done.
- Data starvation due to FIFO full – During a read data transmission and when the FIFO becomes full, the card clock is stopped. If the FIFO remains full for data timeout clocks, a data starvation error is signaled to the BIU (Data Starvation by Host Timeout bit is set in RINTSTS Register) and the data path continues to wait for the FIFO to start to empty.

18.3.3 Internal Direct Memory Access Controller (IDMAC)

The Internal Direct Memory Access Controller (IDMAC) has a Control and Status Register (CSR) and a single Transmit/Receive engine, which transfers data from host memory to the device port and vice versa. The controller utilizes a descriptor to efficiently move data from source to destination with minimal Host CPU intervention. You can program the controller to interrupt the Host CPU in situations such as data Transmit and Receive transfer completion from the card, as well as other normal or error conditions.
The IDMAC and the Host driver communicate through a single data structure. CSR addresses 0x80 to 0x98 are reserved for host programming.
The IDMAC transfers the data received from the card to the Data Buffer in the Host memory, and it transfers Transmit data from the Data Buffer in the Host memory to the FIFO. Descriptors that reside in the Host memory act as pointers to these buffers.
A data buffer resides in physical memory space of the Host and consists of complete data or partial data. Buffers contain only data, while buffer status is maintained in the descriptor. Data chaining refers to data that spans multiple data buffers. However, a single descriptor cannot span multiple data.
A single descriptor is used for both reception and transmission. The base address of the list is written into Descriptor List Base Address Register (DBADDR @0x88). A descriptor list is forward linked. The Last Descriptor can point back to the first entry in order to create a ring structure. The descriptor list resides in the physical memory address space of the Host. Each descriptor can point to a maximum of two data buffers.

1. IDMAC CSR Access
When an IDMAC is introduced, an additional CSR space resides in the IDMAC that controls the IDMAC functionality. The host accesses the new CSR space in addition to the existing control register set in the BIU. The IDMAC CSR primarily contains descriptor information. For a write operation to the CSR, the respective CSR logic of the IDMAC and BIU decodes the address before accepting. For a read operation from the CSR, the appropriate CSR read path is enabled.
You can enable or disable the IDMAC operation by programming bit[25] in the CTRL register of the BIU. This allows the data transfer by accessing the slave interface on the AMBA bus if the IDMAC is present but disabled. When IDMAC is enabled, the FIFO cannot be accessed through the slave interface.

2. Descriptors
- Descriptor structures

The IDMAC uses these types of descriptor structures:
  - Dual-Buffer Structure – The distance between two descriptors is determined by the Skip Length value programmed in the Descriptor Skip Length (DSL) field of the Bus Mode Register (BMOD @0x80).

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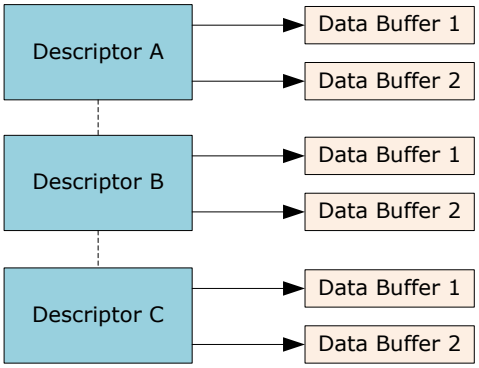

Fig. 18-6 Dual-Buffer Descriptor Structure

- Chain Structure – Each descriptor points to a unique buffer and the next descriptor.

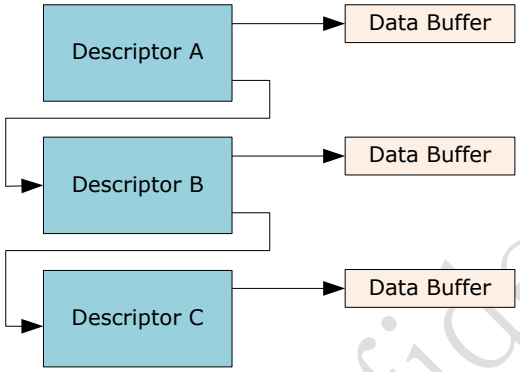

Fig. 18-7 Chain Descriptor Structure

- Descriptor formats

Following figure illustrates the internal formats of a descriptor. The descriptor addresses must be aligned to the bus width used for 32-bit AHB data buses. Each descriptor contains 16 bytes of control and status information. DES0 is a notation used to denote the [31:0] bits, DES1 to denote [63:32] bits, DES2 to denote [95:64] bits, DES3 to denote [127:96] bits.

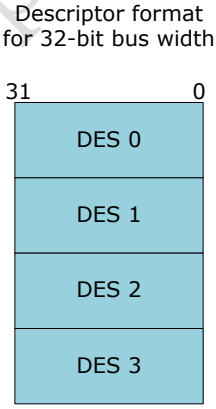

Fig. 18-8 Descriptor Formats for 32-bit AHB Address Bus Width

- The DES0 element in the IDMAC contains control and status information.

Table 18-4 Bits in IDMAC DES0 Element

Bits	Name	Description
31	OWN	When set, this bit indicates that the descriptor is owned by the IDMAC. When this bit is reset, it indicates that the descriptor is owned by the Host. The IDMAC clears this bit when it completes the data transfer.

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30	Card Error Summary (CES)	These error bits indicate the status of the transaction to or from the card. These bits are also present in RINTSTS Indicates the logical OR of the following bits: ● EBE: End Bit Error ● RTO: Response Time out ● RCRC: Response CRC ● SBE: Start Bit Error ● DRTO: Data Read Timeout ● DCRC: Data CRC for Receive ● RE: Response Error
29:6	Reserved	-
5	End of Ring (ER)	When set, this bit indicates that the descriptor list reached its final descriptor. The IDMAC returns to the base address of the list, creating a Descriptor Ring. This is meaningful for only a dual-buffer descriptor structure.
4	Second Address Chained (CH)	When set, this bit indicates that the second address in the descriptor is the Next Descriptor address rather than the second buffer address. When this bit is set, BS2 (DES1[25:13]) should be all zeros.
3	First Descriptor (FS)	When set, this bit indicates that this descriptor contains the first buffer of the data. If the size of the first buffer is 0, next Descriptor contains the beginning of the data.
2	Last Descriptor (LD)	This bit is associated with the last block of a DMA transfer. When set, the bit indicates that the buffers pointed to by this descriptor are the last buffers of the data. After this descriptor is completed, the remaining byte count is 0. In other words, after the descriptor with the LD bit set is completed, the remaining byte count should be 0.
1	Disable Interrupt on Completion (DIC)	When set, this bit will prevent the setting of the TI/RI bit of the IDMAC Status Register (IDSTS) for the data that ends in the buffer pointed to by this descriptor.
0	Reserved	-

- The DES1 element contains the buffer size.

Table 18-5 Bits in IDMAC DES1 Element

Bits	Name	Description
31:26	Reserved	-
25:13	Buffer 2 Size (BS2)	These bits indicate the second data buffer byte size. The buffer size must be a multiple of 2, 4, or 8, depending upon the bus widths—16, 32, and 64,respectively. In the case where the buffer size is not a multiple of 2, 4, or 8, the resulting behavior is undefined. If this field is 0, the DMA ignores this buffer and proceeds to the next buffer in case of a dual-buffer structure. This field is not valid for chain structure; that is, if DES0[4] is set.
12:0	Buffer 1 Size (BS1)	Indicates the data buffer byte size, which must be a multiple of 2, 4, or 8 bytes, depending upon the bus widths—16, 32, and 64,

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		respectively. In the case where the buffer size is not a multiple of 2, 4, or 8, the resulting behavior is undefined. This field should not be zero. Note: If there is only one buffer to be programmed, you need to use only the Buffer 1, and not Buffer 2.

-

- The DES2 element contains the address pointer to the data buffer.

Table 18-6 Bits in IDMAC DES2 Element

Bits	Name	Description
31:26	Reserved	
25:13	Buffer 2 Size (BS2)	These bits indicate the second data buffer byte size. The buffer size must be a multiple of 2, 4, or 8, depending upon the bus widths—16, 32, and 64,respectively. In the case where the buffer size is not a multiple of 2, 4, or 8, the resulting behavior is undefined. If this field is 0, the DMA ignores this buffer and proceeds to the next buffer in case of a dual-buffer structure. This field is not valid for chain structure; that is, if DES0[4] is set.
12:0	Buffer 1 Size (BS1)	Indicates the data buffer byte size, which must be a multiple of 2, 4, or 8 bytes, depending upon the bus widths—16, 32, and 64, respectively. In the case where the buffer size is not a multiple of 2, 4, or 8, the resulting behavior is undefined. This field should not be zero. Note: If there is only one buffer to be programmed, you need to use only the Buffer 1, and not Buffer 2.

- The DES3 element contains the address pointer to the next descriptor if the present descriptor is not the last descriptor in a chained descriptor structure or the second buffer address for a dual-buffer structure.

Table 18-7 Bits in IDMAC DES3 Element

Bits	Name	Description
31:0	Buffer Address Pointer 2/ Next Descriptor Address (BAP2)	These bits indicate the physical address of the second buffer when the dual-buffer structure is used. If the Second Address Chained (DES0[4]) bit is set, then this address contains the pointer to the physical memory where the Next Descriptor is present. If this is not the last descriptor, then the Next Descriptor address pointer must be bus-width aligned.

3. Initialization

IDMAC initialization occurs as follows:

- Write to IDMAC Bus Mode Register—BMOD to set Host bus access parameters.
- Write to IDMAC Interrupt Enable Register—IDINTEN to mask unnecessary interrupt causes.
- The software driver creates either the Transmit or the Receive descriptor list. Then it writes to IDMAC Descriptor List Base Address Register (DBADDR), providing the IDMAC with the starting address of the list.
- The IDMAC engine attempts to acquire descriptors from the descriptor lists.

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- Host Bus Burst Access

The IDMAC attempts to execute fixed-length burst transfers on the AHB Master interface if configured using the FB bit of the IDMAC Bus Mode register. The maximum burst length is indicated and limited by the PBL field. The descriptors are always accessed in the maximum possible burst-size for the 16-bytes to be read— 16*8/bus-width.
The IDMAC initiates a data transfer only when sufficient space to accommodate the configured burst is available in the FIFO or the number of bytes to the end of data, when less than the configured burst-length.
The IDMAC indicates the start address and the number of transfers required to the AHB Master Interface. When the AHB Interface is configured for fixed-length bursts, then it transfers data using the best combination of INCR4/8/16 and SINGLE transactions. Otherwise, in no fixed-length bursts, it transfers data using INCR (undefined length) and SINGLE transactions.

- Host Data Buffer Alignment

The Transmit and Receive data buffers in host memory must be aligned, depending on the data width.

- Buffer Size Calculations

The driver knows the amount of data to transmit or receive. For transmitting to the card, the IDMAC transfers the exact number of bytes to the FIFO, indicated by the buffer size field of DES1.
If a descriptor is not marked as last-LS bit of DES0-then the corresponding buffer(s) of the descriptor are full, and the amount of valid data in a buffer is accurately indicated by its buffer size field. If a descriptor is marked as last, then the buffer cannot be full, as indicated by the buffer size in DES1. The driver is aware of the number of locations that are valid in this case.

- Transmission

IDMAC transmission occurs as follows:

- The Host sets up the elements (DES0-DES3) for transmission and sets the OWN bit (DES0[31]). The Host also prepares the data buffer.
- The Host programs the write data command in the CMD register in BIU.
- The Host will also program the required transmit threshold level (TX_WMark field in FIFOTH register).
- The IDMAC determines that a write data transfer needs to be done as a consequence of step 2.
- The IDMAC engine fetches the descriptor and checks the OWN bit. If the OWN bit is not set, it means that the host owns the descriptor. In this case the IDMAC enters suspend state and asserts the Descriptor Unable interrupt in the IDSTS register. In such a case, the host needs to release the IDMAC by writing any value to the poll demand register.
- It will then wait for Command Done (CD) bit and no errors from BIU which indicates that a transfer can be done.
- The IDMAC engine will now wait for a DMA interface request from BIU. This request will be generated based on the programmed transmit threshold value. For the last bytes of data which can't be accessed using a burst, SINGLE transfers are performed on AHB Master Interface.
- The IDMAC fetches the Transmit data from the data buffer in the Host memory and transfers to the FIFO for transmission to card.
- When data spans across multiple descriptors, the IDMAC will fetch the next descriptor and continue with its operation with the next descriptor. The Last Descriptor bit in the descriptor indicates whether the data spans multiple descriptors or not.
- When data transmission is complete, status information is updated in IDSTS register by setting Transmit Interrupt, if enabled. Also, the OWN bit is cleared by the IDMAC by performing a write transaction to DES0.

- Reception

IDMAC reception occurs as follows:

- The Host sets up the element (DES0-DES3) for reception, sets the OWN (DES0[31]).
- The Host programs the read data command in the CMD register in BIU.
- The Host will program the required receive threshold level (RX_WMark field in FIFOTH register).

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- The IDMAC determines that a read data transfer needs to be done as a consequence of step 2.
- The IDMAC engine fetches the descriptor and checks the OWN bit. If the OWN bit is not set, it means that the host owns the descriptor. In this case the DMA enters suspend state and asserts the Descriptor Unable interrupt in the IDSTS register. In such a case, the host needs to release the IDMAC by writing any value to the poll demand register.
- It will then wait for Command Done (CD) bit and no errors from BIU which indicates that a transfer can be done.
- The IDMAC engine will now wait for a DMA interface request from BIU. This request will be generated based on the programmed receive threshold value. For the last bytes of data which can't be accessed using a burst, SINGLE transfers are performed on AHB.
- The IDMAC fetches the data from the FIFO and transfer to Host memory.
- When data spans across multiple descriptors, the IDMAC will fetch the next descriptor and continue with its operation with the next descriptor. The Last Descriptor bit in the descriptor indicates whether the data spans multiple descriptors or not.
- When data reception is complete, status information is updated in IDSTS register by setting Receive Interrupt, if enabled. Also, the OWN bit is cleared by the IDMAC by performing a write transaction to DES0.

- Interrupts

Interrupts can be generated as a result of various events. IDSTS register contains all the bits that might cause an interrupt. IDINTEN register contains an Enable bit for each of the events that can cause an interrupt.
There are two groups of summary interrupts-Normal and Abnormal-as outlined in IDSTS register. Interrupts are cleared by writing a 1 to the corresponding bit position. When all the enabled interrupts within a group are cleared, the corresponding summary bit is cleared. When both the summary bits are cleared, the interrupt signal dmac_intr_o is de-asserted. Interrupts are not queued and if the interrupt event occurs before the driver has responded to it, no additional interrupts are generated. For example, Receive Interrupt—IDSTS[1] indicates that one or more data was transferred to the Host buffer.
An interrupt is generated only once for simultaneous, multiple events. The driver must scan IDSTS register for the interrupt cause.

18.4 Register Description

18.4.1 Register Summary

Name	Offset	Size	Reset Value	Description
SDMMC_CTRL	0x0000	W	0x01000000	Control register
SDMMC_PWREN	0x0004	W	0x00000000	Power-enable register
SDMMC_CLKDIV	0x0008	W	0x00000000	Clock-divider register
SDMMC_CLKSRC	0x000c	W	0x00000000	SD Clock Source Register
SDMMC_CLKENA	0x0010	W	0x00000000	Clock-enable register
SDMMC_TMOUT	0x0014	W	0xffffff40	Time-out register
SDMMC_CTYPE	0x0018	W	0x00000000	Card-type register
SDMMC_BLKSIZ	0x001c	W	0x00000200	Block-size register
SDMMC_BYTCNT	0x0020	W	0x00000200	Byte-count register
SDMMC_INTMASK	0x0024	W	0x00000000	Interrupt-mask register
SDMMC_CMDARG	0x0028	W	0x00000000	Command-argument register
SDMMC_CMD	0x002c	W	0x00000000	Command register
SDMMC_RESP0	0x0030	W	0x00000000	Response-0 register
SDMMC_RESP1	0x0034	W	0x00000000	Response-1 register

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Name	Offset	Size	Reset Value	Description
SDMMC_RESP2	0x0038	W	0x00000000	Response-2 register
SDMMC_RESP3	0x003c	W	0x00000000	Response-3 register
SDMMC_MINTSTS	0x0040	W	0x00000000	Masked interrupt-status registe
SDMMC_RINTSTS	0x0044	W	0x00000000	Raw interrupt-status register
SDMMC_STATUS	0x0048	W	0x00000406	Status register
SDMMC_FIFOTH	0x004c	W	0x00000000	FIFO threshold register
SDMMC_CDETECT	0x0050	W	0x00000000	Card-detect register
SDMMC_WRTPRT	0x0054	W	0x00000000	Write-protect register
SDMMC_TCBCNT	0x005c	W	0x00000000	Transferred CIU card byte count
SDMMC_TBBCNT	0x0060	W	0x00000000	Transferred host/DMA to/from BIU-FIFO byte count
SDMMC_DEBNCE	0x0064	W	0x00ffffff	Card detect debounce register
SDMMC_USRID	0x0068	W	0x07967797	User ID register
SDMMC_VERID	0x006c	W	0x5342270a	Synopsys version ID register
SDMMC_HCON	0x0070	W	0x00000000	Hardware Configuration Register
SDMMC_UHS_REG	0x0074	W	0x00000000	UHS-1 register
SDMMC_RST_N	0x0078	W	0x00000001	Hardware reset register
SDMMC_BMOD	0x0080	W	0x00000000	Bus Mode Register
SDMMC_PLDMND	0x0084	W	0x00000000	Poll Demand Register
SDMMC_DBADDR	0x0088	W	0x00000000	Descriptor List Base Address Register
SDMMC_IDSTS	0x008c	W	0x00000000	Internal DMAC Status Register
SDMMC_IDINTEN	0x0090	W	0x00000000	Internal DMAC Interrupt Enable Register
SDMMC_DSCADDR	0x0094	W	0x00000000	Current Host Descriptor Address Register
SDMMC_BUFADDR	0x0098	W	0x00000000	Current Buffer Descriptor Address Register
SDMMC_CARDTHRCTL	0x0100	W	0x00000000	Card read threshold enable
SDMMC_BACK_END_POWER	0x0104	W	0x00000000	Back-end power
SDMMC_UHS_REG_EXT	0x0108	W	0x00000000	UHS Register
SDMMC_EMMC_DDR_REG	0x010c	W	0x00000000	eMMC 4.5 DDR START Bit Detection Control Register
SDMMC_ENABLE_SHIFT	0x0110	W	0x00000000	Enable Phase Shift Registe
SDMMC_FIFO_BASE	0x0200	W	0x00000000	FIFO Base Address

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

18.4.2 Detail Register Description

SDMMC_CTRL

Address: Operational Base + offset (0x0000)

Control register

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
25	RW	0x0	use_internal_dmac Present only for the Internal DMAC configuration; else, it is reserved. 0: The host performs data transfers through the slave interface 1: Internal DMAC used for data transfe
24	RW	0x1	enable_OD_pullup External open-drain pullup: 0: Disable 1: Enable Inverted value of this bit is output to ccmd_od_pullup_en_n port. When bit is set, command output always driven in open-drive mode; that is, DWC_mobile_storage drives either 0 or high impedance, and does not drive hard 1.
23:20	RW	0x0	Card_voltage_b Card regulator-B voltage setting; output to card_volt_b port. Optional feature; ports can be used as general-purpose outputs.
19:16	RW	0x0	Card_voltage_a Card regulator-A voltage setting; output to card_volt_a port. Optional feature; ports can be used as general-purpose outputs.
15:12	RO	0x0	reserved
11	RW	0x0	ceata_device_interrupt_status 0: Interrupts not enabled in CE-ATA device (nIEN = 1 in ATA control register) 1: Interrupts are enabled in CE-ATA device (nIEN = 0 in ATA control register) Software should appropriately write to this bit after power-on reset or any other reset to CE-ATA device. After reset, usually CE-ATA device interrupt is disabled (nIEN = 1). If the host enables CE-ATA device interrupt, then software should set this bit.

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Bit	Attr	Reset Value	Description
10	RW	0x0	send_auto_stop_ccsd 0: Clear bit if DWC_mobile_storage does not reset the bit. 1: Send internally generated STOP after sending CCSD to CE-ATA device. NOTE: Always set send_auto_stop_ccsd and send_ccsd bits together send_auto_stop_ccsd should not be set independent of send_ccsd. When set, DWC_Mobile_Storage automatically sends internally- generated STOP command (CMD12) to CE-ATA device. After sending internally-generated STOP command, Auto Command Done (ACD) biin RINTSTS is set and generates interrupt to host if Auto Command Done interrupt is not masked. After sending the CCSD, DWC_mobile_storage automatically clears send_auto_stop_ccsd bit.
9	RW	0x0	send_ccsd 0: Clear bit if DWC_mobile_storage does not reset the bit. 1: Send Command Completion Signal Disable (CCSD) to CE-ATA device When set, DWC_mobile_storage sends CCSD to CE-ATA device. Software sets this bit only if current command is expecting CCS (that is, RW_BLK) and interrupts are enabled in CE-ATA device.Once the CCSD pattern is sent to device, DWC_mobile_storage automatically clears send_ccsd bit. It also sets Command Done (CD) bit in RINTSTS register and generates interrupt to host if Command Done interrupt is not masked. NOTE: Once send_ccsd bit is set, it takes two card clock cycles to drive the CCSD on the CMD line. Due to this, during the boundary conditions it may happen that CCSD is sent to the CE-ATA device, even if the device signalled CCS
8	RW	0x0	abort_read_data 0: no change 1: after suspend command is issued during read-transfer, software polls card to find when suspend happened. Once suspend occurs, software sets bit to reset data state-machine, which is waiting for next block of data. Bit automatically clears once data state machine resets to idle. Used in SDIO card suspend sequence.

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Bit	Attr	Reset Value	Description
7	RW	0x0	send_irq_response 0: no change 1: send auto IRQ response Bit automatically clears once response is sent. To wait for MMC card interrupts, host issues CMD40, and SDMMC Controller waits for interrupt response from MMC card(s). In meantime, if host wants SDMMC Controller to exit waiting for interrupt state, it can set this bit, at which time SDMMC Controller command state-machine sends CMD40 response on bus and returns to idle state.
6	RW	0x0	read_wait 0: clear read wait 1: assert read wait For sending read-wait to SDIO cards
5	RW	0x0	dma_enable 0: disable DMA transfer mode 1: enable DMA transfer mode Even when DMA mode is enabled, host can still push/pop data into or from FIFO; this should not happen during the normal operation. If there is simultaneous FIFO access from host/DMA, the data coherency is lost. Also, there is no arbitration inside SDMMC Controller to prioritize simultaneous host/DMA access.
4	RW	0x0	int_enable Global interrupt enable/disable bit: 0: disable interrupts 1: enable interrupts The int port is 1 only when this bit is 1 and one or more unmasked interrupts are set.
3	RO	0x0	reserved
2	W1C	0x0	dma_reset 0: no change 1: reset internal DMA interface control logic To reset DMA interface, firmware should set bit to 1. This bit is auto-cleared after two AHB clocks. The DMA should be reset before it switches from no-dma mode into dma mode.
1	W1C	0x0	fifo_reset 0: no change 1: reset to data FIFO To reset FIFO pointers To reset FIFO, firmware should set bit to 1. This bit is auto-cleared after completion of reset operation

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Bit	Attr	Reset Value	Description
0	W1C	0x0	controller_reset 0: no change 1: reset SDMMC controller To reset controller, firmware should set bit to 1. This bit is auto-cleared after two AHB and two cclk_in clock cycles. This resets: * BIU/CIU interface * CIU and state machines * abort_read_data, send_irq_response, and read_wait bits of Control register * start_cmd bit of Command register Does not affect any registers or DMA interface, or FIFO or host interrupts

SDMMC_PWREN
Address: Operational Base + offset (0x0004)
Power-enable register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	power_enable Power on/off switch for the card. Once power is turned on, firmware should wait for regulator/switch ramp-up time before trying to initialize card. 0: power off 1: power on Bit values output to card_power_en port.

SDMMC_CLKDIV
Address: Operational Base + offset (0x0008)
Clock-divider register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	clk_divider3 Clock divider-3 value. For example, value of 0 means divide by 2*0 = 0 (no division, bypass), value of 1 means divide by 2*1 = 2, value of "ff"means divide by 2*255 = 510, and so on In MMC-Ver3.3-only mode, bits not implemented because only one clock divider is supported. In our design, the divider is 0 or 1 supported.

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Bit	Attr	Reset Value	Description
23:16	RW	0x00	clk_divider2 Clock divider-2 value. Clock division is 2*n. For example, value of 0 means divide by 2*0 = 0 (no division, bypass), value of 1 means divide by 2*1 = 2, value of "ff"means divide by 2*255 = 510, and so on In MMC-Ver3.3-only mode, bits not implemented because only one clock divider is supported. In our design, the divider is 0 or 1 supported.
15:8	RW	0x00	clk_divider1 Clock divider-1 value. Clock division is 2*n. For example, value of 0 means divide by 2*0 = 0 (no division, bypass), value of 1 means divide by 2*1 = 2, value of "ff"means divide by 2*255 = 510, and so on In MMC-Ver3.3-only mode, bits not implemented because only one clock divider is supported. In our design, the divider is 0 or 1 supported.
7:0	RW	0x00	clk_divider0 Clock divider-0 value. Clock division is 2*n. For example, value of 0 means divide by 2*0 = 0 (no division, bypass), value of 1 means divide by 2*1 = 2, value of "ff"means divide by 2*255 = 510, and so on. In our design, the divider is 0 or 1 supported.

SDMMC_CLKSRC
Address: Operational Base + offset (0x000c)
SD Clock Source Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	clk_source Clock divider source for up to 16 SD cards supported. Each card has two bits assigned to it. For example, bits[1:0] assigned for card-0, which maps and internally routes clock divider[3:0] outputs to cclk_out[15:0] pins, depending on bit value. 00 –Clock divider 0 01 –Clock divider 1 10 –Clock divider 2 11 –Clock divider 3 In our design, the cclk_out is always from clock divider 0, and this register is not implemented.

SDMMC_CLKENA
Address: Operational Base + offset (0x0010)
Clock-enable register

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RW	0x0	cclk_low_power Low-power control for SD card clock and MMC card clock supported. 0: non-low-power mode 1: low-power mode; stop clock when card in IDLE (should be normally set to only MMC and SD memory cards; for SDIO cards, if interrupts must be detected, clock should not be stopped).
15:1	RO	0x0	reserved
0	RW	0x0	cclk_enable Clock-enable control for SD card clock and MMC card clock supported. 0: clock disabled 1: clock enabled

SDMMC_TMOUT
Address: Operational Base + offset (0x0014)
Time-out register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:8	RW	0xffffff	data_timeout Value for card Data Read Timeout; same value also used for Data Starvation by Host timeout. Value is in number of card output clocks cclk_out of selected card. Note: The software timer should be used if the timeout value is in the order of 100 ms. In this case, read data timeout interrupt needs to be disabled.
7:0	RW	0x40	response_timeout Response timeout value. Value is in number of card output clocks –cclk_out.

SDMMC_CTYPE
Address: Operational Base + offset (0x0018)
Card-type register

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RW	0x0	card_width_8 Indicates if card is 8-bit: 0: non 8-bit mode 1: 8-bit mode
15:1	RO	0x0	reserved
0	RW	0x0	card_width Indicates if card is 1-bit or 4-bit: 0: 1-bit mode 1: 4-bit mode

SDMMC_BLKSIZ
Address: Operational Base + offset (0x001c)
Block-size register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0200	block_size Block size

SDMMC_BYTCNT
Address: Operational Base + offset (0x0020)
Byte-count register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000200	byte_count Number of bytes to be transferred; should be integer multiple of Block Size for block transfers. For undefined number of byte transfers, byte count should be set to 0. When byte count is set to 0, it is responsibility of host to explicitly send stop/abort command to terminate data transfer.

SDMMC_INTMASK
Address: Operational Base + offset (0x0024)
Interrupt-mask register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	sdio_int_mask Mask SDIO interrupts. When masked, SDIO interrupt detection for that card is disabled. A 0 masks an interrupt, and 1 enables an interrupt.
23:17	RO	0x0	reserved
16	RW	0x0	data_nobusy_int_mask 0: data no busy interrupt not masked 1: data no busy interrupt masked

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	int_mask Bits used to mask unwanted interrupts. Value of 0 masks interrupt; value of 1 enables interrupt. bit 15: End-bit error (read)/Write no CRC (EBE) bit 14: Auto command done (ACD) bit 13: Start-bit error (SBE) bit 12: Hardware locked write error (HLE) bit 11: FIFO underrun/overrun error (FRUN) bit 10: Data starvation-by-host timeout (HTO) /Volt_switch_int bit 9: Data read timeout (DRTO) bit 8: Response timeout (RTO) bit 7: Data CRC error (DCRC) bit 6: Response CRC error (RCRC) bit 5: Receive FIFO data request (RXDR) bit 4: Transmit FIFO data request (TXDR) bit 3: Data transfer over (DTO) bit 2: Command done (CD) bit 1: Response error (RE) bit 0: Card detect (CD)

SDMMC_CMDARG
Address: Operational Base + offset (0x0028)
Command-argument register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cmd_arg Value indicates command argument to be passed to card.

SDMMC_CMD
Address: Operational Base + offset (0x002c)
Command register

Bit	Attr	Reset Value	Description
31	RW	0x0	start_cmd Start command. Once command is taken by CIU, bit is cleared. When bit is set, host should not attempt to write to any command registers. If write is attempted, hardware lock error is set in raw interrupt register. Once command is sent and response is received from SD_MMC cards, Command Done bit is set in raw interrupt register.

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Bit	Attr	Reset Value	Description
30	RO	0x0	reserved
29	RW	0x0	use_hold_reg Use Hold Register 0: CMD and DATA sent to card bypassing HOLD Register 1: CMD and DATA sent to card through the HOLD Register Note: a. Set to 1'b1 for SDR12 and SDR25 (with non-zero phase-shifted cclk_in_drv); zero phase shift is not allowed in these modes. b. Set to 1'b0 for SDR50, SDR104, and DDR50 (with zero phase-shifted cclk_in_drv) c. Set to 1'b1 for SDR50, SDR104, and DDR50 (with non-zero phase-shifted cclk_in_drv)
28	RW	0x0	volt_switch Voltage switch bit. 0: no voltage switching 1: voltage switching enabled; must be set for CMD11 only
27	RW	0x0	boot_mode Boot Mode. 0: mandatory Boot operation 1: alternate Boot operation
26	RW	0x0	disable_boot Disable Boot. When software sets this bit along with start_cmd, CIU terminates the boot operation. Do NOT set disable_boot and enable_boot together.
25	RW	0x0	expect_boot_ack Expect Boot Acknowledge. When Software sets this bit along with enable_boot, CIU expects a boot acknowledge start pattern of 0-1-0 from the selected card.
24	RW	0x0	enable_boot Enable Boot—this bit should be set only for mandatory boot mode.When Software sets this bit along with start_cmd, CIU starts the boot sequence for the corresponding card by asserting the CMD line low. Do NOT set disable_boot and enable_boot together.

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Bit	Attr	Reset Value	Description
23	RW	0x0	ccs_expected 0: Interrupts are not enabled in CE-ATA device (nIEN = 1 in ATA control register), or command does not expect CCS from device 1: Interrupts are enabled in CE-ATA device (nIEN = 0), and RW_BLK command expects command completion signal from CE-ATA device If the command expects Command Completion Signal (CCS) from the CE-ATA device, the software should set this control bit. DWC_mobile_storage sets Data Transfer Over (DTO) bit in RINTSTS register and generates interrupt to host if Data Transfer Over interrupt is not masked.
22	RW	0x0	read_ceata_device 0: Host is not performing read access (RW_REG or RW_BLK) towards CE-ATA device 1: Host is performing read access (RW_REG or RW_BLK) towards CE-ATA device Software should set this bit to indicate that CE-ATA device is being accessed for read transfer. This bit is used to disable read data timeout indication while performing CE-ATA read transfers. Maximum value of I/O transmission delay can be no less than 10 seconds. DWC_mobile_storage should not indicate read data timeout while waiting for data from CE-ATA device.

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Bit	Attr	Reset Value	Description
21	RW	0x0	update_clock_registers_only 0: normal command sequence 1: do not send commands, just update clock register value into card clock domain Following register values transferred into card clock domain: CLKDIV, CLRSRC, CLKENA. Changes card clocks (change frequency, truncate off or on, and set low-frequency mode); provided in order to change clock frequency or stop clock without having to send command to cards. During normal command sequence, when update_clock_registers_only = 0, following control registers are transferred from BIU to CIU: CMD, CMDARG, TMOUT, CTYPE, BLKSIZ, BYTCNT. CIU uses new register values for new command sequence to card. When bit is set, there are no Command Done interrupts because no command is sent to SD_MMC_CEATA cards.
20:16	RW	0x00	card_number Card number in use. Represents physical slot number of card being accessed. In MMC-Ver3.3-only mode, up to 30 cards are supported; in SD-only mode, up to 16 cards are supported. Registered version of this is reflected on dw_dma_card_num and ge_dma_card_num ports, which can be used to create separate DMA requests, if needed. In addition, in SD mode this is used to mux or demux signals from selected card because each card is interfaced to DWC_mobile_storage by separate bus.

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Bit	Attr	Reset Value	Description
15	RW	0x0	send_initialization 0: do not send initialization sequence (80 clocks of 1) before sending this command 1: send initialization sequence before sending this command After power on, 80 clocks must be sent to card for initialization before sending any commands to card. Bit should be set while sending first command to card so that controller will initialize clocks before sending command to card. This bit should not be set for either of the boot modes (alternate or mandatory).
14	RW	0x0	stop_abort_cmd 0: neither stop nor abort command to stop current data transfer in progress. If abort is sent to function-number currently selected or not in data-transfer mode, then bit should be set to 0. 1: stop or abort command intended to stop current data transfer in progress. When open-ended or predefined data transfer is in progress, and host issues stop or abort command to stop data transfer, bit should be set so that command/data state-machines of CIU can return correctly to idle state. This is also applicable for Boot mode transfers. To Abort boot mode, this bit should be set along with CMD[26] = disable_boot.
13	RW	0x0	wait_prvdata_complete 0: send command at once, even if previous data transfer has not completed 1: wait for previous data transfer completion before sending command The wait_prvdata_complete = 0 option typically used to query status of card during data transfer or to stop current data transfer; card_number should be same as in previous command.

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Bit	Attr	Reset Value	Description
12	RW	0x0	send_auto_stop 0: no stop command sent at end of data transfer 1: send stop command at end of data transfer When set, SDMMC Controller sends stop command to SD_MMC cards at end of data transfer. * when send_auto_stop bit should be set, since some data transfers do not need explicit stop commands * open-ended transfers that software should explicitly send to stop command Additionally, when "resume" is sent to resume –suspended memory access of SD-Combo card –bit should be set correctly if suspended data transfer needs send_auto_stop. Don't care if no data expected from card.
11	RW	0x0	transfer_mode 0: block data transfer command 1: stream data transfer command Don't care if no data expected.
10	RW	0x0	wr 0: read from card 1: write to card Don't care if no data expected from card.
9	RW	0x0	data_expected 0: no data transfer expected (read/write) 1: data transfer expected (read/write)
8	RW	0x0	check_response_crc 0: do not check response CRC 1: check response CRC Some of command responses do not return valid CRC bits. Software should disable CRC checks for those commands in order to disable CRC checking by controller
7	RW	0x0	response_length 0: short response expected from card 1: long response expected from card
6	RW	0x0	response_expect 0: no response expected from card 1: response expected from card
5:0	RW	0x00	cmd_index Command index

SDMMC_RESP0

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Address: Operational Base + offset (0x0030)
Response-0 register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	response0 Bit[31:0] of response

SDMMC_RESP1
Address: Operational Base + offset (0x0034)
Response-1 register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	response Register represents bit[63:32] of long response. When CIU sends auto-stop command, then response is saved in register. Response for previous command sent by host is still preserved in Response 0 register. Additional auto-stop issued only for data transfer commands, and response type is always "short" for them.

SDMMC_RESP2
Address: Operational Base + offset (0x0038)
Response-2 register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	response2 Bit[95:64] of long response

SDMMC_RESP3
Address: Operational Base + offset (0x003c)
Response-3 register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	response3 Bit[127:96] of long response

SDMMC_MINTSTS
Address: Operational Base + offset (0x0040)
Masked interrupt-status registe

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
24	RO	0x0	sdio_interrupt Interrupt from SDIO card; SDIO interrupt for card enabled only if corresponding sdio_int_mask bit is set in Interrupt mask register (mask bit 1 enables interrupt; 0 masks interrupt). 0: no SDIO interrupt from card 1: SDIO interrupt from card
23:17	RO	0x0	reserved
16	RW	0x0	data_nobusy_int_status Data no busy Interrupt Status
15:0	RO	0x0000	int_status Interrupt enabled only if corresponding bit in interrupt mask register is set. bit 15: End-bit error (read)/Write no CRC (EBE) bit 14: Auto command done (ACD) bit 13: Start-bit error (SBE) bit 12: Hardware locked write error (HLE) bit 11: FIFO underrun/overrun error (FRUN) bit 10: Data starvation-by-host timeout (HTO) /Volt_switch_int bit 9: Data read timeout (DRTO) bit 8: Response timeout (RTO) bit 7: Data CRC error (DCRC) bit 6: Response CRC error (RCRC) bit 5: Receive FIFO data request (RXDR) bit 4: Transmit FIFO data request (TXDR) bit 3: Data transfer over (DTO) bit 2: Command done (CD) bit 1: Response error (RE) bit 0: Card detect (CD)

SDMMC_RINTSTS

Address: Operational Base + offset (0x0044)

Raw interrupt-status register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RO	0x0	sdio_interrupt Interrupt from SDIO card; Writes to these bits clear them. Value of 1 clears bit and 0 leaves bit intact. 0: no SDIO interrupt from card 1: SDIO interrupt from card
23:17	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
16	RW	0x0	data_nobusy_int_status Data no busy interrupt status
15:0	RO	0x0000	int_status Writes to bits clear status bit. Value of 1 clears status bit, and value of 0 leaves bit intact. Bits are logged regardless of interrupt mask status. bit 15: End-bit error (read)/Write no CRC (EBE) bit 14: Auto command done (ACD) bit 13: Start-bit error (SBE) bit 12: Hardware locked write error (HLE) bit 11: FIFO underrun/overrun error (FRUN) bit 10: Data starvation-by-host timeout (HTO) /Volt_switch_int bit 9: Data read timeout (DRTO) bit 8: Response timeout (RTO) bit 7: Data CRC error (DCRC) bit 6: Response CRC error (RCRC) bit 5: Receive FIFO data request (RXDR) bit 4: Transmit FIFO data request (TXDR) bit 3: Data transfer over (DTO) bit 2: Command done (CD) bit 1: Response error (RE) bit 0: Card detect (CD)

SDMMC_STATUS
Address: Operational Base + offset (0x0048)
Status register

Bit	Attr	Reset Value	Description
31	RO	0x0	dma_req DMA request signal state
30	RO	0x0	dma_ack DMA acknowledge signal state
29:17	RO	0x0000	fifo_count Number of filled locations in FIFO
16:11	RO	0x00	response_index Index of previous response, including any auto-stop sent by core
10	RO	0x1	data_state_mc_busy Data transmit or receive state-machine is busy

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Bit	Attr	Reset Value	Description
9	RO	0x0	data_busy Inverted version of raw selected card_data[0] 0: card data not busy 1: card data busy default value is 1 or 0 depending on cdata_in
8	RO	0x0	data_3_status Raw selected card_data[3]; checks whether card is present 0: card not present 1: card present default value is 1 or 0 depending on cdata_in
7:4	RO	0x0	command_fsm_states Command FSM states: 0: idle 1: send init sequence 2: Tx cmd start bit 3: Tx cmd tx bit 4: Tx cmd index + arg 5: Tx cmd crc7 6: Tx cmd end bit 7: Rx resp start bit 8: Rx resp IRQ response 9: Rx resp tx bit 10: Rx resp cmd idx 11: Rx resp data 12: Rx resp crc7 13: Rx resp end bit 14: Cmd path wait NCC 15: Wait; CMD-to-response turnaround The command FSM state is represented using 19 bits. The STATUS Register[7:4] has 4 bits to represent the command FSM states. Using these 4 bits, only 16 states can be represented. Thus three states cannot be represented in the STATUS[7:4] register. The three states that are not represented in the STATUS Register[7:4] are: * Bit 16 –Wait for CCS * Bit 17 –Send CCSD * Bit 18 –Boot Mode Due to this, while command FSM is in "Wait for CCS state"or "Send CCSD"or "Boot Mode", the Status register indicates status as 0 for the bit field [7:4].

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Bit	Attr	Reset Value	Description
3	RO	0x0	fifo_full FIFO is full status
2	RO	0x1	fifo_empty FIFO is empty status
1	RO	0x1	fifo_tx_watermark FIFO reached Transmit watermark level; not qualified with data transfer
0	RO	0x0	fifo_rx_watermark FIFO reached Receive watermark level; not qualified with data transfer

SDMMC_FIFOTH

Address: Operational Base + offset (0x004c)

FIFO threshold register

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
30:28	RW	0x0	dma_mutiple_transaction_size Burst size of multiple transaction; should be programmed same as DMA controller multiple-transaction-size SRC/DEST_MSIZE. 3'b000: 1 transfers 3'b001: 4 3'b010: 8 3'b011: 16 3'b100: 32 3'b101: 64 3'b110: 128 3'b111: 256 The unit for transfer is the H_DATA_WIDTH parameter. A single transfer (dw_dma_single assertion in case of Non DW DMA interface) would be signalled based on this value. Value should be sub-multiple of (RX_WMark + 1)* (F_DATA_WIDTH/H_DATA_WIDTH) and (FIFO_DEPTH - TX_WMark)* (F_DATA_WIDTH/ H_DATA_WIDTH) For example, if FIFO_DEPTH = 16, FDATA_WIDTH == H_DATA_WIDTH Allowed combinations for MSize and TX_WMark are: MSize = 1, TX_WMARK = 1-15 MSize = 4, TX_WMark = 8 MSize = 4, TX_WMark = 4 MSize = 4, TX_WMark = 12 MSize = 8, TX_WMark = 8 MSize = 8, TX_WMark = 4 Allowed combinations for MSize and RX_WMark are: MSize = 1, RX_WMARK = 0-14 MSize = 4, RX_WMark = 3 MSize = 4, RX_WMark = 7 MSize = 4, RX_WMark = 11 MSize = 8, RX_WMark = 7 Recommended: MSize = 8, TX_WMark = 8, RX_WMark = 7

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Bit	Attr	Reset Value	Description
27:16	RW	0x000	rx_wmark FIFO threshold watermark level when receiving data to card. When FIFO data count reaches greater than this number, DMA/FIFO request is raised. During end of packet, request is generated regardless of threshold programming in order to complete any remaining data. In non-DMA mode, when receiver FIFO threshold (RXDR) interrupt is enabled, then interrupt is generated instead of DMA request. During end of packet, interrupt is not generated if threshold programming is larger than any remaining data. It is responsibility of host to read remaining bytes on seeing Data Transfer Done interrupt. In DMA mode, at end of packet, even if remaining bytes are less than threshold, DMA request does single transfers to flush out any remaining bytes before Data Transfer Done interrupt is set. 12 bits-1 bit less than FIFO-count of status register, which is 13 bits. Limitation: RX_WMark <= FIFO_DEPTH-2 Recommended: (FIFO_DEPTH/2) - 1; (means greater than (FIFO_DEPTH/2) - 1) NOTE: In DMA mode during CCS time-out, the DMA does not generate the request at the end of packet, even if remaining bytes are less than threshold. In this case, there will be some data left in the FIFO. It is the responsibility of the application to reset the FIFO after the CCS timeout.
15:12	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
11:0	RW	0x000	tx_wmark FIFO threshold watermark level when transmitting data to card. When FIFO data count is less than or equal to this number, DMA/FIFO request is raised. If Interrupt is enabled, then interrupt occurs. During end of packet, request or interrupt is generated, regardless of threshold programming. In non-DMA mode, when transmit FIFO threshold (TXDR) interrupt is enabled, then interrupt is generated instead of DMA request. During end of packet, on last interrupt, host is responsible for filling FIFO with only required remaining bytes (not before FIFO is full or after CIU completes data transfers, because FIFO may not be empty). In DMA mode, at end of packet, if last transfer is less than burst size, DMA controller does single cycles until required bytes are transferred. 12 bits -1 bit less than FIFO-count of status register, which is 13 bits. Limitation: TX_WMark >= 1; Recommended: FIFO_DEPTH/2; (means less than or equal to FIFO_DEPTH/2)

SDMMC_CDETECT
Address: Operational Base + offset (0x0050)
Card-detect register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RO	0x0	card_detect_n Value on card_detect_n input ports; read-only bits. 0 represents presence of card.

SDMMC_WRTPRT
Address: Operational Base + offset (0x0054)
Write-protect register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	write_protect Value on card_write_prt input port. 1 represents write protection.

SDMMC_TCBCNT
Address: Operational Base + offset (0x005c)
Transferred CIU card byte count

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	trans_card_byte_count Number of bytes transferred by CIU unit to card. In 32-bit or 64-bit AMBA data-bus-width modes, register should be accessed in full to avoid read-coherency problems. In 16-bit AMBA data-bus-width mode, internal 16-bit coherency register is implemented. User should first read lower 16 bits and then higher 16 bits. When reading lower 16 bits, higher 16 bits of counter are stored in temporary register. When higher 16 bits are read, data from temporary register is supplied. Both TCBCNT and TBBCNT share same coherency register. When AREA_OPTIMIZED parameter is 1, register should be read only after data transfer completes; during data transfer, register returns 0.

SDMMC_TBBCNT
Address: Operational Base + offset (0x0060)
Transferred host/DMA to/from BIU-FIFO byte count

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	trans_fifo_byte_count Number of bytes transferred between Host/DMA memory and BIU FIFO. In 32-bit or 64-bit AMBA data-bus-width modes, register should be accessed in full to avoid read-coherency problems. In 16-bit AMBA data-bus-width mode, internal 16-bit coherency register is implemented. User should first read lower 16 bits and then higher 16 bits. When reading lower 16 bits, higher 16 bits of counter are stored in temporary register. When higher 16 bits are read, data from temporary register is supplied. Both TCBCNT and TBBCNT share same coherency register.

SDMMC_DEBNCE
Address: Operational Base + offset (0x0064)
Card detect debounce register

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RW	0xffffff	debounce_count Number of host clocks (clk) used by debounce filter logic; typical debounce time is 5-25 ms.

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SDMMC_USRID
Address: Operational Base + offset (0x0068)
User ID register

Bit	Attr	Reset Value	Description
31:0	RW	0x07967797	usrid User identification register; value set by user. Default reset value can be picked by user while configuring core before synthesis. Can also be used as scratch pad register by user. The default value is determined by Configuration Value.

SDMMC_VERID
Address: Operational Base + offset (0x006c)
Synopsys version ID register

Bit	Attr	Reset Value	Description
31:0	RO	0x5342270a	verid Version identification register; register value is hard-wired. Can be read by firmware to support different versions of core.

SDMMC_HCON
Address: Operational Base + offset (0x0070)
Hardware Configuration Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HCON Configuration Dependent. Hardware configurations selected by user before synthesizing core. Register values can be used to develop configuration-independent software drivers. [0]: CARD_TYPE 0: MMC_ONLY 1: SD_MMC [5:1]: NUM_CARDS - 1 [6]: H_BUS_TYPE 0: APB 1: AHB [9:7]: H_DATA_WIDTH 000: 16 bits 001: 32 bits 010: 64 bits others: reserved [15:10]: H_ADDR_WIDTH 0 to 7: reserved 8: 9 bits 9: 10 bits … 31: 32 bits 32 to 63: reserved [17:16]: DMA_INTERFACE 00: none 01: DW_DMA 10: GENERIC_DMA 11: NON-DW-DMA [20:18]: GE_DMA_DATA_WIDTH 000: 16 bits 001: 32 bits 010: 64 bits others: reserved [21]: FIFO_RAM_INSIDE 0: outside 1: inside [22]: IMPLEMENT_HOLD_REG 0: no hold register 1: hold register [23]: SET_CLK_FALSE_PATH 0: no false path 1: false path set [25:24]: NUM_CLK_DIVIDER-1 [26]: AREA_OPTIMIZED 0: no area optimization 1: Area optimization

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SDMMC_UHS_REG
Address: Operational Base + offset (0x0074)
UHS-1 register

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RW	0x0	ddr_reg DDR mode. Determines the voltage fed to the buffers by an external voltage regulator. 0: non-DDR mode 1: DDR mode UHS_REG [16] should be set for card.
15:1	RO	0x0	reserved
0	RW	0x0	volt_reg High Voltage mode. Determines the voltage fed to the buffers by an external voltage regulator. 0: buffers supplied with 3.3V Vdd 1: buffers supplied with 1.8V Vdd These bits function as the output of the host controller and are fed to an external voltage regulator. The voltage regulator must switch the voltage of the buffers of a particular card to either 3.3V or 1.8V, depending on the value programmed in the register. VOLT_REG[0] should be set to 1'b1 for card in order to make it operate for 1.8V.

SDMMC_RST_N
Address: Operational Base + offset (0x0078)
Hardware reset register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x1	card_reset Hardware reset. 0: active mode 1: reset These bits cause the cards to enter pre-idle state, which requires them to be re-initialized. CARD_RESET[0] should be set to 1'b1 to reset card.

SDMMC_BMOD
Address: Operational Base + offset (0x0080)
Bus Mode Register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
10:8	RO	0x0	PBL Programmable Burst Length. These bits indicate the maximum number of beats to be performed in one IDMAC transaction. The IDMAC will always attempt to burst as specified in PBL each time it starts a Burst transfer on the host bus. The permissible values are 1, 4, 8, 16, 32, 64, 128 and 256. This value is the mirror of MSIZE of FIFOTH register. In order to change this value, write the required value to FIFOTH register. This is an encode value as follows. 000 –1 transfers 001 –4 transfers 010 –8 transfers 011 –16 transfers 100 –32 transfers 101 –64 transfers 110 –128 transfers 111 –256 transfers Transfer unit is either 16, 32, or 64 bits, based on HDATA_WIDTH. PBL is a read-only value and is applicable only for Data Access; it does not apply to descriptor accesses.
7	RW	0x0	DE IDMAC Enable. When set, the IDMAC is enabled.
6:2	RW	0x00	DSL Descriptor Skip Length. Specifies the number of HWord/Word/Dword (depending on 16/32/64-bit bus) to skip between two unchained descriptors. This is pplicable only for dual buffer structure.
1	RW	0x0	FB Fixed Burst. Controls whether the AHB Master interface performs fixed burst transfers or not. When set, the AHB will use only SINGLE, INCR4, INCR8 or INCR16 during start of normal burst transfers. When reset, the AHB will use SINGLE and INCR burst transfer operations.
0	RW	0x0	SWR Software Reset. When set, the DMA Controller resets all its internal registers It is automatically cleared after 1 clock cycle

SDMMC_PLDMND

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Address: Operational Base + offset (0x0084)
Poll Demand Register

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	PD Poll Demand. If the OWN bit of a descriptor is not set, the FSM goes to the Suspend state. The host needs to write any value into this register for the IDMAC FSM to resume normal descriptor fetch operation. This is a write only register.

SDMMC_DBADDR
Address: Operational Base + offset (0x0088)
Descriptor List Base Address Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	SDL Start of Descriptor List. Contains the base address of the First Descriptor. The LSB bits [0/1/2:0] for 16/32/64-bit bus-width) are ignored and taken as all-zero by the IDMAC internally. Hence these LSB bits are read-only.

SDMMC_IDSTS
Address: Operational Base + offset (0x008c)
Internal DMAC Status Register

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:13	RO	0x0	FSM DMAC FSM present state. 0: DMA_IDLE 1: DMA_SUSPEND 2: DESC_RD 3: DESC_CHK 4: DMA_RD_REQ_WAI 5: DMA_WR_REQ_WAI 6: DMA_RD 7: DMA_WR 8: DESC_CLOSE

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Bit	Attr	Reset Value	Description
12:10	RO	0x0	EB Error Bits. Indicates the type of error that caused a Bus Error. Valid only with atal Bus Error bit—IDSTS[2] (IDSTS64[2], in case of 64-bit address configuration) set. This field does not generate an interrupt. 1: Host Abort received during transmission 2: Host Abort received during reception Others: Reserved
9	RW	0x0	AIS Abnormal Interrupt Summary. Logical OR of the following: IDSTS[2] Fatal Bus Interrupt IDSTS[4] DU bit Interrupt Only unmasked bits affect this bit. This is a sticky bit and must be cleared each time a corresponding bit that causes AIS to be set is cleared. Writing a 1 clears this bit.
8	RW	0x0	NIS Normal Interrupt Summary. Logical OR of the following: IDSTS[0] Transmit Interrupt IDSTS[1] Receive Interrupt Only unmasked bits affect this bit. This is a sticky bit and must be cleared each time a corresponding bit that causes NIS to be set is cleared. Writing a 1 clears this bit.
7:6	RO	0x0	reserved
5	RW	0x0	CES Card Error Summary. Indicates the status of the transaction to/from the card; also present in RINTSTS. Indicates the logical OR of the following bits: ■EBE –End Bit Error ■RTO –Response Timeout/Boot Ack Timeout ■RCRC –Response CRC ■SBE –Start Bit Error ■DRTO –Data Read Timeout/BDS timeout ■DCRC –Data CRC for Receive ■RE –Response Error Writing a 1 clears this bit. The abort condition of the IDMAC depends on the setting of this CES bit. If the CES bit is enabled, then the IDMAC aborts on a "response error"; however, it will not abort if the CES bit is cleared

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Bit	Attr	Reset Value	Description
4	RW	0x0	DU Descriptor Unavailable Interrupt. This bit is set when the descriptor is unavailable due to OWN bit = 0 (DES0[31] =0). Writing a 1 clears this bit.
3	RO	0x0	reserved
2	RW	0x0	FBE Fatal Bus Error Interrupt. Indicates that a Bus Error occurred (IDSTS[12:10]) (IDSTS64[12:10], in case of 64-bit address configuration). When this bit is set, the DMA disables all its bus accesses. Writing a 1 clears this bit.
1	RW	0x0	RI Receive Interrupt. Indicates the completion of data reception for a descriptor. Writing a 1 clears this bit.
0	RW	0x0	TI Transmit Interrupt. Indicates that data transmission is finished for a descriptor. Writing 1 clears this bit

SDMMC_IDINTEN

Address: Operational Base + offset (0x0090)

Internal DMAC Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9	RW	0x0	AI Abnormal Interrupt Summary Enable. When set, an abnormal interrupt is enabled. This bit enables the following bits: ■IDINTEN[2] Fatal Bus Error Interrupt ■IDINTEN[4] DU Interrupt
8	RW	0x0	NI Normal Interrupt Summary Enable. When set, a normal interrupt is enabled. When reset, a normal interrupt is disabled. This bit enables the following bits: ■IDINTEN[0] Transmit Interrupt ■IDINTEN[1] Receive Interrup
7:6	RO	0x0	reserved
5	RW	0x0	CES Card Error summary Interrupt Enable. When set, it enables the Card Interrupt summary

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Bit	Attr	Reset Value	Description
4	RW	0x0	DU Descriptor Unavailable Interrupt. When set along with Abnormal Interrupt Summary Enable, the DU interrupt is enabled
3	RO	0x0	reserved
2	RW	0x0	FBE Fatal Bus Error Enable. When set with Abnormal Interrupt Summary Enable, the Fatal Bus Error Interrupt is enabled. When reset, Fatal Bus Error Enable Interrupt is disabled.
1	RW	0x0	RI Receive Interrupt Enable. When set with Normal Interrupt Summary Enable, Receive Interrupt is enabled. When reset, Receive Interrupt is disabled
0	RW	0x0	TI Transmit Interrupt Enable. When set with Normal Interrupt Summary Enable, Transmit Interrupt is enabled. When reset, Transmit Interrupt is disabled.

SDMMC_DSCADDR
Address: Operational Base + offset (0x0094)
Current Host Descriptor Address Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	HDA Host Descriptor Address Pointer. Cleared on reset. Pointer updated by IDMAC during operation. This register points to the start address of the current descriptor read by the IDMAC

SDMMC_BUFADDR
Address: Operational Base + offset (0x0098)
Current Buffer Descriptor Address Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	HBA Host Buffer Address Pointer. Cleared on Reset. Pointer updated by IDMAC during operation. This register points to the current Data Buffer Address being accessed by the IDMAC

SDMMC_CARDTHRCTL

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Address: Operational Base + offset (0x0100)
Card read threshold enable

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:16	RW	0x000	CardRdThreshold Card Read Threshold size
15:2	RO	0x0	reserved
1	RW	0x0	BsyClrIntEn Busy Clear Interrupt generation: 0: Busy Clear Interrupt disabled 1: Busy Clear Interrupt enabled Note: The application can disable this feature if it does not want to wait for a Busy Clear Interrupt. For example, in a multi-card scenario, the application can switch to the other card without waiting for a busy to be completed. In such cases, the application can use the polling method to determine the status of busy. By default this feature is disabled and backward-compatible to the legacy drivers where polling is used.
0	RW	0x0	CardRdThrEn Card Read Threshold Enable. 0: Card Read Threshold disabled 1: Card Read Threshold enabled. Host Controller initiates Read Transfer only if CardRdThreshold amount of space is available in receive FIFO.

SDMMC_BACK_END_POWER
Address: Operational Base + offset (0x0104)
Back-end power

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	back_end_power Back end power 0: Off; Reset 1: Back-end Power supplied to card application

SDMMC_UHS_REG_EXT
Address: Operational Base + offset (0x0108)
UHS Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	Reserved

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SDMMC_EMMC_DDR_REG
Address: Operational Base + offset (0x010c)
eMMC 4.5 DDR START Bit Detection Control Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	HALF_START_BIT Control for start bit detection mechanism inside DWC_mobile_storage based on duration of start bit; each bit refers to one slot. For eMMC 4.5, start bit can be: 0: Full cycle (HALF_START_BIT = 0) 1: Less than one full cycle (HALF_START_BIT = 1) Set HALF_START_BIT=1 for eMMC 4.5 and above; set to 0 for SD applications.

SDMMC_ENABLE_SHIFT
Address: Operational Base + offset (0x0110)
Enable Phase Shift Registe

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	Reserved

SDMMC_FIFO_BASE
Address: Operational Base + offset (0x0200)
FIFO Base Address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	fifo_base_addr FIFO base address.

18.5 Interface Description

The interface and IOMUX setting for SDMMC, SDIO, EMMC are shown as follows.

18.5.1 SDMMC IOMUX

Table 18-8 IOMUX Settings for SDMMC

Module Pin	Direction	Pad Name	IOMUX Setting
sdmmc_cclk	O	IO_SDMMC0clkout_MCUJTAGtck_SDCARDgpio2b1	GRF_GPIO2B_IOMUX[3:2]=2'b01
sdmmc_ccmd	I/O	IO_SDMMC0cmd_MCUJTAGtms_SDCARDgpio2b2	GRF_GPIO2B_IOMUX[5:4]=2'b01
sdmmc_cdata0	I/O	IO_SDMMC0data0_UART2DBGsout_SDCARDgpio2a5	GRF_GPIO2A_IOMUX[11:10]=2'b01
sdmmc_cdata1	I/O	IO_SDMMC0data1_UART2DBGsin_SDCARDgpio2a6	GRF_GPIO2A_IOMUX[13:12]=2'b01
sdmmc_cdata2	I/O	IO_SDMMC0data2_JTAGtck_SDCARDgpio2a7	GRF_GPIO2A_IOMUX[15:14]=2'b01
sdmmc_cdata3	I/O	IO_SDMMC0data3_JTAGtms_SDCARDgpio2b0	GRF_GPIO2B_IOMUX[1:0]=2'b01
sdmmc_cdetect	I	IO_SDMMC0dectn_SDCARDgpi	GRF_GPIO2B_IOMUX[7:6]

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n		o2b3	=2'b01

18.5.2 SDIO IOMUX

Table 18-9 IOMUX Settings for SDIO

Module Pin	Direction	Pad Name	IOMUX Setting
sdio_cclk	O	IO_SDIO0clkout_WIFIgpio3a1	GRF_GPIO3A_IOMUX[3:2]=2'b01
sdio_ccmd	I/O	IO_SDIO0cmd_WIFIgpio3a0	GRF_GPIO3A_IOMUX[1:0]=2'b01
sdio_cdata0	I/O	IO_SDIO0data0_WIFIgpio2d4	GRF_GPIO2D_IOMUX[9:8]=2'b01
sdio_cdata1	I/O	IO_SDIO0data1_WIFIgpio2d5	GRF_GPIO2D_IOMUX[11:10]=2'b01
sdio_cdata2	I/O	IO_SDIO0data2_WIFIgpio2d6	GRF_GPIO2D_IOMUX[13:12]=2'b01
sdio_cdata3	I/O	IO_SDIO0data3_WIFIgpio2d7	GRF_GPIO2D_IOMUX[15:14]=2'b01
sdio_cdetectn	I	IO_SDIO0detectn_WIFIgpio3a2	GRF_GPIO3A_IOMUX[5:4]=2'b01
sdio_wprt	I	IO_SDIO0wrprt_WIFIgpio3a3	GRF_GPIO3A_IOMUX[7:6]=2'b01
sdio_int_n	I	IO_SDIO0intn_WIFIgpio3a6	GRF_GPIO3A_IOMUX[13:12]=2'b01
sdio_pwren	O	IO_SDIO0pwren_WIFIgpio3a4	GRF_GPIO3A_IOMUX[9:8]=2'b01
sdio_bkpwr	O	IO_SDIO0bkpwr_WIFIgpio3a5	GRF_GPIO3A_IOMUX[11:10]=2'b01

18.5.3 EMMC IOMUX

Table 18-10 IOMUX Settings for eMMC

Module Pin	Direction	Pad Name	IOMUX Setting
emmc_cclk	O	IO_FLASHdqs_EMMCclkout_FLASHgpio2a4	GRF_GPIO2A_IOMUX[9:8]=2'b10
emmc_ccmd	I/O	IO_FLASHrdy_EMMCcmd_SFCclk_FLASHgpio1d2	GRF_GPIO2A_IOMUX[5:4]=2'b10
emmc_cdata0	I/O	IO_FLASHdata0_EMMCdata0_SFCsio0_FLASHgpio1c2	GRF_GPIO1C_IOMUX[5:4]=2'b10
emmc_cdata1	I/O	IO_FLASHdata1_EMMCdata1_SFCsio1_FLASHgpio1c3	GRF_GPIO1C_IOMUX[7:6]=2'b10
emmc_cdata2	I/O	IO_FLASHdata2_EMMCdata2_SFCsio2_FLASHgpio1c4	GRF_GPIO1C_IOMUX[9:8]=2'b10
emmc_cdata3	I/O	IO_FLASHdata3_EMMCdata3_SFCsio3_FLASHgpio1c5	GRF_GPIO1C_IOMUX[11:10]=2'b10
emmc_cdata4	I/O	IO_FLASHdata4_EMMCdata4_SPI0rxd_FLASHgpio1c6	GRF_GPIO1C_IOMUX[13:12]=2'b10

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emmc_cdata5	I/O	IO_FLASHdata5_EMMCdata5_SPI0txd_FLASHgpio1c7	GRF_GPIO1C_IOMUX[15:14]=2'b10
emmc_cdata6	I/O	IO_FLASHdata6_EMMCdata6_SPI0csn0_FLASHgpio1d0	GRF_GPIO1D_IOMUX[1:0]=2'b10
emmc_cdata7	I/O	IO_FLASHdata7_EMMCdata7_SPI0csn1_FLASHgpio1d1	GRF_GPIO1D_IOMUX[3:2]=2'b10
emmc_rstn	O	IO_FLASHcsn3_EMMCrstnout_FLASHgpio2a3	GRF_GPIO2A_IOMUX[7:6]=2'b10
emmc_pwren	O	IO_FLASHwp_EMMCpwren_FLASHgpio1d3	GRF_GPIO1D_IOMUX[7:6]=2'b10

*Notes: Direction: **I**– Input, **O**– Output, **I/O**– Input/Output*

18.6 Application Notes

18.6.1 Card-Detect and Write-Protect Mechanism

Following figure illustrates how the SD/MMC card detection and write-protect signals are connected. Most of the SD/MMC sockets have card-detect pins. When no card is present, card_detect_n is 1 due to the pull-up. When the card is inserted, the card-detect pin is shorted to ground, which makes card_detect_n go to 0. Similarly in SD cards, when the write-protect switch is toward the left, it shorts the write_protect port to ground.

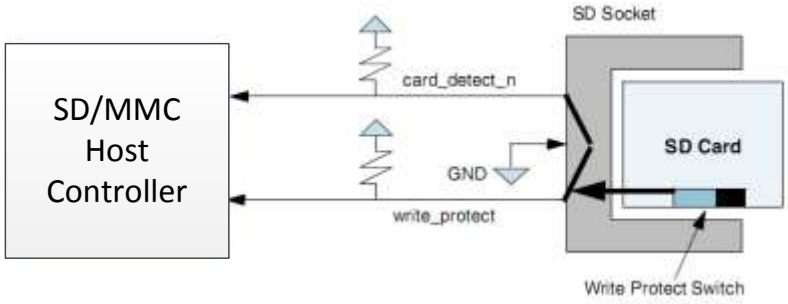

Fig. 18-9 SD/MMC Card-Detect and Write-Protect

18.6.2 SD/MMC Termination Requirement

Following Figure illustrates the SD/MMC termination requirements, which is required to pull up ccmd and cdata lines on the device bus. The recommended specification for pull-up on the ccmd line (Rcmd) is 4.7K - 100K for MMC, and 10K - 100K for an SD. The recommended pull-up on the cdata line (Rdat) is 50K - 100K.

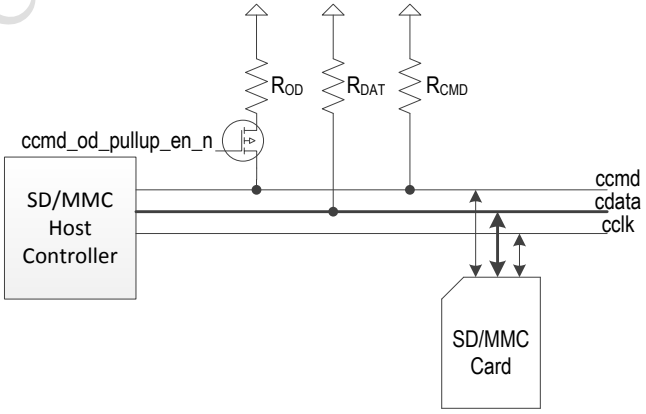

Fig. 18-10 SD/MMC Card Termination

1. Rcmd and Rod Calculation

The SD/MMC card enumeration happens at a very low frequency – 100-400KHz. Since the

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MMC bus is a shared bus between multiple cards, during enumeration open-drive mode is used to avoid bus conflict. Cards that drive 0 win over cards that drive "z". The pull-up in the command line pulls the bus to 1 when all cards drive "z". During normal data transfer, the host chooses only one card and the card driver switches to push-pull mode.
For example, if enumeration is done at 400KHz and the total bus capacitance is 200 pf, the pull-up needed during enumeration is:

2.2 RC = rise-time = 1/400KHz
R = 1/(2.2 * C * 100KHz)
= 1/(2.2 x 200 x 10**-12 x 400 x 10**3)
= 1/(17.6 x 10**-5)
= 5.68K

The R_{OD} and R_{CMD} should be adjusted in such a way that the effective pull-up is at the maximum 5.68K during enumeration. If there are only a few cards in the bus, a fixed R_{CMD} resister is sufficient and there is no need for an additional R_{OD} pull-up during enumeration. You should also ensure the effective pull-up will not violate the Iol rating of the drivers.
In SD mode, since each card has a separate bus, the capacitance is less, typically in the order of 20-30pf (host capacitance + card capacitance + trace + socket capacitance). For example, if enumeration is done at 400KHz and the total bus capacitance is 20pf, the pull-up needed during enumeration is:

2.2 RC = rise-time = 1/400KHz
R = 1/(2.2 * C * 100KHz)
= 1/(2.2 x 20 x 10**-12 x 400 x 10**3)
= 1/(1.76 x 10**-5)
= 56.8K

Therefore, a fixed 56.8K permanent Rcmd is sufficient in SD mode to enumerate the cards. The driver of the SD/MMC on the "command" port needs to be only a push-pull driver. During enumeration, the SD/MMC emulates an open-drain driver by driving only a 0 or a"z" by controlling the ccmd_out and ccmd_out_en signals.

18.6.3 Software/Hardware Restriction

Before issuing a new data transfer command, the software should ensure that the card is not busy due to any previous data transfer command. Before changing the card clock frequency, the software must ensure that there are no data or command transfers in progress.
If the card is enumerated in SDR50, or DDR50 mode, then the application must program the use_hold_reg bit[29] in the CMD register to 1'b0 (phase shift of cclk_in_drv = 0) or 1'b1 (phase shift of cclk_in_drv>0). If the card is enumerated in SDR12 or SDR25 mode, the application must program the use_hold_reg bit[29] in the CMD register to 1'b1.
This programming should be done for all data transfer commands and non-data commands that are sent to the card. When the use_hold_reg bit is programmed to 1'b0, the Host Controller bypasses the Hold Registers in the transmit path. The value of this bit should not be changed when a Command or Data Transfer is in progress. For more details on using use_hold_reg and the implementation requirements for meeting the Card input hold time, refer to "Recommended Usage" in following table.

Table 18-11 Recommended Usage of use_hold_reg

No.	Speed Mode	use_hold_reg	cclk_in (MHz)	clk_in_drv (MHz)	clk_divider	Phase shift
1	SDR104	1'b0	200	200	0	0
2	SDR104	1'b1	200	200	0	Tunable> 0
3	SDR50	1'b0	100	100	0	0
4	SDR50	1'b1	100	100	0	Tunable> 0
5	DDR50 (8bit)	1'b0	100	100	1	0
6	DDR50 (8bit)	1'b1	100	100	1	Tunable> 0

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7	DDR50 (4bit)	1'b0	50	50	0	0
8	DDR50 (4bit)	1'b1	50	50	0	Tunable> 0
9	SDR25	1'b1	50	50	0	Tunable> 0
10	SDR12	1'b1	50	50	1	Tunable> 0

To avoid glitches in the card clock outputs, the software should use the following steps when changing the card clock frequency:

1) Before disable the clocks, ensure that the card is not busy due to any previous data command. To determine this, check for 0 in bit9 of STATUS register.
2) Update the Clock Enable register to disable all clocks. To ensure completion of any previous command before this update, send a command to the CIU to update the clock registers by setting:
   - start_cmd bit
   - "update clock registers only" bits
   - "wait_previous data complete" bit

   Wait for the CIU to take the command by polling for 0 on the start_cmd bit.
3) Set the start_cmd bit to update the Clock Divider and/or Clock Source registers, and send a command to the CIU in order to update the clock registers; wait for the CIU to take the command.
4) Set start_cmd to update the Clock Enable register in order to enable the required clocks and send a command to the CIU to update the clock registers; wait for the CIU to take the command.

In non-DMA mode, while reading from a card, the Data Transfer Over (RINTSTS[3]) interrupt occurs as soon as the data transfer from the card is over. There still could be some data left in the FIFO, and the RX_WMark interrupt may or may not occur, depending on the remaining bytes in the FIFO. Software should read any remaining bytes upon seeing the Data Transfer Over (DTO) interrupt. While using the external DMA interface for reading from a card, the DTO interrupt occurs only after all the data is flushed to memory by the DMA interface unit.

While writing to a card in external DMA mode, if an undefined-length transfer is selected by setting the Byte Count Register to 0, the DMA logic will likely request more data than it will send to the card, since it has no way of knowing at which point the software will stop the transfer. The DMA request stops as soon as the DTO is set by the CIU.

If the software issues a controller_reset command by setting control register bit[0] to 1, all the CIU state machines are reset; the FIFO is not cleared. The DMA sends all remaining bytes to the host. In addition to a card-reset, if a FIFO reset is also issued, then:

- Any pending DMA transfer on the bus completes correctly
- DMA data read is ignored
- Write data is unknown(x)

Additionally, if dma_reset is also issued, any pending DMA transfer is abruptly terminated. When the DW-DMA is used, the DMA controller channel should also be reset and reprogrammed.

If any of the previous data commands do not properly terminate, then the software should issue the FIFO reset in order to remove any residual data, if any, in the FIFO. After asserting the FIFO reset, you should wait until this bit is cleared.

One data-transfer requirement between the FIFO and host is that the number of transfers should be a multiple of the FIFO data width (32bits). For example, you want to write only 15 bytes to an SD/MMC card (BYTCNT), the host should write 16 bytes to the FIFO or program the DMA to do 16-byte transfers. The software can still program the Byte Count register to only 15, at which point only 15 bytes will be transferred to the card. Similarly, when 15 bytes are read from a card, the host should still read all 16 bytes from the FIFO.

It is recommended that you not change the FIFO threshold register in the middle of data transfers.

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18.6.4 Programming Sequence

1. Initialization

Following figure illustrates the initialization flow.

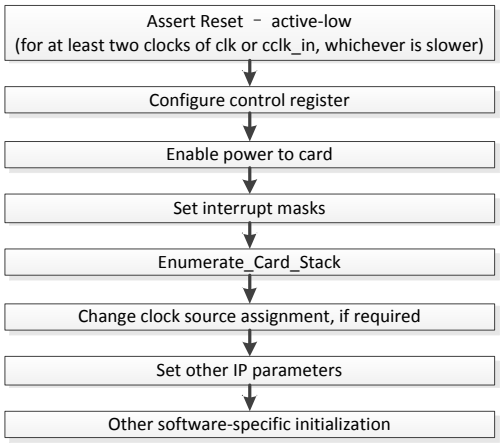

Fig. 18-11 Host Controller Initialization Sequence

Once the power and clocks are stable, reset_n should be asserted(active-low) for at least two clocks of clk or cclk_in, whichever is slower. The reset initializes the registers, ports, FIFO-pointers, DMA interface controls, and state-machines in the design. After power-on reset, the software should do the following:

- Configure control register – For MMC mode, enable the open-drain pullup by setting enable_OD_pullup(bit24) in the control register.
- Enable power to cards – Before enabling the power, confirm that the voltage setting to the voltage regulators is correct. Enable power to the connected cards by setting the corresponding bit to 1 in the Power Enable register. Wait for the power ramp-up time.
- Set masks for interrupts by clearing appropriate bits in the Interrupt Mask register. Set the global int_enable bit of the Control register. It is recommended that you write 0xffff_ffff to the Raw Interrupt register in order to clear any pending interrupts before setting the int_enable bit.
- Enumerate card stack – Each card is enumerated according to card type; for details, refer to "Enumerated Card Stack". For enumeration, you should restrict the clock frequency to 400KHz.
- Changing clock source assignment – set the card frequency using the clock-divider and clock-source registers; for details, refer to "Clock Programming". MMC cards operate at a maximum of 20MHz (at maximum of 52MHz in high-speed mode). SD mode operates at a maximum of 25MHz (at maximum of 50MHz in high-speed mode).
- Set other parameters, which normally do not need to be changed with every command, with a typical value such as timeout values in cclk_out according to SD/MMC specifications.
- ResponseTimeOut = 0x64
- DataTimeOut = highest of one of the following:
- (10*((TAAC*Fop)+(100*NSAC))
- Host FIFO read/write latency from FIFO empty/full
- Set the debounce value to 25ms(default:0x0fffff) in host clock cycle units in the DEBNCE register.
- FIFO threshold value in bytes in the FIFOTH register.

2. Enumerated Card Stack

The card stack does the following:

- Enumerates all connected cards
- Sets the RCA for the connected cards
- Reads card-specific information
- Stores card-specific information locally

Enumeration depends on the operating mode of the SD/MMC card; the card type is first identified and the appropriate card enumeration routine is called.

1) Check if the card is connected.
2) Clear the card type register to set the card width as a single bit. For the given card number,

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clear the corresponding bits in the card_type register. Clear the register bit for a 1-bit, 4-bit bus width. For example, for card number=1, clear bit 0 and bit 16 of the card_type register.

3) Set clock frequency to F_{OD}=400KHz, maximum – Program clock divider0 (bits 0-7 in the CLKDIV register) value to one-half of the cclk_in frequency divided by 400KHz. For example, if cclk_in is 20MHz, then the value is 20, 000/(2*400)=25.
4) Identify the card type; that is, SD, MMC, or SDIO.
   a. Send CMD5 first. If a response is received, then the card is SDIO
   b. If not, send CMD8 with the following Argument
      Bit[31:12] = 20'h0 //reserved bits
      Bit[11:8] = 4'b0001 //VHS value
      Bit[7:0] = 8'b10101010 //Preferred Check Pattern by SD2.0
   c. If Response is received the card supports High Capacity SD2.0 then send ACMD41 with the following Argument
      Bit[31] = 1'b0; //Reserved bits
      Bit[30] = 1'b1; //High Capacity Status
      Bit[29:24] = 6'h0; //Reserved bits
      Bit[23:0] = Supported Voltage Range
   d. If Response is received for ACMD41 then the card is SD. Otherwise the card is MMC.
   e. If response is not received for initial CMD8 then card does not support High Capacity SD2.0, then issue CMD0 followed by ACMD41 with the following Argument
      Bit[31] = 1'b0; //Reserved bits
      Bit[30] = 1'b0; //High Capacity Status
      Bit[29:24] = 6'h0; //Reserved bits
      Bit[23:0] = Supported Voltage Range

5) Enumerate the card according to the card type.
6) Use a clock source with a frequency = Fod (that is, 400KHz) and use the following enumeration command sequence:
   - SD card – Send CMD0, CMD8, ACMD41, CMD2, CMD3.
   - MMC – Send CMD0, CMD1, CMD2, CMD3.

3. Power Control

You can implement power control using the following registers, along with external circuitry:

- Control register bits card_voltage_a and card_voltage_b – Status of these bits is reflected at the IO pins. The bits can be used to generate or control the supply voltage that the memory cards require.
- Power enable register – Control power to individual cards.

Programming these two register depends on the implemented external circuitry. While turning on or off the power enable, you should confirm that power supply settings are correct. Power to all cards usually should be disabled while switching off the power.

4. Clock Programming

The Host Controller supports one clock sources. The clock to an individual card can be enabled or disabled. Registers that support this are:

- CLKDIV – Programs individual clock source frequency. CLKDIV limited to 0 or 1 is recommended.
- CLKSRC – Assign clock source for each card.
- CLKENA – Enables or disables clock for individual card and enables low-power mode, which automatically stops the clock to a card when the card is idle for more than 8 clocks.

The Host Controller loads each of these registers only when the start_cmd bit and the Update_clk_regs_only bit in the CMD register are set. When a command is successfully loaded, the Host Controller clears this bit, unless the Host Controller already has another command in the queue, at which point it gives an HLE(Hardware Locked Error).
Software should look for the start_cmd and the Update_clk_regs_only bits, and should also set the wait_prvdata_complete bit to ensure that clock parameters do not change during data

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transfer. Note that even though start_cmd is set for updating clock registers, the Host Controller does not raise a command_done signal upon command completion.
The following shows how to program these registers:

- Confirm that no card is engaged in any transaction; if there is a transaction, wait until it finishes.
- Stop all clocks by writing xxxx0000 to the CLKENA register. Set the start_cmd, Update_clk_regs_only, and wait_prvdata_complete bits in the CMD register. Wait until start_cmd is cleared or an HLE is set; in case of an HLE, repeat the command.
- Program the CLKDIV and CLKSRC registers, as required. Set the start_cmd, Update_clk_regs_only, and wait_prvdata_complete bits in the CMD register. Wait until start_cmd is cleared or an HLE is set; in case of an HLE, repeat the command.
- Re-enable all clocks by programming the CLKENA register. Set the start_cmd, Update_clk_regs_only, and wait_prvdata_complete bits in the CMD register. Wait until start_cmd is cleared or an HLE is set; in case of an HLE, repeat the command.

5. No-Data Command With or Without Response Sequence

To send any non-data command, the software needs to program the CMD register @0x2C and the CMDARG register @0x28 with appropriate parameters. Using these two registers, the Host Controller forms the command and sends it to the command bus. The Host Controller reflects the errors in the command response through the error bits of the RINTSTS register. When a response is received – either erroneous or valid – the Host Controller sets the command_done bit in the RINTSTS register. A short response is copied in Response Register0, while along response is copied to all four response registers @0x30, 0x34, 0x38, and 0x3C. The Response3 register bit 31 represents the MSB, and the Response0 register bit 0 represents the LSB of a long response.
For basic commands or non-data commands, follow these steps:

- Program the Command register @0x28 with the appropriate command argument parameter.
- Program the Command register @0x2C with the settings in following table.

Table 18-12 Command Settings for No-Data Command

Parameter	Value	Description
Default		
start_cmd	1	-
use_hold_reg	1/0	Choose value based on speed mode being used;ref to "use_hold_reg" on CMD register
update_clk_regs_only	0	No clock parameters update command
data_expected	0	No data command
card number	0	Actual card number(one controller only connect one card, the num is No. 0)
cmd_index	command-index	-
send_initialization	0	Can be 1, but only for card reset commands, such as CMD0
stop_abort_cmd	0	Can be 1 for commands to stop data transfer, such as CMD12
response_length	0	Can be 1 for R2(long) response
response_expect	1	Can be 0 for commands with no response; for example, CMD0, CMD4, CMD15, and so on
User-selectable		
wait_prvdata_complete	1	Before sending command on command line, host should wait for completion of any data command in process, if any (recommended to always set this bit, unless the current command is to query status or stop data transfer when transfer is in progress)
check_response_crc	1	If host should crosscheck CRC of response received

- Wait for command acceptance by host. The following happens when the command is

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loaded into the Host Controller:

- Host Controller accepts the command for execution and clears the start_cmd bit in the CMD register, unless one command is in process, at which point the Host Controller can load and keep the second command in the buffer.
- If the Host Controller is unable to load the command – that is, a command is already in progress, a second command is in the buffer, and a third command is attempted – then it generates an HLE (hardware-locked error).
- Check if there is an HLE.
- Wait for command execution to complete. After receiving either a response from a card or response timeout, the Host Controller sets the command_done bit in the RINTSTS register. Software can either poll for this bit or respond to a generated interrupt.
- Check if response_timeout error, response_CRC error, or response error is set. This can be done either by responding to an interrupt raised by these errors or by polling bits 1, 6, and 8 from the RINTSTS register @0x44. If no response error is received, then the response is valid. If required, the software can copy the response from the response registers @0x30-0x3C.

Software should not modify clock parameters while a command is being executed.

- Data Transfer Commands

Data transfer commands transfer data between the memory card and the Host Controller. To send a data command, the Host Controller needs a command argument, total data size, and block size. Software can receive or send data through the FIFO.

Before a data transfer command, software should confirm that the card is not busy and is in a transfer state, which can be done using the CMD13 and CMD7 commands, respectively.

For the data transfer commands, it is important that the same bus width that is programmed in the card should be set in the card type register @0x18.

The Host Controller generates an interrupt for different conditions during data transfer, which are reflected in the RINTSTS register @0x44 as:

- Data_Transfer_Over (bit 3) – When data transfer is over or terminated. If there is a response timeout error, then the Host Controller does not attempt any data transfer and the "Data Transfer Over" bit is never set.
- Transmit_FIFO_Data_request (bit 4) – FIFO threshold for transmitting data was reached; software is expected to write data, if available, in FIFO.
- Receive_FIFO_Data_request (bit 5) – FIFO threshold for receiving data was reached; software is expected to read data from FIFO.
- Data starvation by Host timeout (bit 10) – FIFO is empty during transmission or is full during reception. Unless software writes data for empty condition or reads data for full condition, the Host Controller cannot continue with data transfer. The clock to the card has been stopped.
- Data read timeout error (bit 9) – Card has not sent data within the timeout period.
- Data CRC error (bit 7) – CRC error occurred during data reception.
- Start bit error (bit 13) – Start bit was not received during data reception.
- End bit error (bit 15) – End bit was not received during data reception or for a write operation; a CRC error is indicated by the card.

Conditions 6, 7, and 8 indicate that the received data may have errors. If there was a response timeout, then no data transfer occurred.

- Single-Block or Multiple-Block Read

Steps involved in a single-block or multiple-block read are:

- Write the data size in bytes in the BYTCNT register @0x20.
- Write the block size in bytes in the BLKSIZ register @0x1C. The Host Controller expects data from the card in blocks of size BLKSIZ each.
- Program the CMDARG register @0x28 with the data address of the beginning of a data read.
- Program the Command register with the parameters listed in following table. For SD and MMC cards, use CMD17 for a single-block read and CMD18 for a multiple-block

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read. For SDIO cards, use CMD53 for both single-block and multiple-block transfers.

Table 18-13 Command Setting for Single or Multiple-Block Read

Parameter	Value	Description
Default		
start_cmd	1	-
use_hold_reg	1/0	Choose value based on speed mode being used;ref to "use_hold_reg" on CMD register
update_clk_regs_only	0	No clock parameters update command
card number	0	Actual card number(one controller only connect one card, the num is No.0)
send_initialization	0	Can be 1, but only for card reset commands, such as CMD0
stop_abort_cmd	0	Can be 1 for commands to stop data transfer, such as CMD12
send_auto_stop	0/1	-
transfer_mode	0	Block transfer
read_write	0	Read from card
data_expected	1	Data command
response_length	0	Can be 1 for R2(long) response
response_expect	1	Can be 0 for commands with no response; for example, CMD0, CMD4, CMD15, and so on
User-selectable		
cmd_index	command -index	-
wait_prvdata_complete	1	0- Sends command immediately 1- Sends command after previous data transfer ends
check_response_crc	1	0- Host Controller should not check response CRC 1- Host Controller should check response CRC

After writing to the CMD register, the Host Controller starts executing the command; when the command is sent to the bus, the command_done interrupt is generated.

- Software should look for data error interrupts; that is, bits 7, 9, 13, and 15 of the RINTSTS register. If required, software can terminate the data transfer by sending a STOP command.
- Software should look for Receive_FIFO_Data_request and/or data starvation by host timeout conditions. In both cases, the software should read data from the FIFO and make space in the FIFO for receiving more data.
- When a Data_Transfer_Over interrupt is received, the software should read the remaining data from the FIFO.

- Single-Block or Multiple-Block Write

Steps involved in a single-block or multiple-block write are:

- Write the data size in bytes in the BYTCNT register @0x20.
- Write the block size in bytes in the BLKSIZ register @0x1C; the Host Controller sends data in blocks of size BLKSIZ each.
- Program CMDARG register @0x28 with the data address to which data should be written.
- Write data in the FIFO; it is usually best to start filling data the full depth of the FIFO.
- Program the Command register with the parameters listed in following table.

Table 18-14 Command Settings for Single or Multiple-Block Write

Parameter	Value	Description
Default		
start_cmd	1	-
use_hold_reg	1/0	Choose value based on speed mode being used;ref to "use_hold_reg" on CMD register
update_clk_regs_only	0	No clock parameters update command
card number	0	Actual card number(one controller only connect one card, the num is No. 0)
send_initialization	0	Can be 1, but only for card reset commands, such as CMD0
stop_abort_cmd	0	Can be 1 for commands to stop data transfer, such as CMD12
send_auto_stop	0/1	-

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transfer_mode	0	Block transfer
read_write	1	Write to card
data_expected	1	Data command
response_length	0	Can be 1 for R2(long) response
response_expect	1	Can be 0 for commands with no response; for example, CMD0, CMD4, CMD15, and so on
User-selectable		
cmd_index	command-index	-
wait_prvdata_complete	1	0- Sends command immediately 1- Sends command after previous data transfer ends
check_response_crc	1	0- Host Controller should not check response CRC 1- Host Controller should check response CRC

After writing to the CMD register, Host Controller starts executing a command; when the command is sent to the bus, a command_done interrupt is generated.

- Software should look for data error interrupts; that is, for bits 7, 9, and 15 of the RINTSTS register. If required, software can terminate the data transfer by sending the STOP command.
- Software should look for Transmit_FIFO_Data_Request and/or timeout conditions from data starvation by the host. In both cases, the software should write data into the FIFO.
- When a Data_Transfer_Over interrupt is received, the data command is over. For an open-ended block transfer, if the byte count is 0, the software must send the STOP command. If the byte count is not 0, then upon completion of a transfer of a given number of bytes, the Host Controller should send the STOP command, if necessary. Completion of the AUTO-STOP command is reflected by the Auto_command_done interrupt – bit 14 of the RINTSTS register. A response to AUTO_STOP is stored in RESP1 @0x34.
- Stream Read

A stream read is like the block read mentioned in "Single-Block or Multiple-Block Read", except for the following bits in the Command register:

transfer_mode = 1; //Stream transfer

cmd_index = CMD20;

A stream transfer is allowed for only a single-bit bus width.

- Stream Write

A stream write is exactly like the block write mentioned in "Single-Block or Multiple-Block Write", except for the following bits in the Command register:

transfer_mode = 1;//Stream transfer

cmd_index = CMD11;

In a stream transfer, if the byte count is 0, then the software must send the STOP command. If the byte count is not 0, then when a given number of bytes completes a transfer, the Host Controller sends the STOP command. Completion of this AUTO_STOP command is reflected by the Auto_command_done interrupt. A response to an AUTO_STOP is stored in the RESP1 register@0x34.

A stream transfer is allowed for only a single-bit bus width.

- Packed Commands

In order to reduce overhead, read and write commands can be packed in groups of commands—either all read or all write—that transfer the data for all commands in the group in one transfer on the bus.

Packed commands can be of two types:

■ Packed Write: CMD23 →CMD25

■ Packed Read: CMD23 → CMD25 → CMD23 → CMD18

Packed commands are put in packets by the application software and are transparent to the core.

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- Sending Stop or Abort in Middle of Transfer

The STOP command can terminate a data transfer between a memory card and the Controller, while the ABORT command can terminate an I/O data transfer for only the SDIO_IOONLY and SDIO_COMBO cards.

- Send STOP command – Can be sent on the command line while a data transfer is in progress; this command can be sent at any time during a data transfer.

You can also use an additional setting for this command in order to set the Command register bits (5-0) to CMD12 and set bit 14 (stop_abort_cmd) to 1. If stop_abort_cmd is not set to 1, the Controller does not know that the user stopped a data transfer. Reset bit 13 of the Command register (wait_prvdata_complete) to 0 in order to make the Controller send the command at once, even though there is a data transfer in progress.

- Send ABORT command – Can be used with only an SDIO_IOONLY or SDIO_COMBO card. To abort the function that is transferring data, program the function number in ASx bits (CCCR register of card, address 0x06, bits (0-2) using CMD52.
- Suspend or Resume Sequence

In an SDIO card, the data transfer between an I/O function and the Controller can be temporarily halted using the SUSPEND command; this may be required in order to perform a high-priority data transfer with another function. When desired, the data transfer can be resumed using the RESUME command.

The following functions can be implemented by programming the appropriate bits in the CCCR register (Function 0) of the SDIO card. To read from or write to the CCCR register, use the CMD52 command.

- SUSPEND data transfer – Non-data command
- Check if the SDIO card supports the SUSPEND/RESUME protocol; this can be done through the SBS bit in the CCCR register @0x08 of the card.
- Check if the data transfer for the required function number is in process; the function number that is currently active is reflected in bits 0-3 of the CCCR register @0x0D. Note that if the BS bit (address 0xc::bit 0) is 1, then only the function number given by the FSx bits is valid.
- To suspend the transfer, set BR (bit 2) of the CCCR register @0x0C.
- Poll for clear status of bits BR (bit 1) and BS (bit 0) of the CCCR @0x0C. The BS (Bus Status) bit is 1 when the currently-selected function is using the data bus; the BR (Bus Release) bit remains 1 until the bus release is complete. When the BR and BS bits are 0, the data transfer from the selected function has been suspended.
- RESUME data transfer – This is a data command
- Check that the card is not in a transfer state, which confirms that the bus is free for data transfer.
- If the card is in a disconnect state, select it using CMD7. The card status can be retrieved in response to CMD52/CMD53 commands.
- Check that a function to be resumed is ready for data transfer; this can be confirmed by reading the RFx flag in CCCR @0x0F. If RF = 1, then the function is ready for data transfer.
- To resume transfer, use CMD52 to write the function number at FSx bits (0-3) in the CCCR register @0x0D. Form the command argument for CMD52 and write it in CMDARG @0x28.
- Write the block size in the BLKSIZ register @0x1C; data will be transferred in units of this block size.
- Write the byte count in the BYTCNT register @0x20. This is the total size of the data; that is, the remaining bytes to be transferred. It is the responsibility of the software to handle the data.
- Program Command register; similar to a block transfer.
- When the Command register is programmed, the command is sent and the function resumes data transfer. Read the DF flag (Resume Data Flag). If it is 1, then the function has data for the transfer and will begin a data transfer as soon as the function or memory is resumed. If it is 0, then the function has no data for the transfer.
- If the DF flag is 0, then in case of a read, the Host Controller waits for data. After the

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data timeout period, it gives a data timeout error.

- Read_Wait Sequence

Read_wait is used with only the SDIO card and can temporarily stall the data transfer—either from function or memory—and allow the host to send commands to any function within the SDIO device. The host can stall this transfer for as long as required. The Host Controller provides the facility to signal this stall transfer to the card. The steps for doing this are:

- Check if the card supports the read_wait facility; read SRW (bit 2) of the CCCR register @0x08. If this bit is 1, then all functions in the card support the read_wait facility. Use CMD52 to read this bit.
- If the card supports the read_wait signal, then assert it by setting the read_wait (bit 6) in the CTRL register @0x00.
- Clear the read_wait bit in the CTRL register.
- Controller/DMA/FIFO Reset Usage
- Controller reset – Resets the controller by setting the controller_reset bit (bit 0) in the CTRL register; this resets the CIU and state machines, and also resets the BIU-to-CIU interface. Since this reset bit is self-clearing, after issuing the reset, wait until this bit is cleared.
- FIFO reset - Resets the FIFO by setting the fifo_reset bit (bit 1) in the CTRL register; this resets the FIFO pointers and counters of the FIFO. Since this reset bit is self-clearing, after issuing the reset, wait until this bit is cleared.

In external DMA transfer mode, even when the FIFO pointers are reset, if there is a DMA transfer in progress, it could push or pop data to or from the FIFO; the DMA itself completes correctly. In order to clear the FIFO, the software should issue an additional FIFO reset and clear any FIFO underrun or overrun errors in the RAWINTS register caused by the DMA transfers after the FIFO was reset.

- Card Read Threshold

When an application needs to perform a Single or Multiple Block Read command, the application must program the CardThrCtl register with the appropriate Card Read Threshold size (CardRdThreshold) and set the Card Read Threshold Enable (CardRdThrEnable) bit to 1'b1. This additional programming ensures that the Host controller sends a Read Command only if there is space equal to the CardRDThreshold available in the Rx FIFO. This in turn ensures that the card clock is not stopped in the middle a block of data being transmitted from the card. The Card Read Threshold can be set to the block size of the transfer, which guarantees that there is a minimum of one block size of space in the RxFIFO before the controller enables the card clock.The Card Read Threshold is required when the Round Trip Delay is greater than 0.5cclk_in period.

- Error Handling

The Host Controller implements error checking; errors are reflected in the RAWINTS register@0x44 and can be communicated to the software through an interrupt, or the software can poll for these bits. Upon power-on, interrupts are disabled (int_enable in the CTRL register is 0), and all the interrupts are masked (bits 0-31 of the INTMASK register; default is 0).

Error handling:

- Response and data timeout errors – For response timeout, software can retry the command. For data timeout, the Host Controller has not received the data start bit – either for the first block or the intermediate block – within the timeout period, so software can either retry the whole data transfer again or retry from a specified block onwards. By reading the contents of the TCBCNT later, the software can decide how many bytes remain to be copied.
- Response errors – Set when an error is received during response reception. In this case, the response that copied in the response registers is invalid. Software can retry the command.

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- Data errors – Set when error in data reception are observed; for example, data CRC, start bit not found, end bit not found, and so on. These errors could be set for any block-first block, intermediate block, or last block. On receipt of an error, the software can issue a STOP or ABORT command and retry the command for either whole data or partial data.
- Hardware locked error – Set when the Host Controller cannot load a command issued by software. When software sets the start_cmd bit in the CMD register, the Host Controller tries to load the command. If the command buffer is already filled with a command, this error is raised. The software then has to reload the command.
- FIFO underrun/overrun error – If the FIFO is full and software tries to write data in the FIFO, then an overrun error is set. Conversely, if the FIFO is empty and the software tries to read data from the FIFO, an underrun error is set. Before reading or writing data in the FIFO, the software should read the fifo_empty or fifo_full bits in the Status register.
- Data starvation by host timeout – Raised when the Host Controller is waiting for software intervention to transfer the data to or from the FIFO, but the software does not transfer within the stipulated timeout period. Under this condition and when a read transfer is in process, the software should read data from the FIFO and create space for further data reception. When a transmit operation is in process, the software should fill data in the FIFO in order to start transferring data to the card.
- CRC Error on Command – If a CRC error is detected for a command, the CE-ATA device does not send a response, and a response timeout is expected from the Host Controller. The ATA layer is notified that an MMC transport layer error occurred.

Notes: During a multiple-block data transfer, if a negative CRC status is received from the device, the data path signals a data CRC error to the BIU by setting the data CRC error bit in the RINTSTS register. It then continues further data transmission until all the bytes are transmitted.

18.6.5 Voltage Switching

The Host Controller supports SD 3.0 Ultra High Speed (UHS-1) and is capable of voltage switching in SD-mode, which can be applied to SD High-Capacity (SDHC) and SD Extended Capacity (SDXC) cards. UHS-1 supports only 4-bit mode.
SD 3.0 UHS-1 supports the following transfer speed modes for UHS-50 and/or UHS-104 cards:

- DS – default-speed up to 25MHz, 3.3V signaling
- HS – high-speed up to 50MHz, 3.3V signaling
- SDR12 – SDR up to SDR 25MHz, 1.8V signaling
- SDR25 – SDR up to 50MHz, 1.8V signaling
- SDR50 – SDR up to 100MHz, 1.8V signaling
- DDR50 – DDR up to 50MHz, 1.8V signaling

Voltage selection can be done in only SD mode. The first CMD0 selects the bus mode-either SD mode or SPI mode. The card must be in SD mode in order for 1.8V signaling mode to apply, during which time the card cannot be switched to SPI mode or 3.3V signaling without a power cycle.
If the System BIOS in an embedded system already knows that it is connected to an SD 3.0 card, then the driver programs the Controller to initiate ACMD41. The software knows from the response of ACMD41 whether or not the card supports voltage switching to 1.8V.

- If bit 32 of ACMD41 response is 1'b1: card supports voltage switching and next command-CMD11-invokes voltage switching sequence. After CMD11 is started, the software must program the VOLT_REG register in the CSR space with the appropriate card number.
- If bit 32 of ACMD41 response is 1'b0: card does not support voltage switching and CMD11 should not be started.

If the card and host controller accept voltage switching, then they support UHS-1 modes of data transfer. After the voltage switch to 1.8V, SDR12 is the default speed.
Since the UHS-1 can be used in only 4-bit mode, the software must start ACMD6 and change

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the card data width to 4-bit mode; ACMD6 is driven in any of the UHS-1 speeds. If the host wants to select the DDR mode of data transfer, then the software must program the DDR_REG register in the CSR space with the appropriate card number.
To choose from any of the SDR or DDR modes, appropriate values should be programmed in the CLKDIV register.

- Voltage Switch Operation

The Voltage Switch operation must be performed in SD mode only.

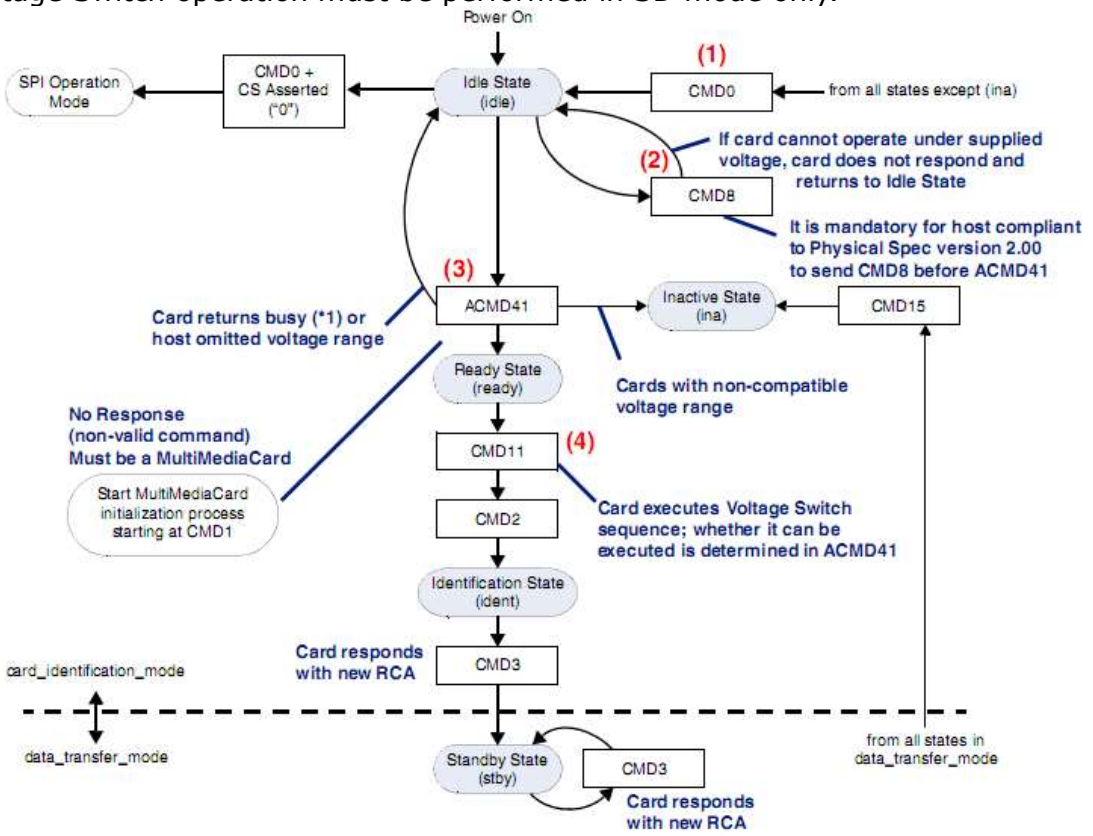

Fig. 18-12 Voltage Switching Command Flow Diagram

The following outlines the steps for the voltage switch programming sequence

- Software Driver starts CMD0, which selects the bus mode as SD.
- After the bus is in SD card mode, CMD8 is started in order to verify if the card is compatible with the SD Memory Card Specification, Version 2. 00. CMD8 determines if the card is capable of working within the host supply voltage specified in the VHS (19:16) field of the CMD; the card supports the current host voltage if a response to CMD8 is received.
- ACMD 41 is started. The response to this command informs the software if the card supports voltage switching; bits 38, 36, and 32 are checked by the card argument of ACMD41; refer to following figure.

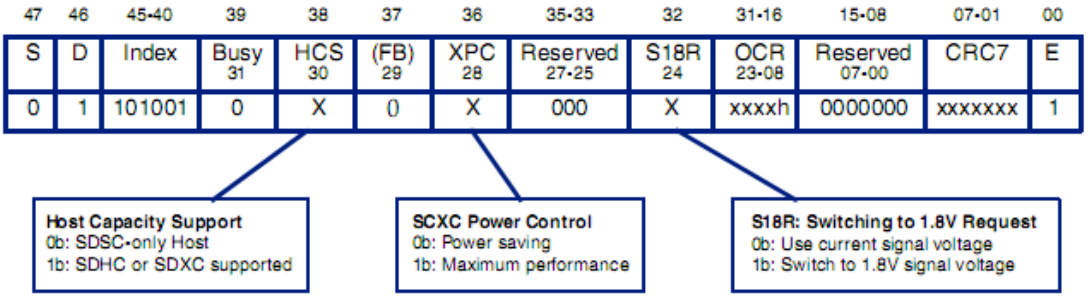

Fig. 18-13 ACMD41 Argument

- Bit 30 informs the card if host supports SDHC/SDXC or not; this bit should be set to 1'b1.
- Bit 28 can be either 1 or 0.
- Bit 24 should be set to 1'b1, indicating that the host is capable of voltage switching;

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refer to Figure 17-16.

47	46	45-40	39	38	37	36-33	32	31-16	15-08	07-01	00
S	D	Index	Busy 31	CCS 30	Rsvd 29	Reserved 28-25	S18R 24	OCR 23-08	Reserved 07-00	CRC7	E
0	0	111111	X	X	0	0000	X	xxxxh	0000000	1111111	1

Busy Status
0b: On Initialization
1b: Initialization complete

Card Capacity Status
0b: SDSC
1b: SCHC or SCXC

S18R: Switching to 1.8V Accepted
0b: Continues current voltage signalling
1b: Ready for switching signal voltage

Fig. 18-14 ACMD41 Response(R3)

- Bit 30 – If set to 1'b1, card supports SDHC/SDXC; if set to 1'b0, card supports only SDSC
- Bit 24 – If set to 1'b1, card supports voltage switching and is ready for the switch
- Bit 31 – If set to 1'b1, initialization is over; if set to 1'b0, means initialization in process
- If the card supports voltage switching, then the software must perform the steps discussed for either the "Voltage Switch Normal Scenario" or the "Voltage Switch Error Scenario".

- Voltage Switch Normal Scenario

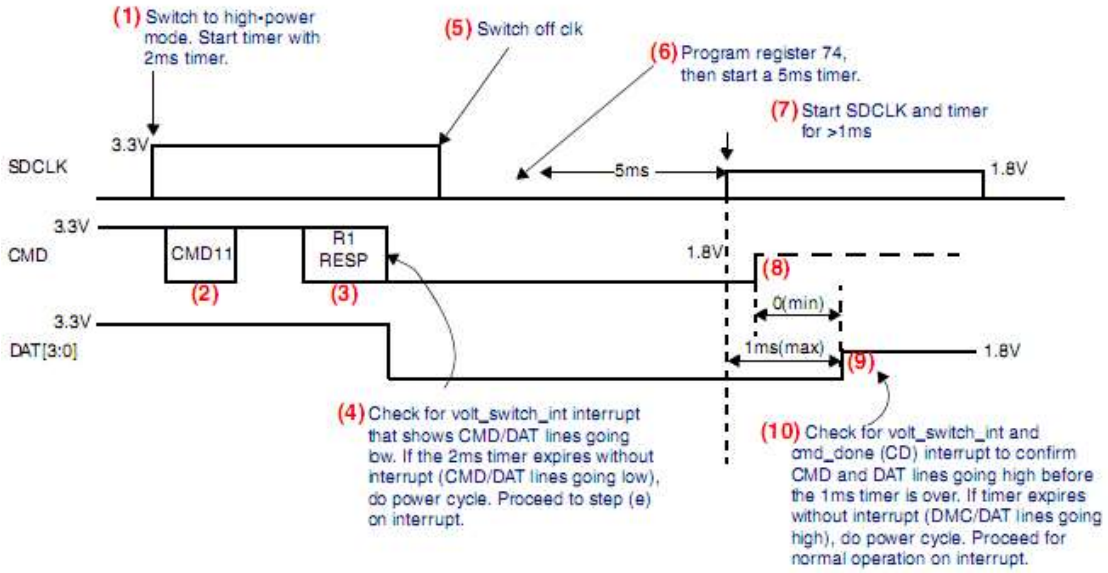

Fig. 18-15 Voltage Switch Normal Scenario

- The host programs CLKENA—cclk_low_power register—with zero (0) for the corresponding card, which makes the host controller move to high-power mode. The application should start a timer with a recommended value of 2ms; this value of 2 ms is determined as below:Total clk required for CMD11 = 48 clks
- Total clk required for RESP R1 = 48 clks
- Maximum clk delay between MCD11 end to start of RESP1 = 60 clks
- Total = 48+48 + 60 = 160
- Minimum frequency during enumeration is 100 KHz; that is, 10us
- Total time = 160 * 10us = 1600us = 1. 6ms ~ 2ms
- The host issues CMD11 to start the voltage switch sequence. Set bit 28 to 1'b1 in CMD when setting CMD11; for more information on setting bits, refer to "Boot Operation".
- The card returns R1 response; the host controller does not generate cmd_done interrupt on receiving R1 response.
- The card drives CMD and DAT [3:0] to low immediately after the response. The host controller generates interrupt (VOLT_SWITCH_INT) once the CMD or DAT [3:0] line goes low. The application should wait for this interrupt. If the 2ms timer expires

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without an interrupt (CMD/DAT lines going low), do a power cycle.

Note: Before doing a power cycle, switch off the card clock by programming CLKENA register

Proceed to step (5) on getting an interrupt (VOLT_SWITCH_INT).

Note: This interrupt must be cleared once this interrupt is received. Additionally, this interrupt should not be masked during the voltage switch sequence.

If the timer expires without interrupt (CMD/DAT lines going low), perform a power cycle. Proceed to step (5) on interrupt.

- Program the CLKENA, cclk_enable register, with 0 for the corresponding card; the host stops supplying SDCLK.
- Program VOLT_REG to the required values for the corresponding card. The application must program the newly-defined VOLT_REG register to assign 1 for the bit corresponding to the card number. The application should start a timer > 5ms.
- After the 5ms timer expires, the host voltage regulator is stable. Program CLKENA, cclk_enable register, with 1 for the corresponding card; the host starts providing SDCLK at 1. 8V; this can be at zero time after VOLT_REG has been programmed. When the CLKENA register is programmed, the application should start another timer > 1ms.
- By detecting SDCLK, the card drives CMD to high at 1. 8V for at least one clock and then stops driving (tri-state); CMD is triggered by the rising edge of SDCLK (SDR timing).
- If switching to 1. 8V signaling is completed successfully, the card drives DAT [3:0] to high at 1. 8V for at least one clock and then stops driving (tri-state); DAT [3:0] is triggered by the rising edge of SDCLK (SDR timing). DAT[3:0] must be high within 1ms from the start of SDCLK.
- The host controller generates a voltage switch interrupt (VOLT_SWITCH_INT) and a command done (CD) interrupt once the CMD and DAT[3:0] lines go high. The application should wait for this interrupt to confirm CMD and DAT lines going high before the 1ms timer is done.

If the timer expires without the voltage switch interrupt (VOLT_SWITCH_INT), a power cycle should be performed. Program the CLKENA register to stop the clock for the corresponding card number. Wait for the cmd_done (CD) interrupt. Proceed for normal operation on interrupt. After the sequence is completed, the host and the card start communication in SDR12 timing.

- Voltage Switch Error Scenario

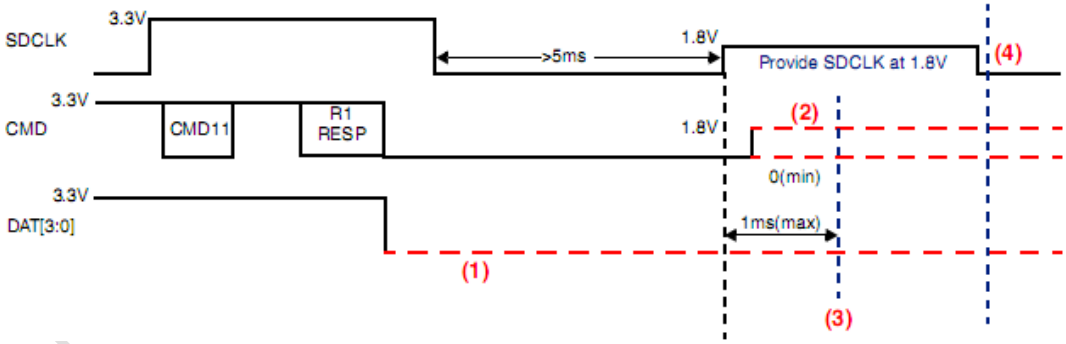

Fig. 18-16 Voltage Switch Error Scenario

- If the interrupt (VOLT_SWITCH_INT) does not come, then the 2 ms timer should time out and a power cycle should be initiated.

Note: Before performing a power cycle, switch off the card clock by programming CLKENA register; no cmd_done (CD) interrupt is generated.

Additionally, if the card detects a voltage error at any point in between steps (5) and (7) in Figure 17-17, the card keeps driving DAT[3:0] to low until card power off.

- CMD can be low or tri-state.
- The host controller generates a voltage switch interrupt once the CMD and DAT[3:0] lines go high. The application should check for an interrupt to confirm CMD and DAT lines going high before the 1 ms timer is done.

If the 1 ms timer expires without interrupt (VOLT_SWITCH_INT) and cmd_done (CD), a power cycle should be performed. Program the CLKENA register to stop SDCLK of the corresponding

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card. Wait for the cmd_done interrupt. Proceed for normal operation on interrupt.

- If DAT[3:0] is low, the host drives SDCLK to low and then stops supplying the card power.

Note: The card checks voltages of its own regulator output and host signals to ensure they are less than 2. 5V. Errors are indicated by (1) and (2) in Figure 7-18.

- If voltage switching is accepted by the card, the default speed is SDR12.
- Command Done is given:
- If voltage switching is properly done, CMD and DAT line goes high.
- If switching is not complete, the 1ms timer expires, and the card clk is switched off.

Note: No other CMD should be driven before the voltage switching operation is completed and Command Done is received.

- The application should use CMD6 to check and select the particular function; the function appropriate-speed should be selected.

After the function switches, the application should program the correct value in the CLKDIV register, depending on the function chosen. Additionally, if Function 0x4 of the Access mode is chosen—that is, DDR50, then the application should also program 1'b1 in DDR_REG for the card number that has been selected for DDR50 mode.

18.6.6 Back-End Power

Each device needs one bit to control the back-end power supply for an embedded device; this bit does not control the VDDH of the host controller. A back_end_power register enables software programming for back-end power. The value on this register is output to the back_end_power signal, which can be used to switch power on and off the embedded device.

18.6.7 DDR Operation

- 4-bit DDR Programming Sequence

DDR programming should be done only after the voltage switch operation has completed. The following outlines the steps for the DDR programming sequence:

- Once the voltage switch operation is complete, the user must program VOLT_REG to the required values for the corresponding card.
- To start a card to work in DDR mode, the application must program a bit of the newly defined VOLT_REG[31:16] register with a value of 1'b1.
- The bit that the user programs depends on which card is to be accessed in DDR mode.
- To move back to SDR mode, a power cycle should be run on the card—putting the card in SDR12 mode—and only then should VOLT_REG[31:16] be set back to 1'b0 for the appropriate card.

- 8-bit DDR Programming Sequence

The following outlines the steps for the 8-bit DDR programming sequence:

- The cclk_in signal should be twice the speed of the required cclk_out. Thus, if the cclk_out signal is required to be 50 MHz, the cclk_in signal should be 100 MHz.
- The CLKDIV register should always be programmed with a value higher than zero (0); that is, a clock divider should always be used for 8-bit DDR mode.
- The application must program the UHS_REG [31:16] register (DDR_REG bits) by assigning it with a value of 1 for the bit corresponding to the card number; this causes the selected card to start working in DDR mode.
- Depending on the card number, the CTYPE [31:16] bits should be set in order to make the host work in the 8-bit mode.
- eMMC4.5 DDR START Bit

The eMMC4.5 changes the START bit definition in the following manner:

- Receiver samples the START bit on the rising edge.
- On the next rising edge after sampling the START bit, the receiver must sample the data.
- Removes requirement of the START bit and END bit to be high for one full cycle.

Notes: The Host Controller does not support a START bit duration higher than one clock cycle. START bit durations of one or less than one clock cycle are supported and can be defined at the time of startup by programming the EMMC_DDR_REG register.

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Following figure illustrates cases for the definition change of the START bit with eMMC4.5; it also illustrates how some of these cases can fail in sampling when higher-value delays are considered for I/O PADs.

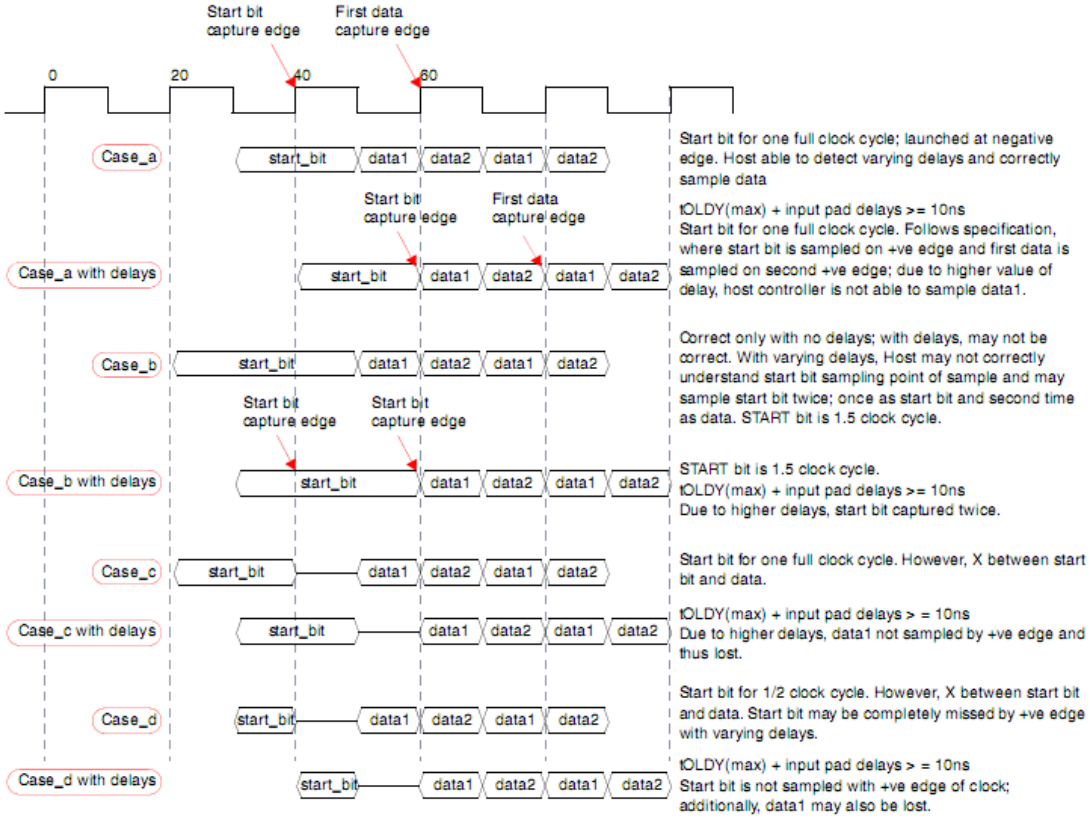

Fig. 18-17 CASES for eMMC 4.5 START bit

- Reset Command/Moving from DDR50 to SDR12

To reset the mode of operation from DDR50 to SDR12, the following sequence of operations has to be done by the application:

- Issue CMD0.

When CMD0 is received, the card changes from DDR50 to SDR12.

- Program the CLKDIV register with an appropriate value.
- Set DDR_REG to 0.

Note: The VOLT_REG register should not be programmed to 0 while switching from DDR50 to SDR12, since the card is still operating in 1.8V mode after receiving CMD0.

18.6.8 H/W Reset Operation

When the RST_n signal goes low, the card enters a pre-idle state from any state other than the inactive state.

H/W Reset Programming Sequence

The following outlines the steps for the H/W reset programming sequence:

- Program CMD12 to end any transfer in process.
- Wait for DTO, even if no response is sent back by the card.
- Set the following resets:
- DMA reset– CTRL[2]
- FIFO reset – CTRL[1] bits

Note: The above steps are required only if a transfer is in process.

- Program the CARD_RESET register with a value of 0; this can be done at any time when the card is connected to the controller. This programming asserts the RST_n signal and resets the card.
- Wait for minimum of 1 µs or cclk_in period, whichever is greater
- After a minimum of 1 µs, the application should program a value of 0 into the CARD_RESET register. This de-asserts the RST_n signal and takes the card out of reset.

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- The application can program a new CMD only after a minimum of 200 μs after the de-assertion of the RST_n signal, as per the MMC 4.41 standard.

Note: For backward compatibility, the RST_n signal is temporarily disabled in the card by default. The host may need to set the signal as either permanently enabled or permanently disabled before it uses the card.

18.6.9 FBE Scenarios

An FBE occurs due to an AHB error response on the AHB bus. This is a system error, so the software driver should not perform any further programming to the Host. The only recovery mechanism from such scenarios is to do one of the following:

- Issue a hard reset by asserting the reset_n signal
- Do a program controller reset by writing to the CTRL[0] register

FIFO Overflow and Underflow

During normal data transfer conditions, FIFO overflow and underflow will not occur. However if there is a programming error, then FIFO overflow/underflow can result. For example, consider the following scenarios.

- For transmit: PBL=4, Tx watermark = 1. For the above programming values, if the FIFO has only one location empty, it issues a dma_req to IDMAC FSM. Due to PBL value=4, the IDMAC FSM performs 4 pushes into the FIFO. This will result in a FIFO overflow interrupt.
- For receive: PBL=4, Rx watermark = 1. For the above programming values, if the FIFO has only one location filled, it issues a dma_req to IDMAC FSM. Due to PBL value=4, the IDMAC FSM performs 4 pops to the FIFO. This will result in a FIFO underflow interrupt.

The driver should ensure that the number of bytes to be transferred as indicated in the descriptor should be a multiple of 4bytes with respect to H_DATA_WIDTH=32. For example, if the BYTCNT = 13, the number of bytes indicated in the descriptor should be 16 for H_DATA_WIDTH=32.

Programming of PBL and Watermark Levels

The DMAC performs data transfers depending on the programmed PBL and threshold values.

Table 18-15 PBL and Watermark Levels

PBL (Number of transfers)	Tx/Rx Watermark Value
1	greater than or equal to 1
4	greater than or equal to 4
8	greater than or equal to 8
16	greater than or equal to 16
32	greater than or equal to 32
64	greater than or equal to 64
128	greater than or equal to 128
256	greater than or equal to 256

18.6.10 Variable Delay/Clock Generation

Variable delay mechanism for the cclk_in_drv is optional, but it can be useful in order to meet a range of hold-time requirements across modes. Variable delay mechanism for the cclk_in_sample is mandatory and is required to achieve the correct sampling point for data. cclk_in/cclk_in_sample/ cclk_in_drv is generated by Clock Generation Unit (CLKGEN) with variable delay mechanism, which includes Phase Shift Unit and Delay Line Unit selectable. The Phase Shift Unit can shift cclk_in_sample/cclk_in_drv by 0/90/180/270-degree relative to cclk_in, controlled by *sample_degree/drv_degree*.

The Delay Line Unit can shift cclk_in_sample/cclk_in_drv in the unit of 40ps~80ps for every delay element. The delay unit number is determined by *sample_delaynum/drv_delaynum*, and enabled by *sample_sel/drv_sel*.

cclk_in is generated by cclkin divided by 2. cclk_in_drv and cclk_in_sample clocks are phase-shifted with delayed versions of cclk_in. All clocks are recommended to have a 50%

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duty cycle; DDR modes must have 50% duty cycles.
The architecture is as follows.

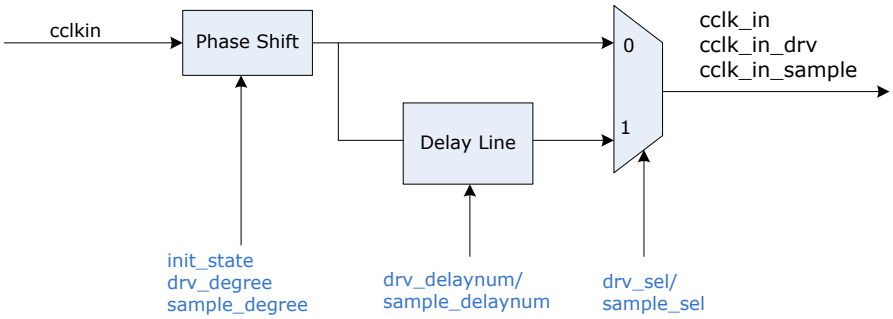

Fig. 18-18 Clock Generation Unit

The control signals for different Host Controller instance are shown as follows:

Table 18-16 Configuration for SDMMC Clock Generation

Signal Name	Source	Default	Description
init_state	CRU_SDMMC_CON0[0]	0	Soft initial state for phase shift.
drv_degree [1:0]	CRU_SDMMC_CON0[2:1]	2	Phase shift for cclk_in_drv. 0: 0-degree 1: 90-degree 2: 180-degree 3: 270-degree
drv_delaynum [7:0]	CRU_SDMMC_CON0[10:3]	0	Element number in delay line for cclk_in_drv.
drv_sel	CRU_SDMMC_CON0[11]	0	cclk_in_drv source selection: 0: use clock after phase_shift 1: use clock after phase_shift and delay line
sample_degree [1:0]	CRU_SDMMC_CON1[1:0]	0	Phase shift for cclk_in_sample. 0: 0-degree 1: 90-degree 2: 180-degree 3: 270-degree
sample_delaynum [7:0]	CRU_SDMMC_CON1[9:2]	0	Element number in delay line for cclk_in_sample
sample_sel	CRU_SDMMC_CON1[10]	0	cclk_in_sample source selection: 0: use clock after phase_shift 1: use clock after phase_shift and delay line

Table 18-17 Configuration for SDIO Clock Generation

Signal Name	Source	Default	Description
init_state	CRU_SDIO_CON0[0]	0	Soft initial state for phase shift.
drv_degree [1:0]	CRU_SDIO_CON0[2:1]	2	Phase shift for cclk_in_drv. 0: 0-degree 1: 90-degree 2: 180-degree 3: 270-degree

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drv_delaynum [7:0]	CRU_SDIO_CON0[10:3]	0	Element number in delay line for cclk_in_drv
drv_sel	CRU_SDIO_CON0[11]	0	cclk_in_drv source selection: 0: use clock after phase_shift 1: use clock after phase_shift and delay line
sample_degree [1:0]	CRU_SDIO_CON1[1:0]	0	Phase shift for cclk_in_sample. 0: 0-degree 1: 90-degree 2: 180-degree 3: 270-degree
sample_delaynum [7:0]	CRU_SDIO_CON1[9:2]	0	Element number in delay line for cclk_in_sample
sample_sel	CRU_SDIO_CON1[10]	0	cclk_in_sample source selection: 0: use clock after phase_shift 1: use clock after phase_shift and delay line

Table 18-18 Configuration for eMMC Clock Generation

Signal Name	Source	Default	Description
init_state	CRU_EMMC_CON0[0]	0	Soft initial state for phase shift.
drv_degree [1:0]	CRU_EMMC_CON0[2:1]	2	Phase shift for cclk_in_drv. 0: 0-degree 1: 90-degree 2: 180-degree 3: 270-degree
drv_delaynum [7:0]	CRU_EMMC_CON0[10:3]	0	Element number in delay line for cclk_in_drv
drv_sel	CRU_EMMC_CON0[11]	0	cclk_in_drv source selection: 0: use clock after phase_shift 1: use clock after phase_shift and delay line
sample_degree [1:0]	CRU_EMMC_CON1[1:0]	0	Phase shift for cclk_in_sample. 0: 0-degree 1: 90-degree 2: 180-degree 3: 270-degree
sample_delaynum [7:0]	CRU_EMMC_CON1[9:2]	0	Element number in delay line for cclk_in_sample
sample_sel	CRU_EMMC_CON1[10]	0	cclk_in_sample source selection: 0: use clock after phase_shift 1: use clock after phase_shift and delay line

The following outlines the steps for clock generation sequence:

- Assert init_state to soft reset the CLKGEN.
- Configure drv_degree/sample_degree.
- If fine adjustment required, delay line can be used by configuring

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drv_delaynum/sample_delaynum and drv_sel/sample_sel.
- Dis-assert init_state to start CLKGEN.

18.6.11 Variable Delay Tuning

Tuning is defined by SD and MMC cards to determine the correct sampling point required for the host, especially for the speed modes SDR104 and HS200 where the output delays from the cards can be up to 2 UI. Tuning is required for other speed modes-such as DDR50-even though the output delay from the card is less than one cycle.
Command for tuning is different for different cards.
- SD Memory Card:
- CMD19 – SD card for SDR50 and SDR104 speed modes. Tuning data is defined by card specifications.
- CMD6 – SD card for speed modes not supporting CMD19. Tuning data is the 64byte SD status.
- Multimedia Card:
- CMD21 – MMC card for HS200 speed mode. Tuning data is defined by card specifications.
- CMD8 – MMC card for speed modes not supporting CMD21. Tuning data is 512 byte ExtCSD data.

The following is the procedure for variable delay tuning:
- Set a phase shift of 0-degree on cclk_in_sample.
- Send the Tuning command to the card; the card in turn sends an R1 response on the CMD line and tuning data on the DAT line.
- If the host sees any of the errors—start bit error, data crc error, end bit error, data read time-out, response crc error, response error—then the sampling point is incorrect.
- Send CMD12 to bring the host controller state machines to idle.
- The card may treat CMD12 as an invalid command because the card has successfully sent the tuning data, and it cannot send a response.
- The host controller may generate a response time-out interrupt that must be cleared by software.
- Repeat steps 2) to 4) by increasing the phase shift value or delay element number on cclk_in_sample until the correct sampling point is received such that the host does not see any of the errors.
- Mark this phase shift value as the starting point of the sampling window.
- Repeat steps 2 to 4 by increasing the phase shift value or delay element number on cclk_in_sample until the host sees the errors starting to come again or the phase shift value reaches 360-degree.
- Mark the last successful phase shift value as the ending point of the sampling window.

A window is established where the tuning block is matched. For example, for a scenario where the tuning block is received correctly for a phase shift window of 90-degree and 180-degree, then an appropriate sampling point is established as 135-degree. Once a sampling point is established, no errors should be visible in the tuning block.

18.6.12 Package Command

In order to reduce overhead, read and write commands can be packed in groups of commands-either all read or all write-that transfer the data for all commands in the group in one transfer on the bus.
Packed commands can be of two types:
- Packed Write: CMD23 → CMD25
- Packed Read: CMD23 → CMD25 → CMD23 → CMD18

Packed commands are put in packets by the application software and are transparent to the core. For more information on packed commands, refer to the eMMC specification.

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18.6.13 Card Detection Method

There are many methods for SDMMC/SDIO card detection.
(1) Method1: Using CDETECT register, which is value on card_detect_n input port. 0 represents presence of card. Commonly for SDMMC/SDIO.
(2) Method2: Using card detection unit, outputting host interrupt (*IRQ_ID[64]/[65]*). The card detection unit looks for any changes in the card-detect signals for card insertion or card removal. It filters out the debounces associated with mechanical insertion or removal, and generates one interrupt to the host. You can program the debounce filter value in DEBNCE[23:0].Following figure illustrates the timing for card-detect signals. Commonly for SDMMC/SDIO.

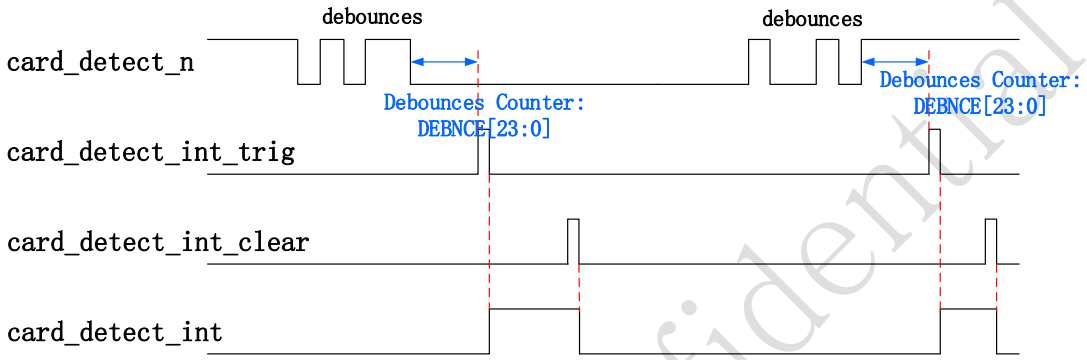

Fig. 18-19 Card Detection Method 2

- Method3: Using card detection unit in GRF, outputting *sdmmc_detect_dual edge_int(IRQ_ID[138])*, only available for SDMMC. Similar to Method2, except that the debounce is selecting from 5ms/15ms/35ms/50ms; and the insertion/removal detection interrupt can be enabled or cleared respectively. The detailed register information is:

Table 18-19 Register for SDMMC Card Detection Method 3

Signal Name	Source	Default	Description
sd_detect_rising_edge_dectect_en	GRF_SIG_DETECT_CON [0]	0	sdmmc detect_n signal rise edge interrupt enable. 1'b1: enable 1'b0: disable
sd_detect_fall_edge_detect_en	GRF_SIG_DETECT_CON [1]	0	sdmmc detect_n signal fall edge interrupt enable. 1'b1: enable 1'b0: disable
sd_detect_filter_time_sel	GRF_SIG_DETECT_CON [15:14]	0	the time select for sd card detect filter: 2'b00: 5ms 2'b01: 15ms 2'b10: 35ms 2'b11: 50ms
sd_detect_rising_edge_dectect_clr	SIG_DETECT_CLR [0]	0	sdmmc detect_n signal rise edge interrupt clear. 1'b1: enable 1'b0: disable
sd_detect_fall_edge_detect_clr	SIG_DETECT_CLR [1]	0	sdmmc detect_n signal fall edge interrupt clear.

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			1'b1: enable 1'b0: disable
sd_detect_rising_edge_dectect_status	SIG_DETECT_STATUS[0]	0	sdmmc detect_n signal rise edge interrupt status. 1'b1: enable 1'b0: disable
sd_detect_fall_edge_detect_status	SIG_DETECT_STATUS[1]	0	sdmmc detect_n signal fall edge interrupt status. 1'b1: enable 1'b0: disable

- Method4: Using card_detect_n for interrupt source, connecting to *IRQ_ID[131]/[132]* directly. Commonly for SDMMC/SDIO.

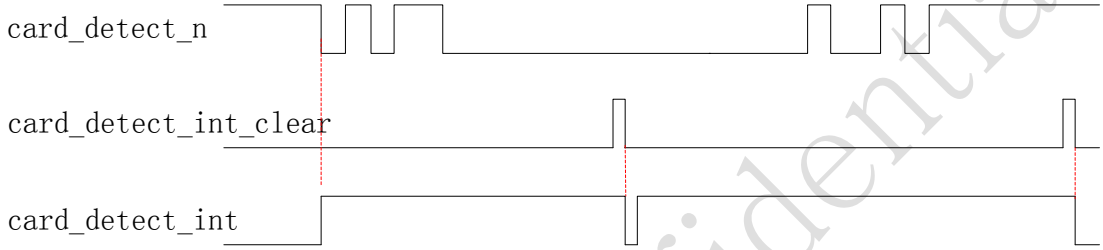

Fig. 18-20 Card Detection Method 4

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Chapter 19 Embedded SRAM

19.1 Overview

The Embedded SRAM is the AXI slave device, which supports read and write access to provide system fast access data storage.

19.1.1 Features supported

- Provide 64KB access space
- Support security and non-security access
- Security or non-security space is software programmable
- Security space is nx4KB(up to whole memory space)
- Support 64bit AXI bus
-

19.1.2 Features not supported

- Don’t support AXI lock transaction
- Don’t support AXI exclusive transaction
- Don’t support AXI cache function
- Don’t support AXI protection function
-

19.2 Block Diagram

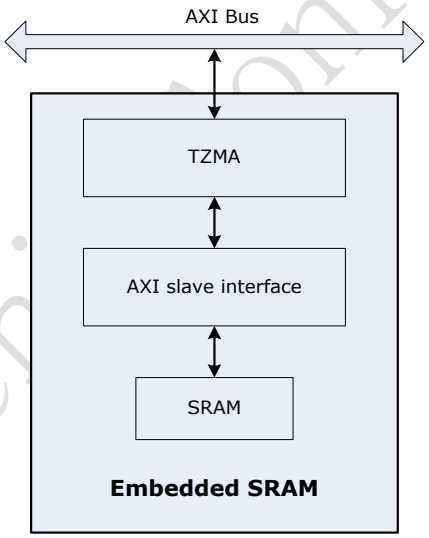

Fig. 19-1 Embedded SRAM block diagram

19.3 Function Description

19.3.1 TZMA

Please refer to 7.3.3 for TZMA functional description.

19.3.2 AXI slave interface

The AXI slave interface is bridge which translate AXI bus access to SRAM interface.

19.3.3 Embedded SRAM access path

The Embedded SRAM can only be accessed by Cortex-A53, DMAC_BUS and CRYPTO.

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19.3.4 Remap

The Embedded SRAM support remap.
Before remap, the Embedded SRAM address range is 0xff8c_0000~0xff8c_ffff,
After set remap, (ref Security GRF register SGRF_SCON3, bit[6]), the system can still access the Embedded SRAM by the old address. at same time, the system also can access the Embedded SRAM by the new address 0xffff_0000 ~ 0xffff_ffff (include the bootaddr)

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Chapter 20 I2S/PCM Controller (8 channel)

20.1 Overview

The I2S/PCM controller is designed for interfacing between the AHB bus and the I2S bus. The I2S bus (Inter-IC sound bus) is a serial link for digital audio data transfer between devices in the system and be invented by Philips Semiconductor. Now it is widely used by many semiconductor manufacturers.

Devices often use the I2S bus are ADC, DAC, DSP, CPU, etc. With the I2S interface, we can connect audio devices and the embedded SoC platform together and provide an audio interface solution for the system.

Not only I2S but also PCM mode surround audio output (up to 7.1channel) and stereo input are supported in I2S/PCM controller.

- Support five internal 32-bit wide and 32-location deep FIFOs, four for transmitting and one for receiving audio data
- Support AHB bus interface
- Support 16 ~ 32 bits audio data transfer
- Support master and slave mode
- Support DMA handshake interface and configurable DMA water level
- Support transmit FIFO empty, underflow, receive FIFO full, overflow interrupt and all interrupts can be masked
- Support configurable water level of transmit FIFO empty and receive FIFO full interrupt
- Support combine interrupt output
- Support 2,4,6,8 channels audio transmitting in I2S and PCM mode
- Support 2 channels audio receiving in I2S and PCM mode
- Support up to 192kHz sample rate
- Support I2S normal, left and right justified mode serial audio data transfer
- Support PCM early, late1, late2, late3 mode serial audio data transfer
- Support MSB or LSB first serial audio data transfer
- Support 16 to 31 bit audio data left or right justified in 32-bit wide FIFO
- Support two 16-bit audio data store together in one 32-bit wide location
- Support 2 independent LRCK signals, one for receiving and one for transmitting audio data
- Support configurable SCLK and LRCK polarity
- Support SCLK is equivalent to MCLK divided by an even number range from 2 to 64 in master mode

20.2 Block Diagram

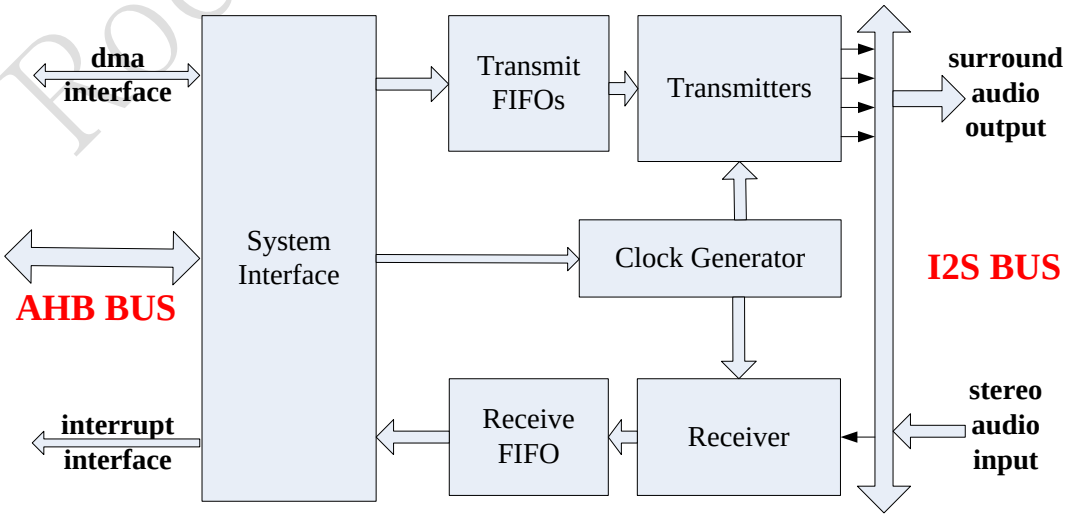

Fig. 20-1 I2S/PCM controller (8 channel) Block Diagram

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System Interface
The system interface implements the AHB slave operation. It contains not only control registers of transmitters and receiver inside but also interrupt and DMA handshake interface.
Clock Generator
The Clock Generator implements clock generation function. The input source clock to the module is MCLK_I2S, and by the divider of the module, the clock generator generates SCLK and LRCK to transmitter and receiver.
Transmitters
The Transmitters implement transmission operation. The transmitters can act as either master or slave, with I2S or PCM mode surround (up to 7.1 channel) serial audio interface.
Receiver
The Receiver implements receive operation. The receiver can act as either master or slave, with I2S or PCM mode stereo serial audio interface.
Transmit FIFOs
The Transmit FIFOs are the buffer to store transmitted audio data. Each of the size of the four FIFOs is 32bits x 32.
Receive FIFO
The Receive FIFO is the buffer to store received audio data. The size of the FIFO is 32bits x 32.

20.3 Function description

In the I2S/PCM controller, there are four conditions: transmitter-master & receiver-master; transmitter-master & receiver-slave; transmitter-slave & receiver-master; transmitter-slave & receiver-slave.

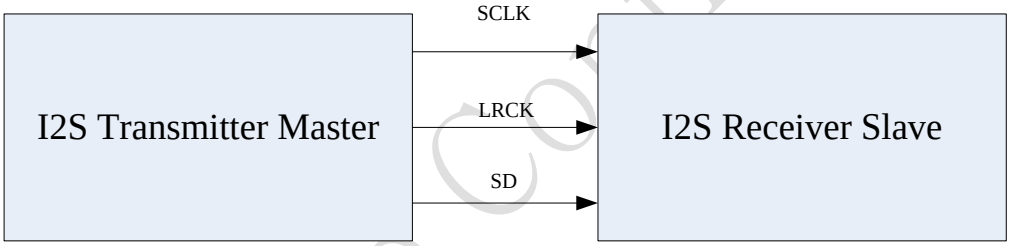

Fig. 20-2 I2S transmitter-master & receiver-slave condition

When transmitter acts as a master, it sends all signals to receiver (slave), and CPU control when to send clock and data to the receiver. When acting as a slave, SD signal still goes from transmitter to receiver, but SCLK and LRCK signals are from receiver (master) to transmitter. Based on three interface specifications, transmitting data should be ready before transmitter receives SCLK and LRCK signals. CPU should know when the receiver to initialize a transaction and when to send data.

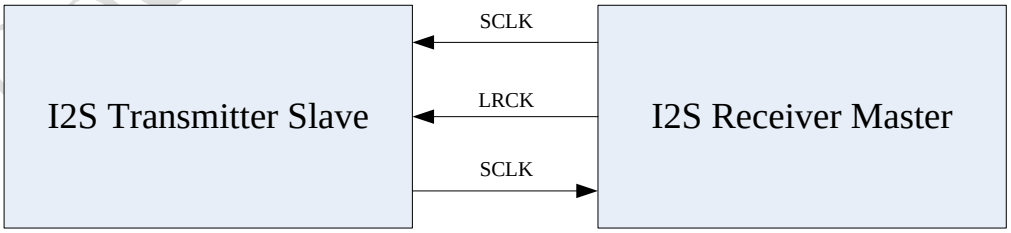

Fig. 20-3 I2S transmitter-slave& receiver-master condition

When the receiver acts as a master, it sends SCLK and LRCK signals to the transmitter (slave) and receives serial data. So CPU must tell the transmitter when to start a transaction for it to prepare transmitting data then start a transfer and send clock and channel-select signals. When the receiver acts as a slave, CPU should only do initial setting and wait for all signals and then start reading data.
Before transmitting or receiving data, CPU need do initial setting to the I2S register. These includes CPU settings, I2S interface registers settings, and maybe the embedded SoC platform settings. These registers must be set before starting data transfer.

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20.3.1 i2s normal mode

This is the waveform of I2S normal mode. For LRCK (i2s_lrck_rx/i2s_lrck_tx0) signal, it goes low to indicate left channel and high to right channel. For SD (i2s_sdo0, i2s_sdo1, i2s_sdo2, i2s_sdo3, i2s_sdi) signal, it transfers MSB or LSB first and sends the first bit one SCLK clock cycle after LRCK changes. The range of SD signal width is from 16 to 32bits.

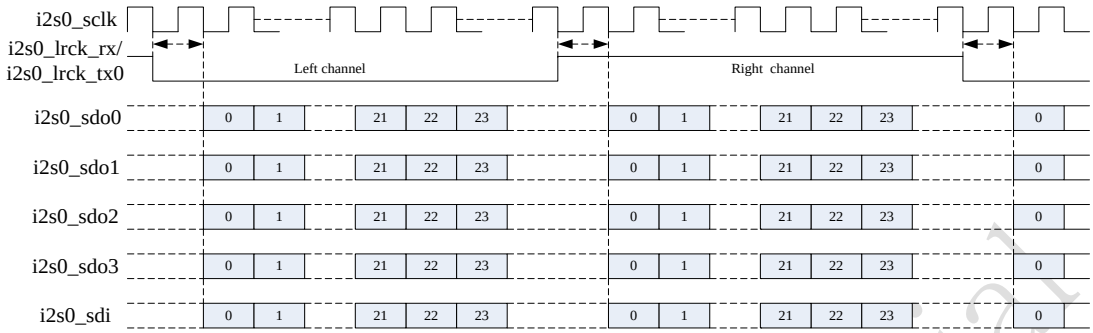

Fig. 20-4 I2S normal mode timing format

20.3.2 i2s left justified mode

This is the waveform of I2S left justified mode. For LRCK (i2s_lrck_rx / i2s_lrck_tx0) signal, it goes high to indicate left channel and low to right channel. For SD (i2s_sdo0, i2s_sdo1, i2s_sdo2, i2s_sdo3, i2s_sdi) signal, it transfers MSB or LSB first and sends the first bit at the same time when LRCK changes. The range of SD signal width is from 16 to 32bits.

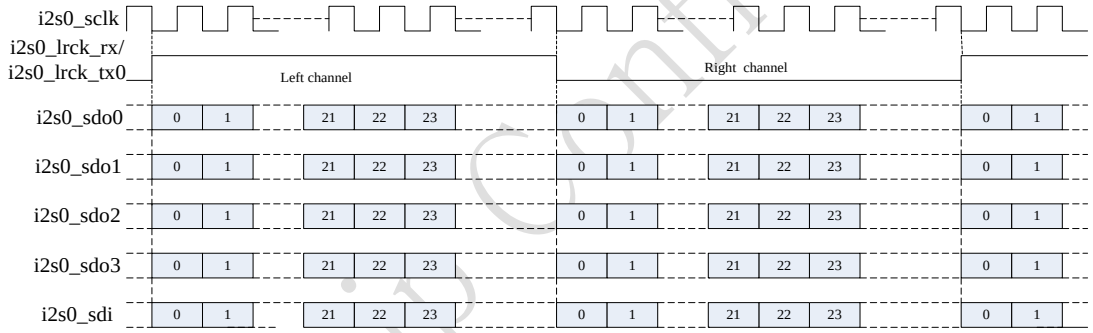

Fig. 20-5 I2S left justified mode timing format

20.3.3 i2s right justified mode

This is the waveform of I2S right justified mode. For LRCK (i2s_lrck_rx / i2s_lrck_tx0) signal, it goes high to indicate left channel and low to right channel. For SD (i2s_sdo0, i2s_sdo1, i2s_sdo2, i2s_sdo3, i2s_sdi) signal, it transfers MSB or LSB first; but different from I2S normal or left justified mode, its data is aligned to last bit at the edge of the LRCK signal. The range of SD signal width is from 16 to 32bits.

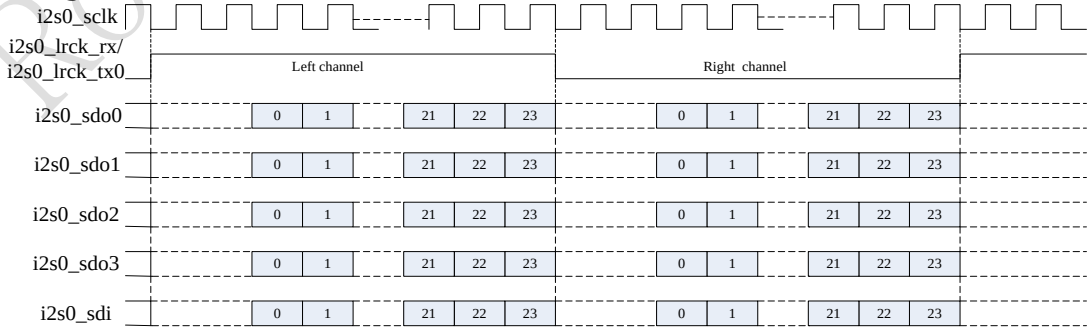

Fig. 20-6 I2S right justified modetiming format

20.3.4 PCM early mode

This is the waveform of PCM early mode. For LRCK (i2s_lrck_rx / i2s_lrck_tx0) signal, it goes high to indicate the start of a group of audio channels. For SD (i2s_sdo0, i2s_sdi) signal, it transfers MSB or LSB first and sends the first bit at the same time when LRCK goes high. The range of SD signal width is from 16 to 32bits.

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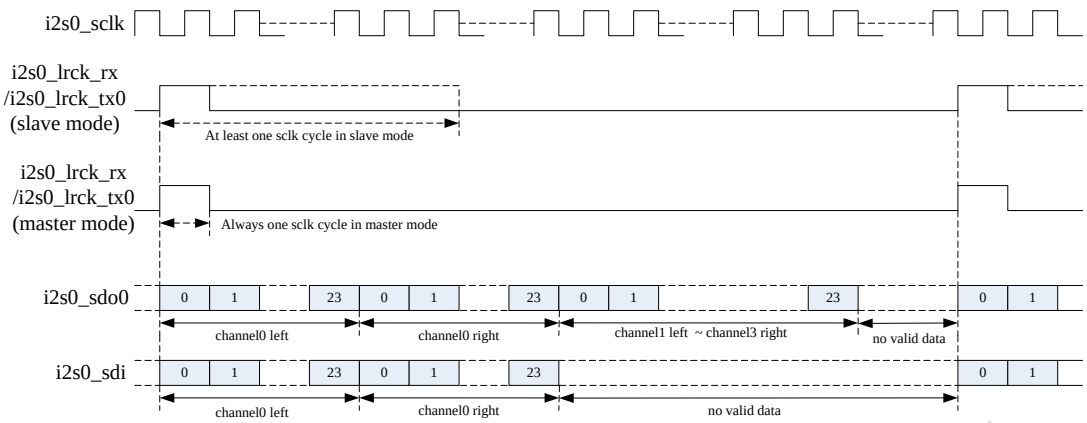

Fig. 20-7 PCM early modetiming format

20.3.5 PCM late1 mode

This is the waveform of PCM early mode. For LRCK (i2s_lrck_rx / i2s_lrck_tx0) signal, it goes high to indicate the start of a group of audio channels. For SD (i2s_sdo0, i2s_sdi) signal, it transfers MSB or LSB first and sends the first bit one SCLK clock cycle after LRCK goes high. The range of SD signal width is from 16 to 32bits.

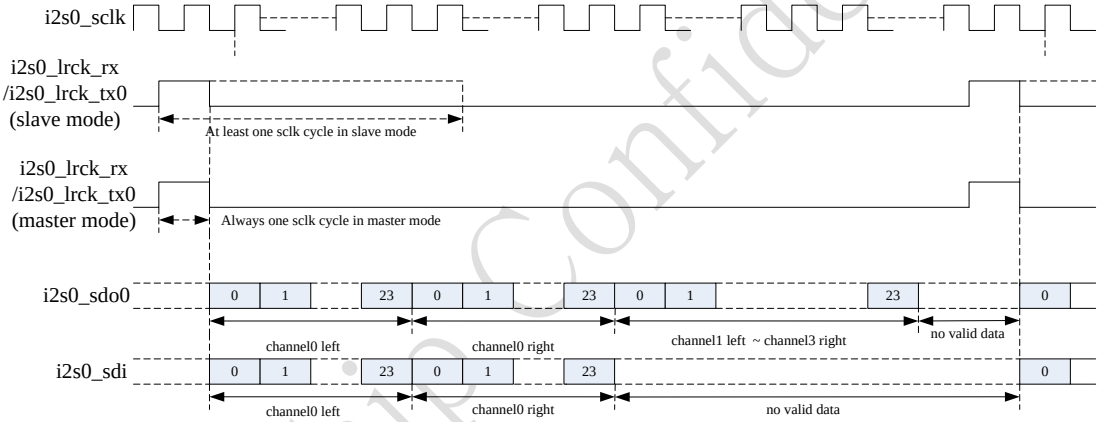

Fig. 20-8 PCM late1 modetiming format

20.3.6 PCM late2 mode

This is the waveform of PCM early mode. For LRCK (i2s_lrck_rx / i2s_lrck_tx0) signal, it goes high to indicate the start of a group of audio channels. For SD (i2s_sdo0, i2s_sdi) signal, it transfers MSB or LSB first and sends the first bit two SCLK clock cycles after LRCK goes high. The range of SD signal width is from 16 to 32bits.

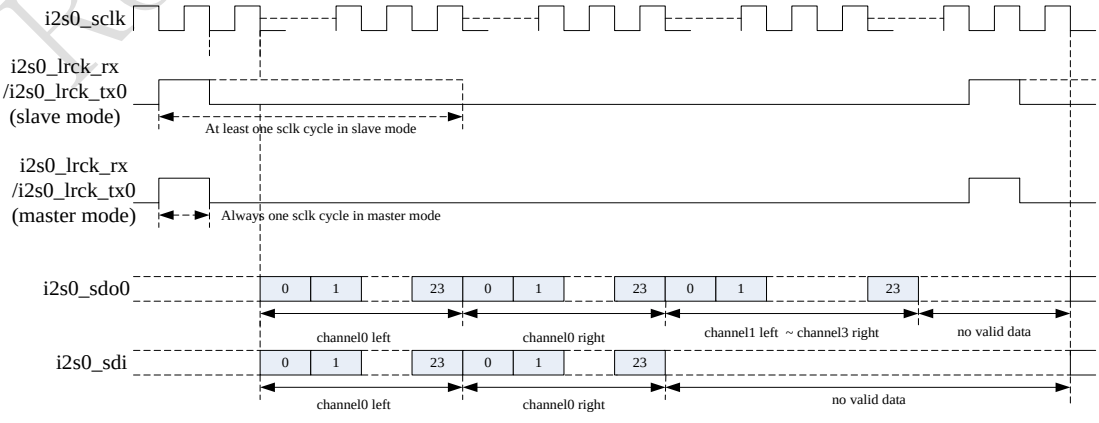

Fig. 20-9 PCM late2 modetiming format

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20.3.7 PCM late3 mode

This is the waveform of PCM early mode. For LRCK (i2s_lrck_rx / i2s_lrck_tx0) signal, it goes high to indicate the start of a group of audio channels. For SD (i2s_sdo0, i2s_sdi) signal, it transfers MSB or LSB first and sends the first bit three SCLK clock cycles after LRCK goes high. The range of SD signal width is from 16 to 32bits.

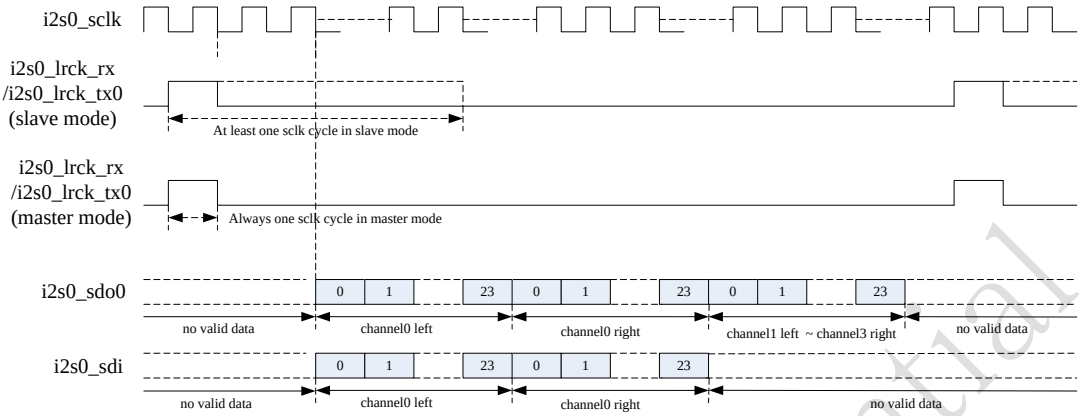

Fig. 20-10 PCM late3 modetiming format

20.4 Register Description

This section describes the control/status registers of the design.

20.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
I2S_TXCR	0x0000	W	0x0000000f	transmit operation control register
I2S_RXCR	0x0004	W	0x0000000f	receive operation control register
I2S_CKR	0x0008	W	0x00071f1f	clock generation register
I2S_FIFOLR	0x000c	W	0x00000000	FIFO level register
I2S_DMACR	0x0010	W	0x001f0000	DMA control register
I2S_INTCR	0x0014	W	0x01f00000	interrupt control register
I2S_INTSR	0x0018	W	0x00000000	interrupt status register
I2S_XFER	0x001c	W	0x00000000	Transfer Start Register
I2S_CLR	0x0020	W	0x00000000	SCLK domain logic clear Register
I2S_TXDR	0x0024	W	0x00000000	Transimt FIFO Data Register
I2S_RXDR	0x0028	W	0x00000000	Receive FIFO Data Register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

20.4.2 Detail Register Description

I2S_TXCR

Address: Operational Base + offset (0x0000)

transmit operation control register

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
22:17	RW	0x00	RCNT right jusitified counter (Can be written only when XFER[0] bit is 0.) Only vailid in I2S Right justified format and slave tx mode is selected. Start to tramsmit data RCNT sclk cycles after left channel valid.
16:15	RW	0x0	CSR Channel select register (Can be written only when XFER[0] bit is 0.) 0:channel 0 enable 1:channel 0 & channel 1 enable 2:channel 0 & channel 1 & channel 2 enable 3:channel 0 & channel 1 & channel 2 & channel 3 enable
14	RW	0x0	HWT Halfword word transform (Can be written only when XFER[0] bit is 0.) Only valid when VDW select 16bit data. 0:32 bit data valid from AHB/APB bus. Low 16 bit for left channel and high 16 bit for right channel. 1:low 16bit data valid from AHB/APB bus, high 16 bit data invalid.
13	RO	0x0	reserved
12	RW	0x0	SJM Store justified mode (Can be written only when XFER[0] bit is 0.) 16bit~31bit DATA stored in 32 bits width fifo. If VDW select 16bit data, this bit is valid only when HWT select 0.Because if HWT is 1, every fifo unit contain two 16bit data and 32 bit space is full, it is impossible to choose justified mode. 0:right justified 1:left justified
11	RW	0x0	FBM First Bit Mode (Can be written only when XFER[0] bit is 0.) 0:MSB 1:LSB
10:9	RW	0x0	IBM I2S bus mode (Can be written only when XFER[0] bit is 0.) 0:I2S normal 1:I2S Left justified 2:I2S Right justified 3:reserved

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Bit	Attr	Reset Value	Description
8:7	RW	0x0	PBM PCM bus mode (Can be written only when XFER[0] bit is 0.) 0:PCM no delay mode 1:PCM delay 1 mode 2:PCM delay 2 mode 3:PCM delay 3 mode
6	RO	0x0	reserved
5	RW	0x0	TFS Transfer format select (Can be written only when XFER[0] bit is 0.) 0: I2S format 1: PCM format
4:0	RW	0x0f	VDW Valid Data width (Can be written only when XFER[0] bit is 0.) 0~14:reserved 15:16bit 16:17bit 17:18bit 18:19bit …… 28:29bit 29:30bit 30:31bit 31:32bit

I2S_RXCR

Address: Operational Base + offset (0x0004)

receive operation control register

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14	RW	0x0	HWT Halfword word transform (Can be written only when XFER[1] bit is 0.) Only valid when VDW select 16bit data. 0:32 bit data valid to AHB/APB bus. Low 16 bit for left channel and high 16 bit for right channel. 1:low 16bit data valid to AHB/APB bus, high 16 bit data invalid.
13	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
12	RW	0x0	SJM Store justified mode (Can be written only when XFER[1] bit is 0.) 16bit~31bit DATA stored in 32 bits width fifo. If VDW select 16bit data, this bit is valid only when HWT select 0.Because if HWT is 1, every fifo unit contain two 16bit data and 32 bit space is full, it is impossible to choose justified mode. 0:right justified 1:left justified
11	RW	0x0	FBM First Bit Mode (Can be written only when XFER[1] bit is 0.) 0:MSB 1:LSB
10:9	RW	0x0	IBM I2S bus mode (Can be written only when XFER[1] bit is 0.) 0:I2S normal 1:I2S Left justified 2:I2S Right justified 3:reserved
8:7	RW	0x0	PBM PCM bus mode (Can be written only when XFER[1] bit is 0.) 0:PCM no delay mode 1:PCM delay 1 mode 2:PCM delay 2 mode 3:PCM delay 3 mode
6	RO	0x0	reserved
5	RW	0x0	TFS Transfer format select (Can be written only when XFER[1] bit is 0.) 0:i2s 1:pcm

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Bit	Attr	Reset Value	Description
4:0	RW	0x0f	VDW Valid Data width (Can be written only when XFER[1] bit is 0.) 0~14:reserved 15:16bit 16:17bit 17:18bit 18:19bit …… 28:29bit 29:30bit 30:31bit 31:32bit

I2S_CKR
Address: Operational Base + offset (0x0008)
clock generation register

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27	RW	0x0	MSS Master/slave mode select (Can be written only when XFER[1] or XFER[0] bit is 0.) 0:master mode(sclk output) 1:slave mode(sclk input)
26	RW	0x0	CKP Sclk polarity (Can be written only when XFER[1] or XFER[0] bit is 0.) 0: sample data at posedge sclk and drive data at negedge sclk 1: sample data at negedge sclk and drive data at posedge sclk
25	RW	0x0	RLP Receive lrck polarity (Can be written only when XFER[1] or XFER[0] bit is 0.) 0:normal polartiy (I2S normal: low for left channel, high for right channel I2S left/right just: high for left channel, low for right channel PCM start signal:high valid) 1:oppsite polarity (I2S normal: high for left channel, low for right channel I2S left/right just: low for left channel, high for right channel PCM start signal:low valid)

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Bit	Attr	Reset Value	Description
24	RW	0x0	TLP Transmit lrck polarity (Can be written only when XFER[1] or XFER[0] bit is 0.) 0:normal polartiy (I2S normal: low for left channel, high for right channel I2S left/right just: high for left channel, low for right channel PCM start signal:high valid) 1:oppsite polarity (I2S normal: high for left channel, low for right channel I2S left/right just: low for left channel, high for right channel PCM start signal:low valid)
23:16	RW	0x07	MDIV mclk divider (Can be written only when XFER[1] or XFER[0] bit is 0.) Serial Clock Divider = Fmclk / Ftxsclk-1.(mclk frequecy / txsclk frequecy-1) 0 :Fmclk=Ftxsclk; 1 :Fmclk=2*Ftxsclk; 2,3 :Fmclk=4*Ftxsclk; 4,5 :Fmclk=6*Ftxsclk; …… 60,61:Fmclk=62*Ftxsclk; 62,63:Fmclk=64*Ftxsclk; …… 252,253:Fmclk=254*Ftxsclk; 254,255:Fmclk=256*Ftxsclk;
15:8	RW	0x1f	RSD Receive sclk divider (Can be written only when XFER[1] or XFER[0] bit is 0.) Receive sclk divider= Fsclk/Frxlrck 0~30:reserved 31: 32fs 32: 33fs 33: 34fs 34: 35fs …… 253: 254fs 254: 255fs 255: 256fs

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Bit	Attr	Reset Value	Description
7:0	RW	0x1f	TSD Transmit sclk divider (Can be written only when XFER[1] or XFER[0] bit is 0.) Transmit sclk divider=Ftxsclk/Ftxlrck 0~30:reserved 31: 32fs 32: 33fs 33: 34fs 34: 35fs …… 253: 254fs 254: 255fs 255: 256fs

I2S_FIFOLR
Address: Operational Base + offset (0x000c)
FIFO level register

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:24	RO	0x00	RFL Receive FIFO Level Contains the number of valid data entries in the receive FIFO.
23:18	RO	0x00	TFL3 Transmit FIFO3 Level Contains the number of valid data entries in the transmit FIFO3.
17:12	RO	0x00	TFL2 Transmit FIFO2 Level Contains the number of valid data entries in the transmit FIFO2.
11:6	RO	0x00	TFL1 Transmit FIFO1 Level Contains the number of valid data entries in the transmit FIFO1.
5:0	RO	0x00	TFL0 Transmit FIFO0 Level Contains the number of valid data entries in the transmit FIFO0.

I2S_DMACR
Address: Operational Base + offset (0x0010)
DMA control register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
24	RW	0x0	RDE Receive DMA Enable 0 : Receive DMA disabled 1 : Receive DMA enabled
23:21	RO	0x0	reserved
20:16	RW	0x1f	RDL Receive Data Level This bit field controls the level at which a DMA request is made by the receive logic. The watermark level = DMARDL+1; that is, dma_rx_req is generated when the number of valid data entries in the receive FIFO is equal to or above this field value + 1.
15:9	RO	0x0	reserved
8	RW	0x0	TDE Transmit DMA Enable 0 : Transmit DMA disabled 1 : Transmit DMA enabled
7:5	RO	0x0	reserved
4:0	RW	0x00	TDL Transmit Data Level This bit field controls the level at which a DMA request is made by the transmit logic. It is equal to the watermark level; that is, the dma_tx_req signal is generated when the number of valid data entries in the TXFIFO(TXFIFO0 if CSR=00;TXFIFO1 if CSR=01,TXFIFO2 if CSR=10,TXFIFO3 if CSR=11)is equal to or below this field value.

I2S_INTCR

Address: Operational Base + offset (0x0014)

interrupt control register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24:20	RW	0x1f	RFT Receive FIFO Threshold When the number of receive FIFO entries is more than or equal to this threshold plus 1, the receive FIFO full interrupt is triggered.
19	RO	0x0	reserved
18	WO	0x0	RXOIC RX overrun interrupt clear Write 1 to clear RX overrun interrupt.

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Bit	Attr	Reset Value	Description
17	RW	0x0	RXOIE RX overrun interrupt enable 0:disable 1:enable
16	RW	0x0	RXFIE RX full interrupt enable 0:disable 1:enable
15:9	RO	0x0	reserved
8:4	RW	0x00	TFT Transmit FIFO Threshold When the number of transmit FIFO (TXFIFO0 if CSR=00; TXFIFO1 if CSR=01, TXFIFO2 if CSR=10, TXFIFO3 if CSR=11) entries is less than or equal to this threshold, the transmit FIFO empty interrupt is triggered.
3	RO	0x0	reserved
2	WO	0x0	TXUIC TX underrun interrupt clear Write 1 to clear TX underrun interrupt.
1	RW	0x0	TXUIE TX underrun interrupt enable 0:disable 1:enable
0	RW	0x0	TXEIE TX empty interrupt enable 0:disable 1:enable

I2S_INTSR
Address: Operational Base + offset (0x0018)
interrupt status register

Bit	Attr	Reset Value	Description
31:18	RO	0x0	reserved
17	RO	0x0	RXOI RX overrun interrupt 0:inactive 1:active
16	RO	0x0	RXFI RX full interrupt 0:inactive 1:active
15:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1	RO	0x0	TXUI TX underrun interrupt 0:inactive 1:active
0	RO	0x0	TXEI TX empty interrupt 0:inactive 1:active

I2S_XFER
Address: Operational Base + offset (0x001c)
Transfer Start Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	RXS RX Transfer start bit 0:stop RX transfer. 1:start RX transfer
0	RW	0x0	TXS TX Transfer start bit 0:stop TX transfer. 1:start TX transfer

I2S_CLR
Address: Operational Base + offset (0x0020)
SCLK domain logic clear Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	RXC RX logic clear This is a self cleard bit. Write 1 to clear all receive logic.
0	RW	0x0	TXC TX logic clear This is a self cleard bit. Write 1 to clear all transmit logic.

I2S_TXDR
Address: Operational Base + offset (0x0400~0x7FC)
Transimt FIFO Data Register

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	TXDR Transimt FIFO Data Register When it is written to, data are moved into the transmit FIFO.

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I2S_RXDR
Address: Operational Base + offset (0x0800~0xBFC)
Receive FIFO Data Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	RXDR Receive FIFO Data Register When the register is read, data in the receive FIFO is accessed.

20.5 Interface description

Module Pin	Dir	Pad Name	IOMUX Setting
i2s_sdi	I	IO_I2Ssdi_AUDIOgpio2b7	GRF_GPIO2B_IOMUX[15:14]=2'b01
i2s_clk	O	IO_I2Sclk_AUDIOgpio2c4	GRF_GPIO2C_IOMUX[9:8]=2'b01
i2s_sclk	I/O	IO_I2Ssclk_AUDIOgpio2b4	GRF_GPIO2B_IOMUX[9:8]=2'b01
i2s_lrck_rx	I/O	IO_I2Slrckrx_PCMsync_AUDIOgpio2b5	GRF_GPIO2B_IOMUX[11:10]=2'b01
i2s_lrck_tx	I/O	IO_I2Slrcktx_AUDIOgpio2b6	GRF_GPIO2B_IOMUX[13:12]=2'b01
i2s_sdo0	O	IO_I2Ssdo0_AUDIOgpio2c0	GRF_GPIO2C_IOMUX[1:0]=2'b01
i2s_sdo1	O	IO_I2Ssdo1_PCMout_AUDIOgpio2c1	GRF_GPIO2C_IOMUX[3:2]=2'b01
i2s_sdo2	O	IO_I2Ssdo2_PCMclk_AUDIOgpio2c2	GRF_GPIO2C_IOMUX[5:4]=2'b01
i2s_sdo3	O	IO_I2Ssdo3_PCMin_AUDIOgpio2c3	GRF_GPIO2C_IOMUX[7:6]=2'b01

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20.6 Application Notes

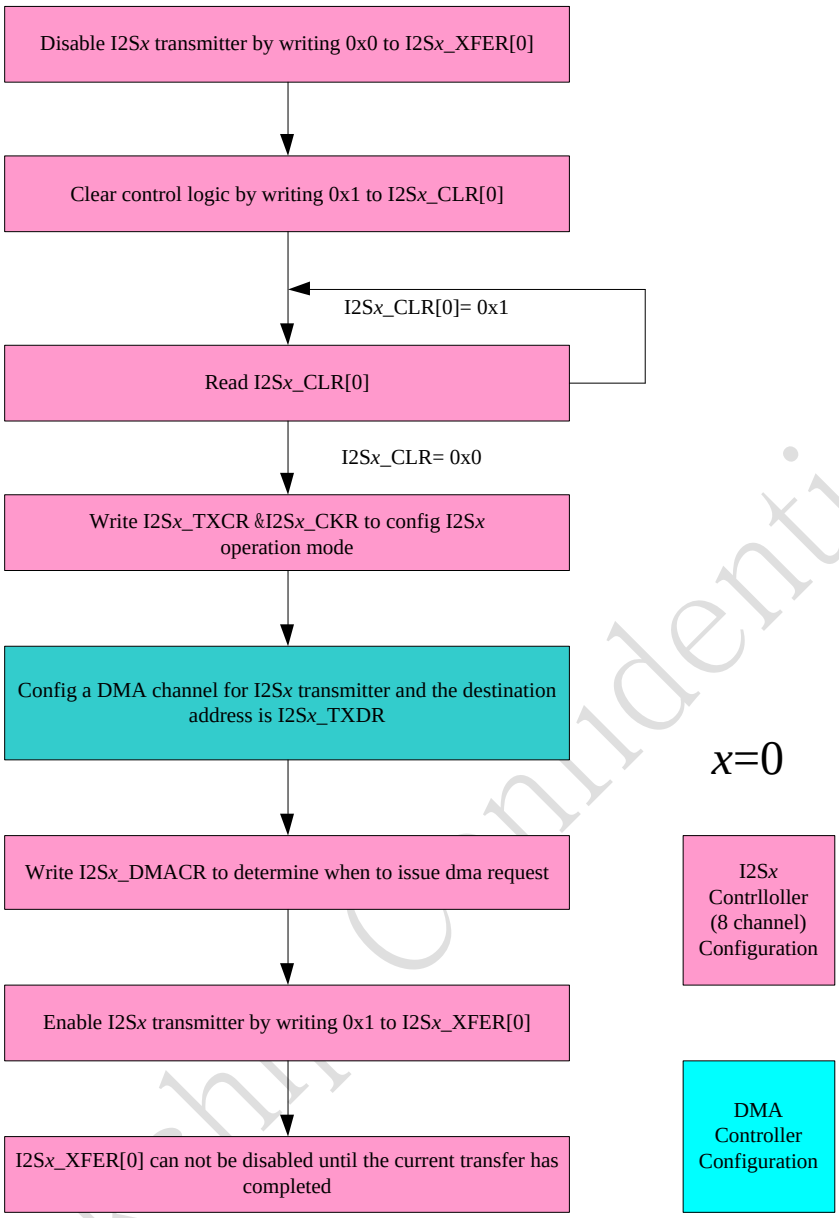

Fig. 20-11 I2S/PCM controller (8 channel) transmit operation flow chart

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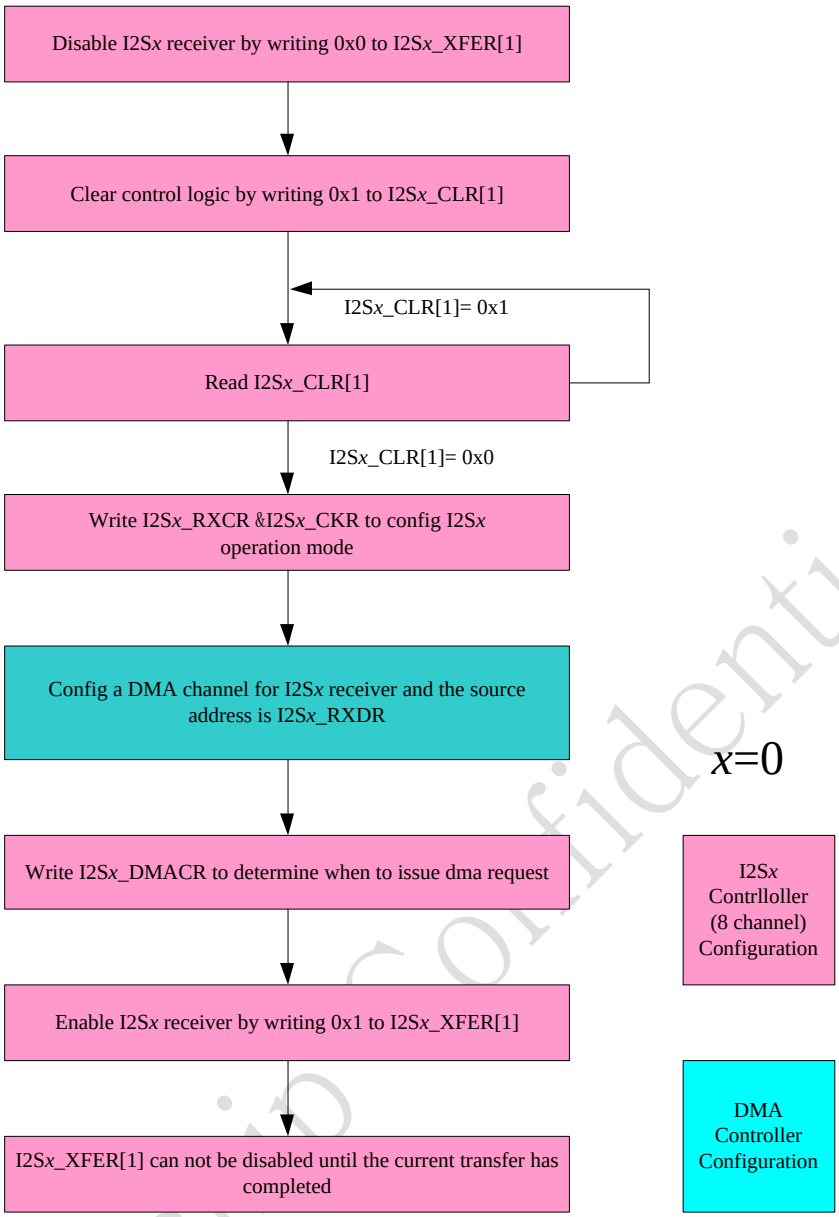

Fig. 20-12 I2S/PCM controller (8 channel) receive operation flow chart

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Chapter 21 SPDIF transmitter

21.1 Overview

The SPDIF transmitter is a self-clocking, serial, unidirectional interface for the interconnection of digital audio equipment for consumer and professional applications, using linear PCM coded audio samples.
It provides the basic structure of the interface. Separate documents define items specific to particular applications.
When used in a professional application, the interface is primarily intended to carry monophonic or stereophonic programmes, at a 48 kHz sampling frequency and with a resolution of up to 24bits per sample; it may alternatively be used to carry signals sampled at 32 kHz or 44.1 kHz.
When used in a consumer application, the interface is primarily intended to carry stereophonic programmes, with a resolution of up to 20 bits per sample, an extension to 24 bits per sample being possible.
When used for other purposes, the interface is primarily intended to carry audio data coded other than as linear PCM coded audio samples. Provision is also made to allow the interface to carry data related to computer software or signals coded using non-linear PCM. The format specification for these applications is not part of this standard.
In all cases, the clock references and auxiliary information are transmitted along with the programme.

- Supports one internal 32-bit wide and 32-location deep sample data buffer
- Supports two 16-bit audio data store together in one 32-bit wide location
- Supports AHB bus interface
- Supports biphase format stereo audio data output
- Supports DMA handshake interface and configurable DMA water level
- Supports sample data buffer empty, block terminate and user data interrupt
- Supports combine interrupt output
- Supports 16 to 31 bit audio data left or right justified in 32-bit wide sample data buffer
- Support 16, 20, 24 bits audio data transfer in linear PCM mode
- Support non-linear PCM transfer

21.2 Block Diagram

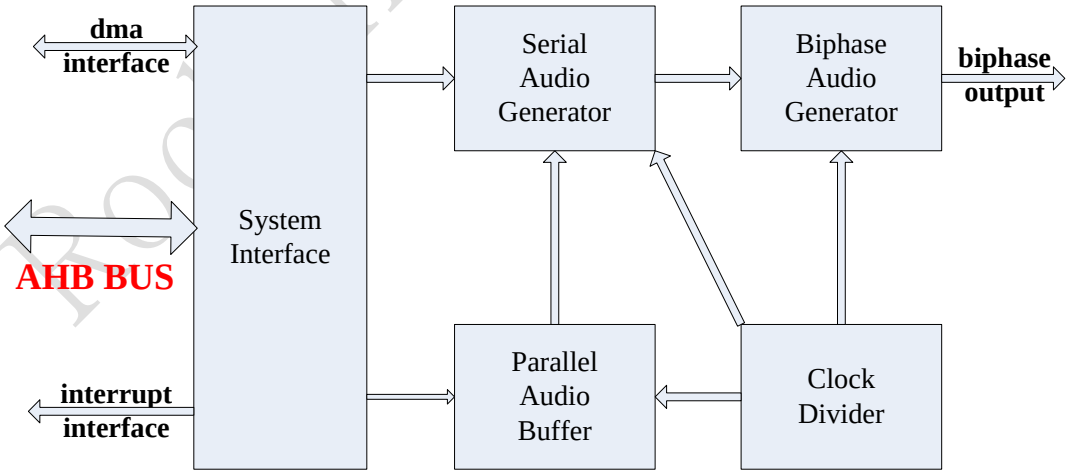

Fig. 21-1 SPDIF transmitter Block Diagram

System Interface
The system interface implements the AHB slave operation. It contains not only control registers of transmitters and receiver inside but also interrupt and DMA handshake interface.
Clock Divider
The Clock Divider implements clock generation function. The input source clock to the module is MCLK, and by the divider of the module, the clock divider generates work clock for digital audio data transformation.

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Parallel Audio Buffer
The Parallel Audio Buffer is the buffer to store transmitted audio data. The size of the FIFO is 32bits x 32.
Serial Audio Converter
The Serial Audio Converter reads parallel audio data from the Parallel Audio Buffer and converts it to serial audio data.
Biphase Audio Generator
The Biphase Audio Generatorreads serial audio data from the Serial Audio Converter and generates biphase audio data based on IEC-60958 standard.

21.3 Function description

21.3.1 Frame Format

A frame is uniquely composed of two sub-frames. For linear coded audio applications, the rate of transmission of frames corresponds exactly to the source sampling frequency.
In the 2-channel operation mode, the samples taken from both channels are transmitted by time multiplexing in consecutive sub-frames. The first sub-frame(left channel in stereophonic operation and primary channel in monophonic operation) normally use preamble M. However, the preamble is changed to preamble B once every 192 frame to identify the start of the block structure used to organize the channel status information. The second sub-frame (right in stereophonic operation and secondary channel in monophonic operation) always use preamble W.

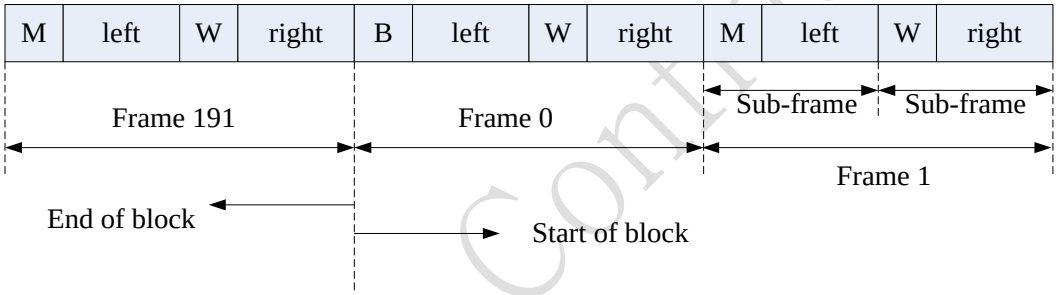

Fig. 21-2 SPDIF Frame Format

In the single channel operation mode in a professional application, the frame format is the same as in the 2-channel mode. Data is carried only in the first sub-frame and may be duplicated in the second sub-frame. If the second sub-frame is not carrying duplicate data, then time slot 28 (validity flag) shall be set to logical '1' (not valid).

21.3.2 Sub-frame Format

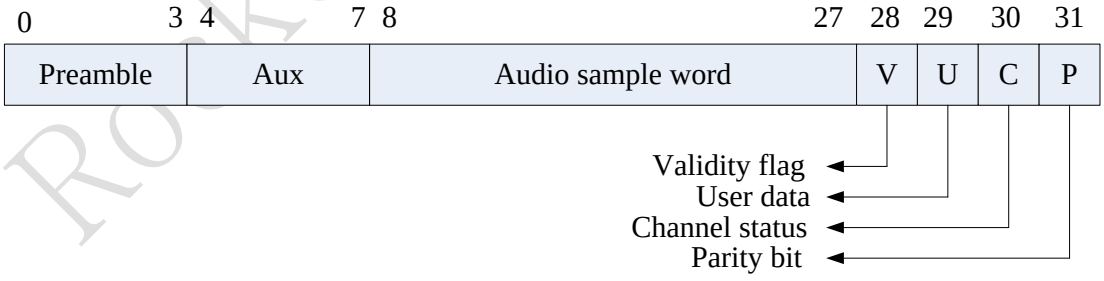

Fig. 21-3 SPDIF Sub-frame Format

Each sub-frame is divided into 32 time slots, numbered from 0 to 31. Time slot 0 to 3 carries one of the three permitted preambles. Time slot 4 to 27 carry the audio sample word in linear 2's complement representation. The MSB is carried by time slot 27. When a 24-bit coding range is used, the LSB is in time slot 4. When a 20-bit coding range is used, time slot 8 to 27 carry the audio sample word with the LSB in time slot 8.Time slot 4 to 7 may be used for other application. Under these circumstances, the bits in the time slot 4 to 7 are designated auxiliary sample bits.
If the source provides fewer bits than the interface allows (either 24 or 20),the unused LSBs are set to a logical '0'. For a non-linear PCM audio application or a data application the main data field may carry any other information. Time slot 28 carries the validity flag associated

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with the main data field. Time slot 29 carries 1 bit of the user data associated with the audio channel transmitted in the same sub-frame. Time slot 30 carries one bit of the channel status words associated with the main data field channel transmitted in the same sub-frame. Time slot 31 carries a parity bit such that time slots 4 to 31 inclusive carries an even number of ones and an even number of zeros.

21.3.3 Channel Coding

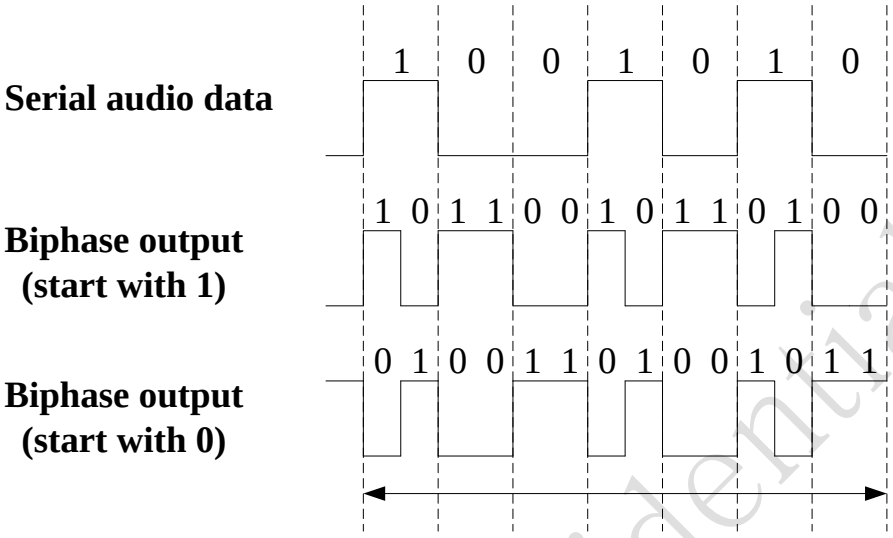

Fig. 21-4 SPDIF Channel Coding

To minimize the direct current component on the transmission line, to facilitate clock recovery from the data stream and to make the interface insensitive to the polarity of connections, time slots 4 to 31 are encoded in biphase-mark.

Each bit to be transmitted is represented by a symbol comprising two consecutive binary states. The first state of a symbol is always different from the second state of the previous symbol. The second state of the symbol is identical to the first if the bit to be transmitted is logical '0'.However, it is different from the first if the bit is logical '1'.

21.3.4 Preamble

Preambles are specific patterns providing synchronization and identification of the sub-frames and blocks.

To achieve synchronization within one sampling period and to make this process completely reliable, these patterns violate the biphase-mark code rules, thereby avoiding the possibility of data imitating the preambles.

A set of three preambles is used. These preambles are transmitted in the time allocated to four time slots (time slots 0 to 3) and are represented by eight successive states. The first state of the preamble is always different from the second state of the previous symbol.

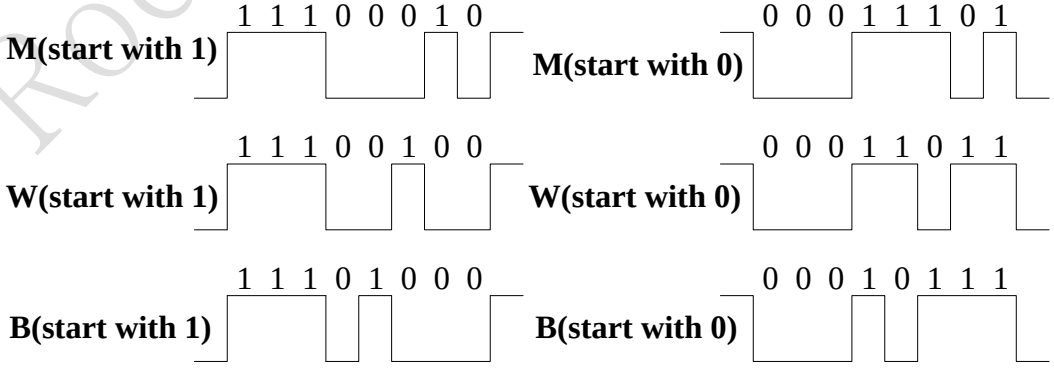

Fig. 21-5 SPDIF Preamble

Like biphase code, these preambles are dc free and provide clock recovery. They differ in at least two states from any valid biphase sequence.

21.3.5 NON-LINEAR PCM ENCODED SOURCE(IEC 61937)

The non-linear PCM encoded audio bitstream is transferred using the basic 16-bit data area of the IEC 60958subframes, i.e. in time slots 12 to 27. Each IEC 60958 frame transfers 32-bit of

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the non-PCM data in consumerapplication mode.
If the SPDIF bitstream conveys linear PCM audio, the symbol frequency is 64 times the PCM sampling frequency(32 time slots per PCM sample times two channels). If a non-linear PCM encoded audio bitstream is conveyed bythe interface, the symbol frequency is 64 times the sampling rate of the encoded audio within that bitstream. Butin the case where a non-linear PCM encoded audio bitstream is conveyed by the interface containing audio withlow sampling frequency, the symbol frequency is 128 times the sampling rate of the encoded audio within thatbitstream.
Each data burst contains a burst-preamble consisting of four 16-bit words (Pa, Pb, Pc, Pd), followed by the burstpayloadwhich contains data of an encoded audio frame.
The burst-preamble consists of four mandatory fields. Pa and Pb represent a synchronization word; Pc givesinformation about the type of data and some information/control for the receiver; Pd gives the length of the burstpayload, the number of bits or number ofbytes according to data-type.
The four preamble words are contained in two sequential SPDIF frames. The frame beginning the data-burstcontains preamble word Pa in subframe 0 and Pb in subframe 1. The next frame contains Pc in subframe 0 andPd in subframe 1. When placed into a SPDIF subframe, the MSB of a 16-bit burst-preamble is placed into timeslot 27 and the LSB is placed into time slot 12.

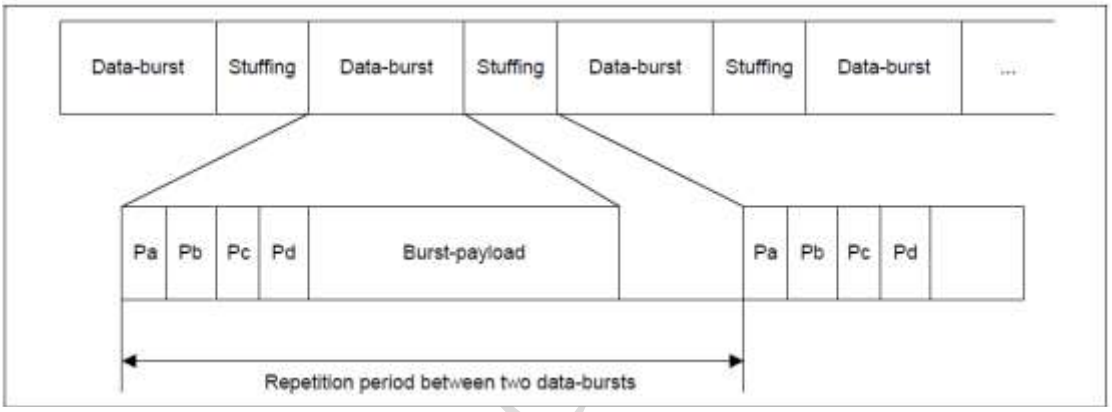

Fig. 21-6 Format of Data-burst

21.4 Register description

21.4.1 Register Summary

Name	Offset	Size	Reset Value	Description
SPDIF_CFGR	0x0000	W	0x00000000	Transfer Configuration Register
SPDIF_SDBLR	0x0004	W	0x00000000	Sample Date Buffer Level Register
SPDIF_DMACR	0x0008	W	0x00000000	DMA Control Register
SPDIF_INTCR	0x000c	W	0x00000000	Interrupt Control Register
SPDIF_INTSR	0x0010	W	0x00000000	Interrupt Status Register
SPDIF_XFER	0x0018	W	0x00000000	Transfer Start Register
SPDIF_SMPDR	0x0020	W	0x00000000	Sample Data Register
SPDIF_VLDFRn	0x0060	W	0x00000000	Validity Flag Register n
SPDIF_USRDRn	0x0090	W	0x00000000	User Data Register n
SPDIF_CHNSRn	0x00c0	W	0x00000000	Channel Status Register n
SPDIF_BURTSINFO	0x0100	W	0x00000000	Channel Burst Info Register
SPDIF_REPETTION	0x0104	W	0x00000000	Channel Repetition Register
SPDIF_BURTSINFO_SHD	0x0108	W	0x00000000	Shadow Channel Burst Info Register
SPDIF_REPETTION_SHD	0x010c	W	0x00000000	Shadow Channel Repetition Register

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Name	Offset	Size	Reset Value	Description
SPDIF_USRDR_SHDn	0x0190	W	0x00000000	Shadow User Data Register n

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

21.4.2 Detail Register Description

SPDIF_CFGR

Address: Operational Base + offset (0x0000)

Transfer Configuration Register

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:16	RW	0x00	MCD mclk divider Fmclk/Fsdo This parameter can be caculated by Fmclk/(Fs*128). Fs=the sample frequency be wanted
15:10	RO	0x0	reserved
9	RW	0x0	PRE_CHANGE Preamble Change The bit only is valid when set to non-linear PCM mode; 0: the Preamble will change when block finish; 1: the Preamble will change every 192 frames just like the linear PCM mode.
8	RW	0x0	PCMTYPE PCM type 0: linear PCM 1: non-linear PCM
7	WO	0x0	CLR mclk domain logic clear Write 1 to clear mclk domain logic. Read return zero.
6	RW	0x0	CSE Channel status enable 0: disable 1: enable The bit should be set to 1 when the channel conveys non-linear PCM
5	RW	0x0	UDE User data enable 0: disable 1: enable
4	RW	0x0	VFE Validity flag enable 0: disable 1: enable

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Bit	Attr	Reset Value	Description
3	RW	0x0	ADJ audio data justified 0: Right justified 1: Left justified
2	RW	0x0	HWT Halfword word transform enable 0: disable 1: enable It is valid only when the valid data width is 16bit.
1:0	RW	0x0	VDW Valid data width 00: 16bit 01: 20bit 10: 24bit 11: reserved The valid data width is 16bit only for non-linear PCM

SPDIF_SDBLR
Address: Operational Base + offset (0x0004)
Sample Date Buffer Level Register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RW	0x00	SDBLR Sample Date Buffer Level Register Contains the number of valid data entries in the sample data buffer.

SPDIF_DMACR
Address: Operational Base + offset (0x0008)
DMA Control Register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	RW	0x0	TDE Transmit DMA Enable 0: Transmit DMA disabled 1: Transmit DMA enabled

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Bit	Attr	Reset Value	Description
4:0	RW	0x00	TDL Transmit Data Level This bit field controls the level at which a DMA request is made by the transmit logic. It is equal to the watermark level; that is, the dma_tx_req signal is generated when the number of valid data entries in the Sample Date Buffer is equal to or below this field value

SPDIF_INTCR

Address: Operational Base + offset (0x000c)

Interrupt Control Register

Bit	Attr	Reset Value	Description
31:18	RO	0x0	reserved
17	W1C	0x0	UDTIC Ueser Data Interrupt Clear Write '1' to clear the user data interrupt.
16	W1C	0x0	BTTIC Block/Data burst transfer finish interrupt clear Write 1 to clear the interrupt.
15:10	RO	0x0	reserved
9:5	RW	0x00	SDBT Sample Date Buffer Threshold Sample Date Buffer Threshold for empty interrupt
4	RW	0x0	SDBEIE Sample Date Buffer empty interrupt enable 0: disable 1: enable
3	RW	0x0	BTTIE Block transfer/repetition period end interrupt enable When enabled, an interrupt will be asserted when the block transfer is finished if the channel conveys linear PCM or when the repetition period is reached if the channel conveys non-linear PCM. 0: disable 1: enable

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Bit	Attr	Reset Value	Description
2	RW	0x0	UDTIE User Data Interrupt 0: disable 1: enable If enabled, an interrupt will be asserted when the content of the user data register is fed into the corresponding shadow register
1:0	RO	0x0	reserved

SPDIF_INTSR
Address: Operational Base + offset (0x0010)
Interrupt Status Register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	SDBEIS Sample Date Buffer empty interrupt status 0: inactive 1: active
3	RW	0x0	BTTIS Block/Data burst transfer interrupt status 0: inactive 1: active
2	RW	0x0	UDTIS User Data Interrupt Status 0: inactive 1: active
1:0	RO	0x0	reserved

SPDIF_XFER
Address: Operational Base + offset (0x0018)
Transfer Start Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	XFER Transfer Start Register Transfer Start Register

SPDIF_SMPDR
Address: Operational Base + offset (0x0020)
Sample Data Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	SMPDR Sample Data Register Sample Data Register

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SPDIF_VLDFRn
Address: Operational Base + offset (0x0060)
Validity Flag Register n

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	VLDFR_SUB_1 Validity Flag Subframe 1 Validity Flag Register 0
15:0	RW	0x0000	VLDFR_SUB_0 Validity Flag Subframe 0 Validity Flag for Subframe 0

SPDIF_USRDRn
Address: Operational Base + offset (0x0090)
User Data Register n

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	USR_SUB_1 User Data Subframe 1 User Data Bit for Subframe 1
15:0	RW	0x0000	USR_SUB_0 User Data Subframe 0 User Data Bit for Subframe 0

SPDIF_CHNSRn
Address: Operational Base + offset (0x00c0)
Channel Status Register n

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	CHNSR_SUB_1 Channel Status Subframe 1 Channel Status Bit for Subframe 1
15:0	RW	0x0000	CHNSR_SUB_0 Channel Status Subframe 0 Channel Status Bit for Subframe 0

SPDIF_BURTSINFO
Address: Operational Base + offset (0x0100)
Channel Burst Info Register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	PD pd Preamble Pd for non-linear pcm, indicating the length of burst payload in unit of bytes or bits.
15:13	RW	0x0	BSNUM Bitstream Number This field indicates the bitstream number. Usually the birstream number is 0.

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Bit	Attr	Reset Value	Description
12:8	RW	0x00	DATAINFO Data-type-dependent info This field gives the data-type-dependent info
7	RW	0x0	ERRFLAG Error Flag 0: indicates a valid burst-payload 1: indicates that the burst-payload may contain errors

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Bit	Attr	Reset Value	Description
6:0	RW	0x00	DATATYPE Data type 0000000: null data 0000001: AC-3 data 0000011: Pause data 0000100: MPEG-1 layer 1 data 0000101: MPEG-1 layer 2 or 3 data or MPEG-2 without extension 0000110: MPEG-2 data with extension 0000111: MPEG-2 AAC 0001000: MPEG-2, layer-1 low sampling frequency 0001001: MPEG-2, layer-2 low sampling frequency 0001010: MPEG-2, layer-3 low sampling frequency 0001011: DTS type I 0001100: DTS type II 0001101: DTS type III 0001110: ATRAC 0001111: ATRAC 2/3 0010000: ATRAC-X 0010001: DTS type IV 0010010: WMA professional type I 0110010: WMA professional type II 1010010: WMA professional type III 1110010: WMA professional type IV 0010011: MPEG-2 AAC low sampling frequency 0110011: MPEG-2 AAC low sampling frequency 1010011: MPEG-2 AAC low sampling frequency 1110011: MPEG-2 AAC low sampling frequency 0010100: MPEG-4 AAC 0110100: MPEG-4 AAC 1010100: MPEG-4 AAC 1110100: MPEG-4 AAC 0010101: Enhanced AC-3 0010110: MAT others: reserved

SPDIF_REPETTION
Address: Operational Base + offset (0x0104)
Channel Repetition Register

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Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	REPETTION Repetition This define the repetition period when the channel conveys non-linear PCM

SPDIF_BURTSINFO_SHD
Address: Operational Base + offset (0x0108)
Shadow Channel Burst Info Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	PD pd Preamble Pd for non-linear pcm, indicating the length of burst payload in unit of bytes or bits.
15:13	RO	0x0	BSNUM Bitstream Number This field indicates the bitstream number. Usually the birstream number is 0.
12:8	RO	0x00	DATAINFO Data-type-dependent info This field gives the data-type-dependent info
7	RO	0x0	ERRFLAG Error Flag 0: indicates a valid burst-payload 1: indicates that the burst-payload may contain errors

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Bit	Attr	Reset Value	Description
6:0	RO	0x00	DATATYPE Data type 0000000: null data 0000001: AC-3 data 0000011: Pause data 0000100: MPEG-1 layer 1 data 0000101: MPEG-1 layer 2 or 3 data or MPEG-2 without extension 0000110: MPEG-2 data with extension 0000111: MPEG-2 AAC 0001000: MPEG-2, layer-1 low sampling frequency 0001001: MPEG-2, layer-2 low sampling frequency 0001010: MPEG-2, layer-3 low sampling frequency 0001011: DTS type I 0001100: DTS type II 0001101: DTS type III 0001110: ATRAC 0001111: ATRAC 2/3 0010000: ATRAC-X 0010001: DTS type IV 0010010: WMA professional type I 0110010: WMA professional type II 1010010: WMA professional type III 1110010: WMA professional type IV 0010011: MPEG-2 AAC low sampling frequency 0110011: MPEG-2 AAC low sampling frequency 1010011: MPEG-2 AAC low sampling frequency 1110011: MPEG-2 AAC low sampling frequency 0010100: MPEG-4 AAC 0110100: MPEG-4 AAC 1010100: MPEG-4 AAC 1110100: MPEG-4 AAC 0010101: Enhanced AC-3 0010110: MAT others: reserved

SPDIF_REPETTION_SHD
Address: Operational Base + offset (0x010c)
Shadow Channel Repetition Register

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Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RO	0x0000	REPETTION Repetition This register provides the repetition of the bitstream when channel conveys non-linear PCM. In the design, it is define the length bwtween Pa of the two consecutive data-burst. For the same audio format, the definition is different. Please convert the actual repetition in order to comply with the design.

SPDIF_USRDR_SHDn
Address: Operational Base + offset (0x0190)
Shadow User Data Register n

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	USR_SUB_1 User Data Subframe 1 User Data Bit for Subframe 1
15:0	RO	0x0000	USR_SUB_0 User Data Subframe 0 User Data Bit for Subframe 0

21.5 Interface description

Table 21-1 SPDIF IOMUX Setting

Module Pin	IO	Pad Name	IOMUX Setting
spdif_tx	O	IO_SPDIFtx_EDPhpd_AUDIOgpio2c7	GPIO2C_IOMUX[15:14]= 2'b01

Notes: 1. O=output

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21.6 Application Notes

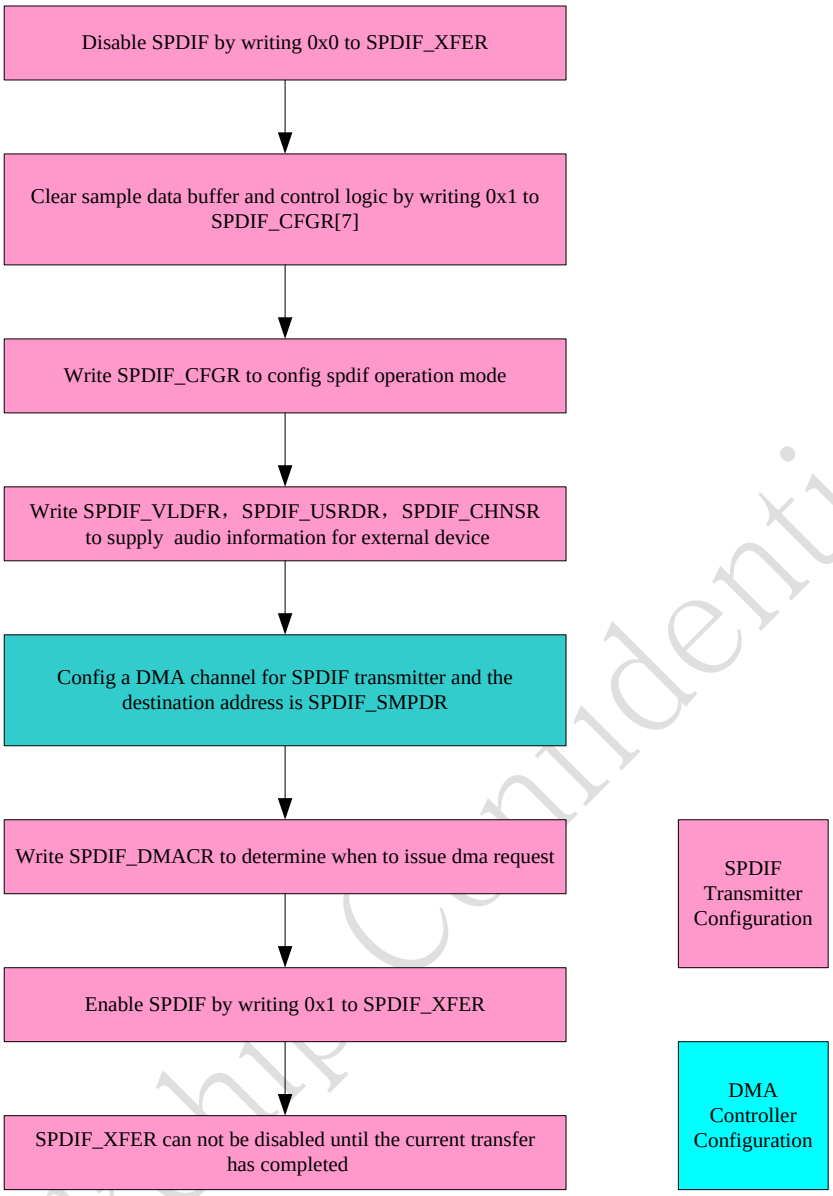

Fig. 21-7 SPDIF transmitter operation flow chart

21.6.1 Channel Status Bit and Validity Flag Bit

Normally the channel status bits and validity flag bits are not necessarily updated frequently. If it is desired to change the channel status bits or validity flag, please write to the corresponding register after a block terminate interrupt is asserted. The new value will take effect immediately.

21.6.2 User Data Bit

As the user data bits are updated frequently, the design takes use of the shadow register mechanism to store and convey the user data bit. When the SPDIF interface is disabled, the values of the shadow user data registers keeps the same with the corresponding user data registers. After the SPDIF starts, any change of the user data register will not go to the corresponding shadow user data registers until an user data interrupt is asserted.

Therefore before the SPDIF transfer starts, prepare the first 384 user data bits by writing them to the SPDIF_USRDR registers. After the SPDIF transfer starts, writing the second 384 user data bits to the SPDIF_USRDR registers. Then wait for the assertion of user data interrupt. The second 384 user data bits goes to the shadow registers, and then third 384 user bits are written to SPDIF_USRDR.

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21.6.3 Burst Info and Repetition

The shadow register mechanism is also applied to the data of burst info and repetition as the user data. The difference is that the update of shadow register will be taken after assertion of the block terminate interrupt.

It is important to note that the repetition defined in the design is a little different from the repetition defined in IEC-61957. The repetition is always defined as the length (measured in IEC-60958 frame) between Pa of two consecutive data-bursts. Therefore the user needs to calculation the new repetition value if the definition of the repetition is different for some audio formats such as AC-3.

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Chapter 22 USB OTG2.0

22.1 Overview

USB OTG 2.0 is a Dual-Role Device controller, which supports both device andhost functions and is fully compliant with OTG Supplement to USB2.0specification,andsupport high-speed(480Mbps),full-speed(12Mbps),low-speed(1.5Mbps) transfer.
USB OTG 2.0 isoptimized for portable electronic devices, point-to-point applications(no hub, directconnection to device) and multi-point applications to devices.USB OTG 2.0 interface supports both device and host functions and is fully compliant with OTG Supplement to USB2.0 specification, and support high-speed(480Mbps), full-speed(12Mbps), low-speed (1.5Mbps) transfer. It is optimized for portable electronic device, point-to-point applications (no hub, direct connection to device) and multi-point applications to devices.

22.1.1 Features

- Compliant with the OTG Supplement to the USB2.0 Specification
- Operates in High-Speed and Full-Speed mode
- Support 9 channels in host mode
- 9 Device mode endpoints in addition to control endpoint 0, 4 in, 3 out and 2 IN/OUT
- Built-in one 1024x35 bits FIFO
- Internal DMA with scatter/gather function
- Supports packet-based, dynamic FIFO memory allocation for endpoints for flexible, efficient use of RAM
- Support dynamic FIFO sizing
- Support Battery Charge in device role
- Support Uart Bypass Mode

22.2 Block Diagram

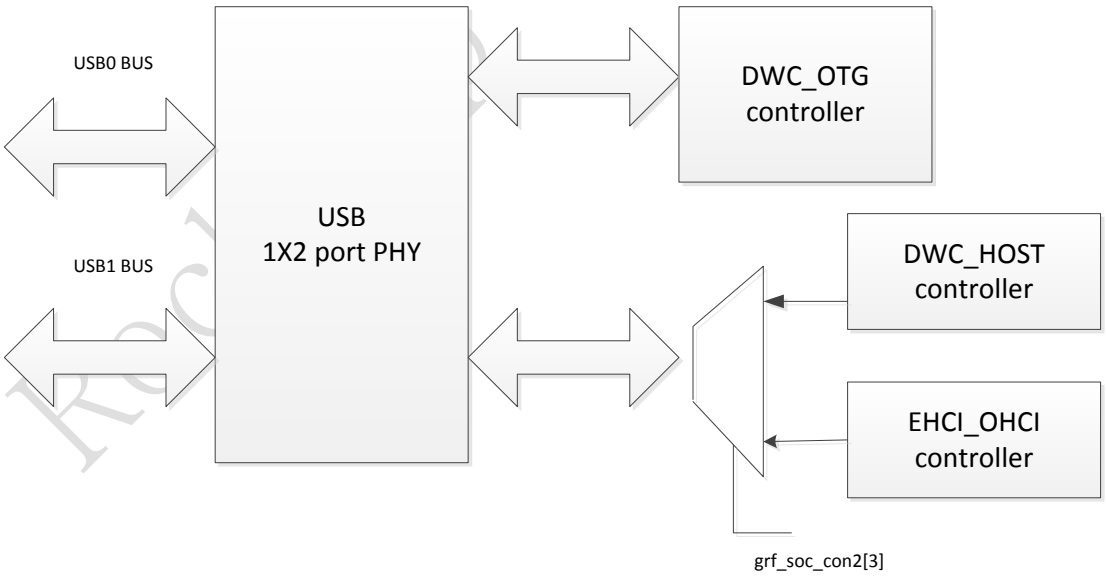

Fig. 22-1 USB OTG 2.0 Architecture

Fig.26-1 shows the architecture of USB OTG 2.0. It is broken up into two separate units: USB OTG 2.0 controller and USB OTG 2.0 PHY. The two units are interconnected with UTMI interface.

22.2.1 USB OTG 2.0 Controller Function

The USB OTG 2.0 Controller controlsSIE(Serial Interface Engine) logic, the endpointlogic, the channel logic and the internal DMA logic.
The SIElogic contains the USB PID and address recognition logic, and other sequencingand

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state machine logic to handle USB packets and transactions. Generally the SIE Logicis required for any USB implementation while the number and types of endpoints will vary as function ofapplication and performance requirements.
The endpoint logic contains the endpointspecific logic: endpoint number recognition, FIFOs and FIFO control, etc.
The channel Logic contains the channel tasks schedule, FIFOs and FIFO control, etc.
The internal DMA logic controls data transaction between system memory and USB FIFOs.

22.2.2 USB OTG 2.0 PHY Function

The USB OTG 2.0 PHY handles the low level USB protocol and signaling. This includes features such as; dataserialization and deserialization, bit stuffing and clock recovery and synchronization. The primary focus ofthis block is to shift the clock domain of the data from the USB 2.0 rate to the frequency of UTMI clock which is 30MHz.

22.2.3 UTMI Interface

- Transmit

Transmit must be asserted to enable any transmissions.
The USB OTG2.0 CONTROLLER asserts TXValid to begin a transmission and negates TXValid to end a transmission.After the USB OTG2.0 CONTROLLER asserts TXValid it can assume that the transmission has started when it detectsTXReady asserted.
The USB OTG2.0 CONTROLLER assumes that the USB OTG2.0 PHYhas consumed a data byte if TXReady and TXValid are asserted.
The USB OTG2.0 CONTROLLER must have valid packet information (PID) asserted on the DataIn bus coincident with theassertion of TXValid. Depending on the USB OTG2.0 PHYimplementation, TXReady may be asserted by theTransmit State Machine as soon as one CLK after the assertion of TXValid.TXValid and TXReady are sampled on the rising edge of CLK.
The Transmit State Machine does NOT automatically generate Packet ID's (PIDs) or CRC. Whentransmitting, the USB OTG2.0 CONTROLLER is always expected to present a PID as the first byte of the data stream and ifappropriate, CRC as the last bytes of the data stream.
The USB OTG2.0 CONTROLLER must use LineState to verify a Bus Idle condition before asserting TXValid in the TX Waitstate.
The state of TXReady in the TX Wait and Send SYNC states is undefined. An MTU implementationmay prepare for the next transmission immediately after the Send EOP state and assert TXReady inthe TX Wait state. An MTU implementation may also assert TXReady in the Send SYNC state. Thefirst assertion of TXReady is Macrocell implementation dependent.
The USB OTG2.0 CONTROLLER must prepare DataIn forthe first byte to be transmitted before asserting TXValid.

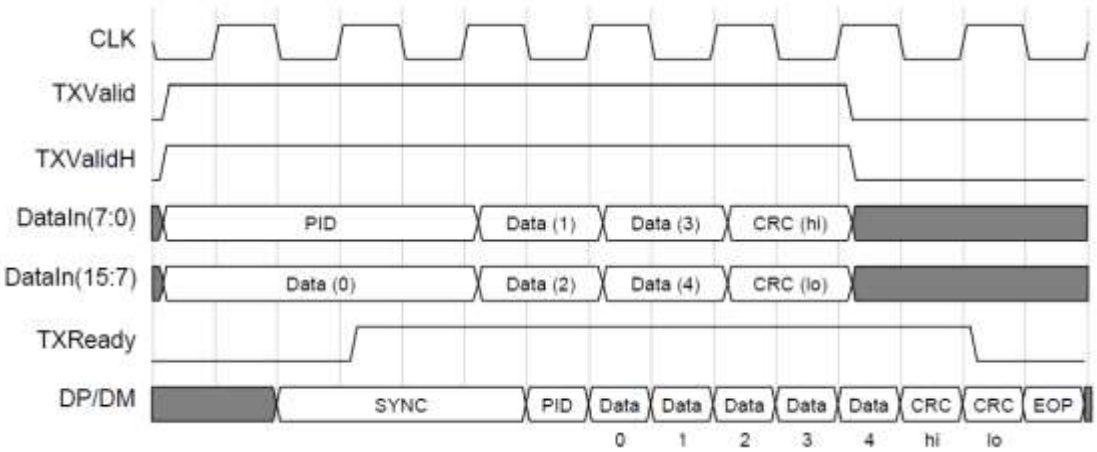

Fig. 22-2 UTMI interface – Transmit timing for a data packet

- Receive

RXActive and RXValid are sampled on the rising edge of CLK.
In the RX Wait state the receiver is always looking for SYNC.
The USB OTG 2.0 PHY asserts RXActive when SYNC is detected (Strip SYNC state).
The USB OTG 2.0 PHY negates RXActive when an EOP is detected (Strip EOP state).

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When RxActive is asserted, RXValid will be asserted if the RX Holding Register is full.
RXValid will be negated if the RX Holding Register was not loaded during the previous byte time.
This will occur if 8 stuffed bits have been accumulated.
The USB OTG2.0 Controller must be ready to consume a data byte if RXActive and RXValid are asserted (RX Data state).
In FS mode, if a bit stuff error is detected then the Receive State Machine will negate RXActive andRXValid, and return to the RXWait state.

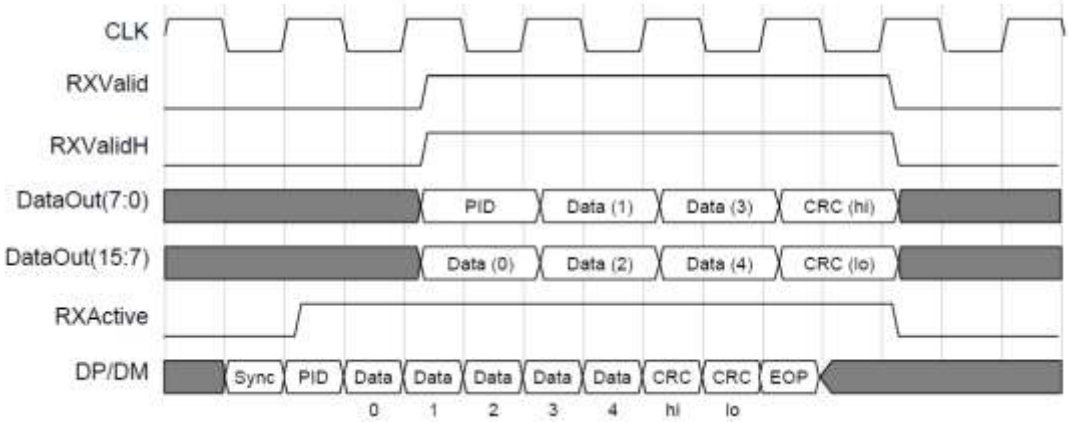

Fig. 22-3 UTMI interface – Receive timing for a data packet

22.3 USB OTG2.0 Controller

Fig.26-4 shows the main components and flow of the USB OTG 2.0 controller system.

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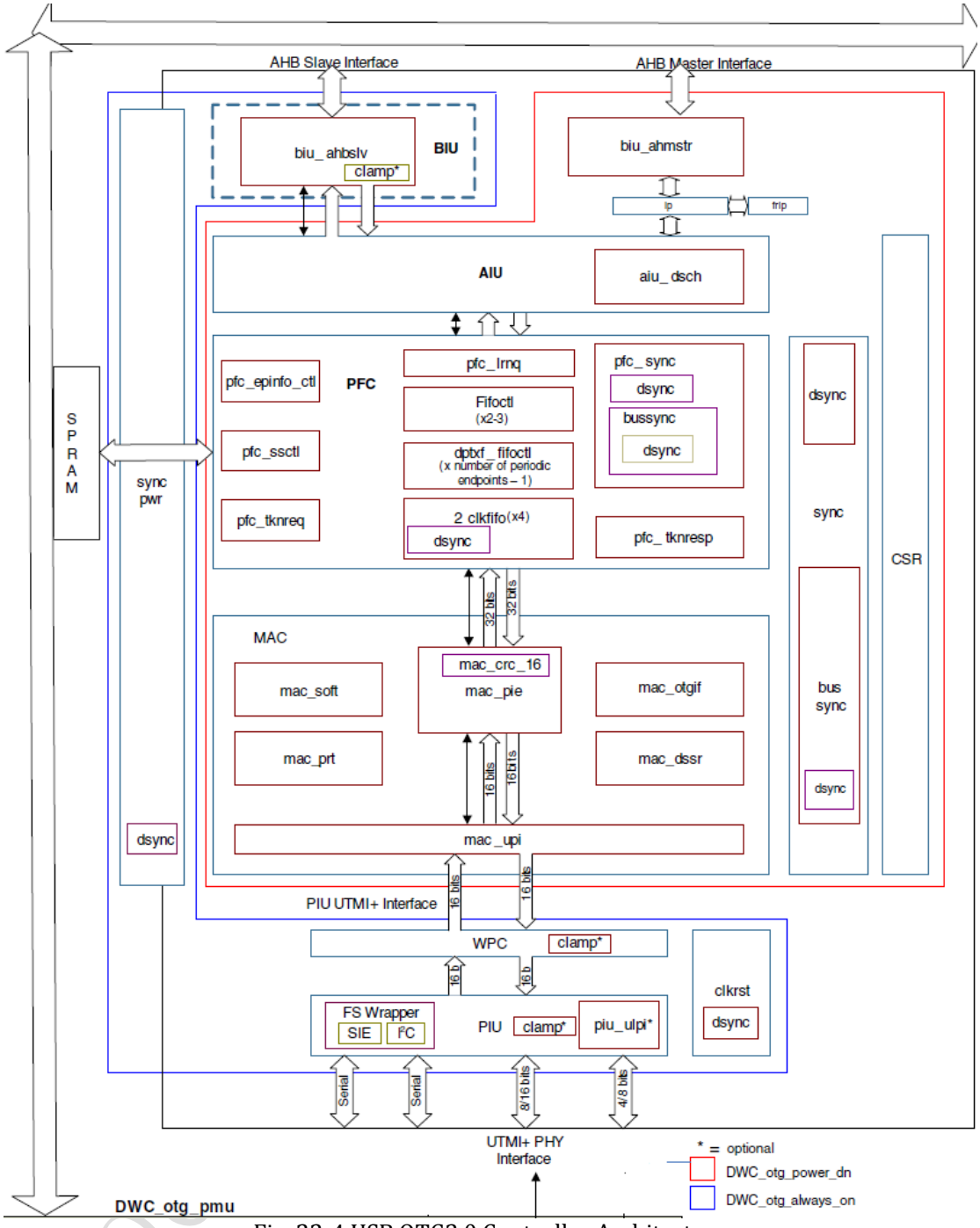

Fig. 22-4 USB OTG2.0 Controller Architecture

1). AHB Slave Bus Interface Unit (BIUS)

The AHB Slave interface unit converts AHB cycles to CSR write/read, Data-FIFO read/write, and DFIFO push/pop signals.

2) Control and Status Registers (CSR)

The CSR block resides in the AHB clock domain, and contains all registers except the Power and Clock Gating Control Register (PCGCCTL) and bits 31:29 of the Core Interrupt register (GINTSTS).

3) Application Interface Unit (AIU)

The application Interface Unit (AIU) consists of the following interfaces:

- AHB Master
- AHB Slave
- Packet FIFO Controller
- Control and Status registers

4). DMA Scheduler (DSCH)

This block is used only in DMA mode. It controls the transfer of data packets between the system memory and the USB OTG 2.0 Controller for both Internal and External DMA.

5).Packet FIFO Controller (PFC)
Several FIFOs are used in Device and Host modes to store data inside the core before transmitting it on either the AHB or the USB. PFC connect the Data FIFO interface to an industry-standard, single-port synchronous SRAM. Address, write data, and control outputs are driven late by the USB OTG 2.0 Controller, but in time to meet the SRAM setup requirements. Input read data is expected late from the SPRAM and registered inside the core before being used.

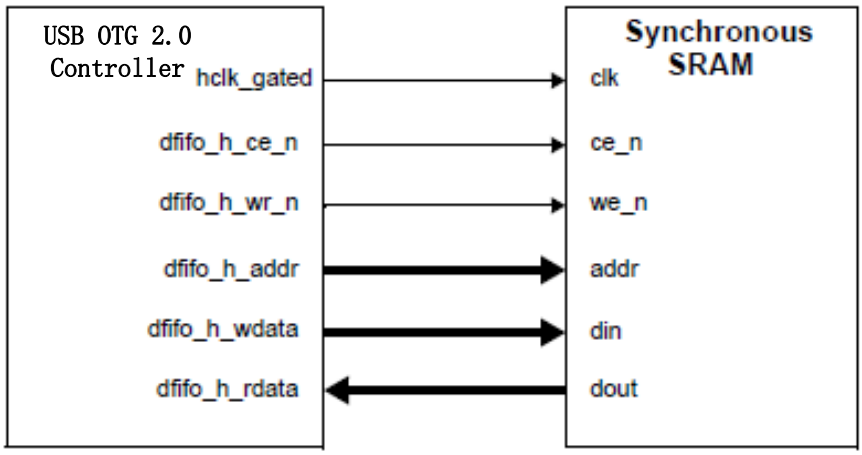

Fig. 22-5 DFIFO single-port synchronous SRAM interface

6).Media Access Controller(MAC)
The Media Access Controller (MAC) module handles USB transactions, and device, host, and OTG protocols.
7) PHY Interface Unit (PIU)
The core uses 16-bit UTMI+ Interface.
8) Wakeup and Power Controller (WPC)
When the USB is suspended or the session is not valid, the PHY is driven into Suspend mode and the PHY clock is stopped to reduce PHY and the core power consumption. To reduce power consumption further, the core also supports AHB clock gating and partial power-down.

22.3.1 Host Architecture

The host uses one transmit FIFO for all non-periodic OUT transactions and one transmit FIFO for all periodic OUT transactions. These transmit FIFOs are used as transmit buffers to hold the data (payload of the transmit packet) to be transmitted over USB.
The host pipes the USB transactions through Request queues (one for periodic and one for non-periodic). Eachentryin the Request - queue holds the IN or OUT channel number along with other information to perform a transaction on the USB. The order in which the requests are written into the queue determines the sequence of transactions on the USB. The host processes the periodic Request queue first, followed by the non-periodic Request queue, at the beginning of each (micro)frame.
The host uses one Receive-FIFO for all periodic and non-periodic transactions. The FIFO is used as a Receive-buffer to hold the received data (payload of the received packet) from the USB until it is transferred to the system memory. The status of each packet received also goes into the FIFO. The status entry holds the IN channel number along with other information, such as received byte count and validity status, to perform a transaction on the AHB.

22.3.2 Device Architecture

The core uses Dedicated Transmit FIFO Operation. In this mode, there are individualtransmit FIFOs for each IN endpoint.
The OTG device uses a single receive FIFO to receive the data for all the OUT endpoints. The receive FIFO holds the status of the received data packet, such as byte count, data PID and the validity of the received data. The DMA or the application reads the data out of the receive FIFO as it is received.

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22.3.3 FIFO Mapping

- Fig.26-6 shows FIFO mapping in Host mode.

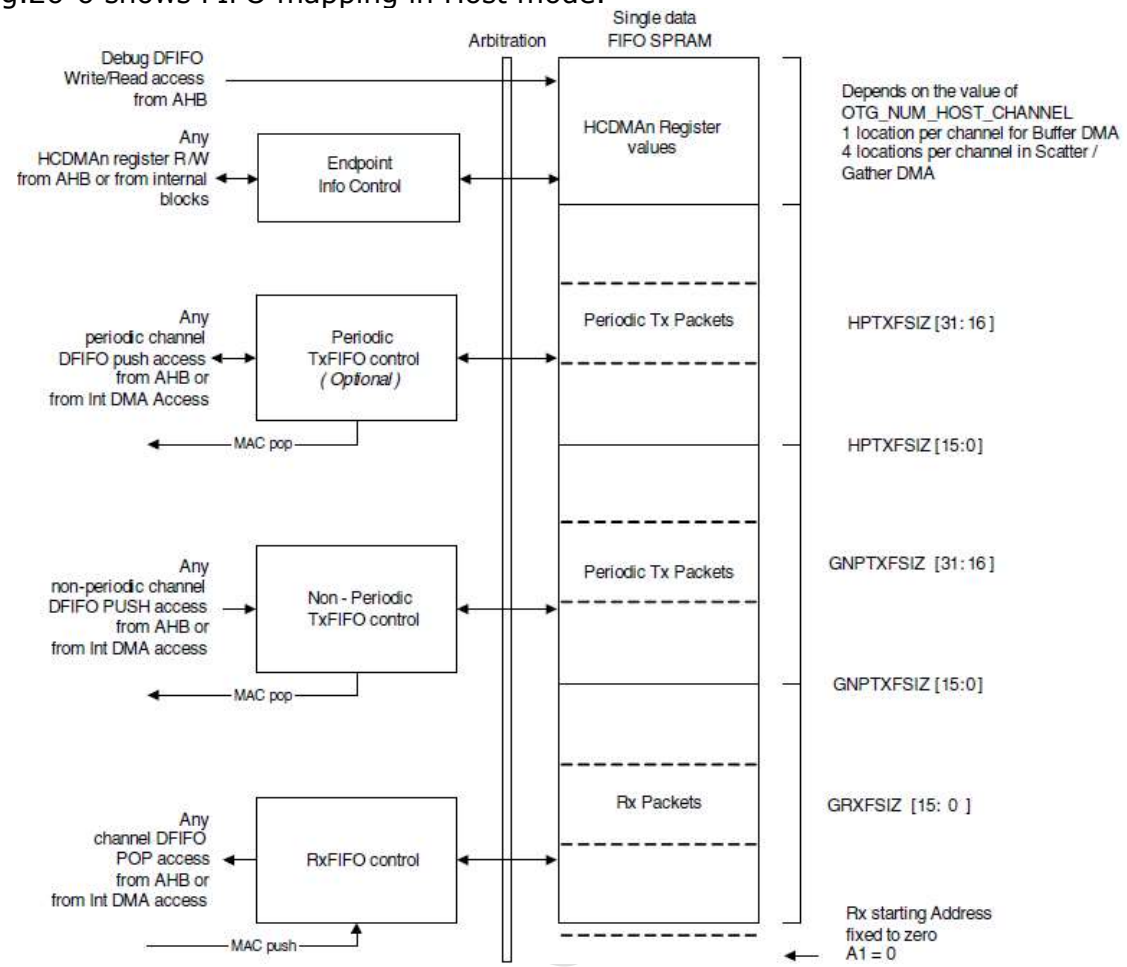

Fig. 22-6 USB OTG 2.0 Controller host mode FIFO address mapping

Note: When the device is operating in Internal DMA mode, the last locations of the SPRAM are used to store the DMAADDR values for each channel.

- Fig.26-7 shows FIFO mapping in Device mode.

When the device is operating in non-Descriptor Internal DMA mode, the last locations of the SPRAM are used to store the DMAADDR values for each channel. When the device is operating in Descriptor mode, then the last locations of the SPRAM store the Base Descriptor address, Current Descriptor address, Current Buffer address, and status quadlet information for each endpoint direction.

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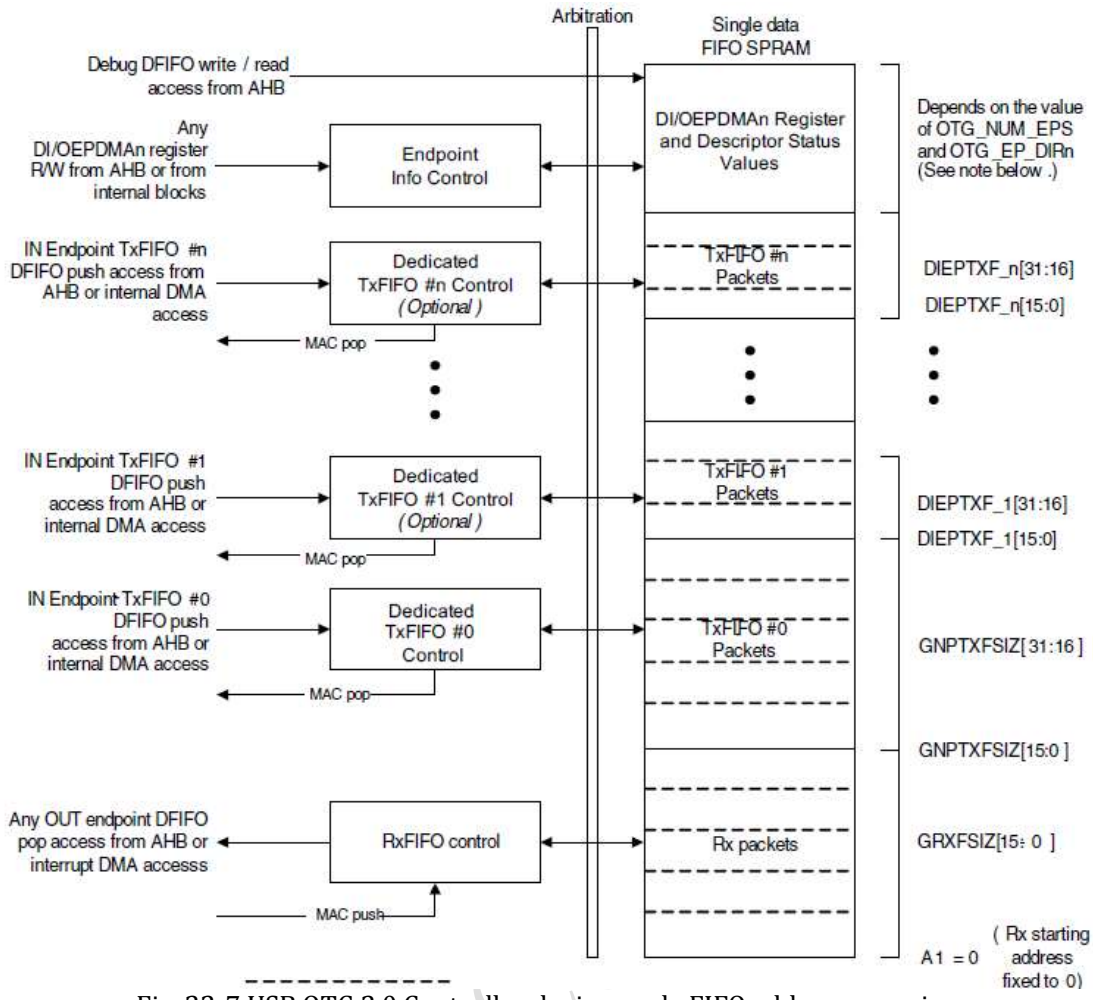

Fig. 22-7 USB OTG 2.0 Controller device mode FIFO address mapping

Note: When the device is operating in non-Scatter Gather Internal DMA mode, the last locations of the SPRAM are used to store the DMAADDR values for each Endpoint (1 location per endpoint).When the device is operating in Scatter Gather mode, then the last locations of the SPRAM store the Base Descriptor address, Current Descriptor address, Current Buffer address, and status quadlet information for each endpoint direction (4 locations per Endpoint). If an Endpoint is bidirectional,then 4 locations will be used for IN, and another 4 for OUT).

22.4 USB OTG2.0 PHY

USB PHY supports dual OTG ports'functions and is fully compliant with USB2.0 specification, andsupport High-speed(480Mbps),full-speed(12Mbps),low-speed(1.5Mbps) transfer. It provides a complete on-chip transceiver physical solution with ESD protection. A minimum number ofexternal components are needed, which include a 45 ohm resistor for resistance calibration purpose. Its feature contains:

- provide dual UTMI ports
- OTG0 Support UART Bypass Function
- Fully compliant with USB specifications Rev 2.0,1.1 HOST/Device and OTG V1.2.
- Supports 480Mbps (HS), 12Mbps (FS)& 1.5Mbps(LS) serial data transmission
- Supports low latency hub mode with 40 bit timeround trip delay
- 8 bit or 16 bit UTMI interface compliant withUTMI+ specification level 3 Rev 1.
- Loop back BIST mode supported
- Built-in I/O and ESD structure
- On-die self-calibrated HS/FS/LS termination
- 12MHz crystal oscillator with integratedphase-locked loop (PLL) oscillator
- Manufactured in SMIC 65/55nm LL process

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• Dual 3.3V / 1.2V supply

22.4.1 Block Diagram

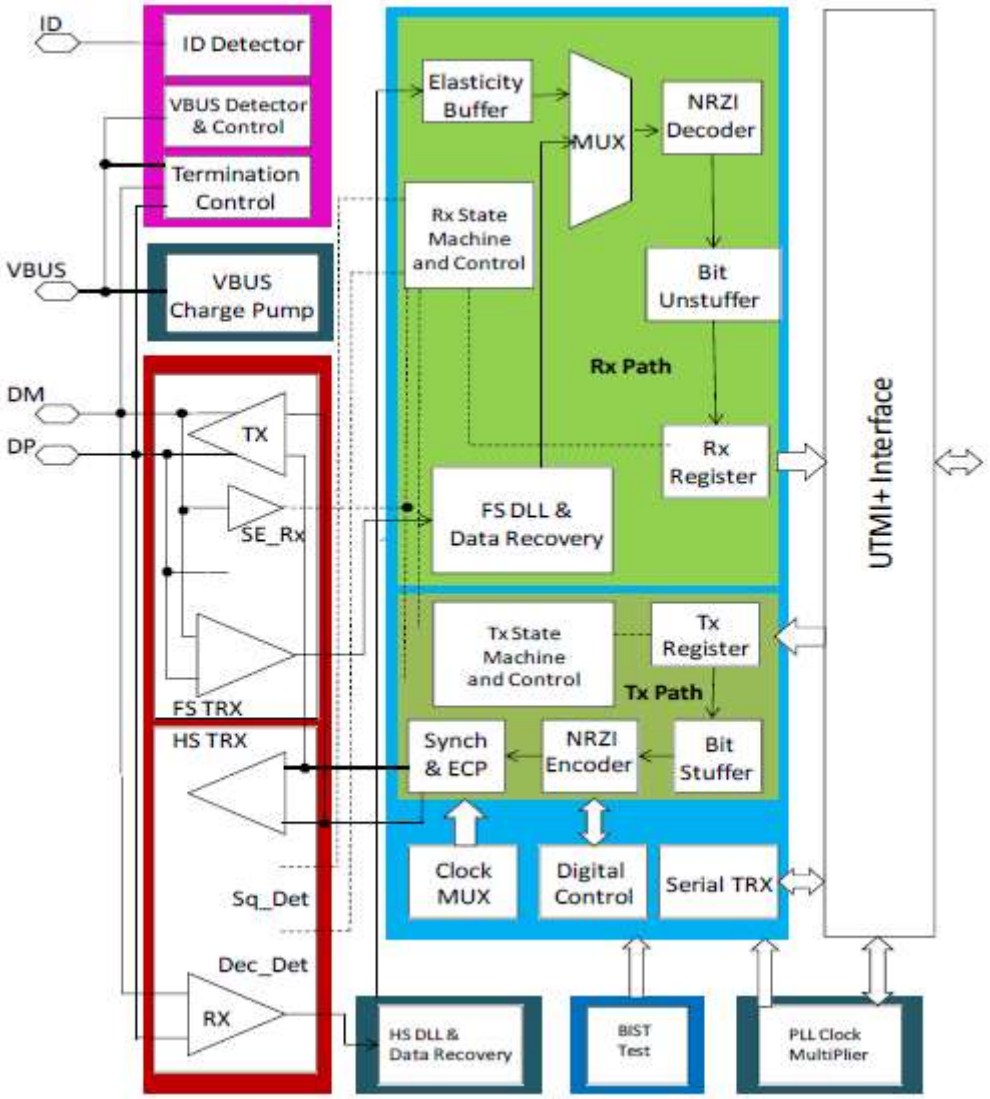

Fig. 22-8 usb phy architecture

HS AFE

The HS AFEcontains the low-level analog circuitry, and also the HS differential data transmitter and receiver, toperform HS transmission envelope detection and host disconnection detection. It works in HS mode only.

HS Transmit driver

The HS transmit driver is active only when transmit is asserted. In HS transceiver enabled mode and thetransmit state machine has data to send, the XCVR selects input. Data from transmit data path will bedriven onto the DP/DM signal lines when enabled.

HS Differential Receiver

When enabled, received HS data will be multiplexed through the receive data path to the receive shift andhold registers. It is active only in HS mode.

transmission envelope detector (Squelch detector)

When the amplitude of the differential signal at a receiver's inputs falls below the squelch threshold, theenvelope detector will indicate the invalid data. It must indicate squelch when the signal drops below 100mV differential amplitude, and also, it must indicate that the line is

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not in the squelch state when the signalexceeds 150mV differential amplitude.

Disconnection envelope detector

In host mode, this envelope detector is active to detect the high speed disconnect state on the line.Disconnection must be indicated when the amplitude of the differential signal at the downstream facingdriver's connector is more than 625 mV, and it must not be indicated when the signal amplitude is less than525 mV.

FS/LS AFE

In FS or LS mode, the FS/LS AFE is active to send and receive the FS or LS data on the USB bus. Also itsupports the reset, suspend and resume detection through the data line single ended receivers.

FS/LS Transmitter

The FS/LHS transmitter is active only when transmit is asserted. In FS or LS transceiver enabled mode andthe transmit state machine has data to send, the XCVR selects input. Data from transmit data path will bedriven onto the DP/DM signal lines when enabled.

FS/LS Differential Receiver

When enabled, received FS or LS data will be multiplexed through the receive data path to the receive shiftand hold registers. It is active only in FS or LS modes.

Single ended receivers

The single ended receivers are used for low-speed and full-speed signaling detection.

Digital Core TX Path

The digital core TX path has some blocks responsible for SYNC and EOP generation, data encoding, bitstuffing and data serialization. And meanwhile, also a TX state machine is involved to manage thecommunication with the controller.

TX Shift/Hold Register

The TX shift/Hold register module consists of an 8-bit primary shift register for parallel/serial conversionand 8-bit hold register used to buffer the next data to serialize. This module is responsible for readingparallel data from the parallel application bus interface upon command and serializing for transmissionover USB.

Bit stuffer

To ensure adequate signal transitions, when sending a packet on USB, a bit stuffer is employed by thetransmitter. A '0' has to be inserted after every six consecutive ones in the data stream before the data inNRZI encoded, to force a transition in the NRZI data stream.

NRZI Encoder

The High speed, Full speed or low speed serial transmitted data are encoded by The NRZI encoder. As astate transition, a '0' is encoded, and as no state transition, a '1' is encoded.

Transmit state machine

The communication between the controller and the PHY in TX path is controlled by the transmit statemachine, which synchronizes the Data with the Sync and the EOP, and also supports the LS, FS and HSModes.

Digital Core RX Path

The digital core RX path includes blocks responsible for SYNC and EOP detection and stripping, datadecoding, bit un-stuffing and data de-serialization. Also a RX state machine is involved to manage thecommunication with the controller. FS/LS data and clock is recovered in this section.

Elasticity buffer

To compensate for differences between transmitting and receiving clocks, the Elasticity Buffer is used tosynchronize the HS extracted data with the PLL internal clock.

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Mux

The Mux block allows the data from the HS or FS/LS receivers to be routed to the shared receive logic.The state of the Mux is determined by the Xcvr Select input.

NRZI Decoder

The NRZI is responsible for decoding the High speed or Full speed received NRZI encoded data. Achange in level is decoded as '0' and no change in level is decoded as '1'.

Bit Un-stuffer

The Bit Un-stuffer not only recognizes the stuffed bits from the data stream, but also discards them. Alsoit detects bit stuff error, which is interpreted as HS EOP.

RX Shift/Hold Register

This module de-serializes received data and transmits 8-bit parallel data to the application bus interface.It consists of an 8-bit primary shit register for serial to parallel conversion and an 8-bit hold register forbuffering the last de-serialized data byte.

Receiver state machine

The receiver state machine controls the communication between the controller and the PHY in the RXpath, strips the SYNC and the EOP from the Data and supports the LS, FS and HS Modes.

PLL Clock Multiplier

This module is composed of the off-chip crystal and the on-chip clock multiplier. It generates theappropriate internal clocks for the UTM and the CLK output signal. All data transfer signals aresynchronized with the CLK signal.

External Crystal

The external crystal is composed of a precise resonance frequency crystal and a crystal oscillator. It isoptional to have this crystal oscillator integrated on-chip or have it off-chip. This crystal/crystal oscillatorprovides a very precise clock of 12 MHz with deviation of ±100 ppm. The oscillator is not a part of thePHY, but external.

Clock Multiplier

The UTM interface is described as an un-directional/bi-directional 8-bit/16-bit parallel interface and theCLK signal is a 60/30 MHz signal. All data transfer signals should be synchronized with the CLK signal.CLK usable signal is internally implemented which blocks any transitions of CLK until it is usable.Meanwhile, the clock multiplier provides another three clocks in addition to the CLK signal. That is a480 MHz and 7.5 MHz clock signals.

Clock MUX

The Clock Multiplexer supplies both the transmitter and receiver paths with the adequate bit clockdepending on the XcvrSelect signal and to ensure smooth clock switching. It also includes clock gatingand power-down features.

Control Logic Block

This block is responsible for controlling, enabling and disabling the different blocks in the system.

OTG Circuitry （optional）

With the OTG circuitry, the system has the capability to dynamically switch between host and peripheral,enable dual role device behavior and point-to-point communication. The OTG circuitry functions asVBUS generation and detection. Both ID detection and terminations control are implemented in it.

ID Detector （optional）

To provide the ID signal that is used to indicate the state of the ID pin on the USB mini receptacle. Thispin makes it able to determine which kind of plug is connected and to confirm

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if the device default stateis A device or B device.

VBUS Detector and termination control

The VBUS detector is a set of comparators, functions to monitor and sense the voltage on USB buspower line. For VBUS signaling and discharging, VBUS pull up and pull-down resistors are alsoimplemented.

Automatic Test Functions

• Loop-back test to address all IP components.

In loop-back test mode, all transmitted data packets are received back in an internal loop to check IPfunctional integrity. There are some digital components that cannot be tested with the scan techniquedue to the high-speed nature of the digital part. To be regarded as a good idea, Loop-back allows testingfull design paths at speed. It should complement the testing suite for digital core to achieve the highestcoverage possible. According to the UTMI specification Section 5.18, version 1.05 Page 34, the 8 bits un-directional data bus can be implemented as 8 bits bi-directional one. This implementation will hinderthe loop-back test functionality.

22.5 UART BYPASS FUNCITON

When in UART bypass mode,UART2 is connect to USB interface; Otherwise, UART2 use normal UART interface。

Signal	CONNECT	I/O	Description
BYPASSDMDATA0	uart2_sout	I	Data for DM0 Transmitter Digital Bypass
BYPASSDMEN0	uoc0_con0[8]	I	DM0 Transmitter Digital Bypass Enable
BYPASSSEL0	uoc0_con0[9]	I	Transmitter Digital Bypass mode Enable
FSVPLUS0	uart2_sin	O	Single-Ended D- Indicator The controller signal indicates the state of the DP during normal operation or UART data reception
OTGDISABLE0	uoc0_con0[4]	I	1’b1: OTG0 disable; 1’b0: OTG0 normal mode
COMMONONN	uoc0_con0[0]	I	Common Block Power-Down Control This signal controls the power-down signals in PLL blocks when the USBPHY is in Suspend Mode. 1: PLL blocks are powered down. 0: PLL blocks remain powered This signal is a strapping option that must be set prior to a power-on reset and remain static during normal operation.

Note: USB OTG2.0 PHY support UART Bypass Function.

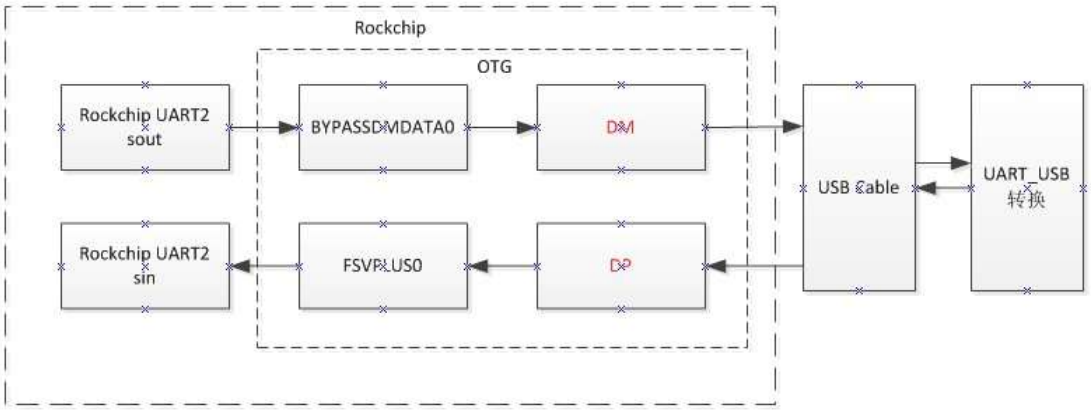

Fig. 22-9 UART Application

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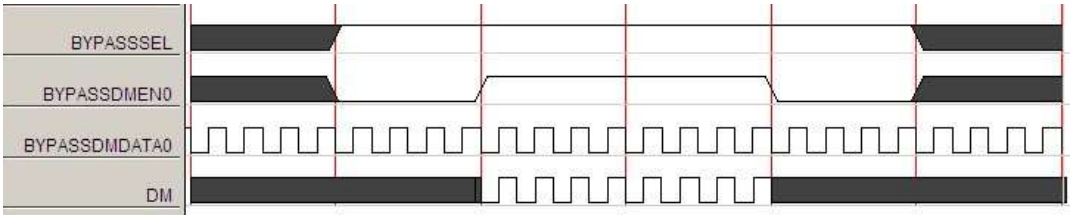

Fig. 22-10 UART Timing Sequence

To use UART and Autoresume functions:
1. Disable the OTG block by setting OTGDISABLE0 to 1’b1.
2. Disable the pull-up resistance on the D+ line by setting OPMODE0[1:0] to 2’b01.
3. To ensure that the XO, Bias, and PLL blocks are powered down in Suspend mode, set COMMONONN to 1’b1.
4. Place the USB PHY in Suspend mode by setting SUSPENDM0 to 1’b0.
5. Set BYPASSSEL0 to 1’b1.
6. To transmit data, controls BYPASSDMEN0, and BYPASSDMDATA0.
To receive data, monitor FSVPLUS0.

To return to normal operating mode:
1. Ensure that there is no activity on the USB.
2. Set BYPASSSEL0 to 1'b0.
3. Set SUSPENDM0 to 1’b1.Resume the USB PHY.
4. Set COMMONONN to 1’b0.
5. set OTGDISABLE0 to 1'b0.

22.6 Register Description

22.6.1 Register Summary

Name	Offset	Size	Reset Value	Description
USBOTG_GOTGCTL	0x0000	W	0x00000000	Control and Status Register
USBOTG_GOTGINT	0x0004	W	0x00000000	Interrupt Register
USBOTG_GAHBCFG	0x0008	W	0x00000000	AHB Configuration Register
USBOTG_GUSBCFG	0x000c	W	0x00001400	USB Configuration Register
USBOTG_GRSTCTL	0x0010	W	0x80000000	Reset Register
USBOTG_GINTSTS	0x0014	W	0x00000000	Interrupt Register
USBOTG_GINTMSK	0x0018	W	0x00000000	Interrupt Mask Register
USBOTG_GRXSTSR	0x001c	W	0x00000000	Receive Status Debug Read Register
USBOTG_GRXSTSP	0x0020	W	0x00000000	Receive Status Read and Pop Register
USBOTG_GRXFSIZ	0x0024	W	0x00000000	Receive FIFO Size Register
USBOTG_GN PTXFSIZ	0x0028	W	0x00000000	Non-Periodic Transmit FIFO Size Register
USBOTG_GNPTXSTS	0x002c	W	0x00000000	Non-Periodic Transmit FIFO/Queue Status Register
USBOTG_GI2CCTL	0x0030	W	0x11000000	I2C Address Register
USBOTG_GPVNDCTL	0x0034	W	0x00000000	PHY Vendor Control Register
USBOTG_GGPIO	0x0038	W	0x00000000	General Purpose Input/Output Register
USBOTG_GUID	0x003c	W	0x00000000	User ID Register
USBOTG_GSNPSID	0x0040	W	0x00004f54	Core ID Register

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Name	Offset	Size	Reset Value	Description
USBOTG_GHWCFG1	0x0044	W	0x00000000	User HW Config1 Register
USBOTG_GHWCFG2	0x0048	W	0x00000000	User HW Config2 Register
USBOTG_GHWCFG3	0x004c	W	0x00000000	User HW Config3 Register
USBOTG_GHWCFG4	0x0050	W	0x00000000	User HW Config4 Register
USBOTG_GLPMCFG	0x0054	W	0x00000000	Core LPM Configuration Register
USBOTG_GPWRDN	0x0058	W	0x00000000	Global Power Down Register
USBOTG_GDFIFOCFG	0x005c	W	0x00000000	Global DFIFO Software Configuration Register
USBOTG_GADPCTL	0x0060	W	0x00000000	ADP Timer,Control and Status Register
USBOTG_HPTXFSIZ	0x0100	W	0x00000000	Host Periodic Transmit FIFO Size Register
USBOTG_DIEPTXFn	0x0104+4*(n-1)	W	0x00000000	Device Periodic Transmit FIFO-n Size Register n = 1 - 15
USBOTG_HCFG	0x0400	W	0x00000000	Host Configuration Register
USBOTG_HFIR	0x0404	W	0x00000000	Host Frame Interval Register
USBOTG_HFNUM	0x0408	W	0x0000ffff	Host Frame Number/Frame Time Remaining Register
USBOTG_HPTXSTS	0x0410	W	0x00000000	Host Periodic Transmit FIFO/Queue Status Register
USBOTG_HAINT	0x0414	W	0x00000000	Host All Channels Interrupt Register
USBOTG_HAINTMSK	0x0418	W	0x00000000	Host All Channels Interrupt Mask Register
USBOTG_HPRT	0x0440	W	0x00000000	Host Port Control and Status Register
USBOTG_HCCHARn	0x0500+0x20*n	W	0x00000000	Host Channel-n Characteristics Register n = 0 - 15
USBOTG_HCSPLTn	0x0504+0x20*n	W	0x00000000	Host Channel-n Split Control Register n = 0 - 15
USBOTG_HCINTn	0x0508+0x20*n	W	0x00000000	Host Channel-n Interrupt Register n = 0 - 15
USBOTG_HCINTMSKn	0x050c+0x20*n	W	0x00000000	Host Channel-n Interrupt Mask Register n = 0 - 15
USBOTG_HCTSIZn	0x0510+0x20*n	W	0x00000000	Host Channel-n Transfer Size Register n = 0 - 15
USBOTG_HCDMAn	0x0514+0x20*n	W	0x00000000	Host Channel-n DMA Address Register n = 0 - 15

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Name	Offset	Size	Reset Value	Description
USBOTG_HCDMABn	0x051c+0x20*n	W	0x00000000	Host Channel-n DMA Buffer Address Register n = 0 - 15
USBOTG_DCFG	0x0800	W	0x08200000	Device Configuration Register
USBOTG_DCTL	0x0804	W	0x00002000	Device Control Register
USBOTG_DSTS	0x0808	W	0x00000000	Device Status Register
USBOTG_DIEPMSK	0x0810	W	0x00000000	Device IN Endpoint common interrupt mask register
USBOTG_DOEPMSK	0x0814	W	0x00000000	Device OUT Endpoint common interrupt mask register
USBOTG_DAINT	0x0818	W	0x00000000	Device All Endpoints interrupt register
USBOTG_DAINTMSK	0x081c	W	0x00000000	Device All Endpoint interrupt mask register
USBOTG_DTKNQR1	0x0820	W	0x00000000	Device IN token sequence learning queue read register1
USBOTG_DTKNQR2	0x0824	W	0x00000000	Device IN token sequence learning queue read register2
USBOTG_DVBUSDIS	0x0828	W	0x00000b8f	Device VBUS discharge time register
USBOTG_DVBUSPULSE	0x082c	W	0x00000000	Device VBUS Pulsing Timer Register
USBOTG_DTHRCTL	0x0830	W	0x08100020	Device Threshold Control Register
USBOTG_DIEPEMPMSK	0x0834	W	0x00000000	Device IN endpoint FIFO empty interrupt mask register
USBOTG_DEACHINT	0x0838	W	0x00000000	Device each endpoint interrupt register
USBOTG_DEACHINTMSK	0x083c	W	0x00000000	Device each endpoint interrupt register mask
USBOTG_DIEPEACHMSKn	0x0840+4*n	W	0x00000000	Device each IN endpoint -n interrupt Register n = 0 - 15
USBOTG_DOEPEACHMSKn	0x0880+4*n	W	0x00000000	Device each out endpoint-n interrupt register n = 0 - 15
USBOTG_DIEPCTL0	0x0900	W	0x00008000	Device control IN endpoint 0 control register
USBOTG_DIEPINTn	0x0908+0x20*n	W	0x00000000	Device Endpoint-n Interrupt Register n = 0 - 15
USBOTG_DIEPTSIZn	0x0910+0x20*n	W	0x00000000	Device endpoint n transfer size register n = 0 - 15

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Name	Offset	Size	Reset Value	Description
USBOTG_DIEPDMAn	0x0914+0x20*n	W	0x00000000	Device endpoint-n DMA address register n = 0 - 15
USBOTG_DTXFSTSn	0x0918+0x20*n	W	0x00000000	Device IN endpoint transmit FIFO status register n = 0 - 15
USBOTG_DIEPDMABn	0x091c	W	0x00000000	Device endpoint-n DMA buffer address register
USBOTG_DIEPCTLn	0x0920+0x20*(n-1)	W	0x00000000	Device endpoint-n control register n = 1 - 15
USBOTG_DOEPCTL0	0x0b00	W	0x00000000	Device control OUT endpoint 0 control register
USBOTG_DOEPINTn	0x0b08+0x20*n	W	0x00000000	Device endpoint-n control register n = 0 - 15
USBOTG_DOEPTSIZn	0x0b10+0x20*n	W	0x00000000	Device endpoint n transfer size register n = 0 - 15
USBOTG_DOEPDMAn	0x0b14+0x20*n	W	0x00000000	Device Endpoint-n DMA Address Register n = 0 - 15
USBOTG_DOEPDMABn	0x0b1c+0x20*n	W	0x00000000	Device endpoint-n DMA buffer address register n = 0 - 15
USBOTG_DOEPCTLn	0x0b20+0x20*(n-1)	W	0x00000000	Device endpoint-n control register n = 1 - 15
USBOTG_PCGCR	0x0b24	W	0x200b8000	Power and clock gating control register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

22.6.2 Detail Register Description

USBOTG_GOTGCTL

Address: Operational Base + offset (0x0000)

Control and Status Register

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27	RW	0x0	ChirpEn Chirp on enable This bit when programmed to 1'b1 results in the core asserting chirp_on before sending an actual Chirp "K" signal on USB. This bit is present only if OTG_BC_SUPPORT = 1. If OTG_BC_SUPPORT != 1, this bit is a reserved bit.

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Bit	Attr	Reset Value	Description
26:22	RO	0x00	MultValidBc Multi Valued ID pin Battery Charger ACA inputs in the following order: Bit 26 - rid_float. Bit 25 - rid_gnd Bit 24 - rid_a Bit 23 - rid_b Bit 22 - rid_c These bits are present only if OTG_BC_SUPPORT = 1. Otherwise, these bits are reserved and will read 5'h0.
21	RO	0x0	reserved
20	RW	0x0	OTGVer OTG version Indicates the OTG revision. 1'b0: OTG Version 1.3. In this version the core supports Data line pulsing and VBus pulsing for SRP. 1'b1: OTG Version 2.0. In this version the core supports only Data line pulsing for SRP.
19	RO	0x0	BSesVld B-session valid Indicates the Device mode transceiver status. 1'b0: B-session is not valid. 1'b1: B-session is valid. In OTG mode, you can use this bit to determine if the device is connected or disconnected. Note: If you do not enabled OTG features (such as SRP and HNP), the read reset value will be 1.The vbus assigns the values internally for non-SRP or non-HNP configurations.
18	RO	0x0	ASesVld A-session valid Indicates the Host mode transceiver status. 1'b0: A-session is not valid 1'b1: A-session is valid Note: If you do not enabled OTG features (such as SRP and HNP), the read reset value will be 1.The vbus assigns the values internally for non-SRP or non-HNP configurations.

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Bit	Attr	Reset Value	Description
17	RO	0x0	DbnTime Long/short debounce time Indicates the debounce time of a detected connection. 1'b0: Long debounce time, used for physical connections (100 ms + 2.5 us) 1'b1: Short debounce time, used for soft connections (2.5 us)
16	RO	0x0	ConIDSts Connector ID Status Indicates the connector ID status on a connect event. 1'b0: The core is in A-Device mode 1'b1: The core is in B-Device mode
15:12	RO	0x0	reserved
11	RW	0x0	DevHNPEn Device HNP Enable The application sets this bit when it successfully receives a SetFeature. SetHNPEnable command from the connected USB host. 1'b0: HNP is not enabled in the application 1'b1: HNP is enabled in the application
10	RW	0x0	HstSetHNPEn Host set HNP enable The application sets this bit when it has successfully enabled HNP (using the SetFeature.SetHNPEnable command) on the connected device. 1'b0: Host Set HNP is not enabled 1'b1: Host Set HNP is enabled
9	RW	0x0	HNPReq HNP request The application sets this bit to initiate an HNP request to the connected USB host. The application can clear this bit by writing a 0 when the Host Negotiation Success Status Change bit in the OTG Interrupt register (GOTGINT.HstNegSucStsChng) is set. The core clears this bit when the HstNegSucStsChng bit is cleared. 1'b0: No HNP request 1'b1: HNP request

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Bit	Attr	Reset Value	Description
8	RO	0x0	HstNegScs Host Negotiation Success The core sets this bit when host negotiation is successful. The core clears this bit when the HNP Request (HNPReq) bit in this register is set. 1'b0: Host negotiation failure 1'b1: Host negotiation success
7:2	RO	0x0	reserved
1	RW	0x0	SesReq Session Request The application sets this bit to initiate a session request on the USB. The application can clear this bit by writing a 0 when the Host Negotiation Success Status Change bit in the OTG Interrupt register (GOTGINT.HstNegSucStsChng) is set. The core clears this bit when the HstNegSucStsChng bit is cleared. If you use the USB 1.1 Full-Speed Serial Transceiver interface to initiate the session request, the application must wait until the VBUS discharges to 0.2 V, after the B-Session Valid bit in this register (GOTGCTL.BSesVld) is cleared. This discharge time varies between different PHYs and can be obtained from the PHY vendor. 1'b0: No session request 1'b1: Session request
0	RO	0x0	SesReqScs Session Request Success The core sets this bit when a session request initiation is successful. 1'b0: Session request failure 1'b1: Session request success

USBOTG_GOTGINT

Address: Operational Base + offset (0x0004)

Interrupt Register

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
20	W1C	0x0	MultiValueChg Multi-Valued input changed This bit when set indicates that there is a change in the value of at least one ACA pin value. This bit is present only if OTG_BC_SUPPORT = 1, otherwise it is reserved.
19	W1C	0x0	DbnceDone Debounce Done The core sets this bit when the debounce is completed after the device connection. The application can start driving USB reset after seeing this interrupt. This bit is only valid when the HNP Capable or SRP Capable bit is set in the Core USB Configuration register (GUSBCFG.HNPCap or GUSBCFG.SRPCap, respectively).
18	W1C	0x0	ADevTOUTChg A-Device Timeout Change The core sets this bit to indicate that the A-device has timed out while waiting for the B-device to connect.
17	W1C	0x0	HstNegDet Host Negotiation Detected The core sets this bit when it detects a host negotiation request on the USB
16:10	RO	0x0	reserved
9	W1C	0x0	HstNegSucStsChng Host Negotiation Success Status Change The core sets this bit on the success or failure of a USB host negotiation request. The application must read the Host Negotiation Success bit of the OTG Control and Status register (GOTGCTL.HstNegScs) to check for success or failure
8	W1C	0x0	SesReqSucStsChng Session Request Success Status Change The core sets this bit on the success or failure of a session request.The application must read the Session Request Success bit in the OTG Control and Status register (GOTGCTL.SesReqScs) to check for success or failure.
7:3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
2	W1C	0x0	SesEndDet Session End Detected The core sets this bit when the utmisrp_bvalid signal is deasserted
1:0	RO	0x0	reserved

USBOTG_GAHBCFG

Address: Operational Base + offset (0x0008)

AHB Configuration Register

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22	RW	0x0	NotiAllDmaWrit Notify All Dma Write Transactions This bit is programmed to enable the System DMA Done functionality for all the DMA write Transactions corresponding to the Channel/Endpoint. This bit is valid only when GAHBCFG.RemMemSupp is set to 1. GAHBCFG.NotiAllDmaWrit = 1. HSOTG core asserts int_dma_req for all the DMA write transactions on the AHB interface along with int_dma_done, chep_last_transact and chep_number signal informations. The core waits for sys_dma_done signal for all the DMA write transactions in order to complete the transfer of a particular Channel/Endpoint. GAHBCFG.NotiAllDmaWrit = 0. HSOTG core asserts int_dma_req signal only for the last transaction of DMA write transfer corresponding to a particular Channel/Endpoint. Similarly, the core waits for sys_dma_done signal only for that transaction of DMA write to complete the transfer of a particular Channel/Endpoint.

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Bit	Attr	Reset Value	Description
21	RW	0x0	RemMemSupp Remote Memory Support This bit is programmed to enable the functionality to wait for the system DMA Done Signal for the DMA Write Transfers. GAHBCFG.RemMemSupp=1. The int_dma_req output signal is asserted when HSOTG DMA starts write transfer to the external memory. When the core is done with the Transfers it asserts int_dma_done signal to flag the completion of DMA writes from HSOTG. The core then waits for sys_dma_done signal from the system to proceed further and complete the Data Transfer corresponding to a particular Channel/Endpoint. GAHBCFG.RemMemSupp=0. The int_dma_req and int_dma_done signals are not asserted and the core proceeds with the assertion of the XferComp interrupt as soon as the DMA write transfer is done at the HSOTG Core Boundary and it does not wait for the sys_dma_done signal to complete the DATA transfers.
20:9	RO	0x0	reserved
8	RW	0x0	PTxFEmpLvl Periodic TxFIFO Empty Level Indicates when the Periodic TxFIFO Empty Interrupt bit in the Core Interrupt register (GINTSTS.PTxFEmp) is triggered. This bit is used only in Slave mode. 1'b0: GINTSTS.PTxFEmp interrupt indicates that the Periodic TxFIFO is half empty 1'b1: GINTSTS.PTxFEmp interrupt indicates that the Periodic TxFIFO is completely empty

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Bit	Attr	Reset Value	Description
7	RW	0x0	NPTxFEmpLvl Non-Periodic TxFIFO Empty Level This bit is used only in Slave mode. In host mode and with Shared FIFO with device mode, this bit indicates when the Non-Periodic TxFIFO Empty Interrupt bit in the Core Interrupt register GINTSTS.NPTxFEmp) is triggered. With dedicated FIFO in device mode, this bit indicates when IN endpoint Transmit FIFO empty interrupt (DIEPINTn.TxFEmp) is triggered. Host mode and with Shared FIFO with device mode: 1'b0: GINTSTS.NPTxFEmp interrupt indicates that the Non-Periodic TxFIFO is half empty 1'b1: GINTSTS.NPTxFEmp interrupt indicates that the Non-Periodic TxFIFO is completely empty Dedicated FIFO in device mode: 1'b0: DIEPINTn.TxFEmp interrupt indicates that the IN Endpoint TxFIFO is half empty 1'b1: DIEPINTn.TxFEmp interrupt indicates that the IN Endpoint TxFIFO is completely empty
6	RO	0x0	reserved
5	RW	0x0	DMAEn DMA Enable 1'b0: Core operates in Slave mode 1'b1: Core operates in a DMA mode This bit is always 0 when Slave-Only mode has been selected.

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Bit	Attr	Reset Value	Description
4:1	RW	0x0	HBstLen Burst Length/Type This field is used in both External and Internal DMA modes. In External DMA mode, these bits appear on dma_burst[3:0] ports, External DMA Mode defines the DMA burst length in terms of 32-bit words: 4'b0000: 1 word 4'b0001: 4 words 4'b0010: 8 words 4'b0011: 16 words 4'b0100: 32 words 4'b0101: 64 words 4'b0110: 128 words 4'b0111: 256 words Others: Reserved Internal DMA Mode AHB Master burst type: 4'b0000: Single 4'b0001: INCR 4'b0011: INCR4 4'b0101: INCR8 4'b0111: INCR16 Others: Reserved
0	RW	0x0	GlblIntrMsk Global Interrupt Mask The application uses this bit to mask or unmask the interrupt line assertion to itself. Irrespective of this bit's setting, the interrupt status registers are updated by the core. 1'b0: Mask the interrupt assertion to the application. 1'b1: Unmask the interrupt assertion to the application.

USBOTG_GUSBCFG

Address: Operational Base + offset (0x000c)

USB Configuration Register

Bit	Attr	Reset Value	Description
31	RW	0x0	CorruptTxpacket Corrupt Tx packet This bit is for debug purposes only. Never set this bit to 1.

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Bit	Attr	Reset Value	Description
30	RW	0x0	ForceDevMode Force Device Mode Writing a 1 to this bit forces the core to device mode irrespective of utmiotg_iddig input pin. 1'b0: Normal Mode 1'b1: Force Device Mode After setting the force bit, the application must wait at least 25 ms before the change to take effect. When the simulation is in scale down mode, waiting for 500 us is sufficient. This bit is valid only when OTG_MODE = 0, 1 or 2. In all other cases, this bit reads 0.
29	RW	0x0	ForceHstMode Force Host Mode Writing a 1 to this bit forces the core to host mode irrespective of utmiotg_iddig input pin. 1'b0: Normal Mode 1'b1: Force Host Mode After setting the force bit, the application must wait at least 25 ms before the change to take effect. When the simulation is in scale down mode, waiting for 500 us is sufficient. This bit is valid only when OTG_MODE =0, 1 or 2. In allother cases, this bit reads 0.
28	RW	0x0	TxEndDelay Tx End Delay Writing a 1 to this bit enables the TxEndDelay timers in the core. 1'b0: Normal mode 1'b1: Introduce Tx end delay timers
27	RW	0x0	IC_USBTrafCtl IC_USB TrafficPullRemove Control When this bit is set, pullup/pulldown resistors are detached from the USB during traffic signaling.This bit is valid only when configuration parameter OTG_ENABLE_IC_USB = 1 and register field GUSBCFG.IC_USBCap is set to 1.

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Bit	Attr	Reset Value	Description
26	RW	0x0	IC_USBCap IC_USB-Capable The application uses this bit to control the IC_USB capabilities. 1'b0: IC_USB PHY Interface is not selected. 1'b1: IC_USB PHY Interface is selected. This bit is writable only if OTG_ENABLE_IC_USB=1 and OTG_FSPHY_INTERFACE!=0.The reset value depends on the configuration parameter OTG_SELECT_IC_USB when OTG_ENABLE_IC_USB = 1. In all other cases, this bit is set to 1'b0 and the bit is read only.
25	RW	0x0	ULPIIfDis ULPI Interface Protect Disable Controls circuitry built into the PHY for protecting the ULPI interface when the link tri-states STP and data. Any pull-ups or pull-downs employed by this feature can be disabled. Please refer to the ULPI Specification for more detail. 1'b0: Enables the interface protect circuit 1'b1: Disables the interface protect circuit
24	RW	0x0	IndPassThrough Indicator Pass Through Controls whether the Complement Output is qualified with the Internal Vbus Valid comparator before being used in the Vbus State in the RX CMD. Please refer to the ULPI Specification for more detail. 1'b0: Complement Output signal is qualified with the Internal VbusValid comparator. 1'b1: Complement Output signal is not qualified with the Internal VbusValid comparator.
23	RW	0x0	IndComple Indicator Complement Controls the PHY to invert the ExternalVbusIndicator input signal,generating the Complement Output. Please refer to the ULPI Specification for more detail 1'b0: PHY does not invert ExternalVbusIndicator signal 1'b1: PHY does invert ExternalVbusIndicator signal
22	RW	0x0	TermSelDLPulse TermSel DLine Pulsing Selection This bit selects utmi_termselect to drive data line pulse during SRP. 1'b0: Data line pulsing using utmi_txvalid (default). 1'b1: Data line pulsing using utmi_termsel.

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Bit	Attr	Reset Value	Description
21	RW	0x0	ULPIExtVbusIndicator ULPI External VBUS Indicator This bit indicates to the ULPI PHY to use an external VBUS over-current indicator. 1'b0: PHY uses internal VBUS valid comparator. 1'b1: PHY uses external VBUS valid comparator. (Valid only when RTL parameter OTG_HSPHY_INTERFACE = 2 or 3)
20	RW	0x0	ULPIExtVbusDrv ULPI External VBUS Drive This bit selects between internal or external supply to drive 5V on VBUS,in ULPI PHY. 1'b0: PHY drives VBUS using internal charge pump (default). 1'b1: PHY drives VBUS using external supply. (Valid only when RTL parameter OTG_HSPHY_INTERFACE = 2 or 3)
19	RW	0x0	ULPIClkSusM ULPI Clock SuspendM This bit sets the ClockSuspendM bit in the Interface Control register on the ULPI PHY. This bit applies only in serial or carkit modes. 1'b0: PHY powers down internal clock during suspend. 1'b1: PHY does not power down internal clock. (Valid only when RTL parameter OTG_HSPHY_INTERFACE = 2 or 3)
18	RW	0x0	ULPIAutoRes ULPI Auto Resume This bit sets the AutoResume bit in the Interface Control register on the ULPI PHY. 1'b0: PHY does not use AutoResume feature. 1'b1: PHY uses AutoResume feature. (Valid only when RTL parameter OTG_HSPHY_INTERFACE = 2 or 3)
17	RW	0x0	ULPIFsLs ULPI FS/LS Select The application uses this bit to select the FS/LS serial interface for the ULPI PHY. This bit is valid only when the FS serial transceiver is selected on the ULPI PHY. 1'b0: ULPI interface 1'b1: ULPI FS/LS serial interface (Valid only when RTL parameters OTG_HSPHY_INTERFACE = 2 or 3 and OTG_FSPHY_INTERFACE = 1, 2, or 3)

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Bit	Attr	Reset Value	Description
16	RW	0x0	OtgI2CSel UTMIFS or I2C Interface Select The application uses this bit to select the I2C interface. 1'b0: UTMI USB 1.1 Full-Speed interface for OTG signals 1'b1: I2C interface for OTG signals This bit is writable only if I2C and UTMIFS were specified for Enable I2C Interface? (parameter OTG_I2C_INTERFACE = 2). Otherwise, reads return 0.
15	RW	0x0	PhyLPwrClkSel PHY Low-Power Clock Select Selects either 480-MHz or 48-MHz (low-power) PHY mode. In FS and LS modes, the PHY can usually operate on a 48-MHz clock to save power. 1'b0: 480-MHz Internal PLL clock 1'b1: 48-MHz External Clock In 480 MHz mode, the UTMI interface operates at either 60 or 30-MHz, depending upon whether 8- or 16-bit data width is selected. In 48-MHz mode, the UTMI interface operates at 48 MHz in FS and LS modes. This bit drives the utmi_fsls_low_power core output signal, and is valid only for UTMI+ PHYs.
14	RO	0x0	reserved
13:10	RW	0x5	USBTrdTim USB Turnaround Time Sets the turnaround time in PHY clocks. Specifies the response time for a MAC request to the Packet FIFO Controller (PFC) to fetch data from the DFIF(SPRAM). This must be programmed to 4'h5: When the MAC interface is 16-bit UTMI+. 4'h9: When the MAC interface is 8-bit UTMI+. Note: The values above are calculated for the minimum AHB frequency of 30 MHz. USB turnaround time is critical for certification where long cables and 5-Hubs are used, so if you need the AHB to run at less than 30 MHz, and if USB turnaround time is not critical, these bits can be programmed to a larger value.
9	RW	0x0	HNPCap HNP-Capable The application uses this bit to control the Otg core's HNP capabilities. 1'b0: HNP capability is not enabled. 1'b1: HNP capability is enabled. This bit is writable only if an HNP mode was specified for Mode of Operation (parameter OTG_MODE). Otherwise, reads return 0.

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Bit	Attr	Reset Value	Description
8	RW	0x0	SRPCap SRP-Capable The application uses this bit to control the Otg core SRP capabilities. If the core operates as a non-SRP-capable B-device, it cannot request the connected A-device (host) to activate VBUS and start a session. 0: SRP capability is not enabled. 1: SRP capability is enabled. This bit is writable only if an SRP mode was specified for Mode of Operation (parameter OTG_MODE). Otherwise, reads return 0.
7	RW	0x0	DDRSel ULPI DDR Select The application uses this bit to select a Single Data Rate (SDR) or Double Data Rate (DDR) or ULPI interface. 1'b0: Single Data Rate ULPI Interface, with 8-bit-wide data bus 1'b1: Double Data Rate ULPI Interface, with 4-bit-wide data bus
6	RW	0x0	PHYSel USB 2.0 High-Speed PHY or USB 1.1 Full-Speed Serial Transceiver The application uses this bit to select either a high-speed UTMI+ or ULPI PHY, or a full-speed transceiver. 1'b0: USB 2.0 high-speed UTMI+ or ULPI PHY 1'b1: USB 1.1 full-speed serial transceiver If a USB 1.1 Full-Speed Serial Transceiver interface was not selected (parameter OTG_FSPHY_INTERFACE = 0), this bit is always 0, with Write Only access. If a high-speed PHY interface was not selected (parameter OTG_HSPHY_INTERFACE = 0), this bit is always 1, with Write Only access. If both interface types were selected (parameters have non-zero values), the application uses this bit to select which interface is active, and access is Read and Write.

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Bit	Attr	Reset Value	Description
5	RW	0x0	FSIntf Full-Speed Serial Interface Select The application uses this bit to select either a unidirectional or bidirectional USB 1.1 full-speed serial transceiver interface. 1'b0: 6-pin unidirectional full-speed serial interface 1'b1: 3-pin bidirectional full-speed serial interface If a USB 1.1 Full-Speed Serial Transceiver interface was not selected (parameter OTG_FSPHY_INTERFACE = 0), this bit is always 0, with Write Only access. If a USB 1.1 FS interface was selected (parameter OTG_FSPHY_INTERFACE! = 0), then the application can set this bit to select between the 3- and 6-pin interfaces, and access is Read and Write.
4	RW	0x0	ULPI_UTMI_Sel ULPI or UTMI+ Select The application uses this bit to select either a UTMI+ interface or ULPI Interface. 1'b0: UTMI+ Interface 1'b1: ULPI Interface This bit is writable only if UTMI+ and ULPI was specified for High-Speed PHY Interface(s) (parameter OTG_HSPHY_INTERFACE = 3). Otherwise, reads return either 0 or 1, depending on the interface selected using the OTG_HSPHY_INTERFACE parameter.
3	RW	0x0	PHYIf PHY Interface The application uses this bit to configure the core to support a UTMI+ PHY with an 8- or 16-bit interface. When a ULPI PHY is chosen, this must be set to 8-bit mode. 1'b0: 8 bits 1'b1: 16 bits This bit is writable only if UTMI+ and ULPI were selected (parameter OTG_HSPHY_DWIDTH = 3). Otherwise, this bit returns the value for the power-on interface selected during configuration.

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Bit	Attr	Reset Value	Description
2:0	RW	0x0	TOutCal HS/FS Timeout Calibration The number of PHY clocks that the application programs in this field is added to the high-speed/full-speed inter-packet timeout duration in the core to account for any additional delays introduced by the PHY. This can be required, because the delay introduced by the PHY in generating the line state condition can vary from one PHY to another. The USB standard timeout value for high-speed operation is 736 to 816 (inclusive) bit times. The USB standard timeout value for full-speedoperation is 16 to 18 (inclusive) bit times. The application must program this field based on the speed of enumeration. The number of bit times added per PHY clock are: High-speed operation: One 30-MHz PHY clock = 16 bit times One 60-MHz PHY clock = 8 bit times Full-speed operation: One 30-MHz PHY clock = 0.4 bit times One 60-MHz PHY clock = 0.2 bit times One 48-MHz PHY clock = 0.25 bit times

USBOTG_GRSTCTL
Address: Operational Base + offset (0x0010)
Reset Register

Bit	Attr	Reset Value	Description
31	RO	0x1	AHBIdle AHB Master Idle Indicates that the AHB Master State Machine is in the IDLE condition.
30	RO	0x0	DMAReq DMA Request Signal Indicates that the DMA request is in progress. Used for debug.
29:11	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
10:6	RW	0x00	TxFNum TxFIFO Number This is the FIFO number that must be flushed using the TxFIFO Flush bit. This field must not be changed until the core clears the TxFIFO Flush bit. 5'h0: Non-periodic TxFIFO flush in Host mode; Non-periodic TxFIFO flush in device mode when in shared FIFO operation. Tx FIFO 0 flush in device mode when in dedicated FIFO mode. 5'h1: Periodic TxFIFO flush in Host mode: Periodic TxFIFO 1 flush in Device mode when in shared FIFO operation; TXFIFO 1 flush in device mode when in dedicated FIFO mode. 5'h2: Periodic TxFIFO 2 flush in Device mode when in shared FIFO operation: TXFIFO 2 flush in device mode when in dedicated FIFO mode. ... 5'hF: Periodic TxFIFO 15 flush in Device mode when in shared FIFO operation: TXFIFO 15 flush in device mode when in dedicated FIFO mode. 5'h10: Flush all the transmit FIFOs in device or host mode.
5	R/WSC	0x0	TxFFlsh TxFIFO Flush This bit selectively flushes a single or all transmit FIFOs, but cannot do so if the core is in the midst of a transaction. The application must write this bit only after checking that the core is neither writing to the TxFIFO nor reading from the TxFIFO. Verify using these registers: Read NAK Effective Interrupt ensures the core is not reading from the FIFO. Write GRSTCTL.AHBIdle ensures the core is not writing anything to the FIFO. Flushing is normally recommended when FIFOs are re-configured or when switching between Shared FIFO and Dedicated Transmit FIFO operation. FIFO flushing is also recommended during device endpoint disable. The application must wait until the core clears this bit before performing any operations. This bit takes eight clocks to clear, using the slower clock of phy_clk or hclk.
4	R/WSC	0x0	RxFFlsh RxFIFO Flush The application can flush the entire RxFIFO using this bit, but must first ensure that the core is not in the middle of a transaction. The application must only write to this bit after checking that the core is neither reading from the RxFIFO nor writing to the RxFIFO. The application must wait until the bit is cleared before performing any other operations. This bit requires 8 clocks (slowest of PHY or AHB clock) to clear.

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Bit	Attr	Reset Value	Description
3	R/WSC	0x0	INTknQFlsh IN Token Sequence Learning Queue Flush This bit is valid only if OTG_EN_DED_TX_FIFO = 0. The application writes this bit to flush the IN Token Sequence Learning Queue.
2	W1C	0x0	FrmCntrRst Host Frame Counter Reset The application writes this bit to reset the (micro)frame number counter inside the core. When the (micro)frame counter is reset, the subsequent SOF sent out by the core has a (micro)frame number of 0.
1	R/WSC	0x0	Reset A write to this bit issues a soft reset to the Otg_power_dn module of the core.

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Bit	Attr	Reset Value	Description
0	R/WSC	0x0	CSftRst Core Soft Reset Resets the hclk and phy_clock domains as follows: Clears the interrupts and all the CSR registers except the following register bits: PCGCCTL.RstPdwnModule PCGCCTL.GateHclk PCGCCTL.PwrClmp PCGCCTL.StopPPhyLPwrClkSelclk GUSBCFG.PhyLPwrClkSel GUSBCFG.DDRSel GUSBCFG.PHYSel GUSBCFG.FSIntf GUSBCFG.ULPI_UTMI_Sel GUSBCFG.PHYIf HCFG.FSLSPclkSel DCFG.DevSpd GGPIO GPWRDN GADPCTL All module state machines (except the AHB Slave Unit) are reset to the IDLE state, and all the transmit FIFOs and the receive FIFO are flushed. Any transactions on the AHB Master are terminated as soon as possible, after gracefully completing the last data phase of an AHB transfer. Any transactions on the USB are terminated immediately. When Hibernation or ADP feature is enabled, the PMU module is not reset by the Core Soft Reset. The application can write to this bit any time it wants to reset the core. This is a self-clearing bit and the core clears this bit after all the necessary logic is reset in the core, which can take several clocks, depending on the current state of the core. Once this bit is cleared software must wait at least 3 PHY clocks before doing any access to the PHY domain (synchronization delay). Software must also must check that bit 31 of this register is 1 (AHB Master is IDLE) before starting any operation. Typically software reset is used during software development and also when you dynamically change the PHY selection bits in the USB configuration registers listed above. When you change the PHY, the corresponding clock for the PHY is selected and used in the PHY domain. Once a new clock is selected, the PHY domain has to be reset for proper operation.

USBOTG_GINTSTS
Address: Operational Base + offset (0x0014)
Interrupt Register

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Bit	Attr	Reset Value	Description
31	W1C	0x0	WkUpInt Resume/Remote Wakeup Detected Interrupt Wakeup Interrupt during Suspend(L2) or LPM(L1) state. During Suspend(L2): Device Mode:This interrupt is asserted only when Host Initiated Resume is detected on USB. Host Mode:This interrupt is asserted only when Device Initiated Remote Wakeup is detected on USB. During LPM(L1): Device Mode:This interrupt is asserted for either Host Initiated Resume or Device Initiated Remote Wakeup on USB. Host Mode:This interrupt is asserted for either Host Initiated Resume or Device Initiated Remote Wakeup on USB.
30	W1C	0x0	SessReqInt Session Request/New Session Detected Interrupt In Host mode, this interrupt is asserted when a session request is detected from the device. In Host mode, this interrupt is asserted when a session request is detected from the device. In Device mode, this interrupt is asserted when the utmisrp_bvalid signal goes high.
29	W1C	0x0	DisconnInt Disconnect Detected Interrupt This interrupt is asserted when a device disconnect is detected.
28	W1C	0x0	ConIDStsChng Connector ID Status Change This interrupt is asserted when there is a change in connector ID status.
27	W1C	0x0	LPM_Int LPM Transaction Received Interrupt Device Mode : This interrupt is asserted when the device receives an LPM transaction and responds with a non-ERRORed response. Host Mode :This interrupt is asserted when the device responds to an LPM transaction with a non-ERRORed response or when the host core has completed LPM transactions for the programmed number of times (GLPMCFG.RetryCnt). This field is valid only if the Core LPM Configuration register's LPMCapable (LPMCap) field is set to 1.
26	RO	0x0	PTxFEmp Periodic TxFIFO Empty This interrupt is asserted when the Periodic Transmit FIFO is either half or completely empty and there is space for at least one entry to be written in the Periodic Request Queue. The half or completely empty status is determined by the Periodic TxFIFO Empty Level bit in the Core AHB Configuration register (GAHBCFG.PTxFEmpLvl).

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Bit	Attr	Reset Value	Description
25	RO	0x0	HChInt Host Channels Interrupt The core sets this bit to indicate that an interrupt is pending on one of the channels of the core (in Host mode). The application must read the Host All Channels Interrupt (HAINT) register to determine the exact number of the channel on which the interrupt occurred, and then read the corresponding Host Channel-n Interrupt (HCINTn) register to determine the exact cause of the interrupt. The application must clear the appropriate status bit in the HCINTn register to clear this bit.
24	RO	0x0	PrtInt Host Port Interrupt The core sets this bit to indicate a change in port status of one of the OTG core ports in Host mode. The application must read the Host Port Control and Status (HPRT) register to determine the exact event that caused this interrupt. The application must clear the appropriate status bit in the Host Port Control and Status register to clear this bit.
23	RW	0x0	ResetDet Reset Detected Interrupt The core asserts this interrupt in Device mode when it detects a reset on the USB in Partial Power-Down mode when the device is in Suspend. This interrupt is not asserted in Host mode.
22	W1C	0x0	FetSusp Data Fetch Suspended This interrupt is valid only in DMA mode. This interrupt indicates that the core has stopped fetching data for IN endpoints due to the unavailability of TxFIFO space or Request Queue space. This interrupt is used by the application for an endpoint mismatch algorithm.
21	W1C	0x0	incomplP Incomplete Periodic Transfer In Host mode, the core sets this interrupt bit when there are incomplete periodic transactions still pending which are scheduled for the current micro-frame. Incomplete Isochronous OUT Transfer (incompISOOUT) The Device mode, the core sets this interrupt to indicate that there is at least one isochronous OUT endpoint on which the transfer is not completed in the current micro-frame. This interrupt is asserted along with the End of Periodic Frame Interrupt (EOPF) bit in this register.

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Bit	Attr	Reset Value	Description
20	W1C	0x0	incompISOIN Incomplete Isochronous IN Transfer The core sets this interrupt to indicate that there is at least one isochronous IN endpoint on which the transfer is not completed in the current micro-frame. This interrupt is asserted along with the End of Periodic Frame Interrupt (EOPF) bit in this register. Note: This interrupt is not asserted in Scatter/Gather DMA mode.
19	RO	0x0	OEPInt OUT Endpoints Interrupt The core sets this bit to indicate that an interrupt is pending on one of the OUT endpoints of the core (in Device mode). The application must read the Device All Endpoints Interrupt (DAINT) register to determine the exact number of the OUT endpoint on which the interrupt occurred, and then read the corresponding Device OUT Endpoint-n Interrupt (DOEPINTn) register to determine the exact cause of the interrupt. The application must clear the appropriate status bit in the corresponding DOEPINTn register to clear this bit.
18	RO	0x0	IEPInt IN Endpoints Interrupt The core sets this bit to indicate that an interrupt is pending on one of the IN endpoints of the core (in Device mode). The application must read the Device All Endpoints Interrupt (DAINT) register to determine the exact number of the IN endpoint on which the interrupt occurred, and then read the corresponding Device IN Endpoint-n Interrupt (DIEPINTn) register to determine the exact cause of the interrupt. The application must clear the appropriate status bit in the corresponding DIEPINTn register to clear this bit.
17	W1C	0x0	EPMis Endpoint Mismatch Interrupt Note: This interrupt is valid only in shared FIFO operation. Indicates that an IN token has been received for a non-periodic endpoint, but the data for another endpoint is present in the top of the Non-periodic Transmit FIFO and the IN endpoint mismatch count programmed by the application has expired.
16	W1C	0x0	RstrDoneInt Restore Done Interrupt The core sets this bit to indicate that the restore command after Hibernation was completed by the core. The core continues from Suspended state into the mode dictated by PCGCCTL.RestoreMode field.This bit is valid only when Hibernation feature is enabled.

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Bit	Attr	Reset Value	Description
15	W1C	0x0	EOPF End of Periodic Frame Interrupt Indicates that the period specified in the Periodic Frame Interval field of the Device Configuration register (DCFG.PerFrInt) has been reached in the current microframe.
14	W1C	0x0	ISOOutDrop Isochronous OUT Packet Dropped Interrupt The core sets this bit when it fails to write an isochronous OUT packet into the RxFIFO because the RxFIFO does not have enough space to accommodate a maximum packet size packet for the isochronous OUT endpoint.
13	W1C	0x0	EnumDone Enumeration Done The core sets this bit to indicate that speed enumeration is complete. The application must read the Device Status (DSTS) register to obtain the enumerated speed.
12	W1C	0x0	USBRst USB Reset The core sets this bit to indicate that a reset is detected on the USB.
11	W1C	0x0	USBSusp USB Suspend The core sets this bit to indicate that a suspend was detected on the USB. The core enters the Suspended state when there is no activity on the utmi_linestate signal for an extended period of time.
10	W1C	0x0	ErlySusp Early Suspend The core sets this bit to indicate that an Idle state has been detected on the USB for 3 ms.
9	W1C	0x0	I2CINT I2C Interrupt The core sets this interrupt when I2C access is completed on the I2C interface. This field is used only if the I2C interface was enabled . Otherwise, reads return 0.

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Bit	Attr	Reset Value	Description
8	W1C	0x0	ULPICKINT ULPI Carkit Interrupt This field is used only if the Carkit interface was enabled . Otherwise, reads return 0. The core sets this interrupt when a ULPI Carkit interrupt is received. The core's PHY sets ULPI Carkit interrupt in UART or Audio mode. I2C Carkit Interrupt (I2CCKINT) This field is used only if the I2C interface was enabled . Otherwise, reads return 0.The core sets this interrupt when a Carkit interrupt is received. The core's PHY sets the I2C Carkit interrupt in Audio mode.
7	RO	0x0	GOUTNakEff Global OUT NAK Effective Indicates that the Set Global OUT NAK bit in the Device Control register DCTL.SGOUTNak), set by the application, has taken effect in the core. This bit can be cleared by writing the Clear Global OUT NAK bit in the Device Control register (DCTL.CGOUTNak).
6	RO	0x0	GINNakEff Global IN Non-Periodic NAK Effective Indicates that the Set Global Non-periodic IN NAK bit in the Device Control register (DCTL.SGNPInNak), set by the application, has taken effect in the core. That is, the core has sampled the Global IN NAK bit set by the application. This bit can be cleared by clearing the Clear Global Nonperiodic IN NAK bit in the Device Control register (DCTL.CGNPInNak). This interrupt does not necessarily mean that a NAK handshake is sent out on the USB. The STALL bit takes precedence over the NAK bit.
5	RO	0x0	NPTxFEmp Non-Periodic TxFIFO Empty This interrupt is valid only when OTG_EN_DED_TX_FIFO = 0. This interrupt is asserted when the Non-periodic TxFIFO is either half or completely empty, and there is space for at least one entry to be written to the Non-periodic Transmit Request Queue. The half or completely empty status is determined by the Non-periodic TxFIFO Empty Level bit in the Core AHB Configuration register(GAHBCFG.NPTxFEmpLvl).
4	RO	0x0	RxFLvl RxFIFO Non-Empty Indicates that there is at least one packet pending to be read from the RxFIFO.

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Bit	Attr	Reset Value	Description
3	W1C	0x0	Sof Start of (micro)Frame In Host mode, the core sets this bit to indicate that an SOF (FS), micro-SOF(HS), or Keep-Alive (LS) is transmitted on the USB. The application must write a 1 to this bit to clear the interrupt. In Device mode, in the core sets this bit to indicate that an SOF token has been received on the USB. The application can read the Device Status register to get the current (micro)frame number. This interrupt is seen only when the core is operating at either HS or FS.
2	RO	0x0	OTGInt OTG Interrupt The core sets this bit to indicate an OTG protocol event. The application must read the OTG Interrupt Status (GOTGINT) register to determine the exact event that caused this interrupt. The application must clear the appropriate status bit in the GOTGINT register to clear this bit.
1	W1C	0x0	ModeMis Mode Mismatch Interrupt The core sets this bit when the application is trying to access: A Host mode register, when the core is operating in Device mode; A Device mode register, when the core is operating in Host mode. The register access is completed on the AHB with an OKAY response, but is ignored by the core internally and does not affect the operation of the core.
0	RO	0x0	CurMod Current Mode of Operation Indicates the current mode. 1'b0: Device mode 1'b1: Host mode

USBOTG_GINTMSK

Address: Operational Base + offset (0x0018)

Interrupt Mask Register

Bit	Attr	Reset Value	Description
31	RW	0x0	WkUpIntMsk Resume/Remote Wakeup Detected Interrupt Mask
30	RW	0x0	SessReqIntMsk Session Request/New Session Detected Interrupt Mask
29	RW	0x0	DisconnIntMsk Disconnect Detected Interrupt Mask

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Bit	Attr	Reset Value	Description
28	RW	0x0	ConIDStsChngMsk Connector ID Status Change Mask
27	RW	0x0	LPM_IntMsk LPM Transaction Received Interrupt Mask
26	RW	0x0	PTxFEmpMsk Periodic TxFIFO Empty Mask
25	RW	0x0	HChIntMsk Host Channels Interrupt Mask
24	RW	0x0	PrtIntMsk Host Port Interrupt Mask
23	RW	0x0	ResetDetMsk Reset Detected Interrupt Mask
22	RW	0x0	FetSuspMsk Data Fetch Suspended Mask
21	RW	0x0	incomplPMsk_incompISOOUTMsk Incomplete Periodic Transfer Mask(Host only) Incomplete Isochronous OUT Transfer Mask(Device only)
20	RW	0x0	incompISOINMsk Incomplete Isochronous IN Transfer Mask
19	RW	0x0	OEPIntMsk OUT Endpoints Interrupt Mask
18	RW	0x0	IEPIntMsk IN Endpoints Interrupt Mask
17	RW	0x0	EPMisMsk Endpoint Mismatch Interrupt Mask
16	RW	0x0	RstrDoneIntMsk Restore Done Interrupt Mask This field is valid only when Hibernation feature is enabled.
15	RW	0x0	EOPFMsk End of Periodic Frame Interrupt Mask
14	RW	0x0	ISOOutDropMsk Isochronous OUT Packet Dropped Interrupt Mask
13	RW	0x0	EnumDoneMsk Enumeration Done Mask
12	RW	0x0	USBRstMsk USB Reset Mask
11	RW	0x0	USBSuspMsk USB Suspend Mask
10	RW	0x0	ErlySuspMsk Early Suspend Mask
9	RW	0x0	I2CIntMsk I2C Interrupt Mask

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Bit	Attr	Reset Value	Description
8	RW	0x0	ULPICKINTMsk_I2CCKINTMsk ULPI Carkit Interrupt Mask (ULPICKINTMsk) I2C Carkit Interrupt Mask (I2CCKINTMsk)
7	RW	0x0	GOUTNakEffMsk Global OUT NAK Effective Mask
6	RW	0x0	GINNakEffMsk Global Non-periodic IN NAK Effective Mask
5	RW	0x0	NPTxFEmpMsk Non-periodic TxFIFO Empty Mask
4	RW	0x0	RxFLvlMsk Receive FIFO Non-Empty Mask
3	RW	0x0	SofMsk Start of (micro)Frame Mask
2	RW	0x0	OTGIntMsk OTG Interrupt Mask
1	RW	0x0	ModeMisMsk Mode Mismatch Interrupt Mask
0	RO	0x0	reserved

USBOTG_GRXSTSR

Address: Operational Base + offset (0x001c)

Receive Status Debug Read Register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24:21	RO	0x0	FN Frame Number (Device Only)This is the least significant 4 bits of the (micro)frame number in which the packet is received on the USB. This field is supported only when isochronous OUT endpoints are supported.

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Bit	Attr	Reset Value	Description
20:17	RO	0x0	PktSts Packet Status Indicates the status of the received packet(Host Only) 4'b0010: IN data packet received 4'b0011: IN transfer completed (triggers an interrupt) 4'b0101: Data toggle error (triggers an interrupt) 4'b0111: Channel halted (triggers an interrupt) Others: Reserved Indicates the status of the received packet(Device only) 4'b0001: Global OUT NAK (triggers an interrupt) 4'b0010: OUT data packet received 4'b0011: OUT transfer completed (triggers an interrupt) 4'b0100: SETUP transaction completed (triggers an interrupt) 4'b0110: SETUP data packet received Others: Reserved
16:15	RO	0x0	DPID Data PID Indicates the Data PID of the received packet 2'b00: DATA0 2'b10: DATA1 2'b01: DATA2 2'b11: MDATA
14:4	RW	0x000	BCnt Byte Count Indicates the byte count of the received data packet.
3:0	RO	0x0	ChNum_EPNum Channel Number(Host) Endpoint Number(Device) (Host Only) Indicates the channel number to which the current received packet belongs. (Device Only) Indicates the endpoint number to which the current received packet belongs.

USBOTG_GRXSTSP

Address: Operational Base + offset (0x0020)

Receive Status Read and Pop Register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
24:21	RO	0x0	FN Frame Number (Device Only) This is the least significant 4 bits of the (micro)frame number in which the packet is received on the USB. This field is supported only when isochronous OUT endpoints are supported.
20:17	RO	0x0	PktSts Packet Status Indicates the status of the received packet(Host Only) 4'b0010: IN data packet received 4'b0011: IN transfer completed (triggers an interrupt) 4'b0101: Data toggle error (triggers an interrupt) 4'b0111: Channel halted (triggers an interrupt) Others: Reserved Indicates the status of the received packet(Device only) 4'b0001: Global OUT NAK (triggers an interrupt) 4'b0010: OUT data packet received 4'b0011: OUT transfer completed (triggers an interrupt) 4'b0100: SETUP transaction completed (triggers an interrupt) 4'b0110: SETUP data packet received Others: Reserved
16:15	RO	0x0	DPID Data PID Indicates the Data PID of the received OUT data packet 2'b00: DATA0 2'b10: DATA1 2'b01: DATA2 2'b11: MDATA
14:4	RO	0x000	BCnt Byte Count Indicates the byte count of the received data packet.
3:0	RO	0x0	ChNum_EPNum Channel Number(Host) Endpoint Number(Device) (Host Only) Indicates the channel number to which the current received packet belongs. (Device Only) Indicates the endpoint number to which the current received packet belongs.

USBOTG_GRXFSIZ

Address: Operational Base + offset (0x0024)

Receive FIFO Size Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	RxFDep RxFIFO Depth This value is in terms of 32-bit words. Minimum value is 16, Maximum value is 32,768. The power-on reset value of this register is specified as the Largest Rx Data FIFO Depth. If Enable Dynamic FIFO Sizing? was deselected, these flops are optimized, and reads return the power-on value. If Enable Dynamic FIFO Sizing? was selected , you can write a new value in this field. You can write a new value in this field. Programmed values must not exceed the power-on value.

USBOTG_GNPTXFSIZ
Address: Operational Base + offset (0x0028)
Non-Periodic Transmit FIFO Size Register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	NPTxFDep Non-periodic TxFIFO For host mode, this field is always valid. For Device mode, this field is valid only when OTG_EN_DED_TX_FIFO==0. This value is in terms of 32-bit words. Minimum value is 16 Maximum value is 32,768 This field is determined by Enable Dynamic FIFO Sizing. OTG_DFIFO_DYNAMIC = 0: These flops are optimized, and reads return the Poweron value. OTG_DFIFO_DYNAMIC = 1: The application can write a new value in this field. Programmed values must not exceed the power-on value. The power-on reset value of this field is specified by OTG_EN_DED_TX_FIFO: OTG_EN_DED_TX_FIFO = 0:The reset value is the Largest Non-periodic Tx Data FIFO Depth parameter, OTG_TX_NPERIO_DFIFO_DEPTH. OTG_EN_DED_TX_FIFO = 1:The reset value is parameter OTG_TX_HNPERIO_DFIFO_DEPTH.

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	NPTxFStAddr Non-periodic Transmit RAM For host mode, this field is always valid. This field contains the memory start address for Non-periodic Transmit FIFO RAM. This field is determined by Enable Dynamic FIFO Sizing?(OTG_DFIFO_DYNAMIC): OTG_DFIFO_DYNAMIC = 0:These flops are optimized, and reads return the power-on value. OTG_DFIFO_DYNAMIC = 1:The application can write a new value in this field. Programmed values must not exceed the power-on value. The power-on reset value of this field is specified by Largest Rx Data FIFO Depth (parameter OTG_RX_DFIFO_DEPTH).

USBOTG_GNPTXSTS

Address: Operational Base + offset (0x002c)

Non-Periodic Transmit FIFO/Queue Status Register

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RO	0x00	NPTxQTop Top of the Non-periodic Transmit Request Queue Entry in the Non-periodic Tx Request Queue that is currently being processed by the MAC. Bits [30:27]: Channel/endpoint number Bits [26:25]: 2'b00: IN/OUT token 2'b01: Zero-length transmit packet (device IN/host OUT) 2'b10: PING/CSPLIT token 2'b11: Channel halt command Bit [24]: Terminate (last entry for selected channel/endpoint)
23:16	RO	0x00	NPTxQSpcAvail Non-periodic Transmit Request Queue Space Available Indicates the amount of free space available in the Non-periodic Transmit Request Queue. This queue holds both IN and OUT requests in Host mode. Device mode has only IN requests. 8'h0: Non-periodic Transmit Request Queue is full 8'h1: 1 location available 8'h2: 2 locations available n: n locations available (0 <=n <= 8) Others: Reserved

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Bit	Attr	Reset Value	Description
15:0	RO	0x0000	NPTxFSpcAvail Non-periodic TxFIFO Space Avail Indicates the amount of free space available in the Non-periodic TxFIFO. Values are in terms of 32-bit words. 16'h0: Non-periodic TxFIFO is full 16'h1: 1 word available 16'h2: 2 words available 16'hn: n words available (where 0 <=n <=32,768) 16'h8000: 32,768 words available Others: Reserved

USBOTG_GI2CCTL
Address: Operational Base + offset (0x0030)
I2C Address Register

Bit	Attr	Reset Value	Description
31	R/WSC	0x0	BsyDne I2C Busy/Done The application sets this bit to 1'b1 to start a request on the I2C interface. When the transfer is complete, the core de-asserts this bit to 1'b0. As long as the bit is set, indicating that the I2C interface is busy, the application cannot start another request on the interface.
30	RW	0x0	RW Read/Write Indicator Indicates whether a read or write register transfer must be performed on the interface.Read/write bursting is not supported for registers. 1'b1: Read 1'b0: Write
29	RO	0x0	reserved
28	RW	0x1	I2CDatSe0 I2C DatSe0 USB Mode Selects the FS interface USB mode. 1'b1: VP_VM USB mode 1'b0: DAT_SE0 USB mode
27:26	RW	0x0	I2CDevAdr I2C Device Address Selects the address of the I2C Slave on the USB 1.1 full-speed serial transceiver that the core uses for OTG signaling. 2'b00: 7'h2C 2'b01: 7'h2D 2'b10: 7'h2E 2'b11: 7'h2F

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Bit	Attr	Reset Value	Description
25	RW	0x0	I2CSuspCtl I2C Suspend Control Selects how Suspend is connected to a full-speed transceiver in I2C mode. 1'b0: Use the dedicated utmi_suspend_n pin 1'b1: Use an I2C write to program the Suspend bit in the PHY register
24	RO	0x1	Ack I2C ACK Indicates whether an ACK response was received from the I2C Slave. This bit is valid when BsyDne is cleared by the core, after application has initiated an I2C access. 1'b0: NAK 1'b1: ACK
23	RW	0x0	I2CEn I2C Enable Enables the I2C Master to initiate I2C transactions on the I2C interface
22:16	RW	0x00	Addr I2C Address This is the 7-bit I2C device address used by software to access any external I2C Slave,including the I2C Slave on a USB 1.1 OTG full-speed serial transceiver. Software can change this address to access different I2C Slaves.
15:8	RW	0x00	RegAddr I2C Register Addr This field programs the address of the register to be read from or written to.
7:0	RW	0x00	RWData I2C Read/Write Data After a register read operation, this field holds the read data for the application. During a write operation, the application can use this register to program the write data to be written to a register. During writes, this field holds the write data.

USBOTG_GPVNDCTL

Address: Operational Base + offset (0x0034)

PHY Vendor Control Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31	R/WSC	0x0	DisUlpiDrvr Disable ULPI Drivers This field is used only if the Carkit interface was enabled (OTG_ULPI_CARKIT = 1). Otherwise, reads return 0. The application sets this bit when it has finished processing the ULPI Carkit Interrupt (GINTSTS.ULPICKINT). When set, the Otg core disables drivers for output signals and masks input signal for the ULPI interface. Otg clears this bit before enabling the ULPI interface.
30:28	RO	0x0	reserved
27	R/WSC	0x0	VStsDone VStatus Done The core sets this bit when the vendor control access is done. This bit is cleared by the core when the application sets the New Register Request bit (bit 25).
26	RO	0x0	VStsBsy VStatus Busy The core sets this bit when the vendor control access is in progress and clears this bit when done.
25	R/WSC	0x0	NewRegReq New Register Request The application sets this bit for a new vendor control access.
24:23	RO	0x0	reserved
22	RW	0x0	RegWr Register Write Set this bit for register writes, and clear it for register reads.
21:16	RW	0x00	RegAddr Register Address The 6-bit PHY register address for immediate PHY Register Set access. Set to 6'h2F for Extended PHY Register Set access.
15:8	RW	0x00	VCtrl UTMI+ Vendor Control Register Address The 4-bit register address a vendor defined 4-bit parallel output bus. Bits 11:8 of this field are placed on utmi_vcontrol[3:0]. ULPI Extended Register Address (ExtRegAddr) The 6-bit PHY extended register address.
7:0	RW	0x00	RegData Register Data Contains the write data for register write. Read data for register read, valid when VStatus Done is set.

USBOTG_GGPIO

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Address: Operational Base + offset (0x0038)
General Purpost Input/Output Register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	GPO General Purpose Output This field is driven as an output from the core, gp_o[15:0]. The application can program this field to determine the corresponding value on the gp_o[15:0] output.
15:0	RO	0x0000	GPI General Purpose Input This field's read value reflects the gp_i[15:0] core input value.

USBOTG_GUID
Address: Operational Base + offset (0x003c)
User ID Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	UserID Application-programmable ID field.

USBOTG_GSNPSID
Address: Operational Base + offset (0x0040)
Core ID Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00004f54	CoreID Release number of the core being used

USBOTG_GHWCFG1
Address: Operational Base + offset (0x0044)
User HW Config1 Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	epdir Endpoint Direction This 32-bit field uses two bits per endpoint to determine the endpoint direction. Endpoint Bits [31:30]: Endpoint 15 direction Bits [29:28]: Endpoint 14 direction ... Bits [3:2]: Endpoint 1 direction Bits[1:0]: Endpoint 0 direction (always BIDIR) Direction 2'b00: BIDIR (IN and OUT) endpoint 2'b01: IN endpoint 2'b10: OUT endpoint 2'b11: Reserved

USBOTG_GHWCFG2
Address: Operational Base + offset (0x0048)
User HW Config2 Register

Bit	Attr	Reset Value	Description
31	RO	0x0	OTG_ENABLE_IC_USB IC_USB mode specified for mode of operation (parameter OTG_ENABLE_IC_USB). To choose IC_USB_MODE, both OTG_FSPHY_INTERFACE and OTG_ENABLE_IC_USB must be 1.
30:26	RO	0x00	TknQDepth Device Mode IN Token Sequence Learning Queue Depth Range: 0-30
25:24	RO	0x0	PTxQDepth Host Mode Periodic Request Queue Depth 2'b00: 2 2'b01: 4 2'b10: 8 Others: Reserved
23:22	RO	0x0	NPTxQDepth Non-periodic Request Queue Depth 2'b00: 2 2'b01: 4 2'b10: 8 Others: Reserved
21	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
20	RO	0x0	MultiProcIntrpt Multi-Processor Interrupt Enabled 1'b0: No 1'b1: Yes
19	RO	0x0	DynFifoSizing Dynamic FIFO Sizing Enabled 1'b0: No 1'b1: Yes
18	RO	0x0	PerioSupport Periodic OUT Channels Supported in Host Mode 1'b0: No 1'b1: Yes
17:14	RO	0x0	NumHstChnl Number of Host Channels Indicates the number of host channels supported by the core in Host mode. The range of this field is 0-15: 0 specifies 1 channel, 15 specifies 16 channels.
13:10	RO	0x0	NumDevEps Number of Device Endpoints Indicates the number of device endpoints supported by the core in Device mode in addition to control endpoint 0. The range of this field is 1-15.
9:8	RO	0x0	FSPhyType Full-Speed PHY Interface Type 2'b00: Full-speed interface not supported 2'b01: Dedicated full-speed interface 2'b10: FS pins shared with UTMI+ pins 2'b11: FS pins shared with ULPI pins
7:6	RO	0x0	HSPhyType High-Speed PHY Interface Type 2'b00: High-Speed interface not supported 2'b01: UTMI+ 2'b10: ULPI 2'b11: UTMI+ and ULPI
5	RO	0x0	SingPnt Point-to-Point 1'b0: Multi-point application (hub and split support) 1'b1: Single-point application (no hub and no split support)

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Bit	Attr	Reset Value	Description
4:3	RO	0x0	OtgArch Architecture 2'b00: Slave-Only 2'b01: External DMA 2'b10: Internal DMA Others: Reserved
2:0	RO	0x0	OtgMode Mode of Operation 3'b000: HNP- and SRP-Capable OTG (Host and Device) 3'b001: SRP-Capable OTG (Host and Device) 3'b010: Non-HNP and Non-SRP Capable OTG (Host and Device) 3'b011: SRP-Capable Device 3'b100: Non-OTG Device 3'b101: SRP-Capable Host 3'b110: Non-OTG Host Others: Reserved

USBOTG_GHWCFG3
Address: Operational Base + offset (0x004c)
User HW Config3 Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	DfifoDepth DFIFO Depth This value is in terms of 32-bit words. Minimum value is 32 Maximum value is 32,768
15	RO	0x0	OTG_ENABLE_LPM LPM mode specified for Mode of Operation (parameter OTG_ENABLE_LPM).
14	RO	0x0	OTG_BC_SUPPORT This bit indicates the HS OTG controller support for Battery Charger. 1'b0: No Battery Charger Support 1'b1: Battery Charger support present.
13	RO	0x0	OTG_ENABLE_HSIC HSIC mode specified for Mode of Operation (parameter OTG_ENABLE_HSIC). Value Range: 0-1 1'b1: HSIC-capable with shared UTMI PHY interface 1'b0: Non-HSIC-capable

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Bit	Attr	Reset Value	Description
12	RO	0x0	OTG_ADP_SUPPORT This bit indicates whether ADP logic is present within or external to the HS OTG controller 1'b0: No ADP logic present with HSOTG controller 1'b1: ADP logic is present along with HSOTG controller.
11	RO	0x0	RstType Reset Style for Clocked always Blocks in RTL 1'b0: Asynchronous reset is used in the core 1'b1: Synchronous reset is used in the core
10	RO	0x0	OptFeature Optional Features Removed Indicates whether the User ID register, GPIO interface ports, and SOF toggle and counter ports were removed for gate count optimization by enabling Remove Optional Features? 1'b0: No 1'b1: Yes
9	RO	0x0	VndctlSupt Vendor Control Interface Support 1'b0: Vendor Control Interface is not available on the core. 1'b1: Vendor Control Interface is available.
8	RO	0x0	I2CIntSel I2C Selection 1'b0: I2C Interface is not available on the core. 1'b1: I2C Interface is available on the core.
7	RO	0x0	OtgEn OTG Function Enabled The application uses this bit to indicate the Otg core's OTG capabilities. 1'b0: Not OTG capable 1'b1: OTG Capable
6:4	RO	0x0	PktSizeWidth Width of Packet Size Counters 3'b000: 4 bits 3'b001: 5 bits 3'b010: 6 bits 3'b011: 7 bits 3'b100: 8 bits 3'b101: 9 bits 3'b110: 10 bits Others: Reserved

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Bit	Attr	Reset Value	Description
3:0	RO	0x0	XferSizeWidth Width of Transfer Size Counters 4'b0000: 11 bits 4'b0001: 12 bits ... 4'b1000: 19 bits Others: Reserved

USBOTG_GHWCFG4
Address: Operational Base + offset (0x0050)
User HW Config4 Register

Bit	Attr	Reset Value	Description
31	RO	0x0	SGDMA Scatter/Gather DMA 1'b1: Dynamic configuration
30	RO	0x0	SGDMACon Scatter/Gather DMA configuration 1'b0: Non-Scatter/Gather DMA configuration 1'b1: Scatter/Gather DMA configuration
29:26	RO	0x0	INEps Number of Device Mode IN Endpoints Including Control Endpoint Range 0 -15 0:1 IN Endpoint 1:2 IN Endpoints .... 15:16 IN Endpoints
25	RW	0x0	DedFifoMode Enable Dedicated Transmit FIFO for device IN Endpoints 1'b0: Dedicated Transmit FIFO Operation not enabled. 1'b1: Dedicated Transmit FIFO Operation enabled.
24	RW	0x0	SessEndFltr session_end Filter Enabled 1'b0: No filter 1'b1: Filter
23	RW	0x0	BValidFltr "b_valid" Filter Enabled 1'b0: No filter 1'b1: Filter
22	RO	0x0	AValidFltr "a_valid" Filter Enabled 1'b0: No filter 1'b1: Filter

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Bit	Attr	Reset Value	Description
21	RO	0x0	VBusValidFltr "vbus_valid" Filter Enabled 1'b0: No filter 1'b1: Filter
20	RO	0x0	IddgFltr "iddig" Filter Enable 1'b0: No filter 1'b1: Filter
19:16	RO	0x0	NumCtlEps Number of Device Mode Control Endpoints in Addition to Endpoint Range: 0-15
15:14	RO	0x0	PhyDataWidth UTMI+ PHY/ULPI-to-Internal UTMI+ Wrapper Data Width When a ULPI PHY is used, an internal wrapper converts ULPI to UTMI+. 2'b00: 8 bits 2'b01: 16 bits 2'b10: 8/16 bits, software selectable Others: Reserved
13:7	RO	0x0	reserved
6	RO	0x0	EnHiber Enable Hibernation 1'b0: Hibernation feature not enabled 1'b1: Hibernation feature enabled
5	RO	0x0	AhbFreq Minimum AHB Frequency Less Than 60 MHz 1'b0: No 1'b1: Yes
4	RO	0x0	EnParPwrDown Enable Partial Power Down 1'b0: Partial Power Down Not Enabled 1'b1: Partial Power Down Enabled
3:0	RO	0x0	NumDevPerioEps Number of Device Mode Periodic IN Endpoints Range: 0-15

USBOTG_GLPMCFG

Address: Operational Base + offset (0x0054)

Core LPM Configuration Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31	RW	0x0	InvSelHsic HSIC-Invert Select HSIC The application uses this bit to control the Otg core HSIC enable/disable. This bit overrides and functionally inverts the if_sel_hsic input port signal. If the core operates as non-HSIC-capable, it can only connect to non-HSIC-capable PHYs. If the core operates as HSIC-capable, it can only connect to HSICcapable PHYs. If the if_sel_hsic input signal is1: 1'b1: HSIC capability is not enabled. 1'b0: HSIC capability is enabled, If InvSelHsic = 1'b0: HSIC capability is enabled. If the if_sel_hsic input signal is 0: 1'b1: HSIC capability is enabled, 1'b0: HSIC capability is not enabled. This bit is writable only if an HSIC mode was specified for Mode of Operation (parameter OTG_ENABLE_HSIC). This bit is valid only if OTG_ENABLE_HSIC is enabled.
30	RW	0x0	HSICCon HSIC-Connect The application must use this bit to initiate the HSIC Attach sequence. Host Mode: Once this bit is set, the host core configures to drive the HSIC Idle state (STROBE = 1 & DATA = 0) on the bus. It then waits for the device to initiate the Connect sequence. Device Mode: Once this bit is set, the device core waits for the HSIC Idle line state on the bus. Upon receiving the Idle line state, it initiates the HSIC Connect sequence. This bit is valid only if OTG_ENABLE_HSIC is 1, if_sel_hsic = 1 and InvSelHSIC is 0. Otherwise, it is read-only.
29:28	RO	0x0	reserved
27:25	RO	0x0	LPM_RetryCnt_Sts LPM Retry Count Status Number of LPM host retries remaining to be transmitted for the current LPM sequence.
24	RW	0x0	SndLPM Send LPM Transaction Host Mode: When the application software sets this bit, an LPM transaction containing two tokens, EXT and LPM, is sent. The hardware clears this bit once a valid response (STALL, NYET, or ACK) is received from the device or the core has finished transmitting the programmed number of LPM retries. Note: This bit must only be set when the host is connected to a local port.

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Bit	Attr	Reset Value	Description
23:21	R/WSC	0x0	LPM_Retry_Cnt LPM Retry Count When the device gives an ERROR response, this is the number of additional LPM retries that the host performs until a valid device response (STALL, NYET, or ACK) is received.
20:17	RW	0x0	LPM_Chnl_Indx LPM Channel Index The channel number on which the LPM transaction must be applied while sending an LPM transaction to the local device. Based on the LPM channel index, the core automatically inserts the device address and endpoint number programmed in the corresponding channel into the LPM transaction.
16	RO	0x0	L1ResumeOK Sleep State Resume OK Indicates that the application or host can start a resume from the Sleep state. This bit is valid in the LPM Sleep (L1) state. It is set in Sleep mode after a delay of 50 us (TL1Residency). The bit is reset when SlpSts is 0 1'b1: The application/core can start resume from the Sleep state 1'b0: The application/core cannot start resume from the Sleep state

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Bit	Attr	Reset Value	Description
15	RO	0x0	SlpSts Port Sleep Status Device Mode: This bit is set as long as a Sleep condition is present on the USB bus. The core enters the Sleep state when an ACK response is sent to an LPM transaction and the timer TL1TokenRetry. has expired. To stop the PHY clock, the application must set the Port Clock Stop bit, which asserts the PHY Suspend input pin. The application must rely on SlpSts and not ACK in CoreL1Res to confirm transition into sleep. The core comes out of sleep: When there is any activity on the USB line_state When the application writes to the Remote Wakeup Signaling bit in the Device Control register (DCTL.RmtWkUpSig) or when the application resets or soft-disconnects the device. Host Mode: The host transitions to the Sleep (L1) state as a sideeffect of a successful LPM transaction by the core to the local port with an ACK response from the device. The read value of this bit reflects the port's current sleep status. The core clears this bit after: The core detects a remote L1 Wakeup signal The application sets the Port Reset bit or the Port L1Resume bit in the HPRT register or The application sets the L1Resume/ Remote Wakeup Detected Interrupt bit or Disconnect Detected Interrupt bit in the Core Interrupt register (GINTSTS.L1WkUpInt or GINTSTS.DisconnInt, respectively). Values: 1'b0: Core not in L1 1'b1: Core in L1
14:13	RO	0x0	CoreL1Res LPM Response Device Mode: The core's response to the received LPM transaction is reflected in these two bits. Host Mode: The handshake response received from the local device for LPM transaction. 2'b11: ACK 2'b10: NYET 2'b01: STALL 2'b00: ERROR (No handshake response)

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Bit	Attr	Reset Value	Description
12:8	RW	0x00	HIRD_Thres HIRD Threshold Device Mode: The core asserts L1SuspendM to put the PHY into Deep Low-Power mode in L1 when the HIRD value is greater than or equal to the value defined in this field (GLPMCFG.HIRD_Thres[3:0]), and HIRD_Thres[4] is set to 1'b1. Host Mode: The core asserts L1SuspendM to put the PHY into Deep Low-Power mode in L1 when HIRD_Thres[4] is set to 1'b1. HIRD_Thres[3:0] specifies the time for which resume signaling is to be reflected by the host TL1HubDrvResume2) on the USB when it detects device-initiated resume. HIRD_Thres must not be programmed with a value greater than 4'b1100 in Host mode, because this exceeds maximum $T_{L1HubDrvResume2}$. No HIRD_Thres[3:0] resumetime(us) 1 4'b0000 60 2 4'b0001 135 3 4'b0010 210 4 4'b0011 285 5 4'b0100 360 6 4'b0101 435 7 4'b0110 510 8 4'b0111 585 9 4'b1000 660 10 4'b1001 735 11 4'b1010 810 12 4'b1011 885 13 4'b1100 960 14 4'b1101 invalid 15 4'b1110 invalid 16 4'b1111 invalid

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Bit	Attr	Reset Value	Description
7	RW	0x0	EnblSlpM Enable utmi_sleep_n For ULPI interface: The application uses this bit to write to the function control [7] in the L1 state, to enable the PHY to go into Low Power mode. For the host, this bit is valid only in Local Device mode. 1'b0: Writes to the ULPI Function Control Bit[7] are disabled. 1'b1: The core is enabled to write to the ULPI Function Control Bit[7], which enables the PHY to enter Low-Power mode. Note: When a ULPI interface is configured, enabling this bit results in a write to Bit 7 of the ULPI Function Control register. The ULPI PHY supports writing to this bit, and in the L1 state asserts SleepM when utmi_l1_suspend_n cannot be asserted. When a ULPI interface is configured, this bit must always be set if you are using the ULPI PHY. Note: For ULPI interface, do not clear this bit during the resume. For all other interfaces: The application uses this bit to control utmi_sleep_n assertion to the PHY in the L1 state. For the host, this bit is valid only in Local Device mode. 1'b0: utmi_sleep_n assertion from the core is not transferred to the external PHY. 1'b1: utmi_sleep_n assertion from the core is transferred to the external PHY when utmi_l1_suspend_n cannot be asserted.
6	RW	0x0	bRemoteWake RemoteWakeEnable Host Mode: The remote wakeup value to be sent in the LPM transaction's wIndex field. Device Mode: This field is updated with the received bRemoteWake LPM token's bmAttribute when an ACK/NYET/STALL response is sent to an LPM transaction.

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Bit	Attr	Reset Value	Description
5:2	RW	0x0	HIRD Host-Initiated Resume Duration Host Mode: The value of HIRD to be sent in an LPM transaction. This value is also used to initiate resume for a duration .$T_{L1HubDrvResume1}$ for host initiated resume. Device Mode: This field is updated with the Received LPM Token HIRD bmAttribute when an ACK/NYET/STALL response is sent to an LPM transaction Sl. No HIRD[3:0] THIRD (us) 1 4'b0000 50 2 4'b0001 125 3 4'b0010 200 4 4'b0011 275 5 4'b0100 350 6 4'b0101 425 7 4'b0110 500 8 4'b0111 575 9 4'b1000 650 10 4'b1001 725 11 4'b1010 800 12 4'b1011 875 13 4'b1100 950 14 4'b1101 1025 15 4'b1110 1100 16 4'b1111 1175
1	RW	0x0	AppL1Res LPM response programmed by application Handshake response to LPM token pre-programmed by device application software. The response depends on GLPMCFG.LPMCap. If GLPMCFG.LPMCap is 1'b0, the core always responds with a NYET. If GLPMCFG.LPMCap is 1'b1, the core responds as follows: 1'b1: ACK. Even though an ACK is pre-programmed, the core responds with an ACK only on a successful LPM transaction. The LPM transaction is successful if: There are no PID/CRC5 errors in both the EXT token and the LPM token (else ERROR); A valid bLinkState = 0001B (L1) is received in the LPM transaction (else STALL); No data is pending in the Transmit queue (else NYET) 1'b0: NYET. The pre-programmed software bit is overridden for response to LPM token when:(1)The received bLinkState is not L1 (STALL response); (2)An error is detected in either of the LPM token packets due to corruption (ERROR response).

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Bit	Attr	Reset Value	Description
0	RW	0x0	LPMCap LPM-Capable The application uses this bit to control the OTG core LPM capabilities. If the core operates as a non-LPM-capable host, it cannot request the connected device/hub to activate LPM mode. If the core operates as a non-LPM-capable device, it cannot respond to any LPM transactions. 1'b0: LPM capability is not enabled. 1'b1: LPM capability is enabled. This bit is writable only if an LPM mode was specified for Mode of Operation (parameter OTG_ENABLE_LPM). Otherwise, reads return 0.

USBOTG_GPWRDN

Address: Operational Base + offset (0x0058)

Global Power Down Register

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:24	RO	0x00	MultValIdBC Multi Valued ID pin Battery Charger ACA inputs in the following order: Bit 26 - rid_float. Bit 25 - rid_gnd Bit 24 - rid_a Bit 23 - rid_b Bit 22 - rid_c These bits are present only if OTG_BC_SUPPORT = 1. Otherwise, these bits are reserved and will read 5'h0.
23	W1C	0x0	ADPInt ADP Interupt This bit is set whenever there is a ADP event.
22	RO	0x0	BSessVld B Session Valid This field reflects the B session valid status signal from the PHY. 1'b0: B-Valid is 0. 1'b1: B-Valid is 1. This bit is valid only when GPWRDN.PMUActv is 1.

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Bit	Attr	Reset Value	Description
21	RO	0x0	IDDIG This bit indicates the status of the signal IDDIG. The application must read this bit after receiving GPWRDN.StsChngInt and decode based on the previous value stored by the application. Indicates the current mode. 1'b1: Device mode 1'b0: Host mode This bit is valid only when GPWRDN.PMUActv is 1.
20:19	RO	0x0	LineState This field indicates the current linestate on USB as seen by the PMU module. 2'b00: DM = 0, DP = 0. 2'b01: DM = 0, DP = 1. 2'b10: DM = 1, DP = 0. 2'b11: Not-defined. This bit is valid only when GPWRDN.PMUActv is 1.
18	RW	0x0	StsChngIntMsk Mask For StsChng Interrupt
17	W1C	0x0	StsChngInt This field indicates a status change in either the IDDIG or BSessVld signal. 1'b0: No Status change 1'b1: status change detected After receiving this interrupt the application should read the GPWRDN register and interpret the change in IDDIG or BSesVld with respect to the previous value stored by the application.
16	RW	0x0	SRPDetectMsk Mask For SRPDetect Interrupt
15	W1C	0x0	SRPDetect This field indicates that SRP has been detected by the PMU.This field generates an interrupt. After detecting SRP during hibernation the application should not restore the core. The application should get into the initialization process. 1'b0: SRP not detected 1'b1: SRP detected
14	RW	0x0	ConnDetMsk Mask for ConnectDet interrupt This bit is valid only when OTG_EN_PWROPT = 2.
13	W1C	0x0	ConnectDet This field indicates that a new connect has been detected 1'b0: Connect not detected 1'b1: Connect detected This bit is valid only when OTG_EN_PWROPT = 2.

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Bit	Attr	Reset Value	Description
12	RW	0x0	DisconnectDetectMsk Mask For DisconnectDetect Interrupt This bit is valid only when OTG_EN_PWROPT = 2.
11	W1C	0x0	DisconnectDetect This field indicates that Disconnect has been detected by the PMU. This field generates an interrupt. After detecting disconnect during hibernation the application must not restore the core, but instead start the initialization process. 1'b0: Disconnect not detected 1'b1: Disconnect detected This bit is valid only when OTG_EN_PWROPT = 2.
10	RW	0x0	ResetDetMsk Mask For ResetDetected interrupt.This bit is valid only when OTG_EN_PWROPT = 2.
9	W1C	0x0	ResetDetected This field indicates that Reset has been detected by the PMU module. This field generates an interrupt. 1'b0: Reset Not Detected 1'b1: Reset Detected This bit is valid only when OTG_EN_PWROPT = 2.
8	RW	0x0	LineStageChangeMsk Mask For LineStateChange interrupt This bit is valid only when OTG_EN_PWROPT = 2.
7	W1C	0x0	LnStsChng Line State Change This interrupt is asserted when there is a Linestate Change detected by the PMU. The application should read GPWRDN.Linestate to determine the current linestate on USB. 1'b0: No LineState change on USB 1'b1: LineState change on USB This bit is valid only when GPWRDN.PMUActv is 1.This bit is valid only when OTG_EN_PWROPT = 2.
6	RW	0x0	DisableVBUS The application should program this bit if HPRT0.PrtPwr was programmed to 0 before entering Hibernation. This is to indicate PMU whether session was ended before entering Hibernation. 1'b0: HPRT0.PrtPwr was not programed to 0. 1'b1: HPRT0.PrtPwr was programmed to 0.

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Bit	Attr	Reset Value	Description
5	RW	0x0	PwrDnSwtch Power Down Switch This bit indicates to the OTG core VDD switch is in ON/OFF state 1'b0: OTG is in ON state 1'b1: OTG is in OFF state Note: This bit must not be written to during normal mode of operation.
4	RW	0x0	PwrDnRst_n Power Down ResetN The application must program this bit to reset the DWC OTG core during the Hibernation exit process or during ADP when powering up the core (in case the DWC OTG core was powered off during ADP process). 1'b1: OTG is in normal operation 1'b0: reset OTG Note: This bit must not be written to during normal mode of operation.
3	RW	0x0	PwrDnClmp Power Down Clamp The application must program this bit to enable or disable the clamps to all the outputs of the DWC OTG core module to prevent the corruption of other active logic. 1'b0: Disable PMU power clamp 1'b1: Enable PMU power clamp
2	RW	0x0	Restore The application should program this bit to enable or disable restore mode from the PMU module. 1'b0: OTG in normal mode of operation 1'b1: OTG in restore mode Note: This bit must not be written to during normal mode of operation. This bit is valid only when OTG_EN_PWROPT = 2.
1	RW	0x0	PMUActv PMU Active This is bit is to enable or disable the PMU logic. 1'b0: Disable PMU module 1'b1: Enable PMU module Note: This bit must not be written to during normal mode of operation.

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Bit	Attr	Reset Value	Description
0	RW	0x0	PMUIntSel PMU Interrupt Select When the hibernation functionality is selected using the configuration option OTG_EN_PWR_OPT = 2, a write to this bit with 1'b1 enables the PMU to generate interrupts to the application. During this state all interrupts from the core module are blocked to the application. Note: This bit must be set to 1'b1 before the core is put into hibernation 1'b0: Internal OTG_core interrupt is selected 1'b1: the external OTG_pmu interrupt is selected Note: This bit must not be written to during normal mode of operation.

USBOTG_GDFIFOCFG
Address: Operational Base + offset (0x005c)
Global DFIFO Software Config Register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	EPInfoBaseAddr This field provides the start address of the EP info controller.
15:0	RW	0x0000	GDFIFOCfg This field is for dynamic programming of the DFIFO Size. This value takes effect only when the application programs a non-zero value to this register.The core does not have any corrective logic if the FIFO sizes are programmed incorrectly.

USBOTG_GADPCTL
Address: Operational Base + offset (0x0060)
ADP Timer,Control and Status Register

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:27	R/WSC	0x0	AR Access Request 2'b00 Read/Write Valid (updated by the core) 2'b01 Read 2'b10 Write 2'b11 Reserved

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Bit	Attr	Reset Value	Description
26	RW	0x0	AdpTmoutMsk ADP Timeout Interrupt Mask When this bit is set, it unmasks the interrupt because of AdpTmoutInt. This bit is valid only if OTG_Ver = 1'b1(GOTGCTL[20]).
25	RW	0x0	AdpSnsIntMsk ADP Sense Interrupt Mask When this bit is set, it unmasks the interrupt due to AdpSnsInt. This bit is valid only if OTG_Ver = 1'b1(GOTGCTL[20]).
24	RW	0x0	AdpPrbIntMsk ADP Probe Interrupt Mask When this bit is set, it unmasks the interrupt due to AdpPrbInt.This bit is valid only if OTG_Ver = 1'b1(GOTGCTL[20]).
23	W1C	0x0	AdpTmoutInt ADP Timeout Interrupt This bit is relevant only for an ADP probe. When this bit is set, it means that the ramp time has completed (GADPCTL.RTIM has reached its terminal value of 0x7FF). This is a debug feature that allows software to read the ramp time after each cycle. This bit is valid only if OTG_Ver = 1'b1.
22	W1C	0x0	AdpSnsInt ADP Sense Interrupt When this bit is set, it means that the VBUS voltage is greater than VadpSns value or VadpSns is reached. This bit is valid only if OTG_Ver = 1'b1 (GOTGCTL[20]).
21	W1C	0x0	AdpPrbInt ADP Probe Interrupt When this bit is set, it means that the VBUS voltage is greater than VadpPrb or VadpPrb is reached. This bit is valid only if OTG_Ver = 1'b1 (GOTGCTL[20]).
20	RW	0x0	ADPEn ADP Enable When set, the core performs either ADP probing or sensing based on EnaPrb or EnaSns. This bit is valid only if OTG_Ver = 1'b1 (GOTGCTL[20]).

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Bit	Attr	Reset Value	Description
19	R/WSC	0x0	ADPRes ADP Reset When set, ADP controller is reset. This bit is auto-cleared after the reset procedure is complete in ADP controller.This bit is valid only if OTG_Ver = 1'b1 (GOTGCTL[20]).
18	RW	0x0	EnaSns Enable Sense When programmed to 1'b1, the core performs a sense operation. This bit is valid only if OTG_Ver = 1'b1 (GOTGCTL[20]).
17	RW	0x0	EnaPrb Enable Probe When programmed to 1'b1, the core performs a probe operation. This bit is valid only if OTG_Ver = 1'b1 (GOTGCTL[20]).
16:6	RO	0x000	RTIM RAMP TIME These bits capture the latest time it took for VBUS to ramp from VADP_SINK to VADP_PRB. The bits are defined in units of 32 kHz clock cycles as follows: 3'b000 - 1 cycles 3'b001 - 2 cycles 3'b010 - 3 cycles and so on till 0x7FF - 2048 cycles A time of 1024 cycles at 32 kHz corresponds to a time of 32 msec. (Note for scaledown ramp_timeout = prb_delta == 2'b00 => 200 cycles prb_delta == 2'b01 => 100 cycles prb_delta == 2'b01 => 50 cycles prb_delta == 2'b01 => 25 cycles.)
5:4	RW	0x0	PrbPer Probe Period These bits sets the TadpPrd as follows: 2'b00 - 0.625 to 0.925 sec (typical 0.775 sec) 2'b01 - 1.25 to 1.85 sec (typical 1.55 sec) 2'b10 - 1.9 to 2.6 sec (typical 2.275 sec) 2'b11 - Reserved (PRB_PER is also scaledown prb_per== 2'b00 => 400 ADP clocks prb_per== 2'b01 => 600 ADP clocks prb_per== 2'b10 => 800 ADP clocks prb_per==2'b11 => 1000 ADP clocks)

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Bit	Attr	Reset Value	Description
3:2	RW	0x0	PrbDelta Probe Delta These bits set the resolution for RTIM value. The bits are defined in units of 32 kHz clock cycles as follows: 2'b00 - 1 cycles 2'b01 - 2 cycles 2'b10 - 3 cycles 2'b11 - 4 cycles For example if this value is chosen to 2'b01, it means that RTIM increments for every three 32Khz clock cycles.
1:0	RW	0x0	PrbDschg Probe Discharge These bits set the times for TadpDschg. These bits are defined as follows: 2'b00 4 msec (Scaledown 2 32Khz clock cycles) 2'b01 8 msec (Scaledown 4 32Khz clock cycles) 2'b10 16 msec (Scaledown 8 32Khz clock cycles) 2'b11 32 msec (Scaledown 16 32Khz clock cycles)

USBOTG_HPTXFSIZ
Address: Operational Base + offset (0x0100)
Host Periodic Transmit FIFO Size Register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	PTxFSize Host Periodic TxFIFO Depth This value is in terms of 32-bit words. Minimum value is 16 Maximum value is 32,768 The power-on reset value of this register is specified as the Largest Host Mode Periodic Tx Data FIFO Depth (parameter OTG_TX_HPERIO_DFIFO_DEPTH). If Enable Dynamic FIFO Sizing? Was deselected (parameter OTG_DFIFO_DYNAMIC = 0), these flops are optimized, and reads returnthe power-on value. If Enable Dynamic FIFO Sizing? was selected (parameter OTG_DFIFO_DYNAMIC = 1), you can write a new value in this field. Programmed values must not exceed the power-on value set .

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	PTxFStAddr Host Periodic TxFIFO Start Address The power-on reset value of this register is the sum of the Largest Rx Data FIFO Depth and Largest Non-periodic Tx Data FIFO Depth specified. These parameters are: In shared FIFO operation: OTG_RX_DFIFO_DEPTH + OTG_TX_NPERIO_DFIFO_DEPTH. In dedicated FIFO mode: OTG_RX_DFIFO_DEPTH + OTG_TX_HNPERIO_DFIFO_DEPTH. If Enable Dynamic FIFO Sizing? was deselected (parameter OTG_DFIFO_DYNAMIC = 0), these flops are optimized, and reads return the power-on value. If Enable Dynamic FIFO Sizing? was selected (parameter OTG_DFIFO_DYNAMIC = 1), you can write a new value in this field.Programmed values must not exceed the power-on value.

USBOTG_DIEPTXFn

Address: Operational Base + offset (0x0104+0x4*(n-1)), n = 1 - 15

Device Periodic Transmit FIFO-n Size Register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	INEP1TxFDep IN Endpoint TxFIFO Depth This value is in terms of 32-bit words. Minimum value is 16 Maximum value is 32,768 The power-on reset value of this register is specified as the Largest IN Endpoint FIFO number Depth (parameter OTG_TX_DINEP_DFIFO_DEPTH_n)(0 < n <= 15). If Enable Dynamic FIFO Sizing? was deselected (parameter OTG_DFIFO_DYNAMIC = 0), these flops are optimized, and reads return the Power-on value. If Enable Dynamic FIFO Sizing? was selected (parameter OTG_DFIFO_DYNAMIC = 1), you can write a new value in this field.Programmed values must not exceed the Power-on value .

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	INEP1TxFStAddr IN Endpoint FIFO1 Transmit RAM Start Address This field contains the memory start address for IN endpoint Transmit FIFOn (0 < n <= 15). The power-on reset value of this register is specified as the Largest Rx Data FIFO Depth (parameter OTG_RX_DFIFO_DEPTH). OTG_RX_DFIFO_DEPTH + SUM 0 to n-1 (OTG_DINEP_TXFIFO_DEPTH_n) For example start address of IN endpoint FIFO 1 is OTG_RX_DFIFO_DEPTH + OTG_DINEP_TXFIFO_DEPTH_0. The start address of IN endpoint FIFO 2 is OTG_RX_DFIFO_DEPTH + OTG_DINEP_TXFIFO_DEPTH_0 + OTG_DINEP_TXFIFO_DEPTH_1. If Enable Dynamic FIFO Sizing? was deselected (parameter OTG_DFIFO_DYNAMIC = 0), these flops are optimized, and reads return the power-on value. If Enable Dynamic FIFO Sizing? was selected (parameter OTG_DFIFO_DYNAMIC = 1), and you have programmed a new value for RxFIFO depth, you can write that value in this field. Programmed values must not exceed the power-on value set .

USBOTG_HCFG
Address: Operational Base + offset (0x0400)
Host Configuration Register

Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved
26	RW	0x0	PerSchedEna Enable Periodic Scheduling Applicable in Scatter/Gather DMA mode only. Enables periodic scheduling within the core. Initially, the bit is reset. The core will not process any periodic channels. As soon as this bit is set, the core will get ready to start scheduling periodic channels and sets HCFG.PerSchedStat. The setting of HCFG.PerSchedStat indicates the core has enabled periodic scheduling. Once HCFG.PerSchedEna is set, the application is not supposed to again reset the bit unless HCFG.PerSchedStat is set. As soon as this bit is reset, the core will get ready to stop scheduling periodic channels and resets HCFG.PerSchedStat. In non-Scatter/Gather DMA mode, this bit is reserved.
25:24	RW	0x0	FrListEn Frame List Entries The value in the register specifies the number of entries in the Frame list. This field is valid only in Scatter/Gather DMA mode.

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Bit	Attr	Reset Value	Description
23	RW	0x0	DescDMA Enable Scatter/gather DMA in Host mode When the Scatter/Gather DMA option selected during configuration of the RTL, theapplication can set this bit during initialization to enable the Scatter/Gather DMA operation. NOTE: This bit must be modified only once after a reset. The following combinations are available for programming: GAHBCFG.DMAEn=0, HCFG.DescDMA=0 => Slave mode GAHBCFG.DMAEn=0, HCFG.DescDMA=1 => Invalid GAHBCFG.DMAEn=1, HCFG.DescDMA=0 => Buffered DMA mode GAHBCFG.DMAEn=1, HCFG.DescDMA=1 => Scatter/Gather DMA mode In non-Scatter/Gather DMA mode, this bit is reserved.
22:16	RO	0x0	reserved
15:8	RW	0x00	ResValid Resume Validation Period This field is effective only when HCFG.Ena32KHzS is set. It controls the resume period when the core resumes from suspend. The core counts the ResValid number of clock cycles to detect a valid resume when this is set.
7	RW	0x0	Ena32KHzS Enable 32-KHz Suspend Mode This bit can only be set if the USB 1.1 Full-Speed Serial Transceiver Interface has been selected. If USB 1.1 Full-Speed Serial Transceiver Interface has not been selected, this bit must be zero. When the USB 1.1 Full-Speed Serial Transceiver Interface is chosen and this bit is set, the core expects the 48-MHz PHY clock to be switched to 32 KHz during a suspend.
6:3	RO	0x0	reserved
2	RW	0x0	FSLSSupp FS- and LS-Only Support The application uses this bit to control the core enumeration speed. Using this bit, the application can make the core enumerate as a FS host, even if the connected device supports HS traffic. Do not make changes to this field after initial programming. 1'b0: HS/FS/LS, based on the maximum speed supported by the connected device 1'b1: FS/LS-only, even if the connected device can support HS

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	FSLSPclkSel FS/LS PHY Clock Select 2'b00: PHY clock is running at 30/60 MHz 2'b01: PHY clock is running at 48 MHz Others: Reserved

USBOTG_HFIR
Address: Operational Base + offset (0x0404)
Host Frame Interval Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	FrInt Frame Interval The value that the application programs to this field specifies the interval between two consecutive SOFs (FS) or micro-SOFs (HS) or Keep-Alive tokens (HS). This field contains the number of PHY clocks that constitute the required frame interval. The default value set in this field for a FS operation when the PHY clock frequency is 60 MHz. The application can write a value to this register only after the Port Enable bit of the Host Port Control and Status register (HPRT.PrtEnaPort) has been set. If no value is programmed, the core calculates the value based on the PHY clock specified in the FS/LS PHY Clock Select field of the Host Configuration register (HCFG.FSLSPclkSel). Do not change the value of this field after the initial configuration. 125 us * (PHY clock frequency for HS) 1 ms * (PHY clock frequency for FS/LS)

USBOTG_HFNUM
Address: Operational Base + offset (0x0408)
Host Frame Number/Frame Time Remaining Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	FrRem Frame Time Remaining Indicates the amount of time remaining in the current micro-frame (HS) or frame (FS/LS), in terms of PHY clocks. This field decrements on each PHY clock. When it reaches zero, this field is reloaded with the value in the Frame Interval register and a new SOF is transmitted on the USB.

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Bit	Attr	Reset Value	Description
15:0	RO	0xffff	FrNum Frame Number This field increments when a new SOF is transmitted on the USB, and is reset to 0 when it reaches 16'h3FFF. This field is writable only if Remove Optional Features? was not selected (OTG_RM_OTG_FEATURES = 0). Otherwise, reads return the frame number value.

USBOTG_HPTXSTS
Address: Operational Base + offset (0x0410)
Host Periodic Transmit FIFO/Queue Status Register

Bit	Attr	Reset Value	Description
31:24	RO	0x00	PTxQTop Top of the Periodic Transmit Request Queue This indicates the entry in the Periodic Tx Request Queue that is currently being processed by the MAC. This register is used for debugging. Bit [31]: Odd/Even (micro)frame 1'b0: send in even (micro)frame 1'b1: send in odd (micro)frame Bits [30:27]: Channel/endpoint number Bits [26:25]: Type 2'b00: IN/OUT 2'b01: Zero-length packet 2'b10: CSPLIT 2'b11: Disable channel command Bit [24]: Terminate (last entry for the selected channel/endpoint)
23:16	RO	0x00	PTxQSpcAvail Periodic Transmit Request Queue Space Available Indicates the number of free locations available to be written in the Periodic Transmit Request Queue. This queue holds both IN and OUT requests. 8'h0: Periodic Transmit Request Queue is full 8'h1: 1 location available 8'h2: 2 locations available n: n locations available (0 <=n <= 16) Others: Reserved

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	PTxFSpcAvail Periodic Transmit Data FIFO Space Available Indicates the number of free locations available to be written to in the Periodic TxFIFO. Values are in terms of 32-bit words . 16'h0: Periodic TxFIFO is full . 16'h1: 1 word available . 16'h2: 2 words available . 16'hn: n words available (where 0 . n . 32,768) . 16'h8000: 32,768 words available . Others: Reserved

USBOTG_HAINT
Address: Operational Base + offset (0x0414)
Host All Channels Interrupt Reigster

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RO	0x0000	HAINT Channel Interrupts One bit per channel: Bit 0 for Channel 0 Bit 1 for Channel 1 …... Bit 15 for Channel 15

USBOTG_HAINTMSK
Address: Operational Base + offset (0x0418)
Host All Channels Interrupt Mask Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	HAINTMsk Channel Interrupt Mask One bit per channel: Bit 0 for channel 0, bit 15 for channel 15

USBOTG_HPRT
Address: Operational Base + offset (0x0440)
Host Port Control and Status Register

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
18:17	RO	0x0	PrtSpd Port Speed Indicates the speed of the device attached to this port. 2'b00: High speed 2'b01: Full speed 2'b10: Low speed 2'b11: Reserved
16:13	RW	0x0	PrtTstCtl Port Test Control The application writes a nonzero value to this field to put the port into a Test mode,and the corresponding pattern is signaled on the port. 4'b0000: Test mode disabled 4'b0001: Test_J mode 4'b0010: Test_K mode 4'b0011: Test_SE0_NAK mode 4'b0100: Test_Packet mode 4'b0101: Test_Force_Enable Others: Reserved
12	R/WSC	0x0	PrtPwr Port Power The application uses this field to control power to this port (write 1'b1 to set to 1'b1and write 1'b0 to set to 1'b0), and the core can clear this bit on an over current condition. 1'b0: Power off 1'b1: Power on
11:10	RO	0x0	PrtLnSts Port Line Status Indicates the current logic level USB data lines Bit [10]: Logic level of D+ Bit [11]: Logic level of D
9	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
8	RW	0x0	PrtRst Port Reset When the application sets this bit, a reset sequence is started on this port. The application must time the reset period and clear this bit after the reset sequence is complete. 1'b0: Port not in reset 1'b1: Port in reset To start a reset on the port, the application must leave this bit set for at least the minimum duration mentioned below, as specified in the USB 2.0 specification, Section 7.1.7.5. The application can leave it set for another 10 ms in addition to the required minimum duration, before clearing the bit, even though there is no maximum limit set by the USB standard. High speed: 50 ms Full speed/Low speed: 10 ms
7	R/WSC	0x0	PrtSusp Port Suspend The application sets this bit to put this port in Suspend mode. The core only stops sending SOFs when this is set. To stop the PHY clock, the application must set the Port Clock Stop bit, which asserts the suspended input pin of the PHY. The read value of this bit reflects the current suspend status of the port. This bit is cleared by the core after a remote wakeup signal is detected or the application sets the Port Reset bit or Port Resume bit in this register or the Resume/Remote Wakeup Detected Interrupt bit or Disconnect Detected Interrupt bit in the Core Interrupt register (GINTSTS.WkUpInt or GINTSTS.DisconnInt, respectively). 1'b0: Port not in Suspend mode 1'b1: Port in Suspend mode

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Bit	Attr	Reset Value	Description
6	R/WSC	0x0	PrtRes Port Resume The application sets this bit to drive resume signaling on the port. The core continues to drive the resume signal until the application clears this bit. If the core detects a USB remote wakeup sequence, as indicated by the Port Resume/Remote Wakeup Detected Interrupt bit of the Core Interrupt register (GINTSTS.WkUpInt), the core starts driving resume signaling without application intervention and clears this bit when it detects a disconnect condition. The read value of this bit indicates whether the core is currently driving resume signaling. 1'b0: No resume driven 1'b1: Resume driven When LPM is enabled and the core is in the L1 (Sleep) state, setting this bit results in the following behavior: The core continues to drive the resume signal until a pre-determined time specified in the GLPMCFG.HIRD_Thres[3:0] field. If the core detects a USB remote wakeup sequence, as indicated by the Port L1 Resume/Remote L1 Wakeup Detected Interrupt bit of the Core Interrupt register (GINTSTS.L1WkUpInt), the core starts driving resume signaling without application intervention and clears this bit at the end of the resume. The read value of this bit indicates whether the core is currently driving resume signaling. 1'b0: No resume driven 1'b1: Resume driven
5	W1C	0x0	PrtOvrCurrChng Port Overcurrent Change The core sets this bit when the status of the Port Overcurrent Active bit (bit 4) in this register changes.
4	RO	0x0	PrtOvrCurrAct Port Overcurrent Active Indicates the overcurrent condition of the port. 1'b0: No overcurrent condition 1'b1: Overcurrent condition

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Bit	Attr	Reset Value	Description
3	W1C	0x0	PrtEnChng Port Enable/Disable Change The core sets this bit when the status of the Port Enable bit [2] of this register changes.
2	W1C	0x0	PrtEna Port Enable A port is enabled only by the core after a reset sequence, and is disabled by an overcurrent condition, a disconnect condition, or by the application clearing this bit.The application cannot set this bit by a register write. It can only clear it to disable the port. This bit does not trigger any interrupt to the application. 1'b0: Port disabled 1'b1: Port enabled
1	W1C	0x0	PrtConnDet Port Connect Detected The core sets this bit when a device connection is detected to trigger an interrupt to the application using the Host Port Interrupt bit of the Core Interrupt register (GINTSTS.PrtInt). The application must write a 1 to this bit to clear the interrupt.
0	RO	0x0	PrtConnSts Port Connect Status 1'b0: No device is attached to the port. 1'b1: A device is attached to the port.

USBOTG_HCCHARn

Address: Operational Base + offset (0x0500)

Host Channel-n Characteristics Register

Bit	Attr	Reset Value	Description
31	R/WSC	0x0	ChEna Channel Enable When Scatter/Gather mode is enabled 1'b0: Indicates that the descriptor structure is not yet ready. 1'b1: Indicates that the descriptor structure and data buffer with data is setup and this channel can access the descriptor. When Scatter/Gather mode is disabled, This field is set by the application and cleared by the OTG host. 1'b0: Channel disabled 1'b1: Channel enabled

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Bit	Attr	Reset Value	Description
30	R/WSC	0x0	ChDis Channel Disable The application sets this bit to stop transmitting/receiving data on a channel, even before the transfer for that channel is complete. The application must wait for the Channel Disabled interrupt before treating the channel as disabled.
29	RW	0x0	OddFrm Odd Frame This field is set (reset) by the application to indicate that the OTG host must perform a transfer in an odd (micro)frame. This field is applicable for only periodic (isochronous and interrupt) transactions. 1'b0: Even (micro)frame 1'b1: Odd (micro)frame This field is not applicable for Scatter/Gather DMA mode and need not be programmed by the application and is ignored by the core.
28:22	RW	0x00	DevAddr Device Address This field selects the specific device serving as the data source or sink.
21:20	RW	0x0	MC_EC Multi Count (MC) / Error Count (EC) When the Split Enable bit of the Host Channel-n Split Control register (HCSPLTn.SpltEna) is reset (1'b0), this field indicates to the host the number of transactions that must be executed per micro-frame for this periodic endpoint. For non periodic transfers, this field is used only in DMA mode, and specifies the number packets to be fetched for this channel before the internal DMA engine changes arbitration. 2'b00: Reserved This field yields undefined results. 2'b01: 1 transaction 2'b10: 2 transactions to be issued for this endpoint per micro-frame 2'b11: 3 transactions to be issued for this endpoint per micro-frame When HCSPLTn.SpltEna is set (1'b1), this field indicates the number of immediate retries to be performed for a periodic split transactions on transaction errors. This field must be set to at least 2'b01.

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Bit	Attr	Reset Value	Description
19:18	RW	0x0	EPType Endpoint Type Indicates the transfer type selected. 2'b00: Control 2'b01: Isochronous 2'b10: Bulk 2'b11: Interrupt
17	RW	0x0	LSpdDev Low-Speed Device This field is set by the application to indicate that this channel is communicating to a low-speed device.
16	RO	0x0	reserved
15	RW	0x0	EPDir Endpoint Direction Indicates whether the transaction is IN or OUT. 1'b0: OUT 1'b1: IN
14:11	RW	0x0	EPNum Endpoint Number Indicates the endpoint number on the device serving as the data source or sink.
10:0	RW	0x000	MPS Maximum Packet Size Indicates the maximum packet size of the associated endpoint.

USBOTG_HCSPLTn
Address: Operational Base + offset (0x0504)
Host Channel-n Split Control Register

Bit	Attr	Reset Value	Description
31	RW	0x0	SpltEna Split Enable The application sets this field to indicate that this channel is enabled to perform split transactions.
30:17	RO	0x0	reserved
16	RW	0x0	CompSplt Do Complete Split The application sets this field to request the OTG host to perform a complete split transaction.

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Bit	Attr	Reset Value	Description
15:14	RW	0x0	XactPos Transaction Position This field is used to determine whether to send all, first, middle, or last payloads with each OUT transaction. 2'b11: All. This is the entire data payload is of this transaction (which is less than or equal to 188 bytes). 2'b10: Begin. This is the first data payload of this transaction (which is larger than 188 bytes). 2'b00: Mid. This is the middle payload of this transaction (which is larger than 188bytes). 2'b01: End. This is the last payload of this transaction (which is larger than 188 bytes).
13:7	RW	0x00	HubAddr Hub Address This field holds the device address of the transaction translator's hub.
6:1	RO	0x0	reserved
0	RW	0x0	PrtAddr Port Address This field is the port number of the recipient transaction translator.

USBOTG_HCINTn

Address: Operational Base + offset (0x0508)

Host Channel-n Interrupt Register

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13	W1C	0x0	DESC_LST_ROLLIntr Descriptor rollover interrupt This bit is valid only when Scatter/Gather DMA mode is enabled. The core sets this bit when the corresponding channel's descriptor list rolls over. For non-Scatter/Gather DMA mode, this bit is reserved.
12	W1C	0x0	XCS_XACT_ERR Excessive Transaction Error This bit is valid only when Scatter/Gather DMA mode is enabled. The core sets this bit when 3 consecutive transaction errors occurred on the USB bus. XCS_XACT_ERR will not be generated for Isochronous channels. For non-Scatter/Gather DMA mode, this bit is reserved.

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Bit	Attr	Reset Value	Description
11	W1C	0x0	BNAIntr BNA (Buffer Not Available) Interrupt This bit is valid only when Scatter/Gather DMA mode is enabled. The core generates this interrupt when the descriptor accessed is not ready for the Core to process. BNA will not be generated for Isochronous channels. For non-Scatter/Gather DMA mode, this bit is reserved.
10	W1C	0x0	DataTglErr Data Toggle Error In Scatter/Gather DMA mode, the interrupt due to this bit is masked in the core.
9	W1C	0x0	FrmOvrun Frame Overrun In Scatter/Gather DMA mode, the interrupt due to this bit is masked in the core
8	W1C	0x0	BblErr Babble Error In Scatter/Gather DMA mode, the interrupt due to this bit is masked in the core.
7	W1C	0x0	XactErr Transaction Error Indicates one of the following errors occurred on the USB: CRC check failure, Timeout, Bit stuff error, False EOP. In Scatter/Gather DMA mode, the interrupt due to this bit is masked in the core.
6	WO	0x0	NYET NYET Response Received Interrupt In Scatter/Gather DMA mode, the interrupt due to this bit is masked in the core.
5	W1C	0x0	ACK ACK Response Received/Transmitted Interrupt In Scatter/Gather DMA mode, the interrupt due to this bit is masked in the core.
4	W1C	0x0	NAK NAK Response Received Interrupt In Scatter/Gather DMA mode, the interrupt due to this bit is masked in the core.
3	W1C	0x0	STALL STALL Response Received Interrupt In Scatter/Gather DMA mode, the interrupt due to this bit is masked in the core.

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Bit	Attr	Reset Value	Description
2	W1C	0x0	AHBErr AHB Error This is generated only in DMA mode when there is an AHB error during AHB read/write.The application can read the corresponding channel's DMA address register to get the error address.
1	W1C	0x0	ChHltd Channel Halted In non-Scatter/Gather DMA mode, it indicates the transfer completed abnormally either because of any USB transaction error or in response to disable request by the application or because of a completed transfer.In Scatter/Gather DMA mode, this indicates that transfer completed due to any of the following: EOL being set in descriptor, AHB error, Excessive transaction errors, In response to disable request by the application,Babble, Stall, Buffer Not Available (BNA)
0	W1C	0x0	XferCompl Transfer Completed For Scatter/Gather DMA mode, it indicates that current descriptor processing got completed with IOC bit set in its descriptor. In non-Scatter/Gather DMA mode, it indicates that Transfer completed normally without any errors.

USBOTG_HCINTMSKn

Address: Operational Base + offset (0x050c)

Host Channel-n Interrupt Mask Register

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13	RW	0x0	DESC_LST_ROLLIntrMsk Descriptor rollover interrupt Mask register This bit is valid only when Scatter/Gather DMA mode is enabled. In non-Scatter/Gather DMA mode, this bit is reserved.
12	RO	0x0	reserved
11	RW	0x0	BNAIntrMsk BNA (Buffer Not Available) Interrupt mask register This bit is valid only when Scatter/Gather DMA mode is enabled. In non-Scatter/Gather DMA mode, this bit is reserved.
10	RW	0x0	DataTglErrMsk Data Toggle Error Mask This bit is not applicable in Scatter/Gather DMA mode.

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Bit	Attr	Reset Value	Description
9	RW	0x0	FrmOvrunMsk Frame Overrun Mask This bit is not applicable in Scatter/Gather DMA mode.
8	RW	0x0	BblErrMsk Babble Error Mask This bit is not applicable in Scatter/Gather DMA mode.
7	RW	0x0	XactErrMsk Transaction Error Mask This bit is not applicable in Scatter/Gather DMA mode
6	RW	0x0	NyetMsk NYET Response Received Interrupt Mask This bit is not applicable in Scatter/Gather DMA mode.
5	RW	0x0	AckMsk ACK Response Received/Transmitted Interrupt Mask This bit is not applicable in Scatter/Gather DMA mode.
4	RW	0x0	NakMsk NAK Response Received Interrupt Mask This bit is not applicable in Scatter/Gather DMA mode.
3	RW	0x0	StallMsk STALL Response Received Interrupt Mask This bit is not applicable in Scatter/Gather DMA mode.
2	RW	0x0	AHBErrMsk AHB Error Mask Note: This bit is only accessible when OTG_ARCHITECTURE = 2
1	RW	0x0	ChHltdMsk Channel Halted Mask
0	RW	0x0	XferComplMsk Transfer Completed Mask This bit is valid only when Scatter/Gather DMA mode is enabled. In non-Scatter/Gather DMA mode, this bit is reserved.

USBOTG_HCTSIZn

Address: Operational Base + offset (0x0510)

Host Channel-n Transfer Size Register

Bit	Attr	Reset Value	Description
31	RW	0x0	DoPng Do Ping This bit is used only for OUT transfers. Setting this field to 1 directs the host to do PING protocol. Note: Do not set this bit for IN transfers. If this bit is set for IN transfers it disables the channel.

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Bit	Attr	Reset Value	Description
30:29	RW	0x0	Pid PID The application programs this field with the type of PID to use for the initial transaction. The host maintains this field for the rest of the transfer. 2'b00: DATA0 2'b01: DATA2 2'b10: DATA1 2'b11: MDATA (non-control)/SETUP (control)
28:19	RW	0x000	PktCnt Packet Count This field is programmed by the application with the expected number of packets to be transmitted (OUT) or received (IN).The host decrements this count on every successful transmission or reception of an OUT/IN packet. Once this count reaches zero, the application is interrupted to indicate normal completion.The width of this counter is specified as Width of Packet Counters (parameter OTG_PACKET_COUNT_WIDTH).
18:0	RW	0x00000	XferSize Transfer Size For an OUT, this field is the number of data bytes the host sends during the transfer. For an IN, this field is the buffer size that the application has Reserved for the transfer. The application is expected to program this field as an integer multiple of the maximum packet size for IN transactions (periodic and non-periodic). The width of this counter is specified as Width of Transfer Size Counters (parameter OTG_TRANS_COUNT_WIDTH).

USBOTG_HCDMAn
Address: Operational Base + offset (0x0514)
Host Channel-n DMA Address Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMAAddr DMA Address This field holds the start address in the external memory from which the data for the endpoint must be fetched or to which it must be stored. This register is incremented on every AHB transaction.

USBOTG_HCDMABn
Address: Operational Base + offset (0x051c)
Host Channel-n DMA Buffer Address Register

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HCDMABn Holds the current buffer address This register is updated as and when the data transfer for the corresponding end point is in progress. This register is present only in Scatter/Gather DMA mode. Otherwise this field is reserved.

USBOTG_DCFG

Address: Operational Base + offset (0x0800)

Device Cconfiguration Register

Bit	Attr	Reset Value	Description
31:26	RW	0x02	ResValid Resume Validation Period This field controls the period when the core resumes from a suspend. When this bit is set, the core counts for the ResValid number of clock cycles to detect a valid resume. This field is effective only when DCFG.Ena32KHzSusp is set.
25:24	RW	0x0	PerSchIntvl Periodic Scheduling Interval PerSchIntvl must be programmed only for Scatter/Gather DMA mode. Description: This field specifies the amount of time the Internal DMA engine must allocate for fetching periodic IN endpoint data. Based on the number of periodic endpoints, this value must be specified as 25,50 or 75% of (micro)frame. When any periodic endpoints are active, the internal DMA engine allocates the specified amount of time in fetching periodic IN endpoint data. When no periodic endpoints are active, then the internal DMA engine services nonperiodic endpoints, ignoring this field. After the specified time within a (micro)frame, the DMA switches to fetching for nonperiodic endpoints. 2'b00: 25% of (micro)frame. 2'b01: 50% of (micro)frame. 2'b10: 75% of (micro)frame. 2'b11: Reserved.

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Bit	Attr	Reset Value	Description
23	RW	0x0	DescDMA Enable Scatter/Gather DMA in Device mode When the Scatter/Gather DMA option selected during configuration of the RTL, the application can set this bit during initialization to enable the Scatter/Gather DMA operation. NOTE: This bit must be modified only once after a reset. The following combinations are available for programming: GAHBCFG.DMAEn=0,DCFG.DescDMA=0 => Slave mode GAHBCFG.DMAEn=0,DCFG.DescDMA=1 => Invalid GAHBCFG.DMAEn=1,DCFG.DescDMA=0 => Buffered DMA mode GAHBCFG.DMAEn=1,DCFG.DescDMA=1 => Scatter/Gather DMA mode
22:18	RW	0x08	EPMisCnt IN Endpoint Mismatch Count This field is valid only in shared FIFO operation.The application programs this filed with a count that determines when the core generates an Endpoint Mismatch interrupt (GINTSTS.EPMis). The core loads this value into an internal counter and decrements it. The counter is reloaded whenever there is a match or when the counter expires. The width of this counter depends on the depth of the Token Queue.
17:13	RO	0x0	reserved
12:11	RW	0x0	PerFrInt Periodic Frame Interval Indicates the time within a (micro)frame at which the application must be notified using the End Of Periodic Frame Interrupt. This can be used to determine if all the isochronous traffic for that (micro)frame is complete. 2'b00: 80% of the (micro)frame interval 2'b01: 85% 2'b10: 90% 2'b11: 95%
10:4	RW	0x00	DevAddr Device Address The application must program this field after every SetAddress control command.

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Bit	Attr	Reset Value	Description
3	RW	0x0	Ena32KHzS Enable 32-KHz Suspend Mode When the USB 1.1 Full-Speed Serial Transceiver Interface is chosen and this bit is set, the core expects the 48-MHz PHY clock to be switched to 32 KHz during a suspend. This bit can only be set if USB 1.1 Full-Speed Serial Transceiver Interface has been selected. If USB 1.1 Full-Speed Serial Transceiver Interface has not been selected, this bit must be zero.
2	RW	0x0	NZStsOUTHShk Non-Zero-Length Status OUT Handshake The application can use this field to select the handshake the core sends on receiving a nonzero-length data packet during the OUT transaction of a control transfer's Status stage. 1'b1: Send a STALL handshake on a nonzero-length status OUT transaction and do not send the received OUT packet to the application. 1'b0: Send the received OUT packet to the application (zero-length or nonzerolength) and send a handshake based on the NAK and STALL bits for the endpoint in the Device Endpoint Control register.
1:0	RW	0x0	DevSpd Device Speed Indicates the speed at which the application requires the core to enumerate, or the maximum speed the application can support. However, the actual bus speed is determined only after the chirp sequence is completed, and is based on the speed of the USB host to which the core is connected. 2'b00: High speed (USB 2.0 PHY clock is 30 MHz or 60 MHz) 2'b01: Full speed (USB 2.0 PHY clock is 30 MHz or 60 MHz) 2'b10: Reserved 2'b11: Full speed (USB 1.1 transceiver clock is 48 MHz)

USBOTG_DCTL

Address: Operational Base + offset (0x0804)

Device Control Register

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
16	RW	0x0	NakOnBble Set NAK automatically on babble The core sets NAK automatically for the endpoint on which babble is received.
15	RW	0x0	IgnrFrmNum Ignore frame number for isochronous endpoints in case of Scatte Do NOT program IgnrFrmNum bit to 1'b1 when the core is operating in Threshold mode. Note: When Scatter/Gather DMA mode is enabled this feature is not applicable to highspeed, high-bandwidth transfers. When this bit is enabled, there must be only one packet per descriptor. 1'b0: The core transmits the packets only in the frame number in which they are intended to be transmitted. 1'b1: The core ignores the frame number, sending packets immediately as the packets are ready. Scatter/Gather: In Scatter/Gather DMA mode, when this bit is enabled, the packets are not flushed when an ISOC IN token is received for an elapsed frame. When Scatter/Gather DMA mode is disabled, this field is used by the application to enable periodic transfer interrupt. The application can program periodic endpoint transfers for multiple (micro)frames. 1'b0: Periodic transfer interrupt feature is disabled; the application must program transfers for periodic endpoints every (micro)frame 1'b1: Periodic transfer interrupt feature is enabled; the application can program transfers for multiple (micro)frames for periodic endpoints. In non-Scatter/Gather DMA mode, the application receives transfer complete interrupt after transfers for multiple (micro)frames are completed.

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Bit	Attr	Reset Value	Description
14:13	RW	0x1	GMC Global Multi Count GMC must be programmed only once after initialization. Applicable only for Scatter/Gather DMA mode. This indicates the number of packets to be serviced for that end point before moving to the next end point. It is only for nonperiodic end points. 2'b00: Invalid. 2'b01: 1 packet. 2'b10: 2 packets. 2'b11: 3 packets. When Scatter/Gather DMA mode is disabled, this field is reserved. and reads 2'b00.
12	RO	0x0	reserved
11	RW	0x0	PWROnPrgDone Power-On Programming Done The application uses this bit to indicate that register programming is completed after a wake-up from Power Down mode.
10	WO	0x0	CGOUTNak Clear Global OUT NAK A write to this field clears the Global OUT NAK.
9	WO	0x0	SGOUTNak Set Global OUT NAK A write to this field sets the Global OUT NAK. The application uses this bit to send a NAK handshake on all OUT endpoints. The application must set this bit only after making sure that the Global OUT NAK Effective bit in the Core Interrupt Register (GINTSTS.GOUTNakEff) is cleared.
8	WO	0x0	CGNPInNak Clear Global Non-periodic IN NAK A write to this field clears the Global Non-periodic IN NAK.
7	WO	0x0	SGNPInNak Set Global Non-periodic IN NAK A write to this field sets the Global Non-periodic IN NAK.The application uses this bit to send a NAK handshake on all non-periodic IN endpoints. The core can also set this bit when a timeout condition is detected on a non-periodic endpoint in shared FIFO operation. The application must set this bit only after making sure that the Global IN NAK Effective bit in the Core Interrupt Register (GINTSTS.GINNakEff) is cleared.

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Bit	Attr	Reset Value	Description
6:4	RW	0x0	TstCtl Test Control 3'b000: Test mode disabled 3'b001: Test_J mode 3'b010: Test_K mode 3'b011: Test_SE0_NAK mode 3'b100: Test_Packet mode 3'b101: Test_Force_Enable Others: Reserved
3	RO	0x0	GOUTNakSts Global OUT NAK Status 1'b0: A handshake is sent based on the FIFO Status and the NAK and STALL bit settings. 1'b1: No data is written to the RxFIFO, irrespective of space availability. Sends a NAK handshake on all packets, except on SETUP transactions. All isochronous OUT packets are dropped
2	RO	0x0	GNPINNakSts Global Non-periodic IN NAK Status 1'b0: A handshake is sent out based on the data availability in the transmit FIFO. 1'b1: A NAK handshake is sent out on all non-periodic IN endpoints, irrespective of the data availability in the transmit FIFO.
1	RW	0x0	SftDiscon Soft Disconnect The application uses this bit to signal the Otg core to do a soft disconnect. As long as this bit is set, the host does not see that the device is connected, and the device does not receive signals on the USB. The core stays in the disconnected state until the application clears this bit. 1'b0: Normal operation. When this bit is cleared after a soft disconnect, the core drives the phy_opmode_o signal on the UTMI+ to 2'b00, which generates a device connect event to the USB host. When the device is reconnected, the USB host restarts device enumeration. 1'b1: The core drives the phy_opmode_o signal on the UTMI+ to 2'b01, which generates a device disconnect event to the USB host.

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Bit	Attr	Reset Value	Description
0	RW	0x0	RmtWkUpSig Remote Wakeup Signaling When the application sets this bit, the core initiates remote signaling to wake the USB host. The application must set this bit to instruct the core to exit the Suspend state. As specified in the USB 2.0 specification, the application must clear this bit 1-15 ms after setting it.If LPM is enabled and the core is in the L1 (Sleep) state, when the application sets this bit, the core initiates L1 remote signaling to wake up the USB host. The application must set this bit to instruct the core to exit the Sleep state. As specified in the LPM specification, the hardware automatically clears this bit 50 us (TL1DevDrvResume) after being set by the application. The application must not set this bit when GLPMCFG bRemoteWake from the previous LPM transaction is zero.

USBOTG_DSTS
Address: Operational Base + offset (0x0808)
Device Status Register

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21:8	RW	0x0000	SOFFN Frame or Micro-frame Number of the Received SOF When the core is operating at high speed, this field contains a micro-frame number. When the core is operating at full or low speed, this field contains a frame number.
7:4	RO	0x0	reserved
3	RW	0x0	ErrticErr Erratic Error The core sets this bit to report any erratic errors (phy_rxvalid_i/phy_rxvldh_i or phy_rxactive_i is asserted for at least 2 ms, due to PHY error) seen on the UTMI+. Due to erratic errors, the Otg core goes into Suspended state and an interrupt is generated to the application with Early Suspend bit of the Core Interrupt register (GINTSTS.ErlySusp). If the early suspend is asserted due to an erratic error, the application can only perform a soft disconnect recover.

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Bit	Attr	Reset Value	Description
2:1	RW	0x0	EnumSpd Enumerated Speed Indicates the speed at which the OTG core has come up after speed detection through a chirp sequence. 2'b00: High speed (PHY clock is running at 30 or 60 MHz) 2'b01: Full speed (PHY clock is running at 30 or 60 MHz) 2'b10: Low speed (PHY clock is running at 48 MHz, internal phy_clk at 6 MHz) 2'b11: Full speed (PHY clock is running at 48 MHz) Low speed is not supported for devices using a UTMI+ PHY.
0	RW	0x0	SuspSts Suspend Status In Device mode, this bit is set as long as a Suspend condition is detected on the USB. The core enters the Suspended state when there is no activity on the utmi_linestate signal for an extended period of time. The core comes out of the suspend: When there is any activity on the utmi_linestate signal, When the application writes to the Remote Wakeup Signaling bit in the Device Control register (DCTL.RmtWkUpSig).

USBOTG_DIEPMSK

Address: Operational Base + offset (0x0810)

Device IN Endpoint common interrupt mask register

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13	RW	0x0	NAKMsk NAK interrupt Mask
12:10	RO	0x0	reserved
9	RW	0x0	BNAInIntrMsk BNA Interrupt Mask
8	RW	0x0	TxfifoUndrnMsk Fifo Underrun Mask
7	RO	0x0	reserved
6	RW	0x0	INEPNakEffMsk IN Endpoint NAK Effective Mask
5	RW	0x0	INTknEPMisMsk IN Token received with EP Mismatch Mask
4	RW	0x0	INTknTXFEmpMsk IN Token Received When TxFIFO Empty Mask

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Bit	Attr	Reset Value	Description
3	RW	0x0	TimeOUTMsk Timeout Condition Mask
2	RW	0x0	AHBErrMsk AHB Error Mask
1	RW	0x0	EPDisbldMsk Endpoint Disabled Interrupt Mask
0	RW	0x0	XferComplMsk Transfer Completed Interrupt Mask

USBOTG_DOEPMSK
Address: Operational Base + offset (0x0814)
Device OUT Endpoint common interrupt mask register

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14	RW	0x0	NYETMsk NYET Interrupt Mask
13	RW	0x0	NAKMsk NAK Interrupt Mask
12	RW	0x0	BbleErrMsk Babble Interrupt Mask
11:10	RO	0x0	reserved
9	RW	0x0	BnaOutIntrMsk BNA interrupt Mask
8	RW	0x0	OutPktErrMsk OUT Packet Error Mask
7	RO	0x0	reserved
6	RW	0x0	Back2BackSETup Back-to-Back SETUP Packets Received Mask Applies to control OUT endpoints only.
5	RO	0x0	reserved
4	RW	0x0	OUTTknEPdisMsk OUT Token Received when Endpoint Disabled Mask Applies to control OUT endpoints only.
3	RW	0x0	SetUPMsk SETUP Phase Done Mask Applies to control endpoints only.
2	RW	0x0	AHBErrMsk AHB Error
1	RW	0x0	EPDisbldMsk Endpoint Disabled Interrupt Mask
0	RW	0x0	XferComplMsk Transfer Completed Interrupt Mask

USBOTG_DAINT

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Address: Operational Base + offset (0x0818)
Device All Endpoints interrupt register

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	OutEPInt OUT Endpoint Interrupt Bits One bit per OUT endpoint: Bit 16 for OUT endpoint 0, bit 31 for OUT endpoint 15
15:0	RO	0x0000	InEpInt IN Endpoint Interrupt Bits One bit per IN Endpoint: Bit 0 for IN endpoint 0, bit 15 for endpoint 15

USBOTG_DAINTMSK
Address: Operational Base + offset (0x081c)
Device All Endpoint interrupt mask register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	OutEpMsk OUT EP Interrupt Mask Bits One per OUT Endpoint: Bit 16 for OUT EP 0, bit 31 for OUT EP 15
15:0	RW	0x0000	InEpMsk IN EP Interrupt Mask Bits One bit per IN Endpoint: Bit 0 for IN EP 0, bit 15 for IN EP 15

USBOTG_DTKNQR1
Address: Operational Base + offset (0x0820)
Device IN token sequence learning queue read register1

Bit	Attr	Reset Value	Description
31:8	RO	0x000000	EPTkn Endpoint Token Four bits per token represent the endpoint number of the token: Bits [31:28]: Endpoint number of Token 5 Bits [27:24]: Endpoint number of Token 4 ..... Bits [15:12]: Endpoint number of Token 1 Bits [11:8]: Endpoint number of Token 0
7	RO	0x0	WrapBit Wrap Bit This bit is set when the write pointer wraps. It is cleared when the learning queue is cleared.
6:5	RO	0x0	reserved
4:0	RO	0x00	INTknWPtr IN Token Queue Write Pointer

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USBOTG_DTKNQR2
Address: Operational Base + offset (0x0824)
Device IN token sequence learning queue read register2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	EPTkn Endpoint Token Four bits per token represent the endpoint number of the token: Bits [31:28]: Endpoint number of Token 13 Bits [27:24]: Endpoint number of Token 12 ..... Bits [7:4]: Endpoint number of Token 7 Bits [3:0]: Endpoint number of Token 6

USBOTG_DVBUSDIS
Address: Operational Base + offset (0x0828)
Device VBUS discharge time register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0b8f	DVBUSDis Device VBUS Discharge Time Specifies the VBUS discharge time after VBUS pulsing during SRP. This value equals:VBUS discharge time in PHY clocks / 1,024.The value you use depends whether the PHY is operating at 30 MHz (16-bit data width) or 60 MHz (8-bit data width). Depending on your VBUS load, this value can need adjustment.

USBOTG_DVBUSPULSE
Address: Operational Base + offset (0x082c)
Device VBUS Pulsing Timer Register

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	DVBUSPulse Device VBUS Pulsing Time Specifies the VBUS pulsing time during SRP. This value equals:VBUS pulsing time in PHY clocks / 1,024.The value you use depends whether the PHY is operating at 30 MHz (16-bit data width) or 60 MHz (8-bit data width).

USBOTG_DTHRCTL
Address: Operational Base + offset (0x0830)
Device Threshold Control Register

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Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27	RW	0x1	ArbPrkEn Arbiter Parking Enable This bit controls internal DMA arbiter parking for IN endpoints. When threshold is enabled and this bit is set to one, then the arbiter parks on the IN endpoint for which there is a token received on the USB. This is done to avoid getting into under-run conditions. By default the parking is enabled.
26	RO	0x0	reserved
25:17	RW	0x008	RxThrLen Receive Threshold Length This field specifies Receive threshold size in DWORDS. This field also specifies the amount of data received on the USB before the core can start transmitting on the AHB. The threshold length has to be at least eight DWORDS. The recommended value for ThrLen is to be the same as the programmed AHB Burst Length (GAHBCFG.HBstLen).
16	RW	0x0	RxThrEn Receive Threshold Enable When this bit is set, the core enables thresholding in the receive direction.
15:13	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
12:11	RW	0x0	AHBThrRatio AHB Threshold Ratio These bits define the ratio between the AHB threshold and the MAC threshold for the transmit path only. The AHB threshold always remains less than or equal to the USB threshold, because this does not increase overhead. Both the AHB and the MAC threshold must be DWORD-aligned. The application needs to program TxThrLen and the AHBThrRatio to make the AHB Threshold value DWORD aligned. If the AHB threshold value is not DWORD aligned, the core might not behave correctly. When programming the TxThrLen and AHBThrRatio, the application must ensure that the minimum AHB threshold value does not go below 8 DWORDS to meet the USB turnaround time requirements. 2'b00: AHB threshold = MAC threshold 2'b01: AHB threshold = MAC threshold / 2 2'b10: AHB threshold = MAC threshold / 4 2'b11: AHB threshold = MAC threshold / 8
10:2	RW	0x008	TxThrLen Transmit Threshold Length This field specifies Transmit threshold size in DWORDS. This field also forms the MAC threshold and specifies the amount of data, in bytes, to be in the corresponding endpoint transmit FIFO before the core can start a transaction on the USB. When the value of AHBThrRatio is 2'h00, the threshold length must be at least 8 DWORDS. If the AHBThrRatio is nonzero, the application must ensure that the AHB threshold value does not go below the recommended 8 DWORDs. This field controls both isochronous and non-isochronous IN endpoint thresholds. The recommended value for ThrLen is to be the same as the programmed AHB Burst Length (GAHBCFG.HBstLen).
1	RW	0x0	ISOThrEn ISO IN Endpoints Threshold Enable When this bit is set, the core enables threshold for isochronous IN endpoints.

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Bit	Attr	Reset Value	Description
0	RW	0x0	NonISOThrEn Non-ISO IN Endpoints Threshold Enable When this bit is set, the core enables threshold for Non Isochronous IN endpoints.

USBOTG_DIEPEMPMSK
Address: Operational Base + offset (0x0834)
Device IN endpoint FIFO empty interrupt mask register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	InEpTxfEmpMsk IN EP Tx FIFO Empty Interrupt Mask Bits These bits act as mask bits for DIEPINTn. TxFEmp interrupt One bit per IN Endpoint: Bit 0 for IN endpoint 0 ... Bit 15 for endpoint 15

USBOTG_DEACHINT
Address: Operational Base + offset (0x0838)
Device each endpoint interrupt register

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	EchOutEPInt OUT Endpoint Interrupt Bits One bit per OUT endpoint: Bit 16 for OUT endpoint 0 ... Bit 31 for OUT endpoint 15
15:0	RO	0x0000	EchInEpInt IN Endpoint Interrupt Bits One bit per IN Endpoint: Bit 0 for IN endpoint 0 ... Bit 15 for endpoint 15

USBOTG_DEACHINTMSK
Address: Operational Base + offset (0x083c)
Device each endpoint interrupt register mask

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	EchOutEpMsk OUT EP Interrupt Mask Bits One per OUT Endpoint: Bit 16 for IN endpoint 0 ... Bit 31 for endpoint 15

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	EchInEpMsk IN EP Interrupt Mask Bits One bit per IN Endpoint: Bit 0 for IN endpoint 0 ... Bit 15 for endpoint 15

USBOTG_DIEPEACHMSKn
Address: Operational Base + offset (0x0840)
Device each IN endpoint -n interrupt Register

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13	RW	0x0	NAKMsk NAK interrupt Mask
12:10	RO	0x0	reserved
9	RW	0x0	BNAInIntrMsk BNA interrupt Mask
8	RW	0x0	TxfifoUndrnMsk Fifo Under run Mask
7	RO	0x0	reserved
6	RW	0x0	INEPNakEffMsk IN Endpoint NAK Effective Mask
5	RW	0x0	INTknEPMisMsk IN Token received with EP Mismatch Mask
4	RW	0x0	INTknTXFEmpMsk IN Token Received When TxFIFO Empty Mask
3	RW	0x0	TimeOUTMsk Timeout Condition Mask(Non-isochronous endpoints)
2	RW	0x0	AHBErrMsk AHB Error Mask
1	RW	0x0	EPDisbldMsk Endpoint Disabled Interrupt Mask
0	RW	0x0	XferComplMsk Transfer Completed Interrupt Mask

USBOTG_DOEPEACHMSKn
Address: Operational Base + offset (0x0880)
Device each out endpoint-n interrupt register

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14	RW	0x0	NYETMsk NYET interrupt Mask
13	RW	0x0	NAKMsk NAK interrupt Mask

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Bit	Attr	Reset Value	Description
12	RW	0x0	BbleErrMsk Babble interrupt Mask
11:10	RO	0x0	reserved
9	RW	0x0	BnaOutIntrMsk BNA interrupt Mask
8	RW	0x0	OutPktErrMsk OUT Packet Error Mask
7	RO	0x0	reserved
6	RW	0x0	Back2BackSETup Back-to-Back SETUP Packets Received Mask Applies to control OUT endpoints only.
5	RO	0x0	reserved
4	RW	0x0	OUTTknEPdisMsk OUT Token Received when Endpoint Disabled Mask Applies to control OUT endpoints only.
3	RW	0x0	SetUPMsk SETUP Phase Done Mask Applies to control endpoints only.
2	RW	0x0	AHBErrMsk AHB Error
1	RW	0x0	EPDisbldMsk Endpoint Disabled Interrupt Mask
0	RW	0x0	XferComplMsk Transfer Completed Interrupt Mask

USBOTG_DIEPCTL0

Address: Operational Base + offset (0x0900)

Device control IN endpoint 0 control register

Bit	Attr	Reset Value	Description
31	R/WSC	0x0	EPEna Endpoint Enable When Scatter/Gather DMA mode is enabled, for IN endpoints this bit indicates that the descriptor structure and data buffer with data ready to transmit is setup. When Scatter/Gather DMA mode is disabled-such as in buffer-pointer based DMA mode-this bit indicates that data is ready to be transmitted on the endpoint. The core clears this bit before setting the following interrupts on this endpoint: Endpoint Disabled; Transfer Completed.

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Bit	Attr	Reset Value	Description
30	R/WSC	0x0	EPDis Endpoint Disable The application sets this bit to stop transmitting data on an endpoint, even before the transfer for that endpoint is complete. The application must wait for the Endpoint Disabled interrupt before treating the endpoint as disabled. The core clears this bit before setting the Endpoint Disabled Interrupt. The application must set this bit only if Endpoint Enable is already set for this endpoint.
29:28	RO	0x0	reserved
27	WO	0x0	SNAK Set NAK A write to this bit sets the NAK bit for the endpoint. Using this bit, the application can control the transmission of NAK handshakes on an endpoint. The core can also set this bit for an endpoint after a SETUP packet is received on that endpoint.
26	WO	0x0	CNAK Clear NAK A write to this bit clears the NAK bit for the endpoint.
25:23	RO	0x0	reserved
22	RW	0x0	TxFNum TxFIFO Number For Shared FIFO operation, this value is always set to 0, indicating that control IN endpoint 0 data is always written in the Non-Periodic Transmit FIFO. For Dedicated FIFO operation, this value is set to the FIFO number that is assigned to IN Endpoint 0.
21	R/WSC	0x0	Stall STALL Handshake The application can only set this bit, and the core clears it, when a SETUP token is received for this endpoint. If a NAK bit, Global Non-periodic IN NAK, or Global OUT NAK is set along with this bit, the STALL bit takes priority.
20	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
19:18	RO	0x0	EPType Endpoint Type Hardcoded to 00 for control
17	RO	0x0	NAKSts NAK Status Indicates the following: 1'b0: The core is transmitting non-NAK handshakes based on the FIFO status 1'b1: The core is transmitting NAK handshakes on this endpoint. When this bit is set, either by the application or core, the core stops transmitting data, even if there is data available in the TxFIFO. Irrespective of this bit's setting, the core always responds to SETUP data packets with an ACK handshake.
16	RO	0x0	reserved
15	RO	0x1	USBActEP USB Active Endpoint This bit is always set to 1, indicating that control endpoint 0 is always active in all configurations and interfaces.
14:11	RW	0x0	NextEp Next Endpoint Applies to non-periodic IN endpoints only. Indicates the endpoint number to be fetched after the data for the current endpoint is fetched. The core can access this field, even when the Endpoint Enable (EPEna) bit is not set. This field is not valid in Slave mode. Note: This field is valid only for Shared FIFO operations.
10:2	RO	0x0	reserved
1:0	RW	0x0	MPS Maximum Packet Size Applies to IN and OUT endpoints. The application must program this field with the maximum packet size for the current logical endpoint. 2'b00: 64 bytes 2'b01: 32 bytes 2'b10: 16 bytes 2'b11: 8 bytes

USBOTG_DIEPINTn

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Address: Operational Base + offset (0x0908)
Device Endpoint-n Interrupt Register

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14	W1C	0x0	NYETIntrpt NYET interrupt The core generates this interrupt when a NYET response is transmitted for a non-isochronous OUT endpoint.
13	W1C	0x0	NAKIntrpt NAK interrupt The core generates this interrupt when a NAK is transmitted or received by the device.In case of isochronous IN endpoints the interrupt gets generated when a zero length packet is transmitted due to un-availability of data in the TXFifo.
12	W1C	0x0	BbleErrIntrpt BbleErr (Babble Error) interrupt The core generates this interrupt when babble is received for the endpoint.
11	W1C	0x0	PktDrpSts Packet Dropped Status This bit indicates to the application that an ISOC OUT packet has been dropped. This bit does not have an associated mask bit and does not generate an interrupt. Dependency: This bit is valid in non-Scatter/Gather DMA mode when periodic transfer interrupt feature is selected.
10	RO	0x0	reserved
9	W1C	0x0	BNAIntr BNA (Buffer Not Available) Interrupt The core generates this interrupt when the descriptor accessed is not ready for the Core to process, such as Host busy or DMA done Dependency: This bit is valid only when Scatter/Gather DMA mode is enabled.

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Bit	Attr	Reset Value	Description
8	W1C	0x0	TxfifoUndrn FIFO Under-run Applies to IN endpoints only. The core generates this interrupt when it detects a transmit FIFO under-run condition for this endpoint. Dependency: This interrupt is valid only when both of the following conditions are true: Parameter OTG_EN_DED_TX_FIFO==1; Threshold is enabled; OUT Packet Error(OutPktErr). Applies to OUT endpoints only. This interrupt is asserted when the core detects an overflow or a CRC error for an OUT packet. Dependency: This interrupt is valid only when both of the following conditions are true: Parameter OTG_EN_DED_TX_FIFO==1; Threshold is enabled.
7	W1C	0x0	TxFEmp Transmit FIFO Empty This bit is valid only for IN Endpoints.This interrupt is asserted when the TxFIFO for this endpoint is either half or completely empty. The half or completely empty status is determined by the TxFIFO Empty Level bit in the Core AHB Configuration register (GAHBCFG.NPTxFEmpLvl)).
6	W1C	0x0	INEPNakEff IN Endpoint NAK Effective Applies to periodic IN endpoints only. This bit can be cleared when the application clears the IN endpoint NAK by writing to DIEPCTLn.CNAK. This interrupt indicates that the core has sampled the NAK bit set (either by the application or by the core). The interrupt indicates that the IN endpoint NAK bit set by the application has taken effect in the core. This interrupt does not guarantee that a NAK handshake is sent on the USB. A STALL bit takes priority over a NAK bit. This bit is applicable only when the endpoint is enabled. Back-to-Back SETUP Packets Received (Back2BackSETup) Applies to Control OUT endpoints only. This bit indicates that the core has received more than three back-to-back SETUP packets for this particular endpoint.

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Bit	Attr	Reset Value	Description
5	W1C	0x0	INTknEPMis IN Token Received with EP Mismatch Applies to non-periodic IN endpoints only. Indicates that the data in the top of the non-periodic TxFIFO belongs to an endpoint other than the one for which the IN token was received. This interrupt is asserted on the endpoint for which the IN token was received. Status Phase Received For Control Write (StsPhseRcvd) This interrupt is valid only for Control OUT endpoints and only in Scatter Gather DMA mode. This interrupt is generated only after the core has transferred all the data that the host has sent during the data phase of a control write transfer, to the system memory buffer. The interrupt indicates to the application that the host has switched from data phase to the status phase of a Control Write transfer. The application can use this interrupt to ACK or STALL the Status phase, after it has decoded the data phase. This is applicable only in case of Scatter Gather DMA mode.
4	W1C	0x0	INTknTXFEmp IN Token Received When TxFIFO is Empty Indicates that an IN token was received when the associated TxFIFO periodic/nonperiodic) was empty. This interrupt is asserted on the endpoint for which the IN token was received. OUT Token Received When Endpoint Disabled (OUTTknEPdis) Indicates that an OUT token was received when the endpoint was not yet enabled. This interrupt is asserted on the endpoint for which the OUT token was received.

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Bit	Attr	Reset Value	Description
3	W1C	0x0	TimeOUT Timeout Condition In shared TX FIFO mode, applies to non-isochronous IN endpoints only. In dedicated FIFO mode, applies only to Control IN endpoints. In Scatter/Gather DMA mode, the TimeOUT interrupt is not asserted. Indicates that the core has detected a timeout condition on the USB for the last IN token on this endpoint. SETUP Phase Done (SetUp) Applies to control OUT endpoints only. Indicates that the SETUP phase for the control endpoint is complete and no more back-to-back SETUP packets were received for the current control transfer. On this interrupt, the application can decode the received SETUP data packet.
2	W1C	0x0	AHBErr AHB Error Applies to IN and OUT endpoints. This is generated only in Internal DMA mode when there is an AHB error during an AHB read/write. The application can read the corresponding endpoint DMA address register to get the error address.
1	W1C	0x0	EPDisbld Endpoint Disabled Interrupt Applies to IN and OUT endpoints. This bit indicates that the endpoint is disabled per the application's request.

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Bit	Attr	Reset Value	Description
0	W1C	0x0	XferCompl Transfer Completed Interrupt Applies to IN and OUT endpoints. When Scatter/Gather DMA mode is enabled: For IN endpoint this field indicates that the requested data from the descriptor is moved from external system memory to internal FIFO. For OUT endpoint this field indicates that the requested data from the internal FIFO is moved to external system memory. This interrupt is generated only when the corresponding endpoint descriptor is closed, and the IOC bit for the corresponding descriptor is set. When Scatter/Gather DMA mode is disabled, this field indicates that the programmed transfer is complete on the AHB as well as on the USB, for this endpoint.

USBOTG_DIEPTSIZn

Address: Operational Base + offset (0x0910)

Device endpoint n transfer size register

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
30:29	RW	0x0	MC Multi Count Applies to IN endpoints only. For periodic IN endpoints, this field indicates the number of packets that must be transmitted per microframe on the USB. The core uses this field to calculate the data PID for isochronous IN endpoints. 2'b01: 1 packet 2'b10: 2 packets 2'b11: 3 packets For non-periodic IN endpoints, this field is valid only in Internal DMA mode. It specifies the number of packets the core must fetch for an IN endpoint before it switches to the endpoint pointed to by the Next Endpoint field of the Device Endpoint-n Control register (DIEPCTLn.NextEp). Received Data PID (RxDPID) Applies to isochronous OUT endpoints only. This is the data PID received in the last packet for this endpoint. 2'b00: DATA0 2'b01: DATA2 2'b10: DATA1 2'b11: MDATA SETUP Packet Count (SUPCnt).Applies to control OUT Endpoints only.This field specifies the number of back-to-back SETUP data packets the endpoint can receive. 2'b01: 1 packet 2'b10: 2 packets 2'b11: 3 packets

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Bit	Attr	Reset Value	Description
28:19	RW	0x000	PktCnt Packet Count Indicates the total number of USB packets that constitute the Transfer Size amount of data for this endpoint. The power-on value is specified for Width of Packet Counters during coreConsultant configuration (parameter OTG_PACKET_COUNT_WIDTH). IN Endpoints: This field is decremented every time a packet (maximum size or short packet) is read from the TxFIFO. OUT Endpoints: This field is decremented every time a packet (maximum size or short packet) is written to the RxFIFO.
18:0	RW	0x00000	XferSize Transfer Size This field contains the transfer size in bytes for the current endpoint. The power-on value is specified for Width of Transfer Size Counters during configuration (parameter OTG_TRANS_COUNT_WIDTH). The core only interrupts the application after it has exhausted the transfer size amount of data. The transfer size can be set to the maximum packet size of the endpoint, to be interrupted at the end of each packet. IN Endpoints: The core decrements this field every time a packet from the external memory is written to the TxFIFO.OUT Endpoints: The core decrements this field every time a packet is read from the RxFIFO and written to the external memory.

USBOTG_DIEPDMAn

Address: Operational Base + offset (0x0914)

Device endpoint-n DMA address register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMAAddr DMA Address Holds the start address of the external memory for storing or fetching endpoint data. Note: For control endpoints, this field stores control OUT data packets as well as SETUP transaction data packets. When more than three SETUP packets are received back-to-back, the SETUP data packet in the memory is overwritten. This register is incremented on every AHB transaction. The application can give only a DWORD-aligned address. When Scatter/Gather DMA mode is not enabled, the application programs the start address value in this field. When Scatter/Gather DMA mode is enabled, this field indicates the base pointer for the descriptor list.

USBOTG_DTXFSTSn
Address: Operational Base + offset (0x0918)
Device IN endpoint transmit FIFO status register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	INEPTxFSpcAvail IN Endpoint TxFIFO Space Avail Indicates the amount of free space available in the Endpoint TxFIFO.Values are in terms of 32-bit words. 16'h0: Endpoint TxFIFO is full 16'h1: 1 word available 16'h2: 2 words available 16'hn: n words available (where 0 . n . 32,768) 16'h8000: 32,768 words available Others: Reserved

USBOTG_DIEPDMABn
Address: Operational Base + offset (0x091c)
Device endpoint-n DMA buffer address register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	DMABufferAddr DMA Buffer Address Holds the current buffer address.This register is updated as and when the data transfer for the corresponding end point is in progress. This register is present only in Scatter/Gather DMA mode. Otherwise this field is reserved.

USBOTG_DIEPCTLn
Address: Operational Base + offset (0x0920)
Device endpoint-n control register

Bit	Attr	Reset Value	Description
31	R/WSC	0x0	EPEna Endpoint Enable Applies to IN and OUT endpoints. When Scatter/Gather DMA mode is enabled, For IN endpoints this bit indicates that the descriptor structure and data buffer with data ready to transmit is setup. For OUT endpoint it indicates that the descriptor structure and data buffer to receive data is setup. When Scatter/Gather DMA mode is enabled-such as for buffer-pointer based DMA mode: For IN endpoints, this bit indicates that data is ready to be transmitted on the endpoint ; For OUT endpoints, this bit indicates that the application has allocated the memory to start receiving data from the USB. The core clears this bit before setting any of the following interrupts on this endpoint: SETUP Phase Done, Endpoint Disabled, Transfer Completed. Note: For control endpoints in DMA mode, this bit must be set to be able to transfer SETUP data packets in memory.

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Bit	Attr	Reset Value	Description
30	R/WSC	0x0	EPDis Endpoint Disable Applies to IN and OUT endpoints. The application sets this bit to stop transmitting/receiving data on an endpoint, even before the transfer for that endpoint is complete. The application must wait for the Endpoint Disabled interrupt before treating the endpoint as disabled. The core clears this bit before setting the Endpoint Disabled interrupt. The application must set this bit only if Endpoint Enable is already set for this endpoint.
29	WO	0x0	SetD1PID Set DATA1 PID Applies to interrupt/bulk IN and OUT endpoints only.Writing to this field sets the Endpoint Data PID (DPID) field in this register to DATA1.This field is applicable both for Scatter/Gather DMA mode and non-Scatter/Gather DMA mode. Set Odd (micro)frame (SetOddFr). Applies to isochronous IN and OUTendpoints only. Writing to this field sets the Even/Odd (micro)frame (EO_FrNum) field to odd (micro)frame. This field is not applicable for Scatter/Gather DMA mode.
28	WO	0x0	SetD0PID Set DATA0 PID Applies to interrupt/bulk IN and OUT endpoints only.Writing to this field sets the Endpoint Data PID (DPID) field in this register to DATA0.This field is applicable both for Scatter/Gather DMA mode and non-Scatter/Gather DMA mode. In non-Scatter/Gather DMA mode: Set Even (micro)frame (SetEvenFr) Applies to isochronous IN and OUT endpoints only. Writing to this field sets the Even/Odd (micro)frame (EO_FrNum) field to even (micro)frame. When Scatter/Gather DMA mode is enabled, this field is reserved. The frame number in which to send data is in the transmit descriptor structure. The frame in which to receive data is updated in recept descriptor structure.

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Bit	Attr	Reset Value	Description
27	WO	0x0	SNAK Set NAK Applies to IN and OUT endpoints. A write to this bit sets the NAK bit for the endpoint. Using this bit, the application can control the transmission of NAK handshakes on an endpoint. The core can also set this bit for OUT endpoints on a Transfer Completed interrupt, or after a SETUP is received on the endpoint.
26	WO	0x0	CNAK Clear NAK Applies to IN and OUT endpoints. A write to this bit clears the NAK bit for the endpoint.
25:22	RW	0x0	TxFNum TxFIFO Number Shared FIFO Operation:non-periodic endpoints must set this bit to zero. Periodic endpoints must map this to the corresponding Periodic TxFIFO number. 4'h0: Non-Periodic TxFIFO; Others: Specified Periodic TxFIFO number. Note: An interrupt IN endpoint can be configured as a non-periodic endpoint for applications such as mass storage. The core treats an IN endpoint as a non-periodic endpoint if the TxFNum field is set to 0. Otherwise, a separate periodic FIFO must be allocated for an interrupt IN endpoint, and the number of this FIFO must be programmed into the TxFNum field. Configuring an interrupt IN endpoint as a non-periodic endpoint saves the extra periodic FIFO area. Dedicated FIFO Operation:these bits specify the FIFO number associated with this endpoint. Each active IN endpoint must be programmed to a separate FIFO number. This field is valid only for IN endpoints.

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Bit	Attr	Reset Value	Description
21	RW	0x0	Stall STALL Handshake Applies to non-control, non-isochronous IN and OUT endpoints only. The application sets this bit to stall all tokens from the USB host to this endpoint. If a NAK bit, Global Non-periodic IN NAK, or Global OUT NAK is set along with this bit, the STALL bit takes priority. Only the application can clear this bit, never the core. Applies to control endpoints only. The application can only set this bit, and the core clears it, when a SETUP token is received for this endpoint. If a NAK bit, Global Non-periodic IN NAK, or Global OUT NAK is set along with this bit, the STALL bit takes priority. Irrespective of this bit's setting, the core always responds to SETUP data packets with an ACK handshake.
20	RW	0x0	Snp Snoop Mode Applies to OUT endpoints only. This bit configures the endpoint to Snoop mode. In Snoop mode, the core does not check the correctness of OUT packets before transferring them to application memory.
19:18	RW	0x0	EPType Endpoint Type Applies to IN and OUT endpoints. This is the transfer type supported by this logical endpoint. 2'b00: Control 2'b01: Isochronous 2'b10: Bulk 2'b11: Interrupt

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Bit	Attr	Reset Value	Description
17	RO	0x0	NAKSts NAK Status Applies to IN and OUT endpoints. Indicates the following: 1'b0: The core is transmitting non-NAK handshakes based on the FIFO status. 1'b1: The core is transmitting NAK handshakes on this endpoint. When either the application or the core sets this bit: The core stops receiving any data on an OUT endpoint, even if there is space in the RxFIFO to accommodate the incoming packet. For non-isochronous IN endpoints: The core stops transmitting any data on an IN endpoint, even if there data is available in the TxFIFO. For isochronous IN endpoints: The core sends out a zero-length data packet, even if there data is available in the TxFIFO. Irrespective of this bit's setting, the core always responds to SETUP data packets with an ACK handshake.

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Bit	Attr	Reset Value	Description
16	RO	0x0	DPID Endpoint Data PID Applies to interrupt/bulk IN and OUT endpoints only. Contains the PID of the packet to be received or transmitted on this endpoint. The application must program the PID of the first packet to be received or transmitted on this endpoint, after the endpoint is activated. The applications use the SetD1PID and SetD0PID fields of this register to program either DATA0 or DATA1 PID. 1'b0: DATA0 1'b1: DATA1 This field is applicable both for Scatter/Gather DMA mode and non-Scatter/Gather DMA mode.Even/Odd (Micro)Frame (EO_FrNum) In non-Scatter/Gather DMA mode: Applies to isochronous IN and OUT endpoints only.Indicates the (micro) frame number in which the core transmits/receives isochronous data for this endpoint. The application must program the even/odd (micro) frame number in which it intends to transmit/receive isochronous data for this endpoint using the SetEvnFr and SetOddFr fields in this register. 1'b0: Even (micro)frame 1'b1: Odd (micro)frame When Scatter/Gather DMA mode is enabled, this field is reserved. The frame number in which to send data is provided in the transmit descriptor structure. The frame in which data is received is updated in receive descriptor structure.
15	R/WSC	0x0	USBActEP USB Active Endpoint Applies to IN and OUT endpoints.Indicates whether this endpoint is active in the current configuration and interface. The core clears this bit for all endpoints (other than EP 0) after detecting a USB reset. After receiving the SetConfiguration and SetInterface commands, the application must program endpoint registers accordingly and set this bit.

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Bit	Attr	Reset Value	Description
14:11	RW	0x0	NextEp Next Endpoint Applies to non-periodic IN endpoints only.Indicates the endpoint number to be fetched after the data for the current endpoint is fetched. The core can access this field, even when the Endpoint Enable (EPEna) bit is low. This field is not valid in Slave mode operation.Note: This field is valid only for Shared FIFO operations.
10:0	RW	0x000	MPS Maximum Packet Size Applies to IN and OUT endpoints.The application must program this field with the maximum packet size for the current logical endpoint. This value is in bytes.

USBOTG_DOEPCTL0
Address: Operational Base + offset (0x0b00)
Device control OUT endpoint 0 control register

Bit	Attr	Reset Value	Description
31	R/WSC	0x0	EPEna Endpoint Enable When Scatter/Gather DMA mode is enabled, for OUT endpoints this bit indicates that the descriptor structure and data buffer to receive data is setup. When Scatter/Gather DMA mode is disabled?such as for buffer-pointer basedDMA mode)-this bit indicates that the application has allocated the memory to start receiving data from the USB. The core clears this bit before setting any of the following interrupts on this endpoint: SETUP Phase Done, Endpoint Disabled,Transfer Completed. Note: In DMA mode, this bit must be set for the core to transfer SETUP data packets into memory.
30	WO	0x0	EPDis Endpoint Disable The application cannot disable control OUT endpoint 0.
29:28	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
27	WO	0x0	SNAK Set NAK A write to this bit sets the NAK bit for the endpoint. Using this bit, the application can control the transmission of NAK handshakes on an endpoint. The core can also set bit on a Transfer Completed interrupt, or after a SETUP is received on the endpoint.
26	WO	0x0	CNAK Clear NAK A write to this bit clears the NAK bit for the endpoint.
25:22	RO	0x0	reserved
21	R/WSC	0x0	Stall STALL Handshake The application can only set this bit, and the core clears it, when a SETUP token is received for this endpoint. If a NAK bit or Global OUT NAK is set along with this bit, the STALL bit takes priority. Irrespective of this bit's setting, the core always responds to SETUP data packets with an ACK handshake.
20	RW	0x0	Snp Snoop Mode This bit configures the endpoint to Snoop mode. In Snoop mode, the core does not check the correctness of OUT packets before transferring them to application memory.
19:18	RO	0x0	EPType Endpoint Type Hardcoded to 2'b00 for control.
17	RO	0x0	NAKSts NAK Status Indicates the following: 1'b0: The core is transmitting non-NAK handshakes based on the FIFO status. 1'b1: The core is transmitting NAK handshakes on this endpoint. When either the application or the core sets this bit, the core stops receiving data, even if there is space in the RxFIFO to accommodate the incoming packet. Irrespective of this bit setting, the core always responds to SETUP data packets with an ACK handshake.

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Bit	Attr	Reset Value	Description
16	RO	0x0	reserved
15	RO	0x0	USBActEP USB Active Endpoint This bit is always set to 1, indicating that a control endpoint 0 is always active in all configurations and interfaces.
14:2	RO	0x0	reserved
1:0	RO	0x0	MPS Maximum Packet Size The maximum packet size for control OUT endpoint 0 is the same as what is programmed in control IN Endpoint 0. 2'b00: 64 bytes 2'b01: 32 bytes 2'b10: 16 bytes 2'b11: 8 bytes

USBOTG_DOEPINTn
Address: Operational Base + offset (0x0b08)
Device endpoint-n control register

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14	W1C	0x0	NYETIntrpt NYET interrupt The core generates this interrupt when a NYET response is transmitted for a non-isochronous OUT endpoint.
13	W1C	0x0	NAKIntrpt NAK interrupt The core generates this interrupt when a NAK is transmitted or received by the device.In case of isochronous IN endpoints the interrupt gets generated when a zero length packet is transmitted due to un-availability of data in the TXFifo.
12	W1C	0x0	BbleErrIntrpt BbleErr (Babble Error) interrupt The core generates this interrupt when babble is received for the endpoint.

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Bit	Attr	Reset Value	Description
11	W1C	0x0	PktDrpSts Packet Dropped Status This bit indicates to the application that an ISOC OUT packet has been dropped. This bit does not have an associated mask bit and does not generate an interrupt. Dependency: This bit is valid in non-Scatter/Gather DMA mode when periodic transfer interrupt feature is selected.
10	RO	0x0	reserved
9	W1C	0x0	BNAIntr BNA (Buffer Not Available) Interrupt The core generates this interrupt when the descriptor accessed is not ready for the Core to process, such as Host busy or DMA done. Dependency: This bit is valid only when Scatter/Gather DMA mode is enabled.
8	W1C	0x0	TxfifoUndrn FIFO Underrun Applies to IN endpoints only. The core generates this interrupt when it detects a transmit FIFO under-run condition for this endpoint. Dependency: This interrupt is valid only when both of the following conditions are true: Parameter OTG_EN_DED_TX_FIFO==1, Threshold is enabled, OUT Packet Error (OutPktErr). Applies to OUT endpoints only. This interrupt is asserted when the core detects an overflow or a CRC error for an OUT packet. Dependency: This interrupt is valid only when both of the following conditions are true: Parameter OTG_EN_DED_TX_FIFO==1, Thresholding is enabled.
7	W1C	0x0	TxFEmp Transmit FIFO Empty This bit is valid only for IN Endpoints. This interrupt is asserted when the TxFIFO for this endpoint is either half or completely empty. The half or completely empty status is determined by the TxFIFO Empty Level bit in the Core AHB Configuration register(GAHBCFG.NPTxFEmpLvl)).

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Bit	Attr	Reset Value	Description
6	W1C	0x0	INEPNakEff IN Endpoint NAK Effective Applies to periodic IN endpoints only. This bit can be cleared when the application clears the IN endpoint NAK by writing to DIEPCTLn.CNAK. This interrupt indicates that the core has sampled the NAK bit set (either by theapplication or by the core). The interrupt indicates that the IN endpoint NAK bit set by the application has taken effect in the core. This interrupt does not guarantee that a NAK handshake is sent on the USB. A STALL bit takes priority over a NAK bit. This bit is applicable only when the endpoint is enabled. Back-to-Back SETUP Packets Received (Back2BackSETup) Applies to Control OUT endpoints only. This bit indicates that the core has received more than three back-to-back SETUP packets for this particular endpoint.
5	W1C	0x0	INTknEPMis IN Token Received with EP Mismatch Applies to non-periodic IN endpoints only. Indicates that the data in the top of the non-periodic TxFIFO belongs to an endpoint other than the one for which the IN token was received. This interrupt is asserted on the endpoint for which the IN token was received. Status Phase Received For Control Write (StsPhseRcvd) This interrupt is valid only for Control OUT endpoints and only in Scatter Gather DMA mode.
4	W1C	0x0	INTknTXFEmp IN Token Received When TxFIFO is Empty Indicates that an IN token was received when the associated TxFIFO periodic/non-periodic) was empty. This interrupt is asserted on the endpoint for which the IN token was received. OUT Token Received When Endpoint Disabled (OUTTknEPdis) Indicates that an OUT token was received when the endpoint was not yet enabled. This interrupt is asserted on the endpoint for which the OUT token was received.

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Bit	Attr	Reset Value	Description
3	W1C	0x0	TimeOUT Timeout Condition In shared TX FIFO mode, applies to non-isochronous IN endpoints only. In dedicated FIFO mode, applies only to Control IN endpoints. In Scatter/Gather DMA mode, the Timeout interrupt is not asserted.Indicates that the core has detected a timeout condition on the USB for the last IN token on this endpoint. SETUP Phase Done (SetUp). Applies to control OUT endpoints only. Indicates that the SETUP phase for the control endpoint is complete and no more back-to-back SETUP packets were received for the current control transfer. On this interrupt, the application can decode the received SETUP data packet.
2	W1C	0x0	AHBErr AHB Error Applies to IN and OUT endpoints. This is generated only in Internal DMA mode when there is an AHB error during an AHB read/write. The application can read the corresponding endpoint DMA address register to get the error address.
1	W1C	0x0	EPDisbld Endpoint Disabled Interrupt Applies to IN and OUT endpoints. This bit indicates that the endpoint is disabled per the application's request.

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Bit	Attr	Reset Value	Description
0	W1C	0x0	XferCompl Transfer Completed Interrupt Applies to IN and OUT endpoints. When Scatter/Gather DMA mode is enabled For IN endpoint this field indicates that the requested data from the descriptor ismoved from external system memory to internal FIFO. For OUT endpoint this field indicates that the requested data from the internal FIFO is moved to external system memory. This interrupt is generated only when the corresponding endpoint descriptor is closed, and the IOC bit for the corresponding descriptor is set. When Scatter/Gather DMA mode is disabled, this field indicates that the programmed transfer is complete on the AHB as well as on the USB, for this endpoint.

USBOTG_DOEPTSIZn

Address: Operational Base + offset (0x0b10)

Device endpoint n transfer size register

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
30:29	RW	0x0	MC Multi Count Applies to IN endpoints only. For periodic IN endpoints, this field indicates the number of packets that must be transmitted per micro-frame on the USB. The core uses this field to calculate the data PID for isochronous IN endpoints. 2'b01: 1 packet 2'b10: 2 packets 2'b11: 3 packets For non-periodic IN endpoints, this field is valid only in Internal DMA mode. It specifies the number of packets the core must fetch for an IN endpoint before it switches to the endpoint pointed to by the Next Endpoint field of the Device Endpoint-n Control register (DIEPCTLn.NextEp). Received Data PID (RxDPID) Applies to isochronous OUT endpoints only. This is the data PID received in the last packet for this endpoint. 2'b00: DATA0 2'b01: DATA2 2'b10: DATA1 2'b11: MDATA SETUP Packet Count (SUPCnt).Applies to control OUT Endpoints only. This field specifies the number of back-to-back SETUP data packets the endpoint can receive. 2'b01: 1 packet 2'b10: 2 packets 2'b11: 3 packets
28:19	RW	0x000	PktCnt Packet Count Indicates the total number of USB packets that constitute the Transfer Size amount of data for this endpoint. The power-on value is specified for Width of Packet Counters (parameter OTG_PACKET_COUNT_WIDTH). IN Endpoints: This field is decremented every time a packet (maximum size or short packet) is read from the TxFIFO. OUT Endpoints: This field is decremented every time a packet (maximum size orshort packet) is written to the RxFIFO.

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Bit	Attr	Reset Value	Description
18:0	RW	0x00000	XferSize Transfer Size This field contains the transfer size in bytes for the current endpoint. The power-on value is specified for Width of Transfer Size Counters (parameter OTG_TRANS_COUNT_WIDTH). The core only interrupts the application after it has exhausted the transfer size amount of data. The transfer size can be set to the maximum packet size of the endpoint, to be interrupted at the end of each packet. IN Endpoints: The core decrements this field every time a packet from the external memory is written to the TxFIFO. OUT Endpoints: The core decrements this field every time a packet is read from the RxFIFO and written to the external memory.

USBOTG_DOEPDMAn
Address: Operational Base + offset (0x0b14)
Device Endpoint-n DMA Address Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMAAddr DMA Address Holds the start address of the external memory for storing or fetching endpoint data. Note: For control endpoints, this field stores control OUT data packets as well as SETUP transaction data packets. When more than three SETUP packets are received back-to-back, the SETUP data packet in the memory is overwritten. This register is incremented on every AHB transaction. The application can giveonly a DWORD-aligned address. WhenScatter/Gather DMA mode is not enabled, the application programs the start address value in this field. When Scatter/Gather DMA mode is enabled, this field indicates the base pointer for the descriptor list.

USBOTG_DOEPDMABn
Address: Operational Base + offset (0x0b1c)
Device endpoint-n DMA buffer address register

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	DMABufferAddr DMA Buffer Address Holds the current buffer address.This register is updated as and when the data transfer for the corresponding end point is in progress. This register is present only in Scatter/Gather DMA mode. Otherwise this field is reserved.

USBOTG_DOEPCTLn

Address: Operational Base + offset (0x0b20)

Device endpoint-n control register

Bit	Attr	Reset Value	Description
31	R/WSC	0x0	EPEna Endpoint Enable Applies to IN and OUT endpoints. When Scatter/Gather DMA mode is enabled, For IN endpoints this bit indicates that the descriptor structure and data buffer with data ready to transmit is setup. For OUT endpoint it indicates that the descriptor structure and data buffer to receive data is setup. When Scatter/Gather DMA mode is enabled-such as for buffer-pointer based DMA mode: For IN endpoints, this bit indicates that data is ready to be transmitted on the endpoint; For OUT endpoints, this bit indicates that the application has allocated the memory to start receiving data from the USB. The core clears this bit before setting any of the following interrupts on this endpoint: SETUP Phase Done, Endpoint Disabled, Transfer Completed. Note: For control endpoints in DMA mode, this bit must be set to be able to transfer SETUP data packets in memory.

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Bit	Attr	Reset Value	Description
30	R/WSC	0x0	EPDis Endpoint Disable Applies to IN and OUT endpoints. The application sets this bit to stop transmitting/receiving data on an endpoint, even before the transfer for that endpoint is complete. The application must wait for the Endpoint Disabled interrupt before treating the endpoint as disabled. The core clears this bit before setting the Endpoint Disabled interrupt. The application must set this bit only if Endpoint Enable is already set for this endpoint.
29	RO	0x0	SetD1PID Field0001 Abstract Applies to interrupt/bulk IN and OUT endpoints only.Writing to this field sets the Endpoint Data PID (DPID) field in this register to DATA1.This field is applicable both for Scatter/Gather DMA mode and non-Scatter/Gather DMA mode. Set Odd (micro)frame (SetOddFr). Applies to isochronous IN and OUT endpoints only. Writing to this field sets the Even/Odd (micro)frame (EO_FrNum) field to odd (micro)frame. This field is not applicable for Scatter/Gather DMA mode.
28	WO	0x0	SetD0PID Set DATA0 PID Applies to interrupt/bulk IN and OUT endpoints only.Writing to this field sets the Endpoint Data PID (DPID) field in this register to DATA0.This field is applicable both for Scatter/Gather DMA mode and non-Scatter/Gather DMA mode. In non-Scatter/Gather DMA mode: Set Even (micro)frame (SetEvenFr) Applies to isochronous IN and OUT endpoints only. Writing to this field sets the Even/Odd (micro)frame (EO_FrNum) field to even (micro)frame. When Scatter/Gather DMA mode is enabled, this field is reserved. The frame number in which to send data is in the transmit descriptor structure. The frame in which to receive data is updated in recept descriptor structure.

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Bit	Attr	Reset Value	Description
27	WO	0x0	SNAK Set NAK Applies to IN and OUT endpoints. A write to this bit sets the NAK bit for the endpoint. Using this bit, the application can control the transmission of NAK handshakes on an endpoint. The core can also set this bit for OUT endpoints on a Transfer Completed interrupt, or after a SETUP is received on the endpoint.
26	WO	0x0	CNAK Clear NAK Applies to IN and OUT endpoints. A write to this bit clears the NAK bit for the endpoint.
25:22	RW	0x0	TxFNum TxFIFO Number Shared FIFO Operation:non-periodic endpoints must set this bit to zero. Periodic endpoints must map this to the corresponding Periodic TxFIFO number. 4'h0: Non-Periodic TxFIFO; Others: Specified Periodic TxFIFO number. Note: An interrupt IN endpoint can be configured as a non-periodic endpoint for applications such as mass storage. The core treats an IN endpoint as a non-periodic endpoint if the TxFNum field is set to 0. Otherwise, a separate periodic FIFO must be allocated for an interrupt IN endpoint using coreConsultant, and the number of this FIFO must be programmed into the TxFNum field. Configuring an interrupt IN endpoint as a non-periodic endpoint saves the extra periodic FIFO area. Dedicated FIFO Operation:these bits specify the FIFO number associated with this endpoint. Each active IN endpoint must be programmed to a separate FIFO number. This field is valid only for IN endpoints.

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Bit	Attr	Reset Value	Description
21	RW	0x0	Stall STALL Handshake Applies to non-control, non-isochronous IN and OUT endpoints only. The application sets this bit to stall all tokens from the USB host to this endpoint. If a NAK bit, Global Non-periodic IN NAK, or Global OUT NAK is set along with this bit, the STALL bit takes priority. Only the application can clear this bit, never the core. Applies to control endpoints only. The application can only set this bit, and the core clears it, when a SETUP token is received for this endpoint. If a NAK bit, Global Non-periodic IN NAK, or Global OUT NAK is set along with this bit, the STALL bit takes priority. Irrespective of this bit's setting, the core always responds to SETUP data packets with an ACK handshake.
20	RW	0x0	Snp Snoop Mode Applies to OUT endpoints only. This bit configures the endpoint to Snoop mode. In Snoop mode, the core does not check the correctness of OUT packets before transferring them to application memory.
19:18	RW	0x0	EPType Endpoint Type Applies to IN and OUT endpoints. This is the transfer type supported by this logical endpoint. 2'b00: Control 2'b01: Isochronous 2'b10: Bulk 2'b11: Interrupt

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Bit	Attr	Reset Value	Description
17	RO	0x0	NAKSts NAK Status Applies to IN and OUT endpoints. Indicates the following: 1'b0: The core is transmitting non-NAK handshakes based on the FIFO status. 1'b1: The core is transmitting NAK handshakes on this endpoint. When either the application or the core sets this bit: The core stops receiving any data on an OUT endpoint, even if there is space in the RxFIFO to accommodate the incoming packet.
16	RO	0x0	DPID Endpoint Data PID Applies to interrupt/bulk IN and OUT endpoints only. Contains the PID of the packet to be received or transmitted on this endpoint. The application must program the PID of the first packet to be received or transmitted on this endpoint, after the endpoint is activated. The applications use the SetD1PID and SetD0PID fields of this register to program either DATA0 or DATA1 PID. 1'b0: DATA0 1'b1: DATA1 This field is applicable both for Scatter/Gather DMA mode and non-Scatter/Gather DMA mode. Even/Odd (Micro)Frame (EO_FrNum). In non-Scatter/Gather DMA mode: Applies to isochronous IN and OUT endpoints only. Indicates the (micro)frame number in which the core transmits/receives isochronous data for this endpoint. The application must program the even/odd (micro) frame number in which it intends to transmit/receive isochronous data for this endpoint using the SetEvnFr and SetOddFr fields in this register. 1'b0: Even (micro)frame 1'b1: Odd (micro)frame When Scatter/Gather DMA mode is enabled, this field is reserved. The frame number in which to send data is provided in the transmit descriptor structure. The frame in which data is received is updated in receive descriptor structure.

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Bit	Attr	Reset Value	Description
15	R/WSC	0x0	USBActEP USB Active Endpoint Applies to IN and OUT endpoints. Indicates whether this endpoint is active in the current configuration and interface. The core clears this bit for all endpoints (other than EP 0) after detecting a USB reset. After receiving the SetConfiguration and SetInterface commands, the application must program endpoint registers accordingly and set this bit.
14:11	RW	0x0	NextEp Next Endpoint Applies to non-periodic IN endpoints only. Indicates the endpoint number to be fetched after the data for the current endpoint is fetched. The core can access this field, even when the Endpoint Enable (EPEna) bit is low. This field is not valid in Slave mode operation. Note: This field is valid only for Shared FIFO operations.
10:0	RW	0x000	MPS Maximum Packet Size Applies to IN and OUT endpoints. The application must program this field with the maximum packet size for the current logical endpoint. This value is in bytes.

USBOTG_PCGCR

Address: Operational Base + offset (0x0b24)

Power and clock gating control register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:14	RW	0x0802e	RestoreValue Restore Value (Applicable only when Hibernation is enabled (OTG_EN_PWROPT=2). Defines port clock select for different speeds. [31] if_dev_mode - 1: Device mode, core restored as device - 0: Host mode, core restored as host [30:29] p2hd_prt_spd (PRT speed) - 00: HS - 01: FS - 10: LS - 11: Reserved [28:27] p2hd_dev_enum_spd (Device enumerated speed) - 00: HS - 01: FS (30/60 MHz clock) - 10: LS - 11: FS (48 MHz clock) [26:20] mac_dev_addr (MAC device address) Device address [19] mac_termselect (Termination selection) - 0: HS_TERM (Program for High Speed) - 1: FS_TERM (Program for Full Speed) [18:17] mac_xcvrselect (Transceiver select) - 00: HS_XCVR (High Speed) - 01: FS_XCVR (Full Speed) - 10: LS_XCVR (Low Speed) - 11: LFS_XCVR (Reserved) [16] sh2pl_prt_ctl[0] - 1: port_power enabled - 0: port_power disabled [15:14] prt_clk_sel (Refer prt_clk_sel table)
13	RW	0x0	EssRegRestored Essential Register Values Restored (Applicable only when Hibernation is enabled (OTG_EN_PWROPT=2). When a value of 1 is written to this field, it indicates that register values of essential registers have been restored.
12:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9	RO	0x0	RestoreMode Restore Mode (Applicable only when Hibernation is enabled (OTG_EN_PWROPT=2). The application should program this bit to specify the restore mode during RESTORE POINT before programming PCGCCTL.EssRegRest bit is set. Host Mode: 1'b0: Host Initiated Resume, Host Initiated Reset 1'b1: Device Initiated Remote Wake up Device Mode: 1'b0: Device Initiated Remote Wake up 1'b1: Host Initiated Resume, Host Initiated Reset
8	RW	0x0	ResetAfterSusp Reset After Suspend Applicable in Partial power-down mode. In partial power-down mode of operation, this bit needs to be set in host mode before clamp is removed if the host needs to issue reset after suspend. If this bit is not set,then the host issues resume after suspend. This bit is not applicable in device mode and non-partial power-down mode.In Hibernation mode, this bit needs to be set at RESTORE_POINT before PCGCCTL.EssRegRestored is set. In this case, PCGCCTL.restore_mode needs to be set to wait_restore.
7	RO	0x0	L1Suspended Deep Sleep This bit indicates that the PHY is in deep sleep when in L1 state.
6	RO	0x0	PhySleep PHY in Sleep This bit indicates that the PHY is in the Sleep state.
5	RW	0x0	Enbl_L1Gating Enable Sleep Clock Gating When this bit is set, core internal clock gating is enabled in Sleep state if the core cannot assert utmi_l1_suspend_n. When this bit is not set, the PHY clock is not gated in Sleep state.
4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3	RW	0x0	RstPdwnModule Reset Power-Down Modules This bit is valid only in Partial Power-Down mode. The application sets this bit when the power is turned off. The application clears this bit after the power is turned on and the PHY clock is up.
2	RW	0x0	PwrClmp Power Clamp This bit is valid only in Partial Power-Down mode (OTG_EN_PWROPT = 1). The application sets this bit before the power is turned off to clamp the signals between the power-on modules and the power-off modules. The application clears the bit to disable the clamping before the power is turned on.
1	RW	0x0	GateHclk Gate Hclk The application sets this bit to gate hclk to modules other than the AHB Slave and Master and wakeup logic when the USB is suspended or the session is not valid. The application clears this bit when the USB is resumed or a new session starts.
0	RW	0x0	StopPclk Stop Pclk The application sets this bit to stop the PHY clock (phy_clk) when the USB is suspended, the session is not valid, or the device is disconnected. The application clears this bit when the USB is resumed or a new session starts.

22.7 Interface description

Table 22-1 USB OTG 2.0 Interface Description

Module Pin	Direction	Pad Name	pinmux
VSSA	AG	VSSA	-
VCCA3P3	AP	VCCA3P3	-
VCCCORE1P2	AP	VCCCORE1P2	-
USB0PN	A	USB0PN	-
USBRBIAS	A	USBRBIAS	-
USB0PP	A	USB0PP	-
VBUS_0	A	VBUS_0	-

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USB0ID	A	USB0ID	-

***Note**: **A**—Analog pad ; **AP**—Analog power; **AG**—Analog ground ;**DP**—Digital power ;**DG**–Digital ground;*

22.8 Application Note

22.8.1 Resume from Suspend Mode

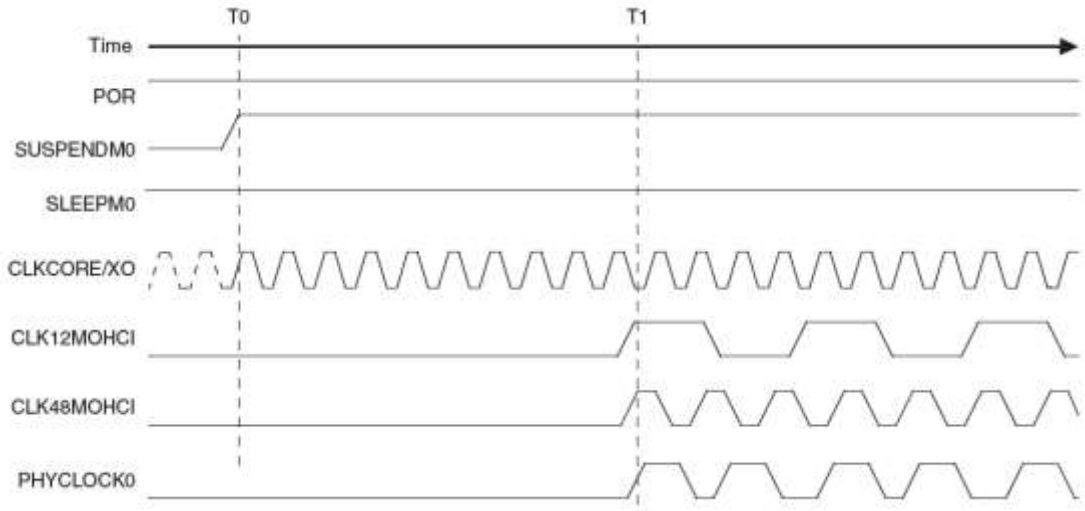

Fig. 22-11 Resume Timing Sequence

When COMMONONN = 1'b1, T1 < T0 + 805 us
When COMMONONN = 1'b0, T1 < T0 + 16 us

22.8.2 Reset a port

Because the assertion of PORTRESETcan occur during data reception or transmission, PORTRESETmust be de-asserted as follows:

❖Reception:

✦FS device: After a minimum of 3 µs of stable SE0 on LINESTATE[1:0]

✦FS/LS host: After a minimum of 8 bit times of J state on LINESTATE[1:0]

✦HS host/device: A minimum of 150 µs after asserting PORTRESET

❖Transmission:

✦FS device: After the controller sets both TXVALID and TXVALIDH to 1'b0, followed by a minimum of 3 µs of stable SE0 on LINESTATE[1:0]

✦FS/LS host: After the controller sets both TXVALID0 and TXVALIDH0 to 1'b0, followed by a minimum of 8 bit times of J state on LINESTATE[1:0]

✦HS host/device: A minimum of 150 µs after the controller sets both TXVALID0 and TXVALIDH0 to 1'b0. The preceding requirements ensure that there is no activity on the USB when PORTRESET0 is de-asserted.

To avoid any data glitches during port reset, the controller must place the USB 2.0 PHY into a safe state. A safe state for host and device ports is defined as follows:

❖Host: The USB 2.0 PHY is set to Non-Driving (OPMODE[1:0] = 2'b01), and the 15-kΩ pull-down resistors are enabled (DPPULLDOWN and DMPULLDOWN = 1'b1).

❖Device: The USB 2.0 PHY is set to Non-Driving (OPMODE[1:0] = 2'b01), which disconnects the 1.5-kΩ resistor from the D+ line.

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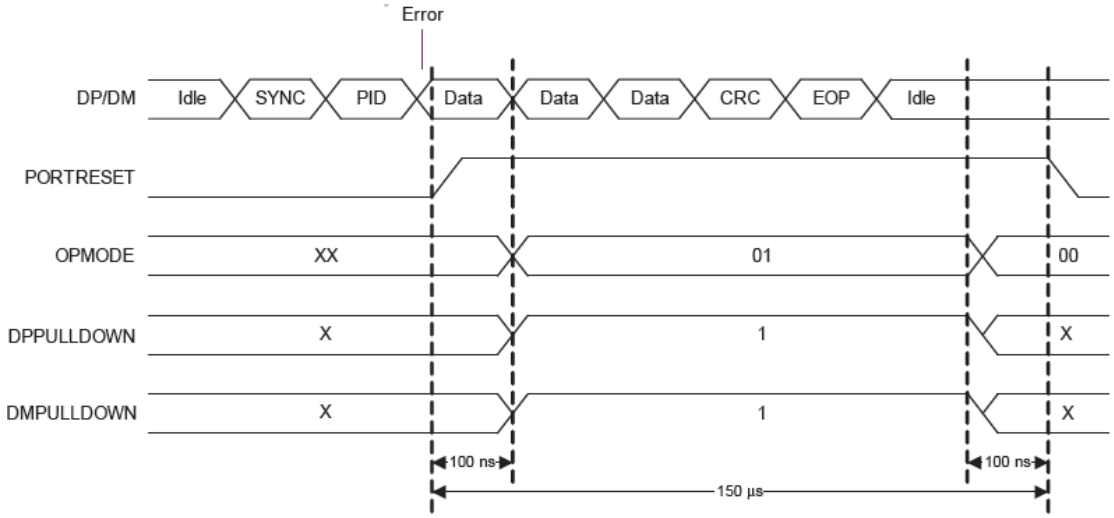

Fig. 22-12 Reset a port when receiving

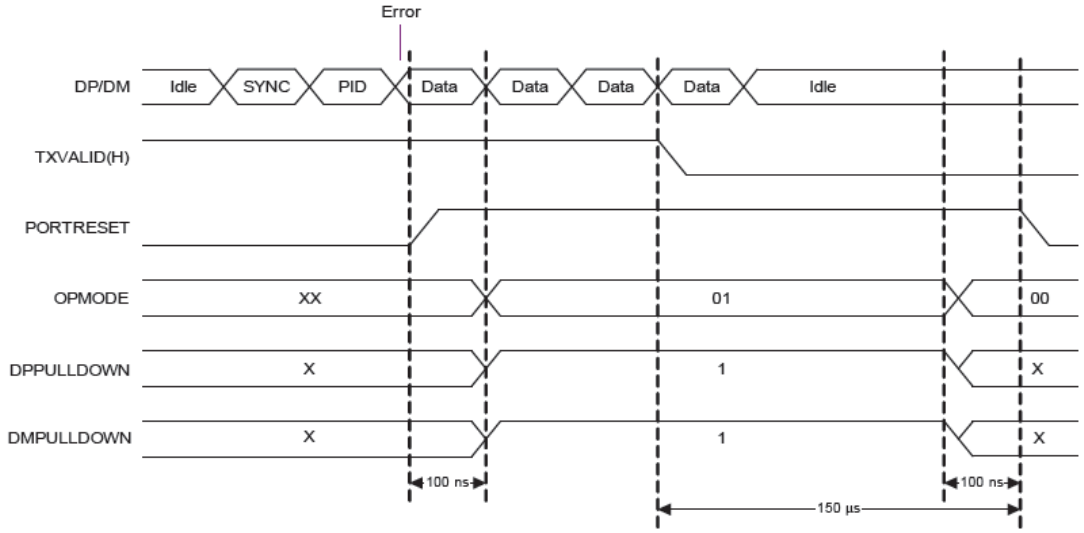

Fig. 22-13 Reset a port when transmitting

CRU_SOFTRST4_CON contains the OTG reset signal. Please refer to "Chapter CRU" for more details.

22.8.3 Relative GRF Registers

GRF_UOC0_CON0 ~ GRF_UOC0_CON2 is OTG PHY register.
GRF_UOC0_CON3 is OTG Controller register.
Please refer to "Chapter GRF" for more details.

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Chapter 23 USB Host 2.0

23.1 Overview

USB HOST2.0 supports Non_OTG Host functions and is fully compliant with USB2.0 specification, andsupporthigh-speed(480Mbps),full-speed(12Mbps),low-speed(1.5Mbps) transfer. It is optimized for point-to-point applications (no hub, direct connection to device).

23.1.1 Features

- Compliant with the USB2.0 Specification
- Operates in Non_OTG Host mode
- Operates in High-Speed, Full-Speed, Low-speed mode
- Support 16 channels in host mode
- Built-in one 840x35 bits FIFO
- Internal DMA with no scatter/gather function
- Support 2 optional controller. 0: EHCI-OHCI controller; 1: DWC controller.

23.2 Block Diagram

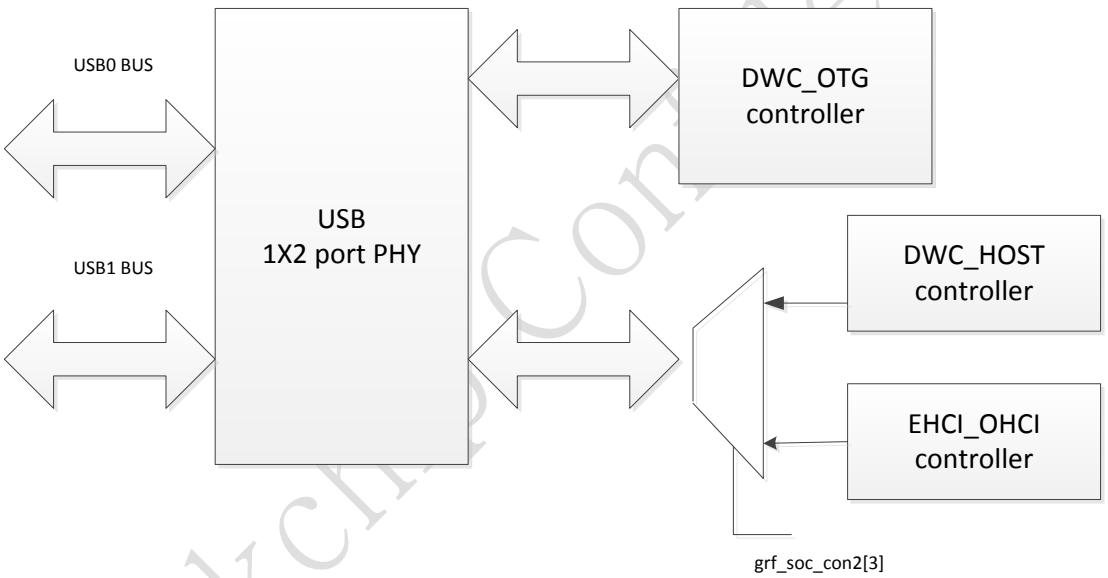

Fig. 23-1 USB HOST 2.0 Architecture

Fig.25-1 shows the architecture of USB. There are 2 separate USB HOST Controllers, one is DWC_HOST, and the other is EHCI_OHCI. The related PHY is USB PHY port1.

23.3 USB Host2.0 Controller

Much the same as USB OTG with no Device Mode supported. See Chapter OTG for more information.
Note：There is another option for host controller: EHCI-OHCI Controller.

23.4 USB Host2.0 PHY

Much the same as USB OTG with no Device Mode supported. See Chapter OTG for more information.
USB Host2.0 PHY doesn’t support UART-DEBUG function.

23.5 Register Description

The Registers are much the same as OTG with no Device-Mode supported. The registers of Device are not available. See Chapter OTG for more information.

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23.6 Interface description

Table 23-1 USB HOST 2.0 Interface Description

Module Pin	Direction	Pad Name	pinmux
VSSA	AG	VSSA	-
VCCA3P3	AP	VCCA3P3	-
VCCCORE1P1	AP	VCCCORE1P1	-
USB1PN	A	USB0PN	-
USBRBIAS	A	USBRBIAS	-
USB1PP	A	USB0PP	-

***Note**: **A**—Analog pad ;**AP**—Analog power; **AG**—Analog ground ;**DP**—Digital power ;**DG**–Digital ground;*

23.7 Application Note

See Chapter OTG for more information.

23.7.1 Reset a port

CRU_SOFTRST4_CON contains HOST reset signal description. Please refer to "Chapter CRU" for more details.

23.7.2 Relative GRF Registers

GRF_UOC0_CON0 ~ GRF_UOC0_CON2 is OTG PHY register.
GRF_UOC0_CON3 is OTG Controller register.
Please refer to "Chapter GRF" for more details.

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Chapter 24 Crypto

24.1 Overview

Crypto is a hardware accelerator of encrypting or decrypting. It supports the most commonly used algorithm: DES/3DES, AES, SHA1, SHA256, MD5 and RSA.
The Crypto supports following features:

- Support AES 128/192/256 bits key mode, ECB/CBC/CTR chain mode, Slave/FIFO mode
- Support DES/3DES (ECB and CBC chain mode) , 3DES (EDE/ EEE key mode), Slave/FIFO mode
- Support SHA1/SHA256/MD5 (with hardware padding) HASH function, FIFO mode only
- Support 160 bit Pseudo Random Number Generator (PRNG)
- Support PKA 512/1024/2048 bit Exp Modulator
- Support up to 150M clock frequency

24.2 Block Diagram

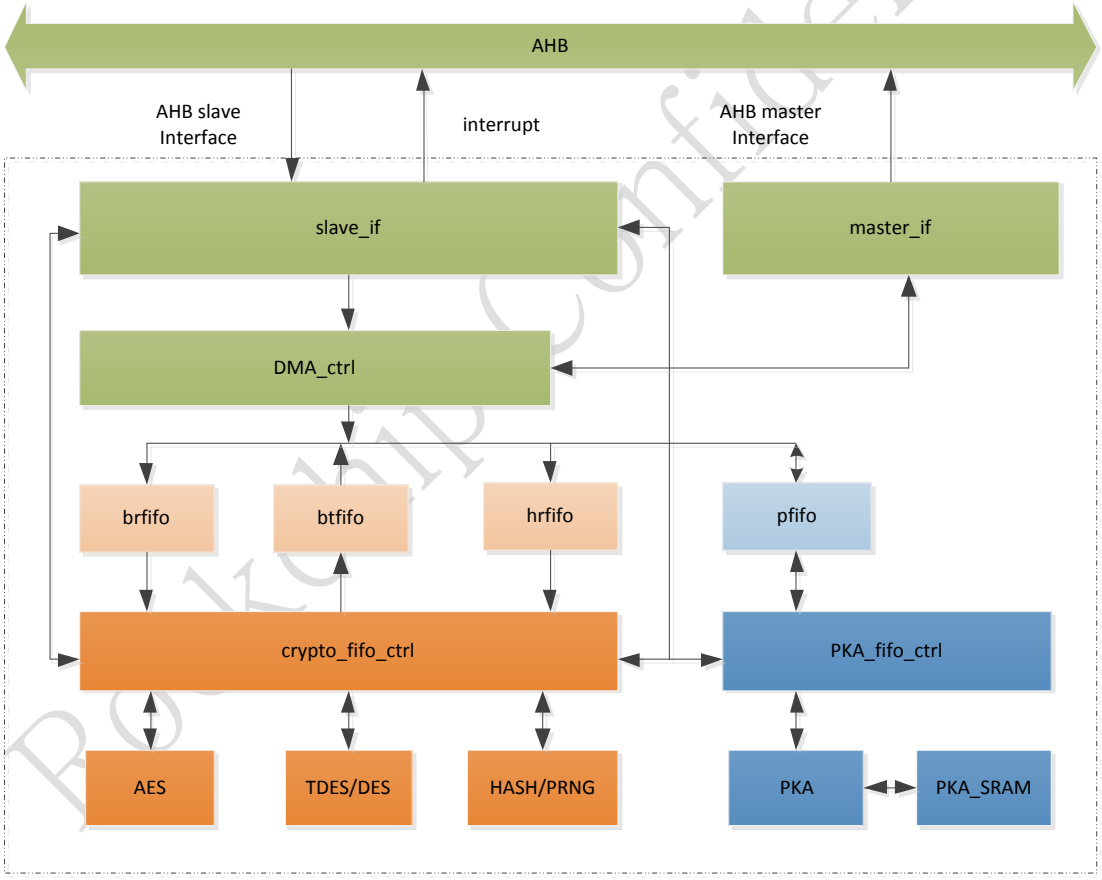

Fig. 24-1 Crypto Architecture

Figure above shows the architecture of Crypto.

24.3 Register description

24.3.1 Register Summary

Name	Offset	Size	Reset Value	Description
CRYPTO_INTSTS	0x0000	W	0x00000000	Interrupt Status Register
CRYPTO_INTENA	0x0004	W	0x00000000	Interrupt Set Register

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Name	Offset	Size	Reset Value	Description
CRYPTO_CTRL	0x0008	W	0x00000000	Control Register
CRYPTO_CONF	0x000c	W	0x00000000	
CRYPTO_BRDMAS	0x0010	W	0x00000000	Block Receiving DMA Start Address Register
CRYPTO_BTDMAS	0x0014	W	0x00000000	Block Transmiting DMA Start Address Register
CRYPTO_BRDMAL	0x0018	W	0x00000000	Block Receiving DMA Length Register
CRYPTO_HRDMAS	0x001c	W	0x00000000	Hash Receiving DMA Start Address Register
CRYPTO_HRDMAL	0x0020	W	0x00000000	Hash Receiving DMA Length Register
CRYPTO_AES_CTRL	0x0080	W	0x00000000	AES Control Register
CRYPTO_AES_STS	0x0084	W	0x00000000	Status Register
CRYPTO_AES_DIN_0	0x0088	W	0x00000000	AES Input Data 0 Register
CRYPTO_AES_DIN_1	0x008c	W	0x00000000	AES Input Data 1 Register
CRYPTO_AES_DIN_2	0x0090	W	0x00000000	AES Input Data 2 Register
CRYPTO_AES_DIN_3	0x0094	W	0x00000000	AES Input Data 3 Register
CRYPTO_AES_DOUT_0	0x0098	W	0x00000000	AES Output Data 0 Register
CRYPTO_AES_DOUT_1	0x009c	W	0x00000000	AES Output Data 1 Register
CRYPTO_AES_DOUT_2	0x00a0	W	0x00000000	AES Output Data 2 Register
CRYPTO_AES_DOUT_3	0x00a4	W	0x00000000	AES Output Data 3 Register
CRYPTO_AES_IV_0	0x00a8	W	0x00000000	AES IV data 0 Register
CRYPTO_AES_IV_1	0x00ac	W	0x00000000	AES IV data 1 Register
CRYPTO_AES_IV_2	0x00b0	W	0x00000000	AES IV data 2 Register
CRYPTO_AES_IV_3	0x00b4	W	0x00000000	AES IV data 3 Register
CRYPTO_AES_KEY_0	0x00b8	W	0x00000000	AES Key data 0 Register
CRYPTO_AES_KEY_1	0x00bc	W	0x00000000	AES Key data 1 Register
CRYPTO_AES_KEY_2	0x00c0	W	0x00000000	AES Key data 2 Register
CRYPTO_AES_KEY_3	0x00c4	W	0x00000000	AES Key data 3 Register
CRYPTO_AES_KEY_4	0x00c8	W	0x00000000	AES Key data 4 Register
CRYPTO_AES_KEY_5	0x00cc	W	0x00000000	AES Key data 5 Register
CRYPTO_AES_KEY_6	0x00d0	W	0x00000000	AES Key data 6 Register
CRYPTO_AES_KEY_7	0x00d4	W	0x00000000	AES Key data 7 Register
CRYPTO_AES_CNT_0	0x00d8	W	0x00000000	AES Input Counter 0 Register
CRYPTO_AES_CNT_1	0x00dc	W	0x00000000	AES Input Counter 1 Register
CRYPTO_AES_CNT_2	0x00e0	W	0x00000000	AES Input Counter 2 Register
CRYPTO_AES_CNT_3	0x00e4	W	0x00000000	AES Input Counter 3 Register
CRYPTO_TDES_CTRL	0x0100	W	0x00000000	TDES Control Register
CRYPTO_TDES_STS	0x0104	W	0x00000000	Status Register
CRYPTO_TDES_DIN_0	0x0108	W	0x00000000	TDES Input Data 0 Register
CRYPTO_TDES_DIN_1	0x010c	W	0x00000000	TDES Input Data 1 Register
CRYPTO_TDES_DOUT_0	0x0110	W	0x00000000	TDES Output Data 0 Register
CRYPTO_TDES_DOUT_1	0x0114	W	0x00000000	TDES Output Data 1 Register
CRYPTO_TDES_IV_0	0x0118	W	0x00000000	TDES IV data 0 Register

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Name	Offset	Size	Reset Value	Description
CRYPTO_TDES_IV_1	0x011c	W	0x00000000	TDES IV data 1 Register
CRYPTO_TDES_KEY1_0	0x0120	W	0x00000000	TDES Key1 data 1 Register
CRYPTO_TDES_KEY1_1	0x0124	W	0x00000000	TDES Key1 data 1 Register
CRYPTO_TDES_KEY2_0	0x0128	W	0x00000000	TDES Key2 data 0 Register
CRYPTO_TDES_KEY2_1	0x012c	W	0x00000000	TDES Key2 data 1 Register
CRYPTO_TDES_KEY3_0	0x0130	W	0x00000000	TDES Key3 data 0 Register
CRYPTO_TDES_KEY3_1	0x0134	W	0x00000000	TDES Key3 data 1 Register
CRYPTO_HASH_CTRL	0x0180	W	0x00000000	Hash Control Register
CRYPTO_HASH_STS	0x0184	W	0x00000000	Hash Status Register
CRYPTO_HASH_MSG_LEN	0x0188	W	0x00000000	Hash Message Len
CRYPTO_HASH_DOUT_0	0x018c	W	0x00000000	Hash Result Register 0
CRYPTO_HASH_DOUT_1	0x0190	W	0x00000000	Hash Result Register 1
CRYPTO_HASH_DOUT_2	0x0194	W	0x00000000	Hash Result Register 2
CRYPTO_HASH_DOUT_3	0x0198	W	0x00000000	Hash Result Register 3
CRYPTO_HASH_DOUT_4	0x019c	W	0x00000000	Hash Result Register 4
CRYPTO_HASH_DOUT_5	0x01a0	W	0x00000000	Hash Result Register 4
CRYPTO_HASH_DOUT_6	0x01a4	W	0x00000000	Hash Result Register 6
CRYPTO_HASH_DOUT_7	0x01a8	W	0x00000000	Hash Result Register 7
CRYPTO_HASH_SEED_0	0x01ac	W	0x00000000	PRNG Seed/HMAC Key Register 0
CRYPTO_HASH_SEED_1	0x01b0	W	0x00000000	PRNG Seed/HMAC Key Register 1
CRYPTO_HASH_SEED_2	0x01b4	W	0x00000000	PRNG Seed/HMAC Key Register 2
CRYPTO_HASH_SEED_3	0x01b8	W	0x00000000	PRNG Seed/HMAC Key Register 3
CRYPTO_HASH_SEED_4	0x01bc	W	0x00000000	PRNG Seed/HMAC Key Register 4
CRYPTO_TRNG_CTRL	0x0200	W	0x00000000	TRNG Control
CRYPTO_TRNG_DOUT_0	0x0204	W	0x00000000	TRNG Output Data 0
CRYPTO_TRNG_DOUT_1	0x0208	W	0x00000000	TRNG Output Data 1
CRYPTO_TRNG_DOUT_2	0x020c	W	0x00000000	TRNG Output Data 2
CRYPTO_TRNG_DOUT_3	0x0210	W	0x00000000	TRNG Output Data 3
CRYPTO_TRNG_DOUT_4	0x0214	W	0x00000000	TRNG Output Data 4
CRYPTO_TRNG_DOUT_5	0x0218	W	0x00000000	TRNG Output Data 5
CRYPTO_TRNG_DOUT_6	0x021c	W	0x00000000	TRNG Output Data 6
CRYPTO_TRNG_DOUT_7	0x0220	W	0x00000000	TRNG Output Data 7
CRYPTO_PKA_CTRL	0x0280	W	0x00000000	PKA Control Register
CRYPTO_PKA_M	0x0400	W	0x00000000	
CRYPTO_PKA_C	0x0500	W	0x00000000	
CRYPTO_PKA_N	0x0600	W	0x00000000	
CRYPTO_PKA_E	0x0700	W	0x00000000	

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

24.3.2 Detail Register Description

CRYPTO_INTSTS

Address: Operational Base + offset (0x0000)

Interrupt Status Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	RW	0x0	PKA_DONE_INT PKA Done Interrupt
4	W1C	0x0	HASH_DONE_INT Hash Done Interrupt
3	W1C	0x0	HRDMA_ERR_INT Specifies the interrupt of hash receiving DMA Error
2	W1C	0x0	HRDMA_DONE_INT Specifies the interrupt of hash receiving DMA DONE
1	W1C	0x0	BCDMA_ERR_INT Specifies the interrupt of block cipher Error
0	W1C	0x0	BCDMA_DONE_INT Specifies the interrupt of block cipher DONE

CRYPTO_INTENA
Address: Operational Base + offset (0x0004)
Interrupt Set Register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	RW	0x0	PKA_DONE_ENA Set the interrupt Enable of PKA done 1'b1: enable 1'b0: disable
4	RW	0x0	HASH_DONE_ENA Set the interrupt Enable of hash done 1'b1: enable 1'b0: disable
3	RW	0x0	HRDMA_ERR_ENA Set the interrupt Enable of hash receiving DMA Error 1'b1: enable 1'b0: disable
2	RW	0x0	HRDMA_DONE_ENA Set the interrupt Enable of hash receiving DMA DONE 1'b1: enable 1'b0: disable
1	RW	0x0	BCDMA_ERR_ENA Set the interrupt Enable of block cipher DMA Error 1'b1: enable 1'b0: disable

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Bit	Attr	Reset Value	Description
0	RW	0x0	BCDMA_DONE_ENA Set the interrupt Enable of block cipher DMA DONE 1'b1: enable 1'b0: disable

CRYPTO_CTRL
Address: Operational Base + offset (0x0008)
Control Register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	Write_Mask
15:10	RO	0x0	reserved
9	RW	0x0	TRNG_FLUSH FLUSH TRNG Software write 1 to start. When finishes, the core will clear it.
8	R/WSC	0x0	TRNG_START Start TRNG Software write 1 to start. When finishes, the core will clear it.
7	R/WSC	0x0	PKA_FLUSH Software write 1 to start Flush Process.The process will clear BRFIFO, BTFIFO, and state machine. Then Software should write 0 to end FLUSH Process
6	RW	0x0	HASH_FLUSH Software write 1 to start Flush Process.The process will clear BRFIFO, BTFIFO, and state machine. Then Software should write 0 to end FLUSH Process
5	RW	0x0	BLOCK_FLUSH Software write 1 to start Flush Process.The process will clear BRFIFO, BTFIFO, and state machine. Then Software should write 0 to end FLUSH Process. It must last for at least 20 cycles to clean registers and FSM
4	R/WSC	0x0	PKA_START Starts/initializes PKA Software write 1 to start. When finishes, the core will clear it.
3	R/WSC	0x0	HASH_START Starts/initializes HASH/PRNG/HMAC Software write 1 to start. When finishes, the core will clear it.

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Bit	Attr	Reset Value	Description
2	R/WSC	0x0	BLOCK_START Starts/initializes Block Cipher Software write 1 to start. When finishes, the core will clear it.
1	R/WSC	0x0	TDES_START Starts/initializes TDES Software write 1 to start. When finishes, the core will clear it.
0	R/WSC	0x0	AES_START Starts/initializes AES Software write 1 to start. When finishes, the core will clear it. Software can also write 0 to clear it.

CRYPTO_CONF
Address: Operational Base + offset (0x000c)

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8	RW	0x0	HR_ADDR_MODE Hash Receive DMA Address Mode 1'b1: fix 1'b0: increment
7	RW	0x0	BT_ADDR_MODE Block Transmit DMA Address Mode 1'b1: fix 1'b0: increment
6	RW	0x0	BR_ADDR_MODE Block Receive DMA Address Mode 1'b1: fix 1'b0: increment
5	RW	0x0	Byteswap_HRFIFO If this bit is high, then the data read from the bus is byte-swapped in a word boundary. If this bit is low (default), then the data is handed over to the FIFO without byte-swap. For little endian bus, this bit should be 1'b1.
4	RW	0x0	Byteswap_BTFIFO If this bit is high, then the data read from the bus is byte-swapped in a word boundary. If this bit is low (default), then the data is handed over to the FIFO without byte-swap. For little endian bus, this bit should be 1'b1.

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Bit	Attr	Reset Value	Description
3	RW	0x0	Byteswap_BRFIFO If this bit is high, then the data read from the bus is byte-swapped in a word boundary. If this bit is low (default), then the data is handed over to the FIFO without byte-swap. For little endian bus, this bit should be 1'b1.
2	RW	0x0	DESSEL Specifies the Destination block cipher of FIFO. AES(=0)/DES(=1)
1:0	RW	0x0	HASHINSEL Specifies the following Data from independent source (0) Data from block cipher input (1) Data from block cipher output (2) Reserved (3)

CRYPTO_BRDMAS
Address: Operational Base + offset (0x0010)
Block Receiving DMA Start Address Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	STARTADDR Specifies the Start Address of DMA The address should be aligned by 32-bit.

CRYPTO_BTDMAS
Address: Operational Base + offset (0x0014)
Block Transmiting DMA Start Address Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	STARTADDR Specifies the Start Address of DMA The address needs to be aligned by 32-bit.

CRYPTO_BRDMAL
Address: Operational Base + offset (0x0018)
Block Receiving DMA Length Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	LENGTH Specifies the Block length of DMA. The length unit is WORD.

CRYPTO_HRDMAS
Address: Operational Base + offset (0x001c)
Hash Receiving DMA Start Address Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	STARTADDR Specifies the Start Address of DMA The address needs to be aligned by 32-bit.

CRYPTO_HRDMAL
Address: Operational Base + offset (0x0020)
Hash Receiving DMA Length Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	LENGTH Specifies the Block length of DMA. The length unit is BYTE.

CRYPTO_AES_CTRL
Address: Operational Base + offset (0x0080)
AES Control Register

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11	RW	0x0	AES_ByteSwap_CNT Change the Big-endian and Little-endian by swapping the byte oder. 0 = Disables Counter data byte swap 1 = Enables Counter data byte swap
10	RW	0x0	AES_ByteSwap_Key Change the Big-endian and Little-endian by swapping the byte oder. 0 = Disables Key byte swap 1 = Enables Key byte swap
9	RW	0x0	AES_ByteSwap_IV Change the Big-endian and Little-endian by swapping the byte oder. 0 = Disables Initial value byte swap 1 = Enables Initial value byte swap
8	RW	0x0	AES_ByteSwap_DO Change the Big-endian and Little-endian by swapping the byte oder. 0 = Disables Output data byte swap 1 = Enables Output data byte swap
7	RW	0x0	AES_ByteSwap_DI Change the Big-endian and Little-endian by swapping the byte oder. 0 = Disables Input data byte swap 1 = Enables Input data byte swap

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Bit	Attr	Reset Value	Description
6	RW	0x0	AES_KeyChange Specifies the AES key change mode selection signal. When the bit is asserted, it will not do key-expansion function to calculate new sub-key. So it is a faster way, when several times of calculation use the same key. But if the keys are different, asserting this bit will have the wrong result. 0 = Key is not changed 1 = Key is changed
5:4	RW	0x0	AES_ChainMode Specifies AES chain mode selection 00 = ECB mode 01 = CBC mode 10 = CTR mode
3:2	RW	0x0	AES_KeySize Specifies the AES key size selection signal 00 : 128-bit key 01 : 192-bit key 10 : 256-bit key
1	RW	0x0	AES_FifoMode Specify AES Fifo Mode 1'b0: Slave mode 1'b1: fifo mode
0	RW	0x0	AES_Enc Specifies the Encryption/ Decryption mode selection signal 0 : Encryption 1 : Decryption

CRYPTO_AES_STS
Address: Operational Base + offset (0x0084)
Status Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	AES_DONE When AES finish, it will be HIGH, And it will not be LOW until it restart . 1: done 0: not done

CRYPTO_AES_DIN_0
Address: Operational Base + offset (0x0088)
AES Input Data 0 Register

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_DIN_0 Specifies AES Input data [127:96].

CRYPTO_AES_DIN_1
Address: Operational Base + offset (0x008c)
AES Input Data 1 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_DIN_1 Specifies AES Input data [95:64].

CRYPTO_AES_DIN_2
Address: Operational Base + offset (0x0090)
AES Input Data 2 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_DIN_2 Specifies AES Input data [63:32]

CRYPTO_AES_DIN_3
Address: Operational Base + offset (0x0094)
AES Input Data 3 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_DIN_3 Specifies AES Input data [31:0]

CRYPTO_AES_DOUT_0
Address: Operational Base + offset (0x0098)
AES Output Data 0 Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	AES_DOUT_0 Specifies AES Output data [127:96].

CRYPTO_AES_DOUT_1
Address: Operational Base + offset (0x009c)
AES Output Data 1 Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	AES_DOUT_1 Specifies the Output data [95:64].

CRYPTO_AES_DOUT_2
Address: Operational Base + offset (0x00a0)
AES Output Data 2 Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	AES_DOUT_2 Specifies AES Output data [63:32].

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CRYPTO_AES_DOUT_3
Address: Operational Base + offset (0x00a4)
AES Output Data 3 Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	AES_DOUT_3 Specifies AES Output data [31:0].

CRYPTO_AES_IV_0
Address: Operational Base + offset (0x00a8)
AES IV data 0 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_IV_0 Specifies AES Initialization vector [127:96]

CRYPTO_AES_IV_1
Address: Operational Base + offset (0x00ac)
AES IV data 1 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_IV_1 Specifies AES Initialization vector [95:64]

CRYPTO_AES_IV_2
Address: Operational Base + offset (0x00b0)
AES IV data 2 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_IV_2 Specifies AES Initialization vector [63:32]

CRYPTO_AES_IV_3
Address: Operational Base + offset (0x00b4)
AES IV data 3 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_IV_3 Specifies AES Initialization vector [31:0]

CRYPTO_AES_KEY_0
Address: Operational Base + offset (0x00b8)
AES Key data 0 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_KEY_0 Specifies AES key data [255:224]

CRYPTO_AES_KEY_1
Address: Operational Base + offset (0x00bc)
AES Key data 1 Register

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_KEY_1 Specifies AES key data [223:192]

CRYPTO_AES_KEY_2
Address: Operational Base + offset (0x00c0)
AES Key data 2 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_KEY_2 Specifies AES key data [191:160]

CRYPTO_AES_KEY_3
Address: Operational Base + offset (0x00c4)
AES Key data 3 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_KEY_3 Specifies AES key data [159:128]

CRYPTO_AES_KEY_4
Address: Operational Base + offset (0x00c8)
AES Key data 4 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_KEY_4 Specifies AES key data [127:96]

CRYPTO_AES_KEY_5
Address: Operational Base + offset (0x00cc)
AES Key data 5 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_KEY_5 Specifies the key data [95:64]

CRYPTO_AES_KEY_6
Address: Operational Base + offset (0x00d0)
AES Key data 6 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_KEY_6 Specifies AES key data [63:32]

CRYPTO_AES_KEY_7
Address: Operational Base + offset (0x00d4)
AES Key data 7 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_KEY_7 Specifies the key data [31:0]

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CRYPTO_AES_CNT_0
Address: Operational Base + offset (0x00d8)
AES Input Counter 0 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_CNT_0 Specifies AES Input Counter [127:96].

CRYPTO_AES_CNT_1
Address: Operational Base + offset (0x00dc)
AES Input Counter 1 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_CNT_1 Specifies AES Input Counter [95:64].

CRYPTO_AES_CNT_2
Address: Operational Base + offset (0x00e0)
AES Input Counter 2 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_CNT_2 Specifies AES Input Counter[63:32]

CRYPTO_AES_CNT_3
Address: Operational Base + offset (0x00e4)
AES Input Counter 3 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_CNT_3 Specifies AES Input Counter [31:0]

CRYPTO_TDES_CTRL
Address: Operational Base + offset (0x0100)
TDES Control Register

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8	RW	0x0	TDES_ByteSwap_Key 0 = Disables Key byte swap 1 = Enables Key byte swap
7	RW	0x0	TDES_ByteSwap_IV 0 = Disables Initial value byte swap 1 = Enables Initial value byte swap
6	RW	0x0	TDES_ByteSwap_DO 0 = Disables Output data byte swap 1 = Enables Output data byte swap
5	RW	0x0	TDES_ByteSwap_DI 0 = Disables Input data byte swap 1 = Enables Input data byte swap

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Bit	Attr	Reset Value	Description
4	RW	0x0	TDES_ChainMode Specifies TDES chain mode selection 0 : ECB mode 1 : CBC mode
3	RW	0x0	TDES_EEE Specifies the TDES key mode selection 1'b0 : EDE 1'b1 : EEE
2	RW	0x0	TDES_Select Specify DES or TDES cipher 1'b0 : DES 1'b1 : TDES
1	RW	0x0	TDES_FifoMode Specify TDES Fifo Mode 1'b0: Slave mode 1'b1: Fifo mode
0	RW	0x0	TDES_Enc Specifies the Encryption/ Decryption mode selection signal 0 : Encryption 1 : Decryption

CRYPTO_TDES_STS
Address: Operational Base + offset (0x0104)
Status Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	TDES_DONE When DES/TDES finishes, it will be HIGH, And it will not be LOW until it restart . 1: done 0: not done

CRYPTO_TDES_DIN_0
Address: Operational Base + offset (0x0108)
TDES Input Data 0 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TDES_DIN_0 Specifies TDES Input data [63:32].

CRYPTO_TDES_DIN_1
Address: Operational Base + offset (0x010c)
TDES Input Data 1 Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TDES_DIN_1 Specifies TDES Input data [31:0].

CRYPTO_TDES_DOUT_0
Address: Operational Base + offset (0x0110)
TDES Output Data 0 Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	TDES_DOUT_0 Specifies TDES Output data [63:32].

CRYPTO_TDES_DOUT_1
Address: Operational Base + offset (0x0114)
TDES Output Data 1 Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	TDES_DOUT_1 Specifies TDES Output data [31:0].

CRYPTO_TDES_IV_0
Address: Operational Base + offset (0x0118)
TDES IV data 0 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TDES_IV_0 Specifies TDES Initialization vector [63:32]

CRYPTO_TDES_IV_1
Address: Operational Base + offset (0x011c)
TDES IV data 1 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TDES_IV_1 Specifies TDES Initialization vector [31:0]

CRYPTO_TDES_KEY1_0
Address: Operational Base + offset (0x0120)
TDES Key1 data 1 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TDES_KEY1_0 Specifies TDES key1 data [63:32]

CRYPTO_TDES_KEY1_1
Address: Operational Base + offset (0x0124)
TDES Key1 data 1 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TDES_KEY1_1 Specifies TDES key1 data [31:0]

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CRYPTO_TDES_KEY2_0
Address: Operational Base + offset (0x0128)
TDES Key2 data 0 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TDES_KEY2_0 Specifies TDES key2 data [63:32]

CRYPTO_TDES_KEY2_1
Address: Operational Base + offset (0x012c)
TDES Key2 data 1 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TDES_KEY_1 Specifies TDES key data [31:0]

CRYPTO_TDES_KEY3_0
Address: Operational Base + offset (0x0130)
TDES Key3 data 0 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TDES_KEY3_0 Specifies TDES key3 data [63:32]

CRYPTO_TDES_KEY3_1
Address: Operational Base + offset (0x0134)
TDES Key3 data 1 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_KEY3_1 Specifies TDES key3 data [31:0]

CRYPTO_HASH_CTRL
Address: Operational Base + offset (0x0180)
Hash Control Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	HASH_SWAP_DO Specifies the Byte swap of data output (hash result) 0 = Does not swap (default) 1 = Swap
2	RW	0x0	reserved
1:0	RW	0x0	Engine_Selection 2'b00: SHA1_HASH 2'b01: MD5_HASH 2'b10: SHA256_HASH 2'b11: PRNG

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CRYPTO_HASH_STS
Address: Operational Base + offset (0x0184)
Hash Status Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	HASH_DONE Hash Done Signal When HASH finishes, it will be HIGH, And it will not be LOW until it restart 1'b1 : done 1'b0 : not done

CRYPTO_HASH_MSG_LEN
Address: Operational Base + offset (0x0188)
Hash Message Len

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	Msg_size Hash total byte.

CRYPTO_HASH_DOUT_0
Address: Operational Base + offset (0x018c)
Hash Result Register 0

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HASH_RESULT_0 Specifies the HASH Result [159:128]

CRYPTO_HASH_DOUT_1
Address: Operational Base + offset (0x0190)
Hash Result Register 1

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HASH_RESULT_1 Specifies the HASH Result [127:96]

CRYPTO_HASH_DOUT_2
Address: Operational Base + offset (0x0194)
Hash Result Register 2

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HASH_RESULT_2 Specifies the HASH Result [95:64]

CRYPTO_HASH_DOUT_3
Address: Operational Base + offset (0x0198)
Hash Result Register 3

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HASH_RESULT_3 Specifies the HASH Result [63:32]

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CRYPTO_HASH_DOUT_4
Address: Operational Base + offset (0x019c)
Hash Result Register 4

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HASH_RESULT_4 Specifies the HASH Result [31:0]

CRYPTO_HASH_DOUT_5
Address: Operational Base + offset (0x01a0)
Hash Result Register 4

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HASH_RESULT_5 Specifies the HASH Result [31:0]

CRYPTO_HASH_DOUT_6
Address: Operational Base + offset (0x01a4)
Hash Result Register 6

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HASH_RESULT_6 Specifies the HASH Result [31:0]

CRYPTO_HASH_DOUT_7
Address: Operational Base + offset (0x01a8)
Hash Result Register 7

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HASH_RESULT_7 Specifies the HASH Result [31:0]

CRYPTO_HASH_SEED_0
Address: Operational Base + offset (0x01ac)
PRNG Seed/HMAC Key Register 0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	HASH_SEED_0 Specifies PRNG Seed/HMAC Key buffer [159:128]

CRYPTO_HASH_SEED_1
Address: Operational Base + offset (0x01b0)
PRNG Seed/HMAC Key Register 1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	HASH_SEED_1 Specifies PRNG Seed/HMAC Key buffer [127:96]

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CRYPTO_HASH_SEED_2
Address: Operational Base + offset (0x01b4)
PRNG Seed/HMAC Key Register 2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	HASH_SEED_2 Specifies PRNG Seed/HMAC Key buffer [95:64]

CRYPTO_HASH_SEED_3
Address: Operational Base + offset (0x01b8)
PRNG Seed/HMAC Key Register 3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	HASH_SEED_3 Specifies PRNG Seed/HMAC Key buffer [63:32]

CRYPTO_HASH_SEED_4
Address: Operational Base + offset (0x01bc)
PRNG Seed/HMAC Key Register 4

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	HASH_SEED_4 Specifies PRNG Seed/HMAC Key buffer [31:0]

CRYPTO_TRNG_CTRL
Address: Operational Base + offset (0x0200)
TRNG Control

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RW	0x0	osc_enable osc_ring enable It controll the running of osc_ring. And it is indepent of clock and flush signal. This mean that it can run even when clock is gating or flush is asserted as long as osc_enable is asserted. Before it is used to get TRNG result , please run osc_ring first to get enough entropy. 1'b1: Enable ; 1'b0: Disable ;
15:0	RW	0x0000	period sample period TRNG use clock_crypto to sample ring osc output, this parameter is specify how many cycles to generate 1 bit random data.

CRYPTO_TRNG_DOUT_0

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Address: Operational Base + offset (0x0204)
TRNG Output Data 0

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	TRNG_DOUT_0

CRYPTO_TRNG_DOUT_1
Address: Operational Base + offset (0x0208)
TRNG Output Data 1

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	TRNG_DOUT_1

CRYPTO_TRNG_DOUT_2
Address: Operational Base + offset (0x020c)
TRNG Output Data 2

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	TRNG_DOUT_2

CRYPTO_TRNG_DOUT_3
Address: Operational Base + offset (0x0210)
TRNG Output Data 3

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	TRNG_DOUT_3

CRYPTO_TRNG_DOUT_4
Address: Operational Base + offset (0x0214)
TRNG Output Data 4

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	TRNG_DOUT_4

CRYPTO_TRNG_DOUT_5
Address: Operational Base + offset (0x0218)
TRNG Output Data 5

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	TRNG_DOUT_5

CRYPTO_TRNG_DOUT_6
Address: Operational Base + offset (0x021c)
TRNG Output Data 6

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	TRNG_DOUT_6

CRYPTO_TRNG_DOUT_7
Address: Operational Base + offset (0x0220)
TRNG Output Data 7

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	TRNG_DOUT_7

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CRYPTO_PKA_CTRL
Address: Operational Base + offset (0x0280)
PKA Control Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x0	block_size PKA Size It specifies the bits of N in PKA calculation. 2'b00: 512 bit 2'b01: 1024 bit 2'b10: 2048 bit

CRYPTO_PKA_M
Address: Operational Base + offset (0x0400)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	m PKA input or output data. PKA result = (M ^ E) mod N. When it finishes, the result data is in M position. Start from PKA_M base address, and may contain 512/1024/2048 bits data.

CRYPTO_PKA_C
Address: Operational Base + offset (0x0500)

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	c PKA pre-calculate data, C = 2 ^(2n+2) mod N

CRYPTO_PKA_N
Address: Operational Base + offset (0x0600)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	n PKA modular

CRYPTO_PKA_E
Address: Operational Base + offset (0x0700)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	e PKA exponent.

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24.4 Application Note

24.4.1 Reset a port

CRU_SOFTRST3_CON. crypto_srstn_req is used to do a soft reset to crypto . Please refer to "Chapter CRU" for more details.

24.4.2 Overall Performance

Use CLKSEL24_CON.crypto_div_con to select crypto frequency: $F_{crypto} = F_{aclk} / (div + 1)$. Make sure F_{crypto} do not exceed 150M.
The performance of crypto FIFO mode is list below.

algorithm	cycle	block size	frequency	throughput rate
DES	17	64 bit	100M	376 M bps
TDES	51	64 bit	100M	125 M bps
AES	11/13/15	128 bit	100M	1160/984/853Mbps
SHA-1	81	512 bit	100M	632 Mbps
MD5	65	512 bit	100M	787 Mbps

24.4.3 Usage

1. Symmetric algorithm
DES/3DES, AES are symmetric algorithms. There are two ways of using these algorithms: Slave mode and FIFO mode.
In Slave mode, you can calculate 1 block size of data by starting the engine. Take AES-128 for example, you should

- Program Input 128 bit Data to AES_DIN_0~AES_DIN_3
- Program Input 128 bit Key to AES_KEY_0~AES_KEY_3
- Program control mode to AES_CTRL to run in different mode
- Program CTRL.AES_START to run
- wait AES_STS.DONE High
- Read AES_DOUT_0 ~ AES_DOUT_3 to get result.

In FIFO mode,

- Program the source address to BRDMAS, the destination address to BTDMAS, program the length in word unit to BRDMAL;
- Program Input 128 bit Key to AES_KEY_0~AES_KEY_3;
- Program control mode to AES_CTRL to run in different mode;
- Program INTENA to enable interrupt;
- Program CTRL.BLOCK_START to start;
- wait interrupt asserted;
- Program INTSTS to clear interrupt status;
- Read the destination address which BTDMA points to.

FIFO mode get much higher throughput rate.

2. HASH
HASH is used to get digest of data. Only support FIFO mode.
There are three source: (1) hr_fifo; (2) br_fifo; (3) bt_fifo.
Take hr_fifo for example

- Program CTRL.HASH_FLUSH 1'b1 to clear, wait several cycle (>= 10 cycles) , and Program CTRL.HASH_FLUSH 1'b0
- Program data source address to HRDMAS, program 1 time data length in word unit to HRDMAL, program total length in byte unit to HASH_MSG_LEN
- Program HASH_CTRL to choose algorithm, for example SHA-256
- Program INTENA to enable interrupt;
- Program CTRL.HASH_START 1'b1 to start;

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- Wait interrupt asserted; Only if HRDMAL length meets can this interrupt be asserted
- If you have another section of data to hash, then go to (2), HASH_MSG_LEN need not to be programmed;
- else go to (8)
- wait HASH_STS.done asserted. Only if Hash_MSG_LEN meet can this bit status register asserted.
- Read HASH_DOUT_0 – HASH_DOUT_7 to get result.

3. Asymmetric Algorithm
Support 512/1024/2048 bit RSA calculation. It provide the big number calculation. Result = M^E mod N
Program CTRL.PKA_FLUSH 1'b1 to flush RSA module;
Wait CTRL.PKA_FLUSH to be LOW. It is self-cleared;
Program input_data(M) to PKA_M; Program pre_caculated C to PKA_C; Program Key(N) to PKA_N; Program Key(E) to PKA_E. C = $2^{(2n+2)}$ mod N. n is the required bit of N. For example 2048 bit N, n = 2048;
Program PKA_CTRL to select RSA size: 512/1024/2048
Program INTENA to enable interrupt;
Program CTRL.PKA_START to start;
Wait interrupt asserted.
Read PKA_M to get results.

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Chapter 25 GraphicsProcessing Unit (GPU)

25.1 Overview

The Rogue core is the sixth generation of programmable PowerVR™ GPU cores. The Rogue architecture builds on the PowerVR™ SGX™ architecture, adding higher compute density and a baseline of Microsoft® DirectX™ features and OpenGL® (OGL) features to handheld variants of the core.
The PowerVR Rogue architecture is scalable and targets market segments from mainstream mobile devices to high-end desktop graphics.
GPU supports the following features:

- Base handheld architecture fully OpenGL ES 1.1/2.0/3.0/3.1, OpenGL 3.2 and DirectX 9.3, OpenCL1.2 compliant
- Support for pull-model attribute evaluation
- Tile-based deferred rendering architecture with concurrent processing of multiple tiles
- Multi-threaded Unified Shading Cluster (USC) engine incorporating pixel shader, vertex shader, and GP-GPU(compute shader) functionality
- USC incorporates an ALU architecture with high SIMD efficiency
- Fully virtualized memory addressing (up to 1 TB address space), supporting unified memory architecture
- Fine-grained task switching, workload balancing and power management
- Advanced DMA driven operation for minimum host CPU interaction
- Dedicated texture load DMA engine and basic ROP engine
- Programmable high quality image anti-aliasing
- Lossless texture and parameter data compression (only supported by some variants)
- Dedicated processor for Rogue core firmware execution

25.2 Block Diagram

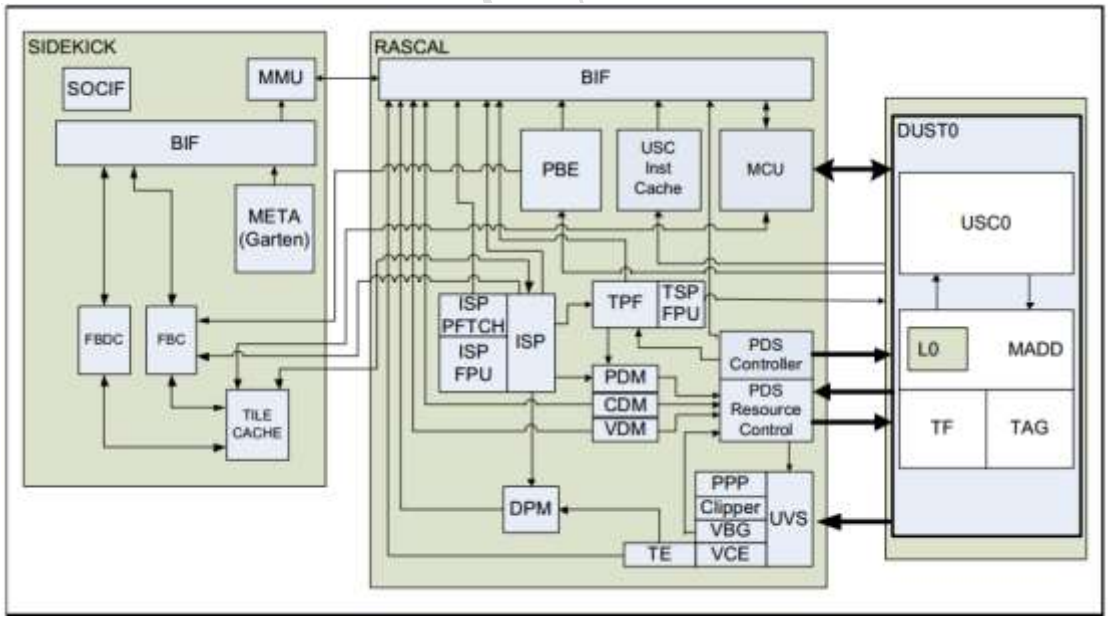

Fig. 25-1 GPU Block Diagram

25.3 Interface Description

Table 25-1GPU jtag interface description

Module Pin	Dir	Pad Name	IOMUX Setting
gpujtag_tck	I/O	IO_SCvcc18v_I2C2SENSORsda_GPUJTAGtck_GPIO1830gpio3d7	GRF_GPIO3D_IOMUX[15:14]=11
gpujtag_t	I/O	IO_SCvcc33v_I2C2SENSORscl_GPUJTAGtrstn	PMUGRF_GPIO0B_IOMUX

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rstn		_GPIO1830gpio0b1	[3:2]=11
gpujtag_tms	O	IO_SCrst_SPI2rxd_GPUJTAGtms_GPIO1830gpio0b2	GRF_GPIO0B_IOMUX[5:4]=11
gpujtag_tdi	I	IO_SCclk_SPI2txd_GPUJTAGtdi_GPIO1830gpio0b3	GRF_GPIO0B_IOMUX[7:6]=11
gpujtag_tdo	O	IO_Scio_SPI2clk_GPUJTAGtdo_GPIO1830gpio0b4	GRF_GPIO0B_IOMUX[9:8]=11

25.4 Application Notes

25.4.1 Clock and Reset control

We can conifig CRU_CLKSEL14_CON[7:6] to select GPU core clock source between CODEC PLL and GENERAL PLL, and then set CRU_CLKSEL14_CON[4:0] to divide clock source from 1:1 to 1:32.
We can conifig CRU_CLKSEL14_CON[14] to select GPU AXI clock source between CODEC PLL and GENERAL PLL, and then set CRU_CLKSEL14_CON[12:8] to divide clock source from 1:1 to 1:32.
We can config CRU_CLKGATE18_CON[0] to gate the gpu config clock, and config CRU_CLKGATE18_CON[1] to gate the gpu AXI clock, and config CRU_CLKGATE18_CON[2] to gate the gpu core clock. However, we strongly recommend that this internal gating way is not adopted because of some design issue of IP itself. Instead, we will implement this function by PMU.

25.4.2 GPU interrupt and idle

The GPU generates only one Rogue hardware interrupt : The GPU_IRQ (GIC interrupt 40), such as 3D phase terminate event.
A GPU_IDLE signal is generated when GPU is not working.

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Chapter 26 VPU_Combo

26.1 Overview

VPU_Combo is composed by the H.265 (HEVC) decoder and the H.264 encoder/decoder to realize the high quality video decoding. VPU_Combo is connected to the AHB bus through an AHB slave and the AXI bus through an AXI master. The register configuration is fed into the VPU_Combo through the AHB slave interface while the stream data is transacted between DDR and VPU_Combo through the AXI master interface.

To reduce the area, H.265 decoder and H.264 encoder/decoder not only share several pieces of the internal memories, but also share the bus master and slave interfaces. Therefore VPU_Combo has no any possibility to have the H.265 video decoding and H.264 video encoding/decoding to work simultaneously

In order to improve large data transaction performance, VPU embeds MMU (memory management unit) and supports the cacheable bus operation.

VPU_Combo supports the next-generation video coding standard HEVC (High Efficiency Video Coding, aka H.265) full-HD decoding up to 60fps. With HEVC standard, the data compression ratio can be doubled compared to H.264/MEPG-4 at the same video quality or alternatively to provide substantially improved video at the same bit rate.

26.1.1 Features

- Supports HEVC Main Profile up to Level 5.1 High Tier: 4096x2304 @60 fps
  - MMU embedded
  - Supports frame timeout interrupt , frame finish interrupt and bitstream error interrupt
  - Supports RLC write mode, RLC mode and Normal Mode
- Supports decoding of the following standards
  - Error detection and concealment support for all video formats
  - Output data structure after decoder is YCbCr 4:2:0 semi-planar to have more efficient bus usage
  - H.264 up to HP level 5.1 including Baseline Profile, Main Profile and High Profile
  - SVC: base layer only for Baseline/High Profile
  - MVC: Stereo High
  - MPEG-4: Simple Profile up to Level 6; Advanced Profile up to Level 5
  - MPEG-2: Main Profile up to High Level
  - JPEG: Baseline interleaved
  - VP8: version 2(webM)
  - For H.264, Image cropping not supported
- Built-in post processor in H.264 decoder supports:
  - Stand-alone mode: rotation, RGB conversion, scaling, dithering
  - Pipe-ling mode:, RGB conversion, scaling, dithering
- Supports encoding of the following standards:
  - H.264: up to HP level 4.1
  - JPEG: Baseline (DCT sequential)
- Built-in pre-processor in H.264 encoder supports:
  - Cropping, rotation, YCbCr conversion

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26.2 Block Diagram

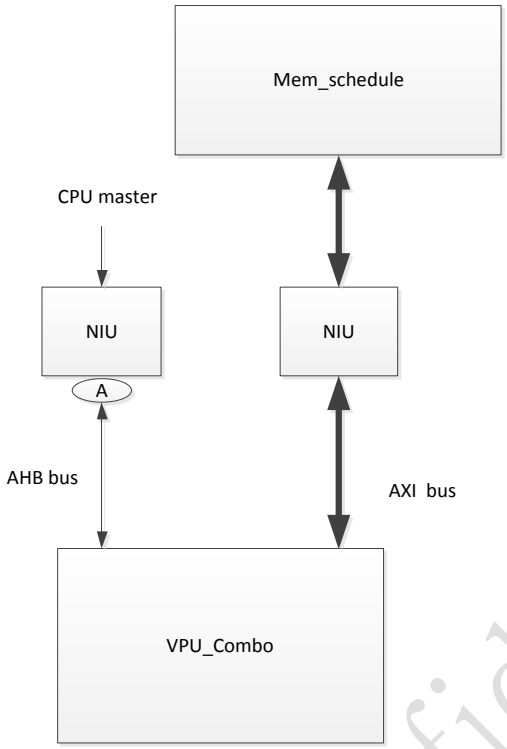

Fig. 26-1VPU Combo in SOC

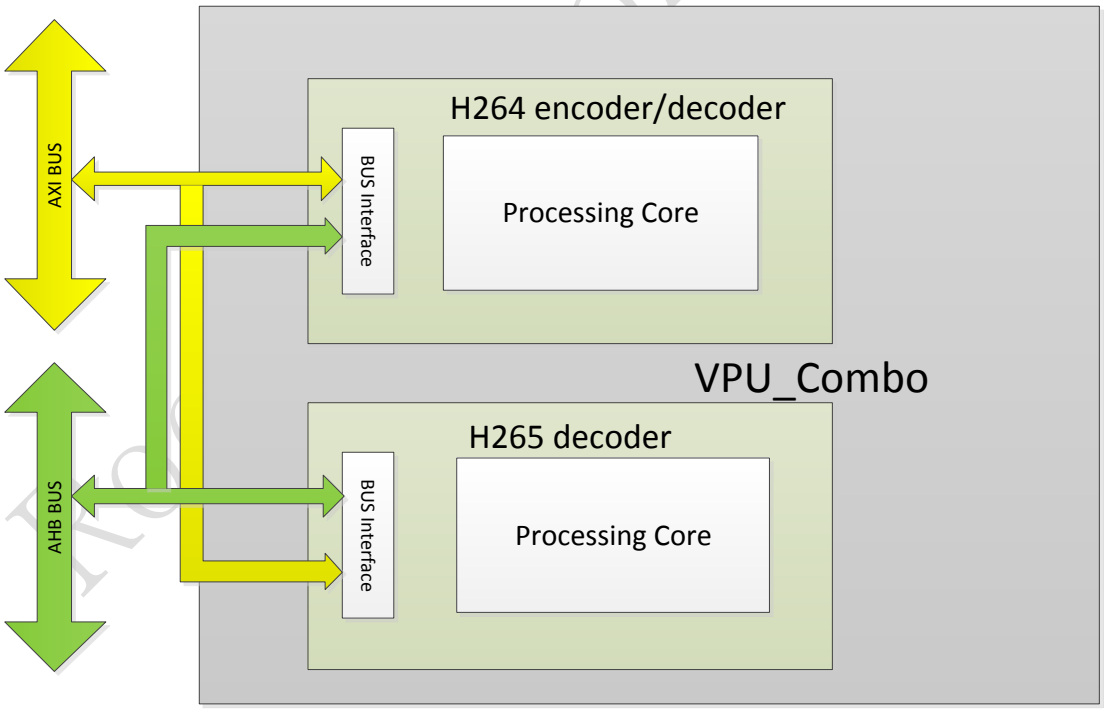

Fig. 26-2 VPU Combo Block Diagram

As shown in the figures above, CPU accesses to HEVC register bank through 32-bit AHB bus. Bitstream and compressed video data are fed into HEVC core though 128-bit AXI read channel, and after several steps of decoding process, decoded pictures are transferred to designated location in the DDR through 64-bit AXI write channel.

CPU accesses to H264 encoder/decoder register bank through 32-bit AHB bus. Video data are

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fed into H.264 core though 64-bit AXI read channel, and after several steps of decoding process, process results are transferred to designated location in the DDR through 64-bit AXI write channel.

26.3 Function Description

26.3.1 HEVC Standard

High Efficiency Video Coding (HEVC) is a video compression standard, a successor to H.264/MPEG-4 AVC (Advanced Video Coding), that was jointly developed by the ISO/IEC Moving Picture Experts Group (MPEG) and ITU-T Video Coding Experts Group (VCEG) as ISO/IEC 23008-2 MPEG-H Part 2 and ITU-T H.265. MPEG and VCEG established a Joint Collaborative Team on Video Coding (JCT-VC) to develop the HEVC standard.
HEVC was designed to substantially improve coding efficiency compared to H.264/MPEG-4 AVC HP, i.e. to reduce bitrate requirements by half with comparable image quality, at the expense of increased computational complexity. HEVC was designed with the goal of allowing video content to have a data compression ratio of up to 1000:1. Depending on the application requirements HEVC encoders can trade off computational complexity, compression rate, robustness to errors, and encoding delay time.Two of the key features where HEVC was improved compared to H.264/MPEG-4 AVC was support for higher resolution video and improved parallel processing methods.

26.3.2 HEVC Coding Tools

1. Coding tree unit

HEVC replaces macroblocks, which were used with previous standards, with Coding Tree Units (CTUs) which can use a larger block structures of up to 64x64 pixels and can better sub-partition the picture into variable sized structures. HEVC initially divides the picture into CTUs which can be 64x64, 32x32, or 16x16 with a larger pixel block size usually increasing the coding efficiency.

2. Parallel processing tools

Tiles allow for the picture to be divided up into a grid of rectangular regions that can independently be decoded/encoded and the main purpose of tiles is to allow for parallel processing.Tiles can be independently decoded and can even allow for random access to specific regions of a picture in a video stream.

Wavefront parallel processing (WPP) is when a slice is divided into rows of CTUs in which the first row is decoded normally but each additional row requires that decisions be made in the previous row. WPP has the entropy encoder use information from the preceding row of CTUs and allows for a method of parallel processing that may allow for better compression than tiles.

Tiles and WPP are allowed but are optional.If tiles are present they must be at least 64 pixels high and 256 pixels wide with a level specific limit on the number of tiles allowed.

Slices can for the most part be decoded independently from each other with the main purpose of tiles being re-synchronization in case of data loss in the video stream. Slices can be defined as self-contained in that prediction is not made across slice boundaries. When in-loop filtering is done on a picture though information across slice boundaries may be required.[1] Slices are CTUs decoded in the order of the raster scan and different coding types can be used for slices such as I types, P types, or B types.

Dependent slices can allow for data related to tiles or WPP to be accessed more quickly by the system than if the entire slice had to be decoded.[1] The main purpose of dependent slices is to allow for low delay video encoding due to its lower latency.

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3. Entropy coding

HEVC uses a context-adaptive binary arithmetic coding (CABAC) algorithm that is fundamentally similar to CABAC in H.264/MPEG-4 AVC. CABAC is the only entropy encoder method that is allowed in HEVC while there are two entropy encoder methods allowed by H.264/MPEG-4 AVC. CABAC and the entropy coding of transform coefficients in HEVC were designed for a higher throughput than H.264/MPEG-4 AVC. For instance, the number of context coded bins have been reduced by 8x and the CABAC bypass-mode has been improved in terms of its design to increase throughput. Another improvement with HEVC is that the dependencies between the coded data has been changed to further increase throughput. Context modeling in HEVC has also been improved so that CABAC can better select a context that increases efficiency when compared to H.264/MPEG-4 AVC.

4. Intra prediction

HEVC specifies 33 directional modes for intra prediction compared to the 8 directional modes for intra prediction specified by H.264/MPEG-4 AVC. HEVC also specifies planar and DC intra prediction modes.[1] The intra prediction modes use data from neighboring prediction blocks that have been previously decoded.

5. Motion compensation

For the interpolation of fractional luma sample positions HEVC uses separable application of one-dimensional half-sample interpolation with an 8-tap filter or quarter-sample interpolation with a 7-tap filter while, in comparison, H.264/MPEG-4 AVC uses a two-stage process that first derives values at half-sample positions using separable one-dimensional 6-tap interpolation followed by integer rounding and then applies linear interpolation between values at nearby half-sample positions to generate values at quarter-sample positions.[1] HEVC has improved precision due to the longer interpolation filter and the elimination of the intermediate rounding error. For 4:2:0 video, the chroma samples are interpolated with separable one-dimensional 4-tap filtering to generate eighth-sample precision, while in comparison H.264/MPEG-4 AVC uses only a 2-tap bilinear filter (also with eighth-sample precision).

As in H.264/MPEG-4 AVC, weighted prediction in HEVC can be used either with uni-prediction (in which a single prediction value is used) or bi-prediction (in which the prediction values from two prediction blocks are combined).

6. Motion vector prediction

HEVC defines a signed 16-bit range for both horizontal and vertical motion vectors (MVs). This was added to HEVC at the July 2012 HEVC meeting with the mvLX variables. HEVC horizontal/vertical MVs have a range of −32768 to 32767 which given the quarter pixel precision used by HEVC allows for a MV range of −8192 to 8191.75 luma samples. This compares to H.264/MPEG-4 AVC which allows for a horizontal MV range of −2048 to 2047.75 luma samples and a vertical MV range of −512 to 511.75 luma samples.

HEVC allows for two MV modes which are Advanced Motion Vector Prediction (AMVP) and merge mode. AMVP uses data from the reference picture and can also use data from adjacent prediction blocks. The merge mode allows for the MVs to be inherited from neighboring prediction blocks. Merge mode in HEVC is similar to "skipped" and "direct" motion inference modes in H.264/MPEG-4 AVC but with two improvements. The first improvement is that HEVC uses index information to select one of several available candidates. The second improvement is that HEVC uses information from the reference picture list and reference picture index.

7. Inverse transforms

HEVC specifies four transform units (TUs) sizes of 4x4, 8x8, 16x16, and 32x32 to code the

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prediction residual. A CTB may be recursively partitioned into 4 or more TUs.[1] TUs use integer basis functions that are similar to the discrete cosine transform (DCT). In addition 4x4 luma transform blocks that belong to an intra coded region are transformed using an integer transform that is derived from discrete sine transform (DST). This provides a 1% bit rate reduction but was restricted to 4x4 luma transform blocks due to marginal benefits for the other transform cases. Chroma uses the same TU sizes as luma so there is no 2x2 transform for chroma.

8. Loop filters

HEVC specifies two loop filters that are applied sequentially, with the deblocking filter (DBF) applied first and the sample adaptive offset (SAO) filter applied afterwards. Both loop filters are applied in the inter-picture prediction loop, i.e. the filtered image is stored in the decoded picture buffer (DPB) as a reference for inter-picture prediction.

8.1 Deblocking filter

The DBF is similar to the one used by H.264/MPEG-4 AVC but with a simpler design and better support for parallel processing.[1] In HEVC the DBF only applies to a 8x8 sample grid while with H.264/MPEG-4 AVC the DBF applies to a 4x4 sample grid. DBF uses a 8x8 sample grid since it causes no noticeable degradation and significantly improves parallel processing because the DBF no longer causes cascading interactions with other operations. Another change is that HEVC only allows for three DBF strengths of 0 to 2. HEVC also requires that the DBF first apply horizontal filtering for vertical edges to the picture and only after that does it apply vertical filtering for horizontal edges to the picture. This allows for multiple parallel threads to be used for the DBF.

8.2 Sample adaptive offset

The SAO filter is applied after the DBF and is designed to allow for better reconstruction of the original signal amplitudes by applying offsets stored in a lookup table in the bitstream. Per CTB the SAO filter can be disabled or applied in one of two modes: edge offset mode or band offset mode. The edge offset mode operates by comparing the value of a sample to two of its eight neighbors using one of four directional gradient patterns. Based on a comparison with these two neighbors, the sample is classified into one of five categories: minimum, maximum, an edge with the sample having the lower value, an edge with the sample having the higher value, or monotonic. For each of the first four categories an offset is applied. The band offset mode applies an offset based on the amplitude of a single sample. A sample is categorized by its amplitude into one of 32 bands (histogram bins). Offsets are specified for four consecutive of the 32 bands, because in flat areas which are prone to banding artifacts, sample amplitudes tend to be clustered in a small range.[1][135] The SAO filter was designed to increase picture quality, reduce banding artifacts, and reduce ringing artifacts.

26.3.3 MMU

The MMU divides memory into 4KB pages, where each page can be individually configured. For each page the following parameters are specified:

- Address translation of virtual memory, this enables the processor to work using address that differ from the physical address in the memory system.
- The permitted types of accesses to that page. Each page can permit read, write, both, or none.

The MMU use 2-level page table structure:
1. The first level, the page directory consists of 1024 directory table entries(DTEs), each pointing to a page table.
2. The second level, the page table consists of 1024 page table entries(PTEs), each pointing to a page in memory.

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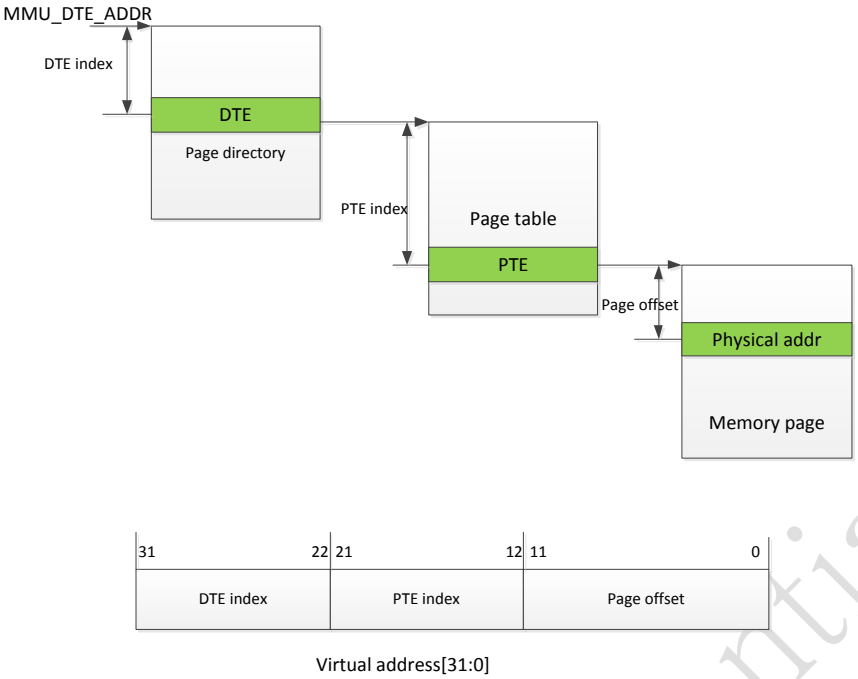

Fig. 26-3 structure of two-level page table

26.3.4 HEVC Working Mode

There are there working modes to be selected for HEVC decoder: RLC Mode, RLC Write Mode, Normal Mode.
The key differences among three working modes are whether CABAC module and Post-CABAC module are involved into the hardware decoding process.
For RLC mode, CABAC are bypassed and the input bitstream to the Post-CABAC module should be already decoded.
For RLC write mode, the decoded results by CABAC are output to the DDR, and the following decoding processes are stopped.
As for the normalmode , all the modules are involved into the decoding process, and complete decoding results are output. Normally, this mode should be selected.

26.3.5 H.264 decoder

The input of the decoder is H.264 standard bit stream in either plain NAL unit format or byte stream format. The input format in use will be automatically detected. The H.264 video encoding allows the use of multiple reference pictures, which means that the decoding order of the pictures may be different from their display order. The decoder can perform internally the display reordering of the decoded pictures or it can skip this and output all the pictures as soon as they are decoded.
The decoder has two operating modes: in the primary mode the HW performs entropy decoding, and in the secondary mode SW performs entropy decoding. Secondary mode is used in H.264 ASO or Slice Group stream decoding.

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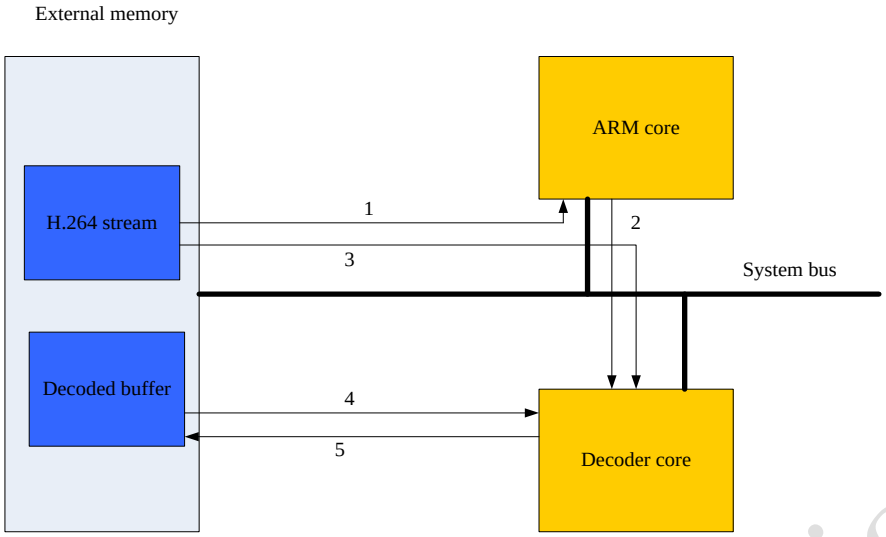

Fig. 26-4 Dataflow of HW performs entropy decoding in video decoder

The dataflow of HW performs entropy decoding is as Fig. 26-4 shown. The decoder software (SW) starts decoding the first picture by parsing the stream headers (1). Software then setups the hardware control registers (picture size, stream start address etc.) and enables the hardware (2).Hardware decodes the picture by reading stream (3) and the reference pictures (required for inter picture decoding)(4) from the external memory. Hardware writes the decoded output picture to memory one macroblock at a time (5). When the picture has been fully decodedor the hardware has run out of stream data, it gives an interrupt with a proper status flag and provides stream and address for software to continue and returns to initial state.

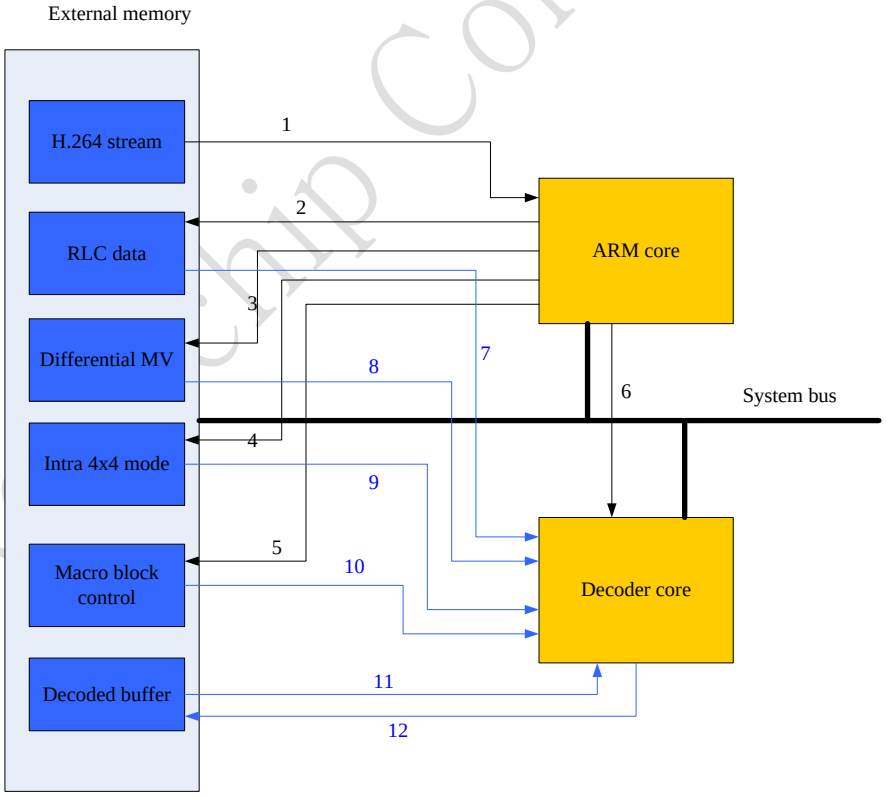

Fig. 26-5 Dataflow of SW performs entropy decoding in video decoder

SW entropy decoding mode (RLC mode) changes the input data format that is transferred from SW to HW. The dataflow of this mode is asFig. 26-5.In this case the decoder software starts decoding the first picture by parsing the stream headers (1), and by performing entropy decoding. Software then writes the following items to external memory:
Run-length-code (RLC) data (2)

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Differential motion vectors (3)
Intra 4x4 prediction modes (4)
Macroblock control data (5)
Last step for the software is to write the hardware control registers and to enable the hardware (6).
Hardware decodes the picture by buffering control data for several macroblocks at a time, and reading then appropriate amount of RLC data, differential motion vectors and intra modes depending on each macroblock type (7)-(10).For the rest of the decoding process (11)-(12), the functionality is identical to the HW entropy decoding mode.When the picture has been fully decoded, hardware can raise an interrupt and write the status bits in the status register.

26.4 Register description

This section describes the control/status registers of the VPU_Combo. Both HEVC and H.264 decoder/encoder shares the same base register address of VPU_Combo.
If HEVC is chosen to work, HEVC register base address is the base address of the VPU_Combo. The HEVC reading MMU master register base address is hevc_base+0x440, the writing MMU register base address is vpu_combo_base + 0x480, and the cache control register base address is vpu_combo_base + 0x400.
If H.264 decoder/encoder is chosen to work, VEPU(encoder) register base address is vpu_combo_base, and VDPU(decoder) base address is vpu_combo_base + 0x400. MMU base address is vpu_combo_base +0x800, and VDPU cache control base address is vpu_combo_base + 0xc00.

26.4.1 VDP_HEVC Register Summary

Name	Offset	Size	Reset Value	Description
hevc_swreg0_id	0x0000	W	0x68761100	ID register (read only)
hevc_swreg1_int	0x0004	W	0x00200022	interrupt and decoder enable register
hevc_swreg2_sysctrl	0x0008	W	0x00000000	Data input and output endian setting and sys ctrl
hevc_swreg3_picpar	0x000c	W	0x00000000	picture parameters
hevc_swreg4_strm_rlc_base	0x0010	W	0x00000000	the stream or rlc data base address
hevc_swreg5_stream_rlc_len	0x0014	W	0x00000000	amount of stream bytes or rlc data byte in the input buffer or the
hevc_swreg6_cabactbl_base	0x0018	W	0x00000000	the base address of cabac table
hevc_swreg7_decout_base	0x001c	W	0x00000000	base address of decoder output picture base address
hevc_swreg8_y_virstride	0x0020	W	0x00000000	the ouput picture y virtual stride
hevc_swreg9_yuv_virstride	0x0024	W	0x00000000	the ouput picture yuv virtual stride
hevc_swreg10_refer0_base	0x0028	W	0x00000000	base address for reference picture index 0
hevc_swreg11_refer1_base	0x002c	W	0x00000000	base address for reference picture index 1
hevc_swreg12_refer2_base	0x0030	W	0x00000000	base address for reference picture index 2
hevc_swreg13_refer3_base	0x0034	W	0x00000000	base address for reference picture index 3

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Name	Offset	Size	Reset Value	Description
hevc_swreg14_refer4_base	0x0038	W	0x00000000	base address for reference picture index 4
hevc_swreg15_refer5_base	0x003c	W	0x00000000	base address for reference picture index 5
hevc_swreg16_refer6_base	0x0040	W	0x00000000	base address for reference picture index 6
hevc_swreg17_refer7_base	0x0044	W	0x00000000	base address for reference picture index 7
hevc_swreg18_refer8_base	0x0048	W	0x00000000	base address for reference picture index 8
hevc_swreg19_refer9_base	0x004c	W	0x00000000	base address for reference picture index 9
hevc_swreg20_refer10_base	0x0050	W	0x00000000	base address for reference picture index 10
hevc_swreg21_refer11_base	0x0054	W	0x00000000	base address for reference picture index 11
hevc_swreg22_refer12_base	0x0058	W	0x00000000	base address for reference picture index 12
hevc_swreg23_refer13_base	0x005c	W	0x00000000	base address for reference picture index 13
hevc_swreg24_refer14_base	0x0060	W	0x00000000	base address for reference picture index 14
hevc_swreg25_refer0_poc	0x0064	W	0x00000000	the poc of reference picture index 0
hevc_swreg26_refer1_poc	0x0068	W	0x00000000	the poc of reference picture index 1
hevc_swreg27_refer2_poc	0x006c	W	0x00000000	the poc of reference picture index 2
hevc_swreg28_refer3_poc	0x0070	W	0x00000000	the poc of reference picture index 3
hevc_swreg29_refer4_poc	0x0074	W	0x00000000	the poc of reference picture index 4
hevc_swreg30_refer5_poc	0x0078	W	0x00000000	the poc of reference picture index 5
hevc_swreg31_refer6_poc	0x007c	W	0x00000000	the poc of reference picture index 6
hevc_swreg32_refer7_poc	0x0080	W	0x00000000	the poc of reference picture index 7
hevc_swreg33_refer8_poc	0x0084	W	0x00000000	the poc of reference picture index 8
hevc_swreg34_refer9_poc	0x0088	W	0x00000000	the poc of reference picture index 9
hevc_swreg35_refer10_poc	0x008c	W	0x00000000	the poc of reference picture index 10
hevc_swreg36_refer11_poc	0x0090	W	0x00000000	the poc of reference picture index 11
hevc_swreg37_refer12_poc	0x0094	W	0x00000000	the poc of reference picture index 12
hevc_swreg38_refer13_poc	0x0098	W	0x00000000	the poc of reference picture index 13
hevc_swreg39_refer14_poc	0x009c	W	0x00000000	the poc of reference picture index 14
hevc_swreg40_cur_poc	0x00a0	W	0x00000000	the poc of cur picture

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Name	Offset	Size	Reset Value	Description
hevc_swreg41_rlcwrite_base	0x00a4	W	0x00000000	the base address or rlcwrite base addr
hevc_swreg42_pps_base	0x00a8	W	0x00000000	the base address of pps
hevc_swreg43_rps_base	0x00ac	W	0x00000000	the base address of rps
hevc_swreg44_cabac_error_en	0x00b0	W	0x00000000	cabac error enable config
hevc_swreg45_cabac_error_status	0x00b4	W	0x00000000	cabac error status
hevc_swreg46_cabac_error_ctu	0x00b8	W	0x00400000	cabac error ctu
hevc_swreg47_sao_ctu_position	0x00bc	W	0x00000000	sao ctu position
hevc_swreg64_performance_cycle	0x0100	W	0x00000000	hevc performance cycle
hevc_swreg65_axi_ddr_rdata	0x0104	W	0x00000000	axi ddr read data num
hevc_swreg66_axi_ddr_wdata	0x0108	W	0x00000000	axi ddr write data number

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

26.4.2 VDP_HEVC Detail Register Description

hevc_swreg0_id
Address: Operational Base + offset (0x0000)
ID register (read only)

Bit	Attr	Reset Value	Description
31:16	RO	0x6876	prod_num product number The ascii code of "hv", which is 0x6876
15	RO	0x0	reserved
14	RW	0x0	codec_flag codec flag 0: only dec 1: dec + enc
13	RO	0x0	reserved
12	RW	0x1	profile hevc profile 0: Main 1: Main10
11:9	RO	0x0	reserved
8	RO	0x1	level level 0: FHD 1: UHD

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Bit	Attr	Reset Value	Description
7:0	RO	0x00	minor_ver minor version minor version

hevc_swreg1_int
Address: Operational Base + offset (0x0004)
interrupt and decoder enable register

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22	RW	0x0	sw_softreset_rdy when it is 1'b1, it says that softreset has been done
21	RW	0x1	sw_force_softreset_valid when sw_force_softreset_valid is 1'b1, sw_softrst_en will always be valid to the system no matter that whether the axi bus is idle; when sw_force_softreset_valid is 1'b0, sw_softrst_en will only be valid when the axi bus is idle.
20	RW	0x0	sw_softrst_en_p softreset enable softreset enable signal write 1 to soft reset, write 0 invalid puls register
19	RO	0x0	reserved
18	RW	0x0	sw_cabu_end_sta cabac decode end status cabac decode end status
17	RW	0x0	sw_colmv_ref_error_sta colmv ref error status colmv ref error status when it is 1'b1, it means that inter module read the invalid dpb frame
16	RO	0x0	reserved
15	RW	0x0	sw_dec_timeout_sta decoder timeout interrupt status When high the decoder has been idling for too long. it will self reset the hardware only when sw_dec_timeout_e is 1'b1, this bit is valid

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Bit	Attr	Reset Value	Description
14	RW	0x0	sw_dec_error_sta status bit of input stream error when high, an error is found in input data stream decoding. It will self reset the hardware
13	RW	0x0	sw_dec_bus_sta bus error status When this bit is high, there is error on the axi bus, it will self reset hardware
12	RW	0x0	sw_dec_rdy_sta decoder ready status when this bit is high, decoder has decoded a picture
11:10	RO	0x0	reserved
9	RW	0x0	sw_dec_irq_raw the raw status of sw_dec_irq the raw status of sw_dec_irq,SW should reset this bit after interrupt is handled
8	RO	0x0	sw_dec_irq decoder IRQ when high, decoder requests an interrrupt. sw_dec_irq = sw_dec_irq_raw && (sw_dec_irq_dis == 1'b0)
7	RW	0x0	sw_stmerror_waitdecfifo_empty whether the stream error process wait the decfifo empty when it is 1'b0, the stream error process will no wait the ca2decfifo empty when it is 1'b1, the stream error process will wait the ca2decfifo empty
6	RO	0x0	reserved
5	RW	0x1	sw_dec_timeout_e Timeout interrupt enable If enabled HW may return timeout interrupt in case HW gets stucked while decoding picture.
4	RW	0x0	sw_dec_irq_dis decoder IRQ disable When hight, there are no interrupts concerning decoder from HW. Polling must be used to see the interrupt status
3:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1	RW	0x1	sw_dec_clkgate_e decoder dynamic clock gating enable 0 = clock is running for all structures 1 = clock is gated for decoder structures that are not used
0	RW	0x0	sw_dec_e decoder enable Decoder enable. Setting this bit high will start the decoding operation. HW will reset this when picture is processed or stream error is detected or bus error or time out interrupt is given

hevc_swreg2_sysctrl
Address: Operational Base + offset (0x0008)
Data input and output endian setting and sys ctrl

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:12	RW	0x00	sw_strm_start_bit exact bit of stream start exact bit of streamd start word where decoding can be started (asosiates with sw_str_rlc_base)
11	RW	0x0	sw_rlc_mode rlc mode enable 0 = HW decodes video from bit stream 1 = HW decodes video from RLC input data
10	RW	0x0	sw_rlc_mode_direct_write cabac decode output direct write cabac decode output direct write enable when this bit is enable , all the module other than cabac and busifd are not work
9	RO	0x0	reserved
8	RW	0x0	sw_out_cbcr_swap output cbcr swap 1'b0: cb(u) is in the lower address, cr(v) is in the higher address 1'b1: cb(u) is in the higher address,cr(v) is in the lower address sw_in_cbcr_swap is the same with sw_out_cbcr_swap

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Bit	Attr	Reset Value	Description
7	RW	0x0	sw_out_swap32_e decoder output data and dpb input data 32bit swap may be used for 64 or 128 bit environment 0 = no swapping of 32 bit words 1 = 32 bit data words are swapped
6	RW	0x0	sw_out_endian dec output data and colmv , dpb data and colmv input endian 0 = little endian 1 = big endian for litter enadian , a data 0x12345678, 0x78 is stored in lower address, 0x12 is stored in higher address
5	RW	0x0	sw_str_swap64_e stream 64bit data swap may be used for 128 bit environment 0 = no swapping of 64 bit words 1 = 64 bit data words are swapped
4	RW	0x0	sw_str_swap32_e stream 32bit data swap may be used for 64 or 128 bit environment 0 = no swapping of 32 bit words 1 = 32 bit data words are swapped
3	RW	0x0	sw_str_endian stream data input endian mode 0 = little endian 1 = big endian for litter enadian , a data 0x12345678, 0x78 is stored in lower address, 0x12 is stored in higher address
2	RW	0x0	sw_in_swap64_e input 64bit data swap for other than stream and dpb data may be used for 128 bit environment 0 = no swapping of 64 bit words 1 = 64 bit data words are swapped
1	RW	0x0	sw_in_swap32_e input 32bit data swap for other than stream and dpb data may be used for 64 or 128 bit environment 0 = no swapping of 32 bit words 1 = 32 bit data words are swapped

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Bit	Attr	Reset Value	Description
0	RW	0x0	sw_in_endian decoder input endian mode for other than stream and dpb data 0 = little endian 1 = big endian for litter enadian , a data 0x12345678, 0x78 is stored in lower address, 0x12 is stored in higher address

hevc_swreg3_picpar
Address: Operational Base + offset (0x000c)
picture parameters

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:21	RW	0x000	sw_slice_num slice number in a frame slice number in a frame (0~199, when it is 0, it real means 1 slice in a frame) just only used for rps read. 2013.11.27 change the meaning from count from 1, so it will be in 1~200 2013.11.30 sw_slice_num max value is change to 600, so sw_slice_num expand to 10bit
20:12	RW	0x000	sw_uv_hor_virstride Field0000 Abstract picture horizontal virtual stride (the unit is 128bit) the max is (4096x1.5 + 128) /16 = 0x188 suggest this register to config to even for advance ddr performance
11:9	RO	0x0	reserved
8:0	RW	0x000	sw_y_hor_virstride picture horizontal virtual stride picture horizontal virtual stride (the unit is 128bit) the max is (4096x1.5 + 128) /16 = 0x188 suggest this register to config to even for advance ddr performance

hevc_swreg4_strm_rlc_base
Address: Operational Base + offset (0x0010)
the stream or rlc data base address

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_strm_rlc_base the stream or rlc data base address when swreg2.sw_rlc_mode =1, it is base address for rlc data when swreg2.sw_rlc_mode =0, it is base address for stream , after a frame is decoded ready or error (stream error , time out , bus error) , it is the last address of the stream the address should 128bit align
3:0	RO	0x0	reserved

hevc_swreg5_stream_rlc_len
Address: Operational Base + offset (0x0014)
amount of stream bytes or rlc data byte in the input buffer or the

Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved
26:0	RW	0x0000000	sw_stream_len amount of stream (unit is 8bit) in the input buffer amount of stream 8bits in the input buffer the max of sw_stream_len : 4096x2304x1.5x1.5 = 0x1440000 128bits unit: 0x1440000/16 = 0x144000 it is count from 0 2013.10.15 change to 23bit for zty's suggestion 2013.10.28, amount of stream data bytes in input buffer. it is count from 1, change to 27bits

hevc_swreg6_cabactbl_base
Address: Operational Base + offset (0x0018)
the base address of cabac table

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_cabactbl_base the base address of cabac table the base address of cabac table the address should 128bit align
3:0	RO	0x0	reserved

hevc_swreg7_decout_base
Address: Operational Base + offset (0x001c)
base address of decoder output picture base address

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_decout_base base address of decoder output picture addr base address of decoder output picture the address should be 128bit align
3:0	RO	0x0	reserved

hevc_swreg8_y_virstride
Address: Operational Base + offset (0x0020)
the ouput picture y virtual stride

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:0	RW	0x00000	sw_y_virstride the output picture y virtual stride the output picture y virtual stride (the unit is 128bit) the max: (4096x1.5 +128) x 2304 = 0xdc8000 we can know the sw_uvout_base = sw_decout_base + (sw_y_virstride <<4)

hevc_swreg9_yuv_virstride
Address: Operational Base + offset (0x0024)
the ouput picture yuv virtual stride

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved
20:0	RW	0x000000	sw_yuv_virstride the ouput picture yuv virtual stride the output picture yuv virtual stride (the unit is 128bit) the max : (4096x1.5 +128) x 2304 x1.5 = 0x14ac000 we can know the sw_mvout_base = sw_decout_base + (sw_yuv_virstride <<4)

hevc_swreg10_refer0_base
Address: Operational Base + offset (0x0028)
base address for reference picture index 0

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer0_base base address for reference picture index0 base address for reference picture index 0 (the address should be 128bit align)
3:0	RW	0x0	sw_ref_valid_0_3 valid flag for picture index 0 ~3 valid flag for picture index 0 ~3

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hevc_swreg11_refer1_base
Address: Operational Base + offset (0x002c)
base address for reference picture index 1

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer1_base base address for reference picture index 1 base address for reference picture index 1 (the address should be 128bit align)
3:0	RW	0x0	sw_ref_valid_4_7 valid flag for picture index 4 ~7 valid flag for picture index 4 ~7

hevc_swreg12_refer2_base
Address: Operational Base + offset (0x0030)
base address for reference picture index 2

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer2_base base address for reference picture index 2 base address for reference picture index 2 (the address should be 128bit align)
3:0	RW	0x0	sw_ref_valid_8_11 valid flag for picture index 8~11 valid flag for picture index 8~11

hevc_swreg13_refer3_base
Address: Operational Base + offset (0x0034)
base address for reference picture index 3

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer3_base base address for reference picture index 3 base address for reference picture index 3 (the address should be 128bit align)
3	RO	0x0	reserved
2:0	RW	0x0	sw_ref_valid_12_14 valid flag for picture index 12~14 valid flag for picture index 12~14

hevc_swreg14_refer4_base
Address: Operational Base + offset (0x0038)
base address for reference picture index 4

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer4_base base address for reference picture index 4 base address for reference picture index 4(the address should be 128bit align)

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Bit	Attr	Reset Value	Description
3:0	RO	0x0	reserved

hevc_swreg15_refer5_base
Address: Operational Base + offset (0x003c)
base address for reference picture index 5

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer5_base base address for reference picture index 5 base address for reference picture index 5(the address should be 128bit align)
3:0	RO	0x0	reserved

hevc_swreg16_refer6_base
Address: Operational Base + offset (0x0040)
base address for reference picture index 6

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer6_base base address for reference picture index 6 base address for reference picture index 6(the address should be 128bit align)
3:0	RO	0x0	reserved

hevc_swreg17_refer7_base
Address: Operational Base + offset (0x0044)
base address for reference picture index 7

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer7_base base address for reference picture index 7 base address for reference picture index 7(the address should be 128bit align)
3:0	RO	0x0	reserved

hevc_swreg18_refer8_base
Address: Operational Base + offset (0x0048)
base address for reference picture index 8

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer8_base base address for reference picture index 8 base address for reference picture index 8(the address should be 128bit align)
3:0	RO	0x0	reserved

hevc_swreg19_refer9_base
Address: Operational Base + offset (0x004c)
base address for reference picture index 9

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer9_base base address for reference picture index 9 base address for reference picture index 9(the address should be 128bit align)
3:0	RO	0x0	reserved

hevc_swreg20_refer10_base
Address: Operational Base + offset (0x0050)
base address for reference picture index 10

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer10_base base address for reference picture index 10 base address for reference picture index 10(the address should be 128bit align)
3:0	RO	0x0	reserved

hevc_swreg21_refer11_base
Address: Operational Base + offset (0x0054)
base address for reference picture index 11

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer11_base base address for reference picture index 11 base address for reference picture index 11(the address should be 128bit align)
3:0	RO	0x0	reserved

hevc_swreg22_refer12_base
Address: Operational Base + offset (0x0058)
base address for reference picture index 12

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer12_base base address for reference picture index 12 base address for reference picture index 12(the address should be 128bit align)
3:0	RO	0x0	reserved

hevc_swreg23_refer13_base
Address: Operational Base + offset (0x005c)
base address for reference picture index 13

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer13_base base address for reference picture index 13 base address for reference picture index 13(the address should be 128bit align)
3:0	RO	0x0	reserved

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hevc_swreg24_refer14_base
Address: Operational Base + offset (0x0060)
base address for reference picture index 14

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer14_base base address for reference picture index 14 base address for reference picture index 14(the address should be 128bit align)
3:0	RO	0x0	reserved

hevc_swreg25_refer0_poc
Address: Operational Base + offset (0x0064)
the poc of reference picture index 0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer0_poc the poc of reference picture index 0 the poc of reference picture index 0

hevc_swreg26_refer1_poc
Address: Operational Base + offset (0x0068)
the poc of reference picture index 1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer1_poc the poc of reference picture index 1 the poc of reference picture index 1

hevc_swreg27_refer2_poc
Address: Operational Base + offset (0x006c)
the poc of reference picture index 2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer2_poc the poc of reference picture index 2 the poc of reference picture index 2

hevc_swreg28_refer3_poc
Address: Operational Base + offset (0x0070)
the poc of reference picture index 3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer3_poc the poc of reference picture index 3 the poc of reference picture index 3

hevc_swreg29_refer4_poc
Address: Operational Base + offset (0x0074)
the poc of reference picture index 4

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer4_poc the poc of reference picture index 4 the poc of reference picture index 4

hevc_swreg30_refer5_poc
Address: Operational Base + offset (0x0078)
the poc of reference picture index 5

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer5_poc the poc of reference picture index 5 the poc of reference picture index 5

hevc_swreg31_refer6_poc
Address: Operational Base + offset (0x007c)
the poc of reference picture index 6

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer6_poc the poc of reference picture index 6 the poc of reference picture index 6

hevc_swreg32_refer7_poc
Address: Operational Base + offset (0x0080)
the poc of reference picture index 7

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer7_poc the poc of reference picture index 7 the poc of reference picture index 7

hevc_swreg33_refer8_poc
Address: Operational Base + offset (0x0084)
the poc of reference picture index 8

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer8_poc the poc of reference picture index 8 the poc of reference picture index 8

hevc_swreg34_refer9_poc
Address: Operational Base + offset (0x0088)
the poc of reference picture index 9

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer9_poc the poc of reference picture index 9 the poc of reference picture index 9

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hevc_swreg35_refer10_poc
Address: Operational Base + offset (0x008c)
the poc of reference picture index 10

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer10_poc the poc of reference picture index 10 the poc of reference picture index 10

hevc_swreg36_refer11_poc
Address: Operational Base + offset (0x0090)
the poc of reference picture index 11

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer11_poc the poc of reference picture index 11 the poc of reference picture index 11

hevc_swreg37_refer12_poc
Address: Operational Base + offset (0x0094)
the poc of reference picture index 12

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer12_poc the poc of reference picture index 12 the poc of reference picture index 12

hevc_swreg38_refer13_poc
Address: Operational Base + offset (0x0098)
the poc of reference picture index 13

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer13_poc the poc of reference picture index 13 the poc of reference picture index 13

hevc_swreg39_refer14_poc
Address: Operational Base + offset (0x009c)
the poc of reference picture index 14

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer14_poc the poc of reference picture index 14 the poc of reference picture index 14

hevc_swreg40_cur_poc
Address: Operational Base + offset (0x00a0)
the poc of cur picture

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_cur_poc the poc of the cur picture the poc of the cur picture

hevc_swreg41_rlcwrite_base
Address: Operational Base + offset (0x00a4)
the base address or rlcwrite base addr

Bit	Attr	Reset Value	Description
31:3	RW	0x00000000	sw_rlcwrite_base the base address of rlcwrite the base address of rlcwrite(the address should 64bit align) cabac output write to this rlcwrite base address when sw_rlc_mode_direct_write in swreg2_sysctrl is valid
2:0	RO	0x0	reserved

hevc_swreg42_pps_base
Address: Operational Base + offset (0x00a8)
the base address of pps

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_pps_base the base address of pps the base address of pps (the address should 128bit align) it is for storing sps(sequence parameter set) and pps(picture parameter set)
3:0	RO	0x0	reserved

hevc_swreg43_rps_base
Address: Operational Base + offset (0x00ac)
the base address of rps

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_rps_base rps base address rps(reference picture set) base address (the address should 128bit align)
3:0	RO	0x0	reserved

hevc_swreg44_cabac_error_en
Address: Operational Base + offset (0x00b0)
cabac error enable config

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
27:0	RW	0x0000000	sw_cabac_error_e cabac error enable regs cabac error enable regs

hevc_swreg45_cabac_error_status
Address: Operational Base + offset (0x00b4)
cabac error status

Bit	Attr	Reset Value	Description
31:28	RW	0x0	sw_colmv_error_ref_picidx colmv error ref picidx when sw_colmv_ref_error_sta is 1'b1, these bits are used for tell which dpb frame is invalid but is read by inter module
27:0	RW	0x0000000	sw_cabac_error_status cabac error status cabac error status

hevc_swreg46_cabac_error_ctu
Address: Operational Base + offset (0x00b8)
cabac error ctu

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:16	RW	0x40	sw_streamfifo_space2full stream fifo space to full It is for debug use, to tell the stream fifo space to full
15:8	RW	0x00	sw_cabac_error_ctu_yoffset cabac error ctu yoffset cabac error ctu yoffset
7:0	RW	0x00	sw_cabac_error_ctu_xoffset cabac error ctu xoffset cabac error ctu xoffset

hevc_swreg47_sao_ctu_position
Address: Operational Base + offset (0x00bc)
sao ctu position

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25:16	RW	0x000	sw_saowr_yoffset saowr y offset saowr y offset , its unit is 4 pixels
15:9	RO	0x0	reserved
8:0	RW	0x000	sw_saowr_xoffet saowr x address offset saowr x address offset, its unit is 128bit

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hevc_swreg64_performance_cycle
Address: Operational Base + offset (0x0100)
hevc performance cycle

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_performance_cycle hevc running cycle hevc running cycle if just want to analys a frame performance cycle, should set the register 0 before start a frame

hevc_swreg65_axi_ddr_rdata
Address: Operational Base + offset (0x0104)
axi ddr read data num

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_axi_ddr_rdata axi ddr rdata num axi ddr rdata num, the unit is byte

hevc_swreg66_axi_ddr_wdata
Address: Operational Base + offset (0x0108)
axi ddr write data number

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_axi_ddr_wdata hevc write data byte num hevc write data byte num

26.4.3 VDP_HEVC MMU Register Summary

Name	Offset	Size	Reset Value	Description
hevc_mmu_DTE_ADDR	0x00000	W	0x00000000	MMU current page Table address
hev**c**_mmu_STATUS	0x00004	W	0x00000018	MMU status register
hevc_mmu_COMMAND	0x00008	W	0x00000000	MMU command register
hevc_mmu_PAGE_FAULT_ADDR	0x0000c	W	0x00000000	MMU logical address of last page fault
hevc_mmu_ZAP_ONE_LINE	0x00010	W	0x00000000	MMU Zap cache line register
hevc_mmu_INT_RAWSTAT	0x00014	W	0x00000000	MMU raw interrupt status register
hevc_mmu_INT_CLEAR	0x00018	W	0x00000000	MMU raw interrupt status register
hevc_mmu_INT_MASK	0x0001c	W	0x00000000	MMU raw interrupt status register
hevc_mmu_INT_STATUS	0x00020	W	0x00000000	MMU raw interrupt status register
hevc_mmu_AUTO_GATING	0x00024	W	0x00000001	mmu auto gating

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*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

26.4.4 VDP_HEVC MMU Detail Register Description

hevc_mmu_DTE_ADDR
Address: Operational Base + offset (0x00000)
MMU current page Table address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	MMU_DTE_ADDR mmu dte base addr mmu dte base addr , the address must be 4kb aligned

hevc_mmu_STATUS
Address: Operational Base + offset (0x00004)
MMU status register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:6	RO	0x00	PAGE_FAULT_BUS_ID Field0000 Abstract Index of master reponsible for last page fault
5	RO	0x0	PAGE_FAULT_IS_WRITE Field0000 Abstract The direction of access for last page fault: 0 = Read 1 = Write
4	RO	0x1	REPLAY_BUFFER_EMPTY Field0000 Abstract The MMU replay buffer is empty
3	RO	0x1	MMU_IDLE Field0000 Abstract The MMU is idle when accesses are being translated and there are no unfinished translated accesses.
2	RO	0x0	STAIL_ACTIVE Field0001 Abstract MMU stall mode currently enabled. The mode is enabled by command
1	RO	0x0	PAGE_FAULT_ACTIVE Field0000 Abstract MMU page fault mode currently enabled . The mode is enabled by command.
0	RO	0x0	PAGING_ENABLED Field0000 Abstract Paging is enabled

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hevc_mmu_COMMAND
Address: Operational Base + offset (0x00008)
MMU command register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2:0	WO	0x0	MMU_CMD Field0000 Abstract MMU_CMD. This can be: 0: MMU_ENABLE_PAGING 1: MMU_DISABLE_PAGING 2: MMU_ENABLE_STALL 3: MMU_DISABLE_STALL 4: MMU_ZAP_CACHE 5: MMU_PAGE_FAULT_DONE 6: MMU_FORCE_RESET

hevc_mmu_PAGE_FAULT_ADDR
Address: Operational Base + offset (0x0000c)
MMU logical address of last page fault

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	PAGE_FAULT_ADDR Field0000 Abstract address of last page fault

hevc_mmu_ZAP_ONE_LINE
Address: Operational Base + offset (0x00010)
MMU Zap cache line register

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	MMU_ZAP_ONE_LINE Field0000 Abstract address to be invalidated from the page table cache

hevc_mmu_INT_RAWSTAT
Address: Operational Base + offset (0x00014)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR Field0000 Abstract read bus error
0	RW	0x0	PAGE_FAULT Field0000 Abstract page fault

hevc_mmu_INT_CLEAR

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Address: Operational Base + offset (0x00018)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	WO	0x0	READ_BUS_ERROR Field0000 Abstract read bus error
0	WO	0x0	PAGE_FAULT Field0000 Abstract page fault

hevc_mmu_INT_MASK
Address: Operational Base + offset (0x0001c)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR Field0000 Abstract read bus error enable an interrupt source if the corresponding mask bit is set to 1
0	RW	0x0	PAGE_FAULT Field0000 Abstract page fault enable an interrupt source if the corresponding mask bit is set to 1

hevc_mmu_INT_STATUS
Address: Operational Base + offset (0x00020)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RO	0x0	READ_BUS_ERROR Field0000 Abstract read bus error
0	RO	0x0	PAGE_FAULT Field0000 Abstract page fault

hevc_mmu_AUTO_GATING
Address: Operational Base + offset (0x00024)
mmu auto gating

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
0	RW	0x1	mmu_auto_gating mmu auto gating when it is 1'b1, the mmu will auto gating it self

26.4.5 VDP_HEVC Pref_cache Register Summary

Name	Offset	Size	Reset Value	Description
pref_cache_VERSION	0x00000	W	0xcac20101	VERSION register
pref_cache_SIZE	0x00004	W	0x07110206	L2 cache SIZE
pref_cache_STATUS	0x00008	W	0x00000000	Status register
pref_cache_COMMAND	0x00010	W	0x00000000	Command setting register
pref_cache_CLEAR_PAGE	0x00014	W	0x00000000	clear page register
pref_cache_MAX_READS	0x00018	W	0x0000001c	maximum read register
pref_cache_ENABLE	0x0001c	W	0x00000003	enables cacheable accesses and cache read allocation
pref_cache_PERFCNT_SRC0	0x00020	W	0x00000000	performance counter 0 source register
pref_cache_PERFCNT_VAL0	0x00024	W	0x00000000	performance counter 0 value register
pref_cache_PERFCNT_SRC1	0x00028	W	0x00000000	performance counter 0 source register
pref_cache_PERFCNT_VAL1	0x0002c	W	0x00000000	performance counter 1 value register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

26.4.6 VDP_HEVC Pref_cache Detail Register Description

pref_cache_VERSION
Address: Operational Base + offset (0x00000)
VERSION register

Bit	Attr	Reset Value	Description
31:16	RO	0xcac2	PRODUCT_ID Field0000 Abstract Field0000 Description
15:8	RO	0x01	VERSION_MAJOR Field0000 Abstract Field0000 Description
7:0	RO	0x01	VERSION_MINOR Field0000 Abstract Field0000 Description

pref_cache_SIZE
Address: Operational Base + offset (0x00004)
L2 cache SIZE

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:24	RO	0x07	External_bus_width Field0000 Abstract Log2 external bus width in bits
23:16	RO	0x11	CACHE_SIZE Field0000 Abstract Log2 cache size in bytes
15:8	RO	0x02	ASSOCIATIVITY Field0000 Abstract Log2 associativity
7:0	RO	0x06	LINE_SIZE Field0000 Abstract Log2 line size in bytes

pref_cache_STATUS
Address: Operational Base + offset (0x00008)
Status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	DATA_BUSY Field0000 Abstract set when the cache is busy handling data
0	RW	0x0	CMD_BUSY Field0000 Abstract set when the cache is busy handling commands

pref_cache_COMMAND
Address: Operational Base + offset (0x00010)
Command setting register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x0	COMMAND Field0000 Abstract The possible command is 1 = Clear entire cache

pref_cache_CLEAR_PAGE
Address: Operational Base + offset (0x00014)
clear page register

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	CLEAR_PAGE Field0000 Abstract writing an address, invlidates all lines in that page from the cache

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pref_cache_MAX_READS
Address: Operational Base + offset (0x00018)
maximum read register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x1c	MAX_READS Field0000 Abstract Limit the number of outstanding read transactions to this amount

pref_cache_ENABLE
Address: Operational Base + offset (0x0001c)
enables cacheable accesses and cache read allocation

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	sw_cache_clk_disgate cache clk disgate cache clk disgate when it is 1'b0, enable cache clk auto clkgating when it is 1'b1, disable cache clk auto clkgating
2	RW	0x0	sw_readbuffer_counter_reject_en counter reject enable default is 1'b0, for enhance cacheable read performnace in readbuffer. 1'b1: normal origin counter reject
1	RW	0x1	permit_cache_read_allocate cache read allocate 1'b1: permit cache read allocate
0	RW	0x1	permit_cacheable_access cacheable access 1'b1: permit cacheable access

pref_cache_PERFCNT_SRC0
Address: Operational Base + offset (0x00020)
performance counter 0 source register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
6:0	RW	0x00	PERFCNT_SRC0 Field0000 Abstract This register holds all the possible source values for Performance Counter 0 0: disabled 1: total clock cycles 2: active clock cycles 3: read transactions, master 4: word reads, master 5: read transactions, slave 6: word reads, slave 7: read hit, slave 8: read misses, slave 9: read invalidates, slave 10: cacheable read transactions, slave 11: bad hit nmber, slave

pref_cache_PERFCNT_VAL0
Address: Operational Base + offset (0x00024)
performance counter 0 value register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PERFCNT_VAL0 Field0000 Abstract Performance counter 0 value

pref_cache_PERFCNT_SRC1
Address: Operational Base + offset (0x00028)
performance counter 0 source register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
6:0	RW	0x00	PERFCNT_SRC1 Field0000 Abstract This register holds all the possible source values for Performance Counter 1 0: disabled 1: total clock cycles 2: active clock cycles 3: read transactions, master 4: word reads, master 5: read transactions, slave 6: word reads, slave 7: read hit, slave 8: read misses, slave 9: read invalidates, slave 10: cacheable read transactions, slave 11: bad hit nmber, slave

pref_cache_PERFCNT_VAL1
Address: Operational Base + offset (0x0002c)
performance counter 1 value register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PERFCNT_VAL1 Field0000 Abstract Performance counter 1 value

26.4.7 VEPU Register Summary

Name	Offset	Size	Reset Value	Description
VEPU_swreg0	0x00000	W	0x48310000	product ID
VEPU_swreg1	0x00004	W	0x00000000	interrrupt control and status
VEPU_swreg2	0x00008	W	0x00000010	axi control
VEPU_swreg3	0x0000c	W	0x00000000	test control register
VEPU_swreg4	0x00010	W	0x00000000	reserverd
VEPU_swreg5	0x00014	W	0x00000000	addr_output_stream
VEPU_swreg6	0x00018	W	0x00000000	base address for output control
VEPU_swreg7	0x0001c	W	0x00000000	base address for reference luma
VEPU_swreg8	0x00020	W	0x00000000	base address for reference chroma
VEPU_swreg9	0x00024	W	0x00000000	base address for reconstructed luma
VEPU_swreg10	0x00028	W	0x00000000	base address for reconstructed chroma
VEPU_swreg11	0x0002c	W	0x00000000	base addr for input luma
VEPU_swreg12	0x00030	W	0x00000000	base address for input cb
VEPU_swreg13	0x00034	W	0x00000000	base address for input cr
VEPU_swreg14	0x00038	W	0x00000000	enc control
VEPU_swreg15	0x0003c	W	0x00000000	input image control
VEPU_swreg16	0x00040	W	0x00000000	encoder control regsiter 0
VEPU_swreg17	0x00044	W	0x00000000	encoder control register 1

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Name	Offset	Size	Reset Value	Description
VEPU_swreg18	0x00048	W	0x00000000	encoder control register 2
VEPU_swreg19	0x0004c	W	0x00000000	encoder control register 3
VEPU_swreg20	0x00050	W	0x00000000	JPEG control regsiter
VEPU_swreg21	0x00054	W	0x00000000	encoder control register 4
VEPU_swreg22	0x00058	W	0x00000000	stream header remainder bits MSB
VEPU_swreg23	0x0005c	W	0x00000000	stream header remainder bits LSB
VEPU_swreg24	0x00060	W	0x00000000	stream buffer limit
VEPU_swreg25	0x00064	W	0x00000000	MAD control register
VEPU_swreg27	0x0006c	W	0x00000000	QP register
VEPU_swreg28	0x00070	W	0x00000001	checkpoint 1 and 2
VEPU_swreg29	0x00074	W	0x00000000	checkpoint 3 and 4
VEPU_swreg30	0x00078	W	0x00000000	checkpoint 5 and 6
VEPU_swreg31	0x0007c	W	0x00000000	checkpoint 7 and 8
VEPU_swreg32	0x00080	W	0x00000000	checkpoint 9 and 10
VEPU_swreg33	0x00084	W	0x00000000	checkpoint word error 1 and 2
VEPU_swreg34	0x00088	W	0x00000000	checkpoint word error 1 and 2
VEPU_swreg35	0x0008c	W	0x00000000	checkpoint word error 1 and 2
VEPU_swreg36	0x00090	W	0x00000000	checkpoint delta QP register
VEPU_swreg37	0x00094	W	0x00000000	rlc control
VEPU_swreg38	0x00098	W	0x00000000	mb control register
VEPU_swreg39	0x0009c	W	0x00000000	Base address for next pic
VEPU_swreg40	0x000a0	W	0x00000000	Stabilization minimum value
VEPU_swreg41	0x000a4	W	0x00000000	Stabilization motion sum
VEPU_swreg42	0x000a8	W	0x00000000	stab_matrix1 and stab_gmv_hor
VEPU_swreg43	0x000ac	W	0x00000000	stab_matrix2 and stab_gmv_ver
VEPU_swreg44	0x000b0	W	0x00000000	stab_matrix3
VEPU_swreg45	0x000b4	W	0x00000000	stab_matrix4
VEPU_swreg46	0x000b8	W	0x00000000	stab_matrix5
VEPU_swreg47	0x000bc	W	0x00000000	stab_matrix6
VEPU_swreg48	0x000c0	W	0x00000000	stab_matrix7
VEPU_swreg49	0x000c4	W	0x00000000	stab_matrix8
VEPU_swreg50	0x000c8	W	0x00000000	stab_matrix9
VEPU_swreg51	0x000cc	W	0x00000000	cabac_table_addr
VEPU_swreg52	0x000d0	W	0x00000000	Base address for MV output
VEPU_swreg53	0x000d4	W	0x00000000	RGB to YUV conversion coefficien A and B
VEPU_swreg54	0x000d8	W	0x00000000	RGB to YUV conversion coefficien C and E
VEPU_swreg55	0x000dc	W	0x00000000	RGB mask MSB bit
VEPU_swreg56	0x000e0	W	0x00000000	intra area control register
VEPU_swreg57	0x000e4	W	0x00000000	CIR intra control reg
VEPU_swreg58	0x000e8	W	0x00000000	Intra slice bitmap for slices 0..31
VEPU_swreg59	0x000ec	W	0x00000000	Intra slice bitmap for slices32..63
VEPU_swreg60	0x000f0	W	0x00000000	1st ROI area register
VEPU_swreg61	0x000f4	W	0x00000000	Register0061 Abstract
VEPU_swreg62	0x000f8	W	0x00000000	MVC control reg

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Name	Offset	Size	Reset Value	Description
VEPU_swreg63	0x000fc	W	0x1f522780	Register0063 Abstract
VEPU_swreg64	0x00100	W	0x00000000	JPEG luma quantization 1
VEPU_swreg65	0x00104	W	0x00000000	JPEG luma quantization 2
VEPU_swreg66	0x00108	W	0x00000000	JPEG luma quantization 3
VEPU_swreg67	0x0010c	W	0x00000000	JPEG luma quantization 4
VEPU_swreg68	0x00110	W	0x00000000	JPEG luma quantization 5
VEPU_swreg69	0x00114	W	0x00000000	JPEG luma quantization 6
VEPU_swreg70	0x00118	W	0x00000000	JPEG luma quantization 7
VEPU_swreg71	0x0011c	W	0x00000000	JPEG luma quantization 8
VEPU_swreg72	0x00120	W	0x00000000	JPEG luma quantization 9
VEPU_swreg73	0x00124	W	0x00000000	Register0073 Abstract
VEPU_swreg74	0x00128	W	0x00000000	JPEG luma quantization 11
VEPU_swreg75	0x0012c	W	0x00000000	JPEG luma quantization 12
VEPU_swreg76	0x00130	W	0x00000000	JPEG luma quantization 13
VEPU_swreg77	0x00134	W	0x00000000	JPEG luma quantization 14
VEPU_swreg78	0x00138	W	0x00000000	JPEG luma quantization 15
VEPU_swreg79	0x0013c	W	0x00000000	JPEG luma quantization 16
VEPU_swreg80	0x00140	W	0x00000000	JPEG chroma quantization 1
VEPU_swreg81	0x00144	W	0x00000000	JPEG chroma quantization 2
VEPU_swreg82	0x00148	W	0x00000000	JPEG chroma quantization 3
VEPU_swreg83	0x0014c	W	0x00000000	JPEG chroma quantization 4
VEPU_swreg84	0x00150	W	0x00000000	JPEG chroma quantization 5
VEPU_swreg85	0x00154	W	0x00000000	JPEG chroma quantization 6
VEPU_swreg86	0x00158	W	0x00000000	JPEG chroma quantization 7
VEPU_swreg87	0x0015c	W	0x00000000	JPEG chroma quantization 8
VEPU_swreg88	0x00160	W	0x00000000	JPEG chroma quantization 9
VEPU_swreg89	0x00164	W	0x00000000	JPEG chroma quantization 10
VEPU_swreg90	0x00168	W	0x00000000	JPEG chroma quantization 11
VEPU_swreg91	0x0016c	W	0x00000000	JPEG chroma quantization 12
VEPU_swreg92	0x00170	W	0x00000000	JPEG chroma quantization 13
VEPU_swreg93	0x00174	W	0x00000000	JPEG chroma quantization 14
VEPU_swreg94	0x00178	W	0x00000000	JPEG chroma quantization 15
VEPU_swreg95	0x0017c	W	0x00000000	JPEG chroma quantization 16
VEPU_swreg96	0x00180	W	0x00000000	DMV 4p/1p penalty values 0-3
VEPU_swreg97	0x00184	W	0x00000000	DMV 4p/1p penalty values 4-7
VEPU_swreg98	0x00188	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg99	0x0018c	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg100	0x00190	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg101	0x00194	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg102	0x00198	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg103	0x0019c	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg104	0x001a0	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg105	0x001a4	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg106	0x001a8	W	0x00000000	DMV 4p/1p penalty values

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Name	Offset	Size	Reset Value	Description
VEPU_swreg107	0x001ac	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg108	0x001b0	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg109	0x001b4	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg110	0x001b8	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg111	0x001bc	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg112	0x001c0	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg113	0x001c4	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg114	0x001c8	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg115	0x001cc	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg116	0x001d0	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg117	0x001d4	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg118	0x001d8	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg119	0x001dc	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg120	0x001e0	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg121	0x001e4	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg122	0x001e8	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg123	0x001ec	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg124	0x001f0	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg125	0x001f4	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg126	0x001f8	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg127	0x001fc	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg128	0x00200	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg129	0x00204	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg130	0x00208	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg131	0x0020c	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg132	0x00210	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg133	0x00214	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg134	0x00218	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg135	0x0021c	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg136	0x00220	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg137	0x00224	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg138	0x00228	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg139	0x0022c	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg140	0x00230	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg141	0x00234	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg142	0x00238	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg143	0x0023c	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg144	0x00240	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg145	0x00244	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg146	0x00248	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg147	0x0024c	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg148	0x00250	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg149	0x00254	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg150	0x00258	W	0x00000000	DMV 4p/1p penalty values

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Name	Offset	Size	Reset Value	Description
VEPU_swreg151	0x0025c	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg152	0x00260	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg153	0x00264	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg154	0x00268	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg155	0x0026c	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg156	0x00270	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg157	0x00274	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg158	0x00278	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg159	0x0027c	W	0x00000000	DMV 4p/1p penalty values
VEPU_swreg160	0x00280	W	0x00000000	vp8 control
VEPU_swreg161	0x00284	W	0x00000000	VP8 bit cost of golden ref frame
VEPU_swreg162	0x00288	W	0x00000000	vp8 loop filter delta registers
VEPU_swreg163	0x0028c	W	0x00000000	vp8 loop filter delta register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

26.4.8 VEPU Detail Register Description

VEPU_swreg0

Address: Operational Base + offset (0x00000)

product ID

Bit	Attr	Reset Value	Description
31:16	RO	0x4831	prod_id Product ID
15:12	RW	0x0	major_num Major number
11:4	RW	0x00	minor_num Minor number
3:0	RW	0x0	synthesis

VEPU_swreg1

Address: Operational Base + offset (0x00004)

interrrupt control and status

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8	RW	0x0	irq_slice_ready IRQ slice ready status bit
7	RO	0x0	reserved
6	RW	0x0	irq_timeout IRQ HW timeout status bit
5	RW	0x0	irq_buffer_full irq buffer full
4	RW	0x0	irq_reset irq SW reset
3	RW	0x0	irq_bus_error

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Bit	Attr	Reset Value	Description
2	RW	0x0	irq_frame_rdy IRQ frame ready status bit. Encoder has finished a frame
1	RW	0x0	irq_dis IRQ disable. No interrupts from HW. SW must use polling
0	RW	0x0	enc_irq HINTenc interrupt from HW. SW resets at IRQ handler.

VEPU_swreg2
Address: Operational Base + offset (0x00008)
axi control

Bit	Attr	Reset Value	Description
31:24	RW	0x00	axi_write_id axi write id axi write id
23:16	RW	0x00	axi_rd_id axi read id axi read id
15	RW	0x0	output_swap16 enable output swap 16-bits
14	RW	0x0	input_swap16 enable input swap 16-bits
13:8	RW	0x00	burst_len burst length
7	RW	0x0	disable_burst disable burst mode for AXI disable burst mode for AXI bus
6	RW	0x0	burst_incr burst increment burst increment. 1: INCR burst allowed 0: use SINGLE burst
5	RW	0x0	burst_discard enable burst data dicard enable burst data dicard. 2 or 3 long reads are using BURST4
4	RW	0x1	clk_gating_en enable clock gating enable clock gating
3	RW	0x0	output_swap32 enable output swap 32-bits enable output swap 32-bits

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Bit	Attr	Reset Value	Description
2	RW	0x0	input_swap32 enable input swap 32-bits enable input swap 32-bits
1	RW	0x0	output_swap8 enable output swap 8-bits enable output swap 8-bits
0	RW	0x0	input_swap8 enable input swap 8-bits enable input swap 8-bits

VEPU_swreg3
Address: Operational Base + offset (0x0000c)
test control register

Bit	Attr	Reset Value	Description
31:28	RW	0x0	test_counter test counter
27:21	RO	0x0	reserved
20:3	RW	0x00000	test_len test data length
2	RW	0x0	test_memory test memory coherency
1	RW	0x0	test_reg test register coherency
0	RW	0x0	test_irq test irq

VEPU_swreg4
Address: Operational Base + offset (0x00010)
reserverd

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	Field0000 Abstract Field0000 Description

VEPU_swreg5
Address: Operational Base + offset (0x00014)
addr_output_stream

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	addr_output_stream base address for output stream

VEPU_swreg6
Address: Operational Base + offset (0x00018)
base address for output control

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	addr_output_control base address for output control base address for output control

VEPU_swreg7
Address: Operational Base + offset (0x0001c)
base address for reference luma

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	addr_reflum base address for reference luma base address for reference luma

VEPU_swreg8
Address: Operational Base + offset (0x00020)
base address for reference chroma

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	addr_refchroma base address for reference chroma base address for reference chroma

VEPU_swreg9
Address: Operational Base + offset (0x00024)
base address for reconstructed luma

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	addr_recon_luma base address for reconstructed luma

VEPU_swreg10
Address: Operational Base + offset (0x00028)
base address for reconstructed chroma

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	addr_recon_chroma base address for reconstructed chroma

VEPU_swreg11
Address: Operational Base + offset (0x0002c)
base addr for input luma

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	addr_input_luma base addr for input luma

VEPU_swreg12
Address: Operational Base + offset (0x00030)
base address for input cb

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	addr_input_cb base address for input cb

VEPU_swreg13
Address: Operational Base + offset (0x00034)
base address for input cr

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	addr_input_cr base address for input cr

VEPU_swreg14
Address: Operational Base + offset (0x00038)
enc control

Bit	Attr	Reset Value	Description
31	RW	0x0	int_timeout_en enable interrupt for timeout
30	RW	0x0	mv_sad_wren enabel writing MV and SAD of each MB to BaseMvWrite
29	RW	0x0	nal_mode NAL size output to base control
28	RW	0x0	slice_rdyint_en enable interruupt for slice ready
27:19	RW	0x000	enc_width encoder width. lumwidth (macroblocks) H264: [9...255) JPEG: [6...511]
18:10	RW	0x000	enc_height encoderd height, lumHeight (macroblocks) H264: [6..255] JPEG: [6..511]
9:7	RO	0x0	reserved
6	RW	0x0	rocon_write_dis disable writing of reconstructed image. recWriteDisable
5	RO	0x0	reserved
4:3	RW	0x0	enc_pic_type encoder picture type. frame type 0: INTER 1: INTRA(IDR) 2: MVC-INTER
2:1	RW	0x0	enc_mode encoding mode. stream type , 1=VP8, 2=JPEG,3=H264

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Bit	Attr	Reset Value	Description
0	RW	0x0	enc_en encoder enable

VEPU_swreg15
Address: Operational Base + offset (0x0003c)
input image control

Bit	Attr	Reset Value	Description
31:29	RW	0x0	input_chroma_offset input chrominance offset (bytes)
28:26	RW	0x0	input_lum_offset input luminance offset(bytes)
25:12	RW	0x0000	input_row_len input luminance row length
11:10	RW	0x0	overfill_right overfill pixels on right edge of image div4[0...3]
9:6	RW	0x0	overfill_bot overfill pixels on bottom edge of image. YFill [0..15]
5:2	RW	0x0	input_format input image format. YUV420P/YUV420SP/YUV422/UYVY422/RGB565/RGB444/RGB888/RGB101010
1:0	RW	0x0	imagein_rotmode input image rotation 0: disabled 1: 90 degress rigth 2: 90 degress left

VEPU_swreg16
Address: Operational Base + offset (0x00040)
encoder control regsiter 0

Bit	Attr	Reset Value	Description
31:26	RW	0x00	init_qp H.264 pic init qp in PPS[0...51]
25:22	RW	0x0	slice_alpha H.264 slice filter alpha c0 offset div2 [-6 ....6]
21:18	RW	0x0	slice_beta h.264 slice filter beta offset div2 [-6 ...6]
17:13	RW	0x00	chroma_qp_offset H264 chroma qp index offset [-12...12]
12:5	RO	0x0	reserved
4:1	RW	0x0	idr_picid IDR pic ID

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Bit	Attr	Reset Value	Description
0	RW	0x0	constr_intra_pred constrained intra prediction

VEPU_swreg17
Address: Operational Base + offset (0x00044)
encoder control register 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	pic_parameter_set_id H.264 pic_parameter_set_id
23:16	RW	0x00	intra_pred_mode H.264 intra prediction previous 4x4 mode favor
15:0	RW	0x0000	frame_num H.264 frame number

VEPU_swreg18
Address: Operational Base + offset (0x00048)
encoder control register 2

Bit	Attr	Reset Value	Description
31:30	RW	0x0	deblocking_filter_mode Deblocking filter mode. 0=enabled. 1=disabled (vp8=simple). 2=disabled on slice
29:23	RW	0x00	h264_slice_size H.264 Slice size. mbRowPerSlice (mb rows) [0..127] 0=one slice per picture
22	RW	0x0	disable_quarter_pixmv H.264 Disable quarter pixel MVs. disableQuarterPixelMv
21	RW	0x0	transform8x8_mode_en H.264 Transform 8x8 enable. High Profile H.264. transform8x8Mode
20:19	RW	0x0	cabac_int_idc H.264 CABAC initial IDC. [0..2]
18	RW	0x0	entropy_coding_mode H.264 CABAC / VP8 boolenc enable. entropyCodingMode. 0=CAVLC (Baseline Profile H.264). 1=CABAC (Main Profile H.264)
17	RW	0x0	h264_inter4x4_mode H.264 Inter 4x4 mode restriction. restricted4x4Mode

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Bit	Attr	Reset Value	Description
16	RW	0x0	h264_stream_mode H.264 Stream mode. 0=NAL unit stream. 1=Byte stream
15:0	RW	0x0000	intra16x16_mode Intra prediction intra 16x16 mode favor

VEPU_swreg19
Address: Operational Base + offset (0x0004c)
encoder control register 3

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30	RW	0x0	mutimv_en Enable using more than 1 MV per macroblock.
29:20	RW	0x000	mv_penalty_1_4p Differential MV penalty for 1/4p ME. DMVPenaltyQp
19:10	RW	0x000	mv_penalty_4p Differential MV penalty for 4p ME. DMVPenalty4p
9:0	RW	0x000	mv_penalty_1p differential MV penalty for 1p

VEPU_swreg20
Address: Operational Base + offset (0x00050)
JPEG control regsiter

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:20	RW	0x000	mv_penalty_16x8_8x16 Penalty for using 16x8 or 8x16 MV.
19:10	RW	0x000	mv_penalty_8x8 Penalty for using 8x8 MV
9:0	RW	0x000	mv_penalty_8x4_4x8 Penalty for using 8x4 or 4x8 MV.

VEPU_swreg21
Address: Operational Base + offset (0x00054)
encoder control register 4

Bit	Attr	Reset Value	Description
31:24	RW	0x00	macroblock_penalty H.264 SKIP macroblock mode / VP8 zero/nearest/near mode penalty
23:16	RW	0x00	completed_slices H.264 amount of completed slices.

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	inter_mode inter MB mode favor in intra/inter selection

VEPU_swreg22
Address: Operational Base + offset (0x00058)
stream header remainder bits MSB

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	strm_hdr_rem stream header remainder bits MSB stream header remainder bits MSB

VEPU_swreg23
Address: Operational Base + offset (0x0005c)
stream header remainder bits LSB

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	strm_hdr_rem2 stream header remainder bits LSB

VEPU_swreg24
Address: Operational Base + offset (0x00060)
stream buffer limit

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	strm_buffer_limit stream buffer limit

VEPU_swreg25
Address: Operational Base + offset (0x00064)
MAD control register

Bit	Attr	Reset Value	Description
31:28	RW	0x0	QP_adjust MAD based QP adjustment. madQpChange [-8..7]
27:22	RW	0x00	MAD_thredhold MAD threshold div256
21	RO	0x0	reserved
20:0	RW	0x000000	qp_sum_div2 QP sum div2 output

VEPU_swreg27
Address: Operational Base + offset (0x0006c)
QP register

Bit	Attr	Reset Value	Description
31:26	RW	0x00	qp_lum H.264 Initial QP. qpLum [0..51]

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Bit	Attr	Reset Value	Description
25:20	RW	0x00	qp_max H.264 Minimum QP. qpMax [0..51]
19:14	RW	0x00	qp_min H.264 Minimum QP. qpMin [0..51]
13	RO	0x0	reserved
12:0	RW	0x0000	checkpoint_distan checkpoint distance checkpoint distance

VEPU_swreg28
Address: Operational Base + offset (0x00070)
checkpoint 1 and 2

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	checkp_1 checkpoint 1 word target/usage
15:0	RW	0x0001	checkp_2 checkpoint 2 word target/usage

VEPU_swreg29
Address: Operational Base + offset (0x00074)
checkpoint 3 and 4

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	checkp_3 checkpoint 3 word target/usage
15:0	RW	0x0000	checkp_4 checkpoint 4 word target/usage checkpoint 4 word target/usage

VEPU_swreg30
Address: Operational Base + offset (0x00078)
checkpoint 5 and 6

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	checkp_5 checkpoint 5 word target/usage
15:0	RW	0x0000	checkp_6 checkpoint 6 word target/usage checkpoint 6 word target/usage

VEPU_swreg31
Address: Operational Base + offset (0x0007c)
checkpoint 7 and 8

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	checkp_7 checkpoint 7 word target/usage
15:0	RW	0x0000	checkp_8 checkpoint 8 word target/usage checkpoint 8 word target/usage

VEPU_swreg32
Address: Operational Base + offset (0x00080)
checkpoint 9 and 10

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	checkp_9 checkpoint 9 word target/usage
15:0	RW	0x0000	checkp_10 checkpoint 10 word target/usage checkpoint 10 word target/usage

VEPU_swreg33
Address: Operational Base + offset (0x00084)
checkpoint word error 1 and 2

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	checkp_error1 checkpoint word error 1
15:0	RW	0x0000	checkp_error2 checkpoint word error 2

VEPU_swreg34
Address: Operational Base + offset (0x00088)
checkpoint word error 1 and 2

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	checkp_error3 checkpoint word error 3
15:0	RW	0x0000	checkp_error4 checkpoint word error 4

VEPU_swreg35
Address: Operational Base + offset (0x0008c)
checkpoint word error 1 and 2

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	checkp_error5 checkpoint word error 5
15:0	RW	0x0000	checkp_error6 checkpoint word error 6

VEPU_swreg36

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Address: Operational Base + offset (0x00090)
checkpoint delta QP register

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:24	RW	0x0	checkp_qp_1 checkpoint delta QP 1
23:20	RW	0x0	checkp_qp_2 checkpoint delta QP 2
19:16	RW	0x0	checkp_qp_3 checkpoint delta QP 3
15:12	RW	0x0	checkp_qp_4 checkpoint delta QP 4
11:8	RW	0x0	checkp_qp_5 checkpoint delta QP 5
7:4	RW	0x0	checkp_qp_6 checkpoint delta QP 6
3:0	RW	0x0	checkp_qp_7 checkpoint delta QP 7

VEPU_swreg37
Address: Operational Base + offset (0x00094)
rlc control

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:23	RW	0x00	stream_st_offset Field0000 Description
22	RO	0x0	reserved
21:0	RW	0x000000	rlc_sum rlc sum rlc sum

VEPU_swreg38
Address: Operational Base + offset (0x00098)
mb control register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	macroblock_count Field0000 Description
15:0	RW	0x0000	mb_count_out Field0000 Description

VEPU_swreg39
Address: Operational Base + offset (0x0009c)
Base address for next pic

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	next_pic_addr Base address for next pic Base address for next pic luminance

VEPU_swreg40
Address: Operational Base + offset (0x000a0)
Stabilization minimum value

Bit	Attr	Reset Value	Description
31:30	RW	0x0	stab_mode Stabilization mode. 0=disabled. 1=stab only. 2=stab+encode
29:24	RO	0x0	reserved
23:0	RW	0x000000	stab_min_value Stabilization minimum value output. max 253*253*255

VEPU_swreg41
Address: Operational Base + offset (0x000a4)
Stabilization motion sum

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	stab_motion_sum Stabilization motion sum div8 output. max 53*253*255*1089/8

VEPU_swreg42
Address: Operational Base + offset (0x000a8)
stab_matrix1 and stab_gmv_hor

Bit	Attr	Reset Value	Description
31:26	RW	0x00	stab_gmv_hor Stabilization GMV horizontal output [-16..16]
25:24	RO	0x0	reserved
23:0	RW	0x000000	stab_matrix1 Stabilization matrix 1 (up-left position) output

VEPU_swreg43
Address: Operational Base + offset (0x000ac)
stab_matrix2 and stab_gmv_ver

Bit	Attr	Reset Value	Description
31:26	RW	0x00	stab_gmv_ver Stabilization GMV vertical output [-16..16]
25:24	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
23:0	RW	0x000000	stab_matrix2 Stabilization matrix 2 (up position) output

VEPU_swreg44
Address: Operational Base + offset (0x000b0)
stab_matrix3

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RW	0x000000	stab_matrix3 Stabilization matrix 3 (up-right position) output

VEPU_swreg45
Address: Operational Base + offset (0x000b4)
stab_matrix4

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RW	0x000000	stab_matrix4 Stabilization matrix 4 (left position) output

VEPU_swreg46
Address: Operational Base + offset (0x000b8)
stab_matrix5

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RW	0x000000	stab_matrix5 Stabilization matrix 5 (GMV position) output

VEPU_swreg47
Address: Operational Base + offset (0x000bc)
stab_matrix6

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RW	0x000000	stab_matrix6 Stabilization matrix 6 (right position) output

VEPU_swreg48
Address: Operational Base + offset (0x000c0)
stab_matrix7

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
23:0	RW	0x000000	stab_matrix7 Stabilization matrix 7 (down-left position) output

VEPU_swreg49
Address: Operational Base + offset (0x000c4)
stab_matrix8

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RW	0x000000	stab_matrix8 Stabilization matrix 8 (down position) output

VEPU_swreg50
Address: Operational Base + offset (0x000c8)
stab_matrix9

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RW	0x000000	stab_matrix9 Stabilization matrix 9 (down-right position) output

VEPU_swreg51
Address: Operational Base + offset (0x000cc)
cabac_table_addr

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cabac_table_addr Base address for cabac context tables (H264) or probability tables (VP8)

VEPU_swreg52
Address: Operational Base + offset (0x000d0)
Base address for MV output

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mv_out_addr Base address for MV output writing

VEPU_swreg53
Address: Operational Base + offset (0x000d4)
RGB to YUV conversion coefficien A and B

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	coeffB_RGB2YUV RGB to YUV conversion B RGB to YUV conversion coefficient B
15:0	RW	0x0000	coeffA_RGB2YUV RGB to YUV conversion coefficient A

VEPU_swreg54
Address: Operational Base + offset (0x000d8)
RGB to YUV conversion coefficien C and E

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	coeffE_RGB2YUV RGB to YUV conversion coefficient E
15:0	RW	0x0000	coeffC_RGB2YUV RGB to YUV conversion coefficient C

VEPU_swreg55
Address: Operational Base + offset (0x000dc)
RGB mask MSB bit

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:26	RW	0x00	Bmask_MST RGB B-component mask MSB bit position [0..31]
25:21	RW	0x00	Gmask_MST RGB G-component mask MSB bit position [0..31]
20:16	RW	0x00	Rmask_MST RGB R-component mask MSB bit position [0..31]
15:0	RW	0x0000	coeffF_RGB2YUV RGB to YUV conversion coefficien F

VEPU_swreg56
Address: Operational Base + offset (0x000e0)
intra area control register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	intra_left_mb Intra area left mb column (inside area) [0..255]
23:16	RW	0x00	intra_right_mb Intra area right mb column (inside area) [0..255]
15:8	RW	0x00	intra_top_mb Intra area top mb row (inside area) [0..255]

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Bit	Attr	Reset Value	Description
7:0	RW	0x00	intra_bot_mb Intra area bottom mb row (inside area) [0..255]

VEPU_swreg57
Address: Operational Base + offset (0x000e4)
CIR intra control reg

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	CIR_first_intra CIR first intra mb. 0=disabled [0..65535]
15:0	RW	0x0000	CIR_intra_mbinterval CIR intra mb interval. 0=disabled [0..65535]

VEPU_swreg58
Address: Operational Base + offset (0x000e8)
Intra slice bitmap for slices 0..31

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	intra_slice_bitmap Intra slice bitmap for slices 0..31. LSB=slice0. MSB=slice31. 1=intra.

VEPU_swreg59
Address: Operational Base + offset (0x000ec)
Intra slice bitmap for slices32..63

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	intra_slice_bitmap Intra slice bitmap for slices32..63. LSB=slice32. MSB=slice63. 1=intra.

VEPU_swreg60
Address: Operational Base + offset (0x000f0)
1st ROI area register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	ROI_1st_leftmb 1st ROI area left mb column (inside area) qp+=Roi1DeltaQp
23:16	RW	0x00	ROI_1st_rightmb 1st ROI area right mb column (outside area) qp-=Roi1DeltaQp
15:8	RW	0x00	ROI_1st_topmb 1st ROI area top mb row (inside area)

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Bit	Attr	Reset Value	Description
7:0	RW	0x00	ROI_1st_botmb 1st ROI area bottom mb row (outside area)

VEPU_swreg61
Address: Operational Base + offset (0x000f4)
Register0061 Abstract

Bit	Attr	Reset Value	Description
31:24	RW	0x00	ROI_2st_leftmb 2st ROI area left mb column (inside area) qp+=Roi1DeltaQp
23:16	RW	0x00	ROI_2st_topmb 2st ROI area top mb row (inside area)
15:8	RW	0x00	ROI_2st_rightmb 2st ROI area right mb column (outside area) qp-=Roi1DeltaQp
7:0	RW	0x00	ROI_2st_botmb 2st ROI area bottom mb row (outside area)

VEPU_swreg62
Address: Operational Base + offset (0x000f8)
MVC control reg

Bit	Attr	Reset Value	Description
31:28	RW	0x0	mv16x16_favor Zero 16x16 MV favor div2.
27:19	RW	0x000	penalty_4x4mv Penalty for using 4x4 MV.
18:16	RW	0x0	mvc_priority_id MVC priority_id [0..7]
15:13	RW	0x0	mvc_view_id MVC view_id [0..7]
12:10	RW	0x0	mvc_temporal_id MVC temporal_id [0..7]
9	RW	0x0	mvc_anchor_pic_flag MVC anchor_pic_flag. Specifies that the picture is part of an anchor access unit.
8	RW	0x0	mvc_inter_view_flag MVC inter_view_flag. Specifies that the picture is used for inter-view prediction.
7:0	RO	0x0	reserved

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VEPU_swreg63
Address: Operational Base + offset (0x000fc)
Register0063 Abstract

Bit	Attr	Reset Value	Description
31:12	RO	0x1f522	HW_CONFIG Field0000 Description
11:0	RO	0x780	MAX_VID_WIDTH Field0000 Description

VEPU_swreg64
Address: Operational Base + offset (0x00100)
JPEG luma quantization 1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_lum_qua1 JPEG luma quantization 1

VEPU_swreg65
Address: Operational Base + offset (0x00104)
JPEG luma quantization 2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_lum_qua2 JPEG luma quantization 2

VEPU_swreg66
Address: Operational Base + offset (0x00108)
JPEG luma quantization 3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_lum_qua3 JPEG luma quantization 3

VEPU_swreg67
Address: Operational Base + offset (0x0010c)
JPEG luma quantization 4

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_lum_qua4 JPEG luma quantization 4

VEPU_swreg68
Address: Operational Base + offset (0x00110)
JPEG luma quantization 5

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_lum_qua5 JPEG luma quantization 5

VEPU_swreg69

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Address: Operational Base + offset (0x00114)
JPEG luma quantization 6

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_lum_qua6 JPEG luma quantization 6 JPEG luma quantization 6

VEPU_swreg70
Address: Operational Base + offset (0x00118)
JPEG luma quantization 7

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_lum_qua7 JPEG luma quantization 7

VEPU_swreg71
Address: Operational Base + offset (0x0011c)
JPEG luma quantization 8

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_lum_qua8 JPEG luma quantization 8

VEPU_swreg72
Address: Operational Base + offset (0x00120)
JPEG luma quantization 9

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_lum_qua9 JPEG luma quantization 9

VEPU_swreg73
Address: Operational Base + offset (0x00124)
Register0073 Abstract

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_lum_qua10 JPEG luma quantization 10 JPEG luma quantization 10

VEPU_swreg74
Address: Operational Base + offset (0x00128)
JPEG luma quantization 11

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_lum_qua11 JPEG luma quantization 11

VEPU_swreg75
Address: Operational Base + offset (0x0012c)
JPEG luma quantization 12

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Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	jpeg_lum_qua12 JPEG luma quantization 12

VEPU_swreg76
Address: Operational Base + offset (0x00130)
JPEG luma quantization 13

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_lum_qua13 JPEG luma quantization 13

VEPU_swreg77
Address: Operational Base + offset (0x00134)
JPEG luma quantization 14

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_lum_qua14 JPEG luma quantization 14

VEPU_swreg78
Address: Operational Base + offset (0x00138)
JPEG luma quantization 15

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_lum_qua15 JPEG luma quantization 15

VEPU_swreg79
Address: Operational Base + offset (0x0013c)
JPEG luma quantization 16

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_lum_qua16 JPEG luma quantization 16

VEPU_swreg80
Address: Operational Base + offset (0x00140)
JPEG chroma quantization 1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_chrom_qua1 JPEG chroma quantization 1

VEPU_swreg81
Address: Operational Base + offset (0x00144)
JPEG chroma quantization 2

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_chrom_qua2 JPEG chroma quantization 2

VEPU_swreg82
Address: Operational Base + offset (0x00148)
JPEG chroma quantization 3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_chrom_qua3 JPEG chroma quantization 3

VEPU_swreg83
Address: Operational Base + offset (0x0014c)
JPEG chroma quantization 4

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_chrom_qua4 JPEG chroma quantization 4

VEPU_swreg84
Address: Operational Base + offset (0x00150)
JPEG chroma quantization 5

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_chrom_qua5 JPEG chroma quantization 5

VEPU_swreg85
Address: Operational Base + offset (0x00154)
JPEG chroma quantization 6

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_chrom_qua6 JPEG chroma quantization 6

VEPU_swreg86
Address: Operational Base + offset (0x00158)
JPEG chroma quantization 7

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_chrom_qua7 JPEG chroma quantization 7

VEPU_swreg87
Address: Operational Base + offset (0x0015c)
JPEG chroma quantization 8

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_chrom_qua8 JPEG chroma quantization 8

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VEPU_swreg88
Address: Operational Base + offset (0x00160)
JPEG chroma quantization 9

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_chrom_qua9 JPEG chroma quantization 9

VEPU_swreg89
Address: Operational Base + offset (0x00164)
JPEG chroma quantization 10

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_chrom_qua10 JPEG chroma quantization 10

VEPU_swreg90
Address: Operational Base + offset (0x00168)
JPEG chroma quantization 11

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_chrom_qua11 JPEG chroma quantization 11

VEPU_swreg91
Address: Operational Base + offset (0x0016c)
JPEG chroma quantization 12

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_chrom_qua12 JPEG chroma quantization 12

VEPU_swreg92
Address: Operational Base + offset (0x00170)
JPEG chroma quantization 13

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_chrom_qua13 JPEG chroma quantization 13

VEPU_swreg93
Address: Operational Base + offset (0x00174)
JPEG chroma quantization 14

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_chrom_qua14 JPEG chroma quantization 14

VEPU_swreg94
Address: Operational Base + offset (0x00178)
JPEG chroma quantization 15

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_chrom_qua15 JPEG chroma quantization 15

VEPU_swreg95
Address: Operational Base + offset (0x0017c)
JPEG chroma quantization 16

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	jpeg_chrom_qua16 JPEG chroma quantization 16

VEPU_swreg96
Address: Operational Base + offset (0x00180)
DMV 4p/1p penalty values 0-3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values 0-3

VEPU_swreg97
Address: Operational Base + offset (0x00184)
DMV 4p/1p penalty values 4-7

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values 4-7

VEPU_swreg98
Address: Operational Base + offset (0x00188)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg99
Address: Operational Base + offset (0x0018c)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg100
Address: Operational Base + offset (0x00190)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

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VEPU_swreg101
Address: Operational Base + offset (0x00194)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg102
Address: Operational Base + offset (0x00198)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg103
Address: Operational Base + offset (0x0019c)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg104
Address: Operational Base + offset (0x001a0)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg105
Address: Operational Base + offset (0x001a4)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg106
Address: Operational Base + offset (0x001a8)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg107
Address: Operational Base + offset (0x001ac)
DMV 4p/1p penalty values

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg108
Address: Operational Base + offset (0x001b0)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg109
Address: Operational Base + offset (0x001b4)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg110
Address: Operational Base + offset (0x001b8)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg111
Address: Operational Base + offset (0x001bc)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg112
Address: Operational Base + offset (0x001c0)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg113
Address: Operational Base + offset (0x001c4)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

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VEPU_swreg114
Address: Operational Base + offset (0x001c8)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg115
Address: Operational Base + offset (0x001cc)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg116
Address: Operational Base + offset (0x001d0)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg117
Address: Operational Base + offset (0x001d4)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg118
Address: Operational Base + offset (0x001d8)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg119
Address: Operational Base + offset (0x001dc)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg120
Address: Operational Base + offset (0x001e0)
DMV 4p/1p penalty values

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg121
Address: Operational Base + offset (0x001e4)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg122
Address: Operational Base + offset (0x001e8)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg123
Address: Operational Base + offset (0x001ec)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg124
Address: Operational Base + offset (0x001f0)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg125
Address: Operational Base + offset (0x001f4)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg126
Address: Operational Base + offset (0x001f8)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

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VEPU_swreg127
Address: Operational Base + offset (0x001fc)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg128
Address: Operational Base + offset (0x00200)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg129
Address: Operational Base + offset (0x00204)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg130
Address: Operational Base + offset (0x00208)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg131
Address: Operational Base + offset (0x0020c)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg132
Address: Operational Base + offset (0x00210)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg133
Address: Operational Base + offset (0x00214)
DMV 4p/1p penalty values

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg134
Address: Operational Base + offset (0x00218)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg135
Address: Operational Base + offset (0x0021c)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg136
Address: Operational Base + offset (0x00220)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg137
Address: Operational Base + offset (0x00224)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg138
Address: Operational Base + offset (0x00228)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg139
Address: Operational Base + offset (0x0022c)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

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VEPU_swreg140
Address: Operational Base + offset (0x00230)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg141
Address: Operational Base + offset (0x00234)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg142
Address: Operational Base + offset (0x00238)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg143
Address: Operational Base + offset (0x0023c)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg144
Address: Operational Base + offset (0x00240)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg145
Address: Operational Base + offset (0x00244)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg146
Address: Operational Base + offset (0x00248)
DMV 4p/1p penalty values

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg147
Address: Operational Base + offset (0x0024c)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg148
Address: Operational Base + offset (0x00250)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg149
Address: Operational Base + offset (0x00254)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg150
Address: Operational Base + offset (0x00258)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg151
Address: Operational Base + offset (0x0025c)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg152
Address: Operational Base + offset (0x00260)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

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VEPU_swreg153
Address: Operational Base + offset (0x00264)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg154
Address: Operational Base + offset (0x00268)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg155
Address: Operational Base + offset (0x0026c)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg156
Address: Operational Base + offset (0x00270)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg157
Address: Operational Base + offset (0x00274)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg158
Address: Operational Base + offset (0x00278)
DMV 4p/1p penalty values

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values

VEPU_swreg159
Address: Operational Base + offset (0x0027c)
DMV 4p/1p penalty values

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMV_4p_1p_penalty DMV 4p/1p penalty values 124-127

VEPU_swreg160
Address: Operational Base + offset (0x00280)
vp8 control

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:12	RW	0x000	vp8_coeff_dmv_penalty VP8 coeff for dmv penalty for intra/inter selection
11:0	RW	0x000	vp8_inter_type VP8 bit cost of inter type

VEPU_swreg161
Address: Operational Base + offset (0x00284)
VP8 bit cost of golden ref frame

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	vp8_ref_frame VP8 bit cost of golden ref frame

VEPU_swreg162
Address: Operational Base + offset (0x00288)
vp8 loop filter delta registers

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:21	RW	0x00	vp8_loopfilter_altref VP8 loop filter delta for alt ref
20:14	RW	0x00	vp8_loopfilter_goldenref VP8 loop filter delta for golden ref
13:7	RW	0x00	vp8_loopfilter_lastref VP8 loop filter delta for last ref
6:0	RW	0x00	vp8_loopfilter_intra VP8 loop filter delta for intra mb

VEPU_swreg163
Address: Operational Base + offset (0x0028c)
vp8 loop filter delta register

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:21	RW	0x00	vp8_loopfilter_splitmv VP8 loop filter delta for SPLITMV

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Bit	Attr	Reset Value	Description
20:14	RW	0x00	vp8_loopfilter_newmv VP8 loop filter delta for NEWMV
13:7	RW	0x00	vp8_loopfilter_zeromv VP8 loop filter delta for ZEROMV
6:0	RW	0x00	vp8_loopfilter_bpred VP8 loop filter delta for BPRED

26.4.9 VDPU Register Summary

Name	Offset	Size	Reset Value	Description
VDPU_SWREG0	0x00000	W	0x67313688	ID register(read only)
VDPU_SWREG1	0x00004	W	0x00000000	interrupt register decoder
VDPU_SWREG2	0x00008	W	0x00000400	device configuration register decoder
VDPU_SWREG3	0x0000c	W	0x00000000	Device control register 0(decmode, picture type etc)
VDPU_SWREG4	0x00010	W	0x00000000	decoder control register 1(picture parameters)
VDPU_SWREG5	0x00014	W	0x00000000	decoder control register2 (stream decoding table selects)
VDPU_SWREG6	0x00018	W	0x00000000	decoder control register 3(stream buffer information)
VDPU_SWREG7	0x0001c	W	0x00000000	decoder control register 4(H264, VC-1,VP6 control)
VDPU_SWREG8	0x00020	W	0x00000000	decoder control register 5(H264, VC-1,VP6 and RV control)
VDPU_SWREG9	0x00024	W	0x00000000	decoder control register 6
VDPU_SREG10	0x00028	W	0x00000000	Base address for differential motion vector base address
VDPU_SWREG11	0x0002c	W	0x00000000	decoder control register 7
VDPU_SWREG12	0x00030	W	0x00000000	Base address for RLC data (RLC) / stream start address/decoded
VDPU_SWREG13	0x00034	W	0x00000000	Base address for decoded picture / base address for JPEG deco
VDPU_SWREG14	0x00038	W	0x00000000	Base address for reference picture index 0 / base address for J
VDPU_SWREG15	0x0003c	W	0x00000000	Base address for reference picture index 1 / JPEG control
VDPU_SWREG15_JPEG_ROI	0x0003c	W	0x00000000	JPEG roi control
VDPU_SWREG16	0x00040	W	0x00000000	base address for reference picture index 2 / List of VLC code len
VDPU_SWREG17	0x00044	W	0x00000000	Base address for reference picture index 3 / List of VLC code le

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Name	Offset	Size	Reset Value	Description
VDPU_SWREG18	0x00048	W	0x00000000	Base address for reference picture index 4 / VC1 control / MPE
VDPU_SWREG19	0x0004c	W	0x00000000	Base address for reference picture index 5
VDPU_SWREG20	0x00050	W	0x00000000	Base address for reference picture index 6
VDPU_SWREG21	0x00054	W	0x00000000	Base address for reference picture index 7
VDPU_SWREG22	0x00058	W	0x00000000	Base address for reference picture index 8
VDPU_SWREG23	0x0005c	W	0x00000000	Base address for reference picture index 9
VDPU_SWREG24	0x00060	W	0x00000000	Base address for reference picture index 10
VDPU_SWREG25	0x00064	W	0x00000000	Base address for reference picture index 11
VDPU_SWREG26	0x00068	W	0x00000000	Base address for reference picture index 12
VDPU_SWREG27	0x0006c	W	0x00000000	Base address for reference picture index 13
VDPU_SWREG28	0x00070	W	0x00000000	Base address for reference picture index14
VDPU_SWREG29	0x00074	W	0x00000000	Base address for reference picture index15
VDPU_SWREG30	0x00078	W	0x00000000	Reference picture numbers for index 0 and 1 (H264 VLC
VDPU_SWREG31	0x0007c	W	0x00000000	Reference picture numbers for index 2 and 3 (H264 VLC) /
VDPU_SWREG32	0x00080	W	0x00000000	Reference picture numbers for index 4 and 5 (H264 VLC)
VDPU_SWREG33	0x00084	W	0x00000000	Reference picture numbers for index 6 and 7 (H264 VLC)
VDPU_SWREG34	0x00088	W	0x00000000	Reference picture numbers for index 8 and 9 (H264 VLC)
VDPU_SWREG35	0x0008c	W	0x00000000	Reference picture numbers for index 10 and 11 (H264 VLC)
VDPU_SWREG35_JPEG_ROI	0x0008c	W	0x00000000	JPEG roi offest/dc base address
VDPU_SWREG36	0x00090	W	0x00000000	Reference picture numbers for index 12 and 13 (H264 VLC)
VDPU_SWREG36_JPEG_ROI	0x00090	W	0x00000000	JPEG roi offset/dc length
VDPU_SWREG37	0x00094	W	0x00000000	Reference picture numbers for index 14 and 15 (H264 VLC)
VDPU_SWREG38	0x00098	W	0x00000000	Reference picture long term flags (H264 VLC) / VPx prediction filt
VDPU_SWREG39	0x0009c	W	0x00000000	Reference picture valid flags (H264 VLC) /VPx prediction filter ta
VDPU_SWREG40	0x000a0	W	0x00000000	Base address for standard dependent tables
VDPU_SWREG41	0x000a4	W	0x00000000	Base address for direct mode motion vectors
VDPU_SWREG42	0x000a8	W	0x00000000	bi_dir initial ref pic list register (0-2)/ VP6 prediction filter taps
VDPU_SWREG43	0x000ac	W	0x00000000	bi-dir initial ref pic list register (3-5)/ VP6 prediction filter taps

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Name	Offset	Size	Reset Value	Description
VDPU_SWREG44	0x000b0	W	0x00000000	bi-dir initial ref pic list register (6-8)/ VP6 prediction filter taps
VDPU_SWREG45	0x000b4	W	0x00000000	bi-dir initial ref pic list register (9- 11) / VP6 prediction filter taps
VDPU_SWREG46	0x000b8	W	0x00000000	bi-dir initial ref pic list register (12- 14) / VP7,VP8 quantization v
VDPU_SWREG47	0x000bc	W	0x00000000	bi-dir and P fwd initial ref pic list register (15 and P 0-3) / VP7,V
VDPU_SWREG48	0x000c0	W	0x00000000	Error concealment register
VDPU_SWREG49	0x000c4	W	0x00000000	Prediction filter tap register for H264, MPEG4, VC1, VP6
VDPU_SWREG50	0x000c8	W	0xfbb56f80	Synthesis configuration register decoder 0 (read only)
VDPU_SWREG51	0x000cc	W	0x00000000	Reference picture buffer control register
VDPU_SWREG52	0x000d0	W	0x00000000	Reference picture buffer information register 1 (read only)
VDPU_SWREG53	0x000d4	W	0x00000000	Reference picture buffer information register 2 (read only)
VDPU_SWREG54	0x000d8	W	0xe5da0000	Synthesis configuration register decoder 1 (read only)
VDPU_SWREG55	0x000dc	W	0x00000000	Reference picture buffer 2 / Advanced prefetch control register
VDPU_SWREG56	0x000e0	W	0x00000000	Reference buffer information register 3 (read only)
VDPU_SWREG57_INTRA_INTER	0x000e4	W	0x00000000	intra_dll3t,intra_dblspeed,inter_dblspeed,stream_len_hi
VDPU_SWREG58	0x000e8	W	0x00000000	Decoder debug register 0 (read only)
VDPU_SWREG59	0x000ec	W	0x00000000	H264 Chrominance 8 pixel interleaved data base
VDPU_SWREG60	0x000f0	W	0x00000000	Interrupt register post-processor
VDPU_SWREG61	0x000f4	W	0x00000000	Device configuration register post-processor
VDPU_SWREG62	0x000f8	W	0x00000000	Deinterlace control register
VDPU_SWREG63	0x000fc	W	0x00000000	base address for reading post-processing input picture uminan
VDPU_SWREG64	0x00100	W	0x00000000	Base address for reading post-processing input picture Cb/Ch
VDPU_SWREG65	0x00104	W	0x00000000	Base address for reading post-processing input picture Cr
VDPU_SWREG66	0x00108	W	0x00000000	Base address for writing post-processed picture luminance/RGB
VDPU_SWREG67	0x0010c	W	0x00000000	Base address for writing post-processed picture Ch
VDPU_SWREG68	0x00110	W	0x00000000	Register for contrast adjusting

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Name	Offset	Size	Reset Value	Description
VDPU_SWREG69	0x00114	W	0x00000000	Register for colour conversion and contrast adjusting
VDPU_SWREG70	0x00118	W	0x00000000	Register for colour conversion 0
VDPU_SWREG71	0x0011c	W	0x00000000	Register for colour conversion 1 + rotation mode
VDPU_SWREG72	0x00120	W	0x00000000	PP input size and -cropping register
VDPU_SWREG73	0x00124	W	0x00000000	PP input picture base address for Y bottom field
VDPU_SWREG74	0x00128	W	0x00000000	PP input picture base for Ch bottom field
VDPU_SWREG79	0x0013c	W	0x00000000	Scaling ratio register 1 & padding for B
VDPU_SWREG80	0x00140	W	0x00000000	Scaling register 0 ratio & padding for R and G
VDPU_SWREG81	0x00144	W	0x00000000	Scaling ratio register 2
VDPU_SWREG82	0x00148	W	0x00000000	Rmask register
VDPU_SWREG83	0x0014c	W	0x00000000	Gmask register
VDPU_SWREG84	0x00150	W	0x00000000	Bmask register
VDPU_SWREG85	0x00154	W	0x00000000	Post-processor control register
VDPU_SWREG86	0x00158	W	0x00000000	Mask 1 start coordinate register
VDPU_SWREG87	0x0015c	W	0x00000000	Mask 2 start coordinate register
VDPU_SWREG88	0x00160	W	0x00000000	Mask 1 size and PP original width register
VDPU_SWREG89	0x00164	W	0x00000000	Mask 2 size register
VDPU_SWREG90	0x00168	W	0x00000000	PiP register 0
VDPU_SWREG91	0x0016c	W	0x00000000	PiP register 1 and dithering control
VDPU_SWREG92	0x00170	W	0x00000000	Display width and PP input size extension register
VDPU_SWREG93	0x00174	W	0x00000000	Display width and PP input size extension register
VDPU_SWREG94	0x00178	W	0x00000000	Base address for alpha blend 2 gui component
VDPU_SWREG95	0x0017c	W	0x00000000	Base address for alpha blend 2 gui component
VDPU_SWREG98	0x00188	W	0x00000000	PP outupt width/height extension
VDPU_SWREG99	0x0018c	W	0xe000f000	PP fuse register (read only)
VDPU_SWREG100	0x00190	W	0xff874000	Synthesis configuration register post-processor (read only)
VDPU_SWREG101	0x00194	W	0x00000000	soft reset signals

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

26.4.10 VDPU Detail Register Description

VDPU_SWREG0

Address: Operational Base + offset (0x00000)

ID register(read only)

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
27:24	RO	0x3	major_version 0:1080p support 1:2160p support
23:16	RO	0x0	reserved
15:0	RW	0x0000	build the rtl's svn num in ic server

VDPU_SWREG1
Address: Operational Base + offset (0x00004)
interrupt register decoder

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	sw_dec_pic_inf B slice detected. This signal is driven high during picture ready interrupt if B-type slice is found. This bit does not launch interrupt but is used to inform SW about h264 tools. DIVX3: For DIVX3 this bit tells the value of extension header flag (flag called FLIPFLOP)
23:19	RO	0x0	reserved
18	RW	0x0	sw_dec_timeout Interrupt status bit decoder timeout. When high the decoder 0 has been idling for too long. HW will self reset. Possible only if timeout interrupt is enabled
17	RW	0x0	sw_dec_slice_int Interrupt status bit dec_slice_decoded. When high SW must set new base addresses for sw_dec_out_base and sw_jpg_ch_out_base before reseting this status bit. Used for JPEG and VP8 snapshot modes
16	RW	0x0	sw_dec_error_int Interrupt status bit input stream error. When high, an error is found in input data stream decoding. HW will self reset.
15	RW	0x0	sw_dec_aso_int Interrupt status bit ASO (Arbitrary Slice Ordering) detected.When high, ASO detected in input data stream decoding. HW will self reset.

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Bit	Attr	Reset Value	Description
14	RW	0x0	sw_dec_buffer_int Interrupt status bit input buffer empty. When high, input stream buffer is empty but picture is not ready. HW will not self reset.
13	RW	0x0	sw_dec_bus_int Interrupt status bit bus. Error response from bus. HW will self reset
12	RW	0x0	sw_dec_rdy_int Interrupt status bit decoder. When this bit is high decoder has decoded a picture. HW will self reset.
11:9	RO	0x0	reserved
8	RW	0x0	sw_dec_irq Decoder IRQ. When high, decoder requests an interrupt. SW will reset this after interrupt is handled.
7:5	RO	0x0	reserved
4	RW	0x0	sw_dec_irq_dis Decoder IRQ disable. When high, there are no interrupts concerning decoder from HW. Polling must be used to see the interrupt statuses.
3:1	RO	0x0	reserved
0	RW	0x0	sw_dec_en decoder enable. Decoder enable. Setting this bit high will start the decoding operation. HW will reset this when picture is processed or ASO or stream error is detected or bus error or timeout interrupt is given.

VDPU_SWREG2
Address: Operational Base + offset (0x00008)
device configuration register decoder

Bit	Attr	Reset Value	Description
31:24	RW	0x00	sw_dec_axi_rd_id Read ID used for decoder reading services in AXI bus (if connected to AXI)

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Bit	Attr	Reset Value	Description
23	RW	0x0	sw_dec_timeout_e Timeout interrupt enable. If enabled HW may return timeout interrupt in case HW gets stucked while decoding picture.
22	RW	0x0	sw_dec_strswap32_e Decoder input 32bit data swap for stream data (may be used for 64 bit environment): 0 = no swapping of 32 bit words 1 = 32 bit data words are swapped (needed in 64 bit environment to achieve 7-6-5-4-3-2-1-0 byte order(also little endian should be enabled))
21	RW	0x0	sw_dec_strendian_e Decoder input endian mode for stream data: 0 = Big endian (0-1-2-3 order) 1 = Little endian (3-2-1-0 order)
20	RW	0x0	sw_dec_inswap32_e Decoder input 32bit data swap for other than stream data (may be used for 64 bit environment): 0 = no swapping of 32 bit words 1 = 32 bit data words are swapped (needed in 64 bit environment to achieve 7-6-5-4-3-2-1-0 byte order(also little endian should be enabled))
19	RW	0x0	sw_dec_outswap32_e Decoder output 32bit data swap (may be used for 64 bit environment): 0 = no swapping of 32 bit words 1 = 32 bit data words are swapped (needed in 64 bit environment to achieve 7-6-5-4-3-2-1-0 byte order(also little endian should be enabled))
18	RW	0x0	sw_dec_data_disc_e Data discard enable. Precise burst lengths are used with reading services. Extra data is discarded internally.

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Bit	Attr	Reset Value	Description
17	RW	0x0	sw_tiled_mode_msb Tiled mode msb. Concanated to Tiled mode lsb which form 2 bit tiled mode. Definition of tiledmode: 0 = Tiled mode not enabled 1 = Tiled mode enabled for 8x4 tile size 2,3 Reserved
16:11	RW	0x00	sw_dec_latency Decoder master interface additional latency. Can be used to slow down decoder HW between services in steps of 8 clock cycles: 0 = no latency 1 = minimum 8 cycles of IDLE between services 2 = minimum 16 cycles of IDLE between services ... 63 = minimum latency of 504 cycles of IDLE between services
10	RW	0x1	sw_dec_clk_gate_e Decoder dynamic clock gating enable: 0 = Clock is running for all structures 1 = Clock is gated for decoder structures that are not used Note: Clock gating value can be changed only when decoder is disabled
9	RW	0x0	sw_dec_in_endian Decoder input endian mode for other than stream data: 0 = Big endian (0-1-2-3 order) 1 = Little endian (3-2-1-0 order)
8	RW	0x0	sw_dec_out_endian Decoder output endian mode: 0 = Big endian (0-1-2-3 order) 1 = Little endian (3-2-1-0 order)
7	RW	0x0	sw_tiled_mode_lsb Tiled mode lsb. Concanated to Tiled mode msb which form 2 bit tiled mode. Defined in tiled_mode_msb
6	RW	0x0	sw_dec_adv_pre_dis Advanced PREFETCH mode disable (advanced reference picture reading mode for video)

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Bit	Attr	Reset Value	Description
5	RW	0x0	sw_dec_scmd_dis AXI Single Command Multiple Data 0 disable. (where only the first addresses of the burst are given from address generator). This bit is used to disable the feature (possible SW workaround if something is not working correctly)
4:0	RW	0x00	sw_dec_max_burst Maximum burst length for decoder bus transactions. Valid values: AXI: 1-16

VDPU_SWREG3
Address: Operational Base + offset (0x0000c)
Device control register 0(decmode, picture type etc)

Bit	Attr	Reset Value	Description
31:28	RW	0x0	sw_dec_mode Decoding mode: 0 = H.264, 1 = MPEG-4, 2 = H.263, 3 = JPEG, 4 = VC-1, 5 = MPEG-2, 6 = MPEG-1, 7 = VP6, 8 = RV, 9 = VP7, 10 = VP8, 11 = AVS, others = reserved
27	RW	0x0	sw_rlc_mode_e RLC mode enable: 1 = HW decodes video from RLC input data + side information (Differential MV's, separate DC coeffs, Intra 4x4 modes, MB control). Valid only for H.264 Baseline and MPEG- 4 SP. 0 = HW decodes video from bit stream (VLC mode) + side information (bitplane data in VC-1)

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Bit	Attr	Reset Value	Description
26	RW	0x0	sw_skip_mode MPEG4: Valid if divx is enabled AVS: 0: special MB type code indicates skipped mbs 1 means that skipped mbs are indicaed using skip_run -syntax elememnt like in H264 VP8: 0 : HW decodes mb_coeff_skip -flag 1 : HW does not decode mb_coeff_skip -flag
25	RW	0x0	sw_divx3_e DIVX3 enable (possible if sw_dec_mode is MPEG4): 0 = disabled 1 = enabled
24	RW	0x0	sw_pjpeg_e Progressive JPEG enable: 0 = baseline JPEG 1 = progressive JPEG
23	RW	0x0	sw_pic_interlace_e Coding mode of the current picture: 0 = progressive 1 = interlaced
22	RW	0x0	sw_pic_fieldmode_e Structure of the current picture (residual structure) 0 = frame structure, this means MBAFF structured picture for interlaced sequence 1 = field structure
21	RW	0x0	sw_pic_b_e B picture enable for current picture: 0=picture type is I or P depending on sw_pic_inter_e 1=picture type is BI (vc1)/D (mpeg1) or B depending on sw_pic_inter_e (not valid for H264 since its slice based information)
20	RW	0x0	sw_pic_inter_e Picture type. 1= Inter type (P) 0= Intra type (I) See also sw_pic_b_e
19	RW	0x0	sw_pic_topfield_e If field structure is enabled this bit informs which one of the fields is being decoded: 0 = bottom field 1 = top field

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Bit	Attr	Reset Value	Description
18	RW	0x0	sw_fwd_interlace_e Coding mode of forward reference picture: 0 = progressive 1 = interlaced Note: for backward reference picture the coding mode is always same as for current picture.
17	RW	0x0	sw_sorenson_e Sorenson Sparc enable (possible if sw_dec_mode is MPEG- 4) 0 = disabled 1 = H.263 compatible stream with Sorenson escape coding
16	RW	0x0	sw_ref_topfield_e Indicates which field should be used as reference if sw_ref_frames = 0 : 0 = bottom field 1 = top field used only in VC-1 mode
15	RW	0x0	sw_dec_out_dis Disable decoder output picture writing: 0 = Decoder output picture is written to external memory 1 = Decoder output picture is not written to external memory
14	RW	0x0	sw_filtering_dis De-block filtering disable 1 = filtering is disabled for current picture 0 = filtering is enabled for current picture
13	RW	0x0	sw_pic_fixed_quant sw_pic_fixed_quant (DEC mode is VC-1 and AVS) 0 = Quantization parameter can vary inside picture 1 = Quantization parameter is fixed (pquant) sw_mvc_e(DEC mode is H264) multi view coding enable. Possible for H264 only

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Bit	Attr	Reset Value	Description
12	RW	0x0	sw_write_mvs_e Direct mode motion vector write enable for current picture / MPEG2 and VP6 motion vector write enable for error concealment purposes: 0 = writing disabled for current picture 1 = the direct mode motion vectors are written to extrenal memory. H264 direct mode motion vectors are written to DPB aside with the corresponding reference picture. Other decoding mode dir mode mvs are written to external memory starting from sw_dir_mv_base
11	RW	0x0	sw_reftopfirst_e Indicates which FWD reference field has been decoded first. 0 = FWD reference bottom field 1 = FWD reference top field
10	RW	0x0	sw_seq_mbaff_e Sequence includes MBAFF coded pictures
9	RW	0x0	sw_picord_count_e h264_high config: Picture order count table read enable. If enabled HW will read picture order counts from memory in the beginning of picture
8	RO	0x0	reserved
7:0	RW	0x00	sw_dec_axi_wr_id Write ID used for decoder writing services in AXI bus (if connected to AXI)

VDPU_SWREG4
Address: Operational Base + offset (0x00010)
decoder control register 1(picture parameters)

Bit	Attr	Reset Value	Description
31:23	RW	0x000	sw_pic_mb_width Picture width in macroblocks = ((width in pixels + 15) /16)
22:19	RW	0x0	sw_mb_width_off The amount of meaningfull horizontal pixels in last MB (width offset) 0 if exactly 16 pixels multiple picture and all the horizontal pixels in last MB are meaningfull

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Bit	Attr	Reset Value	Description
18:11	RW	0x00	sw_pic_mb_height_p Picture height in macroblocks =((height in pixels+15)/16). Picture height is informed as size of the (progressive) frame also for single field (of interlaced content) is being decoded
10:7	RW	0x0	sw_mb_height_off The amount of menaingfull vertical pixels in last MB (height offset 0 if exactly 16 pixels multiple picture and all the vertical pixels in last MB are meaningfull
6	RW	0x0	sw_alt_scan_e indicates alternative vertical scan method used for interlacedd frames
5:3	RW	0x0	sw_pic_mb_w_ext Picture mb width extension. If sw_pic_mb_width does not fit tod 9 bits then these bits are used to increase the range upto 11 bits (used as 3 msb)
2:0	RW	0x0	sw_pic_mb_h_ext Picture mb height extension. If sw_pic_mb_height_p does not fit to 9 bits then these bits are used to increase the range upto 11 bits (used as 3 msb)

VDPU_SWREG5
Address: Operational Base + offset (0x00014)
decoder control register2 (stream decoding table selects)

Bit	Attr	Reset Value	Description
31:26	RW	0x00	sw_strm_start_bit Exact bit of stream start word where decoding can be started (assosiates with sw_rlc_vlc_base)
25	RW	0x0	sw_sync_marker_e Sync markers enable: '0' = synch markers are not used, '1' = synch markers are used. For progressive JPEG this indicates that there are restart markers in the stream after restart interval steps

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Bit	Attr	Reset Value	Description
24	RW	0x0	sw_type1_quant_e MPEG4: Type 1 quantization enable '0' = type 2 inverse Q method '1' = type 1 inverse Q method (Q-tables used) H264 (h264_high config:): scaling matrix enable: '0' = normal transform '1' = use scaling matrix for transform (read from external
23:19	RW	0x00	sw_ch_qp_offset Chroma Qp filter offset. (For H.264 this offset concerns Cb only)
18:14	RW	0x00	sw_ch_qp_offset2 Chroma Qp filter offset for cr type
13:1	RO	0x0	reserved
0	RW	0x0	sw_fieldpic_flag_e Flag for streamd that field_pic_flag exists in stream

VDPU_SWREG6
Address: Operational Base + offset (0x00018)
decoder control register 3(stream buffer information)

Bit	Attr	Reset Value	Description
31	RW	0x0	sw_start_code_e Bit for indicating stream start code existence: '0' = stream doesn't contain start codes '1' = stream contains start codes
30:25	RW	0x00	sw_init_qp Initial value for quantization parameter (picture quantizer).
24	RW	0x0	sw_ch_8pix_ileav_e Enable for additional chrominance data format writing where decoder writes chrominance in group of 8 pixels of Cb and then corresponding 8 pixels of Cr. Data is written to sw_dec_ch8pix_base. Cannot be used if tiled mode is enabled

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Bit	Attr	Reset Value	Description
23:0	RW	0x000000	sw_stream_len Amount of stream data bytes in input buffer. If the given buffer size is not enough for finishing the picture the corresponding interrupt is given and new stream buffer base address and stream buffer size information should be given (assosiates with sw_rlc_vlc_base). For VC-1/VP6 the buffer must include data for one picture/slice of the picture For H264/MPEG4/H263/MPEG2/MPEG1 the buffer must include at least data for one slice/VP of the picture For JPEG the buffer size must be a multiple of 256 bytes or the amount of data for onepicture.

VDPU_SWREG7
Address: Operational Base + offset (0x0001c)
decoder control register 4(H264, VC-1,VP6 control)

Bit	Attr	Reset Value	Description
31	RW	0x0	sw_cabac_e CABAC enable
30	RW	0x0	sw_blackwhite_e '0' = 4:2:0 sampling format '1' = 4:0:0 sampling format (H264 monochroma)
29	RW	0x0	sw_dir_8x8_infer_e Specifies the method to use to derive luma motion vectors in B_skip, B_Direct_16x16 and B_direct_8x8_inference_flag (see direct_8x8_inference flag)
28	RW	0x0	sw_weight_pred_e Weighted prediction enable for P slices
27:26	RW	0x0	sw_weight_bipr_idc weighted prediction specification for B slices: "00" = default weighted prediction is aplied to B slices "01" = explicit weighted prediction shall be applied to B slices "10" = implicit weighted prediction shall be applied to B slices
25:21	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
20:16	RW	0x00	sw_framenum_len H.264: Bit length of frame_num in data stream RV: frame size length. Informs how many bits in stream are used for frame size (HW discards these bits)
15:0	RW	0x0000	sw_framenum current frame_num, used to identify short-term reference frames. Used in reference picture reordering

VDPU_SWREG8
Address: Operational Base + offset (0x00020)
decoder control register 5(H264, VC-1,VP6 and RV control)

Bit	Attr	Reset Value	Description
31	RW	0x0	sw_const_intra_e constrained_intra_pred_flag equal to 1 specifies that intra prediction uses only neighbouring intra macroblocks in prediction. When equal to 0 also neighbouring inter macroblocks are used in intra prediction process.
30	RW	0x0	sw_filt_ctrl_pres deblocking_filter_control_present_flag indicates whether extra variables controlling characteristics of the deblocking filter are present in the slice header.
29	RW	0x0	sw_rdpic_cnt_pres redundant_pic_cnt_present_flag specifies whether redundant_pic_cnt syntax elements
28	RW	0x0	sw_8x8trans_flag_e 8x8 transform flag enable for stream decoding
27:17	RW	0x000	sw_refpic_mk_len Length of decoded reference picture marking bits
16	RW	0x0	sw_idr_pic_e IDR (instantaneous decoding refresh) picture flag.
15:0	RW	0x0000	sw_idr_pic_id idr_pic_id, identifies IDR (instantaneous decoding refresh) picture

VDPU_SWREG9
Address: Operational Base + offset (0x00024)
decoder control register 6

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	sw_pps_id pic_parameter_set_id, identifies the picture parameter set that is referred to in the slice header.
23:19	RW	0x00	sw_refidx1_active pecifies the maximum reference index that can be used while decoding inter predicted macro blocks.
18:14	RW	0x00	sw_refidx0_active Specifies the maximum reference index that can be used while decoding inter predicted macro blocks. This is same as in previous decoders (width increased with q bit)
13:8	RO	0x0	reserved
7:0	RW	0x00	sw_poc_length Length of picture order count field in stream

VDPU_SREG10
Address: Operational Base + offset (0x00028)
Base address for differential motion vector base address

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_diff_mv_base for H264 and MPEG4, RLC mode: Differential motion vector base address.
1:0	RO	0x0	reserved

VDPU_SWREG11
Address: Operational Base + offset (0x0002c)
decoder control register 7

Bit	Attr	Reset Value	Description
31:30	RW	0x0	sw_i4x4_or_dc_base RLC mode: H.264: Intra prediction 4x4 mode base address. RLC mode: MPEG-4: DC component base address.
29:25	RW	0x00	sw_pinit_rlist_f15 Initial reference picture list for P forward picid 15
24:20	RW	0x00	sw_pinit_rlist_f14 Initial reference picture list for P forward picid 14
19:15	RW	0x00	sw_pinit_rlist_f13 Initial reference picture list for P forward picid 13

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Bit	Attr	Reset Value	Description
14:10	RW	0x00	sw_pint_rlist_f12 Initial reference picture list for P forward picid 12
9:5	RW	0x00	sw_pint_rlist_f11 Initial reference picture list for P forward picid 11
4:0	RW	0x00	sw_pint_rlist_f10 Initial reference picture list for P forward picid 10

VDPU_SWREG12
Address: Operational Base + offset (0x00030)
Base address for RLC data (RLC) / stream start address/decoded

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_rlc_vlc_base RLC mode: Base address for RLC data (swreg3.sw_rlc_mode_e = 1). VLC mode: Stream start address / end addr+I288ess with byte precision (swreg4.rlc_mode_en = 0), start bit number in swreg5.stream_start_bit. When sw_dec_buffer_int is high or sw_dec_e is low this register contains HW return value of last_byte_address (not valid for jpeg) where stream has been read (and used) in accuracy of byte. For debug purposses the last_byte_address is also written when stream error/ASO is detected even though it may not be accurate. VP7/VP8: This base address is used as sw_dct_strm0_base including DCT stream for MB rows 0,2n
1:0	RO	0x0	reserved

VDPU_SWREG13
Address: Operational Base + offset (0x00034)
Base address for decoded picture / base address for JPEG deco

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_dec_out_base Video: Base address for decoder output picture. Points directly to start of decoder output picture or field. JPEG/VP8 snapshot: Base address for decoder output luminance picture
1:0	RO	0x0	reserved

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VDPU_SWREG14
Address: Operational Base + offset (0x00038)
Base address for reference picture index 0 / base address for J

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_refer0_base Base address for reference picture index 0. See picture index definition from toplevel_sp
1	RW	0x0	sw_refer0_field_e Refer picture consist of single fields or frame: '0' = reference picture consists of frame '1' = reference picture consists of fields
0	RW	0x0	sw_refer0_topc_e Which field of reference picture is closer to current picture: '0' = bottom field is closer to current picture '1' = top field is closer to current picture

VDPU_SWREG15
Address: Operational Base + offset (0x0003c)
Base address for reference picture index 1 / JPEG control

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_refer1_base Base address for reference picture index 1. See picture index definition from toplevel_sp
1	RW	0x0	sw_refer1_field_e Refer picture consist of single fields or frame: '0' = reference picture consists of frame '1' = reference picture consists of fields
0	RW	0x0	sw_refer1_topc_e Which field of reference picture is closer to current picture: '0' = bottom field is closer to current picture '1' = top field is closer to current picture

VDPU_SWREG15_JPEG_ROI
Address: Operational Base + offset (0x0003c)
JPEG roi control

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19	RW	0x0	sw_jpegroi_in_endian jpeg offset input endian sw_jpegroi_in_endian 0 = big endian (0-1-2-3 order) 1 = little endian (3-2-1-0 order)

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Bit	Attr	Reset Value	Description
18	RW	0x0	sw_jpegroi_in_swap32 jpeg offset input 32-bit swap sw_jpegroi_in_swap32 0: no swapping of 32 bit words 1: 32bit data words are swapped (needed in 64 bit environment to achieve 7-6-5-4-3-2-1 byte order (also little endian should be enabled)
17:16	RW	0x0	sw_roi_sample_size ROI MB num sample each time ROI MB num sample each time 00 01 10 11
15:12	RW	0x0	sw_roi_distance roi distance The distance between the sample MB and ROI start MB
11:10	RW	0x0	sw_roi_out_sel roi output selection ROI output selection 00: output offset/dc 01: output picture 10: output offset/dc and picture 11: output offset/dc
9	RW	0x0	sw_roi_decode roi decode JPEG ROI decode 0: build offset/dc table 1: ROI decode
8	RW	0x0	sw_roi_en roi enable JPEG roi mode enable 0: normal jpeg decode mode 1: JPEG roi mode
7:0	RO	0x0	reserved

VDPU_SWREG16
Address: Operational Base + offset (0x00040)
base address for reference picture index 2 / List of VLC code len

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_refer2_base Base address for reference picture index 2. See picture index definition from toplevel_sp
1	RW	0x0	sw_refer2_field_e Refer picture consist of single fields or frame: '0' = reference picture consists of frame '1' = reference picture consists of fields
0	RW	0x0	sw_refer2_topc_e Which field of reference picture is closer to current picture: '0' = bottom field is closer to current picture '1' = top field is closer to current picture

VDPU_SWREG17
Address: Operational Base + offset (0x00044)
Base address for reference picture index 3 / List of VLC code le

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_refer3_base Base address for reference picture index 3. See picture index definition from toplevel_sp
1	RW	0x0	sw_refer3_field_e Refer picture consist of single fields or frame: '0' = reference picture consists of frame '1' = reference picture consists of fields
0	RW	0x0	sw_refer3_topc_e Which field of reference picture is closer to current picture: '0' = bottom field is closer to current picture '1' = top field is closer to current picture

VDPU_SWREG18
Address: Operational Base + offset (0x00048)
Base address for reference picture index 4 / VC1 control / MPE

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_refer4_base Base address for reference picture index 4. See picture index definition from toplevel_sp
1	RW	0x0	sw_refer4_field_e Refer picture consist of single fields or frame: '0' = reference picture consists of frame '1' = reference picture consists of fields

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Bit	Attr	Reset Value	Description
0	RW	0x0	sw_refer4_topc_e Which field of reference picture is closer to current picture: '0' = bottom field is closer to current picture '1' = top field is closer to current picture

VDPU_SWREG19
Address: Operational Base + offset (0x0004c)
Base address for reference picture index 5

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_refer5_base Base address for reference picture index 5. See picture index definition from toplevel_sp
1	RW	0x0	sw_refer5_field_e Refer picture consist of single fields or frame: '0' = reference picture consists of frame '1' = reference picture consists of fields
0	RW	0x0	sw_refer5_topc_e Which field of reference picture is closer to current picture: '0' = bottom field is closer to current picture '1' = top field is closer to current picture

VDPU_SWREG20
Address: Operational Base + offset (0x00050)
Base address for reference picture index 6

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_refer6_base Base address for reference picture index 6. See picture index definition from toplevel_sp
1	RW	0x0	sw_refer6_field_e Refer picture consist of single fields or frame: '0' = reference picture consists of frame '1' = reference picture consists of fields
0	RW	0x0	sw_refer6_topc_e Which field of reference picture is closer to current picture: '0' = bottom field is closer to current picture '1' = top field is closer to current picture

VDPU_SWREG21
Address: Operational Base + offset (0x00054)
Base address for reference picture index 7

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_refer7_base Base address for reference picture index 7. See picture index definition from toplevel_sp
1	RW	0x0	sw_refer7_field_e Refer picture consist of single fields or frame: '0' = reference picture consists of frame '1' = reference picture consists of fields
0	RW	0x0	sw_refer7_topc_e Which field of reference picture is closer to current picture: '0' = bottom field is closer to current picture '1' = top field is closer to current picture

VDPU_SWREG22
Address: Operational Base + offset (0x00058)
Base address for reference picture index 8

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_refer8_base Base address for reference picture index 8. See picture index definition from toplevel_sp
1	RW	0x0	sw_refer8_field_e Refer picture consist of single fields or frame: '0' = reference picture consists of frame '1' = reference picture consists of fields
0	RW	0x0	sw_refer8_topc_e Which field of reference picture is closer to current picture: '0' = bottom field is closer to current picture '1' = top field is closer to current picture

VDPU_SWREG23
Address: Operational Base + offset (0x0005c)
Base address for reference picture index 9

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_refer9_base Base address for reference picture index 9. See picture index definition from toplevel_sp
1	RW	0x0	sw_refer9_field_e Refer picture consist of single fields or frame: '0' = reference picture consists of frame '1' = reference picture consists of fields

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Bit	Attr	Reset Value	Description
0	RW	0x0	sw_refer9_topc_e Which field of reference picture is closer to current picture: '0' = bottom field is closer to current picture '1' = top field is closer to current picture

VDPU_SWREG24
Address: Operational Base + offset (0x00060)
Base address for reference picture index 10

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_refer10_base Base address for reference picture index 10. See picture index definition from toplevel_sp
1	RW	0x0	sw_refer10_field_e Refer picture consist of single fields or frame: '0' = reference picture consists of frame '1' = reference picture consists of fields
0	RW	0x0	sw_refer10_top_e Which field of reference picture is closer to current picture: '0' = bottom field is closer to current picture '1' = top field is closer to current picture

VDPU_SWREG25
Address: Operational Base + offset (0x00064)
Base address for reference picture index 11

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	sw_refer11_base Base address for reference picture index 11. See picture index definition from toplevel_sp
1	RW	0x0	sw_refer11_field_e Refer picture consist of single fields or frame: '0' = reference picture consists of frame '1' = reference picture consists of fields
0	RW	0x0	sw_refer11_topc_e Which field of reference picture is closer to current picture: '0' = bottom field is closer to current picture '1' = top field is closer to current picture

VDPU_SWREG26
Address: Operational Base + offset (0x00068)
Base address for reference picture index 12

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_refer12_base Base address for reference picture index 12. See picture index definition from toplevel_sp
1	RW	0x0	sw_refer12_field_e Refer picture consist of single fields or frame: '0' = reference picture consists of frame '1' = reference picture consists of fields
0	RW	0x0	sw_refer12_topc_e Which field of reference picture is closer to current picture: '0' = bottom field is closer to current picture '1' = top field is closer to current picture

VDPU_SWREG27
Address: Operational Base + offset (0x0006c)
Base address for reference picture index 13

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_refer13_base Base address for reference picture index 13. See picture index definition from toplevel_sp
1	RW	0x0	sw_refer13_field_e Refer picture consist of single fields or frame: '0' = reference picture consists of frame '1' = reference picture consists of fields
0	RW	0x0	sw_refer13_topc_e Which field of reference picture is closer to current picture: '0' = bottom field is closer to current picture '1' = top field is closer to current picture

VDPU_SWREG28
Address: Operational Base + offset (0x00070)
Base address for reference picture index14

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_refer14_base Base address for reference picture index 14. See picture index definition from toplevel_sp
1	RW	0x0	sw_refer14_field_e Refer picture consist of single fields or frame: '0' = reference picture consists of frame '1' = reference picture consists of fields

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Bit	Attr	Reset Value	Description
0	RW	0x0	sw_refer14_topc_e Which field of reference picture is closer to current picture: '0' = bottom field is closer to current picture '1' = top field is closer to current picture

VDPU_SWREG29
Address: Operational Base + offset (0x00074)
Base address for reference picture index15

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_refer15_base Base address for reference picture index 15. See picture index definition from toplevel_sp
1	RW	0x0	sw_refer15_field_e Refer picture consist of single fields or frame: '0' = reference picture consists of frame '1' = reference picture consists of fields
0	RW	0x0	sw_refer15_topc_e Which field of reference picture is closer to current picture: '0' = bottom field is closer to current picture '1' = top field is closer to current picture

VDPU_SWREG30
Address: Operational Base + offset (0x00078)
Reference picture numbers for index 0 and 1 (H264 VLC

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_refer1_nbr Number for reference picture index 1
15:0	RW	0x0000	sw_refer0_nbr Number for reference picture index 0

VDPU_SWREG31
Address: Operational Base + offset (0x0007c)
Reference picture numbers for index 2 and 3 (H264 VLC) /

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_refer3_nbr Number for reference picture index 3
15:0	RW	0x0000	sw_refer2_nbr Number for reference picture index 2

VDPU_SWREG32
Address: Operational Base + offset (0x00080)
Reference picture numbers for index 4 and 5 (H264 VLC)

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_refer5_nbr Number for reference picture index 5
15:0	RW	0x0000	sw_refer4_nbr Number for reference picture index 4

VDPU_SWREG33
Address: Operational Base + offset (0x00084)
Reference picture numbers for index 6 and 7 (H264 VLC)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_refer7_nbr Number for reference picture index 7
15:0	RW	0x0000	sw_refer6_nbr Number for reference picture index 6

VDPU_SWREG34
Address: Operational Base + offset (0x00088)
Reference picture numbers for index 8 and 9 (H264 VLC)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_refer9_nbr Number for reference picture index 9
15:0	RW	0x0000	sw_refer8_nbr Number for reference picture index 8

VDPU_SWREG35
Address: Operational Base + offset (0x0008c)
Reference picture numbers for index 10 and 11 (H264 VLC)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_refer11_nbr Number for reference picture index 11
15:0	RW	0x0000	sw_refer10_nbr Number for reference picture index 10

VDPU_SWREG35_JPEG_ROI
Address: Operational Base + offset (0x0008c)
JPEG roi offest/dc base address

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_jpegdcoff_base JPEG roi offset/dc base address JPEG roi offest/dc base address
1:0	RO	0x0	reserved

VDPU_SWREG36
Address: Operational Base + offset (0x00090)
Reference picture numbers for index 12 and 13 (H264 VLC)

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_refer13_nbr Number for reference picture index 13
15:0	RW	0x0000	sw_refer12_nbr Number for reference picture index 12

VDPU_SWREG36_JPEG_ROI
Address: Operational Base + offset (0x00090)
JPEG roi offset/dc length

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:0	RW	0x00000	sw_jpegdcoff_len sw_jpegdcoff_len The number of 64bit jpegdcoff, it can be used both when sw_roi_decode is 1'b0 or 1'b1

VDPU_SWREG37
Address: Operational Base + offset (0x00094)
Reference picture numbers for index 14 and 15 (H264 VLC)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_refer15_nbr Number for reference picture index 15
15:0	RW	0x0000	sw_refer14_nbr Number for reference picture index 14

VDPU_SWREG38
Address: Operational Base + offset (0x00098)
Reference picture long term flags (H264 VLC) / VPx prediction filt

Bit	Attr	Reset Value	Description
31:22	RW	0x000	sw_pred_bc_tap_3_3 Prediction filter set 3, tap 3
21:12	RW	0x000	sw_pred_bc_tap_4_0 Prediction filter set 4, tap 0
11:2	RW	0x000	sw_perd_bc_tap_4_1 Prediction filter set 4, tap 1
1:0	RO	0x0	reserved

VDPU_SWREG39
Address: Operational Base + offset (0x0009c)
Reference picture valid flags (H264 VLC) /VPx prediction filter ta

Bit	Attr	Reset Value	Description
31:22	RW	0x000	sw_pred_bc_tap_4_2 Prediction filter set 4, tap 2
21:12	RW	0x000	sw_pred_bc_tap_4_3 Prediction filter set 4, tap 3

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Bit	Attr	Reset Value	Description
11:2	RW	0x000	sw_pred_bc_tap_5_0 Prediction filter set 5, tap 0
1:0	RO	0x0	reserved

VDPU_SWREG40
Address: Operational Base + offset (0x000a0)
Base address for standard dependent tables

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_qtable_base Base address for standard dependent tables: JPEG= AC,DC, QP tables MPEG4=QP table base address if type 1 quantization is used MPEG2=QP table base address H.264=base address for various tables VP6,VP7,VP8=base address for stream decoding tables RV=base address for picture slice sizes
1:0	RO	0x0	reserved

VDPU_SWREG41
Address: Operational Base + offset (0x000a4)
Base address for direct mode motion vectors

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_dir_mv_base Direct mode motion vector write/read base address. For H264 this is used only for direct mode motion vector write base. Progressive JPEG: ACDC coefficient read/write base address. If current round is for DC components this base address is pointing to luminance (separate base adderesses for chrominances), for AC component rounds this base is used for current type
1:0	RO	0x0	reserved

VDPU_SWREG42
Address: Operational Base + offset (0x000a8)
bi_dir initial ref pic list register (0-2)/ VP6 prediction filter taps

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:25	RW	0x00	sw_binit_rlist_b2 Initial reference picture list for bi- direct backward picid 2

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Bit	Attr	Reset Value	Description
24:20	RW	0x00	sw_binit_rlist_f2 Initial reference picture list for bi- direct forward picid 2
19:15	RW	0x00	sw_binit_rlist_b1 Initial reference picture list for bi- direct backward picid 1
14:10	RW	0x00	sw_binit_rlist_f1 Initial reference picture list for bi- direct forward picid 1
9:5	RW	0x00	sw_binit_rlist_b0 Initial reference picture list for bi- direct backward picid 0
4:0	RW	0x00	sw_binit_rlist_f0 Initial reference picture list for bi- direct forward picid 0

VDPU_SWREG43
Address: Operational Base + offset (0x000ac)
bi-dir initial ref pic list register (3-5)/ VP6 prediction filter taps

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:25	RW	0x00	sw_binit_rlist_b5 Initial reference picture list for bi- direct backward picid 5
24:20	RW	0x00	sw_binit_rlist_f5 Initial reference picture list for bi- direct forward picid 5
19:15	RW	0x00	sw_binit_rlist_b4 Initial reference picture list for bi- direct backward picid 4
14:10	RW	0x00	sw_binit_rlist_f4 Initial reference picture list for bi- direct forward picid 4
9:5	RW	0x00	sw_binit_rlist_b3 Initial reference picture list for bi- direct backward picid 3
4:0	RW	0x00	sw_binit_rlist_f3 Initial reference picture list for bi- direct forward picid 3

VDPU_SWREG44
Address: Operational Base + offset (0x000b0)
bi-dir initial ref pic list register (6-8)/ VP6 prediction filter taps

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
29:25	RW	0x00	sw_binit_rlist_b8 Initial reference picture list for bi- direct backward picid 8
24:20	RW	0x00	sw_binit_rlist_f8 Initial reference picture list for bi- direct forward picid 8
19:15	RW	0x00	sw_binit_rlist_b7 Initial reference picture list for bi- direct backward picid 7
14:10	RW	0x00	sw_binit_rlist_f7 Initial reference picture list for bi- direct forward picid 7
9:5	RW	0x00	sw_binit_rlist_b6 Initial reference picture list for bi- direct backward picid 6
4:0	RW	0x00	sw_binit_rlist_f6 Initial reference picture list for bi- direct forward picid 6

VDPU_SWREG45
Address: Operational Base + offset (0x000b4)
bi-dir initial ref pic list register (9- 11) / VP6 prediction filter taps

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:25	RW	0x00	sw_binit_rlist_b11 Initial reference picture list for bi- direct backward picid 11
24:20	RW	0x00	sw_binit_rlist_f11 Initial reference picture list for bi- direct forward picid 11
19:15	RW	0x00	sw_binit_rlist_b10 Initial reference picture list for bi- direct backward picid 10
14:10	RW	0x00	sw_binit_rlist_f10 Initial reference picture list for bi- direct forward picid 10
9:5	RW	0x00	sw_binit_rlist_b9 Initial reference picture list for bi- direct backward picid 9
4:0	RW	0x00	sw_binit_rlist_f9 Initial reference picture list for bi- direct forward picid 9

VDPU_SWREG46
Address: Operational Base + offset (0x000b8)

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bi-dir initial ref pic list register (12- 14) / VP7,VP8 quantization v

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:25	RW	0x00	sw_binit_rlist_b14 Initial reference picture list for bi- direct backward picid 14
24:20	RW	0x00	sw_binit_rlist_f14 Initial reference picture list for bi- direct forward picid 14
19:15	RW	0x00	sw_binit_rlist_b13 Initial reference picture list for bi- direct backward picid 13
14:10	RW	0x00	sw_binit_rlist_f13 Initial reference picture list for bi- direct forward picid 13
9:5	RW	0x00	sw_binit_rlist_b12 Initial reference picture list for bi- direct backward picid 12
4:0	RW	0x00	sw_binit_rlist_f12 Initial reference picture list for bi- direct forward picid 12

VDPU_SWREG47

Address: Operational Base + offset (0x000bc)

bi-dir and P fwd initial ref pic list register (15 and P 0-3) / VP7,V

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:25	RW	0x00	sw_pinit_rlist_f3 Initial reference picture list for P forward picid 3
24:20	RW	0x00	sw_pinit_rlist_f2 Initial reference picture list for P forward picid 2
19:15	RW	0x00	sw_pinit_rlist_f1 Initial reference picture list for P forward picid 1
14:10	RW	0x00	sw_pinit_rlist_f0 Initial reference picture list for P forward picid 0
9:5	RW	0x00	sw_binit_rlist_b15 Initial reference picture list for bi- direct backward picid 15
4:0	RW	0x00	sw_binit_rlist_f15 Initial reference picture list for bi- direct forward picid 15

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VDPU_SWREG48
Address: Operational Base + offset (0x000c0)
Error concealment register

Bit	Attr	Reset Value	Description
31:23	RW	0x000	sw_startmb_x Start MB from SW for X dimension. Used in error concealment case
22:15	RW	0x00	sw_startmb_y Start MB from SW for Y dimension. Used in error concealment case
14:0	RO	0x0	reserved

VDPU_SWREG49
Address: Operational Base + offset (0x000c4)
Prediction filter tap register for H264, MPEG4, VC1, VP6

Bit	Attr	Reset Value	Description
31:22	RW	0x000	sw_pred_bc_tap_0_0 Prediction filter set 0, tap 0
21:12	RW	0x000	sw_pred_bc_tap_0_1 Prediction filter set 0, tap 1
11:2	RW	0x000	sw_pred_bc_tap_0_2 Prediction filter set 0, tap 2
1:0	RO	0x0	reserved

VDPU_SWREG50
Address: Operational Base + offset (0x000c8)
Synthesis configuration register decoder 0 (read only)

Bit	Attr	Reset Value	Description
31	RO	0x1	SW_DEC_MPEG2_PROF Decoding format support, MPEG-2 / MPEG-1 '0' = not supported '1' = supported
30:29	RO	0x3	SW_DEC_VC1_PROF Decoding format support, VC-1 0 = not supported 1 = supported up to simple profile 2 = supported up to main profile 3 = supported up to advanced profile
28	RO	0x1	SW_DEC_JPEG_PROF Decoding format support, JPEG 0 = not supported 1 = supported

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Bit	Attr	Reset Value	Description
27:26	RO	0x2	SW_DEC_MPEG4_PROF Decoding format support, MPEG-4 / H.263 0 = not supported 1 = supported up to simple profile 2 = supported up to advanced simple profile
25:24	RO	0x3	SW_DEC_H264_PROF Decoding format support, H.264 0 = not supported 1 = supported up to baseline profile 2 = supported up to high profile labeled stream with restricted high profile tools
23	RO	0x1	SW_DEC_VP6_PROF Decoding format support, VP6 0 = not supported 1 = supported
22	RO	0x0	SW_DEC_PJEPEG_EXIT Progressive JPEG support: '0' = Not supported '1' = supported
21	RO	0x1	SW_DEC_OBUFF_LEVEL Decoder output buffer level: '0' = 1 MB buffering is used '1' = 4 MB buffering is used
20	RO	0x1	SW_REF_BUFF_EXIST Field0000 Description
19:16	RO	0x5	SW_DEC_BUS_STRD Field0000 Description
15:14	RO	0x1	SW_DEC_SYNTH_LAN
13:12	RO	0x2	SW_DEC_BUS_WIDTH 0 = error 1 = 32 bit bus 2 = 64 bit bus 3 = 128 bit bus
11	RO	0x1	SW_DEC_SOREN_PRO Decoding format support, Sorenson '0' = not supported '1' = supported
10:0	RO	0x780	SW_DEC_MAX_OWIDTH Max configured decoder video resolution that can be decoded. Informed as width of the picture in pixels

VDPU_SWREG51
Address: Operational Base + offset (0x000cc)

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Reference picture buffer control register

Bit	Attr	Reset Value	Description
31	RW	0x0	sw_refbu_e Refer picture buffer enable: '0' = refer picture buffer disabled '1' = refer picture buffer enabled. Valid if picture size is QVGA or more
30:19	RW	0x000	sw_refbu_thr Reference buffer disable threshold value (cache miss amount). Used to buffer shut down (if more misses than allowed)
18:14	RW	0x00	sw_refbu_picid The used reference picture ID for reference buffer usage
13	RW	0x0	sw_refbu_eval_e Enable for HW internal reference ID calculation. If given threshold level is reached by any picture_id after first MB row, that picture_id is used for reference buffer fill for rest of the picture
12	RW	0x0	sw_refbu_fparmod_e Field parity mode enable. Used in refbufferd evaluation mode '0' = use the result field of the evaluation '1' = use the parity mode field
11:9	RO	0x0	reserved
8:0	RW	0x000	sw_refbu_y_offset Y offset for refbufferd. This coordinate is used to compensate the global motion of the video for better buffer hit rate

VDPU_SWREG52

Address: Operational Base + offset (0x000d0)

Reference picture buffer information register 1 (read only)

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_refbu_hit_sum The sum of the refbufferd hits of the picture. Determined for each 8x8 luminance partition of the picture. The proceeding of the HW calculation can be read during HW decoding
15:0	RW	0x0000	sw_refbu_intra_sum The sum of the luminance 8x8 intra partitons of the picture. The proceeding of the HW calculation can be read during HW decoding

VDPU_SWREG53
Address: Operational Base + offset (0x000d4)
Reference picture buffer information register 2 (read only)

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21:0	RW	0x000000	sw_refbu_y_mv_sum The sum of the decoded motion vector y-components of the picture. The first luminance motion vector of each MB is used in calculation. Other motion vectors of the MB are discarded. Each motion vector is saturated between -256 - 255 before calculation. The proceeding of the HW calculation can be read during HW decoding

VDPU_SWREG54
Address: Operational Base + offset (0x000d8)
Synthesis configuration register decoder 1 (read only)

Bit	Attr	Reset Value	Description
31	RO	0x1	SW_DEC_JPEG_EXTENS JPEG sampling support extension for 411 and 444 samplings and support for bigger max resolution than 16 Mpix (up to 67Mpixels): '0' = not supported '1' = supported

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Bit	Attr	Reset Value	Description
30	RO	0x1	SW_DEC_REFBU_ILACE Refbufferd support for interlaced content: '0' = not supported '1' = supported
29	RO	0x1	SW_DEC_DIVX_PROF DIVX Support : '0' = not supported '1' = supported
28	RO	0x0	SW_REF_BUFF2_EXIST Reference picture buffer 2 usage: '0' = not supported '1' = reference buffer 2 is used
27:26	RO	0x1	SW_DEC_RV_PROF Decoding format support, RV 0 = not supported 1 = supported up to 2 = NA
25	RO	0x0	SW_DEC_RTL_ROM ROM implementation type (If design includes ROMs) '0': ROMs are implemented from actual ROM units '1': ROMs are impelemted from RTL
24	RO	0x1	SW_DEC_VP7_PROF Decoding format support, VP7 0 = not supported 1 = supported
23	RO	0x1	SW_DEC_VP8_PROF Decoding format support, VP8 0 = not supported 1 = supported
22	RO	0x1	SW_DEC_AVS_PROF Decoding format support, AVS 0 = not supported 1 = supported
21:20	RO	0x1	SW_DEC_MVC_PROF Decoding format support, MVC 0 = not supported 1 = supported
19	RO	0x1	SW_DEC_VP8SNAP_E Decoding format support, VP8 snapshot 0 = not supported bigger than 1080p resolution 1 = supported upto 16kx16k pixel resolution (defined max)

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Bit	Attr	Reset Value	Description
18:17	RO	0x1	SW_DEC_TILED_L Tiled mode support level 0 = not supported 1 = supported with 8x4 tile size 2,3 = reserved
16:0	RO	0x0	reserved

VDPU_SWREG55
Address: Operational Base + offset (0x000dc)
Reference picture buffer 2 / Advanced prefetch control register

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13:0	RW	0x0000	sw_apf_threshold Advanced prefetch threshold value. If current MB exceeds the threshold the advanced mode is not used. Value 0 disables threshold usage and advanced prefetch usage is restricted by internal memory limitation only

VDPU_SWREG56
Address: Operational Base + offset (0x000e0)
Reference buffer information register 3 (read only)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_refbu_top_sum The sum of the top partitions of the picture
15:0	RW	0x0000	sw_refbu_bot_sum The sum of the bottom partitions of the picture

VDPU_SWREG57_INTRA_INTER
Address: Operational Base + offset (0x000e4)
intra_dll3t,intra_dblspeed,inter_dblspeed,stream_len_hi

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14:8	RW	0x00	debug_service debug_service signals service_wr[2:0], service_rd[3:0]
7	RW	0x0	sw_cache_en cache enable 1'b1: cache enable 1'b0: cache disable when sw_cache_en is 1'b1, sw_pref_sigchan should also be 1'b1

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Bit	Attr	Reset Value	Description
6	RW	0x0	sw_pref_sigchan prefetch single channel enable 1'b1: prefetch single channel enable
5	RW	0x0	sw_axiwr_sel axi write master select 1'b0: auto sel encoder axi signals and decoder axi signals 1'b1: sel decoder axi signals (it only use to set bu_dec_e to 1'b0 in the middle of a frame)
4	RW	0x0	sw_paral_bus paral_bus enable when it is set to 1'b1, the axi support read and write service parallel; when it is set to 1'b0, the axi only support read and write serial
3	RW	0x0	sw_intra_dbl3t sw_intra_dbl3t In chroma dc intra prediction, when this bit is enable, there will 3 cycle enhance for every block
2	RW	0x0	sw_intra_dblspeed intra double speed enable Intra double speed enable
1	RW	0x0	sw_inter_dblspeed inter double speed enable Inter double speed enable
0	RW	0x0	sw_stream_len_hi stream length high bit The extension bit of sw_stream_len

VDPU_SWREG58
Address: Operational Base + offset (0x000e8)
Decoder debug register 0 (read only)

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30	RO	0x0	debug_mv_req mvst_mv_req signal value
29	RO	0x0	debug_rlc_req prtr_res_y_req signal value
28	RO	0x0	debug_res_y_req prtr_res_y_req signal value
27	RO	0x0	debug_res_c_req prtr_res_c_req signal value
26	RO	0x0	debug_strm_da_e strm_da_e signal value

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Bit	Attr	Reset Value	Description
25	RO	0x0	debug_framerdy dfbu_framerdy signal value
24	RO	0x0	debug_filter_req dfbu_req_e signal value
23	RO	0x0	debug_referreq0 prbu_referreq0 signal value
22	RO	0x0	debug_referreq1 prbu_referreq1 signal value
21	RO	0x0	reserved
20:0	RO	0x000000	debug_dec_mb_count HW internal MB counter value

VDPU_SWREG59
Address: Operational Base + offset (0x000ec)
H264 Chrominance 8 pixel interleaved data base

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_dec_ch8pix_base Base address for additional chrominance data format where chrominance is interleaved in group of 8 pixels. The usage is enabled by sw_ch_8pix_ileav_e
1:0	RO	0x0	reserved

VDPU_SWREG60
Address: Operational Base + offset (0x000f0)
Interrupt register post-processor

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13	RW	0x0	sw_pp_bus_int Interrupt status bit bus. Error response from bus. In pipeline mode this bit is not used
12	RW	0x0	sw_pp_rdy_int Interrupt status bit pp. When this bit is high post processor has processed a picture in external mode. In pipeline mode this bit is not used.
11:9	RO	0x0	reserved
8	RW	0x0	sw_pp_irq Post-processor IRQ. SW will reset this after interrupt is handled. HINTpp is not used for pp if IRQ disable pp is high (sw_pp_irq_n_e = 1). In pipeline mode this bit is not used
7:5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4	RW	0x0	sw_pp_irq_dis Post-processor IRQ disable. When high, there are no interrupts from HW concerning post processing. Polling must be used to see the interrupt
3:2	RO	0x0	reserved
1	RW	0x0	sw_pp_pipeline_e Decoder –post-processing pipeline enable: 0 = Post-processing is processing different picture than decoder or is disabled 1 = Post-processing is performed in pipeline with decoder
0	RW	0x0	sw_pp_e External mode post-processing enable. This bit will start the post-processing operation. Not to be used if PP is in pipeline with decoder (sw_pp_pipeline_e = 1). HW will reset this when picture is post-processed.

VDPU_SWREG61
Address: Operational Base + offset (0x000f4)
Device configuration register post-processor

Bit	Attr	Reset Value	Description
31:24	RW	0x00	sw_pp_axi_rd_id Read ID used for AXI PP read services (if connected to AXI)
23:16	RW	0x00	sw_pp_axi_wr_id Write ID used for AXI PP write services (if connected to AXI)
15	RO	0x0	reserved
14	RW	0x0	sw_pp_scmd_dis AXI Single Command Multiple Data disable.
13	RW	0x0	sw_pp_in_a2_endsel Endian/swap select for Alpha blend input source 2: '0' = Use PP in endian/swap definitions (sw_pp_in_endian, sw_pp_in_swap) '1' = Use Ablend source 1 endian/swap definitions

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Bit	Attr	Reset Value	Description
12	RW	0x0	sw_pp_in_a1_swap32 Alpha blend source 1 input 32bit data swap (may be used for 64 bit environment): 0 = no swapping of 32 bit words 1 = 32 bit data words are swapped (needed in 64 bit environment to achieve 7-6-5-4-3-2-1-0 byte order(also little endian should be enabled))
11	RW	0x0	sw_pp_in_a1_endian Alpha blend source 1 input data byte endian mode. 0 = Big endian (0-1-2-3 order) 1 = Little endian (3-2-1-0 order)
10	RW	0x0	sw_pp_in_swap32_e PP input 32bit data swap (may be used for 64 bit environment): 0 = no swapping of 32 bit words 1 = 32 bit data words are swapped (needed in 64 bit environment to achieve 7-6-5-4-3-2-1-0 byte order(also little endian should be enabled))
9	RW	0x0	sw_pp_data_disc_e PP data discard enable. Precise burst lengths are used with reading services. Extra data is discarded internally.
8	RW	0x0	sw_pp_clkgate_e PP dynamic clock gating enable: 1 = Clock is gated from PP structures that are not used 0 = Clock is running for all PP structures Note: Clock gating value can be changed only when PP is not enabled
7	RW	0x0	sw_pp_in_endian PP input picture byte endian mode. Used only if PP is in standalone mode. If PP is running pipelined with the decoder, this bit has no effect. 0 = Big endian (0-1-2-3 order) 1 = Little endian (3-2-1-0 order)

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Bit	Attr	Reset Value	Description
6	RW	0x0	sw_pp_out_endian PP output picture endian mode for YCbCr data or for any data if config value SW_PP_OEN_VERSION=1 0 = Big endian (0-1-2-3 order) 1 = Little endian (3-2-1-0 order) (NOTE: For SW_PP_OEN_VERSION=0 16 bit RGB data this bit works as pixel swapping bit. For 32 bit RGB this bit has no meaning)
5	RW	0x0	sw_pp_out_swap32_e PP output data word swap (may be used for 64 bit environment): 0 = no swapping of 32 bit words 1 = 32 bit data words are swapped (needed in 64 bit environment to achieve 7-6-5-4-3-2-1-0 byte order (also little endian should be enabled))
4:0	RW	0x00	sw_pp_max_burst Maximum burst length for PP bus transactions. 1-16

VDPU_SWREG62
Address: Operational Base + offset (0x000f8)
Deinterlace control register

Bit	Attr	Reset Value	Description
31	RW	0x0	sw_deint_e De-interlace enable. Input data is in interlaced format and deinterlacing needs to be performed
30	RO	0x0	reserved
29:16	RW	0x0000	sw_deint_threshold Threshold value used in deinterlacing
15	RW	0x0	sw_deint_blend_e Blend enable for de-interlacing
14:0	RW	0x0000	sw_deint_edge_det Edge detect value used for deinterlacing

VDPU_SWREG63
Address: Operational Base + offset (0x000fc)
base address for reading post-processing input picture uminan

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_pp_in_lu_base Base address for post-processing input luminance picture. If PP input picture is fetched from fields this base address is used to point to topfield of the picture. Used in external mode only.
1:0	RO	0x0	reserved

VDPU_SWREG64
Address: Operational Base + offset (0x00100)
Base address for reading post-processing input picture Cb/Ch

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_pp_in_cb_base Base address for post-processing input Cb picture or for both chrominance pictures (if chrominances interleaved). If PP input picture is fetched from fields this base address is used to point to topfield of the picture. Used in external mode only
1:0	RO	0x0	reserved

VDPU_SWREG65
Address: Operational Base + offset (0x00104)
Base address for reading post-processing input picture Cr

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_pp_in_cr_base Base address for post-processing input cr picture. Used in external mode only
1:0	RO	0x0	reserved

VDPU_SWREG66
Address: Operational Base + offset (0x00108)
Base address for writing post-processed picture luminance/RGB

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_pp_out_lu_base Base address for post-processing output picture (luminance/YUYV/RGB).

VDPU_SWREG67
Address: Operational Base + offset (0x0010c)
Base address for writing post-processed picture Ch

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_pp_out_ch_base Base address for post-processing output chrominance picture (interleaved chrominance).

VDPU_SWREG68
Address: Operational Base + offset (0x00110)
Register for contrast adjusting

Bit	Attr	Reset Value	Description
31:24	RW	0x00	sw_contrast_thr1 Threshold value 1, used with contrast adjusting
23:20	RO	0x0	reserved
19:10	RW	0x000	sw_contrast_off2 Offset value 2, used with contrast adjusting
9:0	RW	0x000	sw_contrast_off1 Offset value 1, used with contrast adjusting

VDPU_SWREG69
Address: Operational Base + offset (0x00114)
Register for colour conversion and contrast adjusting

Bit	Attr	Reset Value	Description
31	RW	0x0	sw_pp_in_start_ch For YUYV 422 input format. Enable for start_with_chrominance. '0' =the order is Y0CbY0Cr or Y0CrY0Cb '1'= the order is CbY0CrY0 or CrY0CbY0
30	RW	0x0	sw_pp_in_cr_first For YUYV 422 input format and YCbCr 420 semiplanar format. Enable for Cr first (before Cb) '0' =the order is Y0CbY0Cr or CbY0CrY0 (if 420 semiplanar chrominance: CbCrCbCr) '1'= the order is Y0CrY0Cb or CrY0CbY0 (if 420 semiplanar chrominance: CrCbCrCb)
29	RW	0x0	sw_pp_out_start_ch For YUYV 422 output format. Enable for start_with_chrominance. '0' =the order is Y0CbY0Cr or Y0CrY0Cb '1'= the order is CbY0CrY0 or CrY0CbY0
28	RW	0x0	sw_pp_out_cr_first For YUYV 422 output format. Enable for Cr first (beforeCb) '0' =the order is Y0CbY0Cr or CbY0CrY0 '1'= the order is Y0CrY0Cb or CrY0CbY0

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Bit	Attr	Reset Value	Description
27:18	RW	0x000	sw_color_coeffa2 Coefficient a2, used with Y pixel to calculate all color components
17:8	RW	0x000	sw_color_coeffa1 Coefficient a1, used with Y pixel to calculate all color components
7:0	RW	0x00	sw_contrast_thr2 Threshold value 2, used with contrast adjusting

VDPU_SWREG70
Address: Operational Base + offset (0x00118)
Register for colour conversion 0

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:20	RW	0x000	sw_color_coeffd Coefficient d, used with Cb to calculate green component value
19:10	RW	0x000	sw_color_coeffc Coefficient c, used with Cr to calculate green component value
9:0	RW	0x000	sw_color_coeffb Coefficient b, used with Cr to calculate red component value

VDPU_SWREG71
Address: Operational Base + offset (0x0011c)
Register for colour conversion 1 + rotation mode

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:21	RW	0x000	sw_crop_startx Start coordinate x for the cropped area in macroblocks.
20:18	RW	0x0	sw_rotation_mode Rotation mode: 000 = rotation disabled 001 = rotate + 90 010 = rotate –90 011 = horizontal flip (mirror) 100 = vertical flip 101 = rotate 180
17:10	RW	0x00	sw_color_coefff Coefficient f, used with Y to adjust brightness

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Bit	Attr	Reset Value	Description
9:0	RW	0x000	sw_color_coeffe Coefficient e, used with Cb to calculate blue component value

VDPU_SWREG72
Address: Operational Base + offset (0x00120)
PP input size and -cropping register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	sw_crop_starty Start coordinate y for the cropped area in macroblocks.
23	RO	0x0	reserved
22:18	RW	0x00	sw_rangemap_coef_y Range map value for Y component (RANGE_MAPY+9 in VC-1 standard)
17	RO	0x0	reserved
16:9	RW	0x00	sw_pp_in_height PP input picture height in MBs. Can be cropped from a bigger input picture in external mode
8:0	RW	0x000	sw_pp_in_width PP input picture width in MBs. Can be cropped from a bigger input picture in external mode

VDPU_SWREG73
Address: Operational Base + offset (0x00124)
PP input picture base address for Y bottom field

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_pp_bot_yin_base PP input Y base for bottom field
1:0	RO	0x0	reserved

VDPU_SWREG74
Address: Operational Base + offset (0x00128)
PP input picture base for Ch bottom field

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_pp_bot_cin_base PP input C base for bottom field (mixed chrominance)
1:0	RO	0x0	reserved

VDPU_SWREG79

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Address: Operational Base + offset (0x0013c)
Scaling ratio register 1 & padding for B

Bit	Attr	Reset Value	Description
31	RW	0x0	sw_rangemap_y_e Range map enable for Y component (RANGE_MAPY_FLAG in VC-1 standard). For VC1 main profile this bit is used as range expansion enable
30	RW	0x0	sw_rangemap_c_e Range map enable for chrominance component RANGE_MAPUV_FLAG in VC-1 standard)
29	RW	0x0	sw_ycbcr_range Defines the YCbCr range in RGB conversion: 0 = 16 -235 for Y, 16 -240 for Chrominance 1 = 0- 255 for all components
28	RW	0x0	sw_rgb_pix_in32 RGB pixel amount/ 32 bit word 0 = 1 RGB pixel/32 bit 1 = 2 RGB pixels/32 bit
27:23	RW	0x00	sw_rgb_r_padd Amount of ones that will be padded in front of the R- component
22:18	RW	0x00	sw_rgb_g_padd Amount of ones that will be padded in front of the G- component
17:0	RW	0x00000	sw_scale_wratio Scaling ratio for width (outputw-1/inputw-1)

VDPU_SWREG80
Address: Operational Base + offset (0x00140)
Scaling register 0 ratio & padding for R and G

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30	RW	0x0	sw_pp_fast_scale_e 0 = fast downscaling is not enabled 1 = fast downscaling is enabled. The quality of the picture is decreased but performance is improved

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Bit	Attr	Reset Value	Description
29:27	RW	0x0	sw_pp_in_struct PP input data picture structure: 0 = Top field / progressive frame structure: Read input data from top field base address /frame base address and read every line 1 = Bottom field structure: Read input data from bottom field base address and read every line. 2 = Interlaced field structure: Read input data from both top and bottom field base address and take every line from each field. 3 = Interlaced frame structure: Read input data from both top and bottom field base address and take every second line from each field. 4 = Ripped top field structure: Read input data from top field base address and read every second line. 5 = Ripped bottom field structure: Read input data from bottom field base address and read every second line
26:25	RW	0x0	sw_hor_scale_mode Horizontal scaling mode: 00 = Off 01 = Upscale 10 = Downscale
24:23	RW	0x0	sw_ver_scale_mode Vertical scaling mode: 00 = Off 01 = Upscale 10 = Downscale
22:18	RW	0x00	sw_rgb_b_padd Amount of ones that will be padded in front of the B- component
17:0	RW	0x00000	sw_scale_hratio Scaling ratio for height (outputh-1/inputh-1)

VDPU_SWREG81

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Address: Operational Base + offset (0x00144)
Scaling ratio register 2

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_wscale_invra Inverse scaling ratio for width, or ch (inputw-1 / outputw-1)
15:0	RW	0x0000	sw_hscale_invra Inverse scaling ratio for height or cv (inputh-1 / outputh-1)

VDPU_SWREG82
Address: Operational Base + offset (0x00148)
Rmask register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_r_mask Bit mask for R component (and alpha channel)

VDPU_SWREG83
Address: Operational Base + offset (0x0014c)
Gmask register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_g_mask Bit mask for G component (and alpha channel)

VDPU_SWREG84
Address: Operational Base + offset (0x00150)
Bmask register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_b_mask Bit mask for B component (and alpha channel)

VDPU_SWREG85
Address: Operational Base + offset (0x00154)
Post-processor control register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:29	RW	0x0	sw_pp_in_format PP input picture data format 0 = YUYV 4:2:2 interleaved (supported only in external mode) 1 = YCbCr 4:2:0 Semi-planar in linear raster-scan format 2 = YCbCr 4:2:0 planar (supported only in external mode) 3 = YCbCr 4:0:0 (supported only in pipelined mode) 4 = YCbCr 4:2:2 Semi-planar (supported only in pipelined mode) 5 = YCbCr 4:2:0 Semi-planar in tiled format (supported only in external mode (8170 decoder only) 6 = YCbCr 4:4:0 Semi-planar (supported only in pipelined mode, possible for jpeg only) 7 = Escape pp input data format. Defined in swreg86
28:26	RW	0x0	sw_pp_out_format PP output picture data format: 0 = RGB 1 = YCbCr 4:2:0 planar (Not supported) 2 = YCbCr 4:2:2 planar (Not supported) 3 = YUYV 4:2:2 interleaved 4 = YCbCr 4:4:4 planar (Not supported) 5 = YCh 4:2:0 chrominance interleaved 6 = YCh 4:2:2 (Not supported) 7 = YCh 4:4:4 (Not supported)
25:15	RW	0x000	sw_pp_out_height Scaled picture height in pixels (Must be dividable by 2 or by any if Pixel Accurate PP output configuration is enabled) Max scaled picture height is 1920 pixels or maximum three times the input source height minus 8 pixels

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Bit	Attr	Reset Value	Description
14:4	RW	0x000	sw_pp_out_width Scaled picture width in pixels. Must be dividable by 8 or by any if Pixel Accurate PP output configuration is enabled. Max scaled picture width is 1920 pixels or maximum three times the input source width minus 8 pixels
3	RW	0x0	sw_pp_out_tiled_e Tiled mode enable for PP output. Can be used only for YCbYCr 422 output format. Can be used only if correpongind configuration supports this feature. Tile size is 4x4 pixels.
2	RW	0x0	sw_pp_out_swap16_e PP output swap 16 swaps 16 bit halfs inside of 32 bit word. Can be used for 16 bit RGB to change pixel orders but is valid also for any output format NOTE: requires that configuration of SW_PPD_OEN_VERSION=1
1	RW	0x0	sw_pp_crop8_r_e PP input picture width is not 16 pixels multiple. Only 8 pixels of the most right MB of the unrotated input picture is used for PP input.
0	RW	0x0	sw_pp_crop8_d_e PP input picture height is not 16 pixels multiple. Only 8 pixel rows of the most down MB of the unrotated input picture is used for PP input.

VDPU_SWREG86
Address: Operational Base + offset (0x00158)
Mask 1 start coordinate register

Bit	Attr	Reset Value	Description
31:29	RW	0x0	sw_pp_in_format_es Escape PP in format. Used if sw_pp_in_format is defined to 7: 0 0 = YCbCr 4:4:4 1 = YCbCr 4:1:1

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Bit	Attr	Reset Value	Description
28	RO	0x0	reserved
27:23	RW	0x00	sw_rangemap_coef_c Range map value for chrominance component (RANGE_MAPUV+9 in VC-1 standard)
22	RW	0x0	sw_mask1_ablend_e Mask 1 alpha blending enable. Instead of masking the output picture the alpha blending is performed. Alpha blending source can be found from alpha blend 1 base address. Alpha blending can be enabled only for RGB/ YUYV 422 data.
21:11	RW	0x000	sw_mask1_starty Vertical start pixel for mask area 1. Defines the y coordinate. Coordinate 0,0 means the up-left corner in PP output luminance picture. See Table 47 for restrictions
10:0	RW	0x000	sw_mask1_startix Horizontal start pixel for mask area 1. Defines the x coordinate. Coordinate 0,0 means the up-left corner in PP output luminance picture. See Table 47 for restrictions

VDPU_SWREG87
Address: Operational Base + offset (0x0015c)
Mask 2 start coordinate register

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22	RW	0x0	sw_mask2_ablend_e Mask 2 alpha blending enable. Instead of masking the output picture the alpha blending is performed. Alpha blending source can be found from alpha blend 2 base address. Alpha blending can be enabled only for RGB/YUYV 422 data.

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Bit	Attr	Reset Value	Description
21:11	RW	0x000	sw_mask_starty Verical start pixel for mask area 2. Defines the y coordinate. Coordinate 0,0 means the up-left corner in PP ouput Y picture. See Table 47 for restrictions
10:0	RW	0x000	sw_mask2_startx Horizontal start pixel for mask area 2. Defines the x coordinate. Coordinate 0,0 means the up-left corner in PP output Y picture. See Table 47 for restrictions

VDPU_SWREG88
Address: Operational Base + offset (0x00160)
Mask 1 size and PP original width register

Bit	Attr	Reset Value	Description
31:23	RW	0x000	sw_ext_orig_width PP input picture original width in macro blocks.
22	RW	0x0	sw_mask1_e Mask 1 enable. If mask 1 is used this bit is high
21:11	RW	0x000	sw_mask1_endy Mask 1 end coordinate y in pixels (inside of PPD output picture). Range must be between [Mask1StartCoordinateY, ScaledHeight].
10:0	RW	0x000	sw_mask1_endx Mask 1 end coordinate x in pixels (inside of PPD output picture). Range must be between [Mask1StartCoordinateX, ScaledWidth]

VDPU_SWREG89
Address: Operational Base + offset (0x00164)
Mask 2 size register

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22	RW	0x0	sw_mask2_e Mask 2 enable. If mask 1 is used this bit is high

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Bit	Attr	Reset Value	Description
21:11	RW	0x000	sw_mask2_endy Mask 2 end coordinate y in pixels (inside of PP output picture). Range must be between [Mask2StartCoordinateY, ScaledHeight].
10:0	RW	0x000	sw_mask2_endx Mask 2 end coordinate x in pixels (inside of PP output picture). Range must be between [Mask2StartCoordinateX, ScaledWidth].

VDPU_SWREG90
Address: Operational Base + offset (0x00168)
PiP register 0

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29	RW	0x0	sw_right_cross_e Right side overcross enable. 0 = No right side overcross, 1 = right side overcross
28	RW	0x0	sw_left_cross_e Left side overcross enable. 0 = No left side overcross, 1 = left side overcross
27	RW	0x0	sw_up_cross_e Upward overcross enable. 0 = No upward overcross, 1 = upward overcross
26	RW	0x0	sw_down_cross_e Downward overcross enable. 0 = No downward overcross, 1 = downward overcross
25:15	RW	0x000	sw_up_cross Amount of upward overcross (vertical pixels outside of display from the upper side). Range must be between [0, ScaledHeight].
14:11	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
10:0	RW	0x000	sw_down_cross Amount of downward overcross (vertical pixels outside of displayfrom the down side). Range must be between [0, ScaledHeight].

VDPU_SWREG91
Address: Operational Base + offset (0x0016c)
PiP register 1 and dithering control

Bit	Attr	Reset Value	Description
31:30	RW	0x0	sw_dither_select_r Dithering control for R channel: 00 = dithering disabled 01 = use four-bit dither matrix 10 = use five-bit dither matrix 11 = use six-bit dither matrix
29:28	RW	0x0	sw_dither_select_g Dithering control for G channel: 00 = dithering disabled 01 = use four-bit dither matrix 10 = use five-bit dither matrix 11 = use six-bit dither matrix
27:26	RW	0x0	sw_dither_select_b Dithering control for B channel: 00 = dithering disabled 01 = use four-bit dither matrix 10 = use five-bit dither matrix 11 = use six-bit dither matrix
25:24	RO	0x0	reserved
23:22	RW	0x0	sw_pp_tiled_mode Input data is in tiled mode (at the moment valid only for YCbCr 420 data, pipeline or external mode): 0 = Tiled mode not used 1 = Tiled mode enabled for 8x4 sized tiles 2,3 = reserved
21:11	RW	0x000	sw_right_cross Amount of right side overcross (Horizontal pixels outside o display from the right side). Range must be between [0, ScaledWidth].

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Bit	Attr	Reset Value	Description
10:0	RW	0x000	sw_left_cross Amount of left side overcross (Horizontal pixels outside of display from the left side). Range must be between [0, ScaledWidth].

VDPU_SWREG92
Address: Operational Base + offset (0x00170)
Display width and PP input size extension register

Bit	Attr	Reset Value	Description
31:29	RW	0x0	sw_pp_in_h_ext Extended PP input height. Used with JPEG
28:26	RW	0x0	sw_pp_in_w_ext Extended PP input width. Used with JPEG
25:23	RW	0x0	sw_crop_starty_ext Extended PP input crop start coordinate x. Used with JPEG
22:20	RW	0x0	sw_crop_start_ext Extended PP input crop start coordinate y. Used with JPEG
19:12	RO	0x0	reserved
11:0	RW	0x000	sw_display_width Width of the display in pixels. Max HDTV (1920)

VDPU_SWREG93
Address: Operational Base + offset (0x00174)
Display width and PP input size extension register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_abledn1_base Base address for alpha blending input 1 (if mask1 is used in alpha blending mode). Format of data is 24 bit RGB/ YCbCr and endian/swap -mode is as in PP input. Amount of data is informed with mask 1 size or with ablend1_scanline if ablend cropping is supported in configuration.

VDPU_SWREG94
Address: Operational Base + offset (0x00178)
Base address for alpha blend 2 gui component

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_ablend2_base Base address for alpha blending input 2 (if mask2 is used in alpha blending mode). Format of data is 24 bit RGB/ YCbCr and endian/swap -mode is as in PP input. Amount of data is informed with mask 2 size or with ablend2_scanline if ablend cropping is supported in configuration.

VDPU_SWREG95
Address: Operational Base + offset (0x0017c)
Base address for alpha blend 2 gui component

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25:13	RW	0x0000	sw_ablend2_scan Scanline width in pixels for Ablend 2. Usage enabled if corresponding configuration bit is enabled
12:0	RW	0x0000	sw_ablend1_scan Scanline width in pixels for Ablend 1. Usage enabled if corresponding configuration bit is enabled

VDPU_SWREG98
Address: Operational Base + offset (0x00188)
PP outupt width/height extension

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	sw_pp_out_h_ext sw_pp_out_h_ext PP output heightextension
0	RW	0x0	sw_pp_out_w_ext sw_pp_out_w_ext PP output widthextension

VDPU_SWREG99
Address: Operational Base + offset (0x0018c)
PP fuse register (read only)

Bit	Attr	Reset Value	Description
31	RO	0x1	fuse_pp_pp 1 = PP enabled
30	RO	0x1	fuse_pp_deint 1 = Deinterlacing enabled

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Bit	Attr	Reset Value	Description
29	RO	0x1	fuse_pp_ablend 1 = Alpha Blending enabled
28:16	RO	0x0	reserved
15	RO	0x1	fuse_pp_maxw_1920 1 = Max PP output width up to 1920 pixels enabled. Priority coded with priority 1
14	RO	0x1	fuse_pp_maxw_1280 1 = Max PP output width up to 1280 pixels enabled. Priority coded with priority 2
13	RO	0x1	fuse_pp_maxw_720 1 = Max PP output width up to 720 pixels enabled. Priority coded with priority 3
12	RO	0x1	fuse_pp_maxw_352 1 = Max PP output width up to 352 pixels enabled. Priority coded with priority 4
11:0	RO	0x0	reserved

VDPU_SWREG100
Address: Operational Base + offset (0x00190)
Synthesis configuration register post-processor (read only)

Bit	Attr	Reset Value	Description
31	RO	0x1	SW_ABLEND_CROP_E Alpha blending support for input cropping: '0': Not supported. External memory must include the exact image of the area being alpha blended '1' Supported. External memory can include a picture from blended area can be cropped. Requires usage of swreg95
30	RO	0x1	SW_PPD_PIXAC_E Pixel Accurate PP output mode exists: '0'= PIP, Scaling and masks can be adjusted by steps of 8 pixels (width) or 2 pixels (height) '1' = PIP, Scaling and masks can be adjusted by steps of 1 pixel for RGB and 2 pixels for subsampled chroma formats (by using bus specific write strobe functionality)

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Bit	Attr	Reset Value	Description
29	RO	0x1	SW_PPD_TILED_EXIST PP output YCbYCr 422 tiled support (4x4 pixel tiles) '0'=Not supported '1'=Supported
28	RO	0x1	SW_PPD_DITH_EXIST Dithering exists: '0' = no '1' = yes
27:26	RO	0x3	SW_PPD_SCALE_LEVEL Scaling support: 00 = No scaling 01 = Scaling with lo perfomance architecture 10 = Scaling with high performance architecture 11 = Scaling with high performance architecture + fast
25	RO	0x1	SW_PPD_DEINT_EXIST De-interlacing exits: '0' = no '1' = yes
24	RO	0x1	SW_PPD_BLEND_EXIST Alpha blending exists: '0' = no '1' = yes
23	RO	0x1	SW_PPD_IBUFF_LEVEL PP input buffering level: '0' = 1 MB input buffering is used '1' = 4 MB input buffering is used
22:19	RO	0x0	reserved
18	RO	0x1	SW_PPD_OEN_VERSION PP output endian version: '0' = Endian mode supported for other than RGB '1' = Endian mode supported for any output format
17	RO	0x1	SW_PPD_OBUFF_LEVEL PP output buffering level: '0' = 1 unit output buffering is used '1' = 4 unit output buffering is used
16	RO	0x1	SW_PPD_PP_EXIST PPD exists: '0'=no '1'=yes

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Bit	Attr	Reset Value	Description
15:14	RO	0x1	SW_PPD_IN_TILED_L PPD input tiled mode support level 0 = not supported 1 = 8x4 tile size supported
13:11	RO	0x0	reserved
10:0	RO	0x000	SW_PPD_MAX_OWIDTH Max supported PP output width in pixels

VDPU_SWREG101
Address: Operational Base + offset (0x00194)
soft reset signals

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	sw_soft_reset softreset pulse signal softreset pulse signal write to 1'b1, valid; write to 1'b0, invalid;

26.4.11 VPU MMU Register Summary

Name	Offset	Size	Reset Value	Description
VCODEC_MMU_DTE_ADDR	0x00000	W	0x00000000	MMU current page Table address
VCODEC_MMU_STATUS	0x00004	W	0x00000000	MMU status register
VCODEC_MMU_COMMAND	0x00008	W	0x00000000	MMU command register
VCODEC_MMU_PAGE_FAULT_ADDR	0x0000c	W	0x00000000	MMU logical address of last page fault
VCODEC_MMU_ZAP_ONE_LINE	0x00010	W	0x00000000	MMU Zap cache line register
VCODEC_MMU_INT_RAWSTAT	0x00014	W	0x00000000	MMU raw interrupt status register
VCODEC_MMU_INT_CLEAR	0x00018	W	0x00000000	MMU raw interrupt status register
VCODEC_MMU_INT_MASK	0x0001c	W	0x00000000	MMU raw interrupt status register
VCODEC_MMU_INT_STATUS	0x00020	W	0x00000000	MMU raw interrupt status register
VCODEC_MMU_AUTO_GATING	0x00024	W	0x00000001	mmu auto gating

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

26.4.12 VPU MMU Detail Register Description

VCODEC_MMU_DTE_ADDR
Address: Operational Base + offset (0x00000)

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MMU current page Table address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	MMU_DTE_ADDR MMU_DTE_ADDR MMU current page Table address

VCODEC_MMU_STATUS

Address: Operational Base + offset (0x00004)

MMU status register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:6	RO	0x00	PAGE_FAULT_BUS_ID page fault bus id Index of master reponsible for last page fault
5	RO	0x0	PAGE_FAULT_IS_WRITE page fault acess The direction of access for last page fault: 0 = Read 1 = Write
4	RO	0x0	REPLAY_BUFFER_EMPTY replay buffer empty The MMU replay buffer is empty
3	RO	0x0	MMU_IDLE mmu idle The MMU is idle when accesses are being translated and there are no unfinished translated accesses.
2	RO	0x0	STAIL_ACTIVE stall active MMU stall mode currently enabled. The mode is enabled by command
1	RO	0x0	PAGE_FAULT_ACTIVE page fault active MMU page fault mode currently enabled . The mode is enabled by command.
0	RO	0x0	PAGING_ENABLED Paging is enabled Paging is enabled

VCODEC_MMU_COMMAND

Address: Operational Base + offset (0x00008)

MMU command register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
2:0	WO	0x0	MMU_CMD mmu cmd MMU_CMD. This can be: 0: MMU_ENABLE_PAGING 1: MMU_DISABLE_PAGING 2: MMU_ENABLE_STALL 3: MMU_DISABLE_STALL 4: MMU_ZAP_CACHE 5: MMU_PAGE_FAULT_DONE 6: MMU_FORCE_RESET

VCODEC_MMU_PAGE_FAULT_ADDR
Address: Operational Base + offset (0x0000c)
MMU logical address of last page fault

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	PAGE_FAULT_ADDR page fault addr address of last page fault

VCODEC_MMU_ZAP_ONE_LINE
Address: Operational Base + offset (0x00010)
MMU Zap cache line register

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	MMU_ZAP_ONE_LINE zap one line address to be invalidated from the page table cache

VCODEC_MMU_INT_RAWSTAT
Address: Operational Base + offset (0x00014)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR read bus error read bus error status
0	RW	0x0	PAGE_FAULT page fault page fault status

VCODEC_MMU_INT_CLEAR
Address: Operational Base + offset (0x00018)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1	WO	0x0	READ_BUS_ERROR read bus error write 1 to clear read bus error
0	WO	0x0	PAGE_FAULT page fault clear write 1 to page fault clear

VCODEC_MMU_INT_MASK
Address: Operational Base + offset (0x0001c)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR read bus error enable the read bus interrupt souce when this bit is set to 1'b1
0	RW	0x0	PAGE_FAULT page fault mask enable the page fault interrupt souce when this bit is set to 1'b1

VCODEC_MMU_INT_STATUS
Address: Operational Base + offset (0x00020)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RO	0x0	READ_BUS_ERROR read bus error read bus error status
0	RO	0x0	PAGE_FAULT page fault page fault status

VCODEC_MMU_AUTO_GATING
Address: Operational Base + offset (0x00024)
mmu auto gating

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x1	mmu_auto_clkgating mmu_auto_clkgating when it is 1'b1, the mmu will auto gating it self

26.4.13 VDPU Pref_cacheRegister Summary

Name	Offset	Size	Reset Value	Description

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Name	Offset	Size	Reset Value	Description
pref_cache_VERSION	0x00000	W	0xcac20101	VERSION register
pref_cache_SIZE	0x00004	W	0x06170206	L2 cache SIZE
pref_cache_STATUS	0x00008	W	0x00000000	Status register
pref_cache_COMMAND	0x00010	W	0x00000000	Command setting register
pref_cache_CLEAR_PAGE	0x00014	W	0x00000000	clear page register
pref_cache_MAX_READS	0x00018	W	0x0000001c	maximum read register
pref_cache_PERFCNT_SRC0	0x00020	W	0x00000000	performance counter 0 source register
pref_cache_PERFCNT_VAL0	0x00024	W	0x00000000	performance counter 0 value register
pref_cache_PERFCNT_SRC1	0x00028	W	0x00000000	performance counter 0 source register
pref_cache_PERFCNT_VAL1	0x0002c	W	0x00000000	performance counter 1 value register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

26.4.14 VDPU_Pref_cache Detail Register Description

pref_cache_VERSION
Address: Operational Base + offset (0x00000)
VERSION register

Bit	Attr	Reset Value	Description
31:16	RO	0xcac2	PRODUCT_ID Field0000 Abstract Field0000 Description
15:8	RO	0x01	VERSION_MAJOR Field0000 Abstract Field0000 Description
7:0	RO	0x01	VERSION_MINOR Field0000 Abstract Field0000 Description

pref_cache_SIZE
Address: Operational Base + offset (0x00004)
L2 cache SIZE

Bit	Attr	Reset Value	Description
31:24	RO	0x06	External_bus_width Field0000 Abstract Log2 external bus width in bits
23:16	RO	0x17	CACHE_SIZE Field0000 Abstract Log2 cache size in bytes
15:8	RO	0x02	ASSOCIATIVITY Field0000 Abstract Log2 associativity

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Bit	Attr	Reset Value	Description
7:0	RO	0x06	LINE_SIZE Field0000 Abstract Log2 line size in bytes

pref_cache_STATUS
Address: Operational Base + offset (0x00008)
Status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	DATA_BUSY Field0000 Abstract set when the cache is busy handling data
0	RW	0x0	CMD_BUSY Field0000 Abstract set when the cache is busy handling commands

pref_cache_COMMAND
Address: Operational Base + offset (0x00010)
Command setting register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:4	RW	0x0	sw_addrb_sel Field0000 Abstract 2'b00: to sel b[14:6] 2'b01: to sel b[15:9], b[7:6] 2'b10: to sel b[16:10], b[7:6] 2'b11: to sel b[17:11], b[7:6]
3	RO	0x0	reserved
2:0	RW	0x0	COMMAND Field0000 Abstract The possible command is 1 = Clear entire cache

pref_cache_CLEAR_PAGE
Address: Operational Base + offset (0x00014)
clear page register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	CLEAR_PAGE Field0000 Abstract writing an address, invlidates all lines in that page from the cache

pref_cache_MAX_READS
Address: Operational Base + offset (0x00018)

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maximum read register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x1c	MAX_READS Field0000 Abstract Limit the number of outstanding read transactions to this amount

pref_cache_PERFCNT_SRC0

Address: Operational Base + offset (0x00020)

performance counter 0 source register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x0	PERFCNT_SRC0 Field0000 Abstract This register holds all the possible source values for Performance Counter 0 0: total clock cycles 1: active clock cycles 2: read transactions, master 3: word reads, master 4: read transactions, slave 5: word reads, slave 6: read hit, slave 7: read misses, slave 8: read invalidates, slave 9: cacheable read transactions, slave 10: bad hit nmber, slave

pref_cache_PERFCNT_VAL0

Address: Operational Base + offset (0x00024)

performance counter 0 value register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PERFCNT_VAL0 Field0000 Abstract Performance counter 0 value

pref_cache_PERFCNT_SRC1

Address: Operational Base + offset (0x00028)

performance counter 0 source register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3:0	RW	0x0	PERFCNT_SRC1 Field0000 Abstract This register holds all the possible source values for Performance Counter 1 0: total clock cycles 1: active clock cycles 2: read transactions, master 3: word reads, master 4: read transactions, slave 5: word reads, slave 6: read hit, slave 7: read misses, slave 8: read invalidates, slave 9: cacheable read transactions, slave 10: bad hit nmber, slave

pref_cache_PERFCNT_VAL1
Address: Operational Base + offset (0x0002c)
performance counter 1 value register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PERFCNT_VAL1 Field0000 Abstract Performance counter 1 value

26.5 Application Notes

26.5.1 HEVC Configuration flow

1. Write the hevc_vpu_sel bit in the GRF_SOC_CONx register with "1" to choose HEVC to work.
2. Prepare the data in the DDR.
3. Set the HEVC general system configuration in HEVC.swreg2, such as working mode, in/out endian.
4. Set the picture parameters with HEVC.swreg3.
5. Set the input and output data base address and HEVC reference configuration with HEVC.swreg4~HEVC.swreg43.
6. If CABAC error detection is desired, set the HEVC.swreg44 to enable the corresponding error detection.
7. Set the interrupt configuration and start the HEVC with HEVC.swreg1.
8. Wait for the frame interrupt, and then get the processed results in the target DDR
9. Clear all the interrupts, and repeat Process2~Process8 to start a new frame decoding if the decoding is not finished yet.

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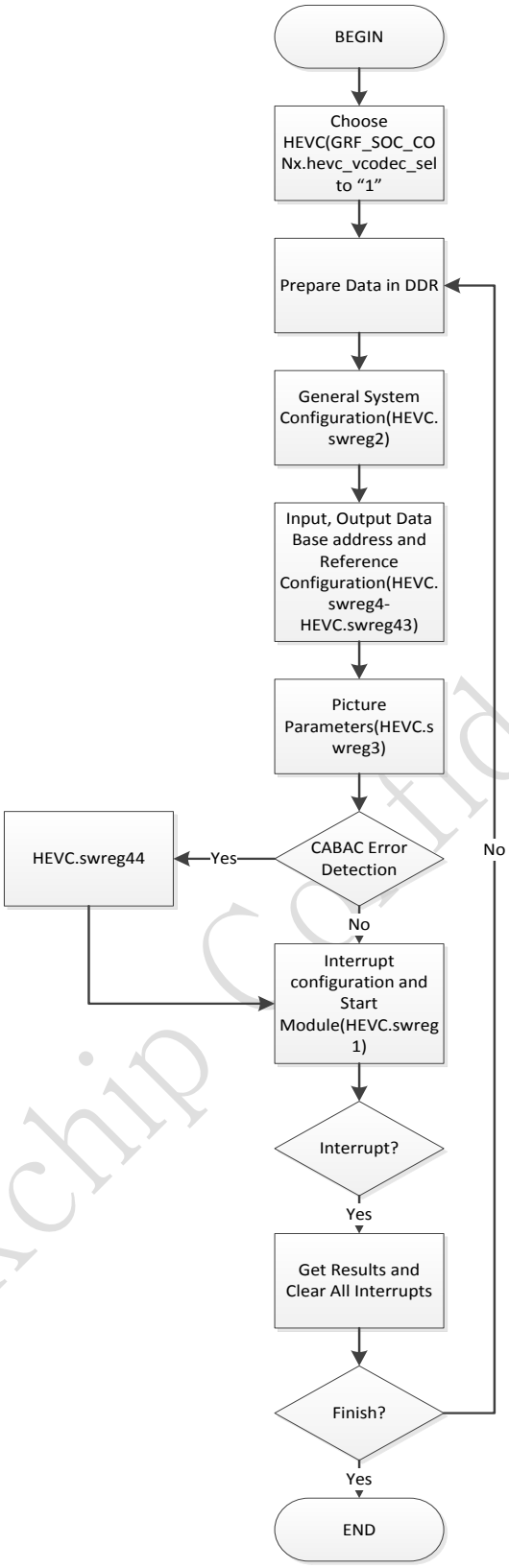

Fig. 26-6 HEVC Common Configuration Flow

26.5.2 VPU Configuration flow

- Because HEVC and VPU share the same memory and ahb bus and some of axi bus, we should set the hevc_vpu_sel bit in GRF_SOC_CONx to 1’b0 to select VPU to work.
- Prepare the decoder data in the DDR memory, And in decoder other than JPEG decoder, the input stream buffer should at least contain a slice or a frame data,

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otherwise the decoder will produce an interrupt and show error and then reset itself.

- Config all the registers will be used. and please notice that which be list as follows:
- In encoder----- We can configure the registers to control the input picture data format (such as endian and swap), but some input data format are fixed, such as cabac_table data. And the register VEPU_SWREG64~95 are JPEG quantization registers. They are write only registers. When you want to write these registers, you should first set VEPU_SWREG14[0] to 1'b0 and VEPU_SWREG14[2:1] to 2'b10(select JPEG mode).
- In decoder---- The decoder can support ref buffer mode or cacheable mode, but they can't be both enabled. We can config the swreg57[6],swreg57[7] to enable cache and config the swreg51 to control the ref buffer.
- You should config VDPU_SWREG1[0] as 1'b1 to enable video decoder. And config VDPU_SWREG60[0] as 1'b1 to enable pp. If pp performed in pipeline with decoder, you should config VDPU_SWREG60[1] as 1'b1 and then config VDPU_SWREG1[0] as 1'b1 to enable decoder and pp. VEPU_SWREG14[0] set to 1'b1 to enable encoder.
- Wait for the frame interrupt,and then check if the frame decoder ready interrupt is right or not, after that,you can get the processed results in the target DDR
- Clear all the interrupts,repeat step 2~5 to start a new frame decoder or encoder.

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Chapter 27 Video Input Processor (VIP)

27.1 Overview

VOP is the display interface from memory frame buffer to display device (LCD panel, LVDS, MIPI, eDP, HDMI and TV set). VOP is connected to an AHB bus through an AHB slave and AXI bus through an AXI master. The register setting is configured through the AHB slave interface and the display frame data is read through the AXI master interface. Furthermore, there is a direct data path between IEP and VOP, which can provide frame data from IEP to VOP.

27.1.1 Features

- Display interface
  - Parallel RGB LCD Interface
    - RGB888,RGB666,RGB565
  - TV Interface
    - ITU-R BT.656(8-bit, 480i/576i/1080i)
    - 3 output mode: valid data in lower 8bit, middle 8bit and higher 8bit
  - HDMI interface
    - support 480p/576p/720p/1080p/2160p/1080i
    - support RGB/YUV420 format
  - Max resolution
    - Max input resolution： 4096x2304
    - Max output resolution： 4096x2160
  - Scaning timing 8192x4096
  - Support configurable polarity of DCLK/HSYNC/VSYNC/DEN
  - 4 groups of scanning output
    - PARALLEL RGB, LVDS, LVDS, HDMI, eDP
    - output pol configurable
  - MIPI control
    - MIPI DCS command mode

- Display process
  - CABC
  - BCSH,8bit
    - Brightness,Contrast,Saturation,Hue adjustment
    - YUV-8bit，RGB-8bit
  - Dither down
    - allegro for RGB565 and RGB666
    - FRC with configurable pattern for RGB666
  - Gamma
    - LUT(lookup table) for R/G/B respectively
    - 8bit RGB look up table
    - gamma before dither
  - Support display data swap
    - BG swap, RB swap, RG swap, dummy swap
  - Support three YUV2RGB transition modes:
    - 8bit-YUV: rec601-mpeg/rec601-jpeg/rec709-hd
  - Support two RGB2YUV transition modes:
    - 8bit-RGB: rec601-full/rec709-full
  - blank display
  - black display
  - standby mode
  - auto dynamic power control

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- X-MIRROR,Y-MIRROR for win0/win1/win2/win3/hwc
- scale down for TV overscan
  - after overlay
  - arbitrary non-integer scaling ratio
  - horizontal scale down using bilinear, 0.5~1.0
  - vertical scale down using bilinear, 0.5~1.0

- Layer process
  - Background layer
    - programmable 24-bit color
  - Win0/Win1 layer
    - Support data format
      - RGB888, ARGB888, RGB565,
      - YCbCr420SP, YCbCr422SP, YCbCr444SP
      - YUV-8bit
    - YUV clip
      - Y-8bit: 16~235;UV-8bit: 16~240
    - Support max input resolution 4096x8192
    - Support max output resolution 4096x2160
    - Support virtual display
    - Support 1/8 to 8 scaling-down and scaling-up engine
      - scale up using bicubic and bilinear
        - Arbitrary non-integer scaling ratio
        - 4 bicubic table for scale up using precise,spline,catrom,Mitchell
      - scale down using bilinear and average
        - Arbitrary non-integer scaling ratio
      - per-pix alpha + scale
    - Support data swap
      - RGB/BPP: rb_swap
      - YUV: mid_swap,uv_swap
    - transparency color key,prior to alpha blending and fading
    - Support fading,prior to alpha blending
    - Support alpha blending
    - Support interlace and de-flicker for interlace output
    - Support IEP direct path input
  - Win2/Win3 layer
    - Support data format
      - RGB888, ARGB888, RGB565
      - 8BPP
      - little endian and big endian for BPP
      - BYPASS and LUT mode(32bit LUT，8bit AA+8bit-RGB) for BPP
    - 4 display regions
      - only one region at one scanning line
    - Support data swap
      - RGB/BPP:rb_swap
    - Support transparency color key,prior to alpha blending and fading
    - Support fading,prior to alpha blending
    - Support alpha blending
    - Support interlace read and interlace output
    - Support IEP direct path input
  - Hardware Cursor layer(HWC for short)
    - Support data format
      - RGB888, ARGB888, RGB565
      - 8BPP
      - little endian and big endian for BPP
      - BYPASS and LUT mode(32bit LUT，8bit AA+8bit-RGB)for BPP
    - Support four hwc size: 32x32,64x64,96x96,128x128
    - Support 2 color modes: normal and reversed color

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- ◆Support fading,prior to alpha blending
- ◆Support alpha blending
- ◆Support displaying out of panel,right or bottom
- ◆Support NORMAL color and REVERSE color mode
- ◆Support interlace read and interlace output

- Overlay
  - ◆ support RGB and YUV domain overlay
  - ◆ Support 6 layers,background/win0/win1/win2/win3/hwc
  - ◆ Win0/Win1/Win2/Win3 overlay position exchangeable
  - ◆ Alpha blending
    - ✧ Support 12 alpha blending modes
    - ✧ Support pre-multiplied alpha
    - ✧ Support global alpha and per_pix alpha
    - ✧ Support 256 level alpha
    - ✧ layer1/layer2/layer3/hwc support alpha

- Bus interface
  - Support AMBA 2.0 AHB slave interface for accessing internal registers and LUT memories，32bit data bus width
  - Support AMBA 3.0 AXI master read interface for loading frame data
    - ◆ 128bit data bus width
  - Support MMU
  - Support two transfer modes
    - ◆ auto outstanding transfer
    - ◆ configuruable outstanding transfer(gather transfer)
  - DMA line mode for YUV
  - Support QoS request for higher bus priority for win2/win3
  - Support NOC hurry for higher bus prioirity for win0/win1
  - Support DMA stop mode
  - Win AXI read ID configurable
  - Max read outstanding number
    - ◆ 32 when MMU disable
    - ◆ 31 when MMU enable

- FBC
  - Support frame buffer compress
    - ◆ Support layer: win0~3
    - ◆ Support format: ARGB888/RGB888/RGB565
    - ◆ Support max size: 2048x1536
    - ◆ Support X-mirror,Y-mirror,rotation 180
    - ◆ Support virtual display

- Interrupt
  - One combined interrupt
    - ◆ high active
    - ◆ raw status readable
    - ◆ combinational with 15 interrupt sources
      - ✧ 2 frame start interrupt
      - ✧ 2 lineflag interrupt
      - ✧ bus error interrupt
      - ✧ win0 empty interrupt
      - ✧ win1 empty interrupt
      - ✧ win2 empty interrupt
      - ✧ win3 empty interrupt
      - ✧ hwc empty interrupt
      - ✧ post empty interrupt
      - ✧ pwm gen interrupt

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- ✧ irq_mmu

27.2 Block Diagram

The architecture is shown in the following figure.

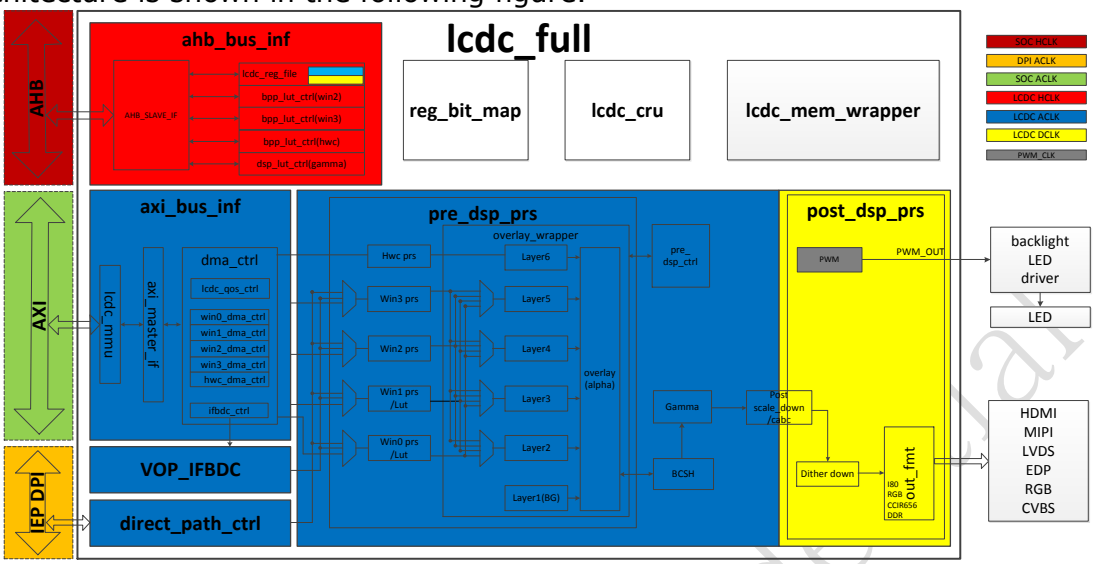

Fig. 27-1 VOP Block Diagram

27.3 Function Description

27.3.1 Pixel format

1.RGB

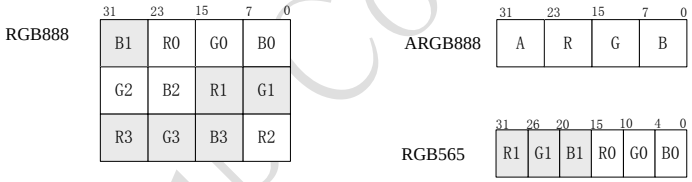

Fig. 27-2 RGB data format

2.YCbCr/YUV(8bit)

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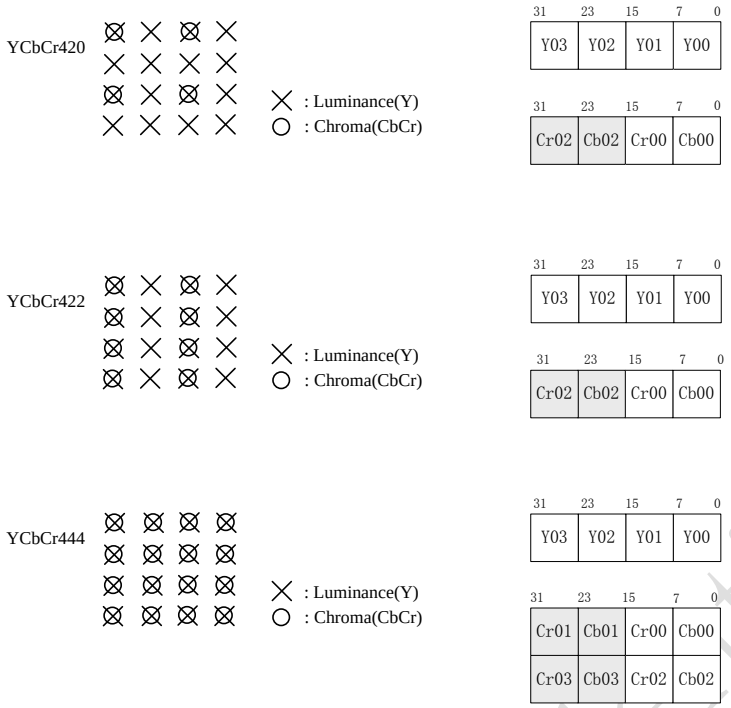

Fig. 27-3 YUV data format

YUV just support SP
YUV-8bit 32bit align

3.BPP

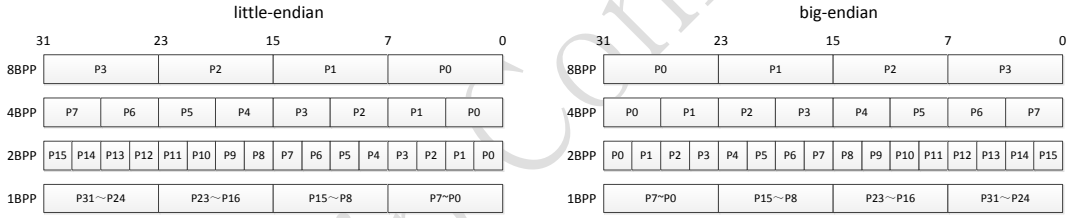

Fig. 27-4 BPP little/big endian data format

27.3.2 Pixel Data Path

There are two data input path for VOP to get display layers' pixel data. One is internal DMA; the other is direction path interface.

1.Internal DMA

Internal DMA can fetch the pixel data through AXI bus from system memory (DDR) for all the display layers. Data fetching is driven by display output requirement.

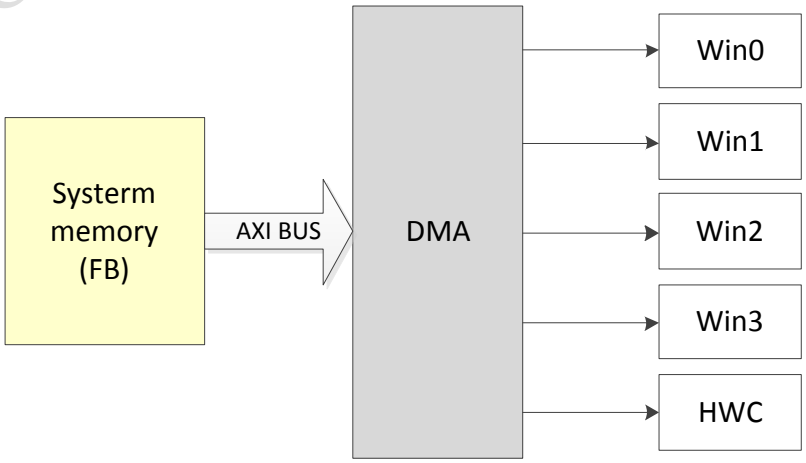

Fig. 27-5 LCDC Internal DMA

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2.Direct Path Interface

Direct path interface (DPI) is used for direct image display of external image processing IP. There is a local bus between VOP and external image processing IP for the data transfer. DPI is connected to WIN0/WIN1/WIN2/WIN3 but can only be configured for One layer use (Win0 or Win1 or Win2 or Win3) in each frame.

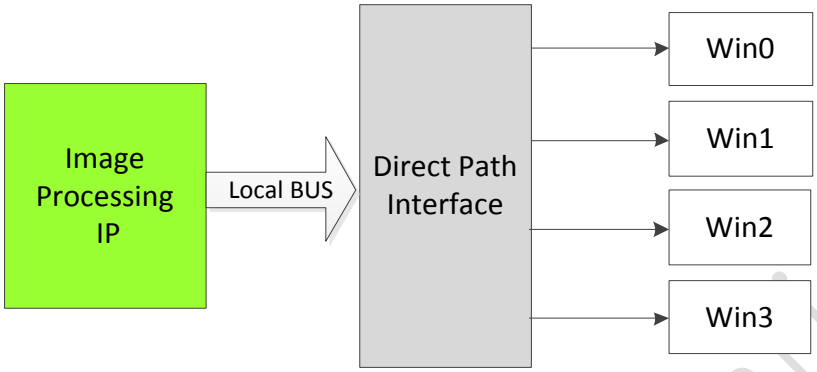

Fig. 27-6 VOP Direct Path Interface

27.3.3 Win Scaling

The scaling operation is the image resizing process by scaling-up or scaling-down the source image from active window size to display window size for displaying on LCD panel or TV set. Horizontal scaling and vertical scaling are realized independently.

1.Scaling factor

（1）scale down or scale up using bilinear

Factor = ((src*2-3)<<11)/(dst-1);

（2）scale up using bicubic

Factor = ((src*2-3)<<15)/(dst-1);

（3）scale down using average

Factor = (dst<<17)/(src*2-1);

2.win scale line buffer mode

For YUV422/YUV420,

(1)LB_YUV_4096X5

If horizontal scale down and dsp_width is greater than 2560 or if horizontal scale up and act_width is greater than 2560,please config win_lb_mode as LB_YUV_4096X5;

(2)LB_YUV_2560X8

If horizontal scale down and dsp_width is less or equal than 2560 or if horizontal scale up and act_width is less or equal than 2560,please config win_lb_mode as LB_YUV_4096X5;

For YUV444/RGB,

(1)LB_RGB_4096X2

If horizontal scale down and dsp_width is greater than 2560 or if horizontal scale up and act_width is greater than 2560,please config win_lb_mode as LB_YUV_4096X2;

This mode does not support vertical scale.

(2)LB_RGB_2560X4

horizontal scale down and dsp_width is greater than 1920 or if horizontal scale up and act_width is greater than 1920,please config win_lb_mode as LB_RGB_2560X4;

This mode only support bilinear for vertical scale up.

(3)LB_RGB_1920X5

horizontal scale down and dsp_width is greater than 1280 or if horizontal scale up and act_width is greater than 1280,please config win_lb_mode as LB_RGB_1920X5;

(4)LB_RGB_1280X8

horizontal scale down and dsp_width is less or equal than 2560 or if horizontal scale up and

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act_width is less or equal than 2560,please config win_lb_mode as LB_RGB_1280X8;

27.3.4 De-flicker

It is necessary to display a non-interlaced video signal on an interlaced display panel (such as TV set). Thus "non-interlaced-to-interlaced conversion" is required.
The easiest approach is to throw away every other active scan line in each non-interlaced frame. Although the cost is minimal, there are problems with this approach. If there is a sharp vertical transition of color or intensity, it will flicker at one-half the refresh rate.
A better solution is to use two lines of non-interlaced data to generate one line of interlace data. Fast vertical transition is smoothed out over several interlace lines.
The vertical filtering of two non-interlaced lines can be done by enabling the vertical scaling offset updated dynamically in different fields, i.e, even field and odd field. The dynamic updated value of scaling offset is half of the scaling factor.

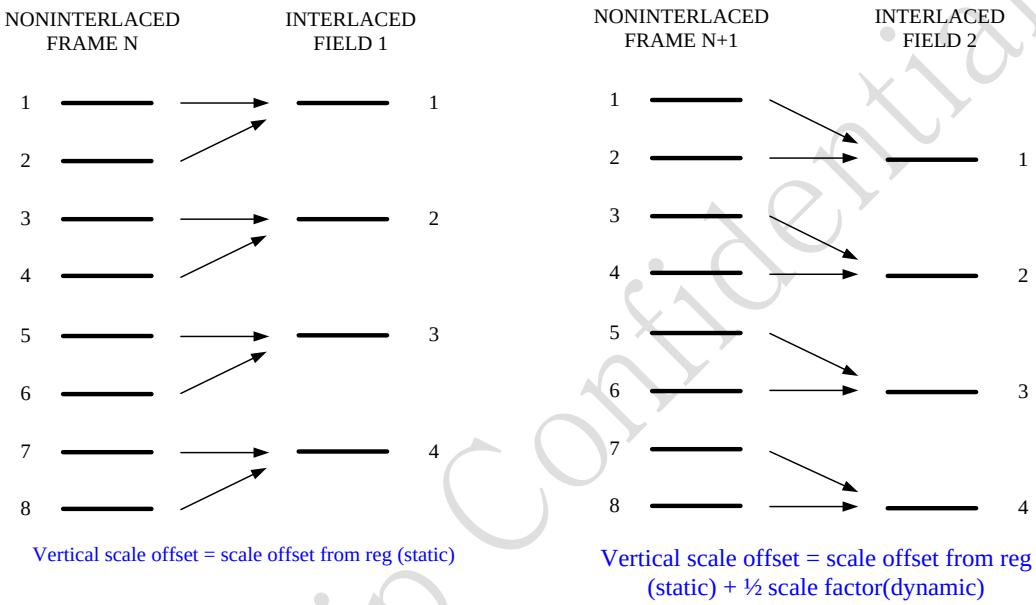

Fig. 27-7 De-flicker

27.3.5 Virtual display

When in virtual display, the active image is part of the virtual (original) image in frame buffer memory.
The virtual width is indicated by setting VIR_STRIDE for different data format. Note that RGB/BPP has one stride(yrgb_vir_stride)； YUV has two virtual stride(yrgb_vir_stride and cbcr_vir_stride).
For RGB-8bit and YUV-8bit，the stride should be multiples of word (32-bit), with dummy bytes in the end of virtual line if the original width is not 32-bit aligned.

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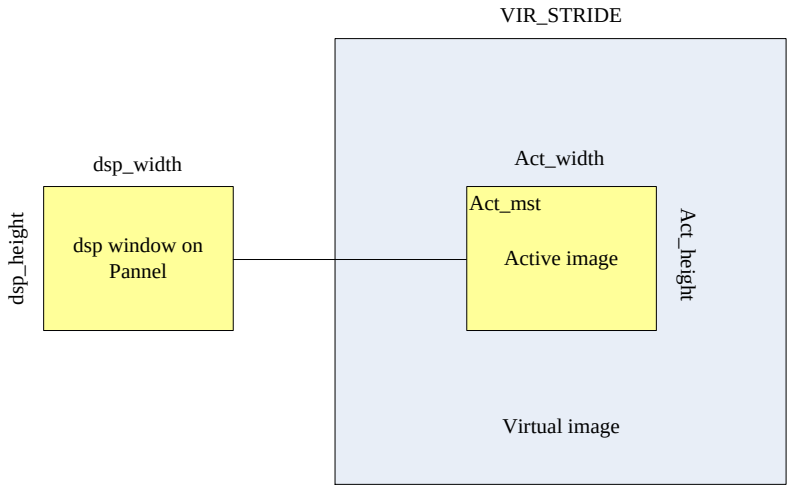

Fig. 27-8 Virtual display

27.3.6 MIRROR display

Mirror display is necessary for the panel with mirror timing interface. There are two types of mirror mode: horizontal mirror(X-mirror) and vertical mirror(Y-mirror).
Win0/1 support X-mirror and Y-mirror;
Win2/3 and HWC support Y-mirror only;
VOP support X-mirror and Y-mirror after overlay.
The default display order is from left to right(L2R) in horizontal direction and from top to bottom(T2B) in vertical direction. However, when X-Mirror is enable, the horizontal display order is from right to left(R2L); when Y-MIRROR is enable, the vertical display order is from bottom to top(B2T).

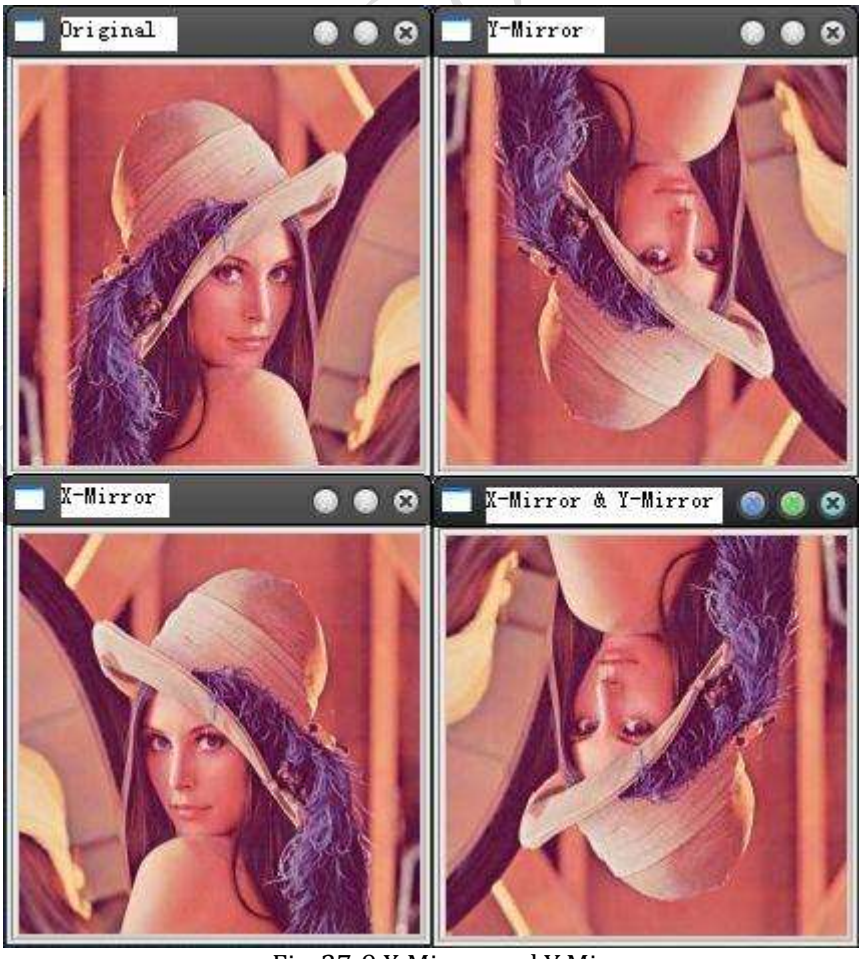

Fig. 27-9 X-Mirror and Y-Mirror

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27.3.7 Display process

1. Overlay display

There are totally 6 layers for overlay display: Background, layer0, layer1, layer2, layer3 and hardware cursor layer(HWC).
Background is a programmable solid color layer, which is always in the bottom of the display screen.
HWC is always on the top of the display screen.

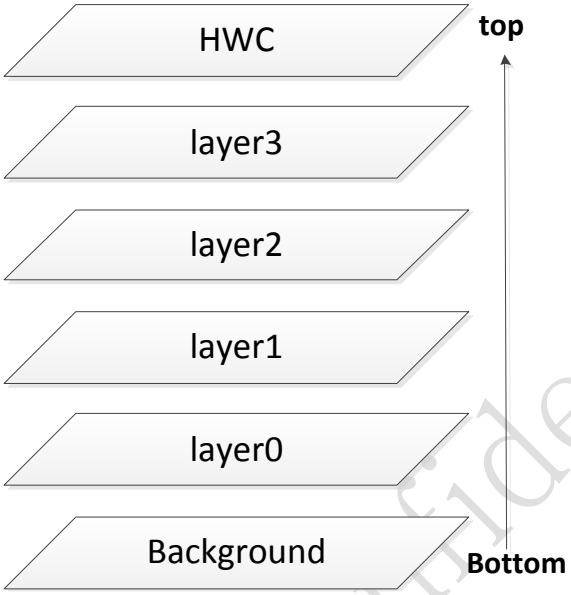
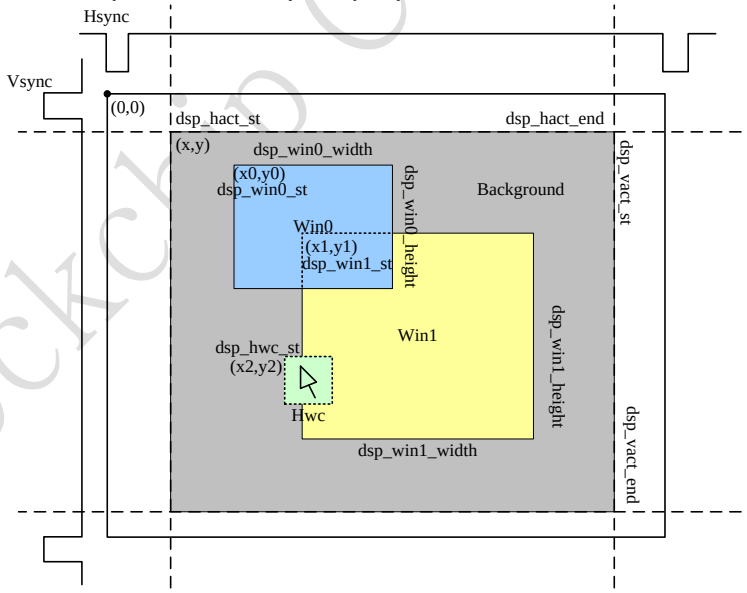


Fig. 27-10 overlay

Following figure is an example of overlay display for win0,win1 and hwc.



2. Post scale down

Post scale down after overlay is supported to fix overscan,that draws the borders of the image beyond the normally visiable area on the screen.
The scale ratio of post scale down is 0.5~1.

Post timing setting

The post scale parameter ,such as,post_dsp_hact_st,post_dsp_hact_end, post_dsp_vact_st,post_dsp_vact_end can be configured.
When post scaling equal "1" ,the post scaler parameter are the same as dsp timing parameter.

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eg:
post_dsp_hact_st = dsp_hact_st
post_dsp_hact_end = dap_hact_end
post_dsp_vact_st = dsp_vact_st
post_dsp_vact_end = dsp_vact_end

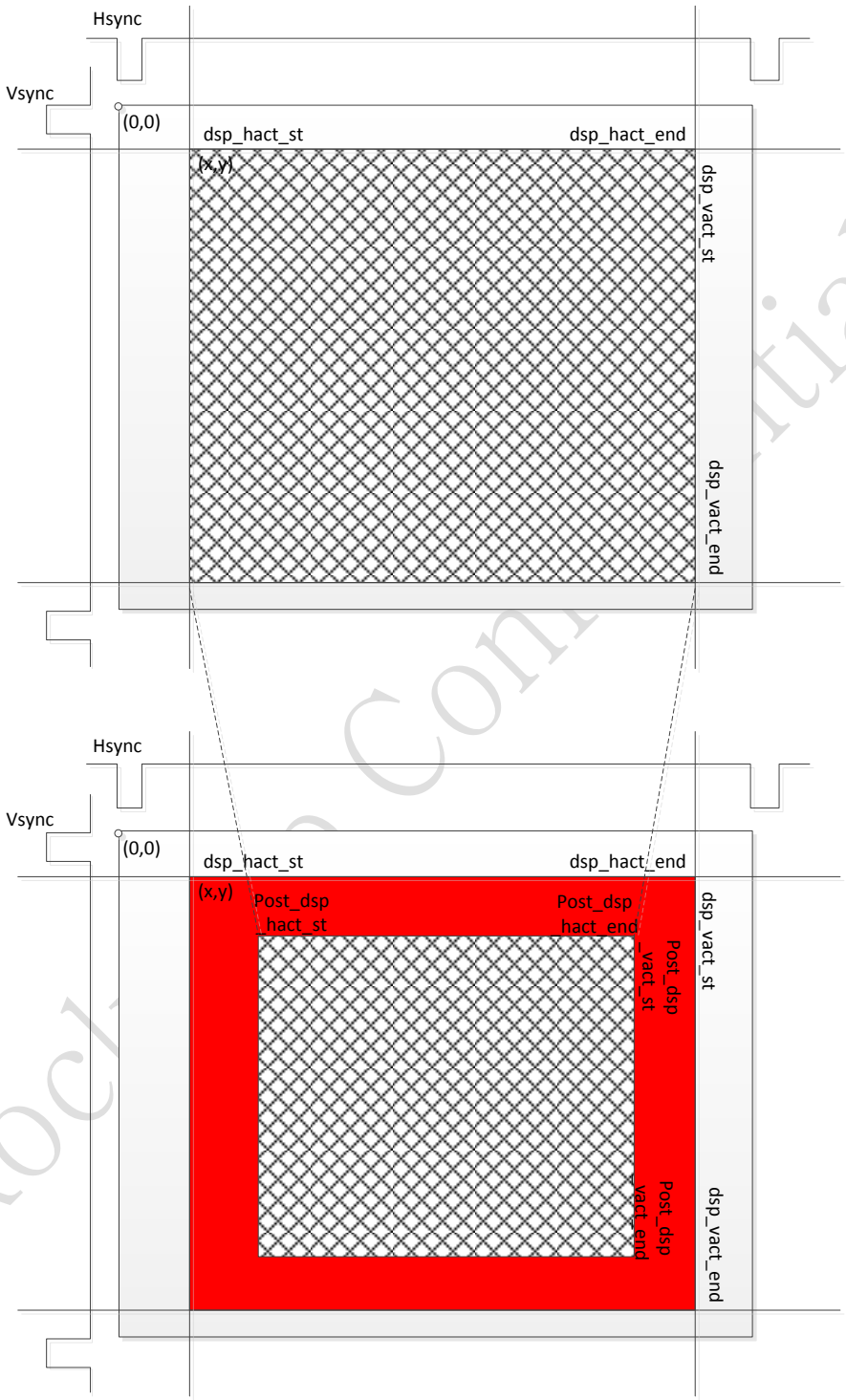

Fig. 27-11 post scaling timing

Post scale down factor
For horizontal scale down,factor = ((src_width*2-3)<<11)/(dst_width-1).
For vertical scale down,factor = ((src_width*2-3)<<11)/(dst_width-1).

3. Transparency color key

The transparency color key value defines the pixel treated as transparent pixel. The pixel

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whose value is equal to the color key value could not be visible on the screen, instead of the pixel in the under layer or solid background color.
There are two transparency color key for win0 layer and win1 layer respectively. When color key is enable, the transparency process is done after scaling but before YUV2RGB color space converter.
Moreover, transparency color key is just available for non-scaling mode.
Following figure is an example of transparency color key for win0 and win1.

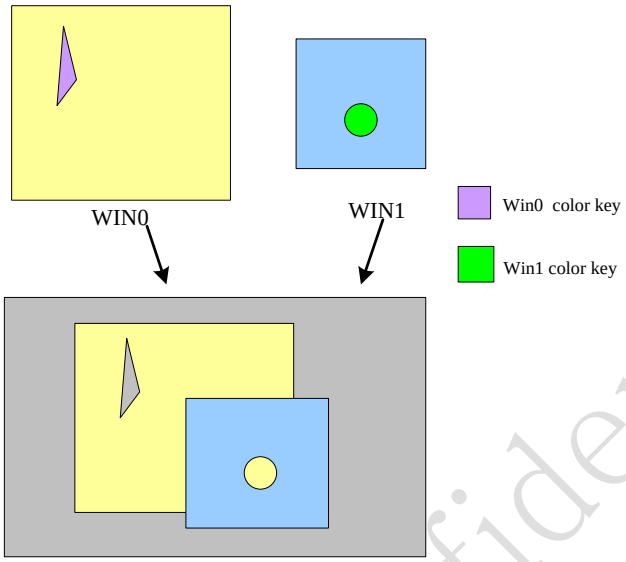
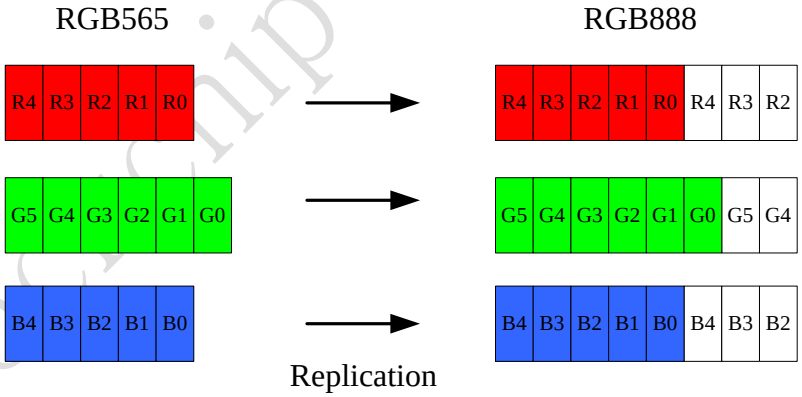


Fig. 27-12 Transparency Color Key

4. Replication(dither up)

If the size of panel data bus is lager than the size of source pixel data,i.e, the source input format is RGB565 and display output format is RGB888, you could do bit replication by replicating MSBs to LSBs if replication is enable (VOP_DSP_CTRL0[9]=1) or filling with "0" to LSBs if replication is disable (VOP_DSP_CTRL0[9]=0).



5. Alpha blending

There are 12 alpha blending mode between two overlay layers for layer1/layer2/layer3/hwc. Layer0 does not support alpha blending with background.VOP supports overlay on RGB or YUV domain.
When in per-pixel mode, the alpha value for every pixel is following with the pixel data. i.e, aRGB，and can be scaled like RGB data. Therefore it is just suitable for win0/win1/win2/win3/hwc layer with ARGB data format.
The alpha blending architechture is shown as follows.

Table 27-1 alpha blending mode settings

Blending Mode	Cs'	Fs	Cd'	Fd
AA_USER_DEFINED	X	User defined	Cd	User defined
AA_CLEAR	X	0	Cd	0
AA_SRC	X	0	Cd	1

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AA_DST	X	1	Cd	1
AA_SRC_OVER	Cs	1	Cd	1-As''
AA_DST_OVER	Cs	1-As''	Cd	1
AA_SRC_IN	Cs	As''	Cd	0
AA_DST_IN	X	0	Cd	As''
AA_SRC_OUT	Cs	1-As''	Cd	0
AA_DST_OUT	X	0	Cd	1-As''
AA_SRC_ATOP	Cs	As''	Cd	1-As''
AA_DST_ATOP	Cs	1-As''	Cd	As''
AA_XOR	Cs	1-As''	Cd	1-As''
AA_SRC_OVER_GLOBAL	Cs*As''	Ags''	Cd	1-As''

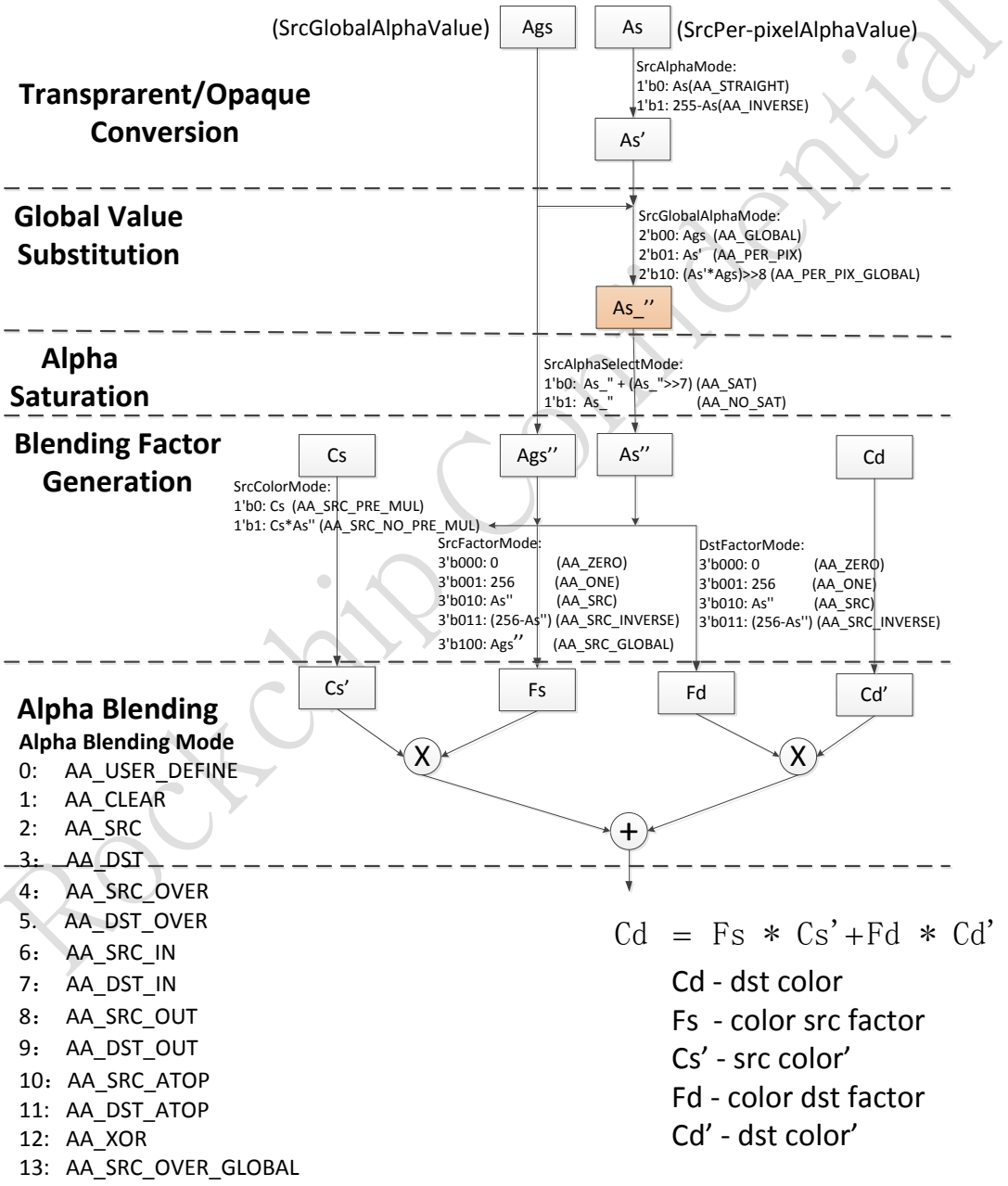

Fig. 27-13 alpha configuration flow

Pseudo Code:

```
switch(alpha_config->alpha_blending_mode)
{
    case AA_USER_DEFINE:
```

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```
break;
case AA_CLEAR:
alpha_config->src_factor_mode=AA_ZERO;
alpha_config->dst_factor_mode=AA_ZERO;
break;
case AA_SRC:
alpha_config->src_factor_mode=AA_ONE;
alpha_config->dst_factor_mode=AA_ZERO;
break;
case AA_DST:
alpha_config->src_factor_mode=AA_ZERO;
alpha_config->dst_factor_mode=AA_ONE;
break;
case AA_SRC_OVER:
alpha_config->src_color_mode=AA_SRC_PRE_MUL;
alpha_config->src_factor_mode=AA_ONE;
alpha_config->dst_factor_mode=AA_SRC_INVERSE;
break;
case AA_DST_OVER:
alpha_config->src_color_mode=AA_SRC_PRE_MUL;
alpha_config->src_factor_mode=AA_SRC_INVERSE;
alpha_config->dst_factor_mode=AA_ONE;
break;
case AA_SRC_IN:
alpha_config->src_color_mode=AA_SRC_PRE_MUL;
alpha_config->src_factor_mode=AA_SRC;
alpha_config->dst_factor_mode=AA_ZERO;
break;
case AA_DST_IN:
alpha_config->src_factor_mode=AA_ZERO;
alpha_config->dst_factor_mode=AA_SRC;
break;
case AA_SRC_OUT:
alpha_config->src_color_mode=AA_SRC_PRE_MUL;
alpha_config->src_factor_mode=AA_SRC_INVERSE;
alpha_config->dst_factor_mode=AA_ZERO;
break;
case AA_DST_OUT:
alpha_config->src_factor_mode=AA_ZERO;
alpha_config->dst_factor_mode=AA_SRC_INVERSE;
break;
case AA_SRC_ATOP:
alpha_config->src_color_mode=AA_SRC_PRE_MUL;
alpha_config->src_factor_mode=AA_SRC;
alpha_config->dst_factor_mode=AA_SRC_INVERSE;
break;
case AA_DST_ATOP:
alpha_config->src_color_mode=AA_SRC_PRE_MUL;
alpha_config->src_factor_mode=AA_SRC_INVERSE;
alpha_config->dst_factor_mode=AA_SRC;
break;
case AA_XOR:
alpha_config->src_color_mode=AA_SRC_PRE_MUL;
alpha_config->src_factor_mode=AA_SRC_INVERSE;
alpha_config->dst_factor_mode=AA_SRC_INVERSE;
break;
case AA_SRC_OVER_GLOBAL:
```

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```
        alpha_config->src_global_alpha_mode=AA_PER_PIX_GLOBAL;
        alpha_config->src_color_mode=AA_SRC_NO_PRE_MUL;
        alpha_config->src_factor_mode=AA_SRC_GLOBAL;
        alpha_config->dst_factor_mode=AA_SRC_INVERSE;
        break;
        default:
            printf("alpha mode error\n");
            break;
    }
```

6. CABC

CABC(Content Adaptive Backlight Control) is used to increase the contrast of such LCD-screens the backlight can be (globally) dimmed when the image to be displayed is dark (i.e. not comprising high intensity image data) while the image data is numerically corrected and adapted to the reduced backlight intensity.
There are 3x7 Gaussian filter tables in reg 0x1d0~0x1e4.
default value as follow:
0x1c8 : 0x15110903
0x1cc : 0x00030911
0x1d0 : 0x1a150b04
0x1d4 : 0x00040b15
0x1d8 : 0x15110903
0x1dc : 0x00030911
program guide：
step1: Config the panel total pixel number to register CABC_CTRL0 [23:1],and config the calc pixel num to register CABC_CTRL1 [23:1].
(typical: calc_pixel_num / total_pixel_num = 90% ~98%).
step2: Config cabc_stage_up and stage up reciprocal to register CABC_CTRL2 [16:0].
step3: Config cabc_stage_down and stage down reciprocal to register CABC_CTRL3 [16:0].
step4: Config cabc global limit signals to register CABC_CTRL2 [31:23] and CABC_CTRL3 [31:23].
step5: write pwm gamma lut to CABC_GAMMA_LUT_ADDR (vop_base_addr + 0x1800),typical gamma value = 2.2
step6: Config cabc_lut_en and cabc_en to register CABC_CTRL0 [0] and CABC_CTRL1 [0].

7. BCSH

BCSH is used to adjust "Brightness,Contrast,Saturation,Hue,like IEP BCSH-8bit. For details,please refer to IEP chapter.

- The brightness adjust support (-32,31).
- The yuv data of color bar are 8bits.

8. Color space conversion

There are three standards for YUV2RGB-8bit.

- YUV2RGB-8bit

1. yuv to rgb (REC-601) range 0 (Y[16:235], UV[16:240], RGB[0:255])
R = 1.164(Y-16) + 1.596(V-128)
G = 1.164(Y-16) - 0.391(U-128) - 0.813(V-128)
B = 1.164(Y-16) + 2.018(U-128)

2. yuv to rgb (REC-601) range 1 (YUV[0:255], RGB[0:255])
R = (Y-16) + 1.402(V-128)
G = (Y-16) - 0.344(U-128) - 0.714(V-128)
B = (Y-16) + 1.772(U-128)

3. yuv to rgb (REC-709) range 0 (Y[16:235], UV[16:240], RGB[0:255])

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R = 1.164(Y-16) + 1.793(V-128)
G = 1.164(Y-16) - 0.213(U-128) - 0.534(V-128)
B = 1.164(Y-16) + 2.115(U-128)

There are two standards for RGB2YUV.

- RGB2YUV-8bit

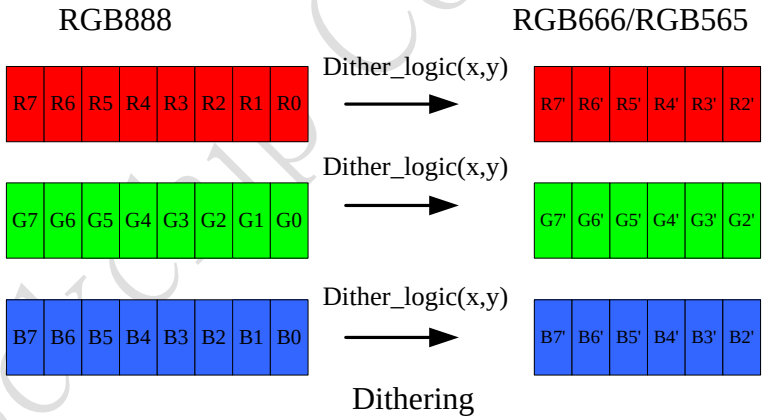
1. rgb to yuv(REC-601) range 0 (Y[16:235], UV[16:240], RGB[0:255])
Y = 0.257R + 0.504G + 0.098B + 16
Cb =-0.148R - 0.291G + 0.439B + 128
Cr = 0.439R - 0.368G - 0.071B + 128

2.rgb to yuv(REC-709) range 0 (Y[16:235], UV[16:240], RGB[0:255])
Y = 0.183R + 0.614G + 0.062B + 16
Cb =-0.101R - 0.338G + 0.439B + 128
Cr = 0.439R - 0.399G - 0.040B + 128

9. Dither Down

Allegro Dither Down

Dithering is an intentional applied form of noise, using to randomize quantization error, and thereby preventing large-scaling patterns such as "banding".
The pixel value is used by dithering process to display the data in a lower color depth on the LCD panel, i.e, the source input format is RGB888 and display output format is RGB565 or RGB666. When dithering is enable(VOP_DSP_CTRL0[2]=1), the output data is generated by dithering algorithm based on the pixel position and the value of removed bits. Otherwise, the MSBs of the pixel color components are output as display data.
There are two dither modes: "RGB888 to RGB666" and "RGB888 to RGB565", which is defined by VOP_DSP_CTRL0[3]. When VOP_DSP_CTRL0[10] is 1, dithering with "RGB888 to RGB666" is available; otherwise, "RGB888 to RGB565" is used.



FRC Dither Down

The pattern of dither frc was configured by vop regfile vop_base+0x1e8~0x1fc.
The following fig is the default pattern picture in vop, you can config different value of regfile 0x1e8~0x1fc to change the pattern picture.

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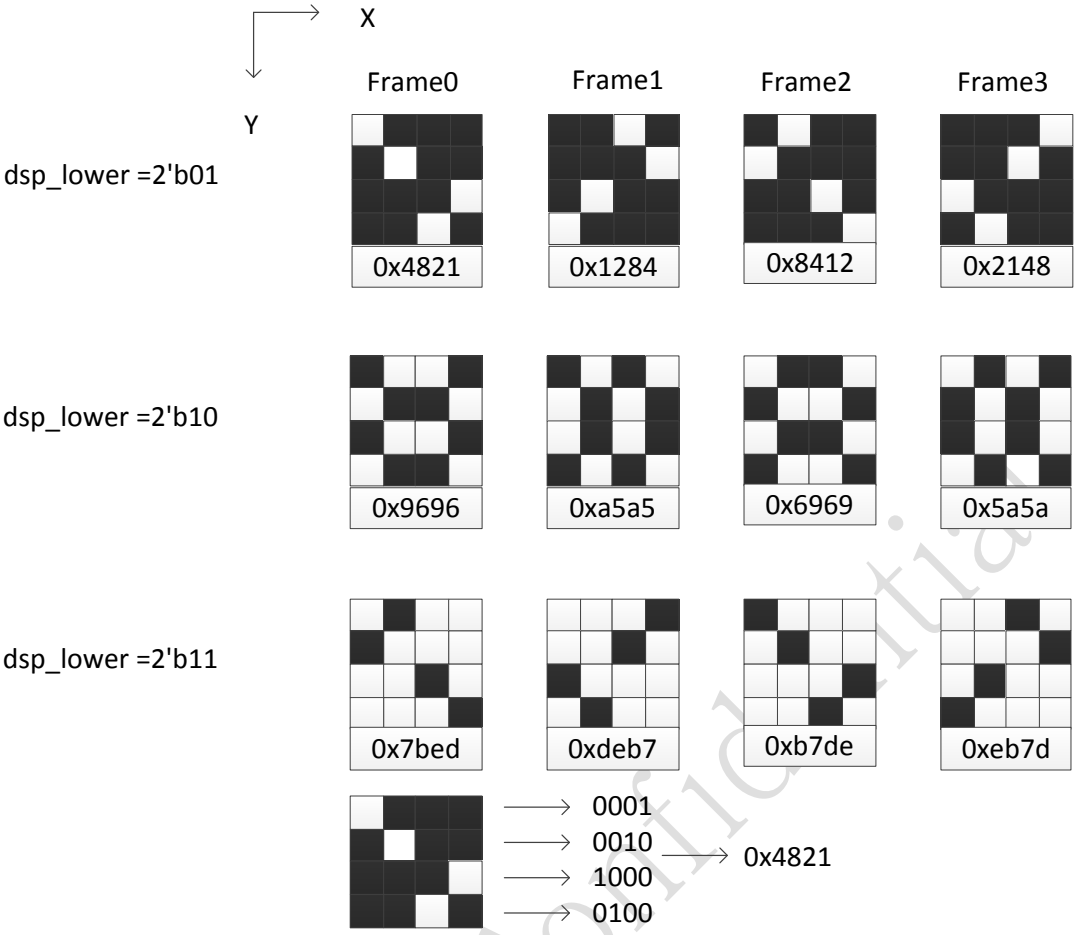

Fig. 27-14 frc pattern diagram

There are four typical pattern as follow ：

(1) default pattern:

0x1e0 : 0x12844821

0x1e4 : 0x21488412

0x1e8 : 0xa55a9696

0x1ec : 0x5aa56969

0x1f0 : 0xdeb77bed

0x1f4 : 0xed7bb7de

(2) for column inversion panel:

0x1e0 : 0x50a00a05

0x1e4 : 0xa050050a

0x1e8 : 0x5a5aa5a5

0x1ec : 0x5a5aa5a5

0x1f0 : 0xaf5ff5fa

0x1f4 : 0x5faffaf5

(3) for 1+2dot panel

0x1e0 : 0x0c308421

0x1e4 : 0x124803c0

0x1e8 : 0xcc339669

0x1ec : 0x33cc9669

0x1f0 : 0xf3cf7bde

0x1f4 : 0xedb7fc3f

(4) default enhance pattern

0x1e0 : 0x12844821

0x1e4 : 0x21488412

0x1e8 : 0x55aaaa55

0x1ec : 0x55aaaa55

0x1f0 : 0xdeb77bed

0x1f4 : 0xed7bb7de

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10. Gamma Correction

Gamma Correction is necessary because most monitors don't have a linear relationship between the voltage and the brightness, which results in your scene looking like it has too much contrast and the light falling off from the source outward, happens too quickly. The result can also be problematic if you are going into a composition program.
You can correct this by "Gamma Correction", which allows you to display the images and textures on your computer in an accurate manner.
Your screen is not linear, in that it displays the brightness unevenly.As a result, the image looks to be more high contrast than it should, you end up adding more lights or turning up the intensity, or you don't use the lighting in a realistic way that matches well with live action scenes. It also creates problems for you if you use compositing software.
There are three 256x8bits line buffers sperately for 8bit-R/G/B gamma correction. You can write gamma correction LUT through register" GAMMA_LUT_ADDR" one by one.

11. Output format

Config dsp_out_mode register to adapt a variety of panel interface. As follow:

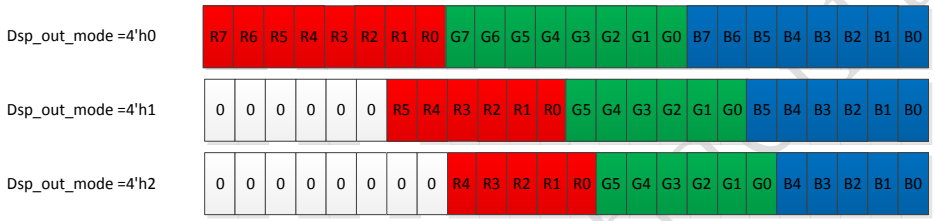

Fig. 27-15 dsp_out_mode description

27.4 Register Description

27.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
VOP_REG_CFG_DONE	0x00000	W	0x00000000	Register config done flag
VOP_VERSION_INFO	0x00004	W	0x00000000	Version for vop
VOP_SYS_CTRL	0x00008	W	0x00821800	System control register0
VOP_SYS_CTRL1	0x0000c	W	0x0003a000	System control register1
VOP_DSP_CTRL0	0x00010	W	0x00000000	Display control register0
VOP_DSP_CTRL1	0x00014	W	0x0000e400	Display control register1
VOP_DSP_BG	0x00018	W	0x00000000	Background color
VOP_MCU_CTRL	0x0001c	W	0x00711c08	MCU mode control register
VOP_LINE_FLAG	0x00020	W	0x00000000	Line flag config register
VOP_INTR_EN	0x00024	W	0x00000000	Interrupt enable register
VOP_INTR_CLEAR	0x00028	W	0x00000000	Interrupt clear register
VOP_INTR_STATUS	0x0002c	W	0x00000000	Interrupt raw status and interrupt status
VOP_WIN0_CTRL0	0x00030	W	0x3a000040	Win0 ctrl register0
VOP_WIN0_CTRL1	0x00034	W	0x00000000	Win0 ctrl register1
VOP_WIN0_COLOR_KEY	0x00038	W	0x00000000	Win0 color key register
VOP_WIN0_VIR	0x0003c	W	0x01400140	Win0 virtual stride
VOP_WIN0_YRGB_MST	0x00040	W	0x00000000	Win0 YRGB memory start address
VOP_WIN0_CBR_MST	0x00044	W	0x00000000	Win0 Cbr memory start address
VOP_WIN0_ACT_INFO	0x00048	W	0x00ef013f	Win0 active window width/height

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Name	Offset	Size	Reset Value	Description
VOP_WIN0_DSP_INFO	0x0004c	W	0x00ef013f	Win0 display width/height on panel
VOP_WIN0_DSP_ST	0x00050	W	0x000a000a	Win0 display start point on panel
VOP_WIN0_SCL_FACTOR_YRGB	0x00054	W	0x10001000	Win0 YRGB scaling factor
VOP_WIN0_SCL_FACTOR_CBR	0x00058	W	0x10001000	Win0 Cbr scaling factor
VOP_WIN0_SCL_OFFSET	0x0005c	W	0x00000000	Win0 scaling start point offset
VOP_WIN0_SRC_ALPHA_CTRL	0x00060	W	0x00000000	Win0 alpha source control register
VOP_WIN0_DST_ALPHA_CTRL	0x00064	W	0x00000000	Win0 alpha destination control register
VOP_WIN0_FADING_CTRL	0x00068	W	0x00000000	Win0 fading contrl register
VOP_WIN0_CTRL2	0x0006c	W	0x00000021	Win0 ctrl register2
VOP_WIN1_CTRL0	0x00070	W	0x3a000040	Win1 ctrl register0
VOP_WIN1_CTRL1	0x00074	W	0x00000000	Win1 ctrl register1
VOP_WIN1_COLOR_KEY	0x00078	W	0x00000000	Win1 color key register
VOP_WIN1_VIR	0x0007c	W	0x01400140	win1 virtual stride
VOP_WIN1_YRGB_MST	0x00080	W	0x00000000	Win1 YRGB memory start address
VOP_WIN1_CBR_MST	0x00084	W	0x00000000	Win1 Cbr memory start address
VOP_WIN1_ACT_INFO	0x00088	W	0x00ef013f	Win1 active window width/height
VOP_WIN1_DSP_INFO	0x0008c	W	0x00ef013f	Win1 display width/height on panel
VOP_WIN1_DSP_ST	0x00090	W	0x000a000a	Win1 display start point on panel
VOP_WIN1_SCL_FACTOR_YRGB	0x00094	W	0x10001000	Win1 YRGB scaling factor
VOP_WIN1_SCL_FACTOR_CBR	0x00098	W	0x10001000	Win1 Cbr scaling factor
VOP_WIN1_SCL_OFFSET	0x0009c	W	0x00000000	Win1 scaling start point offset
VOP_WIN1_SRC_ALPHA_CTRL	0x000a0	W	0x00000000	Win1 alpha source control register
VOP_WIN1_DST_ALPHA_CTRL	0x000a4	W	0x00000000	Win1 alpha destination control register
VOP_WIN1_FADING_CTRL	0x000a8	W	0x00000000	Win1 fading contrl register
VOP_WIN1_CTRL2	0x000ac	W	0x00000043	Win1 ctrl register2
VOP_WIN2_CTRL0	0x000b0	W	0x00000000	win2 ctrl register0
VOP_WIN2_CTRL1	0x000b4	W	0x00501d00	win2 ctrl register1
VOP_WIN2_VIR0_1	0x000b8	W	0x01400140	Win2 virtual stride0 and virtaul stride1
VOP_WIN2_VIR2_3	0x000bc	W	0x01400140	Win2 virtual stride2 and virtaul stride3
VOP_WIN2_MST0	0x000c0	W	0x00000000	Win2 memory start address0
VOP_WIN2_DSP_INFO0	0x000c4	W	0x00ef013f	Win2 display width0/height0 on panel
VOP_WIN2_DSP_ST0	0x000c8	W	0x000a000a	Win2 display start point0 on panel
VOP_WIN2_COLOR_KEY	0x000cc	W	0x00000000	Win2 color key register

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Name	Offset	Size	Reset Value	Description
VOP_WIN2_MST1	0x000d0	W	0x00000000	Win2 memory start address1
VOP_WIN2_DSP_INFO1	0x000d4	W	0x00ef013f	Win2 display width1/height1 on panel
VOP_WIN2_DSP_ST1	0x000d8	W	0x000a000a	Win2 display start point1 on panel
VOP_WIN2_SRC_ALPHA_CTRL	0x000dc	W	0x00000000	Win2 alpha source control register
VOP_WIN2_MST2	0x000e0	W	0x00000000	Win2 memory start address2
VOP_WIN2_DSP_INFO2	0x000e4	W	0x00ef013f	Win2 display width2/height2 on panel
VOP_WIN2_DSP_ST2	0x000e8	W	0x000a000a	Win2 display start point2 on panel
VOP_WIN2_DST_ALPHA_CTRL	0x000ec	W	0x00000000	Win2 alpha destination control register
VOP_WIN2_MST3	0x000f0	W	0x00000000	Win2 memory start address3
VOP_WIN2_DSP_INFO3	0x000f4	W	0x00ef013f	Win2 display width3/height3 on panel
VOP_WIN2_DSP_ST3	0x000f8	W	0x000a000a	Win2 display start point3 on panel
VOP_WIN2_FADING_CTRL	0x000fc	W	0x00000000	Win2 fading contrl register
VOP_WIN3_CTRL0	0x00100	W	0x00000000	Win3 ctrl register0
VOP_WIN3_CTRL1	0x00104	W	0x00601d00	Win3 ctrl register1
VOP_WIN3_VIR0_1	0x00108	W	0x01400140	Win3 virtual stride0 and virtaul stride1
VOP_WIN3_VIR2_3	0x0010c	W	0x01400140	Win3 virtual stride2 and virtaul stride3
VOP_WIN3_MST0	0x00110	W	0x00000000	Win3 memory start address0
VOP_WIN3_DSP_INFO0	0x00114	W	0x00ef013f	Win3 display width0/height0 on panel
VOP_WIN3_DSP_ST0	0x00118	W	0x000a000a	Win3 display start point0 on panel
VOP_WIN3_COLOR_KEY	0x0011c	W	0x00000000	Win3 color key register
VOP_WIN3_MST1	0x00120	W	0x00000000	Win3 memory start address1
VOP_WIN3_DSP_INFO1	0x00124	W	0x00ef013f	Win3 display width1/height1 on panel
VOP_WIN3_DSP_ST1	0x00128	W	0x000a000a	Win3 display start point1 on panel
VOP_WIN3_SRC_ALPHA_CTRL	0x0012c	W	0x00000000	Win3 alpha source control register
VOP_WIN3_MST2	0x00130	W	0x00000000	Win3 memory start address2
VOP_WIN3_DSP_INFO2	0x00134	W	0x00ef013f	Win3 display width2/height2 on panel
VOP_WIN3_DSP_ST2	0x00138	W	0x000a000a	Win3 display start point2 on panel
VOP_WIN3_DST_ALPHA_CTRL	0x0013c	W	0x00000000	Win3 alpha destination control register
VOP_WIN3_MST3	0x00140	W	0x00000000	Win3 memory start address3
VOP_WIN3_DSP_INFO3	0x00144	W	0x00ef013f	Win3 display width3/height3 on panel
VOP_WIN3_DSP_ST3	0x00148	W	0x000a000a	Win3 display start point3 on panel

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Name	Offset	Size	Reset Value	Description
VOP_WIN3_FADING_CTRL	0x0014c	W	0x00000000	Win3 fading contrl register
VOP_HWC_CTRL0	0x00150	W	0x00000000	Hwc ctrl register0
VOP_HWC_CTRL1	0x00154	W	0x00701d00	Hwc ctrl register1
VOP_HWC_MST	0x00158	W	0x00000000	Hwc memory start address
VOP_HWC_DSP_ST	0x0015c	W	0x000a000a	Hwc display start point on panel
VOP_HWC_SRC_ALPHA_CTRL	0x00160	W	0x00000000	Hwc alpha source control register
VOP_HWC_DST_ALPHA_CTRL	0x00164	W	0x00000000	Hwc alpha destination control register
VOP_HWC_FADING_CTRL	0x00168	W	0x00000000	Hwc fading contrl register
VOP_HWC_RESERVED1	0x0016c	W	0x00000000	Hwc reserved
VOP_POST_DSP_HACT_INFO	0x00170	W	0x000a014a	Post scaler down horizontal start and end
VOP_POST_DSP_VACT_INFO	0x00174	W	0x000a00fa	Panel active horizontal scanning start point and end point
VOP_POST_SCL_FACTOR_YRGB	0x00178	W	0x10001000	Post yrgb scaling factor
VOP_POST_RESERVED	0x0017c	W	0x00000000	Post reserved
VOP_POST_SCL_CTRL	0x00180	W	0x00000000	Post scaling start point offset
VOP_POST_DSP_VACT_INFO_F1	0x00184	W	0x000a00fa	Panel active horizontal scanning start point and end point F1
VOP_DSP_HTOTAL_HS_END	0x00188	W	0x014a000a	Panel scanning horizontal width and hsync pulse end point
VOP_DSP_HACT_ST_END	0x0018c	W	0x000a014a	Panel active horizontal scanning start point and end point
VOP_DSP_VTOTAL_VS_END	0x00190	W	0x00fa000a	Panel scanning vertical height and vsync pulse end point
VOP_DSP_VACT_ST_END	0x00194	W	0x000a00fa	Panel active vertical scanning start point and end point
VOP_DSP_VS_ST_END_F1	0x00198	W	0x00000000	Vertical scanning start point and vsync pulse end point of even filed in interlace mode
VOP_DSP_VACT_ST_END_F1	0x0019c	W	0x00000000	Vertical scanning active start point and end point of even filed in interlace mode
VOP_PWM_CTRL	0x001a0	W	0x0000200a	PWM Control Register
VOP_PWM_PERIOD_HPR	0x001a4	W	0x00000000	PWM Period Register/High Polarity Capture Register
VOP_PWM_DUTY_LPR	0x001a8	W	0x00000000	PWM Duty Register/Low Polarity Capture Register
VOP_PWM_CNT	0x001ac	W	0x00000000	PWM Counter Register
VOP_BCSH_COLOR_BAR	0x001b0	W	0x00000000	Color bar config register
VOP_BCSH_BCS	0x001b4	W	0xd0010000	Brightness contrast saturation*contrast config register

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Name	Offset	Size	Reset Value	Description
VOP_BCSH_H	0x001b8	W	0x01000000	Sin hue and cos hue config register
VOP_BCSH_CTRL	0x001bc	W	0x00000000	BCSH contrl register
VOP_CABC_CTRL0	0x001c0	W	0x001db000	Content Adaptive Backlight Control register0
VOP_CABC_CTRL1	0x001c4	W	0x001f4000	Content Adaptive Backlight Control register1
VOP_CABC_CTRL2	0x001c8	W	0x000110f0	Content Adaptive Backlight Control register2
VOP_CABC_CTRL3	0x001cc	W	0x000110f0	Content Adaptive Backlight Control register3
VOP_CABC_GAUSS_LINE0_0	0x001d0	W	0x15110903	CABC gauss line config register00
VOP_CABC_GAUSS_LINE0_1	0x001d4	W	0x00030911	CABC gauss line config register01
VOP_CABC_GAUSS_LINE1_0	0x001d8	W	0x1a150b04	CABC gauss line config register10
VOP_CABC_GAUSS_LINE1_1	0x001dc	W	0x00040b15	CABC gauss line config register11
VOP_CABC_GAUSS_LINE2_0	0x001e0	W	0x15110903	CABC gauss line config register20
VOP_CABC_GAUSS_LINE2_1	0x001e4	W	0x00030911	CABC gauss line config register21
VOP_FRC_LOWER01_0	0x001e8	W	0x12844821	FRC lookup table config register010
VOP_FRC_LOWER01_1	0x001ec	W	0x21488412	FRC lookup table config register011
VOP_FRC_LOWER10_0	0x001f0	W	0xa55a9696	FRC lookup table config register100
VOP_FRC_LOWER10_1	0x001f4	W	0x5aa56969	FRC lookup table config register101
VOP_FRC_LOWER11_0	0x001f8	W	0xdeb77bed	FRC lookup table config register110
VOP_FRC_LOWER11_1	0x001fc	W	0xed7bb7de	FRC lookup table config register111
VOP_IFBDC_CTRL	0x00200	W	0x180e8000	FBDC contrl register
VOP_IFBDC_TILES_NUM	0x00204	W	0x00000000	FBDC tiles num
VOP_IFBDC_FRAME_RST_CYCLE	0x00208	W	0x00000000	FBDC frame reset cycle
VOP_IFBDC_BASE_ADDR	0x0020c	W	0x00000000	FBDC base address
VOP_IFBDC_MB_SIZE	0x00210	W	0x00000000	FBDC macro block size
VOP_IFBDC_CMP_INDEX_INIT	0x00214	W	0x00000000	FBDC compare index initial
VOP_IFBDC_VIR	0x00220	W	0x00000000	FBDC macro block virtual width
VOP_IFBDC_DEBUG0	0x00230	W	0x00000000	FBDC debug0 register
VOP_IFBDC_DEBUG1	0x00234	W	0x00000000	FBDC debug1 register

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Name	Offset	Size	Reset Value	Description
VOP_LATENCY_CTRL0	0x00250	W	0x00000000	Axi performance latency module contrl register0
VOP_RD_MAX_LATENCY_NUM0	0x00254	W	0x00000000	Read max latency number
VOP_RD_LATENCY_THR_NUM0	0x00258	W	0x00000000	The number of bigger than configed threshold value
VOP_RD_LATENCY_SAMP_NUM0	0x0025c	W	0x00000000	Total sample number
VOP_WIN0_DSP_BG	0x00260	W	0x00000000	Win0 layer background color
VOP_WIN1_DSP_BG	0x00264	W	0x00000000	Win1 layer background color
VOP_WIN2_DSP_BG	0x00268	W	0x00000000	Win2 layer background color
VOP_WIN3_DSP_BG	0x0026c	W	0x00000000	Win3 layer background color
VOP_SCAN_LINE_NUM	0x00270	W	0x00000000	Current number of dsp timing
VOP_CABC_DEBUG0	0x00274	W	0x00000000	CABC debug register0
VOP_CABC_DEBUG1	0x00278	W	0x00000000	CABC debug register1
VOP_CABC_DEBUG2	0x0027c	W	0x00000000	CABC debug register2
VOP_DBG_REG_000	0x00280	W	0x00000000	Vop debug register000
VOP_DBG_REG_001	0x00284	W	0x00000000	Vop debug register001
VOP_DBG_REG_002	0x00288	W	0x00000000	Vop debug register002
VOP_DBG_REG_003	0x0028c	W	0x00000000	Vop debug register003
VOP_DBG_REG_004	0x00290	W	0x00000000	Vop debug register004
VOP_DBG_REG_005	0x00294	W	0x00000000	Vop debug register005
VOP_DBG_REG_006	0x00298	W	0x00000000	Vop debug register006
VOP_DBG_REG_007	0x0029c	W	0x00000000	Vop debug register007
VOP_DBG_REG_008	0x002a0	W	0x00000000	Vop debug register008
VOP_DBG_REG_016	0x002c0	W	0x00000000	Vop debug register016
VOP_DBG_REG_017	0x002c4	W	0x00000000	Vop debug register017
VOP_DBG_REG_018	0x002c8	W	0x00000000	Vop debug register018
VOP_DBG_REG_019	0x002cc	W	0x00000000	Vop debug register019
VOP_DBG_REG_020	0x002d0	W	0x00000000	Vop debug register020
VOP_DBG_REG_021	0x002d4	W	0x00000000	Vop debug register021
VOP_DBG_REG_022	0x002d8	W	0x00000000	Vop debug register022
VOP_DBG_REG_023	0x002dc	W	0x00000000	Vop debug register023
VOP_DBG_REG_028	0x002f0	W	0x00000000	Vop debug register028
VOP_MMU_DTE_ADDR	0x00300	W	0x00000000	MMU current page Table address
VOP_MMU_STATUS	0x00304	W	0x00000000	MMU status register
VOP_MMU_COMMAND	0x00308	W	0x00000000	MMU command register
VOP_MMU_PAGE_FAULT_ADDR	0x0030c	W	0x00000000	MMU logical address of last page fault
VOP_MMU_ZAP_ONE_LINE	0x00310	W	0x00000000	MMU Zap cache line register
VOP_MMU_INT_RAWSTAT	0x00314	W	0x00000000	MMU raw interrupt status register
VOP_MMU_INT_CLEAR	0x00318	W	0x00000000	MMU raw interrupt status register
VOP_MMU_INT_MASK	0x0031c	W	0x00000000	MMU raw interrupt status register

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Name	Offset	Size	Reset Value	Description
VOP_MMU_INT_STATUS	0x00320	W	0x00000000	MMU raw interrupt status register
VOP_MMU_AUTO_GATING	0x00324	W	0x00000000	MMU auto gating
VOP_WIN2_LUT_ADDR	0x00400	W	0x00000000	Win2 lut base address
VOP_WIN3_LUT_ADDR	0x00800	W	0x00000000	Win3 lut base address
VOP_HWC_LUT_ADDR	0x00c00	W	0x00000000	Hwc lut base address
VOP_GAMMA_LUT_ADDR	0x01000	W	0x00000000	GAMMA lut base address
VOP_CABC_GAMMA_LUT_ADDR	0x01800	W	0x00000000	CABC GAMMA lut base address
VOP_MCU_BYPASS_WPORT	0x02200	W	0x00000000	Mcu bypass write port
VOP_MCU_BYPASS_RPORT	0x02300	W	0x00000000	Mcu bypass read port

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

27.4.2 Detail Register Description

VOP_REG_CFG_DONE

Address: Operational Base + offset (0x00000)

Register config done flag

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:9	RO	0x0	reserved
8	RW	0x0	reg_load_sys_en vop system register config done flag In the first setting of the register, the new value was saved into the mirror register. When all the system register config finish, writing this register to enable the copyright of the mirror register to real register. Then register would be updated at the start of every frame.
7	RW	0x0	reg_load_fbdc_en vop fbdc register config done flag In the first setting of the register, the new value was saved into the mirror register. When all the fbdc register config finish, writing this register to enable the copyright of the mirror register to real register. Then register would be updated at the start of every frame.

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Bit	Attr	Reset Value	Description
6	RW	0x0	reg_load_iep_en vop iep register config done flag In the first setting of the register, the new value was saved into the mirror register. When all the iep register config finish(only 2 signals direct_path_en,direct_path_layer_sel), writing this register to enable the copyright of the mirror register to real register. Then register would be updated at the start of every frame.
5	RW	0x0	reg_load_hwc_en vop hwc register config done flag In the first setting of the register, the new value was saved into the mirror register. When all the hwc register config finish, writing this register to enable the copyright of the mirror register to real register. Then register would be updated at the start of every frame.
4	RW	0x0	reg_load_win3_en vop win3 register config done flag In the first setting of the register, the new value was saved into the mirror register. When all the win3 register config finish, writing this register to enable the copyright of the mirror register to real register. Then register would be updated at the start of every frame.
3	RW	0x0	reg_load_win2_en vop win2 register config done flag In the first setting of the register, the new value was saved into the mirror register. When all the win2 register config finish, writing this register to enable the copyright of the mirror register to real register. Then register would be updated at the start of every frame.
2	RW	0x0	reg_load_win1_en vop win1 register config done flag In the first setting of the register, the new value was saved into the mirror register. When all the win1 register config finish, writing this register to enable the copyright of the mirror register to real register. Then register would be updated at the start of every frame.

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Bit	Attr	Reset Value	Description
1	RW	0x0	reg_load_win0_en vop win0 register config done flag In the first setting of the register, the new value was saved into the mirror register. When all the win0 register config finish, writing this register to enable the copyright of the mirror register to real register. Then register would be updated at the start of every frame.
0	WO	0x0	reg_load_en vop register config done flag In the first setting of the register, the new value was saved into the mirror register. When all the register config finish, writing this register to enable the copyright of the mirror register to real register. Then register would be updated at the start of every frame.

VOP_VERSION_INFO
Address: Operational Base + offset (0x00004)
Version for vop

Bit	Attr	Reset Value	Description
31:24	RO	0x00	major IP major vertion used for IP structure
23:16	RO	0x00	minor minor vertion big feature change under same structure
15:0	RO	0x0000	svnbuild rtl current svn number

VOP_SYS_CTRL
Address: Operational Base + offset (0x00008)
System control register0

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23	RW	0x1	auto_gating_en LCDC layer axi-clk auto gating enable 1'b0 : disable auto gating 1'b1 : enable auto gating default auto gating enable

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Bit	Attr	Reset Value	Description
22	RW	0x0	vop_standby_en LCDC standby mode Writing "1" to turn LCDC into standby mode, All the layer would disable and the data transfer from frame buffer memory would stop at the end of current frame. The output would be blank. When writing "0" to this bit, standby mode would disable and the LCDC go back to work immediately. 1'b0 : disable 1'b1 : enable * Black display is recommended before setting standby mode enable.
21	RW	0x0	vop_dma_stop VOP DMA stop mode 1'b0 : disable 1'b1 : enable * If DMA is working, the stop mode would not be active until current bus transfer is finished.
20	RW	0x0	vop_mmu_en vop mmu enable signal 1'b0 : bypass mmu 1'b1 : enable mmu
19	RW	0x0	win23_pri_opt_mode 1'b0: win2 win3 dma priority enalbe 1'b1: win2 win3 dma priority disable
18	RW	0x0	post_lb_mode 1'b0 : 4x4096 1'b1 : 8x2048
17	RW	0x1	fs_same_addr_mask_en 1'b0 : disable 1'b1 : enable
16	RW	0x0	overlay_mode 1'b0: RGB overlay 1'b1: YUV overlay
15	RW	0x0	mipi_out_en 1'b0 : gating output clk ,data and control signal 1'b1 : mipi interface enable
14	RW	0x0	edp_out_en 1'b0 : gating output clk ,data and control signal 1'b1 : edp interface enable
13	RW	0x0	hdmi_out_en 1'b0 : gating output clk ,data and control signal 1'b1 : hdmi interface enable

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Bit	Attr	Reset Value	Description
12	RW	0x1	rgb_out_en 1'b0 : gating output clk ,data and control signal 1'b1 : rgb/lvds interface enable
11	RW	0x1	global_regdone_en global config done enalbe 1'b0: DISABLE 1'b1: ENALBE
10	RW	0x0	edpi_wms_fs edpi wms mode ， frame st signal write": edpi_wms_mode frame start (when other register is config done) read : wms mode hold status
9	RW	0x0	edpi_wms_mode 1'b1: mipi command mode
8	RW	0x0	edpi_halt_en mipi flow ctrl enable
7:4	RW	0x0	doub_ch_overlap_num 4'h0: overlap num 0 4'h1: overlap num 2 4'h2: overlap num 4 4'h3: overlap num 6 4'h4: overlap num 8 4'h5: overlap num 10 4'h6: overlap num 12 4'h7: overlap num 14 4'h8: overlap num 16
3	RW	0x0	doub_channel_en mipi double channel enable
2:1	RW	0x0	direct_path_layer_sel direct path layer select 2'b00 : select win0 2'b01 : select win1 2'b10 : select win2 2'b11 : select win3
0	RW	0x0	direct_path_en iep direct path enable signal 1'b0 : disable iep direct path 1'b1 : enable iep direct path

VOP_SYS_CTRL1
Address: Operational Base + offset (0x0000c)
System control register1

Bit	Attr	Reset Value	Description
31:18	RO	0x0	reserved
17:13	RW	0x1d	axi_outstanding_max_num

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Bit	Attr	Reset Value	Description
12	RW	0x0	axi_max_outstanding_en
11:10	RW	0x0	noc_win_qos
9	RW	0x0	noc_qos_en
8:3	RW	0x00	noc_hurry_threshold
2:1	RW	0x0	noc_hurry_value
0	RW	0x0	noc_hurry_en

VOP_DSP_CTRL0
Address: Operational Base + offset (0x00010)
Display control register0

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23	RW	0x0	dsp_y_mir_en 1'b0 : no y_mirror 1'b1 : y_mirror
22	RW	0x0	dsp_x_mir_en 1'b0 : no x_mirror 1'b1 : x_mirror
21	RW	0x0	dsp_yuv_clip YCrCb clip 1'b0 : disable, YCbCr no clip 1'b1 : enable, YCbCr clip before YCbCr2RGB *Y clip: 16~235, CbCr clip: 16~239
20	RW	0x0	dsp_ccir656_avg Cb-Cr filter in CCIR656 mode 1'b0 : drop mode 1'b1 : average mode
19	RW	0x0	dsp_black_en Black display mode When this bit enable, the pixel data output is all black (0x000000)
18	RW	0x0	dsp_blank_en Blank display mode When this bit enable, the Hsync/Vsync/Den output is blank
17	RW	0x0	dsp_out_zero Hsync/Vsync/Den output software ctrl 1'b0 : normal output 1'b1 : all output '0'
16	RW	0x0	dsp_dummy_swap Display dummy swap enable 1'b0 : B+G+R+dummy 1'b1 : dummy+B+G+R

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Bit	Attr	Reset Value	Description
15	RW	0x0	dsp_delta_swap Display delta swap enable 1'b0 : disable 1'b1 : enable *See detail description in Delta display charpter.
14	RW	0x0	dsp_rg_swap Display output red and green swap enable 1'b0 : RGB 1'b1 : GRB
13	RW	0x0	dsp_rb_swap Display output red and blue swap enable 1'b0 : RGB 1'b1 : BGR
12	RW	0x0	dsp_bg_swap Display output blue and green swap enable 1'b0 : RGB 1'b1 : RBG
11	RW	0x0	dsp_field_pol field polarity when interlace dsp 1'b0 : normal 1'b1 : invert
10	RW	0x0	dsp_interlace Interlace display enable 1'b0 : disable 1'b1 : enable *This mode is related to the ITU-R656 output, the display timing of odd field must be set correctly. (lcdc_dsp_vs_st_end_f1/lcdc_dsp_vact_end_f1)
9	RW	0x0	dsp_ddr_phase dclk phase lock 1'b0 : no lock 1'b1 : lock every line
8	RW	0x0	dsp_dclk_ddr dclk output mode 1'b0 : SDR 1'b1 : DDR
7:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3:0	RW	0x0	dsp_out_mode Display output format 4'b0000: Parallel 24-bit RGB888 output R[7:0],G[7:0],B[7:0] 4'b0001: Parallel 18-bit RGB666 output 6'b0,R[5:0],G[5:0],B[5:0] 4'b0010: Parallel 16-bit RGB565 output 8'b0,R[4:0],G[5:0],B[4:0] 4'b0011: Parallel 24-bit RGB888 double pixel mix out phase0:G1[3:0],B1[7:0],G0[3:0],B0[7:0] phase1:R1[7:0],G1[7:4],R0[7:0],G0[7:4] 4'b0100: Serial 2x12-bit 12'b0,G[3:0],B[7:0] + 12'b0,R[7:0],G[7:4] 4'b0101: ITU-656 output mode0 16'b0,pixel_data[7:0] 4'b0110: ITU-656 output mode1 8'b0,pixel_data[7:0],8'b0 4'b0111: ITU-656 output mode2 9'b0,pixel_data[7:0],7'b0 4'b1000: Serial 3x8-bit RGB888 16'b0, B[7:0]+16'b0,G[7:0]+16'b0,R[7:0] 4'b1100: Serial 3x8-bit RGB888 + dummy 16'b0, B[7:0]+16'b0,G[7:0]+16'b0,R[7:0] + dummy 4'b1110: YUV420 output for HDMI 4'b1111: Parallel 30-bit RGBaaa output R[9:0],G[9:0],B[9:0] Others: Reserved.

VOP_DSP_CTRL1
Address: Operational Base + offset (0x00014)
Display control register1

Bit	Attr	Reset Value	Description
31	RW	0x0	mipi_dclk_pol DCLK invert enable 1'b0 : normal 1'b1 : invert default dclk invert
30	RW	0x0	mipi_den_pol DEN polarity 1'b0 : positive 1'b1 : negative

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Bit	Attr	Reset Value	Description
29	RW	0x0	mipi_vsync_pol VSYNC polarity 1'b0 : negative 1'b1 : positive
28	RW	0x0	mipi_hsync_pol HSYNC polarity 1'b0 : negative 1'b1 : positive
27	RW	0x0	edp_dclk_pol DCLK invert enable 1'b0 : normal 1'b1 : invert default dclk invert
26	RW	0x0	edp_den_pol DEN polarity 1'b0 : positive 1'b1 : negative
25	RW	0x0	edp_vsync_pol VSYNC polarity 1'b0 : negative 1'b1 : positive
24	RW	0x0	edp_hsync_pol HSYNC polarity 1'b0 : negative 1'b1 : positive
23	RW	0x0	hdmi_dclk_pol DCLK invert enable 1'b0 : normal 1'b1 : invert default dclk invert
22	RW	0x0	hdmi_den_pol DEN polarity 1'b0 : positive 1'b1 : negative
21	RW	0x0	hdmi_vsync_pol VSYNC polarity 1'b0 : negative 1'b1 : positive
20	RW	0x0	hdmi_hsync_pol HSYNC polarity 1'b0 : negative 1'b1 : positive

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Bit	Attr	Reset Value	Description
19	RW	0x0	rgb_lvds_dclk_pol DCLK invert enable 1'b0 : normal 1'b1 : invert default dclk invert
18	RW	0x0	rgb_lvds_den_pol DEN polarity 1'b0 : positive 1'b1 : negative
17	RW	0x0	rgb_lvds_vsync_pol VSYNC polarity 1'b0 : negative 1'b1 : positive
16	RW	0x0	rgb_lvds_hsync_pol HSYNC polarity 1'b0 : negative 1'b1 : positive
15:14	RW	0x3	dsp_layer3_sel layer3 selection
13:12	RW	0x2	dsp_layer2_sel layer2 selection
11:10	RW	0x1	dsp_layer1_sel layer1 selection
9:8	RW	0x0	dsp_layer0_sel layer0 selection
7	RO	0x0	reserved
6	RW	0x0	dither_up_en
5	RO	0x0	reserved
4	RW	0x0	dither_down_sel dither down mode select 2'b0 : allegro 2'b1 : FRC
3	RW	0x0	dither_down_mode Dither-down mode 1'b0 : RGB888 to RGB565 1'b1 : RGB888 to RGB666
2	RW	0x0	dither_down_en Dither-down enable 1'b0 : disable 1'b1 : enable
1	RW	0x0	pre_dither_down_en 10bit -> 8bit (allegro)

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Bit	Attr	Reset Value	Description
0	RW	0x0	dsp_lut_en Display LUT ram enable 1'b0 : disable 1'b1 : enable *This bit should be "0" when CPU updates the LUT, and should be "1" when Display LUT mode enable.

VOP_DSP_BG
Address: Operational Base + offset (0x00018)
Background color

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:16	RW	0x00	dsp_bg_red Background Red color
15:8	RW	0x00	dsp_bg_green Background Green color
7:0	RW	0x00	dsp_bg_blue Background Blue color

VOP_MCU_CTRL
Address: Operational Base + offset (0x0001c)
MCU mode control register

Bit	Attr	Reset Value	Description
31	RW	0x0	mcu_type MCU LCD output SELECT
30	RW	0x0	mcu_bypass MCU LCD BYPASS MODE Select
29	RW	0x0	mcu_rs MCU LCD RS Select
28	W1C	0x0	mcu_frame_st Write"1" : MCU HOLD Mode Frame Start Read : MCU/LCDC standby HOLD status
27	RW	0x0	mcu_hold_mode MCU HOLD Mode Select
26	RW	0x0	mcu_clk_sel MCU_CLK_SEL for MCU bypass 1'b1 : MCU BYPASS sync with DCLK 1'b0 : MCU BYPASS sync with HCLK
25:20	RW	0x07	mcu_rw_pend MCU_RW signal end point (0-63)
19:16	RW	0x1	mcu_rw_pst MCU_RW signal start point (0-15)
15:10	RW	0x07	mcu_cs_pend MCU_CS signal end point (0-63)

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Bit	Attr	Reset Value	Description
9:6	RW	0x0	mcu_cs_pst MCU_CS signal start point (0-15)
5:0	RW	0x08	mcu_pix_total MCU LCD Interface writing period (1-63)

VOP_LINE_FLAG
Address: Operational Base + offset (0x00020)
Line flag config register

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x0000	dsp_line_flag_num_1 Line number of the Line flag interrupt 1 The display line number when the flag interrupt 1 occur, the range is (0~ DSP_VTOTAL-1).
15:13	RO	0x0	reserved
12:0	RW	0x0000	dsp_line_flag_num_0 Line number of the Line flag interrupt 0 The display line number when the flag interrupt occur, the range is (0~ DSP_VTOTAL-1).

VOP_INTR_EN
Address: Operational Base + offset (0x00024)
Interrupt enable register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RO	0x0	reserved
13	RW	0x0	dsp_hold_valid_intr_en display hold valid interrupt enable 1'b0: DISABLE 1'b1: ENABLE
12	RW	0x0	pwm_gen_intr_en pwm generated interrupt enable 0: Channel 0 Interrupt not generated 1: Channel 0 Interrupt generated
11	RW	0x0	post_buf_empty_intr_en post buffer empty interrupt enable 1'b0: DISABLE 1'b1: ENABLE

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Bit	Attr	Reset Value	Description
10	RW	0x0	hwc_empty_intr_en hwc data empty interrupt enable 1'b0: DISABLE 1'b1: ENABLE
9	RW	0x0	win3_empty_intr_en win3 data empty interrupt enable 1'b0: DISABLE 1'b1: ENABLE
8	RW	0x0	win2_empty_intr_en win2 data empty interrupt enable 1'b0: DISABLE 1'b1: ENABLE
7	RW	0x0	win1_empty_intr_en win1 data empty interrupt enable 1'b0: DISABLE 1'b1: ENABLE
6	RW	0x0	win0_empty_intr_en win0 data empty interrupt enable 1'b0: DISABLE 1'b1: ENABLE
5	RW	0x0	bus_error_intr_en Bus error Interrupt enable 1'b0: DISABLE 1'b1: ENABLE
4	RW	0x0	line_flag1_intr_en Line flag 1 Interrupt enable 1'b0: DISABLE 1'b1: ENABLE
3	RW	0x0	line_flag0_intr_en Line flag 0 Interrupt enable 1'b0: DISABLE 1'b1: ENABLE
2	RW	0x0	addr_same_intr_en memory start addr same interruption enable 1'b0: DISABLE 1'b1: ENABLE
1	RW	0x0	fs_new_intr_en Frame new start interrupt enable 1'b0: DISABLE 1'b1: ENABLE
0	RW	0x0	fs_intr_en Frame start interrupt enable 1'b0: DISABLE 1'b1: ENABLE

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VOP_INTR_CLEAR
Address: Operational Base + offset (0x00028)
Interrupt clear register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RO	0x0	reserved
13	W1C	0x0	dsp_hold_valid_intr_clr display hold valid interrupt clear(Auto clear)
12	W1C	0x0	pwm_gen_intr_clr pwm generated interrupt clear(Auto clear) 0: Channel 0 Interrupt not generated 1: Channel 0 Interrupt generated
11	W1C	0x0	post_buf_empty_intr_clr post buffer empty interrupt clear(Auto clear)
10	W1C	0x0	hwc_empty_intr_clr hwc data empty interrupt clear(Auto clear)
9	W1C	0x0	win3_empty_intr_clr win3 data empty interrupt clear(Auto clear)
8	W1C	0x0	win2_empty_intr_clr win2 data empty interrupt clear(Auto clear)
7	W1C	0x0	win1_empty_intr_clr win1 data empty interrupt clear(Auto clear)
6	W1C	0x0	win0_empty_intr_clr win0 data empty interrupt clear(Auto clear)
5	W1C	0x0	bus_error_intr_clr Bus error Interrupt clear(Auto clear)
4	W1C	0x0	line_flag1_intr_clr Line flag 1 Interrupt clear(Auto clear)
3	W1C	0x0	line_flag0_intr_clr Line flag 0 Interrupt clear(Auto clear)
2	W1C	0x0	addr_same_intr_clr memory start addr same interruption clear(Auto clear)
1	W1C	0x0	fs_new_intr_clr Frame new start interrupt clear (Auto clear)
0	W1C	0x0	fs_intr_clr Frame start interrupt clear (Auto clear)

VOP_INTR_STATUS
Address: Operational Base + offset (0x0002c)
Interrupt raw status and interrupt status

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29	RO	0x0	dsp_hold_valid_intr_raw_sts display hold valid interrupt raw status
28	RO	0x0	pwm_gen_intr_raw_sts pwm generated interrupt raw status 0: Channel 0 Interrupt not generated 1: Channel 0 Interrupt generated
27	RO	0x0	post_buf_empty_intr_raw_sts post buffer empty interrupt raw status
26	RO	0x0	hwc_empty_intr_raw_sts hwc data empty interrupt raw status
25	RO	0x0	win3_empty_intr_raw_sts win3 data empty interrupt raw status
24	RO	0x0	win2_empty_intr_raw_sts win2 data empty interrupt raw status
23	RO	0x0	win1_empty_intr_raw_sts win1 data empty interrupt raw status
22	RO	0x0	win0_empty_intr_raw_sts win0 data empty interrupt raw status
21	RO	0x0	bus_error_intr_raw_sts Bus error Interrupt raw status
20	RO	0x0	line_frag1_intr_raw_sts Line flag 1 Interrupt raw status
19	RO	0x0	line_frag0_intr_raw_sts Line flag 0 Interrupt raw status
18	RW	0x0	addr_same_intr_raw_sts memory start addr same interruption raw status
17	RO	0x0	fs_new_intr_raw_sts Frame start interrupt raw status(when memory start addr are same,no interruption)
16	RO	0x0	fs_intr_raw_sts Frame start raw interrupt status
15:14	RO	0x0	reserved
13	RO	0x0	dsp_hold_valid_intr_sts display hold valid interrupt status
12	RO	0x0	pwm_gen_intr_sts pwm generated interrupt status 0: Channel 0 Interrupt not generated 1: Channel 0 Interrupt generated
11	RO	0x0	post_buf_empty_intr_sts post buffer empty interrupt status
10	RO	0x0	hwc_empty_intr_sts hwc data empty interrupt status

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Bit	Attr	Reset Value	Description
9	RO	0x0	win3_empty_intr_sts win3 data empty interrupt status
8	RO	0x0	win2_empty_intr_sts win2 data empty interrupt status
7	RO	0x0	win1_empty_intr_sts win1 data empty interrupt status
6	RO	0x0	win0_empty_intr_sts win0 data empty interrupt status
5	RO	0x0	bus_error_intr_sts Bus error Interrupt status
4	RO	0x0	line_flag1_intr_sts Line flag 1 Interrupt status
3	RO	0x0	line_flag0_intr_sts Line flag 0 Interrupt status
2	RW	0x0	addr_same_intr_sts memory start addr same interruption status
1	RO	0x0	fs_new_intr_sts Frame start interrupt status(when memory start addr are same,no interruption)
0	RO	0x0	fs_intr_sts Frame start interrupt status

VOP_WIN0_CTRL0
Address: Operational Base + offset (0x00030)
Win0 ctrl register0

Bit	Attr	Reset Value	Description
31:30	RW	0x0	win0_dma_burst_length WIN0 DMA read Burst length 2'b00 : burst16 (burst 15 in rgb888 pack mode) 2'b01 : burst8 (burst 12 in rgb888 pack mode) 2'b10 : burst4 (burst 6 in rgb888 pack mode) 2'b11 : reserved
29:25	RW	0x1d	win0_axi_outstanding_max_num
24	RW	0x0	win0_axi_max_outstanding_en 1'b0 : disable 1'b1 : enable
23	RO	0x0	reserved
22	RW	0x0	win0_y_mir_en 1'b0 : no y_mirror 1'b1 : y_mirror
21	RW	0x0	win0_x_mir_en 1'b0 : no x_mirror 1'b1 : x_mirror

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Bit	Attr	Reset Value	Description
20	RW	0x0	win0_yuv_clip YCrCb clip 1'b0 : disable, YCbCr no clip 1'b1 : enable, YCbCr clip before YCbCr2RGB *Y clip: 16~235, CbCr clip: 16~239
19	RW	0x0	win0_cbr_deflick Win0 Cbr deflick mode 1'b0 : disable 1'b1 : enable
18	RW	0x0	win0_yrgb_deflick win0 YRGB deflick mode 1'b0 : disable 1'b1 : enable
17	RO	0x0	reserved
16	RW	0x0	win0_hw_pre_mul_en 1'b0: no hardware pre multiply mode 1'b1: hardware pre multiply mode
15	RW	0x0	win0_uv_swap Win0 CbCr swap 1'b0 : CrCb 1'b1 : CbCr
14	RW	0x0	win0_mid_swap Win0 Y middle swap 1'b0 : Y3Y2Y1Y0 1'b1 : Y3Y1Y2Y0
13	RW	0x0	win0_alpha_swap win0 alpha swap 1'b0 : ARGB 1'b1 : RGBA
12	RW	0x0	win0_rb_swap win0 RGB RED and BLUE swap 1'b0 : RGB 1'b1 : BGR
11:10	RW	0x0	win0_csc_mode Win0 YUV2RGB or RGB2YUV Color space conversion(YUV2RGB): 2'b00 : mpeg 2'b01 : jpeg 2'b10 : hd 2'b11 : mpeg Color space conversion(RGB2YUV): 2'bx0: BT601 2'bx1: BT709

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Bit	Attr	Reset Value	Description
9	RW	0x0	win0_no_outstanding win0 AXI master read outstanding 1'b0 : enable 1'b1 : disable
8	RW	0x0	win0_interlace_read Win0 interlace read mode 1'b0 : disable 1'b1 : enable
7:5	RW	0x2	win0_lb_mode 3'b000: LB_YUV_4096X5 3'b001: LB_YUV_2560X8 3'b010: LB_RGB_4096X2 3'b011: LB_RGB_2560X4 3'b100: LB_RGB_1920X5 3'b101: LB_RGB_1280X8 others : Reserved
4	RW	0x0	win0_fmt_10 0: yuv 8bit fmt mode 1: yuv 10bit fmt mode
3:1	RW	0x0	win0_data_fmt 3'b000 : ARGB888 3'b001 : RGB888 3'b010 : RGB565 3'b100 : YcbCr420 3'b101 : YcbCr422 3'b110 : YcbCr444
0	RW	0x0	win0_en 1'b0 : disable 1'b1 : enable

VOP_WIN0_CTRL1
Address: Operational Base + offset (0x00034)
Win0 ctrl register1

Bit	Attr	Reset Value	Description
31	RW	0x0	win0_cbr_vsd_mode win0 vertical scaler down mode select 1'b0 : bilinear 1'b1 : average
30	RW	0x0	win0_cbr_vsu_mode win0 vertical scaler down mode select 1'b0 : bilinear 1'b1 : bicubic

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Bit	Attr	Reset Value	Description
29:28	RW	0x0	win0_cbr_hsd_mode win0 horizontal scaler down mode select 2'b00 : bilinear 2'b01 : bicubic 2'b10 : average
27:26	RW	0x0	win0_cbr_ver_scl_mode
25:24	RW	0x0	win0_cbr_hor_scl_mode 2'b00 : no scale 2'b01 : scale up 2'b10 : scale down 2'b11 : no scale
23	RW	0x0	win0_yrgb_vsd_mode win0 vertical scaler down mode select 1'b0 : bilinear 1'b1 : average
22	RW	0x0	win0_yrgb_vsu_mode win0 vertical scaler down mode select 1'b0 : bilinear 1'b1 : bicubic
21:20	RW	0x0	win0_yrgb_hsd_mode win0 horizontal scaler down mode select 2'b00 : bilinear 2'b01 : average
19:18	RW	0x0	win0_yrgb_ver_scl_mode 2'b00 : no scale 2'b01 : scale up 2'b10 : scale down 2'b11 : no scale
17:16	RW	0x0	win0_yrgb_hor_scl_mode 2'b00 : no scale 2'b01 : scale up 2'b10 : scale down 2'b11 : no scale
15	RW	0x0	win0_line_load_mode when yuv fmt,if 1'b0: load data by axi trans 1'b1: load data by lines
14:12	RW	0x0	win0_cbr_axi_gather_num
11:8	RW	0x0	win0_yrgb_axi_gather_num
7	RW	0x0	win0_vsd_cbr_gt2
6	RW	0x0	win0_vsd_cbr_gt4
5	RW	0x0	win0_vsd_yrgb_gt2
4	RW	0x0	win0_vsd_yrgb_gt4

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Bit	Attr	Reset Value	Description
3:2	RW	0x0	win0_bic_coe_sel 2'b00 : PRECISE 2'b01 : SPLINE 2'b10 : CATROM 2'b11 : MITCHELL
1	RW	0x0	win0_cbr_axi_gather_en
0	RW	0x0	win0_yrgb_axi_gather_en

VOP_WIN0_COLOR_KEY
Address: Operational Base + offset (0x00038)
Win0 color key register

Bit	Attr	Reset Value	Description
31	RW	0x0	win0_key_en Win0 transparency color key enable 1'b0 : disable; 1'b1 : enable;
30	RO	0x0	reserved
29:0	RW	0x00000000	win0_key_color Win0 key color

VOP_WIN0_VIR
Address: Operational Base + offset (0x0003c)
Win0 virtual stride

Bit	Attr	Reset Value	Description
31:16	RW	0x0140	win0_vir_stride_uv
15:0	RW	0x0140	win0_vir_stride Win0 Virtual stride Number of words of Win0 Virtual width ARGB888 : win0_vir_width RGB888 : (win0_vir_width*3/4) + (win0_vir_width%3) RGB565 : ceil(win0_vir_width/2) YUV : ceil(win0_vir_width/4)

VOP_WIN0_YRGB_MST
Address: Operational Base + offset (0x00040)
Win0 YRGB memory start address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	win0_yrgb_mst win0 YRGB frame buffer memory start address

VOP_WIN0_CBR_MST
Address: Operational Base + offset (0x00044)
Win0 Cbr memory start address

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	win0_cbr_mst win0 CBR frame buffer memory start address

VOP_WIN0_ACT_INFO
Address: Operational Base + offset (0x00048)
Win0 active window width/height

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x00ef	win0_act_height Win0 active(original) window height win_act_height = (win0 vertical size -1)
15:13	RO	0x0	reserved
12:0	RW	0x013f	win0_act_width Win0 active(original) window width win_act_width = (win0 horizontial size -1)

VOP_WIN0_DSP_INFO
Address: Operational Base + offset (0x0004c)
Win0 display width/height on panel

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:16	RW	0x0ef	win0_dsp_height Win0 display window height win0_dsp_height = (win0 vertical size -1)
15:12	RO	0x0	reserved
11:0	RW	0x13f	win0_dsp_width Win0 display window width win0_dsp_width = (win0 horizontial size -1)

VOP_WIN0_DSP_ST
Address: Operational Base + offset (0x00050)
Win0 display start point on panel

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	win0_dsp_yst Win0 vertical start point(y) of the Panel scanning
15:13	RO	0x0	reserved
12:0	RW	0x000a	win0_dsp_xst Win0 horizontal start point(x) of the Panel scanning

VOP_WIN0_SCL_FACTOR_YRGB
Address: Operational Base + offset (0x00054)
Win0 YRGB scaling factor

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Bit	Attr	Reset Value	Description
31:16	RW	0x1000	win0_vs_factor_yrgb Win0 YRGB vertical scaling factor: factor=((LCDC_WIN0_ACT_INFO[31:16]) /(LCDC_WIN0_DSP_INFO[31:16]))*2^12
15:0	RW	0x1000	win0_hs_factor_yrgb Win0 YRGB horizontal scaling factor: factor=((LCDC_WIN0_ACT_INFO[15:0]) /(LCDC_WIN0_DSP_INFO[15:0]))*2^12

VOP_WIN0_SCL_FACTOR_CBR
Address: Operational Base + offset (0x00058)
Win0 Cbr scaling factor

Bit	Attr	Reset Value	Description
31:16	RW	0x1000	win0_vs_factor_cbr Win0 CBR vertical scaling factor: YCbCr420: factor=((LCDC_WIN0_ACT_INFO[31:16]/ 2) /(LCDC_WIN0_DSP_INFO[31:16]))*2^12 YCbCr422,YCbCr444: factor=((LCDC_WIN0_ACT_INFO[31:16]) /(LCDC_WIN0_DSP_INFO[31:16]))*2^12
15:0	RW	0x1000	win0_hs_factor_cbr Win0 CBR horizontal scaling factor: YCbCr422,YCbCr420: factor=((LCDC_WIN0_ACT_INFO[15:0]/2) /(LCDC_WIN0_DSP_INFO[15:0]))*2^12 YCbCr444: factor=((LCDC_WIN0_ACT_INFO[15:0]) /(LCDC_WIN0_DSP_INFO[15:0]))*2^12

VOP_WIN0_SCL_OFFSET
Address: Operational Base + offset (0x0005c)
Win0 scaling start point offset

Bit	Attr	Reset Value	Description
31:24	RW	0x00	win0_vs_offset_cbr Cbr Vertical scaling start point offset (0x00~0xff)/0x100 = 0~0.99
23:16	RW	0x00	win0_vs_offset_yrgb Y Vertical scaling start point offset (0x00~0xff)/0x100 = 0~0.99
15:8	RW	0x00	win0_hs_offset_cbr Cbr Horizontal scaling start point offset (0x00~0xff)/0x100 = 0~0.99

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Bit	Attr	Reset Value	Description
7:0	RW	0x00	win0_hs_offset_yrgb Y Horizontal scaling start point offset (0x00~0xff)/0x100 = 0~0.99

VOP_WIN0_SRC_ALPHA_CTRL
Address: Operational Base + offset (0x00060)
Win0 alpha source control register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	win0_fading_value
23:16	RW	0x00	win0_src_global_alpha src global alpha
15:9	RO	0x0	reserved
8:6	RW	0x0	win0_src_factor_mode
5	RW	0x0	win0_src_alpha_cal_mode
4:3	RW	0x0	win0_src_blend_mode
2	RW	0x0	win0_src_alpha_mode
1	RW	0x0	win0_src_color_mode
0	RW	0x0	win0_src_alpha_en

VOP_WIN0_DST_ALPHA_CTRL
Address: Operational Base + offset (0x00064)
Win0 alpha destination control register

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8:6	RW	0x0	win0_dst_factor_mode
5:0	RW	0x00	win0_dst_m0_reserved

VOP_WIN0_FADING_CTRL
Address: Operational Base + offset (0x00068)
Win0 fading contrl register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	layer0_fading_en
23:16	RW	0x00	layer0_fading_offset_b
15:8	RW	0x00	layer0_fading_offset_g
7:0	RW	0x00	layer0_fading_offset_r

VOP_WIN0_CTRL2
Address: Operational Base + offset (0x0006c)
Win0 ctrl register2

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x2	win_rid_win0_cbr axi read id of win0 cbr channel

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Bit	Attr	Reset Value	Description
3:0	RW	0x1	win_rid_win0_yrgb axi read id of win0 yrgb channel

VOP_WIN1_CTRL0
Address: Operational Base + offset (0x00070)
Win1 ctrl register0

Bit	Attr	Reset Value	Description
31:30	RW	0x0	win1_dma_burst_length WIN1 DMA read Burst length 2'b00 : burst16 (burst 15 in rgb888 pack mode) 2'b01 : burst8 (burst 12 in rgb888 pack mode) 2'b10 : burst4 (burst 6 in rgb888 pack mode) 2'b11 : reserved
29:25	RW	0x1d	win1_axi_max_outstanding_num
24	RW	0x0	win1_axi_max_outstanding_en 1'b0 : disable 1'b1 : enable
23	RO	0x0	reserved
22	RW	0x0	win1_y_mir_en 1'b0 : no y_mirror 1'b1 : y_mirror
21	RW	0x0	win1_x_mir_en 1'b0 : no x_mirror 1'b1 : x_mirror
20	RW	0x0	win1_yuv_clip YCrCb clip 1'b0 : disable, YCbCr no clip 1'b1 : enable, YCbCr clip before YCbCr2RGB *Y clip: 16~235, CbCr clip: 16~239
19	RW	0x0	win1_cbr_deflick Win1 Cbr deflick mode 1'b0 : disable 1'b1 : enable
18	RW	0x0	win1_yrgb_deflick win1 YRGB deflick mode 1'b0 : disable 1'b1 : enable
17	RO	0x0	reserved
16	RW	0x0	win1_hw_pre_mul_en 1'b0: no hardware pre multiply mode 1'b1: hardware pre multiply mode

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Bit	Attr	Reset Value	Description
15	RW	0x0	win1_uv_swap Win1 CbCr swap 1'b0 : CrCb 1'b1 : CbCr
14	RW	0x0	win1_mid_swap Win1 Y middle 8-bit swap 1'b0 : Y3Y2Y1Y0 1'b1 : Y3Y1Y2Y0
13	RW	0x0	win1_alpha_swap win1 alpha swap 1'b0 : ARGB 1'b1 : RGBA
12	RW	0x0	win1_rb_swap win1 RGB RED and BLUE swap 1'b0 : RGB 1'b1 : BGR
11:10	RW	0x0	win1_csc_mode Win1 YUV2RGB or RGB2YUV Color space conversion(YUV2RGB): 2'b00 : mpeg 2'b01 : jpeg 2'b10 : hd 2'b11 : mpeg Color space conversion(RGB2YUV): 2'bx0: BT601 2'bx1: BT709
9	RW	0x0	win1_no_outstanding win1 AXI master read outstanding 1'b0 : enable 1'b1 : disable
8	RW	0x0	win1_interlace_read Win1 interlace read mode 1'b0 : disable 1'b1 : enable
7:5	RW	0x2	win1_lb_mode 3'b000: LB_YUV_4096X5 3'b001: LB_YUV_2560X8 3'b010: LB_RGB_4096X2 3'b011: LB_RGB_2560X4 3'b100: LB_RGB_1920X5 3'b101: LB_RGB_1280X8 others : Reserved
4	RW	0x0	win1_fmt_10 0: yuv 8bit fmt mode 1: yuv 10bit fmt mode

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Bit	Attr	Reset Value	Description
3:1	RW	0x0	win1_data_fmt 3'b000 : ARGB888 3'b001 : RGB888 3'b010 : RGB565 3'b100 : YcbCr420 3'b101 : YcbCr422 3'b110 : YcbCr444
0	RW	0x0	win1_en 1'b0 : disable 1'b1 : enable

VOP_WIN1_CTRL1
Address: Operational Base + offset (0x00074)
Win1 ctrl register1

Bit	Attr	Reset Value	Description
31	RW	0x0	win1_cbr_vsd_mode win1 vertical scaler down mode select 1'b0 : bilinear 1'b1 : average
30	RW	0x0	win1_cbr_vsu_mode win1 vertical scaler up mode select 1'b0 : bilinear 1'b1 : bicubic
29:28	RW	0x0	win1_cbr_hsd_mode win1 horizontal scaler down mode select 2'b00 : bilinear 2'b01 : bicubic 2'b10 : average
27:26	RW	0x0	win1_cbr_ver_scl_mode 2'b00 : no scale 2'b01 : scale up 2'b10 : scale down 2'b11 : no scale
25:24	RW	0x0	win1_cbr_hor_scl_mode 2'b00 : no scale 2'b01 : scale up 2'b10 : scale down 2'b11 : no scale
23	RW	0x0	win1_yrgb_vsd_mode win1 vertical scaler down mode select 1'b0 : bilinear 1'b1 : average

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Bit	Attr	Reset Value	Description
22	RW	0x0	win1_yrgb_vsu_mode win1 vertical scaler up mode select 1'b0 : bilinear 1'b1 : bicubic
21:20	RW	0x0	win1_yrgb_hsd_mode win1 horizontal scaler down mode select 2'b00 : bilinear 2'b01 : average
19:18	RW	0x0	win1_yrgb_ver_scl_mode 2'b00 : no scale 2'b01 : scale up 2'b10 : scale down 2'b11 : no scale
17:16	RW	0x0	win1_yrgb_hor_scl_mode 2'b00 : no scale 2'b01 : scale up 2'b10 : scale down 2'b11 : no scale
15	RW	0x0	win1_line_load_mode when yuv fmt,if 1'b0: load data by pixels 1'b1: load data by lines
14:12	RW	0x0	win1_cbr_axi_gather_num
11:8	RW	0x0	win1_yrgb_axi_gather_num
7	RW	0x0	win1_vsd_cbr_gt2
6	RW	0x0	win1_vsd_cbr_gt4
5	RW	0x0	win1_vsd_yrgb_gt2
4	RW	0x0	win1_vsd_yrgb_gt4
3:2	RW	0x0	win1_bic_coe_sel 2'b00 : PRECISE 2'b01 : SPLINE 2'b10 : CATROM 2'b11 : MITCHELL
1	RW	0x0	win1_cbr_axi_gather_en
0	RW	0x0	win1_yrgb_axi_gather_en

VOP_WIN1_COLOR_KEY
Address: Operational Base + offset (0x00078)
Win1 color key register

Bit	Attr	Reset Value	Description
31	RW	0x0	win1_key_en Win1 transparency color key enable 1'b0 : disable; 1'b1 : enable;
30	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
29:0	RW	0x00000000	win1_key_color Win1 key color

VOP_WIN1_VIR
Address: Operational Base + offset (0x0007c)
win1 virtual stride

Bit	Attr	Reset Value	Description
31:16	RW	0x0140	win1_vir_stride_uv
15:0	RW	0x0140	win1_vir_stride Win1 Virtual stride Number of words of Win1 Virtual width ARGB888 : win1_vir_width RGB888 : (win1_vir_width*3/4) + (win1_vir_width%3) RGB565 : ceil(win1_vir_width/2) YUV : ceil(win1_vir_width/4)

VOP_WIN1_YRGB_MST
Address: Operational Base + offset (0x00080)
Win1 YRGB memory start address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	win1_yrgb_mst win1 YRGB frame buffer memory start address

VOP_WIN1_CBR_MST
Address: Operational Base + offset (0x00084)
Win1 Cbr memory start address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	win1_cbr_mst win1 CBR frame buffer memory start address

VOP_WIN1_ACT_INFO
Address: Operational Base + offset (0x00088)
Win1 active window width/height

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x00ef	win1_act_height Win1 active(original) window height win_act_height = (win1 vertical size -1)
15:13	RO	0x0	reserved
12:0	RW	0x013f	win1_act_width Win1 active(original) window width win_act_width = (win1 horizontial size -1)

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VOP_WIN1_DSP_INFO
Address: Operational Base + offset (0x0008c)
Win1 display width/height on panel

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:16	RW	0x0ef	win1_dsp_height Win1 display window height win1_dsp_height = (win1 vertical size -1)
15:12	RO	0x0	reserved
11:0	RW	0x13f	win1_dsp_width Win1 display window width win1_dsp_width = (win1 horizontial size -1)

VOP_WIN1_DSP_ST
Address: Operational Base + offset (0x00090)
Win1 display start point on panel

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	win1_dsp_yst Win1 vertical start point(y) of the Panel scanning
15:13	RO	0x0	reserved
12:0	RW	0x000a	win1_dsp_xst Win1 horizontal start point(x) of the Panel scanning

VOP_WIN1_SCL_FACTOR_YRGB
Address: Operational Base + offset (0x00094)
Win1 YRGB scaling factor

Bit	Attr	Reset Value	Description
31:16	RW	0x1000	win1_vs_factor_yrgb Win1 YRGB vertical scaling factor: factor=((LCDC_WIN1_ACT_INFO[31:16]) /(LCDC_WIN1_DSP_INFO[31:16]))*2^12
15:0	RW	0x1000	win1_hs_factor_yrgb Win1 YRGB horizontal scaling factor: factor=((LCDC_WIN1_ACT_INFO[15:0]) /(LCDC_WIN1_DSP_INFO[15:0]))*2^12

VOP_WIN1_SCL_FACTOR_CBR
Address: Operational Base + offset (0x00098)
Win1 Cbr scaling factor

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x1000	win1_vs_factor_cbr Win1 CBR vertical scaling factor: YCbCr420: factor=((LCDC_WIN1_ACT_INFO[31:16]/ 2) /(LCDC_WIN1_DSP_INFO[31:16]))*2^12 YCbCr422,YCbCr444: factor=((LCDC_WIN1_ACT_INFO[31:16]) /(LCDC_WIN1_DSP_INFO[31:16]))*2^12
15:0	RW	0x1000	win1_hs_factor_cbr Win1 Cbr horizontal scaling factor: YCbCr422,YCbCr420: factor=((LCDC_WIN1_ACT_INFO[15:0]/2) /(LCDC_WIN1_DSP_INFO[15:0]))*2^12 YCbCr444: factor=((LCDC_WIN1_ACT_INFO[15:0]) /(LCDC_WIN1_DSP_INFO[15:0]))*2^12

VOP_WIN1_SCL_OFFSET
Address: Operational Base + offset (0x0009c)
Win1 scaling start point offset

Bit	Attr	Reset Value	Description
31:24	RW	0x00	win1_vs_offset_cbr Cbr Vertical scaling start point offset (0x00~0xff)/0x100 = 0~0.99
23:16	RW	0x00	win1_vs_offset_yrgb Y Vertical scaling start point offset (0x00~0xff)/0x100 = 0~0.99
15:8	RW	0x00	win1_hs_offset_cbr Cbr Horizontal scaling start point offset (0x00~0xff)/0x100 = 0~0.99
7:0	RW	0x00	win1_hs_offset_yrgb Y Horizontal scaling start point offset (0x00~0xff)/0x100 = 0~0.99

VOP_WIN1_SRC_ALPHA_CTRL
Address: Operational Base + offset (0x000a0)
Win1 alpha source control register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	win1_fading_value
23:16	RW	0x00	win1_src_global_alpha src global alpha
15:9	RO	0x0	reserved
8:6	RW	0x0	win1_src_factor_mode
5	RW	0x0	win1_src_alpha_cal_mode
4:3	RW	0x0	win1_src_blend_mode

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Bit	Attr	Reset Value	Description
2	RW	0x0	win1_src_alpha_mode
1	RW	0x0	win1_src_color_mode
0	RW	0x0	win1_src_alpha_en

VOP_WIN1_DST_ALPHA_CTRL
Address: Operational Base + offset (0x000a4)
Win1 alpha destination control register

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8:6	RW	0x0	win1_dst_factor_m0
5:0	RW	0x00	win1_dsp_m0_reserved

VOP_WIN1_FADING_CTRL
Address: Operational Base + offset (0x000a8)
Win1 fading contrl register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	win1_fading_en
23:16	RW	0x00	win1_fading_offset_b
15:8	RW	0x00	win1_fading_offset_g
7:0	RW	0x00	win1_fading_offset_r

VOP_WIN1_CTRL2
Address: Operational Base + offset (0x000ac)
Win1 ctrl register2

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x4	win_rid_win1_cbr axi read id of win1 cbr channel
3:0	RW	0x3	win_rid_win1_yrgb axi read id of win1 yrgb channel

VOP_WIN2_CTRL0
Address: Operational Base + offset (0x000b0)
win2 ctrl register0

Bit	Attr	Reset Value	Description
31	RW	0x0	win2_endian_swap3 Win2 region3 8pp palette data Big-endian/ Little-endian select 1'b0 : Big-endian 1'b1 : Little-endian

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Bit	Attr	Reset Value	Description
30	RW	0x0	win2_alpha_swap3 Win2 region3 RGB alpha swap 1'b0 : ARGB 1'b1 : RGBA
29	RW	0x0	win2_rb_swap3 Win2 region3 RGB Red and Blue swap 1'b0 : RGB 1'b1 : BGR
28	RW	0x0	win2_endian_swap2 Win2 region2 8pp palette data Big-endian/ Little-endian select 1'b0 : Big-endian 1'b1 : Little-endian
27	RW	0x0	win2_alpha_swap2 Win2 region2 RGB alpha swap 1'b0 : ARGB 1'b1 : RGBA
26	RW	0x0	win2_rb_swap2 Win2 region2 RGB Red and Blue swap 1'b0 : RGB 1'b1 : BGR
25	RW	0x0	win2_endian_swap1 Win2 region1 8pp palette data Big-endian/ Little-endian select 1'b0 : Big-endian 1'b1 : Little-endian
24	RW	0x0	win2_alpha_swap1 Win2 region1 RGB alpha swap 1'b0 : ARGB 1'b1 : RGBA
23	RW	0x0	win2_rb_swap1 Win2 region1 RGB Red and Blue swap 1'b0 : RGB 1'b1 : BGR
22	RW	0x0	win2_endian_swap0 Win2 region0 8pp palette data Big-endian/ Little-endian select 1'b0 : Big-endian 1'b1 : Little-endian
21	RW	0x0	win2_alpha_swap0 Win2 region0 RGB alpha swap 1'b0 : ARGB 1'b1 : RGBA

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Bit	Attr	Reset Value	Description
20	RW	0x0	win2_rb_swap0 Win2 region0 RGB Red and Blue swap 1'b0 : RGB 1'b1 : BGR
19	RO	0x0	reserved
18:17	RW	0x0	win2_data_fmt3 Win2 region 3 data format 2'b00 : ARGB888 2'b01 : RGB888 2'b10 : RGB565 2'b11 : 8bpp
16	RW	0x0	win2_mst3_en win2 master3 enable 1'b0 : disable 1'b1 : enable
15	RO	0x0	reserved
14:13	RW	0x0	win2_data_fmt2 Win2 region 2 data format 2'b00 : ARGB888 2'b01 : RGB888 2'b10 : RGB565 2'b11 : 8bpp
12	RW	0x0	win2_mst2_en win2 master2 enable 1'b0 : disable 1'b1 : enable
11	RO	0x0	reserved
10:9	RW	0x0	win2_data_fmt1 Win2 region 1 data format 2'b00 : ARGB888 2'b01 : RGB888 2'b10 : RGB565 2'b11 : 8bpp
8	RW	0x0	win2_mst1_en win2 master1 enable 1'b0 : disable 1'b1 : enable
7	RO	0x0	reserved
6:5	RW	0x0	win2_data_fmt0 Win2 region 0 data format 2'b00 : ARGB888 2'b01 : RGB888 2'b10 : RGB565 2'b11 : 8bpp

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Bit	Attr	Reset Value	Description
4	RW	0x0	win2_mst0_en win2 master0 enable 1'b0 : disable 1'b1 : enable
3	RO	0x0	reserved
2	RW	0x0	win2_csc_mode Win2 RGB2YUV conversion mode Color space conversion: 1'b0: BT601 1'b1: BT709
1	RW	0x0	win2_interlace_read Win2 interlace read mode 1'b0 : disable 1'b1 : enable
0	RW	0x0	win2_en 1'b0 : disable 1'b1 : enable

VOP_WIN2_CTRL1
Address: Operational Base + offset (0x000b4)
win2 ctrl register1

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:20	RW	0x5	win_rid_win2 axi read id of win2 channel
19:17	RO	0x0	reserved
16	RW	0x0	win2_lut_en Win2 LUT ram enable 1'b0 : disable 1'b1 : enable *This bit should be "0" when CPU updates the LUT, and should be "1" when Win2 LUT mode enable.
15	RW	0x0	win2_y_mir_en 1'b0 : no y_mirror 1'b1 : y_mirror
14	RW	0x0	win2_no_outstanding Win2 AXI master read outstanding 1'b0 : enable 1'b1 : disable
13	RO	0x0	reserved
12:8	RW	0x1d	win2_axi_max_outstanding_num
7:4	RW	0x0	win2_axi_gather_num

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Bit	Attr	Reset Value	Description
3:2	RW	0x0	win2_dma_burst_length WIN2 DMA read Burst length 2'b00 : burst16 (burst 15 in rgb888 pack mode) 2'b01 : burst8 (burst 12 in rgb888 pack mode) 2'b10 : burst4 (burst 6 in rgb888 pack mode) 2'b11 : reserved
1	RW	0x0	win2_axi_max_outstanding_en 1'b0 : disable 1'b1 : enable
0	RW	0x0	win2_axi_gather_en 1'b0 : disable 1'b1 : enable

VOP_WIN2_VIR0_1
Address: Operational Base + offset (0x000b8)
Win2 virtual stride0 and virtaul stride1

Bit	Attr	Reset Value	Description
31:16	RW	0x0140	win2_vir_stride1 Win2 Virtual stride1 Number of words of Win2 Virtual1 width ARGB888 : win2_vir_width1 RGB888 : (win2_vir_width1 * 3/4) + (win2_vir_width1 % 3) RGB565 : ceil(win2_vir_width1 / 2) 8BPP : ceil(win2_vir_width1 / 4) 4BPP : ceil(win2_vir_width1 / 8) 2BPP : ceil(win2_vir_width1 / 16) 1BPP : ceil(win2_vir_width1 / 32)
15:0	RW	0x0140	win2_vir_stride0 Win2 Virtual stride0 Number of words of Win2 Virtual0 width ARGB888 : win2_vir_width0 RGB888 : (win2_vir_width0 * 3/4) + (win2_vir_width0 % 3) RGB565 : ceil(win2_vir_width0 / 2) 8BPP : ceil(win2_vir_width0 / 4) 4BPP : ceil(win2_vir_width0 / 8) 2BPP : ceil(win2_vir_width0 / 16) 1BPP : ceil(win2_vir_width0 / 32)

VOP_WIN2_VIR2_3
Address: Operational Base + offset (0x000bc)
Win2 virtual stride2 and virtaul stride3

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0140	win2_vir_stride3 Win2 Virtual stride3 Number of words of Win2 Virtual3 width ARGB888 : win2_vir_width3 RGB888 : (win2_vir_width3 * 3/4) + (win2_vir_width3 % 3) RGB565 : ceil(win2_vir_width3 / 2) 8BPP : ceil(win2_vir_width3 / 4) 4BPP : ceil(win2_vir_width3 / 8) 2BPP : ceil(win2_vir_width3 / 16) 1BPP : ceil(win1_vir_width3 / 32)
15:0	RW	0x0140	win2_vir_stride2 Win2 Virtual stride2 Number of words of Win2 Virtual2 width ARGB888 : win2_vir_width2 RGB888 : (win2_vir_width2 * 3/4) + (win2_vir_width2 % 3) RGB565 : ceil(win2_vir_width2 / 2) 8BPP : ceil(win2_vir_width2 / 4) 4BPP : ceil(win2_vir_width2 / 8) 2BPP : ceil(win2_vir_width2 / 16) 1BPP : ceil(win1_vir_width2 / 32)

VOP_WIN2_MST0
Address: Operational Base + offset (0x000c0)
Win2 memory start address0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	win2_mst0 Win2 frame buffer memory start address0 *must be alianed to 8byte address

VOP_WIN2_DSP_INFO0
Address: Operational Base + offset (0x000c4)
Win2 display width0/height0 on panel

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:16	RW	0x0ef	win2_dsp_height0 Win2 display window height0 win2_dsp_height0 = size -1
15:12	RO	0x0	reserved
11:0	RW	0x13f	win2_dsp_width0 Win2 display window width0 win2_dsp_width = size -1

VOP_WIN2_DSP_ST0

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Address: Operational Base + offset (0x000c8)
Win2 display start point0 on panel

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	win2_dsp_yst0 Win2 vertical start point(y) of the Panel scanning
15:13	RO	0x0	reserved
12:0	RW	0x000a	win2_dsp_xst0 Win2 horizontal start point(x) of the Panel scanning

VOP_WIN2_COLOR_KEY
Address: Operational Base + offset (0x000cc)
Win2 color key register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	win2_key_en Win2 transparency color key enable 1'b0 : disable; 1'b1 : enable;
23:0	RW	0x000000	win2_key_color Win2 key color

VOP_WIN2_MST1
Address: Operational Base + offset (0x000d0)
Win2 memory start address1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	win2_mst1 Win2 frame buffer memory start address1 *must be alianed to 8byte address

VOP_WIN2_DSP_INFO1
Address: Operational Base + offset (0x000d4)
Win2 display width1/height1 on panel

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:16	RW	0x0ef	win2_dsp_height1 Win2 display window height1 win2_dsp_height0 = size -1
15:12	RO	0x0	reserved
11:0	RW	0x13f	win2_dsp_width1 Win2 display window width1 win2_dsp_width = size -1

VOP_WIN2_DSP_ST1

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Address: Operational Base + offset (0x000d8)
Win2 display start point1 on panel

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	win2_dsp_yst1 Win2 vertical start point(y) of the Panel scanning
15:13	RO	0x0	reserved
12:0	RW	0x000a	win2_dsp_xst1 Win2 horizontal start point(x) of the Panel scanning

VOP_WIN2_SRC_ALPHA_CTRL
Address: Operational Base + offset (0x000dc)
Win2 alpha source control register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	win2_fading_value
23:16	RW	0x00	win2_src_global_alpha src global alpha
15:9	RO	0x0	reserved
8:6	RW	0x0	win2_src_factor_mode
5	RW	0x0	win2_src_alpha_cal_mode
4:3	RW	0x0	win2_src_blend_mode
2	RW	0x0	win2_src_alpha_mode
1	RW	0x0	win2_src_color_mode
0	RW	0x0	win2_src_alpha_en

VOP_WIN2_MST2
Address: Operational Base + offset (0x000e0)
Win2 memory start address2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	win2_mst2 Win2 frame buffer memory start address2 *must be alianed to 8byte address

VOP_WIN2_DSP_INFO2
Address: Operational Base + offset (0x000e4)
Win2 display width2/height2 on panel

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:16	RW	0x0ef	win2_dsp_height2 Win2 display window height2 win2_dsp_height0 = size -1
15:12	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
11:0	RW	0x13f	win2_dsp_width2 Win2 display window width2 win2_dsp_width = size -1

VOP_WIN2_DSP_ST2
Address: Operational Base + offset (0x000e8)
Win2 display start point2 on panel

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	win2_dsp_yst2 Win2 vertical start point(y) of the Panel scanning
15:13	RO	0x0	reserved
12:0	RW	0x000a	win2_dsp_xst2 Win2 horizontal start point(x) of the Panel scanning

VOP_WIN2_DST_ALPHA_CTRL
Address: Operational Base + offset (0x000ec)
Win2 alpha destination control register

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8:6	RW	0x0	win2_dst_factor_mode
5:0	RW	0x00	win2_dst_m0_reserved

VOP_WIN2_MST3
Address: Operational Base + offset (0x000f0)
Win2 memory start address3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	win2_mst3 Win2 frame buffer memory start address3 *must be alianed to 8byte address

VOP_WIN2_DSP_INFO3
Address: Operational Base + offset (0x000f4)
Win2 display width3/height3 on panel

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:16	RW	0x0ef	win2_dsp_height3 Win2 display window height3 win2_dsp_height0 = size -1
15:12	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
11:0	RW	0x13f	win2_dsp_width3 Win2 display window width3 win2_dsp_width = size -1

VOP_WIN2_DSP_ST3
Address: Operational Base + offset (0x000f8)
Win2 display start point3 on panel

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	win2_dsp_yst3 Win2 vertical start point(y) of the Panel scanning
15:13	RO	0x0	reserved
12:0	RW	0x000a	win2_dsp_xst3 Win2 horizontal start point(x) of the Panel scanning

VOP_WIN2_FADING_CTRL
Address: Operational Base + offset (0x000fc)
Win2 fading contrl register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	win2_fading_en
23:16	RW	0x00	win2_fading_offset_b
15:8	RW	0x00	win2_fading_offset_g
7:0	RW	0x00	win2_fading_offset_r

VOP_WIN3_CTRL0
Address: Operational Base + offset (0x00100)
Win3 ctrl register0

Bit	Attr	Reset Value	Description
31	RW	0x0	win3_endian_swap3 Win3 region3 8pp palette data Big-endian/ Little-endian select 1'b0 : Big-endian 1'b1 : Little-endian
30	RW	0x0	win3_alpha_swap3 Win3 region3 RGB alpha swap 1'b0 : ARGB 1'b1 : RGBA
29	RW	0x0	win3_rb_swap3 Win3 region3 RGB Red and Blue swap 1'b0 : RGB 1'b1 : BGR

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Bit	Attr	Reset Value	Description
28	RW	0x0	win3_endian_swap2 Win3 region2 8pp palette data Big-endian/ Little-endian select 1'b0 : Big-endian 1'b1 : Little-endian
27	RW	0x0	win3_alpha_swap2 Win3 region2 RGB alpha swap 1'b0 : ARGB 1'b1 : RGBA
26	RW	0x0	win3_rb_swap2 Win3 region2 RGB Red and Blue swap 1'b0 : RGB 1'b1 : BGR
25	RW	0x0	win3_endian_swap1 Win3 region1 8pp palette data Big-endian/ Little-endian select 1'b0 : Big-endian 1'b1 : Little-endian
24	RW	0x0	win3_alpha_swap1 Win3 region1 RGB alpha swap 1'b0 : ARGB 1'b1 : RGBA
23	RW	0x0	win3_rb_swap1 Win3 region1 RGB Red and Blue swap 1'b0 : RGB 1'b1 : BGR
22	RW	0x0	win3_endian_swap0 Win3 region0 8pp palette data Big-endian/ Little-endian select 1'b0 : Big-endian 1'b1 : Little-endian
21	RW	0x0	win3_alpha_swap0 Win3 region0 RGB alpha swap 1'b0 : ARGB 1'b1 : RGBA
20	RW	0x0	win3_rb_swap0 Win3 region0 RGB Red and Blue swap 1'b0 : RGB 1'b1 : BGR
19	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
18:17	RW	0x0	win3_data_fmt3 Win3 region 3 data format 2'b00 : ARGB888 2'b01 : RGB888 2'b10 : RGB565 2'b11 : 8bpp
16	RW	0x0	win3_mst3_en win3 master3 enable 1'b0 : disable 1'b1 : enable
15	RO	0x0	reserved
14:13	RW	0x0	win3_data_fmt2 Win3 region 2 data format 2'b00 : ARGB888 2'b01 : RGB888 2'b10 : RGB565 2'b11 : 8bpp
12	RW	0x0	win3_mst2_en win3 master2 enable 1'b0 : disable 1'b1 : enable
11	RO	0x0	reserved
10:9	RW	0x0	win3_data_fmt1 Win3 region 1 data format 2'b00 : ARGB888 2'b01 : RGB888 2'b10 : RGB565 2'b11 : 8bpp
8	RW	0x0	win3_mst1_en win3 master1 enable 1'b0 : disable 1'b1 : enable
7	RO	0x0	reserved
6:5	RW	0x0	win3_data_fmt0 Win3 region 0 data format 2'b00 : ARGB888 2'b01 : RGB888 2'b10 : RGB565 2'b11 : 8bpp
4	RW	0x0	win3_mst0_en Win3 master0 enable 1'b0 : disable 1'b1 : enable
3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
2	RW	0x0	win3_csc_mode Win3 RGB2YUV conversion mode Color space conversion: 1'b0: BT601 1'b1: BT709
1	RW	0x0	win3_interlace_read Win3 interlace read mode 1'b0 : disable 1'b1 : enable
0	RW	0x0	win3_en 1'b0 : disable 1'b1 : enable

VOP_WIN3_CTRL1
Address: Operational Base + offset (0x00104)
Win3 ctrl register1

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:20	RW	0x6	win_rid_win3 axi read id of win3 channel
19:17	RO	0x0	reserved
16	RW	0x0	win3_lut_en Win3 LUT ram enable 1'b0 : disable 1'b1 : enable *This bit should be "0" when CPU updates the LUT, and should be "1" when Win1 LUT mode enable.
15	RW	0x0	win3_y_mir_en 1'b0 : no y_mirror 1'b1 : y_mirror
14	RW	0x0	win3_no_outstanding Win3 AXI master read outstanding 1'b0 : enable 1'b1 : disable
13	RO	0x0	reserved
12:8	RW	0x1d	win3_axi_max_outstanding_num
7:4	RW	0x0	win3_axi_gather_num
3:2	RW	0x0	win3_dma_burst_length WIN3 DMA read Burst length 2'b00 : burst16 (burst 15 in rgb888 pack mode) 2'b01 : burst8 (burst 12 in rgb888 pack mode) 2'b10 : burst4 (burst 6 in rgb888 pack mode) 2'b11 : reserved

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Bit	Attr	Reset Value	Description
1	RW	0x0	win3_axi_max_outstanding_en
0	RW	0x0	win3_axi_gather_en

VOP_WIN3_VIR0_1
Address: Operational Base + offset (0x00108)
Win3 virtual stride0 and virtaul stride1

Bit	Attr	Reset Value	Description
31:16	RW	0x0140	win3_vir_stride1 Win3 Virtual stride1 Number of words of Win3 Virtual1 width ARGB888 : win3_vir_width1 RGB888 : (win3_vir_width1 * 3/4) + (win3_vir_width1 % 3) RGB565 : ceil(win3_vir_width1 / 2) 8BPP : ceil(win3_vir_width1 / 4) 4BPP : ceil(win3_vir_width1 / 8) 2BPP : ceil(win3_vir_width1 / 16) 1BPP : ceil(win3_vir_width1 / 32)
15:0	RW	0x0140	win3_vir_stride0 Win3 Virtual stride0 Number of words of Win3 Virtual1 width ARGB888 : win3_vir_width1 RGB888 : (win3_vir_width1 * 3/4) + (win3_vir_width1 % 3) RGB565 : ceil(win3_vir_width1 / 2) 8BPP : ceil(win3_vir_width1 / 4) 4BPP : ceil(win3_vir_width1 / 8) 2BPP : ceil(win3_vir_width1 / 16) 1BPP : ceil(win3_vir_width1 / 32)

VOP_WIN3_VIR2_3
Address: Operational Base + offset (0x0010c)
Win3 virtual stride2 and virtaul stride3

Bit	Attr	Reset Value	Description
31:16	RW	0x0140	win3_vir_stride3 Win3 Virtual stride3 Number of words of Win3 Virtual1 width ARGB888 : win3_vir_width1 RGB888 : (win3_vir_width1 * 3/4) + (win3_vir_width1 % 3) RGB565 : ceil(win3_vir_width1 / 2) 8BPP : ceil(win3_vir_width1 / 4) 4BPP : ceil(win3_vir_width1 / 8) 2BPP : ceil(win3_vir_width1 / 16) 1BPP : ceil(win3_vir_width1 / 32)

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Bit	Attr	Reset Value	Description
15:0	RW	0x0140	win3_vir_stride2 Win3 Virtual stride2 Number of words of Win3 Virtual1 width ARGB888 : win3_vir_width1 RGB888 : (win3_vir_width1 * 3/4) + (win3_vir_width1 % 3) RGB565 : ceil(win3_vir_width1 / 2) 8BPP : ceil(win3_vir_width1 / 4) 4BPP : ceil(win3_vir_width1 / 8) 2BPP : ceil(win3_vir_width1 / 16) 1BPP : ceil(win3_vir_width1 / 32)

VOP_WIN3_MST0
Address: Operational Base + offset (0x00110)
Win3 memory start address0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	win3_mst0 Win3 frame buffer memory start address0 *must be alianed to 8byte address

VOP_WIN3_DSP_INFO0
Address: Operational Base + offset (0x00114)
Win3 display width0/height0 on panel

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:16	RW	0x0ef	win3_dsp_height0 Win3 display window height0 win3_dsp_height0 = size -1
15:12	RO	0x0	reserved
11:0	RW	0x13f	win3_dsp_width0 Win3 display window width0 win3_dsp_width = size -1

VOP_WIN3_DSP_ST0
Address: Operational Base + offset (0x00118)
Win3 display start point0 on panel

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	win3_dsp_yst0 Win3 vertical start point(y) of the Panel scanning
15:13	RO	0x0	reserved
12:0	RW	0x000a	win3_dsp_xst0 Win3 horizontal start point(x) of the Panel scanning

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VOP_WIN3_COLOR_KEY
Address: Operational Base + offset (0x0011c)
Win3 color key register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	win3_key_en Win3 transparency color key enable 1'b0 : disable; 1'b1 : enable;
23:0	RW	0x000000	win3_key_color Win3 key color

VOP_WIN3_MST1
Address: Operational Base + offset (0x00120)
Win3 memory start address1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	win3_mst1 Win3 frame buffer memory start address1 *must be alianed to 8byte address

VOP_WIN3_DSP_INFO1
Address: Operational Base + offset (0x00124)
Win3 display width1/height1 on panel

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:16	RW	0x0ef	win3_dsp_height1 Win3 display window height1 win3_dsp_height0 = size -1
15:12	RO	0x0	reserved
11:0	RW	0x13f	win3_dsp_width1 Win3 display window width1 win3_dsp_width = size -1

VOP_WIN3_DSP_ST1
Address: Operational Base + offset (0x00128)
Win3 display start point1 on panel

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	win3_dsp_yst1 Win3 vertical start point(y) of the Panel scanning
15:13	RO	0x0	reserved
12:0	RW	0x000a	win3_dsp_xst1 Win3 horizontal start point(x) of the Panel scanning

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VOP_WIN3_SRC_ALPHA_CTRL
Address: Operational Base + offset (0x0012c)
Win3 alpha source control register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	win3_fading_value
23:16	RW	0x00	win3_src_global_alpha src global alpha
15:9	RO	0x0	reserved
8:6	RW	0x0	win3_src_factor_mode
5	RW	0x0	win3_src_alpha_cal_mode
4:3	RW	0x0	win3_src_blend_mode
2	RW	0x0	win3_src_alpha_mode
1	RW	0x0	win3_src_color_mode
0	RW	0x0	win3_src_alpha_en

VOP_WIN3_MST2
Address: Operational Base + offset (0x00130)
Win3 memory start address2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	win3_mst2 Win3 frame buffer memory start address2 *must be alianed to 8byte address

VOP_WIN3_DSP_INFO2
Address: Operational Base + offset (0x00134)
Win3 display width2/height2 on panel

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:16	RW	0x0ef	win3_dsp_height2 Win3 display window height2 win3_dsp_height0 = size -1
15:12	RO	0x0	reserved
11:0	RW	0x13f	win3_dsp_width2 Win3 display window width2 win3_dsp_width = size -1

VOP_WIN3_DSP_ST2
Address: Operational Base + offset (0x00138)
Win3 display start point2 on panel

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	win3_dsp_yst2 Win3 vertical start point(y) of the Panel scanning

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Bit	Attr	Reset Value	Description
15:13	RO	0x0	reserved
12:0	RW	0x000a	win3_dsp_xst2 Win3 horizontal start point(x) of the Panel scanning

VOP_WIN3_DST_ALPHA_CTRL
Address: Operational Base + offset (0x0013c)
Win3 alpha destination control register

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8:6	RW	0x0	win3_dst_factor_mode
5:0	RW	0x00	win3_dst_factor_reserved

VOP_WIN3_MST3
Address: Operational Base + offset (0x00140)
Win3 memory start address3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	win3_mst3 Win3 frame buffer memory start address3 *must be alianed to 8byte address

VOP_WIN3_DSP_INFO3
Address: Operational Base + offset (0x00144)
Win3 display width3/height3 on panel

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:16	RW	0x0ef	win3_dsp_height3 Win3 display window height3 win3_dsp_height0 = size -1
15:12	RO	0x0	reserved
11:0	RW	0x13f	win3_dsp_width3 Win3 display window width3 win3_dsp_width = size -1

VOP_WIN3_DSP_ST3
Address: Operational Base + offset (0x00148)
Win3 display start point3 on panel

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	win3_dsp_yst3 Win3 vertical start point(y) of the Panel scanning
15:13	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
12:0	RW	0x000a	win3_dsp_xst3 Win3 horizontal start point(x) of the Panel scanning

VOP_WIN3_FADING_CTRL
Address: Operational Base + offset (0x0014c)
Win3 fading contrl register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	win3_fading_en
23:16	RW	0x00	win3_fading_offset_b
15:8	RW	0x00	win3_fading_offset_g
7:0	RW	0x00	win3_fading_offset_r

VOP_HWC_CTRL0
Address: Operational Base + offset (0x00150)
Hwc ctrl register0

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14	RW	0x0	hwc_endian_swap hwc 8pp palette data Big-endian/ Little-endian select 1'b0 : Big-endian 1'b1 : Little-endian
13	RW	0x0	hwc_alpha_swap hwc RGB alpha swap 1'b0 : ARGB 1'b1 : RGBA
12	RW	0x0	hwc_rb_swap hwc RGB Red and Blue swap 1'b0 : RGB 1'b1 : BGR
11	RO	0x0	reserved
10	RW	0x0	hwc_csc_mode hwc RGB2YUV Color space conversion: 2'bx0: BT601 2'bx1: BT709
9	RO	0x0	reserved
8	RW	0x0	hwc_interlace_read hwc interlace read mode 1'b0 : disable 1'b1 : enable
7	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
6:5	RW	0x0	hwc_size 2'b00 : 32x32 2'b01 : 64x64 2'b10 : 96x96 2'b11 : 128x128
4	RW	0x0	hwc_mode hwc color mode 1'b0 : normal color mode 1'b1 : reversed color mode
3:1	RW	0x0	hwc_data_fmt 3'b000 : ARGB888 3'b001 : RGB888 3'b010 : RGB565 3'b100: 8bpp 3'b101: 4bpp 3'b110: 2bpp 3'b111: 1bpp
0	RW	0x0	hwc_en

VOP_HWC_CTRL1
Address: Operational Base + offset (0x00154)
Hwc ctrl register1

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:20	RW	0x7	win_rid_hwc axi read id of hwc channel
19:17	RO	0x0	reserved
16	RW	0x0	hwc_lut_en
15	RW	0x0	hwc_y_mir_en 1'b0 : no y_mirror 1'b1 : y_mirror
14	RW	0x0	hwc_no_outstanding hwc AXI master read outstanding 1'b0 : enable 1'b1 : disable
13	RW	0x0	hwc_rgb2yuv_en 1'b0 : enable 1'b1 : disable
12:8	RW	0x1d	hwc_axi_max_outstanding_num
7	RO	0x0	reserved
6:4	RW	0x0	hwc_axi_gather_num

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Bit	Attr	Reset Value	Description
3:2	RW	0x0	hwc_dma_burst_length HWC DMA read Burst length 2'b00 : burst16 (burst 15 in rgb888 pack mode) 2'b01 : burst8 (burst 12 in rgb888 pack mode) 2'b10 : burst4 (burst 6 in rgb888 pack mode) 2'b11 : reserved
1	RW	0x0	hwc_axi_max_outstanding_en 1'b0 : disable 1'b1 : enable
0	RW	0x0	hwc_axi_gather_en

VOP_HWC_MST
Address: Operational Base + offset (0x00158)
Hwc memory start address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	hwc_mst HWC data memory start address

VOP_HWC_DSP_ST
Address: Operational Base + offset (0x0015c)
Hwc display start point on panel

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	hwc_dsp_yst HWC vertical start point(y) of the Panel scanning
15:13	RO	0x0	reserved
12:0	RW	0x000a	hwc_dsp_xst HWC horizontal start point(x) of the Panel scanning

VOP_HWC_SRC_ALPHA_CTRL
Address: Operational Base + offset (0x00160)
Hwc alpha source control register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	hwc_fading_value
23:16	RW	0x00	hwc_src_global_alpha src global alpha
15:9	RO	0x0	reserved
8:6	RW	0x0	hwc_src_factor_mode
5	RW	0x0	hwc_src_alpha_cal_mode
4:3	RW	0x0	hwc_src_blend_mode
2	RW	0x0	hwc_src_alpha_mode

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Bit	Attr	Reset Value	Description
1	RW	0x0	hwc_src_color_mode
0	RW	0x0	hwc_src_alpha_en

VOP_HWC_DST_ALPHA_CTRL
Address: Operational Base + offset (0x00164)
Hwc alpha destination control register

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8:6	RW	0x0	hwc_dst_factor_mode
5:0	RW	0x00	hwc_dst_m0_reserved

VOP_HWC_FADING_CTRL
Address: Operational Base + offset (0x00168)
Hwc fading contrl register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	hwc_fading_en
23:16	RW	0x00	hwc_fading_offset_b
15:8	RW	0x00	hwc_fading_offset_g
7:0	RW	0x00	hwc_fading_offset_r

VOP_HWC_RESERVED1
Address: Operational Base + offset (0x0016c)
Hwc reserved

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_POST_DSP_HACT_INFO
Address: Operational Base + offset (0x00170)
Post scaler down horizontal start and end

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	dsp_hact_st_post Panel display scanning horizontal active start point
15:13	RO	0x0	reserved
12:0	RW	0x014a	dsp_hact_end_post Panel display scanning horizontal active end point

VOP_POST_DSP_VACT_INFO
Address: Operational Base + offset (0x00174)
Panel active horizontal scanning start point and end point

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	dsp_vact_st_post Panel display scanning horizontal active start point
15:13	RO	0x0	reserved
12:0	RW	0x00fa	dsp_vact_end_post Panel display scanning horizontal active end point

VOP_POST_SCL_FACTOR_YRGB
Address: Operational Base + offset (0x00178)
Post yrgb scaling factor

Bit	Attr	Reset Value	Description
31:16	RW	0x1000	post_vs_factor_yrgb post YRGB vertical scaling factor: factor=((src_height[31:16]) /(dst_height[31:16]))*2^12
15:0	RW	0x1000	post_hs_factor_yrgb Post YRGB horizontal scaling factor: factor=((src_width[15:0]) /(dst_width[15:0]))*2^12

VOP_POST_RESERVED
Address: Operational Base + offset (0x0017c)
Post reserved

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_POST_SCL_CTRL
Address: Operational Base + offset (0x00180)
Post scaling start point offset

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	post_ver_sd_en 1'b0 : post ver scl down disable 1'b1 : post ver scl down enable
0	RW	0x0	post_hor_sd_en 1'b0 : post hor scl down disable 1'b1 : post hor scl down enable

VOP_POST_DSP_VACT_INFO_F1
Address: Operational Base + offset (0x00184)
Panel active horizontal scanning start point and end point F1

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
28:16	RW	0x000a	dsp_vact_st_post Panel display scanning horizontal active start point
15:13	RO	0x0	reserved
12:0	RW	0x00fa	dsp_vact_end_post Panel display scanning horizontal active end point

VOP_DSP_HTOTAL_HS_END
Address: Operational Base + offset (0x00188)
Panel scanning horizontal width and hsync pulse end point

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x014a	dsp_htotal Panel display scanning horizontal period
15:13	RO	0x0	reserved
12:0	RW	0x000a	dsp_hs_end Panel display scanning hsync pulse width

VOP_DSP_HACT_ST_END
Address: Operational Base + offset (0x0018c)
Panel active horizontal scanning start point and end point

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	dsp_hact_st Panel display scanning horizontal active start point
15:13	RO	0x0	reserved
12:0	RW	0x014a	dsp_hact_end Panel display scanning horizontal active end point

VOP_DSP_VTOTAL_VS_END
Address: Operational Base + offset (0x00190)
Panel scanning vertical height and vsync pulse end point

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x00fa	dsp_vtotal Panel display scanning vertical period.
15:13	RO	0x0	reserved
12:0	RW	0x000a	dsp_vs_end Panel display scanning vsync pulse width

VOP_DSP_VACT_ST_END

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Address: Operational Base + offset (0x00194)
Panel active vertical scanning start point and end point

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	dsp_vact_st Panel display scanning vertical active start point
15:13	RO	0x0	reserved
12:0	RW	0x00fa	dsp_vact_end Panel display scanning vertical active end point

VOP_DSP_VS_ST_END_F1
Address: Operational Base + offset (0x00198)
Vertical scanning start point and vsync pulse end point of even filed in interlace mode

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x0000	dsp_vs_st_f1 Panel display scanning vertical vsync start point of 2nd field (interlace display mode)
15:13	RO	0x0	reserved
12:0	RW	0x0000	dsp_vs_end_f1 Panel display scanning vertical vsync end point of 2nd field(interlace display mode)

VOP_DSP_VACT_ST_END_F1
Address: Operational Base + offset (0x0019c)
Vertical scanning active start point and end point of even filed in interlace mode

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x0000	dsp_vact_st_f1 Panel display scanning vertical active start point of 2nd field (interlace display mode)
15:13	RO	0x0	reserved
12:0	RW	0x0000	dsp_vact_end_f1 Panel display scanning vertical active end point of 2nd field (interlace display mode)

VOP_PWM_CTRL
Address: Operational Base + offset (0x001a0)
PWM Control Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	rpt Repeat Counter This field defines the repeated effective periods of output waveform in one-shot mode. The value N means N+1 repeated effective periods.
23:16	RW	0x00	scale Scale Factor This fields defines the scale factor applied to prescaled clock. The value N means the clock is divided by 2*N. If N is 0, it means that the clock is divided by 512(2*256).
15	RO	0x0	reserved
14:12	RW	0x2	prescale Prescale Factor This field defines the prescale factor applied to input clock. The value N means that the input clock is divided by 2^N.
11:10	RO	0x0	reserved
9	RW	0x0	clk_sel Clock Source Select 0: non-scaled clock is selected as PWM clock source. It means that the prescale clock is directly used as the PWM clock source 1: scaled clock is selected as PWM clock source
8	RW	0x0	lp_en Low Power Mode Enable 0: disabled 1: enabled When PWM channel is inactive state and Low Power Mode is enabled, the path to PWM Clock prescale module is blocked to reduce power consumption.
7:6	RO	0x0	reserved
5	RW	0x0	output_mode PWM Output mode 0: left aligned mode 1: center aligned mode

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Bit	Attr	Reset Value	Description
4	RW	0x0	inactive_pol Inactive State Output Polarity This defines the output waveform polarity when PWM channel is in inactive state. The inactive state means that PWM finishes the complete waveform in one-shot mode or PWM channel is disabled. 0: negative 1: positive
3	RW	0x1	duty_pol Duty Cycle Output Polarity This defines the polarity for duty cycle. PWM starts the output waveform with duty cycle. 0: negative 1: positive
2:1	RW	0x1	pwm_mode PWM Operation Mode 00: One shot mode. PWM produces the waveform within the repeated times defined by PWMx_CTRL_rpt . 01: Continuous mode. PWM produces the waveform continuously 10: Capture mode. PWM measures the cycles of high/low polarity of input waveform. 11: reserved
0	RW	0x0	pwm_en PWM channel enable 0: disabled 1: enabled. If the PWM is worked the one-shot mode, this bit will be cleared at the end of operation

VOP_PWM_PERIOD_HPR

Address: Operational Base + offset (0x001a4)

PWM Period Register/High Polarity Capture Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pwm_period Output Waveform Period/Input Waveform High Polarity Cycle If PWM is operated at the continuous mode or one-shot mode, this value defines the period of the output waveform. Note that, if the PWM is operated at the center-aligned mode, the period should be an even one, and therefore only the bit [31:1] is taken into account and bit [0] always considered as 0. If PWM is operated at the capture mode, this value indicates the effective high polarity cycles of input waveform. This value is based on the PWM clock. The value ranges from 0 to (2^32-1).

VOP_PWM_DUTY_LPR
Address: Operational Base + offset (0x001a8)
PWM Duty Register/Low Polarity Capture Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pwm_duty Output Waveform Duty Cycle/Input Waveform Low Polarity Cycle If PWM is operated at the continuous mode or one-shot mode, this value defines the duty cycle of the output waveform. The PWM starts the output waveform with duty cycle. Note that, if the PWM is operated at the center-aligned mode, the period should be an even one, and therefore only the [31:1] is taken into account. If PWM is operated at the capture mode, this value indicates the effective low polarity cycles of input waveform. This value is based on the PWM clock. The value ranges from 0 to (2^32-1).

VOP_PWM_CNT
Address: Operational Base + offset (0x001ac)
PWM Counter Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	pwm_cnt Timer Counter The 32-bit indicates current value of PWM Channel 0 counter. The counter runs at the rate of PWM clock. The value ranges from 0 to (2^32-1).

VOP_BCSH_COLOR_BAR
Address: Operational Base + offset (0x001b0)
Color bar config register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	color_bar_v v color value
23:16	RW	0x00	color_bar_u u color value
15:8	RW	0x00	color_bar_y y color value
7:1	RO	0x0	reserved
0	RW	0x0	bcsh_en 1'b0 : bcsh bypass 1'b1 : bcsh enable

VOP_BCSH_BCS
Address: Operational Base + offset (0x001b4)
Brightness contrast saturation*contrast config register

Bit	Attr	Reset Value	Description
31:30	RW	0x3	out_mode video out mode config register 2'b00 : black 2'b01 : blue 2'b10 : color bar 2'b11 : normal video
29:20	RW	0x100	sat_con 1ffSaturation*Contrast*256 : 0,1.992*1.992
19:17	RO	0x0	reserved
16:8	RW	0x100	contrast Contrast*256 : 0,1.992
7:0	RW	0x00	brightness Brightness : -32,31

VOP_BCSH_H
Address: Operational Base + offset (0x001b8)
Sin hue and cos hue config register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
24:16	RW	0x100	cos_hue cos hue value
15:9	RO	0x0	reserved
8:0	RW	0x000	sin_hue sin hue value

VOP_BCSH_CTRL
Address: Operational Base + offset (0x001bc)
BCSH contrl register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	bcsh_r2y_csc_mode Color space conversion: 1'b0: BT601 1'b1: BT709
5	RO	0x0	reserved
4	RW	0x0	bcsh_r2y_en 1'b0:bypass 1'b1:enable
3:2	RW	0x0	bcsh_y2r_csc_mode Color space conversion(YUV2RGB): 2'b00/01 : mpeg 2'b10 : jpeg 2'b11 : hd
1	RO	0x0	reserved
0	RW	0x0	bcsh_y2r_en 1'b0:bypass 1'b1:enable

VOP_CABC_CTRL0
Address: Operational Base + offset (0x001c0)
Content Adaptive Backlight Control register0

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:1	RW	0x0ed800	cabc_calc_pixel_num cabc calc pixel numbers = x % * cabc_total_num
0	RW	0x0	cabc_en 1'b0 : cabc disable 1'b1 : cabc enable

VOP_CABC_CTRL1
Address: Operational Base + offset (0x001c4)
Content Adaptive Backlight Control register1

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:1	RW	0x0fa000	cabc_total_num cabc totala numbers = h_vd * v_vd
0	RW	0x0	cabc_lut_en cabc pwm lut enable

VOP_CABC_CTRL2
Address: Operational Base + offset (0x001c8)
Content Adaptive Backlight Control register2

Bit	Attr	Reset Value	Description
31:24	RW	0x00	cabc_global_su_rec the reciprocal cabc global scale up value,for pwm scale down value.
23	RW	0x0	cabc_global_su_limit_en cabc global scale up limit enable.
22:17	RO	0x0	reserved
16:8	RW	0x110	cabc_stage_up scale stage up (1~1.5 * 256).
7:0	RW	0xf0	cabc_stage_up_rec the reciprocal of scale stage up ,for pwm stage down (0.667 ~ 1 * 256).

VOP_CABC_CTRL3
Address: Operational Base + offset (0x001cc)
Content Adaptive Backlight Control register3

Bit	Attr	Reset Value	Description
31:23	RW	0x000	cabc_global_su cabc global scale up value.
22:17	RO	0x0	reserved
16:8	RW	0x110	cabc_stage_down_rec the reciprocal of scale stage down,as pwm stage up (1~1.5*256).
7:0	RW	0xf0	cabc_stage_down scale stage down (0.667~1 * 256).

VOP_CABC_GAUSS_LINE0_0
Address: Operational Base + offset (0x001d0)
CABC gauss line config register00

Bit	Attr	Reset Value	Description
31:24	RW	0x15	t_line0_3
23:16	RW	0x11	t_line0_2
15:8	RW	0x09	t_line0_1
7:0	RW	0x03	t_line0_0

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VOP_CABC_GAUSS_LINE0_1
Address: Operational Base + offset (0x001d4)
CABC gauss line config register01

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:16	RW	0x03	t_line0_6
15:8	RW	0x09	t_line0_5
7:0	RW	0x11	t_line0_4

VOP_CABC_GAUSS_LINE1_0
Address: Operational Base + offset (0x001d8)
CABC gauss line config register10

Bit	Attr	Reset Value	Description
31:24	RW	0x1a	t_line1_3
23:16	RW	0x15	t_line1_2
15:8	RW	0x0b	t_line1_1
7:0	RW	0x04	t_line1_0

VOP_CABC_GAUSS_LINE1_1
Address: Operational Base + offset (0x001dc)
CABC gauss line config register11

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:16	RW	0x04	t_line1_6
15:8	RW	0x0b	t_line1_5
7:0	RW	0x15	t_line1_4

VOP_CABC_GAUSS_LINE2_0
Address: Operational Base + offset (0x001e0)
CABC gauss line config register20

Bit	Attr	Reset Value	Description
31:24	RW	0x15	t_line2_3
23:16	RW	0x11	t_line2_2
15:8	RW	0x09	t_line2_1
7:0	RW	0x03	t_line2_0

VOP_CABC_GAUSS_LINE2_1
Address: Operational Base + offset (0x001e4)
CABC gauss line config register21

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:16	RW	0x03	t_line2_6
15:8	RW	0x09	t_line2_5
7:0	RW	0x11	t_line2_4

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VOP_FRC_LOWER01_0
Address: Operational Base + offset (0x001e8)
FRC lookup table config register010

Bit	Attr	Reset Value	Description
31:16	RW	0x1284	lower01_frm1
15:0	RW	0x4821	lower01_frm0

VOP_FRC_LOWER01_1
Address: Operational Base + offset (0x001ec)
FRC lookup table config register011

Bit	Attr	Reset Value	Description
31:16	RW	0x2148	lower01_frm3
15:0	RW	0x8412	lower01_frm2

VOP_FRC_LOWER10_0
Address: Operational Base + offset (0x001f0)
FRC lookup table config register100

Bit	Attr	Reset Value	Description
31:16	RW	0xa55a	lower10_frm1
15:0	RW	0x9696	lower10_frm0

VOP_FRC_LOWER10_1
Address: Operational Base + offset (0x001f4)
FRC lookup table config register101

Bit	Attr	Reset Value	Description
31:16	RW	0x5aa5	lower10_frm3
15:0	RW	0x6969	lower10_frm2

VOP_FRC_LOWER11_0
Address: Operational Base + offset (0x001f8)
FRC lookup table config register110

Bit	Attr	Reset Value	Description
31:16	RW	0xdeb7	lower11_frm1
15:0	RW	0x7bed	lower11_frm0

VOP_FRC_LOWER11_1
Address: Operational Base + offset (0x001fc)
FRC lookup table config register111

Bit	Attr	Reset Value	Description
31:16	RW	0xed7b	lower11_frm3
15:0	RW	0xb7de	lower11_frm2

VOP_IFBDC_CTRL
Address: Operational Base + offset (0x00200)
FBDC contrl register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28	RW	0x1	fbdc_rstn
27:24	RW	0x8	fbdc_rid
23	RW	0x0	fbdc_invalidate_pending_o
22	RW	0x0	fbdc_ictrl_notify
21	RW	0x0	fbdc_frame_rst_en
20	RW	0x0	fbdc_dsp_width_ratio 1'b0: 1 pix per word 1'b1: 2 pix per word
19:15	RW	0x1d	fbdc_axi_max_outstanding_num
14	RW	0x0	fbdc_axi_max_outstanding_en 1'b0 : disable 1'b1 : enable
13:7	RW	0x00	fbdc_fmt fbdc input format 0x05: RGB565 0x0c: ARGB888 0x3a: RGB888_pack other: reserved
6:4	RW	0x0	fbdc_rotation_mode fbdc rotation mode 3'b000: no rotation 3'b010: rotation 180 degree 3'b101: x_mirror 3'b110: y_mirror others: reserved
3:2	RW	0x0	vop_fbdc_win_sel vop fbdc output to window 2'b00: win0 2'b01: win1 2'b10: win2 2'b11: win3
1	RW	0x0	fbdc_cor_en 1'b0: disable 1'b1: enable
0	RW	0x0	vop_fbdc_en vop fbdc enable 1'b0: disable 1'b1: enable

VOP_IFBDC_TILES_NUM
Address: Operational Base + offset (0x00204)
FBDC tiles num

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
22:0	RW	0x000000	fbdc_num_tiles Number of compression tiles to be compressed minus 1. It is used in order to allocate the necessary memory space for the header lines required for all tiles of the burst. Depending on the format this value may or may not coincide with the number of pixel tiles. This is essentially the memory size of the texture divided by the compression tile size (256 bytes). The result is then rounded up when necessary

VOP_IFBDC_FRAME_RST_CYCLE
Address: Operational Base + offset (0x00208)
FBDC frame reset cycle

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25:16	RW	0x000	dma_fbdc_frame_rst_cycle
15:10	RO	0x0	reserved
9:0	RW	0x000	fbdc_frame_rst_cycle

VOP_IFBDC_BASE_ADDR
Address: Operational Base + offset (0x0020c)
FBDC base address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	fbdc_base_addr

VOP_IFBDC_MB_SIZE
Address: Operational Base + offset (0x00210)
FBDC macro block size

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24:16	RW	0x000	fbdc_mb_height fbdc macro block height
15:7	RO	0x0	reserved
6:0	RW	0x00	fbdc_mb_width fbdc macro block width

VOP_IFBDC_CMP_INDEX_INIT
Address: Operational Base + offset (0x00214)
FBDC compare index initial

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
22:0	RW	0x000000	fbdc_cmp_index_init

VOP_IFBDC_VIR
Address: Operational Base + offset (0x00220)
FBDC macro block virtual width

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	fbdc_mb_vir_width

VOP_IFBDC_DEBUG0
Address: Operational Base + offset (0x00230)
FBDC debug0 register

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_IFBDC_DEBUG1
Address: Operational Base + offset (0x00234)
FBDC debug1 register

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_LATENCY_CTRL0
Address: Operational Base + offset (0x00250)
Axi performance latency module contrl register0

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24:20	RW	0x00	rd_latency_st_num
19:8	RW	0x000	rd_latency_thr
7:4	RW	0x0	rd_latency_id0
3	RO	0x0	reserved
2	RW	0x0	rd_latency_mode read latency mode 1'b0: clear by frame end 1'b1: clear by software configuration
1	RW	0x0	hand_latency_clr read max latencyu software clear 1'b0: software clear disable 1'b1: software clear enalbe
0	RW	0x0	rd_latency_en vop_axi_rd_perf enable signal 1'b0: disable 1'b1: enable

VOP_RD_MAX_LATENCY_NUM0

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Address: Operational Base + offset (0x00254)
Read max latency number

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RO	0x0	rd_latency_overflow_ch0 overflow flag(bigger than configed threshold value)
15:12	RO	0x0	reserved
11:0	RO	0x000	rd_max_latency_num_ch0 read max latency value of channel 0

VOP_RD_LATENCY_THR_NUM0
Address: Operational Base + offset (0x00258)
The number of bigger than configed threshold value

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RO	0x000000	rd_latency_thr_num_ch0 read latency thr number channel 0

VOP_RD_LATENCY_SAMP_NUM0
Address: Operational Base + offset (0x0025c)
Total sample number

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RO	0x000000	rd_latency_samp_num_ch0

VOP_WIN0_DSP_BG
Address: Operational Base + offset (0x00260)
Win0 layer background color

Bit	Attr	Reset Value	Description
31	RW	0x0	win0_bg_en Win0 layer background enable 1'b0 : disable 1'b1 : enable
30:24	RO	0x0	reserved
23:16	RW	0x00	win0_dsp_bg_red Win0 layer Background Red color
15:8	RW	0x00	win0_dsp_bg_green Win0 layer Background Green color
7:0	RW	0x00	win0_dsp_bg_blue Win0 layer Background Blue color

VOP_WIN1_DSP_BG
Address: Operational Base + offset (0x00264)
Win1 layer background color

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31	RW	0x0	win1_bg_en Win1 layer background enable 1'b0 : disable 1'b1 : enable
30:24	RO	0x0	reserved
23:16	RW	0x00	win1_dsp_bg_red Win1 layer Background Red color
15:8	RW	0x00	win1_dsp_bg_green Win1 layer Background Green color
7:0	RW	0x00	win1_dsp_bg_blue Win1 layer Background Blue color

VOP_WIN2_DSP_BG
Address: Operational Base + offset (0x00268)
Win2 layer background color

Bit	Attr	Reset Value	Description
31	RW	0x0	win2_bg_en Win2 layer background enable 1'b0 : disable 1'b1 : enable
30:24	RO	0x0	reserved
23:16	RW	0x00	win2_dsp_bg_red Win2 layer Background Red color
15:8	RW	0x00	win2_dsp_bg_green Win2 layer Background Green color
7:0	RW	0x00	win2_dsp_bg_blue Win2 layer Background Blue color

VOP_WIN3_DSP_BG
Address: Operational Base + offset (0x0026c)
Win3 layer background color

Bit	Attr	Reset Value	Description
31	RW	0x0	win3_bg_en Win3 layer background enable 1'b0 : disable 1'b1 : enable
30:24	RO	0x0	reserved
23:16	RW	0x00	win3_dsp_bg_red Win3 layer Background Red color
15:8	RW	0x00	win3_dsp_bg_green Win3 layer Background Green color
7:0	RW	0x00	win3_dsp_bg_blue Win3 layer Background Blue color

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VOP_SCAN_LINE_NUM
Address: Operational Base + offset (0x00270)
Current number of dsp timing

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RW	0x0	mmu_idle
15:13	RO	0x0	reserved
12:0	RO	0x0000	rd_latency_samp_num_ch0

VOP_CABC_DEBUG0
Address: Operational Base + offset (0x00274)
CABC debug register0

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_CABC_DEBUG1
Address: Operational Base + offset (0x00278)
CABC debug register1

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_CABC_DEBUG2
Address: Operational Base + offset (0x0027c)
CABC debug register2

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:8	RW	0x00	pwm_mul_post_value
7:0	RO	0x0	reserved

VOP_DBG_REG_000
Address: Operational Base + offset (0x00280)
Vop debug register000

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_DBG_REG_001
Address: Operational Base + offset (0x00284)
Vop debug register001

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_DBG_REG_002
Address: Operational Base + offset (0x00288)
Vop debug register002

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

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VOP_DBG_REG_003
Address: Operational Base + offset (0x0028c)
Vop debug register003

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_DBG_REG_004
Address: Operational Base + offset (0x00290)
Vop debug register004

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_DBG_REG_005
Address: Operational Base + offset (0x00294)
Vop debug register005

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_DBG_REG_006
Address: Operational Base + offset (0x00298)
Vop debug register006

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x0000	dbg_win0_yrgb_cmd_line_cnt
15:0	RO	0x0	reserved

VOP_DBG_REG_007
Address: Operational Base + offset (0x0029c)
Vop debug register007

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_DBG_REG_008
Address: Operational Base + offset (0x002a0)
Vop debug register008

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved
-1:0	RW	RW	dbg

VOP_DBG_REG_016
Address: Operational Base + offset (0x002c0)
Vop debug register016

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

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VOP_DBG_REG_017
Address: Operational Base + offset (0x002c4)
Vop debug register017

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_DBG_REG_018
Address: Operational Base + offset (0x002c8)
Vop debug register018

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_DBG_REG_019
Address: Operational Base + offset (0x002cc)
Vop debug register019

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_DBG_REG_020
Address: Operational Base + offset (0x002d0)
Vop debug register020

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_DBG_REG_021
Address: Operational Base + offset (0x002d4)
Vop debug register021

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_DBG_REG_022
Address: Operational Base + offset (0x002d8)
Vop debug register022

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_DBG_REG_023
Address: Operational Base + offset (0x002dc)
Vop debug register023

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_DBG_REG_028
Address: Operational Base + offset (0x002f0)

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Vop debug register028

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_MMU_DTE_ADDR
Address: Operational Base + offset (0x00300)
MMU current page Table address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	MMU_DTE_ADDR Field0000 Abstract Field0000 Description

VOP_MMU_STATUS
Address: Operational Base + offset (0x00304)
MMU status register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:6	RO	0x00	PAGE_FAULT_BUS_ID Index of master reponsible for last page fault
5	RO	0x0	PAGE_FAULT_IS_WRITE Field0000 Abstract The direction of access for last page fault: 0 = Read 1 = Write
4	RO	0x0	REPLAY_BUFFER_EMPTY Field0000 Abstract The MMU replay buffer is empty
3	RO	0x0	MMU_IDLE Field0000 Abstract The MMU is idle when accesses are being translated and there are no unfinished translated accesses.
2	RO	0x0	STAIL_ACTIVE Field0001 Abstract MMU stall mode currently enabled. The mode is enabled by command
1	RO	0x0	PAGE_FAULT_ACTIVE Field0000 Abstract MMU page fault mode currently enabled . The mode is enabled by command.
0	RO	0x0	PAGING_ENABLED Field0000 Abstract Paging is enabled

VOP_MMU_COMMAND
Address: Operational Base + offset (0x00308)

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MMU command register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2:0	WO	0x0	MMU_CMD Field0000 Abstract MMU_CMD. This can be: 0: MMU_ENABLE_PAGING 1: MMU_DISABLE_PAGING 2: MMU_ENABLE_STALL 3: MMU_DISABLE_STALL 4: MMU_ZAP_CACHE 5: MMU_PAGE_FAULT_DONE 6: MMU_FORCE_RESET

VOP_MMU_PAGE_FAULT_ADDR
Address: Operational Base + offset (0x0030c)
MMU logical address of last page fault

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	PAGE_FAULT_ADDR Field0000 Abstract address of last page fault

VOP_MMU_ZAP_ONE_LINE
Address: Operational Base + offset (0x00310)
MMU Zap cache line register

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	MMU_ZAP_ONE_LINE Field0000 Abstract address to be invalidated from the page table cache

VOP_MMU_INT_RAWSTAT
Address: Operational Base + offset (0x00314)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR Field0000 Abstract read bus error
0	RW	0x0	PAGE_FAULT Field0000 Abstract page fault

VOP_MMU_INT_CLEAR
Address: Operational Base + offset (0x00318)
MMU raw interrupt status register

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Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	WO	0x0	READ_BUS_ERROR Field0000 Abstract read bus error
0	WO	0x0	PAGE_FAULT Field0000 Abstract page fault

VOP_MMU_INT_MASK
Address: Operational Base + offset (0x0031c)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR Field0000 Abstract read bus error
0	RW	0x0	PAGE_FAULT Field0000 Abstract page fault

VOP_MMU_INT_STATUS
Address: Operational Base + offset (0x00320)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RO	0x0	READ_BUS_ERROR Field0000 Abstract read bus error
0	RO	0x0	PAGE_FAULT Field0000 Abstract page fault

VOP_MMU_AUTO_GATING
Address: Operational Base + offset (0x00324)
MMU auto gating

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	mmu_auto_gating mmu auto gating when it is 1'b1, the mmu will auto gating it self

VOP_WIN2_LUT_ADDR
Address: Operational Base + offset (0x00400)
Win2 lut base address

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	win2_lut_addr the head of win2 lut address

VOP_WIN3_LUT_ADDR
Address: Operational Base + offset (0x00800)
Win3 lut base address

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:0	RW	0x00000000	win3_lut_addr the head of win3 lut address

VOP_HWC_LUT_ADDR
Address: Operational Base + offset (0x00c00)
Hwc lut base address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	hwc_lut_addr the head of hwc lut address

VOP_GAMMA_LUT_ADDR
Address: Operational Base + offset (0x01000)
GAMMA lut base address

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RW	0x000000	gamma_lut_addr the head of gamma lut address

VOP_CABC_GAMMA_LUT_ADDR
Address: Operational Base + offset (0x01800)
CABC GAMMA lut base address

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RW	0x000000	gamma_lut_addr the head of gamma lut address

VOP_MCU_BYPASS_WPORT
Address: Operational Base + offset (0x02200)
Mcu bypass write port

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	Field0000 Abstract Field0000 Description

VOP_MCU_BYPASS_RPORT
Address: Operational Base + offset (0x02300)

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Mcu bypass read port

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	Field0000 Abstract Field0000 Description

27.5 Timing Diagram

27.5.1 RGB LCD interface timing

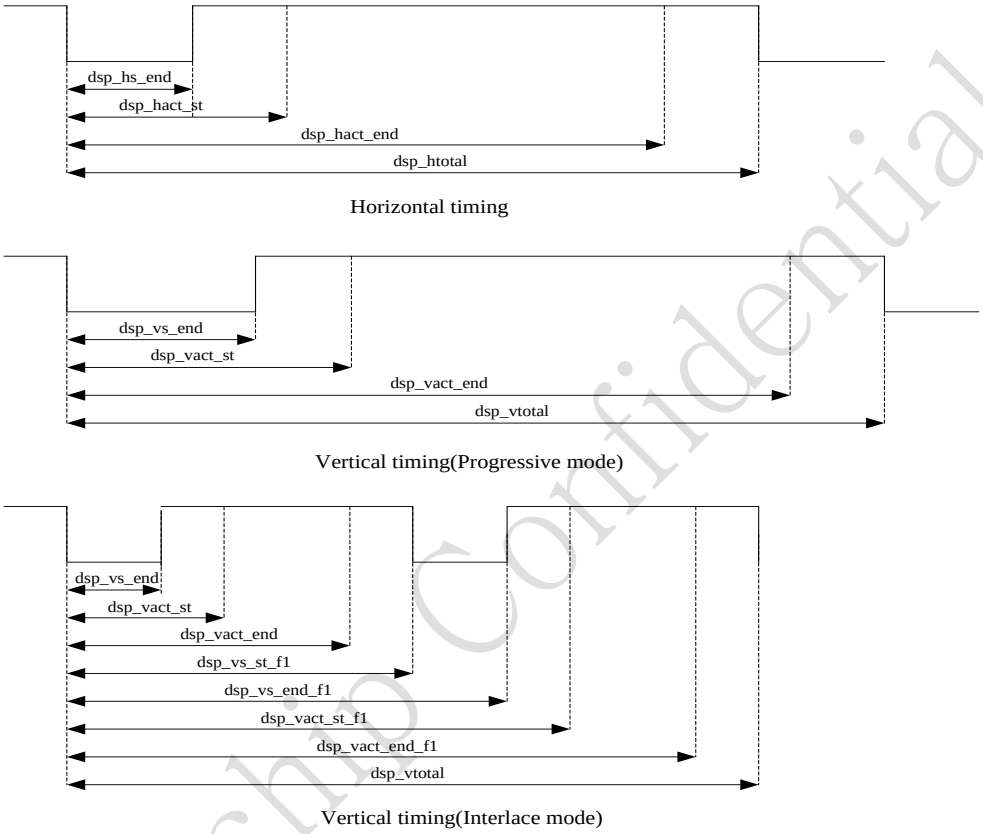
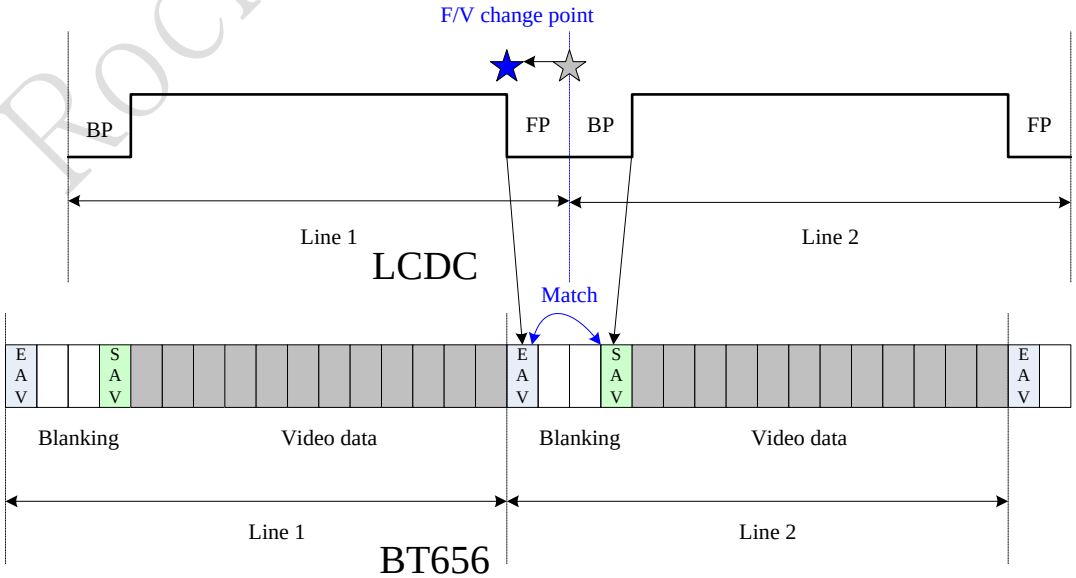


Fig. 27-16 LCDC RGB interface timing (SDR)

27.5.2 ITU656 interface timing



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27.6 Interface Description

27.6.1 Display interface description

The VOP is suitable for different display mode by different usage, which is shown as follows.

Display mode	RGB Parallel 24-bit	RGB Parallel 18-bit	RGB Parallel 16-bit	RGB Serial 2x12-bit	RGB Serial 3x8-bit	RGB Serial 3x8-bit + dummy
DCLK	DCLK	DCLK	DCLK	DCLK	DCLK	DCLK
VSYNC	VSYNC	VSYNC	VSYNC	VSYNC	VSYNC	VSYNC
HSYNC	HSYNC	HSYNC	HSYNC	HSYNC	HSYNC	HSYNC
DEN	DEN	DEN	DEN	DEN	DEN	DEN
DATA	DATA[23:0]	DATA[17:0]	DATA[15:0]	DATA[11:0]	DATA[7:0]	DATA[7:0]

Display mode	MCU Parallel 24-bit	MCU Parallel 18-bit	MCU Parallel 16-bit	MCU Serial 2x12-bit	MCU Serial 3x8-bit
DCLK	RS	RS	RS	RS	RS
VSYNC	CSN	CSN	CSN	CSN	CSN
HSYNC	WEN	WEN	WEN	WEN	WEN
DEN	REN	REN	REN	REN	REN
DATA	DATA[23:0]	DATA[17:0]	DATA[15:0]	DATA[11:0]	DATA[7:0]

Display mode	ITU656 Mode0	ITU656 Mode1	ITU656 Mode2
DCLK	DCLK	DCLK	DCLK
VSYNC	-	-	-
HSYNC	-	-	-
DEN	-	-	-
DATA	DATA[7:0]	DATA[15:8]	DATA[23:16]

In the case of "RGB serial 3x8-bit", there are four scanning modes for the RGB delta data when delta swap is enable, shown as follows.

RGB delta LCD scanning mode	*delta_en*	*dsp_rg_swap*	*dsp_rb_swap*	*dsp_bg_swap*
CSH=1,CSV=1	1	0	1	0
CSH=0,CSV=1	1	0	0	0
CSH=1,CSV=0	1	0	0	1
CSH=0,CSV=0	1	0	1	1

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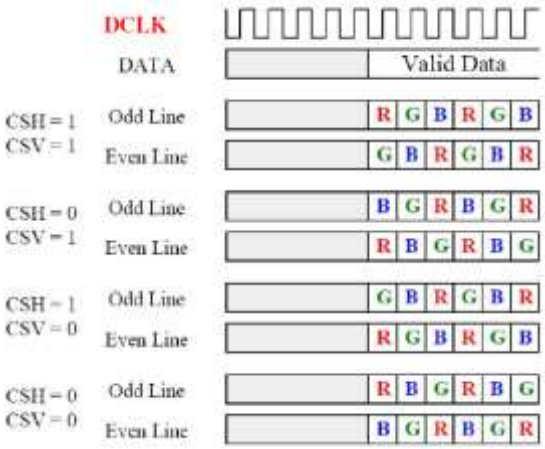

27.6.2 IOMUX description

Table 27-2 RGB Interface Description

Module pin name	Dir	Pad name	IOMUX Setting
lcdc_data0	O	IO_LCDCdata0_LVDSdatan3_MIPIDSIdatan3	See MIPI DSI-DPHY register of chapter 36
lcdc_data1	O	IO_LCDCdata1_LVDSdatap3_MIPIDSIdatap3	See MIPI DSI-DPHY register of chapter 36
lcdc_data2	O	IO_LCDCdata2_LVDSdatan2_MIPIDSIdatan2	See MIPI DSI-DPHY register of chapter 36
lcdc_data3	O	IO_LCDCdata3_LVDSdatap2_MIPIDSIdatap2	See MIPI DSI-DPHY register of chapter 36
lcdc_data4	O	IO_LCDCdata4_LVDSclkp_MIPIDSIclkp	See MIPI DSI-DPHY register of chapter 36
lcdc_data5	O	IO_LCDCdata5_LVDSclkn_MIPIDSIclkn	See MIPI DSI-DPHY register of chapter 36
lcdc_data6	O	IO_LCDCdata6_LVDSdatap1_MIPIDSIdatap1	See MIPI DSI-DPHY register of chapter 36
lcdc_data7	O	IO_LCDCdata7_LVDSdatan1_MIPIDSIdatan1	See MIPI DSI-DPHY register of chapter 36
lcdc_data8	O	IO_LCDCdata8_LVDSdatap0_MIPIDSIdatap0	See MIPI DSI-DPHY register of chapter 36
lcdc_data9	O	IO_LCDCdata9_LVDSdatan0_MIPIDSIdatan0	See MIPI DSI-DPHY register of chapter 36
lcdc_data10	O	IO_LCDCdata10_TRACEdata0_JTAGtrstn_LCDCgpio0b6	PMUGRF_GPIO0B_IOMUX[13:12]=2'b01
lcdc_data11	O	IO_LCDCdata11_TRACEdata1_JTAGtdi_LCDCgpio0b7	PMUGRF_GPIO0B_IOMUX[15:14]=2'b01
lcdc_data12	O	IO_LCDCdata12_TRACEdata2_JTAGtdo_LCDCgpio0c0	PMUGRF_GPIO0C_IOMUX[1:0]=2'b01
lcdc_data13	O	IO_LCDCdata13_TRACEdata3_MCUJTAGtrstn_LCDCgpio0c1	PMUGRF_GPIO0C_IOMUX[3:2]=2'b01
lcdc_data14	O	IO_LCDCdata14_TRACEdata4_MCUJTAGtdi_LCDCgpio0c2	PMUGRF_GPIO0C_IOMUX[5:4]=2'b01
lcdc_data15	O	IO_LCDCdata15_TRACEdata5_MCUJTAGtdo_LCDCgpio0c3	PMUGRF_GPIO0C_IOMUX[7:6]=2'b01
lcdc_data16	O	IO_LCDCdata16_TRACEdata6_UART1BBsin_LCDCgpio0c4	PMUGRF_GPIO0C_IOMUX[9:8]=2'b01
lcdc_data17	O	IO_LCDCdata17_TRACEdata7_UART1BBsout_LCDCgpio0c5	PMUGRF_GPIO0C_IOMUX[11:10]=2'b01
lcdc_data18	O	IO_LCDCdata18_TRACEdata8_UART1BBctsn_LCDCgpio0c6	PMUGRF_GPIO0C_IOMUX[13:12]=2'b01
lcdc_data19	O	IO_LCDCdata19_TRACEdata9_UART1BBrtsn_LCDCgpio0c7	PMUGRF_GPIO0C_IOMUX[15:14]=2'b01
lcdc_data20	O	IO_LCDCdata20_TRACEdata10_UART4EXPctsn_LCDCgpio0d0	PMUGRF_GPIO0D_IOMUX[1:0]=2'b01
lcdc_data21	O	IO_LCDCdata21_TRACEdata11_UART4EXPrtsn_LCDCgpio0d1	PMUGRF_GPIO0D_IOMUX[3:2]=2'b01
lcdc_data22	O	IO_LCDCdata22_TRACEdata12_UART4EXPsout_LCDCgpio0d2	PMUGRF_GPIO0D_IOMUX[5:4]=2'b01
lcdc_data23	O	IO_LCDCdata23_TRACEdata13_UART4EXPsin_LCDCgpio0d3	PMUGRF_GPIO0D_IOMUX[7:6]=2'b01
lcdc_hsync	O	IO_LCDChsync_TRACEdata14_PMUdebug2_LCDCgpio0d4	PMUGRF_GPIO0D_IOMUX[9:8]=2'b01
lcdc_vsync	O	IO_LCDCvsync_TRACEdata15_PMUdebug3_LCDCgpio0d5	PMUGRF_GPIO0D_IOMUX[11:10]=2'b01
lcdc_den	O	IO_LCDCden_TRACEclk_PMUdebug4_LCDCgpio0d6	PMUGRF_GPIO0D_IOMUX[13:12]=2'b01
lcdc_dclk	O	IO_LCDCdclk_TRACEctl_PMUdebug5_LCDCgpio0d7	PMUGRF_GPIO0D_IOMUX[15:14]=2'b01

27.7 Application Notes

27.7.1 DMA transfer mode

There are three DMA transfer modes for loading win0 or win1 frame data determined by following parameters(X=0,1,2,3):
dma_burst_length
winX_no_outstanding
winX_gather_en
winX_gather_thres

- **auto outstanding transfer mode(random transfer)**

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When winX_no_outstanding is 0, multi-bursts transfer command could be sent out to AXI master interface continuously if the internal memory has enough space to store new data. The continuous random burst number is in the range of 1 to 4, mainly depending on the empty level of internal memory, dma_burst_length, data format and active image width.

- **configured outstanding transfer mode(fixed transfer)**

When winX_gather_en is 1, fixed-number of bursts transfer command should be sent out to AXI master interface continuously if the internal memory has enough space to store new data. The fixed-number is determined by winX_gather_thres. Since the internal memory size is limited, there is some restriction for the winX_gather_thres as follows.

Table 27-3 Gather configuration for all format

Gather Threshold	**dma_burst_length =2'b00(burst16)**	**dma_burst_length =2'b01(burst8)**	**dma_burst_length =2'b10(burst4)**
YUV420 YUV422 YUV444	0	0,1,2	0,1,2,3
ARGB888 RGB888 RGB565	0,1,2,3	0,1,2,3	0,1,2,3
8BPP 4BPP 2BPP 1BPP	0,1,2,3	0,1,2,3	0,1,2,3

27.7.2 Win0/Win1 dma load mode

If you want to improve the efficiency of accessing external memory for loading winX frame data, you could assert winX_dma_load. When winX_dma_load is high, winX frame data is loaded in the unit of line composing with one or more burst transfers; otherwise, loaded in the unit of burst transfer. However, it is not suitable for data format YUV420, no-scaling and active width less than 256.

27.7.3 IEP direct path

There are two data source for win0/win1win2/win3: external memory and IEP internal memory. However, the IEP data is just active for one layer at one frame time when IEP data path is enable, determined by dsp_layer0/1/2/3_sel. Moreover, it is not suitable for win0/1 with scaling, reverse display.

Direct path interface (DPI) can be used for LCDC to display images from other image processing IPs, which also has DPI (slave).

There are programming flows for both DPI display on sequence and DPI display off sequence.

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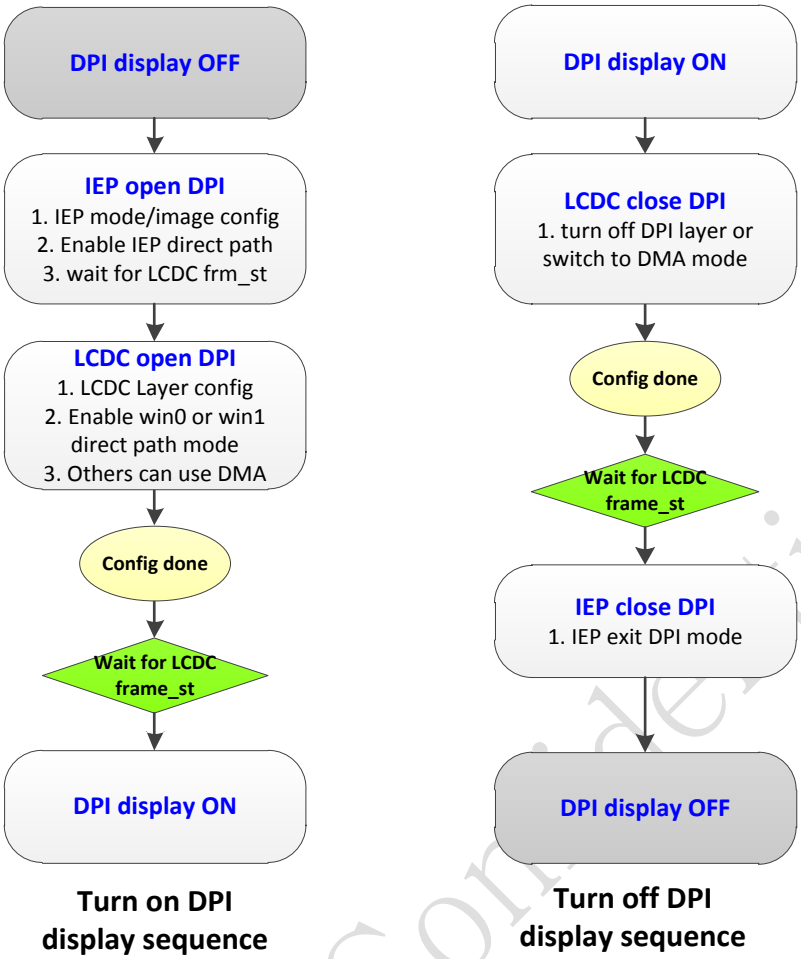

Fig. 27-17 LCDC DPI Programming flow

1.Turn on DPI display

First, configure IEP into DPI mode. After doing image information and image processing mode configuration, IEP DPI mode can be turn on for display. IEP is in idle mode only if LCDC's frame start input signal is valid.

Second, configure LCDC for DPI display. Note that only one layer (Win0 or Win1) can use DPI in same frame. Other layers still can use internal DMA.

Finally, set LCDC "config_done" to confirm all the new configuration and waiting frame sync to start DPI display actually.

2.Turn off DPI display

First, close LCDC layer's DPI mode by turning off DPI layer or switching it to DMA mode. Then set LCDC "config_done" to confirm new configuration.

Second, wait for LCDC's frame synchronization to close DPI display in LCDC.

Finally, turn off IEP's DPI mode.

27.7.4 WIN BPP LUT/GAMMA LUT

WIN1 LUT/DSP LUT should be configured before displaying if win2/3_lut_en/dsp_lut_en is high. You could only update these LUTs by software.

When win1_lut_load_en is 0, the WIN LUT data should be refreshed by software,i.e, writing win1 lut data to the internal memory with the start address WIN1_LUT_MST. The memory size is 256x25, i.e, lower 25bits valid, and the writing data number is determined by software, .

When dsp_lut_load_en is 0, the DSP LUT data should be refreshed by software,i.e, writing dsp lut data to the internal memory with the start address DSP_LUT_MST. The memory size is 256x24, i.e, lower 25bits valid, and the writing data number is determined by software.The program model for CABC gamma LUT is similar.

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27.7.5 DMA QoS request

If you want to get higher priority for VOP to access external memory when the frame data is urgent, a QoS request can be generated and sent out basing on the configured values:
noc_hurry_en
noc_hurry_value
noc_qos_en
noc_win_qos
If noc_qos_en is enable, a win0/1_qos_req is asserted when the empty level of win0/1's linebuffer is greater than the threshold configured in noc_win_qos. And it will be disserted when the empty level is smaller than the threshold or noc_qos_en is disable.
If noc_qos_en is enable, a win0/1_hurry_req is asserted when the empty level of win2/3's fifo is greater than the threshold configured in noc_win_qos. And it will be disserted when the empty level is smaller than the threshold or noc_qos_en is disable.
Either win0/1_qos_req or win2/3_hurry_req is high, a QoS request will be sent out for VOP.

27.7.6 Mirror display

If Y-Mirror display is enable, the frame data is loaded from last line to first line, where the start address of first pixel in last line is defined in WIN0/1_YRGB0_MST/WIN0/1_CBR0_MST/WIN0/1_YRGB1_MST/WIN0/1_CBR_MST/WIN2/3_MST for win0/1/2/3 respectively.
Otherwise, the win's frame line data width and virtual stride should be 64bit-aligned for 8bit-RGB/YUV or 128bit-aligned for 10bit YUV if X-Mirror or Y-Mirror display is enable.

27.7.7 DDR interface

LCD DDR interface is just suitable for Parallel RGB LCD panel and Serial RGB LCD 2x12 panel.
If LCD DDR interface is enable, the timing parameters for LCD panel should be even.
Otherwise, you can synchronize output clock with VSYNC or HSYNC depending on dclk_ddr_sync.

27.7.8 Interrupt

VOP interrupt is comprised of 15 interrupt sources:
2 frame start interrupt
same frame address interrupt
2 line flag interrupt
hold interrupt
bus error interrupt
win0 empty interrupt
win1 empty interrupt
win2 empty interrupt
win3 empty interrupt
hwc empty interrupt
post empty interrupt
pwm gen interrupt
irq_mmu
Every interrupt has independent interrupt enable (VOP_INT_EN,with bit write mask), interrupt clear (VOP_INT_CLR,with bit write mask), interrupt status (VOP_INT_STATUS).

27.7.9 RGB display mode

RGB display mode is used for RGB panel display and CCIR656 output. It is a continuous frames display mode.

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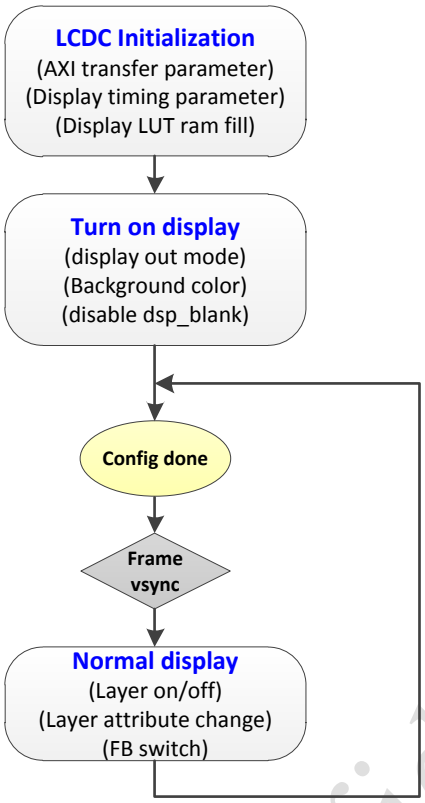

Fig. 27-18 LCDC RGB mode Programming flow

1.LCDC initialization

LCDC initialization should be done before turning display on.
First, AXI bus parameter (LCDC_SYS_CTRL) should be set for DMA transfer.
Second, display panel/interface timing should be set for display output. The registers are: VOP_DSP_HTOTAL_HS_END/ VOP_DSP_HACT_ST_END/ VOP_DSP_VTOTAL_HS_END/ VOP_DSP_VACT_ST_END/ VOP_DSP_VS_ST_END_F1/ VOP_DSP_VACT_ST_END_F1

2.Background display

Before normal display, the background display could be turn on.
First, set display output mode (VOP_DSP_CTRL0/1) according to display device.
Second, disable dsp_blank mode, which would not be enable until frame synchronization.
Finally, writing '1' to "VOP_REG_CFG_DONE" register then all the frame-sync registers will be enable at the beginning of next frame.

3.Normal display

In normal display, all the display layers' attribute could be different according display scenario. So there is a programming loop in this mode.
First, configure all the display layers' attribute registers for the change of image format, location, size, scaling factor, alpha and overlay and so on. Those register would not be enable until frame synchronization.
Finally, write '1' to "VOP_REG_CFG_DONE" register then all the frame-sync registers will be enable at the beginning of next frame.

27.7.10 Immediately control register

There are two type register in VOP , one type is effective immediately,the other is effective by frame sync.
Effective immediately registers list as follows,other registers are all effective by frame sync.

Table 27-4 effective immediately register table

register address	Description
0x008[23:21],0x008[15:8]	some dsp ctrl function bit
0x00c	sys ctrl1 register
0x018	background color register

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0x01c	mcu ctrl register
0x038	win0 color key register
0x078	win1 color key register
0x0cc	win2 color key register
0x11c	win3 color key register
0x188~0x19c	dsp_timing ctrl registers
0x1a0~0x1a8	pwm ctrl registers
0x1c8~0x1dc	cabc_gauss_parameter registers
0x1ec~0x1f4	frc pattern parameter registers

27.7.11 FBDC

VOP supports decoding compressed frame buffer by GPU.It supports virtual display.It only supports 16x4 mode,thus GPU should only generate 16x4 compressed frame buffer for VOP.
For ARGB8888(U8U8U8U8 in GPU),RGB888pack(U8U8U8 in GPU),
the width needs to be 16 pixel align;
the height needs to be 4 pixel align.

For RGB565,
the width needs to be 32 pixel align;
the height needs to be 4 pixel align;
fbdc_dsp_width_ratio should be set "1".

The program model is showed as below:
1. config FBDC;
If GPU enables correlation for FB(frame buffer), set fbdc_cor_en as "1".
Config vop_fbdc_win_sel to select win0/1/2/3 using FBC.
If fbdc_axi_max_outstanding_en=1, the max value of fbdc_axi_max_outstanding_num is 29.
If fbdc_fmt is RGB565, set fbdc_dsp_width_ratio as "1".
Config fbdc_cmp_index_init according to fbdc_rotation_mode.See the following figure. The unit of FBDC width and height registers is macro block.

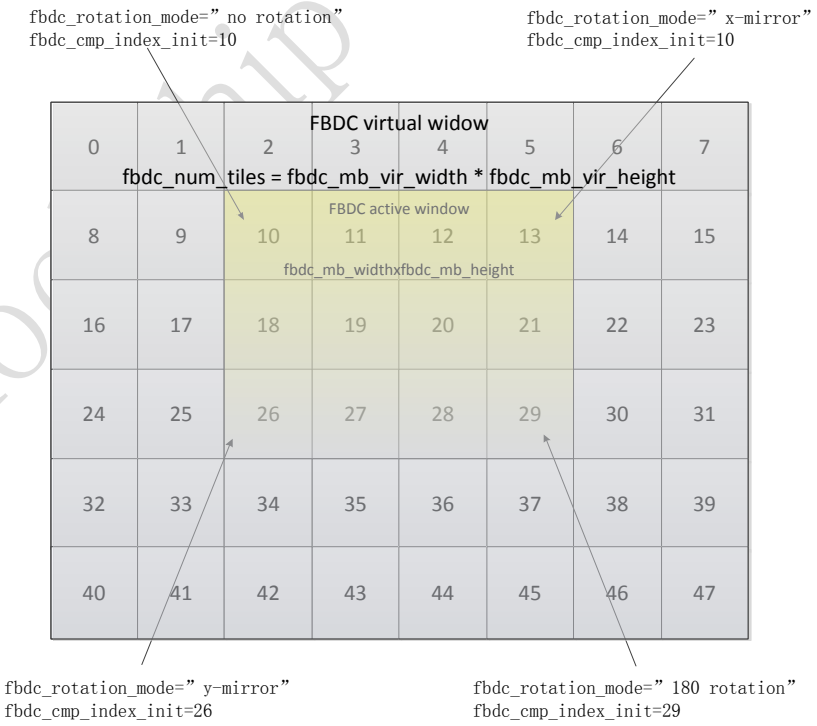

Fig. 27-19 FBDC virtual display

2. config window using FBDC;
   - The active window resolution shoud be the same as the active window of FBDC.

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- VOP does not support vetical scale down ratio>=2 using bilinear.

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Chapter 28 RGA2

28.1 Overview

RGA is a separate 2D raster graphic acceleration unit. It accelerates 2D graphics operations, such as colorpalette, colorfill, image scaling, rotation/mirror, BitBLT, alpha blending.

28.1.1 Features

- Data format
  - Input data: ARGB/RGB888/RGB565/RGB4444/RGB5551/YUV420/YUV422
  - Output data: ARGB/RGB888/RGB565/RGB4444/RGB5551/YUV420/YUV422
  - Pixel Format conversion, BT.601/BT.709
  - Dither up/down operation
  - Max resolution: 8192x8192 source, 4096x4096 destination
- Scaling
  - Down-scaling: Average filter
  - Up-scaling: Bi-cubic filter(source>2048 would use Bi-linear)
  - Arbitrary non-integer scaling ratio，from 1/8 to 8
- Rotation
  - 0, 90, 180, 270 degree rotation
  - x-mirror, y-mirror & rotation operation
- BitBLT
  - Block transfer
  - Color fill, support pattern and solid mode, support with alpha
  - Transparency mode (color keying/stencil test, specified value/value range)
  - Two source BitBLT:
  - A+B=B only BitBLT, not support scale/rotate mode
  - A+B=C second source (B) has same attribute with (C) plus rotation function
- Alpha Blending
  - New comprehensive per-pixel alpha(color/alpha channel separately)
  - Fading
- Raster operation
  - ROP2/ROP3/ROP4
- MMU
  - 4k/64k page size
  - Four channel: SRC/SRC1/DST/CMD, individual base address and enable control bit
  - TLB pre-fetch
- ColorPalette
  - ColorPalette support 1bpp/2bpp/4bpp/8bpp
  - 8bpp ColorPalette support scale, support rotation & alpha

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28.2 Block Diagram

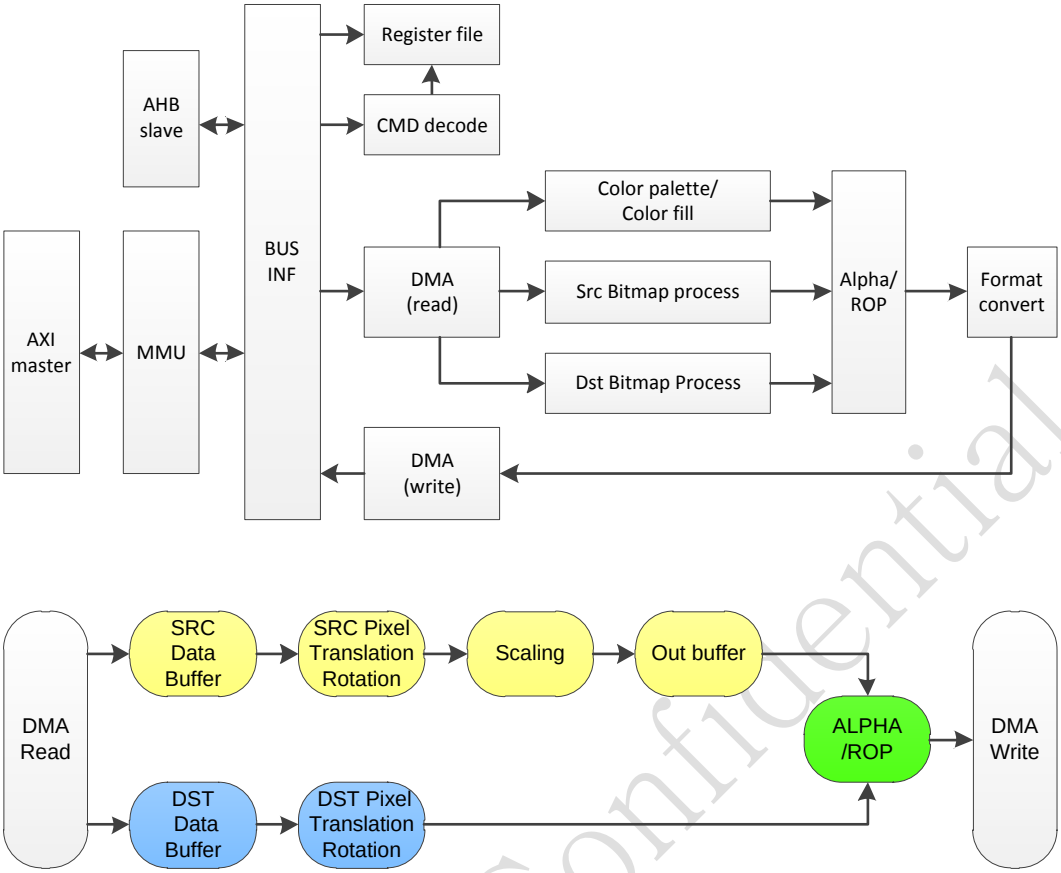

Fig. 28-1 RGA2 Block Diagram

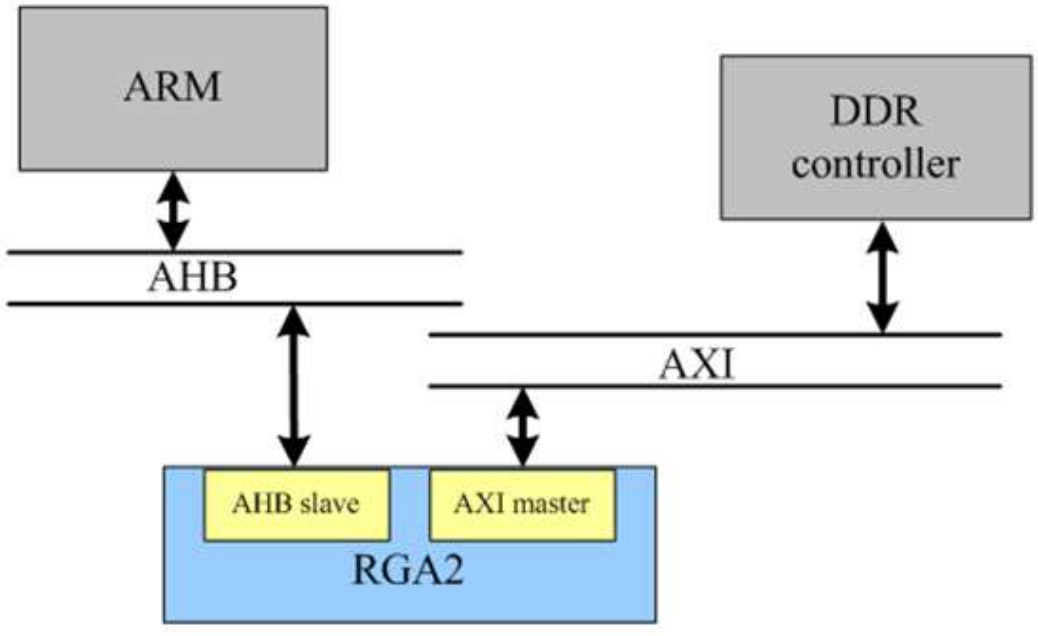

Fig. 28-2 RGA2 in SOC

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28.3 Function Description

28.3.1 Data Format

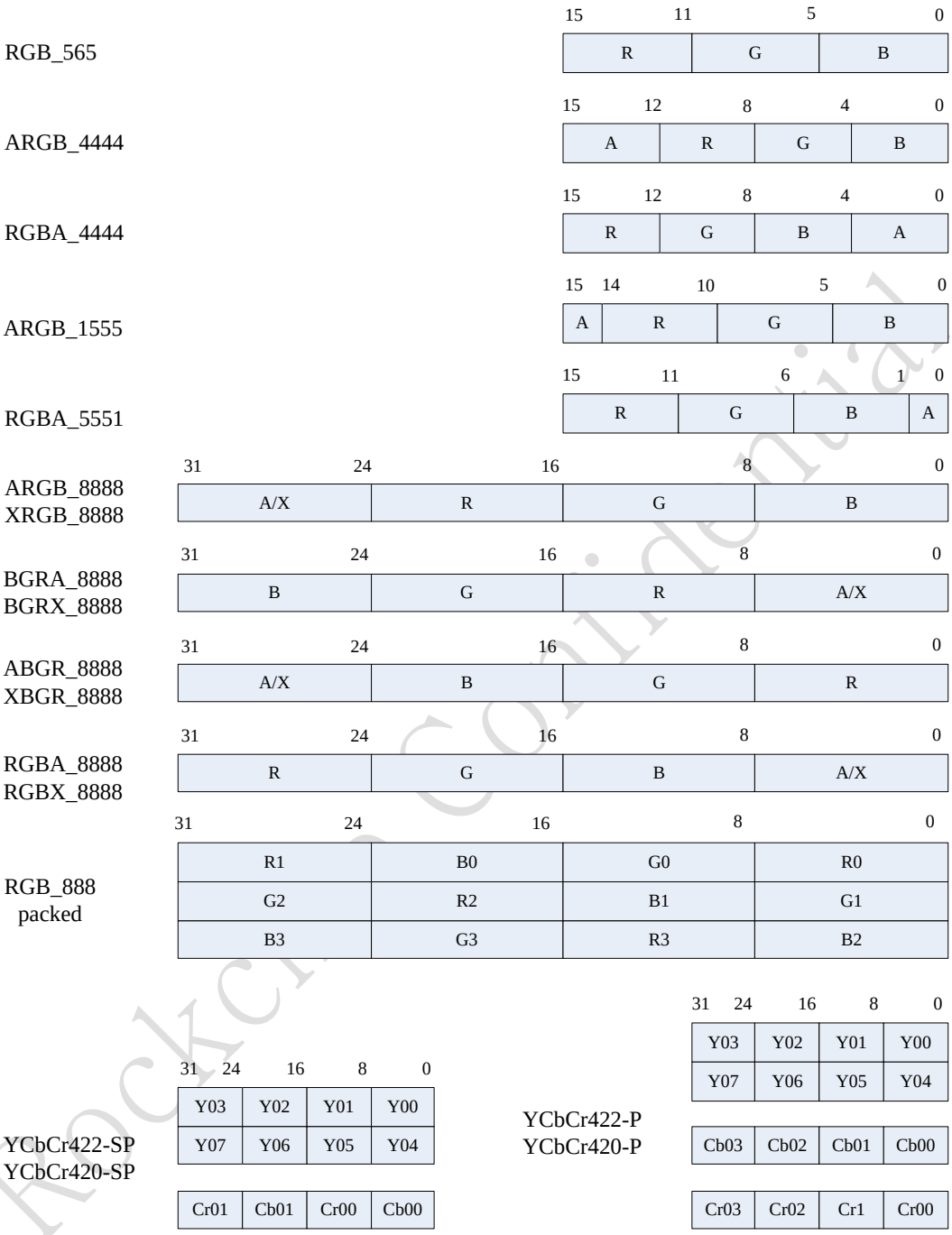

Fig. 28-3 RGA Input Data Format

All input datas (defined by SRC_IN_FMT/DST_IN_FMT) are converted to ABGR8888. The results are converted to the output data format (defined by DST_OUT_FMT).

28.3.2 Dithering

There could have dithering operation for source image when the source image format is not RGB565 and the destination format is RGB565.

The down-dithering is done using Dither Allegro.

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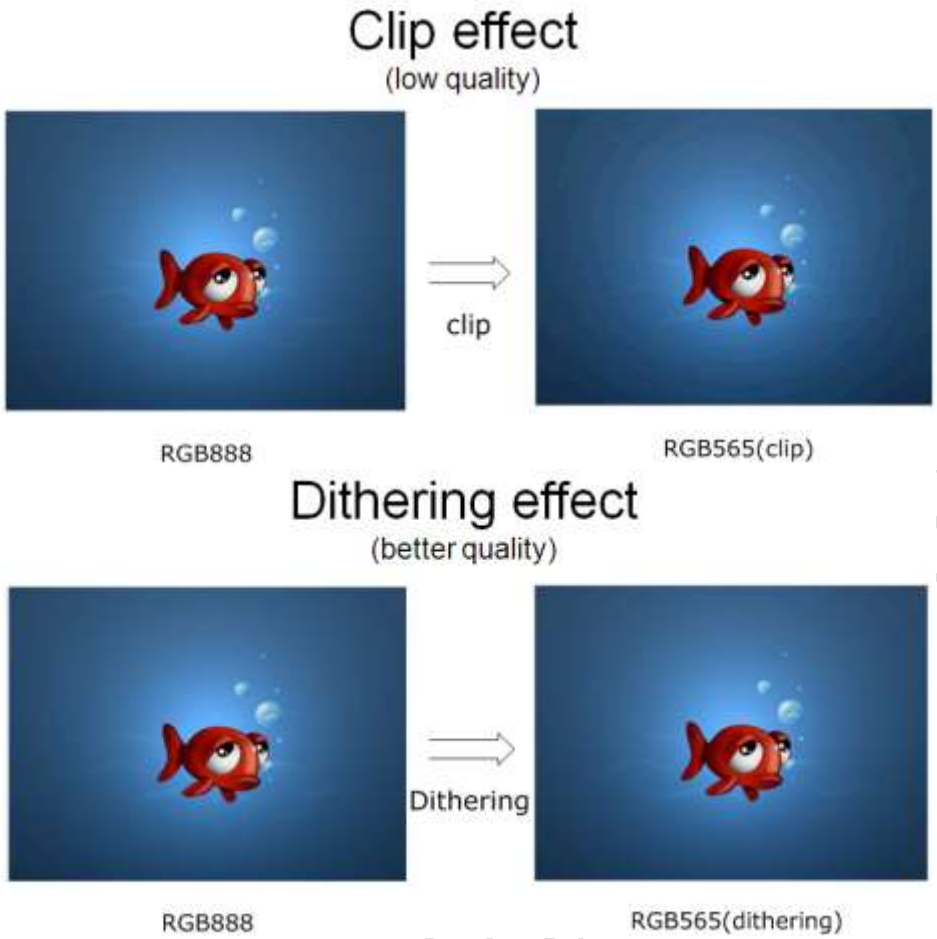

Fig. 28-4 RGA Dither effect

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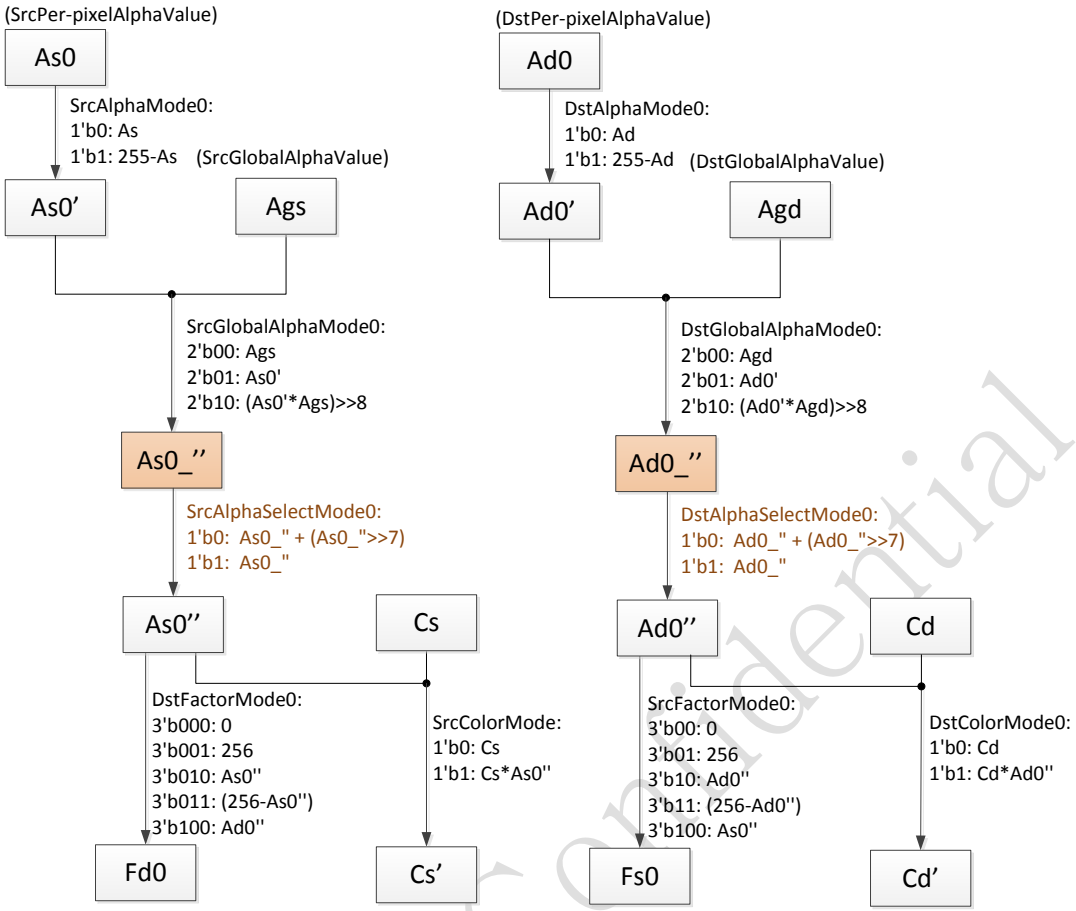
28.3.3 Alpha mode



$$Cd = Fs0 * Cs' + Fd0 * Cd' \quad (1)$$

(Cd – dst color, Fs0 – color src factor0, Cs' – src color', Fd0 – color dst factor0, Cd' – dst color')

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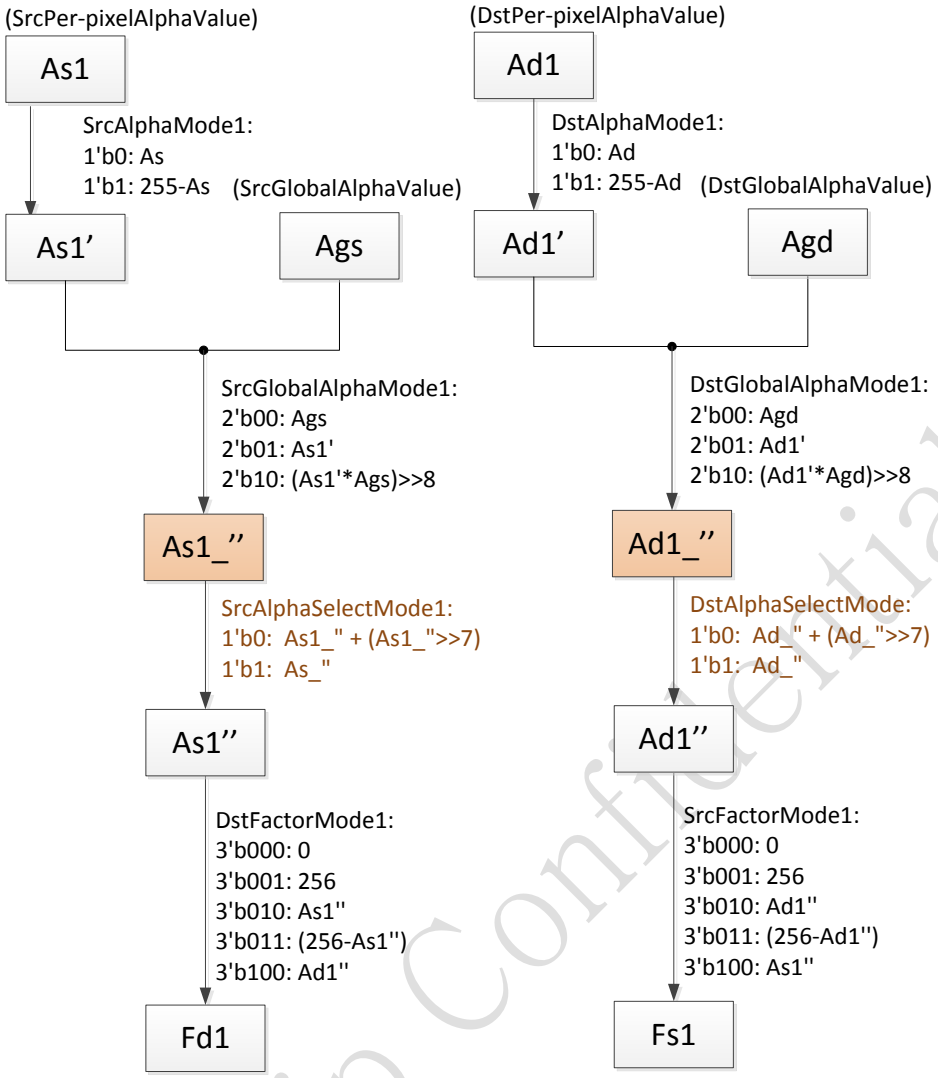

$$Ad = Fs1 * As1'' + Fd1 * Ad1'' \quad (2)$$

(Ad – dst alpha, Fs1 – alpha src factor1, As1'' – src alpha'', Fd1 – alpha dst factor1, Ad1' – dst alpha'')

28.3.4 Color fill

Two modes of color fill can be done by RGA: solid fill and gradient fill.

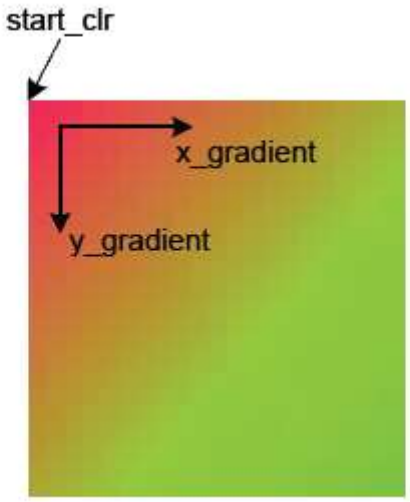

Fig. 28-5 RGA Gradient Fill

Gradient fill using following equations for ARGB calculation of every pixel in different

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coordinary.

A_cur = (A_start + x*x_A_gradient) +y*y_A_gradient;
R_cur = (R_start + x*x_R_gradient) +y*y_R_gradient;
G_cur = (G_start + x*x_G_gradient) +y*y_G_gradient;
B_cur = (B_start + x*x_B_gradient) +y*y_B_gradient;

A_start, R_start, G_start, B_start is the ARGB value of start point. There are four pairs of values for horizontal and vertical gradient. Saturation operation could be enabled or disabled if the color overflows 255 or underflows 0.

28.3.5 Raster Operation (ROP)

Raster operation (ROP) is a Boolean operation between operands, which involve AND, OR, XOR, and NOT operations. For ROP2, operands are P (select pan) and D (Destination bitmap). For ROP3, operands are P (pattern), S (source bitmap) and D (Destination bitmap). For ROP4, operands are P (pattern), S (source bitmap), D (Destination bitmap) and MASK.

Table 28-1 RGA ROP Boolean operations

Operator	Meaning
a	Bitwise AND
n	Bitwise NOT (inverse)
o	Bitwise OR
x	Bitwise exclusive OR (XOR)

28.3.6 Scaling

The scaling operation is the imageresizing processing of source image. Scaling is done base on ARGB8888 format.

There are three sampling modes: scale down (Average); scale up(Bi-cubic);

28.4 Register description

28.4.1 Register Summary

Name	Offset	Size	Reset Value	Description
RGA2_RGA_SYS_CTRL	0x0000	W	0x00000044	RGA system control register
RGA2_RGA_CMD_CTRL	0x0004	W	0x00000000	RGA command control register
RGA2_RGA_CMD_BASE	0x0008	W	0x00000000	RGA command codes base address register
RGA2_RGA_STATUS	0x000c	W	0x00000000	RGA status register
RGA2_RGA_INT	0x0010	W	0x00000000	RGA interrupt register
RGA2_RGA_MMU_CTRL0	0x0014	W	0x00000000	RGA MMU control 0 register
RGA2_RGA_MMU_CMD_BASE	0x0018	W	0x00000000	Register0000 Abstract
RGA_VERSION	0x0028	W	0x04018632	RGA version num
RGA2_RGA_MODE_CTRL	0x0100	W	0x00000000	RGA mode control register
RGA2_RGA_SRC_INFO	0x0104	W	0x00000000	RGA source information register
RGA2_RGA_SRC_BASE0	0x0108	W	0x00000000	source image Y/RGB base address
RGA2_RGA_SRC_BASE1	0x010c	W	0x00000000	RGA source image Cb/Cbr base address register
RGA2_RGA_SRC_BASE2	0x0110	W	0x00000000	RGA source image Cr base address register
RGA2_RGA_SRC_BASE3	0x0114	W	0x00000000	RGA source image 1 base address register

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Name	Offset	Size	Reset Value	Description
RGA2_RGA_SRC_VIR_INFO	0x0118	W	0x00000000	RGA source image virtual stride / RGA source image tile number
RGA2_RGA_SRC_ACT_INFO	0x011c	W	0x00000000	RGA source image active width/height register
RGA2_RGA_SRC_X_FACTOR	0x0120	W	0x00000000	RGA source image horizontal scaling factor
RGA2_RGA_SRC_Y_FACTOR	0x0124	W	0x00000000	RGA source image vertical scaling factor
RGA2_RGA_SRC_BG_COLOR	0x0128	W	0x00000000	RGA source image background color
RGA2_RGA_SRC_FG_COLOR	0x012c	W	0x00000000	RGA source image foreground color
RGA2_RGA_CP_GR_A	0x0130	W	0x00000000	RGA source image transparency color min value
RGA2_RGA_SRC_TR_COLOR0	0x0130	W	0x00000000	RGA source image transparency color min value
RGA2_RGA_CP_GR_B	0x0134	W	0x00000000	RGA source image transparency color max value
RGA2_RGA_SRC_TR_COLOR1	0x0134	W	0x00000000	Register0000 Abstract
RGA2_RGA_DST_INFO	0x0138	W	0x00000000	RGA destination format register
RGA2_RGA_DST_BASE0	0x013c	W	0x00000000	RGA destination image base address 0 register
RGA2_RGA_DST_BASE1	0x0140	W	0x00000000	RGA destination image base address 1 register
RGA2_RGA_DST_BASE2	0x0144	W	0x00000000	RGA destination image base address 2 register
RGA2_RGA_DST_VIR_INFO	0x0148	W	0x00000000	RGA destination image virtual width/height register
RGA2_RGA_DST_ACT_INFO	0x014c	W	0x00000000	RGA destination image active width/height register
RGA2_RGA_ALPHA_CTRL0	0x0150	W	0x00000000	Alpha control register 0
RGA2_RGA_ALPHA_CTRL1	0x0154	W	0x00000000	Register0000 Abstract
RGA2_RGA_FADING_CTRL	0x0158	W	0x00000000	Fading control register
RGA2_RGA_PAT_CON	0x015c	W	0x00000000	Pattern size/offset register
RGA2_RGA_CP_GR_G	0x0160	W	0x00000000	RGA color gradient fill step register (color fill mode)
RGA2_RGA_ROP_CON0	0x0160	W	0x00000000	ROP code 0 control register
RGA2_RGA_CP_GR_R	0x0164	W	0x00000000	RGA color gradient fill step register (color fill mode)
RGA2_RGA_ROP_CON1	0x0164	W	0x00000000	ROP code 1 control register
RGA2_RGA_MASK_BASE	0x0168	W	0x00000000	RGA mask base address register
RGA2_RGA_MMU_CTRL1	0x016c	W	0x00000000	RGA MMU control register 1
RGA2_RGA_MMU_SRC_BASE	0x0170	W	0x00000000	RGA source MMU TLB base address

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Name	Offset	Size	Reset Value	Description
RGA2_RGA_MMU_SRC1_BASE	0x0174	W	0x00000000	RGA source1 MMU TLB base address
RGA2_RGA_MMU_DST_BASE	0x0178	W	0x00000000	RGA destination MMU TLB base address
RGA2_RGA_MMU_ELS_BASE	0x017c	W	0x00000000	RGA ELSE MMU TLB base address

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

28.4.2 Detail Register Description

RGA2_RGA_SYS_CTRL
Address: Operational Base + offset (0x0000)
RGA system control register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	RW	0x0	sw_auto_rst it would auto-resetn after one frame finish. 0: disable 1: enable
4	RW	0x0	sw_cclk_sreset_p RGA core clk domain Soft reset, write '1' to this would reset the RGA engine except config registers.
3	WO	0x0	sw_aclk_sreset_p RGA aclk domain Soft reset, write '1' to this would reset the RGA engine except config registers.
2	WO	0x1	sw_auto_ckg RGA auto clock gating enable bit 0：disable 1：enable
1	WO	0x0	sw_cmd_mode RGA command mode 0：slave mode 1：master mode
0	WO	0x0	sw_cmd_op_st_p RGA operation start bit Only used in passive (slave) control mode

RGA2_RGA_CMD_CTRL
Address: Operational Base + offset (0x0004)
RGA command control register

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:3	RW	0x000	sw_cmd_incr_num RGA command increment number

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Bit	Attr	Reset Value	Description
2	WO	0x0	sw_cmd_stop RGA command stop mode Command execution would stop after the current graphic operation finish if set this bit to 1
1	WO	0x0	sw_cmd_incr_valid_p RGA command increment valid (Auto cleared) When setting this bit, 1. The total cmd number would increase by the RGA_INCR_CMD_NUM. 2. RGA would continue running if idle.
0	RW	0x0	sw_cmd_line_st_p RGA command line fetch start (command line reset) (Auto cleared) When fetch start, the total cmd number would reset to RGA_INCR_CMD_NUM.

RGA2_RGA_CMD_BASE
Address: Operational Base + offset (0x0008)
RGA command codes base address register

Bit	Attr	Reset Value	Description
31:0	RW	0x12345678	sw_cmd_base RGA command codes base address

RGA2_RGA_STATUS
Address: Operational Base + offset (0x000c)
RGA status register

Bit	Attr	Reset Value	Description
31:20	RO	0x000	sw_cmd_total_num RGA command total number
19:8	RO	0x000	sw_cmd_cur_num RGA command current number
7:1	RW	0x00	Reserved Reserved
0	RO	0x0	sw_rga_sta RGA engine status 0: idle 1: working

RGA2_RGA_INT
Address: Operational Base + offset (0x0010)
RGA interrupt register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
10	RW	0x0	sw_intr_af_e All command finished interrupt enable
9	RW	0x0	sw_intr_mmu_e MMU interrupt enable
8	RW	0x0	sw_intr_err_e Error interrupt enable
7	WO	0x0	sw_intr_cf_clr Current command finished interrupt clear
6	WO	0x0	sw_intr_af_clr All command finished interrupt clear
5	WO	0x0	sw_intr_mmu_clr MMU interrupt clear
4	WO	0x0	sw_intr_err_clr Error interrupt clear
3	RO	0x0	sw_intr_cf Current command finished interrupt flag
2	RO	0x0	sw_intr_af All command finished interrupt flag
1	RO	0x0	sw_intr_mmu MMU interrupt
0	RO	0x0	sw_intr_err Error interrupt flag

RGA2_RGA_MMU_CTRL0
Address: Operational Base + offset (0x0014)
RGA MMU control 0 register

Bit	Attr	Reset Value	Description
31:11	RW	0x000000	Reserved
10:9	RW	0x0	sw_els_ch_priority
8:7	RW	0x0	sw_dst_ch_priority
6:5	RW	0x0	sw_src1_ch_priority
4:3	RW	0x0	sw_src_ch_priority
2	RW	0x0	sw_cmd_mmu_flush RGA CMD channel MMU TLB flush: Set 1 to this bit to flush MMU TLB, auto clear
1	RW	0x0	sw_cmd_mmu_en RGA CMD channel MMU enable 0: disable 1: enable
0	RW	0x0	sw_mmu_page_size RGA MMU Page table size 0: 4KB page 1: 64KB page

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RGA2_RGA_MMU_CMD_BASE
Address: Operational Base + offset (0x0018)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:0	RW	0x0000000	sw_mmu_cmd_base RGA command MMU TLB base address (word)

RGA_VERSION
Address: Operational Base + offset (0x0028)
RGA Version num

Bit	Attr	Reset Value	Description
31:24	RO	0x0004	MAJOR: RGA1: 1; RGA1_Plus: 2; RGA2: 3; RGA2_Lite: 4;
RO	0x0	reserved	RO
15:0	RO	0x????	ASCII code: mmip svn num=????

RGA2_RGA_MODE_CTRL
Address: Operational Base + offset (0x0100)
RGA mode control register

Bit	Attr	Reset Value	Description
31:8	RW	0x000000	Reserved
7	RW	0x0	sw_intr_cf_e Current command finished interrupt enable
6	RW	0x0	sw_gradient_sat Gradient saturation calculation mode 0:clip 1:not-clip
5	RW	0x0	sw_alpha_zero_key ARGB888 alpha zero key mode 0x000000 would be changed to 0x000100(RGB888)/0x0020(RGB565)for ARGB888 to RGBX/RGB565 color key 0: disable 1: enable
4	RW	0x0	sw_cf_rop4_pat Color fill/ROP4 pattern 0: solid color 1: pattern color
3	RW	0x0	sw_bb_mode Bitblt mode 0: SRC + DST => DST 1: SRC + SRC1 => DST

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Bit	Attr	Reset Value	Description
2:0	RW	0x0	sw_render_mode RGA 2D render mode 000: Bitblt 001: Color palette 010: Rectangle fill 011: Update palette LUT/pattern ram

RGA2_RGA_SRC_INFO
Address: Operational Base + offset (0x0104)
RGA source information register

Bit	Attr	Reset Value	Description
31:26	RW	0x00	Reserved
25:24	RW	0x0	sw_bic_coe_sel SRC bicubic scaling coefficient select 00: CATROM 01: MITCHELL 10: HERMITE 11: B-SPLINE
23	RW	0x0	sw_src_dither_up SRC dither up enable 0:disable 1:enable
22:19	RW	0x0	sw_src_trans_e Source transparency enable bits [3]: A value stencil test enable bit [2]: B value stencil test enable bit [1]: G value stencil test enable bit [0]: R value stencil test enable bit
18	RW	0x0	sw_src_trans_mode Source transparency mode 0: normal stencil test (color key) 1: inverted stencil test
17:16	RW	0x0	sw_src_vscl_mode SRC vertical scaling mode 00：no scaling 01: down-scaling 10: up-scaling
15:14	RW	0x0	sw_src_hscl_mode SRC horizontal scaling mode 00：no scaling 01: down-scaling 10: up-scaling

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Bit	Attr	Reset Value	Description
13:12	RW	0x0	sw_src_mir_mode SRC mirror mode 00: no mirror 01: x mirror 10: y mirror 11: x mirror + y mirror
11:10	RW	0x0	sw_src_rot_mode SRC rotation mode 00: 0 degree 01: 90 degree 10: 180 degree 11: 270 degree
9:8	RW	0x0	sw_src_csc_mode Source bitmap YUV2RGB conversion mode 00: bypass 01: BT.601-range0 10: BT.601-range1 11: BT.709-range0
7	RW	0x0	sw_cp_endian Source Color palette endian swap 0: big endian 1: little endian
6	RW	0x0	sw_src_uvswap Source Cb-Cr swap 0: CrCb 1: CbCr
5	RW	0x0	sw_src_alpha_swap Source bitmap data alpha swap 0: ABGR 1: BGRA
4	RW	0x0	sw_src_rbswap Source bitmap data RB swap 0: BGR 1: RGB

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Bit	Attr	Reset Value	Description
3:0	RW	0x0	sw_src_fmt Source bitmap data format 0000: ABGR888 0001: XBGR888 0010: BGR packed 0100: RGB565 0101: ARGB1555 0110: ARGB4444 1000: YUV422SP 1001: YUV422P 1010: YUV420SP 1011: YUV420P 1100: 1BPP (color palette) 1101: 2BPP (color palette) 1110: 4BPP (color palette) 1111: 8BPP (color palette)

RGA2_RGA_SRC_BASE0
Address: Operational Base + offset (0x0108)
source image Y/RGB base address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_src_base0 source image Y/RGB base address

RGA2_RGA_SRC_BASE1
Address: Operational Base + offset (0x010c)
RGA source image Cb/Cbr base address register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_src_base1 source image Cb base address (YUV422/420-P) source image Cb/Cr base address (YU,V422/420-SP)

RGA2_RGA_SRC_BASE2
Address: Operational Base + offset (0x0110)
RGA source image Cr base address register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	sw_src_base2 source image Cr base address (YUV422/420-P)

RGA2_RGA_SRC_BASE3
Address: Operational Base + offset (0x0114)

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RGA source image 1 base address register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	sw_src_base3 source image 1 RGB base address (source bitblt mode1)

RGA2_RGA_SRC_VIR_INFO

Address: Operational Base + offset (0x0118)

RGA source image virtual stride / RGA source image tile number

Bit	Attr	Reset Value	Description
31:26	RW	0x00	Reserved
25:16	RW	0x000	sw_mask_vir_stride mask image virtual stride (words)
15	RW	0x0	Reserved
14:0	RW	0x0000	sw_src_act_width source image active width count from 1

RGA2_RGA_SRC_ACT_INFO

Address: Operational Base + offset (0x011c)

RGA source image active width/height register

Bit	Attr	Reset Value	Description
31:29	RW	0x0	Reserved2
28:16	RW	0x0000	sw_src_act_height source image active height
15:13	RW	0x0	Reserved1
12:0	RW	0x0000	sw_src_act_width source image active width

RGA2_RGA_SRC_X_FACTOR

Address: Operational Base + offset (0x0120)

RGA source image horizontal scaling factor

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_src_hsp_factor Source image horizontal up-scaling factor =(DST_ACT_WIDTH/SRC_ACT_WIDTH) * 65536
15:0	RW	0x0000	sw_src_hsd_factor Source image horizontal down-scaling factor =(SRC_ACT_WIDTH/DST_ACT_WIDTH) * 65536

RGA2_RGA_SRC_Y_FACTOR

Address: Operational Base + offset (0x0124)

RGA source image vertical scaling factor

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_src_vsp_factor Source image vertical up-scaling factor (DST_ACT_HEIGHT/SRC_ACT_HEIGHT) * 65536
15:0	RW	0x0000	sw_src_vsd_factor Source image vertical down-scaling factor (SRC_ACT_HEIGHT/DST_ACT_HEIGHT) * 65536

RGA2_RGA_SRC_BG_COLOR
Address: Operational Base + offset (0x0128)
RGA source image background color

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_src_bg_color Source image background color ("0" bit color for mono expansion.)

RGA2_RGA_SRC_FG_COLOR
Address: Operational Base + offset (0x012c)
RGA source image foreground color

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_src_fg_color Source image foreground color Source image foreground color ("1" bit color for mono expansion.) Color fill color, Pan color

RGA2_RGA_CP_GR_A
Address: Operational Base + offset (0x0130)
RGA source image transparency color min value

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_gradient_y_a Y gradient value of Alpha (signed 8.8)
15:0	RW	0x0000	sw_gradient_x_a X gradient value of Alpha (signed 8.8)

RGA2_RGA_SRC_TR_COLOR0
Address: Operational Base + offset (0x0130)
RGA source image transparency color min value

Bit	Attr	Reset Value	Description
31:24	RW	0x00	sw_src_trans_amin source image transparency color A min value
23:16	RW	0x00	sw_src_trans_bmin source image transparency color B min value

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	sw_src_trans_gmin source image transparency color G min value
7:0	RW	0x00	sw_src_trans_rmin source image transparency color R min value

RGA2_RGA_CP_GR_B
Address: Operational Base + offset (0x0134)
RGA source image transparency color max value

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_gradient_y_b Y gradient value of Blue (signed 8.8)
15:0	RW	0x0000	sw_gradient_x_b X gradient value of Blue (signed 8.8)

RGA2_RGA_SRC_TR_COLOR1
Address: Operational Base + offset (0x0134)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:24	RW	0x00	sw_src_trans_amax source image transparency color A max value
23:16	RW	0x00	sw_src_trans_bmax source image transparency color B max value
15:8	RW	0x00	sw_src_trans_gmax source image transparency color G max value
7:0	RW	0x00	sw_src_trans_rmax source image transparency color R max value

RGA2_RGA_DST_INFO
Address: Operational Base + offset (0x0138)
RGA destination format register

Bit	Attr	Reset Value	Description
31:19	RW	0x0000	Reserved
18	RW	0x0	sw_dst_csc_clip BGR2YUV Clip mode(from 0~255 clip to 36~235) 1: clip enable; 0: unclip
17:16	RW	0x0	sw_dst_csc_mode DST bitmap RGB2YUV conversion mode 00: Bypass 01: BT.601-range0 10: BT.601-range1 11: BT.709-range0

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Bit	Attr	Reset Value	Description
15:14	RW	0x0	sw_dither_mode DST dither down bit mode 00: 888 to 666 01: 888 to 565 10: 888 to 555 11: 888 to 444
13	RW	0x0	sw_dither_down DST dither down enable 0:disable 1:enable
12	RW	0x0	sw_src1_dither_up DST/SRC1 dither up enable 0:disable 1:enable
11	RW	0x0	sw_src1_alpha_swap Source 1 bitmap data alpha swap 0: ABGR 1: BGRA
10	RW	0x0	sw_src1_rbswap Source 1 bitmap data RB swap 0: BGR 1: RGB
9:7	RW	0x0	sw_src1_fmt Source 1 bitmap data format 000: ABGR888 001: XBGR888 010: BGR packed 100: RGB565 101: ARGB1555 110: ARGB4444
6	RW	0x0	sw_dst_uvswap Destination Cb-Cr swap 0: CrCb 1: CbCr
5	RW	0x0	sw_dst_alpha_swap Destination bitmap data alpha swap 0: ABGR 1: BGRA
4	RW	0x0	sw_dst_rbswap Destination bitmap data RB swap 0: BGR 1: RGB

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Bit	Attr	Reset Value	Description
3:0	RW	0x0	sw_dst_fmt Destination bitmap data format 0000: ABGR888 0001: XBGR888 0010: BGR packed 0100: RGB565 0101: ARGB1555 0110: ARGB4444 1000: YUV422SP 1001: YUV422P 1010: YUV420SP 1011: YUV420P

RGA2_RGA_DST_BASE0
Address: Operational Base + offset (0x013c)
RGA destination image base address 0 register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_dst_base0 destination image Y/RGB base address

RGA2_RGA_DST_BASE1
Address: Operational Base + offset (0x0140)
RGA destination image base address 1 register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_dst_base1 destination image Cb/CbCr base address

RGA2_RGA_DST_BASE2
Address: Operational Base + offset (0x0144)
RGA destination image base address 2 register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_dst_base2 destination image Cr base address

RGA2_RGA_DST_VIR_INFO
Address: Operational Base + offset (0x0148)
RGA destination image virtual width/height register

Bit	Attr	Reset Value	Description
31:28	RW	0x0	Reserved2
30:16	RW	0x000	sw_src1_vir_stride source image 1 virtual stride (words)
15:12	RW	0x0	Reserved1
14:0	RW	0x000	sw_dst_vir_stride destination image virtual stride(words)

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RGA2_RGA_DST_ACT_INFO
Address: Operational Base + offset (0x014c)
RGA destination image active width/height register

Bit	Attr	Reset Value	Description
31:28	RW	0x0	Reserved2
27:16	RW	0x000	sw_dst_act_height Destination image active height
15:12	RW	0x0	Reserved1
11:0	RW	0x000	sw_dst_act_width Destination image active width

RGA2_RGA_ALPHA_CTRL0
Address: Operational Base + offset (0x0150)
Alpha control register 0

Bit	Attr	Reset Value	Description
31:21	RW	0x000	Reserved
20	RW	0x0	sw_mask_endian ROP4 mask endian swap 0: big endian 1: little endian
19:12	RW	0x00	sw_dst_global_alpha global alpha value of DST(Agd)
11:4	RW	0x00	sw_src_global_alpha global alpha value of SRC(Ags) fading value in fading mod
3:2	RW	0x0	sw_rop_mode ROP mode select 00: ROP 2 01: ROP 3 10: ROP 4
1	RW	0x0	sw_alpha_rop_sel Alpha or ROP select 0: alpha 1: ROP
0	RW	0x0	sw_alpha_rop_e Alpha or ROP enable 0: disable 1: enable

RGA2_RGA_ALPHA_CTRL1
Address: Operational Base + offset (0x0154)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:30	RW	0x0	Reserved Reserved

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Bit	Attr	Reset Value	Description
29	RW	0x0	sw_src_alpha_m1 Src Transparent/opaque of alpha channel (As1') 0: As 1: 255-As
28	RW	0x0	sw_dst_alpha_m1 Dst Transparent/opaque of alpha channel (Ad1') 0: Ad 1: 255-Ad
27:26	RW	0x0	sw_src_blend_m1 Alpha src blend mode select of alpha channel (As1_'') 00: Ags 01: As1' 10: (As1'*Ags)>>8 11: reserved
25:24	RW	0x0	sw_dst_blend_m1 Alpha dst blend mode select of alpha channel(Ad1_'') 00: Agd 01: Ad1' 10: (Ad1'*Agd)>>8 11: reserved
23	RW	0x0	sw_src_alpha_cal_m1 Alpha src calculate mode of alpha channel(As1'') 0: As1''= As1_''+ (As1_''>>7) 1: As1''= As1 _''
22	RW	0x0	sw_dst_alpha_cal_m1 Alpha dst calculate mode of alpha channel(Ad1'') 0: Ad1''= Ad1_'' + (Ad1_''>>7) 1: Ad1''= Ad1_''
21:19	RW	0x0	w_src_factor_m1 Src factore mode of alpha channel(Fs1) 000: 0 001: 256 010: Ad1'' 011: 256-Ad1'' 100: As1''

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Bit	Attr	Reset Value	Description
18:16	RW	0x0	sw_dst_factor_m1 Dst factore mode of alpha channel(Fd1) 000: 0 001: 256 010: As1'' 011: 256-As1'' 100: Ad1''
15	RW	0x0	sw_src_alpha_m0 Src Transparent/opaque of color channel (As0') 0: As 1: 255-As
14	RW	0x0	sw_dst_alpha_m0 Dst Transparent/opaque of color channel (Ad0') 0: Ad 1: 255-Ad
13:12	RW	0x0	sw_src_blend_m0 Alpha src blend mode select of color channel (As0_'') 00: Ags 01: As0' 10: (As0'*Ags)>>8 11: reserved
11:10	RW	0x0	sw_dst_blend_m0 Alpha dst blend mode select of color channel(Ad0_'') 00: Agd 01: Ad0' 10: (Ad0'*Agd)>>8 11: reserved
9	RW	0x0	sw_src_alpha_cal_m0 Alpha src calculate mode of color channel(As0'') 0: As0''= As0_''+ (As0_''>>7) 1: As0''= As0 _''
8	RW	0x0	sw_dst_alpha_cal_m0 Alpha dst calculate mode of color channel(Ad0'') 0: Ad0''= Ad0_'' + (Ad0_''>>7) 1: Ad0''= Ad0_''

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Bit	Attr	Reset Value	Description
7:5	RW	0x0	sw_src_factor_m0 Src factore mode of color channel(Fs0) 000: 0 001: 256 010: Ad0'' 011: 256-Ad0'' 100: As0''
4:2	RW	0x0	sw_dst_factor_m0 Dst factore mode of color channel(Fd0) 000: 0 001: 256 010: As0'' 011: 256-As0'' 100: Ad0''
1	RW	0x0	sw_src_color_m0 SRC color select(Cs') 0: Cs 1: Cs * As0''
0	RW	0x0	sw_dst_color_m0 SRC color select(Cd') 0: Cd 1: Cd * Ad0''

RGA2_RGA_FADING_CTRL
Address: Operational Base + offset (0x0158)
Fading control register

Bit	Attr	Reset Value	Description
31:25	RW	0x00	Reserved
24	RW	0x0	sw_fading_en Fading enable
23:16	RW	0x00	sw_fading_offset_b Fading offset B value
15:8	RW	0x00	sw_fading_offset_g Fading offset G value (Pattern total number when pattern loading)
7:0	RW	0x00	sw_fading_offset_r Fading offset R value (Start point of pattern ram in pattern mode)

RGA2_RGA_PAT_CON
Address: Operational Base + offset (0x015c)
Pattern size/offset register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	sw_pat_offset_y Pattern y offset

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Bit	Attr	Reset Value	Description
23:16	RW	0x00	sw_pat_offset_x Pattern x offset
15:8	RW	0x00	sw_pat_height Pattern height
7:0	RW	0x00	sw_pat_width Pattern width

RGA2_RGA_CP_GR_G
Address: Operational Base + offset (0x0160)
RGA color gradient fill step register (color fill mode)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_gradient_y_g Y gradient value of Green (signed 8.8)
15:0	RW	0x0000	sw_gradient_x_g X gradient value of Green (signed 8.8)

RGA2_RGA_ROP_CON0
Address: Operational Base + offset (0x0160)
ROP code 0 control register

Bit	Attr	Reset Value	Description
31:25	RW	0x00	Reserved
24:0	RW	0x0000000	sw_rop3_code0 Rop3 code 0 control bits

RGA2_RGA_CP_GR_R
Address: Operational Base + offset (0x0164)
RGA color gradient fill step register (color fill mode)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_gradient_y_r Y gradient value of Red(signed 8.8)
15:0	RW	0x0000	sw_gradient_x_r X gradient value of Red(signed 8.8)

RGA2_RGA_ROP_CON1
Address: Operational Base + offset (0x0164)
ROP code 1 control register

Bit	Attr	Reset Value	Description
31:25	RW	0x00	Reserved
24:0	RW	0x0000000	sw_rop3_code1 Rop3 code 1 control bits

RGA2_RGA_MASK_BASE
Address: Operational Base + offset (0x0168)
RGA mask base address register

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_mask_base mask base address in ROP4 mode LUT/ pattern load base address

RGA2_RGA_MMU_CTRL1
Address: Operational Base + offset (0x016c)
RGA MMU control register 1

Bit	Attr	Reset Value	Description
31:14	RW	0x00000	Reserved
13	RW	0x0	sw_els_mmu_flush RGA ELSE channel MMU TLB flush: Set 1 to this bit to flush MMU TLB, auto clear
12	RW	0x0	sw_els_mmu_en RGA ELSE channel MMU enable 0: disable 1: enable
11	RW	0x0	sw_dst_mmu_prefetch_dir 0:forward 1:backward
10	RW	0x0	sw_dst_mmu_prefetch_en 0:disable 1:enable
9	RW	0x0	sw_dst_mmu_flush RGA DST channel MMU TLB flush: Set 1 to this bit to flush MMU TLB, auto clear
8	RW	0x0	sw_dst_mmu_en RGA DST channel MMU enable 0: disable 1: enable
7	RW	0x0	sw_src1_mmu_prefetch_dir 0:forward 1:backward
6	RW	0x0	sw_src1_mmu_prefetch_en 0:disable 1:enable
5	RW	0x0	sw_src1_mmu_flush RGA SRC1 channel MMU TLB flush: Set 1 to this bit to flush MMU TLB, auto clear
4	RW	0x0	sw_src1_mmu_en RGA SRC1 channel MMU enable 0: disable 1: enable
3	RW	0x0	sw_src_mmu_prefetch_dir 0:forward 1:backward

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Bit	Attr	Reset Value	Description
2	RW	0x0	sw_src_mmu_prefetch_en 0:disable 1:enable
1	RW	0x0	sw_src_mmu_flush RGA SRC channel MMU TLB flush: Set 1 to this bit to flush MMU TLB, auto clear
0	RW	0x0	sw_src_mmu_en RGA SRC channel MMU enable 0: disable 1: enable

RGA2_RGA_MMU_SRC_BASE
Address: Operational Base + offset (0x0170)
RGA source MMU TLB base address

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:0	RW	0x0000000	sw_mmu_src_base RGA source MMU TLB base address (128-bit)

RGA2_RGA_MMU_SRC1_BASE
Address: Operational Base + offset (0x0174)
RGA source1 MMU TLB base address

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:0	RW	0x0000000	sw_mmu_src1_base RGA source1 MMU TLB base address (128-bit)

RGA2_RGA_MMU_DST_BASE
Address: Operational Base + offset (0x0178)
RGA destination MMU TLB base address

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:0	RW	0x0000000	sw_mmu_dst_base RGA destination MMU TLB base address (128-bit)

RGA2_RGA_MMU_ELS_BASE
Address: Operational Base + offset (0x017c)
RGA ELSE MMU TLB base address

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:0	RW	0x0000000	sw_mmu_els_base RGA destination MMU TLB base address (128-bit)

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28.5 Programming Guide

28.5.1 Register Partition

There are two types of register in RGA. The first 8 registers (0x0 - 0x1C) are general registers for system configuration including command mode, command parameter, RGA status, general interrupts. The other registers (from 0x100) are command registers for command codes.

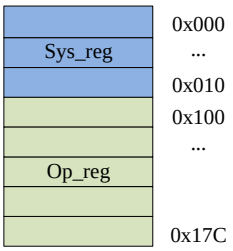

Fig. 28-6 RGA software main register-region

28.5.2 Command Modes

RGA has two command modes: slave mode and master mode. In salve mode (RGA_SYS_CTRL[1] = 1'b0), 2D graphic command only could be run one by one. CPU set all the command registers in RGA and then start RGA running by setting RGA_SYS_CTRL[1] to '1'. In master mode (RGA_SYS_CTRL[1] = 1'b1), 2D graphic commands could be run sequentially. After setting command's number to RGA_CMD_CTRL[12:3], writing '1' to RGA_CMD_CTRL[0] will start the command fetch, then Internal command DMA fetch commands from external command line.

Command line is a collection of several command codes with continuous address. At the first start, the command start address (RGA_CMD_ADDR) and command number (RGA_CMD_CTRL[12:3]) should be set, then write '1' to cmd_line_st (RGA_CMD_CTRL[0]) to start the command line fetch. Incremental command is supported by setting cmd_incr_num (RGA_CMD_CTRL[12:3]) and cmd_incr_valid (RGA_CMD_CTRL[1]=1'b1)

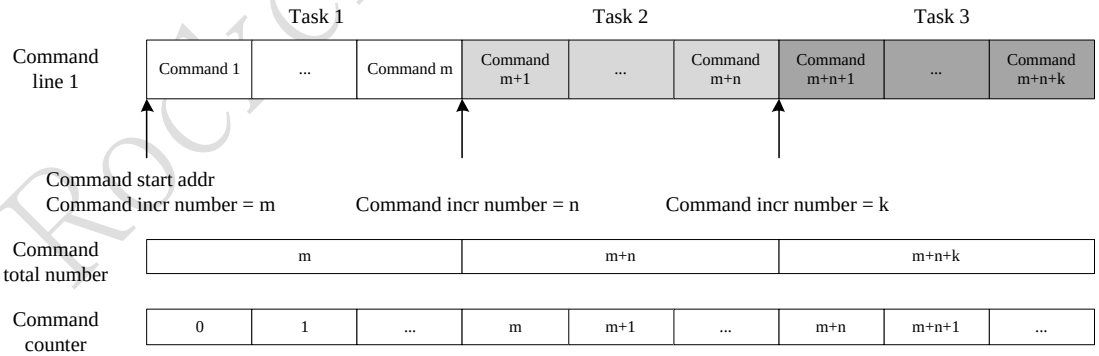

Fig. 28-7 RGA command line and command counter

28.5.3 Command Sync

In slave command mode, command sync is controlled by CPU.

In master command mode, user can enable the current_cmd_int (sw_intr_cf), command by command to generate a interrupt at the end point of target command operation.

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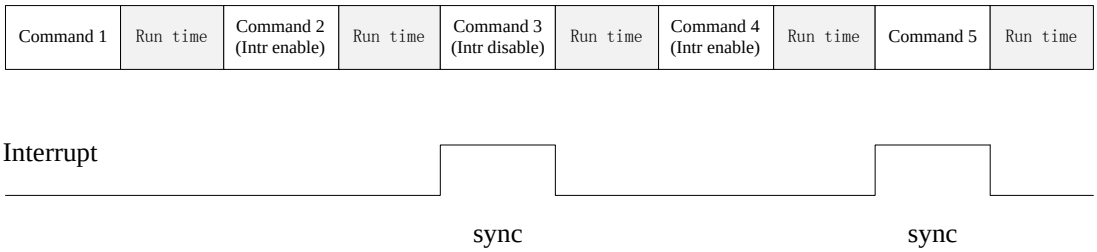

Fig. 28-8 RGA command sync generation

28.5.4 ColorPalette Application Notes

1. Palette/LUT Load into both SRC0 UV-INBUF ram and SRC0 SCL ram;

2. ColorPalette/Pattern interval operations need to initial LUT/pattern ram whether LUT/pattern content update or no;

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Chapter 29 Image Enhancement Processor (IEP)

29.1 Overview

The Image Enhancement Processor (IEP) receives data from and transmits data to system main memory by AXI bus, or output the data to Video Output Processor (VOP) directly.
The features of IEP are as follow:

- Image format
  - Input data: XRGB/RGB565/YUV420/YUV422
  - Output data: ARGB/RGB565/YUV420/YUV422
  - ARGB/XRGB/RGB565/YUV swap
  - YUV semi-planar/planar
  - BT601_l/BT601_f/BT709_l/BT709_f color space conversion
  - RGB dither up/down conversion
  - YUV up/down sampling conversion
  - Max resolution for static image up to 8192x8192
  - Max resolution for dynamic image
    - Deinterlace: 1920x1080
    - Sampling noise reduction: 1920x1080
    - Compression noise reduction: 4096x2304
    - Enhancement: 4096x2304
- Enhancement
  - Gamma adjustment with programmable mapping table
  - Hue/Saturation/Brightness/Contrast enhancement
  - Color enhancement with programmable coefficient
  - Detail enhancement with filter matrix up to 7x7
  - Edge enhancement with filter matrix up to 7x7
  - Programmable difference table for detail enhancement
  - Programmable distance table for detail and edge enhancement
- Noise reduction
  - Compression noise reduction with filter matrix up to 7x7
  - Programmable difference table for compression noise reduction
  - Programmable distance table for compression noise reduction
  - Spatial sampling noise reduction
  - Temporal sampling noise reduction
  - Optional coefficient for sampling noise reduction
- De-interlace
  - Input 4 fields, output 2 frames mode
  - Input 4 fields, output 1 frames mode
  - Input 2 fields, output 1 frames mode
  - Programmable motion detection coefficient
  - Programmable high frequency factor
  - Programmable edge interpolation parameter
- Interface
  - Programmable direct path to VOP
  - 32bit AHB bus slave
  - 64bit AXI bus master
  - Combined interrupt output

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29.2 Block Diagram

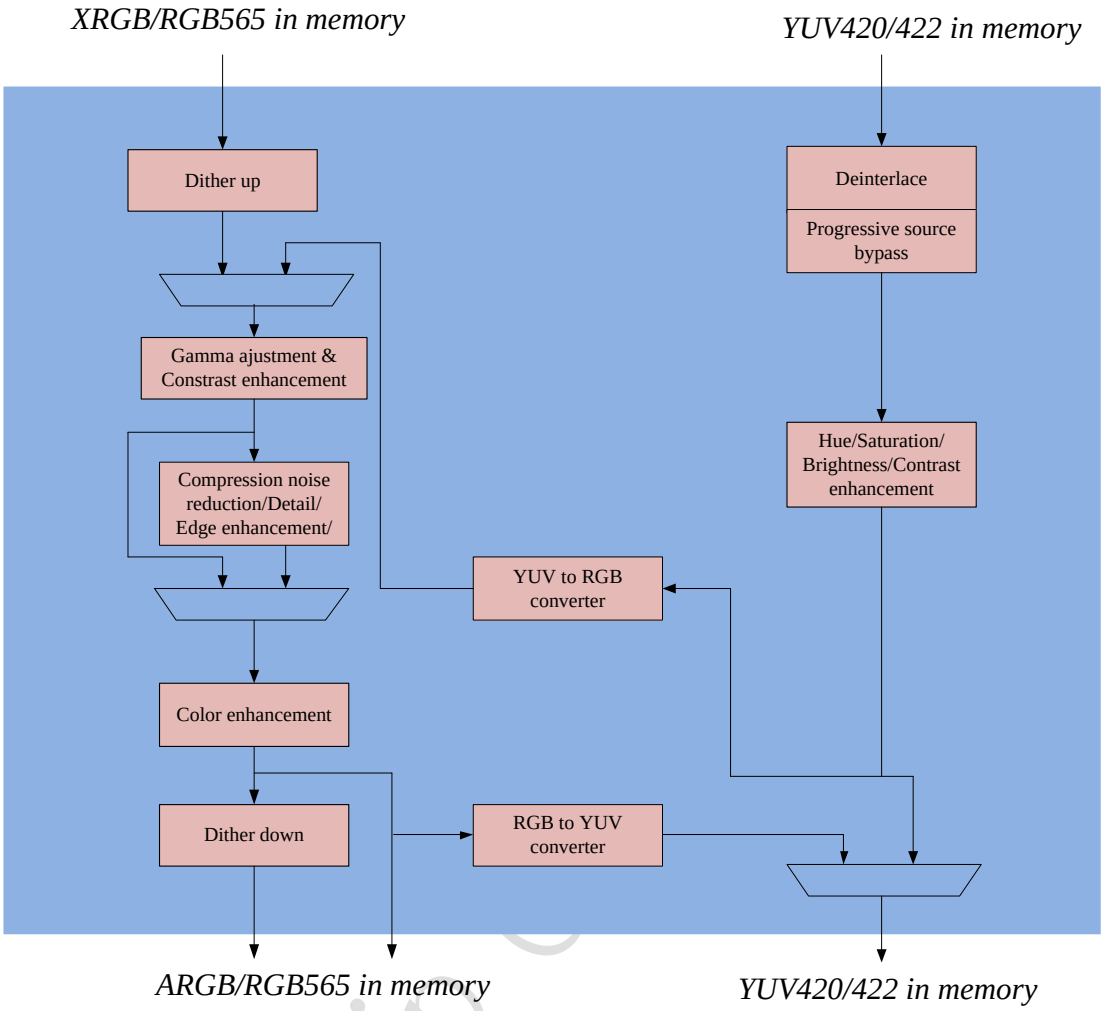

Fig. 29-1 IEP block diagram

The data path in IEP is in the previously diagram. The IEP comprises with:

- Deinterlace

There are five deinterlace mode including I4O2 (input 4 fields and output 2 frames once), I4O1B, I4O1T, I2O1B, I2O1T in the deinterlace block. YUV bypass is also supported.

- Enhancement

Not only hue, saturation, brightness, contrast enhancement, but also blue screen, black screen and color bar are supported in YUV domain enhancement block. Gamma adjustment, edge enhancement, detail enhancement and color enhancement are supported in RGB domain enhancement block.

- Noise Reduction

Spatial and temporal sampling noise can be reduced in YUV domain noise reduction block. Compression noise can be reduced in RGB domain noise reduction block.

29.3 Function description

29.3.1 Deinterlace

There are five deinterlace mode including I4O2, I4O1B, I4O1T, I2O1B and I2O1T in the deinterlace block. The I4O2 mode represents for 4 fields of input images and 2 frames of output images, so all of the two groups of source address registers and two groups of destination address registers need to be configured. For example, if source and destination format are both YUV420, the source address register IEP_SRC_ADDR_YRGB,

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IEP_SRC_ADDR_CBCR are used for source field0 and field 1, the source address register IEP_SRC_ADDR_Y1, IEP_SRC_ADDR_CBCR1 are used for source field2 and field3. The I4O1B and I4O1T mode have the same input images as the I4O2 mode, but only one frame output is generated once. The I2O1B and I2O1T mode have the same output as I4O1B and I4O1T mode, but only two fields input are needed. If bypass mode is selected, there are not any deinterlace operations. The parameter dil_ei_sel, dil_ei_radius, dil_ei_smooth, dil_ei_mode, dil_hf_en and dil_hf_fct in register IEP_CONFIG0 and registers IEP_DIL_MTN_TAB0~7 may have different influence in deinterlace effect depend on the type of the image source.

29.3.2 Noise reduction

Both of spatial and temporal sampling noise reduction are enabled when the 3D denoise bit is set. This function is used for reducing the noise generated at video or picture capturing in the camera sensor. There are four groups of optional noise reduction effect coefficients for luminance and chrominance in spatial and temporal segment.
Compression noise reduction is used for reducing the noise after the decompression of picture or video. Before the compression noise reduction is enabled, the IEP_ENH_DDE_COE0/1 from address 0x400 to 0x5FC for difference and distance coefficients must be written firstly. The filter matrix can be selected from 3x3/5x5/7x7/9x9 and the filter weight can be programmed by configuring IEP_ENH_RGB_CNFG.

29.3.3 Enhancement

Not only hue, saturation, brightness, contrast enhancement, but also blue screen, black screen and color bar are supported in this block. IEP_ENH_YUV_CNFG_0/1/2 registers can be configured to modify the YUV enhance parameters to satisfied with the requirement.
Before the gamma adjustment or contrast enhancement is enabled in RGB domain, the IEP_ENH_CG_TAB from address 0x100 to 0x3FC for B, G, R mapping must be written firstly.
If the color enhancement is enabled, the IEP_ENH_C_COE must be written the required value.
Before the edge or detail enhancement is enabled, the IEP_ENH_DDE_COE0/1 from address 0x400 to 0x5FC for difference and distance coefficients must be written firstly. The filter matrix can be selected from 3x3/5x5/7x7/9x9 and the filter weight can be programmed by configuring IEP_ENH_RGB_CNFG.

29.3.4 Format conversion

The color space conversion either from RGB to YUV or from YUV to RGB has the selections including BT601/709_L/F mode, and the input can be clipped or not.
If the source format is RGB565, dither up must be enabled. In contrary to the destination format is RGB565, dither down must be enabled.

29.3.5 Shadow registers

The configuration registers can be configured at any time, but they cannot have any effect immediately unless config_done is available and a new frame_start is enabled. The registers IEP_RAW_CONFIG0/1, IEP_RAW_VIR_IMG_WIDTH, IEP_RAW_IMG_SCL_FCT, IEP_RAW_SRC_IMG_SIZE, IEP_RAW_ENH_YUV_CNFG_0/1/2 corresponding to the registers have the similar names but without letters _RAW. They are used for raw register value reading before the configurations really have effect on the new frame.

29.3.6 VOP direct path

The IEP_DST_ADDR for DMA writing is useless if vop_path_en bit is set, because all RGB or YUV data is supplied for VOP directly from local bus via VOP and IEP.

29.4 Register description

29.4.1 Register Summary

Name	Offset	Size	Reset Value	Description
IEP_CONFIG0	0x00000	W	0x00000000	configuration register0
IEP_CONFIG1	0x00004	W	0x00000000	configuration register1
IEP_STATUS	0x00008	W	0x00000000	status register

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Name	Offset	Size	Reset Value	Description
IEP_INT	0x0000c	W	0x00000000	interrupt register
IEP_FRM_START	0x00010	W	0x00000000	frame start
IEP_CONFIG_DONE	0x00018	W	0x00000000	configuration done
IEP_FRM_CNT	0x0001c	W	0x00000000	frame counter
IEP_VIR_IMG_WIDTH	0x00020	W	0x01400140	Image virtual width
IEP_SRC_IMG_SIZE	0x00028	W	0x00ef013f	Source image width/height
IEP_DST_IMG_SIZE	0x0002c	W	0x00ef013f	Destination image width/height
IEP_DST_IMG_WIDTH_TILE0	0x00030	W	0x00000000	Destination image tile0 width
IEP_DST_IMG_WIDTH_TILE1	0x00034	W	0x00000000	Destination image tile1 width
IEP_DST_IMG_WIDTH_TILE2	0x00038	W	0x00000000	Destination image tile2 width
IEP_DST_IMG_WIDTH_TILE3	0x0003c	W	0x00000000	Destination image tile3 width
IEP_ENH_YUV_CNFG_0	0x00040	W	0x00000000	brightness,contrast,saturation adjustment
IEP_ENH_YUV_CNFG_1	0x00044	W	0x00000000	Hue configuration
IEP_ENH_YUV_CNFG_2	0x00048	W	0x00000000	color bar configuration
IEP_ENH_RGB_CNFG	0x0004c	W	0x00000000	enhancement RGB configuration
IEP_ENH_C_COE	0x00050	W	0x00000000	rgb color enhancement coefficient
IEP_RAW_CONFIG0	0x00058	W	0x00000000	configuration register0
IEP_RAW_CONFIG1	0x0005c	W	0x00000000	configuration register1
IEP_RAW_VIR_IMG_WIDTH	0x00060	W	0x01400140	Image virtual width
IEP_RAW_SRC_IMG_SIZE	0x00068	W	0x00ef013f	Source image width/height
IEP_RAW_DST_IMG_SIZE	0x0006c	W	0x00ef013f	Destination image width/height
IEP_RAW_ENH_YUV_CNFG_0	0x00070	W	0x00000000	brightness,contrast,saturation adjustment
IEP_RAW_ENH_YUV_CNFG_1	0x00074	W	0x00000000	Hue configuration
IEP_RAW_ENH_YUV_CNFG_2	0x00078	W	0x00000000	color bar configuration
IEP_RAW_ENH_RGB_CNFG	0x0007c	W	0x00000000	enhancement RGB configuration
IEP_SRC_ADDR_YRGB	0x00080	W	0x00000000	Start address of source image(Y/RGB)
IEP_SRC_ADDR_CBCR	0x00084	W	0x00000000	Start address of source image(Cb/Cr)
IEP_SRC_ADDR_CR	0x00088	W	0x00000000	Start address of source image(Cr)
IEP_SRC_ADDR_Y1	0x0008c	W	0x00000000	Start address of source image(Y)
IEP_SRC_ADDR_CBCR1	0x00090	W	0x00000000	Start address of source image(Cb/Cr)
IEP_SRC_ADDR_CR1	0x00094	W	0x00000000	Start address of source image(Cr)
IEP_SRC_ADDR_Y_ITEMP	0x00098	W	0x00000000	Start address of source image(Y integer part)
IEP_SRC_ADDR_CBCR_ITEMP	0x0009c	W	0x00000000	Start address of source image(CBCR integer part)
IEP_SRC_ADDR_CR_ITEMP	0x000a0	W	0x00000000	Start address of source image(CR integer part)
IEP_SRC_ADDR_Y_FTEMP	0x000a4	W	0x00000000	Start address of source image(Y fraction part)

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Name	Offset	Size	Reset Value	Description
IEP_SRC_ADDR_CBCR_FTEMP	0x000a8	W	0x00000000	Start address of source image(CBCR fraction part)
IEP_SRC_ADDR_CR_FTEMP	0x000ac	W	0x00000000	Start address of source image(CR fraction part)
IEP_DST_ADDR_YRGB	0x000b0	W	0x00000000	Start address of destination image(Y/RGB)
IEP_DST_ADDR_CBCR	0x000b4	W	0x00000000	Start address of destination image(Cb/Cr)
IEP_DST_ADDR_CR	0x000b8	W	0x00000000	Start address of destination image(Cr)
IEP_DST_ADDR_Y1	0x000bc	W	0x00000000	Start address of destination image(Y)
IEP_DST_ADDR_CBCR1	0x000c0	W	0x00000000	Start address of destination image(Cb/Cr)
IEP_DST_ADDR_CR1	0x000c4	W	0x00000000	Start address of destination image(Cr)
IEP_DST_ADDR_Y_ITEMP	0x000c8	W	0x00000000	Start address of destination image(Y integer part)
IEP_DST_ADDR_CBCR_ITEMP	0x000cc	W	0x00000000	Start address of destination image(CBCR integer part)
IEP_DST_ADDR_CR_ITEMP	0x000d0	W	0x00000000	Start address of destination image(CR integer part)
IEP_DST_ADDR_Y_FTEMP	0x000d4	W	0x00000000	Start address of destination image(Y fraction part)
IEP_DST_ADDR_CBCR_FTEMP	0x000d8	W	0x00000000	Start address of destination image(CBCR fraction part)
IEP_DST_ADDR_CR_FTEMP	0x000dc	W	0x00000000	Start address of destination image(CR fraction part)
IEP_DIL_MTN_TAB0	0x000e0	W	0x00000000	Deinterlace motion table0
IEP_DIL_MTN_TAB1	0x000e4	W	0x00000000	Deinterlace motion table1
IEP_DIL_MTN_TAB2	0x000e8	W	0x00000000	Deinterlace motion table2
IEP_DIL_MTN_TAB3	0x000ec	W	0x00000000	Deinterlace motion table3
IEP_DIL_MTN_TAB4	0x000f0	W	0x00000000	Deinterlace motion table4
IEP_DIL_MTN_TAB5	0x000f4	W	0x00000000	Deinterlace motion table5
IEP_DIL_MTN_TAB6	0x000f8	W	0x00000000	Deinterlace motion table6
IEP_DIL_MTN_TAB7	0x000fc	W	0x00000000	Deinterlace motion table7
IEP_ENH_CG_TAB	0x00100	W	0x00000000	contrast and gamma enhancement table
IEP_ENH_DDE_COE0	0x00400	W	0x00000000	denoise,detail and edge enhancement coefficient
IEP_ENH_DDE_COE1	0x00500	W	0x00000000	denoise,detail and edge enhancement coefficient
IEP_MMU_DTE_ADDR	0x00800	W	0x00000000	MMU current page table address
IEP_MMU_STATUS	0x00804	W	0x00000018	MMU status register

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Name	Offset	Size	Reset Value	Description
IEP_MMU_CMD	0x00808	W	0x00000000	MMU command register
IEP_MMU_PAGE_FAULT_ADDR	0x0080c	W	0x00000000	MMU logic address of last page fault
IEP_MMU_ZAP_ONE_LINE	0x00810	W	0x00000000	MMU zap cache line register
IEP_MMU_INT_RAWSTAT	0x00814	W	0x00000000	MMU raw interrupt status register
IEP_MMU_INT_CLEAR	0x00818	W	0x00000000	MMU interrupt clear register
IEP_MMU_INT_MASK	0x0081c	W	0x00000000	MMU interrupt mask register
IEP_MMU_INT_STATUS	0x00820	W	0x00000000	MMU interrupt status register
IEP_MMU_AUTO_GATING	0x00824	W	0x00000001	MMU clock auto gating register

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

29.4.2 Detail Register Description

IEP_CONFIG0

Address: Operational Base + offset (0x00000)

configuration register0

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23	RW	0x0	dil_ei_sel deinterlace edge interpolation select
22:21	RW	0x0	dil_ei_radius deinterlace edge interpolation radius
20	RW	0x0	rgb_con_gam_order RGB contrast enhancement and gamma adjustment operation order select. 0:CG prior to DDE 1:DDE prior to CG (CG represent for contrast & gamma operation, and DDE represent for denoise, detail or edge enhancement operation)
19:18	RW	0x0	rgb_enh_sel RGB enhancement select 00: no operation 01: denoise 10: detail enhancement 11: edge enhancement
17	RW	0x0	rgb_con_gam_en RGB contrast enhancement and gamma adjustment enable 0:disable 1:enable
16	RW	0x0	rgb_color_enh_en RGB color enhancement enable 0:disable 1:enable

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Bit	Attr	Reset Value	Description
15	RW	0x0	dil_ei_smooth deinterlace edge interpolation for smooth effect 0: disable 1: enable
14	RW	0x0	yuv_enh_en yuv enhancement enable 0:disable 1:enable
13	RW	0x0	yuv_dns_en YUV 3D denoise enable 0:disable 1:enable
12	RW	0x0	dil_ei_mode deinterlace edge interpolation 0: disable 1: enable
11	RW	0x0	dil_hf_en deinterlace high frequency calculation enable 0: disable 1: enable
10:8	RW	0x0	dil_mode Deinterlace mode select: 000: YUV deinterlace and bypass path disable; 001: I4O2 mode 010: I4O1B mode 011: I4O1T mode 100: I2O1B mode 101: I2O1T mode 110: bypass mode
7:1	RW	0x00	dil_hf_fct deinterlace high frequency factor
0	RW	0x0	vop_path_en VOP direct path enable 0:disable 1:enable

IEP_CONFIG1
Address: Operational Base + offset (0x00004)
configuration register1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	glb_alpha global alpha value only valid when destination format is ARGB

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Bit	Attr	Reset Value	Description
23	RW	0x0	rgb2yuv_input_clip RGB to YUV input range 0:R/G/B=[0,255] 1:R/G/B=[16,235]
22	RW	0x0	yuv2rgb_input_clip YUV to RGB input range 0:Y/U/V=[0,255] 1:Y=[16,235],U/V=[16,240]
21	RW	0x0	rgb_to_yuv_en RGB to YUV conversion enable 0: disable 1: enable
20	RW	0x0	yuv_to_rgb_en YUV to RGB conversion enable 0: disable 1: enable
19:18	RW	0x0	rgb2yuv_coe_sel rgb2yuv coefficient select 00:bt601_1 01:bt601_f 10:bt709_1 11:bt709_f
17:16	RW	0x0	yuv2rgb_coe_sel yuv2rgb coefficient select 00:bt601_1 01:bt601_f 10:bt709_1 11:bt709_f
15	RW	0x0	dthr_down_en dither down enable 0: disable 1: enable
14	RW	0x0	dthr_up_en dither up enable 0: disable 1: enable
13:12	RW	0x0	dst_yuv_swap destination YUV swap 00:SP UV 01:SP VU 10, 11:P

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Bit	Attr	Reset Value	Description
11:10	RW	0x0	dst_rgb_swap destination RGB swap ARGB destination 00:ARGB 01:ABGR 10:RGBA 11:BGRA RGB565 destination 00,10:RGB 01,11:BGR
9:8	RW	0x0	dst_fmt Ouput image Format 00 : ARGB 01 : RGB565 10 : YUV422 11 : YUV420
7:6	RO	0x0	reserved
5:4	RW	0x0	src_yuv_swap source YUV swap 00:SP UV 01:SP VU 10, 11:P
3:2	RW	0x0	src_rgb_swap source RGB swap XRGB source 00:XRGB 01:XBGR 10:RGBX 11:BGRX RGB565 source 00,10:RGB 01,11:BGR
1:0	RW	0x0	src_fmt Input image Format 00 : XRGB 01 : RGB565 10 : YUV422 11 : YUV420

IEP_STATUS
Address: Operational Base + offset (0x00008)
status register

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
19	RW	0x0	rrgb_idle_ack RGB read DMA idle acknowlege
18	RW	0x0	wrgb_idle_ack RGB write DMA idle acknowlege
17	RW	0x0	ryuv_idle_ack YUV read DMA idle acknowlege
16	RW	0x0	wyuv_idle_ack YUV write DMA idle acknowlege
15:9	RO	0x0	reserved
8	RO	0x0	voi_sts vop direct path status 00:idle 01:working
7	RO	0x0	rrgb_sts RGB DMA read status 00:idle 01:working
6	RO	0x0	wrgb_sts RGB DMA write status 00:idle 01:working
5	RO	0x0	ryuv_sts YUV DMA read status 00:idle 01:working
4	RO	0x0	wyuv_sts YUV DMA write status 00:idle 01:working
3	RO	0x0	dde_sts RGB denoise/enhancement status 00:idle 01:working
2	RO	0x0	dil_sts de-interlace or yuv bypass status 00:idle 01:working
1	RO	0x0	reserved
0	RO	0x0	dns_sts YUV 3D denoise status 00:idle 01:working

IEP_INT
Address: Operational Base + offset (0x0000c)

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interrupt register

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	W1C	0x0	frm_done_int_clr Frame process done interrupt clear After be set to 1, this bit will be clear automatically.
15:9	RO	0x0	reserved
8	RW	0x0	frm_done_int_en Frame process done interrupt enable: 0: disable; 1: enable;
7:1	RO	0x0	reserved
0	RO	0x0	frm_done_int Frame process done interrupt 0: inactive; 1: active;

IEP_FRM_START

Address: Operational Base + offset (0x00010)

frame start

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	W1C	0x0	frm_start frame start Write 1, self clear.

IEP_CONFIG_DONE

Address: Operational Base + offset (0x00018)

configuration done

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	config_done configuration done Wait for frame start to update raw register configuration to really used registers.

IEP_FRM_CNT

Address: Operational Base + offset (0x0001c)

frame counter

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	frm_cnt frame counter Self increase one after a frame operation is finished. Write arbitrary value to clear to zero.

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IEP_VIR_IMG_WIDTH
Address: Operational Base + offset (0x00020)
Image virtual width

Bit	Attr	Reset Value	Description
31:16	RW	0x0140	dst_vir_image_width Destination virtual image width
15:0	RW	0x0140	src_vir_image_width Source virtual image width

IEP_SRC_IMG_SIZE
Address: Operational Base + offset (0x00028)
Source image width/height

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x00ef	src_image_height source image height
15:13	RO	0x0	reserved
12:0	RW	0x013f	src_image_width source image width

IEP_DST_IMG_SIZE
Address: Operational Base + offset (0x0002c)
Destination image width/height

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x00ef	dst_image_height Destination image height
15:13	RO	0x0	reserved
12:0	RW	0x013f	dst_image_width Destination image width

IEP_DST_IMG_WIDTH_TILE0
Address: Operational Base + offset (0x00030)
Destination image tile0 width

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	dst_image_width_tile0 Destination image tile0 width

IEP_DST_IMG_WIDTH_TILE1
Address: Operational Base + offset (0x00034)
Destination image tile1 width

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
10:0	RW	0x000	dst_image_width_tile1 Destination image tile1 width

IEP_DST_IMG_WIDTH_TILE2
Address: Operational Base + offset (0x00038)
Destination image tile2 width

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:0	RW	0x000	dst_image_width_tile2 Destination image tile2 width

IEP_DST_IMG_WIDTH_TILE3
Address: Operational Base + offset (0x0003c)
Destination image tile3 width

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:0	RW	0x000	dst_image_width_tile3 Destination image tile3 width

IEP_ENH_YUV_CNFG_0
Address: Operational Base + offset (0x00040)
brightness,contrast,saturation adjustment

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24:16	RW	0x000	sat_con YUV saturation and contrast adjustment saturation * contrast range from 0 to 1.992*1.992, and this value is saturation* contrast * 128
15:8	RW	0x00	contrast YUV contrast adjustment contrast value range from 0 to 1.992, and this value is contrast*128.
7:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5:0	RW	0x00	brightness YUV brightness adjustment range from -32 to 31 000000:0; 000001:1; …… 011111:31; 100000:-32; 100001:-31; …… 111110:-2; 111111:-1;

IEP_ENH_YUV_CNFG_1
Address: Operational Base + offset (0x00044)
Hue configuration

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:8	RW	0x00	cos_hue the cos function value for hue adjustment sin function value range from 0.866 to 1 ,and this value is cos * 128 ,no sign bit
7:0	RW	0x00	sin_hue the sin function value for hue adjustment sin function value range from -0.5 to 0.5 ,and this value is sin * 128 ,and the high bit is sign bit

IEP_ENH_YUV_CNFG_2
Address: Operational Base + offset (0x00048)
color bar configuration

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25:24	RW	0x0	video_mode video mode 00:black screen 01:blue screen 10:color bars 11:normal video
23:16	RW	0x00	color_bar_v color bar v value
15:8	RW	0x00	color_bar_u color bar u value
7:0	RW	0x00	color_bar_y color bar y value

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IEP_ENH_RGB_CNFG
Address: Operational Base + offset (0x0004c)
enhancement RGB configuration

Bit	Attr	Reset Value	Description
31:30	RW	0x0	luma_spat_sel 3D denoise luma spatial coefficient select
29:28	RW	0x0	luma_temp_sel 3D denoise luma temporal coefficient select
27:26	RW	0x0	chroma_spat_sel 3D denoise chroma spatial coefficient select
25:24	RW	0x0	chroma_temp_sel 3D denoise chroma temporal coefficient select
23:16	RW	0x00	enh_threshold enhancement threshold In denoise and detail enhancement operation, more than the threshold, considering as detail; but if less than the threshold, considering as noise, need to to be filtered.
15	RO	0x0	reserved
14:8	RW	0x00	enh_alpha enhancement alpha value 0000000:0 0000001:1/16 0000010:2/16 …… 0001111:15/16 0010000:1 0010001:1+1/16; 0010010:1+2/16; 0010011:1+3/16; …… 0100000:2; …… 0110000:3; …… 1000000:4; …… 1010000:5; …… 1100000:6; other : reserved
7:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	enh_radius enhancement radius 00:R=1 01:R=2 10:R=3 11:R=4

IEP_ENH_C_COE
Address: Operational Base + offset (0x00050)
rgb color enhancement coefficient

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:5	RW	0x0	c_int_coe color enhancement integer coefficient
4:0	RW	0x00	c_frac_coe color enhancement fraction coefficient

IEP_RAW_CONFIG0
Address: Operational Base + offset (0x00058)
configuration register0

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23	RW	0x0	dil_ei_sel deinterlace edge interpolation select
22:21	RW	0x0	dil_ei_radius deinterlace edge interpolation radius
20	RW	0x0	rgb_con_gam_order RGB contrast enhancement and gamma adjustment operation order select. 0:CG prior to DDE 1:DDE prior to CG (CG represent for contrast & gamma operation, and DDE represent for de-noise, detail or edge enhancement operation)
19:18	RW	0x0	rgb_enh_sel RGB enhancement select 00: no operation 01: denoise 10: detail enhancement 11: edge enhancement
17	RW	0x0	rgb_con_gam_en RGB contrast enhancement and gamma adjustment enable 0:disable 1:enable

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Bit	Attr	Reset Value	Description
16	RW	0x0	rgb_color_enh_en RGB color enhancement enable 0:disable 1:enable
15	RW	0x0	dil_ei_smooth deinterlace edge interpolation for smooth effect 0: disable 1: enable
14	RW	0x0	yuv_enh_en yuv enhancement enable 0:disable 1:enable
13	RW	0x0	yuv_dns_en YUV 3D denoise enable 0:disable 1:enable
12	RW	0x0	dil_ei_mode deinterlace edge interpolation 0: disable 1: enable
11	RW	0x0	dil_hf_en deinterlace high frequency calculation enable 0: disable 1: enable
10:8	RW	0x0	dil_mode Deinterlace mode select: 000: YUV deinterlace and bypass path disable; 001: I4O2 mode 010: I4O1B mode 011: I4O1T mode 100: I2O1B mode 101: I2O1T mode 110: bypass mode
7:1	RW	0x00	dil_hf_fct deinterlace high frequency factor
0	RW	0x0	vop_path_en VOP direct path enable 0:disable 1:enable

IEP_RAW_CONFIG1
Address: Operational Base + offset (0x0005c)
configuration register1

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Bit	Attr	Reset Value	Description
31:24	RO	0x00	glb_alpha global alpha value only valid when destination format is ARGB
23	RO	0x0	rgb2yuv_input_clip RGB to YUV input range 0:R/G/B=[0,255] 1:R/G/B=[16,235]
22	RO	0x0	yuv2rgb_input_clip YUV to RGB input range 0:Y/U/V=[0,255] 1:Y=[16,235],U/V=[16,240]
21	RO	0x0	rgb_to_yuv_en RGB to YUV conversion enable 0: disable 1: enable
20	RO	0x0	yuv_to_rgb_en YUV to RGB conversion enable 0: disable 1: enable
19:18	RO	0x0	rgb2yuv_coe_sel rgb2yuv coefficient select 00:bt601_1 01:bt601_f 10:bt709_1 11:bt709_f
17:16	RO	0x0	yuv2rgb_coe_sel yuv2rgb coefficient select 00:bt601_1 01:bt601_f 10:bt709_1 11:bt709_f
15	RO	0x0	dthr_down_en dither down enable 0: disable 1: enable
14	RO	0x0	dthr_up_en dither up enable 0: disable 1: enable
13:12	RO	0x0	dst_yuv_swap destination YUV swap 00:SP UV 01:SP VU 10, 11:P

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Bit	Attr	Reset Value	Description
11:10	RO	0x0	dst_rgb_swap destination RGB swap ARGB destination 00:ARGB 01:ABGR 10:RGBA 11:BGRA RGB565 destination 00,10:RGB 01,11:BGR
9:8	RO	0x0	dst_fmt Ouput image Format 00 : ARGB 01 : RGB565 10 : YUV422 11 : YUV420
7:6	RO	0x0	reserved
5:4	RO	0x0	src_yuv_swap source YUV swap 00:SP UV 01:SP VU 10, 11:P
3:2	RO	0x0	src_rgb_swap source RGB swap XRGB source 00:XRGB 01:XBGR 10:RGBX 11:BGRX RGB565 source 00,10:RGB 01,11:BGR
1:0	RO	0x0	src_fmt Input image Format 00 : XRGB 01 : RGB565 10 : YUV422 11 : YUV420

IEP_RAW_VIR_IMG_WIDTH
Address: Operational Base + offset (0x00060)
Image virtual width

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RO	0x0140	dst_vir_image_width Destination virtual image width
15:0	RO	0x0140	src_vir_image_width Source virtual image width

IEP_RAW_SRC_IMG_SIZE
Address: Operational Base + offset (0x00068)
Source image width/height

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RO	0x00f0	src_image_height source image height
15:13	RO	0x0	reserved
12:0	RO	0x0140	src_image_width source image width

IEP_RAW_DST_IMG_SIZE
Address: Operational Base + offset (0x0006c)
Destination image width/height

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RO	0x00f0	dst_image_height Destination image height
15:13	RO	0x0	reserved
12:0	RO	0x0140	dst_image_width Destination image width

IEP_RAW_ENH_YUV_CNFG_0
Address: Operational Base + offset (0x00070)
brightness,contrast,saturation adjustment

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24:16	RO	0x000	sat_con YUV saturation and contrast adjustment saturation * contrast range from 0 to 1.992*1.992, and this value is saturation* contrast * 128
15:8	RO	0x00	contrast YUV contrast adjustment contrast value range from 0 to 1.992, and this value is contrast*128.
7:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5:0	RO	0x00	brightness YUV brightness adjustment range from -32 to 31 000000:0; 000001:1; …… 011111:31; 100000:-32; 100001:-31; …… 111110:-2; 111111:-1;

IEP_RAW_ENH_YUV_CNFG_1
Address: Operational Base + offset (0x00074)
Hue configuration

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:8	RO	0x00	cos_hue the cos function value for hue adjustment sin function value range from 0.866 to 1 ,and this value is cos * 128 ,no sign bit
7:0	RO	0x00	sin_hue the sin function value for hue adjustment sin function value range from -0.5 to 0.5 ,and this value is sin * 128 ,and the high bit is sign bit

IEP_RAW_ENH_YUV_CNFG_2
Address: Operational Base + offset (0x00078)
color bar configuration

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25:24	RO	0x0	video_mode video mode 00:black screen 01:blue screen 10:color bars 11:normal video
23:16	RO	0x00	color_bar_v color bar v value
15:8	RO	0x00	color_bar_u color bar u value
7:0	RO	0x00	color_bar_y color bar y value

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IEP_RAW_ENH_RGB_CNFG
Address: Operational Base + offset (0x0007c)
enhancement RGB configuration

Bit	Attr	Reset Value	Description
31:30	RW	0x0	luma_spat_sel 3D denoise luma spatial coefficient select
29:28	RW	0x0	luma_temp_sel 3D denoise luma temporal coefficient select
27:26	RW	0x0	chroma_spat_sel 3D denoise chroma spatial coefficient select
25:24	RW	0x0	chroma_temp_sel 3D denoise chroma temporal coefficient select
23:16	RW	0x00	enh_threshold enhancement threshold In denoise and detail enhancement operation, more than the threshold, considering as detail; but if less than the threshold, considering as noise, need to to be filtered.
15	RO	0x0	reserved
14:8	RW	0x00	enh_alpha enhancement alpha value 0000000:0 0000001:1/16 0000010:2/16 …… 0001111:15/16 0010000:1 0010001:1+1/16; 0010010:1+2/16; 0010011:1+3/16; …… 0100000:2; …… 0110000:3; …… 1000000:4; …… 1010000:5; …… 1100000:6; other : reserved
7:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	enh_radius enhancement radius 00:R=1 01:R=2 10:R=3 11:R=4

IEP_SRC_ADDR_YRGB
Address: Operational Base + offset (0x00080)
Start address of source image(Y/RGB)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_yrgb_mst Source image data YRGB start address in Memory

IEP_SRC_ADDR_CBCR
Address: Operational Base + offset (0x00084)
Start address of source image(Cb/Cr)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_cbcr_mst Source image data CbCr start address in Memory

IEP_SRC_ADDR_CR
Address: Operational Base + offset (0x00088)
Start address of source image(Cr)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_cr_mst Source image data Cr start address in Memory

IEP_SRC_ADDR_Y1
Address: Operational Base + offset (0x0008c)
Start address of source image(Y)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_y_mst Source image data Y start address in Memory

IEP_SRC_ADDR_CBCR1
Address: Operational Base + offset (0x00090)
Start address of source image(Cb/Cr)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_cbcr_mst Source image data CbCr start address in Memory

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IEP_SRC_ADDR_CR1
Address: Operational Base + offset (0x00094)
Start address of source image(Cr)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_cr_mst Source image data Cr start address in Memory

IEP_SRC_ADDR_Y_ITEMP
Address: Operational Base + offset (0x00098)
Start address of source image(Y integer part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_y_mst_itemp Interger part source image data Y start address in Memory

IEP_SRC_ADDR_CBCR_ITEMP
Address: Operational Base + offset (0x0009c)
Start address of source image(CBCR integer part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_cbcr_mst_cbcr_itemp Interger part source image data CBCR start address in Memory

IEP_SRC_ADDR_CR_ITEMP
Address: Operational Base + offset (0x000a0)
Start address of source image(CR integer part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_cr_mst_cr_itemp Interger part source image data CR start address in Memory

IEP_SRC_ADDR_Y_FTEMP
Address: Operational Base + offset (0x000a4)
Start address of source image(Y fraction part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_y_mst_ftemp Fraction part source image data Y start address in Memory

IEP_SRC_ADDR_CBCR_FTEMP
Address: Operational Base + offset (0x000a8)
Start address of source image(CBCR fraction part)

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_cbcr_mst_ftemp Fraction part source image data CBCR start address in Memory

IEP_SRC_ADDR_CR_FTEMP
Address: Operational Base + offset (0x000ac)
Start address of source image(CR fraction part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_cr_mst_ftemp Fraction part source image data CR start address in Memory

IEP_DST_ADDR_YRGB
Address: Operational Base + offset (0x000b0)
Start address of destination image(Y/RGB)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_yrgb_mst Destination image data YRGB start address in Memory

IEP_DST_ADDR_CBCR
Address: Operational Base + offset (0x000b4)
Start address of destination image(Cb/Cr)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_cbcr_mst Destination image data CBCR start address in Memory

IEP_DST_ADDR_CR
Address: Operational Base + offset (0x000b8)
Start address of destination image(Cr)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_cr_mst Destination image data CR start address in Memory

IEP_DST_ADDR_Y1
Address: Operational Base + offset (0x000bc)
Start address of destination image(Y)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_y_mst Destination image data Y start address in Memory

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IEP_DST_ADDR_CBCR1
Address: Operational Base + offset (0x000c0)
Start address of destination image(Cb/Cr)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_cbcr_mst Destination image data CbCr start address in Memory

IEP_DST_ADDR_CR1
Address: Operational Base + offset (0x000c4)
Start address of destination image(Cr)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_cr_mst Destination image data Cr start address in Memory

IEP_DST_ADDR_Y_ITEMP
Address: Operational Base + offset (0x000c8)
Start address of destination image(Y integer part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_y_mst_itemp Interger part destination image data Y start address in Memory

IEP_DST_ADDR_CBCR_ITEMP
Address: Operational Base + offset (0x000cc)
Start address of destination image(CBCR integer part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_cbcr_mst_itemp Int part destination image data CBCR start address in Memory

IEP_DST_ADDR_CR_ITEMP
Address: Operational Base + offset (0x000d0)
Start address of destination image(CR integer part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_cr_mst_itemp Interger part destination image data CR start address in Memory

IEP_DST_ADDR_Y_FTEMP
Address: Operational Base + offset (0x000d4)
Start address of destination image(Y fraction part)

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_y_mst_ftemp Fraction part destination image data Y start address in Memory

IEP_DST_ADDR_CBCR_FTEMP
Address: Operational Base + offset (0x000d8)
Start address of destination image(CBCR fraction part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_cbcr_mst_ftemp Fraction part destination image data CBCR start address in Mem

IEP_DST_ADDR_CR_FTEMP
Address: Operational Base + offset (0x000dc)
Start address of destination image(CR fraction part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_cr_mst_ftemp Fraction part destination image data CR start address

IEP_DIL_MTN_TAB0
Address: Operational Base + offset (0x000e0)
Deinterlace motion table0

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x00	mtn_sub_tab3 motion sub table3
23	RO	0x0	reserved
22:16	RW	0x00	mtn_sub_tab2 motion sub table2
15	RO	0x0	reserved
14:8	RW	0x00	mtn_sub_tab1 motion sub table1
7	RO	0x0	reserved
6:0	RW	0x00	mtn_sub_tab0 motion sub table0

IEP_DIL_MTN_TAB1
Address: Operational Base + offset (0x000e4)
Deinterlace motion table1

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x00	mtn_sub_tab3 motion sub table3

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Bit	Attr	Reset Value	Description
23	RO	0x0	reserved
22:16	RW	0x00	mtn_sub_tab2 motion sub table2
15	RO	0x0	reserved
14:8	RW	0x00	mtn_sub_tab1 motion sub table1
7	RO	0x0	reserved
6:0	RW	0x00	mtn_sub_tab0 motion sub table0

IEP_DIL_MTN_TAB2
Address: Operational Base + offset (0x000e8)
Deinterlace motion table2

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x00	mtn_sub_tab3 motion sub table3
23	RO	0x0	reserved
22:16	RW	0x00	mtn_sub_tab2 motion sub table2
15	RO	0x0	reserved
14:8	RW	0x00	mtn_sub_tab1 motion sub table1
7	RO	0x0	reserved
6:0	RW	0x00	mtn_sub_tab0 motion sub table0

IEP_DIL_MTN_TAB3
Address: Operational Base + offset (0x000ec)
Deinterlace motion table3

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x00	mtn_sub_tab3 motion sub table3
23	RO	0x0	reserved
22:16	RW	0x00	mtn_sub_tab2 motion sub table2
15	RO	0x0	reserved
14:8	RW	0x00	mtn_sub_tab1 motion sub table1
7	RO	0x0	reserved
6:0	RW	0x00	mtn_sub_tab0 motion sub table0

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IEP_DIL_MTN_TAB4
Address: Operational Base + offset (0x000f0)
Deinterlace motion table4

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x00	mtn_sub_tab3 motion sub table3
23	RO	0x0	reserved
22:16	RW	0x00	mtn_sub_tab2 motion sub table2
15	RO	0x0	reserved
14:8	RW	0x00	mtn_sub_tab1 motion sub table1
7	RO	0x0	reserved
6:0	RW	0x00	mtn_sub_tab0 motion sub table0

IEP_DIL_MTN_TAB5
Address: Operational Base + offset (0x000f4)
Deinterlace motion table5

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x00	mtn_sub_tab3 motion sub table3
23	RO	0x0	reserved
22:16	RW	0x00	mtn_sub_tab2 motion sub table2
15	RO	0x0	reserved
14:8	RW	0x00	mtn_sub_tab1 motion sub table1
7	RO	0x0	reserved
6:0	RW	0x00	mtn_sub_tab0 motion sub table0

IEP_DIL_MTN_TAB6
Address: Operational Base + offset (0x000f8)
Deinterlace motion table6

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x00	mtn_sub_tab3 motion sub table3
23	RO	0x0	reserved
22:16	RW	0x00	mtn_sub_tab2 motion sub table2
15	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
14:8	RW	0x00	mtn_sub_tab1 motion sub table1
7	RO	0x0	reserved
6:0	RW	0x00	mtn_sub_tab0 motion sub table0

IEP_DIL_MTN_TAB7
Address: Operational Base + offset (0x000fc)
Deinterlace motion table7

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x00	mtn_sub_tab3 motion sub table3
23	RO	0x0	reserved
22:16	RW	0x00	mtn_sub_tab2 motion sub table2
15	RO	0x0	reserved
14:8	RW	0x00	mtn_sub_tab1 motion sub table1
7	RO	0x0	reserved
6:0	RW	0x00	mtn_sub_tab0 motion sub table0

IEP_ENH_CG_TAB
Address: Operational Base + offset (0x00100)
contrast and gamma enhancement table

Bit	Attr	Reset Value	Description
31:24	RW	0x00	cg_tab_3 cg table 3 pixel value 3,7,11,15,……mapping
23:16	RW	0x00	cg_tab_2 cg table 2 pixel value 2,6,10,14,……mapping
15:8	RW	0x00	cg_tab_1 cg table 1 pixel value 1,5,9,13,……mapping
7:0	RW	0x00	cg_tab_0 cg table 0 256x8bit contrast & gamma mapping table pixel value 0,4,8,12,……mapping

IEP_ENH_DDE_COE0
Address: Operational Base + offset (0x00400)
denoise,detail and edge enhancement coefficient

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:24	RW	0x00	dde_coe_3 dde coefficient 3 coefficient number 3,7,11,15,……
23:22	RO	0x0	reserved
21:16	RW	0x00	dde_coe_2 dde coefficient 2 coefficient number 2,6,10,14,……
15:14	RO	0x0	reserved
13:8	RW	0x00	dde_coe_1 dde coefficient 1 coefficient number 1,5,9,13,……
7:6	RO	0x0	reserved
5:0	RW	0x00	dde_coe_0 dde coefficient 0 256x6bit cofficient for denoise and detail enhancement coefficient number 0,4,8,12,……

IEP_ENH_DDE_COE1
Address: Operational Base + offset (0x00500)
denoise,detail and edge enhancement coefficient

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:24	RW	0x00	dde_coe_3 dde coefficient 3 coefficient number 3,7,11,15,……
23:22	RO	0x0	reserved
21:16	RW	0x00	dde_coe_2 dde coefficient 3 coefficient number 2,6,10,14,……
15:14	RO	0x0	reserved
13:8	RW	0x00	dde_coe_1 dde coefficient 1 coefficient number 1,5,9,13,……
7:6	RO	0x0	reserved
5:0	RW	0x00	dde_coe_0 dde coefficient 1 81x6bit cofficient for denoise and detail enhancement coefficient number 0,4,8,12,……

IEP_MMU_DTE_ADDR
Address: Operational Base + offset (0x00800)
MMU current page table address

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mmu_dte_addr page table address

IEP_MMU_STATUS
Address: Operational Base + offset (0x00804)
MMU status register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:6	RW	0x00	mmu_page_fault_bus_id Index of master responsible for the last page fault
5	RW	0x0	mmu_page_fault_is_write The direction of access for last page fault: 0: read 1:write
4	RW	0x1	mmu_replay_buffer_empty The MMU replay buffer is empty.
3	RW	0x1	mmu_idle the MMU is idle when accesses are being translated and there are no unfinished translated access. The MMU_IDLE signal only reports idle when the MMU processor is idle and accesses are active on the external bus. Note: the MMU can be idle in page fault mode.
2	RW	0x0	mmu_stall_active MMU stall mode currently enabled. The mode is enabled by command.
1	RW	0x0	mmu_page_fault_active MMU page fault mode currently enabled.The mode is enabled by command
0	RW	0x0	mmu_paging_enabled mmu paging is enabled

IEP_MMU_CMD
Address: Operational Base + offset (0x00808)
MMU command register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
2:0	RW	0x0	mmu_cmd 0: MMU_ENABLE_PAGING. enable paging. 1: MMU_DISABLE_PAGING. disable paging. 2: MMU_ENABLE_STALL. turn on stall mode. 3: MMU_DISABLE_STALL. turn off stall mode. 4: MMU_ZAP_CACHE. zap the entire page table cache. 5: MMU_PAGE_FAULT_DONE. leave page fault mode. 6: MMU_FORCE_RESET. reset the mmu. The MMU_ENABLE_STALL command can always be issued. Other commands are ignored unless the MMU is idle or stalled.

IEP_MMU_PAGE_FAULT_ADDR
Address: Operational Base + offset (0x0080c)
MMU logic address of last page fault

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mmu_page_fault_addr address of last page fault

IEP_MMU_ZAP_ONE_LINE
Address: Operational Base + offset (0x00810)
MMU zap cache line register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	mmu_zap_one_line address to be invalidated from the page table cache.

IEP_MMU_INT_RAWSTAT
Address: Operational Base + offset (0x00814)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	read_bus_error read bus error
0	RW	0x0	page_fault page fault

IEP_MMU_INT_CLEAR
Address: Operational Base + offset (0x00818)
MMU interrupt clear register

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Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	read_bus_error_clear read bus error interrupt clear. write 1 to this register can clear read bus error interrupt.
0	RW	0x0	page_fault_clear page fault interrupt clear, write 1 to this register can clear page fault interrupt.

IEP_MMU_INT_MASK
Address: Operational Base + offset (0x0081c)
MMU interrupt mask register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	read_bus_error_int_en read bus error interrupt enable
0	RW	0x0	page_fault_int_en page fault interrupt enable

IEP_MMU_INT_STATUS
Address: Operational Base + offset (0x00820)
MMU interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	read_bus_error read bus error interrupt
0	RW	0x0	page_fault page fault interrupt

IEP_MMU_AUTO_GATING
Address: Operational Base + offset (0x00824)
MMU clock auto gating register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x1	mmu_auto_gating mmu clock auto gating when it is 1, the mmu will auto gating itself

29.5 Application Notes

29.5.1 VOP path disabled configure flow

1. Keep IEP direct path disabled.
2. Configure all registers which are needed at any time.
3. Configure IEP_CONFIG_DONE.

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4. Configure IEP_FRM_START.

29.5.2 VOP path enabled configure flow

1. Keep IEP direct path enabled.
2. Configure all IEP registers which are needed.
3. Configure VOP related registers which are needed.
4. Configure CONFIG_DONE register in VOP only.
5. Wait for frame start from VOP and IEP direct path.

29.5.3 VOP path turn on flow

1. Configure all IEP registers which are needed.
2. Configure VOP related registers which are needed.
3. Enable IEP direct path.
4. Enable VOP direct path.
5. Configure CONFIG_DONE register in VOP only.
6. Wait for frame start from VOP and IEP direct path.

29.5.4 VOP path turn off flow

1. Disable VOP direct path.
2. Disable IEP direct path, so IEP do not receive any other CONFIG_DONE and frame start from VOP immediately.
3. Configure CONFIG_DONE register in VOP.
4. Wait for frame start from VOP and IEP direct path, so VOP quit direct path mode completely.
5. Configure IEP registers which are needed at any time.
6. Configure IEP_CONFIG_DONE.
7. Configure IEP_FRM_START, IEP is working at write back mode now.

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Chapter 30 Video Input Processor (VIP)

30.1 Overview

The Video Input Processor, receives the data from Camera or CCIR656 encoder, and transfers the data into system main memory by AXI bus.
The Video Input Processor supports following features:

- Support YCbCr422 input
- Support Raw8bit input
- Support CCIR656(PAL/NTSC) input
- Support JPEG input
- Support YCbCr422/420 output
- Support UYVY/VYUY/YUYV/YVYU configurable
- Support up to 8192x8192 resolution source
- Support picture in picture
- Support arbitrary size window crop
- Support error/terminate interrupt and combined interrupt output
- Support clk/vsync/href polarity configurable
- Support one frame stop/ping-pong mode

30.2 Block Diagram

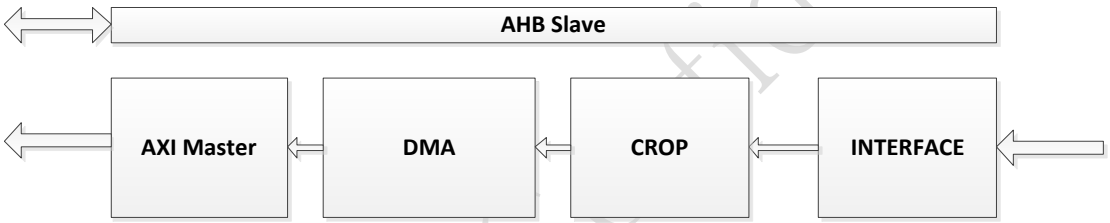

Fig. 30-1 VIP block diagram

The VIP comprises with:

- AHB Slave
  Host configure the registers via the AHB Slave
- AXI Master
  Transmit the data to chip memory via the AXI Master
- INTERFACE
  Translate the input video data into the requisite data format
- CROP
- Bypass or crop the source video data to a smaller size destination
- DMA
  Control the operation of AXI Master

30.3 Function description

This chapter is used to illustrate the operational behavior of how VIP works. If YUV422 or ccir656 signal is received from external devices, VIP translate it into YUV422/420 data, and separate the data to Y and UV data, then store them to different memory via AXI bus separately. But if raw data is received, there are not any translations happened, the 8 data is considered as 16bit data and write directly to memory.

30.3.1 Support Vsync high active or low active

- Vsync Low active as below

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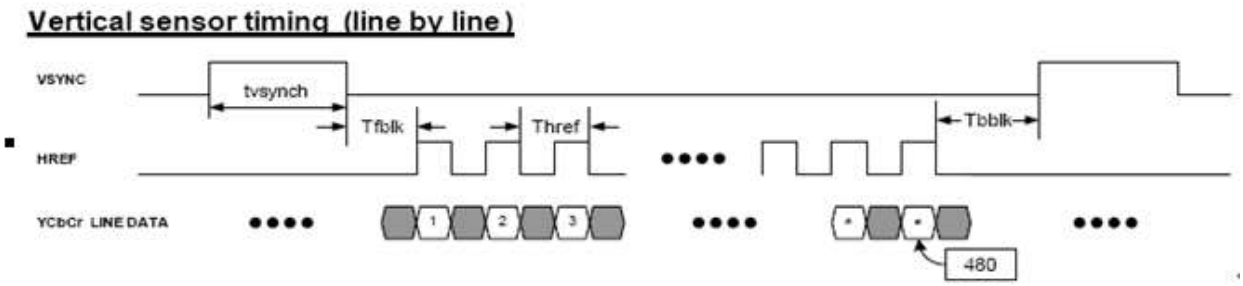

Fig. 30-2 Timing diagram for VIP when vsync low active

- Vsync High active

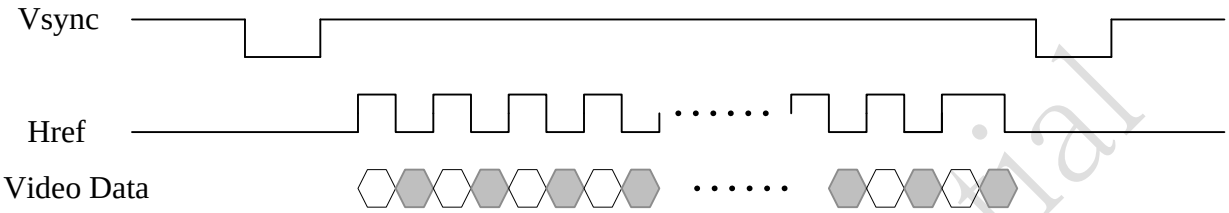

Fig. 30-3 Timing diagram for VIP when vsync high active

30.3.2 Support href high active or low active

- Href high active

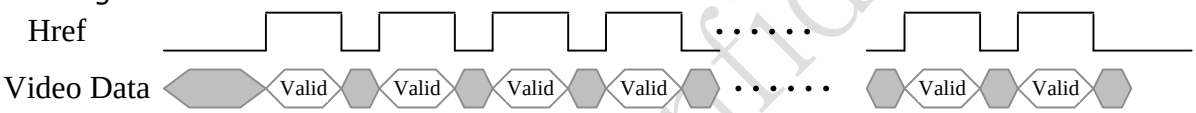

Fig. 30-4 Timing diagram for VIP when href high active

- Href Low active

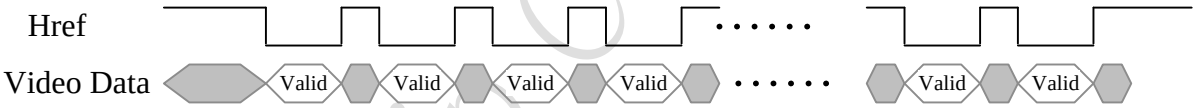

Fig. 30-5 Timing diagram for VIP when href low active

- Y first

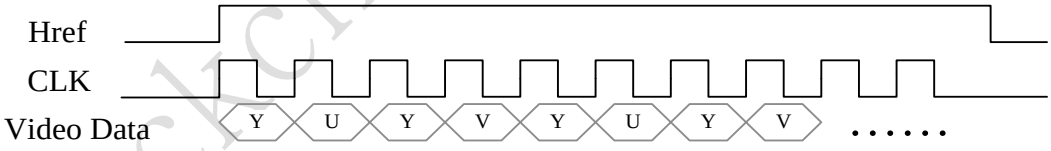

Fig. 30-6 Timing diagram for VIP when Y data first

- U first

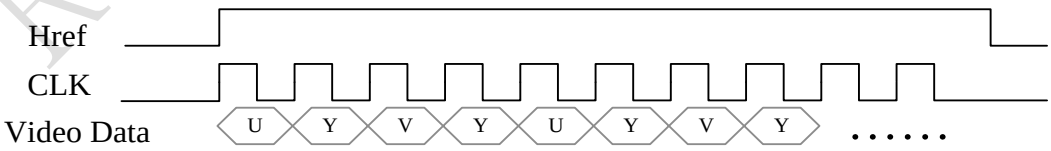

Fig. 30-7 Timing diagram for VIP when U data first

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30.3.3 Support CCIR656 (NTSC and PAL)

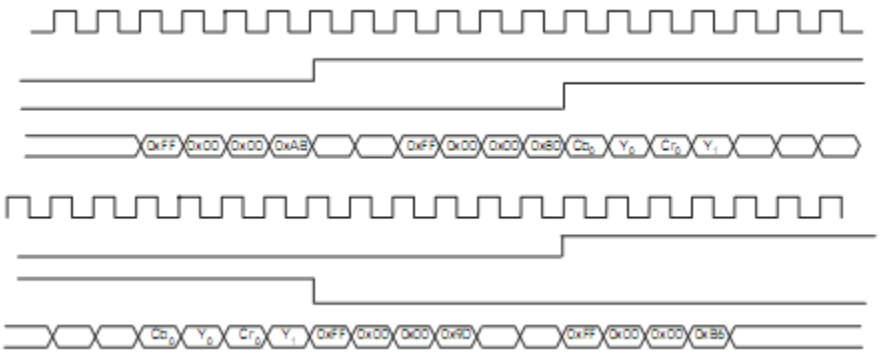

Fig. 30-8 CCIR656 timing

30.3.4 Support Raw data (8-bit) or JPEG

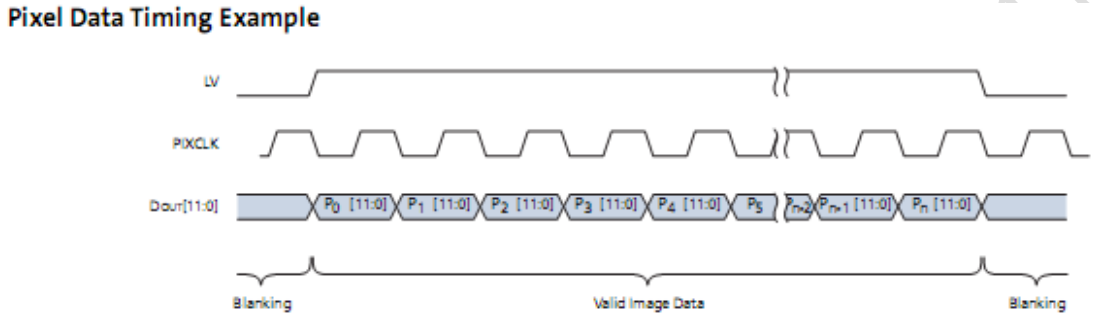

Fig. 30-9 Raw Data or JPEG Timing

VIP module can work in three modes: one frame stop mode, ping-pong mode.

One frame stop mode

In this mode, configure the parameter WORK_MODE to one frame stop mode. After one frame captured, VIP will automatic stop. After capturing, the image Y, UV data will be stored at main memory location defined by VIP_FRM0_ADDR_Y, FRM0_ADDR_UV separately.

Ping-Pong mode

After one frame(F1) captured, VIP will start to capture the next frame(F2) automatically, and host must assign new address pointer of frame1 and clear the frame1 status, thus VIP will capture the third frame automatically(by new F1 address) without any stop and so on for the following frames. But if host did not update the frame buffer address, the VIP will cover the pre-frame data stored in the memory with the following frame data.

Storage

Difference between the YUV mode and raw mode is that in the YUV mode or ccir656 mode, data will be storage in the Y data buffer and UV data buffer; but in the raw or jpeg mode, RGB data will be storage in the same buffer. In addition, in the yuv mode, the width of Y, U or V data is a byte in memory; in Raw or JPEGE mode, the width is a halfword no matter the data source is 8 bit.

CROP

The parameter START_Y and START_X defines the coordinate of crop start point. And the frame size after cropping is following the value of SET_WIDTH and SET_HEIGHT.

30.4 Register description

30.4.1 Register Summary

Name	Offset	Size	Reset Value	Description
VIP_CTRL	0x00000	W	0x00007000	VIP control
VIP_INTEN	0x00004	W	0x00000000	VIP interrupt status

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Name	Offset	Size	Reset Value	Description
VIP_INTSTAT	0x00008	W	0x00000000	VIP interrupt status
VIP_FOR	0x0000c	W	0x00000000	VIP format
VIP_DMA_IDLE_REQ	0x00010	W	0x00000000	DMA idle request
VIP_FRM0_ADDR_UV	0x00018	W	0x00000000	VIP frame0 uv address
VIP_FRM1_ADDR_Y	0x0001c	W	0x00000000	VIP frame1 y address
VIP_FRM1_ADDR_UV	0x00020	W	0x00000000	VIP frame1 uv address
VIP_VIR_LINE_WIDTH	0x00024	W	0x00000000	VIP virtual line width
VIP_SET_SIZE	0x00028	W	0x01e002d0	VIP frame set size
VIP_CROP	0x00044	W	0x00000000	VIP crop start point
VIP_SCL_CTRL	0x00048	W	0x00000000	VIP scale control
VIP_FIFO_ENTRY	0x00054	W	0x00000000	VIP FIFO entry
VIP_FRAME_STATUS	0x00060	W	0x00000004	VIP frame status
VIP_CUR_DST	0x00064	W	0x00000000	VIP current destination address
VIP_LAST_LINE	0x00068	W	0x00000000	VIP last frame line number
VIP_LAST_PIX	0x0006c	W	0x00000000	VIP last line pixel number

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

30.4.2 Detail Register Description

VIP_CTRL
Address: Operational Base + offset (0x00000)
VIP control

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:12	RW	0x7	AXI_BURST_TYPE axi master burst type 0-15 : burst1~16
11:3	RO	0x0	reserved
2:1	RW	0x0	WORK_MODE Working Mode 00 : one frame stop mode 01 : ping pong mode 02,03 : reserved
0	RW	0x0	CAP_EN capture enable 0 : disable 1 : enable

VIP_INTEN
Address: Operational Base + offset (0x00004)
VIP interrupt status

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9	RW	0x0	PST_INF_FRAME_END_EN frame end after interface FIFO interrupt enable 0 : disable 1 : enable
8	RW	0x0	PRE_INF_FRAME_END_EN frame end before interface FIFO interrupt enable 0 : disable 1 : enable
7	RO	0x0	reserved
6	W1C	0x0	BUS_ERR_EN axi master or ahb slave response error interrupt enable 0 : disable 1 : enable
5	RW	0x0	DFIFO_OF_EN DMA FIFO overflow interrupt enable 0 : disable 1 : enable
4	RW	0x0	IFIFO_OF_EN interface FIFO overflow interrupt enable 0 : disable 1 : enable
3	W1C	0x0	PIX_ERR_EN the pixel number of last line not equal to the set height interrupt enable 0 : disable 1 : enable
2	W1C	0x0	LINE_ERR_EN the line number of last frame not equal to the set height interrupt enable 0 : disable 1 : enable
1	W1C	0x0	LINE_END_EN line end interrupt enable 0 : disable 1 : enable
0	W1C	0x0	DMA_FRAME_END_EN dma frame end interrupt enable 0 : disable 1 : enable

VIP_INTSTAT
Address: Operational Base + offset (0x00008)

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VIP interrupt status

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9	RW	0x0	PST_INF_FRAME_END frame end after interface FIFO interrupt 0 : no interrupt 1 : interrupt
8	RW	0x0	PRE_INF_FRAME_END frame end before interface FIFO interrupt 0 : no interrupt 1 : interrupt
7	RO	0x0	reserved
6	W1C	0x0	BUS_ERR axi master or ahb slave response error interrupt 0 : no interrupt 1 : interrupt
5	RW	0x0	DFIFO_OF DMA FIFO overflow interrupt 0 : no interrupt 1 : interrupt
4	RW	0x0	IFIFO_OF interface FIFO overflow interrupt 0 : no interrupt 1 : interrupt
3	W1C	0x0	PIX_ERR the pixel number of last line not equal to the set height interrupt 0 : no interrupt 1 : interrupt
2	W1C	0x0	LINE_ERR the line number of last frame not equal to the set height interrupt 0 : no interrupt 1 : interrupt
1	W1C	0x0	LINE_END line end interrupt 0 : no interrupt 1 : interrupt
0	W1C	0x0	DMA_FRAME_END dma frame end interrupt 0 : no interrupt 1 : interrupt

VIP_FOR

Address: Operational Base + offset (0x0000c)

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VIP format

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19	RW	0x0	UV_STORE_ORDER UV storage order 0 : UVUV 1 : VUVU
18	RW	0x0	RAW_END raw data endian 0 : little end 1 : big end
17	RW	0x0	OUT_420_ORDER output 420 order 0 : UV in the even line 1 : UV in the odd line Note: The first line is even line(line 0).
16	RW	0x0	OUTPUT_420 output 420 or 422 0 : output is 422 1 : output is 420
15	RO	0x0	reserved
14:13	RW	0x0	MIPI_MODE mipi mode 00 : 32 bit bypass mode; 01 : rgb mode; 10 : yuv mode; 11 : reserve;
12:11	RW	0x0	RAW_WIDTH raw data width 00 : 8bit raw data; 01 : 10bit raw data; 10 : 12bit raw data; 11 : reserve;
10	RW	0x0	JPEG_MODE JPEG mode 0 : other mode 1 : mode1
9	RW	0x0	FIELD_ORDER ccir input order 0 : odd field first 1 : even field first
8	RW	0x0	IN_420_ORDER 420 input order 0 : UV in the even line 1 : UV in the odd line Note: The first line is even line(line 0).

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Bit	Attr	Reset Value	Description
7	RW	0x0	INPUT_420 input 420 or 422 0 : 422 1 : 420
6:5	RW	0x0	YUV_IN_ORDER YUV input order 00 : UYVY 01 : YVYU 10 : VYUY 11 : YUYV
4:2	RW	0x0	INPUT_MODE input mode 000 : YUV 010 : PAL 011 : NTSC 100 : RAW 101 : JPEG 110 : MIPI Other : invalid
1	RW	0x0	HREF_POL href input polarity 0 : high active 1 : low active
0	RW	0x0	VSYNC_POL vsync input polarity 0 : low active 1 : high active

VIP_DMA_IDLE_REQ
Address: Operational Base + offset (0x00010)
DMA idle request

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	DMA_IDLE_REQ DMA idle request Write 1 to request stopping transfer in the AXI bus. Self clear when DMA transfer stopped.

VIP_FRM0_ADDR_Y
Address: Operational Base + offset (0x00014)
VIP frame0 y address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	FRM0_ADDR_Y frame0 y address

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VIP_FRM0_ADDR_UV
Address: Operational Base + offset (0x00018)
VIP frame0 uv address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	FRM0_ADDR_UV frame0 uv address

VIP_FRM1_ADDR_Y
Address: Operational Base + offset (0x0001c)
VIP frame1 y address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	FRM1_ADDR_Y frame1 y address

VIP_FRM1_ADDR_UV
Address: Operational Base + offset (0x00020)
VIP frame1 uv address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	FRM1_ADDR_UV frame1 uv address

VIP_VIR_LINE_WIDTH
Address: Operational Base + offset (0x00024)
VIP virtual line width

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14:0	RW	0x0000	VIR_LINE_WIDTH virtual line width

VIP_SET_SIZE
Address: Operational Base + offset (0x00028)
VIP frame set size

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x01e0	SET_HEIGHT set height
15:13	RO	0x0	reserved
12:0	RW	0x02d0	SET_WIDTH set width

VIP_CROP
Address: Operational Base + offset (0x00044)
VIP crop start point

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
28:16	RW	0x0000	START_Y start y point
15:13	RO	0x0	reserved
12:0	RW	0x0000	START_X start x point

VIP_SCL_CTRL
Address: Operational Base + offset (0x00048)
VIP scale control

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	MIPI_32B_BP mipi 32bit bypass 0 : no bypass 1 : bypass
5	RW	0x0	RAW_16B_BP raw 16 bit bypass 0 : no bypass 1 : bypass
4	RW	0x0	YUV_16B_BP YUV 16 bit bypass 0 : no bypass 1 : bypass
3:0	RO	0x0	reserved

VIP_FIFO_ENTRY
Address: Operational Base + offset (0x00054)
VIP FIFO entry

Bit	Attr	Reset Value	Description
31:18	RO	0x0	reserved
17:9	RW	0x000	UV_FIFO_ENTRY valid UV double word in FIFO write 0 clear
8:0	RO	0x000	Y_FIFO_ENTRY valid Y double word in FIFO write 0 clear

VIP_FRAME_STATUS
Address: Operational Base + offset (0x00060)
VIP frame status

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	FRAME_NUM complete frame number write 0 to clear
15:3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
2	RW	0x1	IDLE controller status if idle, can be soft reseted safely 0: busy 1: idle
1	RO	0x0	F1_STS frame 0 status 0 : frame 1 not ready 1 : frame 1 ready write 0 clear
0	RO	0x0	F0_STS frame 0 status 0 : frame 0 not ready 1 : frame 0 ready write 0 clear

VIP_CUR_DST
Address: Operational Base + offset (0x00064)
VIP current destination address

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	CUR_DST current destination address maybe not the current, because the clock synchronization.

VIP_LAST_LINE
Address: Operational Base + offset (0x00068)
VIP last frame line number

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RO	0x0000	LAST_LINE_NUM line number of last frame

VIP_LAST_PIX
Address: Operational Base + offset (0x0006c)
VIP last line pixel number

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x0000	LAST_UV_NUM y number of last line
15:13	RO	0x0	reserved
12:0	RO	0x0000	LAST_Y_NUM y number of last line

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30.5 Interface description

Module Pin	Dir	Pad Name	IOMUX Setting
vip_clkout	O	IO_CIFclkout_TSfail_DVPgpio1b3	GRF_GPIO1B_MUX[5:4]==2'b1
vip_clkin	I	IO_CIFclkin_TSclk_DVPgpio1b2	GRF_GPIO1B_MUX[7:6]==2'b1
vip_href	I	IO_CIFhref_TSvalid_DVPgpio1b1	GRF_GPIO1B_MUX[3:2]==2'b1
vip_vsync	I	IO_CIFvsync_TSsync_DVPgpio1b0	GRF_GPIO1B_MUX[1:0]==2'b1
vip_data0	I	IO_CIFdata0_DVPgpio1b4	GRF_GPIO1B_MUX[9:8]==2'b1
vip_data1	I	IO_CIFdata1_DVPgpio1b5	GRF_GPIO1B_MUX[11:10]==2'b1
vip_data2	I	IO_CIFdata2_TSdata0_DVPgpio1a0	GRF_GPIO1A_MUX[1:0]==2'b1
vip_data3	I	IO_CIFdata3_TSdata1_DVPgpio1a1	GRF_GPIO1A_MUX[3:2]==2'b1
vip_data4	I	IO_CIFdata4_TSdata2_DVPgpio1a2	GRF_GPIO1A_MUX[5:4]==2'b1
vip_data5	I	IO_CIFdata5_TSdata3_DVPgpio1a3	GRF_GPIO1A_MUX[7:6]==2'b1
vip_data6	I	IO_CIFdata6_TSdata4_DVPgpio1a4	GRF_GPIO1A_MUX[9:8]==2'b1
vip_data7	I	IO_CIFdata7_TSdata5_DVPgpio1a5	GRF_GPIO1A_MUX[11:10]==2'b1
vip_data8	I	IO_CIFdata8_TSdata6_DVPgpio1a6	GRF_GPIO1A_MUX[13:12]==2'b1
vip_data9	I	IO_CIFdata9_TSdata7_DVPgpio1a7	GRF_GPIO1A_MUX[15:14]==2'b1
vip_data10	I	IO_CIFdata10_SPI1clk_DVPgpio1b6	GRF_GPIO1B_MUX[13:12]==2'b1
vip_data11	I	IO_CIFdata11_SPI1csn0_DVPgpio1b7	GRF_GPIO1B_MUX[15:14]==2'b1

30.6 Application Notes

The biggest configuration requirement of all operations is the CAP_EN bit must be set after all the mode selection is ready. The configuration order of the input/output data format, YUV order, the address, frame size/width, AXI burst length and other options do not need to care. There are many debug registers to make it easy to read the internal operation information of VIP. The valid pixel number of scale result in FIFO can be known by read VIP_SCL_VALID_NUM. The line number of last frame and the pixel number of last line can be also known by read the VIP_LAST_LINE and VIP_LAST_PIX.

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Chapter 31 Image Signal Processing (ISP)

31.1 Overview

The Image Signal Processing (ISP) represents a complete video and still picture input unit. It contains image processing and scaling functions. The integrated image processing unit supports simple CMOS sensors delivering RGB Bayer pattern without any integrated image processing and also image sensors with integrated YCbCr processing.
Scaling is used for downsizing the sensor data for either displaying them on the LCD, or for generating data stream for MPEG-4 compression. Scaling also can be used for digital zoom effects, because the scalers are capable of up-scaling as well.
An image effects block is present which can create images with sepia, black&white, color selection, negative, emboss and sketch effects.
The camera interface provides SMIA and/or MIPI support, so that ISP can be connected to PHY devices or IP blocks directly.
All data is transmitted via the memory interface to a BVCI/AXI bus system using a bus master interface.
Programming is done by register read/write transactions using an PVCI slave interface.
ISP supports the following features:

- Generic Sensor Interface with programmable polarity for synchronization signals
- ITU-R BT 601/656 compliant video interface supporting YCbCr or RGB Bayer data
- 12 bit camera interface
- 12 bit resolution per color component internally
- YCbCr 4:2:2 processing
- Flash light control
- Mechanical shutter support
- Windowing and frame synchronization
- Frame skip support for video (e.g. MPEG-4) encoding
- Macro block line, frame end, capture error, data loss interrupts and sync. (h_start, v_start) interrupts
- Luminance/chrominance and chrominance blue/red swapping for YUV input signals
- Continuous resize support
- Buffer in system memory organized as ring-buffer
- Buffer overflow protection for raw data
- Asynchronous reset input, software reset for the entire IP and separate software resets for all sub-modules
- Interconnect test support
- Semi planar storage format
- Color processing (contrast, saturation, brightness, hue, offset, range)
- Power management by software controlled clock disabling of currently not needed sub-modules
- Read port provided to read back a picture from system memory
- Black level compensation
- Four channel Lens shade correction (Vignetting)
- Auto focus measurement
- White balancing and black level measurement
- Auto exposure support by brightness measurement in 5x5 sub windows
- Defect pixel cluster correction unit (DPCC) supports on the fly and table based pixel correction
- De-noising pre filter (DPF)
- Enhanced color interpolation (RGB Bayer demosaicing)
- Chromatic aberration correction
- Combined edge sensitive Sharpening / Blurring filter (Noise filter)
- Color correction matrix (cross talk matrix)
- Global Tone Mapping with wide dynamic range unit (WDR)
- Image Stabilization support and Video Stabilization Measurement
- Flexible Histogram calculation

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- Digital image effects (Emboss, Sketch, Sepia, B/W (Grayscale), Color Selection, Negative image, sharpening)
- Solarize effect through gamma correction
- AXI 64 bit interface 32Bit Address range (two DMA-write ports and one DMA-read port)
- Up to 16 Beat Bursts depending on configured FIFO size
- 32 bit AHB programming interface
- Maximum input resolution of 3264x2448 pixels
- Main scaler with pixel-accurate up- and down-scaling to any resolution between 3264x2448 and 32x16 pixel in processing mode
- Support of semiplanar NV21 color storage format
- Support of image cropping

31.2 Block Diagram

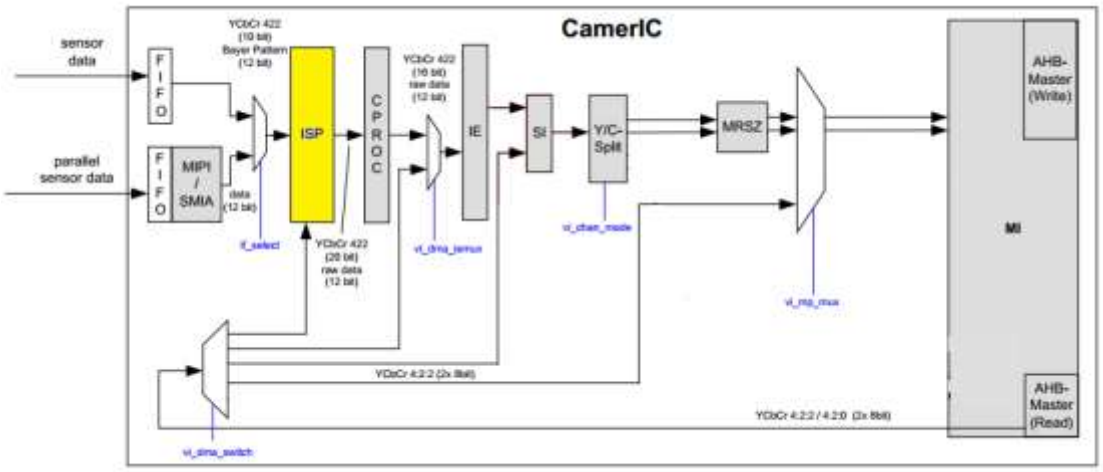

Fig. 32-1 ISP Block Diagram

ISP comprises with:

- MIPI serial camera interface
- Image Signal Processing (ISP)
- Color Processing (CPROC)
- Image Effects (IE)
- Superimpose (SI)
- Luminance /Chrominance Splitter (Y/C Split)
- Crop
- Main Resize (MRSZ)
- MPMUX for selection of main path data flow
- Control Unit

31.3 Function Description

31.3.1 MIPI

The MIPI interface is the second optional serial camera interface of ISP Controller. The interface is implemented according to the CSI2 specification defined by the MIPI Alliance, connecting a PPI data interface to a MIPI_D-PHY physical layer with a 12bits ALOMICs interface.

Features and Standard Compliance

Compliant to MIPI Alliance Standard for Camera Serial Interface 2 (CSI2)

- PPI Interface according to D-PHY specification, Annex A
- Supports up to 4 data lanes (extendable by request)
- Number of lanes is programmable and hardware configurable
- Provides lane merging, error detection and correction, virtual channel detection, programmable data extraction and embedded data separation
- Supported data types are:
  - Generic 8bit data

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- Non-legacy YUV 4:2:0 8bit / 10bit with cosited chroma sampling
- Non-legacy YUV 4:2:0 8bit / 10bit with non-cosited chroma sampling
- Legacy YUV 4:2:0 8bit
- YUV 4:2:2 8bit / 10bit
- RGB 444 / 555 / 565 / 666 / 888 image data
- RAW 6-bit / 7-bit / 8-bit / 10-bit / 12-bit image data
- User-defined 8-bit data

- PVCI similar output interface
- PVCI control interface

RAW 14-bit image data types are not supported. The MIPI D-PHY-Layer features Escape Modus and Low Power Data Transfer are not needed for the Protocol Receiver. According to the CSI2 specification only unidirectional high-speed data transfer is mandatory for the camera interface.

31.3.2 ISP Block

The ISP block includes the interface to the attached parallel sensor device and the MIPI interfaces. It accepts either ITU-R BT 601 YCbCr, well as raw Bayer data or ITU-R BT 656 YCbCr data. The order of Y and Cx as well as Cb and Cr is programmable. It contains the interpolation filter for the raw Bayer to plain RGB conversion.
An input acquisition window is programmable supporting detection of smaller than programmed input images. An error IRQ is generated in case of input error conditions.
The ISP allows programming of a continuous image sampling mode, or a mode where the number of images to be sampled subsequently can be programmed (1…1023).
Additionally it incorporates image quality improvements (gamma correction, black level subtraction, white balancing, etc.).
The ISP block supports also auto exposure capabilities by providing image color statistics information to the system processor for correct sensor device programming.
The input part of the ISP block is fully programmable in terms of signal polarities, active video data positions, and luminance/chrominance order.
The ISP block always delivers YCbCr 4:2:2 data at its output port. Data provided by the RGB channel are downsampled to YCbCr 4:2:2, supporting both co-sited and non co-sited calculation. Luminance and chrominance data are provided in parallel using line and frame end signals. In raw data mode (unprocessed data) these data are transferred via the Y port. Handshaking is used for data qualification. As the sensor device cannot be stopped delivering data the backward handshake (acknowledge) is only used for pixel drop detection which is signaled using an IRQ. To prevent pixel dropping a latency FIFO is implemented. The FIFO depth can be option of customization.
The ISP block also contains its own programming registers to be accessed by a 32 bit PVCI compliant interface supporting single transfers only.
The incoming hsync and vsync control signals from the camera are connected to interrupt logic. It is possible to trigger on these signals for an event triggered configuration during processing.

1. DPCC

The ISP core is designed to operate with high-end, mid-range and low-end image sensors, which mainly differ in the number of pixels and module cost. While high-end sensors aim at a high number of pixels with sufficiently large pixel area, low-cost modules often have a large number of defects. Additionally, for low cost sensors in the manufacturing process no time is available for determining the defect pixels locations.
Another problem is that hot pixels get visible at long integration times and can get a high density up to 5%. This means 250000 defect pixels for a 5 Megapixel sensor.
An improved algorithm has been developed, named Defect Pixel Cluster Correction (DPCC). This is an on-the-fly detection and replacement processing as well as a table based replacement method

Defect Pixel Detection

For each pixel threshold values are calculated by several methods, using the correlation of neighbor pixel of the same color (red, green or blue) with exception of the peak gradient estimation for red and blue that also uses the green pixel values in the 5x5 neighborhood of the raw Bayer image. These methods use statistical properties and linear prediction to

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determine if a pixel needs to be marked as defect. A 3x3 sorting algorithm with rank estimation including the calculation of median values of some pixel groups is a central unit. Output of the detection unit is a marker signal for the following correction stage to indicate, if the current pixel is defect. The detection can be controlled by programmable threshold values, factors and options which methods should be used. The correct setting of the thresholds is important for a good separation between defects and keeping high resolution and detail features in the image.

Defect Pixel Table control

A defect pixel table is implemented as SRAM of user defined size containing entries with defect pixel coordinates. The table control generates a replace flag independently from the on-the-fly detection, so that table correction and on-the-fly correction can be operated in parallel.

Defect Pixel Replacement

Basic algorithm is a switching median filter, which performs sorting and rank ordering. The replacement unit takes the information of the Defect Pixel Detection as input. It uses a statistical sorting filter (median filter) separately for each color to determine the nearest neighbor value for replacement. The filter size is 4 (upper, lower, left and right neighbor for red/blue or diagonal neighbors for green) optionally 5, including the center pixel. The condition for successfull replacement is that NOT more than 4 of 9 pixels are defect. So correction of single and small cluster defects of 1x3 (3x1) and 2x2 pixels is possible (please note these clusters belong to a single color in the bayer pattern).

2. Auto Focus Measurement

An auto-focus measurement block is implemented to support auto-focus control. A substantial part of auto-focus control will be done software supported: The search algorithm which looks for maximum sharpness in the image is implemented using software and the movement of the lens is controlled by software. The auto-focus module which is implemented using hardware delivers measurement values of image sharpness via a register interface.

The module measures the sharpness in 3 windows of selectable size via register settings. The auto-focus measurement block uses the line buffer of the emosaicing block. The data in the buffer are stored in Bayer format and read 3-line-wise (top, middle, bottom).

3. Filter (Noise Reduction, Sharpness, Blurring)

As mentioned above, high-end, mid-range and low-end image sensors mainly differ in the number of pixels and module cost. While high end sensors aim at a high number of pixels with sufficiently large pixel area, low-cost modules have small pixels and lens with small diameter. This combination of small lens diameter and small pixels results in higher pixel noise at a given level of illumination than in high end sensors.

To improve the visual image quality, noise and artifacts of the Bayer Matrix should be eliminated and sharpness should be increased. These are requirements which cannot be easily combined.

Noise and the artifacts of the Bayer pattern must be eliminated by averaging or blurring filters. Sharpness could be improved by a high pass or Laplace filter.

Texture detection allows detecting planes or edges, high or low density of details.

With this texture information an adaptive filter is controlled which reduces noise in planes and improves sharpness in detailed regions. If there are less details below or near the noise level two-dimensional blurring is applied. With a higher detail level, the blurring is done along detected edges or lines. If improvement of sharpness is required, the sharpness will be enhanced orthogonally to the direction of the blurring operation.

For example if a horizontal edge is detected, blurring will be done in horizontal direction and sharpness will be improved in vertical direction. In regions with highest contrast and details the filter can be bypassed or if sharpness improvement is required, a two dimensional sharpness filter is realized.

Additionally if no noise reduction is required a fixed blurring or sharpness filter can be selected. The sharpness can be improved by a fixed sharpness filter or edge depending as described beyond.

The noise reduction level can be adapted by the filter coefficients which determine the weightiness of averaging, and by the threshold values which are compared with the texture detection results.

For an optimal noise reduction it is important to know the effective noise level. If the noise

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level is too high, the noise cannot be eliminated at the best possible rate. If the noise level is low and the noise reduction level is too high, too much details of the image will be lost unnecessarily.
If the light conditions are known, the noise level can be estimated using a table from the exposure settings of the image sensor. If the noise level must be calculated e.g. from the variance of the image RGB data, it is difficult to decide if the variance is caused by the noise or by image patterns.
In addition the hardware effort for separate filters is high because of the necessary line buffers. Therefore the filter algorithm is combined with demosaicing, so that the same line buffer can be used for demosaicing, noise filtering and blurring/sharpening.
The combined demosaicing filter module is designed to operate with linear RGB white balanced Bayer Data. So for best results of interpolation for demosaicing and of texture detection the RGB data should be optimal white balanced.

4. Video Stabilization

The ISP Controller Video Stabilization consists of the following components

- Video Stabilization Measurement
- Video Stabilization Software
- ISP Image Stabililzer

The Video Stabilization Measurement is done in hardware. It calculates per image the horizontal and vertical displacement vector for global motion in comparison to the previous image.
The Video Stabilization Software reads these measurements values, applies some smoothing algorithms and provides the respective cropping window to the ISP Image Stabilization.
The ISP Image Stabilization crops the image with respect to the given cropping window.
The cropping of an image has to be done after the video stabilization measurement for the image has been completed. This involves the usage of at least 1 frame buffer in system memory.
The Image Stabilizer unit delivers YUV data (either unmodified YUV data received as ITU-R BT.601 or ITU-R BT.656 data or color-processed image data) to the following ISP Controller environment.
The Video Stabilization process is currently limited to input line sizes of 2048 pixels and output line sizes of 1280 pixels.

5. Mechanical Shutter Control

With increasing sensor resolution the expectation of high quality pictures also rises. Therefore the exposure control must be decoupled from the frame rate and data read-out. This can be achieved by a mechanical shutter. The mechanical shutter control supports shutter speeds from 1/4000 sec to 10.7 sec by 48-100MHz.

6. Flash Light Control

The sensor interface supports triggering of a LED or tube flash light. The flash light output and the prelight output can be used to control a flash light device. Both the flash light and the prelight are activated by a trigger event. This event may either be a positive or negative edge from the camera or a positive edge from any other trigger source at the input port 'vds_vsync'.
Signal polarity, flash delay time and flash light time are determined by programming respective configuration register.

7. Histogram Calculation

A histogram function is implemented which counts the number of pixels with the same value. In general this histogram is a graphical representation of the pattern of variation that exists in the intensity values of the color or luminance planes.
Usually it is displayed by vertical bars drawn to indicate frequency levels of data collected within specific ranges. This measurement block can be used for different purposes. The most obvious application is an informative display for the end user. It is used for improving the exposure control, which is done by a software control loop.
Histogram Calculation is done independently for 5x5 windows which is important for advanced and fast auto exposure algorithms with complex scene detection.

8. Lens Shading Correction

The lens shading correction deals with the problems of vignetting and lens shading. It is done

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during input data processing: If the lens shading correction is enabled, each pixel is processed and corrected according to the stored settings. The lens shading correction is done by multiplying each input pixel with its respective correction value.
Only the correction factors at predefined sector corners as well as the sector positions are stored. The pixel position specific correction values are calculated using bilinear interpolation.
The correction factors at the sector corners are calculated during a calibration process which uses one or more reference frames which have to be captured under dedicated light conditions and at a dedicated position of the sensor. The captured frames are evaluated by software and the calculated parameters for lens shading correction are stored in multiple illumination specific tables e.g. in external memory or on a flash device. The software controls the lens shading process by loading or updating the correct tables into the hardware module. It is also possible to use different lens shading correction parameters for different environment conditions, e.g. lightness, light direction or sensor position.

9. Wide Dynamic Range (WDR)

ISP contains a global tone and color mapping unit for Wide Dynamic Range (WDR) compression. Compared with a standard camera, the dynamic range of input intensities appears to be widened, since more structure becomes visible out of the dark and bright image regions.
The dynamic range of real-word scenes is much higher than the available dynamic range of low cost CMOS image sensors. The image sensor thus captures a small range of the real word's scene radiance and maps it to the available output range of the sensor. Radiance levels above or below the sensor's value range are clipped to black or white in the sensor.
The auto-exposure control (AEC) controls which portion of the scene radiance is mapped to the sensor value range. The AEC uses a model-driven scene evaluation to determine the best exposure value.
Nevertheless, there is a chance that portions in the scene are mapped to the dark grey tones or near white tones. This is the case especially in high contrast scenes, when there is anyhow not the chance to perfectly reproduce the full scene radiance range.
Global tone mapping can be used to reduce this effect. It aims at shifting textures in dark grey or near white tones into the mid tone range and thus allows to optimize the perceptual reproduction of the scene. Compared with a standard camera, the dynamic range of input intensities appears to be widened, since more structure becomes visible out of the dark and bright image regions.
This step is being performed directly before the Gamma-Correction. Basically by applying a scene dependent tone curve the required intensity shift is being performed.
During this step the following constraints have to be considered:

- Shifting of textures from dark grey into mid tones increases noise. Thus this step should only be performed for images with a sufficiently low noise level in dark grey tones.
- Changing the intensity level of a pixel also effects the color saturation. In order to avoid color clipping a correction of the color saturation is to be performed. This ensures that after tone mapping the colors have the same hue as before.

The tone and color clipping correction unit for wide dynamic range applications (WDR) performs scene dependent correction such as brightening of dark texture tones. Suitably, the denoising pre-filter is being used for edge-preserving noise reduction especially in dark textures. Additionally, color clipping compensation is being performed with tone mapping.

31.3.3 Color Processing

The Color Processing block is responsible for color processing functions, i.e. hue, contrast, brightness, and saturation adjustment. It operates at YCbCr 4:2:2 data.

31.3.4 Image Effects (IE)

The Image Effects block modifies an image by pixel modifications. A set of different modifications can be applied: grayscale-, sepia-, color selection-, negative-, emboss-, sketch and sharpening effects
In addition a solarize effect can be created by using the gamma block of ISP Controller (for more information an application note is available).
The Image Effects module gets YCbCr 4:2:2 data via a 16 bit ([15:8]: Y, [7:0]: Cb/Cr) data

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interface.

IE Feature

- Data input- and output handshake interface
- Supports YCbCr 4:2:2 format
- 3x3 Laplace Filter (for picture edge extraction used for emboss, sketch and sharpening effects)
- ITU-R BT.601 compliant YCbCr to RGB conversion (used for the color selection effect to compare red green blue components of a pixel with defined threshold values)

31.3.5 Super Impose (SI)

The Super Impose module overlays an image with a bitmap from the main memory

- Color of the transparent area in superimpose bitmap is configurable. So the camera picture interfuses through the transparent area (A).
- Furthermore the Superimpose block is able to position a bitmap with the appropriate coordinates over the camera image range (B).

The Super Impose module gets picture data in YCbCr 4:2:2 formats via a 16 bit data interface from the Image Effects module. The Memory Interface module delivers the Y, Cb and Cr pixel components of the superimpose bitmap through three independent handshake data interfaces. Within the common area of the two pictures, output pixel data is determined by the bitmap from main memory or by the image from the Image Effects module.

31.3.6 Luminance/Chrominance Splitter (YC_Split)

The Luminance / Chrominance Splitter is responsible for providing component separated YCbCr 4:2:2 pixel data for further processing. Therefore it has to split the data path for chrominance and luminance values.

31.3.7 Resize

The Resize Modules get pictures in YCbCr 4:2:2 format, and scale them by an arbitrary factor up or down to a new format. See following Figure for a module overview:

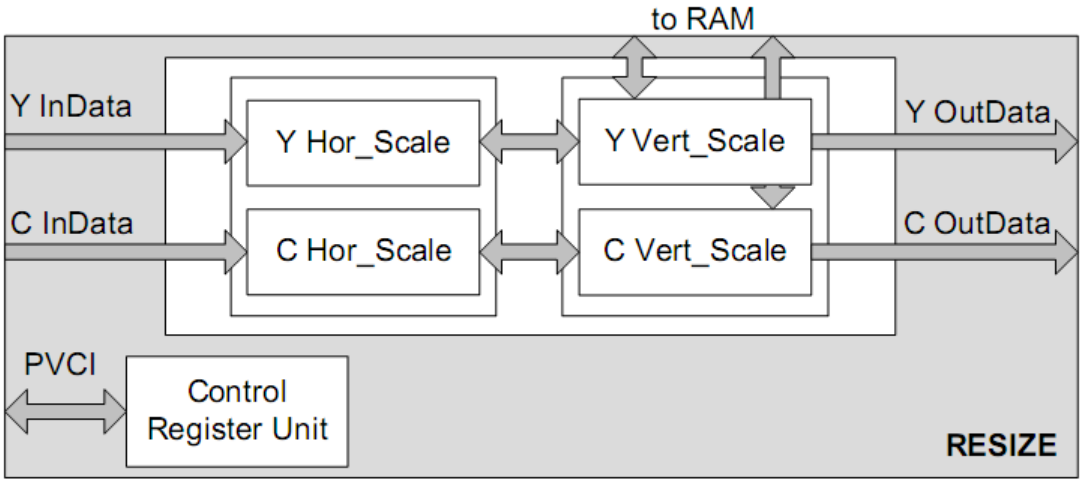

Fig. 32-2 Block Diagram of the Resize Module

Feature and Standard Compliance

PVCI compliant control port

- Data input and output handshake interfaces
- Supports 4:2:2 YCbYCr input format
- Different output formats (4:2:2, 4:2:0, 4:1:1, 4:1:0) possible by choosing from different scaling factors for luminance and chrominance components
- Support of cosited and non-cosited output formats via programmable phase offsets
- Discrete bypassing of each submodule is possible
- Output frame size for Main Picture Scaler (MRSZ) is up to 64 Mpixels.

The Resize module is configurable for horizontal and vertical up- or down-scaling.
Discrete values for the scaling factors of the luminance and the two chrominance components allow conversion between YUV4:2:2 and YUV4:2:0 color format and support of uneven line width.

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Phase shift registers are provided to shift the output pixel positions with respect to the input pixel positions. This allows for e.g. format conversion between cosited and non-cosited color schemes.
In sensor mode the MRSZ block supports only down-scaling. This is because the sensor cannot be stopped from delivering data during one frame.
The Resize module is able to process luminance and chrominance data independently, i.e. there are separate pipelines for luminance and chrominance processing using dedicated scale factors and phase offsets. This allows format conversion to be done by the Resize block (YCbCr 4:2:2 to 4:2:0, 4:1:1, 4:1:0).

31.3.8 Memory Interface (MI)

The Memory Interface block provides three data bus master ports to the system memory, two write ports and one read port.

Write Port

The Memory Interface is responsible for collecting the internal data streams and writing them into system memory. Therefore it is attached to data bus master wrappers to access (write) the system bus.
The following types of data streams are supported:

- Raw 8 or 12 bit data
- 2x 8 bit Y, Cb/Cr main image data

The following modes of operation must be supported:

- Raw data only
- Main image data only
- Semi-planar mode

The image data has to be split into Y, Cb and Cr data to be separately written into system memory. So the Memory Interface (write port) consists of six FIFOs for the data.
The FIFOs are necessary for the component separated data streams of main path. Some of the FIFOs have to be re-used for the raw data. See Fig.32-3 for the definition of memory buffers and Fig.32-4 for the storage scheme in planar and semi-planar mode.

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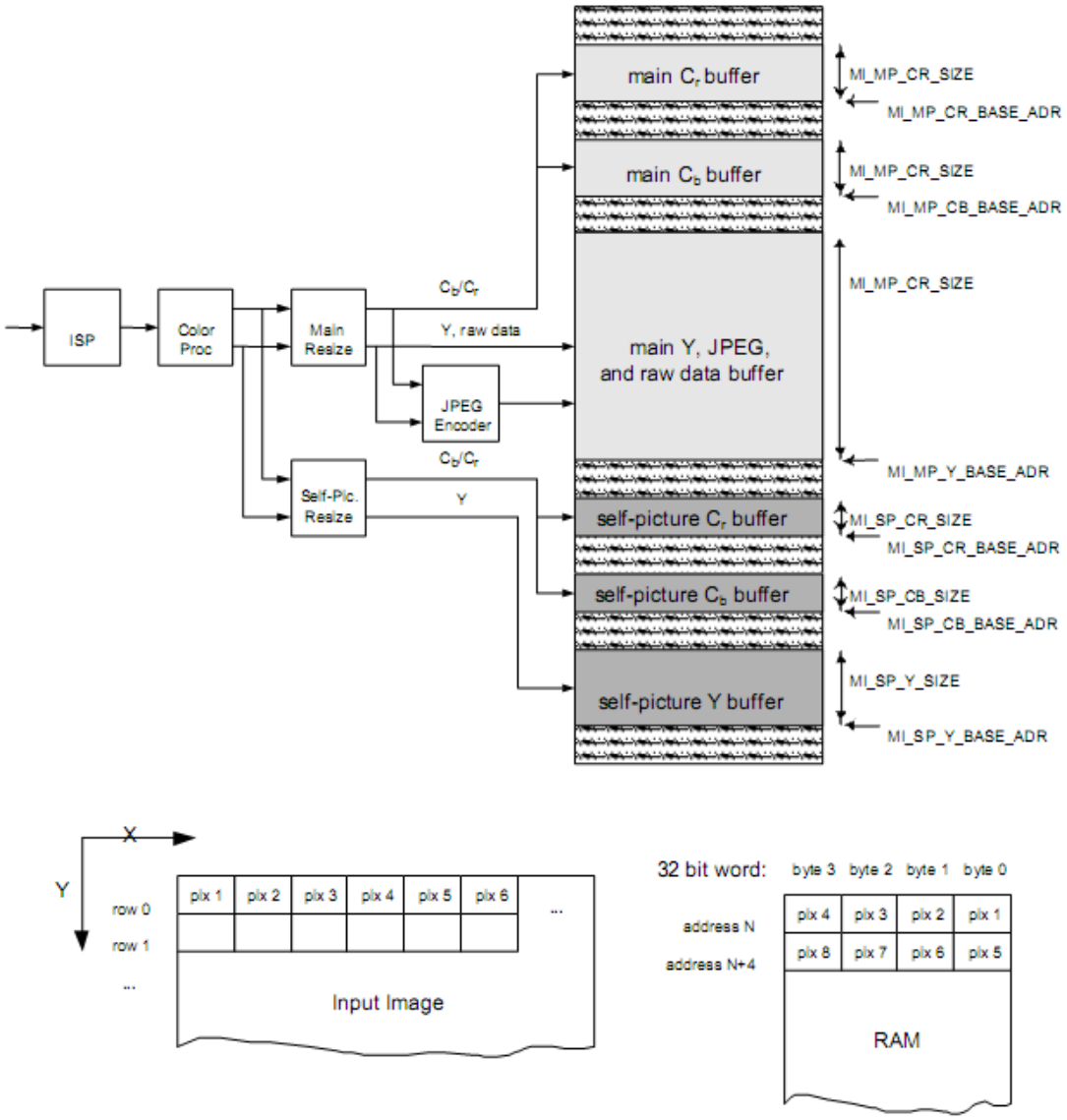

Fig. 32-3 Definition of Memory Buffers

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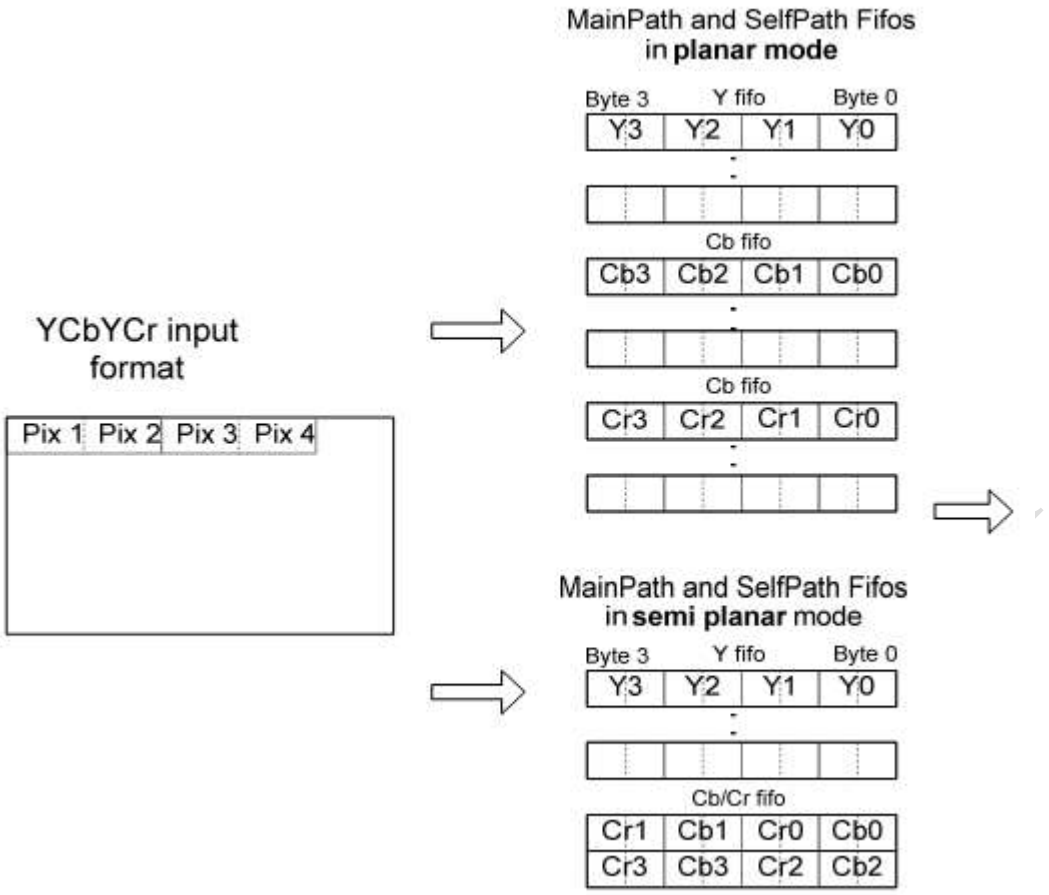

Fig. 32-4 Storage Scheme in Planar and Semi-planar Mode

Read port (DMA)

The Memory Interface also supports reading back picture data from the system memory. Therefore it is attached to a second data bus master wrapper to access (read) the system data bus.

Three independent read channels are provided to accommodate the three color components of a picture. The picture has to be stored component separated (planar) in system memory. If one or two components are not used they can be turned off by programming the respective component size to zero. So it's also possible to read back raw data through one single channel.

The Memory Interface (read port) consists of three FIFOs, one FIFO per channel. Each FIFO features a PVCI interface at the output, so data can be easily halted by de-asserting the acknowledge line.

31.3.9 Control Unit

The Control Unit serves two purposes:

1. Interface o the local configuration register blocks of the other modules
2. Clock and reset control registers for ISP Controller core

The Control Unit has one 32 bit PVCI input interface and multiple 32 bit PVCI output interfaces. All transfers from the PVCI input interface are 32 bit wide.

Each block inside the ISP Controller IP core uses a dedicated clock signal that can be controlled by a programming register inside the Control Unit.

Existing software resets in the ISP Controller blocks can also be controlled by a programming register inside the Control Unit. An asynchronous reset for the processing clock domain has to be generated from the system reset. A soft reset for all registers in the IP core is provided. It works like an asynchronous reset.

All sub module processing clocks can be switched on and off. All sub module configuration clocks are enabled only when the dedicated address is active. All gated processing and configuration clocks are phase synchronous to the processing clock.

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31.4 Register Description

The ISP uses a distributed configuration register scheme. So there is no central unit containing all programming registers, but all sub-modules contain their own programming registers. An address space is reserved for each sub-module inside the total ISP Controller address space.

31.4.1 Registers Summary

No.	Module	Description	Base Name	Offset Address
1	MX_main_control	ISP Main Control Registers	MRV_AFM_BASE	0x0000
2	MX_image_effects	Image Effects	MRV_IMGEFF_BASE	0x0200
3	MX_superimpose	Superimpose	MRV_SI_BASE	0x0300
4	MX_isp	ISP main registers	MRV_ISP_BASE	0x0400
5	MX_isp_flash	FLASH_LIGHT registers	MRV_FLASH_BASE	0x0660
6	MX_isp_shutter	SHUTTER registers	MRV_SHUT_BASE	0x0680
7	MX_cproc	COLOR PROCESSING registers	MRV_CPROC_BASE	0x0800
8	MX_main_resize	MAIN RESIZE registers	MRV_MRSZ_BASE	0x0c00
9	MX_mi	Memory Interface registers	MRV_MI_BASE	0x1400
10	MX_smia	SMIA Interface registers	MRV_SMIA_BASE	0x1a00
11	MX_mipi	MIPI Interface registers	MRV_MIPI_BASE	0x1c00
12	MX_isp_afm	ISP Auto Focus Measurement	MRV_AFM_BASE	0x2000
13	MX_isp_lsc	ISP Lens Shade Correction	MRV_LSC_BASE	0x2200
14	MX_isp_is	ISP Image Stabilization	MRV_IS_BASE	0x2300
15	MX_isp_hist	ISP Histogram	MRV_HIST_BASE	0x2400
16	MX_isp_filter	ISP Filter	MRV_FILT_BASE	0x2500
17	MX_isp_cac	ISP Chromatic Aberration Correction	MRV_CAC_BASE	0x2600
18	MX_isp_exposure	ISP Auto Exposure Measurement	MRV_AE_BASE	0x2700
19	MX_isp_bls	ISP Black Level Subtraction	MRV_BLS_BASE	0x2800
20	MX_isp_dpf	ISP De-noising Pre-filter	MRV_DPF_BASE	0x2900
21	MX_isp_dpcc	ISP Defect Pixel Cluster Correction	MRV_DPCC_BASE	0x2900
22	MX_isp_wdr	ISP Wide Dynamic Range	MRV_WDR_BASE	0x2a00

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

31.4.2 Detail Register Description

The description of the ISP register reference **5.1 Register Interface of CamerIC2D_user_manual_M14v2.pdf.**

31.5 Interface Description

Table 31-1 ISP Interface Description

Module Pin	IO	Pad Name	IOMUX Setting
isp_clkin	I	IO_CIFclkin_TSclk_DVPgpio1b2	GRF_GPIO1B_MUX[7:6]==2'b1
isp_href	I	IO_CIFhref_TSvalid_DVPgpio1b1	GRF_GPIO1B_MUX[3:2]==2'b1
isp_vsync	I	IO_CIFvsync_TSsync_DVPgpio1b0	GRF_GPIO1B_MUX[1:0]==2'b1
isp_data0	I	IO_CIFdata0_DVPgpio1b4	GRF_GPIO1B_MUX[9:8]==2'b1
isp_data1	I	IO_CIFdata1_DVPgpio1b5	GRF_GPIO1B_MUX[11:10]==2'b1
isp_data2	I	IO_CIFdata2_TSdata0_DVPgpio1a0	GRF_GPIO1A_MUX[1:0]==2'b1

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isp_data3	I	IO_CIFdata3_TSdata1_DVPgpio1a1	GRF_GPIO1A_MUX[3:2]==2'b1
isp_data4	I	IO_CIFdata4_TSdata2_DVPgpio1a2	GRF_GPIO1A_MUX[5:4]==2'b1
isp_data5	I	IO_CIFdata5_TSdata3_DVPgpio1a3	GRF_GPIO1A_MUX[7:6]==2'b1
isp_data6	I	IO_CIFdata6_TSdata4_DVPgpio1a4	GRF_GPIO1A_MUX[9:8]==2'b1
isp_data7	I	IO_CIFdata7_TSdata5_DVPgpio1a5	GRF_GPIO1A_MUX[11:10]==2'b1
isp_data8	I	IO_CIFdata8_TSdata6_DVPgpio1a6	GRF_GPIO1A_MUX[13:12]==2'b1
isp_data9	I	IO_CIFdata9_TSdata7_DVPgpio1a7	GRF_GPIO1A_MUX[15:14]==2'b1
isp_data01	I	IO_CIFdata10_SPI1clk_DVPgpio1b6	GRF_GPIO1B_MUX[13:12]==2'b1
isp_data11	I	IO_CIFdata11_SPI1csn0_DVPgpio1b7	GRF_GPIO1B_MUX[15:14]==2'b1

Notes: 1. I=input, O=output, I/O=input/output, bidirectional

31.6 Application Notes

The description of the Application Notes reference **5.1 Register Interface of CamerIC2D_user_manual_M14v2.pdf.**

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Chapter 32 HDMI TX

32.1 Overview

HDMI TX is fully compliant with HDMI 1.4a and 2.0a specification. It offers a simple implementation for consumer electronics like DVD/player/recorder and camcorder. HDMI TX consists of one HMDI transmitter controller and one HMDI transmitter PHY.
It supports following features:

- Video formats:
  - All CEA-861-E video formats up to 1080p at 60 Hz and 720p/1080i at 120 Hz
  - Optional HDMI 1.4b video formats (configuration dependent)
    - All CEA-861-E video formats up to 1080p at 120 Hz
    - HDMI 1.4b 4K x 2K video formats
    - HDMI 1.4b 3D video modes with up to 340 MHz (TMDS clock)
  - HDMI 2.0 video formats, All CEA-861-F video formats
- Colorimetry, 24-bit RGB 4:4:4
- Pixel clock from 13.5 MHz up to 600 MHz
- Up to 192 kHz IEC60958 audio sampling rate
- Flexible synchronous enable per clock domain to set functional power down modes
- AMBA APB 3.0 register access
- I2C DDC, EDID block read mode
- SCDC I2C DDC access
- TMDS Scrambler to enable support for 2160p@60Hz with RGB 4:4:4
- Integrated CEC hardware engine

32.2 Block Diagram

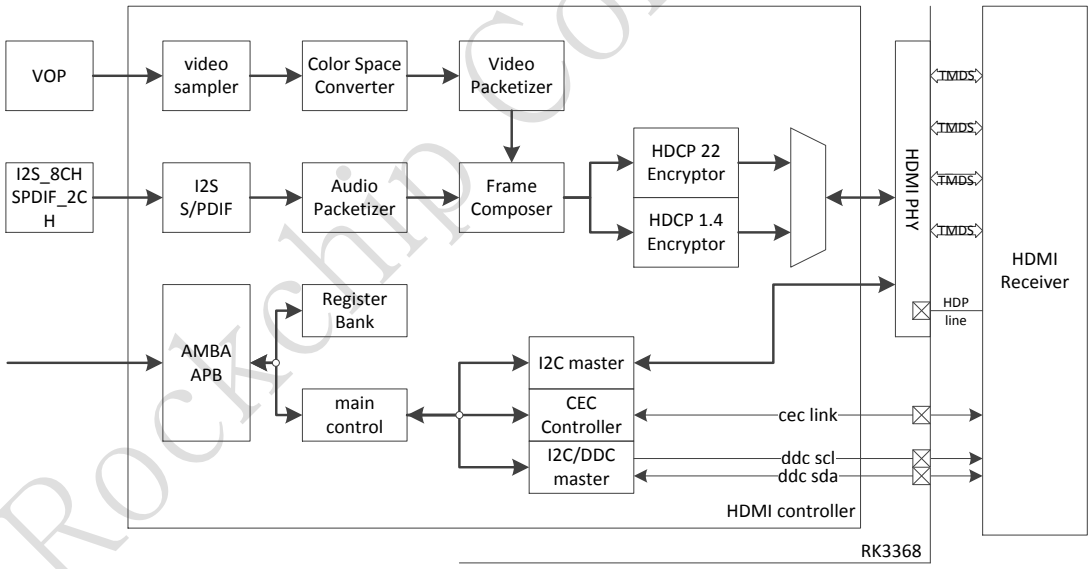

Fig. 32-1 HDMI TX Block Diagram

32.3 Function Description

32.3.1 IOMUX

Customer should notice that CEC channel use IO_EDPHDMIcecinout_ISPflashtrigin_GPIO30gpio3c7 and DDC channel use PAD IO_HDMII2Cscl_I2C5HDMIscl_GPIO30gpio3d3 and IO_HDMII2Csda_I2C5HDMIsda_GPIO30gpio3d2.
Note: PAD IO_HDMI_DDCCEC is Ground reference for the Hot Plug Detect signal.

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32.3.2 Pixel Clock

When using HDMI as display solution, customers must set pixel clock source to NEW_PLL for VOP and HDMI controller.

32.3.3 Video Pixel Sampler

In PX5, HDMI data comes from VOP. The Video pixel sampler block is responsible for the video data synchronization, according to the video data input mapping defined by the Color Depth (Deep Color) and format configuration. Register must be set correctly or cohesively with VOP output.

32.3.4 Video Data Processing

The video processing contains video format timings, pixel encodings (RGB to YCbCr, or YCbCr to RGB), colorimetry and corresponding requirements. This function is implemented by some functional blocks, Video Capture block, Color Space Conversion block, and Deep Color block. The input video pixels can be encoded in either RGB, YCBCR 4:4:4 or YCBCR 4:2:2 formats by Color Space Conversion block.

The input Video data can have a pixel size of 24bits. The deep color block is used to deal with different pixel size. Video at the default 24-bit color depth is carried at a TMDS clock rate equal to the pixel clock rate. Higher color depths are carried using a correspondingly higher TMDS clock rate. HDMI Transmitter support video formats with TMDS rates below 25MHz (e.g. 13.5MHz for 480i/NTSC) that can be transmitted using a pixel-repetition scheme by setting relative registers.

Color Space Conversion

HDMI Transmitter Color space conversion (CSC) is responsible for carrying out the following video color space conversion functions:

- RGB to/from YCbCr
- 4:2:2 to/from 4:4:4 up (pixel repetition or linear interpolation)/down-converter
- Limited to/from full quantization range conversion
- The CSC supports all the timings reported in the CEA-861-D specification and the following pixel modes:
- RGB 444 and YCbCr 444: 24, 30, 36, and 48 bits
- YCbCr 422: 16, 20, and 24 bits

The color space conversion matrix is ruled by the following equations listed in below figure.

$$out_1 = (X_1 \times in_1/4096 + X_2 \times in_2/4096 + X_3 \times in_3/4096 + X_4) \times 2^{scale}$$
$$out_2 = (Y_1 \times in_1/4096 + Y_2 \times in_2/4096 + Y_3 \times in_3/4096 + Y_4) \times 2^{scale}$$
$$out_3 = (Z_1 \times in_1/4096 + Z_2 \times in_2/4096 + Z_3 \times in_3/4096 + Z_4) \times 2^{scale}$$

Fig. 32-2 HDMI Color Space Conversion Matrix Equations

Note: Color Space Conversion to and from YCrCb 4:2:0 is not supported.

32.3.5 Audio Data Processing

The HDMI TX audio process contain audio clock regeneration, placement of audio samples within packets, packet timing control, audio sample rates setting, and channel/speaker assignments. This function is implemented by Audio Capture blocks

The Audio Capture support either SPDIF or four channel I2S input. SPDIF input supports audio sampling rates from 32 to 192 KHz. The I2S input supports from 2-channel to 8-channel audio up to 192 KHz.

The scheme of audio processing as shown in the figure below:

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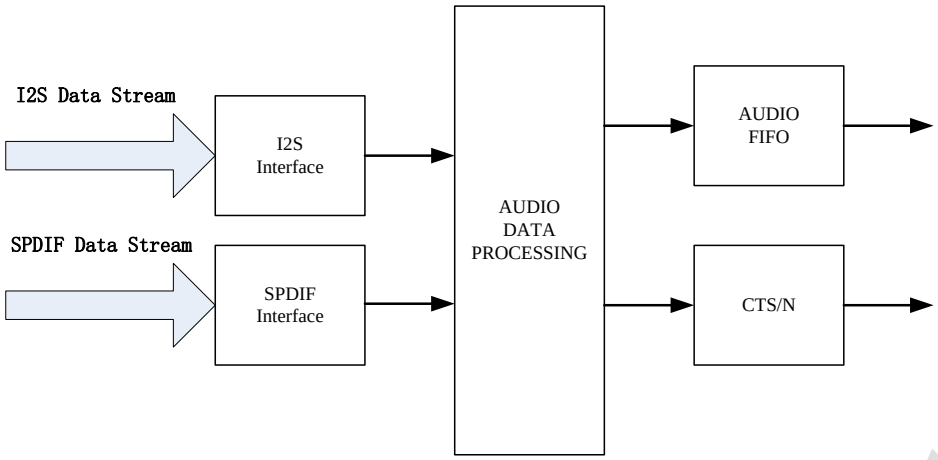

Fig. 32-3 HDMI Audio Data Processing Diagram

1.I2S

The function of this module is to implement I2S audio input feature. The incoming audio stream is captured, processed then transmitted into the TMDS link. Four I2S inputs also allow transmission of DVD-Audio and decoded Dolby Digital to A/V Receivers and high-end displays. The interface supports from 2-channel to 8-channel audio up to 192 kHz. The I2S pins must also be coherent with mclk. The appropriate registers must be configured to describe the format of audio being input. This information is passed over the HDMI link in the CEA-861D Audio Info (AI) packets. Table shows the I2S 8 channel audio formats that are supported for each of the video formats.

Table 32-1 HDMI TX I2S 2 Channel Audio Sampling Frequency

Video Format	32kHz	44.1kHz	48kHz	88.2kHz	96kHz	176.4kHz	192kHz
720x480p /720x576p	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1440x480i/ 1440x576i	Yes	Yes	Yes	Yes	Yes	Yes	Yes
720p	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1080i	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1080p	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 32-2 HDMI TX I2S 8 Channel Audio Sampling Frequency

Video Format	32kHz	44.1kHz	48kHz	88.2kHz	96kHz	176.4kHz	192kHz
720x480p /720x576p	Yes	Yes	Yes	No	No	No	No
1440x480i/ 1440x576i	Yes	Yes	Yes	Yes	No	No	No
720p	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1080i	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1080p	Yes	Yes	Yes	Yes	Yes	Yes	Yes

2.SPDIF

The function of this module is to implement SPDIF audio input feature. The incoming audio stream is captured, processed then transmitted into the TMDS link. SPDIF stream can carry 2-channel uncompressed PCM data (IEC 60958) or a compressed bit stream for multi-channel (IEC 61937) formats. The audio data capture logic forms the audio data into packets in accordance with the HDMI specification. SPDIF input supports audio sampling rates from 32 to 192 KHz. The following shows the SPDIF audio formats that are supported for each of the video formats

Table 32-3 HDMI SPDIF Sampling Frequency at Each Video Format

Video Format	32kHz	44.1kHz	48kHz	88.2kHz	96kHz	176.4kHz	192kHz
720x480p /720x576p	Yes	Yes	Yes	Yes	Yes	No	No
1440x480i/ 1440x576i	Yes	Yes	Yes	Yes	Yes	No	No

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720p	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1080i	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1080p	Yes	Yes	Yes	Yes	Yes	Yes	Yes

3.Audio Sample Clock Capture and Regeneration

Audio data is carried across the HDMI link, which is driven by a TMDS clock running at a rate corresponding to the video pixel rate, does not retain the original audio sample clock. The task of recreating this clock at the Sink is called Audio Clock Regeneration.

The HDMI Transmitter determine the fractional relationship between the TMDS clock and an audio reference clock (128 audio sample rate [fs]) and pass the numerator and denominator of that fraction to the HDMI Sink across the HDMI link. The Sink then re-create the audio clock from the TMDS clock by using a clock divider and a clock multiplier.

The exact relationship between the two clocks will be.

$128 \cdot f_s = f_{TMDS_clock} * N / CTS$.

The scheme of the Audio Sample Clock Capture and Regeneration as shown below:

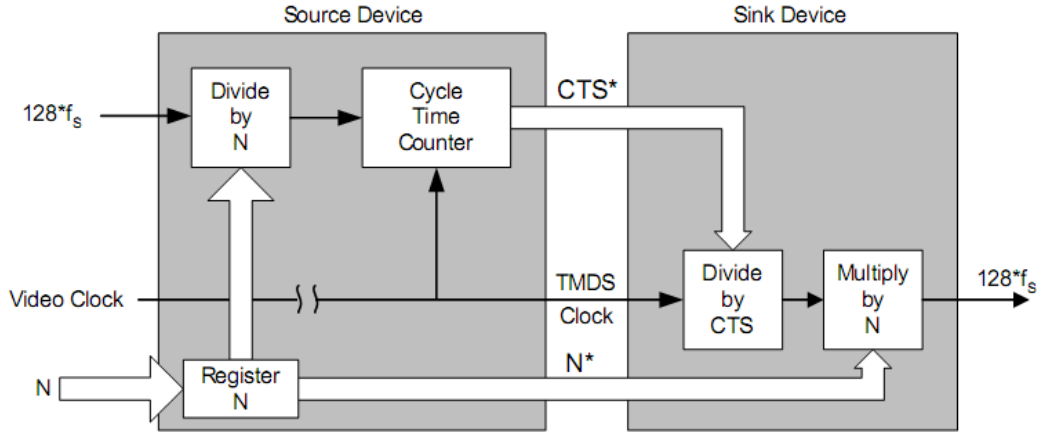

Fig. 32-4 HDMI Audio Clock Regeneration Model

Because there is no audio clock carried through the HDMI link, only the TMDS clock is used. Software sets the CTS/N with a value taken form the below table, which shows the CTS and N value for the supported standard. All other TMDS clocks are not supported; The TMDS clocks divided or multiplied by 1,001 coefficients are not supported.

Table 32-4 HDMI CTS and N table

	TMDS Clock (MHz)													
	25.2		27		54		74.25		148.5		297		597	
Fs (kHZ)	N	CTS	N	CTS	N	CTS	N	CTS	N	CTS	N	CTS	N	CTS
32	4096	25200	4096	27000	4096	54000	4096	74250	4096	148500	3072	222750	3072	445500
44.1	6272	28000	6272	30000	6272	60000	6272	82500	6272	165000	4704	247500	9408	990000
48	6144	25200	6144	27000	6144	54000	6144	74250	6144	148500	5120	247500	6144	495000
88.2	12544	28000	12544	30000	12544	60000	12544	82500	12544	165000	9408	247500	18816	990000
96	12288	25200	12288	27000	12288	54000	12288	74250	12288	148500	10240	247500	12288	495000
176.4	25088	28000	25088	30000	25088	60000	25088	82500	25088	165000	18816	247500	37632	990000
192	24576	25200	24576	27000	24576	54000	24576	74250	24576	148500	20480	247500	24576	4950000

32.3.6 DDC

The DDC functional block is used for configuration and status exchange between the HDMI Source and HDMI Sink. HMDI Transmitter Controller has I2C Master Interface for DDC transactions. It enables for host controller to read EDID, HDCP authentication by issuing simple register access. The I2C bus speed is limited by DDC specification. DDC bus access frequency can be controlled.

32.3.7 EDID

Extended Display Identification Data (EDID) was created by VESA to enable plug and play

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capabilities of monitors. This data, which is stored in the sink device, describes video formats that the DTV Monitor is capable of receiving and rendering. The information is supplied to the source device, over the interface, upon the request of the source device. The source device then chooses its output format, taking into account the format of the original video stream and the formats supported by the DTV Monitor. The function of this module is to implement EDID feature.

32.3.8 HDCP

HDMI Transmitter has a capability for HDCP authentication by hardware. The function of this module is to implement HDCP encryption feature. This feature can be turned on or off depending on register setting.
PX5 supports up to HDCP 2.2. HDCP 1.4 is done by HDMI Transmitter itself. HDCP22 is descripted in next chapter.

32.3.9 Hot Plug Detect

HDMI Transmitter has a capability for detecting the Sink plug in or plug out, and launch an interrupt and registers state indicating for software controlling.

32.3.10 TMDS encoder

The TMDS encoder converts the 2/4/8 bits data into the 10 bit DC-balanced TMDS data. HDMI TX put the TMDS encoding on the audio /video /aux data received from the HDCP XOR mask. This data is output onto three TMDS differential data lines along with a TMDS differential clock.

32.3.11 CEC

The CEC functional block provides high-level control functions between all of the various audiovisual products in a user's environment through one line.

32.4 HDMI PHY

The HDMI Tx PHY is the physical layer of an HDMI digital transmitter (source), capable of encoding and transmitting high-speed data streams carrying RGB video, audio, and control information.
The HDMI Tx PHY includes one PLL that synthesizes the high-speed serial bit clock (required by the transmitter) from a reference pixel clock with a frequency that can vary between 13.5-600 MHz for HDMI operation. The transmitter is capable of transmitting up to 18 Gbps using three lanes for HDMI operation.
The HDMI Tx PHY drives audio and video across the TMDS data channels. For HDMI operation, each serial TMDS link has a data-rate range of 0.25-6 Gbps. The HDMI Tx PHY also drives the TMDS clock at 1/10th of the serial data rate with a frequency range of 25-340 MHz for data rates below 3.4Gbps. For a serial data rate between 3.4Gbps and 6Gbps, the HDMI Tx PHY drives the TMDS clock at 1/40th of the serial data rate.
Within the HDMI Tx PHY, additional support blocks exist:

- Bandgap block (for blocks biasing)
- Resistor Calibration block
- ADC
- Analog test bus

An external, 1.62-kΩ reference resistor (connected to ground) is required for precise current and biasing (within each block).

32.4.1 I2C interface

The HDMI Tx PHY's status and configuration is accessed through internal I2C interface. The internal I2C interface is mapped to PHYConfiguration Registers in HDMI TX registers at address 0x3020. Customers can access these registers though HDMI controller's Register. Operating to these register can trigger one i2c write or i2c read.

32.4.2 PLL Operation

The PLL is responsible for the generation of the high-speed serialization clocks, as well as for the other clocks involved on the data transmission (clocks on the TMDS lanes) or involved on the supported testability.

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The PLL can be configured for the video modes required for each of the supported modes of operation.

32.4.3 HPD/RXSENSE Operation

For HDMI operation, to detect the presence of an HDMI sink system, the HDMI Tx PHY supports two different mechanisms to determine whether a Hot Plug Detect (HPD) signal is available and whether HDMI sink terminations are present and powered.
If voltage is present on the HPD line provided by the HDMI sink system, the HPD circuit identifies and asserts the SNKDET signal on the PHY's digital interface, provided the voltage level is 2.4-5.3 V, as specified by the HDMI specification. The HPD circuit enables the HDMI Tx PHY to identify whether the HDMI sink system is connected and is providing a correct HPD voltage level.
The RXSENSE circuit determines, for each TMDS data and clock lane for the HDMI operation, whether the HDMI sink terminations are present. Users can observe RXSENSE that reflects the state of the TMDS lanes evaluated from phy_stat0 in HDMI controller register.

32.4.4 Power Collapsing

The HDMI Tx PHY supports power collapsing. This feature enables the HDMI Tx PHY to stay—in any given combination of power supply availability (present, floating, or grounded)—in a known state with low current consumption. This low current consumption remains similar to, or below, the current consumption in power-down mode.
Power collapsing provides customers who require control over the power lines with independent control of the PHY power supplies.

32.4.5 Operating Modes

The HDMI Tx PHY can be placed in two different operating modes: Power-Down and Active. Customers can assert a soft reset through the HDMI control registers. This soft reset clears all system FSMs except I2C, JTAG, and control registers.
For each separate video mode in which the HDMI Tx PHY is set to transmit, due to different operating frequency, color depth and pixel repetition that characterizes each one, you need to configure the HDMI Tx PHY block for correct operation and optimized performance. It is recommended for the PHY configuration through the I2C or JTAG interface to be done while the PHY is in power-down mode.
Configuration involves programming the PLL dividers and analog configuration as well as the analog drivers' source termination value, signals voltage level, pre-emphasis and slope boosting. This programming is done through the I2C interface.

32.5 Register Description

The address offset of the HDMI TX is 0xff980000,it contains 16 address section. The offset of the table of Register Summary must multiple with 4 when software configure it。Like the Interrupt registers，its base address is 0x0100.If we want to configure it, its real address is 0xff980000+0x0100*4.

32.5.1 Register Summary

Name	Offset	Size	Reset Value	Description
Identification Registers	0x0000	B		Identification releated registers
Interrupt registers	0x0100	B		Interrupt releated registers
Video Sampler registers	0x0200	B		Video Sampler registers
Video Packetizer registers	0x0800	B		Video Packetizer registers
Frame Composer Registers	0x1000	B		Frame Composer Registers
HDMI Source PHY Registers	0x3000	B		HDMI Source PHY Registers
I2C Master PHY Registers	0x3020	B		I2C Master PHY Registers
Audio Sampler Registers	0x3100	B		Audio Sampler Registers

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Name	Offset	Size	Reset Value	Description
Main Controller Registers	0x4000	B		Main Controller Registers
Color Space Converter Registers	0x4100	B		Color Space Converter Registers
HDCP Encryption Engine Registers	0x5000	B		HDCP Encryption Engine Registers
HDCP BKSV Registers	0x7800	B		HDCP BKSV Registers
HDCP AN Registers	0x7805	B		HDCP AN Registers
Encrypted DPK Embedded Storage Registers	0x780E	B		Encrypted DPK Embedded Storage Registers
CEC Engine Registers	0x7D00	B		CEC Engine Registers
I2C Master Registers	0x7E00	B		I2C Master Registers for E-DDC/SCDC

32.6 Application Notes

This chapter describes how to bring up HDMI transmitter in your system. As shown few examples below, these introduce the basically HDMI transmitter application, likes, the Hot Plug Detect, EDID read back, multiple audio format input and different video resolution displaying.

32.6.1 Initial Operation

The default HDMI transmitter is configured to 24bit RGB 1080P resolution video with 8 channel 48K sample I2S format audio input. It is easily for customer to turn on HDMI transmitter without doing more complex operation. Just do the step, reset the HDMI TX.

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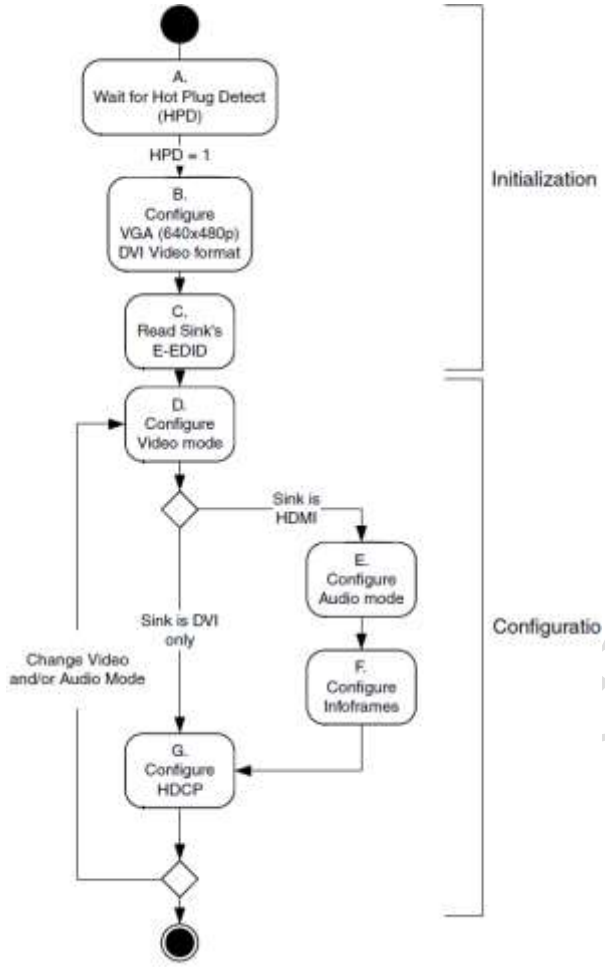

Fig. 32-5 HDMI programming sequence

32.6.2 Step A: Hot Plug Detection

The Hot Plug Detect information notifies the HDMI TX controller when the cable is plugged into a Sink (Receiver) device and when the device is ready to receive video content.
The Hot-Plug Detection indication is a +5V signal on the HDMI cable. The HDMI TX PHY performs level-shifting to a low-level digital signal.
For this step, registers are used on the HDMI TX PHY.
To check the Hot Plug Detect status, the following steps have to be followed:
1. Power-on the HDMI TX PHY HPD Detector. Write 1'b1 in the phy_conf0.enhpdrxsense bit field register.
2. Check the HPD status bit.

- Read the phy_stat0.HPD bit field register.
  - If HPD = 0, the Hot Plug signal is low (no Sink (Receiver) detected).
  - If HPD = 1, the Hot Plug signal is high (Sink (Receiver) detected).

Note: Some receivers may turn off the Hot-Plug Detect indication when the HDMI input is not selected, even when the cable is plugged in.

You can control this function by using the interrupt signal and proper registers from the HDMI transmitter with few operations.

32.6.3 Step B: Configure Video Mode

Once a receiver is detected, it is necessary to start sending video content in DVI mode. Because some receivers need to receive a video signal (TMDS clock more precisely), it is important to clock all the digital logic and be able to reply on an E-EDID read access (Step C). Following flow is a configuration example of VGA (640x480p).

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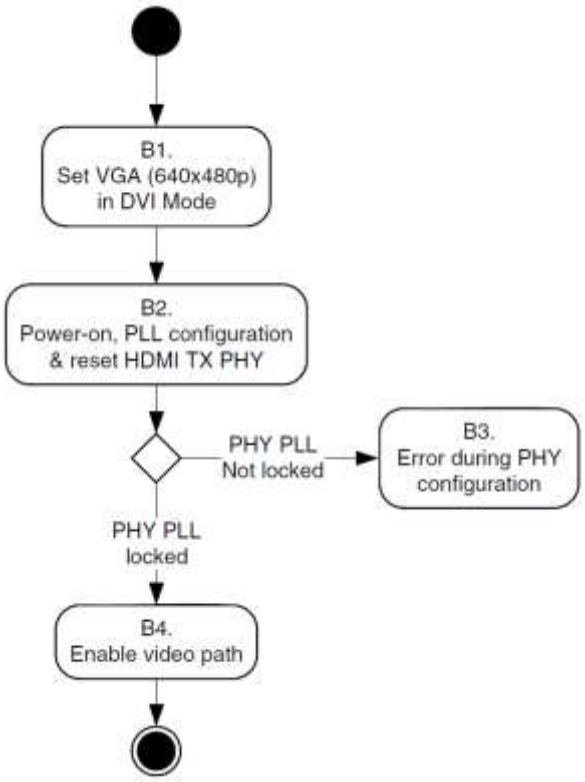

Fig. 32-6 Configure Video Mode

The VGA (640x480p) mode is mandatory for all receivers and transmitters, because it requires the lowest data rate possible on the HDMI specification.

1. Set VGA (640x480p) in DVI mode.

- To select the Video Mapping input RGB444, 8-bits per color components (24-bits RGB), write "1" in the tx_invid0.video_mapping register.
- To configure VGA timing information (CEA mode 1):
  - Write "0" in the fc_invidconf.vsync_in_polarity register.
  - Write "0" in the fc_invidconf.hsync_in_polarity register.
  - Write "1" in the fc_invidconf.de_in_polarity register.
  - Write "0" in the fc_invidconf.r_v_blank_in_osc register.
  - Write "0" in the fc_invidconf.in_I_P register.
  - 640 H active pixels: 0x280
    - Write "2" in the fc_inhactiv1.H_in_activ register.
    - Write "0x80" in the fc_inhactiv0.H_in_activ register.
  - 480 V active pixels: 0x1E0
    - Write "1" in the fc_invactiv1.V_in_activ register.
    - Write "0xE0" in the fc_invactiv0.V_in_activ register.
  - 160 H blanking pixels: 0xA0
    Write "0xA0" in thefc_inhblank0.H_in_blank.
  - 45 V blanking pixels: 0x2D
    Write "0x2D" in the fc_invblank.V_in_blank register.
  - 16 Sync offset: 0x10
    Write "0x10" in the fc_hsyncindelay0.H_in_delay register.
  - 10 Vsync offset: 0x0A
    Write "0x0A" in the fc_vsyncindelay0.V_in_delay register.
  - 96 HSync pulse width: 0x60
    Write "0x60" in the fc_hsyncinwidth0.H_in_width register.
  - 2 VSync pulse width: 0x02

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Write "0x02" in the fc_vsyncinwidth0.V_in_width register.

- To select the HDMI mode:

  Write "1" in the fc_invidconf.DVI_modez register.

2. Power-on, configure the PLL, and reset the HDMI TX PHY.

The HDMI TX PHY has an internal PLL that generates TMDS clock from Pixel clock input. The power-on sequence and PLL configuration is part of the HDMI TX PHY documentation.

To configure HDMI TX PHY, you are generally required to:

a. Place the PHY in reset by writing 0x01 in the mc_phyrstz register.

b. Write the desired color depth and the pixel repetition in the vp_pr_cd register.

c. After a PHY-dependent time, it is required to lift the reset by writing 0x00 to the mc_phyrstz register.

d. Clear the pddq, txpwron configuration bits, define the data enable polarity and clear the interface control selection by writing these values in the phy_cfg0 register.

e. Set the PHY slave address by writing 0x69 in the phy_i2cm_slave register.

f. You are required to look up the configuration for your intended video mode and write those values on the PHY I2C interface. The baseline flow to write to the PHY through the I2C interface is:

- Write register address in the phy_i2cm_address register.
- Write data in the phy_i2cm_datao_1 (MSB, [15:8]) and phy_i2cm_datao_0 (LSB, [7:0]) registers.
- Initialize the write operation by writing 8'h10 in the phy_i2cm_operation register.
- Wait for a done interruption from the I2C master.

g. After all of the required PHY I2C registers have been configured, you now need to set the txpwron bit in the PHY_CONF0 register, leaving the remaining bits in the state defined previously.

3. Handle errors during PHY configuration.

If the PHY is not correctly configured, the PLL does not lock and no activity can happen on the HDMI output.

4. Enable the video path.

a. Set default parameters in the HDMI TX Controller, Control period duration: 12: 0x0C.

Write "0x0C" in the fc_ctrldur.ctrlperiodduration bit field register.

b. Set default parameters in the HDMI TX Controller, Extended Control period duration: 32: 0x20.

Write "0x20" in the fc_exctrldur.exctrlperiodduration bit field register.

c. Set default parameters in the HDMI TX Controller, Max spacing between extended Control period duration: 1: 0x1.

Write "0x01" in the fc_exctrlspac.exctrlperiodspacing bit field register.

d. Set default parameters in the HDMI TX Controller, Preamble filters to fill TMDS data channels.

- Write "0x0B" in the fc_ch0pream.ch0_preamble_filter bit field register.
- Write "0x16" in the fc_ch1pream.ch1_preamble_filter bit field register.
- Write "0x21" in the fc_ch2pream.ch2_preamble_filter bit field register.

e. Enable pixel clock data path:

Write "0" in the mc_clkdis.pixelclk_disable bit field register.

f. Enable TMDS clock data path:

i. Write "0" in the mc_clkdis.tmdsclk_disable bit field register.

ii. Re-Write the VSync pulse width in the fc_vsyncinwidth0.V_in_width bit field register.

After these steps, it is expected to observe a video picture in VGA mode on the receiver side.

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32.6.4 Step C: Reading E-EDID

The E-EDID is a memory present on the receiver side. It contains the video and audio capabilities of the receiver. It is important to read this information and parse it in order to configure the transmitter system in accordance to what the receiver supports. The E-EDID is read using the E-DDC channel present on the HDMI cable.
The E-DDC channel protocol is based on I2C, with segment pointer addressing extension (such as, Extended Read). The hdmi tx controller has an I2C Master controller for this purpose.
1. To perform a "normal" read operation
- Set I2C slave address.
  Write i2cm_slave.slaveaddr[6:0] bit field register.
- Set I2C register address.
  Write i2cm_address.address[7:0] bit field register.
- Activate Read operation.
  Write "1" in the i2cm_operation.rd bit field register.
- Wait for interruption.
  Wait for the ii2cmasterdone interrupt in the ih_i2cm_stat0 register.
- Read data result.
  Read data in the i2cm_datai.datai[7:0] bit field.

2. To perform an E-DDC extended read operation
- Set I2C slave address.
  Write i2cm_slave.slaveaddr[6:0] bit field register.
- Set I2C segment address.
  Write the i2cm_segaddr.seg_addr bit field register.
- Set I2C segment pointer.
  Write i2cm_segptr.segptr bit field register.
- Activate Read operation.
  Write "1"in the i2cm_operation.rd_ext bit field register.
- Wait for interruption.
  Wait for the ii2cmasterdone interrupt in the ih_i2cm_stat0 register.
- Read data result.
  Read data in the i2cm_datai.datai[7:0] bit field register.

32.6.5 Step D: Configure Video Mode

The video mode selected has to match what is supported by the source of the video (Transmitter side) and the sink of the video (receiver side). The video capabilities of the receiver are extracted from the E-EDID information (refer to step C).
To configure any video mode, a similar procedure in step B has to be followed.

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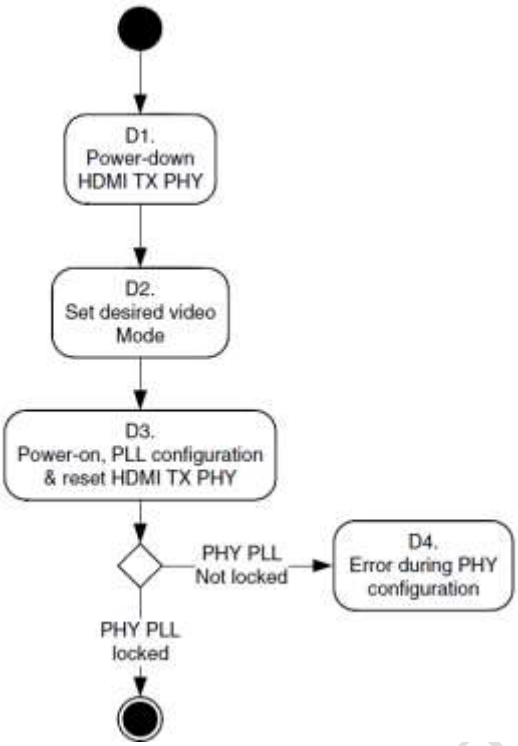

Fig. 32-7 Video Mode Configuration

Before any change is made on the video modes, is it recommended you power-down the HDMI TX PHY to avoid any unexpected behavior on the receiver side.

1. Power-down HDMI TX PHY Detector

- Place the PHY in reset by writing 0x01 in the mc_phyrstz register.
- Then proceed to set SVSRET_MODEZ to 1'b0, PDDQ to 1'b1 and TX_PWRON to 1b0, by writing 2'h12 in the phy_conf0 register.

2. Set desired video mode.

- To select the Video Mapping input mode (RGB444, YCC444, YCC422). Write the video code in the tx_invid0.video_mapping bit field register.
- Set video timing information configuration:
  - Write the fc_invidconf.vsync_in_polarity register.
  - Write the fc_invidconf.hsync_in_polarity register.
  - Write the fc_invidconf.de_in_polarity register.
  - Write the fc_invidconf.r_v_blank_in_osc register.
  - Write the fc_invidconf.in_I_P register.
  - H active pixels
    - Write the fc_inhactiv1.H_in_activ register.
    - Write the fc_inhactiv0.H_in_activ register.
  - V active pixels

    If the desired video mode is 3D, the fc_invact_2d_1, fc_invact_2d_0, and all registers in the following table must be written.

Table 32-5 HDMI 3D structure table

3D STRUCTURE	fc_actspc_hdlr_cfg	
	fc_actspc_hdlr_tgl	fc_actspc_hdlr_en
Frame Packing for interlaced format	1	1
Frame Packing for progressive format	0	1
Field alternative	0	0
Line Alternative	0	0
Side-by-Side(FULL)	0	0
L+depth	0	1

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L+depth+graphics+graphics-depth	0	1
Top-and-Bottom	0	0
Reserved for Future Use	0	0
Side-by-Side(FULL)	0	0
Reserved for Future Use	0	0
Not in use	0	0

- Write the fc_invact_2D_1 register with the 2D Vertical video active[11:8].
- Write the fc_invact_2D_0 register with the 2D Vertical video active[7:0]

The following registers must always be written with the full V active of the desired video
regardless of the type of the video mode.

- Write the fc_invactiv1.V_in_activ register.
- Write the fc_invactiv0.V_in_activ register.

■ H blanking pixels

Write the fc_inhblank0.H_in_blank register.

■ V blanking pixels

Write the fc_invblank.V_in_blank register.

■ HSync offset

Write the fc_hsyncindelay0.H_in_delay register.

■ VSync offset

Write the fc_vsyncindelay0.V_in_delay register.

■ HSync pulse width

Write the fc_hsyncinwidth0.H_in_width register.

■ VSync pulse width

Write the fc_vsyncinwidth0.V_in_width register.

- Select DVI or HDMI mode:
  Write "1" for HDMI in the fc_invidconf.DVI_modez bit field register.

3. Power-on, configure the PLL, and reset the HDMI TX PHY (same as Step B2).

The HDMI TX PHY has an internal PLL that generates TMDS clock from Pixel clock input. All the power-on sequence and PLL configuration is part of the HDMI TX PHY documentation.

The HDMI TX Controller provides all registers necessary for the PHY sequence implementation, which are presented in the "HDMI PHY Registers".

- Place the PHY in reset by writing 0x01 in the mc_phyrstz register.
- Write in the desired color depth and the pixel repetition in the vp_pr_cd register.
- After a PHY-dependent time, it is required to lift the reset by writing 0x00 to the mc_phyrstz register.
- Clear the pddq, txpwron configuration bits, define the data enable polarity and clear the interface control selection by writing these values in the phy_conf0 register.
- Set the PHY slave address by writing 0x69 to the phy_i2cm_slave register.
- Look up the configuration for your intended video mode and write those values on the PHY I2C interface. The baseline flow to write in the PHY through the I2C interface is
  ■ Write the register address in the phy_i2cm_address register.
  ■ Write data into phy_i2cm_datao_1 (MSB, [7:0]) and phy_i2cm_datao_0 (LSB, [7:0]) registers.
  ■ Initialize the write operation by writing 8'h10 in the phy_i2cm_operation register.
  ■ Wait for a done interrupt from the I2C master.
- After all of the required PHY I2C registers have been configured, you now need to set the phy_conf0.txpwron register, leaving the remaining bits in the state defined previously.

32.6.6 Step E: Audio input configuration

HDMI transmitter audio support either SPDIF or four channel I2S input. SPDIF input supports

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audio sampling rates from 32 to 192 KHz. The I2S input supports from 2-channel to 8-channel audio up to 192 KHz. The default audio format is I2S input with 8 channels. The audio sample rate is 48K.

- Configure Audio Input Format with I2S Steps:
  - ■ Select I2S input.
  Write "1" in the aud_conf0.i2s_select bit field register.
  - ■ Enable I2S inputs:
  Write "1" in the aud_conf0.i2s_in_en[3:0] bit field register.
  - ■ Set I2S Mode [Standard | Right-justified | Left-justified | Burst1 | Burst2]:
  Write the aud_conf1.i2s_mode[2:0] bit field register.
  - ■ Set I2S data width [16 bits up to 24 bits]:
  Write the aud_conf1.i2s_width[4:0] bit field.
- Configure Audio Input Format with SPDIF Steps:
  - ■ Select SPDIF input.
  Write "0" in the aud_conf0.i2s_select bit field register.
  - ■ Set S/PDIF Linear-PCM or Non-Linear PCM audio samples:
  Write the aud_spdif1.setnlpcm bit field register.
  - ■ Set SPDIF data width [16 bits up to 24 bits]:
  Write the aud_spdif1.spdif_width[4:0] bit field.
- Configure Audio Parameters Steps:
  - ■ Set Audio input frequency clock FS ratio factor [128 Fs | 256 Fs |512 Fs]:
  Write the aud_inputclkfs.lfsfactor bit field register.
  - ■ Set Audio fixed N factor for Audio Clock Regeneration. This factor depends on the audio sampling rate and video mode.
  Write the aud_n1.audN, aud_n2.audN, and aud_n3.audN bit field registers.
  - ■ Set Audio CTS factor for Audio Clock Regeneration. This factor can be generated automatically or manually.
  For Automatic CTS generation
  Write "0" on the bit field "CTS_manual", Register 0x3205: AUD_CTS3
  For Manual CTS setting
  Write "1" in the aud_cts3.CTS_manual register bit field.
  Write the aud_cts1.audCTS, aud_cts2.audCTS, aud_cts3.audCTS bit field registers.
  - ■ Enable Audio sampler block:
  Write "0" in the mc_clkdis.audclk_disable bit field register.

32.6.7 Step F: Configure Infoframes

This step is only relevant when the HDMI TX controller is configured in HDMI mode (refer to "Step D: Configure Video Mode"). In DVI mode, no Infoframes are transmitted. The configuration of the Infoframes is essential to correctly inform the HDMI Receiver about the content of the video and audio format been transmitted. All the detailed information is provided in the HDMI 1.4b Specification.

32.6.8 Step G: Configure HDCP 1.4

Following flow is an initialization of HDCP 1.4. To detect if the HDMI TX Controller is HDCP capable:

1. Read the product_id1.product_id1 bit field register:
- If product_id1 = 01, HDCP is not present.
- If product_id1 = C1, HDCP is present.

2. Select HDMI mode:

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- Write "1" for HDMI in the a_hdcpcfg0.hdmidvi bit field register.

3. Set the Data enable, Hsync and VSync polarity:

Write the dataenpol, vsyncpol, and hsyncpol bit fields of the a_vidpolcfg register.

4. Set encryption on:

- Write "64 (0x40)" for "OesswindowSize" in the a_oesswcfg register.
- Write "0" for "no HDCP bypass" a_hdcpcfg0.BypassEncryption bit field register.
- Write "0" for "HDCP enable" in the a_hdcpcfg1.encryptiondisable bit field register.

5. Reset HDCP engine:

- Write "0" in the a_hdcpcfg1.swreset bit field register.

6. Configure Device Private Keys, which is illustrated in the flow chart Figure 3-6

7. Enable the encryption:

- Write "1" in the a_hdcpcfg0.rxdetect bit field register.

8. When connected to a repeater the SHA-1 calculation is needed to be done by software, which are indicated in the following steps

a. Wait for an interrupt to be triggered (a_apiintstat.KSVSha1calcint).

b. Request access to KSV memory through setting a_ksvmemctrl.KSVMEMrequest to 1'b1 and pool a_ksvmemctrl.KSVMEMaccess until this value is 1'b1 (access granted).

c. Read VH', M0, Bstatus, and the KSV FIFO.

The data is stored in the revocation memory.

d. Calculate the SHA-1 checksum (VH) over M0, Bstatus, and the KSV FIFO.

e. If the calculated VH equals the VH‵, set a_ksvmemctrl.SHA1fail to 0 and set a_ksvmemctrl.KSVCTRLupd to 1. If the calculated VH is different from VH‵ then set a_ksvmemctrl.SHA1fail to 1 and set a_ksvmemctrl.KSVCTRLupd to 1, forcing the controller to re-authenticate from the beginning.

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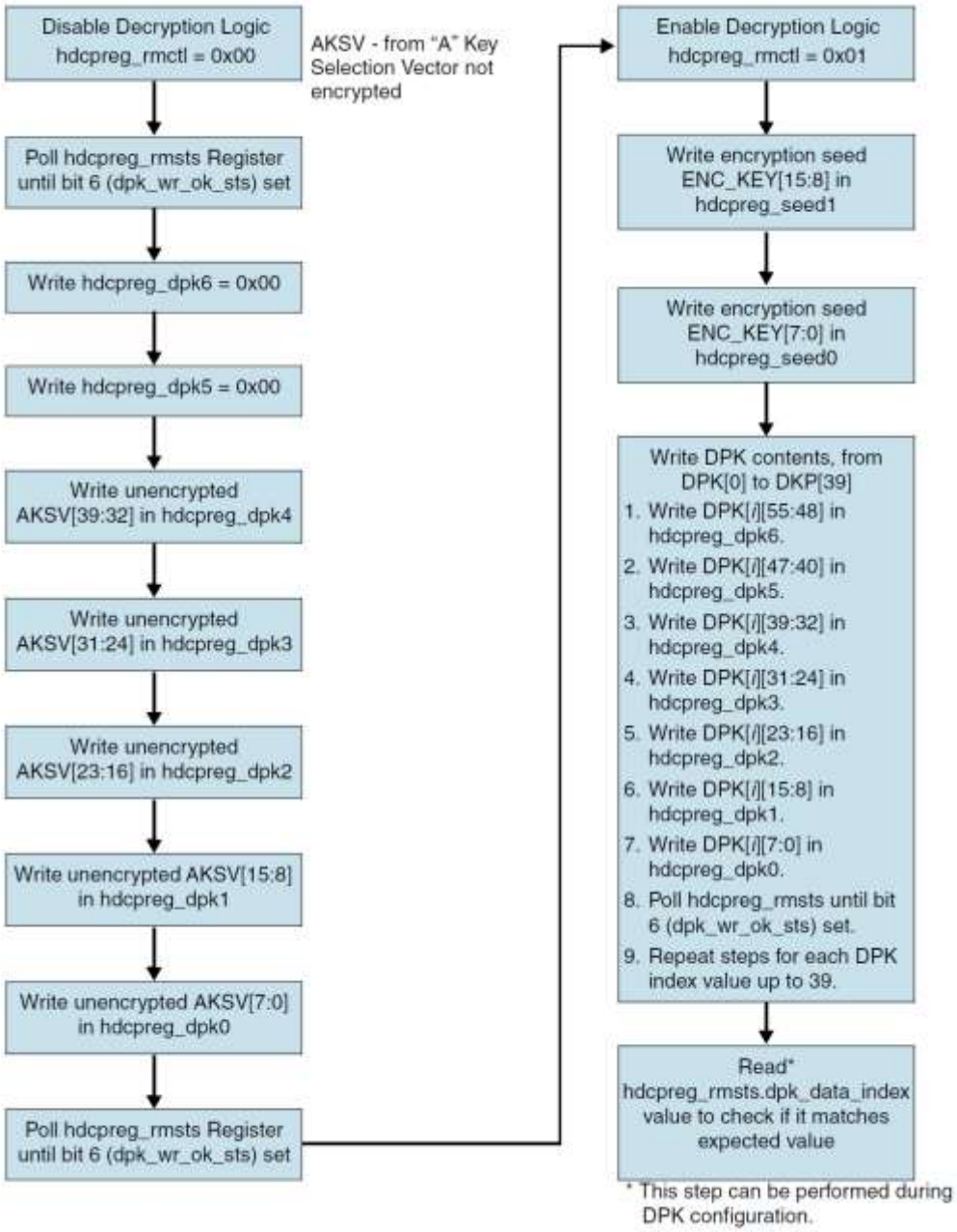

Fig. 32-8 Configuring Device Protection Key Process

32.6.9 CEC OPERATION

The CEC line is used for high-level user control of HDMI-connected devices.The HDMI TX contain CEC TX operations and CEC RX operations.

You can control this function by using the interrupt signal and proper registers from the HDMI transmitter with few operations. The register offset is from 0x7D00.

- Configure The CEC Step:
  - ■ Write the CEC logical address to cec_addr_l,cec_addr_h register
  - ■ Write the size of the frame in bytes which are available in the transmitter data buffer to cec_tx_cnt register
  - ■ Write the desired CEC data(including header and data blocks) to cec_tx_data0 to cec_tx_data15
  - ■ Write 1 to cec_ctrl.send register, to start the cec transmit.

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32.6.10 HDMI PHY MPLL configuration

Table 32-6 HDMI PHY MPLL Generic Configuration Settings

Pixel Clock (MHz)	Pixel Repetition	Color depth [bits]	OPMODE(HDMI1.4 OR 2.0)	Divider Settings				opmode		Divider settings								MPLL Charge pump settings								TMDSCLKSRC (MHz)
				prep_div		tmdsmhl_cntrl				fbdiv2_cntrl			fbdiv1_cntrl		ref_cntrl		n_cntrl	prop_cntrl			int_cntrl			gmp_cntrl		
				0006:Bit 14	0006:Bit 13	0006:Bit 12	0006:Bit 11	0006:Bit 10	0006:Bit 9	0006:Bit 8	0006:Bit 7	0006:Bit 6	0006:Bit 5	0006:Bit 4	0006:Bit 3	0006:Bit 2	0006:Bit 1	0006:Bit 0	0010:Bit 5	0010:Bit 4	0010:Bit 3	0010:Bit 2	0010:Bit 1	0010:Bit 0	0015:Bit 1	0015:Bit 0	
13.5	1	8	1.4	0	0	0	0	0	0	1	0	0	1	1	0	0	1	1	0	1	1	0	0	0	0	0	27
13.5	1	10	1.4	0	1	0	0	0	0	1	0	1	1	1	0	0	1	1	0	1	1	0	0	0	0	0	33.75
13.5	1	12	1.4	1	0	0	0	0	0	1	1	0	1	1	0	0	1	1	0	1	1	0	0	0	0	0	40.5
13.5	1	16	1.4	1	1	0	0	0	0	1	0	0	1	1	0	0	1	0	1	0	1	0	0	0	0	1	54
13.5	3	8	1.4	0	0	0	0	0	0	1	0	0	1	1	0	0	1	0	1	0	1	0	0	0	0	1	54
13.5	3	10	1.4	0	1	0	0	0	0	1	0	1	1	1	0	0	1	0	1	0	1	0	0	0	0	1	67.5
13.5	3	12	1.4	1	0	0	0	0	0	1	1	0	1	1	0	0	1	0	1	0	1	0	0	0	0	1	81
13.5	3	16	1.4	1	1	0	0	0	0	1	0	0	1	1	0	0	0	1	1	1	1	0	0	0	1	0	108
13.5	7	8	1.4	0	0	0	0	0	0	1	0	0	1	1	0	0	0	1	1	1	1	0	0	0	1	0	108
13.5	7	10	1.4	0	1	0	0	0	0	1	0	1	1	1	0	0	0	1	1	1	1	0	0	0	1	0	135
13.5	7	12	1.4	1	0	0	0	0	0	1	1	0	1	1	0	0	0	1	1	1	1	0	0	0	1	0	162
13.5	7	16	1.4	1	1	0	0	0	0	1	0	0	1	1	0	0	0	0	1	1	1	0	0	0	1	1	216
18	2	8	1.4	0	0	0	0	0	0	0	1	1	1	1	0	0	1	0	1	0	1	0	0	0	0	1	54
18	2	10	1.4	0	1	0	0	0	0	1	0	1	1	0	0	0	1	0	1	0	1	0	0	0	0	1	67.5
18	2	12	1.4	1	0	0	0	0	0	1	1	0	1	0	0	0	1	0	1	0	1	0	0	0	0	1	81
18	2	16	1.4	1	1	0	0	0	0	0	1	1	1	1	0	0	0	1	1	1	1	0	0	0	1	0	108
18	5	8	1.4	0	0	0	0	0	0	0	1	1	1	1	0	0	0	1	1	1	1	0	0	0	1	0	108
18	5	10	1.4	0	1	0	0	0	0	1	0	1	1	0	0	0	0	1	1	1	1	0	0	0	1	0	135
18	5	12	1.4	1	0	0	0	0	0	1	1	0	1	0	0	0	0	1	1	1	1	0	0	0	1	0	162
18	5	16	1.4	1	1	0	0	0	0	0	1	1	1	1	0	0	0	0	1	1	1	0	0	0	1	1	216
21.6	4	8	1.4	0	0	0	0	0	0	1	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	108
21.6	4	12	1.4	1	0	0	0	0	0	1	0	1	1	0	0	0	0	1	1	1	1	0	0	0	1	0	162
21.6	4	16	1.4	1	1	0	0	0	0	1	0	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	216
21.6	9	8	1.4	0	0	0	0	0	0	1	0	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	216
21.6	9	12	1.4	1	0	0	0	0	0	1	0	1	1	0	0	0	0	0	1	1	1	0	0	0	1	1	324
21.6	9	16	2.0	1	1	1	1	0	1	1	0	1	1	1	0	0	0	0	1	1	1	0	0	0	1	1	432
24	8	8	1.4	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0	1	1	1	0	0	0	1	1	216
24	8	16	2.0	1	1	1	1	0	1	1	1	0	1	0	0	0	0	0	1	1	1	0	0	0	1	1	432
25.175	No	8	1.4	0	0	0	0	0	0	0	1	0	1	1	0	0	1	1	0	1	1	0	0	0	0	0	25.175
25.175	No	10	1.4	0	1	0	0	0	0	1	0	1	0	1	0	0	1	1	0	1	1	0	0	0	0	0	31.46875
25.175	No	12	1.4	1	0	0	0	0	0	0	1	1	1	1	0	0	1	1	0	1	1	0	0	0	0	0	37.7625

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Pixel Clock (MHz)	Pixel Repetition	Color depth [bits]	OPMODE(HDMI1.4 OR 2.0)	Divider Settings				opmode		Divider settings								MPLL Charge pump settings								TMDSCLKSRC (MHz)
				prep_div		tmdsmhl_cntrl				fbdiv2_cntrl			fbdiv1_cntrl		ref_cntrl		n_cntrl	prop_cntrl			int_cntrl			gmp_cntrl		
				0006:Bit 14	0006:Bit 13	0006:Bit 12	0006:Bit 11	0006:Bit 10	0006:Bit 9	0006:Bit 8	0006:Bit 7	0006:Bit 6	0006:Bit 5	0006:Bit 4	0006:Bit 3	0006:Bit 2	0006:Bit 1	0006:Bit 0	0010:Bit 5	0010:Bit 4	0010:Bit 3	0010:Bit 2	0010:Bit 1	0010:Bit 0	0015:Bit 1	0015:Bit 0	
25.175	No	16	1.4	1	1	0	0	0	0	0	1	0	1	1	0	0	1	0	1	0	1	0	0	0	0	1	50.35
27	No	8	1.4	0	0	0	0	0	0	0	1	0	1	1	0	0	1	1	0	1	1	0	0	0	0	0	27
27	No	10	1.4	0	1	0	0	0	0	1	0	1	0	1	0	0	1	1	0	1	1	0	0	0	0	0	33.75
27	No	12	1.4	1	0	0	0	0	0	0	1	1	1	1	0	0	1	1	0	1	1	0	0	0	0	0	40.5
27	No	16	1.4	1	1	0	0	0	0	0	1	0	1	1	0	0	1	0	1	0	1	0	0	0	0	1	54
27	1	8	1.4	0	0	0	0	0	0	0	1	0	1	1	0	0	1	0	1	0	1	0	0	0	0	1	54
27	1	10	1.4	0	1	0	0	0	0	1	0	1	0	1	0	0	1	0	1	0	1	0	0	0	0	1	67.5
27	1	12	1.4	1	0	0	0	0	0	0	1	1	1	1	0	0	1	0	1	0	1	0	0	0	0	1	81
27	1	16	1.4	1	1	0	0	0	0	0	1	0	1	1	0	0	0	1	1	1	1	0	0	0	1	0	108
27	3	8	1.4	0	0	0	0	0	0	0	1	0	1	1	0	0	0	1	1	1	1	0	0	0	1	0	108
27	3	10	1.4	0	1	0	0	0	0	1	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	135
27	3	12	1.4	1	0	0	0	0	0	0	1	1	1	1	0	0	0	1	1	1	1	0	0	0	1	0	162
27	3	16	1.4	1	1	0	0	0	0	0	1	0	1	1	0	0	0	0	1	1	1	0	0	0	1	1	216
27	7	8	1.4	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0	1	1	1	0	0	0	1	1	216
27	7	10	1.4	0	1	0	0	0	0	1	0	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	270
27	7	12	1.4	1	0	0	0	0	0	0	1	1	1	1	0	0	0	0	1	1	1	0	0	0	1	1	324
27	7	16	2.0	1	1	1	1	0	1	1	0	0	1	1	0	0	0	0	1	1	1	0	0	0	1	1	432
31.5	No	8	1.4	0	0	0	0	0	0	0	1	0	1	1	0	0	1	1	0	1	1	0	0	0	0	0	31.5
33.75	No	8	1.4	0	0	0	0	0	0	0	1	0	1	1	0	0	1	1	0	1	1	0	0	0	0	0	33.75
35.5	No	8	1.4	0	0	0	0	0	0	0	1	0	1	1	0	0	1	1	0	1	1	0	0	0	0	0	35.5
36	No	8	1.4	0	0	0	0	0	0	0	1	0	1	1	0	0	1	1	0	1	1	0	0	0	0	0	36
36	2	8	1.4	0	0	0	0	0	0	0	1	0	1	0	0	0	0	1	1	1	1	0	0	0	1	0	108
36	2	10	1.4	0	1	0	0	0	0	1	0	1	1	0	0	1	0	1	1	1	1	0	0	0	1	0	135
36	2	12	1.4	1	0	0	0	0	0	0	1	1	1	0	0	0	0	1	1	1	1	0	0	0	1	0	162
36	2	16	1.4	1	1	0	0	0	0	0	1	0	1	0	0	0	0	0	1	1	1	0	0	0	1	1	216
36	5	8	1.4	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	1	1	1	0	0	0	1	1	216
36	5	10	1.4	0	1	0	0	0	0	1	0	1	1	0	0	1	0	0	1	1	1	0	0	0	1	1	270
36	5	12	1.4	1	0	0	0	0	0	0	1	1	1	0	0	0	0	0	1	1	1	0	0	0	1	1	324
36	5	16	2.0	1	1	1	1	0	1	0	1	1	1	1	0	0	0	0	1	1	1	0	0	0	1	1	432
40	No	8	1.4	0	0	0	0	0	0	0	1	0	1	1	0	0	1	1	0	1	1	0	0	0	0	0	40
43.2	4	8	1.4	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	216
43.2	4	12	1.4	1	0	0	0	0	0	1	0	1	1	0	0	1	0	0	1	1	1	0	0	0	1	1	324
43.2	4	16	2.0	1	1	1	1	0	1	1	0	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	432

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Pixel Clock (MHz)	Pixel Repetition	Color depth [bits]	OPMODE(HDMI1.4 OR 2.0)	Divider Settings				opmode		Divider settings								MPLL Charge pump settings								TMDSCLKSRC (MHz)
				prep_div		tmdsmhl_cntrl				fbdiv2_cntrl			fbdiv1_cntrl		ref_cntrl		n_cntrl	prop_cntrl			int_cntrl			gmp_cntrl		
				0006:Bit 14	0006:Bit 13	0006:Bit 12	0006:Bit 11	0006:Bit 10	0006:Bit 9	0006:Bit 8	0006:Bit 7	0006:Bit 6	0006:Bit 5	0006:Bit 4	0006:Bit 3	0006:Bit 2	0006:Bit 1	0006:Bit 0	0010:Bit 5	0010:Bit 4	0010:Bit 3	0010:Bit 2	0010:Bit 1	0010:Bit 0	0015:Bit 1	0015:Bit 0	
44.9	No	8	1.4	0	0	0	0	0	0	0	1	0	1	1	0	0	1	1	0	1	1	0	0	0	0	44.9
49.5	No	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	1	49.5
50	No	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	1	50
50.35	No	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	1	50.35
50.35	No	10	1.4	0	1	0	0	0	0	1	0	1	0	0	0	0	1	0	1	0	1	0	0	0	1	62.9375
50.35	No	12	1.4	1	0	0	0	0	0	0	1	0	1	0	0	0	1	0	1	0	1	0	0	0	1	75.525
50.35	No	16	1.4	1	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	1	1	0	0	0	0	100.7
54	No	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	1	54
54	No	10	1.4	0	1	0	0	0	0	1	0	1	0	0	0	0	1	0	1	0	1	0	0	0	1	67.5
54	No	12	1.4	1	0	0	0	0	0	0	1	0	1	0	0	0	1	0	1	0	1	0	0	0	1	81
54	No	16	1.4	1	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	1	1	0	0	0	1	108
54	1	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	0	1	1	1	1	0	0	0	1	108
54	1	10	1.4	0	1	0	0	0	0	1	0	1	0	0	0	0	0	1	1	1	1	0	0	0	1	135
54	1	12	1.4	1	0	0	0	0	0	0	1	0	1	0	0	0	0	1	1	1	1	0	0	0	1	162
54	1	16	1.4	1	1	0	0	0	0	0	0	1	1	1	0	0	0	0	1	1	1	0	0	0	1	216
54	3	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	0	0	1	1	1	0	0	0	1	216
54	3	10	1.4	0	1	0	0	0	0	1	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	270
54	3	12	1.4	1	0	0	0	0	0	0	1	0	1	0	0	0	0	0	1	1	1	0	0	0	1	324
54	3	16	2.0	1	1	1	1	0	1	0	1	0	1	1	0	0	0	0	1	1	1	0	0	0	1	432
56.25	No	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	0	56.25
59.4	No	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	0	59.4
59.4	No	10	1.4	0	1	0	0	0	0	1	0	1	0	0	0	0	1	0	1	0	1	0	0	0	0	74.25
59.4	No	12	1.4	1	0	0	0	0	0	0	1	0	1	0	0	0	1	0	1	0	1	0	0	0	0	89.1
59.4	No	16	1.4	1	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	1	1	0	0	0	1	118.8
65	No	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	1	65
68.25	No	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	1	68.25
71	No	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	1	71
72	No	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	1	72
72	No	10	1.4	0	1	0	0	0	0	1	0	1	0	0	0	0	1	0	1	0	1	0	0	0	1	90
72	No	12	1.4	1	0	0	0	0	0	0	0	1	1	0	0	0	0	1	1	1	1	0	0	0	1	108
72	No	16	1.4	1	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	1	1	0	0	0	1	144
72	2	8	1.4	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	1	1	1	0	0	0	1	216
72	2	10	1.4	0	1	0	0	0	0	1	0	1	1	0	1	1	0	0	1	1	1	0	0	0	1	270

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Pixel Clock (MHz)	Pixel Repetition	Color depth [bits]	OPMODE(HDMI1.4 OR 2.0)	Divider Settings				opmode		Divider settings								MPLL Charge pump settings								TMDSCLKSRC (MHz)
				prep_div		tmdsmhl_cntrl				fbdiv2_cntrl			fbdiv1_cntrl		ref_cntrl		n_cntrl	prop_cntrl			int_cntrl			gmp_cntrl		
				0006:Bit 14	0006:Bit 13	0006:Bit 12	0006:Bit 11	0006:Bit 10	0006:Bit 9	0006:Bit 8	0006:Bit 7	0006:Bit 6	0006:Bit 5	0006:Bit 4	0006:Bit 3	0006:Bit 2	0006:Bit 1	0006:Bit 0	0010:Bit 5	0010:Bit 4	0010:Bit 3	0010:Bit 2	0010:Bit 1	0010:Bit 0	0015:Bit 1	0015:Bit 0	
72	2	12	1.4	1	0	0	0	0	0	0	1	1	1	0	0	1	0	0	1	1	1	0	0	0	1	1	324
72	2	16	2.0	1	1	1	1	0	1	0	1	0	1	0	0	0	0	0	1	1	1	0	0	0	1	1	432
73.25	No	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	0	1	73.25
74.25	No	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	0	1	74.25
74.25	No	10	1.4	0	1	0	0	0	0	1	0	1	0	0	0	1	0	1	1	1	1	0	0	0	1	0	92.8125
74.25	No	12	1.4	1	0	0	0	0	0	0	0	1	1	0	0	0	0	1	1	1	1	0	0	0	1	0	111.375
74.25	No	16	1.4	1	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	1	1	0	0	0	1	0	148.5
75	No	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	0	1	75
78.75	No	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	0	1	78.75
79.5	No	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	0	1	79.5
82.5	No	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	0	1	82.5
82.5	No	10	1.4	0	1	0	0	0	0	1	0	1	0	0	0	1	0	1	1	1	1	0	0	0	1	0	103.125
82.5	No	12	1.4	1	0	0	0	0	0	0	0	1	1	0	0	0	0	1	1	1	1	0	0	0	1	0	123.75
82.5	No	16	1.4	1	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	1	1	0	0	0	1	0	165
83.5	No	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	0	1	83.5
85.5	No	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	0	1	85.5
88.75	No	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	0	1	88.75
90	No	8	1.4	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1	0	0	0	0	1	90
90	No	10	1.4	0	1	0	0	0	0	1	0	1	0	0	0	1	0	1	1	1	1	0	0	0	1	0	112.5
90	No	12	1.4	1	0	0	0	0	0	0	0	1	1	0	0	0	0	1	1	1	1	0	0	0	1	0	135
90	No	16	1.4	1	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	1	1	0	0	0	1	0	180
94.5	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	94.5
99	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	99
99	No	10	1.4	0	1	0	0	0	0	1	0	1	0	0	0	1	0	1	1	1	1	0	0	0	1	0	123.75
99	No	12	1.4	1	0	0	0	0	0	0	0	1	1	0	0	0	0	1	1	1	1	0	0	0	1	0	148.5
99	No	16	1.4	1	1	0	0	0	0	0	0	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	198
100.7	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	100.7
100.7	No	10	1.4	0	1	0	0	0	0	1	0	1	0	0	0	1	0	1	1	1	1	0	0	0	1	0	125.875
100.7	No	12	1.4	1	0	0	0	0	0	0	0	1	1	0	0	0	0	1	1	1	1	0	0	0	1	0	151.05
100.7	No	16	1.4	1	1	0	0	0	0	0	0	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	201.4
101	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	101
102.25	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	102.25
106.5	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	106.5

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Pixel Clock (MHz)	Pixel Repetition	Color depth [bits]	OPMODE(HDMI1.4 OR 2.0)	Divider Settings				opmode		Divider settings								MPLL Charge pump settings								TMDSCLKSRC (MHz)
				prep_div		tmdsmhl_cntrl				fbdiv2_cntrl			fbdiv1_cntrl		ref_cntrl		n_cntrl	prop_cntrl			int_cntrl			gmp_cntrl		
				0006:Bit 14	0006:Bit 13	0006:Bit 12	0006:Bit 11	0006:Bit 10	0006:Bit 9	0006:Bit 8	0006:Bit 7	0006:Bit 6	0006:Bit 5	0006:Bit 4	0006:Bit 3	0006:Bit 2	0006:Bit 1	0006:Bit 0	0010:Bit 5	0010:Bit 4	0010:Bit 3	0010:Bit 2	0010:Bit 1	0010:Bit 0	0015:Bit 1	0015:Bit 0	
108	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	108
108	No	10	1.4	0	1	0	0	0	0	1	0	1	0	0	0	1	0	1	1	1	1	0	0	0	1	0	135
108	No	12	1.4	1	0	0	0	0	0	0	0	1	1	0	0	0	0	1	1	1	1	0	0	0	1	0	162
108	No	16	1.4	1	1	0	0	0	0	0	0	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	216
108	1	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	216
108	1	10	1.4	0	1	0	0	0	0	1	0	1	0	0	0	1	0	0	1	1	1	0	0	0	1	1	270
108	1	12	1.4	1	0	0	0	0	0	0	0	1	1	0	0	0	0	0	1	1	1	0	0	0	1	1	324
108	1	16	2.0	1	1	1	1	0	1	0	0	1	1	1	0	0	0	0	1	1	1	0	0	0	1	1	432
115.5	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	115.5
117.5	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	117.5
118.8	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	118.8
118.8	No	10	1.4	0	1	0	0	0	0	1	0	1	0	0	0	1	0	1	1	1	1	0	0	0	1	0	148.5
118.8	No	12	1.4	1	0	0	0	0	0	0	0	1	1	0	0	0	0	1	1	1	1	0	0	0	1	0	178.2
118.8	No	16	1.4	1	1	0	0	0	0	0	0	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	237.6
119	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	119
121.75	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	121.75
122.5	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	122.5
135	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	135
136.75	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	136.75
140.25	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	140.25
144	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	144
144	No	10	1.4	0	1	0	0	0	0	1	0	1	0	0	0	1	0	1	1	1	1	0	0	0	1	0	180
144	No	12	1.4	1	0	0	0	0	0	0	0	1	1	0	0	1	0	0	1	1	1	0	0	0	1	1	216
144	No	16	1.4	1	1	0	0	0	0	0	0	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	288
146.25	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	146.25
148.25	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	148.25
148.5	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	148.5
148.5	No	10	1.4	0	1	0	0	0	0	1	0	1	0	0	1	1	0	0	1	1	1	0	0	0	1	1	185.625
148.5	No	12	1.4	1	0	0	0	0	0	0	0	1	1	0	0	1	0	0	1	1	1	0	0	0	1	1	222.75
148.5	No	16	1.4	1	1	0	0	0	0	0	0	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	297
154	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	154
156	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	156
157	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	157

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Pixel Clock (MHz)	Pixel Repetition	Color depth [bits]	OPMODE(HDMI1.4 OR 2.0)	Divider Settings				opmode		Divider settings								MPLL Charge pump settings								TMDSCLKSRC (MHz)
				prep_div		tmdsmhl_cntrl				fbdiv2_cntrl			fbdiv1_cntrl		ref_cntrl		n_cntrl	prop_cntrl			int_cntrl			gmp_cntrl		
				0006:Bit 14	0006:Bit 13	0006:Bit 12	0006:Bit 11	0006:Bit 10	0006:Bit 9	0006:Bit 8	0006:Bit 7	0006:Bit 6	0006:Bit 5	0006:Bit 4	0006:Bit 3	0006:Bit 2	0006:Bit 1	0006:Bit 0	0010:Bit 5	0010:Bit 4	0010:Bit 3	0010:Bit 2	0010:Bit 1	0010:Bit 0	0015:Bit 1	0015:Bit 0	
157.5	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	157.5
162	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	162
165	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	165
165	No	10	1.4	0	1	0	0	0	0	1	0	1	0	0	1	1	0	0	1	1	1	0	0	0	1	1	206.25
165	No	12	1.4	1	0	0	0	0	0	0	0	1	1	0	0	1	0	0	1	1	1	0	0	0	1	1	247.5
165	No	16	1.4	1	1	0	0	0	0	0	0	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	330
175.5	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	175.5
179.5	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	179.5
180	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	180
180	No	10	1.4	0	1	0	0	0	0	1	0	1	0	0	1	1	0	0	1	1	1	0	0	0	1	1	225
180	No	12	1.4	1	0	0	0	0	0	0	0	1	1	0	0	1	0	0	1	1	1	0	0	0	1	1	270
180	No	16	2.0	1	1	1	1	0	1	0	0	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	360
182.75	No	8	1.4	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	182.75
185.625	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	185.625
185.625	No	10	1.4	0	1	0	0	0	0	1	0	1	0	0	1	1	0	0	1	1	1	0	0	0	1	1	232.03125
185.625	No	12	1.4	1	0	0	0	0	0	0	0	1	1	0	0	1	0	0	1	1	1	0	0	0	1	1	278.4375
185.625	No	16	2.0	1	1	1	1	0	1	0	0	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	371.25
187	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	187
187.25	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	187.25
189	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	189
193.25	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	193.25
198	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	198
198	No	10	1.4	0	1	0	0	0	0	1	0	1	0	0	1	1	0	0	1	1	1	0	0	0	1	1	247.5
198	No	12	1.4	1	0	0	0	0	0	0	0	1	1	0	0	1	0	0	1	1	1	0	0	0	1	1	297
198	No	16	2.0	1	1	1	1	0	1	0	0	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	396
202.5	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	202.5
204.75	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	204.75
208	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	208
214.75	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	214.75
216	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	216
216	No	10	1.4	0	1	0	0	0	0	1	0	1	0	0	1	1	0	0	1	1	1	0	0	0	1	1	270
216	No	12	1.4	1	0	0	0	0	0	0	0	1	1	0	0	1	0	0	1	1	1	0	0	0	1	1	324
216	No	16	2.0	1	1	1	1	0	1	0	0	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	432

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Pixel Clock (MHz)	Pixel Repetition	Color depth [bits]	OPMODE(HDMI1.4 OR 2.0)	Divider Settings				opmode		Divider settings								MPLL Charge pump settings								TMDSCLKSRC (MHz)
				prep_div		tmdsmhl_cntrl				fbdiv2_cntrl			fbdiv1_cntrl		ref_cntrl		n_cntrl	prop_cntrl			int_cntrl			gmp_cntrl		
				0006:Bit 14	0006:Bit 13	0006:Bit 12	0006:Bit 11	0006:Bit 10	0006:Bit 9	0006:Bit 8	0006:Bit 7	0006:Bit 6	0006:Bit 5	0006:Bit 4	0006:Bit 3	0006:Bit 2	0006:Bit 1	0006:Bit 0	0010:Bit 5	0010:Bit 4	0010:Bit 3	0010:Bit 2	0010:Bit 1	0010:Bit 0	0015:Bit 1	0015:Bit 0	
218.25	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	218.25
229.5	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	229.5
234	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	234
237.6	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	237.6
237.6	No	10	1.4	0	1	0	0	0	0	1	0	1	0	0	1	1	0	0	1	1	1	0	0	0	1	1	297
237.6	No	12	2.0	1	0	1	1	0	1	0	0	1	1	0	0	1	0	0	1	1	1	0	0	0	1	1	356.4
237.6	No	16	2.0	1	1	1	1	0	1	0	0	1	0	1	0	0	0	0	1	1	1	0	0	0	1	1	475.2
245.25	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	245.25
245.5	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	245.5
261	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	261
268.25	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	268.25
268.5	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	268.5
281.25	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	281.25
288	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	288
288	No	10	2.0	0	1	1	1	0	1	1	0	1	0	0	1	1	0	0	1	1	1	0	0	0	1	1	360
288	No	12	2.0	1	0	1	1	0	1	0	0	1	1	0	0	1	0	0	1	1	1	0	0	0	1	1	432
288	No	16	2.0	1	1	1	1	0	1	0	0	1	0	1	0	0	0	0	1	0	1	0	0	0	1	1	576
297	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	297
297	No	10	2.0	0	1	1	1	0	1	1	0	1	0	0	1	1	0	0	1	1	1	0	0	0	1	1	371.25
297	No	12	2.0	1	0	1	1	0	1	0	0	1	1	0	0	1	0	0	1	1	1	0	0	0	1	1	445.5
297	No	16	2.0	1	1	1	1	0	1	0	0	1	0	1	0	0	0	0	1	0	1	0	0	0	1	1	594
317	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	317
330	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	330
330	No	10	2.0	0	1	1	1	0	1	1	0	1	0	0	1	1	0	0	1	1	1	0	0	0	1	1	412.5
330	No	12	2.0	1	0	1	1	0	1	0	0	1	1	0	0	1	0	0	1	1	1	0	0	0	1	1	495
333.25	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	333.25
340	No	8	1.4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	1	1	340
348.5	No	8	2.0	0	0	1	1	0	1	0	0	1	0	0	0	0	0	0	0	1	1	0	0	0	1	1	348.5
356.5	No	8	2.0	0	0	1	1	0	1	0	0	1	0	0	0	0	0	0	0	1	1	0	0	0	1	1	356.5
360	No	8	2.0	0	0	1	1	0	1	0	0	1	0	0	0	0	0	0	0	1	1	0	0	0	1	1	360
360	No	10	2.0	0	1	1	1	0	1	1	0	1	0	0	1	1	0	0	1	1	1	0	0	0	1	1	450
360	No	12	2.0	1	0	1	1	0	1	0	0	1	1	0	0	1	0	0	1	1	1	0	0	0	1	1	540
371.25	No	8	2.0	0	0	1	1	0	1	0	0	1	0	0	0	0	0	0	0	1	1	0	0	0	1	1	371.25

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Pixel Clock (MHz)	Pixel Repetition	Color depth [bits]	OPMODE(HDMI1.4 OR 2.0)	Divider Settings				opmode		Divider settings								MPLL Charge pump settings								TMDSCLKSRC (MHz)
				prep_div		tmdsmhl_cntrl				fbdiv2_cntrl			fbdiv1_cntrl		ref_cntrl		n_cntrl	prop_cntrl			int_cntrl			gmp_cntrl		
				0006:Bit 14	0006:Bit 13	0006:Bit 12	0006:Bit 11	0006:Bit 10	0006:Bit 9	0006:Bit 8	0006:Bit 7	0006:Bit 6	0006:Bit 5	0006:Bit 4	0006:Bit 3	0006:Bit 2	0006:Bit 1	0006:Bit 0	0010:Bit 5	0010:Bit 4	0010:Bit 3	0010:Bit 2	0010:Bit 1	0010:Bit 0	0015:Bit 1	0015:Bit 0	
371.25	No	10	2.0	0	1	1	1	0	1	1	0	1	0	0	1	1	0	0	1	1	1	0	0	0	1	1	464.0625
371.25	No	12	2.0	1	0	1	1	0	1	0	0	1	1	0	0	1	0	0	1	1	1	0	0	0	1	1	556.875
380.5	No	8	2.0	0	0	1	1	0	1	0	0	1	0	0	0	0	0	0	0	1	1	0	0	0	1	1	380.5
396	No	8	2.0	0	0	1	1	0	1	0	0	1	0	0	0	0	0	0	0	1	1	0	0	0	1	1	396
396	No	10	2.0	0	1	1	1	0	1	1	0	1	0	0	1	1	0	0	1	1	1	0	0	0	1	1	495
396	No	12	2.0	1	0	1	1	0	1	0	0	1	1	0	0	1	0	0	1	1	1	0	0	0	1	1	594
432	No	8	2.0	0	0	1	1	0	1	0	0	1	0	0	0	0	0	0	0	1	1	0	0	0	1	1	432
432	No	10	2.0	0	1	1	1	0	1	1	0	1	0	0	1	1	0	0	1	1	1	0	0	0	1	1	540
443.25	No	8	2.0	0	0	1	1	0	1	0	0	1	0	0	0	0	0	0	0	1	1	0	0	0	1	1	443.25
475.2	No	8	2.0	0	0	1	1	0	1	0	0	1	0	0	0	0	0	0	0	1	1	0	0	0	1	1	475.2
475.2	No	10	2.0	0	1	1	1	0	1	1	0	1	0	0	1	1	0	0	1	1	1	0	0	0	1	1	594
495	No	8	2.0	0	0	1	1	0	1	0	0	1	0	0	0	0	0	0	0	1	1	0	0	0	1	1	495
505.25	No	8	2.0	0	0	1	1	0	1	0	0	1	0	0	0	0	0	0	0	1	1	0	0	0	1	1	505.25
552.75	No	8	2.0	0	0	1	1	0	1	0	0	1	0	0	0	0	0	0	0	1	1	0	0	0	1	1	552.75
594	No	8	2.0	0	0	1	1	0	1	0	0	1	0	0	0	0	0	0	0	1	1	0	0	0	1	1	594

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Chapter 33 HDCP22 Controller

33.1 Overview

The HDCP22 controller is an Embedded Security Module (ESM) inside HDMI TX controller. It is a self-contained module pre-integrated with the HDMI controller to ensure DCP robustness rules for HDCP. The ESM has a small MCU inside. It reads ESM Image for operation through the AXI interface.

The HDCP22 Controller supports following features:

- Supports HDCP Revision 2.2
- Bus Interface Features:
  - Supports AXI interface with MMU
  - Supports AHB interface for MMU configuration
  - Supports APB interface for processor to send commands and get status from the ESM (Host Port Interface, HPI)
  - A Secure Key Port (SKP) for cryptographic keys and random nonce
- MMU:
  - 4k/64k page size
  - TLB pre-fetch

33.2 Block Diagram

The Host Controller consists of the following main functional blocks.

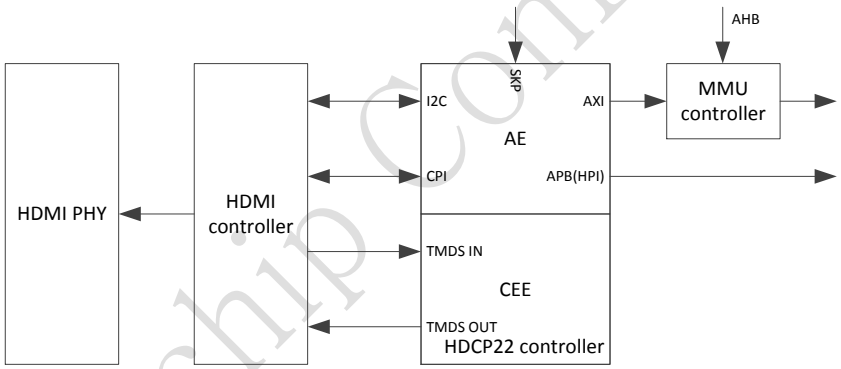

Fig. 33-1 hdcp22 Controller Block Diagram

The embedded security module is composed of two parts:

- The Authentication Engine (AE) contains a controller that runs the HDCP authentication process.
- The Content Encryption Engine (CEE) is an ESM component that encrypts data for the HDCP transmitter.

33.3 Function Description

33.3.1 AXI interface

The ESM requires continuous access to its ESM Image for operation, which is accessed through the AXI interface and usually placed a reserved portion of DDR memory. It also uses this interface for communicating messages with the external host processor to pass content such as SRM data, pairing information, and debug/logging information.

The clock used by ESM requires a minimum frequency: minimum ratio of 1:2.3 (hdcp axi clock: VOP Pixel clock) for proper operation. For an hdmi output display 4K resolution at 600 MHz, this clock can be no less than 261 MHz.

Following Table summarizes the various requirements the ESM places on the PX5's memory, assuming an AXI interface clock of 261 MHz and an average latency of 1000 ns.

Table 33-1 ESM average memory bandwidth usage

Phase	Duration	Appropriate	Description

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	(ms)	Bandwidth (Mbytes/sec)	
Bootup	50	5.0	Initial startup of ESM to Idle state.
Tx Authentication	525ms	1.5	ESM transmitter authentication phase including capability check.
Idle, authenticated	n/a	< 1.0, 4.0	ESM is either in the Idle state waiting to start authentication or in the authenticated state. The second number is when the ESM has debug logging enabled.

The external host memory required by the ESM must be physically contiguous and allocated before the ESM can be started. The ESM Image and R/W memory buffer addresses must be 4K aligned otherwise the ESM cannot properly access the memory. Following table describes the requirements.

Table 33-2 ESM average memory bandwidth usage

Size (bytes)	ESM Access	Purpose
~160K	Read Only	Contains the ESM Image. Note this size is approximate and varies depending on the image type (Rx/Tx) and the current version of the ESM Image.
16K + n*256 (Tx) (n is the number of pairs supported)	Read / Write	Bidirectional communication buffer to exchange data with the Host. Messages such as SRM updates, pairing data, and logging information are placed in this area by the host and the ESM. The number of pairs supported by this parameter is defined in the Tx configuration file ([MEM_PAIRING_SIZE]).

In PX5 the provided ESM host application library allocates a small amount of the processor's memory space, which is accessible by the ESM. Communication with this interface is provided by the ESM host application library.

33.3.2 AHB interface

MMU register is accessed from AHB interface.

33.3.3 APB interface (HPI)

The external host issues commands to the ESM through the HPI. The ESM supports a limited set of commands such as:

- Enabling and disabling of security on the link
- Enabling debugging/logging
- Retrieving pairing information
- Updating the SRM
- ESM status

The external host must issue a SRM list to the ESM when one is available, according to the requirements of DCP LLC.

Communication with this interface is provided by the ESM host application library.

33.3.4 SKP Interface

The SKP is a secure interface that contains the secret keys for the ESM along with an input for a True Random Number Generator (TRNG). The keys are used by the ESM to create session keys and other cryptographic secrets required for operation. Customers should put following

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keys in EFUSE for secure reason.

Table 33-3 Secure Key for ESM

Description	Purpose
Platform Key (PKf)	This key is used to decrypt the ESM Image. The PKf is intended to be common across a deployed model of a product, which allows field upgrades of the ESM Image should it be necessary.
Device Unique Key (DUK)	This key is used to decrypt the DCP Rx key data and to encrypt and decrypt content unique to the ESM such as the pairing message content. This value should be unique for every unit.

Note: *The PKf and DUK must be kept confidential as exposing either one compromises the security of the ESM.*

33.3.5 HDCP22 Controller Behavior

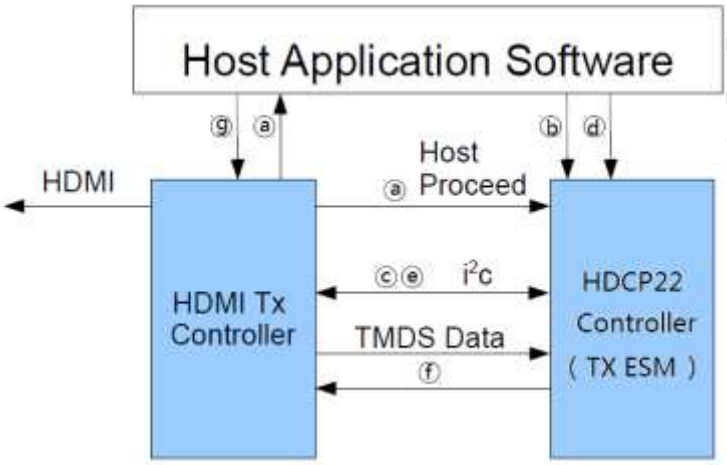

Fig. 33-2 Transmitter Authentication

Hdcp22 controller in PX5 is a transmitter, it works as following steps:
a. The controller asserts Host Proceed to the ESM. The application software receives the notification from the controller.
b. The application software requests the ESM to perform a capability check with the receiver.
c. The ESM queries the receiver. If the receiver does not support HDCP 2.2 then the ESM idles and the application software asks the HDMI controller to check if the receiver supports HDCP 1.4 (or no encryption) and takes appropriate action. The ESM is no longer required.
d. If the receiver supports HDCP 2.2, the application software requests the ESM initiate authentication.
e. The ESM initiates authentication with the receiver and successfully completes authentication (or not). The application software and the HDMI controller are appropriately notified.
f. If authentication is successful the ESM starts to encrypt the TMDS data.
g. The application software can now send high value content securely on the HDMI link.

33.3.6 HDCP22 Software Initialization

An external host controller (ARM Core or MCU) allocates memory, initializes the ESM, and enables the ESM controller. The general reset and startup sequence is:

a. Configure clocks for hdcp controller.
b. Load PKF & DUK from effuse to sgrf.
c. The ESM reset is deasserted and the ESM enters a reset state.
d. The host allocates contiguous memory for the ESM to use. The ESM Image and R/W area buffers must be 4K aligned.
e. The host configures the registers of the ESM with the physical memory pointers for the ESM image and communications buffer.
f. The host copies the ESM Image from a persistent storage location (or other memory region) to the memory.

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g. The host configures itself to support interrupts from the APB Interface (HPI) of the ESM for processing commands.
h. The host enables the ESM controller to a running state. At this point the ESM initiates transactions on the AXI bus to fetch its Image.
i. ESM reports a ready status to the host when it can accept commands.

The ESM host application library provides the startup sequence from steps d through i.

33.3.7 Key Usage

The ESM uses the DUK to protect specific content such as pairing data. When requesting pairing data from the ESM, the information is encrypted with the unique DUK thus preventing its use with any other ESM—the information is unique to the issuing unit, preventing the key pairing data from being used by another one.
The ESM uses the PKf to decrypt the Image it uses for operation. The ESM Image must be processed with tools that are provided with the ESM software package. Following figure illustrates the tools and the flow that are required to produce a deployable ESM Image.

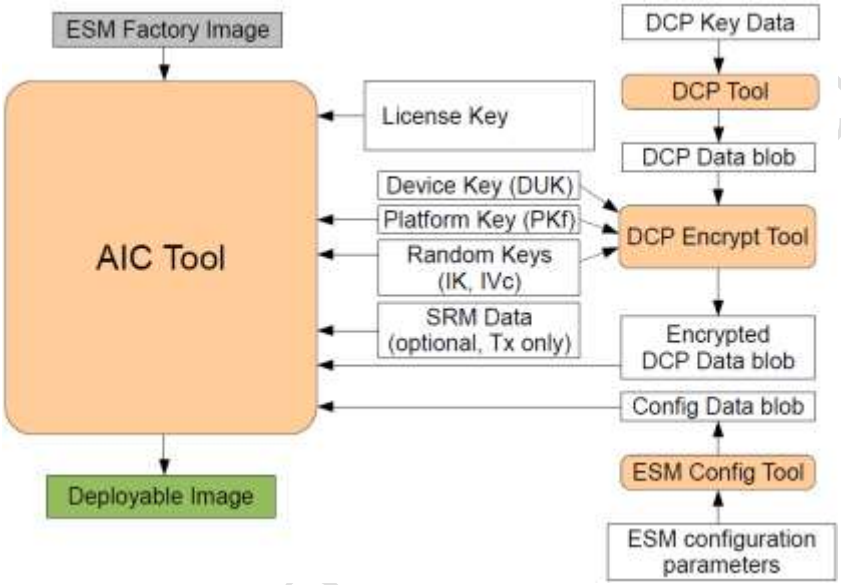

Fig. 33-3 ESM Image tool flow

For a PX5, the DCP key data is common. Note that while the image has an option where a default SRM list can be optionally inserted as part of the manufacturing process, the SRM update can also be sent to the ESM as a separate command after the ESM has been booted. In some cases it may be necessary to provide a field update of the ESM Image after deployment, either because of a factory update or a change to the ESM configuration file. Updated image can be issued using the process noted above as all field Tx units receive the same DCP key.

33.4 Register Description

The registers listed following are used for MMU interface. Any operation to hdcp22 controller is done through Run Time Library (See next chaper: Application Nots).

33.4.1 Register Summary

Name	Offset	Size	Reset Value	Description
HDCP22_MMU_DTE_ADDR	0x00000	W	0x00000000	MMU current page Table address
HDCP22_MMU_STATUS	0x00004	W	0x00000018	MMU status register
HDCP22_MMU_COMMAND	0x00008	W	0x00000000	MMU command register
HDCP22_MMU_PAGE_FAULT_ADDR	0x0000c	W	0x00000000	MMU logical address of last page fault

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Name	Offset	Size	Reset Value	Description
HDCP22_MMU_ZAP_ONE_LINE	0x00010	W	0x00000000	MMU Zap cache line register
HDCP22_MMU_INT_RAWSTAT	0x00014	W	0x00000000	MMU raw interrupt status register
HDCP22_MMU_INT_CLEAR	0x00018	W	0x00000000	MMU raw interrupt status register
HDCP22_MMU_INT_MASK	0x0001c	W	0x00000000	MMU raw interrupt status register
HDCP22_MMU_INT_STATUS	0x00020	W	0x00000000	MMU raw interrupt status register
HDCP22_MMU_AUTO_GATING	0x00024	W	0x00000001	mmu auto gating
HDCP22_MMU_mmu_miss_cnt	0x00028	W	0x00000000	Register0000 Abstract
HDCP22_MMU_mmu_burst_cnt	0x0002c	W	0x00000000	Register0001 Abstract

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

33.4.2 Detail Register Description

HDCP22_MMU_DTE_ADDR
Address: Operational Base + offset (0x00000)
MMU current page Table address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	MMU_DTE_ADDR mmu dte base addr mmu dte base addr , the address must be 4kb aligned

HDCP22_MMU_STATUS
Address: Operational Base + offset (0x00004)
MMU status register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:6	RO	0x00	PAGE_FAULT_BUS_ID Field0000 Abstract Index of master reponsible for last page fault
5	RO	0x0	PAGE_FAULT_IS_WRITE Field0000 Abstract The direction of access for last page fault: 0 = Read 1 = Write
4	RO	0x1	REPLAY_BUFFER_EMPTY Field0000 Abstract The MMU replay buffer is empty

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Bit	Attr	Reset Value	Description
3	RO	0x1	MMU_IDLE Field0000 Abstract The MMU is idle when accesses are being translated and there are no unfinished translated accesses.
2	RO	0x0	STAIL_ACTIVE Field0001 Abstract MMU stall mode currently enabled. The mode is enabled by command
1	RO	0x0	PAGE_FAULT_ACTIVE Field0000 Abstract MMU page fault mode currently enabled . The mode is enabled by command.
0	RO	0x0	PAGING_ENABLED Field0000 Abstract Paging is enabled

HDCP22_MMU_COMMAND
Address: Operational Base + offset (0x00008)
MMU command register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2:0	WO	0x0	MMU_CMD Field0000 Abstract MMU_CMD. This can be: 0: MMU_ENABLE_PAGING 1: MMU_DISABLE_PAGING 2: MMU_ENABLE_STALL 3: MMU_DISABLE_STALL 4: MMU_ZAP_CACHE 5: MMU_PAGE_FAULT_DONE 6: MMU_FORCE_RESET

HDCP22_MMU_PAGE_FAULT_ADDR
Address: Operational Base + offset (0x0000c)
MMU logical address of last page fault

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	PAGE_FAULT_ADDR Field0000 Abstract address of last page fault

HDCP22_MMU_ZAP_ONE_LINE
Address: Operational Base + offset (0x00010)
MMU Zap cache line register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	MMU_ZAP_ONE_LINE Field0000 Abstract address to be invalidated from the page table cache

HDCP22_MMU_INT_RAWSTAT
Address: Operational Base + offset (0x00014)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR Field0000 Abstract read bus error
0	RW	0x0	PAGE_FAULT Field0000 Abstract page fault

HDCP22_MMU_INT_CLEAR
Address: Operational Base + offset (0x00018)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	WO	0x0	READ_BUS_ERROR Field0000 Abstract read bus error
0	WO	0x0	PAGE_FAULT Field0000 Abstract page fault

HDCP22_MMU_INT_MASK
Address: Operational Base + offset (0x0001c)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR Field0000 Abstract read bus error enable an interrupt source if the corresponding mask bit is set to 1
0	RW	0x0	PAGE_FAULT Field0000 Abstract page fault enable an interrupt source if the corresponding mask bit is set to 1

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HDCP22_MMU_INT_STATUS
Address: Operational Base + offset (0x00020)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RO	0x0	READ_BUS_ERROR Field0000 Abstract read bus error
0	RO	0x0	PAGE_FAULT Field0000 Abstract page fault

HDCP22_MMU_AUTO_GATING
Address: Operational Base + offset (0x00024)
mmu auto gating

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x1	mmu_auto_gating mmu auto gating when it is 1'b1, the mmu will auto gating it self

HDCP22_MMU_mmu_miss_cnt
Address: Operational Base + offset (0x00028)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31	RW	0x0	cnt_ctrl_sel sel the counter for mmu_miss or mmu_real_miss 1'b0: mmu_real_miss 1'b1: mmu_miss When sel 1'b1, an axi command miss may count for several times; when sel 1'b0, an axi command only count for a time
30	RW	0x0	miss_cnt_overflow_flag miss cnt overflow flag miss cnt overflow flag
29:0	RW	0x00000000	miss_cnt count for miss AXI command count for miss AXI command

HDCP22_MMU_mmu_burst_cnt
Address: Operational Base + offset (0x0002c)
Register0001 Abstract

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
30	RW	0x0	bust_cnt_overflow_flag burst cnt overflow flag burst cnt overflow flag
29:0	RW	0x00000000	burst_cnt The AXI input burst counter The AXI input burst counter

33.5 Application Notes

33.5.1 Host Application Software

The ESM package contains a series of tools and a run-time library as bellow:

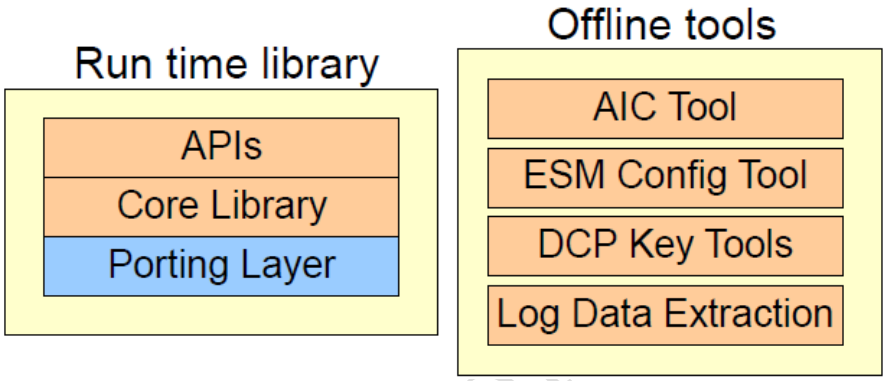

Fig. 33-4 ESM software

33.5.2 Run Time Library

This library is provided as source code and is used by the host application processor to control the ESM.
Setting the configurable options of the ESM is not provided through APIs for security reasons. It is instead embedded into the ESM's Image as part of the process of building a deployable image.
The APIs provide a minimal interface to the ESM, which include the following:

- Load/Initialize firmware image
- Start authentication to secure a link
- Terminate an authenticated link
- Collect logging data
- Get/Set pairing information
- Update SRM data
- Status updates

Sample code is provided and demonstrates how to use the various APIs. In addition, a software API reference guide is included and provides details on the API calls and their parameters.

33.5.3 Offline Tools

There are three separate offline tools that are required to build a deployable image for the ESM as shown in Figure ESM Image tool flow and one tool for logging information. All of these tools are developed to run on a 32 bit Linux system.

33.5.4 ESM Config Tool

This tool is used to insert configuration parameters into the ESM Image before deployment, including debugging parameters, which can be set to assist in-house development and debugging. These parameters cannot be changed in the field with software API calls for security reasons, and any debugging parameters must be disabled before production

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deployment as they can be used to disable certain security features.
This tool reads a configuration file containing parameters. The purpose of the parameter is documented in the commented sections. A sample set of user adjustable parameters are:

- Disable timeouts (debug only)
- Bypass (debug only)
- Pairing limit
- Enable logging and set logging levels (min to max)

Sample configuration files are located at samples/tx_config.txt
The following sample command lines are used to generate TX configuration.
.esm_tx_config -i ../samples/tx_config.txt -o data_tx.cfg
Where the data_tx.cfg are the formatted configuration files that must be passed to the AIC tool to generate the final image.
There is no need to run this tool separately. A makefile is provided that runs all three tools in sequence to create the proper output.

ESM Configuration File

The [CPU_FREQ] parameter in the configuration file must be changed to permit the ESM to operate properly in your system. This value must be set to the frequency of the AXI clock input to the ESM.
The [CEE_BSOD] is a 32 bit decimal value which parameter represents fixed data output on the TMDS data bus if the ESM encounters an authentication issue or detects attempts have been made to tamper with it. This 32 bit value is output as follows:

Fig. 33-5 ESM BSOD Output Mapping

BSOD Bit Positions	HDMI Controller TMDS Data Channel
7..0	ch0[7..0]
15..8	ch1[7..0]
23..16	ch2[7..0]
32..24	Not used

33.5.5 DCP Key Tools

These tools are used to create a DCP key data file, which is required to build the encrypted DCP data blob needed for building the final ESM Image. One tool is used to create the obfuscated data blob and another tool is used to encrypt the DCP key blob before its inclusion into the final ESM Image. (This tool also reads the configuration file firmware.aic to obtain the secret key values).
A sample file containing the DCP test keys is provided for reference.
The information shown in following table is required:

Fig. 33-6 Requirements for DCP Tools

Confidential Elements	Tag	Size
lc128	[LC_128]	16
3072 RSA DCP Public Key (base)	[KPUB_DCP_MOD]	384
3072 RSA DCP Public Key (exp)	[KPUB_DCP_E]	3

33.5.6 AIC Tool

This tool is used to create a deployable ESM Image. To build an ESM image, you must first create the data file for the DCP keys and the configuration. These files, along with an input configuration file for this tool are used to build an image that can be used by the ESM. In addition to specifying the Platform and Device Unique Keys in the configuration file, there are additional keys that also must be specified as noted below.

Content Randomization

There are two separate values, IK and IVc (128 bits and 96 bits respectively), which must be randomly generated for each creation of a deployed ESM Image. It is mandatory that any time an ESM Image is generated to be field deployed that new random values for each are created and used. Failure to do this puts secret data in the ESM Image at risk of being compromised.

Log Data Extraction

The logging information from the ESM is output in a proprietary format. This tool is used to

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produce readable output of captured logging data content.
The AICTOOL program is used to assemble the following files:

- ESM Factory Image file (.rom)
- Configuration file (*.cfg)
- HDCP KEY file (*.out).

The tool takes these three files and builds an encrypted image that can be used by an ESM (the ESM Image). Output ROM file located at bin/firmware_tx.rom
This tool reads an input configuration file (firmware/firmware.aic) to obtain the key values required to build the encrypted ESM Image. Table below describes the user adjustable parameters in this file. Any parameter not documented must not be modified.

Fig. 33-7 Description of user adjustable parameters

Parameter	Description
PKf	The 128 bit value present on the PKf input of the SKP
DUK	The 128 bit value present on the DUK input of the SKP
IK, IVc	Content randomization values as described in this section

33.5.7 Sample Application Software

The Host Library contains a core library component which includes all the functions necessary to use the ESM. The library is built using portable C code and uses various abstraction functions to allow for migration to different platforms. Before using the ESM in your system you must properly implement the abstraction components specific to your platform.
There are two distinct abstraction components used by the Host Library, the System Abstraction Layer and the Host Library Driver (HLD) component. The System Abstraction Layer (or Common Component) defines system level functions used by the Host Library, such as malloc, memcpy, etc. These functions are linked into the Host Library when it is built.
The Host Library Driver (HLD) component establishes a communication between the library and the physical ESM hardware, including managing the memory required for the ESM Image and the read/write area. This is a run-time plug-in module that is defined in your application and specified as a parameter to the ESM initialization function.
The figure below illustrates the abstraction components used by the Host Library package. This figure demonstrates one example where there are three separate ESMs in the system, requiring (typically) three separate HLD instantiations. Consult the ESM Host Library API guide for more details on the HLD implementation.

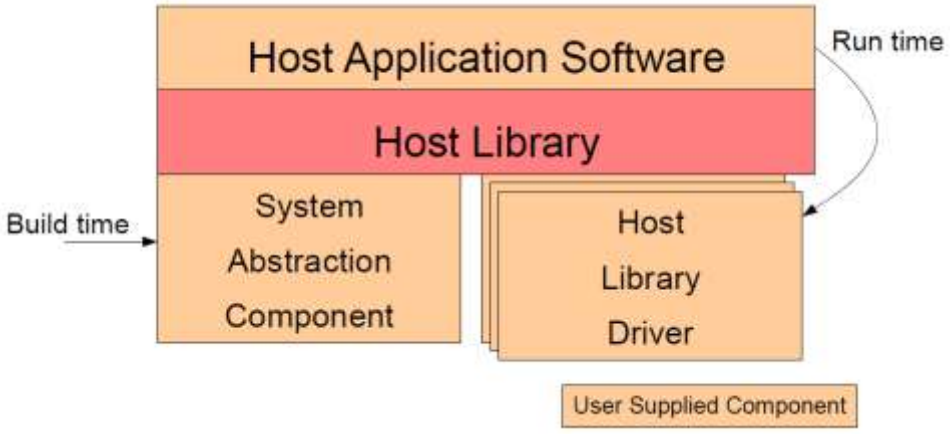

Fig. 33-8 Host Library Layers

The Host Library provides a number of sample applications using the ESM with the library. The first set of applications demonstrates using the library in a User Space Linux application environment and the other applications demonstrate a receiver and transmitter application in a bare metal (non-OS) system.
The sample applications demonstrate API usage, loading the ESM firmware, and running a specific ESM application.
The sample applications provided as a Linux User Space application communicate to the ESM hardware through a Linux Kernel driver which is implemented as a Host Library Driver (HLD) plug-in.

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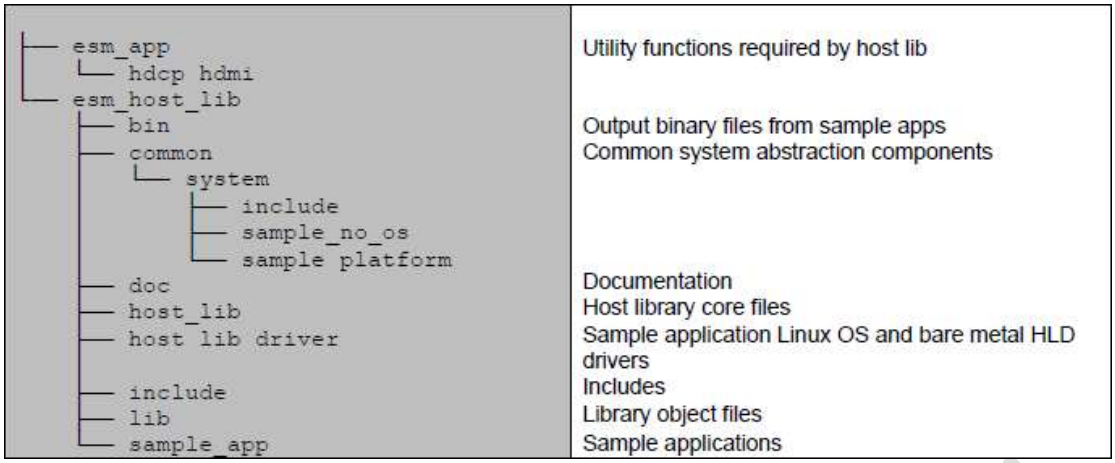

Fig. 33-9 The development tree overview

There are two sample transmitter applications provided in the sample_app folder:
hdcp_tx.c is a Linux based example and utilizes file IO and threads.
hdcp_tx_bm.c is a bare metal example.

Execution Environments

The provided development environments come with two sample platform targets: Linux OS and a bare metal platform as shown in the following table.

Fig. 33-10 Common Platform Target Examples

Common Samples Provided	Description
sample_platform	Demonstrates the system API implementations when targeting a Linux OS platform
sample_no_os	Demonstrates the system API implementations when targeting a platform with no operating system (also known as bare-metal).

Linux OS

These sample applications are targeted for a Linux User Space application environment. In order to use them a custom kernel driver was created and the HLD plug-in for the host library was made to call IOCTLs which mapped to the appropriate interfaces in the kernel driver module to access the ESM. To build for this platform run the makefile from the esm_host_lib folder:
make samples
The implementation of the platform specific APIs (for the Linux example) are in the following files:
./common/system/sample_platform/elliptic_common.c
./common/system/sample_platform/elliptic_log.c
./common/system/sample_platform/include/elliptic_platform_specific_system_type.h
./common/system/sample_platform/include/elliptic_platform_specific_system.h
You can change these implementations as needed for your specific platform.

No OS (Bare Metal)

This sample demonstrates how to build an application for a no OS (bare metal) environment with direct access to memory and system functions. In the supplied example there are no implementations for the HLD and you must first create them appropriate to your system. When you run the make example above it also build this reference application as well.
The implementation of the platform specific APIs are in the following files:
./common/system/sample_no_os/elliptic_common.c
./common/system/sample_no_os/elliptic_log.c
./common/system/sample_no_os/include/elliptic_platform_specific_system_type.h
./common/system/sample_no_os/include/elliptic_platform_specific_system.h
There are currently no implementations and you must create them as appropriate to your platform.

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Chapter 34 eDP TX Controller

34.1 Overview

This eDP TX IP is compliant with DisplayPort standard 1.2a and eDP 1.3. DisplayPort is an industry standard to accommodate the growing broad adoption of digital display technology within the PC and consumer electronics (CE) industries. It consolidates the internal and external connection methods to reduce device complexity and cost, supports necessary features for key cross industry applications, and provides performance scalability to enable the next generation of displays featuring higher color depths, refresh rates, and display resolutions.

This DisplayPort 1.2 specification defines a scalable digital display interface with optional content protection capability for broad application within PC and CE devices. The interface is designed to support both internal chip-to-chip and external box-to-box digital display connections. Potential internal chip-to-chip applications include usage within a notebook PC for driving a panel from a graphics controller, and usage within a monitor or TV for driving the display component from a display controller. Examples of box-to-box applications for DisplayPort include display connections between PCs and monitors, projectors, and TV displays. DisplayPort is also suitable for display connections between consumer electronics devices such as high definition optical disc players, set top boxes, and TV displays.

It supports following features:

- Compliant with DisplayPortTM Specification, Version 1.2.
- Compliant with eDPTM Specification, Version 1.3.
- Main link containing 4 physical lanes of 2.7/1.62 Gbps/lane
- TX PHY lanes, control pins and hot-plug pins are shared by the DisplayPort Source
- Bi-directional auxiliary link with up to 1Mbps speed.
- RGB, YCbCr 4:4:4, YCbCr 4:2:2 and 8/10/12 bit per component video format.
- Video and audio slave mode
- Support PSR
- APB slave bus interface
- Hot plug and unplug detection and link status monitor.
- Support VESA DMT and CVT timing standards.
- Fully support EIA/CEA-861D video timing and Info Frame structure.
- Supports reading of the display EDID whenever the display is connected to power, even an AC-trickle power.
- Up to 0.5% down-spreading support at high-speed link.
- PRBS or programmable transmitter pattern for main link quality test.
- 24 Mhz crystal clock input.
- Built-in video and audio BIST patterns.
- 28nm LP CMOS process with Core voltage 0.9V (min)/ 1.0V (typ)/ 1.08V (max) @ global corner.

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34.2 Block Diagram

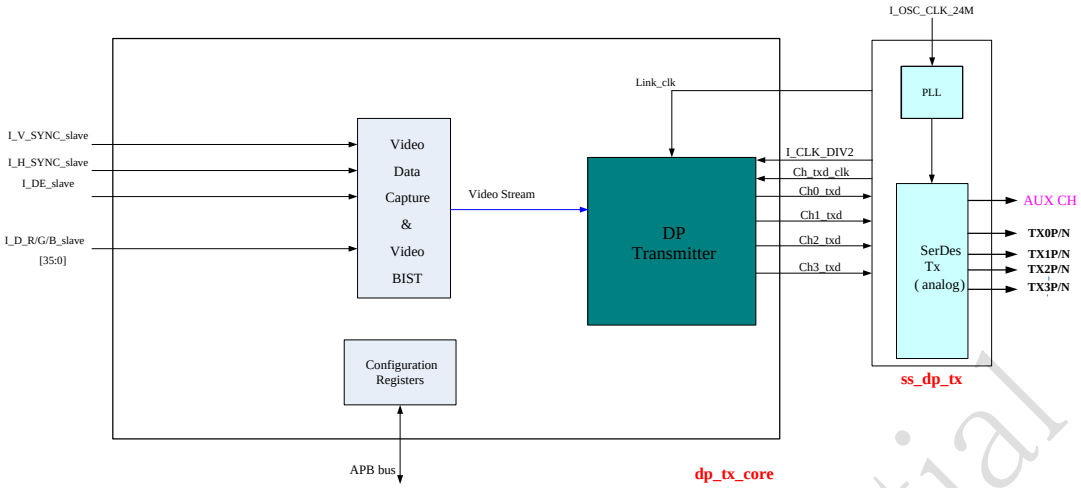

Fig. 34-1 eDP TX controller Block Diagram

Fig.35-1 shows the block diagram of eDP TX controller in top level. The video data and clock are sent directly from the VOP.

The video data capture & video BIST block is separated as video_capture and display_bist module. The block before SerDes is DP_TX main module. Following Table shows the brief function description of each sub-module.

Table 34-1 Brief function description of each module in top level

Module Name in Top Level	Brief Module Function Description
video_capture	Capture block of video data.
display_bist	Generation of arbitrary video format with three types of video data. The output of display_bist module will input to video_capture module directly if display BIST mode is active.
dp_tx	DisplayPort transmitter block.
apb_slave_top	APB slave Bus interface

34.3 Function Description

34.3.1 eDP in SoC

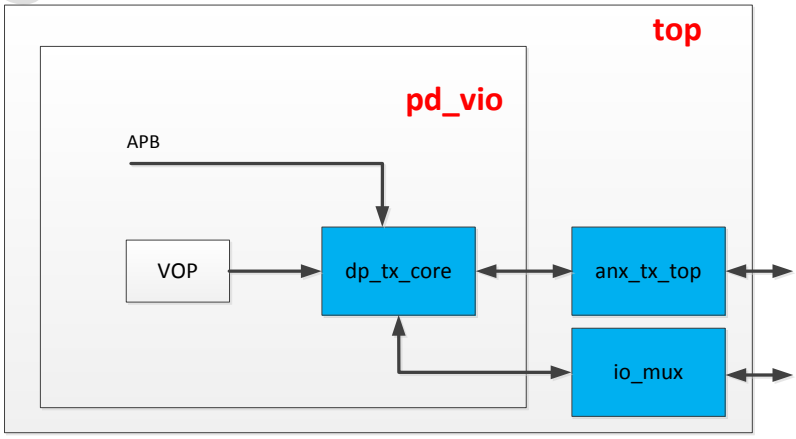

Fig. 34-2 eDP in SoC

Please refer to "eDP TX IP Rockchip Datasheet.docx" for detail information.

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34.4 Register Description

Please refer to "eDP TX IP Rockchip Datasheet.docx" for detail information.

34.5 Interface Description

34.5.1 Video Input Source

In PX5, the eDP TX video source comes from vop.

34.5.2 Hot plug

There is a hot plug input signal to eDP TX. This signal is muxed with GPIO2_C[7], and is enabled by "GPIO2C_IOMUX[15:14] = 2'b01".

34.6 Application Notes

Please refer to "eDP TX IP Rockchip Datasheet.docx" for detail information.

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Chapter 35 MIPI DSI D-PHY

35.1 Overview

The MIPI DSI D-PHY is compliant with the MIPI D-PHY interface specification, revision 1.0. The MIPI DSI D-PHY is used for master applications. The MIPI DSI D-PHY is targeted for transmitting the video digital data for DSI CTRL in mobile applications, supporting a maximum effective bit rate of 1.0Gbps per lane. The assembled four-data-lane system enables up to 4Gbps aggregate communication throughputs, delivering the bandwidth needed for high-throughput data transfer.

The MIPI D-PHY supports the following features:

- Mixed-signal D-PHY mixed-signal hard-macro- HS/LS Transmitter and LS/HS Receiver solution
- Designed to MIPI® v1.0 Specifications and LVDS IEEE Std1596.3-1996
- Integrated PHY Protocol Interface (PPI) supports interface to DSI and UniPro™ MIPI® protocol
- 1.0GHz maximum data transfer rate per lane
- Expandable to support 4 data lanes, providing up to 4Gbps transfer rate
- MIPI-HS, MIPI-LP , LVDS,TTL modes supported
- 10Mbps per lane in low-power mode
- Unidirectional and bi-directional modes supported
- Automatic termination control for HS and LP modes
- Low-Power dissipation: less than 3mA/Lane in HS TX mode
- Tx Buffers with tunable On-Die-Termination and advanced equalization.
- Embedded ESD, boundary scan support logic.

35.2 Block Diagram

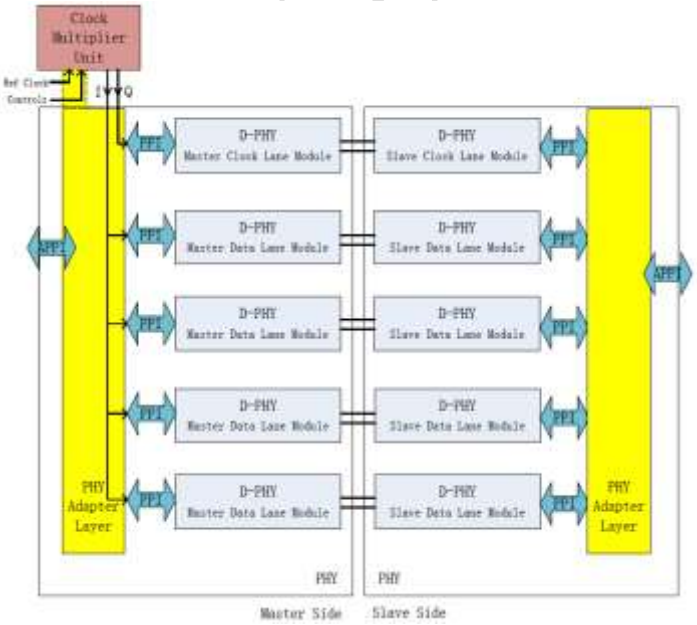

Fig. 35-1 MIPI DSI D-PHY detailed block diagram

MIPI DSI D-PHY configuration contains one Clock Lane Module and four Data Lane Modules. Each of these PHY Lane Modules communicates via two Lines to a complementary part at the other side of the Lane Interconnect.

35.3 Function Description

35.3.1 System Connection

The MIPI DSI DPHY in PX5 supports MIPI/TTL/LVDS mode. VOP outputs data use different ports for MIPI and LVDS/TTL mode. The connection with VOP of three mode shows in following figure:

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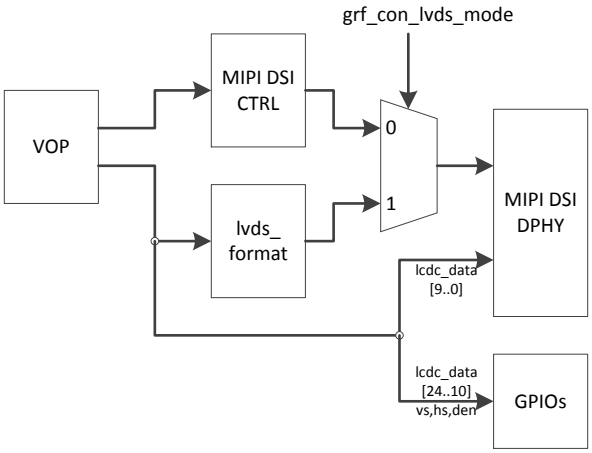

Fig. 35-2 MIPI DSI D-PHY and VOP connection

Because these three modes share same IOs, only one mode should be enabled at the same time. LVDS and MIPI mode use same data ports in MIPI DSI DPHY, they can be selected by grf_con_lvds_mode selects.

35.3.2 MIPI MODE

In MIPI mode, there are some functions of DPHY controlled by grf registers. These register bit and function is show in following table:

Table 35-1 function of grf bits in MIPI mode

register bit	function
grf_con_dsi_phy_lane0_frctxstpm	Force Lane Module Into Transmit mode / Generate Stop state. This signal allows the protocol to force a Lane Module into transmit mode and Stop state during initialization.
grf_con_dsi_phy_lane1_frctxstpm	Force Lane Module Into Transmit mode / Generate Stop state. This signal allows the protocol to force a Lane Module into transmit mode and Stop state during initialization.
grf_con_dsi_phy_lane2_frctxstpm	Force Lane Module Into Transmit mode / Generate Stop state. This signal allows the protocol to force a Lane Module into transmit mode and Stop state during initialization.
grf_con_dsi_phy_lane3_frctxstpm	Force Lane Module Into Transmit mode / Generate Stop state. This signal allows the protocol to force a Lane Module into transmit mode and Stop state during initialization.
grf_con_dsi_phy_lane0_turndisable	Disable Turn-around. This signal is used to prevent a (bi-directional) Lane from going into transmit mode – even if it observes a turn-around request on the Lane interconnect.
grf_con_dsi_phy_forcerxmode	Force Lane Module Into Receive mode / Wait for Stop state. This signal allows the protocol to initialize a Lane Module, or force a bi-directional Lane Module, into receive mode.

35.3.3 LVDS MODE

After video data is sent from VOP, it first goes into lvds_format block. Lvds_format block first

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maps video data to 4 types as show in Table 35-2 LVDS map setting. This function is controlled by register grf_con_lvds_select[1:0]. After video data map, you can use grf_con_lvds_msbsel to revert data bits into MSB.

Table 35-2 LVDS map setting

grf_con_lvds_select	LVDS map when (grf_con_lvds_msbsel == 0) lvds_lane1 use IO_MIPI_DSI_DATAN0 and IO_MIPI_DSI_DATAP0 lvds_lane2 use IO_MIPI_DSI_DATAN1 and IO_MIPI_DSI_DATAP1 lvds_lane3 use IO_MIPI_DSI_DATAN2 and IO_MIPI_DSI_DATAP2 lvds_lane4 use IO_MIPI_DSI_DATAN3 and IO_MIPI_DSI_DATAP3	
0	lvds_lane1	{g_data[0],r_data[5:0]}
	lvds_lane2	{b_data[1:0],g_data[5:1]}
	lvds_lane3	{lcdc_den,lcdc_vsyncn,lcdc_hsyncn,b_data[5:2]}
	lvds_lane4	{1'b0,b_data[7:6],g_data[7:6],r_data[7:6]}
1	lvds_lane1	{g_data[2],r_data[7:2]}
	lvds_lane2	{b_data[3:2],g_data[7:3]}
	lvds_lane3	{lcdc_den,lcdc_vsyncn,lcdc_hsyncn,b_data[7:4]}
	lvds_lane4	{1'b0,b_data[1:0],g_data[1:0],r_data[1:0]}
2	lvds_lane1	{g_data[2],r_data[7:2]}
	lvds_lane2	{b_data[3:2],g_data[7:3]}
	lvds_lane3	{lcdc_den,lcdc_vsyncn,lcdc_hsyncn,b_data[7:4]}
	lvds_lane4	7'b0
3	lvds_lane1	{g_data[0],r_data[5:0]}
	lvds_lane2	{b_data[1:0],g_data[5:1]}
	lvds_lane3	{lcdc_den,lcdc_vsyncn,lcdc_hsyncn,b_data[5:2]}
	lvds_lane4	7'b0

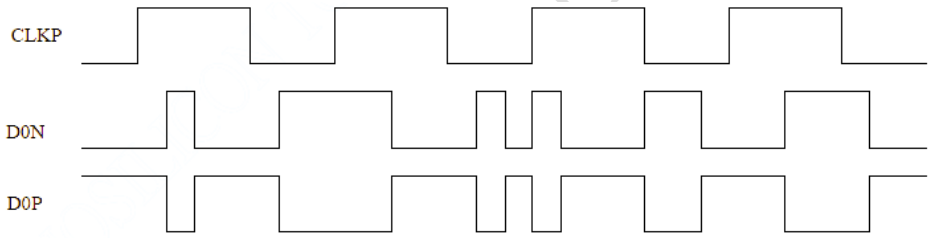

Fig. 35-3 LVDS mode clock timing

, When setting grf_con_lvds_msbsel to 0, the output serial data is transmitted as shown in figure bellow. The rising edge of the output clock is at the transition of bit2 and bit1 of the video_data(r_data,g_data,b_data).

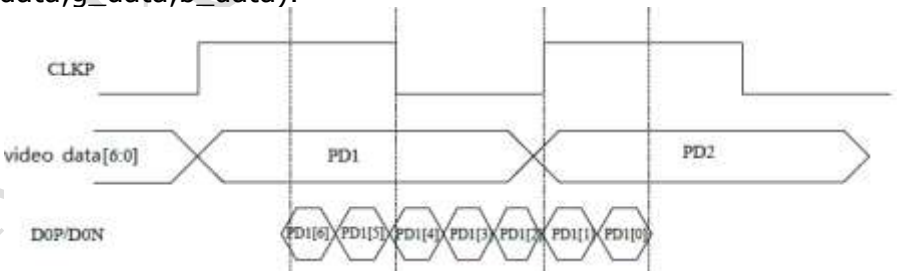

Fig. 35-4 LVDS mode lsb parallel data to serial data timing

35.3.4 TTL MODE

TTL and VOP connection is in table below:

Table 35-3 TTL ports map for VOP

VOP outputs	IO
VSYNC	IO_LCDCvsync_TRACEdata15_PMUdebug3_LCDCgpio0d5
HSYNC	IO_LCDChsync_TRACEdata14_PMUdebug2_LCDCgpio0d4
DCLK	IO_LCDCdclk_TRACEctl_PMUdebug5_LCDCgpio0d7
DEN	IO_LCDCden_TRACEclk_PMUdebug4_LCDCgpio0d6
LCDCdata23	IO_LCDCdata23_TRACEdata13_UART4EXPsin_LCDCgpio0d3

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LCDCdata22	IO_LCDCdata22_TRACEdata12_UART4EXPsout_LCDCgpio0d2
LCDCdata21	IO_LCDCdata21_TRACEdata11_UART4EXPrtsn_LCDCgpio0d1
LCDCdata20	IO_LCDCdata20_TRACEdata10_UART4EXPctsn_LCDCgpio0d0
LCDCdata19	IO_LCDCdata19_TRACEdata9_UART1BBrtsn_LCDCgpio0c7
LCDCdata18	IO_LCDCdata18_TRACEdata8_UART1BBctsn_LCDCgpio0c6
LCDCdata17	IO_LCDCdata17_TRACEdata7_UART1BBsout_LCDCgpio0c5
LCDCdata16	IO_LCDCdata16_TRACEdata6_UART1BBsin_LCDCgpio0c4
LCDCdata15	IO_LCDCdata15_TRACEdata5_M3JTAGtdo_LCDCgpio0c3
LCDCdata14	IO_LCDCdata14_TRACEdata4_M3JTAGtdi_LCDCgpio0c2
LCDCdata13	IO_LCDCdata13_TRACEdata3_M3JTAGtrstn_LCDCgpio0c1
LCDCdata12	IO_LCDCdata12_TRACEdata2_JTAGtdo_LCDCgpio0c0
LCDCdata11	IO_LCDCdata11_TRACEdata1_JTAGtdi_LCDCgpio0b7
LCDCdata10	IO_LCDCdata10_TRACEdata0_JTAGtrstn_LCDCgpio0b6
LCDCdata09	IO_MIPI_DSI_DATAN0
LCDCdata08	IO_MIPI_DSI_DATAP0
LCDCdata07	IO_MIPI_DSI_DATAN1
LCDCdata06	IO_MIPI_DSI_DATAP1
LCDCdata05	IO_MIPI_DSI_CLKN
LCDCdata04	IO_MIPI_DSI_CLKP
LCDCdata03	IO_MIPI_DSI_DATAP2
LCDCdata02	IO_MIPI_DSI_DATAN2
LCDCdata01	IO_MIPI_DSI_DATAP3
LCDCdata00	IO_MIPI_DSI_DATAN3

35.3.5 Program PLL in MIPI mode

Frequency Calculating Formula
The PLL output frequency can be calculated using a simple formula:
PLL_Output_Frequency = (FREF/PREDIV*FBDIV)/2
PLL_Output_Frequency: It is equal to DDR_Clock_Frequency * 2
FREF : 24Mhz
PREDIV : PLL input reference clock divider which can be configured by the register of prediv
FBDIV : Integer value programmed into feedback divider which can be configured by the register of fbdiv
Additional Programming Considerations
1. The divided reference frequency (FREF/PREDIV) should be less than 40MHz.
2. The all possible settings of feedback divider are 12,13,14,16~511.

35.4 Register Description

This section describes the control/status registers of the design.

35.4.1 Register Summary

Name	Offset	Size	Reset Value	Description
MIPI_DSI_DPHY_CTRL_LANE_ENABLE	0x00000	W	0x00000001	Register0000 Abstract
MIPI_DSI_DPHY_CTRL_PWRCTL	0x00004	W	0x00000003	Register0000 Abstract
MIPI_DSI_DPHY_CTRL_PREDIV	0x0000c	W	0x00000003	Register0000 Abstract
MIPI_DSI_DPHY_CTRL_FBDIV	0x00010	W	0x0000007d	Register0000 Abstract

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Name	Offset	Size	Reset Value	Description
MIPI_DSI_DPHY_CTRL_DIG_RST	0x00080	W	0x000000ff	Register0000 Abstract
MIPI_DSI_DPHY_CTRL_SIG_INV	0x00084	W	0x00000002	Register0000 Abstract

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

Name	Offset	Size	Reset Value	Description
DPHY_LVDS_TTL_SETTIING_ctrl	0x00380	W	0x00000001	Register0000 Abstract
DPHY_LVDS_TTL_SETTIING_digital_en	0x00384	W	0x00000012	Register0000 Abstract
DPHY_LVDS_TTL_SETTIING_mode_en	0x0038c	W	0x00000001	Register0000 Abstract
DPHY_LVDS_TTL_SETTIING_lane_en	0x003ac	W	0x00000004	Register0000 Abstract

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

Following table shows registers for 5 lanes. They have same registers but use different address bases. So their final address is lane_base_address + offset.

lane	lane_base_address
clock_lane	0x100
Data0_lane	0x180
Data1_lane	0x200
Data2_lane	0x280
Data3_lane	0x300

Name	Offset	Size	Reset Value	Description
MIPI_DSI_DPHY_LANEX_Tlpx	0x00014	W	0x000000c5	Register0000 Abstract
MIPI_DSI_DPHY_LANEX_S_HSTXTHSPRPR	0x00018	W	0x00000024	Register0001 Abstract
MIPI_DSI_DPHY_LANEX_S_HSTXTHSZERO	0x0001c	W	0x00000000	Register0000 Abstract
MIPI_DSI_DPHY_LANEX_S_HSTX	0x00020	W	0x00000027	Register0000 Abstract
MIPI_DSI_DPHY_LANEX_Ths_exit	0x00024	W	0x0000000a	Register0000 Abstract
MIPI_DSI_DPHY_LANEX_Tclk_post	0x00028	W	0x0000000f	Register0000 Abstract
MIPI_DSI_DPHY_LANEX_LPDT_TX_PPI_SYN_EN	0x00030	W	0x00000000	Register0000 Abstract
MIPI_DSI_DPHY_LANEX_Twakup	0x00034	W	0x00000000	Register0000 Abstract

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Name	Offset	Size	Reset Value	Description
MIPI_DSI_DPHY_LANEX_Tclk_pre	0x00038	W	0x00000006	Register0000 Abstract
MIPI_DSI_DPHY_LANEX_Tta_go	0x00040	W	0x00000004	Register0000 Abstract
MIPI_DSI_DPHY_LANEX_Tta_sure	0x00044	W	0x00000001	Register0000 Abstract
MIPI_DSI_DPHY_LANEX_Tta_wait	0x00048	W	0x00000032	Register0000 Abstract

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

35.4.2 Detail Register Description

MIPI_DSI_DPHY_CTRL_LANE_ENABLE
Address: Operational Base + offset (0x00000)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	lane_en_ck lane_en_ck Field0001 Description
5	RW	0x0	lane_en_3 lane_en_3 Field0001 Description
4	RW	0x0	lane_en_2 lane_en_2 Field0001 Description
3	RW	0x0	lane_en_1 lane_en_1 Field0001 Description
2	RW	0x0	lane_en_0 lane_en_0 Field0001 Description
1:0	RO	0x1	Reserve Reserve Field0000 Description

MIPI_DSI_DPHY_CTRL_PWRCTL
Address: Operational Base + offset (0x00004)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	SYNCRST Field0000 Abstract Field0000 Description

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Bit	Attr	Reset Value	Description
1	RW	0x1	LDO_PD Field0000 Abstract Field0000 Description
0	RW	0x1	PLL_PD Field0000 Abstract Field0000 Description

MIPI_DSI_DPHY_CTRL_PREDIV
Address: Operational Base + offset (0x0000c)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	RW	0x0	FBDIV_HI Field0000 Abstract Field0000 Description
4:0	RW	0x03	PREDIV Field0000 Abstract Field0000 Description

MIPI_DSI_DPHY_CTRL_FBDIV
Address: Operational Base + offset (0x00010)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x7d	FBDIV_LO Field0000 Abstract Field0000 Description

MIPI_DSI_DPHY_CTRL_DIG_RST
Address: Operational Base + offset (0x00080)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:1	RW	0x7f	RESERVE Field0000 Abstract Field0000 Description
0	RW	0x1	dig_rst Field0000 Abstract Field0000 Description

MIPI_DSI_DPHY_CTRL_SIG_INV
Address: Operational Base + offset (0x00084)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1	RW	0x1	PIN_TXCLKESC_0_INV_EN Field0000 Abstract inverting pin_txclkesc_0 enable
0	RW	0x0	PIN_TXBUTECLKHS_INV_EN Field0000 Abstract inverting pin_txbyteclkhs enable

DPHY_LVDS_TTL_SETTIING_ctrl
Address: Operational Base + offset (0x00380)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	digital_internal_reset Field0000 Abstract Field0000 Description
1	RO	0x0	reserved
0	RW	0x1	bit_order Field0000 Abstract selection for MSB and LSB 1: MSB 0: LSB

DPHY_LVDS_TTL_SETTIING_digital_en
Address: Operational Base + offset (0x00384)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x0	digital_internal_en Field0000 Abstract digital internal enable Active high. Default low.
6:0	RO	0x12	Reserved Field0000 Abstract Field0000 Description

DPHY_LVDS_TTL_SETTIING_mode_en
Address: Operational Base + offset (0x0038c)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	ttl_mode_en Field0002 Abstract TTL mode enable 1: enable TTL mode 0: disable TTL mode

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Bit	Attr	Reset Value	Description
1	RW	0x0	lvds_mode_en Field0001 Abstract LVDS mode enable 1: enable LVDS mode 0: disable LVDS mode
0	RW	0x1	mipi_mode_en Field0000 Abstract mipi mode enable 1: enable mipi mode 0: disable mipi mode

DPHY_LVDS_TTL_SETTIING_lane_en
Address: Operational Base + offset (0x003ac)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x0	data_lane0_en Field0000 Abstract Data Lane0 enable for LVDS mode 1: enable Data Lane0 1: disable Data Lane0
6	RW	0x0	data_lane1_en Field0000 Abstract Data Lane1 enable for LVDS mode 1: enable Data Lane1 1: disable Data Lane1
5	RW	0x0	data_lane2_en Field0000 Abstract Data Lane2 enable for LVDS mode 1: enable Data Lane2 1: disable Data Lane2
4	RW	0x0	data_lane3_en Field0000 Abstract Data Lane3 enable for LVDS mode 1: enable Data Lane3 1: disable Data Lane3
3	RW	0x0	clk_lane_en Field0000 Abstract Clock Lane enable for LVDS mode 1: enable Clock Lane 1: disable Clock Lane

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Bit	Attr	Reset Value	Description
2	RW	0x1	pll_powr_off Field0000 Abstract PLL power off enable for LVDS mode 1: power off 0: power on
1:0	RO	0x0	reserved

MIPI_DSI_DPHY_LANEX_Tlpx
Address: Operational Base + offset (0x00014)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:6	RO	0x3	Reserved Field0000 Abstract Field0000 Description
5:0	RW	0x05	Tlpx Field0000 Abstract The value of counter for HS Tlpx Time （>=Tlpx） = Tpin_txbyteclkhs * (2+value)

MIPI_DSI_DPHY_LANEX_S_HSTXTHSPRPR
Address: Operational Base + offset (0x00018)
Register0001 Abstract

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
6:0	RW	0x24	S_HSTXTHSPRPR Field0000 Abstract The value of counter for HS Ths-prepare For clock lane, Ths-prepare(38ns~95ns) For data lane, Ths-prepare(40ns+4UI~85ns+6UI) For clock lane,S_HSTXTHSPRPR[6:0] = 7'b0100100 For data lane, S_HSTXTHSPRPR[6:0] = 7'b0100100 Frequency(1/UI) Value(HEX) 80-110MHz 20 110-150 MHz 06 150-200 MHz 18 200-250 MHz 05 250-300 MHz 51 300-400 MHz 64 400-500 MHz 59 500-600 MHz 6A 600-700 MHz 3E 700-800 MHz 21 800-1000 MHz 09

MIPI_DSI_DPHY_LANEX_S_HSTXTHSZERO
Address: Operational Base + offset (0x0001c)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5:0	RW	0x00	S_HSTXTHSZERO Field0000 Abstract For clock lane, Ths-prepare+Ths-zero (>=300ns) For data lane, Ths-prepare+Ths-zero (>= 145 ns + 10*UI) = Tpin_txbyteclkhs*(5+value) For clock lane, S_HSTXTHSZERO[5:0] = 6'b100000 Frequency(1/UI) Value(HEX) 80 -110 MHz 16 110-150 MHz 16 150-200 MHz 17 200-250 MHz 17 250-300 MHz 18 300-400 MHz 19 400-500 MHz 1B 500-600 MHz 1D 600-700 MHz 1E 700-800 MHz 1F 800-1000 MHz 20 For data lane,S_HSTXTHSZERO[5:0] = 6'b001001 Frequency(1/UI) Value(HEX) 80 -110 MHz 2 110-150 MHz 3 150-200 MHz 4 200-250 MHz 5 250-300 MHz 6 300-400 MHz 7 400-500 MHz 7 500-600 MHz 8 600-700 MHz 8 700-800 MHz 9 800-1000 MHz 9

MIPI_DSI_DPHY_LANEX_S_HSTX
Address: Operational Base + offset (0x00020)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
6:0	RW	0x27	S_HSTX Field0000 Abstract The value of counter for HS Ths-trail For clock lane, Ths-trail (>=60ns) For data lane,Ths-trail (>=max(8UI, 60ns+4UI) For clock lane, S_HXTXTHSTRAIL[6:0] = 7'b0100111 For data lane, S_HXTXTHSTRAIL[6:0] = 7'b0100111 Frequency(1/UI) Value(HEX) 80 -110 MHz 22 110-150 MHz 45 150-200 MHz 0B 200-250 MHz 16 250-300 MHz 2C 300-400 MHz 33 400-500 MHz 4E 500-600 MHz 3A 600-700 MHz 6A 700-800 MHz 29 800-1000 MHz 27

MIPI_DSI_DPHY_LANEX_Ths_exit
Address: Operational Base + offset (0x00024)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x0a	Ths_exit Field0000 Abstract The value of counter for HS Ths-exit Ths-exit = Tpin_txbyteclkhs*value

MIPI_DSI_DPHY_LANEX_Tclk_post
Address: Operational Base + offset (0x00028)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0xf	Tclk_post Field0000 Abstract The value of counter for HS Ths-exit Ths-exit = Tpin_txbyteclkhs*value

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MIPI_DSI_DPHY_LANEX_LPDT_TX_PPI_SYN_EN
Address: Operational Base + offset (0x00030)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	LPDT_TX_PPI_SYN_EN Field0000 Abstract LPDT TX PPI signals internal synchronization enable. 1: enable 0: disable
1:0	RW	0x0	Twakup_H Field0000 Abstract The value[9:8] of counter for HS Twakup

MIPI_DSI_DPHY_LANEX_Twakup
Address: Operational Base + offset (0x00034)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	Twakup Field0000 Abstract The value[7:0] of counter for HS Twakup Twakup for ulpm, Twakup = Tpin_sys_clk*value[9:0]

MIPI_DSI_DPHY_LANEX_Tclk_pre
Address: Operational Base + offset (0x00038)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x6	Tclk_pre Field0000 Abstract Field0000 Description

MIPI_DSI_DPHY_LANEX_Tta_go
Address: Operational Base + offset (0x00040)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5:0	RW	0x04	Tta_go Field0000 Abstract The value of counter for HS Tta-go Tta-go for turnaround Tta-go = Ttxclkesc*value

MIPI_DSI_DPHY_LANEX_Tta_sure
Address: Operational Base + offset (0x00044)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RW	0x01	Tta_sure Field0000 Abstract The value of counter for HS Tta-sure Tta-sure for turnaround

MIPI_DSI_DPHY_LANEX_Tta_wait
Address: Operational Base + offset (0x00048)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RW	0x32	Tta_wait Field0000 Abstract The value of counter for HS Tta-wait Tta-wait for turnaround Interval from receiving ppi turnaround request to sending esc request. Tta-wait = Ttxclkesc*value

35.5 Application Notes

35.5.1 COMMON CONFIGURATION (DEFAULT IN MIPI MODE)

Step1: Enable reference clock.
Step2:Wait a period after reference clock have been enabled.
Step3: Send 0xe4 to register MIPI_DSI_DPHY_CTRL_PWRCTL. Enable PLL and LDO.
Step4: Send 0x7d to register MIPI_DSI_DPHY_CTRL_LANE_ENABLE. Enable all lanes on analog part.
Step5: Send 0xe0 to register MIPI_DSI_DPHY_CTRL_PWRCTL. Reset analog.
Step6:Wait a period after analog has been reset.
Step7: Send 0x1e to register MIPI_DSI_DPHY_CTRL_DIG_RST. Reset digital.
Step8: Send 0x1f to register MIPI_DSI_DPHY_CTRL_DIG_RST. Reset digital.
Step9: Wait a period until pll locked. Run in MIPI mode.

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35.5.2 CONFIGURATION FOR LVDS MODE

Step1: Firstly perform common configuration.
Step2: Configure PLL.
Write 8'h01 to the register MIPI_DSI_DPHY_CTRL_PREDIV;
Write 8'h0e to the register MIPI_DSI_DPHY_CTRL_FBDIV;
Step3: (optional) Select lsb mode, default msb mode.
Write 8'h44 to the register DPHY_LVDS_TTL_SETTIING_ctrl;
Step4: Enable LVDS digital logic.
Write 8'h92 to the register DPHY_LVDS_TTL_SETTIING_digital_en;
Step5: Enable LVDS analog driver.
Write 8'h f8 to the register DPHY_LVDS_TTL_SETTIING_lane_en;
Step6: Select LVDS mode.
Write 8'h 02 to the register DPHY_LVDS_TTL_SETTIING_mode_en;
Step7: Wait a period until PLL locked. Run in LVDS mode.

35.5.3 CONFIGURATION FOR TTL MODE

Step1: Select TTL mode.
Write 8'h04 to the register DPHY_LVDS_TTL_SETTIING_mode_en;
Step2: Enable analog driver.
Write 8'hfd to the register DPHY_LVDS_TTL_SETTIING_lane_en;
Step3: Run in TTL mode.

35.5.4 LOW POWER MODE (FOR DSI ONLY)

Low Power Mode is a special feature for D-PHY. You can control this function by using proper registers from the D-PHY with few operations. The following is a step by step instruction for low power mode in and out.

Low Power in Steps:
Step1: Send 0x01 to register MIPI_DSI_DPHY_CTRL_LANE_ENABLE. Disable all lanes on analog part.
Step2: Send 0xe3 to register MIPI_DSI_DPHY_CTRL_PWRCTL.Disable PLL and LDO.
Step3:Wait a period before reference clock have been disabled.
Step4: Disable reference clock.

Low Power out Steps:
Step1: Enable reference clock.
Step2:Wait a period after reference clock have been enabled.
Step3: Send 0xe4 to register MIPI_DSI_DPHY_CTRL_PWRCTL. Enable PLL and LDO.
Step4: Send 0x7d to register MIPI_DSI_DPHY_CTRL_LANE_ENABLE. Enable all lanes on analog part.
Step5: Send 0xe0 to register MIPI_DSI_DPHY_CTRL_PWRCTL. Reset analog.
Step6:Wait a period after analog has been reset.
Step7: Send 0x1e to register MIPI_DSI_DPHY_CTRL_DIG_RST. Reset digital.
Step8: Send 0x1f to register MIPI_DSI_DPHY_CTRL_DIG_RST. Reset digital.
Step9: Wait a period before normal transmission.

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Chapter 36 MIPI CSI D-PHY

36.1 Overview

The MIPI CSI D-PHY is compliant with the MIPI D-PHY interface specification, revision 1.0. The MIPI CSI D-PHY is used for slave applications. The MIPI CSI D-PHY is targeted for receive the camera video digital data in mobile applications, supporting a maximum effective bit rate of 1.0Gbps per lane. The assembled four-data-lane system enables up to 4Gbps aggregate communication throughputs, delivering the bandwidth needed for high-throughput data transfer.

The MIPI D-PHY supports the following features:

- Mixed-signal D-PHY mixed-signal hard-macro- HS/LS Transmitter and LS/HS Receiver solution
- Designed to MIPI® v1.0 Specifications
- Integrated PHY Protocol Interface (PPI) supports interface to CSI and UniPro™ MIPI® protocol
- 1.0GHz maximum data transfer rate per lane
- Expandable to support 4 data lanes, providing up to 4Gbps transfer rate
- 10Mbps per lane in low-power mode
- Unidirectional and bi-directional modes supported
- Automatic termination control for HS and LP modes
- Low-Power dissipation: less than 3mA/Lane in HS RX mode
- Rx Buffers with tunable On-Die-Termination and advanced equalization.
- Embedded ESD, boundary scan support logic.

36.2 Block Diagram

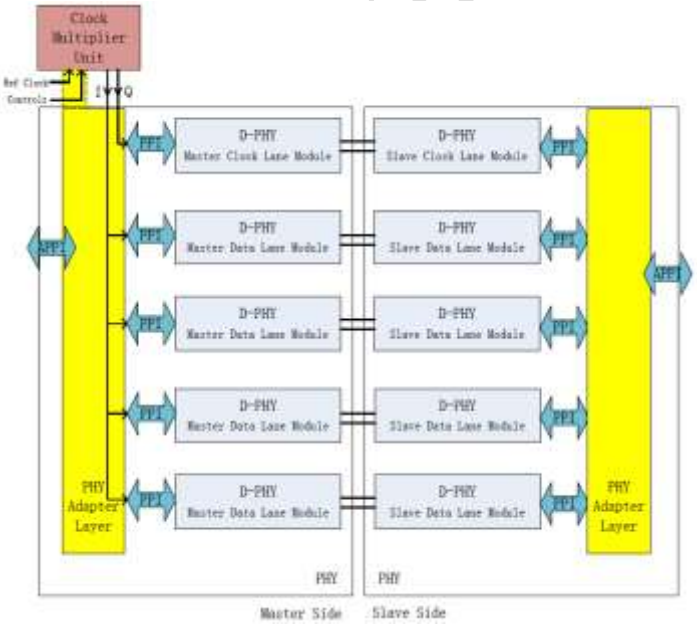

Fig. 36-1 MIPI CSI D-PHY detailed block diagram

MIPI CSI D-PHY configuration contains one Clock Lane Module and four Data Lane Modules. Each of these PHY Lane Modules communicates via two Lines to a complementary part at the other side of the Lane Interconnect.

36.3 Function Description

36.3.1 System Connection

MIPI CSI D-PHY registers are configured by software, but its input ports phy_rst, phy_en and phy_shutdn are selected by grf_conf_isp_mipi_csi_host_sel register from ISP or VIP. D-PHY output video data feeds both VIP and ISP. The connection shows in following figure:

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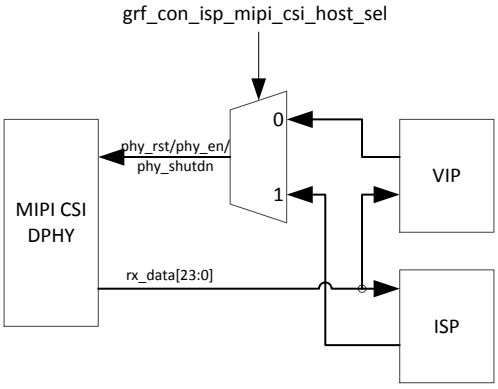

Fig. 36-2 MIPI CSI D-PHY and VIP/ISP connection

In MIPI mode, there are some functions of DPHY controlled by grf registers. These register bit and function is show in following table:

Table 36-1 function of grf bits in MIPI mode

register bit	function
grf_con_dphy_rx_clk_inv_sel	Reverse for receive clock
grf_stat_dphy_rx_rxclkactivehs	
grf_stat_dphy_rx_direction	
grf_con_dphy_rx_forcerxmode[0]	
grf_stat_dphy_rx_ulpsactivenot_0	
grf_con_dphy_rx_forcerxmode[1]	
grf_stat_dphy_rx_ulpsactivenot_1	
grf_con_dphy_rx_forcerxmode[2]	
grf_stat_dphy_rx_ulpsactivenot_2	
grf_con_dphy_rx_forcerxmode[3]	
grf_stat_dphy_rx_ulpsactivenot_4	

36.4 Register Description

This section describes the control/status registers of the design.

36.4.1 Register Summary

Name	Offset	Size	Reset Value	Description
MIPI_CSI_DPHY_CTRL_LANE_ENABLE	0x00000	W	0x00000001	Register0000 Abstract
MIPI_CSI_DPHY_CTRL_PWRCTL	0x00004	W	0x00000003	Register0000 Abstract
MIPI_CSI_DPHY_CTRL_DIG_RST	0x00080	W	0x000000ff	Register0000 Abstract
MIPI_CSI_DPHY_CTRL_SIG_INV	0x00084	W	0x00000002	Register0000 Abstract

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

Following table shows registers for 5 lanes. They have same registers but use different address bases. So their final address is lane_base_address + offset.

lane	lane_base_address

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clock_lane	0x100
Data0_lane	0x180
Data1_lane	0x200
Data2_lane	0x280
Data3_lane	0x300

Name	Offset	Size	Reset Value	Description
MIPI_CSI_DPHY_LANEX_Tlpx	0x00000	W	0x000000c5	Register0000 Abstract
MIPI_CSI_DPHY_LANEX_Tclk_pre	0x00038	W	0x00000006	Register0000 Abstract

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

36.4.2 Detail Register Description

MIPI_CSI_DPHY_CTRL_LANE_ENABLE
Address: Operational Base + offset (0x00000)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	lane_en_ck lane_en_ck Field0001 Description
5	RW	0x0	lane_en_3 lane_en_3 Field0001 Description
4	RW	0x0	lane_en_2 lane_en_2 Field0001 Description
3	RW	0x0	lane_en_1 lane_en_1 Field0001 Description
2	RW	0x0	lane_en_0 lane_en_0 Field0001 Description
1:0	RO	0x1	Reserve Reserve Field0000 Description

MIPI_CSI_DPHY_CTRL_PWRCTL
Address: Operational Base + offset (0x00004)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
2	RW	0x0	SYNCRST Field0000 Abstract Field0000 Description
1	RW	0x1	LDO_PD Field0000 Abstract Field0000 Description
0	RW	0x1	PLL_PD Field0000 Abstract Field0000 Description

MIPI_CSI_DPHY_CTRL_DIG_RST
Address: Operational Base + offset (0x00080)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:1	RW	0x7f	RESERVE Field0000 Abstract Field0000 Description
0	RW	0x1	dig_rst Field0000 Abstract Field0000 Description

MIPI_CSI_DPHY_CTRL_SIG_INV
Address: Operational Base + offset (0x00084)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	PIN_RXBYTECLKHS_INV_EN Field0000 Abstract inverting pin_rxbyteclkhs enable
2:0	RO	0x0	reserved

MIPI_CSI_DPHY_LANEX_THS_SETTLE
Address: Operational Base + offset (0x00000)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RO	0x4	Reserved Field0000 Abstract Field0000 Description

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Bit	Attr	Reset Value	Description
3:0	RW	0xb	Tlpx Field0000 Abstract Configure the count time of the THS-SETTLE by protocol. After the count done, D-PHY will begin to receive the high speed data. (Can be configured from 4'h0 to 4'hd) 4'b0000 80-110 MHz 4'b0001 110-150 MHz 4'b0010 150-200 MHz 4'b0011 200-250 MHz 4'b0100 250-300 MHz 4'b0101 300-400 MHz 4'b0110 400-500 MHz 4'b0111 500-600 MHz 4'b1000 600-700 MHz 4'b1001 700-800 MHz 4'b1010 800-1000 MHz 4'b1011 additional adjust 4'b1100 additional adjust 4'b1101 additional adjust 4'b1110 additional adjust

MIPI_CSI_DPHY_LANEX_MSB_EN
Address: Operational Base + offset (0x00038)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	MSB_EN Field0000 Abstract MSB enable for pin_rxdatahs_* 1: enable 0: disable
5:0	RO	0x0	reserved

36.5 Application Notes

36.5.1 LOW POWER MODE

Low Power Mode is a special feature for D-PHY. You can control this function by using proper registers from the D-PHY with few operations. The following is a step by step instruction for low power mode in and out.

Low Power in Steps:

Step1: Send 0x01 to register MIPI_CSI_DPHY_CTRL_LANE_ENABLE. Disable all lanes on analog part.
Step2: Send 0xe3 to register MIPI_CSI_DPHY_CTRL_PWRCTL.Disable PLL and LDO.
Step3:Wait a period before reference clock have been disabled.

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Step4: Disable reference clock.

Low Power out Steps:

Step1: Enable reference clock.
Step2:Wait a period after reference clock have been enabled.
Step3: Send 0xe4 to register MIPI_CSI_DPHY_CTRL_PWRCTL. Enable PLL and LDO.
Step4: Send 0x7d to register MIPI_CSI_DPHY_CTRL_LANE_ENABLE. Enable all lanes on analog part.
Step5: Send 0xe0 to register MIPI_CSI_DPHY_CTRL_PWRCTL. Reset analog.
Step6:Wait a period after analog has been reset.
Step7: Send 0x1e to register MIPI_CSI_DPHY_CTRL_DIG_RST. Reset digital.
Step8: Send 0x1f to register MIPI_CSI_DPHY_CTRL_DIG_RST. Reset digital.
Step9: Wait a period before normal transmission.

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Chapter 37 MIPI CSI-2 Host Controller

37.1 Overview

The CSI-2 Host Controller implements the CSI-2 protocol on the host side. The CSI-2 link protocol specification is a part of communication protocols defined by MIPI Alliance standards intended for mobile system chip-to-chip communications. The CSI-2 specification is for the image application processor communication in cameras.
The CSI-2 Host Controller is designed to receive data from a CSI-2 compliant camera sensor. A D-PHY configured as a Slave acts as the physical layer.
The MIPI CSI-2 Host Controller supports the following features:

- Compliant with MIPI Alliance Specification for CSI-2, Version 1.01.00-9 November 2010
- Interface with MIPI D-PHY following PHY Protocol Interface, as defined in MIPI Alliance Specification for D-PHY, Version 1.1-7 November 2011
- Up to four D-PHY RX data lanes
- Dynamically configurable multi-lane merging
- Long and Short packet decoding
- Timing accurate signaling of Frame and Line synchronization packets
- Several Frame formats
  - General Frame or Digital Interlaced Video with or without accurate sync timing
  - Data Type (Packet or Frame Level) and Virtual Channel interleaving
- 32-bit Image Data Interface delivering data formatted as recommended in CSI-2 Specification
- All primary and secondary data formats
  - RGB, YUV, and RAW color space definitions
  - From 24-bit down to 6-bit per pixel
  - Generic or user-defined byte-based data types
- Error detection and correction
  - PHY level
  - Packet level
  - Line level
  - Frame level

37.2 Block Diagram

The following diagram shows the MIPI CSI-2 Host Controller architecture.

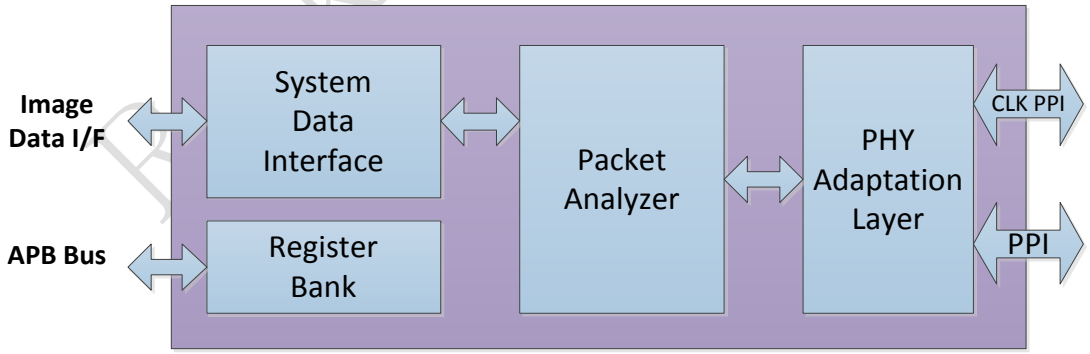

Fig. 37-1 MIPI CSI-2 Host Controller architecture

PHY Adaptation Layer: Manages the D-PHY PPI interface
Packet Analyzer: Merges the data from the different lanes
Image Data Interface: Reorders pixels into 32-bit data for memory storage and generates timing accurate video synchronization signals
AMBA-APB Register Bank: Provides access to configuration and control registers

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37.3 Function Description

37.3.1 Supported Resolutions and Frame Rates

The CSI-2 specification does not define the supported standard resolutions or frame rates. Camera sensor resolution, blanking periods, synchronization events, frame rates, and pixel color depth play a fundamental role in the required bandwidth. All these variables make it difficult to define a standard procedure to estimate the minimum lane rate and the minimum number of lanes that support a specific CSI-2 device.

Table 37-1 presents some predefined and supported camera settings, assuming the following:

- Clock lane frequency is 500 MHz or 750 MHz that results in a bandwidth of 1 Gbps or 1.5 Gbps respectively, for each data lane.
- No significant control/reserved traffic is present on the link when pixel data is being transmitted.

The last column of Table 37-1 presents the minimum number of lanes required for each configuration.

Table 37-1 Supported Camera Settings

Mega Pixels	Mega Pixels with Overhead	Refresh Rate (Hz)	Color Depth (bpp)	CSI2 BW (Mbits)	D-PHY at 1 Gbps Number of Lanes	D-PHY at 1.5Gbps Number of Lanes
2MP	2560000	15	24	922	1	1
2MP	2560000	30	24	1843	2	2
3MP	3840000	15	16	922	1	1
3MP	3840000	30	16	1843	2	2
3MP	3840000	30	24	2765	3	2
5MP	6400000	15	16	1536	2	2
5MP	6400000	15	24	2304	3	2
5MP	6400000	30	16	3072	4	3
8MP	10240000	15	16	2458	3	2
8MP	10240000	15	24	3686	4	3
8MP	10240000	30	12	3686	4	3
12MP	15360000	15	12	2765	3	2
12MP	15360000	15	16	3686	4	3
14MP	17920000	15	12	3226	4	3
16MP	20480000	15	12	3686	4	3
Video Formats						
1280x720 pixels(720p)	921600	30	24	664	1	1
1280x720 pixels(720p)	921600	60	24	1327	2	1
1920x1080 pixels(1080p)	2073600	60	24	2986	3	2

37.3.2 Error Detection

The CSI-2 Host Controller analyzes the received packets and determines if there are protocol errors. It is possible to monitor the following errors:

- Frame errors such as incorrect Frame sequence, reception of a CRC error in the most recent frame, and the mismatch between Frame Start and Frame End
- Line errors such as incorrect line sequence and mismatch between Line Start and Line End
- Packet errors such as ECC or CRC mismatch
- D-PHY errors such as synchronization pattern mismatch

Table 37-2 shows all the errors that CSI-2 Host Controller can identify.

Table 37-2 Errors Identified by the CSI-2 Host Controller

Error	Description	Level	Action
phy_errsotsynchs_*	Start of transmission error on data lane* with no synchronization achieved	PHY	Packets with this error are not delivered in IDI interface
phy_erresc_*	Escape entry error (ULPM) on data lane*	PHY	Informative only. Error is acknowledged in the register and the interrupt pin is raised.
phy_errsoths_*	Start of transmission error on data lane* but synchronization can still be achieved	PHY	Informative only since PHY can recover from this error. Error is acknowledged in register and the interrupt pin is raised.

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vc*_err_crc	Checksum error detected on virtual channel*	Packet	Informative only. Error is acknowledged in the register and Interrupt pin is raised.
vc*_err_crc	Header ECC contains one error detected on virtual channel*	Packet	Informative only since controller can recover the correct header. Error is acknowledged in the register and the interrupt pin is raised.
err_ecc_double	Header ECC contains two errors. Unrecoverable.	Packet	Packets with this error are not delivered in IDI.s
err_id_vc*	Unrecognized or unimplemented data type detected in virtual channel*	Packet	Informative only. Error is acknowledged in the register and the interrupt pin is raised
err_f_bndry_match_vc*	Error matching Frame Start with Frame End for virtual channel*	Frame	Informative only. Error is acknowledged in register and the interrupt pin is raised if not masked.
err_f_seq_vc*	Incorrect Frame Sequence detected in virtual channel*	Frame	Informative only. Error is acknowledged in register and the interrupt pin is raised if not masked.
err_frame_data_vc*	Last received frame, in virtual channel*, had at least one CRC error	Frame	Informative only. Error is acknowledged in the register and the interrupt pin is raised.

37.4 Register Description

This section describes the control/status registers of the design.

37.4.1 Register Summary

Name	Offset	Size	Reset Value	Description
CSIHOST_VERSION	0x0000	W	0x00000000	Version of the CSI2 Host
CSIHOST_N_LANES	0x0004	W	0x00000001	Number of active data lanes
CSIHOST_PHY_SHUTDOWNZ	0x0008	W	0x00000000	PHY shutdown control
CSIHOST_DPHY_RSTZ	0x000c	W	0x00000000	DPHY reset control
CSIHOST_CSI2_RESETN	0x0010	W	0x00000000	CSI-2 Controller reset
CSIHOST_PHY_STATE	0x0014	W	0x00000000	General settings for all blocks
CSIHOST_ERR1	0x0020	W	0x00000000	Error state register 1
CSIHOST_ERR2	0x0024	W	0x00000000	Error state register 2
CSIHOST_MSK1	0x0028	W	0x00000000	Masks for errors 1
CSIHOST_MSK2	0x002c	W	0x00000000	Masks for errors 2
CSIHOST_PHY_TEST_CTRL0	0x0030	W	0x00000000	D-PHY test interface control 0
CSIHOST_PHY_TEST_CTRL1	0x0034	W	0x00000000	D-PHY test interface control 1

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

37.4.2 Detail Register Description

CSIHOST_VERSION

Address: Operational Base + offset (0x0000)

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	VERSION Version of the mipi csi2 host

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CSIHOST_N_LANES
Address: Operational Base + offset (0x0004)

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x1	N_LANES Number of active data lanes 00: 1 data lane (lane 0) 01: 2 data lanes (lanes 0 and 1) 10: 3 data lanes (lanes 0, 1, and 2) 11: 4 data lanes (All) Can only be updated when the D-PHY lane is in Stop state.

CSIHOST_PHY_SHUTDOWNZ
Address: Operational Base + offset (0x0008)

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	PHY_SHUTDOWNZ D-PHY shutdown input. active low

CSIHOST_DPHY_RSTZ
Address: Operational Base + offset (0x000c)

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	DPHY_RSTZ D-PHY reset output. active low

CSIHOST_CSI2_RESETN
Address: Operational Base + offset (0x0010)

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	CSI2_RESETN CSI-2 controller reset output. active low

CSIHOST_PHY_STATE
Address: Operational Base + offset (0x0014)

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10	RO	0x0	PHY_STOPSTATECLK Clock lane in stop state
9	RO	0x0	PHY_RXULPSCLKNOT This signal indicates that the clock lane module has entered the Ultra Low Power state, active low
8	RO	0x0	PHY_RXCLKACTIVEHS Indicates that the clock lane is actively receiving a DDR clock

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Bit	Attr	Reset Value	Description
7	RO	0x0	PHY_STOPSTATEDATA_3 Data lane 3 in stop state
6	RO	0x0	PHY_STOPSTATEDATA_2 Data lane 2 in stop state
5	RO	0x0	PHY_STOPSTATEDATA_1 Data lane 1 in stop state
4	RO	0x0	PHY_STOPSTATEDATA_0 Data lane 0 in stop state
3	RO	0x0	PHY_RXULPSESC_3 lane module 3 has entered the Ultra Low Power mode
2	RO	0x0	PHY_RXULPSESC_2 lane module 2 has entered the Ultra Low Power mode
1	RO	0x0	PHY_RXULPSESC_1 lane module 1 has entered the Ultra Low Power mode
0	RO	0x0	PHY_RXULPSESC_0 lane module 0 has entered the Ultra Low Power mode

CSIHOST_ERR1

Address: Operational Base + offset (0x0020)

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28	RO	0x0	err_ecc_double Header ECC contains 2 errors, unrecoveralbe
27	RO	0x0	vc3_err_crc Checksum error detected on virtual channel 3
26	RO	0x0	vc2_err_crc Checksum error detected on virtual channel 2
25	RO	0x0	vc1_err_crc Checksum error detected on virtual channel 1
24	RO	0x0	vc0_err_crc Checksum error detected on virtual channel 0
23	RO	0x0	err_l_seq_di3 Error in the sequence of lines for vc3 and dt3
22	RO	0x0	err_l_seq_di2 Error in the sequence of lines for vc2 and dt2
21	RO	0x0	err_l_seq_di1 Error in the sequence of lines for vc1 and dt1
20	RO	0x0	err_l_seq_di0 Error in the sequence of lines for vc0 and dt0

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Bit	Attr	Reset Value	Description
19	RO	0x0	err_l_bndry_match_di3 Error matching line start with line end for vc3 and dt3
18	RO	0x0	err_l_bndry_match_di2 Error matching line start with line end for vc2 and dt2
17	RO	0x0	err_l_bndry_match_di1 Error matching line start with line end for vc1 and dt1
16	RO	0x0	err_l_bndry_match_di0 Error matching line start with line end for vc0 and dt0
15	RO	0x0	err_frame_data_vc3 Last received frame, in virtual channel 3, had at least one CRC error
14	RO	0x0	err_frame_data_vc2 Last received frame, in virtual channel 2 had at least one CRC error
13	RO	0x0	err_frame_data_vc1 Last received frame, in virtual channel 1, had at least one CRC error
12	RO	0x0	err_frame_data_vc0 Last received frame, in virtual channel 0, had at least one CRC error
11	RO	0x0	err_f_seq_vc3 Error in the sequence of lines for vc3 and dt3
10	RO	0x0	err_f_seq_vc2 Error in the sequence of lines for vc2 and dt2
9	RO	0x0	err_f_seq_vc1 Error in the sequence of lines for vc1 and dt1
8	RO	0x0	err_f_seq_vc0 Error in the sequence of lines for vc0 and dt0
7	RO	0x0	err_f_bndry_match_vc3 Error matching frame start with frame end for virtual channel 3
6	RO	0x0	err_f_bndry_match_vc2 Error matching frame start with frame end for virtual channel 2
5	RO	0x0	err_f_bndry_match_vc1 Error matching frame start with frame end for virtual channel 1
4	RO	0x0	err_f_bndry_match_vc0 Error matching frame start with frame end for virtual channel 0

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Bit	Attr	Reset Value	Description
3	RO	0x0	phy_errsotsynchs_3 Start of transmission error on data lane 3(no synchronization achieved)
2	RO	0x0	phy_errsotsynchs_2 Start of transmission error on data lane 2 (no synchronization achieved)
1	RO	0x0	phy_errsotsynchs_1 Start of transmission error on data lane 1 (no synchronization achieved)
0	RO	0x0	phy_errsotsynchs_0 Start of transmission error on data lane 0 (no synchronization achieved)

CSIHOST_ERR2

Address: Operational Base + offset (0x0024)

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23	RO	0x0	err_l_seq_di7 Error in the sequence of lines for vc7 and dt7
22	RO	0x0	err_l_seq_di6 Error in the sequence of lines for vc6 and dt6
21	RO	0x0	err_l_seq_di5 Error in the sequence of lines for vc5 and dt5
20	RO	0x0	err_l_seq_di4 Error in the sequence of lines for vc4 and dt4
19	RO	0x0	err_l_bndry_match_di7 Error matching line start with line end for vc7 and dt7
18	RO	0x0	err_l_bndry_match_di6 Error matching line start with line end for vc6 and dt6
17	RO	0x0	err_l_bndry_match_di5 Error matching line start with line end for vc5 and dt5
16	RO	0x0	err_l_bndry_match_di4 Error matching line start with line end for vc4 and dt4
15	RO	0x0	err_id_vc3 Unrecognized or unimplemented data type detected in virtual channel 3
14	RO	0x0	err_id_vc2 Unrecognized or unimplemented data type detected in virtual channel 2
13	RO	0x0	err_id_vc1 Unrecognized or unimplemented data type detected in virtual channel 1

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Bit	Attr	Reset Value	Description
12	RO	0x0	err_id_vc0 Unrecognized or unimplemented data type detected in virtual channel 0
11	RO	0x0	vc3_err_ecc_corrected Header error detected and corrected on virtual channel 3
10	RO	0x0	vc2_err_ecc_corrected Header error detected and corrected on virtual channel 2
9	RO	0x0	vc1_err_ecc_corrected Header error detected and corrected on virtual channel 1
8	RO	0x0	vc0_err_ecc_corrected Header error detected and corrected on virtual channel 0
7	RO	0x0	phy_errsoths_3 Start of transmission error on data lane 3 (synchronization can still be achieved)
6	RO	0x0	phy_errsoths_2 Start of transmission error on data lane 2 (synchronization can still be achieved)
5	RO	0x0	phy_errsoths_1 Start of transmission error on data lane 1 (synchronization can still be achieved)
4	RO	0x0	phy_errsoths_0 Start of transmission error on data lane 0 (synchronization can still be achieved)
3	RO	0x0	phy_erresc_3 Escape entry error (ULPM) on data lane 3
2	RO	0x0	phy_erresc_2 Escape entry error (ULPM) on data lane 2
1	RO	0x0	phy_erresc_1 Escape entry error (ULPM) on data lane 1
0	RO	0x0	phy_erresc_0 Escape entry error (ULPM) on data lane 0

CSIHOST_MSK1

Address: Operational Base + offset (0x0028)

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28	RW	0x0	mask_err_ecc_double Mask for err_ecc_double
27	RW	0x0	mask_vc3_err_crc Mask for vc3_err_crc

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Bit	Attr	Reset Value	Description
26	RW	0x0	mask_vc2_err_crc Mask for vc2_err_crc
25	RW	0x0	mask_vc1_err_crc Mask for vc1_err_crc
24	RW	0x0	mask_vc0_err_crc Mask for vc0_err_crc
23	RW	0x0	mask_err_l_seq_di3 Mask fro err_l_seq_di3
22	RW	0x0	mask_err_l_seq_di2 Mask fro err_l_seq_di2
21	RW	0x0	mask_err_l_seq_di1 Mask fro err_l_seq_di1
20	RW	0x0	mask_err_l_seq_di0 Mask fro err_l_seq_di0
19	RW	0x0	mask_err_l_bndry_match_di3 Mask for err_I_bndry_match_di3
18	RW	0x0	mask_err_l_bndry_match_di2 Mask for err_I_bndry_match_di2
17	RW	0x0	mask_err_l_bndry_match_di1 Mask for err_I_bndry_match_di1
16	RW	0x0	mask_err_l_bndry_match_di0 Mask for err_I_bndry_match_di0
15	RW	0x0	mask_err_frame_data_vc3 Mask for err_frame_data_vc3
14	RW	0x0	mask_err_frame_data_vc2 Mask for err_frame_data_vc2
13	RW	0x0	mask_err_frame_data_vc1 Mask for err_frame_data_vc1
12	RW	0x0	mask_err_frame_data_vc0 Mask for err_frame_data_vc0
11	RW	0x0	mask_err_f_seq_vc3 Mask for err_f_seq_vc3
10	RW	0x0	mask_err_f_seq_vc2 Mask for err_f_seq_vc2
9	RW	0x0	mask_err_f_seq_vc1 Mask for err_f_seq_vc1
8	RW	0x0	mask_err_f_seq_vc0 Mask for err_f_seq_vc0
7	RW	0x0	mask_err_f_bndry_match_vc3 Mask for err_f_bndry_match_vc1
6	RW	0x0	mask_err_f_bndry_match_vc2 Mask for err_f_bndry_match_vc1
5	RW	0x0	mask_err_f_bndry_match_vc1 Mask for err_f_bndry_match_vc1

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Bit	Attr	Reset Value	Description
4	RW	0x0	mask_err_f_bndry_match_vc0 Mask for err_f_bndry_match_vc0
3	RW	0x0	mask_phy_errsotsynchs_3 Mask for phy_errostsynchs_3
2	RW	0x0	mask_phy_errsotsynchs_2 Mask for phy_errostsynchs_2
1	RW	0x0	mask_phy_errsotsynchs_1 Mask for phy_errostsynchs_1
0	RW	0x0	mask_phy_errsotsynchs_0 Mask for phy_errostsynchs_0

CSIHOST_MSK2

Address: Operational Base + offset (0x002c)

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23	RO	0x0	err_l_seq_di7 Error in the sequence of lines for vc7 and dt7
22	RO	0x0	err_l_seq_di6 Error in the sequence of lines for vc6 and dt6
21	RO	0x0	err_l_seq_di5 Error in the sequence of lines for vc5 and dt5
20	RO	0x0	err_l_seq_di4 Error in the sequence of lines for vc4 and dt4
19	RO	0x0	err_l_bndry_match_di7 Error matching line start with line end for vc7 and dt7
18	RO	0x0	err_l_bndry_match_di6 Error matching line start with line end for vc6 and dt6
17	RO	0x0	err_l_bndry_match_di5 Error matching line start with line end for vc5 and dt5
16	RO	0x0	err_l_bndry_match_di4 Error matching line start with line end for vc4 and dt4
15	RO	0x0	err_id_vc3 Unrecognized or unimplemented data type detected in virtual channel 3
14	RO	0x0	err_id_vc2 Unrecognized or unimplemented data type detected in virtual channel 2
13	RO	0x0	err_id_vc1 Unrecognized or unimplemented data type detected in virtual channel 1

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Bit	Attr	Reset Value	Description
12	RO	0x0	err_id_vc0 Unrecognized or unimplemented data type detected in virtual channel 0
11	RO	0x0	vc3_err_ecc_corrected Header error detected and corrected on virtual channel 3
10	RO	0x0	vc2_err_ecc_corrected Header error detected and corrected on virtual channel 2
9	RO	0x0	vc1_err_ecc_corrected Header error detected and corrected on virtual channel 1
8	RO	0x0	vc0_err_ecc_corrected Header error detected and corrected on virtual channel 0
7	RO	0x0	phy_errsoths_3 Start of transmission error on data lane 3 (synchronization can still be achieved)
6	RO	0x0	phy_errsoths_2 Start of transmission error on data lane 2 (synchronization can still be achieved)
5	RO	0x0	phy_errsoths_1 Start of transmission error on data lane 1 (synchronization can still be achieved)
4	RO	0x0	phy_errsoths_0 Start of transmission error on data lane 0 (synchronization can still be achieved)
3	RW	0x0	mask_phy_erresc_3 Mask for phy_erresc_3
2	RW	0x0	mask_phy_erresc_2 Mask for phy_erresc_2
1	RW	0x0	mask_phy_erresc_1 Mask for phy_erresc_1
0	RW	0x0	mask_phy_erresc_0 Mask for phy_erresc_0

CSIHOST_PHY_TEST_CTRL0

Address: Operational Base + offset (0x0030)

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	phy_testclk D-PHY test interface strobe signal It is used to clock TESTDIN bus into the D-PHY. In conjunction with TESTEN signal controls controls the operation selection

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Bit	Attr	Reset Value	Description
0	RW	0x0	phy_testclr D-PHY test interface clear It is used when active performs vendor specific interface initialization (active high)

CSIHOST_PHY_TEST_CTRL1

Address: Operational Base + offset (0x0034)

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RW	0x0	phy_testen D-PHY test interface operation selector: 1: configures address write operation on the falling edge of TESTCLK 0: configures a data write operation on the rising edge of TESTCLK
15:8	RO	0x00	phy_testdout D-PHY output 8-bit data bus for read-back and internal probing functionalities
7:0	RW	0x00	phy_testdin D-PHY test interface input 8-bit data bus for internal register programming and test functionlities access

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Chapter 38 MIPI DSI Controller

38.1 Overview

The Display Serial Interface (DSI) is part of a group of communication protocols defined by the MIPI Alliance. The MIPI Controller is a digital core that implements all protocol functions defined in the MIPI DSI Specification. The MIPI Controller provides an interface between the system and the MIPI D-PHY, allowing the communication with a DSI-compliant display. The MIPI Controller supports one to four lanes for data transmission with MIPI D-PHY.
The MIPI Controller supports the following features:

- Compliant with MIPI Alliance standards
- Support the DPI interface color coding mappings into 24-bit Interface
  - 16 bits per pixel, configurations 1,2,and 3
  - 18 bits per pixel, configurations 1 and 2
  - 24 bits per pixel
- Programmable polarity of all DPI interface signals
- Extended resolutions beyond the DPI standard maximum resolution of 800x480 pixels:
  - Up to 2047 vertical active lines
  - Up to 63 vertical back porch lines
  - Up to 63 vertical front porch lines
  - Maximum resolution is limited by available DSI Physical link bandwidth which depends on the number of lanes and maximum speed per lane
- All commands defined in MIPI Alliance Specification for Display Command Set (DCS)
- Interface with MIPI D-PHY following PHY Protocol Interface (PPI), as defined in MIPI Alliance Specification for D-PHY
- Up to four D-PHY Data Lanes
- Bidirectional communication and escape mode support through data lane 0
- Transmission of all generic commands
- ECC and Checksum capabilities
- End of Transmission Packet(EOTp)
- Ultra Low-Power mode
- Fault recovery schemes

38.2 Block Diagram

The following diagram shows the MIPI Controller architecture.

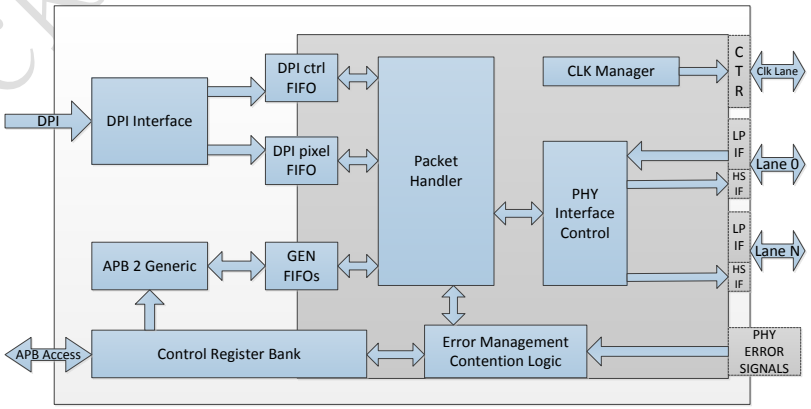

Fig. 38-1 MIPI Controller architecture

The DPI interface captures the data and control signals and conveys them to a FIFO for video control signals and another one for pixel data. This data is then used to build Video packets, hen in Video mode.
The Register Bank is accessible through a standard AMBA-APB slave interface, providing access to the MIPI Controller registers for configuration and control. There is also a fully programmable interrupt generator to inform the system about certain events.

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The PHY Interface Control is responsible for managing the D-PHY PPI interface. It acknowledges the current operation and enables low-power transmission/reception or a high-speed transmission. It also performs data splitting between available D-PHY lanes for high-speed transmission.
The Packet Handler schedules the activities inside the link. It performs several functions based on the interfaces that are currently DPI and the video transmission mode that is used (burst mode or non-burst mode with sync pulse or sync events). It builds long or short packet generating correspondent ECC and CRC codes. This block also performs the following functions: Packet reception, Validation of packet header by checking the ECC, Header correction and notification for single-bit errors, Termination of reception, Multiple header error notification.
The APB-to-Generic block bridges the APB operations into FIFOs holding the Generic commands. The block interfaces with the following FIFOS: Command FIFO, Write payload FIFO, Read payload FIFO.
The Error Management notifies and monitors the error conditions on the DSI link. It controls the timers used to determine if a timeout condition occurred, performing an internal soft reset and triggering an interruption notification.

38.3 Function Description

38.3.1 DPI interface function

The DPI interface follows the MIPI DPI specification with pixel data bus width up to 24 bits. It is used to transmit the information in Video mode in which the transfers from the host processor to the peripheral take the form of a real-time pixel stream. This interface allows sending ShutDown (SD) and ColorMode (CM) commands, which are triggered directly by writing to the register of CFG_MISC_CON[2:1]. To transfer additional commands(for example, to initialize the display), use another interface such as APB Slave Generic Interface to complement the DPI interface.
The DPI interface captures the data and control signals and conveys them to the FIFO interfaces that transmit them to the DSI link. Two different streams of data are presented at the interface; video control signals and pixel data. Depending on the interface color coding, the pixel data is disposed differently throughout the dpipixdata bus. The following table shows the Interface pixel color coding.

Table 38-1 Color table

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Signal Line	16-bit			18-bit		24-bit
	Config1	Config2	Config3	Config1	Config2	
dpipixdata23	Not used	Not used	Not used	Not used	Not used	R7
dpipixdata22	Not used	Not used	Not used	Not used	Not used	R6
dpipixdata21	Not used	Not used	R4	Not used	R5	R5
dpipixdata20	Not used	R4	R3	Not used	R4	R4
dpipixdata19	Not used	R3	R2	Not used	R3	R3
dpipixdata18	Not used	R2	R1	Not used	R2	R2
dpipixdata17	Not used	R1	R0	R5	R1	R1
dpipixdata16	Not used	R0	Not used	R4	R0	R0
dpipixdata15	R4	Not used	Not used	R3	Not used	G7
dpipixdata14	R3	Not used	Not used	R2	Not used	G6
dpipixdata13	R2	G5	G5	R1	G5	G5
dpipixdata12	R1	G4	G4	R0	G4	G4
dpipixdata11	R0	G3	G3	G5	G3	G3
dpipixdata10	G5	G2	G2	G4	G2	G2
dpipixdata9	G4	G1	G1	G3	G1	G1
dpipixdata8	G3	G0	G0	G2	G0	G0
dpipixdata7	G2	Not used	Not used	G1	Not used	B7
dpipixdata6	G1	Not used	Not used	G0	Not used	B6
dpipixdata5	G0	Not used	B5	B5	B5	B5
dpipixdata4	B4	B4	B4	B4	B4	B4
dpipixdata3	B3	B3	B3	B3	B3	B3
dpipixdata2	B2	B2	B2	B2	B2	B2
dpipixdata1	B1	B1	B1	B1	B1	B1
dpipixdata0	B0	B0	Not used	B0	B0	B0

The DPI interface can be configured to increase flexibility and promote correct usage of this interface for several systems. These configuration options are as follows:Polarity control: All the control signals are programmable to change the polarity depending on system requirements.
After the MIPI Controller reset, DPI waits for the first VSYNC active transition to start signal sampling, including pixel data, and preventing image transmission in the middle of a frame.
If interface pixel color coding is 18 bits and the 18-bit loosely packed stream is disabled, the number of lines programmed in the pixels per lines configuration is a multiple of four. This means that in this mode, the two LSBs in the configuration are always inferred as zero. The specification states that in this mode, the pixel line size should be a multiple of four.

38.3.2 APB Slave Generic Interface

The APB Slave interface allows the transmission of generic information in Command mode, and follows the proprietary register interface. Commands sent through this interface are not constrained to comply with the DCS specification, and can include generic commands described in the DSI specification as manufacturer-specific.
The MIPI Controller supports the transmission or write and read command mode packets as described in the DSI specification. These packets are built using the APB register access. The GEN_PLD_DATA register has two distinct functions based on the operation. Writing to this register sends the data as payload when sending a Command mode packet. Reading this register returns the payload of a read back operation. The GEN_HDR register contains the Command mode packet header type and header data. Writing to this register triggers the transmission of the packet implying that for a long Command mode packet, the packet's payload needs to be written in advance in the GEN_PLD_DATA register.
The valid packets available to be transmitted through the Generic interface are as follows:
Generic Write Short Packet 0 Parameters
Generic Write Short Packet 1 Parameters
Generic Write Short Packet 2 Parameter

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Generic Write Short Packet 0 Parameter
Generic Write Short Packet 1 Parameters
Generic Write Short Packet 2 Parameter
Maximum Read Packet Configuration
Generic Long Write Packet
DCS Write Short Packet 0 Parameter
DCS Write Short Packet 1 Parameter
DCS Write Short Packet 0 Parameter
DCS Write Long Packet
A set of bits in the CMD_PKT_STATUS register report the status of the FIFOs associated with APB interface support.
Generic interface packets are always transported using one of the DSI transmission modes; Video mode or Command mode. If neither of these mode are selected, the packets are not transmitted through the link and the released FIFOs eventually get overflowed.
The transfer of packets through the APB bus is based on the following conditions:
The APB protocol defines that the write and read procedure takes two clock cycles each to be executed. This means that the maximum input data rate through the APB interfaces is always half the speed of the APB clock.
The data input bus has a maximum width of 32 bits. This allows for a relation to be defined between the input APB clock frequency and maximum bi rate achievable by the APB interface. The DSI link bit rate when using solely APB is equal to (APB clock frequency) *16 Mbps.
The bandwidth is dependent on the APB clock frequency; the available bandwidth increases with the clock frequency.
To drive the APB interface to achieve high bandwidth Command mode traffic transported by the DSI link, the MIPI Controller should operate in the Command mode only and the APB interface should be the only data source that is currently in use. Thus, the APB interface has the entire bandwidth of the DSI link and does not share it with any another input interface source.
The memory write commands require maximum throughout from the APB interface, because they contain the most amount of data conveyed by the DSI link. While writing the packet information, first write the payload of a given packet into the payload FIFO using the GEN_PLD_DATA register. When the payload data is for the command parameters, place the first byte to be transmitted in the least sgnificant byte position of the APB data bus.
After writing the payload, write the packet header into the command FIFO. For more information and it should follow the pixel to byte conversion organization referred in the Annexure A of the DCS specification. The follow figures show how the pixel data should be orgavized in the APB data write bus. The memory write commands are conveyed in DCS long packets. DCS long packets are encapsulated in a DSI packet. The DSI included in the diagrams.In the follow figures, the Write Memory Command can be replaced by the DCS command Write Memory Start and Write Memory Continue.

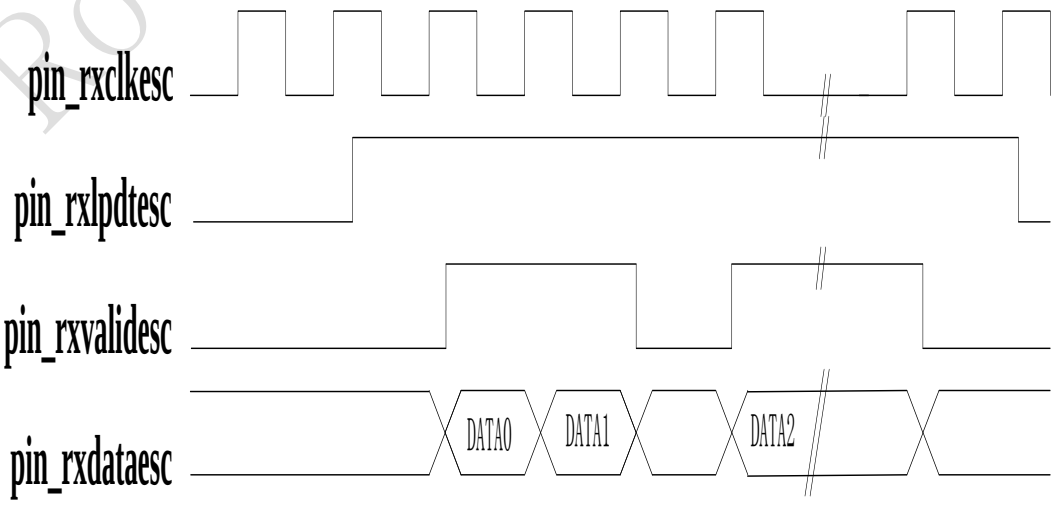

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Fig. 38-2 24bpp APB Pixel to Byte Organization

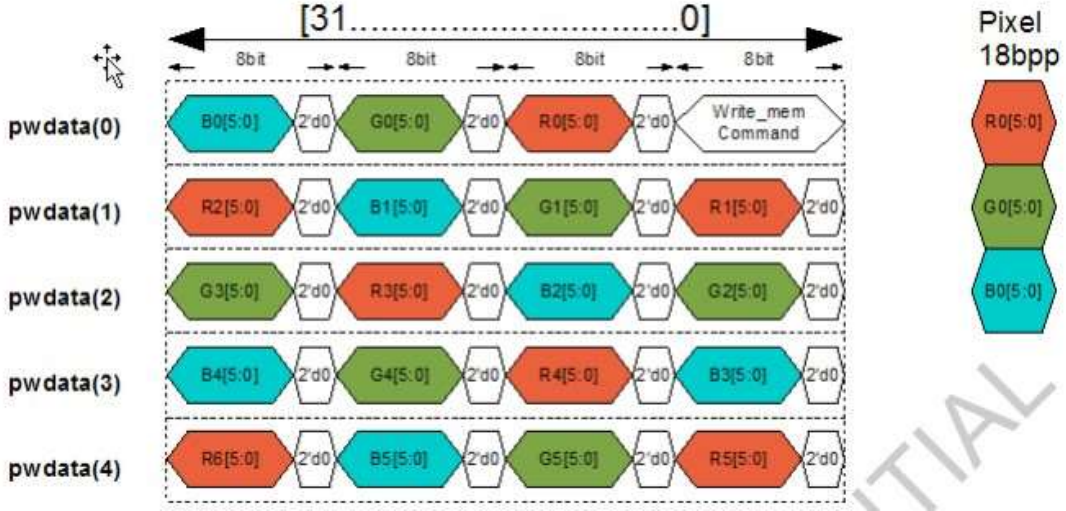

Fig. 38-3 18 bpp APB Pixel to Byte Organization

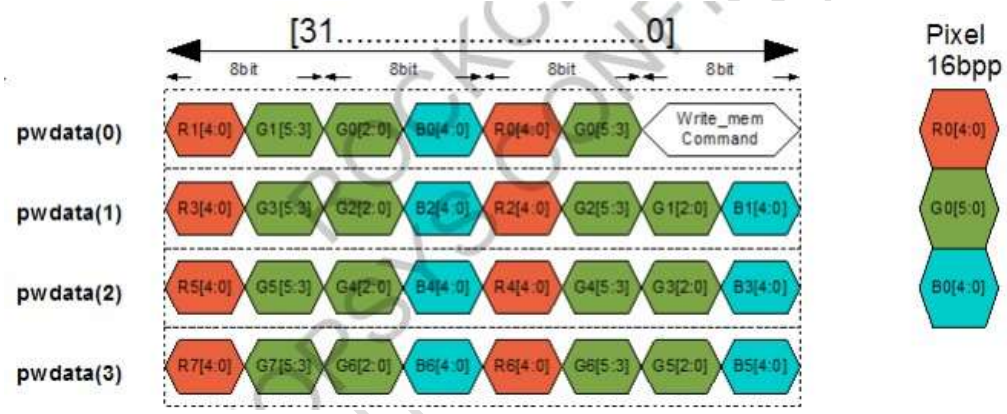

Fig. 38-4 16 bpp APB Pixel to Byte Organization

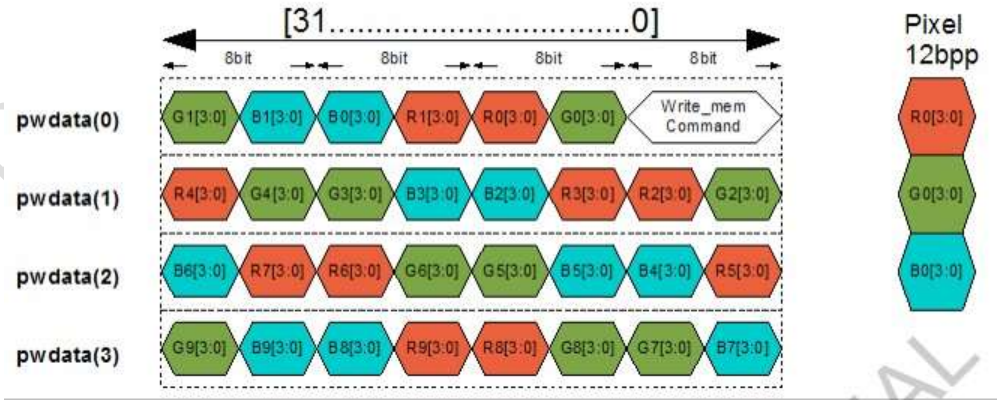

Fig. 38-5 12 bpp APB Pixel to Byte Organization

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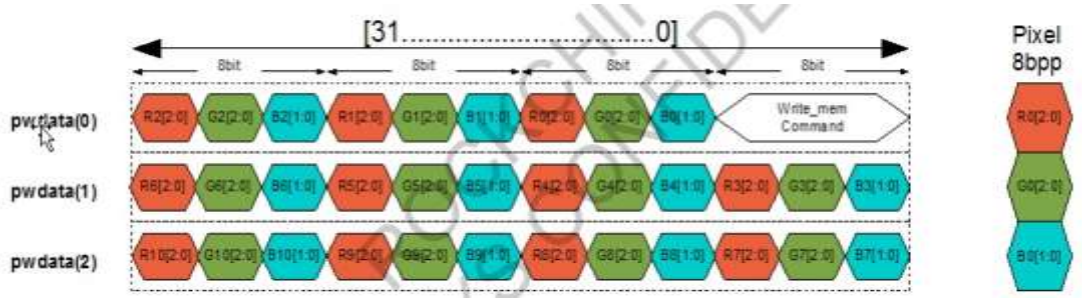

Fig. 38-6 8bpp APB Pixel to Byte Organization

38.3.3 Transmission of Commands in Video Mode

The MIPI Controller supports the transmission of commands, both in high-speed and low-power, while in Video mode. The DSI controller uses Blanking or Low-Power(BLLP) periods to transmit commands inserted through the APB Generic interface. Those periods correspond to the shaded areas of the following figure.

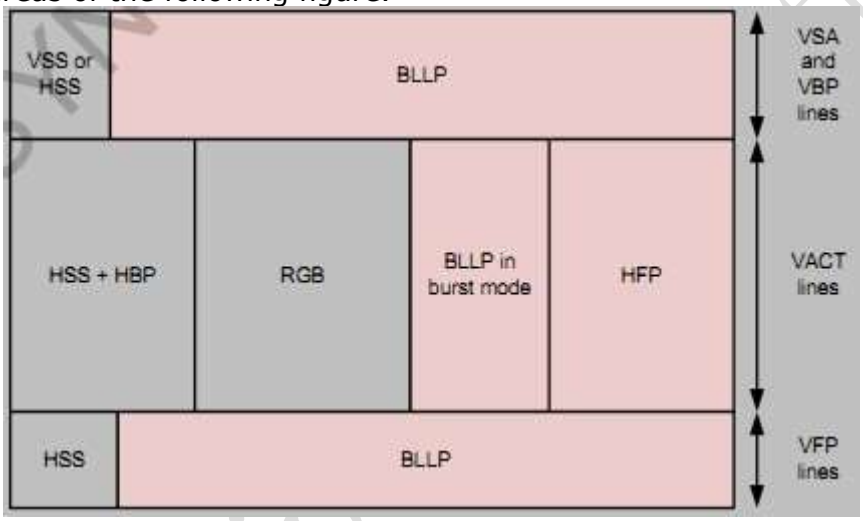

Fig. 38-7 Command Transmission Periods within the Image Area

Commands are transmitted in the blanking periods after the following packet/states:

- Bertical Sync Start (VSS) packets, if the Video Sync pulses are not enabled
- Horizontal Sync End (HSE) packets, in the VSA, VBP, and VFP regions
- Horizontal Sync Start (HSS) packets, if the Video Sync pulses are not enabled in the VSA, VBP, and VFP regions
- Horizontal Active (HACT) state

Only one command is transmitted per line, even in the case of the last line of a frame but one command is possible for each line.

The MIPI Controller avoids sending commands in the last line because it is possible that the last line is shorter than the other ones. For instance, the line time (tL) could be half a cycle longer than the tL on the DPI interface, that is, each line in the frame taking half a cycle from time for the last line. This results in the last line being (1/2 cycle) * (number of lines -1) shorter than tL.

The dpicolorm and dpishutdn input signals are also able to trigger the sending of command packets. The commands are DSI data types Color Mode On, Color Mode Off, Shut Down Peripheral, and Turn on Peripheral. These commands are not sent in the VACT region. If the lpcmden bit of the VID_MODE_CFG register is 1, these commands are sent in LP mode. In LP mode, the ouvact_lpcmd_time field of the LP_CMD_TIM register is used to determine if these commands can be transmitted. It is assumed that outvact_lpcmd_time is greater than or equal to 4 bytes (number of bytes in a short packet), because the DWC_mipi_dsi_host does not transmit these commands on the last line.

If the frame_BTA_ack field is set in the VID_MODE_CFG register, a BTA is generated by

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DWC_mipi_dsi_host after the last line of a frame. This may coincide with a write command or a read command. In either case, the edpihalt signal is held asserted until an acknowledge has been received (control of the DSI bus is returned to the host).
If the lpcmden bit of the VID_MODE_CFG register is set to 1, the commands are sent in low-power in Video mode. In this case, it is necessary to calculate the time available, in bytes, to transmit a command in LP mode for Horizontal Front Porch (HFP), Vertical Sync Active (VSA), Vertical Back Porch (VBP), and Vertical Front Porch(VFP) regions.
The outvact_lpcmd_time field of the LP_CMD_TIM register indicates the time available (in bytes) to transmit a command in LP mode, based on the escape clock, on a line during the VSA, VBP, and the VFP
Outvact_lpcmd_time=(tL-(Time to transmit HSS and HSE frames +tHSA+Time to enter and leave LP mode+Time to send the D-PHY LPDT command)) / escape clock period / 8 / 2
Where,
tL=Line time
tHSA=Time to send a short packet (for sync events) or time of the HAS pulse (for sync pulses)
In the above equation, division by eight is done to convert the time available to bytes and division by two is done because one bit is transmitted once in every two escape clock cycles.
The outvact_lpcmd_time filed can be compared directly with the size of the command to be transmitted to determine if there is enough time to transmit the command. The maximum size of a command that can be transmitted in LP mode is limited to 255 bytes by this field. This register must be programmed to a value greater than or equal to 4 bytes for the transmission of the DCTRL commands such as shutdown and colorm in LP mode.
Consider an example with 12.6 μs per line and assume an escape clock frequency of 15 MHz. In this case, 189 escape clock cycles are available to enter and exit LP mode and transmit command. The following are assumed:
Sync pulses are not being transmitted
Two lane byte clock ticks are required to transmit a short packet
phy_lp2hs_time=16
phy_lp2p_time=20
In this example, a 11-byte command can be transmitted as follows:
outvact_lpcmd_time=(12.6μs-(2*10 ns)-(16*10 ns)-(20*10 ns)-(8*66 ns)) / 66 ns / 8 / 2 = 11 bytes
The invact_lpcmd_time field of the LP_CMD_TIM register indicates the time available (in bytes) to transmite a command in LP mode (based on the escape clock) in the Vertical Active (VACT) region. This time is calculated as follows:
Invact_lpcmd_time=((tHFP-Time to enter and leave low-power mode + Blanking period before the HFP when in Burst mode- Time to send the D-PHY LPDT command) / escape clock period) / 8
Where,
tHFP=line time-tHSA-tHBP-tHACT
tHACT=vid_pkt_size*bits_per_pixel*lane_byte_clock_period / num_lanes
The invact_lpcmd_time field can be compared directly with the size of the command to be transmitted to determined if there is time to transmit the command.
Consider an example where the refresh rate is 60 Hz. The number of lines is 1320 (typical). The tL in this case is 12.6μs. With a lane byte clock of 100 MHz, 1260 clock ticks are availabel to transmit a single frame. If 800 ticks are used for pixel data then 460 ticks (4.6μs) are available for Horizontal Sync Start (HSS), HFP, and HBP. Assuming that 2.3μs is available for HFP and the escape clock is 15MHz, only 34 LP clock ticks are available to enter LP, transmit a command, and return from LP mode. Approximately 12 escape clock ticks are required to enter and leave LP mode. Therefore, only 1 byte could be transmitted in this period.
A short packet (for example, generic short write) requires a minimum of 4 bytes. Therefore, in this exampled, commands are not sent in the VACT region. If Burst mode is enabled, more time is available to transmit commands in the VACT region. The following are assumed:
The controller is not in Burst mode
phy_lp2hs_time=16
phy_lp2hs_time=16

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In this examplem invact_lpcmd_time is calculated as follows:
Invact_lpcmd_time = (2.3μs –(16*10 ns)-(20*10 ns)-(8*66 ns)) /66 ns / 8 = 2 bytes
The outvact_lpcmd_time and invact_lpcmd_time fields allow a simple comparision to determine if a command can be transmitted in any of the BLLP periods.
Fig. 38-8 illustrates the meaning of invact_lpcmd_time and outvact_lpcmd_time, matching them with the shaded areas and the VACT region.

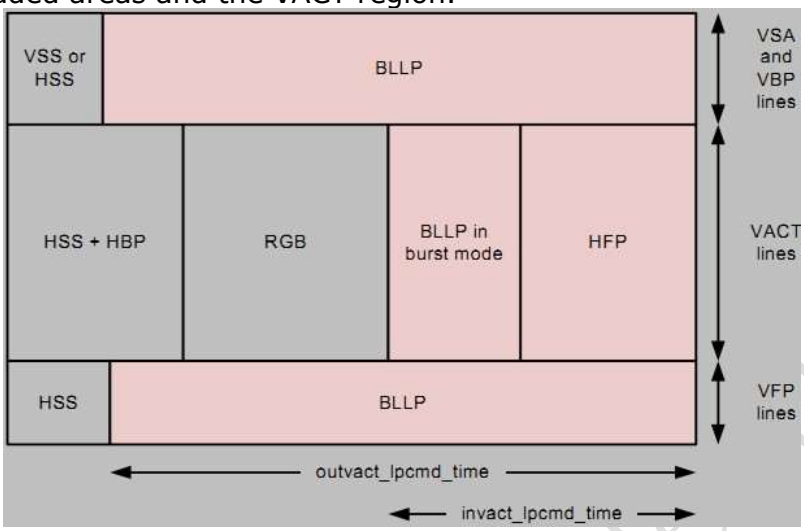

Fig. 38-8 Location in the Image Area

If the lpcmden bit of the VID_MODE_CFG register is 0, the commands are sent in high_speed in Video Mode. In this case, the DWC_mipi_dsi_host automatically determines the area where each command can be sent and no programming or calculation is requied.
On read command Transmission, the max_rd_time field of the PHY_TMR_CFG register configures the maximum amount of time required to perform a read command in lane byte clock cycles.
The maximum time required to perform a read command in Lane byte clock cycles (max_rd_time) =Time to transmit the read command in LP mode + Time to enter and leave LP mode + Time to return the read data packet from the peripheral device.
The time to return the read data packet from the pripheral depends on the number of bytes read and the escape clock frequency of the peripheal; not the escape clock of the host. The max_rd_time field is used in both HS and LP mode to determine if there is time to complete a read command in a BLLP period.
In high-speed mode (lpcmden=0), max_rd_time is calculated as follows:
max_rd_time = phy_hs2lp_time + Time to return the read data packet from the peripheral device + phy_hs2hs_time
In low-power mode (lpcmden = 1), max_rd_time is calculated as follows:
max_rd_time = phy_hs2lp_time +LPDT command time + Read command time in LP mode + Time to return the data read from the peripheral device + phy_lp2hs_time
Where,
LPDT command time = (8*Host escape clock period) / Lane byte clock period
Read command time in LP mode = (32 * host escape clock period) / lane byte clock period
It is recommended to keep the maximum number of bytes read from the peripheral to a minimum to have sufficient time available to issue the read commands on a line. Ensure that max_rd_time* Lane byte clock period is less than outvact_lpcmd_time *8*Escape clock period of the host.
Otherwise, the read commands are serviced on the last line of a frame and the edpihalt signal may be asserted. If it is necessary to read a large number of parameters (＞16), increase the max_rd_time while the read command is being executed. When the read has completed, decrease the max_rd_time to a lower value.

38.4 Register Description

This section describes the control/status registers of the design.

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38.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
MIPIC_VERSION	0x00000	W	0x3133302a	Version of the mipi controller
MIPIC_PWR_UP	0x00004	W	0x00000000	Core power-up
MIPIC_CLKMGR_CFG	0x00008	W	0x00000000	Configuration of the internal clock dividers
MIPIC_DPI_VCID	0x0000c	W	0x00000000	The DPI interface configuration.
MIPIC_DPI_COLOR_CODING	0x00010	W	0x00000000	
MIPIC_DPI_CFG_POL	0x00014	W	0x00000000	
MIPIC_LP_CMD_TIM	0x00018	W	0x00000000	Low-power Command Timing Configuration Register.
MIPIC_PCKHDL_CFG	0x0002c	W	0x00000000	Packet handler configuration
MIPIC_GEN_VCID	0x00030	W	0x00000000	
MIPIC_MODE_CFG	0x00034	W	0x00000000	
MIPIC_VID_MODE_CFG	0x00038	W	0x00000000	Video mode configuration.
MIPIC_VID_PKT_SIZE	0x0003c	W	0x00000000	
MIPIC_VID_NUM_CHUNKS	0x00040	W	0x00000000	
MIPIC_VID_NULL_SIZE	0x00044	W	0x00000000	
MIPIC_VID_HSA_TIME	0x00048	W	0x00000000	Line timing configuration.
MIPIC_VID_HBP_TIME	0x0004c	W	0x00000000	
MIPIC_VID_HLINE_TIME	0x00050	W	0x00000000	
MIPIC_VID_VSA_LINES	0x00054	W	0x00000000	Vertical timing configuration.
MIPIC_VID_VBP_LINES	0x00058	W	0x00000000	
MIPIC_VID_VFP_LINES	0x0005c	W	0x00000000	
MIPIC_VID_VACTIVE_LINES	0x00060	W	0x00000000	
MIPIC_EDPI_CMD_SIZE	0x00064	W	0x00000000	
MIPIC_CMD_MODE_CFG	0x00068	W	0x00000000	Command mode configuration
MIPIC_GEN_HDR	0x0006c	W	0x00000000	Generic packet header configuration.
MIPIC_GEN_PLD_DATA	0x00070	W	0x00000000	Generic payload data in and out.
MIPIC_CMD_PKT_STATUS	0x00074	W	0x00000000	Command packet status
MIPIC_TO_CNT_CFG	0x00078	W	0x00000000	Timeout timers configuration
MIPIC_HS_RD_TO_CNT	0x0007c	W	0x00000000	
MIPIC_LP_RD_TO_CNT	0x00080	W	0x00000000	
MIPIC_HS_WR_TO_CNT	0x00084	W	0x00000000	
MIPIC_LP_WR_TO_CNT	0x00088	W	0x00000000	
MIPIC_BTA_TO_CNT	0x0008c	W	0x00000000	
MIPIC_LPCLK_CTRL	0x00094	W	0x00000000	
MIPIC_PHY_TMR_LPCLK_CFG	0x00098	W	0x00000000	
MIPIC_PHY_TMR_CFG	0x0009c	W	0x00000000	D-PHY timing configuration
MIPIC_PHY_RSTZ	0x000a0	W	0x00000000	D-PHY reset control

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Name	Offset	Size	Reset Value	Description
MIPIC_PHY_IF_CFG	0x000a4	W	0x00000000	D-PHY interface configuration
MIPIC_PHY_ULPS_CTRL	0x000a8	W	0x00000000	D-PHY PPI interface control
MIPIC_PHY_TX_TRIGGERS	0x000ac	W	0x00000000	
MIPIC_PHY_STATUS	0x000b0	W	0x00000000	D-PHY PPI status interface
MIPIC_PHY_TST_CTRL0	0x000b4	W	0x00000001	
MIPIC_PHY_TST_CTRL_1	0x000b8	W	0x00000000	
MIPIC_ERROR_ST0	0x000bc	W	0x00000000	Interrupt status register 0
MIPIC_ERROR_ST1	0x000c0	W	0x00000000	Interrupt status register 1
MIPIC_MSK0	0x000c4	W	0x00000000	Masks the interrupt generation triggerd by the ERROR_ST0 reg
MIPIC_MSK1	0x000c8	W	0x00000000	Masks the interrupt generation triggerd by the ERROR_ST1 reg
MIPIC_INT_FORCE0	0x000d8	W	0x00000000	
MIPIC_INT_FORCE1	0x000dc	W	0x00000000	
MIPIC_VID_SHADOW_CTRL	0x00100	W	0x00000000	
MIPIC_Copy0 DPI_VCID	0x0010c	W	0x00000000	The DPI interface configuration.
MIPIC_Copy0 DPI_COLOR_CODING	0x00110	W	0x00000000	
MIPIC_Copy0 LP_CMD_TIM	0x00118	W	0x00000000	Low-power Command Timing Configuration Register.
MIPIC_Copy0 VID_MODE_CFG	0x00138	W	0x00000000	Video mode configuration.
MIPIC_Copy0 VID_PKT_SIZE	0x0013c	W	0x00000000	
MIPIC_Copy0 VID_NUM_CHUNKS	0x00140	W	0x00000000	
MIPIC_Copy0 VID_NULL_SIZE	0x00144	W	0x00000000	
MIPIC_Copy0 VID_HSA_TIME	0x00148	W	0x00000000	Line timing configuration.
MIPIC_Copy0 VID_HBP_TIME	0x0014c	W	0x00000000	
MIPIC_Copy0 VID_HLINE_TIME	0x00150	W	0x00000000	
MIPIC_Copy0 VID_VSA_LINES	0x00154	W	0x00000000	Vertical timing configuration.
MIPIC_Copy0 VID_VBP_LINES	0x00158	W	0x00000000	
MIPIC_Copy0 VID_VFP_LINES	0x0015c	W	0x00000000	
MIPIC_Copy0 VID_VACTIVE_LINES	0x00160	W	0x00000000	

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*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

38.4.2 Detail Register Description

MIPIC_VERSION
Address: Operational Base + offset (0x00000)
Version of the mipi controller

Bit	Attr	Reset Value	Description
31:0	RO	0x3133302a	version indicates the version of the mipi_controller

MIPIC_PWR_UP
Address: Operational Base + offset (0x00004)
Core power-up

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	shutdownz This bit indicates the core power-up or the reset 0-Reset 1-Power-up

MIPIC_CLKMGR_CFG
Address: Operational Base + offset (0x00008)
Configuration of the internal clock dividers

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:8	RW	0x00	TO_CLK_DIVISION This field indicates the division factor for the Time Out clock used as the timing unit in the configuration of HS to LP and LP to HS transition error.
7:0	RW	0x00	TX_ESC_CLK_DIVISION Field0000 Abstract This field indicates the division factor for the TX_Escape clock source(lanebyteclk).The value 0 and 1 stop the TX_ESC clock generation

MIPIC_DPI_VCID
Address: Operational Base + offset (0x0000c)
The DPI interface configuration.

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x0	dpi_vid This field configures the DPI virtual channel id that is indexed to the Video mode packets.

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MIPIC_DPI_COLOR_CODING
Address: Operational Base + offset (0x00010)

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8	RW	0x0	en18_loosely When set to 1,this bit enables 18 loosely packed pixel stream.
7:4	RO	0x0	reserved
3:0	RW	0x0	dpi_color_coding This field configures the DPI color coding as follows: 000:16bit configuration 1 001:16bit configuration 2 010:16bit configuration 3 011:18bit configuration 1 100:18bit configuration 2 101:24bit

MIPIC_DPI_CFG_POL
Address: Operational Base + offset (0x00014)

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	colorm_active_low When set to 1,this bit configures the color mode pin as active low
3	RW	0x0	shutd_active_low When set to 1,this bit configures the shut down pin as active low
2	RW	0x0	hsync_active_low When set to 1,this bit configures the horizontal synchronism pin as active low.
1	RW	0x0	vsync_active_low When set to 1,this bit configures the vertical synchronism pin as active low
0	RW	0x0	dataen_active_low When set to 1,this bit configures the data enable pin as active low

MIPIC_LP_CMD_TIM
Address: Operational Base + offset (0x00018)
Low-power Command Timing Configuration Register.

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
23:16	RW	0x00	outvact_lpcmd_time outside VACT region command time.This field configures the time available to transmit a command in low-power mode.The time value is expressed in a number of bytes format.The number of bytes represents the maximum size of a packet that can fit in a line during the VSA,VBP,and VFP region. This field must be configured with a value greater than or equal to four bytes to allow the transmission of the DCTRL commands such as shutdown and colorm in low-power mode.
15:8	RO	0x0	reserved
7:0	RW	0x00	invact_lpcmd_time Inside VACT region command time.This field configures the time available to transmit a command in low-power mode.The time value is expressed in a number of bytes format.The number of bytes represents the maximum size of the packet that can fit a line during the VACT region.

MIPIC_PCKHDL_CFG
Address: Operational Base + offset (0x0002c)
Packet handler configuration

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	en_CRC_rx When set to 1,this bit enables the CRC reception and error reporting
3	RW	0x0	en_ECC_rx When set to 1,this bit enables the ECC reception,error correction,and reporting
2	RW	0x0	en_BTA When set to 1,this bit enables the Bus Turn-Around(BTA) request.
1	RW	0x0	en_EOTp_rx Field0000 Abstract When set to 1,this bit enables the EOTp reception
0	RW	0x0	en_EOTp_tx Field0000 Abstract When set to 1,this bit enables the EOTp transmission

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MIPIC_GEN_VCID
Address: Operational Base + offset (0x00030)

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x0	gen_vcid_rx the Generic interface read-back virtual channel identification

MIPIC_MODE_CFG
Address: Operational Base + offset (0x00034)

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	en_video_mode When set to 1,this bit enables the DPI Video mode transmission.

MIPIC_VID_MODE_CFG
Address: Operational Base + offset (0x00038)
Video mode configuration.

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	vpg_orientation This field indicates the color bar orientation as follows: 0:Vertical mode 1:Horizontal mode
23:21	RO	0x0	reserved
20	RW	0x0	vpg_mode This field is to select the pattern 0:Color bar(horizontal or vertical) 1:BER pattern(vertical only)
19:17	RO	0x0	reserved
16	RW	0x0	vpg_en When set to 1,this bit enables the video mode pattern generator
15	RW	0x0	lpcmden When set to 1,this bit enables the command transmission only in low-power mode
14	RW	0x0	frame_BTA_ack When set to 1,this bit enables the request for an acknowledge response at the end of a frame

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Bit	Attr	Reset Value	Description
13	RW	0x0	en_lp_hfp When set to 1,this bit enables the return to low-power inside the HFP period when timing allows.
12	RW	0x0	en_lp_hbp When set to 1,this bit enables the return to low-power inside the HBP period when timing allows.
11	RW	0x0	en_lp_vact When set to 1,this bit enables the return to low-power inside the VACT period when timing allows.
10	RW	0x0	en_lp_vfp When set to 1,this bit enables the return to low-power inside the VFP period when timing allows.
9	RW	0x0	en_lp_vbp When set to 1,this bit enables the return to low-power inside the VBP period when timing allows.
8	RW	0x0	en_lp_vsa When set to 1,this bit enables the return to low-power inside the VSA period when timing allows.
7:2	RO	0x0	reserved
1:0	RW	0x0	vid_mode_type This field indicates the video mode transmission type as follows: 00:Non-burst with sync pulses 01:Non-burst with sync events 10 and 11:Burst with sync pulses

MIPIC_VID_PKT_SIZE
Address: Operational Base + offset (0x0003c)

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13:0	RW	0x0000	vid_pkt_size This field configures the number of pixels on a single vedio packet.if you use the 18-bit mode and do not enable loosely packed stream,this vaule must be a multiple of 4.

MIPIC_VID_NUM_CHUNKS
Address: Operational Base + offset (0x00040)

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Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	num_chunks This field configures the number of chunks to be transmitted during a line period(a chunk is a video packet or a null packet)

MIPIC_VID_NULL_SIZE
Address: Operational Base + offset (0x00044)

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	null_pkt_size This field configures the number of bytes in a null packet

MIPIC_VID_HSA_TIME
Address: Operational Base + offset (0x00048)
Line timing configuration.

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	hsa_time This field configures the Horizontal Synchronism Active period in lane byte clock cycles.

MIPIC_VID_HBP_TIME
Address: Operational Base + offset (0x0004c)

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	hbp_time This field configures the Horizontal Back Porch period in lane byte clock cycles

MIPIC_VID_HLINE_TIME
Address: Operational Base + offset (0x00050)

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14:0	RW	0x0000	hline_time This field configures the size of the total lines counted in lane byte cycles.

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MIPIC_VID_VSA_LINES
Address: Operational Base + offset (0x00054)
Vertical timing configuration.

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:0	RW	0x000	vsa_lines This field configures the Vertical Synchronism Active period measured in number of horizontal lines.

MIPIC_VID_VBP_LINES
Address: Operational Base + offset (0x00058)

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:0	RW	0x000	vbp_lines This field configures the Vertical Back Porch period measured in horizontal lines.

MIPIC_VID_VFP_LINES
Address: Operational Base + offset (0x0005c)

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:0	RW	0x000	vfp_lines This field configures the Vertical Front Porch period measured in horizontal lines.

MIPIC_VID_VACTIVE_LINES
Address: Operational Base + offset (0x00060)

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13:0	RW	0x0000	v_active_line This field configures the Vertical Active period measured in horizontal lines.

MIPIC_EDPI_CMD_SIZE
Address: Operational Base + offset (0x00064)

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	edpi_allowed_cmd_size This field configures the maximum allowed size for an eDPI write memory command,measured in pixels.Automatic partitioning of data obtained from eDPI is permanently enabled.

MIPIC_CMD_MODE_CFG
Address: Operational Base + offset (0x00068)
Command mode configuration

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	max_rd_pkt_size This bit configures the maximum read packet size command transmission type: 0:High-speed 1:Low-power
23:20	RO	0x0	reserved
19	RW	0x0	dcs_lw_tx This bit configures the DCS long write packet command transmission type: 0:high-speed 1:low-power
18	RW	0x0	dcs_sr_0p_tx This bit configures the DCS short read packet with zero parameter command transmission type: 0:High-speed 1:Low-power
17	RW	0x0	dcs_sw_1p_tx This bit configures the DCS short write packet with one parameter command transmission type: 0:High-speed 1:Low-power
16	RW	0x0	dcs_sw_0p_tx This bit configures the DCS short write packet with zero parameter command transmission type: 0:High-speed 1:Low-power
15	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
14	RW	0x0	gen_lw_tx This bit configures the Generic long write packet command transmission type: 0:high-speed 1:low-power
13	RW	0x0	gen_sr_2p_tx This bit configures the Generic short read packet with two parameter command transmission type: 0:High-speed 1:Low-power
12	RW	0x0	gen_sr_1p_tx This bit configures the Generic short read packet with one parameter command transmission type: 0:High-speed 1:Low-power
11	RW	0x0	gen_sr_0p_tx This bit configures the Generic short read packet with zero parameter command transmission type: 0:High-speed 1:Low-power
10	RW	0x0	gen_sw_2p_tx This bit configures the Generic short write packet with two parameter command transmission type: 0:High-speed 1:Low-power
9	RW	0x0	gen_sw_1p_tx This bit configures the Generic short write packet with one parameter command transmission type: 0:High-speed 1:Low-power
8	RW	0x0	gen_sw_0p_tx This bit configures the Generic short write packet with zero parameter command transmission type: 0:High-speed 1:Low-power
7:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1	RW	0x0	ack_rqst_en When set to 1, this bit enables the acknowledge request after each packet transmission
0	RW	0x0	tear_fx_en When set to 1,this bit enables the tearing effect acknowledge request

MIPIC_GEN_HDR
Address: Operational Base + offset (0x0006c)
Generic packet header configuration.

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:16	RW	0x00	gen_WC_MSbyte This field configures the most significant byte of the header packet's Word count for long packets or data 1 for short packets.
15:8	RW	0x00	gen_WC_LSbyte This field configures the least significant byte of the header packet's Word count for long packets or data 0 for short packets.
7:6	RW	0x0	gen_VC This field configures the virtual channel id of the header packet.
5:0	RW	0x00	gen_DT This field configures the packet data type of the header packet

MIPIC_GEN_PLD_DATA
Address: Operational Base + offset (0x00070)
Generic payload data in and out.

Bit	Attr	Reset Value	Description
31:24	RW	0x00	gen_pld_b4 This field indicates byte 4 of the packet payload.
23:16	RW	0x00	gen_pld_b3 This field indicates byte 3 of the packet payload.
15:8	RW	0x00	gen_pld_b2 This field indicates byte 2 of the packet payload.
7:0	RW	0x00	gen_pld_b1 This field indicates byte 1 of the packet payload.

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MIPIC_CMD_PKT_STATUS
Address: Operational Base + offset (0x00074)
Command packet status

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x0	reserved reserved
6	RW	0x0	gen_rd_cmd_busy This bit is set when a read command is issued and cleared when the entire response is stored in the FIFO
5	RW	0x0	gen_pld_r_full This bit indicates the full status of the generic read payload FIFO Vaule after reset:0x0
4	RO	0x0	gen_pld_r_empty This bit indicates the empty status of the generic read payload FIFO Vaule after reset:0x1
3	RO	0x0	gen_pld_w_full This bit indicates the full status of the generic write payload FIFO Vaule after reset:0x0
2	RO	0x0	gen_pld_w_empty This bit indicates the empty status of the generic write payload FIFO Vaule after reset:0x1
1	RO	0x0	gen_cmd_full This bit indicates the full status of the generic command FIFO Vaule after reset:0x0
0	RO	0x0	gen_cmd_empty This bit indicates the empty status of the generic command FIFO Vaule after reset:0x1

MIPIC_TO_CNT_CFG
Address: Operational Base + offset (0x00078)
Timeout timers configuration

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	hstx_to_cnt This field configures the timeout counter that triggers a high-speed transmission timeout contention detection(measured in TO_CLK_DIVISION cycles)

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	lprx_to_cnt This field configures the timeout counter that triggers a low-power reception timeout contention detection(measured in TO_CLK_DIVISION cycles)

MIPIC_HS_RD_TO_CNT
Address: Operational Base + offset (0x0007c)

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	hs_rd_to_cnt This field sets a period for which the MIPI Controller keeps the link still,after sending a high-speed read operation.This period is measured in cycles of lanebyteclk.The counting starts when the D-PHY enters the Stop state and causes no interrupts.

MIPIC_LP_RD_TO_CNT
Address: Operational Base + offset (0x00080)

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	lp_rd_to_cnt This field sets a period for which MIPI Controller keeps the link still,after sending a low-power read operation.This period is measured in cycles of lanebyteclk.The counting starts when the D-PHY enters the Stop state and causes no interrupts.

MIPIC_HS_WR_TO_CNT
Address: Operational Base + offset (0x00084)

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
24	RW	0x0	presp_to_mode When set to 1, this bit ensures that the peripheral respone timeout caused by hs_wr_to_cnt is used only once per eDPI frame,when both the following conditions are met: .dpivsync_edpiwms has risen and fallen .packets originated from eDPI have been transmitted and its FIFO is empty again.
23:16	RO	0x0	reserved
15:0	RW	0x0000	hs_wr_to_cnt This field sets a period for which the MIPI Controller keeps the link inactive after sending a high-speed write operation.This period is measured in cycles of lanebyteclk.The counting starts when the D-PHY enters the Stop state and causes no interrupts.

MIPIC_LP_WR_TO_CNT
Address: Operational Base + offset (0x00088)

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	lp_wr_to_cnt This field sets a period for which the DSI Controller keeps the link still,after sending a low-power write operation.This period is measured in cycles of lanebyteclk.The counting starts when the D-PHY enters the Stop state and causes no interrupts.

MIPIC_BTA_TO_CNT
Address: Operational Base + offset (0x0008c)

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	bta_to_cnt This field sets a period for which the DSI Controller keeps the link still,after completing a Bus Turn-Around.This period is measured in cycles of lanebyteclk.The counting starts when the D-PHY enters the Stop state and causes no interrupts.

MIPIC_LPCLK_CTRL

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Address: Operational Base + offset (0x00094)

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	auto_clklane_ctrl This bit enables the automatic mechanism to stop providing clock in the clock lane when time allows.
0	RW	0x0	phy_txrequestclkhs This bit controls the D-PHY PPI tx requestclkhs signal

MIPIC_PHY_TMR_LPCLK_CFG
Address: Operational Base + offset (0x00098)

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25:16	RW	0x000	phy_hs2lp_time This field configures the maximum time that the PHY takes to go from high-speed to low-power transmission measured in lane byte clock cycles.(clock lane)
15:10	RO	0x0	reserved
9:0	RW	0x000	phy_lp2hs_time This field configures the maximum time that the PHY takes to go from low-power to high-speed transmission measured in lane byte clock cycles.(clock lane)

MIPIC_PHY_TMR_CFG
Address: Operational Base + offset (0x0009c)
D-PHY timing configuration

Bit	Attr	Reset Value	Description
31:24	RW	0x00	phy_hs2lp_time This field configures the maximum time that the PHY takes to go from high-speed to low-power transmission measured in lane byte clock cycles.
23:16	RW	0x00	phy_lp2hs_time This field configures the maximum time that the PHY takes to go from low-power to high-speed transmission measured in lane byte clock cycles.
15	RW	0x0	reserved reserved for future use

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Bit	Attr	Reset Value	Description
14:0	RW	0x0000	max_rd_time This field configures the maximum time required to perform a read command in lane byte clock cycles.This register can only by modified when read commands are not in progress.

MIPIC_PHY_RSTZ
Address: Operational Base + offset (0x000a0)
D-PHY reset control

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	phy_forcepll When the D-PHY is in ULPS,this bit enables the D-PHY PLL
2	RW	0x0	phy_enableclk When set to 1,this bit enables the D-PHY Clock Lane Module
1	RW	0x0	phy_rstz When set to 0,this bit places the digital section of the D-PHY in the reset state
0	RW	0x0	phy_shutdownz When set to 0,this bit places the D-PHY macro in power-down state

MIPIC_PHY_IF_CFG
Address: Operational Base + offset (0x000a4)
D-PHY interface configuration

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:8	RW	0x00	phy_stop_wait_time This field configures the minimum wait period to request a high-speed transmission after the Stop state is accounted in clock lane cycles.
7:2	RO	0x0	reserved
1:0	RW	0x0	n_lanes This field configures the number of active data lanes: 00:One data lane(lane 0) 01:Two data lane(lanes 0 and 1) 10:Three data lanes(lanes 0,1,and 2) 11:Four data lanes(lanes 0,1,2,and 3)

MIPIC_PHY_ULPS_CTRL
Address: Operational Base + offset (0x000a8)

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D-PHY PPI interface control

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	phy_txexitulpslan ULPS mode Exit on all active data lanes
2	RW	0x0	phy_txrequlpslan ULPS mode Request on all active data lanes
1	RW	0x0	phy_txexitulpsclk ULPS mode Exit on clock lane
0	RW	0x0	phy_txrequlpsclk ULPS mode Request on clock lane

MIPIC_PHY_TX_TRIGGERS
Address: Operational Base + offset (0x000ac)

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x0	phy_tx_triggers This field controls the trigger transmissions.

MIPIC_PHY_STATUS
Address: Operational Base + offset (0x000b0)
D-PHY PPI status interface

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12	RO	0x0	ulpsactivenot3lane This bit indicates the status of ulpsactivenot3lane D-PHY signal
11	RO	0x0	phystopstate3lane This bit indicates the status of phystopstate3lane D-PHY signal
10	RO	0x0	ulpsactivenot2lane This bit indicates the status of ulpsactivenot2lane D-PHY signal
9	RO	0x0	phystopstate2lane This bit indicates the status of phystopstate2lane D-PHY signal
8	RO	0x0	ulpsactivenot1lane This bit indicates the status of ulpsactivenot1lane D-PHY signal
7	RO	0x0	phystopstate1lane This bit indicates the status of phystopstate1lane D-PHY signal
6	RW	0x0	rxulpsesc0lane This bit indicates the status of rxulpsesc0lane D-PHY signal

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Bit	Attr	Reset Value	Description
5	RO	0x0	ulpsactivenot0lane This bit indicates the status of ulpsactivenot0lane D-PHY signal
4	RO	0x0	phystopstate0lane This bit indicates the status of phystopstate0lane D-PHY signal
3	RO	0x0	phyulpsactivenotclk This bit indicates the status of phyulpsactivenotclk D-PHY signal
2	RO	0x0	phystopstateclklane This bit indicates the status of phystopstateclklane D-PHY signal
1	RO	0x0	phydirection This bit indicates the status of phydirection D-PHY signal
0	RO	0x0	phylock This bit indicates the status of phylock D-PHY signal

MIPIC_PHY_TST_CTRL0
Address: Operational Base + offset (0x000b4)

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	phy_testclk This bit is used to clock the TESTDIN bus into the D-PHY
0	RW	0x1	phy_testclr PHY test interface clear(active high)

MIPIC_PHY_TST_CTRL_1
Address: Operational Base + offset (0x000b8)

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RW	0x0	phy_testen PHY test interface operation selector: 1:The address write operation is set on the falling edge of the testclk signal 0:The data write operation is set on the rising edge of the testclk signal
15:8	RW	0x00	phy_testdout PHY output 8-bit data bus for read-back and internal probing functionalities

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Bit	Attr	Reset Value	Description
7:0	RW	0x00	phy_testdin PHY test interface input 8-bit data bus for internal register programming and test functionalities access

MIPIC_ERROR_ST0
Address: Operational Base + offset (0x000bc)
Interrupt status register 0

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved
20	RO	0x0	dphy_errors_4 This bit indicates LP1 contention error ErrContentionLP1 from Lane 0
19	RO	0x0	dphy_errors_3 This bit indicates LP0 contention error ErrContentionLP0 from Lane 0
18	RO	0x0	dphy_errors_2 This bit indicates control error ErrControl from Lane 0
17	RO	0x0	dphy_errors_1 This bit indicates ErrSyncEsc low-power data transmission synchronization error from Lane 0
16	RO	0x0	dphy_errors_0 This bit indicates ErrEsc escape entry error from Lane 0
15	RO	0x0	ack_with_err_15 This bit retrieves the DSI protocol violation from the Display Acknowledge error report
14	RO	0x0	ack_with_err_14 This bit retrieves the reserved(specific to device) from the Display Acknowledge error report
13	RO	0x0	ack_with_err_13 This bit retrieves the invalid transmission length from the Display Acknowledge error report
12	RO	0x0	ack_with_err_12 This bit retrieves the DSI VC ID Invalid from the Display Acknowledge error report
11	RO	0x0	ack_with_err_11 This bit retrieves the not recognized DSI data type from the Display Acknowledge error report

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Bit	Attr	Reset Value	Description
10	RO	0x0	ack_with_err_10 This bit retrieves the checksum error(long packet only) from the Display Acknowledge error report
9	RO	0x0	ack_with_err_9 This bit retrieves the ECC error,multi-bit(detected and corrected) from the Display Acknowledge error report
8	RO	0x0	ack_with_err_8 This bit retrieves the ECC error,single-bit(detected and corrected) from the Display Acknowledge error report
7	RO	0x0	ack_with_err_7 This bit retrieves the reserved(specific to device) error from the Display Acknowledge error report
6	RO	0x0	ack_with_err_6 This bit retrieves the False Control error from the Display Acknowledge error report
5	RO	0x0	ack_with_err_5 This bit retrieves the HS Receive Timeout error from the Display Acknowledge error report
4	RO	0x0	ack_with_err_4 This bit retrieves the LP Transmit Sync error error from the Display Acknowledge error report
3	RO	0x0	ack_with_err_3 This bit retrieves the Escape Mode Entry command error from the Display Acknowledge error report
2	RO	0x0	ack_with_err_2 This bit retrieves the EoT sync error from the Display Acknowledge error report
1	RO	0x0	ack_with_err_1 This bit retrieves the SoT Sync error from the Display Acknowledge error report
0	RO	0x0	ack_with_err_0 This bit retrieves the SoT error from the Display Acknowledge error report

MIPIC_ERROR_ST1
Address: Operational Base + offset (0x000c0)
Interrupt status register 1

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12	RO	0x0	gen_pld_recv_err This bit indicates that during a generic interface packet read back,the payload FIFO becomes full and the received data is corrupted.
11	RO	0x0	gen_pld_rd_err This bit indicates that during a DCS read data,the payload FIFO becomes empty and the data sent to the interface is corrupted
10	RO	0x0	gen_pld_send_err This bit indicates that during a Generic interface packet build,the payload FIFO becomes empty and corrupt data is sent.
9	RO	0x0	gen_pld_wr_err This bit indicates that the system tried to write a payload data through the Generic interface and the FIFO is full.Therefore,the command is not written.
8	RO	0x0	gen_cmd_wr_err This bit indicates that the system tried to write a command through the Generic interface and the FIFO is full.Therefore,the command is not written.
7	RO	0x0	dpi_pld_wr_err This bit indicates that during a DPI pixel line storage,the payload FIFO becomes full and the data stored is corrupted.
6	RO	0x0	eopt_err This bit indicates that the EOTp packet is not received at the end of the incoming peripheral transmission.
5	RO	0x0	pkt_size_err This bit indicates that the packet size error is detected during the packet reception.
4	RO	0x0	crc_err This bit indicates that the CRC error is detected in a received packet.
3	RO	0x0	ecc_multi_err This bit indicates that the ECC multiple error is detected and corrected in a received packet.
2	RO	0x0	ecc_single_err This bit indicates that the ECC single error is detected and corrected in a received packet.

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Bit	Attr	Reset Value	Description
1	RO	0x0	to_lp_rx This bit indicates that the low-power reception timeout counter reached the end and contention detection is detected.
0	RO	0x0	to_hs_tx This bit indicates that the high-speed transmission timeout counter reached the end and contention detection is detected.

MIPIC_MSK0

Address: Operational Base + offset (0x000c4)

Masks the interrupt generation triggerd by the ERROR_ST0 reg

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved
20	RW	0x0	dphy_errors_4 This bit indicates LP1 contention error ErrContentionLP1 from Lane 0
19	RW	0x0	dphy_errors_3 This bit indicates LP0 contention error ErrContentionLP0 from Lane 0
18	RW	0x0	dphy_errors_2 This bit indicates control error ErrControl from Lane 0
17	RW	0x0	dphy_errors_1 This bit indicates ErrSyncEsc low-power data transmission synchronization error from Lane 0
16	RW	0x0	dphy_errors_0 This bit indicates ErrEsc escape entry error from Lane 0
15	RW	0x0	ack_with_err_15 This bit retrieves the DSI protocol violation from the Display Acknowledge error report
14	RW	0x0	ack_with_err_14 This bit retrieves the reserved(specific to device) from the Display Acknowledge error report
13	RW	0x0	ack_with_err_13 This bit retrieves the invalid transmission length from the Display Acknowledge error report
12	RW	0x0	ack_with_err_12 This bit retrieves the DSI VC ID Invalid from the Display Acknowledge error report

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Bit	Attr	Reset Value	Description
11	RW	0x0	ack_with_err_11 This bit retrieves the not recognized DSI data type from the Display Acknowledge error report
10	RW	0x0	ack_with_err_10 This bit retrieves the checksum error(long packet only) from the Display Acknowledge error report
9	RW	0x0	ack_with_err_9 This bit retrieves the ECC error,multi-bit(detected and corrected) from the Display Acknowledge error report
8	RW	0x0	ack_with_err_8 This bit retrieves the ECC error,single-bit(detected and corrected) from the Display Acknowledge error report
7	RW	0x0	ack_with_err_7 This bit retrieves the reserved(specific to device) error from the Display Acknowledge error report
6	RW	0x0	ack_with_err_6 This bit retrieves the False Control error from the Display Acknowledge error report
5	RW	0x0	ack_with_err_5 This bit retrieves the HS Receive Timeout error from the Display Acknowledge error report
4	RW	0x0	ack_with_err_4 This bit retrieves the LP Transmit Sync error error from the Display Acknowledge error report
3	RW	0x0	ack_with_err_3 This bit retrieves the Escape Mode Entry command error from the Display Acknowledge error report
2	RW	0x0	ack_with_err_2 This bit retrieves the EoT sync error from the Display Acknowledge error report
1	RW	0x0	ack_with_err_1 This bit retrieves the SoT Sync error from the Display Acknowledge error report
0	RW	0x0	ack_with_err_0 This bit retrieves the SoT error from the Display Acknowledge error report

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MIPIC_MSK1
Address: Operational Base + offset (0x000c8)
Masks the interrupt generation triggerd by the ERROR_ST1 reg

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12	RO	0x0	gen_pld_recv_err This bit indicates that during a generic interface packet read back,the payload FIFO becomes full and the received data is corrupted.
11	RO	0x0	gen_pld_rd_err This bit indicates that during a DCS read data,the payload FIFO becomes empty and the data sent to the interface is corrupted
10	RO	0x0	gen_pld_send_err This bit indicates that during a Generic interface packet build,the payload FIFO becomes empty and corrupt data is sent.
9	RO	0x0	gen_pld_wr_err This bit indicates that the system tried to write a payload data through the Generic interface and the FIFO is full.Therefore,the command is not written.
8	RO	0x0	gen_cmd_wr_err This bit indicates that the system tried to write a command through the Generic interface and the FIFO is full.Therefore,the command is not written.
7	RO	0x0	dpi_pld_wr_err This bit indicates that during a DPI pixel line storage,the payload FIFO becomes full and the data stored is corrupted.
6	RO	0x0	eopt_err This bit indicates that the EOTp packet is not received at the end of the incoming peripheral transmission.
5	RO	0x0	pkt_size_err This bit indicates that the packet size error is detected during the packet reception.
4	RO	0x0	crc_err This bit indicates that the CRC error is detected in a received packet.
3	RO	0x0	ecc_multi_err This bit indicates that the ECC multiple error is detected and corrected in a received packet.

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Bit	Attr	Reset Value	Description
2	RO	0x0	ecc_single_err This bit indicates that the ECC single error is detected and corrected in a received packet.
1	RO	0x0	to_lp_rx This bit indicates that the low-power reception timeout counter reached the end and contention detection is detected.
0	RO	0x0	to_hs_tx This bit indicates that the high-speed transmission timeout counter reached the end and contention detection is detected.

MIPIC_INT_FORCE0
Address: Operational Base + offset (0x000d8)

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved
20	RO	0x0	dphy_errors_4 This bit indicates LP1 contention error ErrContentionLP1 from Lane 0
19	RO	0x0	dphy_errors_3 This bit indicates LP0 contention error ErrContentionLP0 from Lane 0
18	RO	0x0	dphy_errors_2 This bit indicates control error ErrControl from Lane 0
17	RO	0x0	dphy_errors_1 This bit indicates ErrSyncEsc low-power data transmission synchronization error from Lane 0
16	RO	0x0	dphy_errors_0 This bit indicates ErrEsc escape entry error from Lane 0
15	RO	0x0	ack_with_err_15 This bit retrieves the DSI protocol violation from the Display Acknowledge error report
14	RO	0x0	ack_with_err_14 This bit retrieves the reserved(specific to device) from the Display Acknowledge error report
13	RO	0x0	ack_with_err_13 This bit retrieves the invalid transmission length from the Display Acknowledge error report

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Bit	Attr	Reset Value	Description
12	RO	0x0	ack_with_err_12 This bit retrieves the DSI VC ID Invalid from the Display Acknowledge error report
11	RO	0x0	ack_with_err_11 This bit retrieves the not recognized DSI data type from the Display Acknowledge error report
10	RO	0x0	ack_with_err_10 This bit retrieves the checksum error(long packet only) from the Display Acknowledge error report
9	RO	0x0	ack_with_err_9 This bit retrieves the ECC error,multi-bit(detected and corrected) from the Display Acknowledge error report
8	RO	0x0	ack_with_err_8 This bit retrieves the ECC error,single-bit(detected and corrected) from the Display Acknowledge error report
7	RO	0x0	ack_with_err_7 This bit retrieves the reserved(specific to device) error from the Display Acknowledge error report
6	RO	0x0	ack_with_err_6 This bit retrieves the False Control error from the Display Acknowledge error report
5	RO	0x0	ack_with_err_5 This bit retrieves the HS Receive Timeout error from the Display Acknowledge error report
4	RO	0x0	ack_with_err_4 This bit retrieves the LP Transmit Sync error error from the Display Acknowledge error report
3	RO	0x0	ack_with_err_3 This bit retrieves the Escape Mode Entry command error from the Display Acknowledge error report
2	RO	0x0	ack_with_err_2 This bit retrieves the EoT sync error from the Display Acknowledge error report
1	RO	0x0	ack_with_err_1 This bit retrieves the SoT Sync error from the Display Acknowledge error report

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Bit	Attr	Reset Value	Description
0	RO	0x0	ack_with_err_0 This bit retrieves the SoT error from the Display Acknowledge error report

MIPIC_INT_FORCE1
Address: Operational Base + offset (0x000dc)

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12	RO	0x0	gen_pld_recv_err This bit indicates that during a generic interface packet read back,the payload FIFO becomes full and the received data is corrupted.
11	RO	0x0	gen_pld_rd_err This bit indicates that during a DCS read data,the payload FIFO becomes empty and the data sent to the interface is corrupted
10	RO	0x0	gen_pld_send_err This bit indicates that during a Generic interface packet build,the payload FIFO becomes empty and corrupt data is sent.
9	RO	0x0	gen_pld_wr_err This bit indicates that the system tried to write a payload data through the Generic interface and the FIFO is full.Therefore,the command is not written.
8	RO	0x0	gen_cmd_wr_err This bit indicates that the system tried to write a command through the Generic interface and the FIFO is full.Therefore,the command is not written.
7	RO	0x0	dpi_pld_wr_err This bit indicates that during a DPI pixel line storage,the payload FIFO becomes full and the data stored is corrupted.
6	RO	0x0	eopt_err This bit indicates that the EOTp packet is not received at the end of the incoming peripheral transmission.
5	RO	0x0	pkt_size_err This bit indicates that the packet size error is detected during the packet reception.

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Bit	Attr	Reset Value	Description
4	RO	0x0	crc_err This bit indicates that the CRC error is detected in a received packet.
3	RO	0x0	ecc_multi_err This bit indicates that the ECC multiple error is detected and corrected in a received packet.
2	RO	0x0	ecc_single_err This bit indicates that the ECC single error is detected and corrected in a received packet.
1	RO	0x0	to_lp_rx This bit indicates that the low-power reception timeout counter reached the end and contention detection is detected.
0	RO	0x0	to_hs_tx This bit indicates that the high-speed transmission timeout counter reached the end and contention detection is detected.

MIPIC_VID_SHADOW_CTRL
Address: Operational Base + offset (0x00100)

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RW	0x0	vid_shadow_pin_req When set to 1,the video request is done by external pin. In this mode, vid_shadow_req is ignored
15:9	RO	0x0	reserved
8	RW	0x0	vid_shadow_req When set to 1,the DPI registers are copied to the auxiliary registers.after coping,this bit is auto cleard.
7:1	RO	0x0	reserved
0	RW	0x0	vid_shadow_en When set to 1,DPI receives the active configuration from the auxiliary registers.When this bit is set along with the vid_shadow_req bit,the auxiliary registers are automatically updated.

MIPIC_Copy0 DPI_VCID
Address: Operational Base + offset (0x0010c)
The DPI interface configuration.

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	dpi_vid This field configures the DPI virtual channel id that is indexed to the Video mode packets.

MIPIC_Copy0 DPI_COLOR_CODING
Address: Operational Base + offset (0x00110)

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8	RW	0x0	en18_loosely When set to 1,this bit enables 18 loosely packed pixel stream.
7:4	RO	0x0	reserved
3:0	RW	0x0	dpi_color_coding This field configures the DPI color coding as follows: 000:16bit configuration 1 001:16bit configuration 2 010:16bit configuration 3 011:18bit configuration 1 100:18bit configuration 2 101:24bit

MIPIC_Copy0 LP_CMD_TIM
Address: Operational Base + offset (0x00118)
Low-power Command Timing Configuration Register.

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:16	RW	0x00	outvact_lpcmd_time outside VACT region command time.This field configures the time available to transmit a command in low-power mode.The time value is expressed in a number of bytes format.The number of bytes represents the maximum size of a packet that can fit in a line during the VSA,VBP,and VFP region. This field must be configured with a value greater than or equal to four bytes to allow the transmission of the DCTRL commands such as shutdown and colorm in low-power mode.
15:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:0	RW	0x00	invact_lpcmd_time Inside VACT region command time.This field configures the time available to transmit a command in low-power mode.The time value is expressed in a number of bytes format.The number of bytes represents the maximum size of the packet that can fit a line during the VACT region.

MIPIC_Copy0 VID_MODE_CFG
Address: Operational Base + offset (0x00138)
Video mode configuration.

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	vpg_orientation This field indicates the color bar orientation as follows: 0:Vertical mode 1:Horizontal mode
23:21	RO	0x0	reserved
20	RW	0x0	vpg_mode This field is to select the pattern 0:Color bar(horizontal or vertical) 1:BER pattern(vertical only)
19:17	RO	0x0	reserved
16	RW	0x0	vpg_en When set to 1,this bit enables the video mode pattern generator
15	RW	0x0	lpcmden When set to 1,this bit enables the command transmission only in low-power mode
14	RW	0x0	frame_BTA_ack When set to 1,this bit enables the request for an acknowledge response at the end of a frame
13	RW	0x0	en_lp_hfp When set to 1,this bit enables the return to low-power inside the HFP period when timing allows.
12	RW	0x0	en_lp_hbp When set to 1,this bit enables the return to low-power inside the HBP period when timing allows.

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Bit	Attr	Reset Value	Description
11	RW	0x0	en_lp_vact When set to 1,this bit enables the return to low-power inside the VACT period when timing allows.
10	RW	0x0	en_lp_vfp When set to 1,this bit enables the return to low-power inside the VFP period when timing allows.
9	RW	0x0	en_lp_vbp When set to 1,this bit enables the return to low-power inside the VBP period when timing allows.
8	RW	0x0	en_lp_vsa When set to 1,this bit enables the return to low-power inside the VSA period when timing allows.
7:2	RO	0x0	reserved
1:0	RW	0x0	vid_mode_type This field indicates the video mode transmission type as follows: 00:Non-burst with sync pulses 01:Non-burst with sync events 10 and 11:Burst with sync pulses

MIPIC_Copy0 VID_PKT_SIZE
Address: Operational Base + offset (0x0013c)

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13:0	RW	0x0000	vid_pkt_size This field configures the number of pixels on a single vedio packet.if you use the 18-bit mode and do not enable loosely packed stream,this vaule must be a multiple of 4.

MIPIC_Copy0 VID_NUM_CHUNKS
Address: Operational Base + offset (0x00140)

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	num_chunks This field configures the number of chunks to be transmitted during a line period(a chunk is a video packet or a null packet)

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MIPIC_Copy0 VID_NULL_SIZE
Address: Operational Base + offset (0x00144)

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	null_pkt_size This field configures the number of bytes in a null packet

MIPIC_Copy0 VID_HSA_TIME
Address: Operational Base + offset (0x00148)
Line timing configuration.

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	hsa_time This field configures the Horizontal Synchronism Active period in lane byte clock cycles.

MIPIC_Copy0 VID_HBP_TIME
Address: Operational Base + offset (0x0014c)

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	hbp_time This field configures the Horizontal Back Porch period in lane byte clock cycles

MIPIC_Copy0 VID_HLINE_TIME
Address: Operational Base + offset (0x00150)

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14:0	RW	0x0000	hline_time This field configures the size of the total lines counted in lane byte cycles.

MIPIC_Copy0 VID_VSA_LINES
Address: Operational Base + offset (0x00154)
Vertical timing configuration.

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:0	RW	0x000	vsa_lines This field configures the Vertical Synchronism Active period measured in number of horizontal lines.

MIPIC_Copy0 VID_VBP_LINES
Address: Operational Base + offset (0x00158)

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:0	RW	0x000	vbp_lines This field configures the Vertical Back Porch period measured in horizontal lines.

MIPIC_Copy0 VID_VFP_LINES
Address: Operational Base + offset (0x0015c)

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:0	RW	0x000	vfp_lines This field configures the Vertical Front Porch period measured in horizontal lines.

MIPIC_Copy0 VID_VACTIVE_LINES
Address: Operational Base + offset (0x00160)

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13:0	RW	0x0000	v_active_line This field configures the Vertical Active period measured in horizontal lines.

38.5 Application Notes

Low Power Mode is a special feature for D-PHY. You can control this function by using proper registers from the Innosilicon D-PHY with few operations. The following is a step by step instruction for low power mode in and out.
Perform the following steps to configure the DPI packet transmission:
Step1:Global configuration:
Configure n_lanes (PHY_IF_CFG-[1:0]) to define the number of lanes in which the controller has to perform high-speed transmissions.
Step2:Configure the DPI Interface to define how the DPI interface interacts with the controller.
Configure dpi_vid (DPI_CFG-[1:0]): This field configures the virtual channel that the packet generated by the DPI interface is indexed to.
Configure dpi_color_coding (DPI_CFG-[4:2]): This field configures the bits per pixels that the interface transmits and also the variant configuration of each bpp. If you select 18 bpp, and

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the Enable_18_loosely_packed is not active, the number or pixels per line should be a multiple of four.
Configure dataen_active_low (DPI_CFG-[5]): This bit configures the polarity of the dpidataen signal and enables if it is active low.
Configure vsync_active_low(DPI_CFG-[6]): This bit configures the polarity of the dpivsync signal and enables if it is active low.
Configure vsync_active_low(DPI_CFG-[7]): This bit configures the polarity of the dpivsync signal and enables if it is active low.
Configure vsync_active_low(DPI_CFG-[8]): This bit configures the polarity of the dpishutdn signal and enables if it is active low.
Configure vsync_active_low(DPI_CFG-[9]): This bit configures the polarity of the dpicolorm signal and enables if it is active low.
Configure en18_loosely(DPI_CFG-[10]): This bit configures if the pixel packing is done loosely or packed when dpi_color_coding is 18 bpp. This bit enables loosely packing.
Step3: Select the Video Transmission Mode to define how the processor requires the video line to be transported through the DSI link.
Configure low-power transitions (VID_MODE_CFG-[8:3]): This defines the video line to be transported through the DSI link.
Configure low-power transitions (VID_MODE_CFG-[8:3]): This defines the video periods which are permitted to go to low-power if there is available time to do so.
Configure frame_BTA_ack (VID_MODE_CFG-[11]): This specifies if the controller should request the peripheral acknowledge message at the end of frames.
Burst mode: In this mode, the entire active pixel line is buffered into a FIFO and transmitted in a single packed with no interruptions. This transmission mode requires that the DPI Pixel FIFO has the capacity to store a full line of active pixel data inside it. This mode is optimally used if the difference between pixel required bandwidth and DSI link bandwidth is very different. This enables the DWC_mipi_dsi_host to quickly dispatch the entire active video line in a single burst of data and then return to low-power mode.
Configure the register fiedl vid_mode_type (VID_MODE_CFG-[10]), num_chunks (VID_PKT_CFG-[20:11]), and null_pkt_size (VID_PKT_CFG-[30:21]) are automatically ignored by the DWC_mipi_dsi_host.
Non-Burst mode: In this mode, the processor uses the partitioning properties of the DWC_mipi_dsi_host to divide the video line transmission into several DSI packets. This is done to match th pixel required bandwidth with the DSI link bandwidth. With this mode, the controller configuration dose not require a full line of pixel data to be stored inside the DPI Pixel FIFO. It requires only the content of one video packet.
Configure the vid_mode_type field (VID_MODE_CFG-[2:1]) with 2’b0x.
Configure the vid_mode_type field (VID_MODE_CFG-[2:1]) with 2’b00x to enable the transmission of sync pulses.
Configure the vid_mode_type field (VID_MODE_CFG-[2:1]) with 2’b01to enable the transmission of sync events.
Configure the vid_mode_type field (VID_MODE_CFG-[10:0]) with the number of pixels to be transmitted in a single packet.
Configure the en_multi_pkt field (VID_MODE_CFG-[9]) to enable the division of the active video transmission into more than one packet.
Configure the num_chunks field (VID_MODE_CFG-[20:11]) with the number of video chunks that the active video transmission is divided into.
Configure the en_null_pkt field (VID_MODE_CFG-[10]) to enable the insertion of null packets between video packets.
The field is effective only when en_multi_pkt field is activated, otherwise the controller ignores it and does not sent the null packets.
Configure the null_pkt_size field (VID_MODE_CFG-[30:21]) with the actual size of the inserted null packet.
Step4:Define the DPI Horizontal timing configuration as follows:
Configure the hline_time field (TMR_LINE_CFG-[31:18]) with the time taken by a DPI video line accounted in Clock Lane bytes clock cycles (for a clock lane at 500 MHz the Lane byte clock period is 8 ns). When the DPI clock and Clock Lane clock are not multiples, the

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hline_time is a result of a round of a number. If the DWC_mipi_dsi_host is configured to go to low-power, it is possible that the error included in a line is incremented with the next one. At the end of several lines, the DWC_mipi_dsi_host can have a number of errors that can cause a malfunction of the video transmission.
Configure the hsa_time field (TMR_LINE_CFG-[8:0]) with the time taken by a DPI Horizontal Sync Active period accounted in Clock Lane byte clock cycles (normally a period of 8ns).
Configure the hbp_time field (TMR_LINE_CFG-[17:9]) with the time taken by a DPI Horizontal Sync Active period accounted in Clock Lane byte clock cycles (normally a period of 8ns).
Special attention should be given to the calculation of this parameter.
Step5:Define the Vertical line configuration:
Configure the vsa_lines field (VTIMING_CFG-[3:0]) with the number of lines existing in the DPI Vertical Sync Active period.
Configure the vbp_lines field (VTIMING_CFG-[9:4]) with the number of lines existing in the DPI Vertical Back Porch period.
Configure the vfp_lines field (VTIMING_CFG-[15:10]) with the number of lines existing in the DPI Vertical Front Porch period.
Configure the v_active_lines field (VTIMING_CFG-[26:16]) with the number of lines existing in the DPI Vertical Active period.

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Chapter 39 TSP(Transport Stream Processing Module)

39.1 Overview

The Transport Stream Processing Module(TSP) is designed for processing Transport Stream Packets, including receiving TS packets, PID filtering, TS descrambling, De-multiplexing and TS outputting. Processed data are transferred to memory buffer which are continued to be processing by software.

TPS supports the following features:

- Supports 1 TS input channels
- Supports 4 TS Input Mode: sync/valid mode in the case of serial TS input; nosync/valid mode, sync/valid, sync/burst mode in the case of parallel TS input
- Supports 2 TS sources: demodulators and local memory
- Supports 1 Built-in PTIs(Programmable Transport Interface) to process TS simultaneously
- Supports 1 PVR(Personal Video Recording) output channel
- 1 built-in multi-channel DMA Controller
- Support DMA LLP transfer
- Each PTI supports
  - 64 PID filters
  - TS descrambling with 16 sets of Control Word under CSA v2.0 standard, up to 104Mbps
  - 16 PES/ES filters with PTS/DTS extraction and ES start code detection
  - 4/8 PCR extraction channels
  - 64 Section filters with CRC check, and three interrupt mode: stop per unit, full-stop, recycle mode with version number check
  - PID done and error interrupts for each channel
  - PCR/DTS/PTS extraction interrupt for each channel

39.2 Block Diagram

The TSP comprises of following components:

- AMBA AHB slave interface
- Register block
- PTI
- DMAC
- TS Out Interface

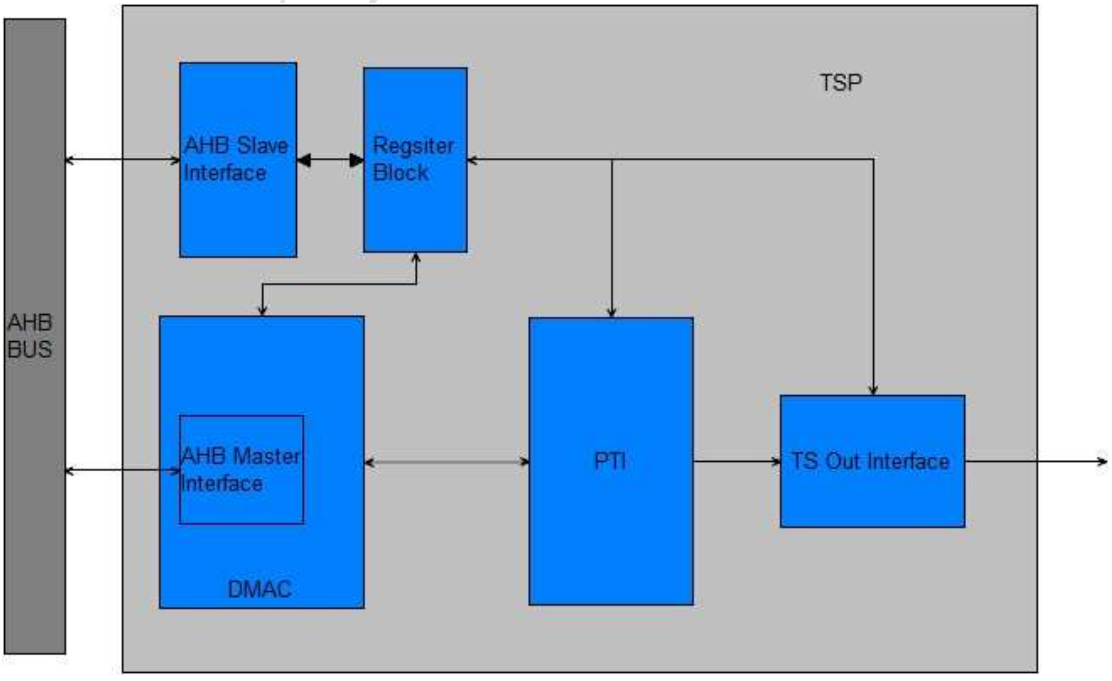

Fig. 39-1 TSP architecture

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AHB Slave INTERFACE

The host processor can get access to the register block through AHB slave interface. The slave interface supports 32bit access.

Register block

All registers in the TSP are addressed at 32-bit boundaries to remain consistent with the AHB bus. Where the physical size of any register is less than 32-bits wide, the upper unused bits of the 32-bit boundary are reserved. Writing to these bits has no effect; reading from these bits returns 0.

PTI

Most of the TS processing are dealt with PTI. TS packets are re-synchronized, filtered, descrambled and demultiplexing, and the processed packets are transferred to memory buffer to be processed further by software. The embedded TS in interface can receive TS packets by connecting to a compliant TS demodulator. TS stream stored in the local memory is another source to fed into PTI through by using LLP DMA mode.

TS Out Interface

TS out interface can output either PID-filtered or non-PID-filtered TS packets from one PTI channel in a certain stream mode as configured. The TS receiver conforms to the stream mode to receive the TS packets.

DMAC

The DMAC performs all DMA transfers which get access to memory.

39.3 Function Description

39.3.1 TS Stream of TS_IN Interface

TS_IN interface supports 4 input TS stream mode: sync/valid serial mode, sync/valid parallel mode, sync/burst parallel mode, nosync/valid parallel mode.

A. Sync/Valid Serial Mode

In this mode, TS_IN interface takes use of TSI_SYNC and TSI_VALID clocked with TSI_CLK signal to sample input serial TS packet data.

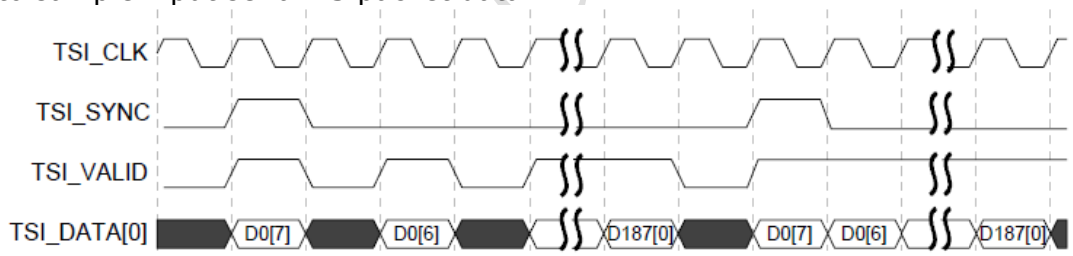

Fig. 39-2 Sync/Valid Serial Mode with Msb-Lsb Bit Ordering

TSI_SYNC must be active high together with TSI_VALID when indicating the first valid bit of a TS packet, and TSI_VALID indicates the 188*8 valid bits of a TS packet. TSI supports both msb-lsb and lsb-msb bit ordering.

B. Sync/Valid Parallel Mode

In this mode, TS_IN interface takes use of TSI_SYNC and TSI_VALID clocked with TSI_CLK signal to sample input parallel TS packet data.

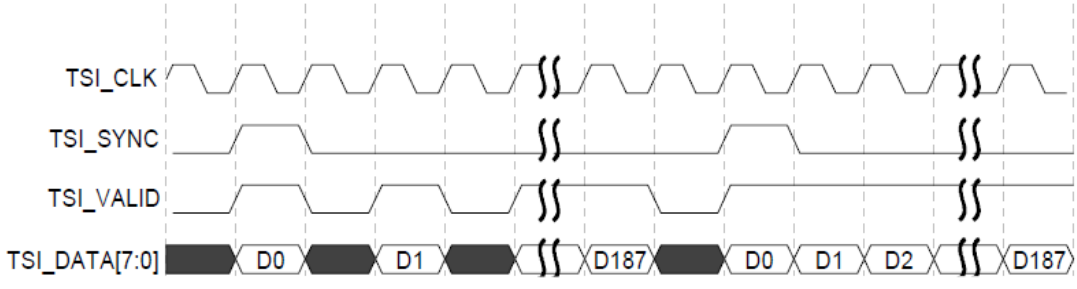

Fig. 39-3 Sync/valid Parallel Mode

TSI_SYNC must be active high together with TSI_VALID when indicating the first valid byte of a TS packet, and TSI_VALID indicates the 188 valid byte of a TS packet.

C. Sync/Burst Parallel Mode

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In this mode, TSI only takes use of TSI_SYNC to sample input parallel TS packet data.

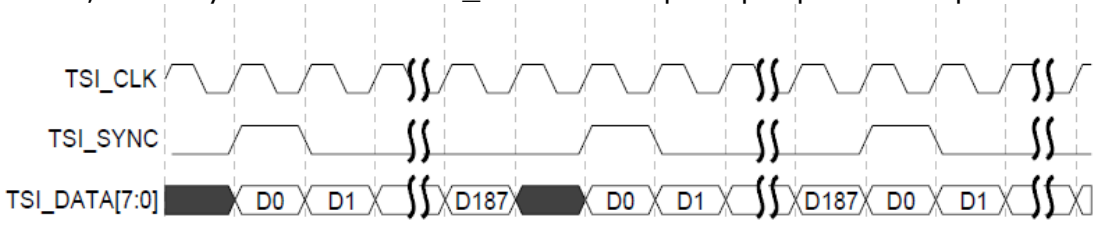

Fig. 39-4 Sync/Burst Parallel Mode

When active high, TSI_SYNC implies the first valid byte of a TS packet and remaining 187 valid bytes of a TS packet are upcoming within the following successive 187 clock cycles.

D. Nosync/Valid Parallel Mode

In this mode, TSI only takes uses of TSI_VALID to sample input parallel TS packet data.

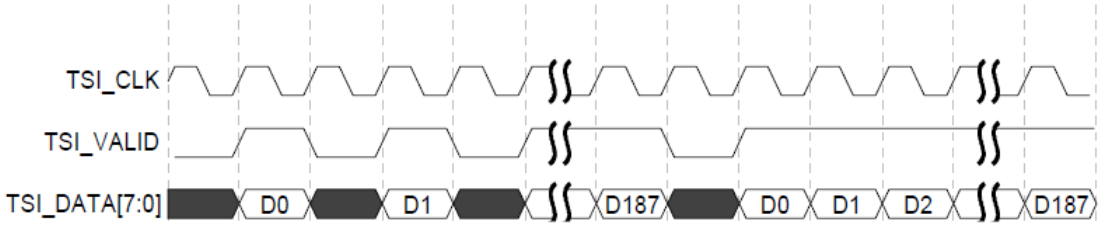

Fig. 39-5 Nosync/Valid Parallel Mode

When active high, TSI_VALID implies a valid byte of a TS packet.

39.3.2 TS output of TS Out Interface

TS out interface transmit the TS data in two mode: serial mode and parallel mode. In the serial mode, the bit order can be lsb-msb or msb-lsb.

The TS_SYNC will be active high when indicating the header of the TS packets, and it only lasts for one cycle. TS_VALID will be active high when the output TS data is valid. The output data is 188 byte TS packet data.

TS out interface also stamp the TS output stream with new PCR value, making PCR adjustment. PCR is used to measure the transport rate.

$$PCR(i) = PCR_base(i) \times 300 + PCR_ext(i)$$

where:

$$PCR_base(i) = ((system_clock_frequency \times t(i)) \; DIV \; 300) \; \% \; 2^{33}$$

$$PCR_ext(i) = ((system_clock_frequency \times t(i)) \; DIV \; 1) \; \% \; 300$$

$$transport_rate(i) = \frac{((i' - i'') \times system_clock_frequency)}{PCR(i') - PCR(i'')}$$

Where

i' is the index of the byte containing the last bit of the immediately following program_clock_reference_base field applicable to the program being decoded.

i is the is the index of any byte in the Transport Stream for $i'' < i < i'$.

i'' is the index of the byte containing the last bit of the most recent program_clock_reference_base field applicable to the program being decoded.

System clock is 27Mhz.

39.3.3 Demux and descrambling

Each PTI has 64 PID channels to deal with demultiplexing and descrambling operation.

The PTI can descramble the TS Packets which are scrambled with CSA v2.0 standard. The TS packets can be scrambled either in TS level or PES level.

The demux module can do the section filtering, pes filtering and es filtering, or directly output

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TS packets.

39.4 Register Description

39.4.1 Register Summary

Name	Offset	Size	Reset Value	Description
TSP_GCFG	0x0000	W	0x00000000	Global Configuration Register
TSP_PVR_CTRL	0x0004	W	0x00000000	PVR Control Register
TSP_PVR_LEN	0x0008	W	0x00000000	PVR DMA Transaction Length
TSP_PVR_BASE_ADDR	0x000c	W	0x00000000	PVR DMA transaction starting address
TSP_PVR_INT_STS	0x0010	W	0x00000000	PVR DMA Interrupt Status Register
TSP_PVR_INT_ENA	0x0014	W	0x00000000	DMA Interrupt Enable Register
TSP_TSOUT_CTRL	0x0018	W	0x00000000	TS Out Control Register
TSP_PVR_TOP_ADDR	0x001c	W	0x00000000	
TSP_PVR_WRITE_ADDR	0x0020	W	0x00000000	
TSP_PTIx_CTRL	0x0100	W	0x00000000	PTI Channel Control Register
TSP_PTIx_LLP_CFG	0x0104	W	0x00000000	LLP DMA Control Register
TSP_PTIx_LLP_BASE	0x0108	W	0x00000000	LLP Descriptor BASE Address
TSP_PTIx_LLP_WRITE	0x010c	W	0x00000000	LLP DMA Writing Software Descriptor Counter
TSP_PTIx_LLP_READ	0x0110	W	0x00000000	LLP DMA Reading Hardware Descriptor Counter
TSP_PTIx_PID_STS0	0x0114	W	0x00000000	PTI PID Channel Status 0 Register
TSP_PTIx_PID_STS1	0x0118	W	0x00000000	PTI PID Channel Status 1 Register
TSP_PTIx_PID_STS2	0x011c	W	0x00000000	PTI PID Channel Status 2 Register
TSP_PTIx_PID_STS3	0x0120	W	0x00000000	PTI PID Channel Status 3 Register
TSP_PTIx_PID_INT_ENA0	0x0124	W	0x00000000	PID Interrupt Enable Register 0
TSP_PTIx_PID_INT_ENA1	0x0128	W	0x00000000	PID Interrupt Enable Register 1
TSP_PTIx_PID_INT_ENA2	0x012c	W	0x00000000	PID Interrupt Enable Register 2
TSP_PTIx_PID_INT_ENA3	0x0130	W	0x00000000	PID Interrupt Enable Register 3
TSP_PTIx_PCR_INT_STS	0x0134	W	0x00000000	PTI PCR Interrupt Status Register
TSP_PTIx_PCR_INT_ENA	0x0138	W	0x00000000	PTI PCR Interrupt Enable Register
TSP_PTIx_PCRn_CTRL	0x013c	W	0x00000000	PID PCR Control Register
TSP_PTIx_PCRn_H	0x015c	W	0x00000000	High Order PCR value
TSP_PTIx_PCRn_L	0x0160	W	0x00000000	Low Order PCR value
TSP_PTIx_DMA_STS	0x019c	W	0x00000000	LLP DMA Interrupt Status Register
TSP_PTIx_DMA_ENA	0x01a0	W	0x00000000	DMA Interrupt Enable Register
TSP_PTIx_DATA_FLAG0	0x01a4	W	0x00000000	PTI_PID_WRITE Flag 0
TSP_PTIx_DATA_FLAG1	0x01a8	W	0x00000000	PTI_PID_WRITE Flag 1
TSP_PTIx_LIST_FLAG	0x01ac	W	0x00000000	PTIx_LIST_WRITE Flag
TSP_PTIx_DST_STS0	0x01b0	W	0x00000000	PTI Destination Status Register
TSP_PTIx_DST_STS1	0x01b4	W	0x00000000	PTI Destination Status Register
TSP_PTIx_DST_ENA0	0x01b8	W	0x00000000	PTI Destination Interrupt Enable Register

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Name	Offset	Size	Reset Value	Description
TSP_PTIx_DST_ENA1	0x01bc	W	0x00000000	PTI Destination Interrupt Enable Register
TSP_PTIx_ECWn_H	0x0200	W	0x00000000	The Even Control Word High Order
TSP_PTIx_ECWn_L	0x0204	W	0x00000000	The Even Control Word Low Order
TSP_PTIx_OCWn_H	0x0208	W	0x00000000	The Odd Control Word High Order
TSP_PTIx_OCWn_L	0x020c	W	0x00000000	The Odd Control Word Low Order
TSP_PTIx_PIDn_CTRL	0x0300	W	0x00000000	PID Channel Control Register
TSP_PTIx_PIDn_BASE	0x0400	W	0x00000000	PTI Data Memory Buffer Base Address
TSP_PTIx_PIDn_TOP	0x0404	W	0x00000000	PTI Data Memory Buffer Top Address
TSP_PTIx_PIDn_WRITE	0x0408	W	0x00000000	PTI Data Memory Buffer Hardware Writing Address
TSP_PTIx_PIDn_READ	0x040c	W	0x00000000	PTI Data Memory Buffer Software Reading Address
TSP_PTIx_LISTn_BASE	0x0800	W	0x00000000	PTI List Memory Buffer Base Address
TSP_PTIx_LISTn_TOP	0x0804	W	0x00000000	PTI List Memory Buffer Top Address
TSP_PTIx_LISTn_WRITE	0x0808	W	0x00000000	PTI List Memory Buffer Hardware Writing Address
TSP_PTIx_LISTn_READ	0x080c	W	0x00000000	PTI List Memory Buffer Software Reading Address
TSP_PTIx_PIDn_CFG	0x0900	W	0x00000008	PID Demux Configure Register
TSP_PTIx_PIDn_FILT_0	0x0904	W	0x00000000	Fliter Word 0
TSP_PTIx_PIDn_FILT_1	0x0908	W	0x00000000	Fliter Word 1
TSP_PTIx_PIDn_FILT_2	0x090c	W	0x00000000	Fliter Word 2
TSP_PTIx_PIDn_FILT_3	0x0910	W	0x00000000	Fliter Word 3

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

39.4.2 Detail Register Description

TSP_GCFG

Address: Operational Base + offset (0x0000)

Global Configuration Register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
6:4	RW	0x0	arbit_cnt DMA channel arbiter counter This field is used to adjust the priority of DMA channels to prevent one channel holds the highest priority for a long time. The 3-bit field sets the largest times for a DMA channel to hold the highest priority to send the bus request. After requested times reach this limit, the highest priority is passed to next DMA channel in order.
3	RW	0x0	tsout_on TS Output Module Switch 1: TS output module switched on 0: TS output module switched off
2	RW	0x0	pvr_on PVR Module Switch 1: PVR function turned on ; 0: PVR function turned off ;
1	RW	0x0	pti1_on PTI0 channel switch 1: PTI1 channel switched on 0: PTI1 channel switched off
0	RW	0x0	pti0_on PTI0 channel switch 1: PTI0 channel switched on 0: PTI1 channel switched off

TSP_PVR_CTRL
Address: Operational Base + offset (0x0004)
PVR Control Register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	fixaddr_en Fix Address Mode Select 1: fixed address mode; 0: incrementing address mode;
5:4	RW	0x0	burst_mode PVR burst mode PVR DMA burst mode 2'b00: INCR4 2'b01: INCR8 2'b10: INCR16 2'b11: Reserverd

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Bit	Attr	Reset Value	Description
3:2	RW	0x0	source PVR Source Select TS source for PVR output. 00: non-PID-filtered TS packets in PTI0; 01: PID filtered TS packets in PTI0; 10: non-PID-filtered TS packets in PTI1; 11: PID-filtered TS packets in PTI1;
1	R/WSC	0x0	stop PVR stop Write 1 to stop DMA channel. DMA will complete current burst transfer and then stop. It may takes several cycles. 1: PVR Stop ; 0: no effect ;
0	R/WSC	0x0	start PVR start Write 1 to start PVR. This bit will be cleared if PVR is stopped or PVR transaction is completed. 1: start PVR 0: no effect.

TSP_PVR_LEN
Address: Operational Base + offset (0x0008)
PVR DMA Transaction Length

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	len Transaction Length Transaction Length

TSP_PVR_BASE_ADDR
Address: Operational Base + offset (0x000c)
PVR DMA transaction starting address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	addr PVR DMA transaction starting address PVR DMA transaction starting address

TSP_PVR_INT_STS
Address: Operational Base + offset (0x0010)
PVR DMA Interrupt Status Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
2	W1C	0x0	pvr_update_flag pvr address pageover flag When write_addr >= (base + top_addr/2), or write addr >= top_addr, the pvr_update_flag will assert HIGH. The application can write 1 to this bit to clear it.
1	W1C	0x0	pvr_error PVR DMA transaction error 1: error response during PVR DMA transaction; 0: no error response during PVR DMA transaction;
0	W1C	0x0	pvr_done PVR DMA transaction done 1: PVR DMA transaction completed; 0: PVR DMA transaction not completed;

TSP_PVR_INT_ENA
Address: Operational Base + offset (0x0014)
DMA Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	pvr_update_ena 1: pvr_update interrupt enable 0: pvr_update interrupt disable
1	RW	0x0	pvr_error_ena PVR DMA Transcation Error Interrupt Enable 1: Error Interrupt Enabled 0: Error Interrupt Disabled
0	RW	0x0	pvr_done_ena PVR DMA Transaction Done Interrupt Enable 1: Done Interrupt Enabled 0: Done Interrupt Disabled

TSP_TSOUT_CTRL
Address: Operational Base + offset (0x0018)
TS Out Control Register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	tso_sdo_sel TS serial data output 1: bit[0] use as serial data output ; 0: bit[7] use as serial data output ;

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Bit	Attr	Reset Value	Description
5	RW	0x0	tso_clk_phase TS output clock phase 0: ts output clock; 1: inverse of ts output clock.
4	RW	0x0	mode TS Output mode Selection Output mode select: 0: Serial Mode 1: Parallel Mode
3	RW	0x0	bit_order ts output serial data byte order Indicates that the output serial data byte order, ignored in the parallel: 0: MSB to LSB 1: LSB to MSB
2:1	RW	0x0	source TS Output Source Select TS source for TS out. 00: non-PID-filtered TS packets in PTI0; 01: PID filtered TS packets in PTI0; 10: non-PID-filtered TS packets in PTI1; 11: PID-filtered TS packets in PTI1;
0	RW	0x0	start TS out start 1: to start TS out function ; 0: to stop TS out function;

TSP_PVR_TOP_ADDR
Address: Operational Base + offset (0x001c)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pvr_top_addr

TSP_PVR_WRITE_ADDR
Address: Operational Base + offset (0x0020)

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	pvr_write_addr The core will update this register to show the PVR write addr

TSP_PTIx_CTRL
Address: Operational Base + offset (0x0100)
PTI Channel Control Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21	RW	0x0	tsi_sdi_sel TS Serial Data Input Select 1: bit[0] use as serial input data 0: bit[7] use as serial input data
20:19	RW	0x0	tsi_error_handle TS ERROR Handle 00: don't output 01: set the error indicator to 1 10: don't care
18	RW	0x0	clk_phase_sel ts input clock phase select 1'b0: ts input clock 1'b1: inverse of ts input clock
17:16	RW	0x0	demux_burst_mode Demux DMA Burst Mode Demux DMA Mode 2'b00: INCR4 2'b01: INCR8 2'b10: INCR16 2'b11: Reserved
15	RW	0x0	sync_bypass Bypass mode Selection 1'b1: Bypass mode, indicating that input TS packets will not be resynchronized and directly fed into the following modules; 1'b0: Synchronous mode, default, indicating that input TS packets will be resynchronized;
14	RW	0x0	cw_byteorder Control Word format Configuration 0: Default: first byte of the word is the highest byte 1: first byte of the word is the lowest byte
13	RW	0x0	cm_on CSA Conformance Mechanism Configuration CSA Conformance Mechanism 0: CM turned off 1: CM turned on
12:11	RW	0x0	tsi_mode TSI Input Mode Selection Input mode selection: 00: Serial Sync/valid Mode 01: Parallel Sync/valid Mode 10: Parallel Sync/burst Mode 11: Parallel Nosync/valid Mode

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Bit	Attr	Reset Value	Description
10	RW	0x0	tsi_bit_order input serial data order Indicates that the input serial data byte order, ignored in the parallel mode: 0: MSB to LSB 1: LSB to MSB
9	RW	0x0	tsi_sel TS Input Source Select Select input TS source 1'b1: HSADC ; 1'b0: internel memory ;
8	RW	0x0	out_byteswap Output byteswap function When enabled, the word to be transferred to memory buffer "B4B3B2B1" is performed byteswapping to "B1B2B3B4".
7	RW	0x0	in_byteswap Input TS Word Byteswap When enabled, the input TS word "B4B3B2B1" is perfomed byteswapping to "B1B2B3B4".
6:4	RW	0x0	unsync_times TS Header Unsynchronized Times If synchronous mode is selected. This field sets the successive times of TS packet header error to re-lock TS header when TS is in locked status;
3:1	RW	0x0	sync_times TS Header Synchronized Times If synchronous mode is selected. This field sets the successive times of finding TS packet header to lock the TS header when TS is in unlocked status;
0	R/WSC	0x0	clear Software clear signal It will reset the core register . It will table several cycles. After reset done, soft_reset will be low. 1. reset; 0. no effect.

TSP_PTIx_LLP_CFG
Address: Operational Base + offset (0x0104)
LLP DMA Control Register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:8	RW	0x0	threshold LLP Transfer Threshold The depth for LLP descriptors is 64. An interrupt will be asserted when transfer reaches the threshold set if DMA transfer interrupt is enabled. 00: 1/1 depth 01: 1/2 depth 10: 1/4 depth 11: 1/8 depth
7:6	RW	0x0	burst_mode LLP DMA Burst Mode LLP DMA Burst Mode 2'b00: INCR4 2'b01: INCR8 2'b10: INCR16 2'b11: Reserverd
5	RW	0x0	hw_trigger Hardware Trigger Select 1. hardware trigger; 0. software trigger;
4	RW	0x0	fix_addr_en Fix Address Mode Select 1: fixed address mode; 0: incrementing address mode;
3	W1C	0x0	cfg_done LLP DMA Configuration Done When all descriptors of LLP are configured, write 1 to to this bit. The core will clear this bit when llp transction is finished ;
2	RW	0x0	pause LLP DMA Pause Write 1 to Pause DMA channel . DMA will complete current burst transfer and then pause. All register stay unchange. If software write 0 later , It will continue to work. It may take several cycles to pause. 1: pause; 0: continue to work ;

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Bit	Attr	Reset Value	Description
1	W1C	0x0	stop LLP DMA Stop Write 1 to stop DMA channel. DMA will complete current burst transter and then stop. It may takes several cycles. 1: stop ; 0: no effect ;
0	W1C	0x0	start LLP DMA start Write 1 to start DMA Channel , self clear after 1 cycle. 1: start ; 0: no effect

TSP_PTIx_LLP_BASE
Address: Operational Base + offset (0x0108)
LLP Descriptor BASE Address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	addr LLP Descriptor BASE Address LLP Descriptor BASE address

TSP_PTIx_LLP_WRITE
Address: Operational Base + offset (0x010c)
LLP DMA Writing Software Descriptor Counter

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	counter LLP DMA Writing Software Descriptor Counter LLP DMA Writing Software Descriptor Counter

TSP_PTIx_LLP_READ
Address: Operational Base + offset (0x0110)
LLP DMA Reading Hardware Descriptor Counter

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RO	0x00	counter LLP DMA Reading Hardware Descriptor Counter LLP DMA Reading Hardware Descriptor Counter

TSP_PTIx_PID_STS0
Address: Operational Base + offset (0x0114)
PTI PID Channel Status 0 Register

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Bit	Attr	Reset Value	Description
31	RW	0x0	pid31_done PID31 Channel Status 1 means done
30	W1C	0x0	pid30_done PID30 Channel Status 1 means done
29	W1C	0x0	pid29_done PID29 Channel Status 1 means done
28	W1C	0x0	pid28_done PID28 Channel Status 1 means done
27	W1C	0x0	pid27_done PID27 Channel Status 1 means done
26	W1C	0x0	pid26_done PID26 Channel Status 1 means done
25	W1C	0x0	pid25_done PID25 Channel Status 1 means done
24	W1C	0x0	pid24_done PID24 Channel Status 1 means done
23	W1C	0x0	pid23_done PID23 Channel Status 1 means done
22	W1C	0x0	pid22_done PID22 Channel Status 1 means done
21	W1C	0x0	pid21_done PID21 Channel Status 1 means done
20	W1C	0x0	pid20_done PID20 Channel Status 1 means done
19	W1C	0x0	pid19_done PID19 Channel Status 1 means done
18	W1C	0x0	pid18_done PID18 Channel Status 1 means done
17	W1C	0x0	pid17_done PID17 Channel Status 1 means done

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Bit	Attr	Reset Value	Description
16	W1C	0x0	pid16_done PID16 Channel Status 1 means done
15	W1C	0x0	pid15_done PID15 Channel Status 1 means done
14	W1C	0x0	pid14_done PID14 Channel Status 1 means done
13	W1C	0x0	pid13_done PID13 Channel Status 1 means done
12	W1C	0x0	pid12_done PID12 Channel Status 1 means done
11	W1C	0x0	pid11_done PID11 Channel Status 1 means done
10	W1C	0x0	pid10_done PID10 Channel Status 1 means done
9	W1C	0x0	pid9_done PID9 Channel Status 1 means done
8	W1C	0x0	pid8_done PID8 Channel Status 1 means done
7	W1C	0x0	pid7_done PID7 Channel Status 1 means done
6	W1C	0x0	pid6_done PID6 Channel Status 1 means done
5	W1C	0x0	pid5_done PID5 Channel Status 1 means done
4	W1C	0x0	pid4_done PID4 Channel Status 1 means done
3	W1C	0x0	pid3_done PID3 Channel Status 1 means done
2	RW	0x0	pid2_done PID2 Channel Status 1 means done

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Bit	Attr	Reset Value	Description
1	W1C	0x0	pid1_done PID1 Channel Status 1 means done
0	W1C	0x0	pid0_done PID0 Channel Status 1 means done

TSP_PTIx_PID_STS1
Address: Operational Base + offset (0x0118)
PTI PID Channel Status 1 Register

Bit	Attr	Reset Value	Description
31	W1C	0x0	pid63_done PID63 Channel Status 1 means done
30	W1C	0x0	pid62_done PID62 Channel Status 1 means done
29	W1C	0x0	pid61_done PID61 Channel Status 1 means done
28	W1C	0x0	pid60_done PID60 Channel Status 1 means done
27	W1C	0x0	pid59_done PID59 Channel Status 1 means done
26	W1C	0x0	pid58_done PID58 Channel Status 1 means done
25	W1C	0x0	pid57_done PID57 Channel Status 1 means done
24	W1C	0x0	pid56_done PID56 Channel Status 1 means done
23	W1C	0x0	pid55_done PID55 Channel Status 1 means done
22	W1C	0x0	pid54_done PID54 Channel Status 1 means done
21	W1C	0x0	pid53_done PID53 Channel Status 1 means done

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Bit	Attr	Reset Value	Description
20	W1C	0x0	pid52_done PID52 Channel Status 1 means done
19	W1C	0x0	pid51_done PID51 Channel Status 1 means done
18	W1C	0x0	pid50_done PID51 Channel Status 1 means done
17	W1C	0x0	pid49_done PID49 Channel Status 1 means done
16	W1C	0x0	pid48_done PID48 Channel Status 1 means done
15	W1C	0x0	pid47_done PID47 Channel Status 1 means done
14	W1C	0x0	pid46_done PID46 Channel Status 1 means done
13	W1C	0x0	pid45_done PID45 Channel Status 1 means done
12	W1C	0x0	pid44_done PID44 Channel Status 1 means done
11	W1C	0x0	pid43_done PID43 Channel Status 1 means done
10	W1C	0x0	pid42_done PID42 Channel Status 1 means done
9	W1C	0x0	pid41_done PID41 Channel Status 1 means done
8	W1C	0x0	pid40_done PID40 Channel Status 1 means done
7	W1C	0x0	pid39_done PID39 Channel Status 1 means done
6	W1C	0x0	pid38_done PID38 Channel Status 1 means done

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Bit	Attr	Reset Value	Description
5	W1C	0x0	pid37_done PID37 Channel Status 1 means done
4	W1C	0x0	pid36_done PID36 Channel Status 1 means done
3	RW	0x0	pid35_done PID35 Channel Status 1 means done
2	W1C	0x0	pid34_done PID34 Channel Status 1 means done
1	W1C	0x0	pid33_done PID33 Channel Status 1 means done
0	RW	0x0	pid32_done PID32 Channel Status 1 means done

TSP_PTIx_PID_STS2
Address: Operational Base + offset (0x011c)
PTI PID Channel Status 2 Register

Bit	Attr	Reset Value	Description
31	RW	0x0	pid31_error PID31 Error Interrupt Status 1 means error detected
30	W1C	0x0	pid30_error PID30 Error Interrupt Status 1 means error detected
29	W1C	0x0	pid29_error PID29 Error Interrupt Status 1 means error detected
28	W1C	0x0	pid28_error PID28 Error Interrupt Status 1 means error detected
27	W1C	0x0	pid27_error PID27 Error Interrupt Status 1 means error detected
26	W1C	0x0	pid26_error PID26 Error Interrupt Status 1 means error detected
25	W1C	0x0	pid25_error PID25 Error Interrupt Status 1 means error detected

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Bit	Attr	Reset Value	Description
24	W1C	0x0	pid24_error PID24 Error Interrupt Status 1 means error detected
23	W1C	0x0	pid23_error PID23 Error Interrupt Status 1 means error detected
22	W1C	0x0	pid22_error PID22 Error Interrupt Status 1 means error detected
21	W1C	0x0	pid21_error PID21 Error Interrupt Status 1 means error detected
20	W1C	0x0	pid20_error PID20 Error Interrupt Status 1 means error detected
19	W1C	0x0	pid19_error PID19 Error Interrupt Status 1 means error detected
18	W1C	0x0	pid18_error PID18 Error Interrupt Status 1 means error detected
17	W1C	0x0	pid17_error PID17 Error Interrupt Status 1 means error detected
16	W1C	0x0	pid16_error PID16 Error Interrupt Status 1 means error detected
15	W1C	0x0	pid15_error PID15 Error Interrupt Status 1 means error detected
14	W1C	0x0	pid14_error PID14 Error Interrupt Status 1 means error detected
13	W1C	0x0	pid13_error PID13 Error Interrupt Status 1 means error detected
12	W1C	0x0	pid12_error PID12 Error Interrupt Status 1 means error detected
11	W1C	0x0	pid11_error PID11 Error Interrupt Status 1 means error detected
10	W1C	0x0	pid10_error PID10 Error Interrupt Status 1 means error detected

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Bit	Attr	Reset Value	Description
9	W1C	0x0	pid9_error PID9 Error Interrupt Status 1 means error detected
8	W1C	0x0	pid8_error PID8 Error Interrupt Status 1 means error detected
7	W1C	0x0	pid7_error PID7 Error Interrupt Status 1 means error detected
6	W1C	0x0	pid6_error PID6 Error Interrupt Status 1 means error detected
5	W1C	0x0	pid5_error PID5 Error Interrupt Status 1 means error detected
4	W1C	0x0	pid4_error PID4 Error Interrupt Status 1 means error detected
3	W1C	0x0	pid3_error PID3 Error Interrupt Status 1 means error detected
2	W1C	0x0	pid2_error PID2 Error Interrupt Status 1 means error detected
1	W1C	0x0	pid1_error PID1 Error Interrupt Status 1 means error detected
0	W1C	0x0	pid0_error PID0 Error Interrupt Status 1 means error detected

TSP_PTIx_PID_STS3

Address: Operational Base + offset (0x0120)

PTI PID Channel Status 3 Register

Bit	Attr	Reset Value	Description
31	W1C	0x0	pid63_error PID63 Error Interrupt Status
30	W1C	0x0	pid62_error PID62 Error Interrupt Status
29	W1C	0x0	pid61_error PID61 Error Interrupt Status
28	W1C	0x0	pid60_error PID60 Error Interrupt Status
27	W1C	0x0	pid59_error PID59 Error Interrupt Status

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Bit	Attr	Reset Value	Description
26	W1C	0x0	pid58_error PID58 Error Interrupt Status
25	W1C	0x0	pid57_error PID57 Error Interrupt Status
24	W1C	0x0	pid56_error PID56 Error Interrupt Status
23	W1C	0x0	pid55_error PID55 Error Interrupt Status
22	W1C	0x0	pid54_error PID54 Error Interrupt Status
21	W1C	0x0	pid53_error PID53 Error Interrupt Status
20	W1C	0x0	pid52_error PID52 Error Interrupt Status
19	W1C	0x0	pid51_error PID51 Error Interrupt Status
18	W1C	0x0	pid50_error PID50 Error Interrupt Status
17	W1C	0x0	pid49_error PID49 Error Interrupt Status
16	W1C	0x0	pid48_error PID48 Error Interrupt Status
15	W1C	0x0	pid47_error PID47 Error Interrupt Status
14	W1C	0x0	pid46_error PID46 Error Interrupt Status
13	W1C	0x0	pid45_error PID45 Error Interrupt Status
12	W1C	0x0	pid44_error PID44 Error Interrupt Status
11	W1C	0x0	pid43_error PID43 Error Interrupt Status
10	W1C	0x0	pid42_error PID42 Error Interrupt Status
9	W1C	0x0	pid41_error PID41 Error Interrupt Status
8	W1C	0x0	pid40_error PID40 Error Interrupt Status
7	W1C	0x0	pid39_error PID39 Error Interrupt Status
6	W1C	0x0	pid38_error PID38 Error Interrupt Status
5	W1C	0x0	pid37_error PID37 Error Interrupt Status

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Bit	Attr	Reset Value	Description
4	W1C	0x0	pid36_error PID36 Error Interrupt Status
3	W1C	0x0	pid35_error PID35 Error Interrupt Status
2	W1C	0x0	pid34_error PID34 Error Interrupt Status
1	W1C	0x0	pid33_error PID33 Error Interrupt Status
0	W1C	0x0	pid32_error PID32 Error Interrupt Status

TSP_PTIx_PID_INT_ENA0
Address: Operational Base + offset (0x0124)
PID Interrupt Enable Register 0

Bit	Attr	Reset Value	Description
31	RW	0x0	pid31_done_ena PID31 Done Enable 1:enabled 0:disabled
30	RW	0x0	pid30_done_ena PID30 Done Enable 1:enabled 0:disabled
29	RW	0x0	pid29_done_ena PID29 Done Enable 1:enabled 0:disabled
28	RW	0x0	pid28_done_ena PID28 Done Enable 1:enabled 0:disabled
27	RW	0x0	pid27_done_ena PID27 Done Enable 1:enabled 0:disabled
26	RW	0x0	pid26_done_ena PID26 Done Enable 1:enabled 0:disabled
25	RW	0x0	pid25_done_ena PID25 Done Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
24	RW	0x0	pid24_done_ena PID24 Done Enable 1:enabled 0:disabled
23	RW	0x0	pid23_done_ena PID23 Done Enable 1:enabled 0:disabled
22	RW	0x0	pid22_done_ena PID22 Done Enable 1:enabled 0:disabled
21	RW	0x0	pid21_done_ena PID21 Done Enable 1:enabled 0:disabled
20	RW	0x0	pid20_done_ena PID20 Done Enable 1:enabled 0:disabled
19	RW	0x0	pid19_done_ena PID19 Done Enable 1:enabled 0:disabled
18	RW	0x0	pid18_done_ena PID18 Done Enable 1:enabled 0:disabled
17	RW	0x0	pid17_done_ena PID17 Done Enable
16	RW	0x0	pid16_done_ena PID16 Done Enable 1:enabled 0:disabled
15	RW	0x0	pid15_done_ena PID15 Done Enable 1:enabled 0:disabled
14	RW	0x0	pid14_done_ena PID14 Done Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
13	RW	0x0	pid13_done_ena PID13 Done Enable 1:enabled 0:disabled
12	RW	0x0	pid12_done_ena PID12 Done Enable 1:enabled 0:disabled
11	RW	0x0	pid11_done_ena PID11 Done Enable 1:enabled 0:disabled
10	RW	0x0	pid10_done_ena PID10 Done Enable 1:enabled 0:disabled
9	RW	0x0	pid9_done_ena PID9 Done Enable 1:enabled 0:disabled
8	RW	0x0	pid8_done_ena PID8 Done Enable 1:enabled 0:disabled
7	RW	0x0	pid7_done_ena PID7 Done Enable 1:enabled 0:disabled
6	RW	0x0	pid6_done_ena PID6 Done Enable 1:enabled 0:disabled
5	RW	0x0	pid5_done_ena PID5 Done Enable 1:enabled 0:disabled
4	RW	0x0	pid4_done_ena PID4 Done Enable 1:enabled 0:disabled
3	RW	0x0	pid3_done_ena PID3 Done Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
2	RW	0x0	pid2_done_ena PID2 Done Enable 1:enabled 0:disabled
1	RW	0x0	pid1_done_ena PID1 Done Enable 1:enabled 0:disabled
0	RW	0x0	pid0_done_ena PID0 Done Enable 1:enabled 0:disabled

TSP_PTIx_PID_INT_ENA1
Address: Operational Base + offset (0x0128)
PID Interrupt Enable Register 1

Bit	Attr	Reset Value	Description
31	RW	0x0	pid63_done PID63 Done Enable 1:enabled 0:disabled
30	RW	0x0	pid62_done PID62 Done Enable 1:enabled 0:disabled
29	RW	0x0	pid61_done PID61 Done Enable 1:enabled 0:disabled
28	RW	0x0	pid60_done PID60 Done Enable 1:enabled 0:disabled
27	RW	0x0	pid59_done PID59 Done Enable 1:enabled 0:disabled
26	RW	0x0	pid58_done PID58 Done Enable 1:enabled 0:disabled
25	RW	0x0	pid57_done PID57 Done Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
24	RW	0x0	pid56_done PID56 Done Enable 1:enabled 0:disabled
23	RW	0x0	pid55_done PID55 Done Enable 1:enabled 0:disabled
22	RW	0x0	pid54_done PID54 Done Enable 1:enabled 0:disabled
21	RW	0x0	pid53_done PID53 Done Enable 1:enabled 0:disabled
20	RW	0x0	pid52_done PID52 Done Enable 1:enabled 0:disabled
19	RW	0x0	pid51_done PID51 Done Enable 1:enabled 0:disabled
18	RW	0x0	pid50_done PID50 Done Enable 1:enabled 0:disabled
17	RW	0x0	pid49_done PID49 Done Enable 1:enabled 0:disabled
16	RW	0x0	pid48_done PID48 Done Enable 1:enabled 0:disabled
15	RW	0x0	pid47_done PID47 Done Enable 1:enabled 0:disabled
14	RW	0x0	pid46_done PID46 Done Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
13	RW	0x0	pid45_done PID45 Done Enable 1:enabled 0:disabled
12	RW	0x0	pid44_done PID44 Done Enable 1:enabled 0:disabled
11	RW	0x0	pid43_done PID43 Done Enable 1:enabled 0:disabled
10	RW	0x0	pid42_done PID42 Done Enable 1:enabled 0:disabled
9	RW	0x0	pid41_done PID41 Done Enable 1:enabled 0:disabled
8	RW	0x0	pid40_done PID40 Done Enable 1:enabled 0:disabled
7	RW	0x0	pid39_done PID39 Done Enable 1:enabled 0:disabled
6	RW	0x0	pid38_done PID38 Done Enable 1:enabled 0:disabled
5	RW	0x0	pid37_done PID37 Done Enable 1:enabled 0:disabled
4	RW	0x0	pid36_done PID36 Done Enable 1:enabled 0:disabled
3	RW	0x0	pid35_done PID35 Done Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
2	RW	0x0	pid34_done PID34 Done Enable 1:enabled 0:disabled
1	RW	0x0	pid33_done PID33 Done Enable 1:enabled 0:disabled
0	RW	0x0	pid32_done PID32 Done Enable 1:enabled 0:disabled

TSP_PTIx_PID_INT_ENA2
Address: Operational Base + offset (0x012c)
PID Interrupt Enable Register 2

Bit	Attr	Reset Value	Description
31	RW	0x0	pid31_error PID31 Error Interrupt Enable 1:enabled 0:disabled
30	RW	0x0	pid30_error PID30 Error Interrupt Enable 1:enabled 0:disabled
29	RW	0x0	pid29_error PID29 Error Interrupt Enable 1:enabled 0:disabled
28	RW	0x0	pid28_error PID28 Error Interrupt Enable 1:enabled 0:disabled
27	RW	0x0	pid27_error PID27 Error Interrupt Enable 1:enabled 0:disabled
26	RW	0x0	pid26_error PID26 Error Interrupt Enable 1:enabled 0:disabled
25	RW	0x0	pid25_error PID25 Error Interrupt Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
24	RW	0x0	pid24_error PID24 Error Interrupt Enable 1:enabled 0:disabled
23	RW	0x0	pid23_error PID23 Error Interrupt Enable 1:enabled 0:disabled
22	RW	0x0	pid22_error PID22 Error Interrupt Enable 1:enabled 0:disabled
21	RW	0x0	pid21_error PID21 Error Interrupt Enable 1:enabled 0:disabled
20	RW	0x0	pid20_error PID20 Error Interrupt Enable 1:enabled 0:disabled
19	RW	0x0	pid19_error PID19 Error Interrupt Enable 1:enabled 0:disabled
18	RW	0x0	pid18_error PID18 Error Interrupt Enable 1:enabled 0:disabled
17	RW	0x0	pid17_error PID17 Error Interrupt Enable 1:enabled 0:disabled
16	RW	0x0	pid16_error PID16 Error Interrupt Enable 1:enabled 0:disabled
15	RW	0x0	pid15_error PID15 Error Interrupt Enable 1:enabled 0:disabled
14	RW	0x0	pid14_error PID14 Error Interrupt Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
13	RW	0x0	pid13_error PID13 Error Interrupt Enable 1:enabled 0:disabled
12	RW	0x0	pid12_error PID12 Error Interrupt Enable 1:enabled 0:disabled
11	RW	0x0	pid11_error PID11 Error Interrupt Enable 1:enabled 0:disabled
10	RW	0x0	pid10_error PID10 Error Interrupt Enable 1:enabled 0:disabled
9	RW	0x0	pid9_error PID9 Error Interrupt Enable 1:enabled 0:disabled
8	RW	0x0	pid8_error PID8 Error Interrupt Enable 1:enabled 0:disabled
7	RW	0x0	pid7_error PID7 Error Interrupt Enable 1:enabled 0:disabled
6	RW	0x0	pid6_error PID6 Error Interrupt Enable 1:enabled 0:disabled
5	RW	0x0	pid5_error PID5 Error Interrupt Enable 1:enabled 0:disabled
4	RW	0x0	pid4_error PID4 Error Interrupt Enable 1:enabled 0:disabled
3	RW	0x0	pid3_error PID3 Error Interrupt Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
2	RW	0x0	pid2_error PID2 Error Interrupt Enable 1:enabled 0:disabled
1	RW	0x0	pid1_error PID1 Error Interrupt Enable 1:enabled 0:disabled
0	RW	0x0	pid0_error PID0 Error Interrupt Enable 1:enabled 0:disabled

TSP_PTIx_PID_INT_ENA3
Address: Operational Base + offset (0x0130)
PID Interrupt Enable Register 3

Bit	Attr	Reset Value	Description
31	RW	0x0	pid63_error PID63 Error Interrupt Enable 1:enabled 0:disabled
30	RW	0x0	pid62_error PID62 Error Interrupt Enable 1:enabled 0:disabled
29	RW	0x0	pid61_error PID61 Error Interrupt Enable 1:enabled 0:disabled
28	RW	0x0	pid60_error PID60 Error Interrupt Enable 1:enabled 0:disabled
27	RW	0x0	pid59_error PID59 Error Interrupt Enable 1:enabled 0:disabled
26	RW	0x0	pid58_error PID58 Error Interrupt Enable 1:enabled 0:disabled
25	RW	0x0	pid57_error PID57 Error Interrupt Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
24	RW	0x0	pid56_error PID56 Error Interrupt Enable 1:enabled 0:disabled
23	RW	0x0	pid55_error PID55 Error Interrupt Enable 1:enabled 0:disabled
22	RW	0x0	pid54_error PID54 Error Interrupt Enable 1:enabled 0:disabled
21	RW	0x0	pid53_error PID53 Error Interrupt Enable 1:enabled 0:disabled
20	RW	0x0	pid52_error PID52 Error Interrupt Enable 1:enabled 0:disabled
19	RW	0x0	pid51_error PID51 Error Interrupt Enable 1:enabled 0:disabled
18	RW	0x0	pid50_error PID50 Error Interrupt Enable 1:enabled 0:disabled
17	RW	0x0	pid49_error PID49 Error Interrupt Enable 1:enabled 0:disabled
16	RW	0x0	pid48_error PID48 Error Interrupt Enable 1:enabled 0:disabled
15	RW	0x0	pid47_error PID47 Error Interrupt Enable 1:enabled 0:disabled
14	RW	0x0	pid46_error PID46 Error Interrupt Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
13	RW	0x0	pid45_error PID45 Error Interrupt Enable 1:enabled 0:disabled
12	RW	0x0	pid44_error PID44 Error Interrupt Enable 1:enabled 0:disabled
11	RW	0x0	pid43_error PID43 Error Interrupt Enable 1:enabled 0:disabled
10	RW	0x0	pid42_error PID42 Error Interrupt Enable 1:enabled 0:disabled
9	RW	0x0	pid41_error PID41 Error Interrupt Enable 1:enabled 0:disabled
8	RW	0x0	pid40_error PID40 Error Interrupt Enable 1:enabled 0:disabled
7	RW	0x0	pid39_error PID39 Error Interrupt Enable 1:enabled 0:disabled
6	RW	0x0	pid38_error PID38 Error Interrupt Enable 1:enabled 0:disabled
5	RW	0x0	pid37_error PID37 Error Interrupt Enable 1:enabled 0:disabled
4	RW	0x0	pid36_error PID36 Error Interrupt Enable 1:enabled 0:disabled
3	RW	0x0	pid35_error PID35 Error Interrupt Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
2	RW	0x0	pid34_error PID34 Error Interrupt Enable 1:enabled 0:disabled
1	RW	0x0	pid33_error PID33 Error Interrupt Enable 1:enabled 0:disabled
0	RW	0x0	pid32_error PID32 Error Interrupt Enable 1:enabled 0:disabled

TSP_PTIx_PCR_INT_STS
Address: Operational Base + offset (0x0134)
PTI PCR Interrupt Status Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	W1C	0x0	pcr7_done PCR7 Status 1: done; 0: not done;
6	W1C	0x0	pcr6_done PCR6 Status 1: done; 0: not done;
5	W1C	0x0	pcr5_done PCR5 Status 1: done; 0: not done;
4	W1C	0x0	pcr4_done PCR4 Status 1: done; 0: not done;
3	W1C	0x0	pcr3_done PCR3 Status 1: done; 0: not done;
2	W1C	0x0	pcr2_done PCR2 Status 1: done; 0: not done;

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Bit	Attr	Reset Value	Description
1	W1C	0x0	pcr1_done PCR1 Status 1: done; 0: not done;
0	W1C	0x0	pcr0_done PCR0 Status 1: done; 0: not done;

TSP_PTIx_PCR_INT_ENA
Address: Operational Base + offset (0x0138)
PTI PCR Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x0	pcr7_done_ena pcr7 done interrupt enable 1: enabled; 0: disabled;
6	RW	0x0	pcr6_done_ena pcr6 done interrupt enable 1: enabled; 0: disabled;
5	RW	0x0	pcr5_done_ena pcr5 done interrupt enable 1: enabled; 0: disabled;
4	RW	0x0	pcr4_done_ena pcr4 done interrupt enable 1: enabled; 0: disabled;
3	RW	0x0	pcr3_done_ena pcr3 done interrupt enable 1: enabled; 0: disabled;
2	RW	0x0	pcr2_done_ena pcr2 done interrupt enable 1: enabled; 0: disabled;
1	RW	0x0	pcr1_done_ena pcr1 done interrupt enable 1: enabled; 0: disabled;

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Bit	Attr	Reset Value	Description
0	RW	0x0	pcr0_done_ena pcr0 done interrupt enable 1: enabled; 0: disabled;

TSP_PTIx_PCRn_CTRL
Address: Operational Base + offset (0x013c)
PID PCR Control Register

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13:1	RW	0x0000	pid PCR Extraction PID number This 13-bit field sets the PID number that needs PCR extraction.
0	RW	0x0	on PCR Extraction Switch 1'b1: PCR extraction switched on ; 1'b0: PCR extraction switched off ;

TSP_PTIx_PCRn_H
Address: Operational Base + offset (0x015c)
High Order PCR value

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RO	0x0	pcr PCR[32] pcr[32]

TSP_PTIx_PCRn_L
Address: Operational Base + offset (0x0160)
Low Order PCR value

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	pcr pcr[31:0] pcr[31:0]

TSP_PTIx_DMA_STS
Address: Operational Base + offset (0x019c)
LLP DMA Interrupt Status Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	W1C	0x0	llp_error LLP DMA Error Status 1: error response during DMA transaction; 0: no error response during DMA transaction;

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Bit	Attr	Reset Value	Description
0	W1C	0x0	llp_done LLP DMA Done Status 1: DMA transaction completed; 0: DMA transaction not completed;

TSP_PTIx_DMA_ENA
Address: Operational Base + offset (0x01a0)
DMA Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	llp_error_ena LLP DMA Error Interrupt Enable 1: enabled 0: disabled
0	RW	0x0	llp_done_ena LLP DMA Done Interrupt Enable 1: enabled 0: disabled

TSP_PTIx_DATA_FLAG0
Address: Operational Base + offset (0x01a4)
PTI_PID_WRITE Flag 0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	data_write_flag_0 From PID0 TO PID31

TSP_PTIx_DATA_FLAG1
Address: Operational Base + offset (0x01a8)
PTI_PID_WRITE Flag 1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	data_write_flag_1 From PID32 TO PID63

TSP_PTIx_LIST_FLAG
Address: Operational Base + offset (0x01ac)
PTIx_LIST_WRITE Flag

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	list_write_flag From PID0 TO PID15

TSP_PTIx_DST_STS0
Address: Operational Base + offset (0x01b0)
PTI Destination Status Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	W1C	0x00000000	demux_dma_status_0 From 0 to 31 channel

TSP_PTIx_DST_STS1
Address: Operational Base + offset (0x01b4)
PTI Destination Status Register

Bit	Attr	Reset Value	Description
31:0	W1C	0x00000000	demux_dma_status_0 From 32 to 63 channel

TSP_PTIx_DST_ENA0
Address: Operational Base + offset (0x01b8)
PTI Destination Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	demux_dma_enable_0 From 0 to 31 channel

TSP_PTIx_DST_ENA1
Address: Operational Base + offset (0x01bc)
PTI Destination Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	demux_dma_enable_1 From 32 to 63 channel

TSP_PTIx_ECWn_H
Address: Operational Base + offset (0x0200)
The Even Control Word High Order

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ecw_h The Even Control Word High Order ECW[63:32]

TSP_PTIx_ECWn_L
Address: Operational Base + offset (0x0204)
The Even Control Word Low Order

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ecw_l The Even Control Word Low Order ECW[31:0]

TSP_PTIx_OCWn_H
Address: Operational Base + offset (0x0208)
The Odd Control Word High Order

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ocw_h The Odd Control Word High order OCW[63:32]

TSP_PTIx_OCWn_L
Address: Operational Base + offset (0x020c)
The Odd Control Word Low Order

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ocw_l The Odd Control Word Low Order OCW[31:0]

TSP_PTIx_PIDn_CTRL
Address: Operational Base + offset (0x0300)
PID Channel Control Register

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	cw_num Control Word Order Number This fields indicates the corresponding order number of control word to be used to descramble TS packets.
15:3	RW	0x0000	pid PID number This 13-bit sets the desired PID number to be processed by PTI channel.
2	RW	0x0	csa_on Descrambling Switch 1'b1: Descrambling function turned on; 1'b0: Descrambling function turned off;
1	R/WSC	0x0	clear PID Channel Clear Write 1 to clear PID channel. This bit will be set to 0 if the channel is clear.
0	R/WSC	0x0	en PID Channel Enable Write 1 to enable channel. Write 0 to this bit will not take any effect. This bit will be 0 when channel is cleared.

TSP_PTIx_PIDn_BASE
Address: Operational Base + offset (0x0400)
PTI Data Memory Buffer Base Address

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	address PTI Data Memory Buffer Base Address PTI Data Memory Buffer Base Address

TSP_PTIx_PIDn_TOP
Address: Operational Base + offset (0x0404)
PTI Data Memory Buffer Top Address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	address PTI Data Memory Buffer Top Address PTI Data Memory Buffer Top Address

TSP_PTIx_PIDn_WRITE
Address: Operational Base + offset (0x0408)
PTI Data Memory Buffer Hardware Writing Address

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	address PTI Data Memory Buffer Hardware Writing Address PTI Data Memory Buffer Hardware Writing Address

TSP_PTIx_PIDn_READ
Address: Operational Base + offset (0x040c)
PTI Data Memory Buffer Software Reading Address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	address PTI Data Memory Buffer Software Reading Address PTI Data Memory Buffer Software Reading Address

TSP_PTIx_LISTn_BASE
Address: Operational Base + offset (0x0800)
PTI List Memory Buffer Base Address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	address PTI Data Memory Buffer Software Reading Address PTI Data Memory Buffer Software Reading Address

TSP_PTIx_LISTn_TOP
Address: Operational Base + offset (0x0804)
PTI List Memory Buffer Top Address

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	address PTI List Memory Buffer Top Address PTI List Memory Buffer Top Address

TSP_PTIx_LISTn_WRITE
Address: Operational Base + offset (0x0808)
PTI List Memory Buffer Hardware Writing Address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	address PTI List Memory Buffer Hardware Writing Address PTI List Memory Buffer Hardware Writing Address

TSP_PTIx_LISTn_READ
Address: Operational Base + offset (0x080c)
PTI List Memory Buffer Software Reading Address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	address PTI List Memory Buffer Software Reading Address PTI List Memory Buffer Software Reading Address

TSP_PTIx_PIDn_CFG
Address: Operational Base + offset (0x0900)
PID Demux Configure Register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	filter_en Filter Byte Enable The proper position of filter byte Enable. For Section filter. the 1st,4th,5th,..18th byte of section header are used to be filtered; For PES filter, the 4th,7th,8th...21th byte of pes header are used to be filtered.
15:12	RO	0x0	reserved
11	RW	0x0	scd_en Start Code Detection Switch Start code detection 1: enabled; 0: disabled; This bit is only valid when n < 16.

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Bit	Attr	Reset Value	Description
10	RW	0x0	cni_on Current Next Indicator Abort when current_next_indicator == 1'b1, 1'b1: abort ; 1'b0: do nothing ;
9:8	RW	0x0	filt_mode Section Filter Mode Filter Mode when the filter mode is configured as section filter. 2'b00: stop per unit; 2'b01: full stop; 2'b10: recycle, update when version number change 2'b11: reserverd
7:6	RW	0x0	video_type Video filtering Type 2'b00: MPEG2 2'b01: H264 2'b10: VC-1 2'b11: Reserved
5:4	RW	0x0	filt_type Filter Type 2'b00: section filtering; 2'b01: pes filtering; 2'b10: es filtering; 2'b11: ts filtering; if n>=16, it is reserved as only section filtering, other values are invalid.
3	RW	0x1	cc_abort Continue Counter Error Abort when continuity counter error happens: 1: abort; 0: do nothing;
2	RW	0x0	tei_abort Ts_error_indicator Abort when ts_error_indicator == 1: 1'b1: abort ; 1'b0: do nothing;
1	RW	0x0	crc_abort CRC Error Abort This bit is valid only when crc_on == 1'b1. When crc error happens, 1'b1: abort ; 1'b0: do nothing.

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Bit	Attr	Reset Value	Description
0	RW	0x0	crc_on CRC Check 1'b1: CRC check function turned on 1'b0: CRC check function turned off

TSP_PTIx_PIDn_FILT_0
Address: Operational Base + offset (0x0904)
Fliter Word 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	filt_byte_3 Fliter Byte 2 This byte refers to 6th byte of section header or 9th byte of pes header
23:16	RW	0x00	filt_byte_2 Fliter Byte 2 This byte refers to 5th byte of section header or 8th byte of pes header
15:8	RW	0x00	filt_byte_1 Fliter Byte 1 This byte refers to 4th byte of section header or 7th byte of pes header
7:0	RW	0x00	filt_byte_0 Fliter Byte 0 This byte refers to 1st byte of section header or 4th byte of pes header

TSP_PTIx_PIDn_FILT_1
Address: Operational Base + offset (0x0908)
Fliter Word 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	filt_byte_3 Fliter Byte 2 This byte refers to 10th byte of section header or 13rd byte of pes header
23:16	RW	0x00	filt_byte_2 Fliter Byte 2 This byte refers to 9th byte of section header or 12nd byte of pes header
15:8	RW	0x00	filt_byte_1 Fliter Byte 1 This byte refers to 8th byte of section header or 11st byte of pes header

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Bit	Attr	Reset Value	Description
7:0	RW	0x00	filt_byte_0 Fliter Byte 0 This byte refers to 7th byte of section header or 10th byte of pes header

TSP_PTIx_PIDn_FILT_2
Address: Operational Base + offset (0x090c)
Fliter Word 2

Bit	Attr	Reset Value	Description
31:24	RW	0x00	filt_byte_3 Fliter Byte 2 This byte refers to 14th byte of section header or 17th byte of pes header
23:16	RW	0x00	filt_byte_2 Fliter Byte 2 This byte refers to 13rd byte of section header or 16th byte of pes header
15:8	RW	0x00	filt_byte_1 Fliter Byte 1 This byte refers to 12nd byte of section header or 15th byte of pes header
7:0	RW	0x00	filt_byte_0 Fliter Byte 0 This byte refers to 11st byte of section header or 14th byte of pes header

TSP_PTIx_PIDn_FILT_3
Address: Operational Base + offset (0x0910)
Fliter Word 3

Bit	Attr	Reset Value	Description
31:24	RW	0x00	filt_byte_3 Fliter Byte 2 This byte refers to 18th byte of section header or 21st byte of pes header
23:16	RW	0x00	filt_byte_2 Fliter Byte 2 This byte refers to 17th byte of section header or 20th byte of pes header
15:8	RW	0x00	filt_byte_1 Fliter Byte 1 This byte refers to 16th byte of section header or 19th byte of pes header

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Bit	Attr	Reset Value	Description
7:0	RW	0x00	filt_byte_0 Fliter Byte 0 This byte refers to 15th byte of section header or 18th byte of pes header

Notes: I=input, O=output, I/O=input/output, bidirectional

39.5 Interface Description

Table 39-1 IOMUX configuration for TSP

Module pin name	Dir	Pad name	IOMUX
ts_clk	I	IO_CIFclkin_TSclk_DVPgpio1b2	GPIO1B_IOMUX[5:4]=2'b10
ts_data0	I	IO_CIFdata2_TSdata0_DVPgpio1a0	GPIO1A_IOMUX[1:0]=2'b10
ts_data1	I	IO_CIFdata3_TSdata1_DVPgpio1a1	GPIO1A_IOMUX[3:2]=2'b10
ts_data2	I	IO_CIFdata4_TSdata2_DVPgpio1a2	GPIO1A_IOMUX[5:4]=2'b10
ts_data3	I	IO_CIFdata5_TSdata3_DVPgpio1a3	GPIO1A_IOMUX[7:6]=2'b10
ts_data4	I	IO_CIFdata6_TSdata4_DVPgpio1a4	GPIO1A_IOMUX[9:8]=2'b10
ts_data5	I	IO_CIFdata7_TSdata5_DVPgpio1a5	GPIO1A_IOMUX[11:10]=2'b10
ts_data6	I	IO_CIFdata8_TSdata6_DVPgpio1a6	GPIO1A_IOMUX[13:12]=2'b10
ts_data7	I	IO_CIFdata9_TSdata7_DVPgpio1a7	GPIO1A_IOMUX[15:14]=2'b10
ts_sync	I	IO_CIFvsync_TSsync_DVPgpio1b0	GPIO1B_IOMUX[1:0]=2'b10
ts_valid	I	IO_CIFhref_TSvalid_DVPgpio1b1	GPIO1B_IOMUX[3:2]=2'b10
ts_fail	O	IO_CIFclkout_TSfail_DVPgpio1b3	GPIO1B_IOMUX[7:6]=2'b10

39.6 Application Notes

39.6.1 Overall Operation Sequence

- Enable desired modules to work by writing correspond bit with '1' in TSP_GCFG. Note: it is important to do this step at first, otherwise writing the corresponding registers will not take effect.
- Set up TS configuration by writing corresponding registers.
- Wait for the interrupts to pick up the desired TS packets following the rules detailed in the following section.

39.6.2 TS Source

TS source can be chosen by writing the bit 9 of TSP_PTIx_CTRL(x=0,1), '1' for demodulator, '0' for local memory.

1.TS_IN Interface

Writing bit 10 of TSP_PTIx_CTRL to choose bit ordering, and writing bit [12:11] to choose input TS mode.

TS_IN interface supports 4 input TS stream mode: sync/valid serial mode, sync/valid parallel mode, sync/burst parallel mode, nosync/valid parallel mode.

2.Local Memory

PTI also can process the TS data read from local memory by using LLP DMA mode.

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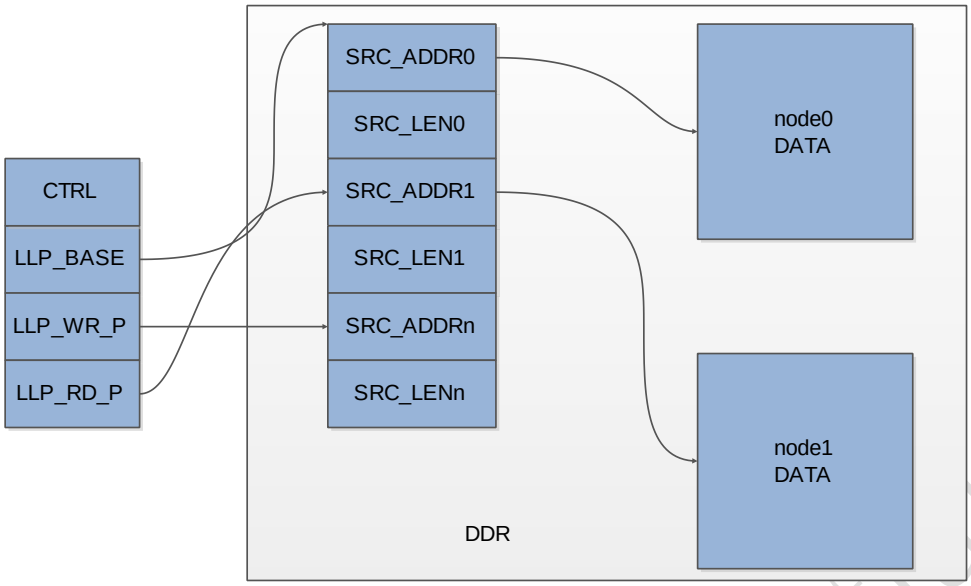

(1) Write PTIx_LLP_BASE with the list base address;
(2) Starting from the list base address, write the list nodes. One list node comprised of two words. The first word describes the TS data base address, the second one describes the length of TS data in unit of word.
(3) Write the PTIx_LLP_WRITE with the number of words that you have written in list memory. Note it is not the number of LLP nodes, so that the number you are writing should be an even one.
(4) Write PTIx_LLP_CFG with the configuration you want. Write the bit 0 with 1 to start LLP DMA. If all the list nodes are written, don't forget to write 1 to bit 3 to tell DMAC that the configuration is finished.

Note:

- *The MSB(bit7) of the 8-bit pointer in the PTIx_LLP_Write and PTIx_LLP_Read is used as the flag bit, and remaining 7 bits are used for addressing. Therefore the the pointer is referred to 7-bit space, not 8-bit space, and remember write the pointer with the correct flag bit. For example, if you have configured 63 LLP nodes and then you have to write the 64th LLP node starting from the list base address,*
- *PTIx_LLP_READ informs that how many words has been processed by LLP DMA. An interrupt may be generated when number of the processed words has reach to the threshold set in the PTIx_LLP_CFG.*
- *If you write the PTIx_LLP_Write several times in a complete DMA transaction, it is important to notice the flag bit of PTIx_LLP_Write, and never make the writing pointer catch up with the reading pointer.*

39.6.3 TS Synchronous Operation

Synchronous mode and Bypass mode can be switched by writing bit 15 of TSP_PTIx_CTRL. In the synchronous mode, 188/192/204 byte TS packets are supported and self-adjusted. Set up locked times in TSP_PTIx_CTRL to inform the successive times of TS packet header detection needs to lock the header of TS packets when in the unlocked mode, and set up unlocked times to informs the successive times of TS packet header error needs to re-lock header of TS packets in the locked mode. It is recommended to use 2-3 as the locked times to quickly and correctly locked the header, and 2-3 as unlocked times to avoid unnecessarily entering into unlocked searching mode.
In the bypass mode, the input TS data will not be re-synchronized and directly fed into the PTI channel.

39.6.4 Descrambling Operation

Descrambler can achieve PES or TS level descrambling which conforms to the CSA v2.0.
■ Enable the channel you want by writing 1 to bit 0 of TSP_PTIx_PIDn_CTRL (x=0~1, n= 0~64);

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- Set the desired PID number
- Turn on descrambling function by setting 1 to bit 2. If the corresponding CW is available or TS is required to be left undescrambled, CSA_ON bit is set to 0;
- Choose corresponding Control Word by setting bit[19:16], and 16 set Control Word are available to be chosen. Don't forget Control Word should be preprared before the descrambling function is enabled.

Note: If the enabled channel is needed to be disabled, write the CLEAR bit to disabled the channel rather than write '0' to EN bit.

39.6.5 Demux Operation

Refer to TSP_PTIx_PIDn_CFG for Demux operation. The software users should be familiar with the demux knowledge.

Users should create a separate memory buffer to receive the processed data for each desired PID channel, and write the base and top address information of the memory buffer into TSP_PTIx_PIDn_BASE and TSP_PTIx_PIDn respectively. Also initial writing address and reading address, normally the same as base address, are also needed to be written into TSP_PTIx_PIDn_WRITE and TSP_PTIx_PIDn_READ respectively. For ES/PES filter, another separate memory needs to be created to store list data, which is used to assist obtaining PES/ES data. List base address, top address, initial writing address and reading address are also needed to write into corresponding registers.

Note:

For channel whose PID channel number larger than 15, the channels can only be used section filter. For others, there is no such limit. They can be configured as section filter, pes filter, es filter or ts filter.

Data memory address boundary should be aligned with word-size, and list memory address boundary should be aligned with word size. If the memory buffer is not larger to store processed data so that writing address reaches the top address, TSP will return to the base address to write data. So fetch the data in time, don't make the writing address catches up with reading address. The list memory buffer has the same issue.

1.Demux data obtain

- **TS filter**

To obtain TS data and section data, when an desired PID done interrupt is generated, read TSP_PTIx_PIDn_READ firstly to know the address that last reading stops, and then read TSP_PTIx_PIDn_WRITE to know the address that hardware has reached. For ts data, start from the TSP_PTIx_PIDn_READ address to get the TS packet data, and stop at the address you want. However, the ending address should not catch up with writing address. It is recommended to obtain the TS data in the unit of TS packet which is 47-word size. At last, don't forget to write the ending address into TSP_PTIx_PIDn_READ to leave a hint where current reading stops.

B. Section filter

Section filter can run three mode to meet different needs: stop-per-unit; full stop; recycle , update when version number change. The PID done interrupt will be generated after each part of a complete section is processed in the first mode, and the PID done will be generated only after the whole section is completed in the last two modes. In the frist two mode, the PID channel will be disabled after the whole section is completed. In the recycle mode, the channel will remain active and start a new section processing when the version number changes. Section filter also supports 16-byte filtering function, which can assign 1st , 4th to 18th byte to be filtered.

The process to obtain section data is similar to the process for TS data. After a PID done interrupt done is generated, refer to the corresponding PID error status register to check if the section data is correct. Read the frist word of the section start address to know the total length of the section according to the format of section data.

Section Length = {First Word[11:8], First Word[23:16]};

Total Length = Section Length;

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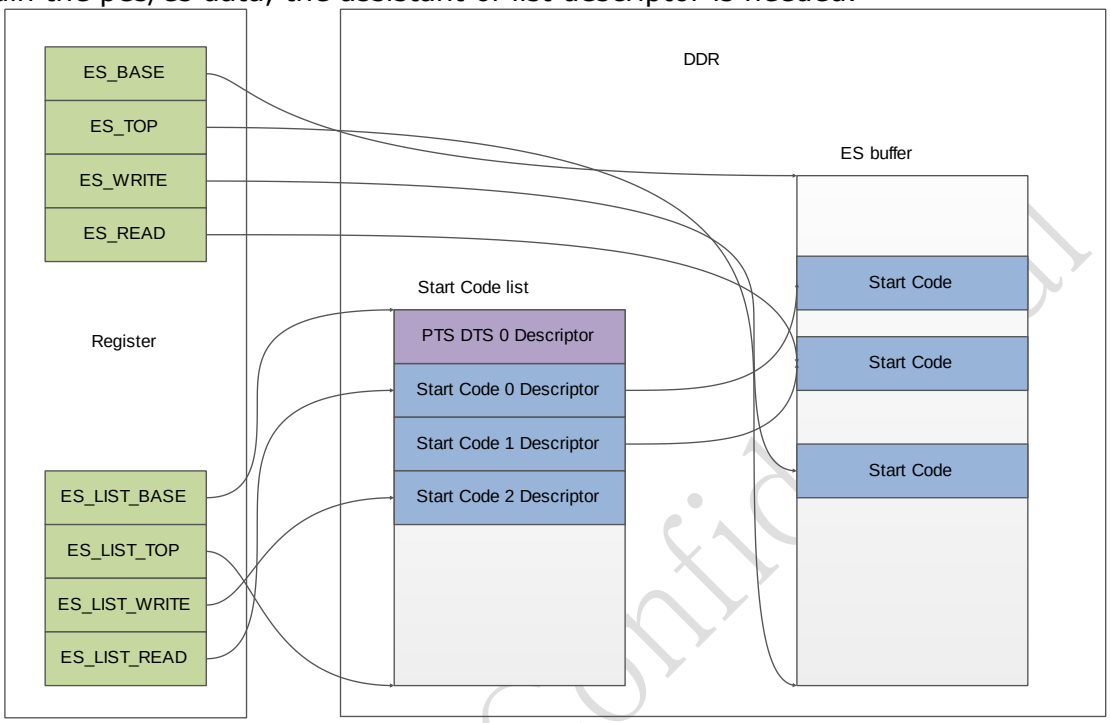
Then start to fetch section data according to the total length. Again don't forget to write the stopped address.

C. PES/ES filter

PES filter supports 16-byte filtering function, which can assign 4th, 7th to 21st byte to be filtered.

ES filter supports start code detection, including MPEG2 start code 0x000001b3, 0x00000100, VC-1 start code 0x0000010d, 0x000010f, H264 start code 0x00001.

To obtain the pes/es data, the assistant of list descriptor is needed.



List memory buffer contains descriptors which contains information to obtain es/pes data which are stored in data memory buffer.

The descriptor stored in list memory buffer can be separated into two groups: PTS_DTS Descriptor and Start Code Descriptor. The descriptor is composed by 4 word content, word_0, word_1, word_2 and word_3. The word_x (x means the sequence number in a descriptor, and they are stored in the memory in sequence order). The format of the 4 words are listed as follows:

(1) start code descriptor

Word_0:

Word_0[29:28] indicates the attributes of the bytes of the pointed word. 2'b00 means the whole word belongs to the new ES/PES packet; 2'b01 means that word[7:0] belongs to the previous packet, and the remaining bytes belong to the new packet; 2'b10 means means that word[15:0] belongs to the previous packet, and the remaining bytes belong to the new packet; 2'b11 means 'b10 means means that word[23:0] belongs to the previous packet, and the remaining bytes belong to the new packet. This pointed word is the word where start code starts, word_2 describes the location of start code.

Word_0[27:24] is equal to 0x0 in the start code descriptor. Users can used to tell two kinds of descriptor.

If the video type is H.264, word_0[23:8] means first_mb_in slice, and word_0 means nal_nuit_type.

Word_1:

the start code of stream.

Word_2:

DDR offset address in the DDR of the word where the start code is located.

Word_3:

0x0

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(2) PTS_DTS Descriptor

Word_0:

Word_0[29:28]: the same as start code descriptor

Word_0[27:24]: 0x1 in PTS_DTS descriptor.

Word_0[3] : PTS[32];

Word_0[2] : DTS[32];

Word_0[1:0] : pts_dts_flag;

Word_1:

DDR offset address of the word that valid data starts.

Word_2:

PTS[31:0]

Word_3

DTS[31:0]

To obtain PES data or ES data when start code detection is disabled, use PTS_DTS descriptor. To obtain ES data when start code detection is enabled, use start code descriptor.

When a PID done interrupt is generated, make sure there is no corresponding PID error generated. Read the TSP_PTIx_LISTn_READ to know the list reading address in the last time. Start from here, read the 4-word descriptor one by one to know the offset of the packets. Refer to the offset in the DDR where in the data memory buffer to obtain data. Finally write TSP_PTIx_LISTn_READ and TSP_PTIx_PIDn_READ with corresponding reading address.

39.6.6 TS Out Interface

All the configuration is done by writing TSP_TSOUT_CTRL. Before programming this register, make sure that you have enabled the TS OUT interface. If you want to disable TS out interface, write '0' to the START bit(bit 0) of TSP_TSOUT_CTRL, and then disable it in the TSP_GFCG.

Each PTI channel can provide TS out interface with PID-filtering TS Packets or non-PID-filtering TS packets, and therefore there are totally 4 sources can be chosen for TS out interface.

39.6.7 PVR

PVR module provide you with the function to record the programs you want. The 4 sources can be assigned with PVR, and they are the same as TS out interface.

Assign the PVR length and PVR address, and then configure TSP_PVR_CTRL to start PVR module. If you want to stop PVR function during recording, write '1' to STOP bit (bit 0) to to TSP_PVR_CTRL to stop it. Remember to take care of the status of PVR_ON bit of TSP_GFCG when programming the PVR-related registers.

39.6.8 PCR extraction

PCR extraction can be enabled by configure PTIx_PCRn_CTRL. Then if the PID-matched TS data contain PCR field, the 33-bit PCR_base field will be written corresponding PTIx_PCRn_H and PTIx_PCRn_L registers. An interrupt will be asserted if PCR interrupt is enabled.

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Chapter 40 High-Speed ADC Interface (HS-ADC)

40.1 Overview

HS-ADC Interface Unit is an interface unit for connecting the GPS ADC interface to AMBA AHB bus. It fetches the bus data received by the GPS ADC interface and stores them to internal asynchronous FIFO after the ADC clock is active. The HS-ADC Interface Unit generates the DMA request signal when data length of the asynchronous FIFO over the almost full level or almost empty level.

HS-ADC supports the following features:

- Support GPS interface with 2bits
- Support combined interrupt output, source including: full interrupt, empty interrupt
- Support DMA transfer mode through generating DMA request from the event of almost full or almost empty, etc.
- Support two channel mode: single channel and dual channel
- Support the most significant bit negation or not
- Support sign bit extension
- Support two storage mode: input data are stored to high 8bit or 10bit and stored to low 8bit or 10bit of 16bit when pushed into FIFO.
- Support an asynchronous build-in FIFO with 128x32 size

40.2 Block Diagram

The HS-ADC diagram is as follows.

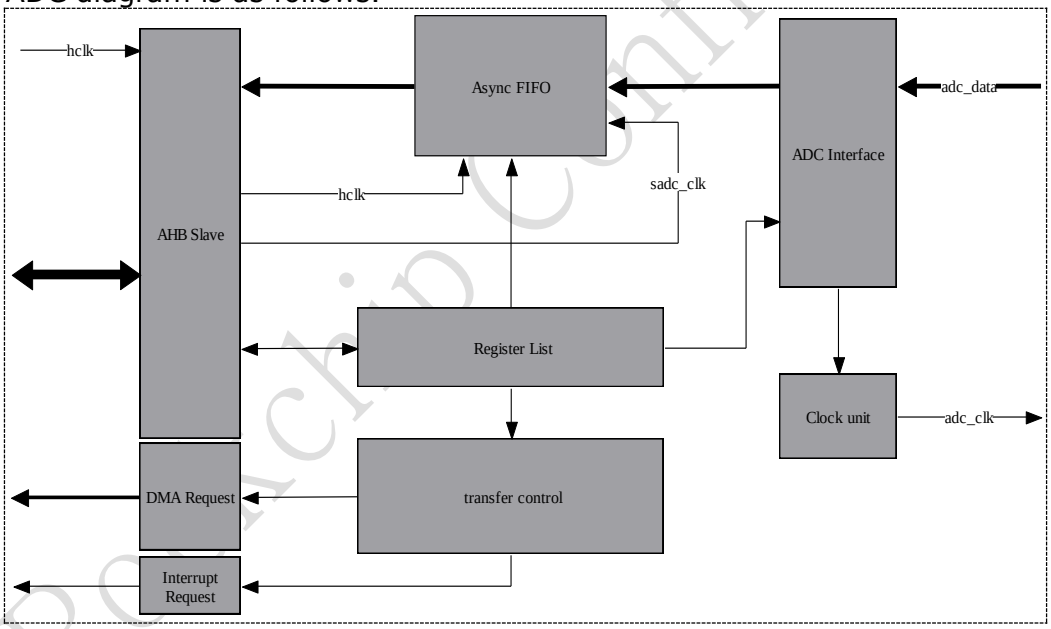

Fig. 40-1 HS-ADC Architecture

40.3 Function Description

This module is designed for GPS interface only.

40.3.1 GPS interface

When this module is used as GPS interface, user should configure GRF register to select 2bits or 4bitsGPS data input and gps_clk as GPS clock input from pad.

Also, user should configure CRU_CLKSEL22_CON[5:4] to select gps_clk as HS-ADC controller working clock source.

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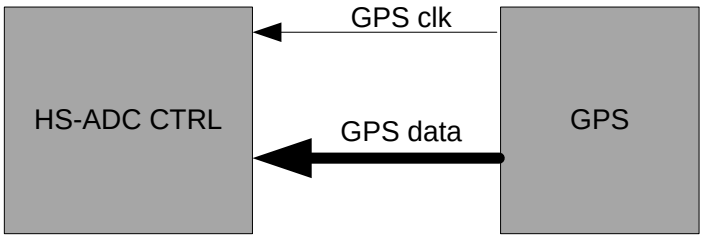

Fig. 40-2 GPS Application Diagram

40.4 Register Description

40.4.1 Register Summary

Name	Offset	Size	Reset Value	Description
HSADC_CTRL	0x00000	W	0x00000000	Control register
HSADC_IER	0x00004	W	0x00000000	Interrupt control register
HSADC_ISR	0x00008	W	0x00000000	Interrupt status register
HSADC_CGCTL	0x00010	W	0x00000000	HSADC Clock Gating control
HSADC_DATA	0x00020	W	0x00000000	Data register

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

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40.4.2 Detail Register Description

HSADC_CTRL
Address: Operational Base + offset (0x00000)
Control register

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:24	RW	0x0	almost_full_level Define almost full trigger level 0x0~"0xf" - configure valid range (Notes: 1 level indicate 4 entries data in the async FIFO. and this configure range mapping to 64 - 124 entries data in the async FIFO.)
23:20	RO	0x0	reserved
19:16	RW	0x0	almost_empty_level Define almost empty trigger level 0x0~"0xf" - configure valid range (Notes: 1 level indicate 4 entries data in the async FIFO. and this configure range mapping to 0 - 60 entries data in the async FIFO.)
15:13	RO	0x0	reserved
12	RW	0x0	sbex Sign Bit extent select Sign bit extent select when store width wider than interface width 0: fill 0 1: fill sign extend bit
11	RW	0x0	gpsw GPS interface data width select 0:2bit data mode 1:4bit data mode
10:8	RO	0x0	reserved
7	RW	0x0	chan data input channel select 0 : single channel, ADC input from adc_data_l 1 : double channel data input from adc_data_l/adc_data_h
6	RW	0x0	stmd Store mode select 0:8bit store mode 1:16bit store mode

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Bit	Attr	Reset Value	Description
5	RW	0x0	dma_req_mode DMA request mode select 1 - almost full generate DMA request signal (Notes: this mode generate DMA request signal from almost full condition and cancel DMA request signal from almost empty condition. so you need configure two level by almost full level and almost empty level) 0 - almost empty generate DMA request signal (Notes: this mode generate DMA request signal from almost empty condition and that only once DMA request.)
4	RW	0x0	smsb MSB negation select 1:negation 0:not negation
3	RW	0x0	sctl store mode select fetch the bus data by AD converter and that store to high 8-bit/10-bit or low 8-bit/10-bit at a haft word width before push to Async FIFO: "1" - store to high 8-bit/10-bit "0" - store to low 8-bit/10-bit (Notes: have sign extend if that configure of store to low 8-bit/10-bit)
2	RW	0x0	bwth The data bus width of AD converter "1" - 10-bit "0" - 8-bit
1	RW	0x0	gps_sel GPS input select 0 : GPS is Not selected 1 : GPS is selected *When gps_sel & ts_sel both equal to 0,then ADC IF is selected
0	RW	0x0	adc_en HS-ADC Interface Unit Enable Bit "1" - enable (Notes: will return 1 when the hardware started transfer) "0" - disable (Notes: other bit can be modify only the hardware return 0)

HSADC_IER
Address: Operational Base + offset (0x00004)
Interrupt control register

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Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	int_empty_en Interrupt en/disable bit for the empty interrupt flag of async FIFO "1" - enable "0" - disable
0	RW	0x0	int_full_en Interrupt en/disable bit for the full interrupt flag of async FIFO "1" - enable "0" - disable

HSADC_ISR
Address: Operational Base + offset (0x00008)
Interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	int_empty_stat_ind Async FIFO empty interrupt flag "1(R)" - This bit will be set to "1" when Async FIFO empty status and that only to read operation. "0(W)" - Write "0" to bit for clear the interrupt flag and that only to wrtie operation.
0	RW	0x0	int_full_stat_ind Async FIFO full interrupt flag . "1(R)" - This bit will be set to "1" when Async FIFO full status and that only to read operation. "0(W)" - Write "0" to bit for clear the interrupt flag and that only to wrtie operation.

HSADC_CGCTL
Address: Operational Base + offset (0x00010)
HSADC Clock Gating control

Bit	Attr	Reset Value	Description
31:1	RW	0x00000000	cycle_cfg clock gated cycles configuration when configure cg_enable to 1 and cycle_cfg to non-zero value,HSADC clock will be gated for cycle_cfg cycles ,then clock recover.

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Bit	Attr	Reset Value	Description
0	RW	0x0	cg_enable clock gating enable control 0:clock gating disable 1:clock gating enable

HSADC_DATA
Address: Operational Base + offset (0x00020)
Data register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	DATA DATA DATA

40.5 Interface Description

40.5.1 GPS mode

Table 40-1 IOMUX configuration in GPS mode

Module pin name	Direction	Pad name	IOMUX
gps_rfclk	I	IO_MACrxd1_UART3GPSctsn_GPSrfclk_GPIO30gpio3c0	GPIO3C_IOMUX[1:0]= 2'b11
gps_mag	I	IO_MACtxd3_GPSmag_GPIO30gpio3b6	GPIO3B_IOMUX[13:12]= 2'b10
gps_sig	I	IO_MACrxd0_GPSsig_GPIO30gpio3b7	GPIO3B_IOMUX[15:14]= 2'b10

40.6 Application Notes

The following sections will describe the operation of DMA requests and DMA transfers.

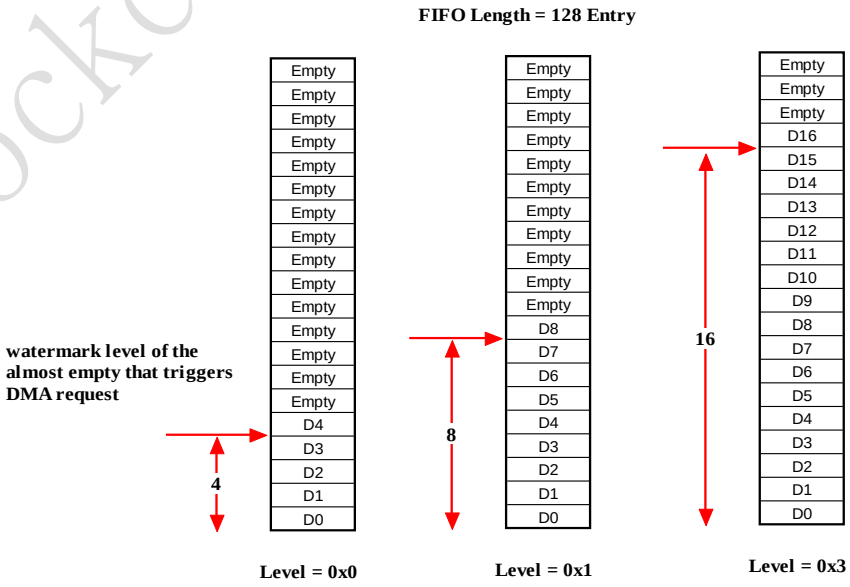

Fig. 40-3 Almost empty triggers a DMA request by DMA request mode

The DMA request signal will be generated from a watermark level trigger when data stored to FIFO over the watermark level of almost empty, where the watermark level can be configured through HSADC_CTRL[19:16] by software. This DMA request mode doesn't care the

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watermark level of almost full. The sample for watermark level configuration is shown in figure above.

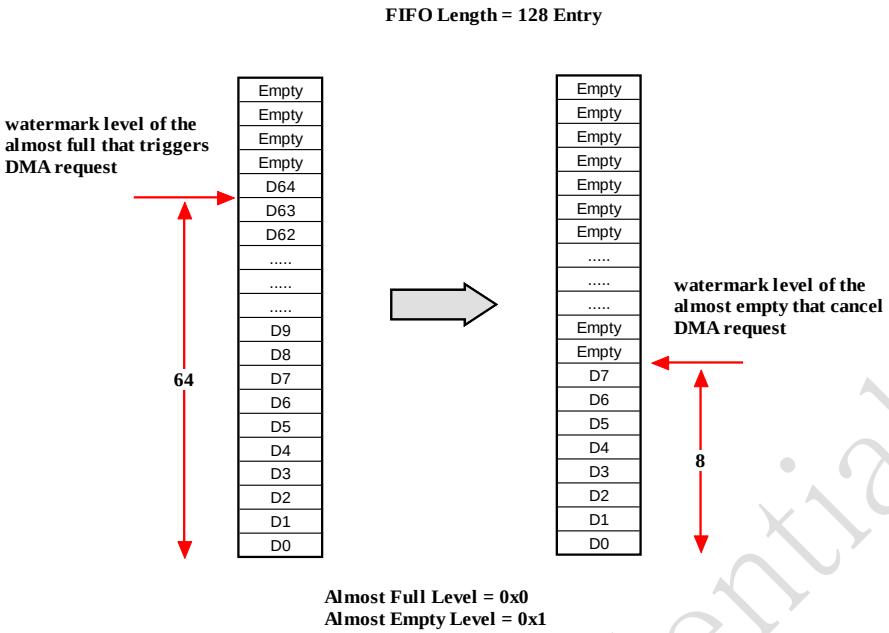

Fig. 40-4 Almost full triggers a DMA request by DMA request mode

The DMA request signal will be generated from a watermark level trigger when data stored to FIFO over the watermark level of almost full. It continues to generate request signal when the number of data in FIFO greater than watermark level of almost empty. This DMA request mode needs configure two watermark levels: watermark level of almost empty at the HSADC_CTRL[19:16] and watermark level of almost full at the HSADC_CTRL[27:24]. The sample for watermark level configuration is shown in figure above. When controller works in TS mode, the interface signal ts_sync should always be used.

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Chapter 41 GMAC Ethernet Interface

41.1 Overview

The GMAC Ethernet Controller provides a complete Ethernet interface from processor to a Reduced Media Independent Interface(RMII) and Reduced Gigabit Media Independent Interface (RGMII)compliant Ethernet PHY.
The GMAC includes a DMA controller. The DMA controller efficiently moves packet data from microprocessor's RAM, formats the data for an IEEE 802.3-2002 compliant packet and transmits the data to an Ethernet Physical Interface(PHY).It also efficiently moves packet data from RXFIFO to microprocessor's RAM.

41.1.1 Features

- Supports 10/100/1000-Mbps data transfer rates with the RGMII interfaces
- Supports 10/100-Mbps data transfer rates with the RMII interfaces
- Supports both full-duplex and half-duplex operation
  - Supports CSMA/CD Protocol for half-duplex operation
  - Supports packet bursting and frame extension in 1000 Mbps half-duplex operation
  - Supports IEEE 802.3x flow control for full-duplex operation
  - Optional forwarding of received pause control frames to the user application in full-duplex operation
  - Back-pressure support for half-duplex operation
  - Automatic transmission of zero-quanta pause frame on deassertion of flow control input in full-duplex operation
- Preamble and start-of-frame data (SFD) insertion in Transmit, and deletion in Receive paths
- Automatic CRC and pad generation controllable on a per-frame basis
- Options for Automatic Pad/CRC Stripping on receive frames
- Programmable frame length to support Standard Ethernet frames
- Programmable InterFrameGap (40-96 bit times in steps of 8)
- Supports a variety of flexible address filtering modes:
  - 64-bit Hash filter (optional) for multicast and uni-cast (DA) addresses
  - Option to pass all multicast addressed frames
  - Promiscuous mode support to pass all frames without any filtering for network monitoring
  - Passes all incoming packets (as per filter) with a status report
- Separate 32-bit status returned for transmission and reception packets
- Supports IEEE 802.1Q VLAN tag detection for reception frames
- MDIO Master interfacefor PHY device configuration and management
- Support detection of LAN wake-up frames and AMD Magic Packet frames
- Support checksum off-load for received IPv4 and TCP packets encapsulated by the Ethernet frame
- Support checking IPv4 header checksum and TCP, UDP, or ICMP checksum encapsulated in IPv4 or IPv6 datagrams
- Comprehensive status reporting for normal operation and transfers with errors
- Support per-frame Transmit/Receive complete interrupt control
- Supports 4-KB receive FIFO depths on reception.
- Supports 2-KB FIFO depth on transmission
- Automatic generation of PAUSE frame control or backpressure signal to the GMAC core based on Receive FIFO-fill (threshold configurable) level
- Handles automatic retransmission of Collision frames for transmission
- Discards frames on late collision, excessive collisions, excessive deferral and underrun conditions
- AXI interface to any CPU or memory
- Software can select the type of AXI burst (fixed and variable length burst) in the AXI Master interface
- Supports internal loopback on the RGMII/RMII for debugging

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- Debug status register that gives status of FSMs in Transmit and Receive data-paths and FIFO fill-levels.

41.2 Block Diagram

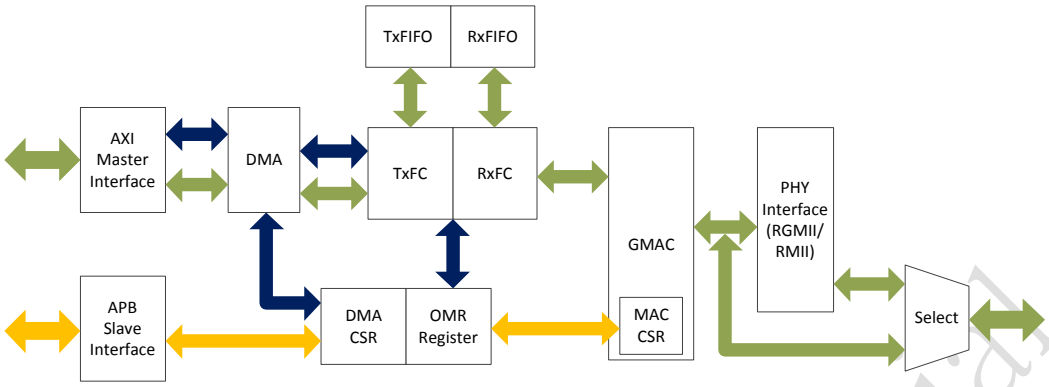

Fig. 41-1 GMAC architecture

The GMAC is broken up into multiple separate functional units. These blocks are interconnected in the MAC module. The block diagram shows the general flow of data and control signals between these blocks.
The GMAC transfers data to system memory through the AXI master interface. The host CPU uses the APB Slave interface to access the GMAC subsystem's control and status registers (CSRs).
The GMAC supports the PHY interfaces of reduced GMII (RGMII) and reduced MII (RMII).
The Transmit FIFO (Tx FIFO) buffers data read from system memory by the DMA before transmission by the GMAC Core. Similarly, the Receive FIFO (Rx FIFO) stores the Ethernet frames received from the line until they are transferred to system memory by the DMA. These are asynchronous FIFOs, as they also transfer the data between the application clock and the GMAC line clocks.

41.3 Function Description

41.3.1 Frame Structure

Data frames transmitted shall have the frame format shown in Fig 32-2.

<inter-frame><preamble><sfd><data><efd>

Fig. 41-2 MAC Frame structure

The preamble <preamble> begins a frame transmission. The bit value of the preamble field consists of 7 octets with the following bit values:
10101010 10101010 10101010 10101010 10101010 10101010 10101010
The SFD (start frame delimiter) <sfd> indicates the start of a frame and follows the preamble.The bit value is 10101011.
The data in a well formed frame shall consist of N octets data.

41.3.2 RMII Interface timing diagram

The Reduced Media Independent Interface (RMII) specification reduces the pin count between Ethernet PHYs and Switch ASICs (only in 10/100 mode). According to the IEEE 802.3u standard, an MII contains 16 pins for data and control. In devices incorporating multiple MAC or PHY interfaces (such as switches), the number of pins adds significant cost with increase in port count. The RMII specification addresses this problem by reducing the pin count to 7 for each port - a 62.5% decrease in pin count.
The RMII module is instantiated between the GMAC and the PHY. This helps translation of the MAC's MII into the RMII. The RMII block has the following characteristics:

- Supports 10-Mbps and 100-Mbps operating rates. It does not support 1000-Mbps operation.
- Two clock references are sourced externally or CRU, providing independent, 2-bit wide

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transmit and receive paths.

Transmit Bit Ordering

Each nibble from the MII must be transmitted on the RMII a di-bit at a time with the order of di-bit transmission shown in Fig.1-3. The lower order bits (D1 and D0) are transmitted first followed by higher order bits (D2 and D3).

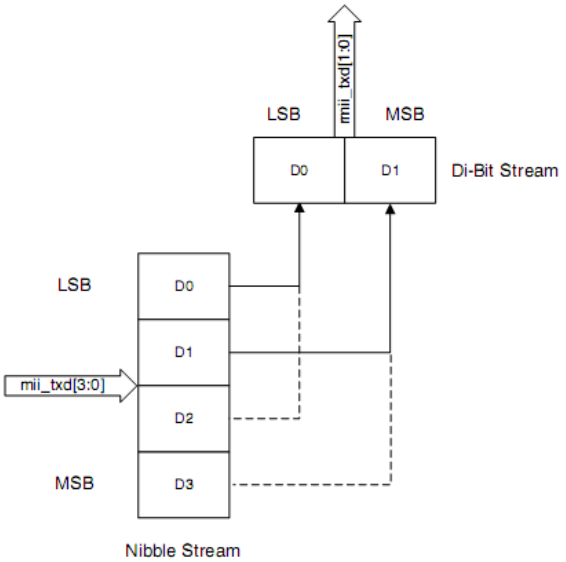

Fig. 41-3 RMII transmission bit ordering

RMII Transmit Timing Diagrams

Fig.1-4 through 1-7show MII-to-RMII transaction timing. The clk_rmii_i(REF_CLK) frequency is 50MHz in RMII interface. In 10Mb/s mode, as the REF_CLK frequency is 10 times as the data rate, the value on rmii_txd_o[1:0] (TXD[1:0]) shall be valid such that TXD[1:0] may be sampled every 10th cycle, regard-less of the starting cycle within the gRup and yield the correct frame data.

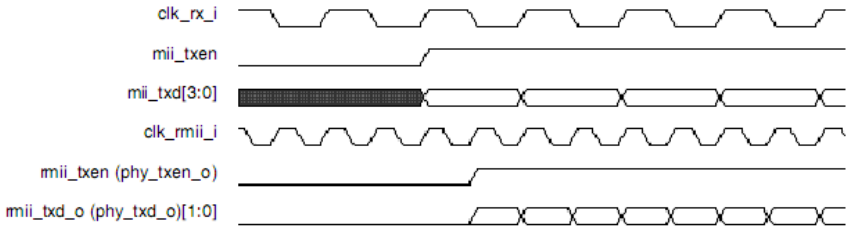

Fig. 41-4 Start of MII and RMII transmission in 100-Mbps mode

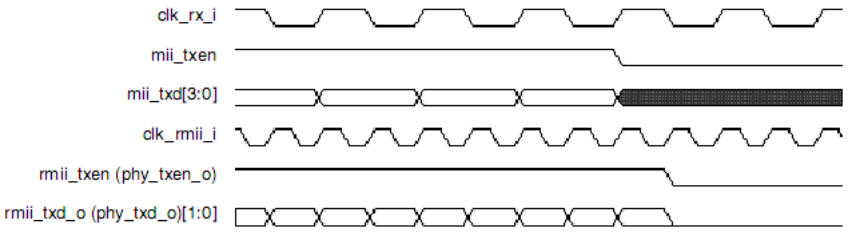

Fig. 41-5 End of MII and RMII Transmission in 100-Mbps Mode

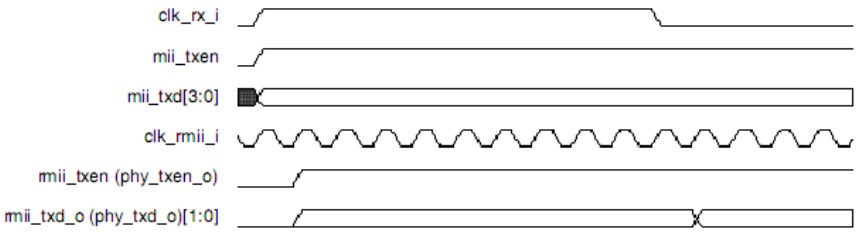

Fig. 41-6 Start of MII and RMII Transmission in 10-Mbps Mode

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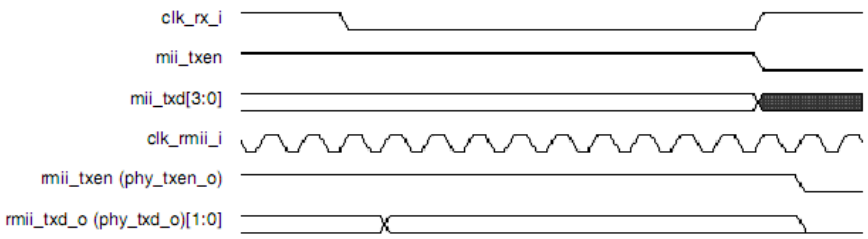

Fig. 41-7 End of MII and RMII Transmission in 10-Mbps Mode

Receive Bit Ordering

Each nibble is transmitted to the MII from the di-bit received from the RMII in the nibble transmission order shown in Fig.1-8. The lower order bits (D0 and D1) are received first, followed by the higher order bits (D2 and D3).

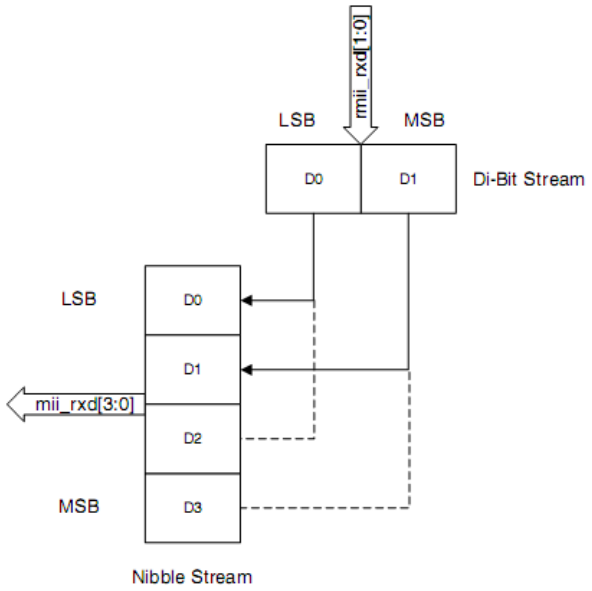

Fig. 41-8 RMII receive bit ordering

41.3.3 RGMII interface

The Reduced Gigabit Media Independent Interface (RGMII) specification reduces the pin count of the interconnection between the GMAC 10/100/1000 controller and the PHY for GMII and MII interfaces. To achieve this, the data path and control signals are reduced and multiplexed together with both the edges of the transmit and receive clocks. For gigabit operation the clocks operate at 125 MHz; for 10/100 operation, the clock rates are 2.5 MHz/25 MHz.
In the GMAC 10/100/1000 controller, the RGMII module is instantiated between the GMAC core's GMII and the PHY to translate the control and data signals between the GMII and RGMII protocols.
The RGMII block has the following characteristics:

- Supports 10-Mbps, 100-Mbps, and 1000-Mbps operation rates.
- For the RGMII block, no extra clock is required because both the edges of the incoming clocks are used.
- The RGMII block extracts the in-band (link speed, duplex mode and link status) status signals from the PHY and provides them to the GMAC core logic for link detection.

41.3.4 Management Interface

The MAC management interface provides a simple, two-wire, serial interface to connect the GMAC and a managed PHY, for the purposes of controlling the PHY and gathering status from the PHY. The management interface consists of a pair of signals that transport the management information across the MII bus: MDIO and MDC.
The GMAC initiates the management write/read operation. The clock gmii_mdc_o(MDC) is a divided clock from the application clock pclk_gmac. The divide factor depends on the clock range setting in the GMII address register. Clock range is set as follows:

Selection pclk_gmacMDC Clock

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0000 60-100 MHz pclk_gmac/42
0001 100-150 MHz pclk_gmac/62
0010 20-35 MHz pclk_gmac/16
0011 35-60 MHz pclk_gmac/26
0100 150-250 MHz pclk_gmac/102
0101 250-300 MHz pclk_gmac/124
0110, 0111 Reserved

The MDC is the derivative of the application clock pclk_gmac. The management operation is performed through the gmii_mdi_i, gmii_mdo_o and gmii_mdo_o_e signals. A three-state buffer is implemented in the PAD.

The frame structure on the MDIO line is shown below.

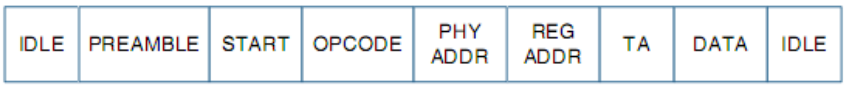

Fig. 41-9 MDIO frame structure

IDLE: The mdio line is three-state; there is no clock on gmii_mdc_o
PREAMBLE: 32 continuous bits of value 1
START: Start-of-frame is 2í01
OPCODE: 2'b10 for read and 2'b01 for write
PHY ADDR: 5-bit address select for one of 32 PHYs
REG ADDR: Register address in the selected PHY
TA: Turnaround is 2'bZ0 for read and 2'b10 for Write
DATA: Any 16-bit value. In a write operation, the GMAC drives mdio; in a read operation, PHY drives it.

41.3.5 Power Management Block

Power management(PMT) supports the reception of network (remote) wake-up frames and Magic Packet frames. PMT does not perform the clock gate function, but generates interrupts for wake-up frames and Magic Packets received by the GMAC. The PMT block sits on the receiver path of the GMAC and is enabled with remote wake-up frame enable and Magic Packet enable. These enables are in the PMT control and status register and are programmed by the application.

When the power down mode is enabled in the PMT, then all received frames are dropped by the core and they are not forwarded to the application. The core comes out of the power down mode only when either a Magic Packet or a Remote Wake-up frame is received and the corresponding detection is enabled.

Remote Wake-Up Frame Detection

When the GMAC is in sleep mode and the remote wake-up bit is enabled in registerGMAC_PMT_CTRL_STA (0x002C), normal operation is resumed after receiving a remote wake-up frame. The application writes all eight wake-up filter registers, by performing a sequential write to address (0028). The application enables remote wake-up by writing a 1 to bit 2 of the registerGMAC_PMT_CTRL_STA.

PMT supports four programmable filters that allow support of different receive frame patterns. If the incoming frame passes the address filtering of Filter Command, and if Filter CRC-16 matches the incoming examined pattern, then the wake-up frame is received.

Filter_offset (minimum value 12, which refers to the 13th byte of the frame) determines the offset from which the frame is to be examined. Filter Byte Mask determines which bytes of the frame must be examined. The thirty-first bit of Byte Mask must be set to zero.

The remote wake-up CRC block determines the CRC value that is compared with Filter CRC-16. The wake-up frame is checked only for length error, FCS error, dribble bit error, GMII error, collision, and to ensure that it is not a runt frame. Even if the wake-up frame is more than 512 bytes long, if the frame has a valid CRC value, it is considered valid. Wake-up frame detection is updated in the registerGMAC_PMT_CTRL_STA for every remote Wake-up frame received. A PMT interrupt to the application triggers a read to theGMAC_PMT_CTRL_STAregister to determine reception of a wake-up frame.

Magic Packet Detection

The Magic Packet frame is based on a method that uses Advanced Micro Device's Magic Packet technology to power up the sleeping device on the network. The GMAC receives a specific

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packet of information, called a Magic Packet, addressed to the node on the network.
Only Magic Packets that are addressed to the device or a broadcast address will be checked to determine whether they meet the wake-up requirements. Magic Packets that pass the address filtering (unicast or broadcast) will be checked to determine whether they meet the remote Wake-on-LAN data format of 6 bytes of all ones followed by a GMAC Address appearing 16 times.
The application enables Magic Packet wake-up by writing a 1 to Bit 1 of the register GMAC_PMT_CTRL_STA. The PMT block constantly monitors each frame addressed to the node for a specific Magic Packet pattern. Each frame received is checked for a 48'hFF_FF_FF_FF_FF_FF pattern following the destination and source address field. The PMT block then checks the frame for 16 repetitions of the GMAC address without any breaks or interruptions. In case of a break in the 16 repetitions of the address, the 48'hFF_FF_FF_FF_FF_FF pattern is scanned for again in the incoming frame. The 16 repetitions can be anywhere in the frame, but must be preceded by the synchronization stream (48'hFF_FF_FF_FF_FF_FF). The device will also accept a multicast frame, as long as the 16 duplications of the GMAC address are detected.
If the MAC address of a node is 48'h00_11_22_33_44_55, then the GMAC scans for the data sequence:
Destination Address Source Address ……. FF FF FF FF FF FF
00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55
00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55
00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55
00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55
…CRC
Magic Packet detection is updated in the PMT Control and Status register for Magic Packet received. A PMT interrupt to the Application triggers a read to the PMT CSR to determine whether a Magic Packet frame has been received.

41.3.6 MAC Management Counters

The counters in the MAC Management Counters (MMC) module can be viewed as an extension of the register address space of the CSR module. The MMC module maintains a set of registers for gathering statistics on the received and transmitted frames. These include a control register for controlling the behavior of the registers, two 32-bit registers containing interrupts generated (receive and transmit), and two 32-bit registers containing masks for the Interrupt register (receive and transmit). These registers are accessible from the Application through the MAC Control Interface (MCI). Non-32-bit accesses are allowed as long as the address is word-aligned.
The organization of these registers is shown in Register Description. The MMCs are accessed using transactions, in the same way the CSR address space is accessed. The Register Description in this chapter describe the various counters and list the address for each of the statistics counters. This address will be used for Read/Write accesses to the desired transmit/receive counter.
The MMC module gathers statistics on encapsulated IPv4, IPv6, TCP, UDP, or ICMP payloads in received Ethernet frames.

41.4 Register description

41.4.1 Register Summary

Name	Offset	Size	Reset Value	Description
GMAC_MAC_CONF	0x0000	W	0x00000000	MAC Configuration Register
GMAC_MAC_FRM_FILT	0x0004	W	0x00000000	MAC Frame Filter
GMAC_HASH_TAB_HI	0x0008	W	0x00000000	Hash Table High Register
GMAC_HASH_TAB_LO	0x000c	W	0x00000000	Hash Table Low Register
GMAC_GMII_ADDR	0x0010	W	0x00000000	GMII Address Register
GMAC_GMII_DATA	0x0014	W	0x00000000	GMII Data Register

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Name	Offset	Size	Reset Value	Description
GMAC_FLOW_CTRL	0x0018	W	0x00000000	Flow Control Register
GMAC_VLAN_TAG	0x001c	W	0x00000000	VLAN Tag Register
GMAC_DEBUG	0x0024	W	0x00000000	Debug register
GMAC_PMT_CTRL_STA	0x002c	W	0x00000000	PMT Control and Status Register
GMAC_INT_STATUS	0x0038	W	0x00000000	Interrupt Status Register
GMAC_INT_MASK	0x003c	W	0x00000000	Interrupt Mask Register
GMAC_MAC_ADDR0_HI	0x0040	W	0x0000ffff	MAC Address0 High Register
GMAC_MAC_ADDR0_LO	0x0044	W	0xffffffff	MAC Address0 Low Register
GMAC_AN_CTRL	0x00c0	W	0x00000000	AN Control Register
GMAC_AN_STATUS	0x00c4	W	0x00000008	AN Status Register
GMAC_AN_ADV	0x00c8	W	0x000001e0	Auto_Negotiation Advertisement Register
GMAC_AN_LINK_PART_AB	0x00cc	W	0x00000000	Auto_Negotiation Link Partner Ability Register
GMAC_AN_EXP	0x00d0	W	0x00000000	Auto_Negotiation Expansion Register
GMAC_INTF_MODE_STA	0x00d8	W	0x00000000	RGMII Status Register
GMAC_MMC_CTRL	0x0100	W	0x00000000	MMC Control Register
GMAC_MMC_RX_INTR	0x0104	W	0x00000000	MMC Receive Interrupt Register
GMAC_MMC_TX_INTR	0x0108	W	0x00000000	MMC Transmit Interrupt Register
GMAC_MMC_RX_INT_MSK	0x010c	W	0x00000000	MMC Receive Interrupt Mask Register
GMAC_MMC_TX_INT_MSK	0x0110	W	0x00000000	MMC Transmit Interrupt Mask Register
GMAC_MMC_TXOCTETCNT_GB	0x0114	W	0x00000000	MMC TX OCTET Good and Bad Counter
GMAC_MMC_TXFRMCNT_GB	0x0118	W	0x00000000	MMC TX Frame Good and Bad Counter
GMAC_MMC_TXUNDFLWERR	0x0148	W	0x00000000	MMC TX Underflow Error
GMAC_MMC_TXCARERR	0x0160	W	0x00000000	MMC TX Carrier Error
GMAC_MMC_TXOCTETCNT_G	0x0164	W	0x00000000	MMC TX OCTET Good Counter
GMAC_MMC_TXFRMCNT_G	0x0168	W	0x00000000	MMC TX Frame Good Counter
GMAC_MMC_RXFRMCNT_GB	0x0180	W	0x00000000	MMC RX Frame Good and Bad Counter
GMAC_MMC_RXOCTETCNT_GB	0x0184	W	0x00000000	MMC RX OCTET Good and Bad Counter
GMAC_MMC_RXOCTETCNT_G	0x0188	W	0x00000000	MMC RX OCTET Good Counter
GMAC_MMC_RXMCFRMCNT_G	0x0190	W	0x00000000	MMC RX Mulitcast Frame Good Counter
GMAC_MMC_RXCRCERR	0x0194	W	0x00000000	MMC RX Carrier
GMAC_MMC_RXLENERR	0x01c8	W	0x00000000	MMC RX Length Error

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Name	Offset	Size	Reset Value	Description
GMAC_MMC_RXFIFOOVRFLW	0x01d4	W	0x00000000	MMC RX FIFO Overflow
GMAC_MMC_IPC_INT_MSK	0x0200	W	0x00000000	MMC Receive Checksum Offload Interrupt Mask Register
GMAC_MMC_IPC_INTR	0x0208	W	0x00000000	MMC Receive Checksum Offload Interrupt Register
GMAC_MMC_RXIPV4GFRM	0x0210	W	0x00000000	MMC RX IPV4 Good Frame
GMAC_MMC_RXIPV4HDERRFRM	0x0214	W	0x00000000	MMC RX IPV4 Head Error Frame
GMAC_MMC_RXIPV6GFRM	0x0224	W	0x00000000	MMC RX IPV6 Good Frame
GMAC_MMC_RXIPV6HDERRFRM	0x0228	W	0x00000000	MMC RX IPV6 Head Error Frame
GMAC_MMC_RXUDPERRFRM	0x0234	W	0x00000000	MMC RX UDP Error Frame
GMAC_MMC_RXTCPERRFRM	0x023c	W	0x00000000	MMC RX TCP Error Frame
GMAC_MMC_RXICMPERRFRM	0x0244	W	0x00000000	MMC RX ICMP Error Frame
GMAC_MMC_RXIPV4HDERROCT	0x0254	W	0x00000000	MMC RX OCTET IPV4 Head Error
GMAC_MMC_RXIPV6HDERROCT	0x0268	W	0x00000000	MMC RX OCTET IPV6 Head Error
GMAC_MMC_RXUDPERROCT	0x0274	W	0x00000000	MMC RX OCTET UDP Error
GMAC_MMC_RXTCPERROCT	0x027c	W	0x00000000	MMC RX OCTET TCP Error
GMAC_MMC_RXICMPERROCT	0x0284	W	0x00000000	MMC RX OCTET ICMP Error
GMAC_BUS_MODE	0x1000	W	0x00020101	Bus Mode Register
GMAC_TX_POLL_DEMAND	0x1004	W	0x00000000	Transmit Poll Demand Register
GMAC_RX_POLL_DEMAND	0x1008	W	0x00000000	Receive Poll Demand Register
GMAC_RX_DESC_LIST_ADDR	0x100c	W	0x00000000	Receive Descriptor List Address Register
GMAC_TX_DESC_LIST_ADDR	0x1010	W	0x00000000	Transmit Descriptor List Address Register
GMAC_STATUS	0x1014	W	0x00000000	Status Register
GMAC_OP_MODE	0x1018	W	0x00000000	Operation Mode Register
GMAC_INT_ENA	0x101c	W	0x00000000	Interrupt Enable Register
GMAC_OVERFLOW_CNT	0x1020	W	0x00000000	Missed Frame and Buffer Overflow Counter Register
GMAC_REC_INT_WDT_TIMER	0x1024	W	0x00000000	Receive Interrupt Watchdog Timer Register
GMAC_AXI_BUS_MODE	0x1028	W	0x00110001	AXI Bus Mode Register
GMAC_AXI_STATUS	0x102c	W	0x00000000	AXI Status Register

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Name	Offset	Size	Reset Value	Description
GMAC_CUR_HOST_TX_DESC	0x1048	W	0x00000000	Current Host Transmit Descriptor Register
GMAC_CUR_HOST_RX_DESC	0x104c	W	0x00000000	Current Host Receive Descriptor Register
GMAC_CUR_HOST_TX_Buf_ADDR	0x1050	W	0x00000000	Current Host Transmit Buffer Address Register
GMAC_CUR_HOST_RX_BUF_ADDR	0x1054	W	0x00000000	Current Host Receive Buffer Adderss Register
GMAC_HW_FEA_REG	0x1058	W	0x000d0f17	The presence of the optional features/functions of the core

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

41.4.2 Detail Register Description

GMAC_MAC_CONF

Address: Operational Base + offset (0x0000)

MAC Configuration Register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	TC Transmit Configuration in RGMII/SGMII/SMII When set, this bit enables the transmission of duplex mode, link speed, and link up/down information to the PHY in the RGMII ports. When this bit is reset, no such information is driven to the PHY.
23	RW	0x0	WD Watchdog Disable When this bit is set, the GMAC disables the watchdog timer on the receiver, and can receive frames of up to 16,384 bytes. When this bit is reset, the GMAC allows no more than 2,048 bytes (10,240 if JE is set high) of the frame being received and cuts off any bytes received after that.
22	RW	0x0	JD Jabber Disable When this bit is set, the GMAC disables the jabber timer on the transmitter, and can transfer frames of up to 16,384 bytes. When this bit is reset, the GMAC cuts off the transmitter if the application sends out more than 2,048 bytes of data (10,240 if JE is set high) during transmission.

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Bit	Attr	Reset Value	Description
21	RW	0x0	BE Frame Burst Enable When this bit is set, the GMAC allows frame bursting during transmission in GMII Half-Duplex mode.
20	RO	0x0	reserved
19:17	RW	0x0	IFG Inter-Frame Gap These bits control the minimum IFG between frames during transmission. 3'b000: 96 bit times 3'b001: 88 bit times 3'b010: 80 bit times ... 3'b111: 40 bit times
16	RW	0x0	DCRS Disable Carrier Sense During Transmission When set high, this bit makes the MAC transmitter ignore the (G)MII CRS signal during frame transmission in Half-Duplex mode. This request results in no errors generated due to Loss of Carrier or No Carrier during such transmission. When this bit is low, the MAC transmitter generates such errors due to Carrier Sense and will even abort the transmissions.
15	RW	0x0	PS Port Select Selects between GMII and MII: 1'b0: GMII (1000 Mbps) 1'b1: MII (10/100 Mbps)
14	RW	0x0	FES Speed Indicates the speed in Fast Ethernet (MII) mode: 1'b0: 10 Mbps 1'b1: 100 Mbps
13	RW	0x0	DO Disable Receive Own When this bit is set, the GMAC disables the reception of frames when the gmii_txen_o is asserted in Half-Duplex mode. When this bit is reset, the GMAC receives all packets that are given by the PHY while transmitting.

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Bit	Attr	Reset Value	Description
12	RW	0x0	LM Loopback Mode When this bit is set, the GMAC operates in loopback mode at GMII/MII. The (G)MII Receive clock input (clk_rx_i) is required for the loopback to work properly, as the Transmit clock is not looped-back internally.
11	RW	0x0	DM Duplex Mode When this bit is set, the GMAC operates in a Full-Duplex mode where it can transmit and receive simultaneously. This bit is RO with default value of 1'b1 in Full-Duplex-only configuration.
10	RW	0x0	IPC Checksum Offload When this bit is set, the GMAC calculates the 16-bit one's complement of the one's complement sum of all received Ethernet frame payloads. It also checks whether the IPv4 Header checksum (assumed to be bytes 25-26 or 29-30 (VLAN-tagged) of the received Ethernet frame) is correct for the received frame and gives the status in the receive status word. The GMAC core also appends the 16-bit checksum calculated for the IP header datagram payload (bytes after the IPv4 header) and appends it to the Ethernet frame transferred to the application (when Type 2 COE is deselected). When this bit is reset, this function is disabled. When Type 2 COE is selected, this bit, when set, enables IPv4 checksum checking for received frame payloads TCP/UDP/ICMP headers. When this bit is reset, the COE function in the receiver is disabled and the corresponding PCE and IP HCE status bits are always cleared.

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Bit	Attr	Reset Value	Description
9	RW	0x0	DR Disable Retry When this bit is set, the GMAC will attempt only 1 transmission. When a collision occurs on the GMII/MII, the GMAC will ignore the current frame transmission and report a Frame Abort with excessive collision error in the transmit frame status. When this bit is reset, the GMAC will attempt retries based on the settings of BL.
8	RW	0x0	LUD Link Up/Down Indicates whether the link is up or down during the transmission of configuration in RGMII interface: 1'b0: Link Down 1'b1: Link Up
7	RW	0x0	ACS Automatic Pad/CRC Stripping When this bit is set, the GMAC strips the Pad/FCS field on incoming frames only if the length's field value is less than or equal to 1,500 bytes. All received frames with length field greater than or equal to 1,501 bytes are passed to the application without stripping the Pad/FCS field. When this bit is reset, the GMAC will pass all incoming frames to the Host unmodified.
6:5	RW	0x0	BL Back-Off Limit The Back-Off limit determines the random integer number (r) of slot time delays (4,096 bit times for 1000 Mbps and 512 bit times for 10/100 Mbps) the GMAC waits before rescheduling a transmission attempt during retries after a collision. This bit is applicable only to Half-Duplex mode and is reserved (RO) in Full-Duplex-only configuration. 2'b00: k = min (n, 10) 2'b01: k = min (n, 8) 2'b10: k = min (n, 4) 2'b11: k = min (n, 1), where n = retransmission attempt. The random integer r takes the value in the range 0 = r < 2^k

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Bit	Attr	Reset Value	Description
4	RW	0x0	DC Deferral Check When this bit is set, the deferral check function is enabled in the GMAC. The GMAC will issue a Frame Abort status, along with the excessive deferral error bit set in the transmit frame status when the transmit state machine is deferred for more than 24,288 bit times in 10/100-Mbps mode. If the Core is configured for 1000 Mbps operation, the threshold for deferral is 155,680 bits times. Deferral begins when the transmitter is ready to transmit, but is prevented because of an active CRS (carrier sense) signal on the GMII/MII. Defer time is not cumulative. If the transmitter defers for 10,000 bit times, then transmits, collides, backs off, and then has to defer again after completion of back-off, the deferral timer resets to 0 and restarts. When this bit is reset, the deferral check function is disabled and the GMAC defers until the CRS signal goes inactive.
3	RW	0x0	TE Transmitter Enable When this bit is set, the transmit state machine of the GMAC is enabled for transmission on the GMII/MII. When this bit is reset, the GMAC transmit state machine is disabled after the completion of the transmission of the current frame, and will not transmit any further frames.
2	RW	0x0	RE Receiver Enable When this bit is set, the receiver state machine of the GMAC is enabled for receiving frames from the GMII/MII. When this bit is reset, the GMAC receive state machine is disabled after the completion of the reception of the current frame, and will not receive any further frames from the GMII/MII.
1:0	RO	0x0	reserved

GMAC_MAC_FRM_FILT
Address: Operational Base + offset (0x0004)
MAC Frame Filter

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Bit	Attr	Reset Value	Description
31	RW	0x0	RA Receive All When this bit is set, the GMAC Receiver module passes to the Application all frames received irrespective of whether they pass the address filter. The result of the SA/DA filtering is updated (pass or fail) in the corresponding bits in the Receive Status Word. When this bit is reset, the Receiver module passes to the Application only those frames that pass the SA/DA address filter.
30:11	RO	0x0	reserved
10	RW	0x0	HPF Hash or Perfect Filter When set, this bit configures the address filter to pass a frame if it matches either the perfect filtering or the hash filtering as set by HMC or HUC bits. When low and if the HUC/HMC bit is set, the frame is passed only if it matches the Hash filter.
9	RW	0x0	SAF Source Address Filter Enable The GMAC core compares the SA field of the received frames with the values programmed in the enabled SA registers. If the comparison matches, then the SAMatch bit of RxStatus Word is set high. When this bit is set high and the SA filter fails, the GMAC drops the frame. When this bit is reset, then the GMAC Core forwards the received frame to the application and with the updated SA Match bit of the RxStatus depending on the SA address comparison.
8	RW	0x0	SAIF SA Inverse Filtering When this bit is set, the Address Check block operates in inverse filtering mode for the SA address comparison. The frames whose SA matches the SA registers will be marked as failing the SA Address filter. When this bit is reset, frames whose SA does not match the SA registers will be marked as failing the SA Address filter.

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Bit	Attr	Reset Value	Description
7:6	RW	0x0	PCF Pass Control Frames These bits control the forwarding of all control frames (including unicast and multicast PAUSE frames). Note that the processing of PAUSE control frames depends only on RFE of Register GMAC_FLOW_CTRL[2]. 2'b00: GMAC filters all control frames from reaching the application. 2'b01: GMAC forwards all control frames except PAUSE control frames to application even if they fail the Address filter. 2'b10: GMAC forwards all control frames to application even if they fail the Address Filter. 2'b11: GMAC forwards control frames that pass the Address Filter.
5	RW	0x0	DBF Disable Broadcast Frames When this bit is set, the AFM module filters all incoming broadcast frames. When this bit is reset, the AFM module passes all received broadcast frames.
4	RW	0x0	PM Pass All Multicast When set, this bit indicates that all received frames with a multicast destination address (first bit in the destination address field is '1') are passed. When reset, filtering of multicast frame depends on HMC bit.
3	RW	0x0	DAIF DA Inverse Filtering When this bit is set, the Address Check block operates in inverse filtering mode for the DA address comparison for both unicast and multicast frames. When reset, normal filtering of frames is performed.

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Bit	Attr	Reset Value	Description
2	RW	0x0	HMC Hash Multicast When set, MAC performs destination address filtering of received multicast frames according to the hash table. When reset, the MAC performs a perfect destination address filtering for multicast frames, that is, it compares the DA field with the values programmed in DA registers.
1	RW	0x0	HUC Hash Unicast When set, MAC performs destination address filtering of unicast frames according to the hash table. When reset, the MAC performs a perfect destination address filtering for unicast frames, that is, it compares the DA field with the values programmed in DA registers.
0	RW	0x0	PR Promiscuous Mode When this bit is set, the Address Filter module passes all incoming frames regardless of its destination or source address. The SA/DA Filter Fails status bits of the Receive Status Word will always be cleared when PR is set.

GMAC_HASH_TAB_HI
Address: Operational Base + offset (0x0008)
Hash Table High Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	HTH Hash Table High This field contains the upper 32 bits of Hash table

GMAC_HASH_TAB_LO
Address: Operational Base + offset (0x000c)
Hash Table Low Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	HTL Hash Table Low This field contains the lower 32 bits of Hash table

GMAC_GMII_ADDR

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Address: Operational Base + offset (0x0010)
GMII Address Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:11	RW	0x00	PA Physical Layer Address This field tells which of the 32 possible PHY devices are being accessed
10:6	RW	0x00	GR GMII Register These bits select the desired GMII register in the selected PHY device

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Bit	Attr	Reset Value	Description
5:2	RW	0x0	CR APB Clock Range The APB Clock Range selection determines the frequency of the MDC clock as per the pclk_gmac frequency used in your design. The suggested range of pclk_gmac frequency applicable for each value below (when Bit[5] = 0) ensures that the MDC clock is approximately between the frequency range 1.0 MHz - 2.5 MHz. Selection pclk_gmac MDC Clock 0000 60-100 MHz pclk_gmac/42 0001 100-150 MHz pclk_gmac/62 0010 20-35 MHz pclk_gmac/16 0011 35-60 MHz pclk_gmac/26 0100 150-250 MHz pclk_gmac/102 0101 250-300 MHz pclk_gmac/124 0110, 0111 Reserved When bit 5 is set, you can achieve MDC clock of frequency higher than the IEEE 802.3 specified frequency limit of 2.5 MHz and program a clock divider of lower value. For example, when pclk_gmac is of frequency 100 Mhz and you program these bits as "1010", then the resultant MDC clock will be of 12.5 Mhz which is outside the limit of IEEE 802.3 specified range. Please program the values given below only if the interfacing chips supports faster MDC clocks. Selection MDC Clock 1000 pclk_gmac/4 1001 pclk_gmac/6 1010 pclk_gmac/8 1011 pclk_gmac/10 1100 pclk_gmac/12 1101 pclk_gmac/14 1110 pclk_gmac/16 1111 pclk_gmac/18

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Bit	Attr	Reset Value	Description
1	RW	0x0	GW GMII Write When set, this bit tells the PHY that this will be a Write operation using register GMAC_GMII_DATA. If this bit is not set, this will be a Read operation, placing the data in register GMAC_GMII_DATA.
0	W1C	0x0	GB GMII Busy This bit should read a logic 0 before writing to Register GMII_ADDR and Register GMII_DATA. This bit must also be set to 0 during a Write to Register GMII_ADDR. During a PHY register access, this bit will be set to 1'b1 by the Application to indicate that a Read or Write access is in progress. Register GMII_DATA (GMII Data) should be kept valid until this bit is cleared by the GMAC during a PHY Write operation. The Register GMII_DATA is invalid until this bit is cleared by the GMAC during a PHY Read operation. The Register GMII_ADDR (GMII Address) should not be written to until this bit is cleared.

GMAC_GMII_DATA
Address: Operational Base + offset (0x0014)
GMII Data Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	GD GMII Data This contains the 16-bit data value read from the PHY after a Management Read operation or the 16-bit data value to be written to the PHY before a Management Write operation.

GMAC_FLOW_CTRL
Address: Operational Base + offset (0x0018)
Flow Control Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	PT Pause Time This field holds the value to be used in the Pause Time field in the transmit control frame. If the Pause Time bits is configured to be double-synchronized to the (G)MII clock domain, then consecutive writes to this register should be performed only after at least 4 clock cycles in the destination clock domain.
15:8	RO	0x0	reserved
7	RW	0x0	DZPQ Disable Zero-Quanta Pause When set, this bit disables the automatic generation of Zero-Quanta Pause Control frames on the deassertion of the flow-control signal from the FIFO layer (MTL or external sideband flow control signal sbd_flowctrl_i/mti_flowctrl_i). When this bit is reset, normal operation with automatic Zero-Quanta Pause Control frame generation is enabled.
6	RO	0x0	reserved
5:4	RW	0x0	PLT Pause Low Threshold This field configures the threshold of the PAUSE timer at which the input flow control signal mti_flowctrl_i (or sbd_flowctrl_i) is checked for automatic retransmission of PAUSE Frame. The threshold values should be always less than the Pause Time configured in Bits[31:16]. For example, if PT = 100H (256 slot-times), and PLT = 01, then a second PAUSE frame is automatically transmitted if the mti_flowctrl_i signal is asserted at 228 (256-28) slot-times after the first PAUSE frame is transmitted. Selection Threshold 00 Pause time minus 4 slot times 01 Pause time minus 28 slot times 10 Pause time minus 144 slot times 11 Pause time minus 256 slot times Slot time is defined as time taken to transmit 512 bits (64 bytes) on the GMII/MII interface.

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Bit	Attr	Reset Value	Description
3	RW	0x0	UP Unicast Pause Frame Detect When this bit is set, the GMAC will detect the Pause frames with the station's unicast address specified in MAC Address0 High Register and MAC Address0 Low Register, in addition to the detecting Pause frames with the unique multicast address. When this bit is reset, the GMAC will detect only a Pause frame with the unique multicast address specified in the 802.3x standard.
2	RW	0x0	RFE Receive Flow Control Enable When this bit is set, the GMAC will decode the received Pause frame and disable its transmitter for a specified (Pause Time) time. When this bit is reset, the decode function of the Pause frame is disabled.
1	RW	0x0	TFE Transmit Flow Control Enable In Full-Duplex mode, when this bit is set, the GMAC enables the flow control operation to transmit Pause frames. When this bit is reset, the flow control operation in the GMAC is disabled, and the GMAC will not transmit any Pause frames. In Half-Duplex mode, when this bit is set, the GMAC enables the back-pressure operation. When this bit is reset, the backpressure feature is disabled.

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Bit	Attr	Reset Value	Description
0	RW	0x0	FCB_BPA Flow Control Busy/Backpressure Activate This bit initiates a Pause Control frame in Full-Duplex mode and activates the backpressure function in Half-Duplex mode if TFE bit is set. In Full-Duplex mode, this bit should be read as 1'b0 before writing to the register GMAC_FLOW_CTRL. To initiate a pause control frame, the application must set this bit to 1'b1. During a transfer of the control frame, this bit will continue to be set to signify that a frame transmission is in progress. After the completion of Pause control frame transmission, the GMAC will reset this bit to 1'b0. The register GMAC_FLOW_CTRL should not be written to until this bit is cleared. In Half-Duplex mode, when this bit is set (and TFE is set), then backpressure is asserted by the GMAC Core. During backpressure, when the GMAC receives a new frame, the transmitter starts sending a JAM pattern resulting in a collision. This control register bit is logically OR'ed with the mti_flowctrl_i input signal for the backpressure function.

GMAC_VLAN_TAG
Address: Operational Base + offset (0x001c)
VLAN Tag Register

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RW	0x0	ETV Enable 12-Bit VLAN Tag Comparison When this bit is set, a 12-bit VLAN identifier, rather than the complete 16-bit VLAN tag, is used for comparison and filtering. Bits[11:0] of the VLAN tag are compared with the corresponding field in the received VLAN-tagged frame. When this bit is reset, all 16 bits of the received VLAN frame's fifteenth and sixteenth bytes are used for comparison.

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	VL VLAN Tag Identifier for Receive Frames This contains the 802.1Q VLAN tag to identify VLAN frames, and is compared to the fifteenth and sixteenth bytes of the frames being received for VLAN frames. Bits[15:13] are the User Priority, Bit[12] is the Canonical Format Indicator (CFI) and bits[11:0] are the VLAN tag's VLAN Identifier (VID) field. When the ETV bit is set, only the VID (Bits[11:0]) is used for comparison. If VL (VL[11:0] if ETV is set) is all zeros, the GMAC does not check the fifteenth and sixteenth bytes for VLAN tag comparison, and declares all frames with a Type field value of 0x8100 to be VLAN frames.

GMAC_DEBUG
Address: Operational Base + offset (0x0024)
Debug register

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25	RW	0x0	TFIFO3 When high, it indicates that the MTL TxStatus FIFO is full and hence the MTL will not be accepting any more frames for transmission.
24	RW	0x0	TFIFO2 When high, it indicates that the MTL TxFIFO is not empty and has some data left for transmission.
23	RO	0x0	reserved
22	RW	0x0	TFIFO1 When high, it indicates that the MTL TxFIFO Write Controller is active and transferring data to the TxFIFO.
21:20	RW	0x0	TFIFOSTA This indicates the state of the TxFIFO read Controller: 2'b00: IDLE state 2'b01: READ state (transferring data to MAC transmitter) 2'b10: Waiting for TxStatus from MAC transmitter 2'b11: Writing the received TxStatus or flushing the TxFIFO

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Bit	Attr	Reset Value	Description
19	RW	0x0	PAUSE When high, it indicates that the MAC transmitter is in PAUSE condition (in full-duplex only) and hence will not schedule any frame for transmission
18:17	RW	0x0	TSAT This indicates the state of the MAC Transmit Frame Controller module: 2'b00: IDLE 2'b01: Waiting for Status of previous frame or IFG/backoff period to be over 2'b10: Generating and transmitting a PAUSE control frame (in full duplex mode) 2'b11: Transferring input frame for transmission
16	RW	0x0	TACT When high, it indicates that the MAC GMII/MII transmit protocol engine is actively transmitting data and not in IDLE state.
15:10	RO	0x0	reserved
9:8	RW	0x0	RFIFO This gives the status of the RxFIFO Fill-level: 2'b00: RxFIFO Empty 2'b01: RxFIFO fill-level below flow-control de-activate threshold 2'b10: RxFIFO fill-level above flow-control activate threshold 2'b11: RxFIFO Full
7	RO	0x0	reserved
6:5	RW	0x0	RFIFORD It gives the state of the RxFIFO read Controller: 2'b00: IDLE state 2'b01: Reading frame data 2'b10: Reading frame status (or time-stamp) 2'b11: Flushing the frame data and Status
4	RW	0x0	RFIFOWR When high, it indicates that the MTL RxFIFO Write Controller is active and transferring a received frame to the FIFO.
3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
2:1	RW	0x0	ACT When high, it indicates the active state of the small FIFO Read and Write controllers respectively of the MAC receive Frame Controller module
0	RW	0x0	RDB When high, it indicates that the MAC GMII/MII receive protocol engine is actively receiving data and not in IDLE state.

GMAC_PMT_CTRL_STA
Address: Operational Base + offset (0x002c)
PMT Control and Status Register

Bit	Attr	Reset Value	Description
31	W1C	0x0	WFFRPR Wake-Up Frame Filter Register Pointer Reset When set, resets the Remote Wake-up Frame Filter register pointer to 3'b000. It is automatically cleared after 1 clock cycle.
30:10	RO	0x0	reserved
9	RW	0x0	GU Global Unicast When set, enables any unicast packet filtered by the GMAC (DAF) address recognition to be a wake-up frame.
8:7	RO	0x0	reserved
6	RC	0x0	WFR Wake-Up Frame Received When set, this bit indicates the power management event was generated due to reception of a wake-up frame. This bit is cleared by a read into this register.
5	RC	0x0	MPR Magic Packet Received When set, this bit indicates the power management event was generated by the reception of a Magic Packet. This bit is cleared by a read into this register.
4:3	RO	0x0	reserved
2	RW	0x0	WFE Wake-Up Frame Enable When set, enables generation of a power management event due to wake-up frame reception.

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Bit	Attr	Reset Value	Description
1	RW	0x0	MPE Magic Packet Enable When set, enables generation of a power management event due to Magic Packet reception.
0	R/WSC	0x0	PD Power Down When set, all received frames will be dropped. This bit is cleared automatically when a magic packet or Wake-Up frame is received, and Power-Down mode is disabled. Frames received after this bit is cleared are forwarded to the application.This bit must only be set when either the Magic Packet Enable or Wake-Up Frame Enable bit is set high.

GMAC_INT_STATUS
Address: Operational Base + offset (0x0038)
Interrupt Status Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RO	0x0	MRCOIS MMC Receive Checksum Offload Interrupt Status This bit is set high whenever an interrupt is generated in the MMC Receive Checksum Offload Interrupt Register. This bit is cleared when all the bits in this interrupt register are cleared.
6	RO	0x0	MTIS MMC Transmit Interrupt Status This bit is set high whenever an interrupt is generated in the MMC Transmit Interrupt Register. This bit is cleared when all the bits in this interrupt register are cleared.This bit is only valid when the optional MMC module is selected during configuration.
5	RO	0x0	MRIS MMC Receive Interrupt Status This bit is set high whenever an interrupt is generated in the MMC Receive Interrupt Register. This bit is cleared when all the bits in this interrupt register are cleared.This bit is only valid when the optional MMC module is selected during configuration.

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Bit	Attr	Reset Value	Description
4	RO	0x0	MIS MMC Interrupt Status This bit is set high whenever any of bits 7:5 is set high and cleared only when all of these bits are low. This bit is valid only when the optional MMC module is selected during configuration.
3	RO	0x0	PIS PMT Interrupt Status This bit is set whenever a Magic packet or Wake-on-LAN frame is received in Power-Down mode). This bit is cleared when both bits[6:5] are cleared due to a read operation to the register GMAC_PMT_CTRL_STA.
2:1	RO	0x0	reserved
0	RO	0x0	RIS RGMII Interrupt Status This bit is set due to any change in value of the Link Status of RGMIIinterface. This bit is cleared when the user makes a read operation the RGMII Status register.

GMAC_INT_MASK
Address: Operational Base + offset (0x003c)
Interrupt Mask Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	PIM PMT Interrupt Mask This bit when set, will disable the assertion of the interrupt signal due to the setting of PMT Interrupt Status bit in Register GMAC_INT_STATUS.
2:1	RO	0x0	reserved
0	RW	0x0	RIM RGMII Interrupt Mask This bit when set, will disable the assertion of the interrupt signal due to the setting of RGMII Interrupt Status bit in Register GMAC_INT_STATUS.

GMAC_MAC_ADDR0_HI
Address: Operational Base + offset (0x0040)
MAC Address0 High Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0xffff	A47_A32 MAC Address0 [47:32] This field contains the upper 16 bits (47:32) of the 6-byte first MAC address. This is used by the MAC for filtering for received frames and for inserting the MAC address in the Transmit Flow Control (PAUSE) Frames.

GMAC_MAC_ADDR0_LO
Address: Operational Base + offset (0x0044)
MAC Address0 Low Register

Bit	Attr	Reset Value	Description
31:0	RW	0xffffffff	A31_A0 MAC Address0 [31:0] This field contains the lower 32 bits of the 6-byte first MAC address. This is used by the MAC for filtering for received frames and for inserting the MAC address in the Transmit Flow Control (PAUSE) Frames.

GMAC_AN_CTRL
Address: Operational Base + offset (0x00c0)
AN Control Register

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12	RW	0x0	ANE Auto-Negotiation Enable When set, will enable the GMAC to perform auto-negotiation with the link partner. Clearing this bit will disable auto-negotiation.
11:10	RO	0x0	reserved
9	R/WSC	0x0	RAN Restart Auto-Negotiation When set, will cause auto-negotiation to restart if the ANE is set. This bit is self-clearing after auto-negotiation starts. This bit should be cleared for normal operation.
8:0	RO	0x0	reserved

GMAC_AN_STATUS
Address: Operational Base + offset (0x00c4)
AN Status Register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5	RO	0x0	ANC Auto-Negotiation Complete When set, this bit indicates that the auto-negotiation process is completed. This bit is cleared when auto-negotiation is reinitiated.
4	RO	0x0	reserved
3	RO	0x1	ANA Auto-Negotiation Ability This bit is always high, because the GMAC supports auto-negotiation.
2	R/WSC	0x0	LS Link Status When set, this bit indicates that the link is up. When cleared, this bit indicates that the link is down.
1:0	RO	0x0	reserved

GMAC_AN_ADV
Address: Operational Base + offset (0x00c8)
Auto_Negotiation Advertisement Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RO	0x0	NP Next Page Support This bit is tied to low, because the GMAC does not support the next page.
14	RO	0x0	reserved
13:12	RW	0x0	RFE Remote Fault Encoding These 2 bits provide a remote fault encoding, indicating to a link partner that a fault or error condition has occurred.
11:9	RO	0x0	reserved
8:7	RW	0x3	PSE Pause Encoding These 2 bits provide an encoding for the PAUSE bits, indicating that the GMAC is capable of configuring the PAUSE function as defined in IEEE 802.3x.

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Bit	Attr	Reset Value	Description
6	RW	0x1	HD Half-Duplex This bit, when set high, indicates that the GMAC supports Half-Duplex. This bit is tied to low (and RO) when the GMAC is configured for Full-Duplex-only operation.
5	RW	0x1	FD Full-Duplex This bit, when set high, indicates that the GMAC supports Full-Duplex.
4:0	RO	0x0	reserved

GMAC_AN_LINK_PART_AB
Address: Operational Base + offset (0x00cc)
Auto_Negotiation Link Partner Ability Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RO	0x0	NP Next Page Support When set, this bit indicates that more next page information is available. When cleared, this bit indicates that next page exchange is not desired.
14	RO	0x0	ACK Acknowledge When set, this bit is used by the auto-negotiation function to indicate that the link partner has successfully received the GMAC's base page. When cleared, it indicates that a successful receipt of the base page has not been achieved.
13:12	RO	0x0	RFE Remote Fault Encoding These 2 bits provide a remote fault encoding, indicating a fault or error condition of the link partner.
11:9	RO	0x0	reserved
8:7	RO	0x0	PSE Pause Encoding These 2 bits provide an encoding for the PAUSE bits, indicating that the link partner's capability of configuring the PAUSE function as defined in IEEE 802.3x.

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Bit	Attr	Reset Value	Description
6	RO	0x0	HD Half-Duplex When set, this bit indicates that the link partner has the ability to operate in Half-Duplex mode. When cleared, the link partner does not have the ability to operate in Half-Duplex mode.
5	RO	0x0	FD Full-Duplex When set, this bit indicates that the link partner has the ability to operate in Full-Duplex mode. When cleared, the link partner does not have the ability to operate in Full-Duplex mode.
4:0	RO	0x0	reserved

GMAC_AN_EXP
Address: Operational Base + offset (0x00d0)
Auto_Negotiation Expansion Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RO	0x0	NPA Next Page Ability This bit is tied to low, because the GMAC does not support next page function.
1	RO	0x0	NPR New Page Received When set, this bit indicates that a new page has been received by the GMAC. This bit will be cleared when read.
0	RO	0x0	reserved

GMAC_INTF_MODE_STA
Address: Operational Base + offset (0x00d8)
RGMII Status Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RO	0x0	LST Link Status Indicates whether the link is up (1'b1) or down (1'b0)

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Bit	Attr	Reset Value	Description
2:1	RO	0x0	LSD Link Speed Indicates the current speed of the link: 2'b00: 2.5 MHz 2'b01: 25 MHz 2'b10: 125 MHz
0	RW	0x0	LM Link Mode Indicates the current mode of operation of the link: 1'b0: Half-Duplex mode 1'b1: Full-Duplex mode

GMAC_MMC_CTRL
Address: Operational Base + offset (0x0100)
MMC Control Register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	RW	0x0	FHP Full-Half preset When low and bit4 is set, all MMC counters get preset to almost-half value. All octet counters get preset to 0x7FFF_F800 (half - 2KBytes) and all frame-counters gets preset to 0x7FFF_FFF0 (half - 16) When high and bit4 is set, all MMC counters get preset to almost-full value. All octet counters get preset to 0xFFFF_F800 (full - 2KBytes) and all frame-counters gets preset to 0xFFFF_FFF0 (full - 16)
4	R/WSC	0x0	CP Counters Preset When set, all counters will be initialized or preset to almost full or almost half as per Bit5 above. This bit will be cleared automatically after 1 clock cycle. This bit along with bit5 is useful for debugging and testing the assertion of interrupts due to MMC counter becoming half-full or full.

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Bit	Attr	Reset Value	Description
3	RW	0x0	MCF MMC Counter Freeze When set, this bit freezes all the MMC counters to their current value. (None of the MMC counters are updated due to any transmitted or received frame until this bit is reset to 0. If any MMC counter is read with the Reset on Read bit set, then that counter is also cleared in this mode.)
2	RW	0x0	ROR Reset on Read When set, the MMC counters will be reset to zero after Read (self-clearing after reset). The counters are cleared when the least significant byte lane (bits[7:0]) is read.
1	RW	0x0	CSR Counter Stop Rollover When set, counter after reaching maximum value will not roll over to zero
0	R/WSC	0x0	CR Counters Reset When set, all counters will be reset. This bit will be cleared automatically after 1 clock cycle

GMAC_MMC_RX_INTR
Address: Operational Base + offset (0x0104)
MMC Receive Interrupt Register

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21	RW	0x0	INT21 The bit is set when the rxfifooverflow counter reaches half the maximum value, and also when it reaches the maximum value.
20:19	RO	0x0	reserved
18	RC	0x0	INT18 The bit is set when the rxlengtherror counter reaches half the maximum value, and also when it reaches the maximum value.
17:6	RO	0x0	reserved
5	RW	0x0	INT5 The bit is set when the rxcrcerror counter reaches half the maximum value, and also when it reaches the maximum value.

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Bit	Attr	Reset Value	Description
4	RC	0x0	INT4 The bit is set when the rxmulticastframes_g counter reaches half the maximum value, and also when it reaches the maximum value.
3	RO	0x0	reserved
2	RC	0x0	INT2 The bit is set when the rxoctetcount_g counter reaches half the maximum value, and also when it reaches the maximum value.
1	RC	0x0	INT1 The bit is set when the rxoctetcount_gb counter reaches half the maximum value, and also when it reaches the maximum value.
0	RC	0x0	INT0 The bit is set when the rxframecount_gb counter reaches half the maximum value, and also when it reaches the maximum value.

GMAC_MMC_TX_INTR
Address: Operational Base + offset (0x0108)
MMC Transmit Interrupt Register

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21	RC	0x0	INT21 The bit is set when the txframecount_g counter reaches half the maximum value, and also when it reaches the maximum value.
20	RC	0x0	INT20 The bit is set when the txoctetcount_g counter reaches half the maximum value, and also when it reaches the maximum value.
19	RC	0x0	INT19 The bit is set when the txcarriererror counter reaches half the maximum value, and also when it reaches the maximum value.
18:14	RO	0x0	reserved
13	RC	0x0	INT13 The bit is set when the txunderflowerror counter reaches half the maximum value, and also when it reaches the maximum value.
12:2	RO	0x0	reserved
1	RC	0x0	INT1 The bit is set when the txframecount_gb counter reaches half the maximum value, and also when it reaches the maximum value.

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Bit	Attr	Reset Value	Description
0	RC	0x0	INT0 The bit is set when the txoctetcount_gb counter reaches half the maximum value, and also when it reaches the maximum value.

GMAC_MMC_RX_INT_MSK

Address: Operational Base + offset (0x010c)

MMC Receive Interrupt Mask Register

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21	RW	0x0	INT21 Setting this bit masks the interrupt when the rxfifooverflow counter reaches half the maximum value, and also when it reaches the maximum value.
20:19	RO	0x0	reserved
18	RW	0x0	INT18 Setting this bit masks the interrupt when the rxlengtherror counter reaches half the maximum value, and also when it reaches the maximum value.
17:6	RO	0x0	reserved
5	RW	0x0	INT5 Setting this bit masks the interrupt when the rxcrcerror counter reaches half the maximum value, and also when it reaches the maximum value.
4	RW	0x0	INT4 Setting this bit masks the interrupt when the rxmulticastframes_g counter reaches half the maximum value, and also when it reaches the maximum value.
3	RO	0x0	reserved
2	RW	0x0	INT2 Setting this bit masks the interrupt when the rxoctetcount_g counter reaches half the maximum value, and also when it reaches the maximum value.
1	RW	0x0	INT1 Setting this bit masks the interrupt when the rxoctetcount_gb counter reaches half the maximum value, and also when it reaches the maximum value.

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Bit	Attr	Reset Value	Description
0	RW	0x0	INT0 Setting this bit masks the interrupt when the rxframecount_gb counter reaches half the maximum value, and also when it reaches the maximum value.

GMAC_MMC_TX_INT_MSK

Address: Operational Base + offset (0x0110)

MMC Transmit Interrupt Mask Register

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21	RW	0x0	INT21 Setting this bit masks the interrupt when the txframecount_g counter reaches half the maximum value, and also when it reaches the maximum value.
20	RW	0x0	INT20 Setting this bit masks the interrupt when the txoctetcount_g counter reaches half the maximum value, and also when it reaches the maximum value.
19	RW	0x0	INT19 Setting this bit masks the interrupt when the txcarriererror counter reaches half the maximum value, and also when it reaches the maximum value.
18:14	RO	0x0	reserved
13	RW	0x0	INT13 Setting this bit masks the interrupt when the txunderflowerror counter reaches half the maximum value, and also when it reaches the maximum value.
12:2	RO	0x0	reserved
1	RW	0x0	INT1 Setting this bit masks the interrupt when the txframecount_gb counter reaches half the maximum value, and also when it reaches the maximum value.
0	RW	0x0	INT0 Setting this bit masks the interrupt when the txoctetcount_gb counter reaches half the maximum value, and also when it reaches the maximum value.

GMAC_MMC_TXOCTETCNT_GB

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Address: Operational Base + offset (0x0114)
MMC TX OCTET Good and Bad Counter

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txoctetcount_gb Number of bytes transmitted, exclusive of preamble and retried bytes, in good and bad frames.

GMAC_MMC_TXFRMCNT_GB
Address: Operational Base + offset (0x0118)
MMC TX Frame Good and Bad Counter

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txframecount_gb Number of good and bad frames transmitted, exclusive of retried frames.

GMAC_MMC_TXUNDFLWERR
Address: Operational Base + offset (0x0148)
MMC TX Underflow Error

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txunderflowerror Number of frames aborted due to frame underflow error.

GMAC_MMC_TXCARERR
Address: Operational Base + offset (0x0160)
MMC TX Carrier Error

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txcarriererror Number of frames aborted due to carrier sense error (no carrier or loss of carrier).

GMAC_MMC_TXOCTETCNT_G
Address: Operational Base + offset (0x0164)
MMC TX OCTET Good Counter

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txoctetcount_g Number of bytes transmitted, exclusive of preamble, in good frames only.

GMAC_MMC_TXFRMCNT_G
Address: Operational Base + offset (0x0168)
MMC TX Frame Good Counter

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txframecount_g Number of good frames transmitted.

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GMAC_MMC_RXFRMCNT_GB
Address: Operational Base + offset (0x0180)
MMC RX Frame Good and Bad Counter

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxframecount_gb Number of good and bad frames received.

GMAC_MMC_RXOCTETCNT_GB
Address: Operational Base + offset (0x0184)
MMC RX OCTET Good and Bad Counter

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxoctetcount_gb Number of bytes received, exclusive of preamble, in good and bad frames.

GMAC_MMC_RXOCTETCNT_G
Address: Operational Base + offset (0x0188)
MMC RX OCTET Good Counter

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxoctetcount_g Number of bytes received, exclusive of preamble, only in good frames.

GMAC_MMC_RXMCFRMCNT_G
Address: Operational Base + offset (0x0190)
MMC RX Mulitcast Frame Good Counter

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxmulticastframes_g Number of good multicast frames received.

GMAC_MMC_RXCRCERR
Address: Operational Base + offset (0x0194)
MMC RX Carrier

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxcrcerror Number of frames received with CRC error.

GMAC_MMC_RXLENERR
Address: Operational Base + offset (0x01c8)
MMC RX Length Error

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxlengtherror Number of frames received with length error (Length type field ≠frame size), for all frames with valid length field.

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GMAC_MMC_RXFIFOOVRFLW
Address: Operational Base + offset (0x01d4)
MMC RX FIFO Overflow

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxfifooverflow Number of missed received frames due to FIFO overflow.

GMAC_MMC_IPC_INT_MSK
Address: Operational Base + offset (0x0200)
MMC Receive Checksum Offload Interrupt Mask Register

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29	RW	0x0	INT29 Setting this bit masks the interrupt when the rxicmp_err_octets counter reaches half the maximum value, and also when it reaches the maximum value.
28	RO	0x0	reserved
27	RW	0x0	INT27 Setting this bit masks the interrupt when the rxtcp_err_octets counter reaches half the maximum value, and also when it reaches the maximum value.
26	RO	0x0	reserved
25	RW	0x0	INT25 Setting this bit masks the interrupt when the rxudp_err_octets counter reaches half the maximum value, and also when it reaches the maximum value.
24:23	RO	0x0	reserved
22	RW	0x0	INT22 Setting this bit masks the interrupt when the rxipv6_hdrerr_octets counter reaches half the maximum value, and also when it reaches the maximum value.
21:18	RO	0x0	reserved
17	RW	0x0	INT17 Setting this bit masks the interrupt when the rxipv4_hdrerr_octets counter reaches half the maximum value, and also when it reaches the maximum value.
16:14	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
13	RW	0x0	INT13 Setting this bit masks the interrupt when the rxicmp_err_frms counter reaches half the maximum value, and also when it reaches the maximum value.
12	RO	0x0	reserved
11	RW	0x0	INT11 Setting this bit masks the interrupt when the rxtcp_err_frms counter reaches half the maximum value, and also when it reaches the maximum value.
10	RO	0x0	reserved
9	RW	0x0	INT9 Setting this bit masks the interrupt when the rxudp_err_frms counter reaches half the maximum value, and also when it reaches the maximum value.
8:7	RO	0x0	reserved
6	RW	0x0	INT6 Setting this bit masks the interrupt when the rxipv6_hdrerr_frms counter reaches half the maximum value, and also when it reaches the maximum value.
5	RW	0x0	INT5 Setting this bit masks the interrupt when the rxipv6_gd_frms counter reaches half the maximum value, and also when it reaches the maximum value.
4:2	RO	0x0	reserved
1	RW	0x0	INT1 Setting this bit masks the interrupt when the rxipv4_hdrerr_frms counter reaches half the maximum value, and also when it reaches the maximum value.
0	RW	0x0	INT0 Setting this bit masks the interrupt when the rxipv4_gd_frms counter reaches half the maximum value, and also when it reaches the maximum value.

GMAC_MMC_IPC_INTR

Address: Operational Base + offset (0x0208)

MMC Receive Checksum Offload Interrupt Register

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
29	RC	0x0	INT29 The bit is set when the rxicmp_err_octets counter reaches half the maximum value, and also when it reaches the maximum value.
28	RO	0x0	reserved
27	RC	0x0	INT27 The bit is set when the rxtcp_err_octets counter reaches half the maximum value, and also when it reaches the maximum value.
26	RO	0x0	reserved
25	RC	0x0	INT25 The bit is set when the rxudp_err_octets counter reaches half the maximum value, and also when it reaches the maximum value.
24:23	RO	0x0	reserved
22	RC	0x0	INT22 The bit is set when the rxipv6_hdrerr_octets counter reaches half the maximum value, and also when it reaches the maximum value.
21:18	RO	0x0	reserved
17	RC	0x0	INT17 The bit is set when the rxipv4_hdrerr_octets counter reaches half the maximum value, and also when it reaches the maximum value.
16:14	RO	0x0	reserved
13	RC	0x0	INT13 The bit is set when the rxicmp_err_frms counter reaches half the maximum value, and also when it reaches the maximum value.
12	RO	0x0	reserved
11	RC	0x0	INT11 The bit is set when the rxtcp_err_frms counter reaches half the maximum value, and also when it reaches the maximum value.
10	RO	0x0	reserved
9	RC	0x0	INT9 The bit is set when the rxudp_err_frms counter reaches half the maximum value, and also when it reaches the maximum value.
8:7	RO	0x0	reserved
6	RC	0x0	INT6 The bit is set when the rxipv6_hdrerr_frms counter reaches half the maximum value, and also when it reaches the maximum value.

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Bit	Attr	Reset Value	Description
5	RC	0x0	INT5 The bit is set when the rxipv6_gd_frms counter reaches half the maximum value, and also when it reaches the maximum value.
4:2	RO	0x0	reserved
1	RC	0x0	INT1 The bit is set when the rxipv4_hdrerr_frms counter reaches half the maximum value, and also when it reaches the maximum value.
0	RC	0x0	INT0 The bit is set when the rxipv4_gd_frms counter reaches half the maximum value, and also when it reaches the maximum value.

GMAC_MMC_RXIPV4GFRM
Address: Operational Base + offset (0x0210)
MMC RX IPV4 Good Frame

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxipv4_gd_frms Number of good IPv4 datagrams received with the TCP, UDP, or ICMP payload

GMAC_MMC_RXIPV4HDERRFRM
Address: Operational Base + offset (0x0214)
MMC RX IPV4 Head Error Frame

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxipv4_hdrerr_frms Number of IPv4 datagrams received with header (checksum, length, or version mismatch) errors

GMAC_MMC_RXIPV6GFRM
Address: Operational Base + offset (0x0224)
MMC RX IPV6 Good Frame

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxipv6_gd_frms Number of good IPv6 datagrams received with TCP, UDP, or ICMP payloads.

GMAC_MMC_RXIPV6HDERRFRM
Address: Operational Base + offset (0x0228)
MMC RX IPV6 Head Error Frame

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxipv6_hdrerr_frms Number of IPv6 datagrams received with header errors (length or version mismatch).

GMAC_MMC_RXUDPERRFRM
Address: Operational Base + offset (0x0234)
MMC RX UDP Error Frame

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxudp_err_frms Number of good IP datagrams whose UDP payload has a checksum error.

GMAC_MMC_RXTCPERRFRM
Address: Operational Base + offset (0x023c)
MMC RX TCP Error Frame

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxtcp_err_frms Number of good IP datagrams whose TCP payload has a checksum error.

GMAC_MMC_RXICMPERRFRM
Address: Operational Base + offset (0x0244)
MMC RX ICMP Error Frame

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxicmp_err_frms Number of good IP datagrams whose ICMP payload has a checksum error.

GMAC_MMC_RXIPV4HDERROCT
Address: Operational Base + offset (0x0254)
MMC RX OCTET IPV4 Head Error

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxipv4_hdrerr_octets Number of bytes received in IPv4 datagrams with header errors (checksum, length, version mismatch). The value in the Length field of IPv4 header is used to update this counter.

GMAC_MMC_RXIPV6HDERROCT
Address: Operational Base + offset (0x0268)
MMC RX OCTET IPV6 Head Error

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxipv6_hdrerr_octets Number of bytes received in IPv6 datagrams with header errors (length, version mismatch). The value in the IPv6 header's Length field is used to update this counter.

GMAC_MMC_RXUDPERROCT
Address: Operational Base + offset (0x0274)
MMC RX OCTET UDP Error

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxudp_err_octets Number of bytes received in a UDP segment that had checksum errors.

GMAC_MMC_RXTCPERROCT
Address: Operational Base + offset (0x027c)
MMC RX OCTET TCP Error

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxtcp_err_octets Number of bytes received in a TCP segment with checksum errors.

GMAC_MMC_RXICMPERROCT
Address: Operational Base + offset (0x0284)
MMC RX OCTET ICMP Error

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxicmp_err_octets Number of bytes received in an ICMP segment with checksum errors.

GMAC_BUS_MODE
Address: Operational Base + offset (0x1000)
Bus Mode Register

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25	RW	0x0	AAL Address-Aligned Beats When this bit is set high and the FB bit equals 1, the AXI interface generates all bursts aligned to the start address LS bits. If the FB bit equals 0, the first burst (accessing the data buffer's start address) is not aligned, but subsequent bursts are aligned to the address.

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Bit	Attr	Reset Value	Description
24	RW	0x0	PBL_Mode 8xPBL Mode When set high, this bit multiplies the PBL value programmed (bits [22:17] and bits [13:8]) eight times. Thus the DMA will transfer data in to a maximum of 8, 16, 32, 64, 128, and 256 beats depending on the PBL value.
23	RW	0x0	USP Use Separate PBL When set high, it configures the RxDMA to use the value configured in bits [22:17] as PBL while the PBL value in bits [13:8] is applicable to TxDMA operations only. When reset to low, the PBL value in bits [13:8] is applicable for both DMA engines.
22:17	RW	0x01	RPBL RxDMA PBL These bits indicate the maximum number of beats to be transferred in one RxDMA transaction. This will be the maximum value that is used in a single block Read/Write. The RxDMA will always attempt to burst as specified in RPBL each time it starts a Burst transfer on the host bus. RPBL can be programmed with permissible values of 1, 2, 4, 8, 16, and 32. Any other value will result in undefined behavior.These bits are valid and applicable only when USP is set high.
16	RW	0x0	FB Fixed Burst This bit controls whether the AXI Master interface performs fixed burst transfers or not. When set, the AHB will use only SINGLE, INCR4, INCR8 or INCR16 during start of normal burst transfers. When reset, the AXI will use SINGLE and INCR burst transfer operations.
15:14	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
13:8	RW	0x01	PBL Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA transaction. This will be the maximum value that is used in a single block Read/Write. The DMA will always attempt to burst as specified in PBL each time it starts a Burst transfer on the host bus. PBL can be programmed with permissible values of 1, 2, 4, 8, 16, and 32. Any other value will result in undefined behavior. When USP is set high,this PBL value is applicable for TxDMA transactions only. The PBL values have the following limitations. The maximum number of beats (PBL) possible is limited by the size of the Tx FIFO and Rx FIFO in the MTL layer and the data bus width on the DMA. The FIFO has a constraint that the maximum beat supported is half the depth of the FIFO, except when specified (as given below). For different data bus widths and FIFO sizes, the valid PBL range (including x8 mode) is provided in the following table. If the PBL is common for both transmit and receive DMA, the minimum Rx FIFO and Tx FIFO depths must be considered. Do not program out-of-range PBL values, because the system may not behave properly. For TxFIFO, valid PBL range in full duplex mode and duplex mode is 128 or less. For RxFIFO, valid PBL range in full duplex mode is all.
7	RO	0x0	reserved
6:2	RW	0x00	DSL Descriptor Skip Length This bit specifies the number of Dword to skip between two unchained descriptors. The address skipping starts from the end of current descriptor to the start of next descriptor. When DSL value equals zero, then the descriptor table is taken as contiguous by the DMA, in Ring mode.
1	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
0	R/WSC	0x1	SWR Software Reset When this bit is set, the MAC DMA Controller resets all GMAC Subsystem internal registers and logic. It is cleared automatically after the reset operation has completed in all of the core clock domains. Read a 0 value in this bit before re-programming any register of the core. Note: The reset operation is completed only when all the resets in all the active clock domains are de-asserted. Hence it is essential that all the PHY inputs clocks (applicable for the selected PHY interface) are present for software reset completion.

GMAC_TX_POLL_DEMAND
Address: Operational Base + offset (0x1004)
Transmit Poll Demand Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	TPD Transmit Poll Demand When these bits are written with any value, the DMA reads the current descriptor pointed to by Register GMAC_CUR_HOST_TX_DESC. If that descriptor is not available (owned by Host), transmission returns to the Suspend state and DMA Register GMAC_STATUS[2] is asserted. If the descriptor is available, transmission resumes.

GMAC_RX_POLL_DEMAND
Address: Operational Base + offset (0x1008)
Receive Poll Demand Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	RPD Receive Poll Demand When these bits are written with any value, the DMA reads the current descriptor pointed to by Register GMAC_CUR_HOST_RX_DESC. If that descriptor is not available (owned by Host), reception returns to the Suspended state and Register GMAC_STATUS[7] is not asserted. If the descriptor is available, the Receive DMA returns to active state.

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GMAC_RX_DESC_LIST_ADDR
Address: Operational Base + offset (0x100c)
Receive Descriptor List Address Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	SRL Start of Receive List This field contains the base address of the First Descriptor in the Receive Descriptor list. The LSB bits [1/2/3:0] for 32/64/128-bit bus width) will be ignored and taken as all-zero by the DMA internally. Hence these LSB bits are Read Only.

GMAC_TX_DESC_LIST_ADDR
Address: Operational Base + offset (0x1010)
Transmit Descriptor List Address Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	STL Start of Transmit List This field contains the base address of the First Descriptor in the Transmit Descriptor list. The LSB bits [1/2/3:0] for 32/64/128-bit bus width) will be ignored and taken as all-zero by the DMA internally. Hence these LSB bits are Read Only.

GMAC_STATUS
Address: Operational Base + offset (0x1014)
Status Register

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28	RO	0x0	GPI GMAC PMT Interrupt This bit indicates an interrupt event in the GMAC core's PMT module. The software must read the corresponding registers in the GMAC core to get the exact cause of interrupt and clear its source to reset this bit to 1'b0. The interrupt signal from the GMAC subsystem (sbd_intr_o) is high when this bit is high.

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Bit	Attr	Reset Value	Description
27	RO	0x0	GMI GMAC MMC Interrupt This bit reflects an interrupt event in the MMC module of the GMAC core. The software must read the corresponding registers in the GMAC core to get the exact cause of interrupt and clear the source of interrupt to make this bit as 1'b0. The interrupt signal from the GMAC subsystem (sbd_intr_o) is high when this bit is high.
26	RO	0x0	GLI GMAC Line interface Interrupt This bit reflects an interrupt event in the GMAC Core's PCS or RGMII interface block. The software must read the corresponding registers in the GMAC core to get the exact cause of interrupt and clear the source of interrupt to make this bit as 1'b0. The interrupt signal from the GMAC subsystem (sbd_intr_o) is high when this bit is high.
25:23	RO	0x0	EB Error Bits These bits indicate the type of error that caused a Bus Error (e.g., error response on the AXI interface). Valid only with Fatal Bus Error bit (Register GMAC_STATUS[13]) set. This field does not generate an interrupt. Bit 23: 1'b1 Error during data transfer by TxDMA 1'b0 Error during data transfer by RxDMA Bit 24: 1'b1 Error during read transfer 1'b0 Error during write transfer Bit 25: 1'b1 Error during descriptor access 1'b0 Error during data buffer access

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Bit	Attr	Reset Value	Description
22:20	RO	0x0	TS Transmit Process State These bits indicate the Transmit DMA FSM state. This field does not generate an interrupt. 3'b000: Stopped; Reset or Stop Transmit Command issued. 3'b001: Running; Fetching Transmit Transfer Descriptor. 3'b010: Running; Waiting for status. 3'b011: Running; Reading Data from host memory buffer and queuing it to transmit buffer (Tx FIFO). 3'b100: TIME_STAMP write state. 3'b101: Reserved for future use. 3'b110: Suspended; Transmit Descriptor Unavailable or Transmit Buffer Underflow. 3'b111: Running; Closing Transmit Descriptor.
19:17	RO	0x0	RS Receive Process State These bits indicate the Receive DMA FSM state. This field does not generate an interrupt. 3'b000: Stopped: Reset or Stop Receive Command issued. 3'b001: Running: Fetching Receive Transfer Descriptor. 3'b010: Reserved for future use. 3'b011: Running: Waiting for receive packet. 3'b100: Suspended: Receive Descriptor Unavailable. 3'b101: Running: Closing Receive Descriptor. 3'b110: TIME_STAMP write state. 3'b111: Running: Transferring the receive packet data from receive buffer to host memory.

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Bit	Attr	Reset Value	Description
16	W1C	0x0	NIS Normal Interrupt Summary Normal Interrupt Summary bit value is the logical OR of the following when the corresponding interrupt bits are enabled in Register OP_MODE: Register GMAC_STATUS[0]: Transmit Interrupt Register GMAC_STATUS[2]: Transmit Buffer Unavailable Register GMAC_STATUS[6]: Receive Interrupt Register GMAC_STATUS[14]: Early Receive Interrupt Only unmasked bits affect the Normal Interrupt Summary bit. This is a sticky bit and must be cleared (by writing a 1 to this bit) each time a corresponding bit that causes NIS to be set is cleared.

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Bit	Attr	Reset Value	Description
15	W1C	0x0	AIS Abnormal Interrupt Summary Abnormal Interrupt Summary bit value is the logical OR of the following when the corresponding interrupt bits are enabled in Register OP_MODE: Register GMAC_STATUS[1]: Transmit Process Stopped Register GMAC_STATUS[3]: Transmit Jabber Timeout Register GMAC_STATUS[4]: Receive FIFO Overflow Register GMAC_STATUS[5]: Transmit Underflow Register GMAC_STATUS[7]: Receive Buffer Unavailable Register GMAC_STATUS[8]: Receive Process Stopped Register GMAC_STATUS[9]: Receive Watchdog Timeout Register GMAC_STATUS[10]: Early Transmit Interrupt Register GMAC_STATUS[13]: Fatal Bus Error Only unmasked bits affect the Abnormal Interrupt Summary bit. This is a sticky bit and must be cleared each time a corresponding bit that causes AIS to be set is cleared.
14	W1C	0x0	ERI Early Receive Interrupt This bit indicates that the DMA had filled the first data buffer of the packet. Receive Interrupt Register GMAC_STATUS[6] automatically clears this bit.
13	W1C	0x0	FBI Fatal Bus Error Interrupt This bit indicates that a bus error occurred, as detailed in [25:23]. When this bit is set, the corresponding DMA engine disables all its bus accesses.
12:11	RO	0x0	reserved
10	W1C	0x0	ETI Early Transmit Interrupt This bit indicates that the frame to be transmitted was fully transferred to the MTL Transmit FIFO.

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Bit	Attr	Reset Value	Description
9	W1C	0x0	RWT Receive Watchdog Timeout This bit is asserted when a frame with a length greater than 2,048 bytes is received.
8	W1C	0x0	RPS Receive Process Stopped This bit is asserted when the Receive Process enters the Stopped state.
7	W1C	0x0	RU Receive Buffer Unavailable This bit indicates that the Next Descriptor in the Receive List is owned by the host and cannot be acquired by the DMA. Receive Process is suspended. To resume processing Receive descriptors, the host should change the ownership of the descriptor and issue a Receive Poll Demand command. If no Receive Poll Demand is issued, Receive Process resumes when the next recognized incoming frame is received. Register GMAC_STATUS[7] is set only when the previous Receive Descriptor was owned by the DMA.
6	W1C	0x0	RI Receive Interrupt This bit indicates the completion of frame reception. Specific frame status information has been posted in the descriptor. Reception remains in the Running state.
5	W1C	0x0	UNF Transmit Underflow This bit indicates that the Transmit Buffer had an Underflow during frame transmission. Transmission is suspended and an Underflow Error TDES0[1] is set.
4	W1C	0x0	OVF Receive Overflow This bit indicates that the Receive Buffer had an Overflow during frame reception. If the partial frame is transferred to application, the overflow status is set in RDES0[11].

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Bit	Attr	Reset Value	Description
3	W1C	0x0	TJT Transmit Jabber Timeout This bit indicates that the Transmit Jabber Timer expired, meaning that the transmitter had been excessively active. The transmission process is aborted and placed in the Stopped state. This causes the Transmit Jabber Timeout TDES0[14] flag to assert.
2	W1C	0x0	TU Transmit Buffer Unavailable This bit indicates that the Next Descriptor in the Transmit List is owned by the host and cannot be acquired by the DMA. Transmission is suspended. Bits[22:20] explain the Transmit Process state transitions. To resume processing transmit descriptors, the host should change the ownership of the bit of the descriptor and then issue a Transmit Poll Demand command.
1	W1C	0x0	TPS Transmit Process Stopped This bit is set when the transmission is stopped.
0	W1C	0x0	TI Transmit Interrupt This bit indicates that frame transmission is finished and TDES1[31] is set in the First Descriptor.

GMAC_OP_MODE
Address: Operational Base + offset (0x1018)
Operation Mode Register

Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved
26	RW	0x0	DT Disable Dropping of TCP/IP Checksum Error Frames When this bit is set, the core does not drop frames that only have errors detected by the Receive Checksum Offload engine. Such frames do not have any errors (including FCS error) in the Ethernet frame received by the MAC but have errors in the encapsulated payload only. When this bit is reset, all error frames are dropped if the FEF bit is reset.

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Bit	Attr	Reset Value	Description
25	RW	0x0	RSF Receive Store and Forward When this bit is set, the MTL only reads a frame from the Rx FIFO after the complete frame has been written to it, ignoring RTC bits. When this bit is reset, the Rx FIFO operates in Cut-Through mode, subject to the threshold specified by the RTC bits.
24	RW	0x0	DFF Disable Flushing of Received Frames When this bit is set, the RxDMA does not flush any frames due to the unavailability of receive descriptors/buffers as it does normally when this bit is reset.
23:22	RO	0x0	reserved
21	RW	0x0	TSF Transmit Store and Forward When this bit is set, transmission starts when a full frame resides in the MTL Transmit FIFO. When this bit is set, the TTC values specified in Register GMAC_OP_MODE[16:14] are ignored. This bit should be changed only when transmission is stopped.
20	W1C	0x0	FTF Flush Transmit FIFO When this bit is set, the transmit FIFO controller logic is reset to its default values and thus all data in the Tx FIFO is lost/flushed. This bit is cleared internally when the flushing operation is completed fully. The Operation Mode register should not be written to until this bit is cleared. The data which is already accepted by the MAC transmitter will not be flushed. It will be scheduled for transmission and will result in underflow and runt frame transmission. Note: The flush operation completes only after emptying the TxFIFO of its contents and all the pending Transmit Status of the transmitted frames are accepted by the host. In order to complete this flush operation, the PHY transmit clock (clk_tx_i) is required to be active.
19:17	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
16:14	RW	0x0	TTC Transmit Threshold Control These three bits control the threshold level of the MTL Transmit FIFO. Transmission starts when the frame size within the MTL Transmit FIFO is larger than the threshold. In addition, full frames with a length less than the threshold are also transmitted. These bits are used only when the TSF bit (Bit 21) is reset. 3'b000: 64 3'b001: 128 3'b010: 192 3'b011: 256 3'b100: 40 3'b101: 32 3'b110: 24 3'b111: 16

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Bit	Attr	Reset Value	Description
13	RW	0x0	ST Start/Stop Transmission Command When this bit is set, transmission is placed in the Running state, and the DMA checks the Transmit List at the current position for a frame to be transmitted. Descriptor acquisition is attempted either from the current position in the list, which is the Transmit List Base Address set by Register GMAC_TX_DESC_LIST_ADDR, or from the position retained when transmission was stopped previously. If the current descriptor is not owned by the DMA, transmission enters the Suspended state and Transmit Buffer Unavailable (Register GMAC_STATUS[2]) is set. The Start Transmission command is effective only when transmission is stopped. If the command is issued before setting DMA Register TX_DESC_LIST_ADDR, then the DMA behavior is unpredictable. When this bit is reset, the transmission process is placed in the Stopped state after completing the transmission of the current frame. The Next Descriptor position in the Transmit List is saved, and becomes the current position when transmission is restarted. The stop transmission command is effective only the transmission of the current frame is complete or when the transmission is in the Suspended state.
12:11	RW	0x0	RFD Threshold for deactivating flow control (in both HD and FD) These bits control the threshold (Fill-level of Rx FIFO) at which the flow-control is deasserted after activation. 2'b00: Full minus 1 KB 2'b01: Full minus 2 KB 2'b10: Full minus 3 KB 2'b11: Full minus 4 KB Note that the deassertion is effective only after flow control is asserted.

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Bit	Attr	Reset Value	Description
10:9	RW	0x0	RFA Threshold for activating flow control (in both HD and FD) These bits control the threshold (Fill level of Rx FIFO) at which flow control is activated. 2'b00: Full minus 1 KB 2'b01: Full minus 2 KB 2'b10: Full minus 3 KB 2'b11: Full minus 4 KB Note that the above only applies to Rx FIFOs of 4 KB or more when the EFC bit is set high.
8	RW	0x0	EFC Enable HW flow control When this bit is set, the flow control signal operation based on fill-level of Rx FIFO is enabled. When reset, the flow control operation is disabled.
7	RW	0x0	FEF Forward Error Frames When this bit is reset, the Rx FIFO drops frames with error status (CRC error, collision error, GMII_ER, giant frame, watchdog timeout, overflow). However, if the frame's start byte (write) pointer is already transferred to the read controller side (in Threshold mode), then the frames are not dropped. When FEF is set, all frames except runt error frames are forwarded to the DMA. But when RxFIFO overflows when a partial frame is wriiten, then such frames are dropped even when FEF is set.
6	RW	0x0	FUF Forward Undersized Good Frames When set, the Rx FIFO will forward Undersized frames (frames with no Error and length less than 64 bytes) including pad-bytes and CRC). When reset, the Rx FIFO will drop all frames of less than 64 bytes, unless it is already transferred due to lower value of Receive Threshold (e.g., RTC = 01).
5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4:3	RW	0x0	RTC Receive Threshold Control These two bits control the threshold level of the MTL Receive FIFO. Transfer (request) to DMA starts when the frame size within the MTL Receive FIFO is larger than the threshold. In addition, full frames with a length less than the threshold are transferred automatically. Note that value of 11 is not applicable if the configured Receive FIFO size is 128 bytes. These bits are valid only when the RSF bit is zero, and are ignored when the RSF bit is set to 1. 2'b00: 64 2'b01: 32 2'b10: 96 2'b11: 128
2	RW	0x0	OSF Operate on Second Frame When this bit is set, this bit instructs the DMA to process a second frame of Transmit data even before status for first frame is obtained.

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Bit	Attr	Reset Value	Description
1	RW	0x0	SR Start/Stop Receive When this bit is set, the Receive process is placed in the Running state. The DMA attempts to acquire the descriptor from the Receive list and processes incoming frames. Descriptor acquisition is attempted from the current position in the list, which is the address set by register GMAC_RX_DESC_LIST_ADDR or the position retained when the Receive process was previously stopped. If no descriptor is owned by the DMA, reception is suspended and Receive Buffer Unavailable (Register GMAC_STATUS[7]) is set. The Start Receive command is effective only when reception has stopped. If the command was issued before setting register GMAC_RX_DESC_LIST_ADDR, DMA behavior is unpredictable. When this bit is cleared, RxDMA operation is stopped after the transfer of the current frame. The next descriptor position in the Receive list is saved and becomes the current position after the Receive process is restarted. The Stop Receive command is effective only when the Receive process is in either the Running (waiting for receive packet) or in the Suspended state.
0	RO	0x0	reserved

GMAC_INT_ENA
Address: Operational Base + offset (0x101c)
Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
16	RW	0x0	NIE Normal Interrupt Summary Enable When this bit is set, a normal interrupt is enabled. When this bit is reset, a normal interrupt is disabled. This bit enables the following bits: Register GMAC_STATUS[0]: Transmit Interrupt Register GMAC_STATUS[2]: Transmit Buffer Unavailable Register GMAC_STATUS[6]: Receive Interrupt Register GMAC_STATUS[14]: Early Receive Interrupt
15	RW	0x0	AIE Abnormal Interrupt Summary Enable When this bit is set, an Abnormal Interrupt is enabled. When this bit is reset, an Abnormal Interrupt is disabled. This bit enables the following bits Register GMAC_STATUS[1]: Transmit Process Stopped Register GMAC_STATUS[3]: Transmit Jabber Timeout Register GMAC_STATUS[4]: Receive Overflow Register GMAC_STATUS[5]: Transmit Underflow Register GMAC_STATUS[7]: Receive Buffer Unavailable Register GMAC_STATUS[8]: Receive Process Stopped Register GMAC_STATUS[9]: Receive Watchdog Timeout Register GMAC_STATUS[10]: Early Transmit Interrupt Register GMAC_STATUS[13]: Fatal Bus Error
14	RW	0x0	ERE Early Receive Interrupt Enable When this bit is set with Normal Interrupt Summary Enable (BIT 16), Early Receive Interrupt is enabled. When this bit is reset, Early Receive Interrupt is disabled.

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Bit	Attr	Reset Value	Description
13	RW	0x0	FBE Fatal Bus Error Enable When this bit is set with Abnormal Interrupt Summary Enable (BIT 15), the Fatal Bus Error Interrupt is enabled. When this bit is reset, Fatal Bus Error Enable Interrupt is disabled.
12:11	RO	0x0	reserved
10	RW	0x0	ETE Early Transmit Interrupt Enable When this bit is set with an Abnormal Interrupt Summary Enable (BIT 15), Early Transmit Interrupt is enabled. When this bit is reset, Early Transmit Interrupt is disabled.
9	RW	0x0	RWE Receive Watchdog Timeout Enable When this bit is set with Abnormal Interrupt Summary Enable (BIT 15), the Receive Watchdog Timeout Interrupt is enabled. When this bit is reset, Receive Watchdog Timeout Interrupt is disabled.
8	RW	0x0	RSE Receive Stopped Enable When this bit is set with Abnormal Interrupt Summary Enable (BIT 15), Receive Stopped Interrupt is enabled. When this bit is reset, Receive Stopped Interrupt is disabled.
7	RW	0x0	RUE Receive Buffer Unavailable Enable When this bit is set with Abnormal Interrupt Summary Enable (BIT 15), Receive Buffer Unavailable Interrupt is enabled. When this bit is reset, the Receive Buffer Unavailable Interrupt is disabled
6	RW	0x0	RIE Receive Interrupt Enable When this bit is set with Normal Interrupt Summary Enable (BIT 16), Receive Interrupt is enabled. When this bit is reset, Receive Interrupt is disabled.
5	RW	0x0	UNE Underflow Interrupt Enable When this bit is set with Abnormal Interrupt Summary Enable (BIT 15), Transmit Underflow Interrupt is enabled. When this bit is reset, Underflow Interrupt is disabled.

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Bit	Attr	Reset Value	Description
4	RW	0x0	OVE Overflow Interrupt Enable When this bit is set with Abnormal Interrupt Summary Enable (BIT 15), Receive Overflow Interrupt is enabled. When this bit is reset, Overflow Interrupt is disabled
3	RW	0x0	TJE Transmit Jabber Timeout Enable When this bit is set with Abnormal Interrupt Summary Enable (BIT 15), Transmit Jabber Timeout Interrupt is enabled. When this bit is reset, Transmit Jabber Timeout Interrupt is disabled.
2	RW	0x0	TUE Transmit Buffer Unavailable Enable When this bit is set with Normal Interrupt Summary Enable (BIT 16), Transmit Buffer Unavailable Interrupt is enabled. When this bit is reset, Transmit Buffer Unavailable Interrupt is disabled.
1	RW	0x0	TSE Transmit Stopped Enable When this bit is set with Abnormal Interrupt Summary Enable (BIT 15), Transmission Stopped Interrupt is enabled. When this bit is reset, Transmission Stopped Interrupt is disabled.
0	RW	0x0	TIE Transmit Interrupt Enable When this bit is set with Normal Interrupt Summary Enable (BIT 16), Transmit Interrupt is enabled. When this bit is reset, Transmit Interrupt is disabled.

GMAC_OVERFLOW_CNT
Address: Operational Base + offset (0x1020)
Missed Frame and Buffer Overflow Counter Register

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28	RC	0x0	FIFO_overflow_bit Overflow bit for FIFO Overflow Counter

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Bit	Attr	Reset Value	Description
27:17	RC	0x000	Frame_miss_number Indicates the number of frames missed by the application This counter is incremented each time the MTL asserts the sideband signal mtl_rxoverflow_o. The counter is cleared when this register is read with mci_be_i[2] at 1'b1.
16	RC	0x0	Miss_frame_overflow_bit Overflow bit for Missed Frame Counter
15:0	RC	0x0000	Frame_miss_number_2 Indicates the number of frames missed by the controller due to the Host Receive Buffer being unavailable. This counter is incremented each time the DMA discards an incoming frame. The counter is cleared when this register is read with mci_be_i[0] at 1'b1.

GMAC_REC_INT_WDT_TIMER
Address: Operational Base + offset (0x1024)
Receive Interrupt Watchdog Timer Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	RIWT RI Watchdog Timer count Indicates the number of system clock cycles multiplied by 256 for which the watchdog timer is set. The watchdog timer gets triggered with the programmed value after the RxDMA completes the transfer of a frame for which the RI status bit is not set due to the setting in the corresponding descriptor RDES1[31]. When the watch-dog timer runs out, the RI bit is set and the timer is stopped. The watchdog timer is reset when RI bit is set high due to automatic setting of RI as per RDES1[31] of any received frame.

GMAC_AXI_BUS_MODE
Address: Operational Base + offset (0x1028)
AXI Bus Mode Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31	RW	0x0	EN_LPI Enable LPI (Low Power Interface) When set to 1, enable the LPI (Low Power Interface) supported by the GMAC and accepts the LPI request from the AXI System Clock controller. When set to 0, disables the Low Power Mode and always denies the LPI request from the AXI System Clock controller.
30	RW	0x0	UNLCK_ON_MGK_RWK Unlock on Magic Packet or Remote Wake Up When set to 1, enables it to request coming out of Low Power mode only when Magic Packet or Remote Wake Up Packet is received. When set to 0, enables it requests to come out of Low Power mode when any frame is received.
29:22	RO	0x0	reserved
21:20	RW	0x1	WR_OSR_LMT AXI Maximum Write Out Standing Request Limit This value limits the maximum outstanding request on the AXI write interface. Maximum outstanding requests = WR_OSR_LMT+1
19:18	RO	0x0	reserved
17:16	RW	0x1	RD_OSR_LMT AXI Maximum Read Out Standing Request Limit This value limits the maximum outstanding request on the AXI read interface. Maximum outstanding requests = RD_OSR_LMT+1
15:13	RO	0x0	reserved
12	RO	0x0	AXI_AAL Address-Aligned Beats This bit is read-only bit and reflects the AAL bit Register0 (register GMAC_BUS_MODE[25]). When this bit set to 1, it performs address-aligned burst transfers on both read and write channels.
11:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3	RW	0x0	BLEN16 AXI Burst Length 16 When this bit is set to 1, or when UNDEF is set to 1, it is allowed to select a burst length of 16.
2	RW	0x0	BLEN8 AXI Burst Length 8 When this bit is set to 1, or when UNDEF is set to 1, it is allowed to select a burst length of 8.
1	RW	0x0	BLEN4 AXI Burst Length 4 When this bit is set to 1, or when UNDEF is set to 1, it is allowed to select a burst length of 4.
0	RO	0x1	UNDEF AXI Undefined Burst Length This bit is read-only bit and indicates the complement (invert) value of FB bit in register GMAC_BUS_MODE[16]. When this bit is set to 1, it is allowed to perform any burst length equal to or below the maximum allowed burst length as programmed in bits[7:1]; When this bit is set to 0, the it is allowed to perform only fixed burst lengths as indicated by BLEN256/128/64/32/16/8/4, or a burst length of 1.

GMAC_AXI_STATUS
Address: Operational Base + offset (0x102c)
AXI Status Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RO	0x0	RD_CH_STA When high, it indicates that AXI Master's read channel is active and transferring data.
0	RO	0x0	WR_CH_STA When high, it indicates that AXI Master's write channel is active and transferring data.

GMAC_CUR_HOST_TX_DESC
Address: Operational Base + offset (0x1048)
Current Host Transmit Descriptor Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HTDAP Host Transmit Descriptor Address Pointer Cleared on Reset. Pointer updated by DMA during operation.

GMAC_CUR_HOST_RX_DESC
Address: Operational Base + offset (0x104c)
Current Host Receive Descriptor Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HRDAP Host Receive Descriptor Address Pointer Cleared on Reset. Pointer updated by DMA during operation.

GMAC_CUR_HOST_TX_Buf_ADDR
Address: Operational Base + offset (0x1050)
Current Host Transmit Buffer Address Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HTBAP Host Transmit Buffer Address Pointer Cleared on Reset. Pointer updated by DMA during operation.

GMAC_CUR_HOST_RX_BUF_ADDR
Address: Operational Base + offset (0x1054)
Current Host Receive Buffer Adderss Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HRBAP Host Receive Buffer Address Pointer Cleared on Reset. Pointer updated by DMA during operation.

GMAC_HW_FEA_REG
Address: Operational Base + offset (0x1058)
The presence of the optional features/functions of the core Register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	Reserved
24	RO	0x0	Alternate (Enhanced Descriptor)
23:20	RO	0x0	Reserved
19	RO	0x1	RxFIFO > 2048 Bytes
18	RO	0x1	IP Checksum Offload (Type 2) in Rx
17	RO	0x0	IP Checksum Offload (Type 1) in Rx
16	RO	0x1	Checksum Offload in Tx
15:14	RO	0x0	Reserved
13	RO	0x0	IEEE 1588-2008 Advanced Time-Stamp

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Bit	Attr	Reset Value	Description
12	RO	0x0	IEEE 1588-2002 Time-Stamp
11	RO	0x1	RMON module
10	RO	0x1	PMT Magic Packet
9	RO	0x1	PMT Remote Wakeup
8	RO	0x1	SMA (MDIO) Interface
7	RO	0x0	Reserved
6	RO	0x0	PCS registers (TBI/SGMII/RTBI PHY interface)
5	RO	0x0	Multiple MAC Address Registers
4	RO	0x1	HASH Filter
3	RO	0x0	Reserved
2	RO	0x1	Half-Duplex support
1	RO	0x1	1000 Mbps support
0	RO	0x1	10/100 Mbps support

41.5 Interface Description

Table 41-1 RMII Interface Description

Module pin name	Dir	Pad name	IOMUX
RMII interface			
mac_clk	I/O	IO_MACclk_ISPshuttertrig_GPIO30gpio3c6	GPIO3C_IOMUX[13:12]=2'b01
mac_txen	O	IO_MACtxen_GPIO30gpio3b5	GPIO3B_IOMUX[11:10]=2'b01
mac_txd1	O	IO_MACtxd1_GPIO30gpio3b1	GPIO3B_IOMUX[3:2]=2'b01
mac_txd0	O	IO_MACtxd0_PWM0_VOPpwm_GPIO30gpio3b0	GPIO3B_IOMUX[1:0]=2'b01
mac_rxdv	I	IO_MACrxdv_ISPflashtrigout_GPIO30gpio3c4	GPIO3C_IOMUX[9:8]=2'b01
mac_rxer	I	IO_MACrxer_ISPprelighttrig_GPIO30gpio3c5	GPIO3C_IOMUX[11:10]=2'b01
mac_rxd1	I	IO_MACrxd1_UART3GPSctsn_GPSrfclk_GPIO30gpio3c0	GPIO3C_IOMUX[1:0]=2'b01
mac_rxd0	I	IO_MACrxd0_GPSsig_GPIO30gpio3b7	GPIO3B_IOMUX[15:14]=2'b01
Management interface			
mac_mdio	I/O	IO_MACmdio_I2C4TPscl_GPIO30gpio3d0	GPIO3D_IOMUX[1:0]=2'b01
mac_mdc	O	IO_MACmdc_ISPshutteren_GPIO30gpio3c3	GPIO3C_IOMUX[7:6]=2'b01

Table 41-2 RGMII Interface Description

Module pin name	Dir	Pad name	IOMUX
RGMII/RMII interface			
mac_clk	I/O	IO_MACclk_ISPshuttertrig_GPIO30gpio3c6	GPIO3C_IOMUX[13:12]=2'b01
mac_txclk	O	IO_MACtxclkout_SPI1csn1_GPIO30gpio3d4	GPIO3D_IOMUX[9:8]=2'b01
mac_txen	O	IO_MACtxen_GPIO30gpio3b5	GPIO3B_IOMUX[11:10]=2'b01
mac_txd3	O	IO_MACtxd3_GPSmag_GPIO30gpio3b6	GPIO3B_IOMUX[13:12]=2'b01
mac_txd2	O	IO_MACtxd2_GPIO30gpio3b2	GPIO3B_IOMUX[5:4]=2'b01
mac_txd1	O	IO_MACtxd1_GPIO30gpio3b1	GPIO3B_IOMUX[3:2]=2'b01
mac_txd0	O	IO_MACtxd0_PWM0_VOPpwm_GPIO30gpio3b0	GPIO3B_IOMUX[1:0]=2'b01
mac_rxclk	I	IO_MACrxclkin_I2C4TPscl_GPIO30gpio3d1	GPIO3D_IOMUX[3:2]=2'b01
mac_rxdv	I	IO_MACrxdv_ISPflashtrigout_GPIO30gpio3c4	GPIO3C_IOMUX[9:8]=2'b01
mac_rxer	I	IO_MACrxer_ISPprelighttrig_GPIO30gpio3c5	GPIO3C_IOMUX[11:10]=2'b01
mac_rxd3	I	IO_MACrxd3_USBdrvvbus1_GPIO30gpio3c2	GPIO3C_IOMUX[5:4]=2'b01
mac_rxd2	I	IO_MACrxd2_UART3GPSrtsn_USBdrvvbus0_GPIO30gpio3c1	GPIO3C_IOMUX[3:2]=2'b01
mac_rxd1	I	IO_MACrxd1_UART3GPSctsn_GPSrfclk_GPIO30gpio3c0	GPIO3C_IOMUX[1:0]=2'b01
mac_rxd0	I	IO_MACrxd0_GPSsig_GPIO30gpio3b7	GPIO3B_IOMUX[15:14]=2'b01
mac_crs	I	IO_MACcrs_GPIO30gpio3b4	GPIO3B_IOMUX[7:6]=2'b01
mac_col	I	IO_MACcol_GPIO30gpio3b3	GPIO3B_IOMUX[9:8]=2'b01
Management interface			
mac_mdio	I/O	IO_MACmdio_I2C4TPscl_GPIO30gpio3d0	GPIO3D_IOMUX[1:0]=2'b01
mac_mdc	O	IO_MACmdc_ISPshutteren_GPIO30gpio3c3	GPIO3C_IOMUX[7:6]=2'b01

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41.6 Application Notes

41.6.1 Descriptors

The DMA in GMAC can communicate with Host driver through descriptor lists and data buffers. The DMA transfers data frames received by the core to the Receive Buffer in the Host memory, and Transmit data frames from the Transmit Buffer in the Host memory. Descriptors that reside in the Host memory act as pointers to these buffers.
There are two descriptor lists; one for reception, and one for transmission. The base address of each list is written into DMA RegistersRX_DESC_LIST_ADDR and TX_DESC_LIST_ADDR, respectively. A descriptor list is forward linked (either implicitly or explicitly). The last descriptor may point back to the first entry to create a ring structure. Explicit chaining of descriptors is accomplished by setting the second address chained in both Receive and Transmit descriptors (RDES1[24] and TDES1[24]). The descriptor lists resides in the Host physical memory address space. Each descriptor can point to a maximum of two buffers. This enables two buffers to be used, physically addressed, rather than contiguous buffers in memory.
A data buffer resides in the Host physical memory space, and consists of an entire frame or part of a frame, but cannot exceed a single frame. Buffers contain only data, buffer status is maintained in the descriptor. Data chaining refers to frames that span multiple data buffers. However, a single descriptor cannot span multiple frames. The DMA will skip to the next frame buffer when end-of-frame is detected. Data chaining can be enabled or disabled
The descriptor ring and chain structure is shown in following figure.

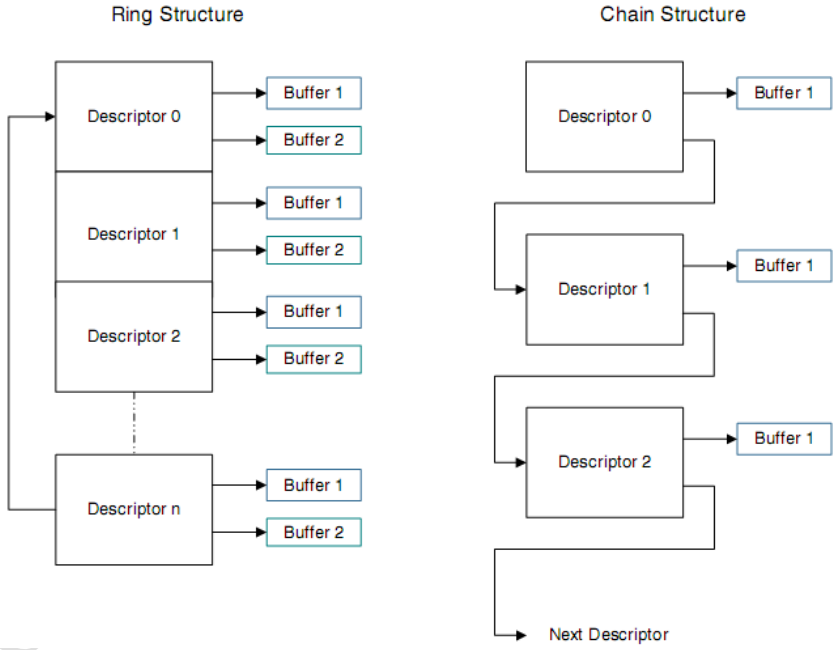

Fig. 41-10 Descriptor Ring and Chain Structure

Each descriptor contains two buffers, two byte-count buffers, and two address pointers, which enable the adapter port to be compatible with various types of memory management schemes. The descriptor addresses must be aligned to the bus width used (Word/Dword/Lword for 32/64/128-bit buses).

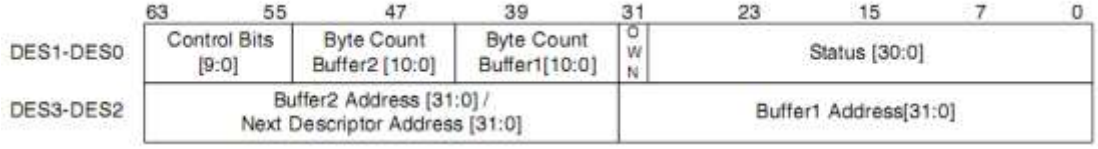

Fig. 41-11 Rx/Tx Descriptors definition

41.6.2 Receive Descriptor

The GMAC Subsystem requires at least two descriptors when receiving a frame. The Receive state machine of the DMA always attempts to acquire an extra descriptor in anticipation of an incoming frame. (The size of the incoming frame is unknown). Before the RxDMA closes a

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descriptor, it will attempt to acquire the next descriptor even if no frames are received.
In a single descriptor (receive) system, the subsystem will generate a descriptor error if the receive buffer is unable to accommodate the incoming frame and the next descriptor is not owned by the DMA. Thus, the Host is forced to increase either its descriptor pool or the buffer size. Otherwise, the subsystem starts dropping all incoming frames.

Receive Descriptor 0 (RDES0)

RDES0 contains the received frame status, the frame length, and the descriptor ownership information.

Table 41-3 Receive Descriptor 0

Bit	Description
31	OWN: Own Bit When set, this bit indicates that the descriptor is owned by the DMA of the GMAC Subsystem. When this bit is reset, this bit indicates that the descriptor is owned by the Host. The DMA clears this bit either when it completes the frame reception or when the buffers that are associated with this descriptor are full.
30	AFM: Destination Address Filter Fail When set, this bit indicates a frame that failed in the DA Filter in the GMAC Core.
29:16	FL: Frame Length These bits indicate the byte length of the received frame that was transferred to host memory (including CRC). This field is valid when Last Descriptor (RDES0[8]) is set and either the Descriptor Error (RDES0[14]) or Overflow Error bits are are reset. The frame length also includes the two bytes appended to the Ethernet frame when IP checksum calculation (Type 1) is enabled and the received frame is not a MAC control frame. This field is valid when Last Descriptor (RDES0[8]) is set. When the Last Descriptor and Error Summary bits are not set, this field indicates the accumulated number of bytes that have been transferred for the current frame.
15	ES: Error Summary Indicates the logical OR of the following bits: • RDES0[0]: Payload Checksum Error • RDES0[1]: CRC Error • RDES0[3]: Receive Error • RDES0[4]: Watchdog Timeout • RDES0[6]: Late Collision • RDES0[7]: IPC Checksum • RDES0[11]: Overflow Error • RDES0[14]: Descriptor Error This field is valid only when the Last Descriptor (RDES0[8]) is set.
14	DE: Descriptor Error When set, this bit indicates a frame truncation caused by a frame that does not fit within the current descriptor buffers, and that the DMA does not own the Next Descriptor. The frame is truncated. This field is valid only when the Last Descriptor (RDES0[8]) is set
13	SAF: Source Address Filter Fail When set, this bit indicates that the SA field of frame failed the SA Filter in the GMAC Core.
12	LE: Length Error When set, this bit indicates that the actual length of the frame received and that the Length/ Type field does not match. This bit is valid only when the Frame Type (RDES0[5]) bit is reset. Length error status is not valid when CRC error is present.
11	OE: Overflow Error When set, this bit indicates that the received frame was damaged due to buffer overflow.
10	VLAN: VLAN Tag When set, this bit indicates that the frame pointed to by this descriptor is a VLAN frame tagged by the GMAC Core.

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9	FS: First Descriptor When set, this bit indicates that this descriptor contains the first buffer of the frame. If the size of the first buffer is 0, the second buffer contains the beginning of the frame. If the size of the second buffer is also 0, the next Descriptor contains the beginning of the frame.
8	LS: Last Descriptor When set, this bit indicates that the buffers pointed to by this descriptor are the last buffers of the frame.
7	IPC Checksum Error/Giant Frame When IP Checksum Engine is enabled, this bit, when set, indicates that the 16-bit IPv4 Header checksum calculated by the core did not match the received checksum bytes. The Error Summary bit[15] is NOT set when this bit is set in this mode.
6	LC: Late Collision When set, this bit indicates that a late collision has occurred while receiving the frame in Half-Duplex mode.
5	FT: Frame Type When set, this bit indicates that the Receive Frame is an Ethernet-type frame (the LT field is greater than or equal to 16'h0600). When this bit is reset, it indicates that the received frame is an IEEE802.3 frame. This bit is not valid for Runt frames less than 14 bytes.
4	RWT: Receive Watchdog Timeout When set, this bit indicates that the Receive Watchdog Timer has expired while receiving the current frame and the current frame is truncated after the Watchdog Timeout.
3	RE: Receive Error When set, this bit indicates that the gmii_rxer_i signal is asserted while gmii_rxdv_i is asserted during frame reception. This error also includes carrier extension error in GMII and Half-duplex mode. Error can be of less/no extension, or error (rxd $\neq$ 0f) during extension.
2	DE: Dribble Bit Error When set, this bit indicates that the received frame has a non-integer multiple of bytes (odd nibbles). This bit is valid only in MII Mode.
1	CE: CRC Error When set, this bit indicates that a Cyclic Redundancy Check (CRC) Error occurred on the received frame. This field is valid only when the Last Descriptor (RDES0[8]) is set.
0	Rx MAC Address/Payload Checksum Error When set, this bit indicates that the Rx MAC Address registers value (1 to 15) matched the frame's DA field. When reset, this bit indicates that the Rx MAC Address Register 0 value matched the DA field. If Full Checksum Offload Engine is enabled, this bit, when set, indicates the TCP, UDP, or ICMP checksum the core calculated does not match the received encapsulated TCP, UDP, or ICMP segment's Checksum field. This bit is also set when the received number of payload bytes does not match the value indicated in the Length field of the encapsulated IPv4 or IPv6 datagram in the received Ethernet frame.

Receive Descriptor 1 (RDES1)

RDES1 contains the buffer sizes and other bits that control the descriptor chain/ring.

Table 41-4 Receive Descriptor 1

Bit	Description
31	Disable Interrupt on Completion When set, this bit will prevent the setting of the RI (CSR5[6]) bit of the GMAC_STATUS Register for the received frame that ends in the buffer pointed to by this descriptor. This, in turn, will disable the assertion of the interrupt to Host due to RI for that frame.
30:26	Reserved.
25	RER: Receive End of Ring When set, this bit indicates that the descriptor list reached its final descriptor. The

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	DMA returns to the base address of the list, creating a Descriptor Ring.
24	RCH: Second Address Chained When set, this bit indicates that the second address in the descriptor is the Next Descriptor address rather than the second buffer address. When RDES1[24] is set, RBS2 (RDES1[21-11]) is a "don't care" value. RDES1[25] takes precedence over RDES1[24].
23:22	Reserved.
21:11	RBS2: Receive Buffer 2 Size These bits indicate the second data buffer size in bytes. The buffer size must be a multiple of 8 depending upon the bus widths (64), even if the value of RDES3 (buffer2 address pointer) is not aligned to bus width. In the case where the buffer size is not a multiple of 8, the resulting behavior is undefined. This field is not valid if RDES1[24] is set.
10:0	RBS1: Receive Buffer 1 Size Indicates the first data buffer size in bytes. The buffer size must be a multiple of 8 depending upon the bus widths (64), even if the value of RDES2 (buffer1 address pointer) is not aligned. In the case where the buffer size is not a multiple of 8, the resulting behavior is undefined. If this field is 0, the DMA ignores this buffer and uses Buffer 2 or next descriptor depending on the value of RCH (Bit 24).

Receive Descriptor 2 (RDES2)

RDES2 contains the address pointer to the first data buffer in the descriptor.

Table 41-5 Receive Descriptor 2

Bit	Description
31:0	Buffer 1 Address Pointer These bits indicate the physical address of Buffer 1. There are no limitations on the buffer address alignment except for the following condition: The DMA uses the configured value for its address generation when the RDES2 value is used to store the start of frame. Note that the DMA performs a write operation with the RDES2[2:0] bits as 0 during the transfer of the start of frame but the frame data is shifted as per the actual Buffer address pointer. The DMA ignores RDES2[2:0] (corresponding to bus width of 64) if the address pointer is to a buffer where the middle or last part of the frame is stored.

Receive Descriptor 3 (RDES3)

RDES3 contains the address pointer either to the second data buffer in the descriptor or to the next descriptor.

Table 41-6 Receive Descriptor 3

Bit	Description
31:0	Buffer 2 Address Pointer (Next Descriptor Address) These bits indicate the physical address of Buffer 2 when a descriptor ring structure is used. If the Second Address Chained (RDES1[24]) bit is set, this address contains the pointer to the physical memory where the Next Descriptor is present. If RDES1[24] is set, the buffer (Next Descriptor) address pointer must be bus width-aligned (RDES3[2:0] = 0, corresponding to a bus width of 64. LSBs are ignored internally.) However, when RDES1[24] is reset, there are no limitations on the RDES3 value, except for the following condition: The DMA uses the configured value for its buffer address generation when the RDES3 value is used to store the start of frame. The DMA ignores RDES3[2:0] (corresponding to a bus width of 64) if the address pointer is to a buffer where the middle or last part of the frame is stored.

41.6.3 Transmit Descriptor

The descriptor addresses must be aligned to the bus width used (64). Each descriptor is

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provided with two buffers, two byte-count buffers, and two address pointers, which enable the adapter port to be compatible with various types of memory-management schemes.

Transmit Descriptor 0 (TDES0)

TDES0 contains the transmitted frame status and the descriptor ownership information.

Table 41-7 Transmit Descriptor 0

Bit	Description
31	OWN: Own Bit When set, this bit indicates that the descriptor is owned by the DMA. When this bit is reset, this bit indicates that the descriptor is owned by the Host. The DMA clears this bit either when it completes the frame transmission or when the buffers allocated in the descriptor are empty. The ownership bit of the First Descriptor of the frame should be set after all subsequent descriptors belonging to the same frame have been set. This avoids a possible race condition between fetching a descriptor and the driver setting an ownership bit.
30:17	Reserved.
16	IHE: IP Header Error When set, this bit indicates that the Checksum Offload engine detected an IP header error and consequently did not modify the transmitted frame for any checksum insertion.
15	ES: Error Summary Indicates the logical OR of the following bits: • TDES0[14]: Jabber Timeout • TDES0[13]: Frame Flush • TDES0[11]: Loss of Carrier • TDES0[10]: No Carrier • TDES0[9]: Late Collision • TDES0[8]: Excessive Collision • TDES0[2]: Excessive Deferral • TDES0[1]: Underflow Error
14	JT: Jabber Timeout When set, this bit indicates the GMAC transmitter has experienced a jabber time-out.
13	FF: Frame Flushed When set, this bit indicates that the DMA/MTL flushed the frame due to a SW flush command given by the CPU.
12	PCE: Payload Checksum Error This bit, when set, indicates that the Checksum Offload engine had a failure and did not insert any checksum into the encapsulated TCP, UDP, or ICMP payload. This failure can be either due to insufficient bytes, as indicated by the IP Header's Payload Length field, or the MTL starting to forward the frame to the MAC transmitter in Store-and-Forward mode without the checksum having been calculated yet. This second error condition only occurs when the Transmit FIFO depth is less than the length of the Ethernet frame being transmitted: to avoid deadlock, the MTL starts forwarding the frame when the FIFO is full, even in Store-and-Forward mode.
11	LC: Loss of Carrier When set, this bit indicates that Loss of Carrier occurred during frame transmission. This is valid only for the frames transmitted without collision and when the GMAC operates in Half-Duplex Mode.
10	NC: No Carrier When set, this bit indicates that the carrier sense signal form the PHY was not asserted during transmission.
9	LC: Late Collision When set, this bit indicates that frame transmission was aborted due to a collision occurring after the collision window (64 byte times including Preamble in RMII Mode and 512 byte times including Preamble and Carrier Extension in RGMII Mode). Not

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	valid if Underflow Error is set.
8	EC: Excessive Collision When set, this bit indicates that the transmission was aborted after 16 successive collisions while attempting to transmit the current frame. If the DR (Disable Retry) bit in the GMAC Configuration Register is set, this bit is set after the first collision and the transmission of the frame is aborted.
7	VF: VLAN Frame When set, this bit indicates that the transmitted frame was a VLAN-type frame.
6:3	CC: Collision Count This 4-bit counter value indicates the number of collisions occurring before the frame was transmitted. The count is not valid when the Excessive Collisions bit (TDES0[8]) is set.
2	ED: Excessive Deferral When set, this bit indicates that the transmission has ended because of excessive deferral of over 24,288 bit times (155,680 bits times in 1000-Mbps mode) if the Deferral Check (DC) bit is set high in the GMAC Control Register.
1	UF: Underflow Error When set, this bit indicates that the GMAC aborted the frame because data arrived late from the Host memory. Underflow Error indicates that the DMA encountered an empty Transmit Buffer while transmitting the frame. The transmission process enters the suspended state and sets both Transmit Underflow (Register GMAC_STATUS[5]) and Transmit Interrupt (Register GMAC_STATUS[0]).
0	DB: Deferred Bit When set, this bit indicates that the GMAC defers before transmission because of the presence of carrier. This bit is valid only in Half-Duplex mode.

Transmit Descriptor 1 (TDES1)

TDES1 contains the buffer sizes and other bits which control the descriptor chain/ring and the frame being transferred.

Table 41-8 Transmit Descriptor 1

Bit	Description
31	IC: Interrupt on Completion When set, this bit sets Transmit Interrupt (Register 5[0]) after the present frame has been transmitted.
30	LS: Last Segment When set, this bit indicates that the buffer contains the last segment of the frame.
29	FS: First Segment When set, this bit indicates that the buffer contains the first segment of a frame.
28:27	CIC: Checksum Insertion Control These bits control the insertion of checksums in Ethernet frames that encapsulate TCP, UDP, or ICMP over IPv4 or IPv6 as described below. • 2'b00: Do nothing. Checksum Engine is bypassed • 2'b01: Insert IPv4 header checksum. Use this value to insert IPv4 header checksum when the frame encapsulates an IPv4 datagram. • 2'b10: Insert TCP/UDP/ICMP checksum. The checksum is calculated over the TCP, UDP, or ICMP segment only and the TCP, UDP, or ICMP pseudo-header checksum is assumed to be present in the corresponding input frame's Checksum field. An IPv4 header checksum is also inserted if the encapsulated datagram conforms to IPv4. • 2'b11: Insert a TCP/UDP/ICMP checksum that is fully calculated in this engine. In other words, the TCP, UDP, or ICMP pseudo-header is included in the checksum calculation, and the input frame's corresponding Checksum field has an all-zero value. An IPv4 Header checksum is also inserted if the encapsulated datagram conforms to IPv4. The Checksum engine detects whether the TCP, UDP, or ICMP segment is encapsulated in IPv4 or IPv6 and processes its data accordingly.

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26	DC: Disable CRC When set, the GMAC does not append the Cyclic Redundancy Check (CRC) to the end of the transmitted frame. This is valid only when the first segment (TDES1[29]).
25	TER: Transmit End of Ring When set, this bit indicates that the descriptor list reached its final descriptor. The returns to the base address of the list, creating a descriptor ring.
24	TCH: Second Address Chained When set, this bit indicates that the second address in the descriptor is the Next Descriptor address rather than the second buffer address. When TDES1[24] is set, TBS2 (TDES1[21–11]) are "don't care" values. TDES1[25] takes precedence over TDES1[24].
23	DP: Disable Padding When set, the GMAC does not automatically add padding to a frame shorter than 64 bytes. When this bit is reset, the DMA automatically adds padding and CRC to a frame shorter than 64 bytes and the CRC field is added despite the state of the DC (TDES1[26]) bit. This is valid only when the first segment (TDES1[29]) is set.
22	Reserved.
21:11	TBS2: Transmit Buffer 2 Size These bits indicate the Second Data Buffer in bytes. This field is not valid if TDES1[24] is set.
10:0	TBS1: Transmit Buffer 1 Size These bits indicate the First Data Buffer byte size. If this field is 0, the DMA ignores this buffer and uses Buffer 2 or next descriptor depending on the value of TCH (Bit 24).

Transmit Descriptor 2 (TDES2)

TDES2 contains the address pointer to the first buffer of the descriptor.

Table 41-9 Transmit Descriptor 2

Bit	Description
31:0	Buffer 1 Address Pointer These bits indicate the physical address of Buffer 1. There is no limitation on the buffer address alignment.

Transmit Descriptor 3 (TDES3)

TDES3 contains the address pointer either to the second buffer of the descriptor or the next descriptor.

Table 41-10 Transmit Descriptor 3

Bit	Description
31:0	Buffer 2 Address Pointer (Next Descriptor Address) Indicates the physical address of Buffer 2 when a descriptor ring structure is used. If the Second Address Chained (TDES1[24]) bit is set, this address contains the pointer to the physical memory where the Next Descriptor is present. The buffer address pointer must be aligned to the bus width only when TDES1[24] is set. (LSBs are ignored internally.)

41.6.4 Programming Guide

DMA Initialization – Descriptors

The following operations must be performed to initialize the DMA.

1. Provide a software reset. This will reset all of the GMAC internal registers and logic. (GMAC_OP_MODE[0]).
2. Wait for the completion of the reset process (poll GMAC_OP_MODE[0], which is only cleared after the reset operation is completed).
3. Program the following fields to initialize the Bus Mode Register by setting values in register GMAC_BUS_MODE
   a. Mixed Burst and AAL

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b. Fixed burst or undefined burst
c. Burst length values and burst mode values.
d. Descriptor Length (only valid if Ring Mode is used)
e. Tx and Rx DMA Arbitration scheme

4. Program the AXI Interface options in the register GMAC_BUS_MODE
a. If fixed burst-length is enabled, then select the maximum burst-length possible on the AXI bus (Bits[7:1])
5. A proper descriptor chain for transmit and receive must be created. It should also ensure that the receive descriptors are owned by DMA (bit 31 of descriptor should be set). When OSF mode is used, at least two descriptors are required.
6. Software should create three or more different transmit or receive descriptors in the chain before reusing any of the descriptors.
7. Initialize receive and transmit descriptor list address with the base address of transmit and receive descriptor (register GMAC_RX_DESC_LIST_ADDR and GMAC_TX_DESC_LIST_ADDR).
8. Program the following fields to initialize the mode of operation by setting values in register GMAC_OP_MODE
a. Receive and Transmit Store And Forward
b. Receive and Transmit Threshold Control (RTC and TTC)
c. Hardware Flow Control enable
d. Flow Control Activation and De-activation thresholds for MTL Receive and Transmit FIFO (RFA and RFD)
e. Error Frame and undersized good frame forwarding enable
f. OSF Mode
9. Clear the interrupt requests, by writing to those bits of the status register (interrupt bits only) which are set. For example, by writing 1 into bit 16 - normal interrupt summary will clear this bit (register GMAC_STATUS).
10. Enable the interrupts by programming the interrupt enable register GMAC_INT_ENA.
11. Start the Receive and Transmit DMA by setting SR (bit 1) and ST (bit 13) of the control register GMAC_OP_MODE.

MAC Initialization

The following MAC Initialization operations can be performed after the DMA initialization sequence. If the MAC Initialization is done before the DMA is set-up, then enable the MAC receiver (last step below) only after the DMA is active. Otherwise, received frames will fill the RxFIFO and overflow. Steps (1) and (2) are to be followed if the TBI/SGMII/RTBI PHY interface is enabled, otherwise follow steps (3) and (4).
1. Program the AN Control register GMAC_AN_CTRL to enable Auto-negotiation ANE (bit-12). Setting ELE (bit-14) of this register will enable the PHY to loop back the transmit data.
2. Check the AN Status Register GMAC_AN_STATUS for completion of the Auto-negotiation process. ANC (bit-5) should be set, and link status (bit-2), when set, indicates that the link is up.
3. Program the register GMAC_GMII_ADDR for controlling the management cycles for external PHY, for example, Physical Layer Address PA (bits 15-11). Also set bit 0 (GMII Busy) for writing into PHY and reading from PHY.
4. Read the 16-bit data of (GMAC_GMII_DATA) from the PHY for link up, speed of operation, and mode of operation, by specifying the appropriate address value in registerGMAC_GMII_ADDR (bits 15-11).
5. Provide the MAC address registers (GMAC_MAC_ADDR0_HI and GMAC_MAC_ADDR0_LO). If more than one MAC address is enabled in your configuration (during configuration in coreConsultant), program them appropriately).
6. If Hash filtering is enabled in your configuration, program the Hash filter register (GMAC_HASH_TAB_HI and GMAC_HASH_TAB_LO).
7. Program the following fields to set the appropriate filters for the incoming frames in register GMAC_MAC_FRM_FILT
a. Receive All
b. Promiscuous mode

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c. Hash or Perfect Filter
d. Unicast, Multicast, broad cast and control frames filter settings etc.
8. Program the following fields for proper flow control in register GMAC_FLOW_CTRL.
a. Pause time and other pause frame control bits
b. Receive and Transmit Flow control bits
c. Flow Control Busy/Backpressure Activate
9. Program the Interrupt Mask register bits, as required, and if applicable, for your configuration.
10. Program the appropriate fields in register GMAC_MAC_CONF for example, Inter-frame gap while transmission, jabber disable, etc. Based on the Auto-negotiation you can set the Duplex mode (bit 11), port select (bit 15), etc.
11. Set the bits Transmit enable (TE bit-3) and Receive Enable (RE bit-2) in register GMAC_MAC_CONF.

Normal Receive and Transmit Operation
For normal operation, the following steps can be followed.
- For normal transmit and receive interrupts, read the interrupt status. Then poll the descriptors, reading the status of the descriptor owned by the Host (either transmit or receive).
- On completion of the above step, set appropriate values for the descriptors, ensuring that transmit and receive descriptors are owned by the DMA to resume the transmission and reception of data.
- If the descriptors were not owned by the DMA (or no descriptor is available), the DMA will go into SUSPEND state. The transmission or reception can be resumed by freeing the descriptors and issuing a poll demand by writing 0 into the Tx/Rx poll demand register (GMAC_TX_POLL_DEMAND and GMAC_RX_POLL_DEMAND).
- The values of the current host transmitter or receiver descriptor address pointer can be read for the debug process (GMAC_CUR_HOST_TX_DESC and GMAC_CUR_HOST_RX_DESC).
- The values of the current host transmit buffer address pointer and receive buffer address pointer can be read for the debug process (GMAC_CUR_HOST_TX_Buf_ADDR and GMAC_CUR_HOST_RX_BUF_ADDR).

Stop and Start Operation
When the transmission is required to be paused for some time then the following steps can be followed.
1. Disable the Transmit DMA (if applicable), by clearing ST (bit 13) of the control register GMAC_OP_MODE.
2. Wait for any previous frame transmissions to complete. This can be checked by reading the appropriate bits of MAC Debug register.
3. Disable the MAC transmitter and MAC receiver by clearing the bits Transmit enable (TE bit-3) and Receive Enable (RE bit-2) in register GMAC_MAC_CONF.
4. Disable the Receive DMA (if applicable), after making sure the data in the RX FIFO is transferred to the system memory (by reading the register GMAC_DEBUG).
5. Make sure both the TX FIFO and RX FIFO are empty.
6. To re-start the operation, start the DMAs first, before enabling the MAC Transmitter and Receiver.

41.6.5 Clock Architecture

In RMII mode, reference clock and TX/RX clock can be from CRU or external OSC as following figure.
The mux select rmii_speed is CRU_CLKSEL43_CON[8].

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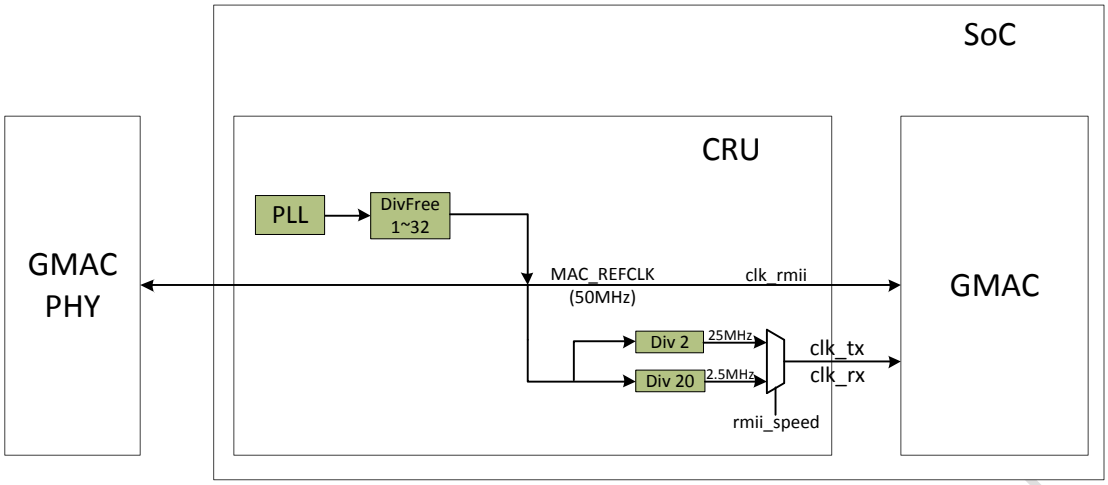

Fig. 41-12 RMII clock architecture when clock source from CRU

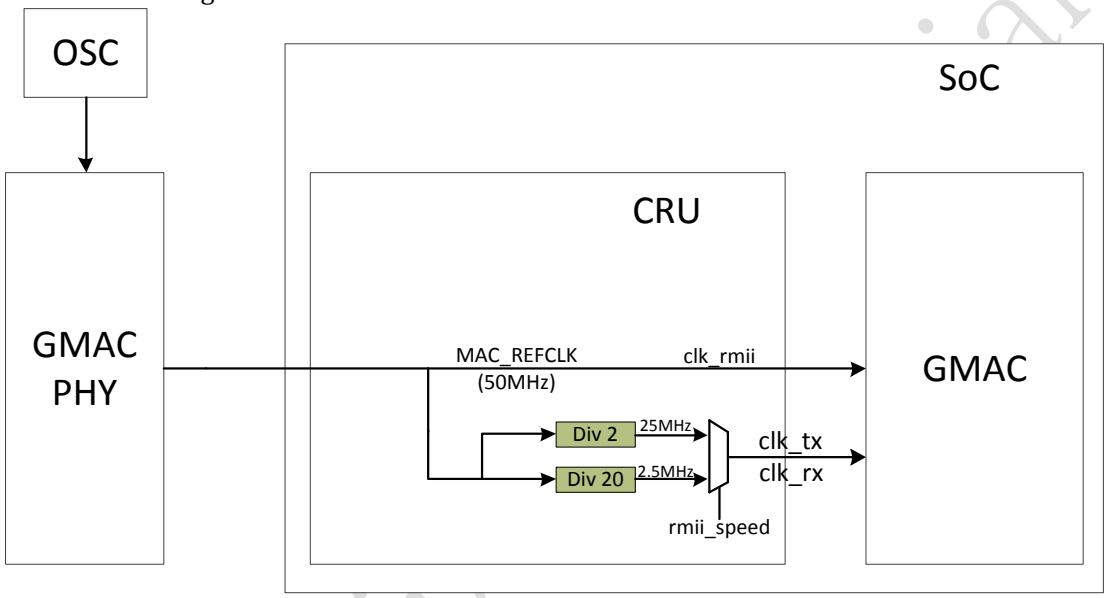

Fig. 41-13 RMII clock architecture when clock source from external OSC

In RGMII mode, clock architecture only supports that TX clock source is from CRU as following figure.

In order to dynamically adjust the timing between TX/RX clock with data, delay line is integrated in TX and RX clock path. Register GRF_SOC_CON16[7] and GRF_SOC_CON16[15] can enable the delay lines, and GRF_SOC_CON16[6:0] and GRF_SOC_CON16[14:8] are used to determine the delay length. There are 100 delay elements in each delay line, and it totally can adjust about 5.1ns typically.

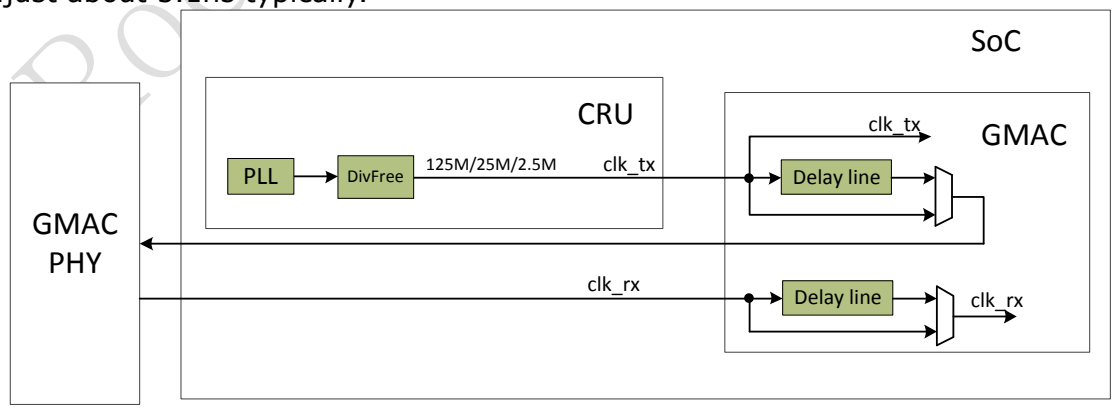

Fig. 41-14 RGMII clock architecture when clock source from CRU

41.6.6 Remote Wake-Up Frame Filter Register

The register wkupfmfilter_reg, address (028H), loads the Wake-up Frame Filter register. To load values in a Wake-up Frame Filter register, the entire register (wkupfmfilter_reg) must be written. The wkupfmfilter_reg register is loaded by sequentially loading the eight register

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values in address (028) for wkupfmfilter_reg0, wkupfmfilter_reg1, ... wkupfmfilter_reg7, respectively. wkupfmfilter_reg is read in the same way.
The internal counter to access the appropriate wkupfmfilter_reg is incremented when lane3 (or lane 0 in big-endian) is accessed by the CPU. This should be kept in mind if you are accessing these registers in byte or half-word mode.

wkupfmfilter_reg0	Filter 0 Byte Mask							
wkupfmfilter_reg1	Filter 1 Byte Mask							
wkupfmfilter_reg2	Filter 2 Byte Mask							
wkupfmfilter_reg3	Filter 3 Byte Mask							
wkupfmfilter_reg4	RSVD	Filter 3 Command	RSVD	Filter 2 Command	RSVD	Filter 1 Command	RSVD	Filter 0 Command
wkupfmfilter_reg5	Filter 3 Offset		Filter 2 Offset		Filter 1 Offset		Filter 0 Offset	
wkupfmfilter_reg6	Filter 1 CRC - 16				Filter 0 CRC - 16			
wkupfmfilter_reg7	Filter 3 CRC - 16				Filter 2 CRC - 16			

Fig. 41-15 Wake-Up Frame Filter Register

Filter i Byte Mask
This register defines which bytes of the frame are examined by filter i (0, 1, 2, and 3) in order to determine whether or not the frame is a wake-up frame. The MSB (thirty-first bit) must be zero. Bit j [30:0] is the Byte Mask. If bit j (byte number) of the Byte Mask is set, then Filter i Offset + j of the incoming frame is processed by the CRC block; otherwise Filter i Offset + j is ignored.
Filter i Command
This 4-bit command controls the filter i operation. Bit 3 specifies the address type, defining the pattern's destination address type. When the bit is set, the pattern applies to only multicast frames; when the bit is reset, the pattern applies only to unicast frame. Bit 2 and Bit 1 are reserved. Bit 0 is the enable for filter i; if Bit 0 is not set, filter i is disabled.
Filter i Offset
This register defines the offset (within the frame) from which the frames are examined by filter i. This 8-bit pattern-offset is the offset for the filter i first byte to examined. The minimum allowed is 12, which refers to the 13th byte of the frame (offset value 0 refers to the first byte of the frame).
Filter i CRC-16
This register contains the CRC_16 value calculated from the pattern, as well as the byte mask programmed to the wake-up filter register block.

41.6.7 System Consideration During Power-Down

GMAC neither gates nor stops clocks when Power-Down mode is enabled. Power saving by clock gating must be done outside the core by the CRU. The receive data path must be clocked with clk_rx_i during Power-Down mode, because it is involved in magic packet/wake-on-LAN frame detection. However, the transmit path and the APB path clocks can be gated off during Power-Down mode.
The pmt interrupt is asserted when a valid wake-up frame is received. This interrupt is generated in the clk_rx domain.
The recommended power-down and wake-up sequence is as follows.
1. Disable the Transmit DMA (if applicable) and wait for any previous frame transmissions to complete. These transmissions can be detected when Transmit Interrupt (TI-Register GMAC_STATUS[0]) is received.
2. Disable the MAC transmitter and MAC receiver by clearing the appropriate bits in the MAC Configuration register.
3. Wait until the Receive DMA empties all the frames from the Rx FIFO (a software timer may be required).
4. Enable Power-Down mode by appropriately configuring the PMT registers.
5. Enable the MAC Receiver and enter Power-Down mode.
6. Gate the APB and transmit clock inputs to the core (and other relevant clocks in the system) to reduce power and enter Sleep mode.

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7. On receiving a valid wake-up frame, the GMAC asserts the pmt interrupt signal and exits Power-Down mode.
8. On receiving the interrupt, the system must enable the APB and transmit clock inputs to the core.
9. Read the register GMAC_PMT_CTRL_STA to clear the interrupt, then enable the other modules in the system and resume normal operation.

41.6.8 GRF Register Summary

GRF Register	Register Description
GRF_SOC_CON15[11:9]	PHY interface select 3'b001: RGMII 3'b100: RMII All others: Reserved
GRF_SOC_CON15[8]	GMAC transmit flow control When set high, instructs the GMAC to transmit PAUSE Control frames in Full-duplex mode. In Half-duplex mode, the GMAC enables the Back-pressure function until this signal is made low again
GRF_SOC_CON15[7]	gmac_speed 1'b1: 100-Mbps 1'b0: 10-Mbps
GRF_SOC_CON15[3]	RMII clock selection 1'b1: 25MHz 1'b0: 2.5MHz
GRF_SOC_CON15[5:4]	RGMII clock selection 2'b00: 125MHz 2'b11: 25MHz 2'b10: 2.5MHz
GRF_SOC_CON15[6]	RMII mode selection 1'b1: RMII mode 1’b0: Reserved
GRF_SOC_CON16[6:0]	RGMII TX clock delay line value
GRF_SOC_CON16[14:8]	RGMII RX clock delay line value
GRF_SOC_CON16[7]	RGMII TX clock delay line enable 1'b1: enable 1'b0: disable
GRF_SOC_CON16[15]	RGMII RX clock delay line enable 1'b1: enable 1'b0: disable

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Chapter 42 Serial Peripheral Interface (SPI)

42.1 Overview

The serial peripheral interface is an APB slave device. A four wire full duplex serial protocol from Motorola. There are four possible combinations for the serial clock phase and polarity. The clock phase (SCPH) determines whether the serial transfer begins with the falling edge of slave select signals or the first edge of the serial clock. The slave select line is held high when the SPI is idle or disabled. This SPI controller can work as either master or slave mode.
SPI Controller supports the following features:

- Support Motorola SPI,TI Synchronous Serial Protocol and National Semiconductor Micro wire interface
- Support 32-bit APB bus
- Support two internal 16-bit wide and 32-location deep FIFOs, one for transmitting and the other for receiving serial data
- Support two chip select signals in master mode
- Support 4,8,16 bit serial data transfer
- Support configurable interrupt polarity
- Support asynchronous APB bus and SPI clock
- Support master and slave mode
- Support DMA handshake interface and configurable DMA water level
- Support transmit FIFO empty, underflow, receive FIFO full, overflow, interrupt and all interrupts can be masked
- Support configurable water level of transmit FIFO empty and receive FIFO full interrupt
- Support combine interrupt output
- Support up to half of SPI clock frequency transfer in master mode and one sixth of SPI clock frequency transfer in slave mode
- Support full and half duplex mode transfer
- Stop transmitting SCLK if transmit FIFO is empty or receive FIFO is full in master mode
- Support configurable delay from chip select active to SCLK active in master mode
- Support configurable period of chip select inactive between two parallel data in master mode
- Support big and little endian, MSB and LSB first transfer
- Support two 8-bit audio data store together in one 16-bit wide location
- Support sample RXD 0~3 SPI clock cycles later
- Support configurable SCLK polarity and phase
- Support fix and incremental address access to transmit and receive FIFO

42.2 Block Diagram

The SPI Controller comprises with:

- AMBA APB interface and DMA Controller Interface
- Transmit and receive FIFO controllers and an FSM controller
- Register block
- Shift control and interrupt

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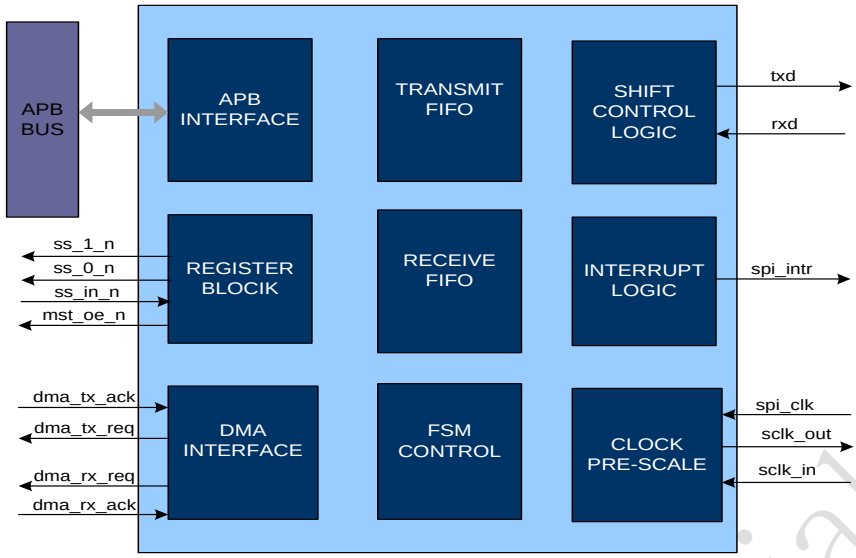

Fig. 42-1 SPI Controller Block diagram

APB INTERFACE

The host processor accesses data, control, and status information on the SPI through the APB interface. The SPI supports APB data bus widths of 8, 16, and 32 bits.

DMA INTERFACE

This block has a handshaking interface to a DMA Controller to request and control transfers. The APB bus is used to perform the data transfer to or from the DMA Controller.

FIFO LOGIC

For transmit and receive transfers, data transmitted from the SPI to the external serial device is written into the transmit FIFO. Data received from the external serial device into the SPI is pushed into the receive FIFO. Both fifos are 32x16bits.

FSM CONTROL

Control the state's transformation of the design.

REGISTER BLOCK

All registers in the SPI are addressed at 32-bit boundaries to remain consistent with the AHB bus. Where the physical size of any register is less than 32-bits wide, the upper unused bits of the 32-bit boundary are reserved. Writing to these bits has no effect; reading from these bits returns 0.

SHIFT CONTROL

Shift control logic shift the data from the transmit fifo or to the receive fifo. This logic automatically right-justifies receive data in the receive FIFO buffer.

INTERRUPT CONTROL

The SPI supports combined and individual interrupt requests, each of which can be masked. The combined interrupt request is the ORed result of all other SPI interrupts after masking.

42.3 Function description

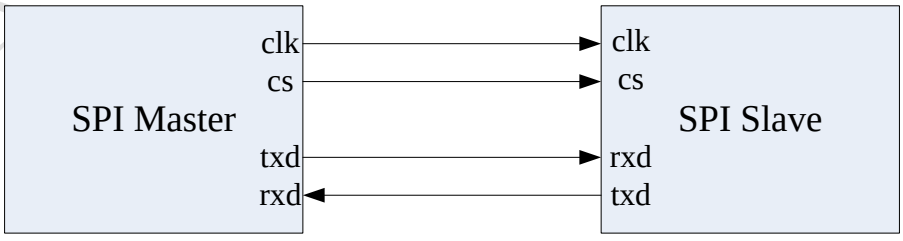

Fig. 42-2 SPI Master and Slave Interconnection

The SPI controller support dynamic switching between master and slave in a system. The diagram show how the SPI controller connects with other SPI devices.

Operation Modes

The SPI can be configured in the following two fundamental modes of operation: Master Mode when SPI_CTRLR0 [20] is 1'b0, Slave Mode when SPI_CTRLR0 [20] is 1'b1.

Transfer Modes

The SPI operates in the following three modes when transferring data on the serial bus.

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1. Transmit and Receive
When SPI_CTRLR0 [19:18]== 2'b00, both transmit and receive logic are valid.
2.Transmit Only
When SPI_CTRLR0 [19:18] == 2'b01, the receive data are invalid and should not be stored in the receive FIFO.
3.Receive Only
When SPI_CTRLR0 [19:18]== 2'b10, the transmit data are invalid.
Clock Ratios
A summary of the frequency ratio restrictions between the bit-rate clock (sclk_out/sclk_in) and the SPI peripheral clock (spi_clk) are described as,
When SPI Controller works as master, the F_{spi_clk}>= 2 × (maximum F_{sclk_out})
When SPI Controller works as slave, the F_{spi_clk}>= 6 × (maximum F_{sclk_in})
With the SPI, the clock polarity (SCPOL) configuration parameter determines whether the inactive state of the serial clock is high or low. To transmit data, both SPI peripherals must have identical serial clock phase (SCPH) and clock polarity (SCPOL) values. The data frame can be 4/8/16 bits in length.
When the configuration parameter SCPH = 0, data transmission begins on the falling edge of the slave select signal. The first data bit is captured by the master and slave peripherals on the first edge of the serial clock; therefore, valid data must be present on the txd and rxd lines prior to the first serial clock edge. The following two figures show a timing diagram for a single SPI data transfer with SCPH = 0. The serial clock is shown for configuration parameters SCPOL = 0 and SCPOL = 1.

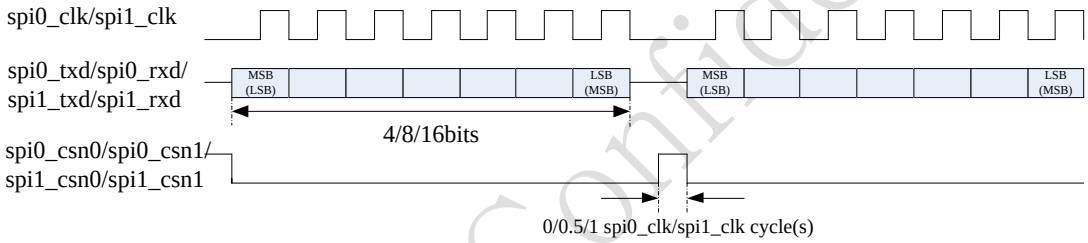

Fig. 42-3 SPI Format (SCPH=0 SCPOL=0)

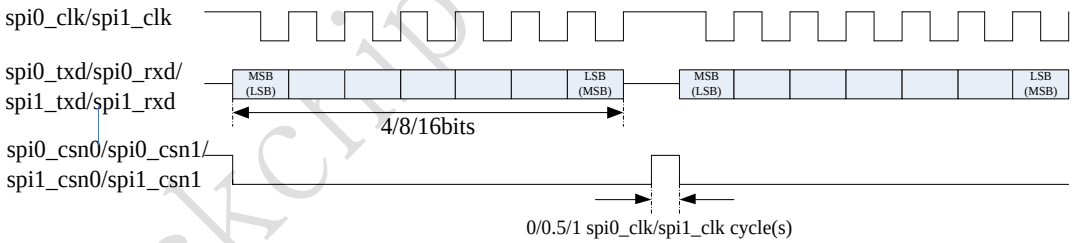

Fig. 42-4 SPI Format (SCPH=0 SCPOL=1)

When the configuration parameter SCPH = 1, both master and slave peripherals begin transmitting data on the first serial clock edge after the slave select line is activated. The first data bit is captured on the second (trailing) serial clock edge. Data are propagated by the master and slave peripherals on the leading edge of the serial clock. During continuous data frame transfers, the slave select line may be held active-low until the last bit of the last frame has been captured. The following two figures show the timing diagram for the SPI format when the configuration parameter SCPH = 1.

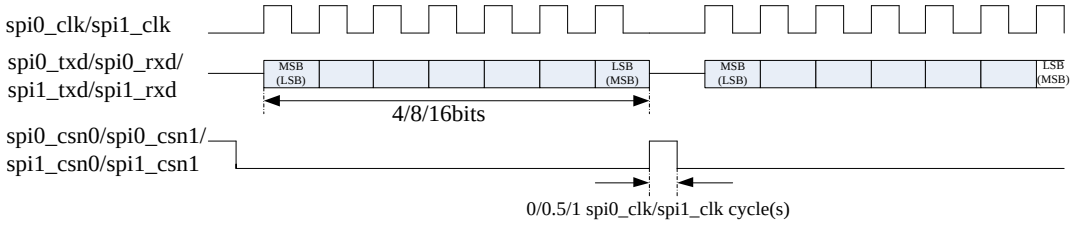

Fig. 42-5 SPI Format (SCPH=1 SCPOL=0)

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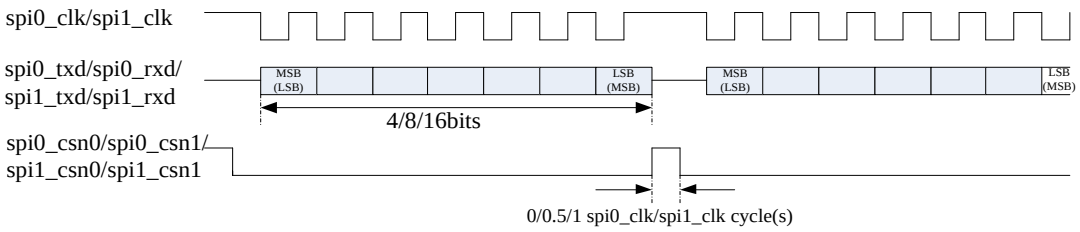

Fig. 42-6 SPI Format (SCPH=1 SCPOL=1)

42.4 Register Description

This section describes the control/status registers of the design. Pay attention that there are two SPI controllers in the chip: spi0 & spi1, so the base address in the following register descriptions can be either spi0 or spi1 base address.

42.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
SPI_CTRLR0	0x0000	W	0x00000002	Control Register 0
SPI_CTRLR1	0x0004	W	0x00000000	Control Register 1
SPI_ENR	0x0008	W	0x00000000	SPI Enable
SPI_SER	0x000c	W	0x00000000	Slave Enable Register
SPI_BAUDR	0x0010	W	0x00000000	Baud Rate Select
SPI_TXFTLR	0x0014	W	0x00000000	Transmit FIFO Threshold Level
SPI_RXFTLR	0x0018	W	0x00000000	Receive FIFO Threshold Level
SPI_TXFLR	0x001c	W	0x00000000	Transmit FIFO Level
SPI_RXFLR	0x0020	W	0x00000000	Receive FIFO Level
SPI_SR	0x0024	W	0x0000000c	SPI Status
SPI_IPR	0x0028	W	0x00000000	Interrupt Polarity
SPI_IMR	0x002c	W	0x00000000	Interrupt Mask
SPI_ISR	0x0030	W	0x00000000	Interrupt Status
SPI_RISR	0x0034	W	0x00000001	Raw Interrupt Status
SPI_ICR	0x0038	W	0x00000000	Interrupt Clear
SPI_DMACR	0x003c	W	0x00000000	DMA Control
SPI_DMATDLR	0x0040	W	0x00000000	DMA Transmit Data Level
SPI_DMARDLR	0x0044	W	0x00000000	DMA Receive Data Level
SPI_TXDR	0x0400~ 0x07fc	W	0x00000000	Transmit FIFO Data
SPI_RXDR	0x0800~ 0x0bfc	W	0x00000000	Receive FIFO Data

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

42.4.2 Detail Register Description

SPI_CTRLR0

Address: Operational Base + offset (0x0000)

Control Register 0

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
21	RW	0x0	MTM Microwire Transfer Mode Valid when frame format is set to National Semiconductors Microwire. 1'b0: non-sequential transfer 1'b1: sequential transfer
20	RW	0x0	OPM Operation Mode 1'b0: Master Mode 1'b1: Slave Mode
19:18	RW	0x0	XFM Transfer Mode 2'b00 :Transmit & Receive 2'b01 : Transmit Only 2'b10 : Receive Only 2'b11 :reserved
17:16	RW	0x0	FRF Frame Format 2'b00: Motorola SPI 2'b01: Texas Instruments SSP 2'b10: National Semiconductors Microwire 2'b11 : Reserved
15:14	RW	0x0	RSD Rxd Sample Delay When SPI is configured as a master, if the rxd data cannot be sampled by the sclk_out edge at the right time, this register should be configured to define the number of the spi_clk cycles after the active sclk_out edge to sample rxd data later when SPI works at high frequency. 2'b00:do not delay 2'b01:1 cycle delay 2'b10:2 cycles delay 2'b11:3 cycles delay
13	RW	0x0	BHT Byte and Halfword Transform Valid when data frame size is 8bit. 1'b0:apb 16bit write/read, spi 8bit write/read 1'b1: apb 8bit write/read, spi 8bit write/read
12	RW	0x0	FBM First Bit Mode 1'b0:first bit is MSB 1'b1:first bit is LSB

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Bit	Attr	Reset Value	Description
11	RW	0x0	EM Endian Mode Serial endian mode can be configured by this bit. Apb endian mode is always little endian. 1'b0:little endian 1'b1:big endian
10	RW	0x0	SSD ss_n to sclk_out delay Valid when the frame format is set to Motorola SPI and SPI used as a master. 1'b0: the period between ss_n active and sclk_out active is half sclk_out cycles. 1'b1: the period between ss_n active and sclk_out active is one sclk_out cycle.
9:8	RW	0x0	CSM Chip Select Mode Valid when the frame format is set to Motorola SPI and SPI used as a master. 2'b00: ss_n keep low after every frame data is transferred. 2'b01:ss_n be high for half sclk_out cycles after every frame data is transferred. 2'b10: ss_n be high for one sclk_out cycle after every frame data is transferred. 2'b11:reserved
7	RW	0x0	SCPOL Serial Clock Polarity Valid when the frame format is set to Motorola SPI. 1'b0: Inactive state of serial clock is low 1'b1: Inactive state of serial clock is high
6	RW	0x0	SCPH Serial Clock Phase Valid when the frame format is set to Motorola SPI. 1'b0: Serial clock toggles in middle of first data bit 1'b1: Serial clock toggles at start of first data bit

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Bit	Attr	Reset Value	Description
5:2	RW	0x0	CFS Control Frame Size Selects the length of the control word for the Microwire frame format. 4'b0000~4'b0010:reserved 4'b0011:4-bit serial data transfer 4'b0100:5-bit serial data transfer 4'b0101:6-bit serial data transfer 4'b0110:7-bit serial data transfer 4'b0111:8-bit serial data transfer 4'b1000:9-bit serial data transfer 4'b1001:10-bit serial data transfer 4'b1010:11-bit serial data transfer 4'b1011:12-bit serial data transfer 4'b1100:13-bit serial data transfer 4'b1101:14-bit serial data transfer 4'b1110:15-bit serial data transfer 4'b1111:16-bit serial data transfer
1:0	RW	0x2	DFS Data Frame Size Selects the data frame length. 2'b00:4bit data 2'b01:8bit data 2'b10:16bit data 2'b11:reserved

SPI_CTRLR1
Address: Operational Base + offset (0x0004)
Control Register 1

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	NDM Number of Data Frames When Transfer Mode is receive only, this register field sets the number of data frames to be continuously received by the SPI. The SPI continues to receive serial data until the number of data frames received is equal to this register value plus 1, which enables you to receive up to 64 KB of data in a continuous transfer.

SPI_ENR
Address: Operational Base + offset (0x0008)
SPI Enable

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Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	ENR SPI Enable Enables and disables all SPI operations. Transmit and receive FIFO buffers are cleared when the device is disabled.

SPI_SER
Address: Operational Base + offset (0x000c)
Slave Enable Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x0	SER Slave Select Enable This register is valid only when SPI is configured as a master device.

SPI_BAUDR
Address: Operational Base + offset (0x0010)
Baud Rate Select

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	BAUDR Baud Rate Select SPI Clock Divider. This register is valid only when the SPI is configured as a master device. The LSB for this field is always set to 0 and is unaffected by a write operation, which ensures an even value is held in this register. If the value is 0, the serial output clock (sclk_out) is disabled. The frequency of the sclk_out is derived from the following equation: Fsclk_out = Fspi_clk/ SCKDV Where SCKDV is any even value between 2 and 65534. For example: for Fspi_clk = 3.6864MHz and SCKDV =2 Fsclk_out = 3.6864/2= 1.8432MHz

SPI_TXFTLR
Address: Operational Base + offset (0x0014)
Transmit FIFO Threshold Level

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x00	TXFTLR Transmit FIFO Threshold Level When the number of transmit FIFO entries is less than or equal to this value, the transmit FIFO empty interrupt is triggered.

SPI_RXFTLR
Address: Operational Base + offset (0x0018)
Receive FIFO Threshold Level

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x00	RXFTLR Receive FIFO Threshold Level When the number of receive FIFO entries is greater than or equal to this value + 1, the receive FIFO full interrupt is triggered.

SPI_TXFLR
Address: Operational Base + offset (0x001c)
Transmit FIFO Level

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RO	0x00	TXFLR Transmit FIFO Level Contains the number of valid data entries in the transmit FIFO.

SPI_RXFLR
Address: Operational Base + offset (0x0020)
Receive FIFO Level

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RO	0x00	RXFLR Receive FIFO Level Contains the number of valid data entries in the receive FIFO.

SPI_SR
Address: Operational Base + offset (0x0024)
SPI Status

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4	RO	0x0	RFF Receive FIFO Full 1'b0: Receive FIFO is not full 1'b1: Receive FIFO is full
3	RO	0x1	RFE Receive FIFO Empty 1'b0: Receive FIFO is not empty 1'b1: Receive FIFO is empty
2	RO	0x1	TFE Transmit FIFO Empty 1'b0: Transmit FIFO is not empty 1'b1: Transmit FIFO is empty
1	RO	0x0	TFF Transmit FIFO Full 1'b0: Transmit FIFO is not full 1'b1: Transmit FIFO is full
0	RO	0x0	BSF SPI Busy Flag When set, indicates that a serial transfer is in progress; when cleared indicates that the SPI is idle or disabled. 1'b0: SPI is idle or disabled 1'b1: SPI is actively transferring data

SPI_IPR
Address: Operational Base + offset (0x0028)
Interrupt Polarity

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	IPR Interrupt Polarity Interrupt Polarity Register 1'b0:Active Interrupt Polarity Level is HIGH 1'b1: Active Interrupt Polarity Level is LOW

SPI_IMR
Address: Operational Base + offset (0x002c)
Interrupt Mask

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	RFFIM Receive FIFO Full Interrupt Mask 1'b0: spi_rxf_intr interrupt is masked 1'b1: spi_rxf_intr interrupt is not masked

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Bit	Attr	Reset Value	Description
3	RW	0x0	RFOIM Receive FIFO Overflow Interrupt Mask 1'b0: spi_rxo_intr interrupt is masked 1'b1: spi_rxo_intr interrupt is not masked
2	RW	0x0	RFUIM Receive FIFO Underflow Interrupt Mask 1'b0: spi_rxu_intr interrupt is masked 1'b1: spi_rxu_intr interrupt is not masked
1	RW	0x0	TFOIM Transmit FIFO Overflow Interrupt Mask 1'b0: spi_txo_intr interrupt is masked 1'b1: spi_txo_intr interrupt is not masked
0	RW	0x0	TFEIM Transmit FIFO Empty Interrupt Mask 1'b0: spi_txe_intr interrupt is masked 1'b1: spi_txe_intr interrupt is not masked

SPI_ISR
Address: Operational Base + offset (0x0030)
Interrupt Status

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RO	0x0	RFFIS Receive FIFO Full Interrupt Status 1'b0: spi_rxf_intr interrupt is not active after masking 1'b1: spi_rxf_intr interrupt is full after masking
3	RO	0x0	RFOIS Receive FIFO Overflow Interrupt Status 1'b0: spi_rxo_intr interrupt is not active after masking 1'b1: spi_rxo_intr interrupt is active after masking
2	RO	0x0	RFUIS Receive FIFO Underflow Interrupt Status 1'b0: spi_rxu_intr interrupt is not active after masking 1'b1: spi_rxu_intr interrupt is active after masking

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Bit	Attr	Reset Value	Description
1	RO	0x0	TFOIS Transmit FIFO Overflow Interrupt Status 1'b0: spi_txo_intr interrupt is not active after masking 1'b1: spi_txo_intr interrupt is active after masking
0	RO	0x0	TFEIS Transmit FIFO Empty Interrupt Status 1'b0: spi_txe_intr interrupt is not active after masking 1'b1: spi_txe_intr interrupt is active after masking

SPI_RISR
Address: Operational Base + offset (0x0034)
Raw Interrupt Status

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RO	0x0	RFFRIS Receive FIFO Full Raw Interrupt Status 1'b0: spi_rxf_intr interrupt is not active prior to masking 1'b1: spi_rxf_intr interrupt is full prior to masking
3	RO	0x0	RFORIS Receive FIFO Overflow Raw Interrupt Status 1'b0: spi_rxo_intr interrupt is not active prior to masking 1'b1: spi_rxo_intr interrupt is active prior to masking
2	RO	0x0	RFURIS Receive FIFO Underflow Raw Interrupt Status 1'b0: spi_rxu_intr interrupt is not active prior to masking 1'b1: spi_rxu_intr interrupt is active prior to masking
1	RO	0x0	TFORIS Transmit FIFO Overflow Raw Interrupt Status 1'b0: spi_txo_intr interrupt is not active prior to masking 1'b1: spi_txo_intr interrupt is active prior to masking

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Bit	Attr	Reset Value	Description
0	RO	0x1	TFERIS Transmit FIFO Empty Raw Interrupt Status 1'b0: spi_txe_intr interrupt is not active prior to masking 1'b1: spi_txe_intr interrupt is active prior to masking

SPI_ICR
Address: Operational Base + offset (0x0038)
Interrupt Clear

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	WO	0x0	CTFOI Clear Transmit FIFO Overflow Interrupt Write 1 to Clear Transmit FIFO Overflow Interrupt
2	WO	0x0	CRFOI Clear Receive FIFO Overflow Interrupt Write 1 to Clear Receive FIFO Overflow Interrupt
1	WO	0x0	CRFUI Clear Receive FIFO Underflow Interrupt Write 1 to Clear Receive FIFO Underflow Interrupt
0	WO	0x0	CCI Clear Combined Interrupt Write 1 to Clear Combined Interrupt

SPI_DMACR
Address: Operational Base + offset (0x003c)
DMA Control

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	TDE Transmit DMA Enable 1'b0: Transmit DMA disabled 1'b1: Transmit DMA enabled
0	RW	0x0	RDE Receive DMA Enable 1'b0: Receive DMA disabled 1'b1: Receive DMA enabled

SPI_DMATDLR
Address: Operational Base + offset (0x0040)
DMA Transmit Data Level

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Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x00	TDL Transmit Data Level This bit field controls the level at which a DMA request is made by the transmit logic. It is equal to the watermark level; that is, the dma_tx_req signal is generated when the number of valid data entries in the transmit FIFO is equal to or below this field value, and Transmit DMA Enable (DMACR[1]) = 1.

SPI_DMARDLR
Address: Operational Base + offset (0x0044)
DMA Receive Data Level

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x00	RDL Receive Data Level This bit field controls the level at which a DMA request is made by the receive logic. The watermark level = DMARDL+1; that is, dma_rx_req is generated when the number of valid data entries in the receive FIFO is equal to or above this field value + 1, and Receive DMA Enable(DMACR[0])=1.

SPI_TXDR
Address: Operational Base + offset (0x0400~0x07fc)
Transmit FIFO Data

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	WO	0x0000	TXDR Transimt FIFO Data Register. When it is written to, data are moved into the transmit FIFO.

SPI_RXDR
Address: Operational Base + offset (0x0800~0x0bfc)
Receive FIFO Data

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	RXDR Receive FIFO Data Register. When the register is read, data in the receive FIFO is accessed.

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42.5 Interface description

Table 42-1 SPI interface description

Module Pin	Dir	Pad Name	IOMUX Setting
spi0_clk	I/O	IO_FLASHale_SPI0clk_FLASHgpio1d5	GRF_GPIO1D_IOMUX[11:10]=10
spi0_csn0	I/O	IO_FLASHdata6_EMMCdata6_SPI0csn0_FLASHgpio1d0	GRF_GPIO1D_IOMUX[1:0]=11
spi0_txd	O	IO_FLASHdata5_EMMCdata5_SPI0txd_FLASHgpio1c7	GRF_GPIO1C_IOMUX[15:14]=11
spi0_rxd	I	IO_FLASHdata4_EMMCdata4_SPI0rxd_FLASHgpio1c6	GRF_GPIO1C_IOMUX[13:12]=11
spi0_csn1	O	IO_FLASHdata7_EMMCdata7_SPI0csn1_FLASHgpio1d1	GRF_GPIO1D_IOMUX[3:2]=11
spi1_clk	I/O	IO_CIFdata10_SPI1clk_DVPgpio1b6	GRF_GPIO1B_IOMUX[13:12]=10
spi1_csn0	I/O	IO_CIFdata11_SPI1csn0_DVPgpio1b7	GRF_GPIO1B_IOMUX[15:14]=10
spi1_rxd	O	IO_I2C3CAMscl_SPI1rxd_DVPgpio1c0	GRF_GPIO1C_IOMUX[1:0]=10
spi1_txd	I	IO_I2C3CAMsda_SPI1txd_DVPgpio1c1	GRF_GPIO1C_IOMUX[3:2]=10
spi2_clk	I/O	IO_Scio_SPI2clk_GPUJTAGtdo_GPIO1830gpio0b4	PMUGRF_GPIO0B_IOMUX[9:8]=10
spi2_csn0	I/O	IO_SCdetect_SPI2csn0_GPIO1830gpio0b5	PMUGRF_GPIO0B_IOMUX[11:10]=10
spi2_rxd	I	IO_SCrst_SPI2rxd_GPUJTAGtms_GPIO1830gpio0b2	PMUGRF_GPIO0BIOMUX[5:4]=10
spi2_txd	O	IO_SCclk_SPI2txd_GPUJTAGtdi_GPIO1830gpio0b3	PMUGRF_GPIO0B_IOMUX[7:6]=10
spi1_csn1	O	IO_MACtxclkout_SPI1csn1_GPIO30gpio3d4	GRF_GPIO3D_IOMUX[9:8]=10

Note: spi0_csn1, spi1_csn1,spi2_csn1 can only be used in master mode

42.6 Application Notes

Clock Ratios

A summary of the frequency ratio restrictions between the bit-rate clock (sclk_out/sclk_in) and the SPI peripheral clock (spi_clk) are described as,

When SPI Controller works as master, the F_{spi_clk}>= 2 × (maximum F_{sclk_out})

When SPI Controller works as slave, the F_{spi_clk}>= 6 × (maximum F_{sclk_in})

Master Transfer Flow

When configured as a serial-master device, the SPI initiates and controls all serial transfers. The serial bit-rate clock, generated and controlled by the SPI, is driven out on the sclk_out line. When the SPI is disabled (SPI_ENR = 0), no serial transfers can occur and sclk_out is held in "inactive" state, as defined by the serial protocol under which it operates.

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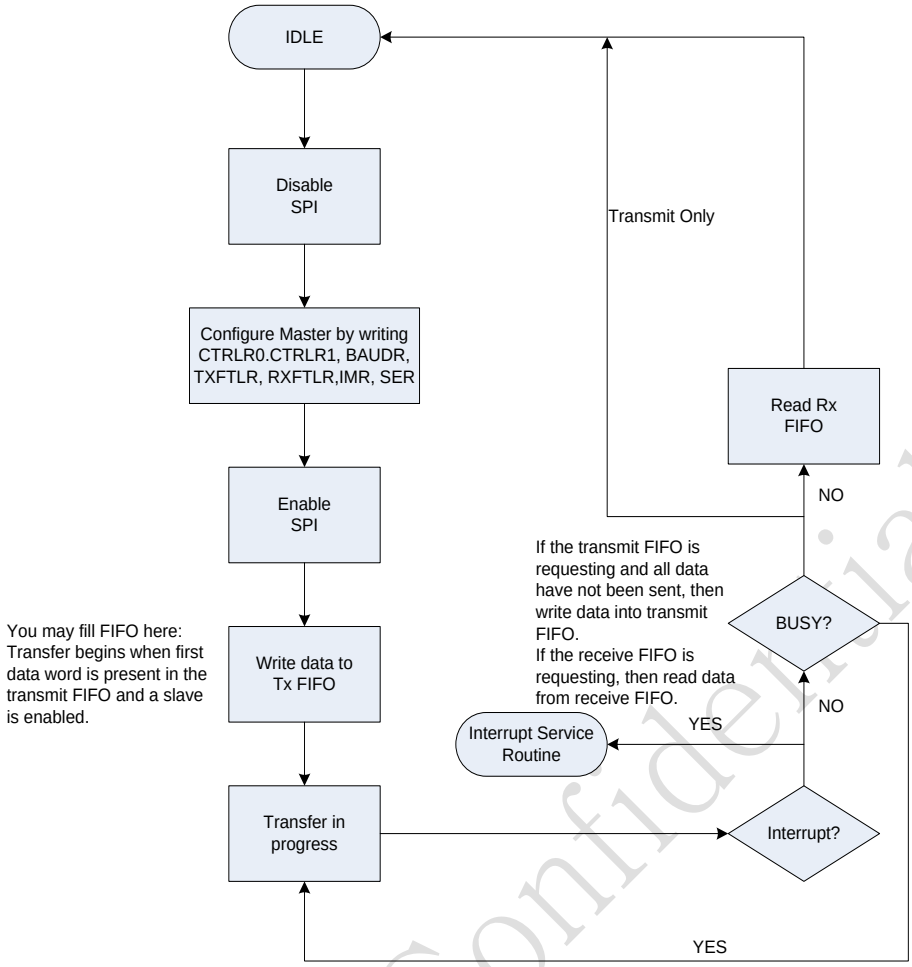

Fig. 42-7 SPI Master transfer flow diagram

Slave Transfer Flow

When the SPI is configured as a slave device, all serial transfers are initiated and controlled by the serial bus master.

When the SPI serial slave is selected during configuration, it enables its txd data onto the serial bus. All data transfers to and from the serial slave are regulated on the serial clock line (sclk_in), driven from the serial-master device. Data are propagated from the serial slave on one edge of the serial clock line and sampled on the opposite edge.

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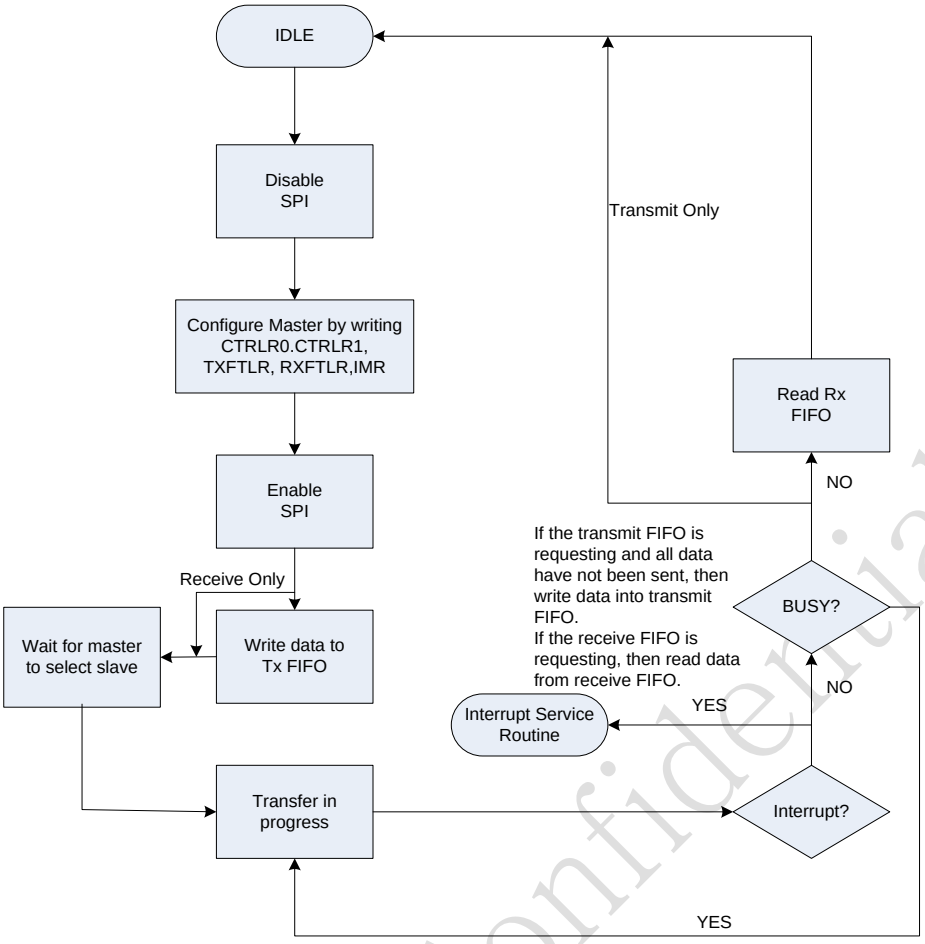

Fig. 42-8 SPI Slave transfer flow diagram

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Chapter 43 Serial Flash Controller (SFC)

43.1 Overview

The serial flash controller (SFC) is used to control the data transfer between the chip system and the serial nor/nand flash device.
The SFC supports the following features:

- Support AHB slave interface to configure register and read/write serial flash
- Support AHB master interface to transfer data from/to spi flash device
- Support two independent clock domain: AHB clock and SPI clock
- Support x1,x2,x4 data bits mode
- Support up to 4 chip select
- Support interrupt output, interrupt maskable
- Support Spansion, MXIC,Gigadevice…vendor's nor flash memory.

43.2 Block Diagram

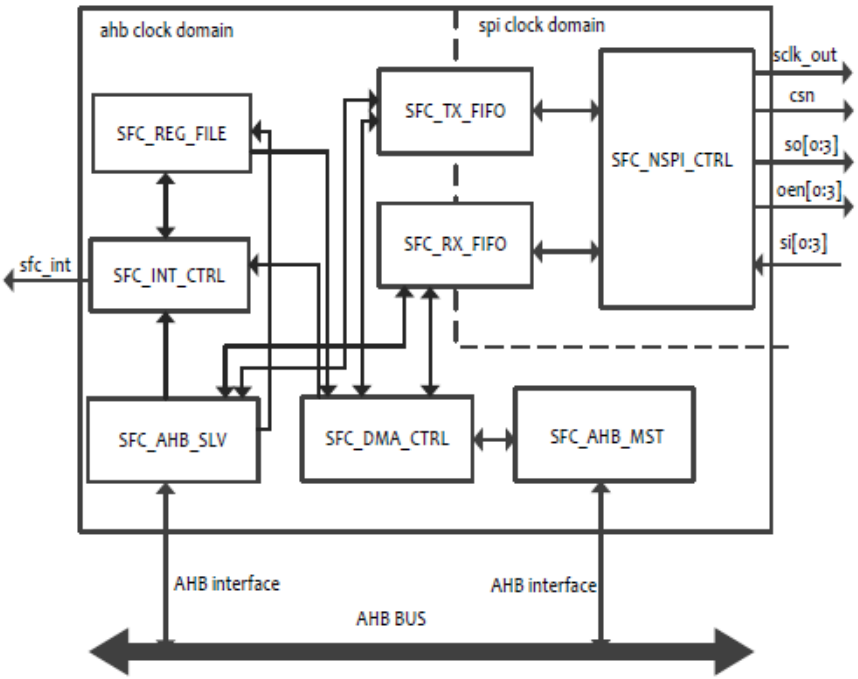

Fig. 43-1 SFC architecture

43.3 Function Description

43.3.1 SFC ahb slave

The ahb slave is used to configure the register, and also write to/read from the serial nor/nand flash device.
The SFC_CTRL register is a global control register, when the controller is in busy state(SFC_SR), SFC_CTRL cannot be set. The field sclk_idle_level_cycles of this register is used to configure the idle level cycles of sclk before read the first bit of the read command. Like the following picture shows: the red line of the sclk is the idle cycles, during these cycles, the chip pad is switched to output. When sclk_idle_level_cycles=0, it means there will be not such idle level.

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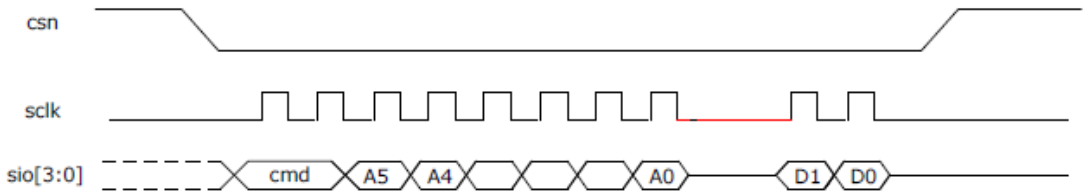
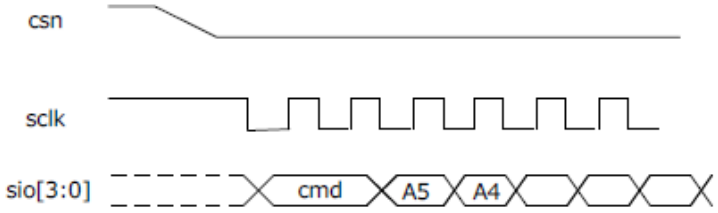


When the field spi mode is set, the transfer waveform will like following, and switch to mode3.



43.4 Register description

43.4.1 Register Summary

Name	Offset	Size	Reset Value	Description
SFC_CTRL	0x0000	W	0x00000000	Control Register 0
SFC_IMR	0x0004	W	0x00000000	Interrupt Mask
SFC_ICLR	0x0008	W	0x00000000	Interrupt Clear
SFC_FTLR	0x000c	W	0x00000000	FIFO Threshold Level
SFC_RCVR	0x0010	W	0x00000000	SFC Recover
SFC_AX	0x0014	W	0x00000000	SFC AX Value
SFC_ABIT	0x0018	W	0x00000000	Flash Address bits
SFC_ISR	0x001c	W	0x00000000	Interrupt Status
SFC_FSR	0x0020	W	0x00000001	FIFO Status
SFC_SR	0x0024	W	0x00000000	SFC Status
SFC_DMAADDR	0x0080	W	0x00000000	DMA Address
SFC_DMATR	0x0084	W	0x00000000	DMA Trigger
SFC_CMD	0x0100	W	0x00000000	SFC CMD
SFC_ADDR	0x0104	W	0x00000000	SFC Address
SFC_DATA	0x0108	W	0x00000000	SFC DATA

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

43.4.2 Detail Register Description

SFC_CTRL

Address: Operational Base + offset (0x0000)

Control Register 0

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
13:12	RW	0x0	DATB Data bits width Data bits width 00: 1bit 01: 2bits 10: 4bits
11:10	RW	0x0	ADRB Address bits width Address bits width 00: 1bit 01: 2bits 10: 4bits
9:8	RW	0x0	CMDB Command bits width Command bits width 00: 1bit 01: 2bits 10: 4bits 11:reserved
7:4	RW	0x0	IDLE_CYCLE Sclk Idle Level Cycles 4'b0: idel hold is disable 4'b1: hold the sclk_out in idle for two cycles when switch to shift in ....
3:2	RO	0x0	reserved
1	RW	0x0	SHIFTPHASE Shift Phase Select 0: shift in the data at posedge sclk_out 1: shift in the data at negedge sclk_out
0	RW	0x0	SPIM SPI MODE Select SPI MODE Select 0: mode 0 1: mode 3

SFC_IMR
Address: Operational Base + offset (0x0004)
Interrupt Mask

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x0	DMAM DMA finish interrupt mask 0: dma_intr interrupt is not masked 1: dma_intr interrupt is masked

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Bit	Attr	Reset Value	Description
6	RW	0x0	NSPIM SPI Error interrupt mask 0: nspi_intr interrupt is not masked 1: nspi_intr interrupt is masked
5	RW	0x0	AHBM AHB Error interrupt mask 0: ahb_intr interrupt is not masked 1: ahb_intr interrupt is masked
4	RW	0x0	TRANSM Transfer finish Interrupt Mask 0: transf_intr interrupt is not masked 1: transf_intr interrupt is masked
3	RW	0x0	TXEM Transmit FIFO Empty Interrupt Mask 0: txe_intr interrupt is not masked 1: txe_intr interrupt is masked
2	RW	0x0	TXOM Transmit FIFO Overflow Interrupt Mask 0: txo_intr interrupt is not masked 1: txo_intr interrupt is masked
1	RW	0x0	RXUM Receive FIFO Underflow Interrupt Mask 0: rxu_intr interrupt is not masked 1: rxu_intr interrupt is masked
0	RW	0x0	RXFM Receive FIFO Full Interrupt Mask 0: rxf_intr interrupt is not masked 1: rxf_intr interrupt is masked

SFC_ICLR
Address: Operational Base + offset (0x0008)
Interrupt Clear

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	W1C	0x0	DMAC DMA finish Interrupt Clear Write and clear
6	W1C	0x0	NSPIC SPI Error Interrupt Clear Write and clear
5	W1C	0x0	AHBC AHB Error Interrupt Clear Write and clear

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Bit	Attr	Reset Value	Description
4	W1C	0x0	TRANSC Transfer finish Interrupt Clear Write and clear
3	W1C	0x0	TXEC Transmit FIFO Empty Interrupt Clear Write and clear
2	W1C	0x0	TXOC Transmit FIFO Overflow Interrupt Clear Write and clear
1	W1C	0x0	RXUC Receive FIFO Underflow Interrupt Clear Write and clear
0	W1C	0x0	RXFC Receive FIFO Full Interrupt Clear Write and clear

SFC_FTLR

Address: Operational Base + offset (0x000c)

FIFO Threshold Level

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:8	RW	0x00	RXFTLR Receive FIFO Threshold Level When the number of receive FIFO entries is bigger than or equal to this value, the receive FIFO full interrupt is triggered.
7:0	RW	0x00	TXFTLR Transmit FIFO Threshold Level When the number of transmit FIFO entries is less than or equal to this value, the transmit FIFO empty interrupt is triggered.

SFC_RCVR

Address: Operational Base + offset (0x0010)

SFC Recover

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RO	0x0	RCVR SFC Recover SFC Recover Write 1 to recover the SFC State Machine, FIFO state and other logic state.

SFC_AX

Address: Operational Base + offset (0x0014)

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SFC AX Value

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RO	0x00	AX AX The AX Value when doing the continuous read(enhance mode).

SFC_ABIT

Address: Operational Base + offset (0x0018)

Flash Address bits

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	ABIT Flash Address bits Flash Address bits

SFC_ISR

Address: Operational Base + offset (0x001c)

Interrupt Status

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x0	DMAS DMA Finish Interrupt Status 0: not active 1: active
6	RW	0x0	NSPIS SPI Error Interrupt Status 0: not active 1: active
5	RW	0x0	AHBS AHB Error Interrupt Status 0: not active 1: active
4	RW	0x0	TRANSS Transfer finish Interrupt Status 0: not active 1: active
3	RW	0x0	TXES Transmit FIFO Empty Interrupt Status 0: not active 1: active
2	RW	0x0	TXOS Transmit FIFO Overflow Interrupt Status 0: not active 1: active

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Bit	Attr	Reset Value	Description
1	RW	0x0	RXUS Receive FIFO Underflow Interrupt Status 0: not active 1: active
0	RW	0x0	RXFS Receive FIFO Full Interrupt Status 0: not active 1: active

SFC_FSR
Address: Operational Base + offset (0x0020)
FIFO Status

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved
20:16	RW	0x00	RXWLVL RX FIFO Water Level 0: fifo is empty 1: 1 entry is taken ...
15:13	RO	0x0	reserved
12:8	RO	0x00	TXWLVL TX FIFO Water Level 0: fifo is full 1: left 1 entry ...
7:4	RO	0x0	reserved
3	RO	0x0	RXFS Receive FIFO Full Status 0: rx fifo is not full 1: rx fifo is full
2	RO	0x0	RXES Receive FIFO Empty Status 0: rx fifo is not empty 1: rx fifo is empty
1	RO	0x0	TXFS Transmit FIFO Full Status 0: tx fifo is not full 1: tx fifo is full
0	RO	0x1	TXES Transmit FIFO Empty Status 0: tx fifo is not empty 1: tx fifo is empty

SFC_SR
Address: Operational Base + offset (0x0024)

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SFC Status

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	WO	0x0	SR SFC Status 0: SFC is idle 1: SFC is busy When busy, don't set the control register.

SFC_DMAADDR

Address: Operational Base + offset (0x0080)

DMA Address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMAADDR DMA Address DMA Address

SFC_DMATR

Address: Operational Base + offset (0x0084)

DMA Trigger

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	W1C	0x0	DMATR DMA Trigger Write 1 to start the dma transfer.

SFC_CMD

Address: Operational Base + offset (0x0100)

SFC CMD

Bit	Attr	Reset Value	Description
31:30	RW	0x0	CS Flash chip select 00: chip select 0 01: chip select 1 10: chip select 2 11: chip select 3
29:16	RW	0x0000	TRB Transfer Bytes number Total Data Bytes number that will write to /read from the flash.
15:14	RW	0x0	ADDRB Address bits number select 00: 0bits 01: 24bits 10: 32bits 11: From the ABIT register

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Bit	Attr	Reset Value	Description
13	RW	0x0	CONT Continuous read mode 0: disable continuous read mode 1: enable continuous read mode
12	RW	0x0	WR Flash Write or Read 0:read 1:write
11:8	WO	0x0	DUMM Dummy Bits Number Dummy Bits Number
7:0	WO	0x00	CMD Flash CMD Flash Command

SFC_ADDR
Address: Operational Base + offset (0x0104)
SFC Address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ADDR SFC Address Flash's address

SFC_DATA
Address: Operational Base + offset (0x0108)
SFC DATA

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ADDR SFC DATA Flash's Data

43.5 Interface Description

Table 43-1 SFC Interface Description

Module pin name	Dir	Pad name	IOMUX
sfc_sclk_out	O	IO_FLASHhrdy_EMMCcmd_SFCclk_FLASHgpio1d2	GPIO1D_IOMUX[5:4]=2'b11
scf_csn0	O	IO_FLASHwrn_SFCcsn0_FLASHgpio1d7	GPIO1D_IOMUX[15:14]=2'b10
scf_csn1	O	IO_FLASHrdn_SFCcsn1_FLASHgpio1d4	GPIO1D_IOMUX[9:8]=2'b10
sfc_si0/ sfc_so0	I/O	IO_FLASHdata0_EMMCdata0_SFCsio0_FLASHgpio1c2	GPIO1C_IOMUX[5:4]=2'b11
sfc_si1/ sfc_so1	I/O	IO_FLASHdata1_EMMCdata1_SFCsio1_FLASHgpio1c3	GPIO1C_IOMUX[7:6]=2'b11
sfc_si2/ sfc_so2	I/O	IO_FLASHdata2_EMMCdata2_SFCsio2_FLASHgpio1c4	GPIO1C_IOMUX[9:8]=2'b11
sfc_si3/ sfc_so3	I/O	IO_FLASHdata3_EMMCdata3_SFCsio3_FLASHgpio1c5	GPIO1C_IOMUX[11:10]=2'b11

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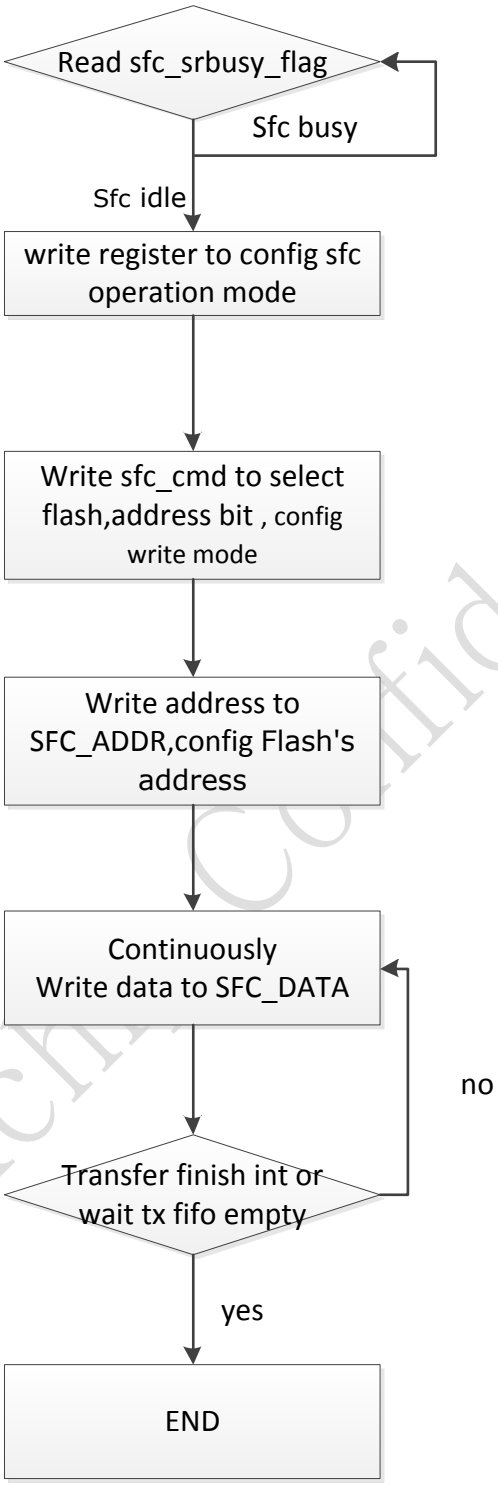
43.6 Application Notes

43.6.1 AHB Slave write flash flow



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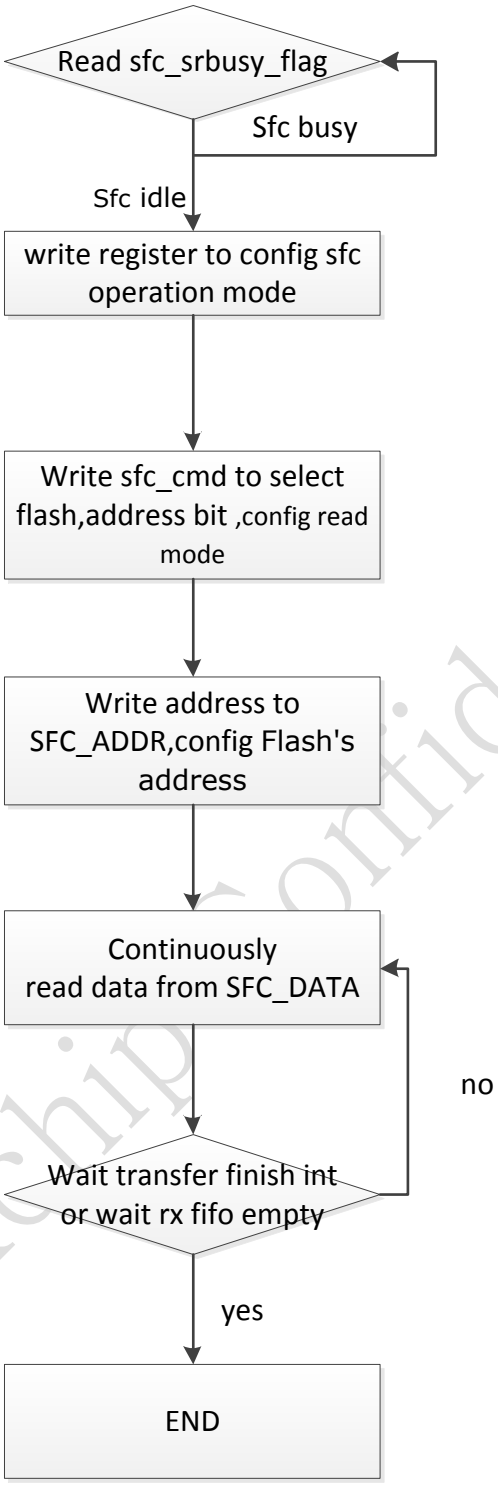
43.6.2 AHB Slave read flash flow



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Chapter 44 Smart Card Controller

44.1 Overview

The Smart Card Reader (SCR) is a communication controller that transmits data between the superior system and the Smart Card. The controller can perform a complete smart card session, including card activation, card deactivation, cold/warm reset, Answer to Reset (ATR) response reception, data transfers, etc.
SCR supports the following features:

- Supports the ISO/IEC 7816-3:1997(E) and EMV2000 (4.0) specifications
- Performs functions needed for complete smart card sessions, including:
  - Card activation and deactivation
  - Cold/warm reset
  - Answer to Reset (ATR) response reception
  - Data transfers to and from the card
- Extensive interrupt support system
- Adjustable clock rate and bit (baud) rate
- Configurable automatic byte repetition
- Handles commonly used communication protocols:
  - T=0 for asynchronous half-duplex character transmission
  - T=1 for asynchronous half-duplex block transmission
- Automatic convention detection
- Configurable timing functions:
  - Smart card activation time
  - Smart card reset time
  - Guard time
  - Timeout timers
- Automatic operating voltage class selection
- Supports synchronous and any other non-ISO 7816 and non-EMV cards
- Advanced Peripheral Bus (APB) slave interface for easy integration with AMBA-based host systems Block Diagram

44.2 Block Diagram

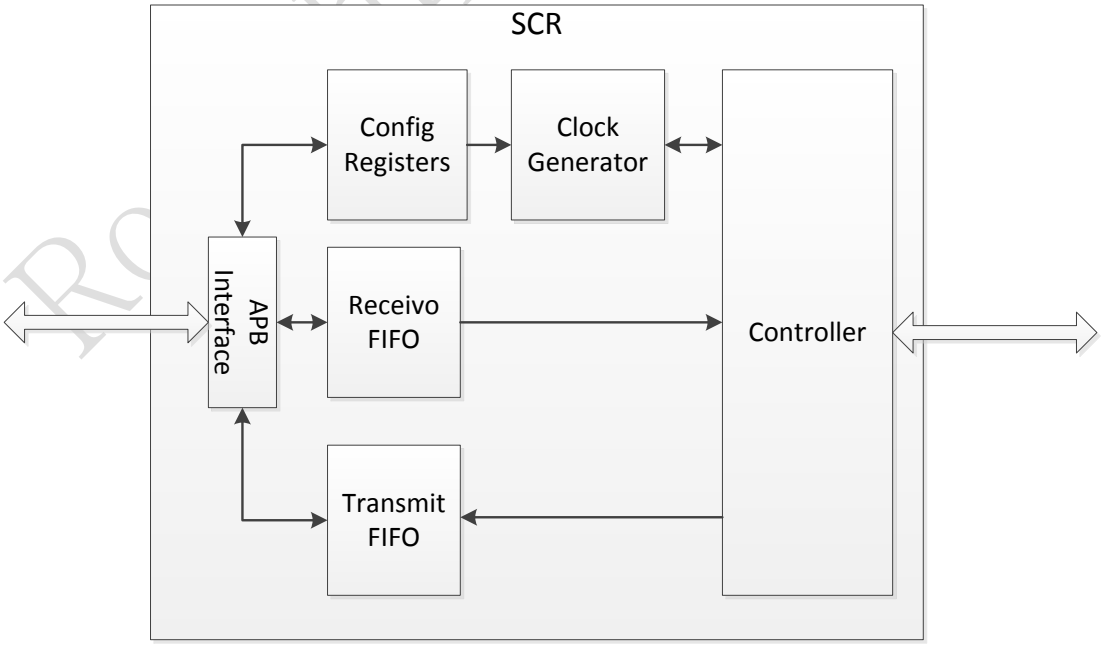

Fig. 44-1 SCR Block Diagram

The host processor gets access to PWM Register Block through the APB slave interface.

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44.2.1 APB Interface

The host processor gets access to PWM Register Block through the APB slave interface.

44.2.2 Configuration Registers

The Configuration Registers block provides control over all functions of the Smart Card Reader

44.2.3 Controller

The Controller is the main block in the SCR core. This block controls receiving characters transmitted by the Smart Card, storing them in the RX FIFO, and transmitting them to the Smart Card. This block also performs card activation, deactivation, and cold and warm reset. After the card is reset, the Answer To Reset (ATR) sequence is received by the controller and stored in RX FIFO.
The parallel to serial conversion needed to transmit data from a Smart Card Reader to a Smart Card and the serial to parallel conversion needed to transmit data in the opposite direction is performed by the UART. The UART also performs the guard time, parity checking and character repeating functions.

44.2.4 Receive FIFO

The Receive FIFO is used to store the data received from the Smart Card until the data is read out by the superior system.

44.2.5 Transmit FIFO

The Transmit FIFO is used to store the data to be transmitted to the Smart Card.

44.2.6 Clock Generator

The Clock Generator generates the Smart Card Clock signal and the Baud Clock Impulse signal, used in timing the Smart Card Reader.)

44.3 Function Description

A Smart Card session consists of following stages:
1. Smart Card insertion
2. Activation of contacts and cold reset sequence
3. Answer To Reset sequence (ATR)
4. Execution of transaction
5. Deactivation of contacts
6. Smart Card removal

44.3.1 Smart Card Insertion

A Smart Card session starts with the insertion of the Smart Card. This event is signaled to the SCR using the SCDETECT input. The SCPRESENT bit is set and also the SCINS interrupt is asserted (if enabled).
When the external card detect switch is not used, the input pin SCDETECT must be tied to inactive state.

44.3.2 Automatic operating voltage class selection

There are three operating classes (1.8V - class C, 3V - class B and 5V - class A) defined in ISO/IEC 7816-3(2006) specification. Only 1.8V and 3.3V are supported by the SCR.
Before the activation of contacts, operating classes have to be enabled via bits VCC18, VCC33 in CTRL2 register. In case that no operating class is enabled, the controller performs activation for all two voltage classes (1.8V, 3V) in sequence.
When Smart Card Reader performs activation of contacts the lowest enabled voltage class is automatically applied first. When the first character start bit of ATR sequence is received, the selected voltage class is correct (even if the ATR is then received with errors). When the ATR sequence reception does not start, ATRFAIL interrupt is not activated, deactivation is performed and next higher enabled voltage class is applied. If the ATR sequence reception does not start and no other higher class is enabled was already applied the ATRFAIL interrupt is activated and the last applied voltage class remains active.
After the automatic voltage class selection is finished the selected class can be read from bits

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VCC18, VCC33 in CTRL2 register. If the automatic voltage class selection fails, these bits remain untouched.
There is a delay applied between deactivation of contacts with lower voltage class and activation of contacts with higher voltage class. This delay should be at least 10 ms according to the ISO/IEC 7816-3 specification.

44.3.3 Activation of Contacts and Cold Reset Sequence

When the Smart Card is properly inserted and the ACT bit in CTRL2 register is asserted, the activation of contacts can be started. The duration of each part of the activation is the time Ta, which is equal to the ADEATIME register value. If no Vpp is necessary, the activation and deactivation part of Vpp can be omitted by clearing the AUTOADEAVPP bit in SCPADS register. The Cold Reset sequence follows immediately after the activation. Time (Tc) is the duration of the Reset. The EMV specification recommends that this value should be between 40000 and 45000. The activation of contacts and cold reset sequence is shown inFig. 44-2.

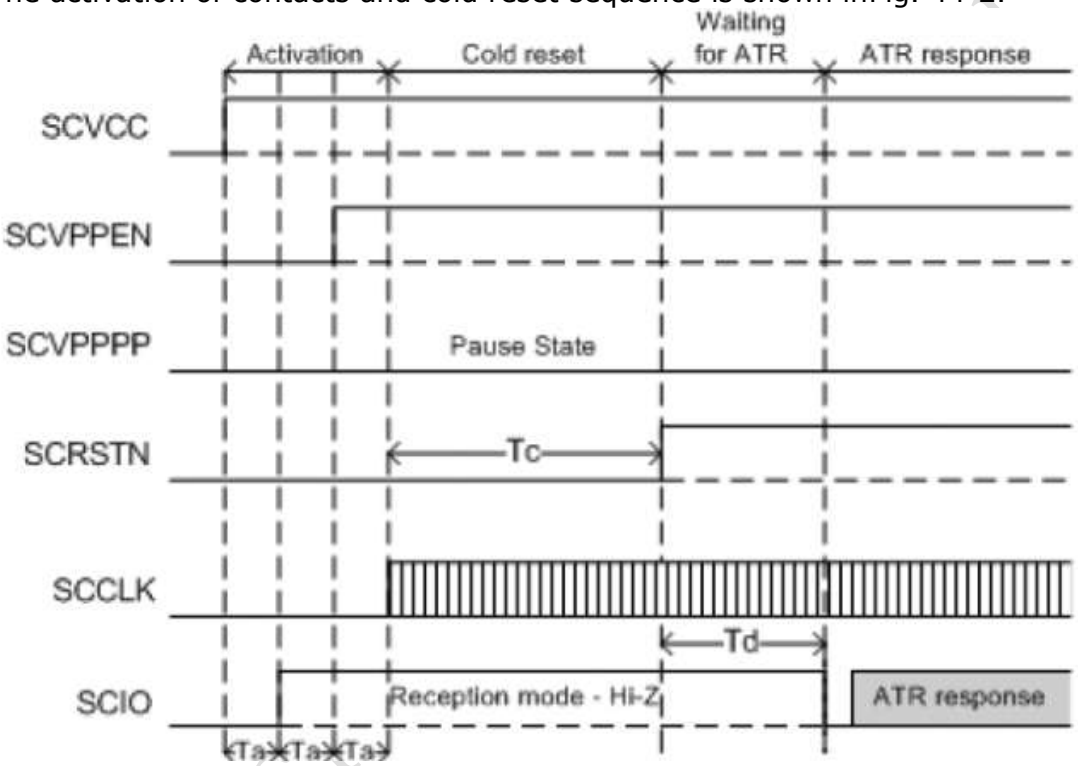

Fig. 44-2 Activation, Cold Reset and ATR

44.3.4 Execution of Transaction

All transfers between the Smart Card Reader and a Smart Card are under the control of the superior system. It controls the number of characters sent to the Smart Card and it knows the number of characters expected to be returned from the Smart Card.

44.3.5 Warm Reset

The Warm Reset sequence is initialized by setting the WRST bit in the CTRL2 register to '1'.Smart Card Reader drives the SCRSTN signal to '0' to perform the Warm Reset as shown in Fig. 44-3. After the SCRSTN assertion, the Warm Reset sequence then continues the same way as the Cold Reset sequence.

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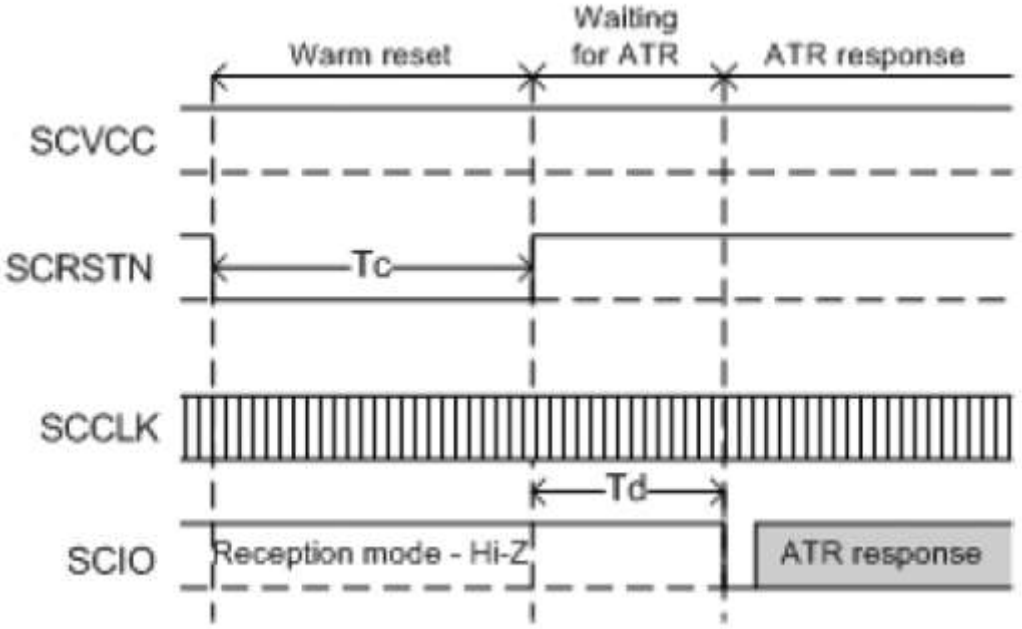

Fig. 44-3 Warm Reset and ATR

44.3.6 Deactivation of Contacts

After the smart card reader detects the removal of the smart card (SCREM interrupt) or the superior system initiates deactivation by setting the DEACT bit in the CTRL2 register to '1', the deactivation is performed immediately as shown in . The duration time (Ta), of each part of the deactivation sequence time is defined in the ADEATIME register.

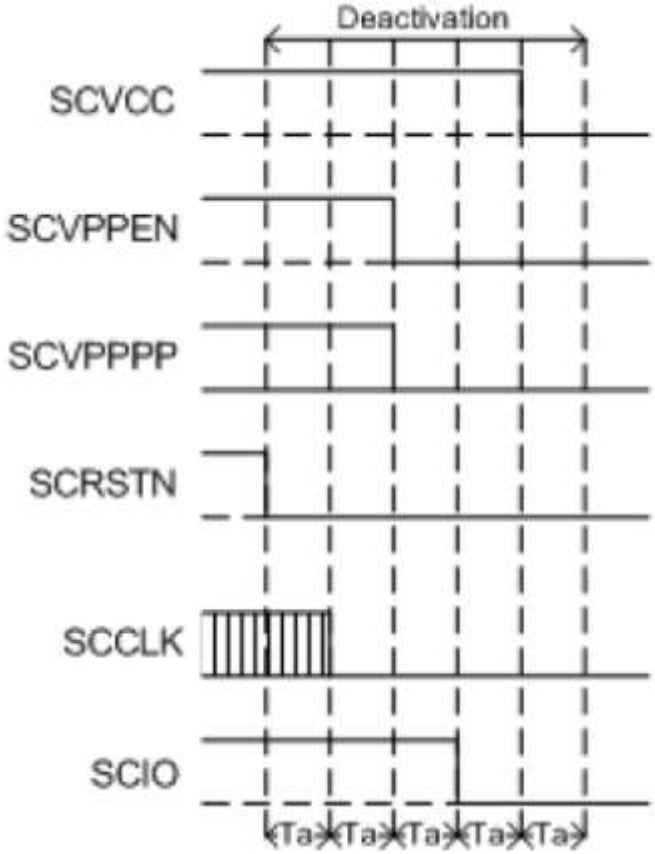

Fig. 44-4 Deactivation Sequence

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44.4 Register description

44.4.1 Register Summary

Name	Offset	Size	Reset Value	Description
SCR_CTRL1	0x0000	HW	0x0000	Control Register 1
SCR_CTRL2	0x0004	HW	0x0000	Control Register 2
SCR_SCPADS	0x0008	HW	0x0000	Smart Card Pads Register
SCR_INTEN1	0x000c	HW	0x0000	Interrupt Enable Register 1
SCR_INTSTAT1	0x0010	HW	0x0000	Interrupt Status Register 1
SCR_FIFOCTRL	0x0014	HW	0x0000	FIFO Control Register
SCR_LEGTXFICNT	0x0018	B	0x00	Legacy TX FIFO Counter
SCR_LEGRXFICNT	0x0019	B	0x00	Legacy RX FIFO Counter
SCR_RXFITH	0x001c	HW	0x0000	RX FIFO Threshold
SCR_REP	0x0020	B	0x00	Repeat
SCR_SCCDDIV	0x0024	HW	0x0000	Smart Card Clock Divisor
SCR_BAUDDIV	0x0028	HW	0x0000	Baud Clock Divisor
SCR_SCGUTIME	0x002c	B	0x00	Smart Card Guardtime
SCR_ADEATIME	0x0030	HW	0x0000	Activation / Deactivation Time
SCR_LOWRSTTIME	0x0034	HW	0x0000	Reset Duration
SCR_ATRSTARTLIMIT	0x0038	HW	0x0000	ATR Start Limit
SCR_C2CLIM	0x003c	HW	0x0000	Two Characters Delay Limit
SCR_INTEN2	0x0040	HW	0x0000	Interrupt Enable Register 2
SCR_INTSTAT2	0x0044	HW	0x0000	Interrupt Status Register 2
SCR_TXFITH	0x0048	HW	0x0000	TX FIFO Threshold
SCR_TXFIFOCNT	0x004c	HW	0x0000	TX FIFO Counter
SCR_RXFIFOCNT	0x0050	HW	0x0000	RX FIFO Counter
SCR_BAUDTUNE	0x0054	B	0x00	Baud Tune Register
SCR_FIFO	0x0200	B	0x00	FIFO

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

44.4.2 Detail Register Description

SCR_CTRL1
Address: Operational Base + offset (0x0000)
Control Register 1

Bit	Attr	Reset Value	Description
15	RW	0x0	GINTEN Global Interrupt Enable When high, INTERRUPT output assertion is enabled.
14	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
13	RW	0x0	TCKEN TCK enable When enabled all ATR bytes beginning from T0 are being XOR-ed. The result must be equal to TCK byte (when present). If the TCK byte does not match the computed value the ATR is considered to be malformed.
12	RW	0x0	ATRSTFLUSH ATR Start Flush FIFO When enabled, both FIFOs are flushed before the ATR is started.
11	RW	0x0	T0T1 T0/T1 Protocol Controls the using of T=0 or T=1 protocol. No character repeating is used when T=1 protocol is selected. The Character Guardtime (minimum delay between the leading edges of two consecutive characters) is reduced to 11 ETU when T=1 protocol is used and Guardtime value N = 255. The delay between the leading edge of the last received character and the leading edge of the first character transmitted is 16 ETU when T=0 protocol is used and 22 ETU when T=1 protocol is used.
10	RW	0x0	TS2FIFO TS to FIFO Enables to store the first ATR character TS in RX FIFO. During ideal card session there is no necessity to store TS character, so it can be disabled
9	RW	0x0	RXEN Receiving enable When enabled the characters sent by the Smart Card are received by the UART and stored in RX FIFO. Receiving is internally disabled while a transmission is in progress.
8	RW	0x0	TXEN Transmission enable When enabled the characters are read from TX FIFO and transmitted through UART to the Smart Card

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Bit	Attr	Reset Value	Description
7	RW	0x0	CLKSTOPVAL Clock Stop Value The value of the scclk output during the clock stop state.
6	RW	0x0	CLKSTOP Clock Stop Clock Stop. When this bit is asserted and the smart card I/O line is in 'Z'state, the SCR core stops driving of the smart card clock signal after the CLKSTOPDELAY time expires. The smart card clock is restarted immediately after the CLKSTOP signal is deasserted. New character transmission can be started by superior system after the CLKSTARTDELAY time expires. The expiration of both times is signaled by the CLKSTOPRUN bit in the Interrupt registers. Reading '1' from this bit signals that the clock is stopped or CLKSTARTDELAY time not expired yet. Reading '0' from this bit signals that the clock is not stopped.
5:3	RO	0x0	reserved
2	RW	0x0	PECH2FIFO Character With Wrong Parity to FIFO Enables storage of the characters received with wrong parity in RX FIFO.
1	RW	0x0	INVORD Inverse Bit Ordering When High, iverse bit ordering convention(MSB-LSB) is used.
0	RW	0x0	INVLEV Inverse Bit Level When high, inverse level convention is used(A= '1', Z='0');

SCR_CTRL2
Address: Operational Base + offset (0x0004)
Control Register 2

Bit	Attr	Reset Value	Description
15:8	RO	0x00	Reserved3 Reserved Reserved bits are hard-wired to zero

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Bit	Attr	Reset Value	Description
7	RW	0x0	VCC50 Control 5V Smart Card Vcc Control 5V Smart Card Vcc. Setting of this bit allows selection of 5V Vcc for Smart Card session (Class A). After the selection of operating class is completed, this bit is in '1'.if this class was selected. Default value after reset is '0'..
6	RW	0x0	VCC33 Control 3V Smart Card Vcc Setting of this bit allows selection of 3V Vcc for Smart Card session (Class B). After the selection of operating class is completed, this bit is in '1'.if this class was selected. Default value after reset is '0'.
5	RW	0x0	VCC18 Control 1.8V Smart Card Vcc Control 1.8V Smart Card Vcc. Setting of this bit allows selection of 1.8V Vcc for Smart Card session (Class C). After the selection of operating class is completed, this bit is in '1'.if this class was selected. Default value after reset is '0'..
4	RW	0x0	DEACT Deactivation Setting of this bit initializes the deactivation sequence. When the deactivation is finished, the DEACT bit is automatically cleared.
3	RW	0x0	ACT Activation Setting of this bit initializes the activation sequence. When the activation is finished, the ACT bit is automatically cleared.
2	WO	0x0	WARMRST Warm Reset Command Writing '1'.to this bit initializes Warm Reset of the Smart Card. This bit is always read as '0'..
1:0	RO	0x0	reserved

SCR_SCPADS
Address: Operational Base + offset (0x0008)
Smart Card Pads Register

Bit	Attr	Reset Value	Description
15:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9	RO	0x0	SCPRESENT Smart Card presented This bit is set to '1'.when the SCDETECT input is active at least for SCDETECTTIME
8	RW	0x0	DSCFCB Direct Smart Card Function Code Bit It provides direct access to SCFCB output
7	RW	0x0	DSCVPPPP Direct Smart Card Vpp Pause/Prog It provides direct access to SCVPPPP output
6	RW	0x0	DSCVPPEN Direct Smart Card Vpp Enable It provides direct access to SCVPPEN output
5	RW	0x0	AUTOADEAVPP Automatic Vpp Handling. When high, it enables automatic handling of DSCVPPEN and DSCVPPPP signals during activation and deactivation sequence.
4	RW	0x0	DSCVCC Direct Smart Card Vcc Direct Smart Card Vcc. When DIRACCPADS = '1'., the DSCVCC bit provides direct access to SCVCCx outputs. The appropriate SCVCC18, SCVCC33 and SCVCC50 outputs are driven according to state of bits VCC18, VCC33 and VCC50 in CTRL2 register.
3	RW	0x0	DSCRST Direct Smart Card Reset When DIRACCPADS = '1'., the DSCRST bit provides direct access to SCRST output
2	RW	0x0	DSCCLK Direct Smart Card Clock When DIRACCPADS = '1'., the DSCCLK bit provides direct access to SCCLK output
1	RW	0x0	DSCIO Direct Smart Card Input/Output When DIRACCPADS = '1'., the DSCIO bit provides direct access to SCIO pad.
0	RW	0x0	DIRACCPADS Direct Access To Smart Card Pads When high, it disables a serial interface functionality and enables direct control of the smart card pads using following 4 bits.

SCR_INTEN1

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Address: Operational Base + offset (0x000c)
Interrupt Enable Register 1

Bit	Attr	Reset Value	Description
15	RW	0x0	SCDEACT Smart Card Deactivation Interrupt When enabled, this interrupt is asserted after the Smart Card deactivation sequence is complete.
14	RW	0x0	SCACT Smart Card Activation Interrupt. When enabled, this interrupt is asserted after the Smart Card activation sequence is complete.
13	RW	0x0	SCINS Smart Card Inserted Interrupt When enabled, this interrupt is asserted after the smart card insertion
12	RW	0x0	SCREM Smart Card Removed Interrupt. When enabled, this interrupt is asserted after the smart card removal.
11	RW	0x0	ATRDONE ATR Done Interrupt When enabled, this interrupt is asserted after the ATR sequence is successfully completed.
10	RW	0x0	ATRFAIL ATR Fail Interrupt When enabled, this interrupt is asserted if the ATR sequence fails.
9	RW	0x0	RXTHRESHOLD RX FIFO Threshold Interrupt When enabled, this interrupt is asserted if the number of bytes in RX FIFO is equal or exceeds the RX FIFO threshold.
8	RW	0x0	C2CFULL Two Consecutive Characters Limit Interrupt When enabled, this interrupt is asserted if the time between two consecutive characters, transmitted between the Smart Card and the Reader in both directions, is equal the Two Characters Delay Limit described below. The C2CFULL interrupt is internally enabled from the ATR start to the deactivation or ATR restart initialization. It is recommended to use this counter to detect unresponsive Smart Cards.

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Bit	Attr	Reset Value	Description
7	RW	0x0	RXPERR Reception Parity Error Interrupt When enabled, this interrupt is asserted after the character with wrong parity was received when the number of repeated receptions exceeds RXREPEAT value or T=1 protocol is used
6	RW	0x0	TXPERR Transmission Parity Error Interrupt. When enabled, this interrupt is asserted if the Smart Card signals wrong character parity during the guardtime after the character transmission was repeated TXREPEAT-times
5	RW	0x0	RXDONE Reception Done Interrupt When enabled, this interrupt is asserted after a character was received from the Smart Card.
4	RW	0x0	TXDONE Transmission Done Interrupt When enabled, this interrupt is asserted after one character was transmitted to the Smart Card.
3	RW	0x0	CLKSTOPRUN Smart Card Clock Stop Interrupt When enabled, this interrupt is asserted in two cases: 1. When the smart card clock is stopped (after CLOCKSTOP assertion). 2. When the new character transfer can be started (the smart card clock is fully running after CLOCKSTOP de-assertion).
2	RW	0x0	RXFIFULL RX FIFO Full Interrupt When enabled, this interrupt is asserted if the RX FIFO is filled up.
1	RW	0x0	TXFIEMPTY TX FIFO Empty Interrupt. When enabled, this interrupt is asserted if the TX FIFO is emptied out.
0	RW	0x0	TXFIDONE TX FIFO Done Interrupt When enabled, this interrupt is asserted after all bytes from TX FIFO were transferred to the Smart Card

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SCR_INTSTAT1
Address: Operational Base + offset (0x0010)
Interrupt Status Register 1

Bit	Attr	Reset Value	Description
15	RW	0x0	SCDEACT Smart Card Deactivation Interrupt When enabled, this interrupt is asserted after the Smart Card deactivation sequence is complete.
14	RW	0x0	SCACT Smart Card Activation Interrupt. When enabled, this interrupt is asserted after the Smart Card activation sequence is complete.
13	RW	0x0	SCINS Smart Card Inserted Interrupt When enabled, this interrupt is asserted after the smart card insertion
12	RW	0x0	SCREM Smart Card Removed Interrupt. When enabled, this interrupt is asserted after the smart card removal.
11	RW	0x0	ATRDONE ATR Done Interrupt When enabled, this interrupt is asserted after the ATR sequence is successfully completed.
10	RW	0x0	ATRFAIL ATR Fail Interrupt When enabled, this interrupt is asserted if the ATR sequence fails.
9	RW	0x0	RXTHRESHOLD RX FIFO Threshold Interrupt When enabled, this interrupt is asserted if the number of bytes in RX FIFO is equal or exceeds the RX FIFO threshold.

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Bit	Attr	Reset Value	Description
8	RW	0x0	C2CFULL Two Consecutive Characters Limit Interrupt When enabled, this interrupt is asserted if the time between two consecutive characters, transmitted between the Smart Card and the Reader in both directions, is equal the Two Characters Delay Limit described below. The C2CFULL interrupt is internally enabled from the ATR start to the deactivation or ATR restart initialization. It is recommended to use this counter to detect unresponsive Smart Cards.
7	RW	0x0	RXPERR Reception Parity Error Interrupt When enabled, this interrupt is asserted after the character with wrong parity was received when the number of repeated receptions exceeds RXREPEAT value or T=1 protocol is used
6	RW	0x0	TXPERR Transmission Parity Error Interrupt. When enabled, this interrupt is asserted if the Smart Card signals wrong character parity during the guardtime after the character transmission was repeated TXREPEAT-times
5	RW	0x0	RXDONE Reception Done Interrupt When enabled, this interrupt is asserted after a character was received from the Smart Card.
4	RW	0x0	TXDONE Transmission Done Interrupt When enabled, this interrupt is asserted after one character was transmitted to the Smart Card.
3	RW	0x0	CLKSTOPRUN Smart Card Clock Stop Interrupt When enabled, this interrupt is asserted in two cases: 1. When the smart card clock is stopped (after CLOCKSTOP assertion). 2. When the new character transfer can be started (the smart card clock is fully running after CLOCKSTOP de-assertion).

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Bit	Attr	Reset Value	Description
2	RW	0x0	RXFIFULL RX FIFO Full Interrupt When enabled, this interrupt is asserted if the RX FIFO is filled up.
1	RW	0x0	TXFIEMPTY TX FIFO Empty Interrupt. When enabled, this interrupt is asserted if the TX FIFO is emptied out.
0	RW	0x0	TXFIDONE TX FIFO Done Interrupt When enabled, this interrupt is asserted after all bytes from TX FIFO were transferred to the Smart Card

SCR_FIFOCTRL
Address: Operational Base + offset (0x0014)
FIFO Control Register

Bit	Attr	Reset Value	Description
15:11	RO	0x0	reserved
10	WO	0x0	RXFIFLUSH Flush RX FIFO RX FIFO is flushed, when '1'.is written to this bit.
9	RO	0x0	RXFIFULL RX FIFO Full RX FIFO Full
8	RO	0x0	RXFIEMPTY RX FIFO Empty Field0000 Description
7:3	RO	0x0	reserved
2	WO	0x0	TXFIFLUSH Flush TX FIFO. TX FIFO is flushed, when '1'.is written to this bit.
1	RO	0x0	TXFIFULL TX FIFO Full TX FIFO Full
0	RO	0x0	TXFIEMPTY TX FIFO Empty. TX FIFO Empty.

SCR_LEGTXFICNT
Address: Operational Base + offset (0x0018)
Legacy TX FIFO Counter

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
7:0	RO	0x00	LEGTXFICNT Legacy TX FIFO Counter It is equal to TX FIFO Counter up to value 255. All values above 255 are read as 255. It is recommended to use the 16-bit TX FIFO Counter instead of this register.

SCR_LEGRXFICNT
Address: Operational Base + offset (0x0019)
Legacy RX FIFO Counter

Bit	Attr	Reset Value	Description
7:0	RO	0x00	LEGRXFICNT Legacy RX FIFO Counter It is equal to RX FIFO Counter up to value 255. All values above 255 are read as 255. It is recommended to use the 16-bit RX FIFO Counter instead of this register.

SCR_RXFITH
Address: Operational Base + offset (0x001c)
RX FIFO Threshold

Bit	Attr	Reset Value	Description
15:0	RW	0x0000	RXFITH RX FIFO Threshold The interrupt is asserted when the number of bytes it receives is equal to, or exceeds the threshold

SCR_REP
Address: Operational Base + offset (0x0020)
Repeat

Bit	Attr	Reset Value	Description
7:4	RW	0x0	RXREP RX Repeat This is a 4-bit, read/write register that specifies the number of attempts to request character re-transmission after wrong parity was detected. The re-transmission of the character is requested using the 1 ETU long error signal during the guardtime

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Bit	Attr	Reset Value	Description
3:0	RW	0x0	TXREP TX Repeat This is a 4-bit, read/write register that specifies the number of attempts to re-transmit the character after the Smart Card signals the wrong parity during the guardtime.

SCR_SCCDDIV
Address: Operational Base + offset (0x0024)
Smart Card Clock Divisor

Bit	Attr	Reset Value	Description
15:0	RW	0x0000	SCCDDIV Smart Card Clock Divisor This is a 16-bit, read/write register that defines the divisor value used to generate the Smart Card Clock from the system clock.

SCR_BAUDDIV
Address: Operational Base + offset (0x0028)
Baud Clock Divisor

Bit	Attr	Reset Value	Description
15:0	RW	0x0000	BAUDDIV Baud Clock Divisor This is a 16-bit, read/write register that defines a divisor value used to generate the Baud Clock impulses from the system clock

SCR_SCGUTIME
Address: Operational Base + offset (0x002c)
Smart Card Guardtime

Bit	Attr	Reset Value	Description
7:0	RW	0x00	SCGUTI Smart Card Guardtime This is an 8-bit, read/write register that sets a delay at the end of each character transmitted from the Smart Card Reader to the Smart Card. The value is in Elementary Time Units (ETU). The parity error is besides signaled during the guardtime

SCR_ADEATIME
Address: Operational Base + offset (0x0030)
Activation / Deactivation Time

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	ADEATIME Activation / Deactivation Time Sets the duration of each part of the activation and deactivation sequence. The value is in Smart Card Clock Cycles.
7:0	RW	0x00	Reserved Reserved Reserved bits are hard-wired to zero.

SCR_LOWRSTTIME
Address: Operational Base + offset (0x0034)
Reset Duration

Bit	Attr	Reset Value	Description
15:8	RW	0x00	LOWRSTTIME Reset Duration Sets the duration of the smart card reset sequence. This value is same for the cold and warm reset. The value is in terms of smart card clock cycles.
7:0	RW	0x00	Reserved Reserved Bits (7:0) of this register are hard-wired to zero.

SCR_ATRSTARTLIMIT
Address: Operational Base + offset (0x0038)
ATR Start Limit

Bit	Attr	Reset Value	Description
15:8	RW	0x00	ATRSTARTLIMIT ATR Start Limit Defines the maximum time between the rising edge of the SCRSTN signal and the start of ATR response. The value is in terms of smart card clock cycles
7:0	RW	0x00	Reserved Reserved Bits (7:0) of this register are hard-wired to zero

SCR_C2CLIM
Address: Operational Base + offset (0x003c)
Two Characters Delay Limit

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	C2CLIM Two Characters Delay Limit This is a 16-bit, read/write register that sets the maximum time between the leading edges of two, consecutive characters. The value is in 16 ETUs.

SCR_INTEN2
Address: Operational Base + offset (0x0040)
Interrupt Enable Register 2

Bit	Attr	Reset Value	Description
15:2	RO	0x0	reserved
1	RW	0x0	TCKERR TCK Error Interrupt. When enabled, this interrupt is asserted if the TCK byte does not match computed value.
0	RW	0x0	TXTHRESHOLD TX FIFO Threshold Interrupt When enabled, this interrupt is asserted if the number of bytes in TX FIFO is equal or less than the TX FIFO threshold.

SCR_INTSTAT2
Address: Operational Base + offset (0x0044)
Interrupt Status Register 2

Bit	Attr	Reset Value	Description
15:2	RO	0x0	reserved
1	RW	0x0	TCKERR TCK Error Interrupt When enabled, this interrupt is asserted if the TCK byte does not match computed value.
0	RW	0x0	TXTHRESHOLD TX FIFO Threshold Interrupt When enabled, this interrupt is asserted if the number of bytes in TX FIFO is equal or less than the TX FIFO threshold.

SCR_TXFITH
Address: Operational Base + offset (0x0048)
TX FIFO Threshold

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	TXFITH TX FIFO Threshold The interrupt is asserted when the number of bytes in TX FIFO is equal or less than the threshold

SCR_TXFIFOCNT
Address: Operational Base + offset (0x004c)
TX FIFO Counter

Bit	Attr	Reset Value	Description
15:0	RO	0x0000	TXFIFOCNT TX FIFO Counter This is a 16-bit, read-only register that provides the number of bytes stored in the RX FIFO

SCR_RXFIFOCNT
Address: Operational Base + offset (0x0050)
RX FIFO Counter

Bit	Attr	Reset Value	Description
15:0	RO	0x0000	RXFIFOCNT RX FIFO Counter This is a 16-bit, read-only register that provides the number of bytes stored in the RX FIFO.

SCR_BAUDTUNE
Address: Operational Base + offset (0x0054)
Baud Tune Register

Bit	Attr	Reset Value	Description
7:4	RO	0x0	reserved
3:0	RW	0x0	BAUDTUNE Baud Tune Register This is a 3-bit, read/write register that defines an additional value used to increase the accuracy of the Baud Clock impulses

SCR_FIFO
Address: Operational Base + offset (0x0200)
FIFO

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
7:0	RW	0x00	FIFO FIFO This is an 8-bit, read/write register that provides access to the receive and transmit FIFO buffers. The TX FIFO is accessed during the APB write transfer. The RX FIFO is accessed during the APB read transfer. All read/write accesses at address range 200h-3ffh are redirected to the FIFO.

44.5 Interface Description

Table 44-1 IOMUX Setting

Module Pin	IO	Pad Name	IOMUX Setting
sc_clk	O	IO_SCclk_SPI2txd_GPUJTAGtdi_GPIO1830gpio0b3	GPIO0B_IOMUX[7:6]= 2'b01
sc_rst	O	IO_SCrst_SPI2rxd_GPUJTAGtms_GPIO1830gpio0b2	GPIO0B_IOMUX[5:4]= 2'b01
sc_vcc18v	O	IO_SCvcc18v_I2CSENSORsda_GPUJTAGtck_GPIO1830gpio3d7	GPIO0B_IOMUX[1:0]= 2'b01
sc_vcc33v	O	IO_SCvcc33v_I2CSENSORscl_GPUJTAGtrstn_GPIO1830gpio0b1	GPIO3D_IOMUX[15:14]= 2'b01
sc_detect	I	IO_SCdetect_SPI2csn0_GPIO1830gpio0b5	GPIO0B_IOMUX[11:10]= 2'b01
sc_io	I/O	IO_Scio_SPI2rxd_GPUJTAGtms_GPIO1830gpio0b4	GPIO8A_IOMUX[9:8]= 2'b01

Notes: 1. I=input, O=output, I/O=input/output, bidirectional
2. Pull up scio pin when data transaction

44.6 Application Notes

44.6.1 SCR Clock

The Smart Card Clock signal is used as the main clock for the smart card. Its frequency can beadjusted using the Smart Card Clock Divisor (SCCDIV). This value is used to divide the system clock.
The SCCLK frequency is given by the following equation:

$$SCCLK_{freq} = \frac{CLK_freq}{2 * (SCCDIV + 1)}, \quad SCCDIV \cong \frac{CLK_{freq}}{2 * SCCLK_{freq}} - 1$$

SCCLK_freq– Smart Card Clock Frequency
CLK_freq- System Clock Frequency
The Baud Clock Impulse signal is used to transmit and receive serial data between the Smart CardReader and the Smart Card. The baud rate can be modified using the Baud Clock Divisor (BAUDDIV)which is used to divide the system clock. The BAUDDIV value must be >= 4. The BAUD rate is given by
the following equation:

$$BAUD_{rate} = \frac{CLK_freq}{2 * (BAUDDIV + 1)}$$

The duration of one bit, Elementary Time Unit (ETU) and parameters F and D are defined in the ISO/IEC7816-3 specification.

$$\frac{1}{BAUD_rate} \cong ETU = \frac{F}{D} * \frac{1}{SCCLK_{freq}}, \frac{F}{D} \cong \frac{BAUDDIV + 1}{SCCDIV + 1}$$

BAUDDIV equation based on SCCDIV value and Smart Card parameters F and D is following:

$$BAUDDIV \cong (SCCDIV + 1) * \frac{F}{D} - 1$$

During the first answer to reset response after the cold reset, the initial ETU must be equal to 372 SmartCard Clock Cycles (given by parameters F=372 and D=1). In this case, the BAUDDIV should be:

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$$BAUDDIV \cong (SCCDIV + 1) * \frac{372}{1} - 1$$

After the ATR is completed, the BAUDDIV register value can be changed according to Smart Cardparameters F and D.

Baud Tune Register (BAUDTUNE) 3-bit value that can be used to increase the accuracy of the BaudClock impulses timing by using the BAUDTUNE Increment from Table listed below in combination with BAUDDIVregister value.

Table 44-2 BAUDTUNE register

BAUDTUNE	000	001	010	011	100	101	110	111
$BAUDTUNE_{INCR}$	+0	+0.125	0.25	+0.375	+0.5	+0.625	+0.75	+0.875

$$BAUDDIV + BAUDTUNE_{INCR} \cong (SCCDIV + 1) * \frac{F}{D} - 1$$

The BAUDDIV register value (nearest integer) can be computed using following equation:

$$BAUDDIV \cong (SCCDIV + 1) * \frac{F}{D} - 1 - BAUDTUNE_{INCR}$$

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Chapter 45 SAR-ADC

45.1 Overview

The ADC is a 3-channel signal-ended 10-bit Successive Approximation Register (SAR) A/D Converter. It uses the supply and ground as it reference which avoid use of any external reference. It converts the analog input signal into 10-bit binary digital codes at maximum conversion rate of 100KSPS with 1MHz A/D converter clock.

45.2 Block Diagram

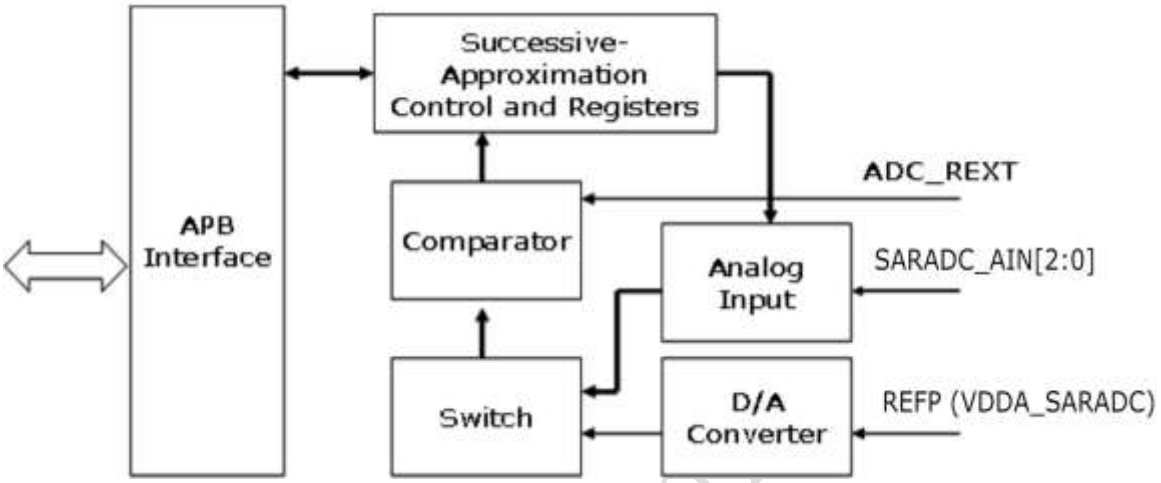

Fig. 45-1 PX5 SAR-ADC block diagram

Successive-Approximate Register and Control Logic Block

This block is exploited to realize binary search algorithm, storing the intermediate result and generate control signal for analog block.

Comparator Block

This block compares the analog input SARADC_AIN[2:0] with the voltage generated from D/A Converter, and output the comparison result to SAR and Control Logic Block for binary search. Three level amplifiers are employed in this comparator to provide enough gain.

45.3 Function description

In PX5, SAR-ADC works at single-sample operation mode.

This mode is useful to sample an analog input when there is a gap between two samples to be converted. In this mode START is asserted only on the rising edge of CLKIN where conversion is needed. At the end of every conversion EOC signal is made high and valid output data is available at the rising edge of EOC.The detailed timing diagram will be shown in the following.

45.4 Register Description

This section describes the control/status registers of the design.

45.4.1 Register Summary

Name	Offset	Size	Reset Value	Description
SARADC_DATA	0x00000	W	0x00000000	This register contains the data after A/D Conversion.
SARADC_STAS	0x00004	W	0x00000000	The status register of A/D Converter.

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Name	Offset	Size	Reset Value	Description
SARADC_CTRL	0x00008	W	0x00000000	The control register of A/D Converter.
SARADC_DLY_PU_SOC	0x0000c	W	0x00000000	delay between power up and start command

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

45.4.2 Detail Register Description

SARADC_DATA

Address: Operational Base + offset (0x00000)

This register contains the data after A/D Conversion.

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:0	RO	0x000	adc_data A/D value of the last conversion (DOUT[9:0]).

SARADC_STAS

Address: Operational Base + offset (0x00004)

The status register of A/D Converter.

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RO	0x0	adc_status ADC status (EOC) 0: ADC stop; 1: Conversion in progress.

SARADC_CTRL

Address: Operational Base + offset (0x00008)

The control register of A/D Converter.

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	int_status Interrupt status. This bit will be set to 1 when end-of-conversion. Set 0 to clear the interrupt.
5	RW	0x0	int_en Interrupt enable. 0: Disable; 1: Enable
4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3	RW	0x0	adc_power_ctrl ADC power down control bit 0: ADC power down; 1: ADC power up and reset. start signal will be asserted (DLY_PU_SOC + 2) sclk clock period later after power up
2:0	RW	0x0	adc_input_src_sel ADC input source selection(CH_SEL[2:0]). 111 : Input source 0 (SARADC_AIN[0]) 110 : Input source 1 (SARADC_AIN[1]) 101 : Input source 2 (SARADC_AIN[2]) Others : Reserved

SARADC_DLY_PU_SOC
Address: Operational Base + offset (0x0000c)
delay between power up and start command

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RW	0x00	DLY_PU_SOC delay between power up and start command The start signal will be asserted (DLY_PU_SOC + 2) sclk clock period later after power up

45.5 Timing Diagram

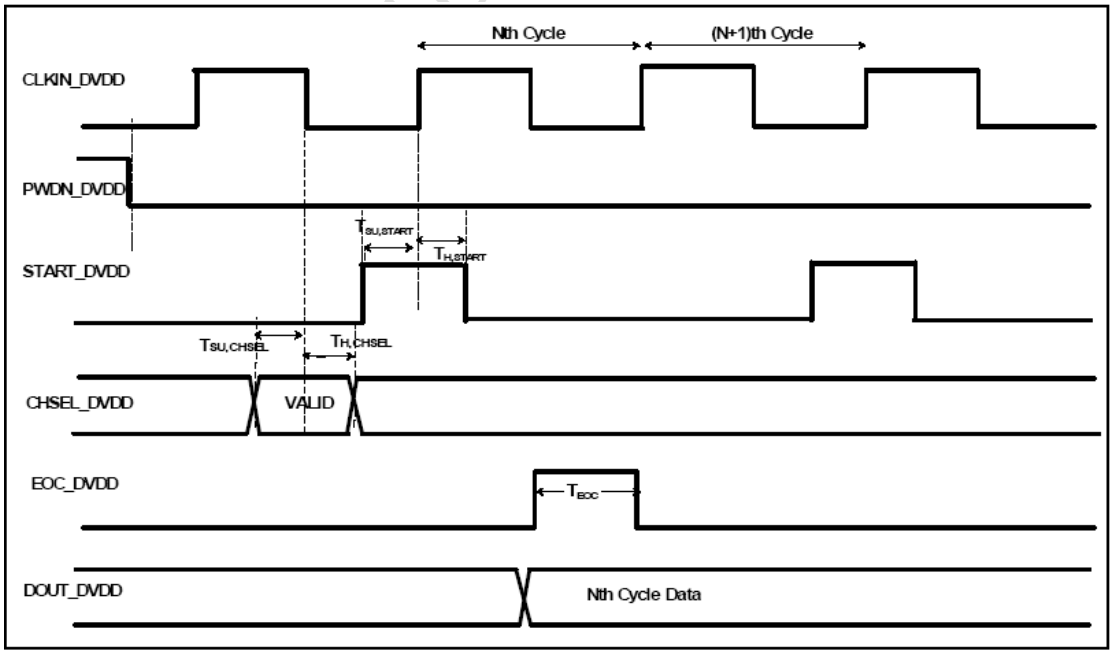

Fig. 45-2 SAR-ADC timing diagram in single-sample conversion mode

The following table has shows the detailed value for timing parameters in the above diagram.

Table 45-1 PX5 SAR-ADC timing parameters list

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Timing	Symbol	Value			Unit	Description
		Min	Typ	Max		
START_OF_CONV Setup time	TSU,START	5			ns	Set Up time for START_OF_CONV w.r.t CLKIN rising edge
START_OF_CONV Hold time	TH,START	5			ns	Hold time for START_OF_CONV w.r.t CLKIN rising edge
CHSEL setup time	TSU,CHSEL	5			ns	Set Up time for CHSEL w.r.t CLKIN falling edge
CHSEL Hold time	TH,CHSEL	5			ns	Hold time for CHSEL w.r.t CLKIN falling edge
Data Setup	TSU,DATA	400		900	ns	Set Up time for output data w.r.t either CLKIN rising edge or END_OF_CONV falling edge
Data Hold	TH,DATA	100		600	ns	Hold time for output data w.r.t either CLKIN rising edge or END_OF_CONV falling edge
Data access time	TDAC	100		600	ns	Valid data w.r.t CLKIN rising edge
Delay time	TDelay			5	ns	Delay between Valid data and EOC_DVDD rising edge
EOC Pulse Width (max frequency)	TEOC	400		900	ns	Pulse width of EOC
CLKIN Rise Time	TCR			2	ns	CLKIN Rise Time
CLKIN Fall Time	TCF			2	ns	CLKIN Fall Time
CLK Pulse Width(Duty Cycle)	TCPW	45		55	%	CLKIN High/Low Time Period
CLK Period	TCP	1			us	CLKIN Time Period

45.6 Application Notes

Steps of adc conversion:

- Write SARADC_CTRL[3] as 0 to power down adc converter.
- Write SARADC_CTRL[2:0] as n to select adc channel(n).
- Write SARADC_CTRL[5] as 1 to enable adc interrupt.
- Write SARADC_CTRL[3] as 1 to power up adc converter.
- Wait for adc interrupt or poll SARADC_STAS register to assert whether the coversion is completed
- Read the conversion result from SARADC_DATA[9:0]

Note: The A/D converter was designed to operate at maximum 1MHZ.

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Chapter 46 Temperature-Sensor ADC(TS-ADC)

46.1 Overview

TS-ADC Controller module supports user-defined mode and automatic mode. User-defined mode refers, TSADC all the control signals entirely by software writing to register for direct control. Automatic mode refers to the module automatically poll TSADC output, and the results were checked. If you find that the temperature High in a period of time, an interrupt is generated to the processor down-measures taken; if the temperature over a period of time High, the resulting TSHUT gave CRU module, let it reset the entire chip, or via GPIO give PMIC.
TS-ADC Controller supports the following features:

- Support User-Defined Mode and Automatic Mode
- In User-Defined Mode, start_of_conversion can be controlled completely by software, and also can be generated by hardware.
- In Automatic Mode, the temperature of alarm interrupt can be configurable
- In Automatic Mode, the temperature of system reset can be configurable
- Support to 2 channel TS-ADC, the temperature criteria of each channel can be configurable
- In Automatic Mode, the time interval of temperature detection can be configurable
- In Automatic Mode, when detecting a high temperature, the time interval of temperature detection can be configurable
- High temperature debounce can be configurable
- -40~125°C temperature range and 5°C temperature resolution
- 8-bit SARADC up to 50KS/s sampling rate

46.2 Block Diagram

TS-ADC controller comprises with:

- APB Interface
- TS-ADC control logic

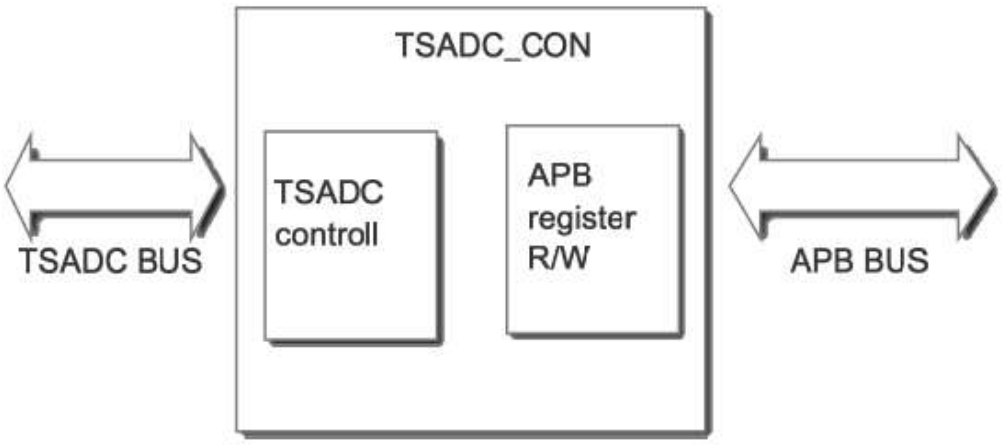

Fig. 46-1 TS-ADC Controller Block Diagram

46.3 Function Description

46.3.1 APB Interface

There is an APB Slave interface in TS-ADC Controller, which is used to configure the TS-ADC Controller registers and look up the temperature from the temperature sensor.

46.3.2 TS-ADC Controller

This block is exploited to realize binary search algorithm, storing the intermediate result and generate control signal for analog block. This block compares the analog input with the voltage generated from D/A Converter, and output the comparison result to SAR and Control

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Logic Block for binary search. Three level amplifiers are employed in this comparator to provide enough gain.

46.4 Register description

46.4.1 Register Summary

Name	Offset	Size	Reset Value	Description
TSADC_USER_CON	0x00000	W	0x00000208	The control register of A/D Converter.
TSADC_AUTO_CON	0x00004	W	0x00000000	TSADC auto mode control register
TSADC_INT_EN	0x00008	W	0x00000000	
TSADC_INT_PD	0x0000c	W	0x00000000	
TSADC_DATA0	0x00020	W	0x00000000	This register contains the data after A/D Conversion.
TSADC_DATA1	0x00024	W	0x00000000	This register contains the data after A/D Conversion.
TSADC_COMP0_INT	0x00030	W	0x00000000	TSADC high temperature level for source 0
TSADC_COMP1_INT	0x00034	W	0x00000000	TSADC high temperature level for source 1
TSADC_COMP0_SHUT	0x00040	W	0x00000000	TSADC high temperature level for source 0
TSADC_COMP1_SHUT	0x00044	W	0x00000000	TSADC high temperature level for source 1
TSADC_HIGHT_INT_DEBOUNCE	0x00060	W	0x00000003	high temperature debounce
TSADC_HIGHT_TSHUT_DEBOUNCE	0x00064	W	0x00000003	high temperature debounce
TSADC_AUTO_PERIOD	0x00068	W	0x00010000	TSADC auto access period
TSADC_AUTO_PERIOD_HT	0x0006c	W	0x00010000	TSADC auto access period when temperature is high

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

46.4.2 Detail Register Description

TSADC_USER_CON

Address: Operational Base + offset (0x00000)

The control register of A/D Converter.

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12	RO	0x0	adc_status ADC status (EOC) 0: ADC stop; 1: Conversion in progress.
11:6	RW	0x08	inter_pd_soc interleave between power down and start of conversion

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Bit	Attr	Reset Value	Description
5	RW	0x0	start When software write 1 to this bit , start_of_conversion will be assert. This bit will be cleared after TSADC access finishing. When TSADC_USER_CON[4] = 1'b1 take effect.
4	RW	0x0	start_mode start mode. 0: tsadc controller will asert start_of_conversion after "inter_pd_soc" cycles. 1: the start_of_conversion will be controlled by TSADC_USER_CON[5].
3	RW	0x1	adc_power_ctrl ADC power down control bit 0: ADC power down; 1: ADC power up and reset.
2:0	RW	0x0	adc_input_src_sel ADC input source selection(CH_SEL[2:0]). 000 : Input source 0 (SARADC_AIN[0]) 001 : Input source 1 (SARADC_AIN[1]) Others : Reserved

TSADC_AUTO_CON
Address: Operational Base + offset (0x00004)
TSADC auto mode control register

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25	RW	0x0	last_tshut_2cru last_tshut_2cru for cru first/second reset TSHUT status. This bit will set to 1 when tshut is valid, and only be cleared when application write 1 to it. This bit will not be cleared by system reset.
24	RW	0x0	last_tshut_2gpio last_tshut_2gpio for hardware reset TSHUT status. This bit will set to 1 when tshut is valid, and only be cleared when application write 1 to it. This bit will not be cleared by system reset.
23:18	RO	0x0	reserved
17	RO	0x0	sample_dly_sel 0: AUTO_PERIOD is used. 1: AUTO_PERIOD_HT is used.

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Bit	Attr	Reset Value	Description
16	RO	0x0	auto_status 0: auto mode stop; 1: auto mode in progress.
15:6	RO	0x0	reserved
5	RW	0x0	src1_en 0: do not care the temperature of source 1 1: if the temperature of source 1 is too high , TSHUT will be valid
4	RW	0x0	src0_en 0: do not care the temperature of source 0 1: if the temperature of source 0 is too high , TSHUT will be valid
3:1	RO	0x0	reserved
0	RW	0x0	auto_en 0: TSADC controller works at user-define mode 1: TSADC controller works at auto mode

TSADC_INT_EN
Address: Operational Base + offset (0x00008)

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12	RW	0x0	eoc_int_en eoc_Interrupt enable. eoc_interrupt enable in user defined mode 0: Disable; 1: Enable
11:10	RO	0x0	reserved
9	RW	0x0	tshut_2cru_en_src1 Field0000 Abstract 0: TSHUT output to cru disabled. TSHUT output will always keep low . 1: TSHUT output works.
8	RW	0x0	tshut_2cru_en_src0 Field0000 Abstract 0: TSHUT output to cru disabled. TSHUT output will always keep low . 1: TSHUT output works.
7:6	RO	0x0	reserved
5	RW	0x0	tshut_2gpio_en_src1 0: TSHUT output to gpio disabled. TSHUT output will always keep low . 1: TSHUT output works.

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Bit	Attr	Reset Value	Description
4	RW	0x0	tshut_2gpio_en_src0 0: TSHUT output to gpio disabled. TSHUT output will always keep low . 1: TSHUT output works.
3:2	RO	0x0	reserved
1	RW	0x0	ht_inten_src1 high temperature interrupt enable for src1 0: disable 1: enable
0	RW	0x0	ht_inten_src0 high temperature interrupt enable for src0 0: disable 1: enable

TSADC_INT_PD
Address: Operational Base + offset (0x0000c)

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8	RW	0x0	eoc_int_pd Interrupt status. This bit will be set to 1 when end-of-conversion. Set 0 to clear the interrupt.
7:6	RO	0x0	reserved
5	RW	0x0	tshut_o_src1 TSHUT output status When TSADC output is bigger than COMP_SHUT, this bit will be valid, which means temperature is VERY high, and the application should in charge of this. write 1 to it , this bit will be cleared.
4	RW	0x0	tshut_o_src0 TSHUT output status When TSADC output is bigger than COMP_SHUT, this bit will be valid, which means temperature is VERY high, and the application should in charge of this. write 1 to it , this bit will be cleared.
3:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1	RW	0x0	ht_irq_src1 When TSADC output is bigger than COMP_INT, this bit will be valid, which means temperature is high, and the application should in charge of this. write 1 to it , this bit will be cleared.
0	RW	0x0	ht_irq_src0 When TSADC output is bigger than COMP_INT, this bit will be valid, which means temperature is high, and the application should in charge of this. write 1 to it , this bit will be cleared.

TSADC_DATA0
Address: Operational Base + offset (0x00020)
This register contains the data after A/D Conversion.

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RO	0x000	adc_data A/D value of the channel 0 last conversion (DOUT[9:0]).

TSADC_DATA1
Address: Operational Base + offset (0x00024)
This register contains the data after A/D Conversion.

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RO	0x000	adc_data A/D value of the channel 1 last conversion (DOUT[9:0]).

TSADC_COMP0_INT
Address: Operational Base + offset (0x00030)
TSADC high temperature level for source 0

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	tsadc_comp_src0 TSADC high temperature level. TSADC output is bigger than tsadc_comp, means the temperature is high. TSADC_INT will be valid.

TSADC_COMP1_INT
Address: Operational Base + offset (0x00034)
TSADC high temperature level for source 1

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Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	tsadc_comp_src1 TSADC high temperature level. TSADC output is bigger than tsadc_comp, means the temperature is high. TSADC_INT will be valid.

TSADC_COMP0_SHUT
Address: Operational Base + offset (0x00040)
TSADC high temperature level for source 0

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	tsadc_comp_src0 TSADC high temperature level. TSADC output is bigger than tsadc_comp, means the temperature is too high. TSHUT will be valid.

TSADC_COMP1_SHUT
Address: Operational Base + offset (0x00044)
TSADC high temperature level for source 1

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	tsadc_comp_src1 TSADC high temperature level. TSADC output is bigger than tsadc_comp, means the temperature is too high. TSHUT will be valid.

TSADC_HIGHT_INT_DEBOUNCE
Address: Operational Base + offset (0x00060)
high temperature debounce

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x03	debounce TSADC controller will only generate interrupt or TSHUT when temperature is higher than COMP_INT for "debounce" times.

TSADC_HIGHT_TSHUT_DEBOUNCE
Address: Operational Base + offset (0x00064)
high temperature debounce

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:0	RW	0x03	debounce TSADC controller will only generate interrupt or TSHUT when temperature is higher than COMP_SHUT for "debounce" times.

TSADC_AUTO_PERIOD
Address: Operational Base + offset (0x00068)
TSADC auto access period

Bit	Attr	Reset Value	Description
31:0	RW	0x00010000	auto_period when auto mode is enabled, this register controls the interleave between every two accessing of TSADC.

TSADC_AUTO_PERIOD_HT
Address: Operational Base + offset (0x0006c)
TSADC auto access period when temperature is high

Bit	Attr	Reset Value	Description
31:0	RW	0x00010000	auto_period This register controls the interleave between every two accessing of TSADC after the temperature is higher than COMP_SHUT or COMP_INT

46.5 Application Notes

46.5.1 Channel Select

The system has two Temperature Sensors, channel 0is for CPU, and channel 1 is for GPU.

46.5.2 Single-sample conversion

To saving power, the TS-ADC used single-sample conversion. The timing as flowing picture:

The typical timing diagram of TS-ADC operation is shown in following figure. Each conversion costs 11 clock cycles. The sample signal denotes the end of previous conversion and the beginning of next conversion.

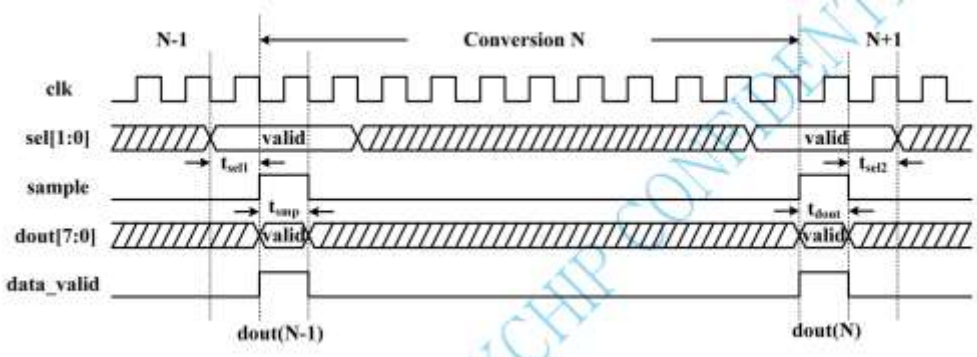

Fig. 46-2 Single-sample conversion

If a conversion is set to be more than 11 clock cycle, dout[7:0] will be valid until the falling edge of

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next sample signal, but the data_valid signal still last one clock cycle.

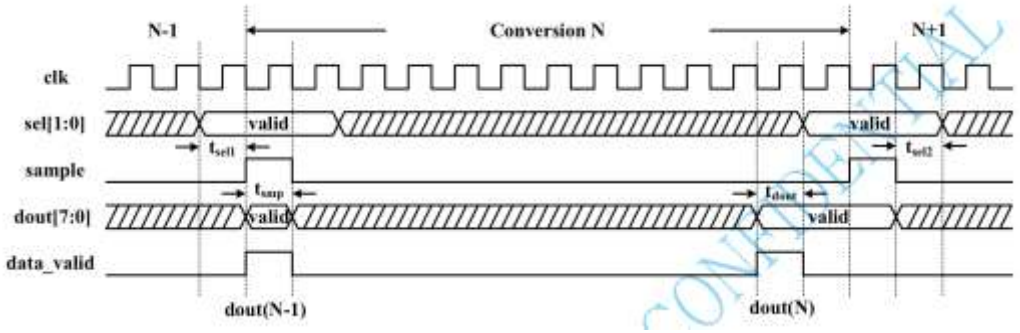

Fig. 46-3 Single-sample conversion – over 11 cycles

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46.5.3 Temperature-to-code mapping

Table 46-1 Temperature Code Mapping

The relation between temperature and TS-ADC output data is shown in following table.

Temperature /℃	ADC Output Data		
	Min	Typ	Max
-40		106	
-35		108	
-30		110	
-25		112	
-20		114	
-15		116	
-10		118	
-5		120	
0		122	
5		124	
10		126	
15		128	
20		130	
25		132	
30		134	
35		136	
40		138	
45		140	
50		142	
55		144	
60		146	
65		148	
70		150	
75		152	
80		154	
85		156	
90		158	
95		160	
100		162	
105		163	
110		165	
115		167	
120		169	
125		171	

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Note:

1. Code to Temperature mapping of the Temperature sensor is a piece wise linear curve. Any temperature, code faling between to 2 give temperatures can be linearly interpolated.
2. Code to Temperature mapping should be updated based on sillcon results.

46.5.4 User-Define Mode

1. In user-define mode, the PD_DVDD and CHSEL_DVDD are generate by setting register TSADC_USER_CON, bit[3] and bit[2:0]. In order to ensure timing between PD_DVDD and CHSEL_DVDD, the CHSEL_DVDD must be set before the PD_DVDD.
2. In user-define mode, you can choose the method to control the START_OF_CONVERSION by setting bit[4] of TSADC_USER_CON. If set to 0, the start_of_conversion will be assert after "inter_pd_soc" cycles, which could be set by bit[11:6] of TSADC_USER_CON. And if start_mode was set 1, the start_of_conversion will be controlled by bit[5] of

TSADC_USER_CON.

3. Software can get the four channel temperature from TSADC_DATAn (n=0,1,2,3).

46.5.5 Automatic Mode

You can use the automatic mode with the following step:

1. Set TSADC_AUTO_PERIOD，configure the interleave between every two accessing of TSADC in normal operation.
2. Set TSADC_AUTO_PERIOD_HT. configure the interleave between every two accessing of TSADC after the temperature is higher than COMP_SHUT or COMP_INT.
3. Set TSADC_COMPn_INT(n=0,1), configure the high temperature level, if tsadc output is smaller than the value, means the temperature is high, tsadc_int will be asserted.
4. Set TSADC_COMPn_SHUT(n=0,1), configure the super high temperature level, if tsadc output is smaller than the value, means the temperature is too high, TSHUT will be asserted.
5. Set TSADC_INT_EN, you can enable the high temperature interrupt for all channel; and you can also set TSHUT output to gpio to reset the whole chip; and you can set TSHUT output to cru to reset the whole chip.
6. Set TSADC_HIGHT_INT_DEBOUNCE and TSADC_HIGHT_TSHUT_DEBOUNCE, if the temperature is higher than COMP_INT or COMP_SHUT for "debounce" times, TSADC controller will generate interrupt or TSHUT.
7. Set TSADC_AUTO_CON, enable the TSADC controller.

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Chapter 47 Timer

47.1 Overview

Timer is a programmable timer peripheral. This component is an APBslave device. Timer0~11 and STimer0~1 count up from zero to a programmed value and generate an interrupt when the count reaches the programmed value.
Timer supports the following features:

- Timer0~11 is used for no-secure, STimer0~1 is used for secure.
- Two operation modes: free-running and user-defined count.
- Maskable for each individual interrupt.

47.2 Block Diagram

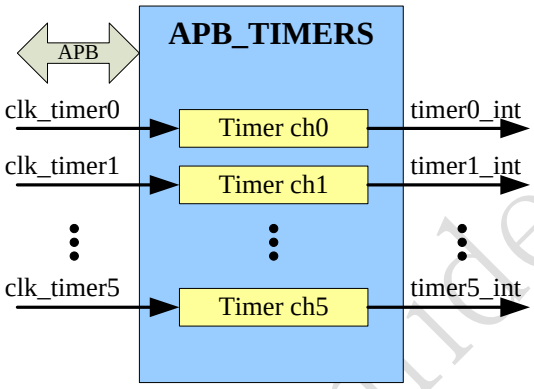

Fig. 47-1 Timers Block Diagram

The above figure shows the architecture of the APB timers (include six programmable timer channel) that in the cpu subsystem. The other APB timers that in the alive subsystem only include one programmable timer channel.

47.3 Function description

47.3.1 Timer clock

TIMER0, TIMER1, TIMER2, Timer3, TIMER4 and TIMER5 are in the CPU subsystem, using timer ch0 ~ ch5 respectively. The timer clock is 24MHz OSC.
TIMER6 and TIMER7are in the ALIVE subsystem, using timer ch0 ~ ch1. The timer clock is 24MHz OSC.

47.3.2 Programming sequence

1. Initialize the timer by the TIMERn_CONTROLREG (0≤n≤11)register:

- Disable the timer by writing a "0" to the timer enable bit (bit 0). Accordingly, the timer_enoutput signal is de-asserted.
- Program the timer mode—user-defined or free-running—by writing a "0" or "1" respectively,to the timer mode bit (bit 1).
- Set the interrupt mask as either masked or not masked by writing a "0" or "1" respectively, tothe timer interrupt mask bit (bit 2).

2. Load the timer count value into the TIMERn_LOAD_COUNT1 (0≤n≤11) and TIMERn_LOAD_COUNT0 (0≤n≤11)register.

3. Enable the timer by writing a "1" to bit 0 of TIMERn_CONTROLREG (0≤n≤11).

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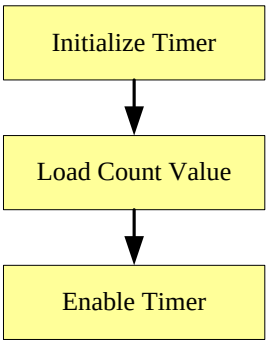

Fig. 47-2 Timer Usage Flow

47.3.3 Loading a timer countvalue

The initial value for each timer—that is, the value from which it counts down—is loaded into the timer using the load count register (TIMERn_LOAD_COUNT1 (0≤n≤11) and TIMERn_ LOAD_COUNT0 (0≤n≤11)). Two events can cause a timer to load the initial value from its load count register:

- Timer is enabled after reset or disabled.
- Timer counts down to 0, when timer is configured into free-running mode.

47.3.4 Timer mode selection

- User-defined count mode – Timer loads TIMERn_LOAD_COUNT1(0≤n≤11)and TIMERn_ LOAD_ COUNT0(0≤n≤11)register as initial value. Timer will not automatically load the count register, when timer counts down to 0. User need to disable timer firstly and follow the programming sequence to make timer work again.
- Free-running mode – Timer loads the TIMERn_LOAD_COUNT1(0≤n≤11)and TIMERn_LOAD_COUNT0(0≤n≤11)register as initial value.Timer will automatically load the count register, when timer counts down to 0.

47.4 Register Description

This section describes the control/status registers of the design. Software should read and write these registers using 32-bits accesses.

47.4.1 Registers Summary

Timer0 - Timer5
Base = 0xff81_0000

Name	Offset	Size	Reset Value	Description
TIMER0_LOAD_COUNT0	0x0000	W	0x00000000	Timer0 Load Count Register
TIMER0_LOAD_COUNT1	0x0004	W	0x00000000	Timer0 Load Count Register
TIMER0_CURRENT_VALUE0	0x0008	W	0x00000000	Timer0 Current Value Register
TIMER0_CURRENT_VALUE1	0x000C	W	0x00000000	Timer0 Current Value Register
TIMER0_CONTROLREG	0x0010	W	0x00000000	Timer0 Control Register
TIMER0_INTSTATUS	0x0018	W	0x00000000	Timer0 Interrupt Status Register
TIMER1_LOAD_COUNT0	0x0020	W	0x00000000	Timer1 Load Count Register
TIMER1_LOAD_COUNT1	0x0024	W	0x00000000	Timer1 Load Count Register
TIMER1_CURRENT_VALUE0	0x0028	W	0x00000000	Timer1 Current Value Register
TIMER1_CURRENT_VALUE1	0x002c	W	0x00000000	Timer1 Current Value Register
TIMER1_CONTROLREG	0x0030	W	0x00000000	Timer1 Control Register
TIMER1_INTSTATUS	0x0038	W	0x00000000	Timer1 Interrupt Status Register
TIMER2_LOAD_COUNT0	0x0040	W	0x00000000	Timer2 Load Count Register
TIMER2_LOAD_COUNT1	0x0044	W	0x00000000	Timer2 Load Count Register
TIMER2_CURRENT_VALUE0	0x0048	W	0x00000000	Timer2 Current Value Register
TIMER2_CURRENT_VALUE1	0x004c	W	0x00000000	Timer2 Current Value Register

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TIMER2_CONTROLREG	0x0050	W	0x00000000	Timer2 Control Register
TIMER2_INTSTATUS	0x0058	W	0x00000000	Timer2 Interrupt Status Register
TIMER3_LOAD_COUNT0	0x0060	W	0x00000000	Timer3 Load Count Register
TIMER3_LOAD_COUNT1	0x0064	W	0x00000000	Timer3 Load Count Register
TIMER3_CURRENT_VALUE0	0x0068	W	0x00000000	Timer3 Current Value Register
TIMER3_CURRENT_VALUE1	0x006c	W	0x00000000	Timer3 Current Value Register
TIMER3_CONTROLREG	0x0070	W	0x00000000	Timer3 Control Register
TIMER3_INTSTATUS	0x0078	W	0x00000000	Timer3 Interrupt Status Register
TIMER4_LOAD_COUNT0	0x0080	W	0x00000000	Timer4 Load Count Register
TIMER4_LOAD_COUNT1	0x0084	W	0x00000000	Timer4 Load Count Register
TIMER4_CURRENT_VALUE0	0x0088	W	0x00000000	Timer4 Current Value Register
TIMER4_CURRENT_VALUE1	0x008c	W	0x00000000	Timer4 Current Value Register
TIMER4_CONTROLREG	0x0090	W	0x00000000	Timer4 Control Register
TIMER4_INTSTATUS	0x0098	W	0x00000000	Timer4 Interrupt Status Register
TIMER5_LOAD_COUNT0	0x00a0	W	0x00000000	Timer5 Load Count Register
TIMER5_LOAD_COUNT1	0x00a4	W	0x00000000	Timer5 Load Count Register
TIMER5_CURRENT_VALUE0	0x00a8	W	0x00000000	Timer5 Current Value Register
TIMER5_CURRENT_VALUE1	0x00ac	W	0x00000000	Timer5 Current Value Register
TIMER5_CONTROLREG	0x00b0	W	0x00000000	Timer5 Control Register
TIMER5_INTSTATUS	0x00b8	W	0x00000000	Timer5 Interrupt Status Register

Timer6 – Timer11
Base = 0xff82_0000

Name	Offset	Size	Reset Value	Description
TIMER6_LOAD_COUNT0	0x0000	W	0x00000000	Timer6 Load Count Register
TIMER6_LOAD_COUNT1	0x0004	W	0x00000000	Timer6 Load Count Register
TIMER6_CURRENT_VALUE0	0x0008	W	0x00000000	Timer6 Current Value Register
TIMER6_CURRENT_VALUE1	0x000C	W	0x00000000	Timer6 Current Value Register
TIMER6_CONTROLREG	0x0010	W	0x00000000	Timer6 Control Register
TIMER6_INTSTATUS	0x0018	W	0x00000000	Timer6 Interrupt Status Register
TIMER7_LOAD_COUNT0	0x0020	W	0x00000000	Timer7 Load Count Register
TIMER7_LOAD_COUNT1	0x0024	W	0x00000000	Timer7 Load Count Register
TIMER7_CURRENT_VALUE0	0x0028	W	0x00000000	Timer7 Current Value Register
TIMER7_CURRENT_VALUE1	0x002c	W	0x00000000	Timer7 Current Value Register
TIMER7_CONTROLREG	0x0030	W	0x00000000	Timer7 Control Register
TIMER7_INTSTATUS	0x0038	W	0x00000000	Timer7 Interrupt Status Register
TIMER8_LOAD_COUNT0	0x0040	W	0x00000000	Timer8 Load Count Register
TIMER8_LOAD_COUNT1	0x0044	W	0x00000000	Timer8 Load Count Register
TIMER8_CURRENT_VALUE0	0x0048	W	0x00000000	Timer8 Current Value Register
TIMER8_CURRENT_VALUE1	0x004c	W	0x00000000	Timer8 Current Value Register
TIMER8_CONTROLREG	0x0050	W	0x00000000	Timer8 Control Register
TIMER8_INTSTATUS	0x0058	W	0x00000000	Timer8 Interrupt Status

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				Register
TIMER9_LOAD_COUNT0	0x0060	W	0x00000000	Timer9 Load Count Register
TIMER9_LOAD_COUNT1	0x0064	W	0x00000000	Timer9 Load Count Register
TIMER9_CURRENT_VALUE0	0x0068	W	0x00000000	Timer9 Current Value Register
TIMER9_CURRENT_VALUE1	0x006c	W	0x00000000	Timer9 Current Value Register
TIMER9_CONTROLREG	0x0070	W	0x00000000	Timer9 Control Register
TIMER9_INTSTATUS	0x0078	W	0x00000000	Timer9 Interrupt Status Register
TIMER10_LOAD_COUNT0	0x0080	W	0x00000000	Timer10 Load Count Register
TIMER10_LOAD_COUNT1	0x0084	W	0x00000000	Timer10 Load Count Register
TIMER10_CURRENT_VALUE0	0x0088	W	0x00000000	Timer10 Current Value Register
TIMER10_CURRENT_VALUE1	0x008c	W	0x00000000	Timer10 Current Value Register
TIMER10_CONTROLREG	0x0090	W	0x00000000	Timer10 Control Register
TIMER10_INTSTATUS	0x0098	W	0x00000000	Timer10 Interrupt Status Register
TIMER11_LOAD_COUNT0	0x00a0	W	0x00000000	Timer11 Load Count Register
TIMER11_LOAD_COUNT1	0x00a4	W	0x00000000	Timer11 Load Count Register
TIMER11_CURRENT_VALUE0	0x00a8	W	0x00000000	Timer11 Current Value Register
TIMER11_CURRENT_VALUE1	0x00ac	W	0x00000000	Timer11 Current Value Register
TIMER11_CONTROLREG	0x00b0	W	0x00000000	Timer11 Control Register
TIMER11_INTSTATUS	0x00b8	W	0x00000000	Timer11 Interrupt Status Register

STimer0-1
Base: 0xff83_0000

Name	Offset	Size	Reset Value	Description
TIMER0_LOAD_COUNT0	0x0000	W	0x00000000	Timer0 Load Count Register
TIMER0_LOAD_COUNT1	0x0004	W	0x00000000	Timer0 Load Count Register
TIMER0_CURRENT_VALUE0	0x0008	W	0x00000000	Timer0 Current Value Register
TIMER0_CURRENT_VALUE1	0x000C	W	0x00000000	Timer0 Current Value Register
TIMER0_CONTROLREG	0x0010	W	0x00000000	Timer0 Control Register
TIMER0_INTSTATUS	0x0018	W	0x00000000	Timer0 Interrupt Status Register
TIMER1_LOAD_COUNT0	0x0020	W	0x00000000	Timer1 Load Count Register
TIMER1_LOAD_COUNT1	0x0024	W	0x00000000	Timer1 Load Count Register
TIMER1_CURRENT_VALUE0	0x0028	W	0x00000000	Timer1 Current Value Register
TIMER1_CURRENT_VALUE1	0x002c	W	0x00000000	Timer1 Current Value Register
TIMER1_CONTROLREG	0x0030	W	0x00000000	Timer1 Control Register
TIMER1_INTSTATUS	0x0038	W	0x00000000	Timer1 Interrupt Status Register

*Notes: Size: **B** – Byte (8 bits) access, **HW** – Half WORD (16 bits) access, **W** –WORD (32 bits) access*

47.4.2 Detail Register Description

TIMERn_LOAD_COUNT0
Address: Operational Base + offset(0x00+n*0x20)
Timer n Load Count Register(0≤n≤11)

Bit	Attr	Reset Value	Description

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31:0	RW	0x0	Low 32 bits Value to be loaded into Timer n. This is the value from which counting commences.

TIMERn_LOAD_COUNT1
Address: Operational Base + offset(0x04+n*0x20)
Timer n Load Count Register(0≤n≤11)

Bit	Attr	Reset Value	Description
31:0	RW	0x0	High 32 bits Value to be loaded into Timer n. This is the value from which counting commences.

TIMERn_CURRENT_VALUE0
Address: Operational Base + offset(0x08+n*0x20)
Timer n Current Value Register(0≤n≤11)

Bit	Attr	Reset Value	Description
31:0	R	0x0	Low 32 bits of Current Value of Timer n.

TIMERn_CURRENT_VALUE1
Address: Operational Base + offset(0x0c+n*0x20)
Timer n Current Value Register(0≤n≤11)

Bit	Attr	Reset Value	Description
31:0	R	0x0	High 32 bits of Current Value of Timer n.

TIMERn_CONTROLREG
Address: Operational Base + offset(0x10+n*0x20)
Timer n Control Register(0≤n≤11)

Bit	Attr	Reset Value	Description
31:3	-	-	Reserved
2	RW	0x0	Timer interrupt mask. 1'b0: mask 1'b1: not mask
1	RW	0x0	Timer mode. 1'b0: free-running mode 1'b1: user-defined count mode
0	RW	0x0	Timer enable. 1'b0: disable 1'b1: enable

TIMERn_INTSTATUS
Address: Operational Base + offset(0x18+n*0x20)
Timer n Interrupt Status Register(0≤n≤11)

Bit	Attr	Reset Value	Description
31:1	-	-	Reserved
0	RW	0x0	This register contains the interrupt status for timer n Write 1 to this register will clear the interrupt

*Notes: Attr: **RW** – Read/writable, **R** – read only, **W** – write only*

47.5 Application Notes

In the chip, the timer_clk is from 24MHz OSC, asynchronous to the pclk. When user disable the timer enable bit (bit 0 of TIMERn_CONTROLREG(0≤n≤11)), the timer_en output signal is de-asserted, and timer_clk will stop. When user enable the timer, the timer_en signal is asserted and timer_clkwill start running.
The application is only allowed to re-config registers when timer_en is low.

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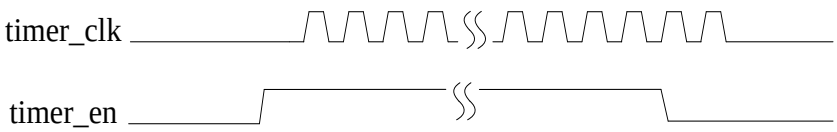

Fig. 47-3 Timing between timer_en and timer_clk

Please refer to funciton description section for the timer usage flow.

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Chapter 48 eFuse

48.1 Overview

In PX5, there are two eFuse. One is organized as 32bits by 8 one-time programmable electrical fuses with random access interface, and the other is organized as 32bits by 32 one-time programmable electrical fuses.
The eFuse can only be accessed by APB bus at secure mode.
It is a type of non-volatile memory fabricated in standard CMOS logic process. The main features are as follows:

- Programming condition : VQPS_EFUSE = 1.5V±10%
- Program time : 10us±0.2us .
- Read condition : VQPS_EFUSE = 0V
- Provide standby mode

48.2 Block Diagram

In the following diagram, all the signals except power supply VDD_EFUSE and VQPS_EFUSE are controlled by registers. For detailed description, please refer to detailed register descriptions.

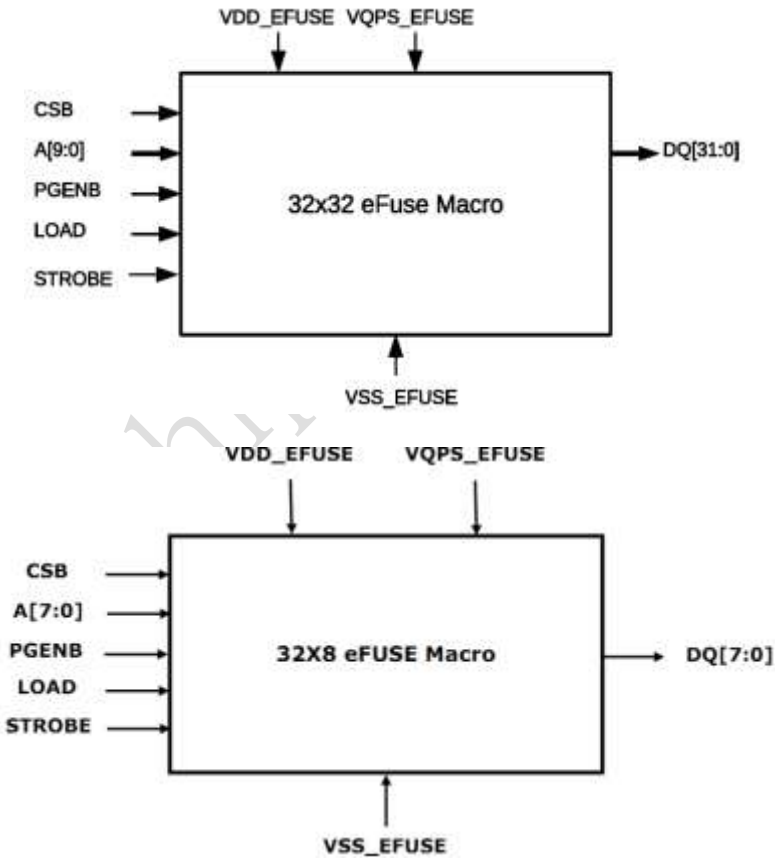

Fig. 48-1 PX5 eFuse block diagram

48.3 Function description

eFuse has three operation modes. They are defined as standby, read and programming.
Program (PGM) Mode
In order to enter programming mode, the following conditions need to be satisfied: VQPS_EFUSE is at high voltage, LOAD signal is low, PGENB signal is low, and CSB signal is low. All bits can be individually programmed (one at a time) with the proper address selected, the STROBE signal high and the address bits satisfying setup and hold time with respect to STROBE.

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Read Mode
In order to enter read mode the following conditions need to be satisfied: VQPS_EFUSE is at ground, the LOAD signal is high, the PGENB signal is high, and the CSB is low. An entire 8-bit word of data can be read in one read operation with STROBE being high and a proper address selected (address signals A5~A7 are "don't cares").
Standby Mode
Standby is defined when the macro is not being programmed or read. The conditions for standby mode are: the LOAD signal is low, the STROBE signal is low, the CSB signal is high and PGENB is high.

48.4 Register Description

This section describes the control/status registers of the design.

48.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
EFUSE_CTRL	0x0000	W	0x00000000	efuse control register
EFUSE_DOUT	0x0004	W	0x00000000	efuse data out register

*Notes: Size: **B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**- WORD (32 bits) access*

48.4.2 Detail Register Description

EFUSE_CTRL
Address: Operational Base + offset (0x0000)
eFuse control register

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
15:6	RW	0x00	efuse_addr efuse address pins : A[7:0] / A[9:0]
5:4	RO	0x0	reserved
3	RW	0x0	efuse_pgenb efuse program enable (active low) :PGENB
2	RW	0x0	efuse_load efuse turn on sense amplifier and load data into latch (active high) : LOAD
1	RW	0x0	efuse_strobe efuse turn on the array for read or program access (active high) : STROBE
0	RW	0x0	efuse_csb efuse chip select enable signal, active low : CSB

EFUSE_DOUT
Address: Operational Base + offset (0x0004)
eFuse data out register

Bit	Attr	Reset Value	Description
31:0	RO	0x00	efuse_dout efuse data output

48.5 Timing Diagram

- When efuse is in program(PGM) mode.

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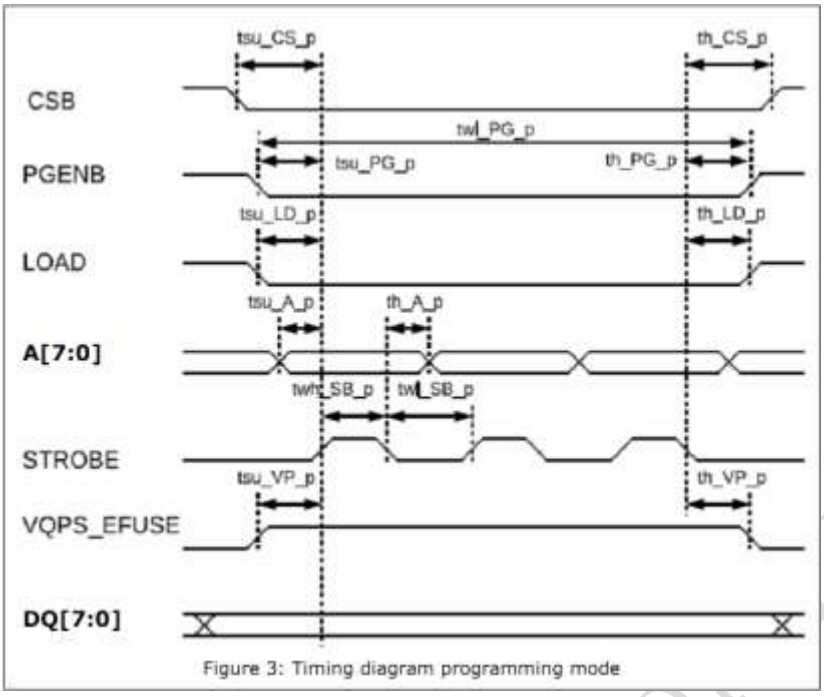

Fig. 48-2 PX5 efuse timing diagram in program mode

- When efuse is in read mode.

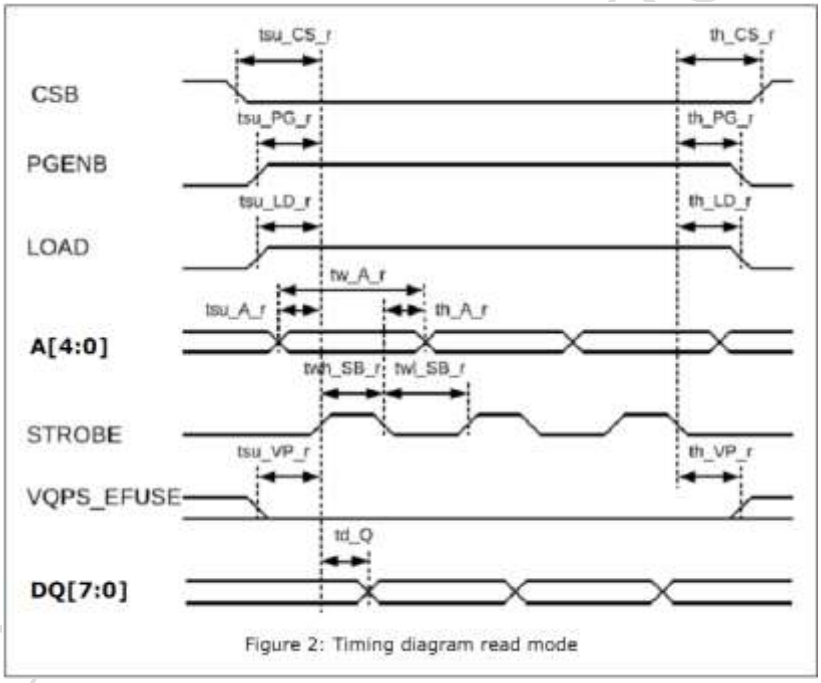

Fig. 48-3 PX5 efuse timing diagram in read mode

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The following table has shows the detailed value for timing parameters in the above diagram.

Table 48-1 PX5 eFuse timing parameters list

Mode	Item	Description	Min	Typ	Max	Unit
Read Mode	twh_SB_r	Pulse width high of STROBE read strobe	20		-	ns
	twl_SB_r	Pulse width low of STROBE read strobe	15		-	ns
	tsu_A_r	A[7:0] to STROBE setup time in read mode	25		-	ns
	th_A_r	A[7:0] to STROBE hold time in read mode	3		-	ns
	tw_A_r	A[7:0] pulse width while LOAD high in read mode	48		100	ns
	tsu_CS_r	CSB to STROBE setup time in read mode	16		-	ns
	th_CS_r	CSB to STROBE hold time in read mode	6		-	ns
	tsu_PG_r	PGENB to STROBE setup time in read mode	14		-	ns
	th_PG_r	PGENB to STROBE hold time in read mode	10		-	ns
	tsu_LD_r	LOAD to STROBE setup time in read mode	10		-	ns
	th_LD_r	LOAD to STROBE hold time in read mode	7		-	ns
	tsu_VP_r	VQPS_EFUSE to STROBE setup time in read mode	20		-	ns
	th_VP_r	VQPS_EFUSE to STROBE hold time in read mode	20		-	ns
	td_Q	DQ[7:0] delay time after STROBE high	0		8	ns
PGM Mode	twh_SB_p	Pulse width high of STROBE PGM strobe	9.8	10	10.2	us
	twl_SB_p	Pulse width low of STROBE PGM strobe	15		-	ns
	tsu_A_p	A[7:0] to STROBE setup time in PGM mode	12		-	ns
	th_A_p	A[7:0] to STROBE hold time in PGM mode	3		-	ns
	tsu_CS_p	CSB to STROBE setup time in PGM mode	16		-	ns
	th_CS_p	CSB to STROBE hold time in PGM mode	6		-	ns
	tsu_PG_p	PGENB to STROBE setup time in PGM mode	14		-	ns
	th_PG_p	PGENB to STROBE hold time in PGM mode	10		-	ns
	twl_PG_p	PGENB pulse width low (cumulative) in PGM mode	-		100	ms
	tsu_LD_p	LOAD to STROBE setup time in PGM mode	10		-	ns
	th_LD_p	LOAD to STROBE hold time in PGM mode	7		-	ns
	tsu_VP_p	VQPS_EFUSE to STROBE setup time in PGM mode	20		-	ns
	th_VP_p	VQPS_EFUSE to STROBE hold time in PGM mode	20		-	ns

48.6 Application Notes

During usage of efuse, customers must pay more attention to the following items:
1. In condition of program(PGM) mode, VQPS_EFUSE= 1.5V±10%.
2. Q0~Q7/Q31 will be reset to "0" once CSB at high.
3. No data access allowed at the rising edge of CSB.
4. All the program timing for each signal must be more than the value defined in the timing table.

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Chapter 49 General-Purpose Input/Output Ports (GPIO)

49.1 Overview

GPIO is a programmable General Purpose Programming I/O peripheral. This component is an APB slave device. GPIO controls the output data and direction of external I/O pads. It also can read back the data on external pads using memory-mapped registers.
GPIO supports the following features:

- 32 bits APB bus width
- 32 independently configurable signals
- Separate data registers and data direction registers for each signal
- Software control for each signal, or for each bit of each signal
- Configurable interrupt mode

49.2 Block Diagram

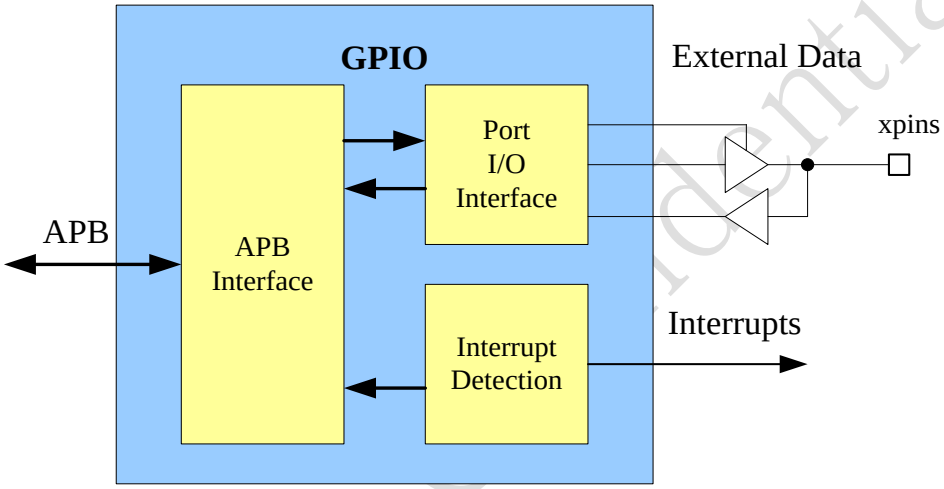

Fig. 49-1 GPIO block diagram

Block descriptions:
APB Interface
The APB Interface implements the APB slave operation. Its data bus width is 32 bits.
Port I/O Interface
External data Interface to or from I/O pads.
Interrupt Detection
Interrupt interface to or from interrupt controller.

49.3 Function description

49.3.1 Operation

Control Mode(software)
Under software control, the data and direction control for the signal aresourced from the data register (GPIO_SWPORTA_DR) and direction control register (GPIO_SWPORTA_DDR).
The direction of the external I/O pad is controlled by a write to the Porta datadirection register (GPIO_SWPORTA_DDR). The data written to this memory-mapped register gets mapped on to an output signal, GPIO_PORTA_DDR, of the GPIO peripheral. This output signal controls thedirection of an external I/O pad.
The data written to the Porta data register (GPIO_SWPORTA_DR) drives the output buffer of the I/O pad.External data are input on the external data signal, GPIO_EXT_PORTA. Reading the external signal register(GPIO_EXT_PORTA) shows the value on the signal, regardless of the direction. This register is read-only,meaning that it cannot be written from the APB

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software interface.

Reading External Signals

The data on the GPIO_EXT_PORTA external signal can always be read. The data on the external GPIO signal is read by an APB read of the memory-mapped register, GPIO_EXT_PORTA.

An APB read to the GPIO_EXT_PORTA register yields a value equal to that which is on the GPIO_EXT_PORTA signal.

Interrupts

Port A can be programmed to accept external signals as interrupt sources on any of the bits of the signal. The type of interrupt is programmable with one of the following settings:

- Active-high and level
- Active-low and level
- Rising edge
- Falling edge

The interrupts can be masked by programming the GPIO_INTMASK register. The interrupt status can be read before masking (called raw status) and after masking.

The interrupts are combined into a single interrupt output signal, which has the same polarity as the individual interrupts. In order to mask the combined interrupt, all individual interrupts have to be masked. The single combined interrupt does not have its own mask bit.

Whenever Port A is configured for interrupts, the data direction must be set to Input. If the data direction register is reprogrammed to Output, then any pending interrupts are not lost.However, no new interrupts are generated.

For edge-detected interrupts, the ISR can clear the interrupt by writing a 1 to the GPIO_PORTA_EOI register for the corresponding bit to disable the interrupt. This write also clears the interrupt status and raw status registers. Writing to the GPIO_PORTA_EOI register has no effect on level-sensitive interrupts. If level-sensitive interrupts cause the processor to interrupt, then the ISR can poll the GPIO_INT_RAWSTATUS register until the interrupt source disappears, or it can write to the GPIO_INTMASK register to mask the interrupt before exiting the ISR. If the ISR exits without masking or disabling the interrupt prior to exiting, then the level-sensitive interrupt repeatedly requests an interrupt until the interrupt is cleared at the source.

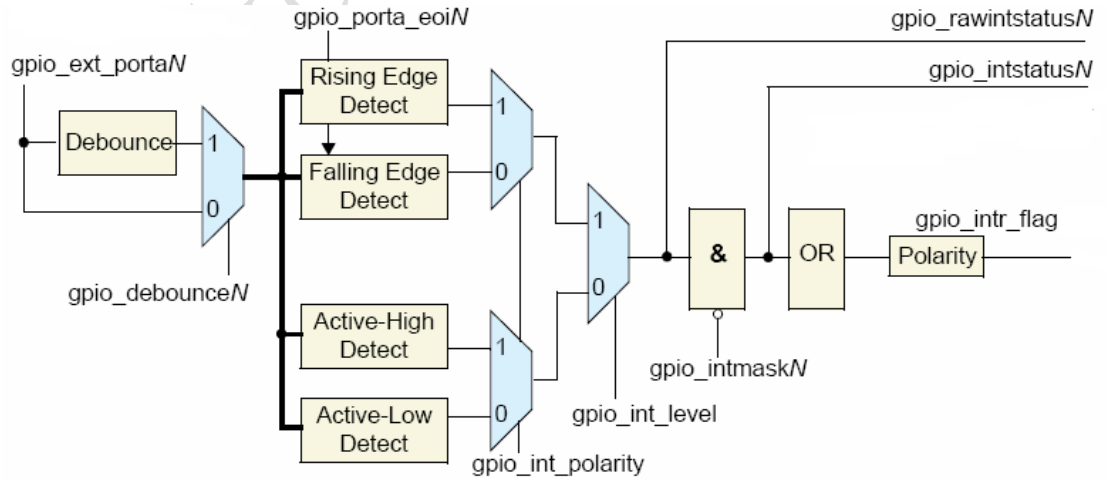

Fig. 49-2 GPIO Interrupt RTL Block Diagram

Debounce operation

Port A has been configured to include the debounce capability interrupt feature. The external signal can be debounced to remove any spurious glitches that are less than one period of the external debouncing clock.

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When input interrupt signals are debounced using a debounce clock(pclk), the signals must be active for a minimum of two cycles of the debounce clock to guarantee that they are registered. Any input pulse widths less than a debounce clock period are bounced. A pulse width between one and two debounce clock widths may or may not propagate, depending on its phase relationship to the debounce clock. If the input pulse spans two rising edges of the debounce clock, it is registered. If it spans only one risingedge, it is not registered.

Synchronization of Interrupt Signals to the System Clock

Interrupt signals are internally synchronized to pclk. Synchronization topclk must occur for edge-detect signals. With level-sensitive interrupts, synchronization is optional andunder software control(GPIO_LS_SYNC).

49.3.2 Programming

Programming Considerations

- Reading from an unused location or unused bits in a particular register always returns zeros. There is no error mechanism in the APB.
- Programming the GPIO registers for interrupt capability, edge-sensitive or level-sensitiveinterrupts, and interrupt polarity should be completed prior to enabling the interrupts on PortA inorder to prevent spurious glitches on the interrupt lines to the interrupt controller.
- Writing to the interrupt clear register clears an edge-detected interrupt and has no effect on alevel-sensitive interrupt.

4 GPIOs' hierarchy in the chip

GPIO0 is in PD_PMU subsystem, GPIO1~3 arein PD_ALIVE subsystem.

49.4 Register Description

This section describes the control/status registers of the design. Software should read and write these registers using 32-bits accesses. There are 4 GPIOs (GPIO0 ~ GPIO3), and each of them has same register group. Therefore, 4 GPIOs' register groups have 4 different base address.

49.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
GPIO_SWPORTA_DR	0x0000	W	0x00000000	Port A data register
GPIO_SWPORTA_DDR	0x0004	W	0x00000000	Port A data direction register
GPIO_INTEN	0x0030	W	0x00000000	Interrupt enable register
GPIO_INTMASK	0x0034	W	0x00000000	Interrupt mask register
GPIO_INTTYPE_LEVEL	0x0038	W	0x00000000	Interrupt level register
GPIO_INT_POLARITY	0x003c	W	0x00000000	Interrupt polarity register
GPIO_INT_STATUS	0x0040	W	0x00000000	Interrupt status of port A
GPIO_INT_RAWSTATUS	0x0044	W	0x00000000	Raw Interrupt status of port A
GPIO_DEBOUNCE	0x0048	W	0x00000000	Debounce enable register
GPIO_PORTA_EOI	0x004c	W	0x00000000	Port A clear interrupt register
GPIO_EXT_PORTA	0x0050	W	0x00000000	Port A external port register
GPIO_LS_SYNC	0x0060	W	0x00000000	Level_sensitive synchronization enable register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

49.4.2 Detail Register Description

GPIO_SWPORTA_DR

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Address: Operational Base + offset (0x0000)
Port A data register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	gpio_swporta_dr Values written to this register are output on the I/O signals for Port A if the corresponding data direction bits for Port A are set to Output mode.The value read back is equal to the last value written to this register.

GPIO_SWPORTA_DDR
Address: Operational Base + offset (0x0004)
Port A data direction register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	gpio_swporta_ddr Values written to this register independently control the direction of the corresponding data bit in Port A. 1'b0: Input (default) 1'b1: Output

GPIO_INTEN
Address: Operational Base + offset (0x0030)
Interrupt enable register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	gpio_int_en Allows each bit of Port A to be configured for interrupts. Whenever a 1 is written to a bit of this register, it configures the corresponding bit on Port A to become an interrupt; otherwise, Port A operates as a normal GPIO signal. Interrupts are disabled on the corresponding bits of Port A if the corresponding data direction register is set to Output. 1'b0: Configure Port A bit as normal GPIO signal (default) 1'b1: Configure Port A bit as interrupt

GPIO_INTMASK
Address: Operational Base + offset (0x0034)
Interrupt mask register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	gpio_int_mask Controls whether an interrupt on Port A can create aninterrupt for the interrupt controller by not masking it. Whenever a 1 is written to a bit in this register, it masks the interrupt generation capability for this signal; otherwise interrupts are allowed through. 1'b0: Interrupt bits are unmasked (default) 1'b1: Mask interrupt

GPIO_INTTYPE_LEVEL
Address: Operational Base + offset (0x0038)
Interrupt level register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	gpio_inttype_level Controls the type of interrupt that can occur on Port A. 1'b0: Level-sensitive (default) 1'b1: Edge-sensitive

GPIO_INT_POLARITY
Address: Operational Base + offset (0x003c)
Interrupt polarity register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	gpio_int_polarity Controls the polarity of edge or level sensitivity that can occur on input of Port A. 1'b0: Active-low (default) 1'b1: Active-high

GPIO_INT_STATUS
Address: Operational Base + offset (0x0040)
Interrupt status of port A

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	gpio_int_status Interrupt status of Port A

GPIO_INT_RAWSTATUS
Address: Operational Base + offset (0x0044)
Raw Interrupt status of port A

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	gpio_int_rawstatus Raw interrupt of status of Port A (premasking bits

GPIO_DEBOUNCE
Address: Operational Base + offset (0x0048)

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Debounce enable register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	gpio_debounce Controls whether an external signal that is the sourceof an interrupt needs to be debounced to remove anyspurious glitches. Writing a 1 to a bit in this registerenables the debouncing circuitry. A signal must bevalid for two periods of an external clock before it isinternally processed. 1'b0: No debounce (default) 1'b1: Enable debounce

GPIO_PORTA_EOI

Address: Operational Base + offset (0x004c)

Port A clear interrupt register

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	gpio_porta_eoi Controls the clearing of edge type interrupts from Port A. When a 1 is written into a corresponding bit of this register, the interrupt is cleared. All interrupts are cleared when Port A is not configured for interrupts. 1'b0: No interrupt clear (default) 1'b1: Clear interrupt

GPIO_EXT_PORTA

Address: Operational Base + offset (0x0050)

Port A external port register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	gpio_ext_porta When Port A is configured as Input, then reading this location reads the values on the signal. When the data direction of Port A is set as Output, reading this location reads the data register for Port A.

GPIO_LS_SYNC

Address: Operational Base + offset (0x0060)

Level_sensitive synchronization enable register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	gpio_ls_sync Writing a 1 to this register results in all level-sensitive interrupts being synchronized to pclk_intr. 1'b0: No synchronization to pclk_intr (default) 1'b1: Synchronize to pclk_intr

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49.5 Interface description

Table 49-1 GPIO interface description

Module Pin	Dir	Pad Name	IOMUX Setting
GPIO0 Interface			
gpio0_porta[7:0]	I/O	GPIO0_A[7:0]	PMUGRF_GPIO0A_IOMUX[15:0]=16'h0000
gpio0_porta[15:8]	I/O	GPIO0_B[7:0]	PMUGRF_GPIO0B_IOMUX[15:0]=16'h0000
gpio0_porta[23:16]	I/O	GPIO0_C[7:0]	PMUGRF_GPIO0C_IOMUX[15:0]=16'h0000
gpio0_porta[31:24]	I/O	GPIO0_D[7:0]	PMUGRF_GPIO0D_IOMUX[15:0]=16'h0000
GPIO1 Interface			
gpio1_porta[7:0]	I/O	GPIO1_A[7:0]	GRF_GPIO1A_IOMUX[15:0]=16'h0000
gpio1_porta[15:8]	I/O	GPIO1_B[7:0]	GRF_GPIO1B_IOMUX[15:0]=16'h0000
gpio1_porta[23:16]	I/O	GPIO1_C[7:0]	GRF_GPIO1C_IOMUX[15:0]=16'h0000
gpio1_porta[31:24]	I/O	GPIO1_D[7:0]	GRF_GPIO1D_IOMUX[15:0]=16'h0000
GPIO2 Interface			
gpio2_porta[7:0]	I/O	GPIO2_A[7:0]	GRF_GPIO2A_IOMUX[15:0]=16'h0000
gpio2_porta[15:8]	I/O	GPIO2_B[7:0]	GRF_GPIO2B_IOMUX[15:0]=16'h0040
gpio2_porta[23:16]	I/O	GPIO2_C[7:0]	GRF_GPIO2C_IOMUX[15:0]=16'h0000
gpio2_porta[31:24]	I/O	GPIO2_D[7:0]	GRF_GPIO2D_IOMUX[15:0]=16'h0000
GPIO3 Interface			
gpio3_porta[7:0]	I/O	GPIO3_A[7:0]	GRF_GPIO3A_IOMUX[15:0]=16'h0000
gpio3_porta[15:8]	I/O	GPIO3_B[7:0]	GRF_GPIO3B_IOMUX[15:0]=16'h0000
gpio3_porta[23:16]	I/O	GPIO3_C[7:0]	GRF_GPIO3C_IOMUX[15:0]=16'h0000
gpio3_porta[31:24]	I/O	GPIO3_D[7:0]	GRF_GPIO3D_IOMUX[15:0]=16'h0000

49.6 Application Notes

Steps to set GPIO's direction

- Write GPIO_SWPORT_DDR[x] as 1 to set this gpio as output direction and Write GPIO_SWPORT_DDR[x] as 0 to set this gpio as input direction.
- Default GPIO's direction is input direction.

Steps to set GPIO's level

- Write GPIO_SWPORT_DDR[x] as 1 to set this gpio as output direction.
- Write GPIO_SWPORT_DR[x] as v to set this GPIO's value.

Steps to get GPIO's level

- Write GPIO_SWPORT_DDR[x] as 0 to set this gpio as input direction.
- Read from GPIO_EXT_PORT[x] to get GPIO's value

Steps to set GPIO as interrupt source

- Write GPIO_SWPORT_DDR[x] as 0 to set this gpio as input direction.
- Write GPIO_INTTYPE_LEVEL[x] as v1 and write GPIO_INT_POLARITY[x] as v2 to set interrupt type
- Write GPIO_INTEN[x] as 1 to enable GPIO's interrupt

Note: Please switch iomux to GPIO mode first!

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Chapter 50 WatchDog

50.1 Overview

Watchdog Timer (WDT) is an APB slave peripheral that can be used to prevent system lockup that may be caused by conflicting parts or programs in a SoC.The WDT would generated interrupt or reset signal when it's counter reaches zero, then a reset controller would reset the system.

WDT supports the following features:

- 32 bits APB bus width
- WDT counter's clock is pclk
- 32 bits WDT counter width
- Counter counts down from a preset value to 0 to indicate the occurrence of a timeout
- WDT can perform two types of operations when timeout occurs:
  - Generate a system reset
  - First generate an interrupt and if this is not cleared by the service routine by the time a second timeout occurs then generate a system reset
- Programmable reset pulse length
- Total 16 defined-ranges of main timeout period

50.2 Block Diagram

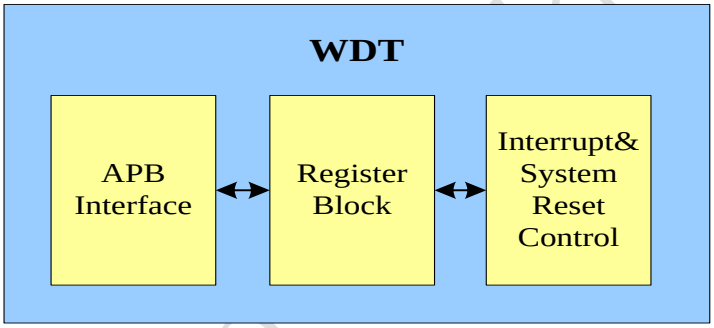

Fig. 50-1 WDT block diagram

Block Descriptions:

APB Interface

The APB Interface implements the APB slave operation. It's data bus width is 32 bits.

Register Block

A register block that read coherence for the current count register.

Interrupt & system reset control

An interrupt/system reset generation block comprising of a decrementing counter and control logic.

50.3 Function description

50.3.1 Operation

Counter

The WDT counts from a preset (timeout) value in descending order to zero. When the counter reaches zero, depending on the output response mode selected, either a system reset or an interrupt occurs. When the counter reaches zero, it wraps to the selected timeout value and continues decrementing. The user can restart the counter to its initial value. This is programmed by writing to the restart register at any time. The process of restarting the watchdog counter is sometimes referred as kicking the dog. As a safety feature to prevent accidental restarts, the value 0x76 must be written to the Current Counter Value Register (WDT_CRR).

Interrupts

The WDT can be programmed to generate an interrupt (and then a system reset) when a timeout occurs. When a 1 is written to the response mode field (RMOD, bit 1) of the Watchdog

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Timer Control Register (WDT_CR), the WDT generates an interrupt. If it is not cleared by the time a second timeout occurs, then it generates a system reset. If a restart occurs at the same time the watchdog counter reaches zero, an interrupt is not generated.

System Resets

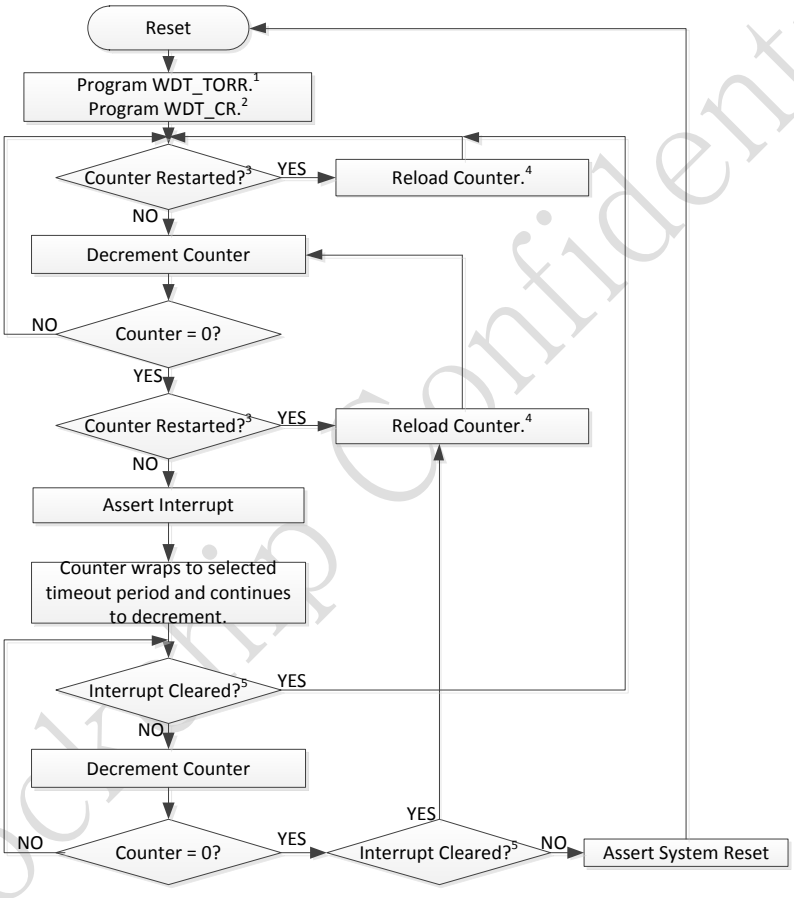
When a 0 is written to the output response mode field (RMOD, bit 1) of the Watchdog Timer Control Register (WDT_CR), the WDT generates a system reset when a timeout occurs.

Reset Pulse Length

The reset pulse length is the number of pclk cycles for which a system reset is asserted. When a system reset is generated, it remains asserted for the number of cycles specified by the reset pulse length or until the system is reset. A counter restart has no effect on the system reset once it has been asserted.

50.3.2 Programming sequence

Operation Flow Chart (Response mode=1)



1. Select required timeout period.
2. Set reset pulse length, response mode, and enable WDT.
3. Write 0x76 to WDT_CRR.
4. Starts back to selected timeout period.
5. Can clear by reading WDT_EOI or restarting (kicking) the counter by writing 0x76 to WDT_CRR.

Fig. 50-2 WDT Operation Flow

50.4 Register Description

This section describes the control/status registers of the design.

50.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
WDT_CR	0x0000	W	0x0000000a	Control Register
WDT_TORR	0x0004	W	0x00000000	Timeout range Register

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Name	Offset	Size	Reset Value	Description
WDT_CCVR	0x0008	W	0x00000000	Current counter value Register
WDT_CRR	0x000c	W	0x00000000	Counter restart Register
WDT_STAT	0x0010	W	0x00000000	Interrupt status Register
WDT_EOI	0x0014	W	0x00000000	Interrupt clear Register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

50.4.2 Detail Register Description

WDT_CR

Address: Operational Base + offset (0x0000)

Control Register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:2	RW	0x2	rst_pluse_lenth Reset pulse length. This is used to select the number of pclk cycles for which the system reset stays asserted. 3'b000: 2 pclk cycles 3'b001: 4 pclk cycles 3'b010: 8 pclk cycles 3'b011: 16 pclk cycles 3'b100: 32 pclk cycles 3'b101: 64 pclk cycles 3'b110: 128 pclk cycles 3'b111: 256 pclk cycles
1	RW	0x1	resp_mode Response mode. Selects the output response generated to a timeout. 1'b0: Generate a system reset 1'b1: First generate an interrupt and if it is not cleared by the time a second timeout occurs then generate a system reset
0	RW	0x0	wdt_en WDT enable 1'b0: WDT disabled 1'b1: WDT enabled

WDT_TORR

Address: Operational Base + offset (0x0004)

Timeout range Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3:0	RW	0x0	timeout_period Timeout period. This field is used to select the timeout period from which the watchdog counter restarts. A change of the timeout period takes effect only after the next counter restart (kick). The range of values available for a 32-bit watchdog counter are: 4'b0000: 0x0000ffff 4'b0001: 0x0001ffff 4'b0010: 0x0003ffff 4'b0011: 0x0007ffff 4'b0100: 0x000fffff 4'b0101: 0x001fffff 4'b0110: 0x003fffff 4'b0111: 0x007fffff 4'b1000: 0x00ffffff 4'b1001: 0x01ffffff 4'b1010: 0x03ffffff 4'b1011: 0x07ffffff 4'b1100: 0x0fffffff 4'b1101: 0x1fffffff 4'b1110: 0x3fffffff 4'b1111: 0x7fffffff

WDT_CCVR
Address: Operational Base + offset (0x0008)
Current counter value Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	cur_cnt Current counter value This register, when read, is the current value of the internal counter. This value is read coherently when ever it is read

WDT_CRR
Address: Operational Base + offset (0x000c)
Counter restart Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:0	W1C	0x00	cnt_restart Counter restart This register is used to restart the WDT counter. As a safety feature to prevent accidental restarts, the value 0x76 must be written. A restart also clears the WDT interrupt. Reading this register returns zero.

WDT_STAT
Address: Operational Base + offset (0x0010)
Interrupt status Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RO	0x0	wdt_status This register shows the interrupt status of the WDT. 1'b1: Interrupt is active regardless of polarity. 1'b0: Interrupt is inactive.

WDT_EOI
Address: Operational Base + offset (0x0014)
Interrupt clear Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RO	0x0	wdt_int_clr Clears the watchdog interrupt. This can be used to clear the interrupt without restarting the watchdog counter.

50.5 Application Notes

Please refer to the function description section.

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Chapter 51 Pulse Width Modulation (PWM)

52.1 Overview

The pulse-width modulator (PWM) feature is very common in embedded systems. It provides a way to generate a pulse periodic waveform for motor control or can act as a digital-to-analog converter with some external components.

The PWM Module supports the following features:

- 4-built-in PWM channels
- Configurable to operate in capture mode
  - Measures the high/low polarity effective cycles of this input waveform
  - Generates a single interrupt at the transition of input waveform polarity
  - 32-bit high polarity capture register
  - 32-bit low polarity capture register
  - 32-bit current value register
  - 4-depth 32bit FIFO built within Channel 3, enabling the Channel 3 make use of DMA to fetch high/low polarity data stored in the FIFO. FIFO full interrupt and overflow interrupt are supported in this mode.
- Configurable to operate in continuous mode or one-shot mode
  - 32-bit period counter
  - 32-bit duty register
  - 32-bit current value register
  - Configurable PWM output polarity in inactive state and duty period pulse polarity
  - Period and duty cycle are shadow buffered. Change takes effect when the end of the effective period is reached or when the channel is disabled
  - Programmable center or left aligned outputs, and change takes effect when the end of the effective period is reached or when the channel is disabled
  - 8-bit repeat counter for one-shot operation. One-shot operation will produce N + 1 periods of the waveform, where N is the repeat counter value, and generates a single interrupt at the end of operation
  - Continuous mode generates the waveform continuously, and do not generates any interrupts
- pre-scaled operation to bus clock and then further scaled
- Available low-power mode to reduce power consumption when the channel is inactive.

52.2 Block Diagram

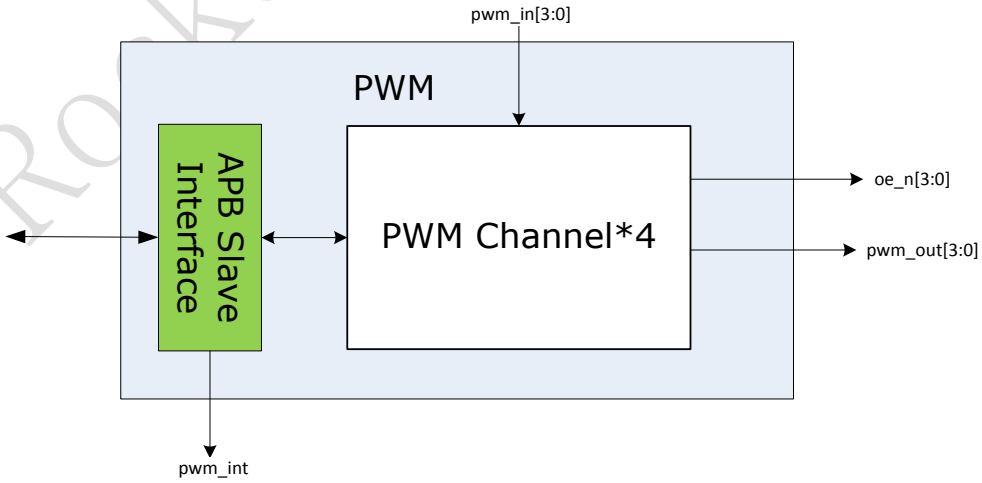

Fig. 51-1 PWM architecture

52.2.1 PWM APB Slave Interface

The host processor gets access to PWM Register Block through the APB slave interface with 32-bit bus width, and asserts the active-high level interrupt.

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52.2.2 PWM Channels

This is the control logic of PWM module, and controls the operation of PWM module according to the configured working mode.

52.3 Functional description

The PWM supports three operation modes: reference mode, one-shot mode and continuous mode. For the one-shot mode and the continuous mode, the PWM output can be configured as the left-aligned mode or the center-aligned mode.

52.3.1 Reference mode

The reference mode is used to measure the PWM channel input waveform high/low effective cycles with the PWM channel clock, and asserts an interrupt when the polarity of the input waveform changes. The number of the high effective cycles is recorded in the PWMx_PERIOD_HPC register, while the number of the low effective cycles is recorded in the PWMx_DUTY_LPC register.

There is one more special working mode for the Channel 3. The Channel 3 can be configured to be working in FIFO mode. The PWM starts the DMA transfer when the 4-depth 32-bit FIFO is half filled with the data, whose MSB indicates the polarity of lower 31-bit effective cycle counter.

Note: the PWM input waveform is doubled buffered when the PWM channel is working in order to filter unexpected shot-time polarity transition, and therefore the interrupt is asserted several cycles after the input waveform polarity changes, and so does the change of the values of PWMx_PERIOD_HPC and PWMx_DUTY_LPC.

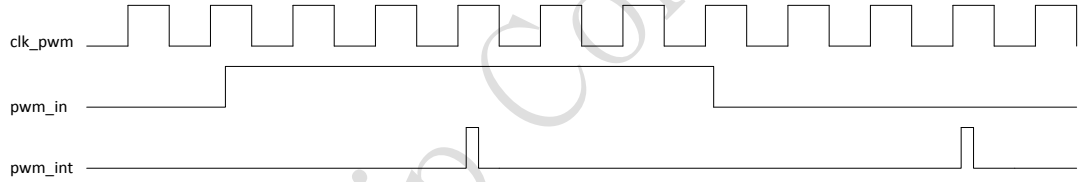

Fig. 51-2 PWM Reference Mode

52.3.2 Continuous Mode

The PWM channel generates a series of the pulses continuously as expected once the channel is enabled with continuous mode.
In the continuous mode, the PWM output waveforms can be in one form of the two output mode: left-aligned mode or center-aligned mode.
For the left-aligned output mode, the PWM channel firstly starts the duty cycle with the configured duty polarity (PWMx_CTRL.duty_pol). Once duty cycle number (PWMx_DUTY_LPC) is reached, the output is switched to the opposite polarity. After the period number (PWMx_PERIOD_HPC) is reached, the output is again switched to the opposite polarity to start another period of desired pulse.

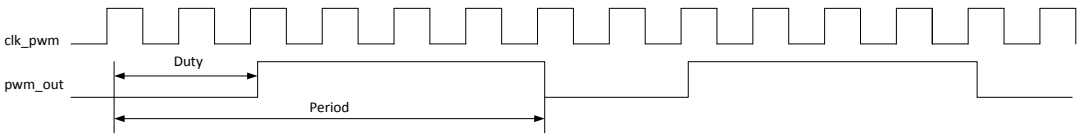

Fig. 51-3 PWM Left-aligned Output Mode

For the center-aligned output mode, the PWM channel firstly starts the duty cycle with the configured duty polarity (PWMx_CTRL.duty_pol). Once one half of duty cycle number (PWMx_DUTY_LPC) is reached, the output is switched to the opposite polarity. Then if there is one half of duty cycle left for the whole period , the output is again switched to the opposite polarity. Finally after the period number (PWMx_PERIOD_HPC) is reached, the output is again switched to the opposite polarity to start another period of desired pulse.

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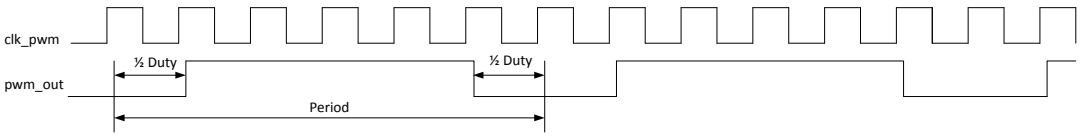

Fig. 51-4 PWM Center-aligned Output Mode

Disable the PWM channel, the channel stops generating the output waveforms and output polarity is fixed as the configured inactive polarity (PWMx_CTRL.inactive_pol).

52.3.3 One-shot Mode

Unlike the continuous mode, the PWM channel generates the output waveforms within the configured periods (PWM_CTRL.rpt + 1), and then stops. At the same times, an interrupt is asserted to inform that the operation has been finished.
There are also two output modes for the one-shot mode: the left-aligned mode and the center-aligned mode.

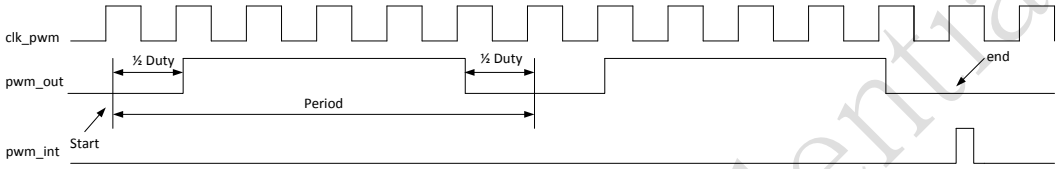

Fig. 51-5 PWM Center-aligned Output Mode

52.4 Register description

52.4.1 Register Summary

Name	Offset	Size	Reset Value	Description
PWM_PWM0_CNT	0x00000	W	0x00000000	PWM Channel 0 Counter Register
PWM_PWM0_PERIOD_HPR	0x00004	W	0x00000000	PWM Channel 0 Period Register/High Polarity Capture Register
PWM_PWM0_DUTY_LPR	0x00008	W	0x00000000	PWM Channel 0 Duty Register/Low Polarity Capture Register
PWM_PWM0_CTRL	0x0000c	W	0x00000000	PWM Channel 0 Control Register
PWM_PWM1_CNT	0x00010	W	0x00000000	PWM Channel 1 Counter Register
PWM_PWM1_PERIOD_HPR	0x00014	W	0x00000000	PWM Channel 1 Period Register/High Polarity Capture Register
PWM_PWM1_DUTY_LPR	0x00018	W	0x00000000	PWM Channel 1 Duty Register/Low Polarity Capture Register
PWM_PWM1_CTRL	0x0001c	W	0x00000000	PWM Channel 1 Control Register
PWM_PWM2_CNT	0x00020	W	0x00000000	PWM Channel 2 Counter Register
PWM_PWM2_PERIOD_HPR	0x00024	W	0x00000000	PWM Channel 2 Period Register/High Polarity Capture Register
PWM_PWM2_DUTY_LPR	0x00028	W	0x00000000	PWM Channel 2 Duty Register/Low Polarity Capture Register
PWM_PWM2_CTRL	0x0002c	W	0x00000000	PWM Channel 2 Control Register
PWM_PWM3_CNT	0x00030	W	0x00000000	PWM Channel 3 Counter Register
PWM_PWM3_PERIOD_HPR	0x00034	W	0x00000000	PWM Channel 3 Period Register/High Polarity Capture Register
PWM_PWM3_DUTY_LPR	0x00038	W	0x00000000	PWM Channel 3 Duty Register/Low Polarity Capture Register

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Name	Offset	Size	Reset Value	Description
PWM_PWM3_CTRL	0x0003c	W	0x00000000	PWM Channel 3 Control Register
PWM_INTSTS	0x00040	W	0x00000000	Interrupt Status Register
PWM_INT_EN	0x00044	W	0x00000000	Interrupt Enable Register
PWM_FIFO_CTRL	0x00050	W	0x00000000	PWM Channel 3 FIFO Mode Control Register
PWM_FIFO_INTSTS	0x00054	W	0x00000000	FIFO Interrupts Status Register
PWM_FIFO	0x00060-0x0007C	W	0x00000000	FIFO Register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

52.4.2 Detail Register Description

PWM_PWM0_CNT
Address: Operational Base + offset (0x00000)
PWM Channel 0 Counter Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	CNT Timer Counter The 32-bit indicates current value of PWM Channel 0 counter. The counter runs at the rate of PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM0_PERIOD_HPR
Address: Operational Base + offset (0x00004)
PWM Channel 0 Period Register/High Polarity Capture Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PERIOD_LPR Output Waveform Period/Input Waveform High Polarity Cycle If PWM is operated at the continuous mode or one-shot mode, this value defines the period of the output waveform. Note that, if the PWM is operated at the center-aligned mode, the period should be an even one, and therefore only the bit [31:1] is taken into account and bit [0] always considered as 0. If PWM is operated at the capture mode, this value indicates the effective high polarity cycles of input waveform. This value is based on the PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM0_DUTY_LPR
Address: Operational Base + offset (0x00008)

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PWM Channel 0 Duty Register/Low Polarity Capture Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DUTY_LPR Output Waveform Duty Cycle/Input Waveform Low Polarity Cycle If PWM is operated at the continuous mode or one-shot mode, this value defines the duty cycle of the output waveform. The PWM starts the output waveform with duty cycle. Note that, if the PWM is operated at the center-aligned mode, the period should be an even one, and therefore only the [31:1] is taken into account. If PWM is operated at the capture mode, this value indicates the effective low polarity cycles of input waveform. This value is based on the PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM0_CTRL

Address: Operational Base + offset (0x0000c)

PWM Channel 0 Control Register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	rpt Repeat Counter This field defines the repeated effective periods of output waveform in one-shot mode. The value N means N+1 repeated effective periods.
23:16	RW	0x00	scale Scale Factor This fields defines the scale factor applied to prescaled clock. The value N means the clock is divided by 2*N. If N is 0, it means that the clock is divided by 512(2*256).
15	RO	0x0	reserved
14:12	RW	0x0	prescale Prescale Factor This field defines the prescale factor applied to input clock. The value N means that the input clock is divided by 2^N.
11:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9	RW	0x0	clk_sel Clock Source Select 0: non-scaled clock is selected as PWM clock source. It means that the prescale clock is directly used as the PWM clock source 1: scaled clock is selected as PWM clock source
8	RW	0x0	lp_en Low Power Mode Enable 0: disabled 1: enabled When PWM channel is inactive state and Low Power Mode is enabled, the path to PWM Clock prescale module is blocked to reduce power consumption.
7:6	RO	0x0	reserved
5	RW	0x0	output_mode PWM Output mode 0: left aligned mode 1: center aligned mode
4	RW	0x0	inactive_pol Inactive State Output Polarity This defines the output waveform polarity when PWM channel is in inactive state. The inactive state means that PWM finishes the complete waveform in one-shot mode or PWM channel is disabled. 0: negative 1: positive
3	RW	0x0	duty_pol Duty Cycle Output Polarity This defines the polarity for duty cycle. PWM starts the output waveform with duty cycle. 0: negative 1: positive
2:1	RW	0x0	pwm_mode PWM Operation Mode 00: One shot mode. PWM produces the waveform within the repeated times defined by PWMx_CTRL_rpt . 01: Continuous mode. PWM produces the waveform continuously 10: Capture mode. PWM measures the cycles of high/low polarity of input waveform. 11: reserved

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Bit	Attr	Reset Value	Description
0	RW	0x0	pwm_en PWM channel enable 0: disabled 1: enabled. If the PWM is worked the one-shot mode, this bit will be cleared at the end of operation

PWM_PWM1_CNT
Address: Operational Base + offset (0x00010)
PWM Channel 1 Counter Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	CNT Timer Counter The 32-bit indicates current value of PWM Channel 1 counter. The counter runs at the rate of PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM1_PERIOD_HPR
Address: Operational Base + offset (0x00014)
PWM Channel 1 Period Register/High Polarity Capture Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PERIOD_LPR Output Waveform Period/Input Waveform High Polarity Cycle If PWM is operated at the continuous mode or one-shot mode, this value defines the period of the output waveform. Note that, if the PWM is operated at the center-aligned mode, the period should be an even one, and therefore only the bit [31:1] is taken into account and bit [0] always considered as 0. If PWM is operated at the capture mode, this value indicates the effective high polarity cycles of input waveform. This value is based on the PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM1_DUTY_LPR
Address: Operational Base + offset (0x00018)
PWM Channel 1 Duty Register/Low Polarity Capture Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DUTY_LPR Output Waveform Duty Cycle/Input Waveform Low Polarity Cycle If PWM is operated at the continuous mode or one-shot mode, this value defines the duty cycle of the output waveform. The PWM starts the output waveform with duty cycle. Note that, if the PWM is operated at the center-aligned mode, the period should be an even one, and therefore only the [31:1] is taken into account. If PWM is operated at the capture mode, this value indicates the effective low polarity cycles of input waveform. This value is based on the PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM1_CTRL
Address: Operational Base + offset (0x0001c)
PWM Channel 1 Control Register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	rpt Repeat Counter This field defines the repeated effective periods of output waveform in one-shot mode. The value N means N+1 repeated effective periods.
23:16	RW	0x00	scale Scale Factor This fields defines the scale factor applied to prescaled clock. The value N means the clock is divided by 2*N. If N is 0, it means that the clock is divided by 512(2*256).
15	RO	0x0	reserved
14:12	RW	0x0	prescale Prescale Factor This field defines the prescale factor applied to input clock. The value N means that the input clock is divided by 2^N.
11:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9	RW	0x0	clk_sel Clock Source Select 0: non-scaled clock is selected as PWM clock source. It means that the prescale clock is directly used as the PWM clock source 1: scaled clock is selected as PWM clock source
8	RW	0x0	lp_en Low Power Mode Enable 0: disabled 1: enabled When PWM channel is inactive state and Low Power Mode is enabled, the path to PWM Clock prescale module is blocked to reduce power consumption.
7:6	RO	0x0	reserved
5	RW	0x0	output_mode PWM Output mode 0: left aligned mode 1: center aligned mode
4	RW	0x0	inactive_pol Inactive State Output Polarity This defines the output waveform polarity when PWM channel is in inactive state. The inactive state means that PWM finishes the complete waveform in one-shot mode or PWM channel is disabled. 0: negative 1: positive
3	RW	0x0	duty_pol Duty Cycle Output Polarity This defines the polarity for duty cycle. PWM starts the output waveform with duty cycle. 0: negative 1: positive
2:1	RW	0x0	pwm_mode PWM Operation Mode 00: One shot mode. PWM produces the waveform within the repeated times defined by PWMx_CTRL_rpt 01: Continuous mode. PWM produces the waveform continuously 10: Capture mode. PWM measures the cycles of high/low polarity of input waveform. 11: reserved

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Bit	Attr	Reset Value	Description
0	RW	0x0	pwm_en PWM channel enable 0: disabled 1: enabled. If the PWM is worked the one-shot mode, this bit will be cleared at the end of operation

PWM_PWM2_CNT
Address: Operational Base + offset (0x00020)
PWM Channel 2 Counter Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	CNT Timer Counter The 32-bit indicates current value of PWM Channel 2 counter. The counter runs at the rate of PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM2_PERIOD_HPR
Address: Operational Base + offset (0x00024)
PWM Channel 2 Period Register/High Polarity Capture Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PERIOD_LPR Output Waveform Period/Input Waveform High Polarity Cycle If PWM is operated at the continuous mode or one-shot mode, this value defines the period of the output waveform. Note that, if the PWM is operated at the center-aligned mode, the period should be an even one, and therefore only the bit [31:1] is taken into account and bit [0] always considered as 0. If PWM is operated at the capture mode, this value indicates the effective high polarity cycles of input waveform. This value is based on the PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM2_DUTY_LPR
Address: Operational Base + offset (0x00028)
PWM Channel 2 Duty Register/Low Polarity Capture Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DUTY_LPR Output Waveform Duty Cycle/Input Waveform Low Polarity Cycle If PWM is operated at the continuous mode or one-shot mode, this value defines the duty cycle of the output waveform. The PWM starts the output waveform with duty cycle. Note that, if the PWM is operated at the center-aligned mode, the period should be an even one, and therefore only the [31:1] is taken into account. If PWM is operated at the capture mode, this value indicates the effective low polarity cycles of input waveform. This value is based on the PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM2_CTRL
Address: Operational Base + offset (0x0002c)
PWM Channel 2 Control Register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	rpt Repeat Counter This field defines the repeated effective periods of output waveform in one-shot mode. The value N means N+1 repeated effective periods.
23:16	RW	0x00	scale Scale Factor This fields defines the scale factor applied to prescaled clock. The value N means the clock is divided by 2*N. If N is 0, it means that the clock is divided by 512(2*256).
15	RO	0x0	reserved
14:12	RW	0x0	prescale Prescale Factor This field defines the prescale factor applied to input clock. The value N means that the input clock is divided by 2^N.
11:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9	RW	0x0	clk_sel Clock Source Select 0: non-scaled clock is selected as PWM clock source. It means that the prescale clock is directly used as the PWM clock source 1: scaled clock is selected as PWM clock source
8	RW	0x0	lp_en Low Power Mode Enable 0: disabled 1: enabled When PWM channel is inactive state and Low Power Mode is enabled, the path to PWM Clock prescale module is blocked to reduce power consumption.
7:6	RO	0x0	reserved
5	RW	0x0	output_mode PWM Output mode 0: left aligned mode 1: center aligned mode
4	RW	0x0	inactive_pol Inactive State Output Polarity This defines the output waveform polarity when PWM channel is in inactive state. The inactive state means that PWM finishes the complete waveform in one-shot mode or PWM channel is disabled. 0: negative 1: positive
3	RW	0x0	duty_pol Duty Cycle Output Polarity This defines the polarity for duty cycle. PWM starts the output waveform with duty cycle. 0: negative 1: positive
2:1	RW	0x0	pwm_mode PWM Operation Mode 00: One shot mode. PWM produces the waveform within the repeated times defined by PWMx_CTRL_rpt. 01: Continuous mode. PWM produces the waveform continuously 10: Capture mode. PWM measures the cycles of high/low polarity of input waveform. 11: reserved

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Bit	Attr	Reset Value	Description
0	RW	0x0	pwm_en PWM channel enable 0: disabled 1: enabled. If the PWM is worked the one-shot mode, this bit will be cleared at the end of operation

PWM_PWM3_CNT
Address: Operational Base + offset (0x00030)
PWM Channel 3 Counter Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	CNT Timer Counter The 32-bit indicates current value of PWM Channel 3 counter. The counter runs at the rate of PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM3_PERIOD_HPR
Address: Operational Base + offset (0x00034)
PWM Channel 3 Period Register/High Polarity Capture Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PERIOD_LPR Output Waveform Period/Input Waveform High Polarity Cycle If PWM is operated at the continuous mode or one-shot mode, this value defines the period of the output waveform. Note that, if the PWM is operated at the center-aligned mode, the period should be an even one, and therefore only the bit [31:1] is taken into account and bit [0] always considered as 0. If PWM is operated at the capture mode, this value indicates the effective high polarity cycles of input waveform. This value is based on the PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM3_DUTY_LPR
Address: Operational Base + offset (0x00038)
PWM Channel 3 Duty Register/Low Polarity Capture Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DUTY_LPR Output Waveform Duty Cycle/Input Waveform Low Polarity Cycle If PWM is operated at the continuous mode or one-shot mode, this value defines the duty cycle of the output waveform. The PWM starts the output waveform with duty cycle. Note that, if the PWM is operated at the center-aligned mode, the period should be an even one, and therefore only the [31:1] is taken into account. If PWM is operated at the capture mode, this value indicates the effective low polarity cycles of input waveform. This value is based on the PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM3_CTRL
Address: Operational Base + offset (0x0003c)
PWM Channel 3 Control Register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	rpt Repeat Counter This field defines the repeated effective periods of output waveform in one-shot mode. The value N means N+1 repeated effective periods.
23:16	RW	0x00	scale Scale Factor This fields defines the scale factor applied to prescaled clock. The value N means the clock is divided by 2*N. If N is 0, it means that the clock is divided by 512(2*256).
15	RO	0x0	reserved
14:12	RW	0x0	prescale Prescale Factor This field defines the prescale factor applied to input clock. The value N means that the input clock is divided by 2^N.
11:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9	RW	0x0	clk_sel Clock Source Select 0: non-scaled clock is selected as PWM clock source. It means that the prescale clock is directly used as the PWM clock source 1: scaled clock is selected as PWM clock source
8	RW	0x0	lp_en Low Power Mode Enable 0: disabled 1: enabled When PWM channel is inactive state and Low Power Mode is enabled, the path to PWM Clock prescale module is blocked to reduce power consumption.
7:6	RO	0x0	reserved
5	RW	0x0	output_mode PWM Output mode 0: left aligned mode 1: center aligned mode
4	RW	0x0	inactive_pol Inactive State Output Polarity This defines the output waveform polarity when PWM channel is in inactive state. The inactive state means that PWM finishes the complete waveform in one-shot mode or PWM channel is disabled. 0: negative 1: positive
3	RW	0x0	duty_pol Duty Cycle Output Polarity This defines the polarity for duty cycle. PWM starts the output waveform with duty cycle. 0: negative 1: positive
2:1	RW	0x0	pwm_mode PWM Operation Mode 00: One shot mode. PWM produces the waveform within the repeated times defined by PWMx_CTRL_rpt 01: Continuous mode. PWM produces the waveform continuously 10: Capture mode. PWM measures the cycles of high/low polarity of input waveform. 11: reserved

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Bit	Attr	Reset Value	Description
0	RW	0x0	pwm_en PWM channel enable 0: disabled 1: enabled. If the PWM is worked the one-shot mode, this bit will be cleared at the end of operation

PWM_INTSTS
Address: Operational Base + offset (0x00040)
Interrupt Status Register

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11	RO	0x0	CH3_Pol Channel 3 Interrupt Polarity Flag This bit is used in capture mode in order to idetifiy the transition of the input waveform when interrupt is generated. When bit is 1, please refer to PWM3_PERIOD_HPR to know the effective high cycle of Channel 0 input waveform. Otherwise, please refer to PWM3_PERIOD_LPR to know the effective low cycle of Channel 3 input waveform. Write 1 to CH3_IntSts will clear this bit.
10	RO	0x0	CH2_Pol Channel 2 Interrupt Polarity Flag This bit is used in capture mode in order to idetifiy the transition of the input waveform when interrupt is generated. When bit is 1, please refer to PWM2_PERIOD_HPR to know the effective high cycle of Channel 0 input waveform. Otherwise, please refer to PWM2_PERIOD_LPR to know the effective low cycle of Channel 2 input waveform. Write 1 to CH2_IntSts will clear this bit.
9	RO	0x0	CH1_Pol Channel 1 Interrupt Polarity Flag This bit is used in capture mode in order to idetifiy the transition of the input waveform when interrupt is generated. When bit is 1, please refer to PWM1_PERIOD_HPR to know the effective high cycle of Channel 0 input waveform. Otherwise, please refer to PWM1_PERIOD_LPR to know the effective low cycle of Channel 1 input waveform. Write 1 to CH1_IntSts will clear this bit.

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Bit	Attr	Reset Value	Description
8	RO	0x0	CH0_Pol Channel 0 Interrupt Polarity Flag This bit is used in capture mode in order to idetify the transition of the input waveform when interrupt is generated. When bit is 1, please refer to PWM0_PERIOD_HPR to know the effective high cycle of Channel 0 input waveform. Otherwise, please refer to PWM0_PERIOD_LPR to know the effective low cycle of Channel 0 input waveform. Write 1 to CH0_IntSts will clear this bit.
7:4	RO	0x0	reserved
3	RW	0x0	CH3_IntSts Channel 3 Interrupt Status 0: Channel 3 Interrupt not generated 1: Channel 3 Interrupt generated
2	RW	0x0	CH2_IntSts Channel 2 Interrupt Status 0: Channel 2 Interrupt not generated 1: Channel 2 Interrupt generated
1	RW	0x0	CH1_IntSts Channel 1 Interrupt Status 0: Channel 1 Interrupt not generated 1: Channel 1 Interrupt generated
0	RW	0x0	CH0_IntSts Channel 0 Raw Interrupt Status 0: Channel 0 Interrupt not generated 1: Channel 0 Interrupt generated

PWM_INT_EN
Address: Operational Base + offset (0x00044)
Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	CH3_Int_en Channel 3 Interrupt Enable 0: Channel 3 Interrupt disabled 1: Channel 3 Interrupt enabled
2	RW	0x0	CH2_Int_en Channel 2 Interrupt Enable 0: Channel 2 Interrupt disabled 1: Channel 2 Interrupt enabled

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Bit	Attr	Reset Value	Description
1	RW	0x0	CH1_Int_en Channel 1 Interrupt Enable 0: Channel 1 Interrupt disabled 1: Channel 1 Interrupt enabled
0	RW	0x0	CH0_Int_en Channel 0 Interrupt Enable 0: Channel 0 Interrupt disabled 1: Channel 0 Interrupt enabled

PWM_FIFO_CTRL
Address: Operational Base + offset (0x00050)
PWM Channel 3 FIFO Mode Control Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	overflow_int_en FIFO Overflow Interrupt Enable When high, an interrupt asserts when the channel 3
1	RW	0x0	full_int_en FIFO Full Interrupt Enable When high, an interrupt asserts when the channel 3 FIFO is full.
0	RW	0x0	fifo_mode_sel FIFO MODE Sel When high, PWM FIFO mode is activated,

PWM_FIFO_INTSTS
Address: Operational Base + offset (0x00054)
FIFO Interrupts Status Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	Field0000 Abstract Field0000 Description
0	RW	0x0	fifo_full_intsts FIFO Full Interrupt Status This bit indicates the FIFO is full

PWM_FIFO
Address: Operational Base + offset (0x00060~0x0007c)
FIFO Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31	RO	0x0	Pol Polarity This bit indicates the polarity of the lower 31-bit counter. 0: Low 1: High
30:0	RO	0x00000000	CYCLE_CNT High/Low Cycle Counter This 31-bit counter indicates the effective cycles of high/low waveform.

52.5 Interface Description

Table 51-1 PWM Interface Description

Module pin	Dir	Pad name	IOMUX
pwm3	O	IO_IRtx_UART3GPSsout_PWM3_GPIO30gpio3d6①	GRF_GPIO3D_IOMUX[13:12]=2'b11
pwm2	O	IO_CLK32_PWM2	PMUGRF_SOC_CON0[7]=1'b1
pwm1	O	IO_TEST_clkout_PWM1_PMUdebug1_PMUgpio0b0	PMUGRF_GPIO0B_IOMUX[1:0]=2'b10
pwm0	O	IO_MACtxd0_PWM0_VOPpwm_GPIO30gpio3b0	GRF_GPIO3B_IOMUX[1:0]=2'b10

Note: IO_IRtx_UART3GPSsout_PWM3_GPIO30gpio3d6 support DMA FIFO mode.

52.6 Application Notes

52.6.1 PWM Common Reference Mode Standard Usage Flow

1. Set PWMx_CTRL.pwm_en to '0' to disable the PWM channel.
2. Choose the prescale factor and the scale factor for pclk by programming PWMx_CTRL.prescale and PWMx_CTRL.scale, and select the clock needed by setting PWMx_CTRL.clk_sel.
3. Configure the channel to work in the reference mode.
4. Enable the INT_EN.chx_int_en to enable the interrupt generation.
5. Enable the channel by writing'1' to PWMx_CTRL.pwm_en bit to start the channel.
6. When an interrupt is asserted, refer to INTSTS register to know the raw interrupt status. If the corresponding polarity flag is set, turn to PWMx_PERIOD_HPC register to know the effective high cycles of input waveforms, otherwise turn to PWMx_DUTY_LPC register to know the effective low cycles.
7. Write '0' to PWMx_CTRL.pwm_en to disable the channel.

52.6.2 PWM Channel 3 Reference Mode with FIFO Standard Usage Flow

1. Set PWMx_CTRL.pwm_en to '0' to disable the PWM channel.
2. Configure the corresponding DMAC.
2. Choose the prescale factor and the scale factor for pclk by programming PWMx_CTRL.prescale and PWMx_CTRL.scale, and select the clock needed by setting PWMx_CTRL.clk_sel.
3. Configure the channel to work in the reference mode.
4. Configure the channel 3 to work in FIFO mode by writing '1' to FIFO_CTRL.fifo_mode_sel.
5. Enable the FIFO interrupt generation by writing'1' to FIFO_CTRL.fifo_int_en or FIFO_CTRL.overflow_int_en.
5. Enable the channel by writing'1' to PWMx_CTRL.pwm_en bit to start the channel.
6. After the DMA transfer complete, refer to the correspond DDR to know the high/low effective cycles.
7. Write '0' to PWMx_CTRL.pwm_en to disable the channel.

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52.6.3 PWM One-shotMode/Continuous Standard Usage Flow

1. Set PWMx_CTRL.pwm_en to '0' to disable the PWM channel.
2. Choose the prescale factor and the scale factor for pclk by programming PWMx_CTRL.prescale and PWMx_CTRL.scale, and select the clock needed by setting PWMx_CTRL.clk_sel.
3. Choose the output mode by setting PWMx_CTRL.output_mode, and set the duty polarity and inactive polarity by programming PWMx_CTRL.duty_pol and PWMx_CTRL.inactive_pol.
4. Set the PWMx_CTRL.rpt if the channel is desired to work in the one-shot mode;
5. Configure the channel to work in the one-shot mode or the continuous mode.
6. Enable the INT_EN.chx_int_en to enable the interrupt generation if if the channel is desired to work in the one-shot mode;
7. If the channel is working in the one-shot mode, an interrupt is asserted after the end of operation, and the PWMx_CTRL.pwm_en is automatically cleared. Whatever mode the channel is working, write '0' to PWMx_CTRL.pwm_en bit to disable the PWM channel.

52.6.4 Low-power mode

Setting PWMx_CTRL.lp_en to '1' makes the channel enter the low-power mode. When then PWM channel is inactive, the APB bus clock to the clock prescale moduleis gated in order to reduce the power consumption. It is recommended to disable the channel before entering the low-power mode, and quit the low-power mode before enabling the channel.

52.6.5 Other notes

When the channel is active to produce waveforms, it is free to programming the PWMx_PERIOD_HPC and PWMx_DUTY_LPC register. The change will not take effect immediately until the current period ends.

An active channel can be changed to another operation mode without disable the PWM channel. However, during the transition of the operation mode there may be some irregular output waveforms. So does changing the clock division factor when the channel is active.

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Chapter 52 I2C Interface

52.1 Overview

The Inter-Integrated Circuit (I2C) is a two wired (SCL and SDA), bi-directional serial bus that provides an efficient and simple method of information exchange between devices. This I2C bus controller supports master mode acting as a bridge between AMBA protocol and generic I2C bus system.

I2C Controller supports the following features:

- Item Compatible with I2C-bus
- AMBA APB slave interface
- Supports master mode of I2C bus
- Software programmable clock frequency and transfer rate up to 、
- 400Kbit/sec
- Supports 7 bits and 10 bits addressing modes
- Interrupt or polling driven multiple bytes data transfer
- Clock stretching and wait state generation
- There are two I2C controller in bus sub-system:I2C PMU, and I2C AUDIO. There are four I2C controller in peripheral sub-system: I2C SENSOR, I2C CAM, I2C_TP, and I2C_HDMI.

52.2 Block Diagram

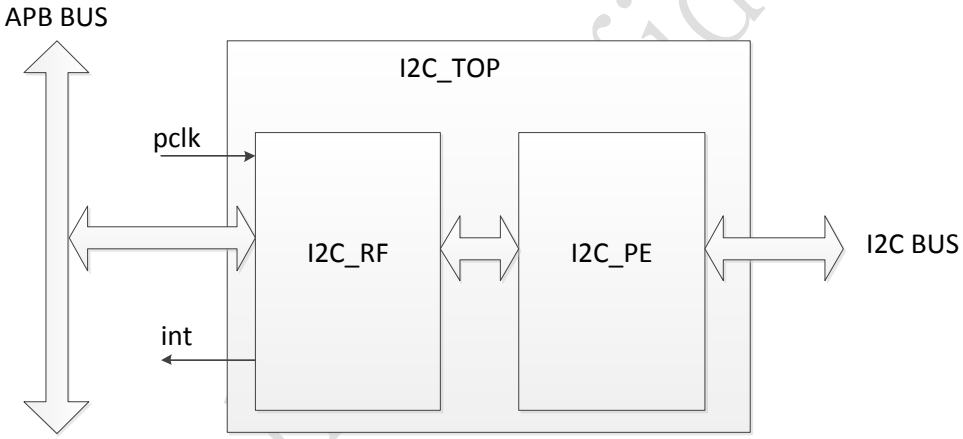

Fig. 52-1 I2C architecture

I2C_RF

I2C_RF module is used to control the I2C controller operation by the host with APB interface. It implements the register set and the interrupt functionality. The CSR component operates synchronously with the pclk clock.

I2C_PE

I2C_PE module implements the I2C master operation for transmit data to and receive data from other I2C devices. The I2C master controller operates synchronously with the pclk.

I2C_TOP

I2C_TOP module is the top module of the I2C controller.

52.3 Function description

This chapter provides a description about the functions and behavior under various conditions. The I2C controller supports only Masterfunction. Itsupports the 7-bits/10-bits addressing mode and support general call address. The maximum clock frequency and transfer rate can be up to 400Kbit/sec.

The operations of I2C controller is divided to 2 parts and described separately: initialization and master mode programming.

52.3.1 Initialization

The I2C controller is based on AMBA APB bus architecture and usually is part of a SOC. So before I2C operates, some system setting & configuration must be conformed, which

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includes:

- I2C interrupt connection type: CPU interrupt scheme should be considered. If the I2C interrupt is connected to extra Interrupt Controller module, we need decide the INTC vector.
- I2C Clock Rate: The I2C controller uses the APB clock as the system clock so the APB clock will determine the I2C bus clock. The correct register setting is subject to the system requirement.

52.3.2 Master Mode Programming

1. SCL Clock: When the I2C controller is programmed in Master mode, the SCL frequency is determine by I2C_CLKDIV register. The SCL frequency is calculated by the following formula：
SCL Divisor = 8*(CLKDIVL + 1 + CLKDIVH + 1)
SCL = PCLK/ SCLK Divisor

2. Data Receiver Register Access
When the i2c controller received MRXCNT bytes data， CPU can get the datas through register RXDATA0 ~ RXDATA7. The controller can receive up to 32 byte data in one transaction.
When MRXCNT register is written, the I2C controller will start to drive SCL to receive data.

3. Transmit Trasmitter Register
Data to transmit are written to TXDATA0~7 by CPU. The controller can transmit up to 32 byte data in one transaction. The lower byte will be transmitted first.
When MTXCNT register is written, the I2C controller will start to transmit data.

4. Start Command
Write 1 to I2C_CON[3], the controller will send I2C start command.

5. Stop Command
Write 1 to I2C_CON[4], the controller will send I2C stop command

6. I2C Operation mode
There are four i2c operation modes.
When I2C_CON[2:1] is 2'b00, the controller transmit all valid data in TXDATA0~TXDATA7 byte by byte. The controller will transmit lower byte first.
When I2C_CON[2:1] is 2'b01,the controller will transmit device address in MRXADDR first (Write/Read bit = 0) and then transmit device register address in MRXRADDR. After that, the controller will assert restart signal and resend MRXADDR (Write/Read bit = 1). At last, the controller enter receive mode.
When I2C_CON[2:1] is 2'b10, the controller is in receive mode, it will triggered clock to read MRXCNT byte data.
When I2C_CON[2:1] is 2'b11,the controller will transmit device address in MRXADDR first (Write/Read bit = 1) and then transmit device register address in MRXRADDR . After that, the controller will assert restart signal and resend MRXADDR (Write/Read bit = 1). At last, the controller enter receive mode.

7. Read/Write Command
When I2C_OPMODE(I2C_CON[2:1]) is 2'b01 or 2'b11, the Read/Write command bit is decided by controller itself.
In RX only mode (I2C_CON[2:1] is 2'b10), the Read/Write command bit is decided by MRXADDR[0].
In TX only mode (I2C_CON[[2:1] is 2'b00), the Read/Write command bit is decided by TXDATA[0].

8. Master Interrupt Condition
There are 7 interrupt bits in I2C_ISR register related to master mode.
Byte transmitted finish interrupt (Bit 0): The bit is asserted when Master complete transmitting a byte.
Byte received finish interrupt (Bit 1): The bit is asserted when Master complete receiving a byte.
MTXCNT bytes data transmitted finish interrupt (Bit 2): The bit is asserted when Master complete transmitting MTXCNT bytes.
MRXCNT bytes data received finish interrupt (Bit 3): The bit is asserted when Master complete receiving MRXCNT bytes.
Start interrupt(Bit 4): The bit is asserted when Master finish asserting start command to I2C bus.

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Stop interrupt (Bit 5): The bit is asserted when Master finish asserting stop command to I2C bus.
NAK received interrupt (Bit 6): The bit is asserted when Master received a NAK handshake.

9. Last byte acknowledge control

If I2C_CON[5] is 1, the I2C controller will transmit NAK handshake to slave when the last byte received in RX only mode.
If I2C_CON[5] is 0, the I2C controller will transmit ACK handshake to slave when the last byte received in RX only mode.

10. How to handle nak handshake received

If I2C_CON[6] is 1, the I2C controller will stop all transactions when NAK handshake received. And the software should take responsibility to handle the problem.
If I2C_CON[6] is 0, the I2C controller will ignore all NAK handshake received.

11. I2C controller data transfer waveform

- Bit transferring

(a) Data Validity
The SDA line must be stable during the high period of SCL, and the data on SDA line can only be changed when SCL is in low state.

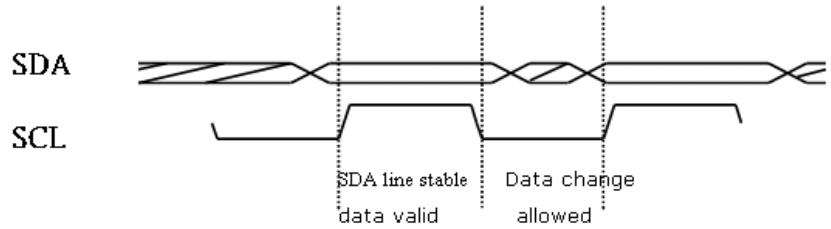

Fig. 52-2 I2C DATA Validity

(b) START and STOP conditions
START condition occurs when SDA goes low while SCL is in high period. STOP condition is generated when SDA line goes high while SCL is in high state.

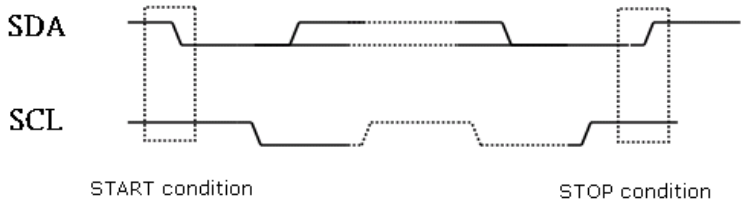

Fig. 52-3 I2C Start and stop conditions

- Data transfer

(a) Acknowledge
After a byte of data transferring (clocks labeled as 1~8), in 9th clock the receiver must assert an ACK signal on SDA line, if the receiver pulls SDA line to low, it means "ACK", on the contrary, it's "NOT ACK".

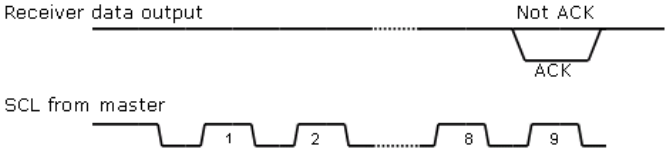

Fig. 52-4 I2C Acknowledge

(b) Byte transfer
The master own I2C bus might initiate multi byte to transfer to a slave.The transfer starts from a "START" command and ends in a "STOP"command. After every byte transfer, the receiver must reply an ACK to transmitter.

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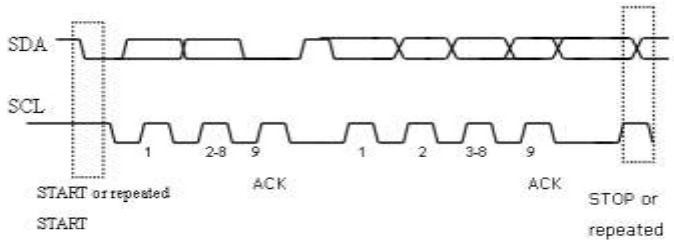

Fig. 52-5 I2C byte transfer

52.4 Register Description

This section describes the control/status registers of the design.

52.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
I2C_CON	0x0000	W	0x00000000	control register
I2C_CLKDIV	0x0004	W	0x00060006	Clock divisor register
I2C_MRXADDR	0x0008	W	0x00000000	the slave address accessed for master receive mode
I2C_MRXRADDR	0x000c	W	0x00000000	the slave register address accessed for master recevie mode
I2C_MTXCNT	0x0010	W	0x00000000	master transmit count
I2C_MRXCNT	0x0014	W	0x00000000	master receive count
I2C_IEN	0x0018	W	0x00000000	interrupt enable register
I2C_IPD	0x001c	W	0x00000000	interrupt pending register
I2C_FCNT	0x0020	W	0x00000000	finished count
I2C_TXDATA0	0x0100	W	0x00000000	I2C transmit data register 0
I2C_TXDATA1	0x0104	W	0x00000000	I2C transmit data register 1
I2C_TXDATA2	0x0108	W	0x00000000	I2C transmit data register 2
I2C_TXDATA3	0x010c	W	0x00000000	I2C transmit data register 3
I2C_TXDATA4	0x0110	W	0x00000000	I2C transmit data register 4
I2C_TXDATA5	0x0114	W	0x00000000	I2C transmit data register 5
I2C_TXDATA6	0x0118	W	0x00000000	I2C transmit data register 6
I2C_TXDATA7	0x011c	W	0x00000000	I2C transmit data register 7
I2C_RXDATA0	0x0200	W	0x00000000	I2C receive data register 0
I2C_RXDATA1	0x0204	W	0x00000000	I2C receive data register 1
I2C_RXDATA2	0x0208	W	0x00000000	I2C receive data register 2
I2C_RXDATA3	0x020c	W	0x00000000	I2C receive data register 3
I2C_RXDATA4	0x0210	W	0x00000000	I2C receive data register 4
I2C_RXDATA5	0x0214	W	0x00000000	I2C receive data register 5
I2C_RXDATA6	0x0218	W	0x00000000	I2C receive data register 6
I2C_RXDATA7	0x021c	W	0x00000000	I2C receive data register 7

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

52.4.2 Detail Register Description

I2C_CON

Address: Operational Base + offset (0x0000)

control register

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Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	act2nak operation when NAK handshake is received 1'b0: ignored 1'b1: stop transaction
5	RW	0x0	ack last byte acknowledge control last byte acknowledge control in master receive mode . 1'b0: ACK 1'b1: NAK
4	W1C	0x0	stop stop enable when this bit is written to 1，I2C will generate stop signal. It cleared itself when stop operation ends.
3	W1C	0x0	start start enable when this bit is written to 1，I2C will generate start signal. It cleared itself when start operation ends.
2:1	RW	0x0	i2c_mode 2'b00: transmit only 2'b01: transmit address (device + register address) --> restart -->transmit address –>receive only 2'b10: receive only 2'b11: transmit address (device + register address, write/read bit is 1) --> restart -->transmit address (device address) -->receive data
0	RW	0x0	i2c_en i2c module enable

I2C_CLKDIV

Address: Operational Base + offset (0x0004)

Clock divisor register

Bit	Attr	Reset Value	Description
31:16	RW	0x0006	CLKDIVH SCL high level clock count T（SCL_HIGH）= T(PCLK) * (CLKDIVH +1)* 8
15:0	RW	0x0006	CLKDIVL SCL low level clock count T（SCL_LOW）= T(PCLK) * (CLKDIVL+1) * 8

I2C_MRXADDR

Address: Operational Base + offset (0x0008)

the slave address accessed for master receive mode

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved
26	RW	0x0	addhvld address high byte valid
25	RW	0x0	addmvld address middle byte valid
24	RW	0x0	addlvld address low byte valid
23:0	RW	0x000000	saddr master address register the lowest bit indicate write or read

I2C_MRXRADDR
Address: Operational Base + offset (0x000c)
the slave register address accessed for master receive mode

Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved
26	RW	0x0	sraddhvld address high byte valid
25	RW	0x0	sraddmvld address middle byte valid
24	RW	0x0	sraddlvld address low byte valid
23:0	RW	0x000000	sraddr slave register address accessed

I2C_MTXCNT
Address: Operational Base + offset (0x0010)
master transmit count

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RW	0x00	mtxcnt master transmit count

I2C_MRXCNT
Address: Operational Base + offset (0x0014)
masterreceive count

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RW	0x00	mrxcnt master receive count

I2C_IEN
Address: Operational Base + offset (0x0018)
interrupt enable register

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Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	nakrcvien NAK handshake received interrupt enable
5	RW	0x0	stopien stop operation finished interrupt enable
4	RW	0x0	startien start operation finished interrupt enable
3	RW	0x0	mbrfien MRXCNT data received finished interrupt enable
2	RW	0x0	mbtfien MTXCNT data transmit finished interrupt enable
1	RW	0x0	brfien byte receive finished interrupt enable
0	RW	0x0	btfien byte transmit finished interrupt enable

I2C_IPD
Address: Operational Base + offset (0x001c)
interrupt pending register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	nakrcvipd NAK handshake received interrupt pending bit
5	RW	0x0	stopipd stop operation finished interrupt pending bit
4	RW	0x0	startipd start operation finished interrupt pending bit
3	RW	0x0	mbrfipd MRXCNT data received finished interrupt pending bit
2	RW	0x0	mbtfipd MTXCNT data transmitfinished interrupt pending bit
1	RW	0x0	brfipd byte receivefinished interrupt pending bit
0	RW	0x0	btfipd byte transmitfinished interrupt pending bit

I2C_FCNT
Address: Operational Base + offset (0x0020)
finished count

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5:0	RW	0x00	fcnt finished count the count of data which has been transmitted or received for debug purpose

I2C_TXDATA0
Address: Operational Base + offset (0x0100)
I2C transmit data register 0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txdata0

I2C_TXDATA1
Address: Operational Base + offset (0x0104)
I2C transmit data register 1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txdata1

I2C_TXDATA2
Address: Operational Base + offset (0x0108)
I2C transmit data register 2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txdata2

I2C_TXDATA3
Address: Operational Base + offset (0x010c)
I2C transmit data register 3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txdata3

I2C_TXDATA4
Address: Operational Base + offset (0x0110)
I2C transmit data register 4

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txdata4

I2C_TXDATA5
Address: Operational Base + offset (0x0114)
I2C transmit data register 5

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txdata5

I2C_TXDATA6
Address: Operational Base + offset (0x0118)

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I2C transmit data register 6

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txdata6

I2C_TXDATA7
Address: Operational Base + offset (0x011c)
I2C transmit data register 7

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txdata7

I2C_RXDATA0
Address: Operational Base + offset (0x0200)
I2C receive data register 0

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rxdata0

I2C_RXDATA1
Address: Operational Base + offset (0x0204)
I2C receive data register 1

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rxdata1

I2C_RXDATA2
Address: Operational Base + offset (0x0208)
I2C receive data register 2

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rxdata2

I2C_RXDATA3
Address: Operational Base + offset (0x020c)
I2C receive data register 3

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rxdata3

I2C_RXDATA4
Address: Operational Base + offset (0x0210)
I2C receive data register 4

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rxdata4

I2C_RXDATA5
Address: Operational Base + offset (0x0214)
I2C receive data register 5

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rxdata5

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I2C_RXDATA6
Address: Operational Base + offset (0x0218)
I2C receive data register 6

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rxdata6

I2C_RXDATA7
Address: Operational Base + offset (0x021c)
I2C receive data register 7

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rxdata7

52.5 Interface description

Table 52-1 I2C Interface Description

Module pin	Dir	Pad name	IOMUX
I2C PMU Interface			
i2c_pmu_sda	I/O	IO_I2C0PMUsda_PMUgpio0a6	PMUGRF_GPIO0A_IOMUX[13:12]=01
i2c_pmu_scl	I/O	IO_I2C0PMUscl_PMUgpio0a7	PMUGRF_GPIO0A_IOMUX[15:14]=01
I2C AUDIO Interface			
i2c_audio_sda	I/O	IO_I2C1AUDIOsda_AUDIOgpio2c5	GRF_GPIO2C_IOMUX[11:10]=01
i2c_audio_scl	I/O	IO_I2C1AUDIOscl_AUDIOgpio2c6	GRF_GPIO2C_IOMUX[13:12]=01
I2C SENSOR Interface			
i2c_sensor_sda	I/O	IO_SCvcc18v_I2C2SENSORsda_GPUJTAGtck_GPIO1830gpio3d7	GRF_GPIO3D_IOMUX[15:14]=10
i2c_sensor_scl	I/O	IO_SCvcc33v_I2C2SENSORscl_GPUJTAGtrstn_GPIO1830gpio0b1	PMUGRF_GPIO0B_IOMUX[3:2]=10
I2C CAM Interface			
i2c_cam_sda	I/O	IO_I2C3CAMsda_SPI1txd_DVPgpio1c1	GRF_GPIO1C_IOMUX[3:2]=01
i2c_cam_scl	I/O	IO_I2C3CAMscl_SPI1rxd_DVPgpio1c0	GRF_GPIO1C_IOMUX[1:0]=01
I2C TP Interface			
i2c_tp_sda	I/O	IO_MACmdio_I2C4TPsda_GPIO30gpio3d0	GRF_GPIO3D_IOMUX[1:0]=10
i2c_tp_scl	I/O	IO_MACrxclkin_I2C4TPscl_GPIO30gpio3d1	GRF_GPIO3D_IOMUX[3:2]=10
I2C HDMI Interface			
i2c_hdmi_sda	I/O	IO_HDMII2Csda_I2C5HDMIsda_GPIO30gpio3d2	GRF_GPIO3D_IOMUX[5:4]=10
i2c_hdmi_scl	I/O	IO_HDMII2Cscl_I2C5HDMIscl_GPIO30gpio3d3	GRF_GPIO3D_IOMUX[7:6]=10

52.6 Application Notes

The I2C controller core operation flow chart below is to describe how the software configures and performs an I2C transaction through this I2C controller core. Descriptions are divided into 3 sections, transmit only mode, receive only mode, and mix mode. Users are strongly advised to following.

- Transmit only mode (I2C_CON[1:0]=2'b00)

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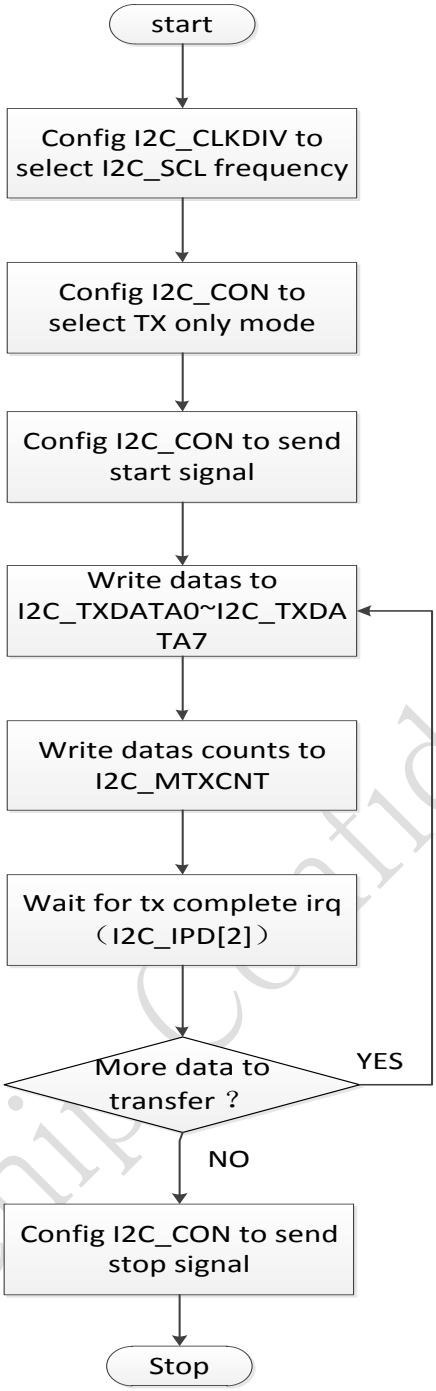

Fig. 52-6 I2C Flow chat for transmit only mode

- Receive only mode (I2C_CON[1:0]=2'b10)

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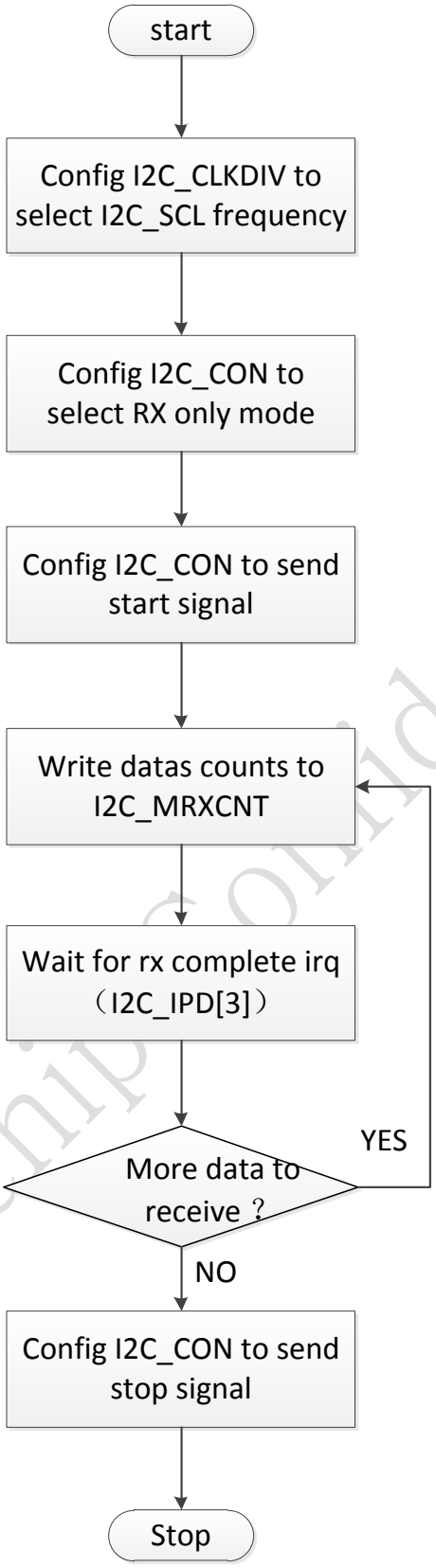

Fig. 52-7 I2C Flow chat for receive only mode

- Mix mode (I2C_CON[1:0]=2'b01 or I2C_CON[1:0]=2'b11)

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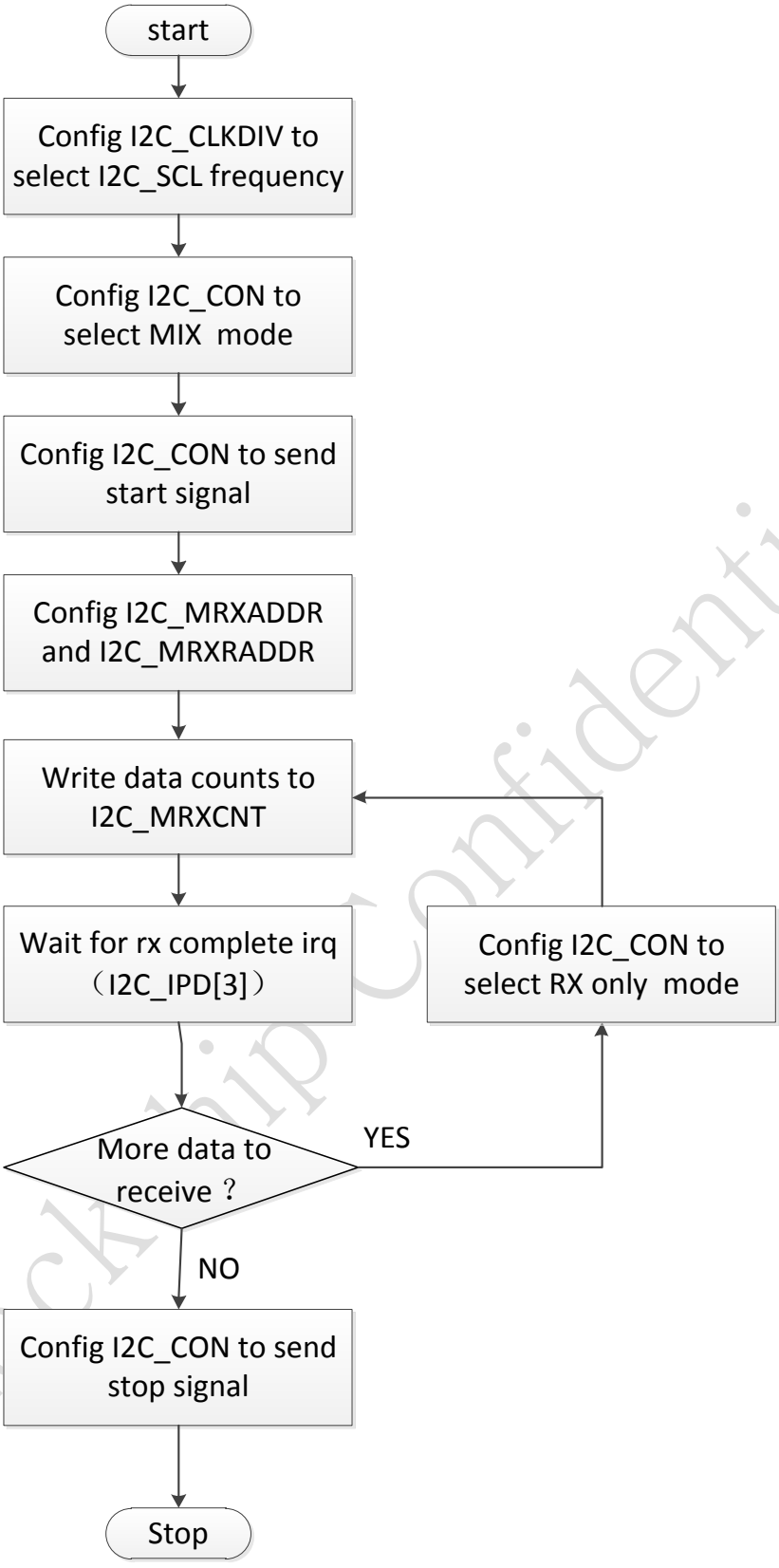

Fig. 52-8 I2C Flow chat for mix mode

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Chapter 53 Universal Asynchronous Receiver/Transmitter (UART)

53.1 Overview

The Universal Asynchronous Receiver/Transmitter (UART) is used for serial communication with a peripheral, modem (data carrier equipment, DCE) or data set. Data is written from a master (CPU) over the APB bus to the UART and it is converted to serial form and transmitted to the destination device. Serial data is also received by the UART and stored for the master (CPU) to read back.

UART Controller supports the following features:

- AMBA APB interface – Allows for easy integration into a Synthesizable Components for AMBA 2 implementation
- Support interrupt interface to interrupt controller
- Contain two 64Bytes FIFOs for data receive and transmit
- Programmable serial data baud rate as calculated by the following: baud rate = (serial clock frequency)/(16×divisor)
- UART_BB/UART_BT/UART_GPS/UART_EXP support auto flow-control, UART_DBG do not support auto flow-control
- UART_GPS support IrDA 1.0 SIR mode with up to 115.2K baud data rate
- UART_BB/UART_BT/UART_GPS/UART_EXP are in peripheral subsystem, UART_DBG is in bus subsystem

53.2 Block Diagram

This section provides a description about the functions and behavior under various conditions.

The UART Controller comprises with:

- AMBA APB interface
- FIFO controllers
- Register block
- Modem synchronization block and baud clock generation block
- Serial receiver and serial transmitter

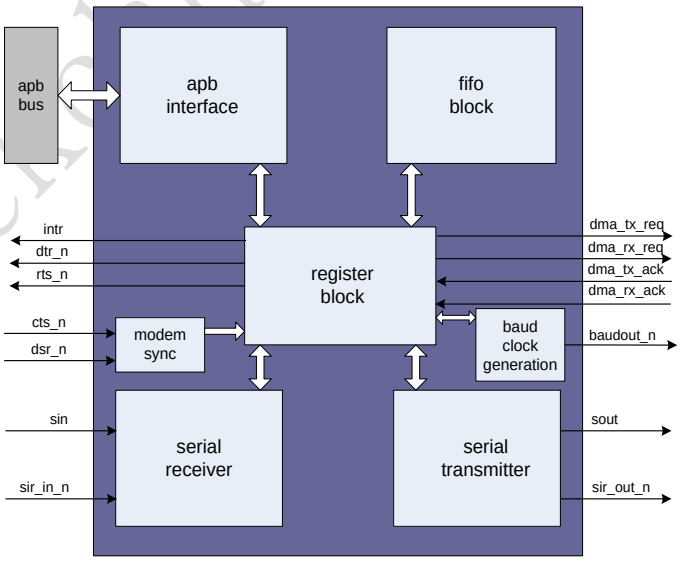

Fig. 53-1 UART Architecture

APB INTERFACE

The host processor accesses data, control, and status information on the UART through the APB interface. The UART supports APB data bus widths of 8, 16, and 32 bits.

Register block

Be responsible for the main UART functionality including control, status and interrupt generation.

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Modem Synchronization block
Synchronizes the modem input signal.
FIFO block
Be responsible for FIFO control and storage (when using internal RAM) or signaling to control external RAM (when used).
Baud Clock Generator
Generate the transmitter and receiver baud clock along with the output reference clock signal (baudout_n).
Serial Transmitter
Converts the parallel data, written to the UART, into serial form and adds all additional bits, as specified by the control register, for transmission. This makeup of serial data, referred to as a character can exit the block in two forms, either serial UART format or IrDA 1.0 SIR format.
Serial Receiver
Converts the serial data character (as specified by the control register) received in either the UART or IrDA 1.0 SIR format to parallel form. Parity error detection, framing error detection and line break detection is carried out in this block.

53.3 Function description

UART (RS232) Serial Protocol
Because the serial communication is asynchronous, additional bits (start and stop) are added to the serial data to indicate the beginning and end. An additional parity bit may be added to the serial character. This bit appears after the last data bit and before the stop bit(s) in the character structure to perform simple error checking on the received data, as shown in Figure.

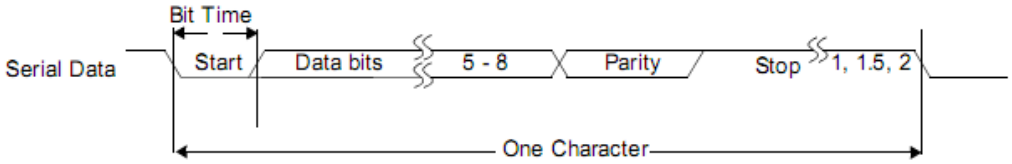

Fig. 53-2 UART Serial protocol

IrDA 1.0 SIR Protocol
The Infrared Data Association (IrDA) 1.0 Serial Infrared (SIR) mode supports bi-directional data communications with remote devices using infrared radiation as the transmission medium. IrDA 1.0 SIR mode specifies a maximum baud rate of 115.2 Kbaud.
Transmitting a single infrared pulse signals a logic zero, while a logic one is represented by not sending a pulse. The width of each pulse is 3/16ths of a normal serial bit time. Data transfers can only occur in half-duplex fashion when IrDA SIR mode is enabled.

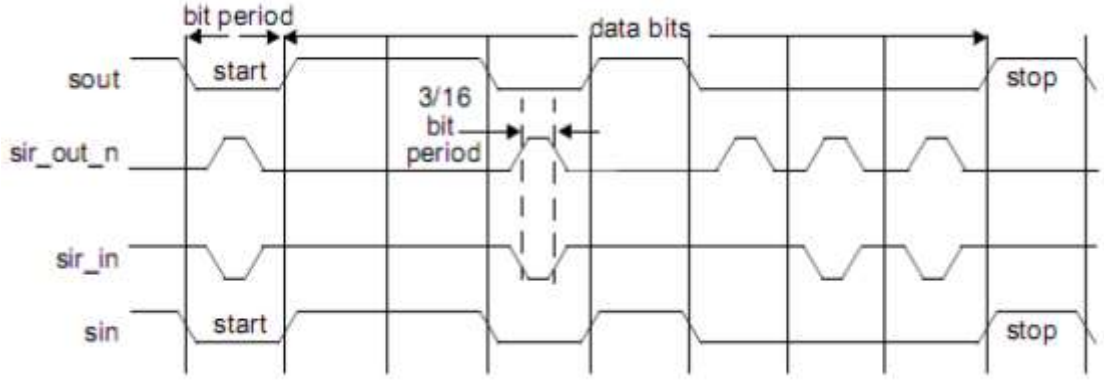

Fig. 53-3 IrDA 1.0

Baud Clock
The baud rate is controlled by the serial clock (sclk or pclk in a single clock implementation) and the Divisor Latch Register (DLH and DLL). As the exact number of baud clocks that each bit was transmitted for is known, calculating the mid-point for sampling is not difficult, that is every 16 baud clocks after the mid point sample of the start bit.

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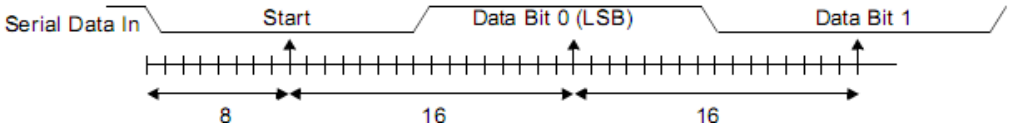

Fig. 53-4 UART baud rate

FIFO Support

1.NONE FIFO MODE

If FIFO support is not selected, then no FIFOs are implemented and only a single receive data byte and transmit data byte can be stored at a time in the RBR and THR.

2.FIFO MODE

The FIFO depth of UART0/UART1/UART2/UART3/UART4 is 64bytes. The FIFO mode of all the UART is enabled by register FCR[0].

Interrupts

The following interrupt types can be enabled with the IER register.

- Receiver Error
- Receiver Data Available
- Character Timeout (in FIFO mode only)
- Transmitter Holding Register Empty at/below threshold (in Programmable THRE Interrupt mode)
- Modem Status

DMA Support

The uart supports DMA signaling with the use of two output signals (dma_tx_req_n and dma_rx_req_n) to indicate when data is ready to be read or when the transmit FIFO is empty.

The dma_tx_req_n signal is asserted under the following conditions:

- When the Transmitter Holding Register is empty in non-FIFO mode.
- When the transmitter FIFO is empty in FIFO mode with Programmable THRE interrupt mode disabled.
- When the transmitter FIFO is at, or below the programmed threshold with Programmable THRE interrupt mode enabled.

The dma_rx_req_n signal is asserted under the following conditions:

- When there is a single character available in the Receive Buffer Register in non-FIFO mode.
- When the Receiver FIFO is at or above the programmed trigger level in FIFO mode.

Auto Flow Control

The UART can be configured to have a 16750-compatible Auto RTS and Auto CTS serial data flow control mode available. If FIFOs are not implemented, then this mode cannot be selected. When Auto Flow Control mode has been selected, it can be enabled with the Modem Control Register (MCR[5]). Following figure shows a block diagram of the Auto Flow Control functionality.

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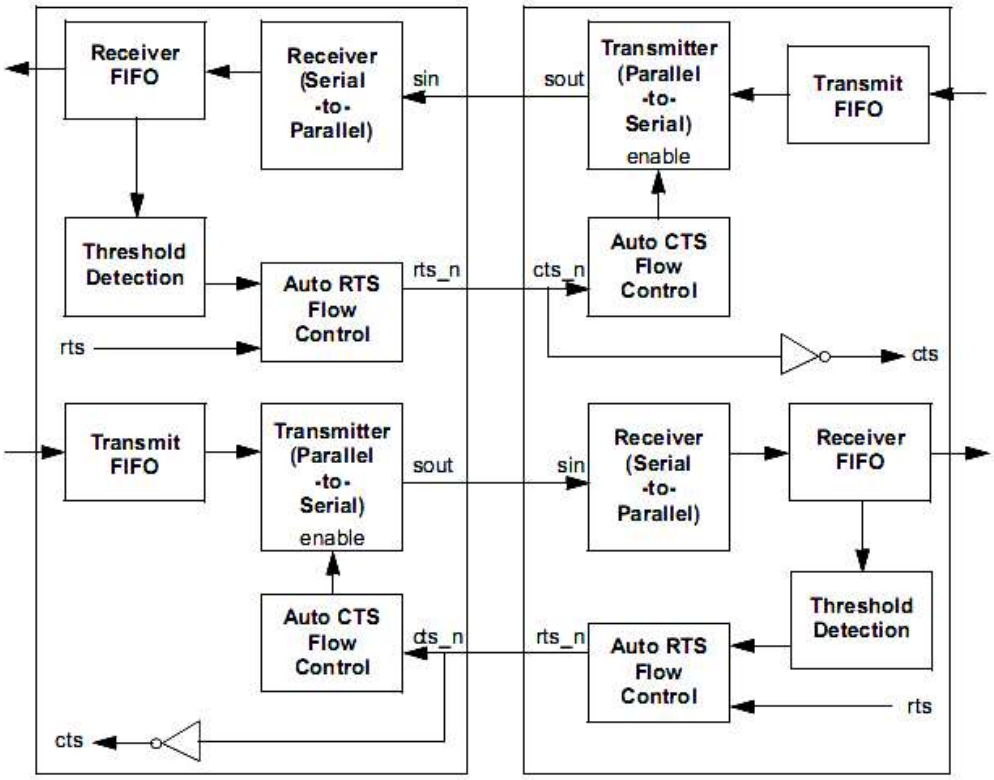

Fig. 53-5 UART Auto flow control block diagram

Auto RTS – Becomes active when the following occurs:

- Auto Flow Control is selected during configuration
- FIFOs are implemented
- RTS (MCR[1] bit and MCR[5]bit are both set)
- FIFOs are enabled (FCR[0]) bit is set)
- SIR mode is disabled (MCR[6] bit is not set)

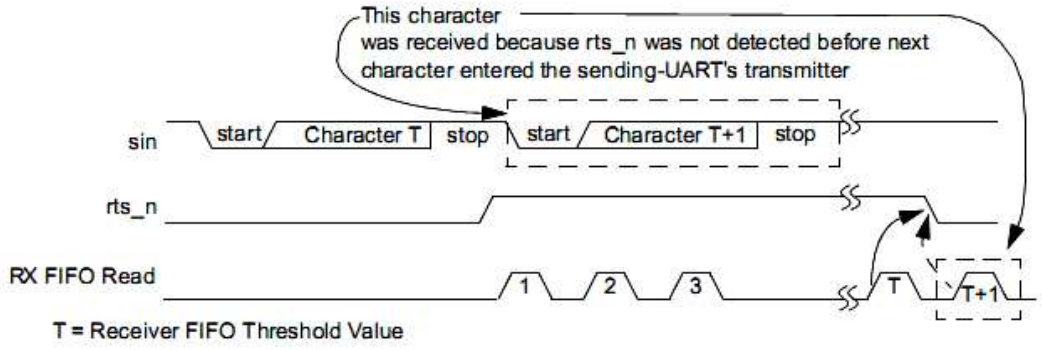

Fig. 53-6 UART AUTO RTS TIMING

Auto CTS – becomes active when the following occurs:

- Auto Flow Control is selected during configuration
- FIFOs are implemented
- AFCE (MCR[5] bit is set)
- FIFOs are enabled through FIFO Control Register FCR[0] bit
- SIR mode is disabled (MCR[6] bit is not set)

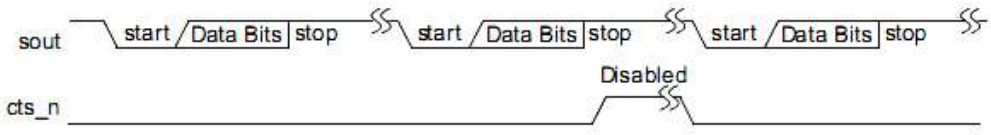

Fig. 53-7 UART AUTO CTS TIMING

53.4 Register Description

This section describes the control/status registers of the design. There are 5 UARTs, and each one has its own base address.

53.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
UART_RBR	0x0000	W	0x00000000	Receive Buffer Register
UART_THR	0x0000	W	0x00000000	Transmit Holding Register
UART_DLL	0x0000	W	0x00000000	Divisor Latch (Low)
UART_DLH	0x0004	W	0x00000000	Divisor Latch (High)
UART_IER	0x0004	W	0x00000000	Interrupt Enable Register
UART_IIR	0x0008	W	0x00000001	Interrupt Identification Register
UART_FCR	0x0008	W	0x00000000	FIFO Control Register
UART_LCR	0x000c	W	0x00000000	Line Control Register
UART_MCR	0x0010	W	0x00000000	Modem Control Register
UART_LSR	0x0014	W	0x00000060	Line Status Register
UART_MSR	0x0018	W	0x00000000	Modem Status Register
UART_SCR	0x001c	W	0x00000000	Scratchpad Register
UART_SRBR	0x0030~0x006c	W	0x00000000	Shadow Receive Buffer Register
UART_STHR	0x0030~0x006c	W	0x00000000	Shadow Transmit Holding Register
UART_FAR	0x0070	W	0x00000000	FIFO Access Register
UART_TFR	0x0074	W	0x00000000	Transmit FIFO Read
UART_RFW	0x0078	W	0x00000000	Receive FIFO Write
UART_USR	0x007c	W	0x00000006	UART Status Register
UART_TFL	0x0080	W	0x00000000	Transmit FIFO Level
UART_RFL	0x0084	W	0x00000000	Receive FIFO Level
UART_SRR	0x0088	W	0x00000000	Software Reset Register
UART_SRTS	0x008c	W	0x00000000	Shadow Request to Send
UART_SBCR	0x0090	W	0x00000000	Shadow Break Control Register
UART_SDMAM	0x0094	W	0x00000000	Shadow DMA Mode
UART_SFE	0x0098	W	0x00000000	Shadow FIFO Enable
UART_SRT	0x009c	W	0x00000000	Shadow RCVR Trigger
UART_STET	0x00a0	W	0x00000000	Shadow TX Empty Trigger
UART_HTX	0x00a4	W	0x00000000	Halt TX
UART_DMASA	0x00a8	W	0x00000000	DMA Software Acknowledge
UART_CPR	0x00f4	W	0x00000000	Component Parameter Register
UART_UCV	0x00f8	W	0x3330382a	UART Component Version
UART_CTR	0x00fc	W	0x44570110	Component Type Register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

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53.4.2 Detail Register Description

UART_RBR
Address: Operational Base + offset (0x0000)
Receive Buffer Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	data_input Data byte received on the serial input port (sin) in UART mode, or the serial infrared input (sir_in) in infrared mode. The data in this register is valid only if the Data Ready (DR) bit in the Line Status Register (LSR) is set. If in non-FIFO mode (FIFO_MODE == NONE) or FIFOs are disabled (FCR[0] set to zero), the data in the RBR must be read before the next data arrives, otherwise it is overwritten, resulting in an over-run error. If in FIFO mode (FIFO_MODE != NONE) and FIFOs are enabled (FCR[0] set to one), this register accesses the head of the receive FIFO. If the receive FIFO is full and this register is not read before the next data character arrives, then the data already in the FIFO is preserved, but any incoming data are lost and an over-run error occurs.

UART_THR
Address: Operational Base + offset (0x0000)
Transmit Holding Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:0	RW	0x00	data_output Data to be transmitted on the serial output port (sout) in UART mode or the serial infrared output (sir_out_n) in infrared mode. Data should only be written to the THR when the THR Empty (THRE) bit (LSR[5]) is set. If in non-FIFO mode or FIFOs are disabled (FCR[0] = 0) and THRE is set, writing a single character to the THR clears the THRE. Any additional writes to the THR before the THRE is set again causes the THR data to be overwritten. If in FIFO mode and FIFOs are enabled (FCR[0] = 1) and THRE is set, x number of characters of data may be written to the THR before the FIFO is full. The number x (default=16) is determined by the value of FIFO Depth that you set during configuration. Any attempt to write data when the FIFO is full results in the write data being lost.

UART_DLL

Address: Operational Base + offset (0x0000)

Divisor Latch (Low)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	baud_rate_divisor_L Lower 8-bits of a 16-bit, read/write, Divisor Latch register that contains the baud rate divisor for the UART. This register may only be accessed when the DLAB bit (LCR[7]) is set and the UART is not busy (USR[0] is zero). The output baud rate is equal to the serial clock (sclk) frequency divided by sixteen times the value of the baud rate divisor, as follows: baud rate = (serial clock frequency) / (16 * divisor). Note that with the Divisor Latch Registers (DLL and DLH) set to zero, the baud clock is disabled and no serial communications occur. Also, once the DLH is set, at least 8 clock cycles of the slowest UART clock should be allowed to pass before transmitting or receiving data.

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UART_DLH
Address: Operational Base + offset (0x0004)
Divisor Latch (High)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	baud_rate_divisor_H Upper 8 bits of a 16-bit, read/write, Divisor Latch register that contains the baud rate divisor for the UART.

UART_IER
Address: Operational Base + offset (0x0004)
Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x0	prog_thre_int_en Programmable THRE Interrupt Mode Enable This is used to enable/disable the generation of THRE Interrupt. 1'b0: disabled 1'b1: enabled
6:4	RO	0x0	reserved
3	RW	0x0	modem_status_int_en Enable Modem Status Interrupt. This is used to enable/disable the generation of Modem Status Interrupt. This is the fourth highest priority interrupt. 1'b0: disabled 1'b1: enabled
2	RW	0x0	receive_line_status_int_en Enable Receiver Line Status Interrupt. This is used to enable/disable the generation of Receiver Line Status Interrupt. This is the highest priority interrupt. 1'b0: disabled 1'b1: enabled
1	RW	0x0	trans_hold_empty_int_en Enable Transmit Holding Register Empty Interrupt.

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Bit	Attr	Reset Value	Description
0	RW	0x0	receive_data_available_int_en Enable Received Data Available Interrupt. This is used to enable/disable the generation of Received Data Available Interrupt and the Character Timeout Interrupt (if in FIFO mode and FIFOs enabled). These are the second highest priority interrupts. 1'b0: disabled 1'b1: enabled

UART_IIR
Address: Operational Base + offset (0x0008)
Interrupt Identification Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:6	RO	0x0	fifos_en FIFOs Enabled. This is used to indicate whether the FIFOs are enabled or disabled. 2'b00: disabled 2'b11: enabled
5:4	RO	0x0	reserved
3:0	RO	0x1	int_id Interrupt ID This indicates the highest priority pending interrupt which can be one of the following types: 4'b0000: modem status 4'b0001: no interrupt pending 4'b0010: THR empty 4'b0100: received data available 4'b0110: receiver line status 4'b0111: busy detect 4'b1100: character timeout

UART_FCR
Address: Operational Base + offset (0x0008)
FIFO Control Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:6	WO	0x0	rcvr_trigger RCVR Trigger. This is used to select the trigger level in the receiver FIFO at which the Received Data Available Interrupt is generated. In auto flow control mode it is used to determine when the rts_n signal is de-asserted. It also determines when the dma_rx_req_n signal is asserted in certain modes of operation. The following trigger levels are supported: 2'b00: 1 character in the FIFO 2'b01: FIFO 1/4 full 2'b10: FIFO 1/2 full 2'b11: FIFO 2 less than full
5:4	WO	0x0	tx_empty_trigger TX Empty Trigger. This is used to select the empty threshold level at which the THRE Interrupts are generated when the mode is active. It also determines when the dma_tx_req_n signal is asserted when in certain modes of operation. The following trigger levels are supported: 2'b00: FIFO empty 2'b01: 2 characters in the FIFO 2'b10: FIFO 1/4 full 2'b11: FIFO 1/2 full
3	WO	0x0	dma_mode DMA Mode This determines the DMA signalling mode used for the dma_tx_req_n anddma_rx_req_n output signals when additional DMA handshaking signals are not selected . 1'b0: mode 0 1'b1: mode 11100 = character timeout.
2	WO	0x0	xmit_fifo_reset XMIT FIFO Reset. This resets the control portion of the transmit FIFO and treats the FIFO as empty. This also de-asserts the DMA TX request and single signals when additional DMA handshaking signals are select. Note that this bit is'self-clearing'. It is not necessary to clear this bit.

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Bit	Attr	Reset Value	Description
1	WO	0x0	rcvr_fifo_reset RCVR FIFO Reset. This resets the control portion of the receive FIFO and treats the FIFO as empty. This also de-asserts the DMA RX request and single signals when additional DMA handshaking signals are selected. Note that this bit is 'self-clearing'. It is not necessary to clear this bit.
0	WO	0x0	fifo_en FIFO Enable. FIFO Enable. This enables/disables the transmit (XMIT) and receive (RCVR) FIFOs. Whenever the value of this bit is changed both the XMIT and RCVR controller portion of FIFOs is reset.

UART_LCR

Address: Operational Base + offset (0x000c)

Line Control Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x0	div_lat_access Divisor Latch Access Bit. Writeable only when UART is not busy (USR[0] is zero), always readable. This bit is used to enable reading and writing of the Divisor Latch register (DLL and DLH) to set the baud rate of the UART. This bit must be cleared after initial baud rate setup in order to access other registers.
6	RW	0x0	break_ctrl Break Control Bit. This is used to cause a break condition to be transmitted to the receiving device. If set to one the serial output is forced to the spacing (logic 0) state. When not in Loopback Mode, as determined by MCR[4], the sout line is forced low until the Break bit is cleared. If MCR[6] set to one, the sir_out_n line is continuously pulsed. When in Loopback Mode, the break condition is internally looped back to thereceiver and the sir_out_n line is forced low.
5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4	RW	0x0	even_parity_sel Even Parity Select. Writeable only when UART is not busy (USR[0] is zero), always readable. This is used to select between even and odd parity, when parity is enabled (PEN set to one). If set to one, an even number of logic 1s is transmitted or checked. If set to zero, an odd number of logic 1s is transmitted or checked.
3	RW	0x0	parity_en Parity Enable. Writeable only when UART is not busy (USR[0] is zero), always readable. This bit is used to enable and disable parity generation and detection in transmitted and received serial character respectively. 1'b0: parity disabled 1'b1: parity enabled
2	RW	0x0	stop_bits_num Number of stop bits. Writeable only when UART is not busy (USR[0] is zero), always readable. This is used to select the number of stop bits per character that the peripheral transmits and receives. If set to zero, one stop bit is transmitted in the serial data.If set to one and the data bits are set to 5 (LCR[1:0] set to zero) one and a half stop bits is transmitted. Otherwise, twostop bits are transmitted. Note that regardless of the number of stop bits select, the receiver checks only the first stop bit. 1'b0: 1 stop bit 1'b1: 1.5 stop bits when DLS (LCR[1:0]) is zero, else 2 stop bit.

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	data_length_sel Data Length Select. Writeable only when UART is not busy (USR[0] is zero), always readable. This is used to select the number of data bits per character that the peripheral transmits and receives. The number of bit that may be selected areas follows: 2'b00: 5 bits 1'b01: 6 bits 1'b10: 7 bits 1'b11: 8 bits

UART_MCR
Address: Operational Base + offset (0x0010)
Modem Control Register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	sir_mode_en SIR Mode Enable. SIR Mode Enable. This is used to enable/disable the IrDA SIR Mode. 1'b0: IrDA SIR Mode disabled 1'b1: IrDA SIR Mode enabled
5	RW	0x0	auto_flow_ctrl_en Auto Flow Control Enable. 1'b0: Auto Flow Control Mode disabled 1'b1: Auto Flow Control Mode enabled
4	RW	0x0	loopback LoopBack Bit. This is used to put the UART into a diagnostic mode for test purposes.
3	RW	0x0	out2 OUT2. This is used to directly control the user-designated Output2 (out2_n) output. The value written to this location is inverted and driven out on out2_n, that is: 1'b0: out2_n de-asserted (logic 1) 1'b1: out2_n asserted (logic 0)
2	RW	0x0	out1 OUT1

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Bit	Attr	Reset Value	Description
1	RW	0x0	req_to_send Request to Send. This is used to directly control the Request to Send (rts_n) output. The Request To Send (rts_n) output is used to inform the modem or data set that the UART is ready to exchange data.
0	RW	0x0	data_terminal_ready Data Terminal Ready. This is used to directly control the Data Terminal Ready (dtr_n) output. The value written to this location is inverted and driven out on dtr_n, that is: 1'b0: dtr_n de-asserted (logic 1) 1'b1: dtr_n asserted (logic 0)

UART_LSR
Address: Operational Base + offset (0x0014)
Line Status Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RO	0x0	receiver_fifo_error Receiver FIFO Error bit. This bit is relevant FIFOs are enabled (FCR[0] set to one). This is used to indicate if there is at least one parity error, framing error, or break indication in the FIFO. 1'b0: no error in RX FIFO 1'b1: error in RX FIFO
6	RO	0x1	trans_empty Transmitter Empty bit. Transmitter Empty bit. If FIFOs enabled (FCR[0] set to one), this bit is set whenever the Transmitter Shift Register and the FIFO are both empty. If FIFOs are disabled, this bit is set whenever the Transmitter Holding Register and the Transmitter Shift Register are both empty.

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Bit	Attr	Reset Value	Description
5	RO	0x1	trans_hold_reg_empty Transmit Holding Register Empty bit. If THRE mode is disabled (IER[7] set to zero) and regardless of FIFO's being implemented/enabled or not, this bit indicates that the THR or TX FIFO is empty. This bit is set whenever data is transferred from the THR or TX FIFO to the transmitter shift register and no new data has been written to the THR or TX FIFO. This also causes a THRE Interrupt to occur, if the THRE Interrupt is enabled. If IER[7] set to one and FCR[0] set to one respectively, the functionality is switched to indicate the transmitter FIFO is full, and no longer controls THRE interrupts, which are then controlled by the FCR[5:4] threshold setting.
4	RO	0x0	break_int Break Interrupt bit. This is used to indicate the detection of a break sequence on the serial input data.
3	RO	0x0	framing_error Framing Error bit. This is used to indicate the occurrence of a framing error in the receiver. A framing error occurs when the receiver does not detect a valid STOP bit in the received data.
2	RO	0x0	parity_eror Parity Error bit. This is used to indicate the occurrence of a parity error in the receiver if the Parity Enable (PEN) bit (LCR[3]) is set.
1	RO	0x0	overrun_error Overrun error bit. This is used to indicate the occurrence of an overrun error. This occurs if a new data character was received before the previous data was read.
0	RO	0x0	data_ready Data Ready bit. This is used to indicate that the receiver contains at least one character in the RBR or the receiver FIFO. 1'b0: no data ready 1'b1: data ready

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UART_MSR
Address: Operational Base + offset (0x0018)
Modem Status Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RO	0x0	data_carrior_detect Data Carrier Detect. This is used to indicate the current state of the modem control line dcd_n.
6	RO	0x0	ring_indicator Ring Indicator. This is used to indicate the current state of the modem control line ri_n.
5	RO	0x0	data_set_ready Data Set Ready. This is used to indicate the current state of the modem control line dsr_n.
4	RO	0x0	clear_to_send Clear to Send. This is used to indicate the current state of the modem control line cts_n.
3	RO	0x0	delta_data_carrier_detect Delta Data Carrier Detect. This is used to indicate that the modem control line dcd_n has changed since the last time the MSR was read.
2	RO	0x0	trailing_edge_ring_indicator Trailing Edge of Ring Indicator. Trailing Edge of Ring Indicator. This is used to indicate that a change on the input ri_n (from an active-low to an inactive-high state) has occurred since the last time the MSR was read.
1	RO	0x0	delta_data_set_ready Delta Data Set Ready. This is used to indicate that the modem control line dsr_n has changed since the last time the MSR was read.
0	RO	0x0	delta_clear_to_send Delta Clear to Send. This is used to indicate that the modem control line cts_n has changed since the last time the MSR was read.

UART_SCR
Address: Operational Base + offset (0x001c)

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Scratchpad Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	temp_store_space This register is for programmers to use as a temporary storage space.

UART_SRBR

Address: Operational Base + offset (0x0030~0x006c)

Shadow Receive Buffer Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RO	0x00	shadow_rbr This is a shadow register for the RBR and has been allocated sixteen 32-bit locations so as to accommodate burst accesses from the master. This register contains the data byte received on the serial input port (sin) in UART mode or the serial infrared input (sir_in) in infrared mode. The data in this register is valid only if the Data Ready (DR) bit in the Line status Register (LSR) is set. If FIFOs are disabled (FCR[0] set to zero), the data in the RBR must be read before the next data arrives, otherwise it is overwritten, resulting in an overrun error. If FIFOs are enabled (FCR[0] set to one), this register accesses the head of the receive FIFO. If the receive FIFO is full and this register is not read before the next data character arrives, then the data already in the FIFO are preserved, but any incoming data is lost. An overrun error also occurs.

UART_STHR

Address: Operational Base + offset (0x0030~0x006c)

Shadow Transmit Holding Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RO	0x00	shadow_thr This is a shadow register for the THR.

UART_FAR

Address: Operational Base + offset (0x0070)

FIFO Access Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	fifo_access_test_en This register is use to enable a FIFO access mode for testing, so that the receive FIFO can be written by the master and the transmit FIFO can be read by the master when FIFOs are implemented and enabled. When FIFOs are not enabled it allows the RBR to be written by the master and the THR to be read by the master. 1'b0: FIFO access mode disabled 1'b1: FIFO access mode enabled

UART_TFR
Address: Operational Base + offset (0x0074)
Transmit FIFO Read

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RO	0x00	trans_fifo_read Transmit FIFO Read. These bits are only valid when FIFO access mode is enabled (FAR[0] is set to one).When FIFOs are implemented and enabled, reading this register gives the data at the top of the transmit FIFO. Each consecutive read pops the transmit FIFO and gives the next data value that is currently at the top of the FIFO.

UART_RFW
Address: Operational Base + offset (0x0078)
Receive FIFO Write

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9	WO	0x0	receive_fifo_framing_error Receive FIFO Framing Error. These bits are only valid when FIFO access mode is enabled (FAR[0] is set to one).
8	WO	0x0	receive_fifo_parity_error Receive FIFO Parity Error. These bits are only valid when FIFO access mode is enabled (FAR[0] is set to one).

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Bit	Attr	Reset Value	Description
7:0	WO	0x00	receive_fifo_write Receive FIFO Write Data. These bits are only valid when FIFO access mode is enabled (FAR[0] is set to one). When FIFOs are enabled, the data that is written to the RFWD is pushed into the receive FIFO. Each consecutive write pushes the new data to the next write location in the receive FIFO. When FIFOs not enabled, the data that is written to the RFWD is pushed into the RBR.

UART_USR
Address: Operational Base + offset (0x007c)
UART Status Register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RO	0x0	receive_fifo_full Receive FIFO Full. This is used to indicate that the receive FIFO is completely full. 1'b0: Receive FIFO not full 1'b1: Receive FIFO Full This bit is cleared when the RX FIFO is no longer full.
3	RO	0x0	receive_fifo_not_empty Receive FIFO Not Empty. This is used to indicate that the receive FIFO contains one or more entries. 1'b0: Receive FIFO is empty 1'b1: Receive FIFO is not empty This bit is cleared when the RX FIFO is empty.
2	RO	0x1	trasn_fifo_empty Transmit FIFO Empty. This is used to indicate that the transmit FIFO is completely empty. 1'b0: Transmit FIFO is not empty 1'b1: Transmit FIFO is empty This bit is cleared when the TX FIFO is no longer empty

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Bit	Attr	Reset Value	Description
1	RO	0x1	trans_fifo_not_full Transmit FIFO Not Full. This is used to indicate that the transmit FIFO in not full. 1'b0: Transmit FIFO is full 1'b1: Transmit FIFO is not full This bit is cleared when the TX FIFO is full.
0	RO	0x0	uart_busy UART Busy. UART Busy. This is indicates that a serial transfer is in progress, when cleared indicates that the UART is idle or inactive. 1'b0:UART is idle or inactive 1'b1:UART is busy (actively transferring data)

UART_TFL
Address: Operational Base + offset (0x0080)
Transmit FIFO Level

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x00	trans_fifo_level Transmit FIFO Level. This is indicates the number of data entries in the transmit FIFO.

UART_RFL
Address: Operational Base + offset (0x0084)
Receive FIFO Level

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RO	0x00	receive_fifo_level Receive FIFO Level. This is indicates the number of data entries in the receive FIFO.

UART_SRR
Address: Operational Base + offset (0x0088)
Software Reset Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	WO	0x0	xmit_fifo_reset XMIT FIFO Reset. This is a shadow register for the XMIT FIFO Reset bit (FCR[2]).

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Bit	Attr	Reset Value	Description
1	WO	0x0	rcvr_fifo_reset RCVR FIFO Reset. This is a shadow register for the RCVR FIFO Reset bit (FCR[1]).
0	WO	0x0	uart_reset UART Reset. This asynchronously resets the UART and synchronously removes the reset assertion. For a two clock implementation both pclk and sclk domains are reset.

UART_SRTS
Address: Operational Base + offset (0x008c)
Shadow Request to Send

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	shadow_req_to_send Shadow Request to Send. This is a shadow register for the RTS bit (MCR[1]), this can be used to remove the burden of having to performing a read-modify-write on the MCR.

UART_SBCR
Address: Operational Base + offset (0x0090)
Shadow Break Control Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	shadow_break_ctrl Shadow Break Control Bit. This is a shadow register for the Break bit (LCR[6]), this can be used to remove the burden of having to performing a read modify write on the LCR.

UART_SDMAM
Address: Operational Base + offset (0x0094)
Shadow DMA Mode

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	shadow_dma_mode Shadow DMA Mode. This is a shadow register for the DMA mode bit (FCR[3]).

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UART_SFE
Address: Operational Base + offset (0x0098)
Shadow FIFO Enable

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	shadow_fifo_en Shadow FIFO Enable. Shadow FIFO Enable. This is a shadow register for the FIFO enable bit (FCR[0]).

UART_SRT
Address: Operational Base + offset (0x009c)
Shadow RCVR Trigger

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
1:0	RW	0x0	shadow_rcvr_trigger Shadow RCVR Trigger. This is a shadow register for the RCVR trigger bits (FCR[7:6]).

UART_STET
Address: Operational Base + offset (0x00a0)
Shadow TX Empty Trigger

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
1:0	RW	0x0	shadow_tx_empty_trigger Shadow TX Empty Trigger. This is a shadow register for the TX empty trigger bits (FCR[5:4]).

UART_HTX
Address: Operational Base + offset (0x00a4)
Halt TX

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	halt_tx_en This register is use to halt transmissions for testing, so that the transmit FIFO can be filled by the master when FIFOs are implemented and enabled. 1'b0: Halt TX disabled 1'b1: Halt TX enabled

UART_DMASA
Address: Operational Base + offset (0x00a8)
DMA Software Acknowledge

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Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	WO	0x0	dma_software_ack This register is use to perform a DMA software acknowledge if a transfer needs to be terminated due to an error condition.

UART_UCV
Address: Operational Base + offset (0x00f8)
UART Component Version

Bit	Attr	Reset Value	Description
31:0	RO	0x3330382a	ver ASCII value for each number in the version

UART_CTR
Address: Operational Base + offset (0x00fc)
Component Type Register

Bit	Attr	Reset Value	Description
31:0	RO	0x44570110	peripheral_id This register contains the peripherals identification code.

53.5 Interface description

Table 53-1 UART Interface Description

Module pin	Dir	Pad name	IOMUX
UART_BT Interface			
uartbt_sin	I	IO_UART0BTsin_WIFIgpio2d0	GPIO2D_IOMUX[1:0]=2'b 01
uartbt_sout		IO_UART0BTsout_WIFIgpio2d1	GPIO2D_IOMUX[3:2]=2'b 01
uartbt_cts_n	I	IO_UART0BTctsn_WIFIgpio2d2	GPIO2D_IOMUX[5:4]=2'b 01
uartbt_rts_n	O	IO_UART0BTrtsn_WIFIgpio2d3	GPIO2D_IOMUX[7:6]=2'b 01
UART_BB Interface			
uartbb_sin	I	IO_LCDCdata16_TRACEdate6_UART1BBsin_LCDCgpio0c4	GPIO0C_IOMUX[9:8]=2'b11
uartbb_sout	O	IO_LCDCdata17_TRACEdate7_UART1BBsout_LCDCgpio0c5	GPIO0C_IOMUX[11:10]=2'b11
uartbb_cts_n	I	IO_LCDCdata18_TRACEdate8_UART1BBctsn_LCDCgpio0c6	GPIO0C_IOMUX[13:12]=2'b11
uartbb_rts_n	O	IO_LCDCdata19_TRACEdate9_UART1BBrtsn_LCDCgpio0c7	GPIO0C_IOMUX[15:14]=2'b11
UART_DBG Interface			
uartdbg_sin	I	IO_SDMMC0data1_UART2DBGsin_SDCARDgpio2a6	GPIO2A_IOMUX[11:10]=2'b11
uartdbg_sout	O	IO_SDMMC0data0_UART2DBGsout_SDCARDgpio2a5	GPIO2A_IOMUX[13:12]=2'b11
UART_GPS Interface			
uartgps_sin	I	IO_IRrx_UART3GPSsin_GPIO30gpio3d5	GPIO3D_IOMUX[11:10]=2'b10
uartgps_sout	O	IO_IRtx_UART3GPSsout_PWM3_GPIO30gpio3d6	GPIO3D_IOMUX[13:12]=2'b10
uartgps_cts_n	I	IO_MACrxd1_UART3GPSctsn_GPSrfclk_GPIO30gpio3c0	GPIO3C_IOMUX[1:0]=2'b10
uartgps_rts_n	O	IO_MACrxd2_UART3GPSrtsn_USBdrvvbus0_GPIO30gpio3c1	GPIO4B_IOMUX[3:2]=2'b11
uartgps_sirsin	I	IO_IRrx_UART3GPSsin_GPIO30gpio3d5	GPIO3D_IOMUX[11:10]=2'b01
uartgps_sirout	O	IO_IRtx_UART3GPSsout_PWM3_GPIO30gpio3d6	GPIO3D_IOMUX[13:12]=2'b01
UART_EXP Interface			
uartexp_sin	I	IO_LCDCdata23_TRACEdate13_UART4EXPsin_LCDCgpio0d3	GPIO0D_IOMUX[7:6]=2'b11
uartexp_sout	O	IO_LCDCdata22_TRACEdate12_UART4EXPsout_LCDCgpio0d2	GPIO0D_IOMUX[5:4]=2'b11
uartexp_cts_n	I	IO_LCDCdata20_TRACEdate10_UART4EXPctsn_LCDCgpio0d0	GPIO0D_IOMUX[1:0]=2'b11
uartexp_rts_n	O	IO_LCDCdata21_TRACEdate11_UART4EXPrtsn_LCDCgpio0d1	GPIO0D_IOMUX[3:2]=2'b11

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53.6 Application Notes

53.6.1 None FIFO Mode Transfer Flow

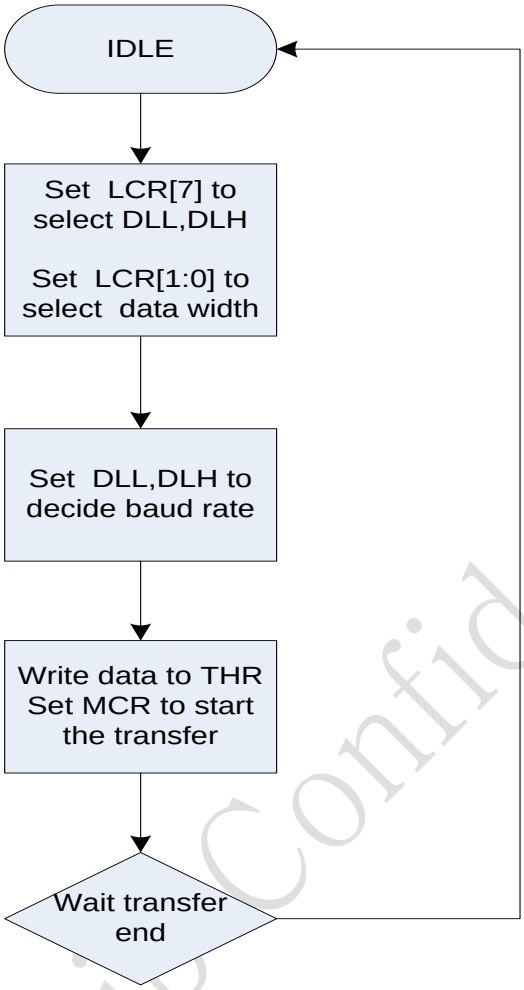

Fig. 53-8 UART none fifo mode

53.6.2 FIFO Mode Transfer Flow

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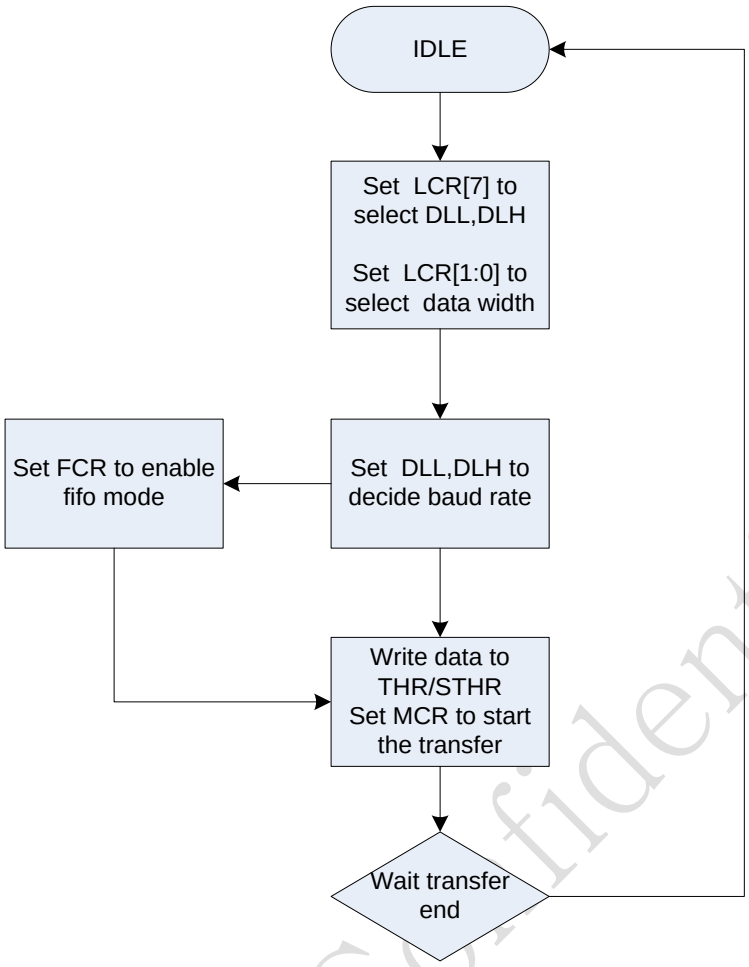

Fig. 53-9 UART fifo mode

The UART is an APB slave performing:
Serial-to-parallel conversion on data received from a peripheral device.
Parallel-to-serial conversion on data transmitted to the peripheral device.
The CPU reads and writes data and control/status information through the APB interface. The transmitting and receiving paths are buffered with internal FIFO memories enabling up to 64-bytes to be stored independently in both transmit and receive modes. A baud rate generator can generate a common transmit and receive internal clock input. The baud rates will depend on the internal clock frequency. The UART will also provide transmit, receive and exception interrupts to system. A DMA interface is implemented for improving the system performance.

53.6.3 Baud Rate Calculation

UART clock generation

The following figures shows the UART clock generation.
UART source clocks can be selected from CODEC PLL and GENERAL PLL outputs. UART_BT source clocks can also be selected from NEW PLL and USBPHY 480M. UART clocks can be generated by 1 to 64 division of its source clock, or can be fractionally divided again, or be provided by XIN24M.

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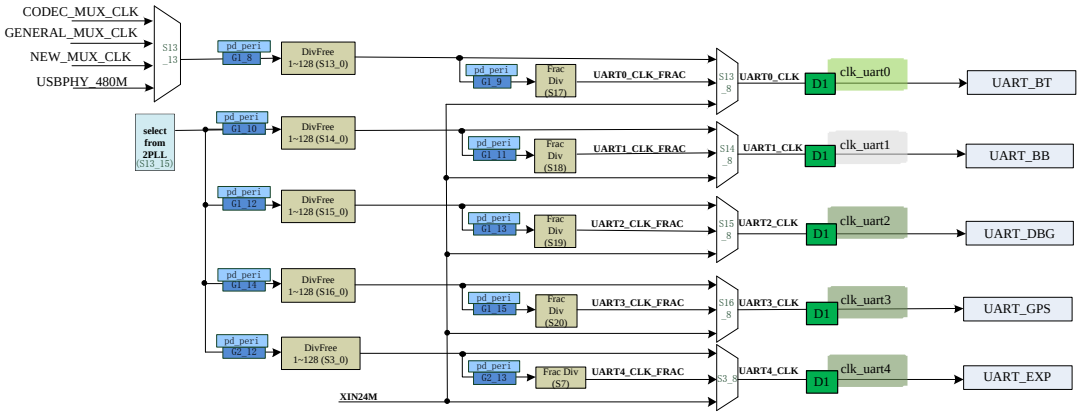

Fig. 53-10 UART clock generation

UART baud rate configuration

The following table provides some reference configuration for different UART baud rates.

Table 53-2 UART baud rate configuration

Baud Rate	Reference Configuration
115.2 Kbps	Configure GENERAL PLL to get 648MHz clock output; Divide 648MHz clock by 1152/50625 to get 14.7456MHz clock; Config UART_DLL to 8.
460.8 Kbps	Configure GENERAL PLL to get 648MHz clock output; Divide 648MHz clock by 1152/50625 to get 14.7456MHz clock; Configure UART_DLL to 2.
921.6 Kbps	Configure GENERAL PLL to get 648MHz clock output; Divide 648MHz clock by 1152/50625 to get 14.7456MHz clock; Configure UART_DLL to 1.
1.5 Mbps	Choose GENERAL PLL to get 384MHz clock output; Divide 384MHz clock by 16 to get 24MHz clock; Configure UART_DLL to 1
3 Mbps	Choose GENERAL PLL to get 384MHz clock output; Divide 384MHz clock by 8 to get 48MHz clock; Configure UART_DLL to 1
4 Mbps	Configure GENERAL PLL to get 384MHz clock output; Divide 384MHz clock by 6 to get 64MHz clock; Configure UART_DLL to 1

53.6.4 CTS_n and RTS_n Polarity Configurable

The polarity of cts_n and rts_n ports can be configured by GRF registers.

- GRF_SOC_CON17[4:0] (grf_uart_cts_sel[4:0]) used to configure the polarity of cts_n. Every bit for one UART, bit4 is for UART_EXP, bit3 is for UART_GPS, bit2 is for UART_DBG, bit1 is for UART_BB, bit0 is for UART_BT.
- GRF_SOC_CON17[9:5] (grf_uart_rts_sel[4:0]) used to configure the polarity of rts_n. Every bit for one UART, bit4 is for UART_EXP, bit3 is for UART_GPS, bit2 is for UART_DBG, bit1 is for UART_BB, bit0 is for UART_BT.
- When grf_uart_cts_sel[*] is configured as 1'b1, cts_n is high active. Otherwise, low active.
- When grf_uart_rts_sel[*] is configured as 1'b1, rts_n is high active. Otherwise, low active.

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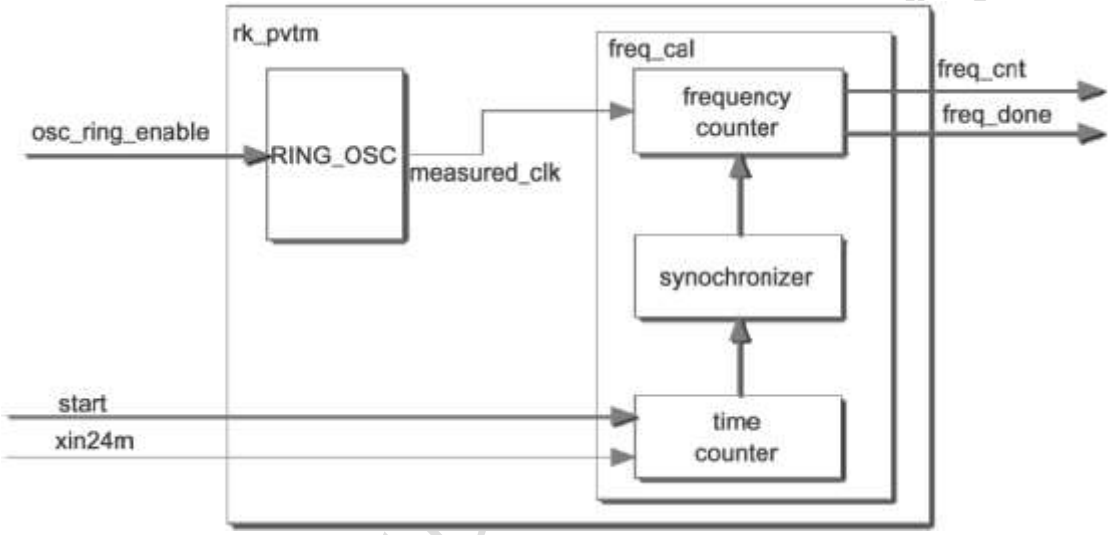
Chapter 54 Process-Voltage-Temperature Monitor (PVTM)

54.1 Overview

The Process-Voltage-Temperature Monitor (PVTM) is used to monitor the chip performance variance caused by chip process, voltage and temperature.
PVTM supports the following features:

- a clock oscillation ring is integrated and used to generate a clock like signal, the frequency of this clock is determined by the cell delay value of clock oscillation ring circuit
- a frequency counter is used to measure the frequency of the clock oscillation ring.
- Three PVTM blocks are supported
  - core_pvtm, used in pd_core power domain
  - gpu_pvtm, used in pd_gpu power domain
  - pmu_pvtm, used in pd_pmu power domain

54.2 Block Diagram



The PVTM include two main blocks

- RING_OSC, it is composed with inverters with odd number, which is used to generate a clock
- Freq_cal, it is used to measure the frequency of clock which generated from the RING_SOC block

54.3 Function description

Frequency Calculation

A frequency fixed clock(24MH) is used to calculate the clock cycles of RING_OSC generated clock. Suppose the time period is 1s, then the clock period of RING_OSC clock is T= 1/2*clock_counter(s)

pmu_pvtm usage

pmu_pvtm is also used as a clock in low power mode, which can replace the function of 32KHz clock source by configure PMUGRF_SOC_CON0[6].

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